HIDDEN FACTS IN

NEW SSCE MATHEMATICS

BY:

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OTUMUDIA PUBLISHERS LIMITED

ISBN-978-978-958-649-3

We are grateful as we acknowledge WAEC, NECO and NABTEB for being the source of questions from past General Mathematics School Certificate Examination papers.

We are grateful to all those who contributed to the success of this book.

PUBLISHED BY:

OTUMUDIA PUBLISHERS LIMITED

NO 3 ADONOVWE STREET, OFF CENTRAL MOTOR PARK.
P. O BOX 540, UGHELLI NORTH,
DELTA STATE.
GSM NO. 08038633394, 08038633392
08022264074, 08022264072

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PREFACE

To boost the numerous effort by Mathematicians in addressing the perennial problems of poor understanding and its attendant low grades in the subject, this book covers the NEW SSCE syllabus directly including oversight areas.

A pool of past questions over the years on each topics were painstakingly collected ; part of which were used as an additional examples to illustrate elaborately the required concepts on each topics while similar ones were left as exercises.

Answers and mathematical tables are provided at the end of the text.

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CHAPTER ONE

Fractions and Approximation

Fractions

Fractions are of the form $\frac{a}{b}$

Where **a** is the numerator and **b** is the denominator ($b \neq 0$)

Types of fraction are:

Proper	*improper*	*mixed numbers*
$\frac{2}{3}, \frac{5}{29}$	$\frac{3}{2}, \frac{29}{5}$	$1\frac{2}{3}, 3\frac{5}{29}$

The basic arithmetic rules of addition, subtraction, division and multiplication apply to them. If we are to perform more than one operation of the aforementioned, then BODMAS pattern must be strictly adhered to.

2014/1

Simplify $10\frac{2}{5} - 6\frac{2}{3} + 3$

A $6\frac{4}{15}$ B $6\frac{11}{15}$ C $7\frac{4}{15}$ D $7\frac{11}{15}$

Solution

We change mixed numbers to improper fractions

$$= \frac{52}{5} - \frac{20}{3} + 3$$

LCM of the denominators 5 and 3 is 15

$$= \frac{156 - 100 + 45}{15}$$

$$= \frac{101}{15} = 6\frac{11}{15} \text{ (B)}$$

2011/36

Simplify $1\frac{3}{4} - \left(2\frac{1}{3} + 4\right)$

A $3\frac{5}{12}$ B $2\frac{7}{12}$ C $-4\frac{7}{12}$ D $3\frac{1}{2}$ E $-5\frac{5}{12}$

Solution

First, we change mixed numbers to improper fractions.

$$= \frac{7}{4} - \left(\frac{7}{3} + 4\right)$$

Applying BODMAS, bracket first

$$= \frac{7}{4} - \frac{19}{3}$$

Applying LCM of 4 and 3 i.e 12

$$= \frac{21 - 76}{12} = -\frac{55}{12} = -4\frac{7}{12} \text{ (C)}$$

2010/8

Simplify $1\frac{1}{2} + 2\frac{1}{3} \times \frac{3}{4} - \frac{1}{2}$

A $-2\frac{1}{3}$ B $-2\frac{1}{4}$ C $2\frac{1}{8}$ D $2\frac{3}{4}$

Solution

We change mixed numbers to improper fractions.

$$= \frac{3}{2} + \frac{7}{3} \times \frac{3}{4} - \frac{1}{2}$$

Applying BODMAS, multiplication comes first

$$= \frac{3}{2} + \frac{7}{4} - \frac{1}{2}$$

Applying LCM to the denominators

$$= \frac{6 + 7 - 2}{4} = \frac{11}{4} \text{ i.e } 2\frac{3}{4} \text{ (D)}$$

1999/3 (Nov)

Simplify $6\frac{3}{4} \div \left(21\frac{1}{3} \times 2\frac{1}{4} - 21\right)$

A $6\frac{3}{4}$ B 4 C $\frac{1}{4}$ D $\frac{4}{27}$

Solution

First, we change mixed numbers to improper fractions.

$$= \frac{27}{4} \div \left(\frac{64}{3} \times \frac{9}{4} - 21\right)$$

Applying BODMAS, bracket first. Inside the bracket, we apply BODMAS also and there multiplication 1st.

$$= \frac{27}{4} \div (48 - 21)$$

$$= \frac{27}{4} \div 27$$

$$= \frac{27}{4} \times \frac{1}{27} = \frac{1}{4} \text{ (C)}$$

2001/1a (Nov)

Simplify $1\frac{2}{3}$ *of* $2\frac{1}{4} \div 3\frac{1}{2} - 1\frac{1}{4}$

Solution

First, we change mixed numbers to improper fractions.

$$= \frac{5}{3} \text{ of } \frac{9}{4} \div \frac{7}{2} - \frac{5}{4}$$

Applying BODMAS, **of** 1st

$$= \frac{5}{3} \times \frac{9}{4} \div \frac{7}{2} - \frac{5}{4}$$

Next **division**, we change $\div \frac{7}{2}$ *to* $\times \frac{2}{7}$

$$= \frac{15}{4} \times \frac{2}{7} - \frac{5}{4}$$

$$= \frac{15}{14} - \frac{5}{4}$$

Applying LCM of 14 and 4

$$= \frac{2}{3} \times 24 = \frac{-5}{28}$$

2004/2 Neco

Simplify $1\frac{3}{7} + 3\frac{3}{4} \times 2\frac{2}{5} \div 5\frac{1}{4}$

A $1\frac{3}{7}$ B $2\frac{18}{49}$ C $3\frac{1}{7}$ D $3\frac{3}{7}$ E $5\frac{1}{4}$

Solution

First, we change mixed numbers to improper fractions.

$$= \frac{10}{7} + \frac{15}{4} \times \frac{12}{5} \div \frac{21}{4}$$

Applying BODMAS, division first, we change $\div \frac{21}{4}$ *to* $\times \frac{4}{21}$

$$= \frac{10}{7} + \frac{15}{4} \times \frac{12}{5} \times \frac{4}{21}$$

Effecting multiplications first

$$= \frac{10}{7} + \frac{12}{7}$$

$$= \frac{10 + 12}{7} = \frac{22}{7} = 3\frac{1}{7} \text{ (C)}$$

2005/1

Simplify $15\frac{1}{2} - 6\frac{3}{4} \times 2\frac{2}{3} + 5$

A $-16\frac{1}{2}$ B $-2\frac{1}{2}$ C $2\frac{1}{2}$ D $16\frac{1}{2}$

Solution

First, we change mixed numbers to improper fractions.

$$= \frac{31}{2} - \frac{27}{4} \times \frac{8}{3} + 5$$

Applying BODMAS, **multiplication** first.

$$= \frac{31}{2} - 18 + 5$$

$$= \frac{31}{2} - 13 = \frac{5}{2} = 2\tfrac{1}{2} \text{ (C)}$$

2005/23 Neco

Simplify $2\frac{1}{2} + 3\frac{3}{4} \div 4\frac{1}{2} - 1\frac{1}{3}$

A $\frac{1}{18}$ B 1 C $1\frac{5}{12}$ D 2 E $2\frac{1}{6}$

Solution

First, we change mixed numbers to improper fractions.

$$= \frac{5}{2} + \frac{15}{4} \div \frac{9}{2} - \frac{4}{3}$$

Applying BODMAS, **division** first, we change $\div \frac{9}{2}$ to $\times \frac{2}{9}$

$$= \frac{5}{2} + \frac{15}{4} \times \frac{2}{9} - \frac{4}{3}$$

Effecting the multiplication

$$= \frac{5}{2} + \frac{5}{6} - \frac{4}{3}$$

Since we are left with addition and subtraction, we apply LCM to the denominators

$$= \frac{15 + 5 - 8}{6} = \frac{12}{6} = 2 \text{ (D)}$$

2005/ 1 (Nov)

Evaluate: $\frac{2}{7} \div \left(1\frac{7}{8} \div 1\frac{1}{14}\right) \times \frac{4}{7}$

A $\frac{4}{7}$ B $\frac{2}{7}$ C $\frac{8}{49}$ D $\frac{32}{343}$

Solution

First, we change mixed numbers to improper fractions.

$$= \frac{2}{7} \div \left(\frac{15}{8} \div \frac{15}{14}\right) \times \frac{4}{7}$$

Applying BODMAS, **bracket** first,

$$= \frac{2}{7} \div \left(\frac{15}{8} \times \frac{14}{15}\right) \times \frac{4}{7}$$

$$= \frac{2}{7} \div \frac{7}{4} \times \frac{4}{7}$$

Applying BODMAS, **division** first, we change $\div \frac{7}{4}$ to $\times \frac{4}{7}$

$$= \frac{2}{7} \times \frac{4}{7} \times \frac{4}{7}$$

$$= \frac{32}{343} \text{ (D)}$$

2006/1 Neco (Dec)

Evaluate: $2\frac{1}{2} \times \frac{3}{4}$ *of* $\frac{1}{3} \div \frac{4}{5}$

A $1\frac{7}{25}$ B $\frac{25}{32}$ C $\frac{1}{2}$ D $\frac{7}{25}$ E $\frac{7}{32}$

Solution

Applying BODMAS, **of** first, while we change the mixed number to improper fraction.

$$= \frac{5}{2} \times \frac{1}{4} \div \frac{4}{5}$$

Applying BODMAS, **division** 1st, we change $\div \frac{4}{5}$ to $\times \frac{5}{4}$

$$= \frac{5}{2} \times \frac{1}{4} \times \frac{5}{4}$$

$$= \frac{25}{32} \text{ (B)}$$

2007/9 Neco

Simplify $2\frac{1}{2} + 1\frac{1}{3}$ *of* $\left(1\frac{5}{6} - \frac{1}{3}\right)$

A $5\frac{3}{4}$ B $4\frac{1}{2}$ C $3\frac{2}{3}$ D $3\frac{1}{2}$ E $2\frac{3}{4}$

Solution

First, we change mixed numbers to improper fractions.

$$= \frac{5}{2} + \frac{4}{3} \text{ of } \left(\frac{11}{6} - \frac{1}{3}\right)$$

Applying BODMAS, **bracket** first

$$= \frac{5}{2} + \frac{4}{3} \text{ of } \left(\frac{11 - 2}{6}\right)$$

$$= \frac{5}{2} + \frac{4}{3} \text{ of } \frac{9}{6}$$

Applying BODMAS, **of** first

$$= \frac{5}{2} + \frac{4}{3} \times \frac{9}{6}$$ Effecting the multiplications

$$= \frac{5}{2} + 2 = \frac{9}{2} = 4\tfrac{1}{2} \text{ (B)}$$

2009/1a (Nov)

Simplify, without using tables or calculator:

$$3\frac{1}{3} \div 1\frac{1}{4} - \frac{4}{9}$$

Solution

We change mixed numbers to improper fractions.

$$= \frac{10}{3} \div \frac{5}{4} - \frac{4}{9}$$

Applying BODMAS, **division** first, we change $\div \frac{5}{4}$ to $\times \frac{4}{5}$

$$= \frac{10}{3} \times \frac{4}{5} - \frac{4}{9}$$

Working with multiplication first

$$= \frac{8}{3} - \frac{4}{9}$$

$$= \frac{24 - 4}{9} = \frac{20}{9} \text{ i.e } 2\frac{2}{9}$$

2009/1 Neco (Dec)

Simplify $4\frac{1}{3} \div 3\frac{1}{4} \times \left(\frac{3}{4} + \frac{1}{3}\right)$

A $\frac{12}{13}$ B $1\frac{1}{12}$ C $1\frac{1}{9}$ D $1\frac{5}{12}$ E $1\frac{4}{9}$

Solution

Change mixed numbers to improper fractions.

$$= \frac{13}{3} \div \frac{13}{4} \times \left(\frac{3}{4} + \frac{1}{3}\right)$$

Applying BODMAS, **bracket** first

$$= \frac{13}{3} \div \frac{13}{4} \times \left(\frac{9+4}{12}\right)$$

$$= \frac{13}{3} \div \frac{13}{4} \times \frac{13}{12}$$

Applying BODMAS, **division** first, we change $\div \frac{13}{4}$ *to* $\times \frac{4}{13}$

$$= \frac{13}{3} \times \frac{4}{13} \times \frac{13}{12} = \frac{13}{9} = 1\frac{4}{9} \quad (\text{E})$$

2012/5b Neco

Simplify $\dfrac{3\frac{2}{3} \times 1\frac{1}{8}}{3\frac{2}{3} - 2\frac{1}{4}} \div \dfrac{5\frac{1}{2}}{3\frac{7}{9}}$

Solution

The whole problem is separated by ÷ . Change mixed numbers to improper fractions

$$= \frac{\frac{11}{3} \times \frac{9}{8}}{\frac{11}{3} - \frac{9}{4}} \div \frac{\frac{11}{2}}{\frac{34}{9}}$$

Next, we solve numerator and denominator differently

$$= \frac{\frac{33}{8}}{\frac{44-27}{12}} \div \frac{11}{2} \times \frac{9}{34} \quad \textit{changed from division to } \times$$

$$= \frac{33}{8} \times \frac{12}{17} \div \frac{11 \times 9}{2 \times 34}$$

Changing from division to multiplication

$$= \frac{33}{8} \times \frac{12}{17} \times \frac{2 \times 34}{11 \times 9} = 2$$

2012/ 1a

Simplify $\dfrac{1\frac{1}{4} + \frac{7}{9}}{1\frac{4}{9} - 2\frac{2}{3} \times \frac{9}{64}}$

Solution

The whole problem is separated by ÷ . Change mixed numbers to improper fractions

$$= \left(\frac{5}{4} + \frac{7}{9}\right) \div \left(\frac{13}{9} - \frac{8}{3} \times \frac{9}{64}\right)$$

We introduced bracket to separate numerator from denominator. Applying BODMAS to the last bracket implies multiplication first

$$= \left(\frac{45+28}{36}\right) \div \left(\frac{13}{9} - \frac{3}{8}\right)$$

$$= \frac{73}{36} \div \left(\frac{104-27}{72}\right)$$

$$= \frac{73}{36} \div \left(\frac{77}{72}\right)$$

Changing from division to multiplication

$$= \frac{73}{36} \times \left(\frac{72}{77}\right) = \frac{146}{77} \text{ i.e } 1\frac{69}{77}$$

2011/ 1a

Simplify $\dfrac{\frac{1}{2} \textit{ of } \frac{1}{4} \div \frac{1}{3}}{\frac{1}{6} - \frac{3}{4} + \frac{1}{2}}$

Solution

We solve numerator and denominator differently.

Applying BODMAS "**of**" first

$$= \frac{\frac{1}{8} \div \frac{1}{3}}{\frac{1}{6} - \frac{3}{4} + \frac{1}{2}}$$

Applying LCM to the denominator and invert $^1/_3$ to $^3/_1$ to change from ÷ to × in numerator

$$= \frac{\frac{1}{8} \times \frac{3}{1}}{\frac{2-9+6}{12}} = \frac{\frac{3}{8}}{-\frac{1}{12}} \text{ i.e } \frac{3}{8} \div \left(-\frac{1}{12}\right)$$

Changing from division to multiplication

$$= \frac{3}{8} \times \left(-\frac{12}{1}\right)$$

$$= -\frac{9}{2} = -4\frac{1}{2}$$

2009/12 NABTEB (Dec)

Arrange $^5/_6$, $^2/_3$ and $^3/_8$ in descending order of magnitude

A $^2/_3$, $^3/_8$, and $^5/_6$ B $^3/_8$, $^2/_3$, and $^5/_6$ C $^5/_6$, $^2/_3$, and $^3/_8$

D $^5/_6$, $^3/_8$, and $^2/_3$

Solution

To know the value of their magnitude, we multiply through by the LCM of the dominators 6, 3, and 8

3	3	6	8
2	1	2	8
2	1	1	4
2	1	1	2
	1	1	1

LCM is 3 × 2 × 2 × 2 = 24

$$= \frac{5}{6} \times 24, \quad \frac{2}{3} \times 24, \quad \frac{3}{8} \times 24$$

= 20, 16 and 9

They are in descending order already

$^5/_6$, $^2/_3$, and $^3/_8$ (C)

2008/2 Neco

Arrange the following numbers in ascending order:

$$\frac{1}{8}, \frac{1}{7}, \frac{1}{4}, \frac{11}{56}, \frac{17}{56}$$

A $\frac{1}{8}, \frac{1}{7}, \frac{1}{4}, \frac{11}{56}, \frac{17}{56}$ B $\frac{1}{8}, \frac{1}{7}, \frac{11}{56}, \frac{1}{4}, \frac{17}{56}$

C $\frac{1}{8}, \frac{11}{56}, \frac{1}{7}, \frac{1}{4}, \frac{17}{56}$ D $\frac{1}{8}, \frac{1}{7}, \frac{11}{56}, \frac{17}{56}, \frac{1}{4}$

E $\frac{1}{8}, \frac{1}{7}, \frac{1}{4}, \frac{17}{56}, \frac{11}{56}$

Solution

We multiply through by the LCM of the denominators 8, 7, 4, 56, i.e 56

$= \frac{1}{8} \times 56, \ \frac{1}{7} \times 56, \ \frac{1}{4} \times 56, \ \frac{11}{56} \times 56, \ \frac{17}{56} \times 56$

$= 7, \ 8, \ 14, \ 11, \ 17$

Arrange ascending order: 7, 8, 11, 14, 17

$\frac{1}{8}, \frac{1}{7}, \frac{11}{56}, \frac{1}{4}, \frac{17}{56}$ (B)

2009/4 (Nov) Exercise 1.1

Simplify $\frac{11}{8} \times \frac{11}{7} - \frac{6}{5} \div \frac{4}{5}$

A $\frac{269}{280}$ B $1\frac{11}{269}$ C $\frac{37}{56}$ D $1\frac{19}{37}$

2010/6b Neco Exercise 1.2

Simplify $\dfrac{4\frac{2}{9} - 1\frac{13}{15}}{2\frac{1}{5} + \frac{4}{7} \text{ of } 2\frac{1}{3}}$

2011/ 4a Neco adjusted Exercise 1.3

Simplify $2\frac{4}{5} \text{ of } \frac{2}{3} \div 4\frac{1}{5}$

2009/1 Exercise 1.4

Simplify: $\left(3\frac{1}{2} + 4\frac{1}{3}\right) \div \left(\frac{5}{6} - \frac{2}{3}\right)$

A $1\frac{1}{4}$ B $8\frac{1}{2}$ C 35 D 47

2007/ 6a Exercise 1.5

Simplify $\dfrac{3\frac{1}{12} + \frac{7}{8}}{2\frac{1}{4} - \frac{1}{6}}$

2006/1a NATEB (Dec) Exercise 1.6

Simplify $\dfrac{1\frac{1}{4}}{2 + \frac{1}{4} \text{ of } 28}$

2006/1a Exercise 1.7

Simplify $\dfrac{4\frac{2}{9} - 1\frac{13}{15}}{2\frac{1}{5} + \frac{4}{7} \times 2\frac{1}{3}}$

2005/40 (Nov) Exercise 1.8

Simplify $\dfrac{2\frac{1}{2} + \frac{1}{3} - 1\frac{3}{4}}{\frac{1}{2} + 1\frac{1}{3} - 1\frac{1}{4}}$

A $\frac{7}{13}$ B $\frac{7}{12}$ C $\frac{12}{13}$ D $1\frac{6}{7}$

2015/ 34 (Nov) Exercise 1.9

Simplify $\frac{1}{8}\left(3\frac{1}{4} - 4\frac{2}{3} + 2\frac{1}{6}\right) \div 3\frac{3}{4}$

A $\frac{1}{40}$ B $\frac{5}{12}$ C $\frac{4}{15}$ D $2\frac{13}{40}$

2015/1a Exercise 1.10

Without using mathematical tables or calculators simplify : $3\frac{4}{9} \div \left(5\frac{1}{3} - 2\frac{3}{4}\right) + 5\frac{9}{10}$

2014/5 Neco (Nov) Exercise 1.11

Simplify $\left(\frac{4}{3} + \frac{3}{2}\right) \times 3\frac{1}{2} \div 3\frac{1}{4}$

A $\frac{2}{39}$ B $2\frac{3}{39}$ C $2\frac{19}{39}$ D $3\frac{2}{39}$ E $4\frac{2}{39}$

Word problems

2014/10

Three quarter of a number added to two and a half of that number gives 13. Find the number.

A 4 B 5 C 6 D 7

Solution

Let the number be x, we are to find x in: $\frac{3}{4}x + 2\frac{1}{2}x = 13$

i.e $\frac{3}{4}x + \frac{5}{2}x = 13$

Multiply through by LCM 4 to clear fractions

$3x + 10x = 52$

$13x = 52$

$x = \frac{52}{13}$ i.e 4

2011/12b

Ade received $\frac{3}{5}$ of a sum of money, Nelly $\frac{1}{3}$ of the remainder while Austin took the rest. If Austin's share is greater than Nelly's share by ₦ 3,000, how much did Ade received?

Solution

Let the money be x

Ade's share: $\frac{3}{5}x$

Nelly's share: $\frac{1}{3}$ of $\left(1x - \frac{3}{5}x\right)$ i.e $\frac{1}{3}$ of $\left(\frac{2}{5}x\right) = \frac{2}{15}x$

Austin's share: $1x - \frac{3}{5}x - \left[\frac{1}{3} \text{ of } (1x - \frac{3}{5}x)\right]$

i.e $\frac{2}{5}x - \frac{2}{15}x = \frac{4}{15}x$

The last statement is

Austin's share – Nelly's share = 3000

$$\frac{4}{15}x - \frac{2}{15}x = 3000$$

$$\left(\frac{4-2}{15}\right)x = 3000$$

$$\frac{2}{15}x = 3000$$

$$x = \frac{3000 \times 15}{2} \text{ i.e ₦ } 22{,}500$$

Thus Ade's share $= \frac{3}{5}x$

$= \frac{3}{5} \times 22{,}500$ i.e ₦ 13,500

2010/ 2Neco
What fraction of 5 weeks is 5 days?
A 7 B 1 C $^1/_6$ D $^1/_7$ E $^1/_{10}$
Solution
One week is 7days
5 week is $7 \times 5 = 35$ days
Thus 5 days fraction of 5 weeks = $\frac{5}{35} = {}^1/_7$ (D)

2008/ 12 Neco (Dec)
A civil servant spent $^2/_7$ of his monthly salary on clothes, $^7/_{20}$ of the remainder on feeding and save the rest. If he spent ₦ 7, 000.00 on feeding, find his monthly salary
A ₦28,000.00 B ₦32,000.00 C ₦40,000.00
D ₦ 48,000.00 E ₦49,000.00
Solution
Let the salary be x
Clothes: $\frac{2}{7}$ of x i.e $\frac{2}{7}x$
Feeding: $\frac{7}{20}$ of $\left(x - \frac{2}{7}x\right)$ i.e $\frac{7}{20} \times \frac{5}{7}x = \frac{1}{4}x$

This step not necessary in this question
Savings: $x - \frac{2}{7}x - \frac{1}{4}x$ i.e $\left(\frac{28-8-7}{28}\right)x = \frac{13}{28}x$

If he spent ₦ 7,000.00 on feeding, find his monthly…
$\frac{1}{4}x = 7000$, find x
$x = 7000 \times 4$
= ₦28, 000.00 (A)

2007/ 7b Neco
A quarter of farmland is too rough to use. If two-fifth of the remainder is kept for poultry farms and this leaves $22^1/_2$ acres for planting cash crops, what is the area of the farm?
Solution
Let the farmland be x
Rough part: $\frac{1}{4}x$
Poultry farm: $\frac{2}{5}$ of $\left(x - \frac{1}{4}x\right)$ i.e $\frac{2}{5}\left(\frac{3}{4}x\right) = \frac{3}{10}x$
Cash crops: $x - \frac{1}{4}x - \frac{3}{10}x$ i.e $\frac{20x-5x-6x}{20} = \frac{9}{20}x$
Next, cash crops part is $22^1/_2$ acres
Thus $\frac{9}{20}x = 22^1/_2$ find x
$$\frac{9}{20}x = \frac{45}{2}$$
$$x = \frac{45 \times 20}{2 \times 9}$$
= 50 acres

2006/9 Exercise 1.12
A baker used 40% of a 50kg bag of flour. If $^1/_8$ of the amount used was for cake, how many kilograms of flour was used for cake?
A $2^1/_2$ B $6^1/_4$ C $15^5/_8$ D $17^1/_2$

2004/6(c)
A man left ₦5,720 to be shared among his son and three daughters. Each daughter's was ¾ of the son's share. How much did the son receive?
Solution
Let the son's share be P
Each daughter share will be $\frac{3}{4}p$.
The three daughter's share will be $3\left(\frac{3}{4}p\right)$ i.e $\frac{9}{4}p$
Sum of shares = total amount
But $P + \frac{9}{4}p = 5720$
$$\frac{13}{4}p = 5720$$
$$P = \frac{5720 \times 4}{13}$$
= ₦ 1,760 son's share

2011/ 11 Neco
A poultry farmer discovered that $^1/_6$ of the eggs laid by the birds in the farm were bad. Find the percentages that were good.
A 83.33% B 16.66% C 0.83% D 0.66% E 0.16%
Solution
Bad eggs percentage = $^1/_6 \times 100$
= 16.66% ≈ 16.67%
Thus good eggs percentage = 100 – 16.67
= 83.33% A
Alternatively
Good eggs percentage = $1 - {}^1/_6\,(100)$
$= \frac{5}{6} \times 100$
= 83.33%
$1 - {}^1/_6$ is used as the whole (Total) egg laid in the farm is in fraction term 1.

2015/4a Exercise 1.13
By how much is the sum of $3\frac{2}{3}$ and $2\frac{1}{5}$ less than 7 ?

2014/4 Neco (Nov) Exercise 1.14
What fraction must be added to the sum of $3\frac{1}{2}$ and $\frac{4}{5}$ to give $\frac{1}{2}$?
A $^{19}/_5$ B $^7/_2$ C $^5/_4$ D $^4/_5$ E – $^{19}/_5$

2014/7a Neco (Nov) Exercise 1.15
Ade, Tayo and Uche jointly owned a business and shared the profit such that Ade received $^1/_3$, Tayo received $^7/_{12}$ and Uche received $^1/_{12}$. If Ade received ₦ 7,000.00 more than Uche, what was the total profit and how much did each received?

2015/39 Exercise 1.16
A farmer uses $^2/_5$ of his land to grow cassava, $^1/_3$ of the remainder for yam and the rest for maize. Find the part of land used for maize
A $^2/_{15}$ B $^2/_5$ C $^2/_3$ D $^4/_5$

2015/2 (Nov) Exercise 1.16b
A man spent $\frac{1}{4}$ of his monthly salary on rent, $\frac{2}{5}$ on food and $\frac{1}{6}$ on his children's education. What fraction of his salary is left?
A $\frac{11}{60}$ B $\frac{13}{60}$ C $\frac{7}{20}$ D $\frac{19}{30}$

Approximation
Decimal and Significant figures

These twin concepts are working tools for presenting most answers in mathematics. The author deliberately fixed them together to avoid the usual confusion associated with them. Once the next digit to the one being considered is any of 5, 6, 7, 8, 9; we round up i.e. adding 1 to our digit.

On the other hand, if the digit is any of 0, 1, 2, 3, 4 we round down, i.e adding nothing to our digit.

Examples.

Approximate the following

(a) 65009.269 to
(i)1d.p (ii) 1s.f

(b) 24.543 to
(i) 2 d.p (ii) 2 s.f.

(c) 0.001658 to
(i) 3d.p (ii) 3 s.f

Analysis and Solution

a. (i) $65009.269 \approx 65009.3$ to 1d.p

Here, we considered the first digit after the decimal point. Since the number after it is 6, we rounded up by adding 1 to 2. The last sign is the approximation sign.

a. (ii) $65009.269 \approx 70000.000$ to 1 s.f

b. (i) $24.543 \approx 24.54$ to 2 d.p

Here, the two digits after decimal point are five four **(Not fifty-four).** The last of the two is 4 and next to it is 3, thus, we round down.

b. (ii) $24.543 \approx 25.000$ to 2 s.f
we rounded up

c. (i) $0.001658 \approx 0.002$ to 3 d.p.

(ii) $0.001658 \approx 0.00166$ to 3 s.f

2010/ 13 Neco

Express 0.00629946 to 3 significant figures

A 0.000 B 0.006 C 0.006210
D 0.00629 E 0.00630

Solution

Significant figures, our starting point is first non-zero digit. Take 629 to approximate to 3s.f

$0.00629946 \approx 0.00630$ E

Since the number after 629 is 9 we round up by adding 1 to 9 making 629 to be 630

2010/3 NABTEB (Dec)

Round off the following 51065 to 3 significant figures

A 51000 B 51100 C 51200 D 51300

Solution

In significant figures, our concern is the first figure, not zero or decimal point. So we take 510 for 3s.f

$51065 \approx 51100$ (B)

We round up by adding 1 to 510 since after 510 we have 6

2010/7

Correct 0.002473 to 3 significant figures

A 0.002 B 0.0024 C 0.00247 D 0.0025

Solution

In significant figures, our concern is the first figure, not zero or decimal point. So we take 247 for 3s.f and approximate

$0.002473 \approx 0.00247$ (C)

We round down by adding nothing 247

Since after 247 in 0.002473 we have 3 which is below 5

2005/3 (Nov)

Which of these numbers is correct to three significant figures?

A 0.003 B 0.0034 C 0.012 D 3.04

Solution

0.003 is correct to 3 d.p and correct to 1 s.f but not 3s.f
0.0034 is correct to 4 d.p and correct to 2s.f
0.012 is correct to 3 d.p and correct to 2 s.f
3.04 is correct to 3 s.f D. (also it is correct to 2 d.p)

2014/34

Approximate 0.0033780 to 3 significant figures

A 338 B 0.338 C 0.00338 D 0.003

Solution

In significant figures, our concem is the first figure, not zero or decimal point. So we take 337 for 3s.f and approximated there

$0.0033780 \approx 0.00338$ to 3s.f C

We round up by adding 1

2012/1

Express 302.10495 correct to five significant figures

A 302.10 B 302.11 C 302.15 D 302.1049

Solution

In significant figures, our concern is the first non zero figure, so we take 302.10 for 5 s.f and approximate

$302.10495 \approx 302.10$ (A)

We rounded down here, reason after the last 0 in 302.10495 is 4 which is not up to 5 and above

2004/1 Neco

Express the product of 0.007 and 0.057 to two significant figures

A 0.00040 B 0.00039 C 0.0040 D 0.0039 E 0.040

Solution

$0.007 \times 0.057 = 0.000399$

≈ 0.00040 to 2 s.f (A)

2006/1

Evaluate $(0.13)^3$ correct to **three** significant figures.

A 0.00219 B 0.00220 C 0.00300 D 0.00390

Solution

$(0.13)^3 = 0.0\,0\,2\;\;1\;\;9\;\;7$

Three significant figures: 1st 2nd 3rd 0

Next to the last digit 9 is 7, so we round up by adding 1 to it.

$0.002197 \approx 0.00220$ to 3s.f (B)

2008/4 Neco (Dec)

Evaluate $\frac{6.42 + 2.13}{4.1 - 2.85}$, correct to two significant figures

A 6.9 B 6.8 C 5.9 D 5.8 E 5.4

Solution

Solving the numerator and denominator differently:

$$\frac{6.42 + 2.13}{4.1 - 2.85} = \frac{8.55}{1.25}$$

$$= 6.84$$

Two significant figures: 1st 2nd 0

Next to the last digit 8 is 4, so we round down by adding nothing to it.

Thus $6.84 \approx 6.8$ to 2 s.f (B)

2008/4 NABTEB (Dec)

Express the difference between $1\frac{7}{25}$ and $\frac{17}{300}$ to 3 significant figures?

A 1.22 B 1.23 C 0.12 D 0.13

Solution

$$1\frac{7}{25} - \frac{17}{300} = \frac{32}{25} - \frac{17}{300}$$

Changing to decimal

$$= 1.28 - 0.05667$$

$$= 1.22333$$

1st 2nd 3rd 0 0 0

Next to the last digit 2 is 3, so we round down by adding nothing to it.

Thus $1.22333 \approx 1.22$ to 3s.f A

2011/1 Neco

Express the sum of 0.0714 and 0.0163 in two significant figures.

A 0.080 B 0.086 C 0.087 D 0.088 E 0.090

Solution

$$\begin{array}{r} 0.0714 \\ +\underline{0.0163} \\ \underline{0.0877} \end{array}$$

In significant figures, our concern is the first figure, not zero or decimal point. So we take 87 for 2 s.f and approximate

$0.0877 \approx 0.088$ to 2s.f (D)

We round up by adding 1 to 7 in 87 since the number after 87 is 7

2008/5 NABTEB (Dec)

Express 42.467 to 2 decimal places

A 424.67 B 42.50 C 42.47 D 42.46

Solution

Decimal place, our starting point is the decimal point. Immediately after decimal point, we start counting. So we take **.46** for approximation to 2 d.p

$42.467 \approx 42.47$ (C)

Since the number after .46 is 7 we round up by adding 1 to 6

2010/1 Neco

Simplify 0.0589 + 7.382 – 0.7953 correct to 2 decimal places.

A 6.60 B 6.64 C 6.65 D 8.20 E 8.24

Solution

$$\begin{array}{r} 0.0589 \\ \underline{7.382*} \\ 7.4409 \\ -\underline{0.7953} \\ \underline{6.6456} \end{array}$$

In decimal places approximation, our concern is the decimal point. Immediately after decimal point, we start counting. So we take **.64** for approximation to 2 d.p

$6.6456 \approx 6.65$ to 2 d.p (C)

Since the number after **.64** is 5 we round up by adding 1 to 4 in **.64** to become **.65**

2013/2 Neco

Sum up the following numbers 0.032, 4.154, 6.0 and 0.3065, to two decimal places

A 10.00 B 10.40 C 10.49 D 10.50 E 11.00

Solution

$$\begin{array}{r} 0.032* \\ +4.154* \\ 6.0*** \\ \underline{0.3065} \\ \underline{10.4925} \end{array}$$

In decimal place, we start counting after decimal point, whether it is zero or not so we take 10.49 for 2d.p and approximate

$10.4925 \approx 10.49$ to 2d.p (C)

2007/6 Neco

Evaluate $\frac{1}{2} + \frac{3}{20} + \frac{5}{200} + \frac{7}{2000}$, correct to three decimal places.

A 0.685 B 0.680 C 0.679 D 0.678 E 0.670

Solution

Since we are working in decimal places we change fractions to decimals having our **3 d.p** in mind

$$\frac{1}{2} + \frac{3}{20} + \frac{5}{200} + \frac{7}{2000} = 0.5 + 0.15 + 0.025 + 0.0035$$

$$= 0.6785$$

Three decimal place : 1st 2nd 3rd 0

Next to the 3rd digit 8 is 5, so we round up by adding 1 to it.

$0.6785 \approx 0.679$ to 3d.p

2005/1 Neco

Express 329,761 to the nearest thousand

A 300,000 B 320,000 C 329,000 D 329,700 E 330,000

Solution

The place value of thousand here

329, 761

000

is 329. The next number to it is 7. Thus we round up by adding 1

$329{,}761 \approx 330{,}000$ to the nearest thousand (E)

2005/ 2

Find, correct to the nearest kobo, 85.23% of ₦ 50

A ₦ 43.00 B ₦ 42.60 C ₦42.61 D ₦ 42.62

Solution

First convert ₦ 50 to kobo

₦ 50 = 5000 kobo

$$85.23\% \text{ of ₦ } 50 = \frac{85.23}{100} \times 500$$

$$= 4261.5 \text{ kobo}$$

Divide by 100 to get ₦

= ₦ 42.615 kobo

≈ ₦ 42.62 (D)

2006/ 23
What is the place value of 9 in the number 3.0492?

A $\frac{9}{10000}$ B $\frac{9}{1000}$ C $\frac{9}{100}$ D $\frac{9}{10}$

Solution

The decimal place value system of numbers is

		1^{st}	2^{nd}	3^{rd}	4^{th} positions
3	**.**	**0**	**4**	**9**	**2**
		Tenth	hundredth	thousandth	

Thus the position of 9 is *thousandth* i.e $\frac{9}{1000}$ (B)

2011/ 21
What is the value of 3 in the number 42.7531?

A $\frac{3}{10000}$ B $\frac{3}{1000}$ C $\frac{3}{100}$ D $\frac{3}{10}$

Solution

The decimal place value system of numbers is

			1^{st}	2^{nd}	3^{rd}	4^{th} positions
4	**2**	**.**	**7**	**5**	**3**	**2**
			Tenth	hundredth	thousandth	

Thus the position of 3 is thousandth i.e $\frac{3}{1000}$ (B)

2015/29 Neco Exercise 1.17
The number 192039 was corrected to 192000 by a student. Which of the following can be the correct approximation of figures taken?
I. to the nearest hundred **II.** to the nearest thousand
III. to 3 significant figures **IV.** to 4 significant figures
A. I and II only B. II and III only
C. III and IV only D. I, II, III & IV

2014/1 (Nov) Exercise 1.18
Correct 0.005854 to 2 significant figures
A. 0.0058 B. 0.0059 C. 0.0060 D. 0.0100

2014/18 NABTEB (Nov) Exercise 1.19
Evaluate 0.8916 ÷ 5.36 and round off to two significant figure
A. 0.19 B. 0.18 C. 0.17 D. 0.16

2014/3 Neco Exercise 1.20
Evaluate $\frac{0.042 \times 0.5}{0.04 \times 0.14}$ and approximate your answer to one decimal places
A. 0.38 B. 3.75 C. 3.80 D. 37.50 E. 375.00

2003/1 Neco (Nov) Exercise 1.21
Express the number 0.0099687 correct to four decimal places
A. 0.0100 B. 0.0099 C. 0.0090 D. 0.0010 E. 0.0009

2003/7 Neco (Nov) Exercise 1.22
The difference between 0.005348 correct to 3 significant figures and 0.005384 correct to 4 decimal places is
A. 0.0001 B. 0.0004 C. 0.0005 D. 0.00004 E. 0.00005

2003/6 Neco (Nov) Exercise 1.23
Correct 5.0962894 to three significant figures
A. 5.96 B. 5.10 C. 5.09 D. 5.02 E. 5.00

Percentage error
This is as a result of little degree of error allowed in measurement

$$PE = \frac{\text{wrong value} - \text{actual value}}{\text{actual value}} \times 100$$

$$PE = \frac{\text{Error}}{\text{Exact}} \times 100$$

2008/11 Neco (Dec)
A rope which is 4.85m long was measured by a boy as 4.95m. What is his percentage error?
A $2^1/_{50}\%$ B $2^3/_{50}\%$ C $2^6/_{97}\%$
D $20^3/_{50}\%$ E $20^6/_{97}\%$

Solution

$$P.E = \frac{error}{exact} \times 100$$
$$= \frac{4.95 - 4.85}{4.85} \times 100$$
$$= \frac{0.1}{4.85} \times 100$$

Working in fraction, since our options are in it

$$= \frac{10}{4.85}$$

To clear decimal, we move decimal point twice at the denominator and do same at the numerator.

$$= \frac{1000}{485} \text{ i.e } \frac{200}{97} = 2^6/_{97}\% \quad (\text{C})$$

2011/20
The length of a piece of stick is 1.75m. A girl measured it as 1.80m. Find the percentage error

A $\frac{284}{7}\%$ B $\frac{29}{7}\%$ C 5% D $\frac{20}{7}\%$

Solution

$$P.E = \frac{error}{exact} \times 100$$
$$= \frac{1.80 - 1.75}{1.75} \times 100$$
$$= \frac{0.05}{1.75} \times 100$$

We are working in fractions

$$= \frac{5}{1.75} = \frac{500}{175} \text{ i.e } \frac{20}{7}\% \quad (\text{D})$$

To remove decimal, we move decimal point twice at the denominator and do same at the numerator.

2004/10 Neco
A sales boy gave a change of ₦ 75.00 to a buyer instead of ₦80.00, calculate his percentage error, correct to one decimal place.
A 6.0% B 6.2% C 6.3% D 6.6% E 6.7%

Solution

$$P.E = \frac{error}{exact} \times 100$$
$$= \frac{80 - 75}{80} \times 100 \quad = \frac{5}{80} \times 100$$
$$= 6.25\% \approx 6.3\% \text{ to 1 d.p}$$

2007/ 3 Neco

A boy when rounding up a number wrote 98 instead of 980 (to 2 significant figures). What is the percentage error?

A 0% B 10% C 11.1% D 90% E 96%

Solution

$$\text{P.E} = \frac{error}{exact} \times 100$$

$$= \frac{980 - 98}{980} \times 100$$

$$= \frac{882}{980} \times 100 = 90\% \text{ (D)}$$

2012/4 Neco

A man estimated his transport fare for a journey as ₦ 210.00 instead of ₦ 220.00. Find the percentage error in his estimate, correct to 3 significant figures.

A 4.54% B 4.55% C 45.4% D 45.5% E 45.6%

Solution

$$\text{P.E} = \frac{error}{exact} \times 100$$

$$= \frac{220 - 210}{220} \times 100$$

$$= \frac{10}{220} \times 100$$

$= 4.54545\% \approx 4.55\%$ to 3 s.f

2005/21 Neco

A woman underestimated the cost of a pair of shoes and her percentage error was $18^3/_4\%$. If the actual price of the shoe was ₦ 64.00, what was her underestimate?

A ₦12.00 B ₦ 17.92 C ₦ 52.00 D ₦52.48 E ₦ 76.00

Solution

Let the underestimation be x

Underestimation here implies wrong price is lower than the actual price.

$$\text{P.E} = \frac{error}{exact} \times 100$$

$$18^3/_4\% = \frac{64 - x}{64}$$

$$\frac{75}{4} \times \frac{1}{100} = \frac{64 - x}{64}$$

$75 \times 64 = 400 (64 - x)$

$4800 = 25600 - 400x$

$400x = 25600 - 4800$

$x = {}^{20800}/_{400}$ i.e ₦ 52.00 (C)

2010/4 Neco

The distance between two points is measured to be 3.62km. If this is more than the actual distance and the percentage error is calculated to be 5. What is the actual distance?

A 3.18km B 3.45km C 8.62km D 3.80km E 3.81km

Solution

$$\text{P.E} = \frac{error}{exact} \times 100$$

$$\text{P. E} = \frac{wrongvalue - actual}{actual\ value} \times 100$$

Let the actual value be x; PE is 5%

$$\frac{5}{100} = \frac{3.62 - x}{x}$$

$5x = 100(3.62 - x)$

$5x = 362 - 100x$

$5x + 100x = 362$

$105x = 362$

$x = \frac{362}{105}$ i.e 3.4476km ≈ 3.45km (B)

2002/38 Neco (Nov) Exercise 1.24

During a chemistry practical, a student recorded his reading as 18.13cm^3 instead of 18.31cm^3. Calculate his percentage error

A 9.8% B 1.8% C 0.98% D 0.18% E 0.098%

2014/41 NABTEB (Nov) Exercise 1.25

A string is 4.8m. A boy measured it to be 4.95m. Find the percentage error

A $^5/_{16}\%$ B $1^5/_{16}\%$ C $3^1/_{33}\%$ D $3^1/_8\%$

2013/7 Exercise 1.26

A sales boy gave a change of ₦ 68 instead of ₦ 72 . Calculate his percentage error

A 4% B $5^5/_9\%$ C $5^{15}/_{17}\%$ D 7%

2004/44 Neco (Nov) Exercise 1.27

A man made a table with a rectangle top of dimension 36cm by 44cm instead of 37cm by 41cm. What is the percentage error in the perimeter of the table correct to 1 decimal place?

A 11.1% B 10.8% C 4.4% D 2.6% E 2.5%

CHAPTER TWO

Commercial mathematics

RATIO

This is the act of comparing like quantities such as weights, lengths, money etc.

The ratio m : n is represented in fraction as $\frac{m}{n}$

In ratios, m : n $\neq$ n : m

Similarly, the ratios m : n : r is expressed in fraction as

$\frac{m}{n} : \frac{n}{r}$ and $\frac{m}{r}$

2009/ 11 Neco

If it costs three shop keepers A, B and C ₦ 918,000.00 to purchase a common generator for their three shops and the amount was shared in the ration 1: 2: 3 respectively. How much would shopkeeper B pay?

A ₦ 316,000.00 B ₦ 312,000.00 C ₦ 309,000.00
D ₦ 306,000.00 E ₦ 303,000.00

Solution

Shop keeper B ratio is 2

Sum of ratios 1 + 2 + 3 = 6

Amount shop keeper B pays = $\frac{2}{6} \times 918000$

= ₦ 306, 000.00 (D)

2006/ 36

A seller allows 20% discount for cash payment on the marked price of his goods. What is the ratio of the cash payment to the marked price?

A 1 : 4 B 1 : 5 C 3 : 4 D 4 : 5

Solution

Cash payment	:	marked price
Discount (100 – 20) %	:	100%
80	:	100
8	:	10
4	:	5 (D)

2012/3

In 1995, the enrolments of two schools X and Y were 1,050 and 1,190 respectively. Find the ratio of the enrolment of x to y.

A 50 : 11 B 15 : 17 C 13 : 55 D 12 :11

Solution

School X : School Y

1,050 : 1, 190

Next we reduce the values to their lowest term possible

105 : 119 i.e division by 10

15 : 17 i.e division by 7

2014/3a Neco

What number must be added to each term of the ratio 13: 16 so that it becomes the ratio 4: 5?

Solution

Let the number be x

Note the question says "what number" and not what *numbers* as this will make us use x and y

13 + x : 16 + x = 4 : 5

and **not** 13 + x : 16 + y = 4 : 5

In division form

$$\frac{13 + x}{16 + x} = \frac{4}{5}$$

$$5(13 + x) = 4(16 + x)$$
$$65 + 5x = 64 + 4x$$
$$5x - 4x = 64 - 65$$
$$x = -1$$

Our new ratio becomes

$$\frac{13 - 1}{16 - 1} = \frac{12}{15} \text{ i.e } {}^4/_5$$

2006/42 Neco

What number must be added to each term of the ratio 11 : 15, so that it becomes the ratio 4 : 8?

A – 7 B 4 C 7 D 8 E 17

Solution

Let the number be x

11 + x : 15 + x = 4 : 8

In division form

$$\frac{11 + x}{15 + x} = \frac{4}{8}$$

$$8(11 + x) = 4(15 + x)$$
$$88 + 8x = 60 + 4x$$
$$8x - 4x = 60 - 88$$
$$x = -\frac{28}{4} = -7 \text{ (A)}$$

2007/ 2 Neco

A, Q, R, and S shared a sum of money in the ratio 3 : 4 : 6 : 7. If S received ₦ 600.00 more than Q, Calculate the amount of money that was shared

A ₦ 3,000.00 B ₦ 4,000.00 C ₦ 6,000.00
D ₦ 8,000.00 E ₦ 12,000.00

Solution

Sum of ratios: 3 + 4 + 6 + 7 = 20

Ratio of S is 7

Ratio of Q is 4

Let the amount of money shared be x

Then $\frac{7}{20}$ of x – 600 = $\frac{4}{20}$ of x

$$\text{i.e } \frac{7x}{20} - 600 = \frac{4x}{20}$$

$$\frac{7x}{20} - \frac{4x}{20} = 600$$

$$\frac{3x}{20} = 600$$

$$x = \frac{600 \times 20}{3}$$

= ₦ 4,000.00 (B)

2007/12b NABTEB

The monthly profit of a transport business was shared between two partners, a husband and wife in the ratio 7 : 5. If the wife received ₦15,000.00 less than the husband, find out how much the husband received.

Solution

Sum of ratios: 7 + 5 = 12

Ratio of wife is 5

Let the amount of money shared be x

Then $\frac{5}{12}$ of x + 15000 = $\frac{7}{12}$ of x

$$\text{i.e } 15000 = \frac{7}{12}x - \frac{5x}{12}$$

$$15000 = \frac{2x}{12}$$

$$\frac{15000 \times 12}{2} = x$$

Amount shared is ₦ 90,000

Husband share is $= \frac{7}{12} \times 90000$

$= $ ₦ 52,500

2006/10 Neco (Dec)

A sum of money is to be shared among Victor, Faith and Debby in ratio $2 : 1^1/_2 : 2^1/_2$ respectively. If Faith receives ₦ 10,000.00, what is the sum of the money?

A ₦ 13,333.00 B ₦ 16,670.00 C ₦ 30,000.00

D ₦ 40,000.00 E ₦ 66,000.00

Solution

Let the sum of money shared be x

the given ratios has two fractions, making them whole will help.

$2 : 1^1/_2 : 2^1/_2$ i.e $2 : {}^3/_2 : {}^5/_2$

it becomes 4 : 3 : 5 (We multiplied through by LCM 2)

Sum of ratios: 4 + 3 + 5 = 12

Faith ratio is 3

Thus $\frac{3}{12}$ of x = 10,000

$3x = 10,000 \times 12$

$x = \frac{10,000 \times 12}{3}$

= ₦ 40,000.00 (D)

2005/ 6 Neco

The ratio of Bimbo's age to her mother is 5 : 7. What is Bimbo's age if her mother is 63 years old?

A 8.5 years B 12 years C 35 years

D 45 year E 52 years

Solution

Bimbo	:	mother
5	:	7
x	:	63

5 : 7 = x : 63

$$\frac{5}{7} = \frac{x}{63}$$

$$x = \frac{5 \times 63}{7}$$

= 45 years (D)

2007/15b NABTEB

Obi and Audu own a shop. The ratio of Obi's share to Audu's is 13: 7. Later Audu sells $\frac{1}{5}$ of his share to Obi for ₦ 6,300.00. Find the value of the shop

Solution

Let the value of the shop be x

Obi	:	Audu
13	:	7

Sum of ratios is 13 + 7 = 20

$$\frac{1}{5} of \left(\frac{7}{20} of\ x\right) = 6300$$

$$\frac{7x}{100} = 6300$$

$$x = \frac{6300 \times 100}{7}$$

= ₦ 90,000

2013/3a

The present ages of a father and his son are in the ratio 10 : 3. If the son is 15years old now, in how many years will the ratio of their age be 2 : 1 ?

Solution

<u>At present</u>

Father	:	son
10	:	3

We let the father age be x

x : 15

Changing ratio to division form

$$\frac{10}{3} = \frac{x}{15}$$

$$x = \frac{10 \times 15}{3}$$

= 50 years

At present we have

Father : son

50 years : 15 years

In the years y, when their ratio is 2: 1

We have

50 + y : 15 + y

2 : 1

$$\frac{50 + y}{15 + y} = \frac{2}{1}$$

50 + y = 2(15 + y)

50 + y = 30 + 2y

50 – 30 = 2y – y

20 = y

2007/8

₦140,000 is shared between Abu, Kayode and Uche. Abu has twice as much as Kayode and Kayode has twice as much as Uche . What is Kayode's share?

A ₦ 80,000 B ₦ 40,000 C ₦ 20,000 D ₦10,000

Solution

Abu	:	Kayode	:	Uche
A	:	K	:	U
A = 2K	:	K = 2U	:	U

Then changing all the ratios to one letter

2(2u)	:	2u	:	u
4u	:	2u	:	u

But 4u + 2u + u = 140,000

7u = 140000

$u = \frac{140000}{7}$

u = ₦ 20,000

Kayode's share is 2u i.e 2 × 20,000 = ₦ 40,000 (B)

2015/27

Ann scores $^1/_2$ as many marks as Brenda and Brenda scores $^1/_2$ as many marks as Catherine. If the sum of their marks is 112, how many marks did each student score?

Solution

A (Ann) B (Brenda) C (Catherine)

$\frac{1}{2}B = A$ $B = \frac{1}{2}c$ c

Then changing all the ratios to one letter

$\frac{1}{2}(\frac{1}{2}c)$ $\frac{1}{2}c$ c

the result is

$\frac{1}{4}c$ $\frac{1}{2}c$ c

But $\frac{1}{4}c + \frac{1}{2}c + c = 112$

To clear fractions, multiply through by 4 LCM of the denominators

$$c + 2c + 4c = 4 \times 112$$
$$7c = 4 \times 112$$
$$c = \frac{4 \times 112}{7} \text{ i.e } 64$$

Marks are A B C

16 : 32 : 64

2007/ 37

If the ratio x : y = 3 : 5 and y : z = 4 : 7,
find the ratio x : y : z

A 15 : 28: 84 B 12 : 20 : 35 C 3 : 5 : 4 D 5 : 4 : 7

Solution

The target here is to make y uniform value similar to what we do when solving simultaneous linear equations

x : y = 3 : 5 ------ (1)

and y : z = 4 : 7 -------(2)

RHS only (1) × 4 and (2) × 5 will make y uniform

x : y = 12 : 20

y : z = 20 : 35

Equating ratios x : y : z becomes 12 : 20 : 35

Alternatively

The target here is to make y uniform value

x : y = 3 : 5 implies $\frac{x}{y} = \frac{3}{5}$

$y = \frac{5x}{3}$ -----(1)

y : z = 4 : 7 implies $\frac{y}{z} = \frac{4}{7}$

$y = \frac{4z}{7}$ -----(2)

Equation terms in y from (1) and (2)

$$\frac{5x}{3} = \frac{4z}{7}$$
$$35x = 12z$$
$$\frac{x}{z} = \frac{12}{35}$$

Equating ratios x : z i.e 12 : 35

Put x = 12 into (1) or z = 35 into (2)

$$y = \frac{5(12)}{3} \text{ i.e } 20$$

Thus x : y: z becomes 12: 20:35 (B)

2009/42 (Nov)

If x : y = a : 2 and y : z = 3 : b, find x : z

A a : 2b B a : b C 2b : 3a D 3a : 2b

Solution

x : y = a : 2 implies $\frac{x}{y} = \frac{a}{2}$

$y = \frac{2x}{a}$ ----- (1)

y : z = 3: b implies $\frac{y}{z} = \frac{3}{b}$

$y = \frac{3z}{b}$ ---- (2)

Equating terms in y from (1) and (2)

$$\frac{2x}{a} = \frac{3z}{b}$$
$$2bx = 3az$$
$$\frac{x}{z} = \frac{3a}{2b}$$

Equation ratio x : z is 3a : 2b (D)

2014/ 30

If x : y = 3 : 2 and y : z = 5 : 4, find the value of x in the ratio x : y : z

A 8 B 10 C 15 D 20

Solution

x : y = 3 : 2 implies $\frac{x}{y} = \frac{3}{2}$

$y = \frac{2x}{3}$ ----- (1)

y : z = 5: 4 implies $\frac{y}{z} = \frac{5}{4}$

$y = \frac{5z}{4}$ ----- (2)

Equating terms in y from (1) and (2)

$$\frac{2x}{3} = \frac{5z}{4}$$
$$8x = 15z$$
$$\frac{x}{z} = \frac{15}{8}$$

Equating ratio x : z is 15 : 18

Thus x is 15 (C)

2014/21 NABTEB (Nov) counter example

A sum of money is shared among ten students so that the first gets ₦ 80.00 , the next gets ₦ 105.00, the next gets the next gets ₦ 130.00 and so on. How much does the tenth student get?

A ₦ 280.00 B ₦ 305.00 C ₦ 330.00 D ₦ 355.00

Solution

The sharing here followed a simple pattern

80, 105, 130, …

Arithmetic Progression (A.P)

Whose first term a is 80, common difference d is 25

Tenth student's share : $T_{10} = 80 + (10 - 1)25$

$= 80 + (9)25$

$= 80 + 225$

$=$ ₦ 305.00 (B)

Exercise 2.1
If A is twice as old as B and B is twice as old as C and the sum of their ages is 63 years, find ages A, B and C

Exercise 2.2
If P : Q = 2 : 3 and Q : R = 4 : 7, find P : Q : R?

Exercise 2.3
If x : y = 2 : 3 and y : z = 3: 4, find x : y : z

2015/5 Neco Exercise 2.4
A manufacturing company is owned by Dr. A and Engr. B and they agreed that their profit will be shared in the ratio 4 : 7. At the end of the year, Dr. A received ₦ 7500.00 less than Engr. B. What is the net profit of the company for the year ended?
A ₦ 7,500 B ₦ 10,000 C ₦ 17,500 D ₦27,500 E ₦ 52,500

2014/14a NABTEB (Nov) Exercise 2.5
Three men provided capitals ₦ 10,000.00 , ₦ 25,000.00 ₦ 30,000.00 for a business on the understanding that the share of the profit would be proportional to the capital provided. If the profit is ₦ 25,000.00, how much does each receive?

2014/45 NABTEB (Nov) Exercise 2.6
Express (0.0425 : 2.5) as a fraction

A. $\frac{17}{10,000}$ B. $\frac{17}{1000}$ C. $\frac{17}{250}$ D. $\frac{17}{100}$

2014/43 NABTEB (Nov) Exercise 2.7
If 3 children share ₦ 10.50 among themselves in the ratio 6 : 7 : 8, how much is the largest share?
A ₦ 3.00 B ₦ 3.30 C ₦ 4.00 D ₦ 4.50

1976/10 (Nov) Exercise 2.8
Chijoke, Adeleke and Martins contributed ₦ 8 000 , ₦ 6 000, ₦ 4 000 respectively as capital to an enterprise. Out of the total profits in a year, Martins received ₦ 500 for his services as a manager. Each of them received 20% on the first ₦ 3 000 of his capital and the remaining profit was shared in the ratio of the remaining parts of their capitals. If at the end of the year Martins received ₦ 1 200 in all, find the total profit and the amounts which Chijoke and Adeleke respectively received

2014/33 Neco (Nov) Exercise 2.9
In a certain wedding, the ratio of men to women is 2 : 7. If there are 124 men, how many women attended the wedding?
A. 389 B. 399 C. 411 D. 434 E. 501

2013/2b Exercise 2.9b
Salem, Sunday and Shaka shared a sum of ₦ 31,100.00 For every ₦ 32.00 that Salem gets, Sunday gets 50kobo and for every ₦ 34.00 Sunday gets, Shaka gets ₦ 32.00. Find Shaka's share

Percentages (%)

The concept of percentage is as old as mathematics. It simply means expression out of 100. 25% means 25/100. A lot of commercial activities are done in percentage: Commission, Discount, Depreciation, Profit and Loss, Taxes etc

2005/ 50
The monthly salary of a man increased from ₦ 2,700 to ₦ 3,200. Find the percentage increase.
A 10% B 15% C 15.6% D 18.5%

Solution

$$\text{Percentage increase} = \frac{3200 - 2700}{2700} \times 100$$
$$= \frac{500}{2700} \times 100 = 18.5\% \quad (\text{D})$$

2005/36
The salary of a man was increased in the ratio 40: 47. Calculate the percentage increase in the salary
A 85.1% B 58.5% C 17.5% D 14.9%

Solution

$$\text{Percentage increase} = \frac{47 - 40}{40} \times 100$$
$$= \frac{7}{40} \times 100 = 17.5\% \quad (\text{C})$$

2009/18 Neco (Dec)
When the salary of a man was increased by 20% he was earning ₦ 48,500.00. What would have been his salary if the increase had been by 26%
A ₦ 60,140.00 B ₦ 50,925.00 C ₦ 50,900.00
D ₦ 49,000.00 E ₦ 48,750.00

Solution

Let the salary be x and in 100%
When increased by 20% it will be 120%

Thus 120% of x = 48,500

$$\frac{120x}{100} = 48,500$$
$$120x = 48,500 \times 100$$
$$x = \frac{48,500 \times 100}{120}$$
$$= 40416.67$$

Next 126% of x $= \frac{126}{100} \times 40416.67$

(Increase is 100 + 26) = ₦ 50, 925.00 (B)

2005/ 7 (Nov)
If 3% of K = M, find 1% of K

A $\frac{M}{3}$ B $\frac{M}{300}$ C $\frac{3M}{100}$ D $\frac{100M}{3}$

Solution

3% of K = M

$$\frac{3K}{100} = M$$
$$3K = 100M$$
$$K = \frac{100M}{3}$$

Thus 1% of K $= \frac{1}{100} \times \frac{100M}{3}$

$$= \frac{M}{3} \quad (\text{A})$$

Discount

2006/ 31 Neco

A buyer paid ₦ 900.00 for an article over which he enjoyed 10% cash discount. How much was the marked price?

A ₦ 810.00 B ₦ 900.00 C ₦ 1,000.00
D ₦ 1,100.00 E ₦ 9,000.00

Solution

Let the marked price be x

Allowing 10% discount on x gives ₦ 900.00

i.e (100 – 10)% of x = ₦ 900

$$\frac{90x}{100} = 900$$

$$90x = 900 \times 100$$

$$x = \frac{900 \times 100}{90}$$

$$= ₦\ 1{,}000.00 \quad (C)$$

2009/ 8 Neco

After allowing a discount of 10% on an article, a seller collected the sum of ₦2,700.00. Find the amount of the discount paid

A ₦ 27.00 B ₦ 270.00 C ₦ 290.00
D ₦ 300.00 E ₦ 2, 400.00

Solution

Let the price of article before discount be x

Allowing 10% discount on x gives ₦ 2,700

i.e (100 – 10)% of x = ₦ 2700

$$\frac{90x}{100} = 2700$$

$$90x = 2700 \times 100$$

$$x = \frac{2700 \times 100}{90}$$

$$= ₦\ 3{,}000$$

Amount of discount paid = ₦ 3,000 – ₦ 2700

= ₦ 300.00 (D)

2005/ 11a (Nov)

A retailer allows a trade discount of 20% and a cash discount of 5% on every article he sells. If a customer pays ₦ 1900 cash for an article, calculate the;

(i) cash discount (ii) marked price

Solution

First, we find the original price of the item

Let it be x

100 – (20 + 5) % of x = ₦ 1900

$$\frac{75x}{100} = 1900$$

$$75x = 1900 \times 100$$

$$x = \frac{1900 \times 100}{75}$$

$$= ₦\ 2{,}533.33$$

(i) Cash discount = 5% of ₦ 2,533.33

$$= \frac{5}{100} \times 2{,}533.33$$

$$= ₦\ 126.67$$

(ii) Marked price is ₦ 2,533.33

2007/14b NABTEB

A trader allows a retailer 20% trade discount and 5% for cash payment. What will be the marked price of an article for which a customer pays ₦ 4,740.00?

Solution

Let the marked price be x

100 – (20 + 5)% of x = ₦ 4,750

$$\frac{75x}{100} = 47500$$

$$75x = 47500 \times 100$$

$$x = \frac{4750 \times 100}{75} = ₦\ 6{,}333.33$$

2011/ 45

A shopkeeper allows a discount of 15% on the marked price of a mobile phone. If a customer paid GH₵ 170.00 for mobile phone, what was the marked price of the phone?

A GH₵ 144.50 B GH₵ 195 .40 C GH₵ 200.00
D GH₵ 255.00

Solution

Let the marked price be x

Discount allowed 15% i.e 85% of x = 170

$$\frac{85}{100}x = 170$$

$$85x = 170 \times 100$$

$$= \frac{170 \times 100}{85} = GH₵\ 200.00 \quad (C)$$

Loss

2005/ 16 (Nov)

In a certain shipment, 2% of the crates were damaged. If the loss per damaged crate was D35 and the total loss due to damages was D 700, how many crates were shipped?

A. 2,000 B. 1,000 C. 200 D. 100

Solution

Loss per damaged crate = D 35

Total loss due to damage = D700

Thus number of crates damaged = $\frac{700}{35}$ i.e 20 crates

Let the shipped crates be x

But 2% of shipped crates (x) got damaged = 20 crates

$$\frac{2}{100}x = 20$$

$$x = \frac{20 \times 100}{2}$$

$$= 1{,}000 \text{ crates}$$

2005 / 9 (Nov)

A woman bought 600 oranges at ₦ 1200. If 20% of the orange got bad and the rest were sold at ₦25 for 12, find the loss

A ₦50 B ₦200 C ₦ 250 D ₦ 288

Solution

Cost per orange = $\frac{1200}{600}$ i.e ₦ 2.00

20% of orange bad = $\frac{20}{100} \times 600$ (oranges)

= 120 orange got bad

Oranges sold = 600 – 120 = 480

Amount of oranges sold = $\frac{480}{12} \times 25$ = ₦ 1,000

Loss = ₦ 1200 – ₦ 1000

= ₦200 (B)

Cost

2014/ 3b Neco

A businessman paid ₦25,000.00 for 12 machines.
Find the: (i) cost of 10 machines of the same type
(ii) Number of such machines ₦ 175,000.00 can buy.

Solution

First we find the cost per machine.
If 12 machines cost ₦ 25,000.00
then one machine will cost $\frac{25000}{12}$ = ₦ 2083.33

(i) Cost of 10 machines of the same type
= 10 × ₦ 2083.33
= ₦ 20,833.30

(ii) Number of such machines ₦ 175, 000.00 can buy
= $\frac{175,000.00}{2083.33}$
= 84 machines

2012/ 3a Neco

The cost of telephone service per month is calculated on three part rates,
₦ 14.00 per unit for the first 25 units,
₦ 18.00 per unit for the next 50 units and
₦ 22.00 per unit for the remainder.
If the total amount paid is ₦ 1,580.00, find the remaining units

Solution

₦ 14.00 per unit for the 1st 25 units
₦ 18.00 per unit for the next 50 units
₦ 22.00 per unit for the remainder
14 × 25 = ₦ 350
18 × 50 = ₦ 900
= ₦ 1,250
Remaining units amount is ₦ 1,580 – ₦ 1,250
= ₦ 330
Thus Remaining units amount is $\frac{330}{22}$ = 15 units

2010/ 1a Neco

Find the difference of the cost of employing 15 females at ₦ 4,860.00 each and 10 males at ₦ 7,050.00 each

Solution

First, we find both costs separately

Female cost	Male cost
15 × 4860	10 × 7050
= ₦ 72,900	= ₦ 70,500

Difference in cost = 72,900 – 70,500
= ₦ 2,400

1976/1b (Nov)

If 8kg of coffee costing ₵20.00 a kg is mixed with 12kg of another kind costing ₵22.00 a kg, what is the cost of the mixture per kg?

Solution

8kg of coffee at ₵20.00 per kg **cost** 8 × 20 = ₵160
12kg of 2nd coffee at ₵22.00 per kg **cost** 12 × 22 = ₵264
Total weight is 8kg + 12kg = 20kg
Cost of mixture per Kg = $\frac{160 + 264}{20}$
= $\frac{424}{20}$ i.e ₵ 21.20

2006/ 11 Neco (Dec)

A pair of shoes cost ₦ 6,500.00 and 2 shirts cost ₦ 4,500.00. How much will 3 pairs of shoes and 3 shirts cost?
A ₦ 9,000.00 B ₦ 13,500.00 C ₦ 22,500.00
D ₦ 26,250.00 E ₦ 33,000.00

Solution

A pair of shoe costs ₦ 6,500.00
2 shirts cost ₦ 4,500.00 then 1 shirt cost $\frac{4500}{2}$ = ₦ 2,250.00
Thus, cost of 3 pairs of shoe is 3 × 6500 = ₦19500
Cost of 3 shirts is 3 × 2250 = ₦ 6,750
Combined cost ₦ 26,250.00 (D)

2005/5

Three bags and five cups cost ₦ 800 altogether. If each of the bags cost ₦150, how much does a cup cost?
A ₦ 50 B ₦ 70 C ₦ 100 D ₦ 160

Solution

Since each (one) bag costs ₦150
3 bags cost is 3 × 150 = ₦ 450
Cost of cups is ₦ 800 – ₦ 450 = ₦ 350
Cost of each cup is $\frac{350}{5}$ = ₦ 70 (B)

2006/ 37 (Nov)

Find the difference in cost per week between employing 16 men at a daily wage of D23.50 each and employing 26 boys at a daily wage of D12 each.
A D124 B D 392 C D 448 D D532

Solution

Daily cost of employing men is 16 × D23.50 = 376
Weekly cost of employing men is 7 × D376.00 = D2632.00
Daily cost of employing boys is 26 × D12 = D312.00
Weekly cost of employing boys is 312 × 7 = D2184.00
Difference in cost per week = 2632 – 2184
= D448 (D)

Income

2010/ 4 Neco

Mr. Kolo is employed as a poultry farm manager on an initial salary of ₦ 300,000.00 per annum. His salary increases annually by ₦ 4,000.00. Find the total money earned by him after working for five years.

Solution

1st year: ₦ 300,000.00
2nd year: ₦ 300,000.00 + ₦ 4,000.00 = ₦ 304,000
3rd year: ₦ 304,000 + ₦ 4,000 = ₦ 308,000
4th year: ₦ 308,000 + ₦ 4,000 = ₦ 312,000
5th year: ₦ 312,000 + ₦ 4,000 = ₦ 316,000
At the end of 5 years, total money earned
= 300,000 + 304,000 + 308,000 + 312,000 + 316,000 = ₦ 1,540,000

2010/9 Neco

An employee earns ₦ 450,000.00 per annum out of which he spends 12% on house rent. How much is left for other expenses?
A ₦ 45,000.00 B ₦54,000.00 C ₦ 396,000.00
D ₦ 449,912.00 E ₦ 449,988.00

Solution

What is left for other expenses = annual salary – House rent
First, we find the house rent i.e 12% of 450, 000
= $\frac{12}{100} \times 450,000$
= ₦ 54,000
Amount left for expenses = ₦ 450,000 – ₦ 54,000= ₦ 396,000.00

Joint cases

2005/ 4(Nov)

A sales clerk is on a monthly basic salary of ₦ 7,500.00. He is given a commission of $2^1/_2$ % of his monthly sales and 3% of his basic salary as utility allowance. He pays 2% of his total income as income tax and 5 kobo in the naira as cooperative society fund.
If his sales for a month is ₦ 25,000.00, calculate his take - home pay for that month.

Solution

Monthly sales is ₦ 25,000.00, find take home pay.
Basic salary : ₦7, 500
Commission: $2^1/_2$ of sales (₦ 25,000.00)

$$\text{i.e } \frac{5}{2} \times \frac{1}{100} \times 25000 = 625$$

Utility allowance: 3% of basic salary

$$\text{i.e } \frac{3}{100} \times 7{,}500 = 225$$

Total income for that month: ₦ 7,500 + ₦ 225 + ₦ 625 = ₦ 8350

Next is Deductions

Tax : 2% of total income = $\frac{2}{100} \times 8350$ i.e ₦167

Cooperative society fund : 5kobo in the naira of his income

$$= \frac{8350}{5} \text{ i.e ₦1,670}$$

Take - home pay = total income – deductions
= 8350 – 167 – 1670
= ₦ 6,513

2012/ 6a Neco

A man earns a total of ₦ 24,000.00 per month. Some of his monthly expenses are listed below:

House keeping	₦ 6,400.00
Petrol and electricity	₦ 1,700.50
Water rate	₦1,000.90
Insurance	₦ 900.30
Income tax	₦ 4,100.10

(i) How much is paid on water rates per year?
(ii) Express the income tax paid as percentage of his total income.
(iii) If petrol and electricity expenses are in the ratio of 16 : 5, by how much is the amount expended on petrol more than that of electricity.

Solution

(i) Water rate per year = ₦ 1,000.90 × 12 = ₦ 12,010.80
i.e 12months

(ii) Income tax % = $\frac{4100.10}{24000} \times 100 = 17.08\%$

Petrol : electricity
16 : 5

Total ratios = 16 + 5 i.e 21
Amount difference between Petrol and electricity

$$= \frac{11}{21} \times 1700.50 = ₦\ 890.74$$

Alternatively

Total ratios = 16 + 5 i.e 21

Petrol amount	electricity amount
$\frac{16}{21} \times 1700.50$	$\frac{5}{21} \times 1700.50$
= ₦1295.62	= ₦ 404.88

Difference in amount = 1295.62 – 404.88 = ₦ 890.74

2010/6 Neco

A married man with four children earns ₦ 240,000.00 per annum. He claims a personal allowance of ₦ 30,000.00 and allowance of ₦15,000.0 for his wife and ₦ 10,000.00 for each child. What is his taxable income if the allowances are not taxable?

A ₦210,000.00 B ₦ 176,000.00 C ₦155,000.00
D ₦ 85,000.00 E ₦ 64,000.00

Solution

Allowance for 4 children = 4 × 10,000
= ₦ 40,000

$$\text{Taxable income} = \underbrace{240{,}000}_{\text{Salary}} - \underbrace{(30{,}000 + 15{,}000 + 40{,}000)}_{\text{allowances}}$$

= 240, 000 – 85, 000
= ₦155,000.00 (C)

2007/14a NABTEB

A married man with 5 children is on an annual salary of ₦ 75, 000.00. The man is given tax relief as follows:

Personal allowance of ₦ 9,000.00
Children allowance of ₦ 1,500.00 per child for a maximum of 4 children.
Dependent relative allowance of $^1/_{10}{}^{th}$ of his salary
Life insurance allowance of ₦ 5,000.00
If tax is paid at 10k in ₦ on the 1st ₦ 20,000.00 and 15k in ₦ on the remaining, calculate the amount of tax he pays:

Solution

Annual salary is **₦ 75,000.00**

Tax Relief are: personal allowance	₦ 9,000.00
Children allowance 4 × 1500 is	₦ 6,000
Dependent relative allowance $^1/_{10}$ × 75000 is	₦ 7,500
Life insurance allowance is	₦ 5,000
Total tax relief	**₦ 27, 500**

Amount taxable is ₦ 75,000 – ₦ 27,500 = *47,500*

First tax is 10 × 20,000 = 200000 kobo i.e	₦ 2,000.00
2nd tax is 15 × 27,500 = 412500 kobo i.e	₦ 4,125.00
Total tax	₦ 6,125.00

1979/8

In 1965, the amount of taxes collected in a country amounted to Le20 000 per 100 persons. In 1975, the population increased by 8% and the revenue from taxation by 25%.
(a) Calculate, to 3 significant figures, the percentage increase in taxation per person
(b) If in 1976, the government decided to reduce taxation per person by 2%, calculate, correct to the nearest 10 Leones, the revenue from taxation per 100 persons

Solution

a. In 1965, 100 persons paid taxes of Le 20 000
1 person paid taxes of $\frac{1}{100}$ × Le20 000 = Le200

In 1975, 100 persons increased by 8% to $\frac{108}{100} \times 100$ persons
= 108 persons

Le20,000 increased by 25% to ($\frac{125}{100}$ × 20,000) = Le25,000
108 persons paid taxes of Le 25,000
1 person paid taxes of $\frac{1}{108}$ × Le25,000 = Le231.48

Thus, increase in taxation per person is Le231.48 – Le200
= Le31.48

The percentage increase in tax = $\frac{Le31.48}{Le200} \times 100\% = 15.74\%$
≈ 15.7% to 3 s.f

b. In 1975, the taxation per person is Le 231.48

In 1976, the taxation per person is reduced by 2% to

$$\frac{98}{100} \times 231.48 = \text{Le } 226.85$$

The taxation per 100 persons = Le226.85 × 100 = Le22685

≈ Le22690 to nearest 10 Leones

Commission

2009/3 Neco (Dec)

A sales representative is entitled to a commission of 6.5% on every radio set sold. If his commission amounts to ₦ 1,250.00k per radio set sold, calculate the price of a radio set (to the nearest ₦)

A ₦ 19210.00 B ₦ 19220.00 C ₦19229.00
D ₦ 19230.00 E ₦19231.00

Solution

Let the radio price be y

Commission of 6.5 % on y gave ₦ 1,250.00k

$$\text{i.e } 6.5\% \text{ of } y = 1250$$
$$\frac{6.5}{100}y = 1250$$
$$6.5y = 1250 \times 100$$
$$y = \frac{1250 \times 100}{6.5}$$
$$= ₦\ 19{,}230.77\text{k}$$
$$\approx ₦\ 19{,}231.00 \text{ to the nearest } ₦$$

2005/3 Neco

Find the marked price of an article, if the ₦ 75.50 commission allowed is 4.5%

A ₦ 1,602.28 B ₦ 1,650.00 C ₦1,653.28
D ₦ 1,677.78 E ₦ 6,899.00

Solution

Let the marked price be y

Allowed 4.5% commission on y gives ₦ 75.50

$$\text{i.e } 4.5\% \text{ of } y = ₦\ 75.50$$
$$\frac{4.5y}{100} = 75.50$$
$$4.5y = 75.50 \times 100$$
$$y = \frac{75.50 \times 100}{4.5}$$
$$= ₦\ 1{,}677.78 \quad (\text{D})$$

2010/ 5 Neco

A sales girl gets a commission of 8% of the value of the things she sells. Find her commission for selling 3 keyboards at ₦ 25,000.00 each and 5 calculators at ₦ 600.00 each

A ₦ 2,048.00 B ₦ 3,200.00 C ₦5,740.80
D ₦ 6,240.00 E ₦ 9,750.00

Solution

Commission on keyboard 8% of ₦ 25,000.00

$$= \frac{8}{100} \times 25{,}000 \text{ i.e } ₦\ 2{,}000$$

Thus 3 keyboards commission = 3 × 2,000 i.e **₦ 6,000**

Commission on calculators 8% of ₦ 600.00

$$= \frac{8}{100} \times 600 \text{ i.e } ₦\ 48.00$$

Thus 5 calculators commission = 5 × 48 i.e **₦ 240**

Total commission = ₦ 6000 + ₦ 240

= ₦ 6240.00 (D)

Hire purchase

2004/ 4b (Nov)

An electrical shop sells a video recorder for D3,200 cash price. Hire purchase deal is offered to Mr. Bah at an extra 20% cost. He pays a deposit of D1,200 and Dx in 6 equal monthly installments. Calculate the value of x

Solution

Cash sales price is D3,200

$$\text{Hire purchase is 20\% of } 3{,}200 = \frac{20}{100} \times 3{,}200$$
$$= \text{D}640$$

Since Mr. Bah bought it at Hire purchase

Cost = 3,200 + 640

= D 3,840

He pays a deposit of D1,200 and left to pay Dx

Thus Dx = D3,840 – D1,200

= D 2, 640

Depreciation

2008/ 7c

A car costs ₦300,000.00. It depreciates by 25% in the first year and 20% in the second year. Find its value after 2 years

Solution

$$1^{st} \text{ year : 25\% of ₦300,000.00} = \frac{25}{100} \times 300{,}000 \text{ i.e ₦}75{,}000$$

Car value after 1st year = ₦ 300,000.00 – ₦ 75,000.00

= ₦ 225,000

$$2^{nd} \text{ year : 20\% ₦225,000} = \frac{20}{100} \times 225{,}000$$
$$= ₦\ 45{,}000$$

Car value after 2nd year = ₦ 225 – ₦ 45, 000

= ₦ 180,000.00

2006/ 25

A machine valued at ₦ 20,000 depreciates by 10% every year. What will be the value of the machine at the end of two years?

A ₦ 16,200 B ₦ 14,200 C ₦12,000 D ₦ 8,000

Solution

$$1^{st} \text{ year : 10\% of ₦ 20,000} = \frac{10}{100} \times 20{,}000 \text{ i.e ₦ } 2{,}000$$

Machine value after 1st year = ₦ 20,000 – ₦ 2,000

= ₦ 18,000

$$2^{nd} \text{ year: 10\% of ₦ 18,000} = \frac{10}{100} \times 18{,}000 \text{ i.e ₦ } 1{,}800$$

Machine value after 2nd year = ₦ 18,000 – 1,800

= ₦ 16, 200 (A)

Tax

2005/ 40

What is the purchase tax, at 86 kobo in the naira on a jewelry valued at ₦ 2,160.00?

A ₦ 1,857.60 B ₦ 2,759.50
C ₦ 3,759.50 D ₦ 3,856.70

Solution

Purchase tax is 2160 × 86 = 185760 kobo

= ₦ 1,857.60 (A)

Exchange

2011/ 1

If ₦112.00 exchanges for D14.95, calculate the value of D1.00 in naira

A 0.13 B 7.49 C 8.00 D 13.00

Solution

If ₦112.00 exchanges for D14.95

Then D1.00 will be $\frac{112.00}{14.95}$ = 7.49 naira (B)

2015/3

A trader bought an engine for $15,000.00 outside Nigeria. If the exchange rate is $0.075 to ₦1, how much did the engine cost in naira?

A ₦ 250,000.00 B ₦ 200,000.00
C ₦150,000.00 D ₦ 100,000.00

Solution

If the exchange rate is $0.075 to ₦1,

Then $15,000 will be $\frac{15000}{0.075}$ = ₦ 200,000 (B)

1977/3 Exercise 2.10

A man made two investments, one at 7% and the other at 8% interest. The yield on the two investments was Le222. If the total amount invested was Le3 000, how much was invested at each rate?

1977/8 Exercise 2.11

The cost of a man's holiday in 1974 was made up of 65% hotel bills, 20% traveling expenses and 15% incidentals. In 1975 the hotel bills had gone up by 40% and the traveling expenses by 60%. He reduced his incidentals by 10% and thereby made a saving of ¢30 on the incidentals. By how much did the cost of his holiday in 1975 exceed that of 1974?

2014/15b NABTEB (Nov) Exercise 2.12

Given that the market exchange rate is ₦ 200.00 to 1 British pounds (£) , convert ₦ 726,840.00 to £

2014/2 (Nov) Exercise 2.13

A man has a wife and 6 children and his total income in a year was GH₵ 850. He was given the following tax free allowances:

Personal GH₵ 120;

Wife GH₵ 30;

Children GH₵ 25 per child, for a maximum of 4 children

Medical GH₵ 40

The rest was taxed as follows:

First GH₵ 200 at 10%

Next GH₵ 200 at 15%

Next GH₵ 200 at 20%

Remainder at 25%

Calculate his : (a) taxable income; (b) monthly tax

2015/8b Neco Exercise 2.14

In 2001, the salary of Mr Edward was ₦ 25,700.00 and this was increased by 15% in 2002. If he paid $12^1/_2$% of his salary as income tax, calculate, correct to 3 significant figures, the amount of tax paid in 2002

2014/15b NABTEB (Nov) Exercise 2.15

A man is allowed the following tax relief:

Personal allowance	$12^1/_2$% of basic salary
Wife allowance	10 % of basic salary
Children allowance	₦ 5,000.00 per child up to maximum of 4 children
Aged parent	₦10,000.00 for only one
Dependent relative	10% up to maximum of 2 relatives

A married man with 5 children has an aged father and two relatives. If the man's basic salary is ₦ 850,000.00 per annum and tax is calculated at the rate of 15% find the man's

(i) taxable income per annum
(ii) monthly tax
(iii) monthly pay

2014/31 (Nov) Exercise 2.16

After allowing a discount of 15% on an article, a seller collected GH₵ 36,000.00 in cash. How much was the discount?

A GH₵ 6,952.94 B GH₵ 6,502.94
C GH₵ 6,352.94 D GH₵ 6,203.94

2008/3 Neco Exercise 2.17

If 15% of a sum of money is given as ₦ 75,000.00, what is the total sum?

A ₦ 750,000.00 B ₦ 500, 000.00 C ₦5,000,000.00
D ₦ 75,000,000.00 E ₦ 750,000,000.00

2013/ 7 Neco Exercise 2.18

The marked price of an item is ₦ 500.00. If a discount of 10% is given, at what price is the item sold?

A ₦ 200.00 B ₦ 250.00 C ₦ 300.00
D ₦ 450.00 E ₦ 500.00

Profit and loss

Any transaction is profitable if selling price is greater than cost price otherwise it is a loss

$$\text{Percentage profit} = \frac{SP - CP}{CP} \times 100$$

Where SP = Selling price

CP = Cost price.

$$\% \text{ loss} = \frac{CP - SP}{CP} \times 100$$

2007/ 10 Neco

By selling a book at a loss of 19%, a man lost ₦11.40. Find the actual price paid for the book

A ₦ 13.57 B ₦ 20.63 C ₦ 21.67
D ₦ 24.00 E ₦ 60.00

Solution

$$\% \text{ loss} = \frac{CP - SP}{CP} \times 100$$

CP = ? % loss is 19% But CP – SP = ₦ 11.40

Substituting

$$19 = \frac{11.40}{CP} \times 100$$

$$19CP = 1140$$

$$CP = \frac{1140}{19}$$

$$= ₦\ 60.00 \quad (E)$$

2006/ 9 Neco

Ngozi sold an article for ₦ 1,755.00 and made a profit of 35%. Find the cost price.

A ₦ 2,370.00 B ₦ 2,369.25 C ₦ 1,300.00
D ₦ 614.25 E ₦ 614.00

Solution

$$\% \text{ profit} = \frac{CP - SP}{CP} \times 100$$

% profit = 35%; SP = ₦ 1,755; CP = ?

Substituting

$$35 = \frac{1755 - CP}{CP} \times 100$$

$$35CP = 175500 - 100CP$$

$$35CP + 100CP = 175500$$

$$135CP = 175500$$

$$CP = \frac{175500}{135}$$

$$= ₦\ 1,300.00 \quad (C)$$

2008/31 NABTEB (Dec)

Joseph bought a car for ₦ 200,000.00 and later sold it for ₦ 170, 000.00, what is his percentage loss?

A 117.6% B 115.6% C15.6% D 15 %

Solution

$$\% \text{ loss} = \frac{CP - SP}{CP} \times 100$$

CP = ₦ 200,000.00; SP = ₦ 170,000.00; loss = ?

Substituting

$$\% \text{ loss} = \frac{200000 - 170000}{200000} \times 100$$

$$= \frac{30000}{200000} \times 100$$

$$= 15\ \% \quad (D)$$

2005/19 Neco

A man bought a certain number of articles, of which 10 cost ₦ 15.00 and sold them at ₦ 19.50 per dozen. Find the percentage gain or loss.

A 7.7% B 8.3% C 23.1% D 30% E 92.9%

Solution

If 10 articles cost ₦ 15.00

Then 1 article will cost $\frac{15}{10}$ = ₦ 1.50 CP

Selling point

If 12 articles was sold for ₦ 19.50

Then 1 article will be sold at $\frac{19.50}{12}$ = ₦ 1.625 SP

Since SP ₦ 1.625 is greater than CP ₦ 1.50, i.e % profit

$$\% \text{ gain} = \frac{SP - CP}{CP} \times 100$$

$$= \frac{1.625 - 1.50}{1.50} \times 100$$

$$= 8.3\% \quad (B)$$

2006/ 11 Neco

Mohammed sold an article which originally cost ₦ 2,500.0 at the rate of ₦ 2,050.00. Find the percentage loss.

A 22% B 18% C 16% D 10% E 8%

Solution

$$\% \text{ loss} = \frac{CP - SP}{CP} \times 100$$

% loss = ? ; CP = ₦ 2,500.0; SP = ₦ 2,050.00

Substituting

$$\% \text{ loss} = \left(\frac{2500 - 2050}{2500}\right) \times 100$$

$$= \frac{450}{25} \quad \text{i.e } 18\% \quad (B)$$

2005/ 49

Ladi sold a car for ₦ 84,000 at a loss of 4%. How much did Ladi buy the car?

A ₦ 80,500 B ₦ 80,640 C ₦ 87,360 D ₦ 87,500

Solution

$$\% \text{ loss} = \frac{CP - SP}{CP} \times 100$$

% loss = 4%; CP = ?; SP = ₦ 84,000

Substituting

$$4 = \frac{CP - 84{,}000}{CP} \times 100$$

$$4CP = (CP - 84{,}000)100$$

$$4CP = 100CP - 8{,}400{,}000$$

$$8400\ 000 = 100CP - 4CP$$

$$8400000 = 96CP$$

$$\frac{8400000}{96} = CP$$

$$= ₦\ 87,500 \quad (D)$$

2008/7

A trader bought 100 tubers of yam at 5 for ₦ 350.00. She sold them in a set of 4 for ₦ 290.00. Find her gain percent.

A 3.6% B 3.5% C 3.4% D 2.5%

Solution

$$\% \text{ gain} = \frac{SP - CP}{CP} \times 100$$

Cost price (CP) : $\frac{100}{5} \times 350$ = ₦ 7,000

Selling price (SP) : $\frac{100}{4} \times 290$ = ₦ 7,250

Substituting

$$\% \text{ gain} = \left(\frac{7250 - 7000}{7000}\right) \times 100$$

$$= \frac{250}{70}$$

$$= 3.57\% \approx 3.6\% \quad (\text{ A })$$

2011/ 5b Neco

A dealer sells an article for ₦ 560.00 and makes a profit of 12%. Find the percentage of the profit he will realize, if he sells the article for ₦ 600.00

Solution

In the **first case**, lets us find CP

$$\% \text{ profit} = \frac{SP - CP}{CP} \times 100$$

% profit = 12%; SP = ₦ 560.00; CP = ?

Substituting

$$12 = \frac{560 - CP}{CP} \times 100$$

12CP = (560 – CP)100

12CP = 56000 – 100CP

12CP + 100CP = 56000

112CP = 56000

CP = $\frac{56000}{112}$ i.e ₦ 500.00

2nd case:

SP = ₦ 600.00; CP = ₦ 500.00; % profit = ?

$$\% \text{ profit} = \frac{600 - 500}{500} \times 100$$

$$= \frac{100}{500} \times 100 = 20\%$$

2011/3 Neco

An item was sold for ₦ 650.00 with a profit of 30%. If the profit had been 20%, what was the selling price?

A ₦ 600.00 B ₦ 550.00 C ₦ 500.00
D ₦ 450.00 E ₦ 400.00

Solution

In the **first case**, lets us find CP

$$\% \text{ profit} = \frac{CP - SP}{CP} \times 100$$

% profit = 30%, SP = ₦ 650.00, CP = ?

Substituting

$$30 = \frac{650 - CP}{CP} \times 100$$

30CP = (650 – CP) 100

30CP = 65000 – 100CP

30P + 100CP = 65000

130CP = 65000

CP = $\frac{65000}{130}$ i.e ₦ 500.00

2nd case: SP = ? ; % profit = 20%; CP = ₦ 500.00

$$20 = \frac{SP - 500}{500} \times 100$$

10000 = (SP – 500) 100

100 = SP – 500,

SP = 100 + 500 i.e ₦ 600.00 (A)

2009/ 23 Neco (Dec)

By selling an article for ₦ 600.00, a man made a profit of 20%. What should be the selling price to make a profit of 30% ?

A ₦ 550.00 B ₦ 580.00 C ₦ 630.00
D ₦ 650.00 E ₦ 780.00

Solution

In the **first case**, lets us find CP

$$\% \text{ profit} = \frac{SP - CP}{CP} \times 100$$

% profit = 20%, SP = ₦ 600.00, CP =?

Substituting

$$20 = \frac{600 - CP}{CP} \times 100$$

20CP = (600 – CP) 100

20CP = 60000 – 100CP

20P + 100CP = 60000

120CP = 60000

CP = $\frac{60000}{120}$ i.e ₦ 500

2nd case: % profit = 30%, SP = ? , CP = ₦ 500

Thus 30 = $\frac{SP - 500}{500} \times 100$

30 × 5 = SP – 500

150 = SP – 500,

SP = 500 + 150 i.e ₦ 650 (D)

2004/7a Neco (Dec)

Victor sold a second – hand television to Funmi at a loss of 20%. Funmi then sold it to Usman for ₦5,200.00 making a loss of 15%. How much did Victor buy it?

Solution

Funmi to Usman transaction at Loss %

$$\% \text{ loss} = \frac{CP - SP}{CP} \times 100$$

SP = ₦ 5, 200.00 CP =?, % loss = 15%

Substituting

$$15 = \frac{CP - 5200}{CP} \times 100$$

15CP = (CP – 5200)100

15CP = 100CP – 520000

520000 = 100CP – 15CP

520000 = 85CP

CP = $\frac{520000}{85}$ i.e ₦ 6,117.65

Victor to Funmi transaction at Loss %

$$\% \text{ loss} = \frac{CP - SP}{CP} \times 100$$

% loss = 20%, CP =?, SP = ₦ 6,117.65(Funmi to Usman cost price)

Substituting

$$20 = \frac{CP - 6117.65}{CP} \times 100$$

20CP = (CP – 6117.65) 100

20CP = 100CP – 611765

611765 = 80CP

CP = $\frac{611765}{80}$

= ₦ 7,647.06

2009/ 6b

For musical shows, x children were present. There were 60 more adults than children. An adult paid D5 and a child D2. If a total of D1280 was collected, calculate the:

(i) value of x;

(ii) ratio of the number of children to the number adults;

(iii) average amount paid per person

(iv) percentage profit it the organizers spent D720 on the show.

Solution

(i) Children — Adult

x — x + 60

$$2 \times x + 5(x + 60) = 1280$$
$$2x + 5x + 300 = 1280$$
$$7x = 1280 - 300$$
$$7x = 980$$
$$x = \frac{980}{7} \text{ i.e. } 140$$

(ii) x : x + 60 (where x is 140)

140 : 200

i.e. 7 : 10

(iii) Average amount paid = $\frac{Total\ amount\ collected}{number\ of\ people}$

$$= \frac{1280}{340} \text{ i.e D3.76}$$

(iv) CP =D720; SP = D 1280; % profit = ?

$$\%\ profit = \frac{SP - CP}{CP} \times 100$$
$$= \frac{1280 - 720}{720}$$
$$= \frac{560}{720} \times 100 = 77.78\%$$

2007/ 13a NABTEB

A trader bought 98 units of an article at ₦ 180.00 each. He sold 42 of them at a profit of 20%, 35 at a loss of 4% and the remainder at a profit of 15%. Find the overall

(i) selling profit to the nearest kobo and

(ii) percentage gain or loss to 2 decimal places

Solution

Total cost price : 98 × 180 = ₦ 17,640.00

42 of the items

Cost price (CP): 42 × 180 = ₦ 7,560.00

SP = ?, % profit = 20%

Thus $\%\ profit = \frac{SP - CP}{CP} \times 100$

$$20 = \frac{SP - 7560}{7560} \times 100$$
$$20 \times 7560 = (SP - 7560)100$$
$$151200 = 100SP - 756000$$
$$151200 + 756000 = 100SP$$
$$\frac{907200}{100} = SP$$

SP = ₦ 9,072.00

35 of the items

Cost price (CP): 35 × 180 = ₦ 6,300.00

SP = ?; % Loss = 4%

$$\%\ Loss = \frac{CP - SP}{CP} \times 100$$
$$4 = \frac{6300 - SP}{6300} \times 100$$
$$4 \times 6300 = (6300 - SP)100$$

100 will cancel out

$$4 \times 63 = 6300 - SP$$
$$252 = 6300 - SP$$
$$SP = 6300 - 252$$

= ₦ 6,048.00

Remainder of the items (98 – 42 – 35 = 21)

Cost price (CP) : 21×180 = ₦ 3,780.00

SP = ?, % profit = 15%

$$15 = \frac{SP - 3780}{3780} \times 100$$
$$15 \times 3780 = (SP - 3780)\ 100$$
$$56700 = 100SP - 378000$$
$$56700 + 378000 = 100SP$$
$$434700 = 100SP$$
$$SP = \frac{434700}{100}$$

= ₦ 4,347.00

(i) Overall selling price:

9,072 + 6,048 + 4,347 = ₦ 19,467.00

(ii) Since overall CP = ₦ 17,640.00 and SP = ₦ 19,467

i.e SP is bigger, we find % gain

$$\%\ gain = \frac{19467 - 17640}{17640} \times 100$$
$$= \frac{1827 \times 100}{17640}$$
$$= 10.357\% \approx 10.36\ \% \text{ to 2d.p}$$

2008/12a NABTEB (Dec)

A trader sold an article for ₦ 4,500.00 and made a profit of 15%. How much must he sell it to make a loss of $7^1/_2$%?

Solution

First, we find CP from the % profit

$$\%\ profit = \frac{SP - CP}{CP} \times 100$$

% profit = 15%, SP = ₦ 4,500.00, CP =?

Substituting

$$15 = \frac{4500 - CP}{CP} \times 100$$
$$15CP = (4500 - CP)100$$
$$15CP = 450000 - 100CP$$
$$15CP + 100CP = 450000$$
$$115CP = 450000$$
$$CP = \frac{450000}{115} \text{ i.e ₦ } 3,913.04$$

Next is the proposal for $7^1/_2$% loss

$$\%\ loss = \frac{CP - SP}{CP} \times 100$$
$$7^1/_2 = \frac{(3913.04 - SP)}{3913.04} \times 100$$
$$\frac{15}{2} = \frac{391304 - 100SP}{3913.04}$$
$$\frac{15 \times 3913.04}{2} = 391304 - 100SP$$
$$29347.8 = 391304 - 100SP$$
$$100SP = 361956.2$$
$$SP = \frac{361956.2}{100} = \text{₦ } 3,619.56$$

2008/13 NABTEB (Dec)

In the beginning of year 2007 a poultry attendant bought 1300 day old birds at ₦150.00each. Later in that year she realized 7038 eggs which she sold at ₦450.00 per crate. She spent ₦18,600.0 on feeding the birds and ₦ 8,500.00 on drugs for the year. By the end of the year, she sold the remaining 112 chickens at the rate of ₦800.00 each. Find the percentage gain or loss

Solution

Let us put the data in the picture of expenditure (cost) and sales

Cost of day old birds: 1300 × ₦150 = ₦195,000.00

Sale of eggs: $\frac{7038}{36}$ (Note 36 eggs in a crate) = 195.5

Thus sales of eggs = 195.5 × ₦ 450 = ₦ 87,975

Cost of drugs: ₦ 8,500

Sales of remaining chickens: 112 × ₦ 800 = ₦ 89,600

Total cost: 195,000 + 18,600 + 8,500 = **₦ 222,100**

Total sales: 87,975 + 89,600 = **₦ 177,575**

Cost is greater than sales; hence % loss

$$\% \text{ loss} = \frac{CP - SP}{CP} \times 100$$
$$= \frac{222100 - 177575}{222100} \times 100$$
$$= \frac{44525}{2221}$$
$$= 20.05\%$$

2006/ 2a

A dealer sold a car to a man and made a profit of 15%. The man then sold it to a woman for ₦120,175.00 at a loss of 5%. How much did the dealer buy the car?

Solution

Let us start with the *man to the woman* transactions, where we have data to work with.

$$\% \text{ loss} = \frac{CP - SP}{CP} \times 100$$

% loss = 5%, SP = ₦ 120, 175.00, CP =?

Substituting

$$5 = \frac{CP - 120175}{CP} \times 100$$
$$5CP = (CP - 120175)100$$
$$5CP = 100 - 12017500$$
$$12017500 = 100CP - 5CP$$
$$12017500 = 95CP$$
$$CP = \frac{12017500}{95} \quad \text{i.e ₦126,500.00}$$

The *dealer to the man* transaction

$$\% \text{ profit} = \frac{SP - CP}{CP} \times 100$$

Here SP = ₦126,500 (i.e CP at man to woman transaction)

% profit = 15%, CP = ?

Substituting

$$15 = \frac{126500 - CP}{CP} \times 100$$
$$15CP = (126500 - CP)100$$
$$15CP = 12650000 - 100CP$$
$$15CP + 100CP = 12650000$$
$$115CP = 12650000$$
$$CP = \frac{12650000}{115} \quad \text{i.e ₦ 110, 000.00}$$

2001/ 8b (Nov) special case

A man bought 250 electric lanterns in the United States of America (USA) for \$5,000. He sold them in Lagos at ₦1,920 each. At the time of his journey back to USA, the exchange rate was ₦ 96 to the dollar.

(i) If by selling the lanterns at ₦ 1,920 each, he made a profit of 20%. Calculate the exchange rate at the time he bought the lanterns

(ii) How much in dollar will the sales amount to at the time of his journey back to USA?

Solution

(i) Bought 250 at \$5,000

Sold ₦ 1,920 × 250 = ₦ 480,000

Return exchange rate ₦ 96

Amount sales in Dollars \$ $\frac{480,000}{96}$ = \$5,000

$$\% \text{ Profit} = \frac{SP - CP}{CP} \times 100$$
$$20 = \frac{480,000 - CP}{CP} \times 100$$
$$20CP = (480000 - CP)100$$
$$20CP + 100CP = 48000000$$
$$120CP = 48,000,000$$
$$CP = \frac{48,000,000}{120} = 400,000$$

Let exchange rate be x

$$\text{Thus } \$ 5,000 = \frac{400,000}{x}$$
$$x = \frac{400,000}{5000}$$
$$= ₦ 80 \text{ per Dollar}$$

(ii) Sales at return to USA

250 × ₦ 1,920 = ₦ 480,000

In Dollars = $\frac{480,000}{96}$ = \$ 5,000

2015/9b

A trader bought 30 baskets of pawpaw and 100 baskets of mangoes for ₦2,450.00. She sold the pawpaw at a profit of 40% and the mangoes at a profit of 30%. If her profit on the entire transaction was ₦855.00, find the:

(i) cost price of a basket of pawpaw

(ii) selling price of the 100 baskets of mangoes

Solution

Let p be pawpaw and m be mangoes

From cost: 30p + 100m = 2450 --------(1)

From profit: 40p + 30m = 855 ----------(2)

(1) × 4 and (2) × 3 and subtract

$$\begin{array}{r} 120p + 400m = 9800 \\ -(120p + 90m = 2565) \\ \hline 310m = 7235 \end{array}$$

$$m = \frac{7235}{310} \quad \text{i.e ₦23.34}$$

Substitute m value into (2)

$$40p + 30(23.34) = 855$$
$$40p + 700.2 = 855$$
$$40p = 855 - 700.2$$
$$p = \frac{154.8}{40} \quad \text{i.e ₦3.87}$$

(i) A basket of pawpaw cost ₦3.87

(ii) $\% \text{ profit} = \frac{SP - CP}{CP} \times 100$

% profit is 30%; CP is 100(23.34) i.e ₦ 2334; SP = ?

$$30 = \frac{SP - 2334}{2334} \times 100$$

$$30 \times 2334 = (SP - 2334)100$$

$$70020 = 100SP - 233400$$

$$233400 + 70020 = 100SP$$

$$\frac{303420}{100} = SP$$

$$₦3,034.20 = SP$$

2010/7a Exercise 2.19

Madam Kwakyewaa imported a quantity of frozen fish costing GH₵ 400.00. The goods attracted an import duty of 15% of its cost. She also paid a sales tax of 10 % of the total cost of the goods including the import duty and then sold the goods for GH₵ 660.00. Calculate her percentage profit.

2010/3b Exercise 2.20

Bala sold an article for ₦6,900.00 and made a profit of 15%. If he sold it for ₦6,600.00 he would have made a:

A profit of 13% B loss of 12%
C profit of 10% D loss of 5%

2012/4b Exercise 2.21

A sales man bought some plates at ₦ 50.00 each. If he sold all of them for ₦ 600.00 and made a profit of 20% on the transaction, how many plates did he buy?

2013/6b Exercise 2.22

A man sold 100 articles at 25 for $66.00 and made a gain of 32%. Calculate his gain or loss percent if he sold them at 20 for $50.00

2009/10 Neco Exercise 2.23

By selling 30 bananas for ₦ 45.00, a trader makes a profit of 5%. What would be the percentage loss if he sells the same 30 bananas for ₦ 42.00?

A 2% B 3% C 5% D 6.7% E 7.14%

2014/6b Exercise 2.24

A television set was marked for sale at GH₵ 760.00 in order to make a profit of 20%, the television set was actually sold at a discount of 5%. Calculate, correct to 2 significant figures, the actual percentage profit.

2006/4(Nov) Exercise 2.25

A trader buys a bag of rice for ₦ 3, 200.00 and sells it at a gain of $2^1/_2$ percent. What is the selling price?

A ₦ 3,120.00 B ₦ 3,280.00 C ₦ 3,360.00 D ₦ 3,840.00

2014/ 8 Neco Exercise 2.26

A trader sold an article for ₦ 2,400.00 at a loss of 25%. Calculate his cost price.

A ₦ 3,000.00 B ₦ 3,200.00 C ₦ 3,500.00
D ₦3,600.00 E ₦ 4,000.00

2005/12a (old) Exercise 2.27

The marked price of radio set is D5,200.

(i) If this includes a 30% profit; calculate the cost price;
Amadu bought this radio set and was offered a 5% discount;
(ii) How much did he pay?
(iii) What is the actual profit made by the seller?

2008/12 Neco Exercise 2.28

A pair of shoes was sold for ₦ 2,250.00 at a loss of 10%. What was the cost price?

A ₦ 750.00 B ₦ 2,500.00 C ₦2,538.00 D₦3,288.00 E ₦4,038.00

SIMPLE INTEREST & COMPOUND INTEREST.

If a certain amount of money is to be paid (interest) per annum on a borrowed or deposited money (principal) on a continuous basis as long as the transaction holds, then the interest is:

Simple interest; if the interest to be paid in one year does not affect the principal for the next year calculation of interest, i.e . the principal is fixed all through the transaction

Mathematically,

$$SI = \frac{PRT}{100}$$

*Where **P** is Principal; **R** is ratio and **T** is time*

Amount = P + interest

Compound interest; if the interest to be paid in the 1st year plus the principal forms the new principal for the next year. Thus the principal continues to increase as well as the interest. Mathematically.

$$CI = A - P$$

*Where **A** is amount and **P** is principal*

But $\mathbf{A} = P\left(1 + \frac{R}{100}\right)^n - P$

*Where **n** is number of years and **R** is ratio*

Problems on simple interest

2009/5

Find the value to which ₦ 3000.00 will amount in 5 years at 6% per annum simple interest

A ₦ 3,900.00 B ₦ 3,750.00 C ₦3,600.00 D ₦ 3,300.00

Solution

Amount = Principal + interest

$$SI = \frac{PTR}{100}$$

$$SI = \frac{3000 \times 5 \times 6}{100}$$

= ₦ 900

Thus amount = ₦ 3000.00 + ₦ 900
= ₦ 3,900.00 (A)

2013/ 6 Neco

What will be the simple interest on ₦ 2500.00 for 2 years at the rate of 5% per annum?

A ₦ 250.00 B ₦ 500.00 C ₦ 625.00
D ₦ 1250.00 E ₦ 6250.00

Solution

$$SI = \frac{PTR}{100}$$

Here P is ₦ 2500; T is 2; R is 5

Substituting

$$SI = \frac{2500 \times 2 \times 5}{100}$$

= ₦ 250.00 (A)

2007/ 36

At what rate percent per annum will ₦ 520.00 yield a simple interest of ₦ 39.00 in three years?

A 4% B $3^1/_2$% C 3% D $2^1/_2$%

Solution

$$SI = \frac{PTR}{100}$$

Here SI is ₦ 39, P is ₦ 520, T is 3, R is?

Substituting

$$39 = \frac{520 \times 3 \times R}{100}$$

$$39 \times 100 = 520 \times 3 \times R$$

$$\frac{39 \times 100}{520 \times 3} = R$$

$$2^1/_2\% = R \text{ (D)}$$

2007/1 Neco

Calculate the rate percent per annum at which ₦5,000.00 doubles itself in 20years

A 25% B 12% C 8% D 5% E 4%

Solution

$$SI = \frac{PTR}{100}$$

Here P is ₦ 5,000; T is 20; R is ?

SI is ₦ 5,000(i.e Double of ₦ 5,000 minus ₦ 5,000)

Substituting

$$5000 = \frac{5000 \times 20 \times R}{100}$$

$$5000 \times 100 = 5000 \times 20 \times R$$

$$\frac{5000 \times 100}{5000 \times 20} = R$$

$$5\% = R \text{ (D)}$$

2007/ 13b NABTEB

A simple interest on a sum of money invested at 4% for 4years was ₦ 4,040.00. How much was invested?

Solution

$$SI = \frac{PTR}{100}$$

Here SI is ₦ 4,040; P is?; T is 4; R is 4

Substituting

$$4{,}040 = \frac{P \times 4 \times 4}{100}$$

$$4{,}040 \times 100 = P \times 4 \times 4$$

$$\frac{4{,}040 \times 100}{4 \times 4} = P$$

₦ 25,250 = P

2008/13 NABTEB(Dec)

How long will it take ₦2,600.00 to earn ₦ 520.00 at 5% per annum simple interest?

A 14 years B 10year C 8years D 4 years

Solution

$$SI = \frac{PTR}{100}$$

Here SI is ₦ 520; P is ₦2,600; T is?; R is 5

Substituting

$$520 = \frac{2600 \times T \times 5}{100}$$

$$520 \times 100 = 2600 \times T \times 5$$

$$\frac{520 \times 100}{2600 \times 5} = T$$

$$4\text{years} = T \text{ (D)}$$

2010/29 NABTEB (Dec)

After how many years will 6 million naira yield an interest of ₦ 860,000 at the rate of 10% per annum?

A 1.4 years B 4 years, 8 months C 48 years D 96 years

Solution

$$SI = \frac{PTR}{100}$$

Here SI is ₦ 860,000; P is ₦ 6,000,000; T is?; R is10

Substituting

$$860\,000 = \frac{6000000 \times T \times 10}{100}$$

$$860\,000 \times 100 = 6\,000\,000 \times T \times 10$$

$$\frac{860000 \times 100}{6000\,000 \times 10} = T$$

$$1.43 = T \quad \text{i.e 1.4years (A)}$$

2010/5 NABTEB (Dec)

Find the amount if simple interest is paid on ₦34320 in 5 years at $6^1/_4$ % per annum.

A ₦ 45045 B ₦ 43043 C ₦ 23595 D ₦ 10725

Solution

Amount = Principal + (simple) interest

$$SI = \frac{PTR}{100}$$

Here SI is? P is ₦ 34,320; T is 5; R is $6^1/_4$ i.e $^{25}/_4$

Substituting

$$SI = \frac{34320 \times 5 \times 25}{100 \times 4}$$

= ₦ 10,725

Thus, Amount = ₦ 34320 + ₦ 10,725

= ₦ 45,045 (A)

2007/ 1b

A man invested ₦ 20,000in bank A and ₦ 25,000 in bank B at the beginning of a year. Bank A pays simple interest at a rate of y% per annum and B pays 1.5y% per annum. If his total interest at the end of the year from the two banks was ₦ 4,600, find the value of y

Solution

Bank A	Bank B
P is ₦20,000	P is ₦25,000
R is y%	R is 1.5y
Time T is 1year	T is 1 year

(Reason beginning of year to the end implies one year)

$$₦\,4{,}600 = SI_A + SI_B \quad \textit{SI is given as } \frac{PTR}{100}$$

$$4600 = \frac{20000 \times 1 \times y}{100} + \frac{25000 \times 1 \times 1.5y}{100}$$

$$4600 = 200y + 250 \times 1.5y$$

$$4600 = 200y + 375y$$

$$4600 = 575y$$

$$y = \frac{4600}{575}$$

$$y = 8\%$$

2008/ 5 Counter example

p naira invested for 4years at r % simple interest per annum yields 0.36p naira interest. Find the value of r

A $1^1/_9$ B $1^4/_9$ C 9 D 11

Solution

$$SI = \frac{PTR}{100}$$

$$0.36p = \frac{p \times 4 \times r}{100}$$

Our target is to make r subject of formula

$$0.36p \times 100 = 4pr$$

$$\frac{36p}{4p} = r$$

$$9 = r \text{ (C)}$$

2005/45

A man took a loan of \$P at the rate of 4% per annum simple interest. If at the end of 5years he paid back \$720, find the value of P?

A \$600 B \$556 C \$500 D \$456

Solution

$$SI = \frac{PTR}{100}$$

Recall that what he paid back is amount.

Amount = Principal + interest

720 = P + interest

Thus interest is 720 – P

Substituting

$$720 - P = \frac{P \times 5 \times 4}{100}$$

$$(720 - P)\,100 = 20P$$

$$72000 - 100P = 20P$$

$$72000 = 20P + 100P$$

$$72000 = 120P$$

$$\frac{72000}{120} = P;$$

$$P = \$600 \text{ (A)}$$

2009/ 6 Neco (Dec)

If the simple interest on ₦ 4,500.00 for 3 years is ₦ 540.00, find the simple interest on ₦ 6,500.00 for 2 years at the same rate.

A ₦ 390.00 B ₦ 520.00 C ₦ 540.00
D ₦ 590.00 E ₦ 780.00

Solution

We are to find *rate* from the *first case* of SI

$$SI = \frac{PTR}{100}$$

Here P is ₦ 4,500; T is 3; R is ? SI is 540

Substituting

$$540 = \frac{4500 \times 3 \times R}{100}$$

$$540 \times 100 = 4500 \times 3 \times R$$

$$\frac{540 \times 100}{4500 \times 3} = R$$

$$4\% = R$$

In the *second case*: P is ₦ 6,500, T is 2 and R is 4.

Substituting

$$SI = \frac{6500 \times 2 \times 4}{100} = ₦\,520 \text{ (B)}$$

2009/ 5 Neco

A man deposited ₦14,600.00 in a Micro- finance Bank on 16th July and on 9th of October the same year he withdrew his money. Find the accrued amount if the bank pays an interest of $8^1/_2$% per annum

Solution

First we find the simple interest

The deposit was kept in the bank for 3 months

16 July to 9 October and not 4 months.

$$SI = \frac{PTR}{100}$$

P is ₦ 14, 600, T is $\frac{3}{12}$ years, R is $8^1/_2$ i.e $\frac{17}{2}$

Substituting

$$SI = \frac{14{,}600 \times 3 \times 17}{100 \times 12 \times 2}$$

$$= ₦\,310.25$$

Accrued Amount = principal + interest

= ₦ 14,600.00 + ₦ 310.25

= ₦14,910.25

2011/7a Neco

A civil servant obtained a loan of ₦350,000.00. He is expected to pay back the loan over 4years with a simple interest at a rate of 5 % per annum.

i. Calculate the simple interest on the loan
ii. How much did he pay back?
iii. If this amount is paid back in monthly installments, how much did he pay back each month?

Solution

(i) $SI = \frac{PTR}{100}$

Here P is ₦ 350,000; T is 4; R is 5

Substituting

$$SI = \frac{350\,000 \times 4 \times 5}{100}$$

$$= ₦\,70{,}000.00$$

(ii) it refer to Amount = principal + interest

= ₦ 350,000 + ₦ 70,000

= ₦ 420,000

(iii) Time T is 4 years.

We have 12 monthly installments payment

$$= \frac{420{,}000}{4 \times 12}$$

= ₦ 8,750 per month

2005/ 41 Neco Exercise 2.29

Find the rate at which ₦ 327.50 will yield ₦ 78.60 in 6 years.

A 7% B 6% C 5% D 4% E 3%

2006/8 Neco (Dec) Exercise 2.30

If ₦10,000.00 is kept in a bank at the interest rate of $12^1/_2$% per annum, how long will it take the money to yield an interest of ₦ 2,500.00?

A 1 yr B $1^1/_2$ yrs C 2yrs D $2^1/_2$ yrs E 3 yrs

2007/5 Neco Exercise 2.31

Find the simple interest on ₦ 2970.00 in 12 years at the rate of 6%

A ₦ 4125.00 B ₦ 2238.40 C ₦ 2138.40
D ₦ 2069.20 E ₦ 1069.50

2010/8 Neco Exercise 2.32

A man borrows ₦16,000.00 on condition that he pays back ₦16,900.0 after 9months. At what rate percent per annum is interest charged?

A $^2/_{15}$ % B $1^7/_8$ % C $2^1/_2$% D $5^5/_8$ % E $7^1/_2$%

2011/10 Neco Exercise 2.33
If ₦15,000.00 amount to ₦20,000.00 in two years at simple interest, what is the rate of interest?
A $12^1/_2$% B $16^2/_3$ % C $37^1/_2$ % D $66^2/_3$ % E 67 %

2014/ 7 Neco Exercise 2.34
Find the simple interest on ₦ 700.00 for 9 years at 3% per annum.
A ₦ 210.00 B ₦ 189.00 C ₦ 100.00
D ₦ 76.00 E ₦ 63.00

2006/ 7 (Nov) Exercise 2.35
At what rate percent will ₦ 4,800.00 amounts to ₦ 5,040.00 for $2^1/_2$ years simple interest?
A 5% B 3% C $2^1/_2$% D 2%

2009/3(Nov) Exercise 2.36
Find the simple interest on ₦ 5400.00 for 10 months at 5% per annum
A ₦ 207.00 B ₦ 208.00 C ₦ 225.00 D ₦ 228.00

2001/8a (Nov) Exercise 2.37
Find the simple interest on ₦ 3000.00 in $3^1/_2$ years at 4% per annum. Give your answer to the nearest Naira.

2012/7 Exercise 2.38
If ₦ 2,500.00 amounts to ₦ 3,500.00 in 4 years at simple interest, find the rate at which the interest was charged.
A 5% B 7 $^1/_2$% C 8% D 10%

Compound interest
It can be calculated in two ways:

Formula method
We use the amount formula stated earlier
Compound interest = amount – principal
$$= P\left(1 + \frac{R}{100}\right)^n - P$$
Where P is principal, R is rate and n is time
$$\textit{Note:}\ A = P\left(1 + \frac{R}{100}\right)^n$$

Without the use of formula (non use of amount formula)
The use of simple interest calculation for each year is vital here. Let us study the preceding examples carefully.

Example 1
The compound interest on ₦ 500.00 for 2 years at 6% per annum is ….
A. ₦ 30.00 B. ₦ 31.00 C. ₦ 61.80
D. ₦ 91.80 E. ₦ 92.80.

Solution

Without the use of formula (non use of amount formula)
We shall have two different calculations since two years was mentioned
Apply the formula $I = \frac{PRT}{100}$, We have that
1^{st} year interest is initial "simple" interest from the question.
$$I_1 = \frac{500 \times 6 \times 1}{100}$$
$= 5 \times 6$
$=$ ₦ 30

Note: The use of 1 for Time not 2 as the conventional simple interest.

Amount for 1^{st} year = Principal + interest
(New principal) = ₦ 500 + ₦ 30
= ₦ 530

2^{nd} year interest
$$I_2 = \frac{530 \times 6 \times 1}{100}$$
Note: the use of 1 year not 2 years.
$= \frac{53 \times 6}{10}$
$= \frac{318}{10}$
= ₦ 31.8.

Amount for 2^{nd} years = ₦ 530 + ₦ 31.8
(Last principal) = ₦ 561.8.
∴ Compound interest = Amount – initial principal
= ₦ 561.8 – ₦ 500
= ₦ 61.80 (C)

Alternatively

1^{st} year :	Principal	₦ 500	
	6% interest	+ 30	($^6/_{100}$ x 500)
2^{nd} year	Principal	₦ 530.00	
	6% interest	+ 31.80	($^6/_{100}$ x 530)
		₦ 561.80	

Compound interest = ₦ 561.80 – ₦ 500
= ₦ 61.80 (C)

Formula method
Compound interest = amount – principal
$$= P\left(1 + \frac{R}{100}\right)^n - P$$
Here : P is 500 , R is 6 , n is 2 Substituting,

Compound interest = 500 (1 + $^6/_{100}$ $)^2$ - 500
=500 (1 + 0.06 $)^2$ - 500
= 500 (1.06 $)^2$ - 500
= 561.8 – 500
= ₦ 61.8 (C)

Example 2
Find the compound interest on ₦ 400 for 2 years at 8% per annum.
A. ₦ 32.00 B. ₦ 34.56 C. ₦ 66.56
D. ₦ 432.00 E. ₦ 466.56

Solution

Without use of the formula (non use of amount formula)
Since rate for two years was mentioned in the question, we shall have two different calculations.
Applying $I = \frac{PRT}{100}$

1^{St} year interest
$$I_1 = \frac{400 \times 8 \times 1}{100}$$
Please, note the use of Time 1 year and not 2 years
= 4 x 8
= ₦ 32
Amount for 1^{st} year = ₦ 400 + ₦32 = ₦ 432
(New Principal)

2nd year Interest:

$$I_2 = \frac{432 \times 8 \times 1}{100}$$

Please, note the use of T = 1 not 2

$$= \frac{3456}{100} = ₦\,34.56$$

Amount for 2nd year = ₦ 432 + ₦ 34.56 = ₦ 466.56
(Last Principal)

∴ Compound interest = ₦ 466.56 – ₦ 400
= ₦ 66.56 (C)

Alternatively,

1st year:	Principal	₦ 400	
	8% interest	+ 32	($^8/_{100}$ x 400)
2nd year:	Principal	₦ 432	
	8% interest	+ 34.56	($^8/_{100}$ x 432)
		₦ 466.56	

Compound interest = ₦ 466.56 – ₦ 400
= ₦ 66.56 (C)

Formula method

Compound interest = amount – principal

$$= P\left(1 + \frac{R}{100}\right)^n - P$$

Here : P is 400 , R is 8 , n is 2 Substituting,

Compound interest = $400 (1 + ^8/_{100})^2 - 400$
$= 400 (1 + 0.08)^2 - 400$
$= 400 (1.08)^2 - 400$
$= 466.56 – 400$
= ₦ 66.56 (C)

If no restriction is given, we can use any of the methods in solving our compound interest problems but not both at the same time

2004/6a (Nov)

Find the compound interest (without use of formula) on ₦ 285.38 in 2years at the rate of $2^1/_2$% per annum. Give your answer to the nearest kobo.

Solution

Here we are restricted to non formula method

We shall have two different calculations since two years were mentioned

Applying the formula $I = \frac{PTR}{100}$, We have that

1st year interest is initial "simple" interest from the question.

$$I_1 = \frac{285.38 \times 1 \times 2.5}{100}$$

= ₦ 7.1345

Note: The use of 1 for Time not 2 as in the conventional simple interest.

Amount for 1st year = Principal + interest
(New principal) = ₦ 285.38 + ₦ 7.1345
= ₦ 292.5145

2nd year interest

$$I_2 = \frac{292.5145 \times 1 \times 2.5}{100}$$

= ₦ 7.3129

Note: the use of 1 year not 2 years.

Amount for 2nd year = ₦ 292.5145 + ₦ 7.3129
(Last principal) = ₦ 299.8274

∴ Compound interest = Amount- initial principal
= ₦ 299.8274 - ₦ 285.38
= ₦ 14.4474
≈ ₦ 14.45 to the nearest kobo

2008/7b

A man inverts f1500 for two years at compound interest. After one year, his money amount to f 1560. Find the;
(i) Rate of interest;
(ii) Interest for the second year.

Solution

Here we apply the non-formula method for compound interest

(i) $I = \frac{PTR}{100}$

I is f 1560 – f 1500 since interest is amount minus principal

P is f1500; R is ? T is 1

Substituting

$$60 = \frac{1500 \times R \times 1}{100}$$

$$60 = 15R$$

$$^{60}/_{15} = R$$

Thus R is 4%

(ii) 2nd year interest

Already, we are told that amount is f1560

$$I_2 = \frac{1560 \times 4 \times 1}{100}$$

$= f\ 62.40$

2011/ 4 Neco

Find the compound interest on ₦ 225.00 invested for 20 years at 4% per annum.

A ₦ 824.00	B ₦ 718.00	C ₦ 688.00
D ₦ 652.00	E ₦ 268.00	

Solution

Compound interest = amount – principal

$$= P\left(1 + \frac{R}{100}\right)^n - P$$

Here P is ₦ 225; R is 4; n is 20

Substituting

$$\text{Compound interest} = 225\left(1 + \frac{4}{100}\right)^{20} - 225$$

$= 225(1 + 0.04)^{20} - 225$
$= 225(1.04)^{20} - 225$
$= 493.00 – 225$
= ₦ 268.00 (E)

2006/4a Neco

By using formula, find the:
(i) Simple interest;
(ii) Compound interest;
On ₦ 2, 800.00 for 3 years at the rate of 4% per annum.

Solution

(i) $I = \frac{PRT}{100}$

$$= \frac{2800 \times 3 \times 4}{100} = ₦\,336.00$$

(ii) Compound interest = amount – principal

$$= P\left(1+\frac{R}{100}\right)^n - P$$

$$= 2800\left(1+\frac{4}{100}\right)^3 - 2800$$

$$= 2800(1 + 0.04)^3 - 2800$$

$$= 2800(1.04)^3 - 2800$$

= 3149.62 – 2800

= ₦ 349.62

2006/38 Neco

Find the compound interest on ₦ 450.00 in 2 years at 5%

A ₦ 497.00 B ₦ 496.13 C ₦ 495.00
D ₦ 46.13 E ₦ 45.00

Solution

Compound interest = amount – principal

$$= P\left(1+\frac{R}{100}\right)^n - P$$

Here P is ₦ 450; R is 5; n is 2

Substituting

$$\text{Compound interest} = 450\left(1+\frac{5}{100}\right)^2 - 450$$

$$= 450(1 + 0.05)^2 - 450$$

$$= 450(1.05)^2 - 450$$

= 496.125 – 450

= ₦ 46.13

2005/ 40 Neco Exercise 2.39

Calculate the compound interest on ₦ 2,000.00 for 2 years at 10% per annum.

A ₦ 420.00 B ₦ 400.00 C ₦ 320.00
D ₦ 300.00 E ₦ 200.00

2015/5b Neco Exercise 2.40

Calculate the compound interest on ₦ 25,000.00 for 3 years at 12% per annum

2014/12b NABTEB (Nov) Exercise 2.41

Find the compound interest on ₦ 450.00 in 3 years at 4% per annum

CHAPTER THREE

Algebraic Process

Algebra is the aspect of mathematics that involves both letters and numbers or letters only. Let's consider some examples below:

(a) $2 + 3 - \sqrt{16(5+4)}$
(b) $9^2 + 26 - {}^1/_4 + 1/20$
(c) $y + 1$
(d) $x + y = 2$
(e) $x^2 + 3x + 4$
(f) $x^2 - y^2$

The expression (a) and (b) are in numerical terms only. Hence, they cannot be called Algebra. Whereas (c) to (f) are algebraic expressions. The process of solving mathematical problems involving algebraic expression is called algebraic process.

Simple Equation (Linear)

Simple linear equations are expressions whose highest degree of the unknown is one. Here we shall concern ourselves with linear equations in *one variable*.

2006/8 (Nov)

Solve for x in the equation: $5(x-7) - 7x = -3(4x-5)$

A ${}^1/_5$ B ${}^5/_6$ C 2 D 5

Solution

Opening brackets

$5(x-7) - 7x = -3(4x-5)$ becomes

$5x - 35 - 7x = -12x + 15$

Collect like terms together

$5x - 7x + 12x = 15 + 35$

$10x = 50$

$x = {}^{50}/_{10}$ i.e 5 (D)

2008/16 Neco

If $5x - 3 = 4x - 7$, what is the value of 6x?

A 26 B6 C 4 D – 4 E – 24

Solution

First, we find x

$5x - 3 = 4x - 7$

$5x - 4x = 3 - 7$

$x = -4$

Thus, $6x = 6(-4)$ i.e –24 (E)

2013/ 18 Neco

If $6x + 7 = 4x - 3$, what is the value of $8x - 4$?

A – 44 B – 5 C – 1 D 36 E 44

Solution

First, we solve for x in $6x + 7 = 4x - 3$

$6x + 7 = 4x - 3$

$6x - 4x = -7 - 3$

$2x = -10$

$x = {}^{-10}/_2$ i.e -5

Substituting for $x = -5$ in $8x - 4$

$8x - 4 = 8(-5) - 4$

$= -40 - 4$ i.e -44 (A)

2006/4a NABTEB (Nov)

Solve the equation: $3(2x-7) = 2(x-8)$

Solution

Opening up brackets on both sides

$6x - 21 = 2x - 16$

$6x - 2x = 21 - 16$

$4x = 5$ Thus $x = {}^5/_4$ i.e $1{}^1/_4$

2010/40 NABTEB (Nov)

Simplify $\frac{3x}{5} + \frac{2x}{15} = \frac{3}{5}$

A $\frac{9}{11}$ B $\frac{9}{7}$ C $\frac{7}{11}$ D $\frac{8}{7}$

Solution

$$\frac{3x}{5} + \frac{2x}{15} = \frac{3}{5}$$

Multiply through by the LCM of 5 and 15 i.e 15

$3(3x) + 2x = 3(3)$

$9x + 2x = 9$

$11x = 9$

$x = \frac{9}{11}$ (A)

2006/23 Neco (Dec)

Find the value of t for which

$$\frac{1}{3}(t+5) = \frac{1}{4}(5t-2)$$

A $-2\frac{4}{11}$ B $-1\frac{3}{11}$ C $\frac{4}{11}$ D $1\frac{3}{11}$ E $2\frac{4}{11}$

Solution

$$\frac{1}{3}(t+5) = \frac{1}{4}(5t-2)$$

Multiply through by 12, LCM of 3 and 4

$4(t + 5) = 3(5t - 2)$

$4t + 20 = 15t - 6$

$20 + 6 = 15t - 4t$

$26 = 11t$

$t = {}^{26}/_{11}$ i.e $2\frac{4}{11}$ (E)

2007/14 Neco

Solve the equation $\frac{3b}{9} + \frac{1}{2} = \frac{3}{4} + \frac{b}{4}$

A – 3 B – 2 C 2 D 3 E 4

Solution

$$\frac{3b}{9} + \frac{1}{2} = \frac{3}{4} + \frac{b}{4}$$

Collect like terms together

$$\frac{3b}{9} - \frac{b}{4} = \frac{3}{4} - \frac{1}{2}$$

Multiply through by the LCM of 9, 4 and 2 i.e 36

$4(3b) - 9(b) = 9(3) - 18(1)$

$12b - 9b = 27 - 18$

$3b = 9$

$b = {}^{9}/_{3}$ i.e 3 (D)

2008/28 Neco (Dec)

Solve the equation $\frac{3}{2(x-2)} - \frac{2}{3(2-x)} = 0$

A 2 B 3 C 6 D 13 E 26

Solution

$$\frac{3}{2(x-2)} - \frac{2}{3(2-x)} = 0$$

$$\frac{3}{2(x-2)} = \frac{2}{3(2-x)}$$

Cross multiply

$9(2 - x) = 4(x - 2)$

$18 - 9x = 4x - 8$

$18 + 8 = 4x + 9x$

$26 = 13x$

$x = {}^{26}/_{13}$ i.e 2 (A)

2008/39

If $P = {}^{1}/_{2}$ and $\frac{1}{p-1} = \frac{2}{p+x}$, find the value of x

A $-2^{1}/_{2}$ B $-1^{1}/_{2}$ C $1^{1}/_{2}$ D $2^{1}/_{2}$

Solution

$$\frac{1}{p-1} = \frac{2}{p+x}$$

Substituting for $P = {}^{1}/_{2}$

$$\frac{1}{\frac{1}{2} - 1} = \frac{2}{\frac{1}{2} + x}$$

Cross multiply

${}^{1}/_{2} + x = 2(-{}^{1}/_{2})$

${}^{1}/_{2} + x = -1$

$x = -1 - {}^{1}/_{2}$

$= -1^{1}/_{2}$ (B)

2014 /6a

If $\frac{3}{2p - \frac{1}{2}} = \frac{\frac{1}{3}}{\frac{1}{4}p + 1}$, find p

Solution

First, we resolve the fraction inside fraction

$$\frac{3}{\frac{4p-1}{2}} = \frac{\frac{1}{3}}{\frac{p+4}{4}}$$

$$3 \div \frac{4p-1}{2} = \frac{1}{3} \div \frac{p+4}{4}$$

Changing division to multiplication

$$3 \times \frac{2}{4p-1} = \frac{1}{3} \times \frac{4}{p+4}$$

$$\frac{6}{4p-1} = \frac{4}{3(p+4)}$$

Cross multiply to clear fraction

$6 \times 3(p + 4) = 4(4p - 1)$

$18p + 72 = 16p - 4$

$18p - 16p = -72 - 4$

$2p = -76$

$p = {}^{-76}/_{2}$ i.e -38

2009/2a (Nov)

Solve the equation $\frac{4x-1}{3} - \frac{3x-1}{2} = \frac{5-2x}{4}$

Solution

$$\frac{4x-1}{3} - \frac{3x-1}{2} = \frac{5-2x}{4}$$

Multiply through by the LCM of 3, 2 and 4 i.e 12

$$4(4x-1)-6(3x-1) = 3(5-2x)$$
$$16x-4-18x+6 = 15-6x$$
Collect like terms together
$$16x-18x+6x = 15+4-6$$
$$4x = 13$$
$$x = {}^{13}/_4 \text{ i.e } 3^1/_4$$

2009/13 (Nov)

Solve for x if $\frac{2}{x} + \frac{3}{2x} = \frac{3}{2}$

A $\frac{7}{6}$ B $\frac{3}{7}$ C $\frac{6}{7}$ D $\frac{7}{3}$

Solution

$$\frac{2}{x} + \frac{3}{2x} = \frac{3}{2}$$

Find the LCM of x, 2x and 2

x	x	2x	2
2	1	2	2
	1	1	1

L C M of x, 2x and 2 is 2x

Multiplying through by 2x

$$2x\frac{2}{x} + 2x\frac{3}{2x} = 2x\frac{3}{2}$$
$$2(2) + 3 = 3x$$
$$4+3 = 3x$$
$$7 = 3x$$
$$x = {}^7/_3 \quad (D)$$

2008/27 NABTEB (Nov)

Solve for x if $\frac{5}{m} + \frac{8}{3m} = 2$

A $3^5/_6$ B $2^1/_6$ C $1^5/_6$ D $1^1/_2$

Solution

$$\frac{5}{m} + \frac{8}{3m} = 2$$

Find the LCM of 3m and m

m	m	3m
3	1	3
	1	1

L C M of m and 3m is 3m

Multiplying through by 3m

$$3m\frac{5}{m} + 3m\frac{8}{3m} = 3m \times 2$$
$$3(5) + 8 = 3m(2)$$
$$15+8 = 6m$$
$$23 = 6m$$
$$m = \frac{23}{6} \text{ i.e } 3^5/_6 \quad (A)$$

2012/16 Neco

If $\frac{5}{x} = 4 - \frac{7}{x}$, find the value of x

A – 1 B $^1/_3$ C $^1/_2$ D 1 E 3

Solution

$$\frac{5}{x} = 4 - \frac{7}{x}$$

Collect like terms together

$$\frac{5}{x} + \frac{7}{x} = 4$$

Multiply through by x (LCM)

$$x\frac{5}{x} + x\frac{7}{x} = x \times 4$$
$$5+7 = 4x$$
$$12 = 4x$$
$$^{12}/_4 = x \text{ i.e } x = 3 \quad (E)$$

2009/10b Neco (Dec) ***counter example***

Solve the equation $\frac{2}{x-1} - \frac{3}{x+1} = 1$

Solution

Apply LCM method to the LHS

$$\frac{2(x+1)-3(x-1)}{(x-1)(x+1)} = 1$$
$$\frac{2x+2-3x+3}{(x-1)(x+1)} = 1$$
$$\frac{-x+5}{(x-1)(x+1)} = 1$$

Cross multiply

$$-x+5 = (x-1)(x+1)$$
$$-x+5 = x^2+x-x-1$$
$$-x+5 = x^2-1$$
$$x^2+x-6 = 0$$

Factorizing

$$x^2-2x+3x-6 = 0$$
$$x(x-2)+3(x-2) = 0$$
$$(x-2)(x+3) = 0$$
$$x-2 = 0 \text{ or } x+3 = 0$$
$$x = 2 \text{ or } -3$$

2005/31 Neco ***Special case***

If $\frac{3y-4x}{4y-3x} = 5$, evaluate $\frac{x}{y}$

A $\frac{17}{11}$ B $\frac{16}{11}$ C $\frac{16}{17}$ D $\frac{14}{17}$ E $\frac{3}{4}$

Solution

$$\frac{3y-4x}{4y-3x} = 5$$

Cross multiply

$$3y-4x = 5(4y-3x)$$
$$3y-4x = 20y-15x$$

Collect like terms together

$$15x-4x = 20y-3y$$
$$11x = 17y$$

Divide through by y

$$\frac{11x}{y} = 17$$
$$\frac{x}{y} = \frac{17}{11} \quad (A)$$

Alternatively

$$\frac{3y-4x}{4y-3x}=5$$

Divide LHS through by y

$$\frac{\frac{3y}{y}-\frac{4x}{y}}{\frac{4y}{y}-\frac{3x}{y}} = 5$$

$$\frac{3-4\frac{x}{y}}{4-3\frac{x}{y}}=5$$

Cross multiply

$$3-4\frac{x}{y} = 5(4-3\frac{x}{y})$$

$$3-4\frac{x}{y} = 20-15\frac{x}{y}$$

Collect like terms together

$$15\frac{x}{y}-4\frac{x}{y} = 20-3$$

$$11\frac{x}{y} = 17$$

$$\frac{x}{y} = \frac{17}{11} \quad (\text{ A })$$

2006/27 Neco (Dec)

If $\frac{2a-3b}{3a-2b}=4$, evaluate $\frac{a^2+b^2}{2ab}$

A 3 B 2 C $1\frac{1}{4}$ D $\frac{2}{3}$ E $\frac{4}{7}$

Solution

$$\frac{2a-3b}{3a-2b}=4$$

Cross multiply

$2a-3b = 4(3a-2b)$

$2a-3b = 12a-8b$

Collect like terms together

$8b-3b = 12a-2a$

$5b = 10a$

$$b = \frac{10a}{5}$$

Substituting, we have

$$\frac{a^2+b^2}{2ab} = \frac{a^2+\left(\frac{10a}{5}\right)^2}{2a\times\left(\frac{10a}{5}\right)}$$

$$= \frac{a^2+\frac{100a^2}{25}}{\frac{20a^2}{5}}$$

$$= \frac{25a^2+100a^2}{25}\div\frac{20a^2}{5}$$

$$= \frac{125a^2}{25}\times\frac{5}{20a^2}$$

$$= \frac{5}{4} \text{ i.e } 1\tfrac{1}{4} \quad (\text{ C })$$

Note: Author deliberately left $^{10a}/_5$ unreduced to 2a as readers may come across problems which will not reduce to whole number

2014/25 Neco

If $\frac{1}{x} + \frac{1}{y} = \frac{1}{3}$, find y if x = 5

A $6\frac{1}{2}$ B $7\frac{1}{2}$ C $8\frac{1}{2}$ D $9\frac{1}{3}$ E $9\frac{1}{2}$

Solution

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{3}$$

Substituting for x = 5

$$\frac{1}{5} + \frac{1}{y} = \frac{1}{3}$$

Collect like terms together

$$\frac{1}{y} = \frac{1}{3} - \frac{1}{5}$$

$$\frac{1}{y} = \frac{5-3}{15}$$

$$\frac{1}{y} = \frac{2}{15}$$

$15 = 2y$

$$\frac{15}{2} = y \quad \text{i.e} \quad y = 7\tfrac{1}{2} \; (\text{ B })$$

2009/1b

If 9 is subtracted from half a certain number, the result is the same as when $\frac{1}{3}$ is added to the number. Find the number.

Solution

Let the number be x

1st statement : $\frac{1}{2}x - 9$

2nd statement : $\frac{1}{2}x - 9 = x + \frac{1}{3}$

Solving: $\frac{1}{2}x - 9 = x + \frac{1}{3}$

Multiply through by the LCM of 2 and 3 i.e 6

$3x - 54 = 6x + 2$

Collect like terms together

$3x - 6x = 54 + 2$

$-3x = 56$

$$x = \frac{56}{-3} \text{ i.e } -18\tfrac{2}{3}$$

2006/7a Neco Exercise 3.1

Solve the equation: $\frac{2}{3}(3x-5)-\frac{3}{5}(2x-3)=3$

2009/35 Exercise 3.2

Solve $\frac{2x+1}{6} - \frac{3x-1}{4} = 0$

A 1 B $\frac{1}{5}$ C $-\frac{1}{5}$ D -1

2011/2 Exercise 3.3

Solve for x in the equation: $\frac{3}{5}(2x-1)=\frac{1}{4}(5x-3)$

A 0 B 1 C 2 D 3

2012/26 Neco Exercise 3.4

Solve $\frac{2x-1}{3} + \frac{4x+1}{5} = \frac{3(x-4)}{4}$

2008/46 NABTEB (Nov) Exercise 3.5

Solve for x if $\frac{x}{4} = \frac{x}{6} + \frac{2}{3}$

A 3 B 4 C 6 D 8

2008/21 NABTEB (Nov) Exercise 3.6

Solve the equation $\left(\frac{m}{2}-\frac{2}{3}\right)\left(m+\frac{2}{3}\right)$

A – 6 B – 5 C – 4 D – 3

2013/4b Exercise 3.7

Solve the equation: $\frac{(x+5)}{3} - \frac{(2x-1)}{5} = \frac{5}{6}$

2009/9 Exercise 3.8

If $\frac{3}{2x} - \frac{2}{3x} = 4$, solve for x

A $\frac{4}{5}$ B $\frac{4}{13}$ C $\frac{5}{24}$ D $\frac{13}{24}$

2012/8 Neco Exercise 3.9

Solve for x in the equation: $\frac{1}{x} + \frac{2}{3x} = \frac{1}{3}$

A 5 B 4 C 3 D 1

2015/2b Neco Exercise 3.10

Solve the equation : $\frac{1}{3}(2x + 1) - \frac{2}{5}(x-2) = 3$

2015/7 Neco Exercise 3.11

If $12x - 5 = 9x - \frac{7}{2}$, find 10x

A. – 5 B. – $\frac{1}{2}$ C. $\frac{1}{2}$ D. 5 E. 12

2014/8b Neco (Nov) Exercise 3.12

Solve the equation : $\frac{2x+5}{3} - \frac{3x+2}{4} = \frac{1}{6}$

Expansion of algebraic terms

The term expansion in algebraic mathematics is the same as in arithmetic. It simply means opening up of barriers (brackets). The expression

(t + s) (v + w)

Means (t + s) x (v + w). it can be expanded as:

$(t+s)(v+w) = t(v+w) + s(v+w)$
$= tv + tw + sv + sw$

on the other hand, we can expand it as:

$(t+s)(v+w) = (t+s)v + (t+s)w$
$= tv + sv + tw + sw$

2005/30 Neco

Expand $2x(x-y) - 3y(4x+3y)$

A $2x^2 - 14xy + 9y^2$ B $2x^2 - 14xy - 9y^2$
C $2x^2 - 10xy + 9y^2$ D $2x^2 + 14xy - 9y^2$ E $2x^2 - 10xy - 9y^2$

Solution

$2x(x-y) - 3y(4x+3y) = 2x^2 - 2xy - 12xy - 9y^2$
$= 2x^2 - 14xy - 9y^2$ (B)

2007/43

Expand the expression $(3a - xy)(3a + xy)$

A $9a^2 - x^2y^2$ B $9a^2 + x^2y^2$ C $9a^2 - xy$ D $9a^2 + x^2y$

Solution

$(3a-xy)(3a+xy) = 3a(3a+xy) - xy(3a+xy)$
$= 9a^2 + 3axy - 3axy - x^2y^2$
$= 9a^2 - x^2y^2$ (A)

2010 /16 Neco

If $x^2 - 5x + c = (x-8)(x+3)$. Find the value of c

A 24 B 5 C – 5 D – 9 E – 24

Solution

$x^2 - 5x + c = (x-8)(x+3)$

Expanding RHS

$= x(x+3) - 8(x+3)$
$= x^2 + 3x - 8x - 24$
$x^2 - 5x + c = x^2 - 5x - 24$

Thus $c = -24$ (E)

2005/5b

If (m + 1) and (m – 3) are factors of $m^2 - km + c$, find the values of k and c

Solution

If (m + 1) and (m – 3) are factors of $m^2 - km + c$, then

$(m+1)(m-3) = m^2 - km + c$
$m(m-3) + 1(m-3) = m^2 - km + c$
$m^2 - 3m + m - 3 = m^2 - km + c$
$m^2 - 2m - 3 = m^2 - km + c$

Thus $-3 = c$ and $-2m = -km$
$2 = k$

2011/17 Neco Exercise 3.13

What is the common factor of the expression $y^2 + y$ and $y^2 - 1$?

A 2y B y C y – 1 D y + 1 E 2y – 1

Coefficient of term in expansion

(a) Find the coefficient of w in the expansion of (8w + 3)(8w – 3)

Solution

$(8w+3)(8w-3) = 8w(8w-3) + 3(8w-3)$
$= 64w^2 - 24w + 24w - 9$
$= 64w^2 - 9$

Hence the coefficient of w is zero.

(b) Find the coefficient of xy in the expansion
$(2x - y + 4)(5x + 3y)$

Solution

Using the smaller bracket of the two terms to multiply,
$(2x - y + 4)(5x + 3y) = 5x(2x - y + 4) + 3y(2x - y + 4)$
$= 10x^2 - 5xy + 20x + 6xy - 3y^2 + 12y$
Collect like terms together
$= 10x^2 - 5xy + 6xy + 20x + 12y - 3y^2$
$= 10x^2 + xy + 20x + 12y - 3y^2$
Thus the coefficient of xy is + 1

2007/15 Neco
What is the coefficient of x in the expansion $(2x - 3)(3x + 8)$?
A – 25 B – 24 C 6 D 7 E 25

Solution

First, we expand as:
$(2x - 3)(3x + 8) = 3x(2x - 3) + 8(2x - 3)$
$= 6x^2 - 9x + 16x - 24$
$= 6x^2 + 7x - 24$
Thus the coefficient of x is 7 (D)

2012/35
Find the coefficient of m in the expansion of

$$\left(\frac{m}{2} - 1\tfrac{1}{2}\right)\left(m + \frac{2}{3}\right)$$

A $-\frac{1}{6}$ B $-\frac{1}{2}$ C –1 D $-1\frac{1}{6}$

Solution

$$\left(\frac{m}{2} - 1\tfrac{1}{2}\right)\left(m + \frac{2}{3}\right) = \left(\frac{m}{2} - \frac{3}{2}\right)\left(m + \frac{2}{3}\right)$$

Expanding, we have

$$= m\left(\frac{m}{2} - \frac{3}{2}\right) + \frac{2}{3}\left(\frac{m}{2} - \frac{3}{2}\right)$$

$$= \frac{m^2}{2} - \frac{3}{2}m + \frac{m}{3} - 1$$

$$= \frac{m^2}{2} + \left(\frac{-9m + 2m}{6}\right) - 1$$

$$= \frac{m^2}{2} - \frac{7}{6}m - 1$$

Thus the coefficient of m is $-\frac{7}{6}$ i.e $-1\frac{1}{6}$ (D)

2008/24 Neco Exercise 3.14
Find the coefficient of xy in the expansion of
$(x - 4y)(3x + 2y)$
A 14 B 12 C 10 D – 10 E – 12

2009/21 Neco (Dec) Exercise 3.15
Find the coefficient of x in the expansion of
$(3x + 2)(5x - 8)$?
A – 16 B – 15 C – 14 D 1 E 14

2013/ 27 Neco Exercise 3.16
In the expansion of $(3x^2 - 2x + 7)(2x + 3)$, what is the coefficient of x^2
A – 5 B 3 C 5 D 6 E 8

Factorization

The term factorization is the reverse of expansion, the subtopic preceding this. The process of factorization involves factoring out a **common factor** from a given expression. This is done based on the given problem. Different strokes for different folks. Some methods are as shown next:

One term factor
When confronted with factorization problems of the forms shown below, it is helpful if we factor out a letter or term common to both sidesz
Examples
a.) $ax + ay = a(x + y)$
b.) $at^2 - a^2t = at(t - a)$
c.) $xy^2z^3 - x^2yz^2 = xyz^2(yz - x)$
d.) $s(t - 1) + 3(1 - t) = s(t - 1) - 3(t - 1)$
$= (t - 1)(s - 3)$

2011/28
Factorize the expression: $am + bn - an - bm$
A $(a - b)(m + n)$ B $(a - b)(m - n)$
C $(a + b)(m - n)$ D $(a + b)(m + n)$

Solution

Grouping like terms together
$am + bn - an - bm = am - bm + bn - an$

$= m(a - b) + n(b - a)$
We will not get the desired result; so we factor – n instead of + n

$= m(a - b) - n(-b + a)$
Same as $= m(a - b) - n(a - b)$
$= (m - n)(a - b)$ B

2010/2
Factorize : $x + y - ax - ay$
A$(x - y)(1 - a)$ B$(x + y)(1 + a)$ C$(x + y)(1 - a)$ D$(x - y)(1 + a)$

Solution

Group like terms together
$x + y - ax - ay = x - ax + y - ay$
$= x(1 - a) + y(1 - a)$
$= (x + y)(1 - a)$ C

2005/11
Factorize: $x - ay + y - ax$
A$(1 - a)(x - y)$ B$(1 + a)(x - y)$ C$(1 + a)(x + y)$ D$(1 - a)(x + y)$

Solution

Group like terms together
$x - ay + y - ax = x - ax - ay + y$

$= x(1 - a) - y(a - 1)$
We will not get the desired result; so we factor + y instead of – y

$= x(1 - a) + y(-a + 1)$
Same as $= x(1 - a) + y(1 - a)$
$= (x + y)(1 - a)$ D

2005/45 (Nov)
Factorize: $3xg - hy + 3hg - xy$
A $(3g + y)(x + h)$ B $(3g - y)(x + h)$
C $(3x - y)(3g + h)$ D $(x - h)(3g - y)$

Solution

Grouping like terms together
$3xg - hy + 3hg - xy = 3xg + 3hg - hy - xy$
$= 3g(x + h) - y(h + x)$
Same as $= 3g(x + h) - y(x + h)$
$= (3g - y)(x + h)$ B

2008/14 NABETEB (Nov) Exercise 3.17

Factorize ab – ad + cb – cd

A (a – c)(b – d) B (a – c)(b + d)

C (a + c)(b – d) D (a + c)(b + d)

2008/45 NABETEB (Nov) Exercise 3.18

Factorize the expression 4abxy – 4acxy + 4dxy

A 4y(ab – ac + d) B 4x(ab – ac + d)

C 4xy(ab – ac + d) D 4xy(b – a + ad)

2015/8 Exercise 3.19

Factorise completely 6ax – 12by – 9ay + 8bx

A.(2a – 3b)(4x + 3y) B. (3a + 4b)(2x – 3y)

C.(3a – 4b)(2x + 3y) D.(2a + 3b)(4x –3y)

2014/18

If (x – a) is a factor of $bx - ax + x^2 - ab$, find the other factor

A (x + b) B (x – b) C (a + b) D (a – b)

Solution

Grouping like terms together

$bx - ax + x^2 - ab = x^2 - ax - ab + bx$

$= x(x-a) - b(a-x)$

Step cannot help us

$= x(x - a) + b(-a + x)$

Same as $= x(x - a) + b(x - a)$

$= (x - a)(x + b)$

Thus, the other factor is (x + b) A.

2011/19 Neco

Factorize $5xy + 90qy - 30y^2 - 15xq$

A 5(y – 3q)(x – 8y) B 5(y – 3q)(x + 6y)

C 5(y + 3q)(x + 8y) D 5(y – 3q)(x – 6y)

E 5(y + 3q)(x – 6y)

Solution

Group like terms together

$5xy + 90qy - 30y^2 - 15xq = 90qy - 30y^2 + 5xy - 15xq$

$= 30y(3q - y) + 5x(y - 3q)$

Our purpose is not served in this step

Retaking the step $= 30y(3q - y) - 5x(-y + 3q)$

Which is the same as $= 30y(3q - y) - 5x(3q - y)$

$= (30y - 5x)(3q - y)$

The first bracket can be factorized as

$= 5(6y - x)(3q - y)$ D

2010/41 NABTEB (Nov)

What result do you get when $6x^2 + 18x + 12xy - 24x^2y^2$ is factorized?

A $(6x + x)(3y + 2 - 4xy^2)$ B $6x(x + 3 + 2y - 4xy^2)$

C $(6x + 3y)(2y - 4x^2y)$ D $(6x + 2y - 4x)(3y - xy)$

Solution

$6x^2 + 18x + 12xy - 24x^2y^2 = 6x(x + 3 + 2y - 4xy^2)$ B

If we try $6x^2 + 18x + 12x - 24x^2y^2$

$= 6x(x + 3) + 12y(x - 2x^2y)$ *dead end*

2009/11 (Nov)

Factorize completely : $x^2 - x - ax + a$

A (x – a)(x – 1) B (x – a)(x + 1)

C (x – a)(x – 1) D (x + a)(x + 1)

Solution

$x^2 - x - ax + a = x(x - 1) - a(x - 1)$

$= (x - a)(x - 1)$ A

2009/14 Exercise 3.20

Factorize $x^2 - 2x - 3xy + 6y$ completely

A (x – 2)(3x – y) B (x – 3y)(x – 2)

C (x – 2)(x + 3y) D (3x + y)(x – 2)

2011/8 Exercise 3.21

One of the factors of $(mn - nq - n^2 + mq)$ is (m – n)

The other factor is

A (n – q) B (q – n) C (n + q) D (q – m)

2014/3 NABTEB (Nov) Exercise 3.22

If (x – a) is a factor of $bx - ax + x^2 - ab$, what is the other factor?

A.(x + b) B.(b – a) C. (x – a) D. (a + b)

2014/1a (Nov) Exercise 3.23

Factorise completely : $m^2 - 2mn + n^2 - 9r^2$

Difference of two squares expressions

These are expressions of the form

$$x^2 - y^2 = (x - y)(x + y)$$

2014/53 Neco

Factorize $(25y^2 - 4)$

A (5y – 2)(5y + 2) B (2y – 5)(5y + 2)

C (5y – 2)(5y – 2) D (5y + 2)(5y + 2) E (5y – 2)(2 – 5y)

Solution

Working with an eye of difference of two squares

$25y^2 - 4 = 5^2y^2 - 2^2$

$= (5y)^2 - 2^2$

$= (5y - 2)(5y + 2)$ A

2012/25 Neco

Factorize $8ab^2 - 32a^2$

A $4a(b - 2a)(b + 2a)$ B $4a(b - 2a)^2$

C $4a(b + 2a)^2$ D $4a(b + 2a)^3$ E $4a(b - 2a)^3$

Solution

Working with an eye of difference of two squares

$8ab^2 - 32a^3 = 2a(4b^2 - 16a^2)$

$= 2a[(2b)^2 - (4a)^2]$

$= 2a(2b - 4a)(2b + 4a)$

Factor out 2

$= 4a(b - 2a)(b + 2a)$ A

2009/22 Neco (Dec)

Factorize $(5d - 2e)^2 - 9e^2$

A (5d – 2e)(3d – 3e) B 5(d + e)(5d + e)

C 5(d – e)(5d + e) D (5d – 3e)(3d + 3e) E (5d – e)(3d – e)

Solution

Working with an eye of difference of two squares

$(5d - 2e)^2 - 9e^2 = (5d - 2e)^2 - (3e)^2$

$= [(5d - 2e) - 3e][(5d - 2e) + 3e]$

$= (5d - 2e - 3e)(5d - 2e + 3e)$

$= (5d - 5e)(5d + e)$

$= 5(d - e)(5d + e)$ *C*

2007/16 Neco

Factorize $50k^4 - 8b^2$

A $(5k^2 - 2b)(5k^2 + 2b)$ B $2(5k^2 - 2b)(5k^2 + 2b)$

C $(5k^2 - 2b)(5k^2 - 2b)$ D $2[(5k^2 + 2b)(5k^2 + 2b)]$

E $2[(5k^2 - 2b)(5k^2 + 2b)]$

Solution

Working with an eye of difference of two squares

$50k^4 - 8b^2 = 2(25k^4 - 4b^2)$

$= 2[(5k^2)^2 - (2b)^2]$

$= 2[(5k^2 - 2b)(5k^2 + 2b)]$ E

2011/3b Neco

Use the idea of difference of two squares to find the exact value of k in the equation : $24k = 106^2 - 94^2$

Solution

$24k = (106 - 94)(106 + 94)$

$24k = (12)(200)$

$24k = 2400$

$k = \frac{2400}{24}$

$k = 100$

2007/6a NABTEB

Factorize completely: $(x^2 + x)^2 - (2x + 2)^2$

Solution

Applying difference of two squares

$(x^2 + x)^2 - (2x + 2)^2$

$= [(x^2 + x) + (2x + 2)] [(x^2 + x) - (2x +2)]$

$= (x^2 + x + 2x + 2)(x^2 + x - 2x - 2)$

$= (x^2 + 3x + 2)(x^2 - x - 2)$

Factorizing each quadratic expression

$= (x^2 + 2x + x +2)(x^2 - 2x + x - 2)$

$= [x(x + 2) + 1(x + 2)] [x(x - 2) + 1(x - 2)]$

$= [(x + 2)(x +1)] [(x - 2)(x + 1)]$

$= (x + 2)(x +1)(x - 2)(x + 1)$

2006/10 (Nov) Exercise 3.24

Factorize completely : $147x^2 - 3y^2$

A $3x(49x - y)$ B $3xy(49x - y)$ C $3(7x - y)^2$

D $3(7x + y)(7x - y)$

2005/2a(i) Neco Exercise 3.25

Factorize $25a^2 - 9b^2$

1977/2a Exercise 3.26

If $13V = 573^2 - 560^2$, find the value of V

2014/39 (Nov) Exercise 3.27

Find the common factor of $(9r^2 - 16s^2)$ and $(12r + 16s)$

A.$4(3r + 4s)$ B.$4(3r - 4s)$ C.$(3r - 4s)$ D.$(3r + 4s)$

2014/52 Neco Exercise 3.27b

Simplify $(4x + 1)^2 - (3x - 1)^2$

A $7x (x - 2)$ B $7x (x + 2)$ C $- 7x (x - 2)$

D $- 7x (x + 2)$ E $49x$

Trinomial expression

These types of expressions are always in three terms. The methods of solving the trinomials which are mostly quadratic ($\mathbf{ax^2 + bx + c}$) in nature is in two ways:

When coefficient x^2 is unity (a = 1)

Examples

(1) Factorize $x^2 + 2x - 3$

Solution

We find the factors of c (–3) that when added together gives us b (+ 2) i.e. comparing the given expression with $ax^2 + bx + c$

Factors of –3	**sum**
- 3 and + 1	- 2
+3 and – 1	+2

The factors + 3 and – 1 satisfy our condition, Thus

$x^2 + 2x - 3 = x^2 + 3x - x -3$

$= x(x + 3) - 1 (x + 3)$

$= (x + 3) (x - 1)$

(2) Factorize completely $x^2 - 4x - 12$

As usual we find the factors of C (- 12) that When added will give b (- 4)

Factors of –12	**sum**
12 and –1	11
3 and – 4	-1
2 and – 6	- 4

The last factors 2 and – 6 satisfied our condition, thus

$x^2 - 4x - 12 = x^2 + 2x - 6x - 12$

$= x(x + 2) - 6(x + 2)$

$= (x + 2) (x - 6)$

Note: To be objective minded, readers need not to go through long process of trial and error before arriving at the needed factors.

2010/42 NABTEB (Nov)

Factorize $x^2 + 7x + 12$

A $(x + 4)(x + 3)$ B $(x - 7)(x - 7)$

C $(x + 7)(x - 7)$ D $(x - 49)(x - 49)$

Solution

Here the factors of 12 that when added gives +7 are 3 and 4

$x^2 + 7x + 12 = x^2 + 4x + 3x + 12$

$= x(x + 4) + 3(x + 4)$

$= (x + 4)(x + 3)$ A

2015/32 Neco Exercise 3.28

Factorise $n^2 + 24n - 81$

A.$(n - 3)(n + 27)$ B. $(n + 3)(n - 27)$

C.$(n - 9)(n + 9)$ D.$(n - 9)(n -9)$ E.$(n - 1)(n + 81)$

When the coefficient of x^2 is not unity. (-1 inclusive)

Here we look for the factors of ac that when added gives us b.

Examples

(1) Factorize $12 +11x - x^2$

Here $a \neq 1$ rather a = - 1

Thus ac = (-1) x 12 i.e - 12 and b = +11

Factors of – 12 that when added gives us +11 are +12 and –1

$12 + 11x - x^2 = 12 + 12x - x - x^2$

$= 12 (1 + x) - x (1 + x)$

$= (1+ x) (12 - x)$

(2) Factorize $2y^2 + 11y +15$

Solution

ac = 30, b = 11, trying the factors of 30 that when added give us + 11, we have + 6 and + 5

$2y^2 + 11y +15 = 2y^2 + 6y +5y +15$

$= 2y (y + 3) +5 (y + 3)$

$= (y + 3) (2y + 5)$

(3) Factorize $5x^2 +17xy + 6y^2$ *special case*

Solution

Here we treat the equation which appears in square of two different functions as one square of function of x.

Thus ac = 30 and b =17

The factors of +30 that when added will give + 17

Are + 2 and +15. Thus,

$5x^2 + 17xy + 6y^2 = 5x^2 + 15xy + 2xy + 6y^2$
$= 5x(x + 3y) + 2y(x + 3y)$
$= (x + 3y)(5x + 2y)$

2005/2a ii
Factorize $6 + x - 2x^2$
Solution
Here ac is –12 and b is + 1, factor of –12 that when added gives +1 are –3 and 4
$6 + x - 2x^2 = 6 - 3x + 4x - 2x^2$
$= 3(2 - x) + 2x(2 - x)$
$= (2 - x)(3 + 2x)$

2010/23 Neco
Factorize the expression $42 - 15x - 3x^2$
A $3(x + 7)(x - 2)$ B $3(7 - x)(2 - x)$
C $3(x - 7)(2 - x)$ D $3(7 + x)(2 - x)$ E $3(7 + x)(2 + x)$
Solution
$42 - 15x - 3x^2 = 3(14 - 5x - x^2)$
Here ac is – 14 and factor of – 14 that add up to give –5 are – 7 and 2
$3(14 - 5x - x^2) = 3(14 - 7x + 2x - x^2)$
$= 3[7(2 - x) + x(2 - x)]$
$= 3(2 - x)(7 + x)$ D

2007/41 Exercise 3.29
Which of the following is a factor of $2 - x - x^2$?
A $1 - x$ B $1 + x$ C $x - 1$ D $2 - x$

2008/28 Neco Exercise 3.30
Factorize $35 - 2x - x^2$
A $(5 + x)(7 + x)$ B $(5 + x)(x - 7)$ C $(5 - x)(7 + x)$
D $(x - 5)(x + 7)$ E $(x - 5)(x - 7)$

2006/22 Neco (Dec) ***Special case***
Factorize $2t^2 - ty - 3y^2$
A $(2t - 3y)(t - y)$ B $(2t - 3y)(t + y)$
C $(2t + 3y)(t - y)$ D $(2t + 3y)(t + y)$
E $(2t + 3y)(t + y)^2$
Solution
Here we treat the equation which appears in squares of two different functions as one square.
Here ac is – 6 and b is –1, factors of – 6 that when added gives –1 are 2 and –3
$2t^2 - ty - 3y^2 = 2t^2 + 2ty - 3ty - 3y^2$
$= 2t(t + y) - 3y(t + y)$
$= (2t - 3y)(t + y)$ B

2008/12 ***special case***
Factorize $5y^2 + 2ay - 3a^2$
A $(a - y)(5y - 3a)$ B $(y - a)(5y - 3a)$
C $(y - a)(5y + 3a)$ D $(y + a)(5y - 3a)$
Solution
Here we treat the equation which appears in squares of two different functions as one square. Coefficient of y^2 is not unity.
ac is –15 and b is 2 the factors of –15 that when added gives 2 are 5 and – 3. Thus
$5y^2 + 2ay - 3a^2 = 5y^2 + 5ay - 3ay - 3a^2$
$= 5y(y + a) - 3a(y + a)$
$= (5y - 3a)(y + a)$ D

2013/16 Neco Exercise 3.31
Factorize $2a^2 + 7ab - 15b^2$
A $(2ab + 3)(ab + 5)$ B $(2ab + 3)(ab - 5)$
C $(5ab + 3)(2ab - 3)$ D $(2a - 3b)(a + 5b)$
E $(2a + 3b)(a - 5b)$

2012/7b ***factor theorem***
Given that $x^2 + bx + 18$ is factorized as $(x + 2)(x + c)$. Find the value of c and b
Solution
If $(x + 2)$ and $(x + c)$ are factor then
$x + 2 = 0$
$x = -2$
Put $x = -2$ into original equation
$x^2 + bx + 18 = 0$ becomes
$(-2)^2 + b(-2) + 18 = 0$
$4 - 2b + 18 = 0$
$4 + 18 = 2b$
$\frac{22}{2} = \frac{2b}{2}$
$b = 11$
Substitute for b and factorize
$x^2 + 11x + 18 = x^2 + 2x + 9x + 18$
$= x(x + 2) + 9(x + 2)$
$= (x + 2)(x + 9)$
Thus c is 9

2010/20 Neco ***factor theorem***
Find the value of P if $y - 2$ is a factor of $y^2 - py - 10$
A – 3 B – 2 C 2 D 3 E 5
Solution
If $(y - 2)$ is a factor, then
$y - 2 = 0$
$y = 2$
Next we put $y = 2$ into $y^2 - py - 10$
$2^2 - p(2) - 10 = 0$
$4 - 2p - 10 = 0$
$4 - 10 = 2p$
$-6 = 2p$
$p = {}^{-6}/_{2}$ i.e – 3 (A)

1992/15 PCE Exercise 3.32
If $x - 2$ is a factor of $2x^2 - x + k$ then k is
A. 6 B. 2 C. – 3 D. – 6

Other simplification of algebraic terms

2005/3b Neco

Simplify 4{x – [2(3 – x) – 3(2x +1) + 1]}

Solution

Starting from the inner brackets

4{x – [2(3 – x) – 3(2x +1) + 1]}

= 4{x – [6 – 2x – 6x – 3 +1]}

= 4{x – [4 – 8x]}

= 4(x – 4 + 8x)

= 4x – 16 + 32x = 36x – 16

2008/20 Neco

Find the HCF of $x^2yz + xy^2z$ and $x^2y - y^3$

A x(x + z) B xy(x + y) C y(x + y)
D yz(z + x) E z(x + z)

Solution

HCF of $x^2yz + xy^2z$ and $x^2y - y^3$

First, we factor out common terms from each item

$x^2yz + xy^2z = xyz(x + y)$

and $x^2y - y^3 = y(x^2 - y^2)$

$= y(x - y)(x + y)$

From our result; y(x + y) is the HCF

2014/31 NABTEB

Find the least common multiples of $x^2y(x - y)$ and $xy^2(x - y)$

A. $xy(x - y)$ B. $x^2y(x - y)$ C. $x^2y^2(x - y)$ D. $x^2y^2(x - y)^2$

Solution

(x – y)	$x^2y(x - y)$	$xy^2(x - y)$
xy	x^2y	xy^2
x	x	y
y	1	y
	1	1

Thus LCM is $(x - y) \times xy \times x \times y = x^2y^2(x - y)$

2014/ 24 Neco

Simply $\frac{1}{3}[5y-(2+3y)+(7y-4)]$

A 6y + 4 B 3y + 2 C 6y + 2 D 3y – 2 E 9y – 6

Solution

Simplifying from the inner bracket

$$\frac{1}{3}[5y-(2+3y)+(7y-4)] = \frac{1}{3}[5y-2-3y+7y-4]$$

$$= \frac{1}{3}[9y-6]$$

$$= 3y - 2 \qquad D$$

2010/30 Neco

Find the sum of 25a – 15b + c, 13a – 10b + 4c and a + 20b – c

A 12a – 5b + 5c B 12a + 5b – 5c C 13a + 5b + 4c
D 39a – 5b + 4c E 39a + 5b + 4c

Solution

Doing it arithmetically, we have

$$\begin{array}{r} 25a - 15b + c \\ +\ 13a - 10b + 4c \\ a + 20b - c \\ \hline 39a - 5b + 4c \\ \hline \end{array} \quad D$$

2014/57 Neco

Divide $x^2yz + xy^2z$ by $x^2y - y^3$

A $\frac{xy}{z(xy-y^2)}$ B $\frac{xz}{x-y}$ C $\frac{xy}{x-z}$

D $\frac{xz}{x+y}$ E $\frac{xy}{x+z}$

Solution

We factor out common terms in the numerator and denominator

$$\frac{x^2yz+xy^2z}{x^2y-y^3} = \frac{xyz(x+y)}{y(x^2-y^2)}$$

Difference of two squares at denominator

$$= \frac{xyz(x+y)}{y(x-y)(x+y)}$$

$$= \frac{xyz}{y(x-y)}$$

$$= \frac{xz}{x-y} \qquad B$$

2008/25 NABTEB (Nov) Exercise 3.33

Calculate the LCM of $2ax^2y$, $8a^2xy^3$ and $6ax^3y^2$

A $16x^2y^3$ B $18a^2x^3y$ C $24ax^2y^3$ D $24a^2x^3y^2$

2014/37 Exercise 3.34

Subtract $^1/_2(a - b - c)$ from the sum of $^1/_2(a - b + c)$ and $^1/_2(a + b - c)$

A $^1/_2(a + b + c)$ B $^1/_2(a - b - c)$
C $^1/_2(a - b + c)$ D $^1/_2(a + b - c)$

2008/25 Neco Exercise 3.35

Simplify 5x – [4y + 3{x – 2(y – x)}]

A 4x + 2y B 4x – 2y C 2(2x – y)
D – 2(2x – y) E – 4(x + y)

Simplification of algebraic fractions

Simplification of algebraic fractions is done by applying the same principles as in the case of fractional numbers only. The basic rules of multiplication, addition and subtraction also apply in this case. For instance, if we are asked to simplify the following :

(1) $\frac{1}{3} \times \frac{81}{7} \times \frac{21}{3}$ or $\frac{1\times 81\times 21}{3\times 7\times 3}$

(2) $\frac{1}{4} + \frac{2}{3}$

(3) $\frac{3}{5} + \frac{1}{6}$

The above problem (1) makes use of canceling out of factors in numerator and denominator while (2) and (3) make use of L.C.M of the denominators,

(a) Division and multiplication of algebraic fractions

Here we apply the principles learnt in the preceding subtopics to factorize the numerator and the denominator " separately" then we reduce the resulting terms where possible; if they are not in their simplest form

2007/10a

Simplify $\dfrac{x^2-y^2}{3x+3y}$

Solution

We simplify numerator and denominator differently

$$\frac{x^2-y^2}{3x+3y} = \frac{(x-y)(x+y)}{3(x+y)}$$

$$= \frac{x-y}{3}$$

2007/2a

Simplify $\dfrac{x^2-8x+16}{x^2-7x+12}$

Solution

We factorize numerator and denominator differently

$$\frac{x^2-8x+16}{x^2-7x+12} = \frac{x^2-4x-4x+16}{x^2-3x-4x+12}$$

$$= \frac{x(x-4)-4(x-4)}{x(x-3)-4(x-3)}$$

$$= \frac{(x-4)(x-4)}{(x-4)(x-3)} = \frac{x-4}{x-3}$$

2008/9

Simplify : $\dfrac{\frac{1}{x}+\frac{1}{y}}{x+y}$

A $\dfrac{1}{x+y}$ B $\dfrac{1}{xy}$ C x + y D xy

Solution

$$\frac{\frac{1}{x}+\frac{1}{y}}{x+y} = \left(\frac{1}{x}+\frac{1}{y}\right) \div (x+y)$$

Simplifying the first bracket

$$= \left(\frac{y+x}{xy}\right) \div (x+y)$$

Changing $\div$ to $\times$

$$= \frac{x+y}{xy} \times \frac{1}{x+y}$$

y + x is same as x + y just like 2 + 3 and 3 + 2

$$= \frac{1}{xy} \quad \text{B}$$

2009/14 (Nov)

Evaluate : $\dfrac{12ab-4b^2}{2b^2-6ab}$

A. – 2 B. – 1 C. 1 D. 2

Solution

Simplifying numerator and denominator differently

$$\frac{12ab-4b^2}{2b^2-6ab} = \frac{4b(3a-b)}{2b(b-3a)}$$

To get a better result, we factor out –2b instead of 2b

$$= \frac{4b(3a-b)}{-2b(3a-b)} = -2 \quad \text{A}$$

2013/9

Simplify: $\dfrac{x^2-y^2}{(x+y)^2} \div \dfrac{(x-y)^2}{(3x+3y)}$

A $\dfrac{x-y}{3}$ B x + y C $\dfrac{3}{x-y}$ D x – y

Solution

Let us take each terms differently as:

$$\frac{x^2-y^2}{(x+y)^2} \div \frac{(x-y)^2}{(3x+3y)} = \frac{(x-y)(x+y)}{(x+y)(x+y)} \div \frac{(x-y)(x-y)}{3(x+y)}$$

$$= \frac{x-y}{x+y} \div \frac{(x-y)(x-y)}{3(x+y)}$$

Changing from $\div$ to $\times$

$$= \frac{x-y}{x+y} \times \frac{3(x+y)}{(x-y)(x-y)}$$

$$= \frac{3}{x-y} \quad \text{C}$$

2008/15 Exercise 3.36

Simplify: $\dfrac{2x^2-5x-12}{4x^2-9}$

A $\dfrac{x+4}{2x+3}$ B $\dfrac{x+4}{2x-3}$ C $\dfrac{x-4}{2x+3}$ D $\dfrac{x-4}{2x-3}$

2012/9 Exercise 3.37

Simplify $\dfrac{54k^2-6}{3k+1}$

A $6(1-3k^2)$ B $6(3k^2-1)$ C $6(3k-1)$ D $6(1-3k)$

2014/2a Neco Exercise 3.38

Simplify $\dfrac{a^2-4b^2}{a^2-5ab+6b^2}$

2014/45 (Nov) Exercise 3.39

Simplify : $\dfrac{m^2-n^2}{n-m}$

A. m + n B. – m – n C. – m + n D. m – n

(b) Addition and subtraction of algebraic fractions

2006/43 (Nov)

Express $\dfrac{x+1}{2} - \dfrac{3x-1}{3}$ as a single fraction.

A $\dfrac{3x-1}{6}$ B $\dfrac{1-3x}{6}$ C $\dfrac{5-3x}{6}$ D $\dfrac{3x-5}{6}$

Solution

The LCM of the denominators 2 and 3 is 6

$$\frac{x+1}{2} - \frac{3x-1}{3} = \frac{3(x+1)-2(3x-1)}{6}$$

$$= \frac{3x+3-6x+2}{6}$$

$$= \frac{5-3x}{6} \quad \text{C}$$

2009/12 (Nov)

Simplify : $\frac{t^2}{12} - \frac{t}{3} + \frac{1}{3}$

A $3(t-2)^2$ B $\frac{1}{12}3(t-2)^2$ C $\frac{t-2}{12}$ D $\frac{1}{12}(t-2)^2$

Solution

LCM of 12 and 3 is 12. Thus

$$\frac{t^2}{12} - \frac{t}{3} + \frac{1}{3} = \frac{t^2-4t+4}{12}$$

$$= \frac{t^2-2t-2t+4}{12}$$

$$= \frac{t(t-2)-2(t-2)}{12}$$

$$= \frac{(t-2)(t-2)}{12}$$

$$= \frac{(t-2)^2}{12} \quad D$$

2012/20 Neco

Simplify $\frac{2x+1}{2} - \frac{3x-7}{9} - \frac{5}{18}$

A 2x + 1 B 2x + 6 C 2x + 1

D $\frac{2x+18}{3}$ E $\frac{2x+3}{3}$

Solution

The LCM of denominators 2, 9 and 18 is 18

$$\frac{2x+1}{2} - \frac{3x-7}{9} - \frac{5}{18} = \frac{9(2x+1)-2(3x-7)-5}{18}$$

$$= \frac{18x+9-6x+14-5}{18}$$

$$= \frac{12x+18}{18}$$

$$= \frac{6(2x+3)}{18}$$

$$= \frac{2x+3}{3} \quad E$$

2007/46

Simplify : $\frac{4}{2x} - \frac{2+x}{x}$

A –1 B – 2x C 2x D $\frac{2-x}{x}$

Solution

LCM of 2x and x is 2x. Thus

$$\frac{4}{2x} - \frac{2+x}{x} = \frac{4-2(2+x)}{2x}$$

$$= \frac{4-4-2x}{2x}$$

$$= \frac{-2x}{2x}$$

$$= -1 \quad A$$

2005/5 (Nov)

Simplify : $\frac{x}{x-2} - \frac{2}{x-3}$, for $x \neq 2$ and $x \neq 3$

A 1 B $\frac{1}{x-3}$ C $\frac{-2x}{(x-3)(x-2)}$ D $\frac{(x-1)(x-4)}{(x-3)(x-2)}$

Solution

The LCM of the denominator x – 2 and x – 3 is (x – 2) (x – 3).

$$\frac{x}{x-2} - \frac{2}{x-3} = \frac{x(x-3)-2(x-2)}{(x-2)(x-3)}$$

$$= \frac{x^2-3x-2x+4}{(x-2)(x-3)}$$

$$= \frac{x^2-5x+4}{(x-2)(x-3)}$$

Factorizing the numerator

$$= \frac{x^2-x-4x+4}{(x-2)(x-3)}$$

$$= \frac{x(x-1)-4(x-1)}{(x-2)(x-3)}$$

$$= \frac{(x-1)(x-4)}{(x-3)(x-2)} \quad D$$

2005/38

Simplify $\frac{2}{3xy} - \frac{3}{4yz}$

A $\frac{2-x}{12xyz}$ B $\frac{2z-3x}{12xyz}$ C $\frac{4z-3x}{12xyz}$ D $\frac{8z-9x}{12xyz}$

Solution

The LCM of the denominators 3xy and 4yz is

3	3xy	4yz
2	xy	4yz
2	xy	2yz
x	xy	yz
y	y	yz
z	1	z
	1	1

LCM 3×2×2 × x × y × z = 12xyz

$$\frac{2}{3xy} - \frac{3}{4yz} = \frac{4z(2)-3(3x)}{12xyz}$$

$$= \frac{8z-9x}{12xyz} \quad D$$

2005/1a

Express as a single fraction, $\frac{1}{y+1} - \frac{2}{y-1} + \frac{3}{y^2-1}$

Solution

The LCM of the denominators y + 1, y – 1 and $y^2 - 1$ is $y^2 - 1$ since $y^2 - 1 = y^2 - 1^2$ i.e (y – 1) (y + 1)

Thus $\frac{1}{y+1} - \frac{2}{y-1} + \frac{3}{y^2-1} = \frac{1(y-)-2(y+1)+3}{(y+1)(y-1)}$

$$= \frac{y-1-2y-2+3}{(y+1)(y-1)}$$

$$= -\frac{y}{(y+1)(y-1)}$$

2005/32 Neco

Simplify $\left(2-\frac{2x}{3y}\right)+\left(3+\frac{1}{y}\right)$

A $\frac{3y-2x+3}{3y}$ B $\frac{9x-2x+1}{3y^2}$ C $\frac{15y-2x-3}{y}$

D $\frac{15y-2x+3}{3y}$ E $\frac{15y+2x+3}{3y^2}$

Solution

First we simplify each bracket

$$\left(2-\frac{2x}{3y}\right)+\left(3+\frac{1}{y}\right) = \left(\frac{6y-2x}{3y}\right)+\left(\frac{3y+1}{y}\right)$$

The LCM of the denominator 3y and y is 3y

$$= \frac{6y-2x+3(3y+1)}{3y}$$

$$= \frac{6y-2x+9y+3}{3y}$$

$$= \frac{15y-2x+3}{3y} \quad \text{D}$$

2006/18 Neco (Dec)

If $\frac{4}{x-5} - \frac{3}{x-6}$ is expressed as $\frac{p}{(x-5)(x-6)}$ then P is equal to

A x + 9 B x + 5 C x – 9 D x – 39

Solution

The LCM of the denominators x – 5 and x – 6 is (x – 5)(x – 6). Thus

$$\frac{4}{x-5} - \frac{3}{x-6} = \frac{4(x-6)-3(x-5)}{(x-5)(x-6)}$$

$$= \frac{4x-24-3x+15}{(x-5)(x-6)}$$

$$= \frac{x-9}{(x-5)(x-6)}$$

Hence P is x – 9 C

2007/17 Neco

Express as a single fraction : $\frac{m-n}{m} - \frac{n-m}{n} - 1$

A $\frac{m^2+n^2-mn}{mn}$ B $\frac{n^2-m^2-mn}{mn}$

C $\frac{m^2+n^2+mn}{mn}$ D $\frac{m^2-n^2-mn}{mn}$ E $\frac{m+n}{mn}$

Solution

LCM of m and n is mn. Thus

$$\frac{m-n}{m} - \frac{n-m}{n} - 1 = \frac{n(m-n)-m(n-m)-mn}{mn}$$

$$= \frac{mn-n^2-mn+m^2-mn}{mn}$$

$$= \frac{m^2-n^2-mn}{mn} \quad \text{D}$$

2008/16 Neco (Dec)

Evaluate $2 + \frac{2}{x} - \frac{x+4}{2}$

A $\frac{2-x^2}{2x}$ B $\frac{8-x^2}{2x}$ C $\frac{4-x^2}{2x}$ D $\frac{4+8x-x^2}{2x}$

E $\frac{8-4x-x^2}{2x}$

Solution

$$2 + \frac{2}{x} - \frac{x+4}{2}$$

LCM of the denominators x and 2 is 2x

$$2 + \frac{2}{x} - \frac{x+4}{2} = \frac{2(2x)+2(2)-x(x+4)}{2x}$$

$$= \frac{4x+4-x^2-4x}{2x}$$

$$= \frac{4-x^2}{2x} \quad \text{C}$$

2011/31

Simplify: $\frac{m}{n} + \frac{(m-1)}{5n} - \frac{(m-2)}{10n}$ when n ≠ o

A $\frac{m-3}{10n}$ B $\frac{11m}{10n}$ C $\frac{m+1}{10n}$ D $\frac{11m+4}{10n}$

Solution

LCM of n, 5n and 10n is 2 × 5 × n i.e 10n

n	n	5n	10n
5	1	5	10
2	1	1	2
	1	1	1

LCM = 2 × 5 × n

$$\frac{m}{n}+\frac{(m-1)}{5n}-\frac{(m-2)}{10n} = \frac{10m+2(m-1)-1(m-2)}{10n}$$

$$= \frac{10m+2m-2-m+2)}{10n}$$

$$= \frac{11m}{10n} \quad \text{B}$$

But if we are asked to solve and not simplify

$$\frac{m}{n}+\frac{(m-1)}{5n}-\frac{(m-2)}{10n} = 0$$

We multiply through by the LCM 10n

10m + 2(m – 1) – 1 (m – 2) = 0

10m + 2m – 2 – m + 2 = 0

Explanation only; not a good example in this direction

2012/28 Neco

Simplify $\frac{2a}{b-1} + \frac{a}{b+2}$

A $\frac{3a(b+1)}{(b-1)(b+2)}$ B $\frac{a(b+1)}{(b-1)(b+2)}$ C $\frac{3b(a+1)}{(b-1)(b+2)}$

D $\frac{ab(b+1)}{(b-1)(b+2)}$ E $\frac{3b(a-1)}{(b-1)(b+2)}$

Solution

The LCM of denominators b – 1 and b + 2 is (b – 1)(b + 2). Thus

$$\frac{2a}{b-1} + \frac{a}{b+2} = \frac{2a(b+2)+a(b-1)}{(b-1)(b+2)}$$
$$= \frac{2ab+4a+ab-a}{(b-1)(b+2)}$$
$$= \frac{3ab+3a}{(b-1)(b+2)}$$
$$= \frac{3a(b+1)}{(b-1)(b+2)} \quad \text{A}$$

2013/41

Express $\frac{2}{x+3} - \frac{1}{x-2}$ as a simple fraction

A $\frac{x-7}{x^2+x-6}$ B $\frac{x-1}{x^2+x-6}$ C $\frac{x-2}{x^2+x-6}$
D $\frac{x+7}{x^2+x-6}$

Solution

The LCM of denominators x + 3 and x – 2 is (x + 3)(x – 2). Thus

$$\frac{2}{x+3} - \frac{1}{x-2} = \frac{2(x-2)-1(x+3)}{(x+3)(x-2)}$$
$$= \frac{2x-4-x-3}{x(x-2)+3(x-2)}$$
$$= \frac{x - 7}{x^2 - 2x + 3x - 6}$$
$$= \frac{x-7}{x^2+x-6} \quad \text{A}$$

2014/36

If $\frac{2}{x-3} - \frac{3}{x-2}$ is equal to $\frac{p}{(x-3)(x-2)}$, find P

A – x – 5 B – (x + 3) C 5x – 13 D 5 – x

Solution

The LCM of denominators x – 3 and x – 2 is (x – 3) (x – 2). Thus

$$\frac{2}{x-3} - \frac{3}{x-2} = \frac{2(x-2)-3(x-3)}{(x-3)(x-2)}$$
$$= \frac{2x-4-3x+9}{(x-3)(x-2)}$$
$$= \frac{5-x}{(x-3)(x-2)}$$

Thus P is 5 – x D

2013/19 Neco

Simplify $\frac{x+y}{x} - \frac{x-y}{y}$

A $\frac{2xy+y^2+x^2}{xy}$ B $\frac{2xy-y^2+x^2}{xy}$

C $\frac{2xy+y^2-x^2}{xy}$ D $\frac{xy+y^2+x^2}{xy}$

E $\frac{x^2y+y^2-x^2}{xy}$

Solution

The LCM of x and y is xy. Thus

$$\frac{x+y}{x} - \frac{x-y}{y} = \frac{y(x+y)-x(x-y)}{xy}$$
$$= \frac{xy+y^2-x^2+xy}{xy}$$
$$= \frac{2xy+y^2-x^2}{xy} \quad \text{C}$$

2005/39 (Old) Exercise 3.40

Simplify $\frac{3}{x+2} - \frac{2}{2x-3}$

A $\frac{4x-1}{(x-2)(2x-3)}$ B $\frac{4x-5}{(x+2)(2x-3)}$

C $\frac{4x-11}{(x+2)(2x-3)}$ D $\frac{4x-13}{(x+2)(2x-3)}$

2006/4 Exercise 3.41

Simplify: $\frac{x-4}{4} - \frac{x-3}{6}$

A $\frac{x-18}{12}$ B $\frac{x-6}{12}$ C $\frac{x-18}{24}$ D $\frac{x-6}{24}$

2010/23 Exercise 3.42

Simplify : $\frac{2}{2+x} - \frac{2}{2-x}$

A $\frac{4}{4-x^2}$ B $\frac{8}{4-x^2}$ C $-\frac{4x}{4-x^2}$ D $\frac{8-4x}{4-x^2}$

2010/24 Neco Exercise 3.43

Write $\frac{2}{2+x} - \frac{1}{2-x}$ as a single fraction

A $\frac{6-3x}{(2-x)^2}$ B $\frac{4-3x}{2-x^2}$ C $\frac{2-3x}{(4-x)^2}$

D $\frac{2-3x}{4-x^2}$ E $\frac{2-3x}{4+x^2}$

2009/7 Neco (Dec) Exercise 3.44

Simplify $\frac{1}{x-1} - \frac{2}{x^2-1}$

A $\frac{1}{x-1}$ B $\frac{1}{x+1}$ C $\frac{-1}{x^2-1}$ D $\frac{1}{x^2-1}$ E $\frac{1}{x^2+1}$

2012/34 Exercise 3.45

Express $3-\left(\frac{x-y}{y}\right)$ as a single fraction

A $\frac{3xy}{y}$ B $\frac{x-4y}{y}$ C $\frac{4y+x}{y}$ D $\frac{4y-x}{y}$

2011/16 Neco Exercise 3.46

Simplify $\frac{5}{x-1} - \frac{6}{x-2}$

A $\frac{-(x+4)}{x^2-3x+2}$ B $\frac{(x+4)}{x^2-3x+2}$

C $\dfrac{(x+4)}{x^2-3x+2}$ D $\dfrac{(x-4)}{x^2-3x+2}$ E $\dfrac{(x-4)}{x^2-3x+2}$

2014/30 Neco Exercise 3.47

Simplify $\dfrac{1}{x-1} - \dfrac{2}{x+2}$

A $\dfrac{-x}{(x-1)(x+2)}$ B $\dfrac{x}{(x-1)(x+2)}$

C $\dfrac{4-x}{(x-1)(x+2)}$ D $\dfrac{4+x}{(x-1)(x+2)}$ E $\dfrac{x-4}{(x-1)(x+2)}$

2015/51 Neco Exercise 3.48

Write $\dfrac{3x+2}{4} - \dfrac{x-1}{4} - \dfrac{5}{12}$ as a single fraction

A. $\dfrac{3x+2}{4}$ B. $\dfrac{x-1}{3}$ C. $\dfrac{x-1}{5}$

D. $\dfrac{3x+2}{6}$ E. $\dfrac{3x+2}{12}$

2014/8 NABTEB (Nov) Exercise 3.49

Simplify $\dfrac{a}{4} + \dfrac{2a}{3} - \dfrac{a}{12}$

A. $\dfrac{2a}{3}$ B. $\dfrac{a}{4}$ C. $\dfrac{5a}{6}$ D. $\dfrac{a}{12}$

Substitution in algebraic expressions

There are two types of substitutions in algebraic expressions:

(a) When the value to be substituted is given
(b) Solving for the value to be substituted under given conditions

(a) When the value to be substituted is given

2006/4b

If $p = \dfrac{2u}{1-u}$ and $q = \dfrac{1+u}{1-u}$,

express $\dfrac{p+q}{p-q}$ in terms of u

Solution

$$p+q = \frac{2u}{1-u} + \frac{1+u}{1-u}$$
$$= \frac{2u+1+u}{1-u} = \frac{3u+1}{1-u}$$

$$p-q = \frac{2u}{1-u} - \frac{1+u}{1-u}$$
$$= \frac{2u-(1+u)}{1-u} = \frac{2u-1-u}{1-u}$$
$$= \frac{u-1}{1-u} = \frac{-(1-u)}{1-u} = -1$$

Thus $\dfrac{p+q}{p-q} = \dfrac{3u+1}{1-u} \div -1$

$$= \frac{3u+1}{(1-u)\times -1} = \frac{3u+1}{u-1}$$

2008/2b

Given that $P = \dfrac{x^2-y^2}{x^2+xy}$

(i) Express P in its simplest form;
(ii) Find the value of P if x = – 4 and y = – 6

Solution

(i) $\dfrac{x^2-y^2}{x^2+xy} = \dfrac{(x+y)(x-y)}{x(x+y)}$

$$= \frac{x-y}{x}$$

(ii) substituting for x and y

$$\frac{x-y}{x} = \frac{-4-(-6)}{-4}$$
$$= \frac{-4+6}{-4}$$
$$= \frac{2}{-4} = -\frac{1}{2}$$

2015/1a Neco

If the numbers x, y, z are in the ratio 6 : 5 : 8,

find $\dfrac{12x-9z}{4y+z}$

Solution

x = 6, y = 5, z = 8
Substituting

$$\frac{12x-9z}{4y+z} = \frac{12(6)-9(8)}{4(5)+8}$$
$$= \frac{72-72}{20+8} = \frac{0}{8} = 0$$

2007/11

If x = 3, y = 2 and z = 4, what is the value of $3x^2 - 2y + z$?

A. 17 B. 27 C. 35 D. 71

Solution

$3x^2 - 2y + z = 3(3^2) - 2(2) + 4$
$= 3 \times 9 - 4 + 4$
$= 27$ B

2006/6a Neco

If $x = \dfrac{3m-2}{m-1}$, express $\dfrac{x+1}{2x-1}$ in terms of m

Solution

$\dfrac{x+1}{2x-1} = (x+1) \div (2x-1)$

Substituting x

$$= \left(\frac{3m-2}{m-1}+1\right) \div \left[2\left(\frac{3m-2}{m-1}\right)-1\right]$$

$$= \left(\frac{3m-2+m-1}{m-1}\right) \div \left[\frac{6m-4}{m-1} - 1\right]$$

$$= \left(\frac{4m-3}{m-1}\right) \div \left[\frac{6m-4-1(m-1)}{m-1}\right]$$

$$= \left(\frac{4m-3}{m-1}\right) \div \left[\frac{6m-4-m+1}{m-1}\right]$$

$$= \left(\frac{4m-3}{m-1}\right) \div \left(\frac{5m-3}{m-1}\right)$$

Changing $\div$ to $\times$

$$= \frac{4m-3}{m-1} \times \frac{m-1}{5m-3} \quad = \frac{4m-3}{5m-3}$$

2015/38 Neco Exercise 3.50

Evaluate $\left(\frac{a + b}{a - b}\right)^3$ for a = -7 , b = 3

A. $-\frac{8}{125}$ B. $-\frac{4}{10}$ C. $\frac{8}{125}$ D. $\frac{16}{125}$ E. $\frac{2}{5}$

2008/11 Exercise 3.51

Given that x = 2 and $y = -\frac{1}{4}$, evaluate $\frac{x^2 y - 2xy}{5}$

A. 0 B. $\frac{1}{5}$ C. 1 D. 2

2006/17 Neco (Dec) Exercise 3.51b

Evaluate $\frac{x^2+x-2}{2x^2+x-3}$ when x = –1

A 2 B 1 C – $^1/_2$ D – 1 E – 2

2014/6b Neco (Nov) Exercise 3.52

Given that Z = $\frac{3x-2}{2x+3}$, express $\frac{2z+3}{3z-2}$ in terms of x

(b) Solving for the value to be substituted under the conditions:

(i) undefined expressions
(ii) expression equals zero

Undefined expressions

For any expression of the form $\frac{a(x)}{b(x)}$ to be undefined

then b (x) = 0

2005/33 Neco

Find the value of x for which the fraction $\frac{x+1}{2x-1}$ is undefined.

A – 2 B – $^1/_2$ C 0 D $^1/_2$ E 2

Solution

The expression cannot reduce further

$\frac{x+1}{2x-1}$ is undefined if 2x – 1 = 0

2x = 1
x = $^1/_2$ D

2005/ 47

For what value of x is the expression $\frac{2x-2}{(x-2)(x+3)}$ not defined?

A x = 4 or – 2 B x = 2 or – 3
C x = 3 or – 2 D x = 2 or – 4

Solution

The expression cannot reduce further
So for it to be undefined

(x – 2)(x + 3) = 0
x – 2 = 0 or x + 3 = 0
x = 2 or –3 B

2014/29 Neco

Find the value of x for which the fraction $\frac{2x^2+y^2}{x^2-4}$ is undefined.

Solution

A x = ± 5 B x = ± 4 C x = ± 3 D x = ± 2 E x = ± 1

Solution

It is clear that the expression will not reduce further hence we impose the undefined condition

$x^2 - 4 = 0$
$x^2 - 2^2 = 0$
(x – 2)(x + 2) = 0
x – 2 = 0 or x + 2 = 0
x = 2 or – 2 D

2006/5

Given that $y = 1 - \frac{2x}{4x-3}$, find the value of x for which y is undefined

A 3 B $^3/_4$ C – $^3/_4$ D – 3

Solution

For any expression of the form $\frac{a(x)}{b(x)}$ to be undefined b(x) = 0; here the expression is not yet in full fraction $\frac{a(x)}{b(x)}$, making it full fraction

$$1 - \frac{2x}{4x-3} = \frac{4x-3-2x}{4x-3}$$

$$= \frac{2x-3}{4x-3}$$

Since the expression cannot reduce further: we apply the undefined condition

4x – 3 = 0
4x = 3
x = $^3/_4$ B

2006/20

Given that y = $\frac{cr - px}{aq - bp}$, then the value of y is undefined if

A cr = px B cr > px C aq = bp D aq< bp
E aq > bp

Solution

Since the given expression cannot reduce further. It is undefined if

aq – bp = 0
aq = bp C

2012/1a Neco ***Good case***

Find the value of x for which the fraction

$\frac{x^2-9}{2x^2-7x+3}$ is undefined.

Solution

First, we simplify the expression to see whether it will reduce further

$$\frac{x^2-9}{2x^2-7x+3} = \frac{x^2-3^2}{2x^2-6x-x+3}$$

$$= \frac{(x-3)\ (x+3}{2x(x-3)-1(x-3)}$$

$$= \frac{(x-3)\ (x+3)}{(2x-1)\ (x-3)}$$

$$= \frac{x+3}{2x-1}$$

Next, we impose the undefined condition

$2x - 1 = 0$

$2x = 1$ thus $x = {}^1/_2$

2007/ 10 Exercise 3.53

For what value of x is the expression $\frac{2x-1}{x+3}$ **not** defined?

A 3 B 2 C ${}^1/_2$ D – 3

2008/21 Neco (Dec) Exercise 3.54

Find the values of x for which $\frac{2x+5}{4x^2-9}$ is not defined.

A $x = {}^3/_2$ or $-{}^3/_2$ B $x = {}^2/_3$ or $-{}^2/_3$ C $x = {}^2/_5$ or $-{}^2/_5$
D $x = {}^5/_2$ or ${}^3/_2$ E $x = {}^5/_2$ or $-{}^3/_2$

2009/10 Neco (Dec) Exercise 3.55

For what value of x is $\frac{2x+1}{12-5x-3x^2}$ undefined?

A x = 5 or 1 B x = 3 or $-{}^3/_4$ C x = – 3 or ${}^4/_3$
D x = –5 or 1 E x = 5 or ${}^3/_4$

2012/24 Neco Exercise 3.56

For what values of x is $\frac{x^2+1}{x^2-1}$ not defined?

A x = –1 or 1 B $x = {}^1/_2$ or 0 C $x = {}^1/_2$ or 2
D $x = -{}^1/_2$ or 2 E x = 0 or –2

(ii) Expression equals zero

For any algebraic expression $\frac{a(x)}{b(x)} = 0$ Then $a(x) = 0$

Examples

(1) For what value (s) of x is the expression

$\frac{x+3}{x^2+10x-25}$ equals zero?

Solution

Since the numerator cannot be factorised further, we impose the condition

$$\frac{x+3}{x^2+10x-25} = 0$$

if $x + 3 = 0$

$x = -3$

(2) For what value (s) of x is the expression

$\frac{x^2-2x-15}{x^2-25}$ equals zero?

First, we factorize the numerator $x^2 - 2x - 15$

$= x^2 + 3x - 5x - 15$

$= x(x+3) - 5(x+3)$

$= (x+3)(x-5)$ Thus

$$\frac{x^2-2x-15}{x^2-25} = \frac{(x+3)(x-5)}{(x-5)(x+5)}$$

$$= \frac{x+3}{x+5}$$

Imposing the given condition on the reduced form

$x + 3 = 0$

$x = -3$

2006/11a Neco (Dec)

Find the value of x for which the expression

$\frac{2x^2+9x-11}{6-4x}$ is zero

Solution

For any algebraic expression $\frac{a(x)}{b(x)} = 0$ then $a(x) = 0$

First, we simplify the expression to see whether it will reduce

$$\frac{2x^2+9x-11}{6-4x} = \frac{2x^2+11x-2x-11}{6-4x}$$

$$= \frac{x(2x+11)-1\ (2x+11)}{6-4x}$$

$$= \frac{(x-1)(2x+11)}{6-4x} = \frac{(x-1)(2x+11)}{2(3-2x)}$$

It can not reduce further; hence we impose the condition of equals zero

$2x^2 + 9x - 11 = 0$

i.e $(x-1)(2x+11) = 0$

$x - 1 = 0$ or $2x + 11 = 0$

$x = 1$ or $2x = -11$

$x = 1$ or $-{}^{11}/_2$

2014/44 Neco (Nov) Exercise 3.57

For what values of x is the fraction $\frac{x^2-9x+14}{x^2-x-2}$ equals to zero?

A. 2 or -1 B. -2 or -7 C. 2 or -1 D. -2 or 1 E. 2 or 7

Change of subject of the Formula

The term change of subject of the formula implies transformation of a given formula to make another quantity (usually a letter) the subject of the equation. This is other than the one in that position.

In doing it, basic mathematical rules must not be violated.

Some Basic Guides

(a) Remove square or square root if present
(b) Clear fractions, clear bracket if they exist
(c) Collect like terms together involving the required subject.
(d) If necessary factorize the term on each side of the equation
(e) Isolate the required subject usually by division by a common factor

Note

The above stated guides are no strict rules to be followed when solving any problem under this topic; rather, they are guides which readers will find useful

Cases involving indices(powers)

2005/4 (Nov)

Make x the subject of the relation $y = \frac{\sqrt{k^2 + x^2}}{t}$

A $x = \sqrt{yt - k^2}$ B $x = \sqrt{(yt)^2 - k^2}$

C $x = (yt)^2 - k^2$ D $x = (yt - k)^2$

Solution

$$y = \frac{\sqrt{k^2 + x^2}}{t}$$

First we clear fraction

$$yt = \sqrt{k^2 + x^2}$$

Since x is under square root, we take square of both sides.

$$(yt)^2 = k^2 + x^2$$

Next, we isolate x

$$(yt)^2 - k^2 = x^2$$

x is squared , so we take square root

$$\sqrt{(yt)^2 - k^2} = x \quad \text{B}$$

2005/1c Neco

If $4a^3 = c - b^2$, find the value of b when $a = -3$ and $c = 24$ (correct to 2 d.p)

Solution

$$4a^3 = c - b^2$$

First we make b^2 the subject of the formula

$$b^2 = c - 4a^3$$

b is under square, we take square root of both sides

$$b = \sqrt{c - 4a^3}$$

Substituting for the given values of a and c

$$= \sqrt{24 - 4(-3)^3}$$

$$= \sqrt{24 - 4(-27)}$$

$$= \sqrt{24 + 108}$$

$$= \sqrt{132} = 11.489 \approx 11.49 \text{ to 2 dp.}$$

2005/ 2a

(i) Make t the subject of relation $k = m\sqrt{\frac{t-3}{t-9}}$

(ii) Find the value of t, if k = 2 and m = 1

Solution

$$k = m\sqrt{\frac{t-3}{t-9}}$$

Since T is under square root, we take square of both sides

$$k^2 = m^2\left(\frac{t-3}{t-9}\right)$$

$$k^2 = \frac{m^2t - 3m^2}{t-9}$$

Cross multiply to clear fraction

$$k^2(t - 9) = m^2t - 3m^2$$

Collect terms in **t** together

$$k^2t - m^2t = 9k^2 - 3m^2$$

$$t(k^2 - m^2) = 9k^2 - 3m^2$$

$$t = \frac{9k^2 - 3m^2}{k^2 - m^2}$$

(ii) If $k = 2$ and $m = 1$ then

$$t = \frac{9(2^2) - 3(1^2)}{2^2 - 1^2}$$

$$= \frac{36-3}{4-1} = \frac{33}{3} = 11$$

2005/34 Neco

If $A = \frac{1}{2}B + \frac{1}{8}\left(\frac{C^2}{B}\right)$, make C the subject of the formula

A $\pm 2\sqrt{2BCA - B}$ B $\pm 2\sqrt{2(2A - B}$

C $\pm 2\sqrt{B(2A - B)}$ D $\pm 2\sqrt{2B(2A - B)}$ E $\pm 2\sqrt{B(2A + B)}$

Solution

$$A = \frac{1}{2}B + \frac{1}{8}\left(\frac{C^2}{B}\right)$$

First, we clear fraction, LCM of 8 and 2 is 8, so we multiply through by 8

$$8A = 4B + \frac{C^2}{B}$$

Isolate term in C

$$8A - 4B = \frac{C^2}{B}$$

$$B(8A - 4B) = C^2$$

$$8AB - 4B^2 = C^2$$

Since C is squared, we take square root

$$\pm\sqrt{8AB - 4B^2} = C$$

$$\pm\sqrt{4B(2A - B} = C$$

$$\text{i.e } \pm 2\sqrt{B(2A - B)} = C$$

2006/6a (Nov)

(i) Make r the subject of the relation $Q = P\left[1+\frac{r}{100}\right]^2$

(ii) Find, correct to 3 s.f, the value of r when Q = 625 and P = 225

Solution

$$Q = P\left[1+\frac{r}{100}\right]^2$$

$$\frac{Q}{P} = \left[1+\frac{r}{100}\right]^2$$

Since **r**'s bracket is squared, we take square root to free it

$$\sqrt{\frac{Q}{P}} = 1 + \frac{r}{100}$$

$$\sqrt{\frac{Q}{P}} - 1 = \frac{r}{100}$$

Multiply through by 100 to clear fraction

$$100\left(\sqrt{\frac{Q}{P}} - 1\right) = r$$

$$r = 100\left(\sqrt{\frac{625}{225}} - 1\right)$$

$$= 100\left(\frac{25}{15} - 1\right)$$

$$= 100\left(\frac{10}{15}\right) = 66.666 \approx 66.7 \text{ to 3 s.f}$$

2014/7b Neco (Dec)

Given the formula $100\left[\left(\frac{A}{P}\right)^{\frac{1}{n}} - 1\right] = r$

(i) Make A the subject of the formula

(ii) Find the value of A when P = 7808, n = 3 and r = 25

Solution

$$100\left[\left(\frac{A}{P}\right)^{\frac{1}{n}} - 1\right] = r$$

$$\left(\frac{A}{P}\right)^{\frac{1}{n}} - 1 = \frac{r}{100}$$

$$\left(\frac{A}{P}\right)^{\frac{1}{n}} = \frac{r}{100} + 1$$

Since A is under root **n**, we take power of **n** of both sides

$$\frac{A}{P} = \left(\frac{r}{100} + 1\right)^n$$

$$A = P\left(1 + \frac{r}{100}\right)^n$$

(ii) Substituting for the given values

$$A = 7808\left(1 + \frac{25}{100}\right)^3$$

$$= 7808(1 + 0.25)^3$$

$$= 15250$$

2014/12b (Nov)

Given the relation $T = \sqrt{\frac{u}{\frac{1}{f} + \frac{1}{g}}}$

(i) make g the subject of the relation

(ii) find g when T = 3, f = 4 and u = 5

Solution

$$T = \sqrt{\frac{u}{\frac{1}{f} + \frac{1}{g}}}$$

Since g is under square root, we take square of both sides

$$T^2 = \frac{u}{\frac{1}{f} + \frac{1}{g}}$$

Cross multiply to clear fraction

$$T^2\left(\frac{1}{f} + \frac{1}{g}\right) = u$$

Next we isolate g,

$$\frac{1}{f} + \frac{1}{g} = \frac{u}{T^2}$$

$$\frac{1}{g} = \frac{u}{T^2} - \frac{1}{f}$$

Apply LCM of fraction to the RHS

$$\frac{1}{g} = \frac{fu - T^2}{T^2 f}$$

Just invert both sides

$$g = \frac{T^2 f}{fu - T^2}$$

(ii) Substituting for the given values

$$g = \frac{3^2 \times 4}{4 \times 5 - 3^2}$$

$$= \frac{9 \times 4}{20 - 9} = \frac{36}{11} = 3\tfrac{3}{11}$$

2008/34

If $P = \left[\frac{Q(R-T)}{15}\right]^{\frac{1}{3}}$, make T the subject of the relation

A $T = \frac{R + p^3}{15Q}$ B $T = \frac{R - 15p^3}{Q}$

C $T = R - \frac{15p^3}{Q}$ D $T = \frac{15R - Q}{P^3}$

Solution

$$P = \left[\frac{Q(R-T)}{15}\right]^{\frac{1}{3}}$$

Since T's bracket is cube – root, we raise both sides to power 3

$$P^3 = \frac{Q(R-T)}{15}$$

Cross multiply to clear fraction

$$15P^3 = Q(R - T)$$

$$15P^3 = QR - QT$$

Next, isolate term in t

$$QT = QR - 15P^3$$

$$T = \frac{QR - 15P^3}{Q} \quad \text{i.e } R - \frac{15P^3}{Q} \quad \text{(C)}$$

2013 /3b Neco

Given that $h = \left(\frac{2p+q}{p-3q}\right)^{\frac{1}{2}}$, express p in terms of q and h

Solution

$$h = \left(\frac{2p+q}{p-3q}\right)^{\frac{1}{2}},$$

Since p is under square root, we take square of both sides

$$h^2 = \frac{2p+q}{p-3q}$$

Next, we clear fraction

$h^2(p-3q) = 2p+q$

$h^2p - 3h^2q = 2p + q$

Collect terms in p together

$h^2p - 2p = 3h^2q + q$

$p(h^2 - 2) = 3h^2q + q$

$$p = \frac{q(3h^2+1)}{h^2-2}$$

2006/12 Neco (Dec)

If $v^2 = u^2 + 2as$, what is/are the value(s) of u when v = 7, a = 10 and s = 2?

A -3 B 3 C ± 3 D -9 E ± 9

Solution

First, we make u the subject of the formula

$v^2 = u^2 + 2as$

$v^2 - 2as = u^2$

Since u is squared, we take square root of both sides

$\sqrt{v^2 - 2as} = u$

Substituting

$$u = \sqrt{7^2 - 2\times10\times2}$$
$$= \sqrt{49-40}$$
$$= \sqrt{9} \text{ i.e } \pm 3 \quad (C)$$

2006/6 Neco (Dec)

If $S = \frac{a^2 \pm \sqrt{b^2 - c}}{2a}$, find s when a = 2, b = – 2 and c = – 5

A $\frac{7}{4}$ or $\frac{1}{4}$ B $\frac{7}{4}$ or $-\frac{1}{4}$

C $-\frac{7}{4}$ or $-\frac{1}{4}$ D $-\frac{7}{4}$ or -4 E -7 or -4

Solution

Substituting for the given values

$$S = \frac{2^2 \pm \sqrt{(-2)^2 - (-5)}}{2\times 2}$$
$$= \frac{4 \pm \sqrt{4+5}}{4}$$
$$= \frac{4\pm\sqrt{9}}{4} = \frac{4\pm 3}{4} = \frac{7}{4} \text{ or } \frac{1}{4} \quad (A)$$

2007/18 Neco Exercise 3.58

If $z = \sqrt{\frac{x}{m} - \frac{n}{y}}$, make y the subject of the formula.

A $y = \frac{x - z^2m}{mn}$ B $y = x - \frac{z^2}{mn}$ C $y = \frac{mn}{x - z^2m}$

D $y = \frac{mn}{nz^2 - x}$ E $y = \frac{z^2}{mn} - x$

2007/6b Exercise 3.59

If $P = \frac{m}{2} - \frac{n^2}{5m}$,

(i) Make n the subject of the relation;

(ii) Find, correct to 3s.f, the value of n when P = 14 and m = – 8

2010/29 Exercise 3.60

If $y = \frac{2(\sqrt{x^2+m}\,)}{3N}$, make x the subject of the formula

A $\frac{\sqrt{9y^2N^2 - 2m}}{2}$ B $\frac{\sqrt{9y^2N^2 + 2m}}{2}$

C $\frac{\sqrt{9y^2N^2 - 4m}}{2}$ D $\frac{\sqrt{9y^2N^2 + 4m}}{2}$

2014/14 Exercise 3.61

Make U the subject of the formula $E = \frac{m}{2g}(v^2 - u^2)$

A $u = \sqrt{v^2 - \frac{2Eg}{m}}$ B $u = \sqrt{\frac{v^2}{m} - \frac{2Eg}{4}}$

C $u = \sqrt{v - \frac{2Eg}{m}}$ D $u = \sqrt{\frac{2v^2Eg}{m}}$

2015/34 Exercise 3.62

Make k the subject of the relation $T = \sqrt{\frac{Tk - H}{k - H}}$

A. $k = \frac{H(T^2-1)}{T^2 - T}$ B. $k = \frac{HT}{(T-1)^2}$

C. $k = \frac{H(T^2+1)}{T}$ D. $k = \frac{H(T-1)}{T}$

2012/23 Neco Exercise 3.63

Make r the subject of the relation $V = P\left(1 + \frac{r}{100}\right)^t$

A $100\left(\sqrt[t]{\frac{v}{p}} - 1\right)$ B $\sqrt[t]{\frac{v}{p}} - 1$ C $\sqrt[t]{\frac{v}{p}}$

D $\sqrt[t]{\frac{v}{p}+2}$ E $100\left(\sqrt[t]{\frac{v}{p}} - 1\right)$

Cases involving non - indices (No powers)

2013/17 Neco

Make t the subject of the formula p = (t – 1)ax – at

A $\frac{p-ax}{ax-a}$ B $\frac{p+ax}{ax-a}$ C p + ax D $\frac{p+at}{at}$

E pxa + t

Solution

p = (t – 1)ax – at

Open up bracket containing t

p = axt – ax – at

Collect terms in t together

p + ax = axt – at

Factor out t at RHS

p + ax = t(ax – a)

$\frac{p+ax}{ax-a}$ = t B

2001/1b (Nov)

If $\frac{2d-3c}{5d+c} = \frac{m}{n}$, express d in term of c, m and n

Solution

We are asked to make d the subject of the formula.

Cross multiply to clear fractions

n(2d – 3c) = m(5d + c)

Opening brackets, we have

2dn – 3cn = 5dm + cm

Collect terms in d together

2dn – 5dm = cm + 3cn

Factor out d

d(2n – 5m) = cm + 3cn

d = $\frac{cm+3cn}{2n-5m}$

2005/9

If $\frac{1}{p} = \frac{1}{q} + \frac{1}{r}$, $p = \frac{2}{5}$ and $q = \frac{4}{7}$ find the value of r

A $\frac{4}{17}$ B $\frac{3}{4}$ C $\frac{4}{3}$ D $\frac{17}{4}$

Solution

In indices, we are told that $1/\frac{1}{a}$ = a i.e $1 \div {}^1/_a = 1 \times \frac{a}{1}$ = a

Thus substituting for the given values

$\frac{1}{p} = \frac{1}{q} + \frac{1}{r}$ becomes

$\frac{5}{2} = \frac{7}{4} + \frac{1}{r}$ and <u>not</u> $\frac{5}{2} = \frac{4}{7} + \frac{1}{r}$ <u>not</u> $\frac{5}{2} = \frac{7}{4} + \left(\frac{1}{r}\right)$

No substituting in **r**

$\frac{5}{2} - \frac{7}{4} = \frac{1}{r}$

$\frac{10-7}{4} = \frac{1}{r}$

$\frac{3}{4} = \frac{1}{r}$

Thus 3r = 4 and r = ${}^4/_3$ (C)

2005/6b(Nov)

If $\frac{1}{a} = \frac{1}{b} + \frac{1}{c}$ and b + c – k = 0, express a in terms of b and k only

Solution

We are to eliminate c form $\frac{1}{a} = \frac{1}{b} + \frac{1}{c}$

From b + c – k = 0, we make c the subject formula

c = k – b

Thus $\frac{1}{a} = \frac{1}{b} + \frac{1}{c}$ becomes

$\frac{1}{a} = \frac{1}{b} + \frac{1}{k-b}$

Next, we make a the subject of the formula,
clear fractions on RHS

$\frac{1}{a} = \frac{k-b+b}{b(k-b)}$

$\frac{1}{a} = \frac{k}{b(k-b)}$

b(k – b) = ak

$\frac{b(k-b)}{k}$ = a

2009/8 Neco (Dec)

Make r the formula in the expression $\frac{r}{y} + \frac{p}{x} = \frac{r}{x}$

A r = $\frac{yp}{x-y}$ B r = $\frac{yp}{y-x}$ C r = $\frac{p}{y(y-x)}$

D r = $\frac{xy}{p}$ E r = $\frac{x+y}{p}$

Solution

$\frac{r}{y} + \frac{p}{x} = \frac{r}{x}$

Collect terms in r together

$\frac{p}{x} = \frac{r}{x} - \frac{r}{y}$

Applying LCM method to RHS

$\frac{p}{x} = \frac{ry-rx}{xy}$

Isolate terms in r

$\frac{pxy}{x} = ry - rx$

py = r(y – x)

$\frac{yp}{y-x}$ = r B

2006/46(Nov)

Given that p = $\frac{x+y}{x-y}$, make y the subject of the relation

A $\frac{x(p-1)}{p+1}$ B $\frac{x(p+1)}{p-1}$ C $\frac{x+P}{x-1}$ D $\frac{p-x}{p+1}$

Solution

$$p = \frac{x+y}{x-y}$$

First we clear fractions

p(x – y) = x + y

px – py = x + y

Collect terms in y together

px – x = y + py

Factor y in the RHS

px – x = y(1 + p)

$$\frac{px-x}{1+p} = y \quad \text{i.e} \quad \frac{x(p-1)}{1+p} = y$$

2007/ 42

Make W the subject of the relation $\frac{a+bc}{wd+f} = g$

A $\frac{a+bc-fg}{dg}$ B $\frac{a-bc+fg}{dg}$ C $\frac{a+bc-f}{dg}$ D $\frac{a+bc-dg}{fg}$

Solution

$$\frac{a+bc}{wd+f} = g$$

Cross multiply to clear fraction

g(wd + f) = a + bc

Opening bracket, we have

gwd + gf = a + bc

Isolate term in w

gwd = a + bc – gf

w = $\frac{a+bc-fg}{dg}$ A

2006/2 Neco (Dec)

Make 'A' the subject of the equation

(a) $y = 1 - \frac{x}{3}\left(B + \frac{5}{7}A\right)$

(b)Hence, find the value of A if y = –3, x = 4 and B= –2

Solution

(a) $y = 1 - \frac{x}{3}\left(B + \frac{5}{7}A\right)$

Isolate term in A

$$\frac{x}{3}\left(B + \frac{5}{7}A\right) = 1 - y$$

Multiply through by 3 to clear fraction outside the bracket

$$x\left(B + \frac{5}{7}A\right) = 3(1-y)$$

$$xB + \frac{5}{7}Ax = 3 - 3y$$

Isolate term in A

$$\frac{5}{7}Ax = 3 - 3y - xB$$

Multiply through by 7 to clear fraction

5Ax = 21 – 21y – 7xB

$$A = \frac{21-21y-7xB}{5x}$$

(b) Substituting for the given values

$$A = \frac{21-21(-3)-7(4)(-2)}{5\times 4}$$

$$A = \frac{21+63+56}{20} = \frac{140}{20} = 7$$

2011/50

If $E = \frac{MN}{S+N}$ and E = 75, M = 120, N = 5000, find S

A 1000 B 2000 C 3000 D 4000

Solution

First, we make S subject of the formula

$E = \frac{MN}{S+N}$ becomes

E(S + N) = MN

ES + EN = MN

ES = MN – EN

$$S = \frac{MN-EN}{E}$$

$$= \frac{N(M-E)}{E}$$

Substituting for the given values

$$= \frac{5000(120-75)}{75}$$

$$= \frac{5000\times 45}{75} = 3000 \quad \text{C}$$

2008/19 Neco (Dec)

Make S the subject of the formula $\frac{V}{S+T} = \frac{T}{S}$

A $S = \frac{T^2}{V+T}$ B $S = \frac{T}{V-T}$ C $S = \frac{T}{V+T}$

D $S = \frac{T^2}{V-T}$ E $S = \frac{VT^2}{V+T}$

Solution

$$\frac{V}{S+T} = \frac{T}{S}$$

Cross multiply to clear fractions

VS = T(S + T)

VS = TS + T^2

Collect terms in S together

VS – TS = T^2

S(V – T) = T^2

$$S = \frac{T^2}{V-T} \quad (\text{D})$$

2008/21 Neco Exercise 3.64

If $A = \frac{1}{2}h(a+b)$, make 'a' the subject of the formula

A $\frac{2A-2bh}{b}$ B $\frac{2A-bh}{h}$ C $\frac{2A-bh}{b}$

D $\frac{A-2bh}{h}$ E $\frac{2A-h}{bh}$

2010/29 Neco Exercise 3.65

Make p the subject of the formula $h = \frac{p-q}{pt-qr}$

A $\frac{hr-q}{ht-1}$ B $\frac{h(qr-1)}{ht-r}$ C $\frac{q(hr-1)}{ht-1}$

D $\frac{q(hr-h)}{rt-h}$ E $\frac{q(r-h)}{ht+r}$

2011/21 Neco Exercise 3.66

An expression is given as $\frac{a}{x} + \frac{b}{y} = \frac{a}{y}$

Make a the subject of the formula

A $\frac{bx}{y(y-x)}$ B $\frac{x}{y-bx}$ C $\frac{bx}{x-y}$

D $\frac{x}{bx-y}$ E $\frac{bx}{y-x}$

2012/11 Exercise 3.67

Make p the subject of the relation : $q = \frac{3p}{r} + \frac{s}{2}$

A $p = \frac{2q-rs}{6}$ B p = 2qr – sr – 3

C $p = \frac{2qr-s}{6}$ D $p = \frac{2qr-rs}{6}$

2014/51 Neco Exercise 3.68

Make u the subject of the formula $S = \frac{2uv-r}{u-2}$

A $u = \frac{2s-r}{s-2v}$ B $u = \frac{2s+r}{s+2v}$ C $u = \frac{2s+1}{s-2v}$

D $u = \frac{s+2v}{2s-r}$ E $u = \frac{s-2v}{2s-r}$

2015/26 Neco Exercise 3.69

If $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$, make v the subject of the formula

A. $v = \frac{fu}{f-u}$ B. $v = \frac{u}{f-u}$

C. $v = \frac{fu+fv}{u}$ D. $v = \frac{u-f}{fu}$ E $v = \frac{fu}{u-f}$

2015/11a Exercise 3.70

Make m the subject of the relation

$$h = \frac{mt}{d(m+p)}$$

2014/42 (Nov) Exercise 3.71

Given that $\frac{1}{m} = t - \frac{1}{n}$, make n the subject of the formula

A. $n = \frac{t-1}{m}$ B. $n = \frac{m}{t-1}$

C. $n = \frac{mt-1}{m}$ D. $n = \frac{m}{mt-1}$

CHAPTER FOUR

Equations

Simultaneous linear equations

Simultaneous linear equation as the name implies, is a pair of linear equation (highest degree of the unknown is one) occurring together. Though, in this case there are two different unknowns unlike in the simple equation, which has only one unknown. Unless otherwise stated, we can apply either **elimination or substitution method** in solving any given simultaneous linear equation. Though a third method of solving exists; this is graphical solution. It is left for discussion in a later chapter.

Uniform coefficient problems

Examples

(1) Solve the equations

x + 2y = 5 and x + 3y = 8

Solution

Applying substitution method

x + 2y = 5 ---- (1)

x + 3y = 8 --- (2)

From (1) x = 5 – 2y

Substitute x value into (2) and not into (1)

x + 3y = 8 will become

(5 – 2y) + 3y = 8

5 – 2y + 3y = 8

3y – 2y = 8 –5

y = 3

Substitute y value into (1)

x + 2y = 5 will become

x + 2(3) = 5

x + 6 = 5

x = 5 – 6

x = -1

Solution set is x = -1 and y =3

Applying elimination method to the same problem

x + 3y = 8 ------- (1)

-(x + 2y = 5) ------- (2)

y = 3

i.e equation (1) - (2)

The equations are arranged to place the one with bigger values on top to enable us subtract the smaller one from it.

Substitute y value into (1)

x + 3y = 8 will become

x + 3 (3) = 8

x + 9 = 8

x = 8 – 9

x = -1

Solution set is x = -1 and y =3

Note: Students are to use any of the methods and not the two at the same time on a particular question

2014/29 NABTEB

Solve for x and y, if x + y = 3 and 3x – y = 1

A 1, 2 B 2, 1 C 3, 1 D 1, 3

Solution

x + y = 3

+ (3x – y = 1)

4x = 4

x = 1

Substitute x = 1 into $x + y = 3$

$$1 + y = 3$$
$$y = 3 - 1$$
$$y = 2$$

x, y i.e 1, 2 A

2011/7

Solve the equations: $3x - 2y = 7$
$x + 2y = -3$

A $x = 1, y = -2$ B $x = 1, y = 3$
C $x = -2, y = -1$ D $x = 4, y = -3$

Solution

$$\begin{array}{r} 3x - 2y = 7 \\ +(\ x + 2y = -3\ ) \\ \hline 4x = 4 \\ x = 1 \end{array}$$

Substitute x = 1 into $3x - 2y = 7$

$$3(1) - 2y = 7$$
$$3 - 2y = 7$$
$$-2y = 7 - 3$$
$$-2y = 4$$
$$y = -2$$

Thus $x = 1$, $y = -2$ (A)

2013/13

If $\frac{1}{2}x + 2y = 3$ and $\frac{3}{2}x + 2y = 1$, find $(x + y)$

A 3 B 2 C 1 D 0

Solution

First, we solve for x and y

$$\begin{array}{r} \frac{1}{2}x + 2y = 3 \\ -(\ \frac{3}{2}x + 2y = 1) \\ \hline -x = 2 \\ x = -2 \end{array}$$

Substitute for $x = -2$ into $\frac{1}{2}x + 2y = 3$

$$\frac{1}{2}(-2) + 2y = 3$$
$$-1 + 2y = 3$$
$$2y = 3 + 1$$
$$2y = 4$$
$$y = 2$$

Thus $x + y = -2 + 2$ i.e 0 (D)

2006/28

If $x + y = 12$ and $3x - y = 20$, find the value of $2x - y$

A 8 B 10 C 12 D 15

Solution

$$\begin{array}{r} x + y = 12 \\ +(\ 3x - y = 20\ ) \\ \hline 4x = 32 \\ x = 8 \end{array}$$

Substitute for x = 8 into $x + y = 12$

$$8 + y = 12$$
$$y = 12 - 8$$
$$y = 4$$

Thus, $2x - y = 2(8) - 4$
$= 16 - 4$
$= 12$ (C)

2011/14 Neco

Solve the simultaneous equations
$3x - y = 7$ and $2x = 3 - y$

A (–1, 2) B (– 2, – 1) C (– 2, 1)
D (2, 1) E (2, – 1)

Solution

$$\begin{array}{r} 2x + y = 3 \\ +(\ 3x - y = 7\ ) \\ \hline 5x = 10 \\ x = 2 \end{array}$$

Substitute for x = 2 into $2x + y = 3$

$$2(2) + y = 3$$
$$4 + y = 3$$
$$y = 3 - 4$$
$$y = -1$$

Solution in coordinate form (x, y) is (2, –1) E

2007/20 Neco Exercise 4.1

Find the valve of $x - y$, if $2x + 2y = 16$ and $8x - 2y = 44$

A 2 B 4 C 6 D 8 E 10

2014/24 NABTEB (Nov) Exercise 4.2

Find the value of $2x - y$. If $x + y = 8$ and $4x - y = 22$

A 2 B 4 C 6 D 10

2014/8 (Nov) Exercise 4.3

If $3x - y = 5$ and $2x + y = 15$, evaluate $x^2 + 2y$

A 29 B 30 C 35 D 42

Exercise 4.4

Find the value of $x - y$, If $2x + y = 8$ and $4x - y = 22$

A 2 B 5 C 7 D 3

Non - uniform coefficient problems

Examples

(1) Solve the equations

$3p + 2q = 7$ and $4p + 3q = 10$

Solution

Applying substitution method

$4p + 3q = 10$ ----------(1)
$3p + 2q = 7$ -----------(2)

from (2) $3p = 7 - 2q$

$$p = \frac{7 - 2q}{3}$$

Substitute p value into (1) **not** (2)

$$4p + 3q = 10$$
$$4\left(\frac{7 - 2q}{3}\right) + 3q = 10$$
$$\left(\frac{28 - 8q}{3}\right) + 3q = 10$$

Multiply through by 3 to clear fractions.

$$28 - 8q + 9q = 30$$
$$9q - 8q = 30 - 28$$
$$q = 2$$

Substitute q value into equation (2)

$3p + 2q = 7$ will become

$$3p + 2(2) = 7$$
$$3p + 4 = 7$$
$$3p = 7 - 4$$
$$3p = 3$$
$$p = 1$$

(p, q) format i.e (1, 2)

Applying Elimination method to the same problem.

$4p + 3q = 10$ -------- (1)

$3p + 2q = 7$--------- (2)

To eliminate q , let us exchange their coefficients i.e (2) x 3 and (1) x 2

$$\begin{array}{l} 9p + 6q = 21 \\ -\underline{(8p + 6q = 20)} \\ p \quad = 1 \end{array}$$

Substitute p value into (2)

$3p + 2q = 7$ will become

$3(1) + 2q = 7$

$3 + 2q = 7$

$2q = 7 - 3$

$2q = 4$

$q = \frac{4}{2} = 2$

Solution set in ordered pair means (x, y) format i.e (1,2)

2009/9 (Nov)

Find the value of x in the simultaneous equations:

$2x + y = 4$;

$x - 2y = 2$

A 6 B 4 C 3 D 2

Solution

$2x + y = 4$ ------ (1)

$x - 2y = 2$ ------ (2)

(1) × 2 and add to eliminate y

$$\begin{array}{l} 4x + 2y = 8 \\ \underline{+(\ x - 2y = 2\ )} \\ 5x \quad = 10 \\ x \quad = 2 \quad (D) \end{array}$$

2007/12

Solve the simultaneous equations:

$x + y = 2$ and $3x - 2y = 1$

A $x = 2, y = 1$ B $x = 1, y = 1$

C $x = 1, y = 2$ D $x = -1, y = 1$

Solution

$x + y = 2$ ----- (1)

$3x - 2y = 1$ ----- (2)

(1) × 3 and subtract to eliminate x

$$\begin{array}{l} 3x + 3y = 6 \\ \underline{-(3x - 2y = 1)} \\ 5y = 5 \\ y = 1 \end{array}$$

Substitute for y = 1 into (1)

$x + y = 2$ becomes

$x + 1 = 2$

$x = 2 - 1$

$x = 1$

Thus $x = 1, y = 1$ (B)

2005/18

If $2x + y = -1$ and $x + 2y = 7$, find the value of $x - y$

A 8 B 2 C – 2 D – 8

Solution

First, we solve for x and y

$2x + y = -1$ ----- (1)

$x + 2y = 7$ ------ (2)

(2) × 2 and subtract

$$\begin{array}{l} 2x + 4y = 14 \\ \underline{-(2x + y = -1)} \\ 3y = 15 \\ y = 5 \end{array}$$

Substitute for y = 5 into (2)

$x + 2y = 7$ becomes

$x + 2(5) = 7$

$x + 10 = 7$

$x = 7 - 10$

$x = -3$

Thus, $x - y = -3 - 5$

$= -8$ (D)

2009/ 12 Neco (Dec)

Solve $5d = 2e - 14$

$5e = d + 12$

A $e = -2, d = -2$ B $e = -2, d = 2$

C $e = 2, d = -2$ D $e = 1, d = -1$ E $e = -1, d = 1$

Solution

Rearranging the equation, we have

$5d - 2e = -14$ ---- (1)

$-d + 5e = 12$ ------ (2)

(2) × 5 and add

$$\begin{array}{l} 5d - 2e = -14 \\ \underline{+(-5d + 25e = 60)} \\ 23e = 46 \\ e = 2 \end{array}$$

Substitute for e = 2 into (2)

$-d + 5e = 12$ becomes

$-d + 5(2) = 12$

$-d = 12 - 10$

$-d = 2$

$d = -2$ Option C

2006/29 Neco Exercise 4.5

Solve the simultaneous equations

$2x = 10 - y$

$3x - 2y = 1$

A $x = 3, y = 4$ B $x = 4, y = 3$

C $x = 4, y = \frac{1}{2}$ D $x = 5, y = -21$ E $x = 10, y = 1$

2010/17 NABTEB (Nov) Exercise 4.6

Solve for x and y in the equations

$2x + 5y = 1$

$3x - 2y = 30$

A $x = 7, y = -2$ B $x = 8, y = -3$

C $x = 6, y = 3$ D $x = 8, y = 3$

2005/21 Exercise 4.7

Solve for y in the equations:

$2x - 3y = 22$

$3x + 2y = 7$

A – 5 B – 4 C 4 D 5

2009/8 Exercise 4.8

Solve the following simultaneous equations:

$2x + 3y = 7$

$x + 5y = 0$

A $x = 5, y = -1$ B $x = \frac{1}{5}, y = -1$

C $x = -\frac{1}{5}, y = 1$ D $x = -5, y = 1$

2006/9 (Nov) Exercise 4.9

Solve the simultaneous equations:

$2x - 3y = 5$

$4x + 5y = -1$

A $x = 4, y = 5$ B $x = 1, y = -1$

C $x = -1, y = -1$ D $x = 1, y = 0$

2007/7c Exercise 4.10

Solve the equations:

$3x - 2y = 21$

$4x + 5y = 5$

2015/37 Neco Exercise 4.11

If $2x + 3y = 19$ and $8x = 11 + y$, by how much is $7x$ greater than $2y$?

A 1 B 2^2 C 3^2 D 4^2 E 5^2

Simultaneous linear equation in fractions.

Examples

(1) Solve the Simultaneous equations

$$\frac{x}{2} - \frac{y}{6} = 1 \quad \text{and} \quad \frac{x}{3} + \frac{y}{2} = \frac{1}{5}$$

Solution

$\frac{x}{2} - \frac{y}{6} = 1$ ------ (1)

$\frac{x}{3} + \frac{y}{2} = \frac{1}{5}$ ------ (2)

To clear fractions multiply through by their LCM

In equation (1) the LCM of the denominators 2,6 and 1 is 6, while the LCM of 3, 2 and 5 in (2) is 30

Thus, (1) x 6 and (2) x 30

$3x - y = 6$ (a) from (1)

$10x + 15y = 6$ (b) from (2)

Applying substitution method, we have from (a)

$y = 3x - 6$

Substitute y value into (b)

$10x + 15y = 6$ will become

$10x + 15(3x - 6) = 6$

$10x + 45x - 90 = 6$

$55x = 6 + 90$

$55x = 96$

$x = \frac{96}{55} = 1\frac{41}{55}$

Substitute x value into (a)

$3x - y = 6$ will become

$3\left(\frac{96}{55}\right) - y = 6$

$\frac{288}{55} - y = 6$ $y = \frac{288}{55} - 6$ i.e $y = -\frac{42}{55}$

(2) Solve the equations

$\frac{2}{y} + \frac{3}{x} = 12$ and $\frac{5}{y} - \frac{3}{x} = 9$

Solution

We change the unknown to another, which will be reconverted after solving.

Let $p = \frac{1}{y}$ and $q = \frac{1}{x}$ then

$2(^1/_y) + 3(^1/_x) = 12$ ----(1)

$5(^1/_y) - 3(^1/_x) = 9$ -----(2)

Changing unknown, we have

$2p + 3q = 12$ ----------- (a)

$+ \underline{(5p - 3q = 9)}$ ------------(b)

$7p = 21$

$p = {}^{21}/_7$ i.e $p = 3$

Substitute p value into (a)

$2p + 3q = 12$ will become

$2(3) + 3q = 12$

$6 + 3q = 12$

$3q = 12 - 6$ i.e 6

$q = {}^6/_3$ i.e $q = 2$

$p = {}^1/_y$, $q = {}^1/_x$

$3 = {}^1/_y$, $2 = {}^1/_x$

$y = {}^1/_3$, $x = ½$

2014/3a

Solve the simultaneous equations:

$$\frac{1}{x} + \frac{1}{y} = 5, \quad \frac{1}{y} - \frac{1}{x} = 1$$

Solution

Let $p = \frac{1}{x}$ and $q = \frac{1}{y}$

$p + q = 5$ and $q - p = 1$

Solving the resulting equations

$$\begin{array}{r} p + q = 5 \\ +(-p + q = 1) \\ \hline 2q = 6 \\ q = 3 \end{array}$$

Substitute $q = 3$ into $p + q = 5$

$p + 3 = 5$

$p = 5 - 3$

$p = 2$

Changing back to our original variables

$p = \frac{1}{x}$ and $q = \frac{1}{y}$

$2 = \frac{1}{x}$ and $3 = \frac{1}{y}$

$2x = 1$ and $3y = 1$

$x = \frac{1}{2}$ and $y = \frac{1}{3}$

2009/9b Neco (Nov)

Solve the simultaneous equations

$$\frac{3}{c} - \frac{4}{d} = \frac{1}{3}, \quad \frac{2}{c} - \frac{5}{d} = 1$$

Hint: put $x = \frac{1}{c}$ and $y = \frac{1}{d}$

Solution

Here the examiner gave us a guide to follow

$\frac{3}{c} - \frac{4}{d} = \frac{1}{3}$ and $\frac{2}{c} - \frac{5}{d} = 1$ becomes

$$3\left(\frac{1}{c}\right) - 4\left(\frac{1}{d}\right) = \frac{1}{3} \; and \; 2\left(\frac{1}{c}\right) - 5\left(\frac{1}{d}\right) = 1$$

Substituting for $x = \frac{1}{c}$ and $y = \frac{1}{d}$

$3x - 4y = \frac{1}{3}$ ------ (1)

$2x - 5y = 1$ ------ (2)

Multiply (i) through by 3 to clear fraction

$9x - 12y = 1$ ---- (3)

Working with equations (2) and (3)

(3) × 2, (2) × 9 and subtract

$$\begin{array}{r} 18x - 24y = 2 \\ -(\ 18x - 45y = 9\ ) \\ \hline 21y = -7 \end{array}$$

$y = -\frac{7}{21}$ i.e $-\frac{1}{3}$

Substitute for $y = -\frac{1}{3}$ into (2)

$2x - 5y = 1$ becomes

$2x - 5(-\frac{1}{3}) = 1$

$2x + \frac{5}{3} = 1$

$2x = 1 - \frac{5}{3}$

$2x = -\frac{2}{3}$

$x = -\frac{2}{3} \div 2$ i.e $-\frac{2}{3} \times \frac{1}{2} = -\frac{1}{3}$

Next, we change c and d into the original equation

$x = \frac{1}{c}$ i.e $c = \frac{1}{x}$.

$= 1 \div \left(\frac{-1}{3}\right)$ i.e $1 \times \left(\frac{-3}{1}\right)$ Thus $c = -3$

$y = \frac{1}{d}$, $d = \frac{1}{y}$ Thus $d = -3$

2006/3a

Solve for x and y in the following equation

$2x - y = \frac{9}{2}$

$x + 4y = 0$

Solution

$2x - y = \frac{9}{2}$ ---- (1)

$x + 4y = 0$ ------ (2)

From (2) $x = -4y$, next substitute $x = -4y$ into (1)

$2(-4y) - y = \frac{9}{2}$

$-8y - y = \frac{9}{2}$

Multiply through by 2 to clear fraction

$-16y - 2y = 9$

$-18y = 9$

$y = \frac{9}{-18}$ i.e $-\frac{1}{2}$

Next, $x = -4y$ becomes

$x = -4(-\frac{1}{2})$

$= 2$

2012/33

If $x + 0.4y = 3$ and $y = \frac{1}{2}x$, find the value of $(x + y)$

A $1\frac{1}{4}$ B $2\frac{1}{2}$ C $3\frac{3}{4}$ D 5

Solution

First, we find value of x and y

$x + 0.4y = 3$ ----- (1)

$y = \frac{1}{2}x$ ------- (2)

Substitute $y = \frac{1}{2}x$ into (1) and work in fraction as shown in the given options

Note that 0.4 is the same as $\frac{4}{10}$ i.e $\frac{2}{5}$

$x + \frac{2}{5}\left(\frac{1}{2}x\right) = 3$

$x + \frac{1}{5}x = 3$

$\frac{6x}{5} = 3$

$6x = 15$

$x = \frac{15}{6}$ i.e $\frac{5}{2}$

Next, $y = \frac{1}{2}\left(\frac{5}{2}\right) = \frac{5}{4}$

Thus $x + y = \frac{5}{2} + \frac{5}{4} = \frac{10+5}{4}$ i.e $\frac{15}{4}$ i.e $3\frac{3}{4}$ (C)

2014/36 (Nov) Exercise 4.12

If $\frac{1}{2}p + q = 1$ and $p - \frac{1}{2}q = 7$, find $(p + q)$

A. -8 B. – 4 C. 4 D. 8

2015/43 Exercise 4.13

Find the value of p if $\frac{1}{4}p + 3q = 10$ and $2p - \frac{1}{3}q = 7$

A. 4 B. 3 C. – 3 D. – 4

Special cases

2012/12

If $x + y = 2y - x + 1 = 5$, find the value of x

A 3 B 2 C 1 D – 1

Solution

$x + y = 2y - x + 1 = 5$

First, we rearrange the equation as

$x + y = 2y - x + 1$ and $2y - x + 1 = 5$

Collect like terms together

$x + x + y - 2y = 1$ and $2y - x = 5 - 1$

$2x - y = 1$ and $2y - x = 4$

Solving the resulting equations

$2x - y = 1$ -------- (1)

$-x + 2y = 4$ ------ (2)

(1) × 2 and add to eliminate y

$$\begin{array}{r} 4x - 2y = 2 \\ +(-x + 2y = 4\ ) \\ \hline 3x = 6 \end{array}$$

$x = 2$ (B)

2005/35 Neco

Solve the equations:

$x + y + 7 = 2x - 3y + 5 = 3x - 4y$

A x = 6, y = 1 B x = 1, y = 6

C x = –1, y = – 6 D x = – 1, y = 6 E x = – 6, y = 1

Solution

Presenting the problem properly

$x + y + 7 = 2x - 3y + 5 = 3x - 4y$ is the same as

$x + y + 7 = 2x - 3y + 5$ and $2x - 3y + 5 = 3x - 4y$

Collect like terms together

$x - 2x + y + 3y = 5 - 7$ and $2x - 3x - 3y + 4y = -5$

$4y - x = -2$ and $y - x = -5$

Solving the resulting simultaneous equations

$$\begin{array}{r} 4y - x = -2 \\ -(\ y - x = -5\ ) \\ \hline 3y = 3 \end{array}$$

$y = 1$

Substitute for $y = 1$ into $y - x = -5$

$$1 - x = -5$$
$$-x = -5 - 1$$
$$-x = -6$$

Thus $x = 6$ and $y = 1$ (A)

2005/2b

The line $y = 3x + 5$ and $y = -4x - 1$ intersect at a point k. find the coordinates of k

Solution

$y = 3x + 5$ and $y = -4x - 1$

We rearrange the two equations and solve

$$\begin{array}{r} y - 3x = 5 \\ -(y + 4x = -1) \\ \hline -7x = 6 \\ x = -\,^6/_7 \end{array}$$

Substitute x value into $y - 3x = 5$

$$y - 3(-\,^6/_7) = 5$$
$$y + \tfrac{18}{7} = 5$$
$$y = 5 - \tfrac{18}{7}$$
$$y = \tfrac{17}{7}$$

Coordinates of k are $(-\,^6/_7,\ ^{17}/_7)$

2008/8 Exercise 4.14

If $p - 2g + 1 = g + 3p$ and $p - 2 = 0$, find g

A – 2 B – 1 C 1 D 2

Word problems

2005/ 39 (Nov)

The sum of two numbers is 31 while their positive difference is 13. Find their product

A 99 B 117 C 198 D 286

Solution

Let the two number be x and y

$x + y = 31$

$x - y = 13$

Find **xy**, first we solve for x and y

$$\begin{array}{r} x + y = 31 \\ +(x - y = 13) \\ \hline 2x = 44 \\ x = 22 \end{array}$$

Substitute for $x = 22$ into $x + y = 31$

$$22 + y = 31$$
$$y = 31 - 22$$
$$y = 9$$

Product $xy = 22 \times 9$ i.e 198 (C)

2009/ 2b (Nov)

From a shop, Kofi bought 2 singlets and 3 shirts for GH¢ 31.00. While Kwasi bought 3singlets and 2 shirts for GH¢ 29.00.How much will Yaw pay for one singlet and one shirt he bought from the same shop?

Solution

$2s + 3t = 31$ ------- (1)

$3s + 2t = 29$ ------- (2)

(1) × 3 and (2) × 2 then subtract

$$\begin{array}{r} 6s + 9t = 93 \\ -(6s + 4t = 58) \\ \hline 5t = 35 \\ t = 7 \end{array}$$

Substituting t value into (1)

$2s + 3t = 31$ becomes

$$2s + 3(7) = 31$$
$$2s + 21 = 31$$
$$2s = 31 - 21$$
$$2s = 10$$
$$s = {}^{10}/_2 \text{ i.e } 5$$

Yaw pay GH¢ 5.00 for singlet and GH¢ 7.00 for shirt.

2009/12

Five bottles of fanta and two packets of biscuits cost GH¢ 6.00. Three bottles of fanta and four packets of biscuits cost GH¢ 5.00. Find the cost of a bottle of fanta

A GH¢ 0.50 B GH¢ 1.00 C GH¢ 1.50 D GH¢ 5.00

Solution

$5f + 2b = 6$ --- (1)

$3f + 4b = 5$ ---- (2)

(1) × 2 and subtract

$$\begin{array}{r} 10f + 4b = 12 \\ -(3f + 4b = 5) \\ \hline 7f = 7 \\ f = 1 \end{array}$$

Thus cost of a bottle of fanta is GH¢ 1.00 (B)

Quadratic equation I

A quadratic function (equation) is noted by the highest power of two among others. Its general form is

$$ax^2 + bx + c = 0 \text{ where } a \neq 0.$$

If we allow a case where $a = 0$ then $ax^2 = 0$ thus, reducing the equation to a linear one since the next power of x is one i.e bx. Other constants b, and c can take the value of zero.

Examples

(i) $x^2 + x + 10 = 0$ (a = 1,b = 1, C = 10)
(ii) $7x^2 - x - 13 = 0$ (a = 7, b = -1 C = -13)
(iii) $19x^2 + 3 = 0$ (a = 19, b = 0, c = 3)
(iv) $14x - x^2 = 0$ (a = - 1, b = 14, c = 0)
(v) $x^2 = 0$ (a = 1, b = 0, c = 0)

Comparison between quadratic and other relations

Examples

(i) $y = 2x + 1$
(ii) $y = 5x^3 + x^2 - 16x + 4$
(iii) $y = 19 - 2x - 3x^2$

Analysis

Example (1) has the highest degree of one thus; it will have only one solution.

Example (ii) has the highest degree of three thus; it will have three solutions.

Example (iii) has the highest power of two thus, it will have two solutions. These solutions are called roots of a quadratic relation, they are also known as the zeros of the relation because when such roots are substituted, the whole equation becomes zero.

2012/21 Neco
Which of the following is **NOT** a quadratic expression?
A $3x^2 – 2x$ B $x(x – 3)$ C $x^2 – 5$
D $5x(x – 2)$ E $5(x – 1)$

Solution

Expanding the unexpanded options i.e B, D and E
B. $x(x – 3) = x^2 – 3x$
D. $5x(x – 2) = 5x^2 – 10x$
E. $5(x – 1) = 5x – 10$
(E) has the highest power of 1; it is not a quadratic expression.

To derive a quadratic equation with known roots.
Given any two roots of a quadratic equation we can derive the required equation

Examples

1. If 2 and +3 are the roots of a quadratic equation, then the equation is ?

Solution

Let x be our variable (unknown)
$x = 2$ or 3
$x = 2$ or $x = 3$
$x – 2 = 0$ or $x – 3 = 0$
$(x – 2)(x – 3) = 0$
$x(x – 3) – 2(x – 3) = 0$
$x^2 – 3x – 2x + 6 = 0$
$x^2 – 5x + 6 = 0$
The method of solving e.g.1 can be generalized as :
let a and b be the roots of any quadratic equation. Then,
$x = a$ or $x = b$
$x – a = 0$ or $x – b = 0$
$(x – a)(x – b) = 0$
$x(x – b) – a(x – b) = 0$
$x^2 – bx – ax + ab = 0$
$x^2 – (b + a)x + ab = 0$
$\underline{x^2 – (a + b)x + ab = 0}$ ---------*
a + b implies **sum of roots**
ab implies **product of roots**
we can restate * as
x^2 – (sum of roots)x + product of roots = 0
Students can always quote any of the two formulae above depending on ability to recall.

2008/40
Find the quadratic equation whose roots are c and – c
A $x^2 + c^2 = 0$ B $x^2 – c^2 = 0$
C $x^2 + 2cx + c^2 = 0$ D $x^2 – 2cx + c^2 = 0$

Solution

Applying the formula :
x^2 – (sum of roots)x + product of roots = 0
Sum of roots: $c + (– c) = 0$
Product of roots: $c \times (– c) = – c^2$
Substituting, we have
$x^2 – (0)x + (– c^2) = 0$
$x^2 – c^2 = 0$ B

2008/27 Neco
Construct a quadratic equation whose roots are $– ^3/_2$ and 7.
A $3x^2 + 11x + 21 = 0$ B $2x^2 + 11x + 21 = 0$
C $2x^2 – 11x – 21 = 0$ D $3x^2 – 11x – 21 = 0$
E $x^2 – 11x – 21 = 0$

Solution

Applying the formula:
x^2 – (sum of roots) x + product of roots = 0
Sum of roots: $= -\frac{3}{2} + 7$ i.e $\frac{11}{2}$
Product of roots $= -\frac{3}{2} \times 7$ i.e $\frac{-21}{2}$
Substituting, we have
$$x^2 - \frac{11}{2}x + \left(\frac{-21}{2}\right) = 0$$
$$x^2 - \frac{11}{2}x - \frac{21}{2} = 0$$
Multiply through by 2 to clear fractions
$2x^2 – 11x – 21 = 0$ C

2009/2a Neco
The roots of the equation $ax^2 + bx + c = 0$ are $x = – ^2/_3$ and $^3/_2$. Find the value of a, b and c

Solution

Applying the formula:
x^2 – (sum of roots)x + product of roots = 0
Sum of roots $= – ^2/_3 + ^3/_2$ i.e $\frac{-4+9}{6} = ^5/_6$
Product of roots $= – ^2/_3 \times ^3/_2$ i.e $– 1$
Substituting
$x^2 – ^5/_6x + (–1) = 0$
Multiply through by 6 to clear fraction
$6x^2 – 5x – 6 = 0$
Comparing the result with $ax^2 + bx + c = 0$
$a = 6$, $b = – 5$ and $c = – 6$

2010/45 NABTEB (Nov)
Given that $x = ^3/_2$ and $x = – 6$, construct a quadratic equation for the above roots
A $x^2 + 12x – 9 = 0$ B $x^2 + 12x – 18 = 0$
C $x^2 – ^9/_2 x – 9 = 0$ D $2x^2 – 9x – 18 = 0$

Solution

Applying the formula: $x^2 – (a + b)x + ab = 0$
$a + b = ^3/_2 – 6$ i.e $–^9/_2$; $ab = – ^3/_2 \times (6)$ i.e $– 9$
Substituting
$x^2 – (–^9/_2)x + (–9) = 0$
$x^2 + ^9/_2x – 9 = 0$
Multiply through by 2 to clear fraction
$2x^2 + 9x – 18 = 0$ D

2009/27 Neco (Dec)
Find the quadratic equation whose roots are $^1/_2$ and $– ^2/_3$
A $6x^2 – x + 2 = 0$ B $6x^2 – x – 2 = 0$
C $6x^2 + x – 2 = 0$ D $2x^2 – 3x – 5 = 0$
E $6x^2 – 7x – 2 = 0$

Solution

Applying the formula: $x^2 – (a + b)x + ab = 0$
$$a + b = \frac{1}{2} + \left(-\frac{2}{3}\right) \text{ i.e } \frac{1}{2} - \frac{2}{3} = \frac{3-4}{6} \text{ i.e } -^1/_6$$
$$ab = \frac{1}{2} \times \left(-\frac{2}{3}\right) \text{ i.e } – ^1/_3$$

Substituting

$x^2 - (-{}^1/_6)x + (-{}^1/_3) = 0$

$x^2 + {}^1/_6 x - {}^1/_3 = 0$

Multiply through by the LCM of 6 and 3 i.e 6 to clear fractions $6x^2 + x - 2 = 0$ C

2009/20 Neco (Dec)

Given that x = ½ and – ½ are the roots of the equation $px^2 + qx + r = 0$, find p, q and r.

A p = 4, q = 0, r = –1 B p = 4,q = 1, r = 0
C p = 1, q = 4, r = 0 D p = 1, q = 0, r = – 1
E p = 1, q = 4, r = – 1

Solution

Applying the formula : $x^2 - (a + b)x + ab = 0$

$a + b = \frac{1}{2} + \left(-\frac{1}{2}\right)$ i.e 0

$ab = \frac{1}{2} \times \left(-\frac{1}{2}\right)$ i.e $-{}^1/_4$

Substituting, we have

$x^2 - ox + (-{}^1/_4) = 0$

$x^2 - {}^1/_4 = 0$

Multiply through by 4

$4x^2 - 1 = 0$

Comparing with $px^2 + qx + r = 0$

p = 4, q = 0, r = –1 A

2011/23 Neco

Find the quadratic equation whose roots are $^1/_2$ and $^2/_3$.

A $6x^2 + 7x - 2 = 0$ B $6x^2 + 7x + 2 = 0$
C $2 - 7x - 6x^2 = 0$ D $6x^2 - 7x - 2 = 0$
E $6x^2 - 7x + 2 = 0$

Solution

Applying the formula: $x^2 - (a + b)x + ab = 0$

$a + b = \frac{1}{2} + \left(\frac{2}{3}\right)$ i.e $\frac{3+4}{6} = \frac{7}{6}$

$ab = \frac{1}{2} \times \left(\frac{2}{3}\right)$ i.e $^1/_3$

Substituting, we have

$x^2 - {}^7/_6 x + {}^1/_3 = 0$

Multiply through by the LCM of 6 and 3 i.e 6 to clear fractions

$6x^2 - 7x + 2 = 0$ E

2014/42 (Nov) Neco

Construct a quadratic equation whose are $-{}^1/_2$ and $2{}^1/_2$.

A $4x^2 + 8x + 5 = 0$ B $4x^2 + 8x - 5 = 0$
C $4x^2 - 8x - 5 = 0$ D $4x^2 - 8x + 5 = 0$
E $4x^2 - 8x = 0$

Solution

Applying the formula:

x^2 – (sum of roots)x + product of roots = 0

Sum of roots = $-{}^1/_2 + 2{}^1/_2$ i.e 2

Product of roots = $-{}^1/_2 \times 2{}^1/_2$ i.e $-{}^1/_2 \times {}^5/_2 = -{}^5/_4$

Substituting, we have

$x^2 - 2x + \left(-\frac{5}{4}\right) = 0$

$x^2 - 2x - \frac{5}{4} = 0$

Multiply through by 4 to clear fraction

$4x^2 - 8x - 5 = 0$ C

2014/45 Neco (Nov)

Find the quadratic equation whose roots are $-{}^3/_2$ and $^1/_4$.

A $8x^2 + 10x = 3$ B $8x^2 - 10x = 3$
C $8x^2 + 10x = -3$ D $8x^2 - 10x = -3$ E $8x^2 + 10x = 5$

Solution

Applying the formula : $x^2 - (a + b)x + ab = 0$

$a + b = -{}^3/_2 \times {}^1/_4 = \frac{-6+1}{4}$ i.e $-{}^5/_4$; $ab = -{}^3/_2 \times {}^1/_4 = -{}^3/_8$

Substituting, we have

$x^2 - \left(-\frac{5}{4}\right)x + \left(-\frac{3}{8}\right) = 0$

$x^2 + \frac{5}{4}x - \frac{3}{8} = 0$

Multiply through by LCM of 4 and 8 i.e 8 to clear fractions

$8x^2 + 10x - 3 = 0$

$8x^2 + 10x = 3$ (A)

2010/45 Counter example

If the sum of the roots of the equation $(x - p)(2x + 1) = 0$ is 2, find the value of P

A $1{}^1/_2$ B $^1/_2$ C $-{}^1/_2$ D $-1{}^1/_2$

Solution

First, we expand as

$(x - p)(2x + 1) = 0$

$x(2x + 1) - p(2x + 1) = 0$

$2x^2 + x - 2px - p = 0$

$2x^2 + (1 - 2p)x - p = 0$

Comparing this with the general formula sum and product of root

x^2 – (sum of root)x + product of roots = 0

The question's statement implies that:

$-(1 - 2p) = 2$

$-1 + 2p = 2$

$2p = 2 + 1$

$2p = 3$

$p = {}^3/_2$ Thus $p = 1{}^1/_2$ A

2009/34 (Nov) Exercise 4.15

Find the equation whose roots are – 3 and 5

A $x^2 + 2x - 15 = 0$ B $x^2 - 2x - 15 = 0$
C $x^2 + 8x - 15 = 0$ D $x^2 - 8x - 15 = 0$

2006/11 (Nov) Exercise 4.16

Find the equation whose roots are –5 and –3

A $x^2 + 2x - 15 = 0$ B $x^2 + 8x + 15 = 0$
C $x^2 - 8x + 15 = 0$ D $x^2 - 8x - 15 = 0$

2014/41 NABTEB Exercise 4.17

Find the equation whose roots are 2 and $^1/_2$

A $x^2 - 5x + 2 = 0$ B $2x^2 - 5x - 2 = 0$
C $x^2 - 2x + 5 = 0$ D $2x^2 - 5x + 2 = 0$

2011/16 Exercise 4.18

Form the equation whose roots are x = $^1/_2$ and $-{}^2/_3$

A $6x^2 - x + 2 = 0$ B $6x^2 - x - 2 = 0$
C $6x^2 + x + 2 = 0$ D $6x^2 + x - 2 = 0$

2005/24 Exercise 4.19

Find the equation whose roots are $-{}^1/_2$ and $^3/_2$

A $2x^2 - 4x - 3 = 0$ B $4x^2 - 4x - 3 = 0$
C $4x^2 - 6x - 3 = 0$ D $2x^2 - 6x - 3 = 0$

2005/5 Exercise 4.20

Find the equation whose roots are 2 and $-3{}^1/_2$

A $2x^2 + 3x + 14 = 0$ B $2x^2 + 5x + 7 = 0$
C $2x^2 + 5x - 7 = 0$ D $2x^2 + 3x - 14 = 0$

Methods for solving quadratic equations

Quadratic equation can be solved, (finding the roots) by the following methods:

(i) Factorization
(ii) Completing the Squares
(iii) Formula
(iv) Graphical

We shall discuss the first three methods under this chapter. Meanwhile, we will treat the graphical solution under graph & charts.

Factorization method

Our discussion shall be based on the general form of a quadratic equation i.e $ax^2 + bx + c = 0$. Thus when the author mention "a" he is referring to the coefficient of x^2 or any other variable as the case maybe. Similarly "b" implies coefficient of x and "c" refers to the constant.

When a is unity (+1)

Examples

(i) $x^2 + 5x + 6 = 0$
(ii) $k^2 + 7k + 10 = 0$
(iii) $d^2 + 10d - 24 = 0$
(iv) $h^2 + 8h + 16 = 0$

Quadratic problems like the ones stated above are factorized and solved by finding two factors of c that when added will give b

Example 1.

Solve by factorization : $x^2 + 5x + 6 = 0$

Solution

C = + 6 and b = + 5

	Factors of + 6	Added values
(i)	+ 6 and +1	+ 7
(ii)	+ 2 and + 3	+ 5
(iii)	- 2 and – 3	- 5

Out of the stated factors of 6; only (ii) fits b which is+ 5

We now replace + 5x by 2x + 3x in the equation. Thus.

$x^2 + 5x + 6 = 0$ becomes
$x^2 + 2x + 3x + 6 = 0$
$x(x + 2) + 3(x + 2) = 0$
$(x + 3)(x + 2) = 0$
$x + 3 = 0$ or $x + 2 = 0$
$x = -3$ or -2

Example 2

Solve $k^2 + 7k + 10 = 0$

Solution

c = +10 and b = +7

Factors of +10	Added values
(i) + 10 and + 1	+ 11
(ii) + 5 and + 2	+ 7

The last one fits in. Thus, we replace +7k with +5k +2k

$k^2 + 7k + 10 = 0$
$k^2 + 5k + 2k + 10 = 0$
$k(k + 5) + 2(k + 5) = 0$
$(k + 2)(k + 5) = 0$
$k + 2 = 0$ or $k + 5 = 0$
$k = -2$ or -5

Example 3

Solve $d^2 + 10d - 24 = 0$

Solution

c = -24 and b = 10

	Factors of – 24	Added values
(i)	-24 and 1	-23
(ii)	+ 8 and – 3	+5
(iii)	+3 and – 8	-5
(iv)	-12 and + 2	-10
(v)	12 and – 2	+10

Factor (v) meets our need; we replace +10d with +12d – 2d

$d^2 + 10d - 24 = 0$ becomes
$d^2 + 12d - 2d - 24 = 0$
$d(d + 12) - 2(d + 12) = 0$
$(d + 12)(d - 2) = 0$
$d + 12 = 0$ or $d - 2 = 0$
$d = -12$ or 2

2010/21 NABTEB (Nov)

Factorize the following equation $x^2 - 8x + 16 = 0$

A x = 4, 4 B x = – 3, 4 C x = – 3, 4 D x = – 4, – 4

Solution

$x^2 - 8x + 16 = 0$

Here coefficient of x^2 is unity. C is 16 and b is –8

Factors of c that when added give us b are – 4 and – 4

$x^2 - 8x + 16 = 0$ becomes
$x^2 - 4x - 4x + 16 = 0$
$x(x - 4) - 4(x - 4) = 0$
$(x - 4)(x - 4) = 0$
$x - 4 = 0$ or $x - 4 = 0$
x = 4 twice (A)

2011/ 39

Find the smaller value of x that satisfies the equation:

$x^2 + 7x + 10 = 0$

A – 5 B – 2 C 2 D 5

Solution

$x^2 + 7x + 10 = 0$

Factors of c (10) that when added give b (+7) are 2 and 5.

Next, we replace + 7x with + 2x + 5x

$x^2 + 7x + 10 = 0$ becomes
$x^2 + 2x + 5x + 10 = 0$
$x(x + 2) + 5(x + 2) = 0$
$(x + 2)(x + 5) = 0$
$x + 2 = 0$ or $x + 5 = 0$
$x = -2$ or -5

Smaller value of x is – 5 A

Note with negative numbers the "so called" bigger numbers are smaller in value than the smaller number, so – 100 is smaller than – 10

2009/19 Neco (Dec)

Solve the equation $6x + 16 = x^2$

A – 2, 8 B 2, 4 C 2, 6 D 2, 8 E 2, – 8

Solution

$6x + 16 = x^2$

Rearranging

$x^2 - 6x - 16 = 0$

Next, we find the factors of c i.e – 16 and when added give b i.e – 6 , they are 2 and – 8

Then we replace – 6x by – 8x + 2x

$$x^2 - 6x - 16 = 0 \text{ becomes}$$
$$x^2 - 8x + 2x - 16 = 0$$
$$x(x - 8) + 2(x - 8) = 0$$
$$(x - 8)(x + 2) = 0$$
$$x - 8 = 0 \text{ or } x + 2 = 0$$
$$x = 8 \text{ or } -2 \qquad A$$

1993/10 Exercise 4.21
Solve the equation $x^2 - 2x - 3 = 0$
A.(-3, 1) B.(-1, -3) C.(3, 1) D.(-3, 0) E.(- 1, 3)

VTR – 11/17 NTI TCII
Exercise 4.22
Find the sum of the roots of the quadratic equation
$$x^2 + x - 9 = 3$$
A.- 12 B. – 7 C. -1 D. 1

1995/14 Exercise 4.23
What is the smaller value of x for which $x^2 - 3x + 2 = 0$
A. 1 B.2 C. 3 D. 4 E.5

2001/91 Neco Exercise 4.24
Solve the equation $x^2 - 3x - 10 = 0$
A – 2 or – 5 B – 3 or -10 C5 or – 2 D2 or 5 E 3 or 10

When a is not Unity (- 1 inclusive)
Examples:
(i) $3 - 2x - x^2 = 0$
(ii) $9 + 8t - t^2 = 0$
(iii) $7w^2 + 10w = -3$

Quadratic problems like the ones listed above are solved by finding the factors of the product of a and c (ac) that will Sum up to give b.

Example 1.
Solve by factorization method; $3 - 2x - x^2 = 0$
Solution
a = -1 and c = 3 thus ac = -3 and b = - 2

Factors of – 3	Added up
(i) 3 and – 1	+ 2
(ii) – 3 and +1	- 2

Factors (ii) meet our need ; thus, we replace – 2x with –3x + x
$$3 - 2x - x^2 = 0 \text{ becomes}$$
$$3 - 3x + x - x^2 = 0$$
$$3(1 - x) + x(1 - x) = 0$$
$$(1 - x)(3 + x) = 0$$
$$1 - x = 0 \text{ or } 3 + x = 0$$
$$x = 1 \text{ or } -3$$

Example 2.
Solve by factorization method $9 + 8t - t^2 = 0$
Solution
a = –1 and c = 9 thus ac = – 9 and b = +8

Factors of – 9	Added up
(i) +3 and – 3	0
(ii) – 9 and +1	- 8
(iii) 9 and – 1	+8

Factors (iii) suit us. Thus, we replace + 8t with + 9t – t
$$9 + 8t - t^2 = 0 \text{ becomes}$$
$$9 + 9t - t - t^2 = 0$$
$$9(1 + t) - t(1 + t) = 0$$
$$(1 + t)(9 - t) = 0$$
$$1 + t = 0 \text{ or } 9 - t = 0$$
$$t = -1 \text{ or } 9$$

Example 3
Solve by factorization $7w^2 + 10w = -3$
Solution
Rearranging, $7w^2 + 10w + 3 = 0$
a = + 7 and c = + 3, thus ac = + 21 and b = + 10

Factors of + 21		Added up
(i)	+ 21 and 1	+ 22
(ii)	– 3 and – 7	- 10
(iii)	+ 3 and + 7	+ 10

Factors (iii) meet our need; we replace +10w with +3w +7w
$$7w^2 + 10w + 3 = 0 \text{ becomes}$$
$$7w^2 + 3w + 7w + 3 = 0$$
$$w(7w + 3) + 1(7w + 3) = 0$$
$$(7w + 3)(w + 1) = 0$$
$$7w + 3 = 0 \text{ or } w + 1 = 0$$
$$7w = -3 \text{ or } w = -1$$
$$w = -{}^3/_7 \text{ or } -1$$

2009/37
If c and k are the roots of $6 - x - x^2 = 0$, find c + k
A 2 B 1 C – 1 D – 3
Solution
$$6 - x - x^2 = 0$$
Here the coefficient of x^2 is –1 i.e not unity
ac i.e – 1 × 6 is – 6 and b is – 1
Factors of ac that when added gives b are 2 and – 3
Next, we replace – x *by* + 2x – 3x
$$6 - x - x^2 = 0 \text{ becomes}$$
$$6 + 2x - 3x - x^2 = 0$$
$$2(3 + x) - x(3 + x) = 0$$
$$(3 + x)(2 - x) = 0$$
$$3 + x = 0 \text{ or } 2 - x = 0$$
$$x = -3 \text{ or } 2 \text{ i.e c and k}$$
Thus c + k = – 3 + 2
= – 1 C

1989/24 Exercise 4.25
Solve the following equation $6x^2 - 7x - 5 = 0$
A. x = ½ or x = -2 $^1/_2$ B. x = $^1/_3$ or x = 2$^1/_2$
C. x = 1 $^2/_3$ or x = -$^1/_2$ D. x = -1$^2/_3$ or x = ½
E. x = $^5/_6$ or x = -1

1990/24 Exercise 4.26
Solve the equation $2a^2 - 3a - 27 = 0$
A.$^3/_2$, 9 B. –$^2/_3$, 9 C. 3, $^9/_2$ D. –3, -$^9/_2$ E. –3, $^9/_2$

1995/17 Exercise 4.27
Solve the equation $3x^2 + 25x - 18 = 0$
A. –3,2 B. – 2, 3 C. – 2,9 D. – 9, $^2/_3$ E. –$^2/_3$, 9

2004/20 Exercise 4.28
Solve the equation $10 - 3x - x^2 = 0$
A x = 2 or –5 B x = -2 or 5 C x = -1 or 10
D x = 2 or 5

2004/25 Exercise 4.29
Find the sum of the roots of the equation
$$2x^2 + 3x - 9 = 0$$
A. – 18 B.- 6 C. $^9/_2$ D. - $^3/_2$

1998/8 Exercise 4.30
Solve the equation $5x^2 - 4x - 1 = 0$
A. –1, $^1/_5$ B. –1, -$^1/_5$ C. –1, $^1/_5$ D. 1, -$^1/_5$ E. – 1, 5

When b = 0

Examples

(i) $x^2 - 4 = 0$

(ii) $c^2 = 9$

(iii) $4q^2 = 36$

(iv) $16k^2 = 49$

(v) $2b^2 = 2$

Quadratic problems like the ones shown above are solved by either applying our knowledge of "difference of two squares" i.e.
$x^2 - y^2$ factorizes into $(x - y)(x + y)$
or making the unknown the subject of formula

(1) Solve the equation $x^2 - 4 = 0$

Solution

To achieve a difference of two squares status we observe that

$x^2 - 4 = 0$
$x^2 - 2^2 = 0$
$(x - 2)(x + 2) = 0$
$x - 2 = 0$ or $x + 2 = 0$
$x = +2$ or -2

Alternatively

$x^2 - 4 = 0$
$x^2 = 4$
$x = \pm\sqrt{4}$ i.e +2 or -2

(2) Solve the equation $c^2 = 9$

Solution

$C^2 = 9$
Rearranging, we have,
$C^2 - 9 = 0$
Expressing in difference of two squares format
$C^2 - 3^2 = 0$
$(C - 3)(C + 3) = 0$
$C - 3 = 0$ or $C + 3 = 0$
$C = +3$ or -3

Alternatively

$C^2 = 9$
$C = \pm\sqrt{9}$
$= +3$ or -3

(3) Solve the equation $4q^2 = 36$

Solution

$4q^2 = 36$
Rearranging, we have
$4q^2 - 36 = 0$
Expressing in difference of two squares format.
$2^2q^2 - 6^2 = 0$
$(2q)^2 - 6^2 = 0$
$(2q - 6)(2q + 6) = 0$
$2q - 6 = 0$ or $2q + 6 = 0$
$2q = 6$ or $2q = -6$
$q = {}^6/_2$ or $-{}^6/_2$
$q = 3$ or -3

Alternatively

$4q^2 = 36$
$q^2 = {}^{36}/_4$
$q = \pm\sqrt{9} = +3$ or -3

(4) Solve the equation $16k^2 = 49$

Solution

$16k^2 = 49$
Rearranging, we have
$16k^2 - 49 = 0$
Expressing in difference of two squares format.
$4^2k^2 - 7^2 = 0$
$(4k)^2 - 7^2 = 0$
$(4k - 7)(4k + 7) = 0$
$4k - 7 = 0$ or $4k + 7 = 0$
$4k = 7$ or $4k = -7$
$k = {}^7/_4$ or $-{}^7/_4$
$k = 1{}^3/_4$ or $-1{}^3/_4$

Alternatively

$16k^2 = 49$
$k^2 = \frac{49}{16}$
$k = \pm\sqrt{\frac{49}{16}}$
$= +{}^7/_4$ or $-{}^7/_4$

(5) Solve the equation $2b^2 = 2$

Solution

$2b^2 = 2$
$b^2 = {}^2/_2$
$b^2 = 1$
$b = \pm\sqrt{1}$
$= +1$ or -1

When C = 0

Examples

(i) $x^2 - 2x = 0$

(ii) $2f^2 + 3f = 0$

(iii) $5m^2 = 10m$

(iv) $16z^2 = 15z$

Cases of Quadratic problems similar to the examples above are solved by factoring out "one" of the variable (unknown) and equating the result to zero. Of course the unknown factored out is equal to zero at the end of the day.

(1) Solve $x^2 - 2x = 0$

Solution

$x^2 - 2x = 0$
Factoring out x, we have
$x(x - 2) = 0$
$x = 0$ or $x - 2 = 0$
$x = 0$ or 2

(2) Solve the equation $2f^2 + 3f = 0$

Solution

$2f^2 + 3f = 0$
Factoring out f,
$f(2f + 3) = 0$
$f = 0$ or $2f + 3 = 0$
$f = 0$ or $2f = -3$
$f = 0$ or $-{}^3/_2$

(3) Solve the equation $5m^2 = 10m$
Solution
$5m^2 = 10m$
Rearranging
$5m^2 - 10m = 0$
$5m(m - 2) = 0$
$5m = 0$ or $m - 2 = 0$
$m = {}^0/_5$ or $m = 2$
$m = 0$ or 2

(4) Solve the equation $6z^2 = 15z$
Solution
$6z^2 = 15z$
Rearranging
$6z^2 - 15z = 0$
$3z(2z - 5) = 0$
$3z = 0$ or $2z - 5 = 0$
$z = {}^0/_3$ or $2z = 5$
$z = 0$ or ${}^5/_2$

2014/40 NABTEB Exercise 4.31
Solve for x, if $x^2 - 2x = 0$
A $x = 0$ B $x = 2$ C $x = 0$ or 2 D $x = -2$ or $+2$

2000/37 Exercise 4.32
Solve the equation $3y^2 = 27y$
A. $y = 0$ or 3 B. $y = 0$ or 9 C. $y = -3$ or 3
D. $y = 3$ or 9

1994/23 Exercise 4.33
Which of the following is a root of the equation
$x^2 + 6x = 0$
A. 0 B. 1 C. 2. D. 3 E. it does not have any root.

Miscellaneous cases

2014/32 NABTEB
Solve the equation $\frac{x+3}{2x-3} = \frac{3x}{4x-6}$
A $-6, {}^3/_2$ B $6, -{}^3/_2$ C $6, {}^2/_3$ D $6, {}^3/_2$

Solution
$$\frac{x+3}{2x-3} = \frac{3x}{4x-6}$$
Cross multiply
$(x + 3)(4x - 6) = 3x(2x - 3)$
$x(4x - 6) + 3(4x - 6) = 3x(2x - 3)$
$4x^2 - 6x + 12x - 18 = 6x^2 - 9x$
$6x^2 - 4x^2 - 9x + 6x - 12x + 18 = 0$
$2x^2 - 15x + 18 = 0$
Factorizing
$2x^2 - 12x - 3x + 18 = 0$
$2x(x - 6) - 3(x - 6) = 0$
$(x - 6)(2x - 3) = 0$
$x - 6 = 0$ or $2x - 3 = 0$
$x = 6$ or $2x = 3$
$x = 6$ or ${}^3/_2$ (D)

2014/10a
Solve: $(x - 2)(x - 3) = 12$
Solution
First, we expand LHS
$x(x - 3) - 2(x - 3) = 12$
$x^2 - 3x - 2x + 6 = 12$
$x^2 - 5x + 6 - 12 = 0$
$x^2 - 5x - 6 = 0$
Factorizing
$x^2 - 6x + x - 6 = 0$
$x(x - 6) + 1(x - 6) = 0$
$(x - 6)(x + 1) = 0$
$x - 6 = 0$ or $x + 1 = 0$
$x = 6$ or -1

2014/42
Solve for x, if $(3x + 2)(2x - 7) = 0$
A ${}^2/_3$ or $-{}^7/_2$ B $-{}^2/_3$ or ${}^7/_2$ C 3 or 2
D 2 or -7

Solution
The given problem is already factorized by presentation
$(3x + 2)(2x - 7) = 0$
$3x + 2 = 0$ or $2x - 7 = 0$
$3x = -2$ or $2x = 7$
$x = -{}^2/_3$ or ${}^7/_2$ (B)

2009/10b
Solve the equation $\frac{2}{x-1} - \frac{3}{x+1} = 1$

Solution
Applying LCM of $x - 1$ and $x + 1$ i.e $(x - 1)(x + 1)$ to LHS
$$\frac{2(x+1)-3(x-1)}{(x-1)(x+1)} = 1$$
$$\frac{2x+2-3x+3}{(x-1)(x+1)} = 1$$
$5 - x = (x - 1)(x + 1)$
Expanding the RHS
$5 - x = x(x - 1) + 1(x - 1)$
$5 - x = x^2 - x + x - 1$
$5 - x = x^2 - 1$
Rearranging
$x^2 + x - 6 = 0$
Factorizing
$x^2 - 2x + 3x - 6 = 0$
$x(x - 2) + 3(x - 2) = 0$
$x - 2 = 0$ or $x + 3 = 0$
$x = 2$ or -3

2014/40 (Nov) Neco
If 4 is a root of the quadratic equation
$x^2 + kx + 17 = 0$, find value of k
A $-{}^{33}/_4$ B ${}^4/_{33}$ C 4 D ${}^{33}/_4$ E 17

Solution
If 4 is a root of the given quadratic equation, then we substitute 4 for x in it
$x^2 + kx + 17 = 0$ becomes
$4^2 + k(4) + 17 = 0$
$16 + 4k + 17 = 0$
$4k = -33$
$k = -{}^{33}/_4$ (A)

2005/26
Given that one of the roots of the equation
$2x^2 + (k + 2)x + k = 0$ is 2, find the value of k
A -4 B -2 C -1 D $-{}^1/_4$

Solution
Since 2 is a root of the equation, we can substitute 2 for x as

$2(2^2) + (k+2)2 + k = 0$

$8 + 2k + 4 + k = 0$

$3k + 12 = 0$

$3k = -12$

$k = -^{12}/_3$ i.e -4 A

Completing the Squares method

This method is applied where factorization method is not possible. It derives the way of application from its name "completing the squares". It seeks to make the given equation a perfect square, which can be factorized.

Examples

1. Solve the equation $2x^2 + 3x - 8 = 0$ by completing the squares method

Solution

$2x^2 + 3x - 8 = 0$

Make the coefficient of x^2 unity i.e dividing through by 2

$x^2 + {^3/_2}x - 4 = 0$

Carry the constant (- 4) to the RHS

$x^2 + {^3/_2}x = 4$

Add the square of half of the coefficient of x to both sides of the equation i.e $({^1/_2} \text{ of } {^3/_2})^2$ becomes $({^3/_4})^2$

$x^2 + {^3/_2}x + ({^3/_4})^2 = 4 + ({^3/_4})^2$

On the LHS, we take two terms under squares while we perform arithmetic operation on RHS

$(x + ¾)^2 = 4 + {^9/_{16}}$

$(x + ¾)^2 = \frac{73}{16}$

Take the square root of both sides

$x + ¾ = \pm\sqrt{\frac{73}{16}}$

$x + ¾ = \pm\frac{\sqrt{73}}{4}$

$x = -{^3/_4} \pm \frac{\sqrt{73}}{4}$

$= -{^3/_4} + \frac{\sqrt{73}}{4}$ or $-{^3/_4} - \frac{73}{4}$

$= -\frac{3}{4} + \frac{8.5}{4}$ or $-\frac{3}{4} - \frac{8.5}{4}$

$= \frac{5.5}{4}$ or $-\frac{11.5}{4}$;

$= 1.4$ or -2.9 to 1 d.p

Example 2

Solve the equation $x^2 - 5x - 1 = 0$

by completing the square method, leaving your roots correct to 1 decimal place.

Solution

The coefficient of x^2 is already unity.

$x^2 - 5x - 1 = 0$

Carry the constant (–1) to the RHS

$x^2 - 5x = 1$

Add the square of half the coefficient of x to both sides.

Here the coefficient of x is (-5). i.e $({^1/_2} \text{ of } -5)^2 = (-{^5/_2})^2$

$x^2 - 5x + (-{^5/_2})^2 = 1 + (-{^5/_2})^2$

The two terms under square on the RHS are x and $-{^5/_2}$

$(x - {^5/_2})^2 = 1 + {^{25}/_4}$

$= \frac{29}{4}$

Take the square root of both sides

$x - {^5/_2} = \pm\sqrt{\frac{29}{4}}$

$x - {^5/_2} = \pm\frac{\sqrt{29}}{2}$

$x = \frac{5}{2} \pm \sqrt{\frac{29}{2}}$

$x = \frac{5}{2} + \sqrt{\frac{29}{2}}$ or $\frac{5}{2} - \sqrt{\frac{29}{2}}$

$= \frac{10.4}{2}$ or $-\frac{0.4}{2} = 5.2$ or -0.2 to 1 dp.

2001/6a (Nov)

Solve the equation $2x^2 + 7x + 2 = 0$, by method of completing the square. Give answer correct to 3 decimal places.

Solution

$2x^2 + 7x + 2 = 0$

Make the coefficient of x^2 unity here divide through by 2

$$x^2 + \frac{7}{2}x + 1 = 0$$

Carry the constant 1 to the RHS

$$x^2 + \frac{7}{2}x = -1$$

Add the square of half of the coefficient of x to both sides of the equation i.e $({^1/_2} \text{ of } {^7/_2})^2$ becomes $({^7/_4})^2$

$$x^2 + \frac{7}{2}x + \left(\frac{7}{4}\right)^2 = -1 + \left(\frac{7}{4}\right)^2$$

On the LHS, we take the two terms *under squares* while we perform arithmetic operation on the RHS

$$\left(x + \frac{7}{4}\right)^2 = -1 + \frac{49}{16}$$

$$\left(x + \frac{7}{4}\right)^2 = \frac{33}{16}$$

Take the square root of both sides

$$x + \frac{7}{4} = \pm\sqrt{\frac{33}{16}}$$

$$x + \frac{7}{4} = \pm\frac{\sqrt{33}}{4}$$

$$x = -\frac{7}{4} \pm \frac{\sqrt{33}}{4}$$

$$= -\frac{7}{4} + \frac{\sqrt{33}}{4} \quad or \quad -\frac{7}{4} - \frac{\sqrt{33}}{4}$$

$$= -\frac{7}{4} + \frac{5.7446}{4} \quad or \quad -\frac{7}{4} - \frac{5.7446}{4}$$

$$= \frac{-1.2554}{4} \quad or \quad \frac{-12.7446}{4}$$

$= -0.31385$ or -3.18615

≈ -0.314 or -3.186 to 3d.p

Making a given expression a perfect square

Some quadratic expressions just require a certain number to be added to make it a perfect square.

We can do this by applying our knowledge of completing the square method.

Examples

(1) What must be added to make the expression $x^2 + 6x$ a perfect square?

Solution

First step we check whether the coefficient of x^2 is unity, if not we make it unity by dividing through by the coefficient of x^2.

2nd step. We take square of half the coefficient of x

i.e $(\frac{1}{2} \text{ of } + 6)^2 = (+3)^2$ i.e 9

Thus, adding +9 and not +3 to make the expression a perfect square.

2. Find the value of t such that the given expression is a perfect square. $y^2 + ty + \frac{9}{4}$

Solution

This is a reverse process of the steps taking in the preceding example.

Starting with last operation, which is squaring.

To remove square we take square root of the constant.

i.e $\sqrt{9/4} = \pm 3/2$ ignore minus values.

$= +3/2$

Next operation to be removed is "$\frac{1}{2}$ of ".

To do this, we multiply the new constant by 2

i.e $2 \times 3/2 = 3$ Thus t = 3

2010/48 counter example

If $x^2 + kx + \dfrac{16}{9}$ is a perfect square, find the value of x

A $8/3$ B $7/3$ C $5/3$ D $2/3$

Solution

This is a reverse process of the steps taken in the preceding example.

Starting with the last operation, which is squaring

To remove square, we take square root of $16/9$

i.e $\sqrt{\dfrac{16}{9}} = \pm\dfrac{4}{3}$ ignore minus value here

$= +4/3$

Next operation to be remove is "$\frac{1}{2}$ of "

To do this, we multiply the new constant by 2 i.e

$$2 \times \frac{4}{3} = \frac{8}{3}$$

Thus k is $8/3$ (A)

2008/17 Neco

What must be added to expression $v^2 - 18v$ to make it a perfect square?

A – 81 B – 18 C 9 D 36 E 81

Solution

We observe that the coefficient of v is – 18

thus square of ½ the coefficient of v is

$$\left(\frac{1}{2} \textit{ of } -18\right)^2 \text{ i.e } \left(\frac{-18}{2}\right)^2 = 81$$

Therefore for $v^2 - 18v$ to be a perfect square, 81 is be added E

2006/20

What must be added to $x^2 - 3x$ to make it a perfect square?

A $\dfrac{9}{4}$ B $\dfrac{9}{2}$ C 6 D 9

Solution

We observe that the coefficient of x is – 3

Then square of ½ the coefficient of x is

$$\left(\frac{1}{2} \textit{ of } -3\right)^2 \text{ i.e } \left(\frac{-3}{2}\right)^2 = \frac{9}{4} \quad \text{A}$$

Therefore for $x^2 - 3x$ to be a perfect square, 9/4 is to be added

2008/25 Neco (Dec)

What must be added to $x^2 + 5x$ to make it a perfect square?

A 10x B $\dfrac{5}{2}$x C $\dfrac{5}{2}$ D $\dfrac{25}{4}$ E 25

Solution

We observe that the coefficient of x is 5

Thus square of ½ the coefficient of x is

$$\left(\frac{1}{2} \textit{ of } 5\right)^2 \text{ i.e } \left(\frac{5}{2}\right)^2 = \frac{25}{4} \quad \text{D}$$

Therefore for $x^2 + 5x$ to be a perfect square, $25/4$ is be added

2008/41 NABTEB (Nov) Exercise 4.34

What value of m makes $x^2 + 8x + m$ perfect square?

A – 16 B – 8 C 8 D 16

2014/7b (Nov) Exercise 4.35

When k is added to the expression $y^2 - 12y$, the expression becomes $(y + p)^2$. Find the values of p and k

2005/33 (Nov) Exercise 4.36

Find the value of t that will make $x^2 - 40x + t$ a perfect square

A 900 B 400 C 200 D 100

Formula Method

Given any quadratic equation $x^2 + bx + c = 0$ the roots are gotten by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Where a, b and c are constants. The general formula (*Almighty formula)* as it is called can be used to solve any quadratic equation.

It is derived from the general form of a quadratic equation: $ax^2 + bx + c = 0$; by completing the square method as:

$ax^2 + bx + c = 0$

Divide through by a to make the coefficient of x^2 unity

$$x^2 + \frac{bx}{a} + \frac{c}{a} = 0$$

$$x^2 + \frac{b}{a}x = -\frac{c}{a}$$

Add the square of ½ coefficient of x to both sides

i.e $(\frac{1}{2} \text{ of } b/a)^2 = (b/2a)^2$

$$x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 = -\frac{c}{a} + \left(\frac{b}{2a}\right)^2$$

$$= -\frac{c}{a} + \frac{b^2}{4a^2}$$

$$= \frac{-4ac + b^2}{4a^2}$$

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

Take square root of both sides

$$x + \frac{b}{2a} = \pm\sqrt{\frac{b^2 - 4ac}{4a^2}}$$

$$= \pm\frac{\sqrt{b^2 - 4ac}}{2a}$$

$$x = -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Examples

1. Solve the equation $2x^2 + 7x - 5 = 0$ correct to 1 dp

Solution

Mere observation shows that the given equation cannot be factorized. Thus we apply the general formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Comparing the given equation with $ax^2 + bx + c = 0$

We have; $a = 2$, $b = 7$, $c = -5$

Substituting

$$= \frac{-7 \pm \sqrt{7^2 - 4 \times 2 \times (-5)}}{2 \times 2}$$

$$= \frac{-7 \pm \sqrt{49 + 40}}{4}$$

$$= \frac{-7 \pm \sqrt{89}}{4}$$

$$= \frac{-7 \pm 9.4}{4}$$

$$= \frac{-7 + 9.4}{4} \text{ or } -\frac{7 - 9.4}{4}$$

$= 0.6$ or -4.1 to 1dp

2. Solve the equation of $t^2 - 2t - 5 = 0$ correct to 2 d.p

Solution

Comparing the given equation with the general form $at^2 + bt + c = 0$; $a = 1$, $b = -2$, $c = -5$

Substituting,

$$t = \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \times 1 \times (-5)}}{2 \times 1}$$

$$= \frac{+2 \pm \sqrt{4 + 20}}{2}$$

$$= \frac{2 \pm \sqrt{24}}{2}$$

$$= \frac{2 + 4.899}{2} \text{ or } \frac{2 - 4.899}{2}$$

$$= \frac{6.899}{2} \text{ or } -\frac{2.899}{2}$$

$= 3.449$ or -1.449

≈ 3.45 or -1.45 to 2 d.p

2008/7a

Solve, correct to two decimal places, the equation $4x^2 = 11x + 21$

Solution

Rearranging the equation

$$4x^2 - 11x - 21 = 0$$

No factor of **ac** (4×-21) that will give **b** (-11)

Thus the equation is not factorizable

Applying the general formula for $ax^2 + bx + c = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Here a is 4, b is -11 and c is -21

$$= \frac{-(-11) \pm \sqrt{(-11)^2 - 4 \times 4 \times (-21)}}{2 \times 4}$$

$$= \frac{11 \pm \sqrt{121 + 336}}{8} = \frac{11 \pm \sqrt{457}}{8}$$

$$= \frac{11 \pm 21.378}{8}$$

$$= \frac{11 + 21.378}{8} \text{ or } \frac{11 - 21.378}{8}$$

$$= \frac{32.378}{8} \text{ or } \frac{-10.378}{8}$$

$= 4.05$ or -1.30 to 2d.p

2006/21 Neco (Dec) Exercise 4.37

Which of the following are roots of the quadratic equation $2x^2 - 8x + 5 = 0$?

A $2 \pm \frac{\sqrt{6}}{2}$ B $-2 \pm \frac{\sqrt{6}}{2}$ C $-4 \pm \frac{\sqrt{6}}{4}$

D $-6 \pm \frac{\sqrt{6}}{4}$ E $-8 \pm \frac{\sqrt{6}}{4}$

2014/13(Nov) Exercise 4.38

Solve the equation : $x^2 - \frac{13}{2}x + \frac{15}{2} = 0$

A. $x = -1$ or $\frac{5}{2}$ B. $x = -\frac{3}{2}$ or 5

C. $x = -\frac{1}{2}$ or $\frac{5}{2}$ D. $x = \frac{3}{2}$ or 5

Word problems

2009/47 (Nov)

If 3 times a certain number is subtracted from twice the square of the number, the result is 5. What are the possible values of the number?

A – 5, – 1 B –1, $2^1/_2$ C –1, – $2^1/_2$ D 5, – 1

Solution

Let the number be x

$2(x^2) - 3x = 5$

i .e $2x^2 - 3x - 5 = 0$

Coefficient of x^2 is not unity.

Here ac, i.e $2 \times (-5)$ is – 10 and b is – 3

Factors of ac that when added gives b are 2 and – 5

Replace – 3x by + 2x – 5x

$2x^2 - 3x - 5 = 0$ becomes

$2x^2 + 2x - 5x - 5 = 0$

$2x(x + 1) - 5(x + 1) = 0$

$(x + 1)(2x - 5) = 0$

$x + 1 = 0$ or $2x - 5 = 0$

$x = -1$ or $2x = 5$

$x = -1$ or $^5/_2$ i.e $-1, 2^1/_2$ B

1995/7c

The product of two consecutive positive odd numbers is 195. By constructing a quadratic equation and solving it find the two numbers.

Solution

Let the two consecutive odd numbers be y and y + 2

$y(y + 2) = 195$

$y^2 + 2y = 195$

$y^2 + 2y - 195 = 0$

Factorising, we have

$y^2 - 13y + 15y - 195 = 0$

$y(y - 13) + 15(y - 13) = 0$

$(y - 13)(y + 15) = 0$

$y - 13 = 0$ or $y + 15 = 0$

$y = 13$ or -15

Since the numbers are positive we disregard –15.Thus, y = 13 and y +2 = 13 + 2 i.e 15

2003/40 NABTEB

Find two consecutive numbers whose product is 156.

A 12 and 13 B 15 and 16

C – 12 and 13 D – 13 and 12

Let the two consecutive numbers be x and x +1

Translating: $x(x + 1) = 156$

i .e $x^2 + x = 156$

$x^2 + x - 156 = 0$

Factorising

$x^2 + 13x - 12x - 156 = 0$

$x(x + 13) - 12(x + 13) = 0$

$(x + 13)(x - 12) = 0$

$x + 13 = 0$ or $x - 12 = 0$

$x = -13$ or 12 (D)

2014/41 Neco (Nov) Exercise 4.39

The sum of two consecutive odd numbers is – 4. Find the two numbers

A. 3 and –1 B. –3 and –1 C. 3 and 1

D. 3 and 4 E.–3 and – 4

2015/11 Exercise 4.40

Adding 42 to a given positive number gives the same result as squaring the number. Find the number

A. 14 B. 13 C. 7 D. 6

Simultaneous Linear and Quadratic equations

The approach given to problems of this nature, is to make one of the unknown in the linear equation the subject of formula then substitute the value into the quadratic equation or factoring the quadratic equation which has unknown of the form $x^2 - y^2$ (difference of two squares) and then substituting into the linear equation

Examples:

1. Solve the equations

$x - y = 2$ and $x^2 + y^2 = 52$

Solution

$x - y = 2$ ------ (1)

$x^2 + y^2 = 52$ ------ (2)

From (1) $x - y = 2$

$x = y + 2$

Substitute x value into (2)

$x^2 + y^2 = 52$ will become

$(y + 2)^2 + y^2 = 52$

$y^2 + 4y + 4 + y^2 = 52$

$2y^2 + 4y + 4 - 52 = 0$

$2y^2 + 4y - 48 = 0$

Reducing the terms, we have

$y^2 + 2y - 24 = 0$

Factorizing,

$y^2 + 6y - 4y - 24 = 0$

$y(y + 6) - 4(y + 6) = 0$

$(y + 6)(y - 4) = 0$

$y + 6 = 0$ or $y - 4 = 0$

$y = -6$ or $+4$

Substitute y values into (1)

$x - y = 2$ will become

$x - (-6) = 2$ or $x - 4 = 2$

$x + 6 = 2$ or $x - 4 = 2$

$x = 2 - 6$ or $x = 2 + 6$

$x = -4$ or 8

The solution sets are

(-4, - 6) and (8, 4) in (x, y) format

2. Solve the equation

$p + q = 3$, and $p^2 - q^2 = 15$

Solution

$p + q = 3$ - ---- (1)

$p^2 - q^2 = 15$ --- (2)

The LHS of (2) can be factorized as :

$(p + q)(p - q) = 15$

Substituting the value of p + q = 3

$3(p - q) = 15$

Divide both sides by 3

$p - q = 5$ ---- (3)

$\underline{p + q = 3}$ ---- (1)

$2p = 8$

$p = ^8/_2 = 4$

Substitute for p value in (1)

$p + q = 3$ will become

$4 + q = 3$ $q = 3 - 4 = -1$

Other Examples

3. If $y = x^2 - 4x - 10$ and $y = 2$

Find the values of x that satisfies both relations

Solution

Since $y = x^2 - 4x - 10$ and $y = 2$ then

$y \Rightarrow x^2 - 4x - 10 = 2$

$x^2 - 4x - 10 - 2 = 0$

$x^2 - 4x - 12 = 0$

Factorising

$x^2 - 6x + 2x - 12 = 0$

$x(x - 6) + 2(x - 6) = 0$

$(x - 6)(x + 2) = 0$

$x = 6$ or -2

2001/22

The graph of the curve $y = 2x^2 - 5x - 1$ and a straight line PQ were drawn to solve the equation $2x^2 - 5x + 2 = 0$ what is the equation of the line PQ?

A. $y = -1$ B. $y = 1$ C. $y = 3$ D. $y = -3$

Solution

Let $y = 2x^2 - 5x - 1$ ------- (1)

$2x^2 - 5x + 2 = 0$ ------- (2)

Then (2) – (1)

Will yield the required equation.

$y = 2x^2 - 5x + 2 - (2x^2 - 5x - 1)$

$= 2x^2 - 5x + 2 - 2x^2 + 5x + 1.$

$y = 3$ (C)

VTR - 13/3A NTI TCII

Find the value of x and y such that $y = ½ (x^2 - 3)$ and $x + y = 6$.

Solution

$y = ½ (x^2 - 3)$ ----- (1)

$x + y = 6$ ---- (2)

From (2) $y = 6 - x$

Substitute y value into (1)

$y = ½ (x^2 - 3)$ will become

$6 - x = ½ (x^2 - 3)$

Multiply through by 2 to clear fraction

$2(6 - x) = x^2 - 3$

$12 - 2x = x^2 - 3$

i.e. $x^2 + 2x - 15 = 0$

Factorizing

$x^2 + 5x - 3x - 15 = 0$

$x(x + 5) - 3(x + 5) = 0$

$(x + 5)(x - 3) = 0$

$x + 5 = 0$ or $x - 3 = 0$

$x = -5$ or 3

Substitute x value into (2)

$x + y = 6$ will become

$-5 + y = 6$ or $3 + y = 6$

$y = 6 + 5$ or $y = 6 - 3$

$y = 11$ or 3

The solution Sets: (– 5, 11) and (3, 3) in (x, y) format.

2002/32 NABTEB

Write down the equation whose roots are the points of intersection of the graphs of $y = x^2 + x - 2$ and $y = x + 1$

A. $x^2 + 3 = 0$ B. $x^2 - 3 = 0$ C. $x^2 - 1 = 0$

D. $x^2 + 2x - 3 = 0$

Solution

$y = x^2 + x - 2$ ------- (1)

$y = x + 1$ -------- (2)

From (2) substitute $y = x + 1$ into (1)

$y = x^2 + x - 2$ will become

$x + 1 = x^2 + x - 2$

$x^2 + x - 2 - x - 1 = 0$

$x^2 - 3 = 0$ (B)

2002/1 Neco Exercise 4.41

If $y = 3x^2 - 5x - 2$ at what values of x is $y = -4$?

2003/9 Neco Exercise 4.42

If $x - y = 3$ and $x^2 - y^2 = 0$, find the values of x and y respectively.

A. $^1/_2$, $-^1/_2$ B. $^1/_2$, $^1/_2$ C. $^{-3}/_2$, $^{-3}/_2$

D. $^{-3}/_2$, $^3/_2$ E. $^3/_2$, $^{-3}/_2$

Inequalities I

It is not always that equality sign serves our purposes. Statements such as Lagos is more populated than Abuja, ₦200 is not equal to ₦500 will not fit with equality. Thus, our use of inequality sign.

$<$ " is less than ''

$>$ "is greater than''

$\geq$ " is greater than or equal to"

$\leq$ " is less than or equal to "

< (less than) and > (greater than) are called strict inequalities

Examples

(1) $x > 3$ means x could be any of 4,5,6 … But x cannot be 3

(2) $y < 5$ means y can take values 4,3,2,1,0, – 1 … but not 5

≥ (greater than or equal to) and ≤ (less than or equal to) are called weak inequalities.

the word "at most" is used for greater than or equal to.

Examples

1. $t \geq 10$ means it can take values 10, 11, 12 …

2. $s \leq 17$ means it accepts values as from 17, 16, 15 …

Linear inequalities in one variable

Basic operational principles of linear equations applies to linear inequalities except when we multiply or divide both sides of the equation by a negative number – then we reverse the inequality symbol concerned

Examples

1. Solve $3x - 4 > 8$

Solution

$3x - 4 > 8$

$3x > 8 + 4$

$3x > 8 + 4$

$3x > 12$

Divide both sides by 3

$x > {}^{12}/_3$

$x > 4$

2. For what values of x is

$\frac{1}{2}(x + 3) < 3$

Solution

$\frac{1}{2}(x + 3) < 3$

Multiply both sides by 2

$x + 3 < 6$

$x < 6 - 3$

$x < 3$

3. Solve the inequality $6x - 14 \geq 13x$

Solution

$6x - 14 \geq 13x$

Collect like terms together

$6x - 13x \geq 14$

$-7x \geq 14$

Divide both sides by the coefficient of x i.e -7

$$\frac{-7x}{-7} \leq \frac{14}{-7}$$

Notice the inequality symbol changes from $\geq$ to $\leq$

$x \leq -2$

4. Solve the inequality $x + 3 \leq 4x$

Solution

$x + 3 \leq 4x$

Collect like terms together

$x - 4x \leq -3$

$-3x \leq -3$

Divide both sides by the coefficient of x i.e -3

$$\frac{-3x}{-3} \geq \frac{-3}{-3}$$

$x \geq 1$

Checking

Lets try the values of x within the solution range: 1, 2, 3…

When x =1

$x + 3 \leq 4x$ will be

$1 + 3 \leq 4(1)$ true

when x = 2

$x + 3 \leq 4x$ will be

$2 + 3 \leq 4(2)$ true

$5 < 8$

when x = 3

$3 + 3 \leq 4(3)$

$6 < 12$

Let's now try the non-reversed inequality. i.e

$-3x \leq -3$

Divide both sides by -3 but do not reverse the sign as some students will argue – that both sides are negative.

$$\frac{-3x}{-3} \leq \frac{-3}{-3}$$

$x \leq 1$

Lets try the values of x within the solution range: 1, 0, -1, -2…

when x = 1

$x + 3 \leq 4x$

$4 = 4$ true

when x = 0

$x + 3 \leq 4x$

$0 + 3 \leq 0$ Is it true?

When x = -1

$x + 3 \leq 4x$

$-1 + 3 \leq 4(-1)$

$2 \leq -4$ Is it true?

Please adhere strictly to sign changes when dividing or multiplying both sides of inequality equation by a negative value

2014/20 NABTEB (Nov)

Find the range of values of x for which $3(x + 8) < 7x$

A $x > -6$ B $x < -6$ C $x > 2$ D $x > 6$

Solution

$3(x + 8) < 7x$

$3x + 24 < 7x$

Collect like terms together

$3x - 7x < -24$

$-4x < -24$

Divide both sides by -4 and reverse the inequality sign

$$\frac{-4x}{-4} > \frac{-24}{4}$$

$x > -6$ (A)

2007/44

For what range of values of x is $4x - 3(2x - 1) > 1$?

A $x > -1$ B $x > 1$ C $x < 1$ D $x < -1$

Solution

$4x - 3(2x - 1) > 1$

$4x - 6x + 3 > 1$

Collect like terms together

$4x - 6x > 1 - 3$

$-2x > -2$

Divide both sides by the coefficient of x i.e -2 and reverse the inequality sign

$$\frac{-2x}{-2} < \frac{-2}{-2}$$

$x < 1$ C

2007/27 Neco

Solve the inequality: $x - 4(x + 2) > 8 + 5x$

A $x > 0$ B $x < 0$ C $x > -2$ D $x < -2$ E $x > 2$

Solution

$x - 4(x + 2) > 8 + 5x$

$x - 4x - 8 > 8 + 5x$

Collect like terms together

$x - 4x - 5x > 8 + 8$

$-8x > 16$

Divide both side by -8 and reverse the inequality sign

$$\frac{-8x}{-8} < \frac{16}{-8}$$

$x < -2$ (D)

2009/36 (Nov)

Solve the inequality: $\frac{1}{3}(2x - 1) < 5$

A $x < -5$ B $x < 7$ C $x > 8$ D $x < 8$

Solution

$\frac{1}{3}(2x - 1) < 5$

Multiply through by 3 to clear fraction

$2x - 1 < 15$

$2x < 15 + 1$

$2x < 16$

$$\frac{2x}{2} < \frac{16}{2}$$

$x < 8$ (D)

2005/50 Neco

Solve the inequality, $\frac{x}{6} - \frac{x}{2} \leq \frac{2}{3}$

A $x < 1$ B $x \leq -2$ C $x < -2$ D $x \leq 2$ E $x \geq -2$

Solution

$$\frac{x}{6} - \frac{x}{2} \leq \frac{2}{3}$$

Multiply through by the LCM of the denominators 6, 2 and 3 i.e 6 to clear fractions

$$x - 3x \leq 2(2)$$
$$x - 3x \leq 4$$
$$-2x \leq 4$$

Divide both sides by – 2 and reverse the inequality sign

$$\frac{-2x}{-2} \geq \frac{4}{-2}$$
$$x \geq -2 \quad (E)$$

2009/1a Neco

Solve the inequalities given below

$$\frac{1}{2}(3x-4) - \frac{1}{3}(2+3x) \geq x+5$$

Solution

$$\frac{1}{2}(3x-4) - \frac{1}{3}(2+3x) \geq x+5$$

Multiply through by the LCM of the denominators 2 and 3 i.e 6 to clear fractions

$$3(3x-4) - 2(2+3x) \geq 6(x+5)$$
$$9x - 12 - 4 - 6x \geq 6x + 30$$

Collect like terms together

$$9x - 6x - 6x \geq 30 + 12 + 4$$
$$-3x \geq 46$$

Divide both side by – 3 and reverse the inequality sign

$$\frac{-3x}{-3} \leq \frac{46}{-3}$$
$$x \leq -15^1/_3$$

2008/ 27 Neco (Dec)

Solve the inequality $(y - 3) < \frac{y}{3}$

A $y < 2$ B $y < 3.5$ C $y < 4.5$ D $y > 4.5$ E $y > 6$

Solution

$$(y-3) < \frac{y}{3}$$

Multiply through by 3 to clear fraction

$$3y - 9 < y$$

Collect like terms together

$$3y - y < 9$$
$$2y < 9$$
$$\frac{2y}{2} < \frac{9}{2}$$
$$y < 4.5 \quad (C)$$

2008/41

Solve the inequality $1 - 2x < -\frac{1}{3}$

A $x < \frac{2}{3}$ B $x < -\frac{2}{3}$ C $x > \frac{2}{3}$ D $x > -\frac{2}{3}$

Solution

$$1 - 2x < -\frac{1}{3}$$

Multiply through by 3 to clear fraction

$$3 - 6x < -1 - 3$$
$$-6x < -1$$
$$-6x < -4$$

Divide both sides by – 6 and reverse the inequality sign

$$\frac{-6x}{-6} > \frac{-4}{-6}$$
$$x > \frac{2}{3} \quad (C)$$

2008/42

For what range of values of x is $\frac{1}{2}x + 2 \geq 2x - 1$?

A $x \leq 2$ B $x \geq 2$ C $x \leq 3$ D $x \geq 3$

Solution

$$\frac{1}{2}x + 2 \geq 2x - 1$$

Multiply through by 2 to clear fraction

$$x + 4 \geq 4x - 2$$

Collect like terms together

$$x - 4x \geq -2 - 4$$
$$-3x \geq -6$$

Divide both side by – 3 and reverse the inequality sign

$$\frac{-3x}{-3} \leq \frac{-6}{-3}$$
$$x \leq 2 \quad (A)$$

2008/ 2a

Solve the inequality: $\frac{2}{5}(x-2) - \frac{1}{6}(x+5) \leq 0$

Solution

$$\frac{2}{5}(x-2) - \frac{1}{6}(x+5) \leq 0$$

Multiply through by the LCM of the denominators 5 and 6 i.e 30 to clear fraction

$$6 \times 2(x-2) - 5(x+5) \leq 0$$
$$12x - 24 - 5x - 25 \leq 0$$

Collect like terms together

$$12x - 5x \leq 24 + 25$$
$$7x \leq 49$$
$$\frac{7x}{7} \leq \frac{49}{7}$$
$$x \leq 7$$

2012/14

Solve the inequality: $\frac{-m}{2} - \frac{5}{4} \leq \frac{5m}{12} - \frac{7}{6}$

A $m \geq \frac{5}{4}$ B $m \leq \frac{5}{4}$ C $m \geq \frac{-1}{11}$ D $x \leq \frac{-1}{11}$

Solution

$$\frac{-m}{2} - \frac{5}{4} \leq \frac{5m}{12} - \frac{7}{6}$$

Multiply through by the LCM of the denominators 2, 4, 12 and 6 i.e 12 to clear fractions

$$-6m - 3(5) \leq 5m - 2(7)$$
$$-6m - 15 \leq 5m - 14$$

Collect like terms together

$-6m - 5m \leq -14 + 15$

$-11m \leq 1$

Divide both sides by – 11 and reverse the inequality sign

$$\frac{-11m}{-11} \geq \frac{1}{-11}$$

$$m \geq \frac{-1}{11} \quad (C)$$

2009/ 43 (Nov)

The time spent in answering a question was recorded by 3 boys as follows: x = 1 min 9 sec ; $y = 1^1/_9$ min and z = 1.19min. Arrange these times in order starting with the least

A $x < y < z$ B $y < x < z$ C $y < z < x$ D $z < y < x$

Solution

x = 1 min 9 sec

$y = 1^1/_9$ min i.e 1min 6.67 sec ($^1/_9$ of 60sec)

z = 1.19 min i.e 1 min 11.4 sec (0.19 × 60sec)

Smallest is y, next x, then z

$y < x < z$ B

2010/16 Exercise 4.43

Solve the inequality: $3(x + 1) \leq 5(x + 2) + 15$

A $x \geq -14$ B $x \leq -14$ C $x \leq -11$ D $x \geq -11$

2012/22 Neco Exercise 4.44

Solve the inequality: $4(2 - 3x) < 2 - \frac{2}{3}x$

A $x \leq -\frac{9}{17}$ B $x \geq \frac{9}{17}$ C $x < \frac{9}{17}$

D $x > \frac{9}{17}$ E $x > \frac{17}{9}$

2012/4b Neco Exercise 4.45

Solve the inequality $\frac{1}{3}(x+8) - \frac{1}{7}(2x-4) \leq 1$

2013/10 Exercise 4.46

Solve the inequality: $\frac{2x-5}{2} < (2-x)$

A $x > 0$ B $x < 0$ C $x > 2^1/_2$ D $x < 2^1/_4$

2013/2a Exercise 4.47

Solve: $7x + 4 < \frac{1}{2}(4x + 3)$

2014/23 Neco Exercise 4.48

Find the solution set of the inequality

$$\frac{x}{3} + \frac{3-2x}{5} \leq \frac{x+1}{5}$$

A $x \geq 1^1/_2$ B $x \geq 1^1/_3$ C $x \leq 1$

D $x \leq -1$ E $x \leq -1^1/_2$

2015/2a Exercise 4.49

Solve the inequality : $4 + \frac{3}{4}(x + 2) \leq \frac{3}{8}x + 1$

2015/4b Neco Exercise 4.50

Solve the inequality : $\frac{5x}{8} - \frac{1}{6} \leq \frac{x}{3} + \frac{7}{24}$

Inequality combination and ranges

Two inequalities about a particular variable can be combined as follows:

1. If $-4 < x$ and $x < 2$, combine the two inequalities and give the range of x to whole numbers

Solution

By observation we know that x is between - 4 and + 2, thus combining

$-4 < x < 2$

Range of x = -3, -2, -1, 0, 1

The two limits (- 4 and 2) are not included because of the strict inequalities at both ends

2. Given that $x > 8$ and $20 > x$, combine the two inequalities and list their ranges that are whole numbers only

Solution

i . Combination

$20 > x$ and $x > 8$

Combined as

$20 > x > 8$

ii. Range = { 19,18 … 11,10,9}

Both extreme values of 20 and 8 are not included because of the strict inequalities

3. If $t \leq 3$ and $0 < t$

the result of combining the two inequalities is:

Solution

$0 < t \leq 3$ and

Range = { 1, 2, 3} in whole numbers only.

0 is not included because of the strict inequality at that end but 3 is included as a result of the weak inequality at that limit.

4. If $r < -5$ and $-9 \leq r$

the result of combining the two inequalities is:

Solution

$-9 \leq r < -5$

Range (whole numbers) = {-9, -8,…-3 , – 4}

-9 is included because of the weak inequality there whereas – 5 is not, due to the strict inequality

2009/36

If x is a positive integer, list the values of x which satisfy the equations $3x - 4 \leq 6$ and $x - 1 > 0$

A {1, 2, 3} B {2, 3} C {2, 3, 4} D {2, 3, 4, 5}

Solution

Solving for x values

$3x - 4 \leq 6$ and $x - 1 > 0$

$3x \leq 6 + 4$ and $x > 1$

$3x \leq 10$ and $x > 1$

$$\frac{3x}{3} \leq \frac{10}{3} \quad \text{and} \quad x > 1$$

$x \leq {}^{10}/_3$ i.e 3 (integer)

Range $x > 1$ but $x \leq 3$ = {2, 3}

2005/19

Find the range of values of x for which

$2x - 1 \leq 3$ and $2 - x \leq 5$

A $-3 \leq x \leq 1$ B $-2 \leq x \leq 3$ C $-3 \leq x \leq 4$ D $-3 \leq x \leq 2$

Solution

Solving both inequalities at the same time

$2x - 1 \le 3$ and $2 - x \le 5$

$2x \le 3 + 1$ and $-x \le 5 - 2$

$2x \le 4$ and $-x \le 3$

Here we reverse inequality

$\frac{2x}{2} \le \frac{4}{2}$ and $\frac{-x}{-1} \ge \frac{3}{-1}$

$x \le 2$ and $x \ge -3$

Combing both results

$x \ge -3$ is same as $-3 \le x$

$-3 \le x \le 2$ (D)

2011/6b Exercise 4.51

If $2 + x < 6$ and $7 + x \ge 4$, what is the range of x satisfying both inequalities

Inequalities involving quadratic expression

Here the basic operational principle of solving quadratic equations by method of factorization is mostly applied.

1993/20 UME

Solve the inequality $y^2 - 3y > 18$

A. $-3 < y < 6$ B. $y < -3$ or $y > 6$

C. $y > -3$ or $y > 6$ D. $y < -3$ or $y < 6$

Solution

$y^2 - 3y > 18$

Rearranging

$y^2 - 3y - 18 > 0$

Factorising the resulting quadratic expression on the LHS,

$y^2 - 6y + 3y - 18 > 0$

$y(y - 6) + 3(y - 6) > 0$

$(y - 6)(y + 3) > 0$

By quadratic principle

$y - 6 > 0$ or $y + 3 > 0$

$y > 6$ or $y > -3$ (C)

1997/18 UME

Find the range of values of x which satisfies the inequality $12x^2 < x + 1$

A $-\frac{1}{4} < x < \frac{1}{3}$ B. $\frac{1}{4} < x < \frac{1}{3}$

C. $\frac{1}{4} < x < \frac{1}{3}$ D. $-\frac{1}{4} < x < -\frac{1}{3}$

Solution

$12x^2 < x + 1$

Rearranging

$12x^2 - x - 1 < 0$

Factorizing the LHS

$12x^2 - 4x + 3x - 1 < 0$

$4x(3x - 1) + 1(3x - 1) < 0$

$(3x - 1)(4x + 1) < 0$

By quadratic format

$3x - 1 < 0$ or $4x + 1 < 0$

$3x < 1$ or $4x < -1$

$x < \frac{1}{3}$ or $x < -\frac{1}{4}$

The values show that x is between $-{}^1/_4$ and ${}^1/_3$. Thus

$-{}^1/_4 < x < {}^1/_3$ A.

2000/21 UME

Solve the inequality $2 - x > x^2$

A. $x < -2$ or $x > 1$ B. $x > 2$ or $x < -1$

C. $-1 < x < 2$ D. $-2 < x < 1$

Solution

$2 - x > x^2$

Rearranging

$2 - x - x^2 > 0$

Factorizing the LHS

$2 - 2x + x - x^2 > 0$

$2(1 - x) + x(1 - x) > 0$

$(1 - x)(2 + x) > 0$

By quadratic principle

$1 - x > 0$ or $2 + x > 0$

$-x > -1$ or $x > -2$

$x < 1$ or $x > -2$

Combining the two equalities from the smallest -2

$-2 < x < 1$ (D)

Note that:

$x > -2$ means $-2 < x$ to enable us combine the two inequalities

2006/33 UME Exercise 4.52

If $y = x^2 - x - 12$, find the range of values of x for which $y \ge 0$.

A. $x \le -3$ or $x \ge 4$ B. $x < -3$ or $x > 4$.

C. $-3 \le x \le 4$ D. $-3 < x \le 4$

2007/19 UME Exercise 4.53

The solution of the quadratic inequality $(x^2 + x - 12) \ge 0$ is

A. $x \ge -3$ or $x \le 4$ B. $x \ge 3$ or $x \ge -4$

C. $x \le 3$ or $x \le -4$ D. $x \ge 3$ or $x \le -4$

2015/35 Neco Exercise 4.54

Solve the inequality $\frac{y - 1}{2} \le \frac{6}{y}$

A. $y \le -3$ B. $y \le 4$ C. $y \le -3$ or $y \le 4$

D. $-3 \le y \le 4$ E. $-4 \le y \le 3$

Representation of inequality in a number line

O – open dot is used to represent strict inequalities i.e < or > while,

● – thick dot represents weak inequality i.e $\ge$ or $\le$

Single inequality number line

2005/9b

Solve the following inequality : $2(x + 3) - 3(x - 1) \le 12$

Represent your answer on the number line

Solution

$2(x + 3) - 3(x - 1) \le 12$

$2x + 6 - 3x + 3 \le 12$

$2x - 3x \le 12 - 6 - 3$

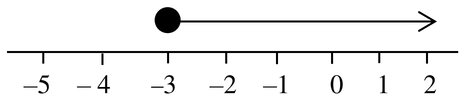
$-x \le 3$

Divide both sides by -1 and reverse the inequality sign

$\frac{-x}{-1} \ge \frac{3}{-1}$

$x \ge -3$

The line starts at -3 with thick dot (weak inequality there) and moves forward



2005/9a

Find the smallest integer that satisfies the inequality

$$x + 8 < 4x - 15$$

Solution

First we solve for the range of x

$$x + 8 < 4x - 15$$

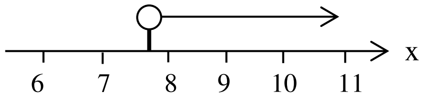
Collect like terms together

$$x - 4x < -15 - 8$$
$$-3x < -23$$

Divide both sides by –3 and reverse the inequality sign

$$\frac{-3x}{-3} > \frac{-23}{-3}$$
$$x > 7.67$$

The smallest <u>integer</u> that satisfies the inequality is 8.
The number line will explain it better

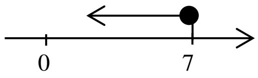


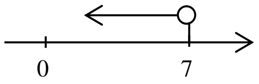
So the first whole number (integer) on the path of the arrow is 8

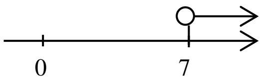
2006/14

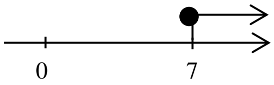
Represent the solution of the inequality

$$\frac{x-2}{2} \geq \frac{x+3}{4}$$ on the number line

A 

B 

C 

D 

Solution

$$\frac{x-2}{2} \geq \frac{x+3}{4}$$

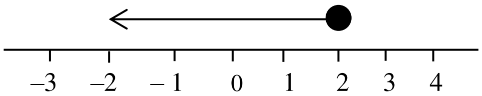
Multiply through by the LCM of 4 and 2 i.e 4 to clear fractions

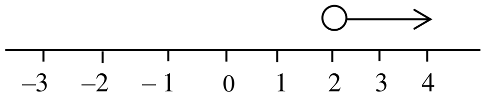
$$2(x-2) \geq x + 3$$
$$2x - 4 \geq x + 3$$
$$2x - x \geq 3 + 4$$
$$x \geq 7 \quad (D)$$

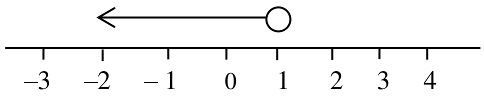
2009/15

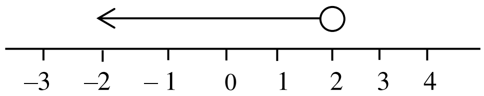
Which of the following illustrates the solution of

$$\frac{(3x+4)}{2} < x + 3 \ ?$$

A. 

B. 

C. 

D. 

Solution

First we solve $\frac{(3x+4)}{2} < x + 3$

Multiply through by 2 to clear fraction

$$3x + 4 < 2(x + 3)$$
$$3x + 4 < 2x + 6$$

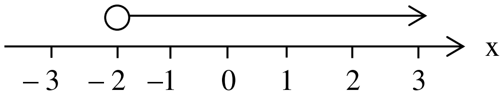
Collect like terms together

$$3x - 2x < 6 - 4$$
$$x < 2$$

The line starts at 2 with open dot and goes backward (D)

2014/21 Neco

Which of the following inequalities is represented by the number line ?



A. $x > -2$ B. $x \geq -2$ C. $x \leq 2$ D. $x < 2$ E. $x = 2$

Solution

The starting point is –2 and it is open dot i.e strict inequality. Since the arrow is pointing forward from –2 to positive axis it implies greater than

Thus $x > -2$ (A)

2014/10c NABTEB

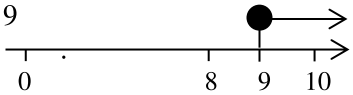
Solve the inequality : $\frac{3}{4}(x-3) \geq \frac{9}{2}$ and represent your answer in a number line

Solution

$$\frac{3}{4}(x-3) \geq \frac{9}{2}$$

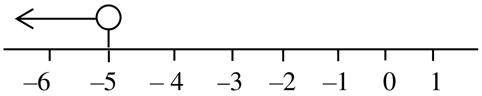
Multiply through by the LCM of 4 and 2 i.e 4 to clear fractions

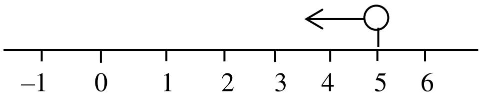
$$3(x-3) \geq 2 \times 9$$
$$3x - 9 \geq 18$$
$$3x \geq 18 + 9$$
$$3x \geq 27$$
$$\frac{3x}{3} \geq \frac{27}{3}$$
$$x \geq 9$$

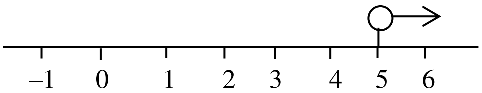
The number line is 

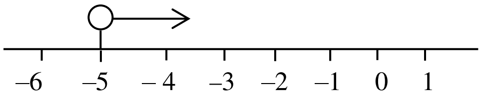
2005/46 Exercise 4.55 option D

Represent the solution of $3x + 15 > 0$ on a number line

A. 

B. 

C. 

D. 

2006/5b Neco (Dec) Exercise 4.56

(i) Find the range of values of x for which

$$2x - 3 > 7x + 12$$

(ii) Represent your answer on a number line

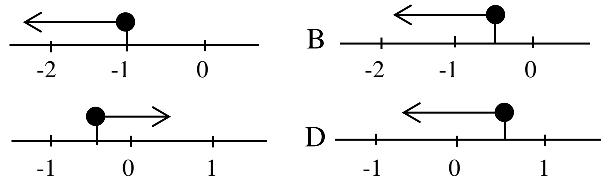
2014/10a (Nov) Exercise 4.57

(i) Solve the inequality : $\frac{1}{2}x - \frac{5}{6}(x+2) \leq 1 + x$

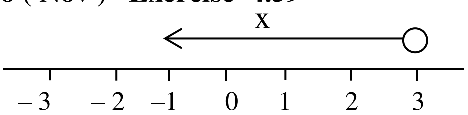
(ii) Illustrate the solution on a number line

2006/41 (Nov) Exercise 4.58

Which of the following represents the solution of the inequality $x - \frac{1}{4} \leq \frac{3x}{2}$ on the number line ?



2006/7 Neco (Nov) Exercise 4.59

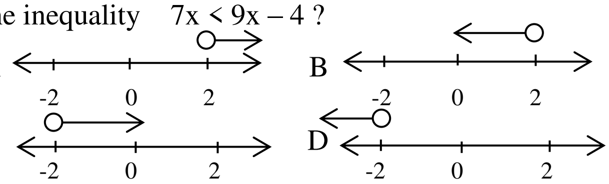


The number line above can be represented by which one of the following inequalities?

A. x < – 3 B. x < 3 C. x > – 3 D. x ≥ 3 E. x > 3

2013/12 Exercise 4.60

Which of the following lines represents the solution of the inequality 7x < 9x – 4 ?



Double inequalities number line

2006/7a

(i) If 4x < 2 + 3x and x – 8 < 3x, what range of values of x satisfies both inequalities?

(ii) Represent your result in (i) on the number line

Solution

4x < 2 + 3x and x – 8 < 3x

4x – 3x < 2 and x – 3x < 8

x < 2 and – 2x < 8

Divide both sides by – 2 and reverse inequality

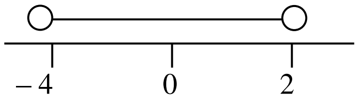
$$\frac{-2x}{-2} > \frac{8}{-2}$$

x > – 4

Combining the two ranges

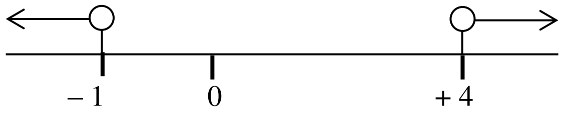
– 4 < x < 2

(ii)



2009/22 NABTEB (Nov)

State the range of values of x for the graph below:



A either x < – 1 or x ≥ 4 B either x < – 1 or x > 4

C either x < – 1 or x ≤ 4 D either x ≥ 1 or x ≥ – 4

Solution

LHS is x < – 1 (open dot and arrow backward)

RHS is x > 4 (open dot and arrow forward) Option B

2009/9 Neco (Nov)

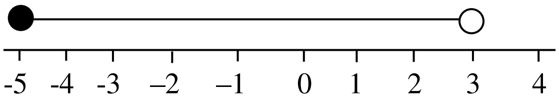
Show on the number line the inequalities – 5 ≤ x < 3

Solution

The range is from – 5 to 3

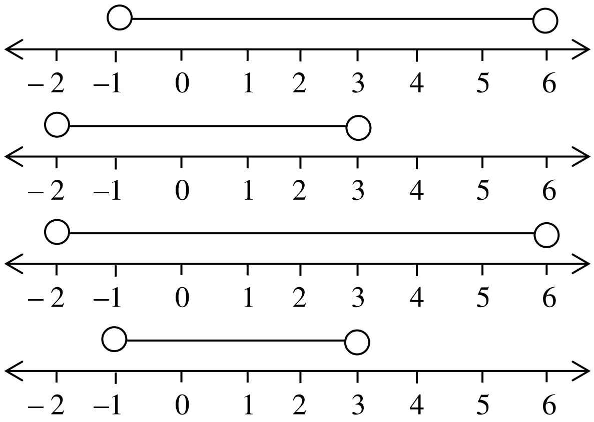
Thick dot at – 5; reason – weak inequality of ≤

Open dot at 3; reason – strict inequality of <



2011/48

Illustrate the inequality –1< 3x + 5 < 14 on a number line



Solution

–1 < 3x + 5 < 14

First, we solve for the range

Applying the split method, we have

– 1 < 3x + 5 and 3x + 5 < 14

Collect like terms together

– 1 – 5 < 3x and 3x < 14 – 5

– 6 < 3x and 3x < 9

$-\frac{6}{3} < \frac{3x}{3}$ and $\frac{3x}{3} < \frac{9}{3}$

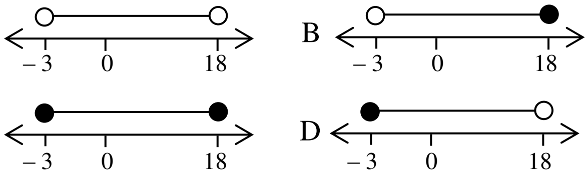
–2 < x and x < 3

Thus combining the result, we have – 2 < x < 3

Open dot at – 2 and open dot at 3 (B)

2014/7

Which of the following number lines represents the solution to the inequality : $-9 \leq \frac{2}{3}x - 7 < 5$



Solution

First, we solve for the range

Applying the split method

$-9 \leq \frac{2}{3}x - 7 < 5$ will become

$-9 \leq \frac{2}{3}x - 7$ and $\frac{2}{3}x - 7 < 5$

$-9 + 7 \leq \frac{2}{3}x$ and $\frac{2}{3}x < 5 + 7$

$-2 \leq \frac{2}{3}x$ and $\frac{2}{3}x < 12$

– 6 ≤ 2x and 2x < 36

– 3 ≤ x and x < 18

Then combining the results, we have

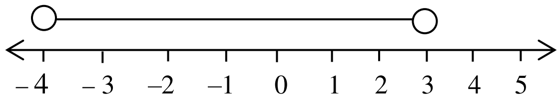
– 3 ≤ x < 18 option D Thick dot at –3 and open dot at 18

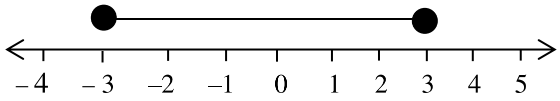
2014/9
Given that x > y and 3< y, which of the following is / are true?
I y > 3   II x < 3   III x > y > 3
A I only   B I and II only   C I and III only   D I, II and III only

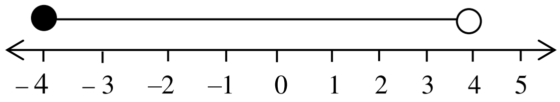
Solution

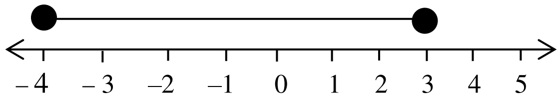
From the statement
$x > y$,
$x > 3$ and $y > 3$
Conveniently, we can take y > 3, then
Combining $x > y > 3$. Thus option C

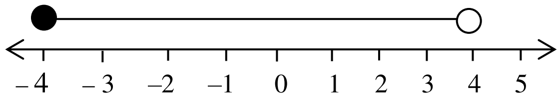
2014/27 Neco
Which of the following number lines correctly represents the inequality $-4 \le x \le 3$?

A.


B.


C.


D.


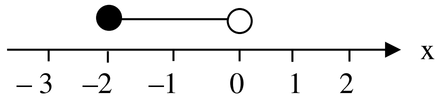
E.


Solution

$-4 \le x \le 3$

The line starts at – 4 and ends at 3.
Both ends are weak inequalities i.e thick dots
Option D

2012/17 Neco
The number line below represents

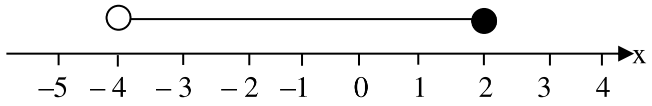


A $-2 > x \ge 0$ B $-2 \ge x > 0$ C $-2 < x < 0$
D $-2 < x \le 0$ E $-2 \le x < 0$

Solution

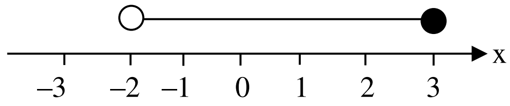
Thick dot at – 2 shows weak inequality
Open dot at 0 shows strict inequality
The range is from – 2 to 0
Thus $-2 \le x < 0$ (E)

2005/25 Neco   Exercise 4.61



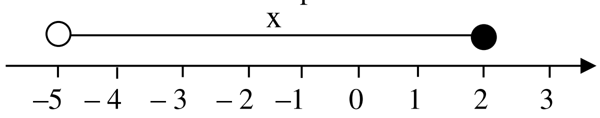
Which of the equations represent the number line above?
A $-4 \le x \le 3$ B $-4 < x < 3$ C $-4 \le x < 3$
D $x \le 3$ E $-4 < x \le 3$

2006/14 Neco   Exercise 4.62
What is the solution of the inequality represented on the number line below?



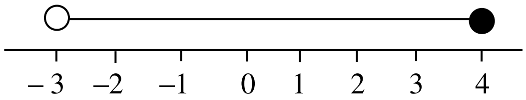
A $-2 \le x \le 3$ B $-2 < x \le 3$ C $-2 > x > 3$
D $-2 \le x < 3$ E $-2 < x < 3$

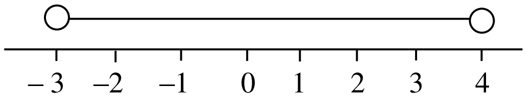
2008/15 Neco   Exercise 4.63
The number line below represents :

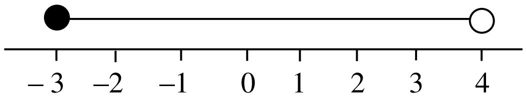


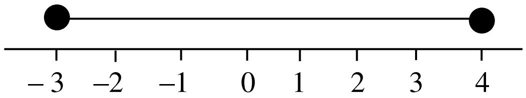
A $0 \le x < -5$ B $0 < x < -5$ C $-5 \le x < 2$
D $-5 < x \le 2$ E $-5 < x < 2$

2008/43   Exercise 4.64
Which of the number lines below best represent the inequality $-3 \le x < 4$?

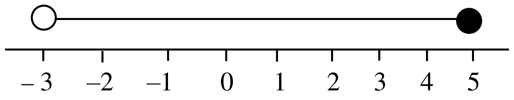
A.


B.


C.


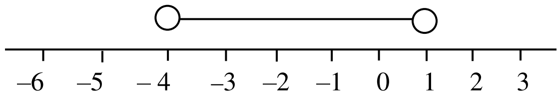
D.


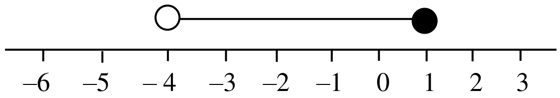
2011/22 Neco   Exercise 4.65
The number line below represents

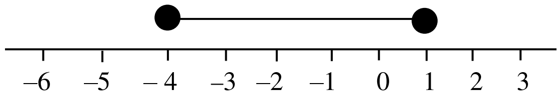


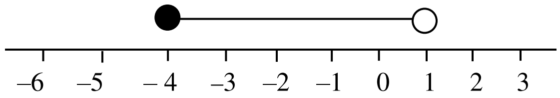
A $-3 \le x < 5$ B $-3 < x \le 5$ C $-3 < x < 5$
D $-3 \le x \le 5$ E $-3 \ge x \ge 5$

2012/10   Exercise 4.66
Represent the inequality $-7 < 4x + 9 \le 14$ on a number line

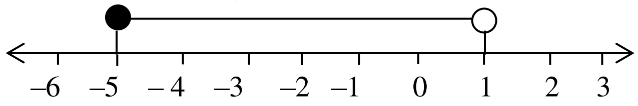
A.


B.


C.


D.


2014/36 Neco (Nov)   Exercise 4.67



Which of the following is illustrated in the diagram above, if x varies over the set of real numbers?
A. $-5 < x < 1$ B. $-5 < x \le 1$ C. $-5 \le x \le 1$ D $-5 \le x < 1$ E $-5 \le x$

CHAPTER FIVE

Indices and logarithm

Indices

Laws of indices

1. $x^a \times x^b = x^{a+b}$ **but** $x^a \times y^b \neq x^{a+b}$

Examples on simple application

i. $3^5 \times 3^1 = 3^{5+1}$ i.e 3^6 **but** $3^5 \times 2^1 \neq 3^{5+1}$

ii. $7^2 \times 7^2 = 7^{2+2}$ i.e 7^4 **but** $7^2 \times 8^2 \neq 7^{2+2}$

2. $x^a \div x^b = x^{a-b}$

OR $\frac{x^a}{x^b} = x^{a-b}$

Examples on simple application

i. $2^4 \div 2^3 = 2^{4-3}$ i.e 2^1

ii. $5^3 \div 5^1 = 5^{3-1}$ i.e 5^2

iii. $\frac{3^4}{3^2} = 3^{4-2}$ i.e 3^2

3. $x^0 = 1$

Examples on simple application

i. $2^0 = 1$

ii. $10^0 = 1$

iii. $(4^2)^0 = 1$

4. $x^{-a} = \frac{1}{x^a}$ and

$\left(\frac{y}{x}\right)^{-1} = \frac{x}{y}$ *Also* $\left(\frac{1}{x}\right)^{-a} = x^a$ *most common* $x^{-1} = \frac{1}{x}$ or

$$\left(\frac{y}{x}\right)^{-a} = \left(\frac{x}{y}\right)^a$$

Examples on simple application

i. $10^{-2} = \frac{1}{10^2}$

ii. $\left(\frac{4}{5}\right)^{-2} = \left(\frac{5}{4}\right)^2$

iii. $\left(\frac{1}{100}\right)^{-2} = 100^2$

5. $(x^a)^b = x^{ab}$

Examples on simple application

i. $(2^3)^2 = 2^{3\times2}$ i.e 2^6

ii. $(5^2)^2 = 5^{2\times2}$ i.e 5^4

6. $x^{1/a} = \sqrt[a]{x}$

Examples on simple application

i. $16^{1/2} = \sqrt{16}$

ii. $27^{1/3} = \sqrt[3]{27}$

ii. $10000^{1/4} = \sqrt[4]{10000}$

7. $x^{b/a} = \left(\sqrt[a]{x}\right)^b$

i. $4^{3/2} = \left(\sqrt{4}\right)^3$

ii. $125^{2/3} = \left(\sqrt[3]{125}\right)^2$

The given examples on simple application were not fully solved; it is intentionally done to allow for step by step understanding of the laws

Indices involving numbers and their powers

2015/33 Neco

Simplify $\left(\frac{1}{4}\right)^{-\frac{1}{2}}$

A. $-\frac{1}{2}$ B. $\frac{1}{4}$ C. $\frac{1}{2}$ D.2 E.4

Solution

Applying indices law

$$\left(\frac{1}{4}\right)^{-\frac{1}{2}} = (4)^{\frac{1}{2}}$$

$$= 2 \quad D$$

2008/8 NABTEB (Nov)

Find the value of $8^{-\frac{2}{3}}$

A $^1/_2$ B $^1/_4$ C $^1/_8$ D $^1/_{16}$

Solution

Applying the law of indices

$$8^{-\frac{2}{3}} = \frac{1}{8^{\frac{2}{3}}}$$

$$= \frac{1}{(2^3)^{\frac{2}{3}}}$$

$$= \frac{1}{2^2} \text{ i.e } \frac{1}{4} \qquad (B)$$

2011/12 Neco

Evaluate $(0.008)^{\frac{1}{3}}$

A 0.15 B 0.8 C 0.5 D 0.2 E 0.1

Solution

$$(0.008)^{\frac{1}{3}} = \left(\frac{8}{1000}\right)^{\frac{1}{3}}$$

We raise items inside the bracket to power 3 to cancel root 3 outside

$$= \left(\frac{2^3}{10^3}\right)^{\frac{1}{3}}$$

$$= \left(\frac{2}{10}\right)^{\frac{3}{3}} = \frac{2}{10} \text{ i.e } 0.2 \quad D$$

2008/3 NABTEB (Nov)

Simplify $(-8)^{\frac{4}{3}}$

A – 32 B – 16 C 16 D 32

Solution

The power 4 which is even will cancel the minus sign

$$(-8)^{\frac{4}{3}} = (8)^{\frac{4}{3}} = (2^3)^{\frac{4}{3}} = 2^4 \text{ i.e } 16 \quad (C)$$

2014/7 NABTEB (Nov)

The value of $\left(4^{\frac{1}{4}}\right)^6$ is

A $^1/_8$ B 4 C 6 D 8

Solution

$$\left(4^{\frac{1}{4}}\right)^6 = 4^{\frac{6}{4}} = 4^{\frac{3}{2}} = (2^2)^{\frac{3}{2}} = 2^3 = 8 \quad D$$

2014/8 NABTEB

Simplify $125^{-\frac{2}{3}} \times 15$

A $^1/_{25}$ B $^3/_{25}$ C $^1/_5$ D $^3/_5$

Solution

$$125^{-\frac{2}{3}} \times 15 = \frac{15}{125^{\frac{2}{3}}}$$

Raise denominator in power 3

$$= \frac{15}{(5^3)^{\frac{2}{3}}} = \frac{15}{(5)^2}$$

$$= \frac{3 \times 5}{5 \times 5} = \frac{3}{5} \quad D$$

2008/1 Neco (Nov)

Simplify $125^{\frac{1}{3}} \times 49^{\frac{1}{2}} \times 10^{-1}$

A $^1/_{35}$ B $^2/_{35}$ C $^1/_{14}$ D $3^1/_2$ E $4^1/_3$

Solution

$$125^{\frac{1}{3}} \times 49^{\frac{1}{2}} \times 10^{-1} = \frac{(5^3)^{\frac{1}{3}} \times (7^2)^{\frac{1}{2}}}{10}$$

$$= \frac{5 \times 7}{10} = \frac{7}{2} \text{ i.e } 3^1/_2 \quad D$$

2005/1b(i) Neco

Simplify $\left(\frac{1}{16}\right)^{-\frac{3}{4}} + 5(9^0)$

Solution

Applying indices laws

$$\left(\frac{1}{16}\right)^{-\frac{3}{4}} + 5(9^0) = (16)^{\frac{3}{4}} + 5 \times 1$$

Express 16 in power that will cancel root 4

$$= (2^4)^{\frac{3}{4}} + 5$$
$$= 2^3 + 5$$
$$= 8 + 5 \text{ i.e } 13$$

2005/2a

Evaluate $2 \div \left(\frac{64}{125}\right)^{-\frac{2}{3}}$

Solution

Applying the laws of indices for minus power

$$2 \div \left(\frac{64}{125}\right)^{-\frac{2}{3}} = 2 \div \left(\frac{125}{64}\right)^{\frac{2}{3}}$$

Next, we seek to express the bracket item in power 3 that will cancel roots 3 outside it

$$= 2 \div \left(\frac{5^3}{4^3}\right)^{\frac{2}{3}}$$

$$= 2 \div \left(\frac{5}{4}\right)^{3 \times \frac{2}{3}} = 2 \div \left(\frac{5}{4}\right)^2$$

$$= 2 \div \frac{25}{16} = 2 \times \frac{16}{25} = \frac{32}{25} \text{ i.e } 1^7/_{25}$$

2006/1b (Nov)

Without using mathematical tables or calculator

Simplify : $16^{\frac{1}{2}} \times 8^{-4} \times 3^{-\frac{2}{3}} \div 81^{-\frac{2}{3}}$

Solution

Express the items in 3 and 2 with their power

$$16^{\frac{1}{2}} \times 8^{-4} \times 3^{-\frac{2}{3}} \div 81^{-\frac{2}{3}} = (2^4)^{\frac{1}{2}} \times (2^3)^{-4} \times 3^{-\frac{2}{3}} \div (3^4)^{-\frac{2}{3}}$$

$$= 2^2 \times 2^{-12} \times 3^{-\frac{2}{3}} \div 3^{-\frac{8}{3}}$$

Applying laws of indices

$$= 2^{2-12} \times 3^{-\frac{2}{3} - \left(-\frac{8}{3}\right)}$$

$$= 2^{-10} \times 3^{-\frac{2}{3}+\frac{8}{3}}$$

$$= 2^{-10} \times 3^{\frac{6}{3}}$$

$$= 2^{-10} \times 3^2$$

$$= \frac{9}{2^{10}} \text{ i.e } \frac{9}{1024}$$

2014/10 Neco

Evaluate $\frac{9^{\frac{1}{2}} \times 27^{\frac{2}{3}}}{64^{\frac{2}{3}}}$

A $11^1/_6$ B $1^3/_4$ C $1^{11}/_{16}$ D $1^7/_{21}$ E $1^{13}/_{63}$

Solution

We should clear root 2 by raising item in it to power 2 similarly; cube-root is cleared by raising its item to power 3

$$\frac{9^{\frac{1}{2}} \times 27^{\frac{2}{3}}}{64^{\frac{2}{3}}} = \frac{(3^2)^{\frac{1}{2}} \times (3^3)^{\frac{2}{3}}}{(4^3)^{\frac{2}{3}}}$$

$$= \frac{3 \times 3^2}{4^2} = {}^{27}/_{16} \text{ i.e } 1^{11}/_{16} \quad (C)$$

2008/3 Neco (Nov)

Evaluate $27 \times \frac{2^0 \times 3^{-2}}{4^{-\frac{1}{2}}}$

A 48 B 12 C 6 D $^3/_2$ E $^1/_3$

Solution

Express 27 in 3 and its powers, 4 in 2 and its power

$$27 \times \frac{2^0 \times 3^{-2}}{4^{-\frac{1}{2}}} = \frac{3^3 \times 2^0 \times 3^{-2}}{(2^2)^{-\frac{1}{2}}}$$

$$= \frac{3^3 \times 2^0 \times 3^{-2}}{2^{-1}}$$

Applying indices laws

$$= 3^{3-2} \times 2^{0+1}$$

$$= 3^1 \times 2^1$$

$$= 6 \quad (C)$$

2008/1 Neco

Evaluate $\frac{2^5 \times 4^{-2}}{2^{-3} \times 2^6}$

A 9 B 8 C 3 D $^8/_9$ E $^1/_4$

Express all items in 3 and its powers

$$\frac{2^5 \times 4^{-2}}{2^{-3} \times 2^6} = \frac{2^5 \times 2^{2(-2)}}{2^{-3} \times 2^6}$$

$$= \frac{2^5 \times 2^{-4}}{2^{-3} \times 2^6}$$

$$= \frac{2^{5-4}}{2^{-3+6}} = \frac{2^1}{2^3} = 2^{-2} = {}^1/_4 \; (E)$$

2006/3

Simplify $\dfrac{25^{\frac{2}{3}} \div 25^{\frac{1}{6}}}{\left(\frac{1}{5}\right)^{-\frac{7}{6}} \times \left(\frac{1}{5}\right)^{\frac{1}{6}}}$

A 25 B 1 C $^1/_5$ D $^1/_{25}$

Solution

Express all items in 5 and its powers

$$\frac{25^{\frac{2}{3}} \div 25^{\frac{1}{6}}}{\left(\frac{1}{5}\right)^{-\frac{7}{6}} \times \left(\frac{1}{5}\right)^{\frac{1}{6}}} = \frac{(5^2)^{\frac{2}{3}} \div (5^2)^{\frac{1}{6}}}{(5^{-1})^{-\frac{7}{6}} \times (5^{-1})^{\frac{1}{6}}}$$

$$= \frac{5^{\frac{4}{3}} \div 5^{\frac{2}{6}}}{5^{\frac{7}{6}} \times 5^{-\frac{1}{6}}}$$

Applying indices laws

$$= \frac{5^{\frac{4}{3}-\frac{2}{6}}}{5^{\frac{7}{6}+\left(-\frac{1}{6}\right)}} = \frac{5^{\frac{8-2}{6}}}{5^{\frac{7-1}{6}}} = \frac{5^{\frac{6}{6}}}{5^{\frac{6}{6}}} = 1 \;\; B$$

2006/34 Neco

Evaluate: $\sqrt{1\frac{9}{16}} + \left(\frac{32}{162}\right)^{\frac{1}{2}} \div \left(\frac{27}{64}\right)^{-\frac{1}{3}}$

A $^7/_{12}$ B $1^1/_2$ C $1^7/_{12}$ D $1^{91}/_{108}$ E $2^7/_{27}$

Solution

By proper presentation

$$\sqrt{1\frac{9}{16}} + \left(\frac{32}{162}\right)^{\frac{1}{2}} \div \left(\frac{27}{64}\right)^{-\frac{1}{3}} =$$

$$\left(\frac{25}{16}\right)^{\frac{1}{2}} + \left(\frac{32}{162}\right)^{\frac{1}{2}} \div \left(\frac{27}{64}\right)^{-\frac{1}{3}}$$

By laws of indices $\left(\frac{y}{x}\right)^{-\mathbf{A}} = \left(\frac{x}{y}\right)^{\mathbf{A}}$

$$= \left(\frac{25}{16}\right)^{\frac{1}{2}} + \left(\frac{32}{162}\right)^{\frac{1}{2}} \div \left(\frac{64}{27}\right)^{\frac{1}{3}}$$

Express all items in the power that cancel their roots. i.e power 2 cancel square root, power 3 cancel cube root

$$= \left(\frac{5^2}{4^2}\right)^{\frac{1}{2}} + \left(\frac{16 \times 2}{81 \times 2}\right)^{\frac{1}{2}} \div \left(\frac{4^3}{3^3}\right)^{\frac{1}{3}}$$

Second bracket item has to be reduced before complying

$$= \left(\frac{5}{4}\right)^{\frac{2}{2}} + \left(\frac{16}{81}\right)^{\frac{1}{2}} \div \left(\frac{4}{3}\right)^{\frac{3}{3}} = \frac{5}{4} + \left(\frac{4^2}{9^2}\right)^{\frac{1}{2}} \div \frac{4}{3}$$

$$= \frac{5}{4} + \frac{4}{9} \div \frac{4}{3}$$

Applying BODMAS to the fractions, division 1st

$$= \frac{5}{4} + \frac{4}{9} \times \frac{3}{4} = \frac{5}{4} + \frac{1}{3} = \frac{15+4}{12} = \frac{19}{12} = 1\frac{7}{12}$$

2013/13 Neco Exercise 5.1

Simplify $(343^{-2})^{\frac{1}{3}}$

A 7^{-2} B 7^{-1} C 7^0 D 7^1 E 7^2

2012/11 Neco Exercise 5.2

Simplify $\dfrac{27^{-\frac{1}{2}}}{81^{\frac{2}{3}}}$

A $3^{\frac{6}{25}}$ B $3^{-\frac{6}{25}}$ C $3^{-\frac{25}{6}}$ D $3^{\frac{25}{6}}$ E $\left(\frac{3}{4}\right)^{-2}$

2008/2 Neco (Nov) Exercise 5.3

Simplify $16^{\frac{3}{4}} \times 2^4 \times 32^{-1}$

A 0 B 2 C 4 D 10 E 20

2010/6 NABTEB (Nov) Exercise 5.4

Simplify $\left(\frac{4}{25}\right)^{-\frac{1}{2}}$

A $3^1/_2$ B $2^1/_2$ C $4^1/_2$ D $6\,^1/_3$

2013/10 Neco Exercise 5.5

Evaluate $\dfrac{3^4 \times 27^{-3}}{3^{-4}}$

A $^1/_3$ B 3 C 9 D 18 E 81

2007/4 Neco Exercise 5.6

Evaluate $\dfrac{16^{-\frac{1}{4}} + 27^{\frac{2}{3}}}{4^{-\frac{3}{2}}}$

A $1^3/_8$ B $2^1/_4$ C $5^1/_2$ D 9 E 76

2012/39 Exercise 5.7

Solve: $\left(\frac{27}{125}\right)^{-\frac{1}{3}} \times \left(\frac{4}{9}\right)^{\frac{1}{2}}$

A $^{10}/_9$ B $^9/_{10}$ C $^2/_5$ D $^{12}/_{125}$

Indices in algebraic terms and algebraic powers

2010/18 Neco

Simplify $\sqrt[3]{27a^{-6}}$

A $3a$ B $3a^2$ C $3a^{-2}$ D $9a^{-2}$ E $9a^2$

Solution

Express 27 in 3 and its powers

$$\sqrt[3]{27a^{-6}} = (3^3 a^{-6})^{\frac{1}{3}}$$

$$= 3^{\frac{3}{3}} a^{\frac{-6}{3}} = 3a^{-2} \quad (C)$$

2008/10

Simplify: $\sqrt[3]{27x^3y^9}$

A $9xy^3$ B $3xy^6$ C $3xy^3$ D $9y^3$

Solution

$$\sqrt[3]{27x^3y^9} = (3^3x^3y^9)^{\frac{1}{3}}$$

$$= 3^{\frac{3}{3}} x^{\frac{3}{3}} y^{\frac{9}{3}}$$

$$= 3xy^3 \quad (C)$$

2014/35

Simplify : $\sqrt{\dfrac{8^2 \times 4^{n+1}}{2^{2n} \times 16}}$

A 16 B 8 C 4 D 1

Solution

$$\sqrt{\frac{8^2 \times 4^{n+1}}{2^{2n} \times 16}} = \left(\frac{8^2 \times 4^{n+1}}{2^{2n} \times 16}\right)^{\frac{1}{2}}$$

Our target will be to make each item that cannot cancel out in powers of 2

$$= \left(\frac{8^2 \times 4^n \times 4^1}{2^{2n} \times 4^2}\right)^{\frac{1}{2}}$$

$$= \left(\frac{8^2 \times 2^{2n} \times 2^2}{2^{2n} \times 4^2}\right)^{\frac{1}{2}}$$

2^{2n} will cancel out, while power ½ will cancel out all power 2

$$= \frac{8 \times 2}{4} = 4 \quad (C)$$

2014/3 Neco (Nov)

Simplify: $(27x^3 \times 64x^6)^{\frac{1}{3}} \times (3x)^{-1}$

A $3x$ B $4x$ C $4x^2$ D $4x^3$ E $12x^3$

Solution

$$(27x^3 \times 64x^6)^{\frac{1}{3}} \times (3x)^{-1} = \frac{(27x^3 \times 64x^6)^{\frac{1}{3}}}{3x}$$

We seek to raise items under power $^1/_3$ to power 3

$$= \frac{(3^3x^3 \times 4^3x^6)^{\frac{1}{3}}}{3x}$$

$$= \frac{3x \times 4x^2}{3x} = 4x^2 \quad (C)$$

Note that : power 6 ÷ 3 is 2

2014/ 11c NABTEB (Nov) ***Adjusted***

Simplify $\dfrac{3^{x-1} \times 9^{x+1} \times 27^{-x}}{81^{5x-3}}$

Solution

The lowest base here is 3 and we can change 9, 27 and 81 to 3 and its power

$$\frac{3^{x-1} \times 9^{x+1} \times 27^{-x}}{81^{5x-3}} = \frac{3^{x-1} \times 3^{2(x+1)} \times (3^3)^{-x}}{3^{4(5x-3)}}$$

$$= \frac{3^{x-1} \times 3^{2x+2} \times 3^{-3x}}{3^{20x-12}}$$

Applying law of indices to the powers at numerator

$$= \frac{3^{x-1+(2x+2)+(-3x)}}{3^{20x-12}}$$

$$= \frac{3^{x-1+2x+2-3x}}{3^{20x-12}}$$

$$= \frac{3^1}{3^{20x-12}} = 3^1 \times 3^{-(20x-12)}$$

$$= 3^1 \times 3^{-20x+12}$$

$$= \frac{3^{1+12}}{3^{20x}} = \frac{3^{13}}{3^{20x}}$$

2008/14 Neco (Nov) Exercise 5.8

Simplify $56a^{-6} \div 7a^{-4}$

A $8a^{-10}$ B $8a^{-8}$ C $8a^{-2}$ D $8a^2$ A $8a^{10}$

2007/7 Exercise 5.9

Simplify: $\dfrac{3x^3}{(3x)^3}$

A 1 B $^1/_3$ C $^1/_9$ D $^1/_{27}$

2012/7 f/maths Exercise 5.10

Simplify $\dfrac{x^{3n+1}}{x^{2n+5/2}\sqrt{x^{2n-3}}}$

A. 0 B. $-\frac{1}{2}$ C. 1 D. 10

2014/11c NABTEB (Nov) Exercise 5.11

Simplify $\dfrac{3^{x-2} \times 9^{x+1} \times 27^{-x}}{81^{5x-3}}$

Indices involving equations

There are cases where the resulting indices of a given problem become equation. These equations are sometimes:

- simple linear or simultaneous linear
- quadratic in nature

Whatever the equation may be, we are required to solve accordingly.

Simple linear cases

2005/12

Solve the equation $2^7 = 8^{5-x}$

A $\frac{5}{8}$ B $\frac{8}{3}$ C $\frac{3}{2}$ D $\frac{15}{4}$

Solution

$$2^7 = 8^{5-x}$$

Express RHS in 2 and its power

$$2^7 = 2^{3(5-x)}$$

Since they are in the same base, we equate powers

$$7 = 3(5-x)$$
$$7 = 15 - 3x$$
$$3x = 15 - 7$$
$$x = \frac{8}{3} \quad (B)$$

2007/21 Neco

If $27^{x-2} = 729$, find x

A 3 B 4 C 5 D 6 E 7

Solution

$$27^{x-2} = 729$$

Our target will be to raise both sides to same base number and its power.

$$3^{3(x-2)} = 3^6$$

Since they are in the same base, we equate powers

$$3(x-2) = 6$$
$$x - 2 = 2$$
$$x = 2 + 2$$
$$x = 4 \quad (B)$$

2005/ 14

If $3^{\frac{1}{y}} = \left(\frac{1}{27}\right)^{\frac{2}{3}}$, find y

A 2 B $^1/_2$ C $-^1/_2$ D -2

Solution

$$3^{\frac{1}{y}} = \left(\frac{1}{27}\right)^{\frac{2}{3}}$$

Express RHS in 3 and its power to align with LHS

$$3^{\frac{1}{y}} = (3^{-3})^{\frac{2}{3}}$$
$$3^{\frac{1}{y}} = 3^{-2}$$

Since they are in the same base, we equate powers

$$\frac{1}{y} = -2$$
$$1 = -2y$$
$$-^1/_2 = y \quad (C)$$

2006/21 Neco

Solve the equation: $3^{2x-1} = \frac{1}{27^x}$

A -3 B -1 C $-^1/_5$ D $^1/_5$ E 3

Solution

$$3^{2x-1} = \frac{1}{27^x}$$

Express RHS in 3 and its power to align with LHS

$$3^{2x-1} = \frac{1}{3^{3x}}$$
$$3^{2x-1} = 3^{-3x}$$

Since they are in the same base, we equate powers

$$2x - 1 = -3x$$
$$2x + 3x = 1$$
$$5x = 1$$
$$x = ^1/_5 \quad (D)$$

2006/19 Neco (Nov)

If $\frac{5^{x-3}}{125^{3x-4}} = 1$, find the value of x

A $1^1/_8$ B $^8/_9$ C $^7/_8$ D $^5/_8$ E 0

Solution

$$\frac{5^{x-3}}{125^{3x-4}} = 1$$

We cross multiply

$$5^{x-3} = 125^{3x-4}$$

Next, we express RHS in 5 and its power to be like LHS

$$5^{x-3} = 5^{3(3x-4)}$$

Since they are in the same base, we equate powers

$$x - 3 = 3(3x-4)$$
$$x - 3 = 9x - 12$$
$$12 - 3 = 9x - x$$
$$9 = 8x$$
$$^9/_8 = x \text{ i.e } 1^1/_8 \quad (A)$$

2006/3a NABTEB (Nov)

If $4^{\frac{3x}{2}} = \frac{\sqrt{8^x}}{4}$, find x

Solution

$$4^{\frac{3x}{2}} = \frac{\sqrt{8^x}}{4}$$

Proper presentation shows

$$4^{\frac{3x}{2}} = \frac{8^{\frac{1}{2}x}}{4}$$

Express all items in 2 and its power for uniformity

$$(2^2)^{\frac{3x}{2}} = \frac{(2^3)^{\frac{1}{2}x}}{2^2}$$
$$2^{3x} = 2^{\frac{3x}{2}} \times 2^{-2}$$
$$2^{3x} = 2^{\frac{3x}{2} - 2}$$

Since they are in the same base, we equate powers

$$3x = \frac{3x}{2} - 2$$

$$3x - \frac{3x}{2} = -2$$

$$\frac{6x - 3x}{2} = -2$$

$3x = -4$

$x = {}^{-4}/_3$

2008/7 Neco (Nov)

If $\left(\frac{4}{3}\right)^{m+2} = \frac{9}{16}$, find the value of m

A – 4 B – 2 C 0 D 2 E 4

Solution

$$\left(\frac{4}{3}\right)^{m+2} = \frac{9}{16}$$

Our target is to make RHS similar to LHS

$$\left(\frac{4}{3}\right)^{m+2} = \frac{3^2}{4^2}$$

$$\left(\frac{4}{3}\right)^{m+2} = \left(\frac{3}{4}\right)^2$$

We are near success, next invert ¾ to $^4/_3$ and reflect it in the power

$$\left(\frac{4}{3}\right)^{m+2} = \left(\frac{4}{3}\right)^{-2}$$

Since they are in the same base, we equate powers

$m + 2 = -2$

$m = -2 - 2$

$m = -4$ (A)

2008/30 Counter example

If $2^n = 128$, find the value of $(2^{n-1})(5^{n-2})$

A $5(10^6)$ B $2(10^6)$ C $5(10^5)$ D $2(10^5)$

Solution

First, we solve for n from

$2^n = 128$

Express the right hand side in 2 and its powers

$2^n = 2^7$

Equating powers

$n = 7$

Next, $(2^{n-1})(5^{n-2}) = (2^{7-1})(5^{7-2})$

$= (2^6)(5^5)$

$= 2 \times 2^5 \times 5^5$

$= 2(2 \times 5)^5$

$= 2(10^5)$ D

2009/1b

If $\frac{2^{1-y} \times 2^{y-1}}{2^{y+2}} = 8^{2-3y}$, find y

Solution

$$\frac{2^{1-y} \times 2^{y-1}}{2^{y+2}} = 8^{2-3y}$$

Express RHS in 2 and its powers

$$\frac{2^{1-y} \times 2^{y-1}}{2^{y+2}} = 2^{3(2-3y)}$$

Applying the law of indices

$2^{1-y+y-1-(y+2)} = 2^{6-9y}$

$2^{1-y+y-1-y-2} = 2^{6-9y}$

$2^{-y-2} = 2^{6-9y}$

Equating powers

$-y - 2 = 6 - 9y$

$9y - y = 6 + 2$

$8y = 8$

$y = 1$-

2010/2b Neco

If $8^{\frac{x}{2}} = 2^{\frac{3}{8}} \times 4^{\frac{3}{4}}$, find x

Solution

$$8^{\frac{x}{2}} = 2^{\frac{3}{8}} \times 4^{\frac{3}{4}}$$

Express all items in 2 and its power

$$2^{3\left(\frac{x}{2}\right)} = 2^{\frac{3}{8}} \times 2^{2\left(\frac{3}{4}\right)}$$

$$2^{\frac{3x}{2}} = 2^{\frac{3}{8}} \times 2^{\frac{6}{4}}$$

Applying the law of indices to RHS power items

$$2^{\frac{3x}{2}} = 2^{\frac{3}{8}+\frac{6}{4}} \text{ i.e } 2^{\frac{3+12}{8}}$$

$$2^{\frac{3x}{2}} = 2^{\frac{15}{8}}$$

Since they are in the same base we equate powers

$$\frac{3x}{2} = \frac{15}{8}$$

$$3x = \frac{15 \times 2}{8}$$

Multiply both sides by $^1/_3$ to get x

$$3x \times \frac{1}{3} = \frac{15 \times 2}{8} \times \frac{1}{3}$$

$x = {}^5/_4$

2011/46

If $27^x = 9^y$, find the value of $^x/_y$

A $^1/_3$ B $^2/_3$ C $1^1/_2$ D 3

Solution

$27^x = 9^y$

Raise both sides to 3 and its powers

$3^{3x} = 3^{2y}$

Now they are in the same base, we equate powers

$3x = 2y$

To get $^x/_y$, divide both sides, by y

$$\frac{3x}{y} = \frac{2y}{y}$$

$$\frac{3x}{y} = 2$$

Next, we divide both sides by coefficient of $^x/_y$ i.e 3

$$\frac{x}{y} = \frac{2}{3} \quad \text{(B)}$$

2014/6a NABTEB (Nov)

Solve the equation: $16 \times 8^x = 32 \times 4^{2x-1}$

Solution

$16 \times 8^x = 32 \times 4^{2x-1}$

Here we cannot work in 4 or 8 or 16 and their powers but in 2 and its power

$$2^4 \times (2^3)^x = 2^5 \times (2^2)^{2x-1}$$
$$2^4 \times 2^{3x} = 2^5 \times 2^{4x-2}$$
Applying law of indices to the powers
$$2^{4+3x} = 2^{4x-2+5}$$
$$2^{4+3x} = 2^{4x+3}$$
Now they are in the same base, we equate powers
$$4 + 3x = 4x + 3$$
$$4 - 3 = 4x - 3x$$
$$1 = x$$

2014/1 Neco

If $4^m = \frac{1}{64}$, find the value of $2m^2$

A – 18 B 16 C 18 D 32 E 256

Solution

$$4^m = \frac{1}{64}$$
We should raise both sides to 4 and its powers
$$4^m = \frac{1}{4^3}$$
$$4^m = 4^{-3}$$
Same base achieved, next we equate powers
$$m = -3$$
$$\text{Thus } 2m^2 = 2(-3)^2$$
$$= 2 \times 9$$
$$= 18 \quad (C)$$

2014/ 1a NABTEB (Nov)

Find n in the equation $36 \times \left(\frac{1}{6}\right)^n = 6 \times 216^{n-1}$

Solution

$$36 \times \left(\frac{1}{6}\right)^n = 6 \times 216^{n-1}$$
We raise items on both sides to 6 and its power
$$6^2 \times 6^{-n} = 6^1 \times (6^3)^{n-1}$$
Applying law of indices to the powers
$$6^{2-n} = 6^1 \times 6^{3n-3}$$
$$6^{2-n} = 6^{3n-3+1}$$
$$6^{2-n} = 6^{3n-2}$$
Same base achieved, next we equate powers
$$2 - n = 3n - 2$$
$$2 + 2 = 3n + n$$
$$4 = 4n$$
$$1 = n$$

2013/2 Exercise 5.12

If $9^{(2-x)} = 3$, find x

A 1 B $^3/_2$ C 2 D $^5/_2$

2014/11 NABTEB (Nov) Exercise 5.13

If $9^{x-1} = 27$, the value of x is

A $2^1/_2$ B $2^1/_4$ C 2 D 3

2012/13 Neco Exercise 5.14

Find the value of x in the equation $4^x = 32$

A 1.5 B 2.0 C 2.5 D 4.0 E 8.0

2009/10 (Nov) Exercise 5.15

If $9^{(2x-1)} = 27$, find x

A 5 B 3 C $^5/_4$ D $^4/_5$

2010/18 Exercise 5.16

Given that $\frac{5^{n+3}}{25^{2n-3}} = 5^0$, find n

A n = 1 B n = 2 C n = 3 D n = 5

2010/12 Neco Exercise 5.17

Find the value of x, given that
$$\frac{1}{3} \text{ of } 9^{2x} = 27^x$$
A – 2 B – 1 C 0 D 1 E 2

2009/28 Neco (Nov) Exercise 5.18

Solve the equation $3^{2x-1} = \frac{1}{243}$

A – 2 B 2 C 3 D 4 E 5

2009/2 NABTEB (Nov) Exercise 5.19

Given $5^{(x+1)} = 125$, find the value of x

A 6 B 3 C 2 D 1

2009/31 Exercise 5.20

If $9^{2x} = \frac{1}{3}(27^x)$, find x

A 2 B1 C – 1 D – 2

Simultaneous linear in nature

2008/6a

If $2^{x+y} = 16$ and $4^{x-y} = \frac{1}{32}$, find the value of x and y

Solution

$$2^{x+y} = 16 \text{ and } 4^{x-y} = \frac{1}{32}$$
Express both equations in 2 and its power
$$2^{x+y} = 2^4 \text{ and } 2^{2(x-y)} = \frac{1}{2^5}$$
$$2^{x+y} = 2^4 \text{ and } 2^{2x-2y} = 2^{-5}$$
Same base achieved, next we equate powers
$$x + y = 4 \text{ -------- } (1)$$
$$2x - 2y = -5 \text{ ------- } (2)$$
From (1) y = 4 – x , substituting into (2)
$$2x - 2y = -5 \text{ becomes}$$
$$2x - 2(4 - x) = -5$$
$$2x - 8 + 2x = -5$$
$$4x = -5 + 8$$
$$4x = 3$$
$$x = {^3/_4}$$
Substitute $x = {^3/_4}$ into (1)

$¾ + y = 4$

Thus $y = 4 - {^3/_4}$ i.e ${^{13}/_4}$

2011/2b

If $9^{(1-x)} = 27^y$ and $x - y = -1^1/_2$, find the value of x + y

Solution

$$9^{(1-x)} = 27^y$$
Raise both sides to 3 and its power
$$3^{2(1-x)} = 3^{3y}$$
Same base achieved, next we equate powers
$$2(1 - x) = 3y$$
$$2 - 2x = 3y$$

$$2 = 3y + 2x$$
i.e $3y + 2x = 2$ ----- (1)

Also given: $x - y = -1\frac{1}{2} \Rightarrow -\frac{3}{2}$ ------ (2)

Multiply (2) through by 2 to clear fractions

$$2x - 2y = -3 \text{ ------ (3)}$$
$$\underline{-(2x + 3y = 2)}$$
$$-5y = -5$$
$$y = 1$$

Substituting for y = 1 into $2x + 3y = 2$

$2x + 3(1) = 2$

$2x = 2 - 3$

$2x = -1$

$x = -\frac{1}{2}$ Thus $x + y = -\frac{1}{2} + 1$ i.e $\frac{1}{2}$

1996/5 PCE Exercise 5.21

If $5^{x-y} = 125$ and $3^{4y-x} = 243$, Find the value of 2^{x-2y}

A. $2^{25/3}$ B. $2^{1/3}$ C. $2^{-1/3}$ D. $2^{-25/3}$

2014/1b (Nov) Exercise 5.22a

Solve simultaneously, the equations

$$5x - 4y = 6 \text{ and } 3^{3(y-x)} = \frac{1}{27}$$

1978/11a Exercise 5.22b

Calculate the values of x and y if $\dfrac{27^x}{81^{x+2y}} = 9$

and $x + 4y = 0$

Quadratic in nature

2009/4 Neco (Nov)

Solve the equation $3^{x^2} = \dfrac{27^x}{9}$

A $-1, 2$ B $1, 2$ C $1, -2$ D $-1, -2$ E $0, 3$

Solution

$$3^{x^2} = \frac{27^x}{9}$$

Express the RHS in 3 and its power

$$3^{x^2} = \frac{3^{3x}}{3^2}$$

Applying the law of indices

$$3^{x^2} = 3^{3x-2}$$

Same base achieved, next we equate powers

$$x^2 = 3x - 2$$

$x^2 - 3x + 2 = 0$

Factorizing

$x^2 - 2x - x + 2 = 0$

$x(x - 2) - 1(x - 2) = 0$

$(x - 2)(x - 1) = 0$

$x - 2 = 0$ or $x - 1 = 0$

$x = 2$ or 1 B

2009/11 Neco (Nov)

Solve $(x + 1)^2 = \dfrac{9}{25}$

A $-\frac{2}{5}, \frac{8}{5}$ B $\frac{2}{5}, -\frac{8}{5}$ C $-\frac{2}{5}, -\frac{8}{5}$

D $-\frac{2}{5}, -\frac{5}{8}$ E $\frac{2}{5}, \frac{5}{8}$

Solution

$$(x + 1)^2 = \frac{9}{25}$$

Raise the RHS to a number in power of 2 to align with LHS

$$(x + 1)^2 = \frac{3^2}{5^2}$$

$$(x+1)^2 = \left(\frac{3}{5}\right)^2$$

Since they are in the same power, we can equate base

$x + 1 = \frac{3}{5}$	or $x + 1 = \pm\frac{3}{5}$
$x = \frac{3}{5} - 1$	or $x = -1 \pm \frac{3}{5}$
$= -\frac{2}{5}$	$= -\frac{2}{5}$ or $-\frac{8}{5}$

2011/4 f/maths

Solve: $2^{(2y+2)} - 9(2^y) = -2$

Solution

Applying law of indices

$2^{(2y+2)} - 9(2^y) = -2$ becomes

$2^{2y} \times 2^2 - 9(2^y) = -2$

i.e $4(2^y)^2 - 9(2^y) = -2$

Let $2^y = p$

$4p^2 - 9p = -2$

i.e $4p^2 - 9p + 2 = 0$

Factorizing : $4p^2 - 8p - p + 2 = 0$

$4p(p - 2) - 1(p - 2) = 0$

$(4p - 1)(p - 2) = 0$

$4p - 1 = 0$ or $p - 2 = 0$

$4p = 1$ or $p = 2$

$p = \frac{1}{4}$ or $p = 2$

Changing back to our original variable

$p = \frac{1}{4}$ implies $2^y = \frac{1}{4}$ ***or*** $p = 2$ implies $2^y = 2$

i.e $2^y = 2^{-2}$ ***or*** $2^y = 2^1$

$y = -2$ ***or*** 1

2013/2a Neco f/maths Exercise 5.23

Solve $5^{2x} + 4 \times 5^{x+1} - 125 = 0$

1997/6 PCE Exercise 5.24

Solve for x in the equation $3^{2x} - 12(3^x) + 27 = 0$

A. 3, 9 B. 2, 1 C. –3, 9 D. –1, 2

1999/9a (Nov) f/maths Exercise 5.25

Solve the equation:

$3^{2x+2} - 10(3^{x+1}) + 9 = 0$

Other cases

2012/10 Neco

Given $x^{\frac{1}{3}} = (0.25)^{\frac{1}{6}}$, find x

A $(0.25)^3$ B 0.05 C 0.50 D 2.00 E 5.00

Solution

$$x^{\frac{1}{3}} = (0.25)^{\frac{1}{6}}$$

Note that 0.25 is same as $\frac{25}{100}$

$$x^{\frac{1}{3}} = \left(\frac{5^2}{10^2}\right)^{\frac{1}{6}}$$

$$x^{\frac{1}{3}} = \left(\frac{5}{10}\right)^{\frac{2}{6}} \text{ i.e } x^{\frac{1}{3}} = \left(\frac{5}{10}\right)^{\frac{1}{3}}$$

The same powers achieved, next we equate bases

Thus $x = \frac{5}{10}$ i.e 0.5 (C)

2005/1a

If $2^{x} + 2^{(x-1)} = 48$, find the value of x

Solution

$2^x + 2^{(x-1)} = 48$

Factor out 2^x at the RHS

$2^x(1 + 2^{-1}) = 48$

$2^x(1 + {}^1/_2) = 48$

$2^x(\frac{3}{2}) = 48$

Multiply both sides by $\frac{2}{3}$ to clear fraction

$2^x = 48 \times \frac{2}{3}$

$2^x = 16 \times 2$

Express the RHS in 2 and it powers

$2^x = 2^5$

Same base achieved, next we equate powers

$x = 5$

2005/29 Neco

Solve the equation $0.2^x = \frac{1}{25}$

A – 5 B – 2 C 1 D 2 E 5

Solution

$0.2^x = \frac{1}{25}$

We raise both sides to the same base

$\left(\frac{2}{10}\right)^x = \frac{1}{25}$

It can't work, but ${}^2/_{10}$ is ${}^1/_5$

$\left(\frac{1}{5}\right)^x = \frac{1}{25}$

Ok now, we can express them in 5 and its power

$5^{-x} = 5^{-2}$

Same base achieved, next we equate powers

$-x = -2$

$x = 2$ (D)

2014/31 Neco (Nov)

Solve for x if $3x^{\frac{3}{5}} = 81$

A. 3 B. 9 C. 27 D. 81 E. 243

Solution

$3x^{\frac{3}{5}} = 81$

Divide both sides by 3

$x^{\frac{3}{5}} = 27$

To get x we raise both sides to the opposite of it power

$\left(x^{\frac{3}{5}}\right)^{\frac{5}{3}} = (27)^{\frac{5}{3}}$

Recall that 27 is 3^3

$x = (3^3)^{\frac{5}{3}}$

$= 3^5$ i.e 243 (E)

2009/13 Exercise 5.26

If $(2x + 3)^3 = 125$, find the value of x

A 1 B 2 C 3 D 4

2006/6 (Nov) Exercise 5.27

Find x if $5x^3 = \frac{1}{25}$

A ${}^1/_{25}$ B ${}^1/_5$ C 1 D 5

2006/7b (Nov) Exercise 5.28

Solve the equation $(3x - 1)^3 = \frac{8}{27}$

Theory of Logarithm

This aspect of logarithm deals with applying the relationship between indices & logarithm and laws of logarithm to solve problems. Here, we will not use log tables.

Relationship between Indices and Logarithm

If $b^x = N$ then ,

$x = \log_b{}^N$

Note: Students should observe that both indices and logarithm make use of the same base when we are converting from one form to another.

Examples on simple applications

(i) Given that $10^2 = 100$,
Write the expression in logarithm form.

Solution

$2 = \log_{10} 100$

(ii) If $2^6 = 64$,
Write the expression in logarithmic form.

Solution

$6 = \log_2 64$

(iii) Express $4^x = 256$ in logarithmic form.

Solution

$4^x = 256$

$x = \log_4 256$

(iv) Write in logarithmic form $7^{-2} = \frac{1}{49}$

Solution

$-2 = \log_7{}^{1/49}$

Laws of Logarithm

(1) $\log_b (MN) = \log_b M + \log_b{}^N$

Examples on simple applications

a) $\log_6 30 = \log_6 (6 \times 5)$
$= \log_6 6 + \log_6 5$

b) $\log_2 72 = \log_2 (8 \times 9)$
$= \log_2{}^8 + \log_2{}^9$

c) $\log_3{}^9 + \log_3{}^{27} = \log_3 (9 \times 27)$
$= \log_3 243$

and not $\log_3 9 + \log_3 27 = \log_3 (9 + 27)$

(i) $\log_{10} 100 = \log_{10}10^2$

$= 2\log_{10}^{10}$

(ii) $\log_7 343 = \log_7 7^3$

$= 3 \log_7 7$

(iii) $\frac{1}{2} \log_5 25$

$= \log_5 25^{1/2}$

(2) $\log_b (M/N) = \log_b M - \log_b N$

Examples on simple applications

(a) $\log_{10}(100/3) = \log_{10} 100 - \log_{10} 3$

(b) $\log_6 \frac{36}{17} = \log_6 36 - \log_6 17$

(c) $\log_3 81 - \log_3 9 = \log_3 (81/9)$

(3) $\log_b N^k = k \log_b N$

Special cases under (3)

(i) $\log_b b = \log_b b^1$

$= 1\text{Log}_b b = 1$

i.e $\log_b b = 1$

Stated as; Any logarithm to its base is one.

Examples

(a) $\log_{10} 10 = 1$

(b) $\log_3 3 = 1$

(c) $\log_6 6 = 1$

(ii) $\log_b 1 = \log_b b^0$ (Since $b^0 = 1$)

$= 0 \log_b b$

$= 0$ i.e $\log_b 1 = 0$

Stated as: Log of one to any base is zero

Note:

The examples given under relationship between indices & logarithm were not fully solved; this was deliberately done to allow for step-by-step comprehension of the stated concepts (principles). Examples of fully solved problems applying the earlier stated principles are given below:

Simplification in logarithm

2009/7 NABTEB (Nov)

Simplify log 4 + log 8

A log32 B log 12 C log 2^3 D log $^1/_2$

Solution

Applying laws of logarithm

$\log 4 + \log 8 = \log 4 \times 8$

$= \log 32$ (A)

2006/4a NABTEB (Nov)

Simplify $\log_3 54 + \log_3 15 - \log_3 10$

Solution

Applying laws of logarithm

$\log_3 54 + \log_3 15 - \log_3 10 = \log_3 \frac{54 \times 15}{10}$

$= \log_3 81$

Next, express 81 in powers of 3

$= \log_3 3^4$

$= 4\log_3 3$

$= 4$

2009/30 NABTEB (Nov)

Evaluate $\log_a a^3$

A 0 B1 C 3 D a

Solution

Applying law of logarithm

$\log_a a^3 = 3\log_a a$

$= 3$ (C)

2008/5 Neco (Nov)

Without using tables, evaluate $2 + \log_{10}20 - \log_{10}2$

A 2 B 3 C 4 D 5 E 6

Solution

Since the log here is $\log_{10}$, we express 2 in $\log_{10}$

$2 + \log_{10}20 - \log_{10}2 = \log_{10}10^2 + \log_{10}20 - \log_{10}2$

$= \log_{10} \frac{10^2 \times 20}{2}$

$= \log_{10} 10^2 \times 10$

$= \log_{10} 10^3$

$= 3\log_{10}10$

$= 3$ (B)

2009/29 Neco (Nov)

Simplify $\log_x 8 - \log_x 2 + \log_x 4$

A $\log_x 2$ B $2\log_x 2$ C $4\log_x 2$ D $8\log_x 2$ E $16\log_x 2$

Solution

Applying logarithm laws

$\log_x 8 - \log_x 2 + \log_x 4 = \log_x \frac{8 \times 4}{2}$

$= \log_x 16$

$= \log_x 2^4$

$= 4\log_x 2$ (C)

2005/ 1b (ii) Neco

Simplify: $\log_{10}\sqrt{1000}$

Solution

$\log_{10}(1000)^{\frac{1}{2}} = \log_{10}(10^3)^{\frac{1}{2}}$

$= \log_{10} 10^{\frac{3}{2}}$

$= \frac{3}{2} \log_{10} 10$

$= \frac{3}{2}$ i.e $1^1/_2$

2006/4 Neco (Nov)

Evaluate $\log_{10}45 + \log_{10}9^{-1} - \log_{10}2^{-1}$ without using table

A 10 B 5 C 2 D 1 E $^1/_2$

Solution

Applying the laws of logarithm

$\log_{10}45 + \log_{10}9^{-1} - \log_{10}2^{-1} = \log_{10} \frac{45 \times \frac{1}{9}}{\frac{1}{2}}$

By simple arithmetic, we have

$= \log_{10} \frac{45 \times 2}{9}$

Some terms will cancel out

$= \log_{10} 10$

$= 1$ (D)

2005/4

Simplify the expression $\log_{10}18 - \log_{10}2.88 + \log_{10}16$

A 31.12 B 3.112 C 2 D 1

Solution

Applying the laws of logarithm

$$\log_{10}18 - \log_{10}2.88 + \log_{10}16 = \log_{10}\frac{18\times16}{2.88}$$

$$= \log_{10}\frac{288}{2.88}$$

Move decimal point forward twice in denominator and numerator

$$= \log_{10}\frac{28800}{288}$$

$= \log_{10}100$

$= \log_{10}10^2$

$= 2\log_{10}10$

$= 2$ (C)

2008/32 *Counter example*

Simplify $\dfrac{\log\sqrt{8}}{\log 4 - \log 2}$

A $^2/_3$ B $\frac{1}{2}\log 2$ C $^3/_2$ D log 2

Solution

The problem at hand seems not to follow our normal pattern

$$\frac{\log\sqrt{8}}{\log 4-\log 2} = \frac{\log\sqrt{8}}{\log 2^2-\log 2}$$

$$= \frac{\log\sqrt{8}}{2\log 2-\log 2}$$

$$= \frac{\log\sqrt{8}}{\log 2}$$

Next, numerator $= \dfrac{\log 8^{\frac{1}{2}}}{\log 2}$

$$= \frac{\log(2^3)^{\frac{1}{2}}}{\log 2} = \frac{\log 2^{\frac{3}{2}}}{\log 2}$$

$$= \frac{\frac{3}{2}\log 2}{\log 2}$$

$= {}^3/_2$ (C)

2010/11 Neco

Simplify $\dfrac{\log 3-\log 2}{\log 9-\log 4}$

A – 2.0 B – 0.5 C 0.0 D 0.5 E 2.0

Solution

Applying laws of logarithm

$$\frac{\log 3-\log 2}{\log 9-\log 4} = \frac{\log\frac{3}{2}}{\log\frac{9}{4}}$$

Proper observation shows the denominator is a multiple of numerator

$$= \frac{\log\frac{3}{2}}{\log\left(\frac{3}{2}\right)^2}$$

$$= \frac{\log\frac{3}{2}}{2\log\frac{3}{2}}$$

$= {}^1/_2$ i.e 0.5 (D)

2011/7 Neco

Simplify $\log_{10}\left(\dfrac{125}{8}\right) - 2\log_{10}\left(\dfrac{5}{4}\right)$

A 125 B 16 C 10 D 4 E 1

Solution

By laws of logarithm

$$\log_{10}\left(\frac{125}{8}\right) - 2\log_{10}\left(\frac{5}{4}\right) = \log_{10}\frac{125}{8} - \log_{10}\left(\frac{5}{4}\right)^2$$

$$= \log_{10}\left(\frac{125}{8}\div\frac{25}{16}\right)$$

Changing division to multiplication

$$= \log_{10}\left(\frac{125}{8}\times\frac{16}{25}\right)$$

Some terms will cancel out

$= \log_{10}10$

$= 1$ (E)

2009/14 Neco (Nov)

Simplify: $\log_{10}\sqrt{65} + \log_{10}\sqrt{2} - \log_{10}\sqrt{13}$

A 1 B $^1/_5$ C $^1/_4$ D $^1/_2$ E $^1/_3$

Solution

Applying logarithm laws

$$\log_{10}\sqrt{65} + \log_{10}\sqrt{2} - \log_{10}\sqrt{13} = \log_{10}\frac{\sqrt{65}\times\sqrt{2}}{\sqrt{13}}$$

Simplify $\sqrt{65}$ into $\sqrt{13}\times\sqrt{5}$

$$= \log_{10}\frac{\sqrt{13}\times\sqrt{5}\times\sqrt{2}}{\sqrt{13}}$$

$= \log_{10}\sqrt{5\times2}$

$= \log_{10}\sqrt{10}$ i.e $\log_{10}10^{\frac{1}{2}}$

$= \frac{1}{2}\log_{10}{}^{10} = \frac{1}{2}$ D

2014/4a (Nov)

Without using tables or calculator, evaluate:

$$\log_{10}\left(\frac{75}{10}\right) - 2\log_{10}\left(\frac{5}{9}\right) + \log_{10}\left(\frac{100}{243}\right)$$

Solution

Applying logarithm laws

$$\log_{10}\left(\frac{75}{10}\right) - 2\log_{10}\left(\frac{5}{9}\right) + \log_{10}\left(\frac{100}{243}\right)$$

$$= \log_{10}\left(\frac{75}{10}\right) - \log_{10}\left(\frac{5}{9}\right)^2 + \log_{10}\left(\frac{100}{243}\right)$$

$$= \log_{10}\left[\frac{75}{10}\div\left(\frac{5}{9}\right)^2\times\frac{100}{243}\right]$$

Changing $\div$ to $\times$ in the bracket, we have

$$= \log_{10}\left(\frac{75}{10}\times\frac{9\times9}{5\times5}\times\frac{100}{243}\right)$$

$= \log_{10}10$

$= 1$

2008/5 Neco (Nov)
Without using tables evaluate $2 + \log_{10}20 - \log_{10}2$
A 2 B 3 C 4 D 5 E 6
Solution
Since the log here is $\log_{10}$, we express 2 in $\log_{10}$
$$2 + \log_{10}20 - \log_{10}2 = \log_{10}10^2 + \log_{10}20 - \log_{10}2$$
$$= \log_{10}\frac{10^2 \times 20}{2}$$
$$= \log_{10}10^2 \times 10$$
$$= \log_{10}10^3$$
$$= 3\log_{10}10$$
$$= 3 \quad (B)$$

2011/17 Exercise 5.29
Simplify $\dfrac{\log\sqrt{27}}{\log\sqrt{81}}$
A 3 B 2 C $^3/_2$ D $^3/_4$

2006/1 (Nov) Exercise 5.30
Simplify $\dfrac{\log 27}{\log 9}$
A $1^1/_2$ B 3 C log 3 D $\log\left(\frac{3}{2}\right)$

2012/6 *f/maths* Exercise 5.31
Evaluate: $\log_{10}\left(\frac{1}{3} + \frac{1}{4}\right) + 2\log_{10}2 + \log_{10}\left(\frac{3}{7}\right)$
A. –3 B. 0 C. $^5/_6$ D. 1

2005/1a Exercise 5.32
Evaluate without using the mathematical table or calculator, $\log_{10}\sqrt{30} - \log_{10}\sqrt{6} + \log_{10}\sqrt{2}$

Change of base

If any given base is not conducive or good enough to work with , we can change base
$$\log_b N = \frac{\log_c N}{\log_c b}$$
Here the new base is not common to the numbers given but it suits our work.
There are cases where the number whose logarithm we are looking for will be used :
$$\log_b c = \frac{\log_c C}{\log_c b} = \frac{1}{\log_c b}$$

2008/ 6 Neco
Find the value of $\log_{\sqrt{3}} 81$
A 81 B 16 C 8 D 3 E $\sqrt{3}$
Solution
Here we change base $\sqrt{3}$ to 3 as shown below:
$$\log_{\sqrt{3}} 81 = \frac{\log_3 81}{\log_3\sqrt{3}}$$
$$= \frac{\log_3 3^4}{\log_3 3^{\frac{1}{2}}}$$
$$= \frac{4\log_3 3}{\frac{1}{2}\log_3 3} = 4 \div \tfrac{1}{2} = 4 \times \tfrac{2}{1} \quad \text{i.e } 8 \quad (C)$$

2005/9 Neco
Calculate the value of $\log_2 x$ if $\log_x 4 = \frac{1}{2}$
A 1 B 2 C 3 D 4 E 8
Solution
We will change base but first we align $\log_x 4$ to $\log_2 x$
$$\log_x 4 = \tfrac{1}{2}$$
$$\log_x 2^2 = \tfrac{1}{2}$$
$$2\log_x 2 = ½$$
Multiply through by ½
$$\log_x 2 = \tfrac{1}{4}$$
Now, we can change base as : $\log_2 x = \dfrac{\log_x x}{\log_x 2}$
$$= \frac{1}{\log_x 2}$$
Substituting for $\log_x 2$ value $= \dfrac{1}{1/4}$ i.e $1 \div \frac{1}{4}$
$$= 1 \times \tfrac{4}{1} = 4 \quad D$$

2013/11 Neco
Simplify the expression $\log_4 3 \times \log_3 4 + \log_2 32$
A 1 B 2 C 3 D 6 E 7
Solution
We change base in $\log_3 4$ to align with $\log_4 3$
$$\log_4 3 \times \log_3 4 + \log_2 32 = \log_4 3 \times \frac{\log_4 4}{\log_4 3} + \log_2 2^5$$
$$= \log_4 3 \times \frac{1}{\log_4 3} + 5\log_2 2$$
$$= 1 + 5 = 6\ (D)$$

2012/11a Neco

Given that $\log_3 27 = \log_{\frac{1}{4}} 64 + n\log_3 \frac{1}{81}$

find the value of n

Solution

A look at the problem shows that $\log_{\frac{1}{4}} 64$ will be an obstacle, so we change its base

$$\log_{\frac{1}{4}} 64 = \log_{4^{-1}} 64$$

$$= \frac{\log_2 64}{\log_2 4^{-1}}$$ and our working proper

$$\log_3 3^3 = \frac{\log_2 2^6}{\log_2 2^{-2}} + \log_3\left(\frac{1}{81}\right)^{\mathbf{n}}$$

$$3\log_3 3 = \frac{6\log_2 2}{-2\log_2 2} + \log_3 3^{-4\mathbf{n}}$$

$$3 = \frac{6}{-2} + -4n\log_3 3$$

$$3 = -3 - 4n$$

$$4n = -3 - 3$$

$$n = {}^{-6}/_4$$

2015/34 Neco Exercise 5.33

Evaluate $\log_{\sqrt[3]{5}} 125$

A. 3 B. 5 C. 6 D. 8 E. 9

2015/1 Neco Exercise 5.34

Evaluate $\log_3 27 + \log_9 3 - \log_{32} 2$

A. $-{}^1/_5$ B. ${}^1/_2$ C. 3 D. $3^3/_{10}$ E. $3^1/_2$

2015/8c Neco Exercise 5.35

Without using mathematical tables or calculator, evaluate: $\log_{16} 0.25$

2015/4 *f/maths* Exercise 5.36

If $\log_3 x = \log_9 3$, find the value of x

A. 3^2 B. $3^{\frac{1}{2}}$ C. $3^{\frac{1}{3}}$ D. $2^{\frac{1}{3}}$

2013/3 Neco *f/maths* Exercise 5.37

Find without using tables, the value of

$$\log_{2/3}\left(\frac{8}{27}\right) \times \log_{1/100} 10$$

A. $-2^1/_2$ B. $-1^1/_2$ C. $-{}^2/_3$ D. $1^1/_2$ E. $2^1/_2$

Substitution in Logarithm

Some logarithmic problems involve substituting for a given log value. By this, we are then constrained or limited to solving the problem by simplifying to get either exactly the given log value or its multiples.

2014/5 NABTEB (Nov)

If $\log_{10} 3 = 0.477$, evaluate $\log_{10} 27$ correct to 2 s.f without using log tables

A 0.55 B 1.4 C 1.43 D 1.8

Solution

Our target will be to express $\log_{10} 27$ *in* $\log_{10} 3$ and its multiples

$$\log_{10} 27 = \log_{10} 3^3$$

$$= 3\log_{10} 3$$

Substituting for the given value

$$= 3(0.477)$$

$$= 1.431 \approx 1.4 \text{ to 2 s.f} \quad (\text{B})$$

2009/1 NABTEB (Nov)

If $\log 2 = 0.301$, what is $\log 64$?

A 1.506 B 1.606 C 1.706 D 1.806

Solution

We express $\log 64$ *in* $\log 2$ and its multiples

$$\log 64 = \log 2^6$$

$$= 6\log 2$$

Substituting for the given value

$$= 6(0.301)$$

$$= 1.806 \quad (\text{D})$$

2006/10b NABTEB (Nov)

Given that $\log_{10} 2 = 0.3010$ and $\log_{10} 7 = 0.8451$, evaluate, without the use of table $\log_{10} 3.92$

Solution

Our target will be to express $\log_{10} 3.92$ *in* $\log_{10} 2$ and $\log_{10} 7$ with its multiples.

$$\log_{10} 3.92 = \log_{10} \tfrac{392}{100}$$

$$= \log_{10} 392 - \log_{10} 100$$

Note: 392 is 28×14 i.e $2 \times 2 \times 7 \times 2 \times 7$

For better working 8×49

$$= \log_{10}(8 \times 49) - \log_{10} 10^2$$

$$= \log_{10} 8 + \log_{10} 49 - \log_{10} 10^2$$

$$= \log_{10} 2^3 + \log_{10} 7^2 - \log_{10} 10^2$$

$$= 3\log_{10} 2 + 2\log_{10} 7 - 2\log_{10} 10$$

Substituting for the given values

$= 3(0.3010) + 2(0.8451) - 2$

$= 0.903 + 1.6902 - 2$

$= 2.5932 - 2 = 0.5932$

2006/7b Neco

If $\log 2 = 0.3010$ and $\log 7 = 0.8451$, evaluate, without using mathematical table or calculator

(i) $\log 0.2$ (ii) $\log 35^2$

Solution

(i) Our target will be to express 0.2 in the form of 2 or 7 and their multiples

$$\log 0.2 = \log\frac{2}{10}$$

$$= \log 2 - \log 10$$

Substituting for log 2, Note log10 is 1

$$= 0.3010 - 1$$

$$= -0.6990$$

(ii) $\log 35^2 = 2\log 35$

Expressing 35 *in* 2 and 7

$$= 2[\log(5 \times 7)]$$

$$= 2[\log 5 + \log 7]$$

We don't have 5, but 5 is $\frac{10}{2}$

$$= 2\left[\log\frac{10}{2} + \log 7\right]$$

$$= 2[\log 10 - \log 2 + \log 7]$$

Substituting

$$= 2[1 - 0.3010 + 0.8451]$$

$$= 2[1.5441]$$

$$= 3.0882$$

2006/12 Neco ***special case***

If $10^n = 0.63$ and $\log_{10}63 = 1.8$, find the value of n

A – 2.63 B – 1.80 C – 0.20 D 0.20 E 2.63

Solution

Here we are dealing with indices and logarithm

$$10^n = 0.63$$

Changing from indices to logarithm

$$\log_{10}0.63 = n$$

What we were given is $\log_{10}63$, so we convert 0.63 to 63

$$\log_{10}0.63 = \log_{10}\frac{63}{100}$$

$$= \log_{10}63 - \log_{10}100$$

$$= \log_{10}63 - \log_{10}10^2 \text{ i.e } \log_{10}63 - 2\log_{10}10$$

Substituting

$$= 1.8 - 2$$

$$= -0.2 \quad (C)$$

2008/1a

Without using calculator or table, find the value of log 3.6 given that log 2 = 0.3010, log 3 = 0.4771 and log 5 = 0.6990

Solution

Our target will be to express 3.6 in 2, 3 and 5 and their multiples

$$\log 3.6 = \log\frac{36}{10}$$

$$= \log 36 - \log 10$$

$$= \log(4 \times 9) - \log 10$$

$$= \log 4 + \log 9 - \log 10$$

$$= \log 2^2 + \log 3^2 - \log 10$$

$$= 2\log 2 + 2\log 3 - \log 10$$

Substituting for the given values

$$= 2(0.3010) + 2(0.4771) - 1$$

$$= 0.602 + 0.9542 - 1$$

$$= 1.5562 - 1$$

$$= 0.5562$$

2009/6a

If log5 = 0.6990, log 7 = 0.8451 and log8 = 0.9031, evaluate $\log\left(\frac{35\times 49}{40\div 56}\right)$

Solution

First, we apply simple arithmetic to simplify the bracket as

$$\log\left(\frac{35\times 49}{40\div 56}\right) = \log\left(35\times 49\div\frac{40}{56}\right)$$

$$= \log\left(35\times 49\times\frac{56}{40}\right)$$

$$= \log(7\times 49\times 7)$$

$$= \log 7^4$$

$$= 4\log 7$$

Substituting

$$= 4(0.8451)$$

$$= 3.3804$$

2009/3 ***Counter example***

If log 2 = x, log 3 = y and log 7 = z,

find in terms of x, y and z, the value of $\log\left(\frac{28}{3}\right)$

A 2x + y – z B 2x + z – y C x + y – 2z

D x + z – y

Solution

$$\log\left(\frac{28}{3}\right) = \log 28 - \log 3$$

Next, we express 28 in 2, 3 and 7

$$= \log(4\times 7) - \log 3$$

$$= \log 2^2 + \log 7 - \log 3$$

$$= 2\log 2 + \log 7 - \log 3$$

Substituting for the given values

$$= 2x + z - y \quad (B)$$

2013/1b

Given that $\log_{10}2 = 0.3010$ and $\log_{10}3 = 0.4771$ evaluate correct to 2 significant figures and without using table or calculator, $\log_{10}1.125$.

Solution

Our target will be to express 1.125 in terms of 2 and 3

$$\log_{10}1.125 = \log_{10}\frac{1125}{100}$$

By log law of division

$$= \log_{10}1125 - \log_{10}100$$

But 1125 is 5×225

is $5 \times 3 \times 75$

is $5 \times 3 \times 3 \times 25$

i.e $5^3 \times 3^2$

Case of 3 is ok, but we have 5 in place of 2;

the way out is $\frac{10}{2}$ as 5

$$= \log_{10}(5^3\times 3^2) - \log_{10}10^2$$

$$= \log_{10}5^3 + \log_{10}3^2 - \log_{10}10^2$$

$$= 3\log_{10}5 + 2\log_{10}3 - 2\log_{10}10$$

$$= 3[\log_{10}\frac{10}{2}] + 2\log_{10}3 - 2\log_{10}10$$

$$= 3\log_{10}10 - 3\log_{10}2 + 2\log_{10}3 - 2$$

Substituting for the given values

$$= 3 - 3(0.3010) + 2(0.4771) - 2$$

$$= 3 - 2 - 0.903 + 0.9542$$

$$= 1 - 0.0512$$

$$= 0.9488$$

2006/13a Exercise 5.38

If $\log_{10}2 = 0.3010$ and $\log_{10}7 = 0.8451$, evaluate without using logarithm table or calculator, $\log_{10}35$

Relationship between indices and logarithm

2014/7 NABTEB

If $\log_n 1 = x$, what is the value of x?

A – 2 B – 1 C 0 D 1

Solution

$\log_n 1 = x$

Changing from log to indices

$n^x = 1$

By law of indices

$x = 0$ (C)

Since only $n^0 = 1$ by indices

2014/28 NABTEB

If $\log_a x = P$, express x in terms of a and p

A $x = a + p$ B $x = {}^a/_p$ C $x = p$ D $x = a^p$

Solution

$\log_a x = P$

Changing from log to indices

$a^p = x$ (D)

2013/13 Neco

If $\log_2 16 = 4x - 1$, what is the value of x?

A $^3/_4$ B $^4/_5$ C 1 D $^5/_4$ E $^4/_3$

Solution

$\log_2 16 = 4x - 1$

Changing from logarithm to indices

$2^{4x-1} = 16$

Express the RHS in 2 and its powers

$2^{4x-1} = 2^4$

Since they are in same base, we can equate powers

$4x - 1 = 4$

$4x = 4 + 1$

$4x = 5$

$x = {}^5/_4$ (D)

2005/24

If $\log_9 x = 1.5$, find x

A 36 B 27 C 24.5 D 13.5

Solution

$\log_9 x = 1.5$

Changing logarithm to indices

$9^{1.5} = x$

Next, we express 1.5 in fractional form 1.5 is $^{15}/_{10}$ i.e $^3/_2$

Thus $9^{\frac{3}{2}} = x$

$\left(3^2\right)^{\frac{3}{2}} = x$

$3^3 = x$

i.e $27 = x$ (B)

2006/34 (Nov)

Solve for x in the equation: $\log_8 32 = x + 2$

A $\frac{11}{3}$ B $-\frac{1}{3}$ C – 1 D $-\frac{11}{3}$

Solution

$\log_8 32 = x + 2$

Change from log to indices

$8^{x+2} = 32$

Expressing both sides in base2 and its powers

$2^{3(x+2)} = 2^5$

Now they are in the same base, we equate powers

$3(x + 2) = 5$

$3x + 6 = 5$

$3x = 5 - 6$

$3x = -1$

$x = {}^{-1}/_3$ (B)

2005/28 Neco

Solve the equation $\log_2 x^2 = -6$

A – 32 B – 12 C $^1/_{64}$ D $^1/_8$ E 8

Solution

$\log_2 x^2 = -6$

Changing from logarithm to indices

$x^2 = 2^{-6}$

Next, we seek to align RHS power to LHS

$x^2 = (2^{-3})^2$

Since they are in the same power, we equate terms

$x = 2^{-3}$

$x = {}^1/_8$ (D)

2005/2 (Nov)

Given that $10^x = 3$, express $\log_{10}300$ in terms of x

A 2x B x + 2 C 2x + 1 D x – 2

Solution

$10^x = 3$

Changing from indices to logarithm

$\log_{10} 3 = x$

Next, we seek to express $\log_{10}300$ in form of $\log_{10}3$

$\log_{10}300 = \log_{10}(3 \times 100)$

By law of log multiplication

$= \log_{10}3 + \log_{10}100$

$= \log_{10}3 + \log_{10}10^2$

$= \log_{10}3 + 2\log_{10}10$

Substituting for x

$= x + 2$ B

2009/32 (Nov) Exercise 5.39

If $\log_2 p = 3$, find the value of p

A 3 B 6 C 8 D 9

2015/6 Exercise 5.40

Given that $\log_x 64 = 3$, evaluate $x\log_2 8$

A. 6 B. 9 C. 12 D. 24

Logarithmic equations

At times, questions in logarithm involve equations, which may be simple linear, simultaneous linear equations or quadratic equation in nature.

Which ever pattern the questions follows; we solve using the knowledge of the method of solving such equation.

Simple equation

2014/1b NABTEB (Nov)

Solve $\log_3(2x + 1) - \log_3(x - 3) = 2$

Solution

Express the RHS in $\log_3$ form

$\log_3(2x + 1) - \log_3(x - 3) = \log_3 3^2$

Applying laws of logarithm

$$\log_3 \frac{2x+1}{x-3} = \log_3 3^2$$

Equating terms

$$\frac{2x+1}{x-3} = 3^2$$

Cross multiply

$$2x + 1 = 9(x - 3)$$
$$2x + 1 = 9x - 27$$
$$1 + 27 = 9x - 2x$$
$$28 = 7x$$
$$\frac{28}{7} = \frac{7x}{7}$$
$$4 = x$$

2014/ 23 Neco

If $\log_3(2x - 1) - \log_3 2 = 1$, find x

A $4\frac{1}{2}$ B $3\frac{1}{2}$ C $2\frac{1}{2}$ D $1\frac{1}{2}$ E 1

Solution

$\log_3(2x - 1) - \log_3 2 = 1$

Applying laws of logarithm

$$\log_3 \frac{2x+1}{2} = \log_3 3$$

Equating terms

$$\frac{2x+1}{2} = 3$$
$$2x - 1 = 2 \times 3$$
$$2x = 6 + 1$$
$$x = \frac{7}{2} \text{ i.e } 3\frac{1}{2} \quad (\text{B})$$

2007/2b

If $4\log a + 5\log a - 6\log a = \log 8$, what is a?

Solution

Applying laws of logarithm

$4\log a + 5\log a - 6\log a = \log 8$ becomes

$\log a^4 + \log a^5 - \log a^6 = \log 8$

$$\log\left(\frac{a^4 \times a^5}{a^6}\right) = \log 8$$

Some terms will cancel out

$$\log a^3 = \log 8$$

Equating terms

$$a^3 = 8 \text{ or } a = \sqrt[3]{8}$$
$$a^3 = 2^3$$
$$a = 2$$

2012/7c Neco

If $\log_2 x = 3 + \log_2 y$, find the value of $\frac{x}{y}$

Solution

Transforming 3 to $\log_2$

$\log_2 x = 3 + \log_2 y$ becomes

$\log_2 x = \log_2 2^3 + \log_2 y$

Collect unknown terms together

$\log_2 x - \log_2 y = \log_2 8$

Applying laws of logarithm

$$\log_2 \frac{x}{y} = \log_2 8$$

Equating terms $\frac{x}{y} = 8$

2007/7b

If $\log_{10}y + 3\log_{10}x = 2$, express y in terms of x

Solution

Since the LHS is in base 10, we align the RHS as well

$\log_{10}y + 3\log_{10}x = \log_{10}10^2$

$\log_{10}y + \log_{10}x^3 = \log_{10}10^2$

Applying the law of addition in log

$\log_{10} y \times x^3 = \log_{10}10^2$

Since both sides are in the same base 10, we equate terms

$yx^3 = 10^2$

Making y the subject of formula

$$y = \frac{100}{x^3}$$

2011/1a Neco ***Counter example***

Given that $\log(y - 2)^3 = \log 64$, without using mathematical tables, find y

Solution

$\log(y - 2)^3 = \log 64$

we can equate terms directly here

$(y - 2)^3 = 64$

Next, we express the RHS in power 3 to align with LHS

$(y - 2)^3 = 4^3$

Since they are in the same power, we can equate terms

$$y - 2 = 4$$
$$y = 4 + 2$$
$$y = 6$$

2009/1b Neco (Nov)

Find x if $\log_3(3x - 2) - \log_3(x + 2) = 2$

Solution

Applying laws of logarithm

$\log_3(3x - 2) - \log_3(x + 2) = 2$ becomes

$\log_3(3x - 2) - \log_3(x + 2) = \log_3 3^2$

i.e $\log_3 \frac{3x-2}{x+2} = \log_3 3^2$

Equating terms

$$\frac{3x-2}{x+2} = 3^2$$

Cross multiply

$$3x - 2 = 9(x + 2)$$
$$3x - 2 = 9x + 18$$
$$3x - 9x = 18 + 2$$
$$-6x = 20$$
$$x = \frac{20}{-6} \text{ i.e } -\frac{10}{3}$$

2007/11 Neco

Solve the equation $3\log_3 x - \log_3 x = 2$

A $\frac{1}{\sqrt{2}}$ B $\sqrt{2}$ C $\sqrt{3}$ D 3 E 9

Solution

$3\log_3 x - \log_3 x = 2$

Expressing both sides in $\log_3$, we have

$\log_3 x^3 - \log_3 x = \log_3 3^2$

Applying the laws of logarithm

$$\log_3 \frac{x^3}{x} = \log_3 3^2$$

Since they are in the same base, we equate terms

$$\frac{x^3}{x} = 3^2$$

i.e $x^2 = 3^2$

Since they are in the same power, we equate terms

$x = 3$ (D)

2006/55 Neco

Solve the equation:

$$\log_{10}(6a+3) + \log_{10}\left(\frac{3}{7}\right) = \log_{10}(2a-1)$$

A – 4 B – $^1/_2$ C $^1/_2$ D 2 E 4

Solution

$$\log_{10}(6a+3) + \log_{10}\left(\frac{3}{7}\right) = \log_{10}(2a-1)$$

Since they are in the same base, we apply the law of logarithm as:

$$\log_{10}(6a+3)\times\frac{3}{7} = \log_{10}(2a-1)$$

Since they are in the same base, we equate terms as:

$$(6a + 3) \times \frac{3}{7} = 2a - 1$$

Cross multiply

$(6a + 3) \times 3 = 7(2a - 1)$

$18a + 9 = 14a - 7$

$18a - 14a = -9 - 7$

$4a = -16$

$a = -\frac{16}{4}$

$= -4$ (A)

2006/ 31 ***counter example***

If $\log_a 270 - \log_a 10 + \log_a \frac{1}{3} = 2$,

what is the value of a?

A 3 B 4 C 5 D 6

Solution

$\log_a 270 - \log_a 10 + \log_a \frac{1}{3} = 2$

Since the LHS is all in the same base a, we do same to the RHS

$\log_a 270 - \log_a 10 + \log_a \frac{1}{3} = \log_a a^2$

Applying the laws of logarithm

$$\log_a \frac{270\times\frac{1}{3}}{10} = \log_a a^2$$

$$\log_a 9 = \log_a a^2$$

Since they are in the same base, we equate terms as:

$9 = a^2$

$3^2 = a^2$

$3 = a$ (A)

2005/ 9a Exercise 5.41

If $\log_{10} Q - \log_{10} \frac{1}{2}(Q-1) = 1$, find the value of Q

2009/3 (Nov) Exercise 5.42

Given that $\log x = 2\log 3 + \log 6 - \log 2$,
find the value of x

A 9 B 18 C 27 D 81

2012/9 Neco Exercise 5.43

Solve the equation $\log_{10}x - \log_{10}(2x - 3.8) = 1$

A 0 B – 1 C – 2 D 1 E 2

2005/34 (Nov) Exercise 5.44

Solve for x in the equation:

$\log 2x + \log 3 = 2 \log\left(\frac{6}{5}\right)$

A $\frac{6}{25}$ B $\frac{39}{25}$ C $\frac{39}{50}$ D $\frac{1}{5}$

2012/32 Exercise 5.45

Which of these statement about $y = 8\sqrt{m}$ is correct?

A $\log y = \log 8 \times \log\sqrt{m}$ B $\log y = 3\log 2 \times \frac{1}{2}\log m$

C $\log y = 3\log 2 - \frac{1}{2}\log m$ D $\log y = 3\log 2 + \frac{1}{2}\log m$

Simultaneous linear equations in nature

1997/8

Solve the simultaneous equations

$\log_{10} x + \log_{10} y = 4$

$\log_{10} x + 2\log_{10} y = 3$

Solution

Expressing both sides of the equations in logarithm to base 10

$\log_{10}x + \log_{10}y = \log_{10}10^4$ ---- (1)

$\log_{10}x + \log_{10}y^2 = \log_{10} 10^3$ ----- (2)

From (1)

$\log_{10}(xy) = \log_{10}10^4$

$xy = 10^4$ ---- (a)

Also from (2)

$\log_{10}(xy^2) = \log_{10}10^3$

$xy^2 = 10^3$

i.e $(xy)y = 1000$

Substituting for xy value from (a)

$(10^4)y = 1000$

$y = \frac{1000}{10000} = \frac{1}{10} = 10^{-1}$

Substituting y values into (a)

$xy = 10^4$

$x(10^{-1}) = 10^4$

$x = 10^4 \times 10^1$

$x = 10^5 = 100{,}000$

Solution set is : $x = 10^5$ and $y = 10^{-1}$

2006/2 f/maths

If $2^{2x-3y} = 32$ and $\log_y x = 2$,
find the values of x and y

Solution

Converting logarithm to indices

$\log_y x = 2$ becomes

$y^2 = x$ ----------------------(1)

Note that $2^{2x-3y} = 32$ is same as

$2^{2x-3y} = 2^5$

Equating powers

$2x - 3y = 5$ ----------------(2)

Substituting for $x = y^2$

$2y^2 - 3y = 5$

Rearranging

$2y^2 - 3y - 5 = 0$

Factorizing

$2y^2 + 2y - 5y - 5 = 0$

$2y(y+1) - 5(y+1) = 0$

$(y+1)(2y-5) = 0$

$y + 1 = 0$ or $2y - 5 = 0$

$y = -1$ or $2y = 5$

$y = -1$ or $^5/_2$

when $y = -1$; $x = y^2$ becomes $x = 1$

when $y = {}^5/_2$; $x = y^2$ becomes $x = {}^{25}/_4$

2013/13 Neco *f/maths* Exercise 5.46

Solve the equations:

If $\log_3(p + q) = 1$ and $\log_2(2p - q) = 2$
find the value of pq

A. $2^1/_3$ B. $1^5/_9$ C. $1^1/_2$ D. $^2/_3$ E. $^3/_7$

2010/9a Neco *f/maths* Exercise 5.47

Solve the equations:

$\log_8(4 + x) + \log_8 y = 2$

$\log_4(x + 52) - \log_4 y = 2$

2009/2 *f/maths* Exercise 5.48

Solve the simultaneous equations

$\log_2 x - \log_2 y = 2$, $\log_2(x - y) = 3$

Quadratic equation in nature

2009/8b Neco

Given that $\log_2(2x-1) + \log_2(x+1) = 3$, find x

Solution

Applying laws of logarithm

$\log_2(2x-1) + \log_2(x+1) = 3$ becomes

$\log_2(2x-1) + \log_2(x+1) = \log_2 2^3$

$\log_2(2x-1)(x+1) = \log_2 8$

Equating terms

$(2x-1)(x+1) = 8$

Expanding the LHS

$2x(x+1) - 1(x+1) = 8$

$2x^2 + 2x - x - 1 = 8$

$2x^2 + x - 9 = 0$

Factorizing, it is not factorizable

Applying the general formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Here a is 2, b is 1 and c is – 9 . Substituting

$$x = \frac{-1 \pm \sqrt{1 - 4 \times 2 \times (-9)}}{2 \times 2}$$

$$= \frac{-1 \pm \sqrt{1 + 72}}{4}$$

$$= \frac{-1 \pm \sqrt{73}}{4}$$

$$= \frac{-1 \pm 8.544}{4}$$

$$= \frac{-1 + 8.544}{4} \quad or \quad \frac{-1 - 8.544}{4}$$

$$= \frac{7.544}{4} \quad or \quad \frac{-9.544}{4}$$

$= 1.886$ or -2.386

1993/6 UME

If $2\log_3 y + \log_3 x^2 = 4$, then y is

A. $\frac{4 - \log_3 x^2}{2}$ B. $\frac{4}{\log_3 x^2}$ C . $\frac{2}{x}$ D. $\frac{\pm 9}{x}$

Solution

Applying the laws of logarithm,

$\log_3 y^2 + \log_3 x^2 = 4$

Since they are in the same base

$\log_3(y^2 \times x^2) = 4$

Converting to indices

$3^4 = y^2 x^2$

$\frac{3^4}{x^2} = y^2$

$\left(\frac{3^2}{x}\right)^2 = y^2$

$y = \pm\sqrt{\left(\frac{9}{x}\right)^2} = \pm\frac{9}{x}$ (D)

2009/6 Neco f/maths

Solve the equations:

$\log(x^2 + 6x) = 1.4314$

A. –9 or 3 B. 9 or –3 C. –9 or 5 D. 5 or 9 E. 9 or 7

Solution

$\log(x^2 + 6x) = 1.4314$

Changing from logarithm to indices

$x^2 + 6x = 10^{1.4314}$

Actually the RHS is the antilog of 1.4314

$x^2 + 6x = 27$

$x^2 + 6x - 27 = 0$

factorizing: $x^2 + 9x - 3x - 27 = 0$

$x(x+9) - 3(x+9) = 0$

$(x+9)(x-3) = 0$

$x + 9 = 0$ or $x - 3 = 0$

$x = -9$ or 3 A.

2013/18 Neco *f/maths* Exercise 5.49

Solve the logarithmic equation :

$\log_5(x^2 - 9) = 0$

A. $\pm\sqrt{3}$ B. $\pm\sqrt{7}$ C. $\pm\sqrt{10}$ D. ± 3 E. ± 9

2014/1 *f/maths* Exercise 5.50

Solve $(\log_2 m)^2 - \log_2 m^3 = 10$

1976/8 ii Exercise 5.51

Without using tables, find the value of n in

$2\log(n + 3) = \log 4$

CHAPTER SIX

Standard form & Logarithm II

Standard Form

This topic like most topics in mathematics is a bit confusing to students due to lack of basic understanding. Standard form is the expression of numbers in scientific form to reduce the stress and space of writing very large or very small numbers.

It is written in the form $A \times 10^n$. Where A is a number that can be any of 1,2,3,4,5,6,7,8,9. and n can be a positive or negative integer.

Examples;

(a) Express the following in standard form

(i) 1,267,000,000 (ii) 0.000 000 00156

Solution

(i) 1,267,000,000 = 1 267 000 000. 0

We have only made the decimal point clearer, then start counting backward (reverse) till you get to the space before the first digit – place your decimal there and note the number of digits or places jumped.

$= 1.267 \times 10^9$

(ii) Similarly, we move forward till the space between first digit and second one. Hence

$0.000\,000\,00156 = 1.56 \times 10^{-9}$

(b) Express the following in standard form

(i) 5 (ii) 50 (iii) 500 (iv) 5000

Solution

(i) 5 is the same as 4.0

$5 = 5.0 \times 10^0$ is the same as 5.0 since $10^0 = 1$

(ii) $50 = 5.0 \times 10^1$

(iii) $500 = 5.0 \times 10^2$

(iv) $5000 = 5.0 \times 10^3$

2009/1 (Nov)

Express 0.00347 in the standard form

A 3.47×10^{-3} B 3.47×10^{-2} C 3.47×10^2 D 3.47×10^3

Solution

$0.00347 = 3.47 \times 10^{-3}$ (A)

2012/1 Neco

Express 0.0462 in standard form

A 0.462×10^{-1} B 0.462×10^{-2} C 4.62×10^{-1}

D 4.62×10^{-2} E 4.62×10^{-3}

Solution

$0.0462 = 4.62 \times 10^{-2}$ (D)

2010/27 NABTEB (Nov)

Express 0.0000043169 in standard form to three significant figures

A 4.3169×10^{-5} B 4.3169×10^{-6}

C 4.317×10^{-6} D 4.31×10^{-4}

Solution

First, we express in standard form

$0.0000043169 = 4.3169 \times 10^{-6}$

Next, approximation to 3 s.f

$= 4.32 \times 10^{-6}$

Note that approximation to 3 decimal places is 4.317×10^{-6}

2014/3 NABTEB Exercise 6.1

Evaluate 64.25×10^{-3}

A 0.06425 B 0.6425 C 6.425 D 64.25

2009/8 NABTEB (Nov) Exercise 6.2

Express 0.000502 in standard form

A 5.02×10^4 B 5.02×10^3

C 5.02×10^{-4} D 5.02×10^{-3}

2010/14 Neco Exercise 6.3

Which of the following has the same value as 0.0162560 ?

A 1.6256×10^2 B 1.6256×10^1 C 1.6256×10^0

D 1.6256×10^{-2} E 1.6256×10^{-3}

Operations with standard form

Addition

Example

i. Express the sum of 6.03×10^6 and $2.17\ 10^5$ in standard form

solution

Firstly, we factor out the lowest power of 10 i.e 10^5

Thus $6.03 \text{ x } 10^6 + 2.17 \text{ x } 10^5$

$= (6.03 \times 10^1 + 2.17) \times 10^5$

$= (60.3 + 2.17) \times 10^5$

$= 62.47 \times 10^5$

Converting to standard form , we have

$= 6.247 \times 10^5 \times 10^1$

$= 6.247 \times 10^6$

Subtraction

Example

Without using a calculator, find the difference between 9.5×10^7 and 3.08×10^6 leaving your answer in standard form

Solution

We factor out the lowest power of 10 i.e 10^6

Thus, $9.5 \times 10^7 - 3.08 \times 10^6$

$= (9.5 \times 10^1 - 3.08) \times 10^6$

$= (95.00 - 3.08) \times 10^6$

$= 91.92 \times 10^6$

Converting to standard form, we have

$= 9.192 \times 10^6 \times 10^1$

$= 9.192 \times 10^7$

2002/2c Exercise 6.3b

Without using calculator, find the difference between 3.4×10^9 and 2.2×10^8. Leave your answer in standard form

Multiplication

2014/2 NABTEB

Simplify 0.0285 × 0.267, leaving the answer in standard form

A 7.6095×10^{-3} B 7.6095×10^{-2}

C 7.6095×10^{-1} D 7.6095×10^{0}

Solution

First, we express them in whole numbers without decimals

$0.0285 \times 0.267 = 285 \times 10^{-4} \times 267 \times 10^{-3}$

$= 285 \times 267 \times 10^{-4} \times 10^{-3}$

$= 76095 \times 10^{-4+(-3)}$

$= 76095 \times 10^{-7}$

Converting to standard form

$= 7.6095 \times 10^{-7} \times 10^{4}$

$= 7.6095 \times 10^{-7+4}$

$= 7.6095 \times 10^{-3}$ A

2006/ 3 (Nov)

Express the product of 1.04 and 0.08 in the form $a \times 10^n$, where $1<a<10$ and n an integer

A 8.32×10^{2} B 8.32×10^{1} C 8.32×10^{-1}

D 8.32×10^{-2}

Solution

We are simply asked to work in standard form expressing the item in whole numbers without decimals

$1.04 \times 0.08 = 104 \times 10^{-2} \times 8 \times 10^{-2}$

$= 104 \times 8 \times 10^{-2-2}$

$= 832 \times 10^{-4}$

Converting to standard form

$= 8.32 \times 10^{-4} \times 10^{2}$

$= 8.32 \times 10^{-4+2}$

$= 8.32 \times 10^{-2}$ (D)

2006/2 (Nov)

Evaluate $(4.5 \times 10^{-2})^2$ leave your answer in the standard form

A 2.025×10^{-4} B 2.025×10^{-3} C 9.0×10^{-4} D 9.0×10^{-3}

Solution

$(4.5 \times 10^{-2})^2 = 4.5 \times 10^{-2} \times 4.5 \times 10^{-2}$

Expressing them in whole numbers

$= 45 \times 10^{-2} \times 10^{-1} \times 45 \times 10^{-2} \times 10^{-1}$

$= 45 \times 45 \times 10^{-2-1-2-1}$

$= 2025 \times 10^{-6}$

Converting to standard form

$= 2.025 \times 10^{3} \times 10^{-6}$

$= 2.025 \times 10^{-3}$ (B)

2011/2 Neco

Express the product of 0.0045 and 0.025 in standard form

A 1.125×10^{-1} B 1.125×10^{-3} C 1.8×10^{1}

D 1.125×10^{2} E 1.8×10^{2}

Solution

First, we express them in whole numbers without decimals

$0.0045 \times 0.025 = 45 \times 10^{-4} \times 25 \times 10^{-3}$

$= 45 \times 25 \times 10^{-4} \times 10^{-3}$

$= 1125 \times 10^{-4+(-3)}$

$= 1125 \times 10^{-4-3}$

$= 1125 \times 10^{-7}$

Converting to standard form

$= 1.125 \times 10^{3} \times 10^{-7}$

$= 1.125 \times 10^{3+(-7)}$

$= 1.125 \times 10^{3-7}$

$= 1.125 \times 10^{-4}$

2014/4 Neco

Express 0.17 × 0.17 in standard form

A 2.89×10^{-4} B 2.89×10^{-2} C 2.89×10^{-1}

D 2.89×10^{1} E 2.89×10^{4}

Solution

We express them in whole numbers without decimals

$0.17 \times 0.17 = 17 \times 10^{-2} \times 17 \times 10^{-2}$

$= 17 \times 17 \times 10^{-2} \times 10^{-2}$

$= 289 \times 10^{-2+(-2)}$

$= 289 \times 10^{-4}$

Converting to standard form

$= 2.89 \times 10^{2} \times 10^{-4}$

$= 2.89 \times 10^{2-4}$

$= 2.89 \times 10^{-2}$ (B)

2010/1 Exercise 6.4

Simplify 0.000215 × 0.000028 and express your answer in the standard form

A 6.03×10^{9} B 6.02×10^{9} C 6.03×10^{-9} D 6.02×10^{-9}

2014/2 NABTEB (Nov) Exercise 6.5

Express the product of 0.06 and 0.09 in standard form

A 5.4×10^{-3} B 5.4×10^{-2} C 5.4×10^{-1} D 5.4×10^{2}

1976/4a (Nov) Exercise 6.6

Multiply 7.37×10^{9} by 3.02×10^{-7}, giving your answer in the form $A \times 10^n$ where n is an integer and A is a number between 1 and 10 to 3 significant figures

Division

2007/3

Simplify $(0.3 \times 10^5) \div (0.4 \times 10^7)$, leaving your answer in the standard form

A 7.5×10^{-4} B 7.5×10^{-3} C 7.5×10^{-2} D 7.5×10^{-1}

Solution

We express terms in whole numbers without decimals

$$(0.3 \times 10^5) \div (0.4 \times 10^7) = \frac{3 \times 10^5 \times 10^{-1}}{4 \times 10^7 \times 10^{-1}}$$

10^{-1} will cancel out

$$= \frac{3}{4} \times \frac{10^5}{10^7}$$

$= 0.75 \times 10^{5-7}$

$= 0.75 \times 10^{-2}$

Converting to standard form

$= 7.5 \times 10^{-2} \times 10^{-1}$

$= 7.5 \times 10^{-3}$ (B)

2007/1a

Evaluate, without using mathematical tables or calculator $(3.69 \times 10^5) \div (1.64 \times 10^{-3})$, leaving your answer in the standard form

Solution

We express them in whole numbers without decimals

$(3.69 \times 10^5) \div (1.64 \times 10^{-3}) = \dfrac{369 \times 10^5 \times 10^{-2}}{164 \times 10^{-3} \times 10^{-2}}$

10^{-2} will cancel out

$= \dfrac{369}{164} \times \dfrac{10^5}{10^{-3}}$

$= 2.25 \times 10^{5-(-3)}$

$= 2.25 \times 10^8$

2013/1 Neco

Evaluate 0.009 ÷ 0.012, leave your answer in the standard form

A 7.5×10^{-2} B 7.5×10^{-1} C 7.5×10^0
D 7.5×10^1 E 7.5×10^2

Solution

We express them in whole numbers without decimals

0.009 ÷ 0.012 is the same as $\dfrac{0.009}{0.012} = \dfrac{9 \times 10^{-3}}{12 \times 10^{-3}}$

$= \dfrac{9}{12} \times 10^{-3-(-3)}$

$= \dfrac{3}{4} \times 10^{-3+3}$

$= 0.75$

Converting to standard form

$= 7.5 \times 10^{-1}$ B

2002/3 NABTEB Exercise 6.7

Evaluate $\dfrac{0.9687}{0.001}$ leaving your answer in standard form

A. 9.687×10^{-4} B. 9.687×10^{-1}
C. 9.687×10^2 D. 9.687×10^3

2004/4 Neco Exercise 6.8

Express the quotient of 0.422 and 0.004in standard form.

A. 1.055×10^2 B.10.55×10^1 C.105.5×10^0
D. 10.55×10^1 E. 1.055×10^{-2}

Multiplication and division joint cases

2014/1a

Without using tables or calculator,

simplify $\dfrac{0.6 \times 32 \times 0.004}{1.2 \times 0.008 \times 0.16}$

leaving your answer in the standard form (scientific notation)

Solution

First, we express in whole numbers without decimals

$\dfrac{0.6 \times 32 \times 0.004}{1.2 \times 0.008 \times 0.16} = \dfrac{6 \times 10^{-1} \times 32 \times 4 \times 10^{-3}}{12 \times 10^{-1} \times 8 \times 10^{-3} \times 16 \times 10^{-2}}$

Some terms will cancel out

$= \dfrac{1}{2} \times \dfrac{10^{-4}}{10^{-6}}$

Power –6 goes to numerator $= \dfrac{1}{2} \times 10^{-4-(-6)}$

$= 0.5 \times 10^{-4+6}$

$= 0.5 \times 10^2$

Converting to standard form

$= 5.0 \times 10^{-1} \times 10^2$

$= 5.0 \times 10^1$

2006/2 Neco (Nov)

Evaluate $2.25 \times 10^{-2} \div 0.225 \times 10^{-3} \times 22.5 \times 10^2$

A 2.25×10^{-3} B 2.25×10^{-2} C 2.25×10^{-1}
D 2.25×10^2 E 2.25×10^5

Solution

We express them in whole numbers without decimals

$2.25 \times 10^{-2} \div 0.225 \times 10^{-3} \times 22.5 \times 10^2$

$= \dfrac{225 \times 10^{-2} \times 10^{-2} \times 225 \times 10^2 \times 10^{-1}}{225 \times 10^{-3} \times 10^{-3}}$

$= \dfrac{225 \times 225 \times 10^{-2-2+2-1}}{225 \times 10^{-3-3}}$

Some terms will cancel out

$= \dfrac{225}{1} \times \dfrac{10^{-3}}{10^{-6}}$

Power –6 goes to numerator

$= 225 \times 10^{-3-(-6)}$

$= 225 \times 10^{-3+6}$

$= 225 \times 10^3$

Converting to standard form

$= 2.25 \times 10^3 \times 10^2$

$= 2.25 \times 10^5$ (E)

2009/8b Neco (Nov)

Simplify without using calculator or tables, $\dfrac{0.0125 \times 0.625}{0.075 \times 0.25}$ correct to 2 decimal places

Solution

First, we express in whole numbers without decimals

$\dfrac{0.0125 \times 0.625}{0.075 \times 0.25} = \dfrac{125 \times 10^{-4} \times 625 \times 10^{-3}}{75 \times 10^{-3} . \times 25 \times 10^{-2}}$

$= \dfrac{125 \times 625}{75 \times 25} \times \dfrac{10^{-4+(-3)}}{10^{-3+(-2)}}$

Some terms will cancel out

$= \dfrac{125}{3} \times \dfrac{10^{-7}}{10^{-5}}$

$= \dfrac{125}{3} \times 10^{-7-(-5)}$

Minus power from denominator becomes plus

$= \dfrac{125}{3} \times 10^{-7+5}$

$= \dfrac{125}{3} \times 10^{-2}$

$= 41.667 \times 10^{-2}$

Here we are **not** working in standard form

$= 0.41667$

≈ 0.42 to 2d.p

2012/8 Neco

Simplify $\dfrac{0.0324 \times 0.0064}{0.048 \times 0.12}$ and give your answer in standard form.

A 36×10^{-2} B 3.6×10^{-2} C 3.6×10^2
D 0.36×10^1 E 0.36×10^{-1}

Solution

First, we express in whole numbers without decimals

$$\frac{0.0324 \times 0.0064}{0.048 \times 0.12} = \frac{324\times10^{-4}\times 64\times10^{-4}}{48\times10^{-3}\times12\times10^{-2}}$$

$$= \frac{324\times64\times10^{-4+(-4)}}{48\times12\times10^{-3+(-2)}}$$

$$= \frac{36}{1} \times \frac{10^{-8}}{10^{-5}}$$

$= 36 \times 10^{-8-(-5)}$

$= 36 \times 10^{-8+5}$

$= 36 \times 10^{-3}$

Converting to standard form

$= 3.6 \times 10^{1} \times 10^{-3}$

$= 3.6 \times 10^{-3+1}$ i.e 3.6×10^{-2} (B)

2009/34 NABTEB (Nov) Exercise 6.9

Simplify $\frac{0.00275\times 0.008}{0.0025\times 0.09}$ and give your answer in standard form.

A 9.7×10^{-2} B 9.7×10^{-1} C 9.7×10^{-3} D 9.7×10^{1}

2014/2a NABTEB (Nov) Exercise 6.10

Evaluate $\frac{4.56\times3.6}{0.12}$, leave your answer in standard form

1977/2b Exercise 6.11

Evaluate $\frac{pq^3}{r^2}$, if $p = 10^8$, $q = 10^{-4}$ and $r = 0.000\ 01$, expressing your answer in powers of 10

Cases involving square root

2009/6

Express the square root of 0.000144 in the standard form

A 1.2×10^{-4} B 1.2×10^{-3} C 1.2×10^{-2} D 1.2×10^{-1}

Solution

We express in numbers without decimal

$$\sqrt{0.000144} = (144\times10^{-6})^{\frac{1}{2}}$$

$$= (12^2\times10^{-6})^{\frac{1}{2}}$$

Taking the square root of the numbers and powers as

$= 12^{\frac{2}{2}} \times 10^{-\frac{6}{2}}$

$= 12 \times 10^{-3}$

Converting to standard form

$= 1.2 \times 10^{1} \times 10^{-3}$

$= 1.2 \times 10^{-2}$ (C)

2005/6a

Without using Mathematical tables or calculator

evaluate: $\sqrt{\frac{0.18\times12.5}{0.05\times0.2}}$

Solution

We express in numbers without decimals

$$\sqrt{\frac{0.18\times12.5}{0.05\times0.2}} = \sqrt{\frac{18\times10^{-2}\times125\times10^{-1}}{5\times10^{-2}\times2\times10^{-1}}}$$

$$= \sqrt{\frac{18\times125\times10^{-2-1}}{5\times2\times10^{-2-1}}}$$

Some terms will cancel out as square cannot be applied directly at this level

$= \sqrt{9\times25\times10^{-3-(-3)}}$

$= \sqrt{3^2\times5^2\times10^{-3+3}}$ i.e $\sqrt{3^2\times5^2}$

$= 3 \times 5$

$= 15$ (not in standard form)

$= 1.5 \times 10^{1}$ (in standard form)

2009/3a

Without using table or calculator evaluates: $\sqrt{\frac{0.14\times3.2}{0.0028\times90}}$

Solution

We express given numbers in whole without decimals

$$\sqrt{\frac{0.14\times3.2}{0.0028\times90}} = \sqrt{\frac{14\times10^{-2}\times32\times10^{-1}}{28\times10^{-4}\times9\times10^{1}}}$$

$$= \sqrt{\frac{14\times32\times10^{-2-1}}{28\times9\times10^{-4+1}}}$$

$$= \sqrt{\frac{16\times10^{-3}}{9\times10^{-3}}}$$

$$= \sqrt{\frac{16}{9}\times10^{-3-(-3)}} = \sqrt{\frac{16}{9}\times10^{-3+3}}$$

$$= \sqrt{\frac{16}{9}} = \frac{4}{3}$$

$= 1.333$

2008/1a Exercise 6.12

Without using tables or calculators, find the value of

$\sqrt{\frac{0.00196\times2.25}{62.5}}$, in standard form

2006/2a Neco Exercise 6.13

Simplify without using mathematical tables or calculator

$$\sqrt{\frac{2.401\times10^{5}}{0.0049\times10^{-2}}}$$

Common Logarithm

Logarithm to base 10 is known as the common logarithm. There are other bases of logarithm e.g base e (natural, Naperian base).

The logarithm of a number consists of two parts called the *Characteristics* and the *mantissa.*
Characteristics is a whole number while Mantissa is a decimal. To get the characteristics of any given number, express the number in standard form, the index is the characteristics.
E.g, Study carefully the table below:

Numbers	Standard form	Logarithms Xtics	mantissa	Answers
(a) 5	5.0×10^0	0	.6990	Log of 5 = 0.6990
(b) 60	6.0×10^1	1	.7782	Log of 60 = 1.7782
(c) 8156	8.156×10^3	3	.9115	Logof 8156 =3.9115
(d) 0.0082	8.2×10^{-3}	-3 or $\bar{3}$	.9138	Log of 0.0082 = $\bar{3}.9138$
(e) 0.2	2.0×10^{-1}	- 1 or $\bar{1}$	.3010	Log of 0.2 = $\bar{1}.3010$

You will agree with the author that your ability to find the characteristics of any given problem depends on basic knowledge of standard form

Logarithm of numbers greater than one

Multiplication (Addition of log values)

E.g. (1) Using log table, evaluate 69.24×8.31

No	Log
69.24	1.8403
8.31	0.9196
575.3	2.7599

Analysis
$69.24 = 6.92 \times 10^1$
$8.31 = 8.31 \times 10^0$
Check 69 under 2 difference of 4
i.e 8401 + difference (2) = 8403
Check 83 under 1
add up log values
Check the Antilog of 75 under 9, diff. of 9
5741 + 12 = 5753
Antilog xtics 2
Movement 3 steps from the imaginary decimal before the first digit i.e 5

Eg (2)
Using logarithm table, evaluate $7031 \div 4.911$.
Solution

No	Log
7031	3.8471
4.911	0.6912
1432	3.1559

Division in log, we subtract the log values and find anti log.

E.g (3) With the aid of tables evaluate $(42.95)^3$
Solution

No	Log
$(42.95)^3$	1.6330×3
79250.0	4.8990

Note; Since antilog xtics is 4, movement 5 steps from the imaginary decimal point before the first digit i.e 7
Hence 7925 = 79250.0

E.g (4) Evaluate the $\sqrt[3]{66.32}$ using tables
Solution

No	Log
$(66.32)^{1/3}$	$1.8216 \div 3$
4.048	0.6072

Note
Log problems involving roots, we divide the log value by the equivalent root e.g

$\sqrt{\ }$, divide by 2

$\sqrt[3]{\ }$, divide by 3

$\sqrt[4]{\ }$, divide by 4 etc

Logarithm of numbers less than one

E.g 1 Evaluate 0.5624×0.0378

No	Log
0. 5624	$\bar{1}.7500$
0.0378	$\bar{2}.5775$
0.02125	$\bar{2}.3275$

Addition $-1 + 1 - 2 = -2$

$\therefore 0.5624 \times 0.0378 = 0.02125$

E.g 2 Evaluate $0.003512 \div 0.6207$

No	Log
0.003512	$\bar{3}.5455$
0.6207	$\bar{1}.7929$
0.005657	$\bar{3}.7526$

Subtraction

$\therefore 0.003512 \div 0.6207 = 0.005657$

E.g. (3) Evaluate $(0.07392)^4$

No	log
$(0.07392)^4$	$\bar{2}.8628 \times 4$
0.00002826	$\bar{5}.4512$

$\therefore (0.07392)^4 = 0.00002826$
Multiply the xtics and the mantissa separately and add the result i.e $0.8628 \times 4 = 3.4512$

and $\bar{2} \times 4 = \bar{8}$

Thus $\bar{8} + 3.4512 = \bar{5}.4512$

E.g (4) Evaluate $\sqrt[4]{0.00692}$

No	Log
$(0.00692)^{1/4}$	$\bar{3}.8401 \div 4$ is expressed as: $(\bar{4} + 1.8401) \div 4$
0.2884	$\bar{1}.4600$

$\therefore (90.00692)^{1/4} = 0.2884$
Think of a number to be added to $\bar{4}$ to give $\bar{3}$ on the xtic side only i.e $\bar{4} + 1 = \bar{3}$
Then divide $\bar{4}$ by 4 and 1.8401 by 4.

Problems on logarithm II

2011/1b

Given that $\sqrt{x} = 10^{\bar{1}.6741}$, without calculator, find the value of x

Solution

$$\sqrt{x} = 10^{\bar{1}.6741}$$

$$x^{\frac{1}{2}} = 10^{\bar{1}.6741}$$

We take the $\log_{10}$ of both sides

$$\log_{10} x^{\frac{1}{2}} = \log_{10} 10^{\bar{1}.6741}$$

$$\tfrac{1}{2}\log_{10} x = \bar{1}.6741 \text{ Log }_{10}10$$

$$\tfrac{1}{2}\log_{10} x = \bar{1}.6741$$

Multiply through by 2 to clear fraction

$$\log_{10} x = [2 \times 0.6741 + 2 \times \bar{1}]$$
$$= 1.3482 + (-2)$$

$$\log_{10} x = \bar{1}.3482$$

Take antilog of $\bar{1}.3482$ to get x

$$x = 0.2229$$

2007/34

Find the value of a if $\log_{10}a + \log_{10}a^2 = 0.9030$
A 4.0 B 2.0 C 1.6 D 0.0

Solution

$\log_{10}a + \log_{10}a^2 = 0.9030$
Applying the law of addition log
$\log_{10} a \times a^2 = 0.9030$
Changing from log to indices
$10^{0.9030} = a^3$
Antilog of **.**90 under 3 is 7998 with characteristic of 0 is decimal place of one step forward and after 7.
$7.998 = a^3$
$\sqrt[3]{7.998} = a$
thus $a = 1.9998$
≈ 2.0 B

2005/ 15

If $\log_{10}{}^{a} = \bar{2}.4472$, find a
A 2.800 B 0.280 C 0.028 D 0.0028

Solution

To get a value, we check the reverse of the log value of a to base 10 i.e antilog of $\bar{2}.4472$ implies .44 under 7 difference 2 then we move two steps backward

Antilog of $\bar{2}.4472$ = 2799
1
2800

Apply the place of decimal $\bar{2}$ to it
2800 = 0.0280 (C)

2014/45

If log 5.957 = 0.7750, find log $\sqrt[3]{0.0005957}$

A $\bar{4}.1986$ B $\bar{2}.9250$ C $\bar{1}.5917$ D $\bar{1}.2853$

Solution

No	log
0.0005957	$\bar{4}.7750 \div 3$ is expressed as: $\bar{6} + 2.7750 \div 3$
	$\bar{2}.9250$ (B)

2007/7a

With the aid of four – figure logarithm table evaluate $(0.004592)^{\frac{1}{3}}$

Solution

No	log
$(0.004592)^{\frac{1}{3}}$	$\bar{3}.6620 \div 3$
0.1663	$\bar{1}.2207$

2005/3a (Nov)

Use logarithm table to evaluate $\sqrt{\dfrac{0.0751}{0.00784}}$

Solution

No	log
0.0751	$\bar{2}.8756$ Subtraction
0.00784	$\bar{3}.8943$
$\left(\frac{0.0751}{0.00784}\right)^{\frac{1}{2}}$	$0.9813 \div 2$
3.095	0.4907

2005/10b Neco

Use logarithm table to evaluate $\dfrac{\sqrt[3]{0.5915}}{17.8^2}$

Solution

No	log
$\sqrt[3]{0.5915}$	$\bar{1}.7720 \div 3$ is express as: $\bar{3} \div 3 + 2.7720 \div 3 = \bar{1}.9240$
17.8^2	1.2504×2 2.5008
0.002650	$\bar{3}.4232$

2007/1 NABTEB

Use logarithm table to evaluate

$$\sqrt[4]{\frac{0.784^3 \times 23.67}{3.479}} \quad = \quad A^{1/4}$$

Solution

No	log	
0.784^3	$\bar{1}.8943 \times 3 = 2.6829 + (\bar{3})$	
	(say 3 × 0.8943 and 3 × $\bar{1}$)	$\bar{1}.6829$
23.67	1.3742	1.3742
		1.0571
3.479	0.5414	0.5414
		0.5157
$A^{1/4}$		0.5157 ÷ 4
1.346		0.1289

2011/8 Neco

Use four figure tables table to evaluate

$$\left(\frac{22.5^{\frac{1}{3}} \times 33.5}{\sqrt[5]{0.00523}}\right)^2 = A^2$$

Solution

No	log		
$22.5^{\frac{1}{3}}$	1.3522 ÷ 3	0.4507	
33.5	1.5250	1.5250	
Numerator		1.9757	1.9757
$(0.00523)^{\frac{1}{5}}$	$\bar{3}.7185 \div 5$ is expressed as:		
	$\bar{5} + 2.7185 \div 5$	$\bar{1}.5437$	$\bar{1}.5437$
A^2			2.4320×2
73110.0			4.8640

2008/6a NABTEB (Nov)

Use logarithm table to evaluate

$$\sqrt[3]{\frac{117.5 \times 2.56}{5.96 \times 39}} \quad = \quad A^{1/3}$$

Solution

No	log	
117.5	2.0701	
2.56	0.4082	
Numerator	2.4783	2.4783
5.96	0.7752	
39	1.5911	
Denominator	2.3663	2.3663
$A^{1/3}$		0.1120 ÷ 3
1.090		0.0373

2006/12 Neco (Nov)

Use logarithm table to evaluate

$$\sqrt{\frac{(0.2514)^2 \times 2.005}{(0.3927)^{\frac{1}{3}}}} \quad = \quad A^{1/2}$$

Solution

No	log		
$(0.2514)^2$	$\bar{1}.4004 \times 2$	$0.8008 + (\bar{2})$	
	(Say 2 × 0.4004 and 2 × $\bar{1}$)	$\bar{2}.8008$	
2.005	0.3021	0.3021	
Numerator		$\bar{1}.1029$	$\bar{1}.1029$
$(0.3927)^{\frac{1}{3}}$	$\bar{1}.5941 \div 3$ is expressed as:		
	$\bar{3} + 2.5941 \div 3$	$\bar{1}.8647$	
$\bar{1}.8647$			
Denominator			$\bar{1}.2382$
$A^{1/2}$	$\bar{1}.2382 \div 2$		
	expressed as $\bar{2} \div 2$ and $1.2382 \div 2$		
0.4160	$\bar{1}.6191$		

2009/9 Neco

Use logarithm table to evaluate $\dfrac{846.2^2 \times \sqrt{54.36 \times 10^{-3}}}{462.4^{\frac{1}{3}}}$

Solution

Note that: $\sqrt{54.36 \times 10^{-3}}$ *is the same as* $\sqrt{0.05436}$

No	log		
846.2^2	2.9275 × 2	5.8550	
$\sqrt{0.05436}$	$\bar{2}.7353 \div 2$	$\bar{1}.3677$	
Numerator		5.2227	5.2227
$462.4^{\frac{1}{3}}$	2.6650 ÷ 3		0.8883
21600.0			4.3344

2014/8 Neco Exercise 6.14

Use logarithm table to evaluate $\dfrac{\sqrt[3]{0.5915} \times (392.8)^3}{39.436 \times (12.03)^3}$

2014/6a Neco(Nov) Exercise 6.15

Use mathematical tables to evaluate $\dfrac{3.267}{0.483 \times 0.385}$

1978/11b Exercise 6.16

If a = 5.732, b = 0.2795 and c = 378.4, use logarithm tables to evaluate $a^2b^3 \div \sqrt{c}$ correct to 3 significant figures

2006/10a NABTEB (Nov) Exercise 6.17

Use logarithm table to evaluate

$$\sqrt{\frac{(3.415)^4 \times 28.91}{0.267}}$$, correct to 3 significant figures

CHAPTER SEVEN

Number bases & Surds

Number bases

Just as we use fuel gauge to measure petrol, baskets to measure garri; similarly we measure (count) numbers in bases. The common one is base 10 (denary/decimal system). Others are base 5 (quinary system), base 8 (octal system) etc

Conversion from base ten to others

This is done by repeated division and we take our result of the remainders from bottom to top

2010/3 Neco

Convert 29_{ten} to a number in base two

A 1011_{two} B 1101_{two} C 1111_{two}
D 10111_{two} E 11101_{two}

Solution

$$\begin{array}{c|l} 2 & 29 \\ \hline 2 & 14 + 1 \\ 2 & 7 + 0 \\ 2 & 3 + 1 \\ 2 & 1 + 1 \\ & 0 + 1 \end{array} \uparrow$$

$29_{ten} = 11101_{two}$ (E)

2011/9 Neco

Convert 59_{ten} to base two

A 1111011_{two} B 1110111_{two} C 111011_{two}
D 110111_{two} E 11110_{two}

Solution

$$\begin{array}{c|l} 2 & 59 \\ \hline 2 & 29 + 1 \\ 2 & 14 + 1 \\ 2 & 7 + 0 \\ 2 & 3 + 1 \\ 2 & 1 + 1 \\ & 0 + 1 \end{array} \uparrow$$

$59_{ten} = 111011_{two}$

2014/1 NABTEB

Express 12_{ten} in binary

A 1000 B 1001 C 1011 D 1100

Solution

Converting from base ten to 2, long division

$$\begin{array}{c|l} 2 & 12 \\ \hline 2 & 6 + 0 \\ 2 & 3 + 0 \\ 2 & 1 + 1 \\ & 0 + 1 \end{array} \uparrow$$

$12_{ten} = 1100_2$ (D)

2012/4 Exercise 7.1

Convert 35_{10} to a number in base 2

A 1011 B 10011 C 100011 D 11001

2005/1a Neco Exercise 7.2

Express 37_{10} in base two

Exercise 7.3

Express 15_{ten} in binary

A 1001 B 1110 C 1011 D 1111

Conversion of a given base to base 10

This is done by method of expansion in the powers of "**old**" base as shown below

2013/3 Neco

Convert 11011_{two} to denary

A 27_{ten} B 29_{ten} C 49_{ten} D 51_{ten} E 53_{ten}

Solution

$$\begin{aligned} 11011_{two} &= 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 \\ &= 16 + 8 + 0 + 2 + 1 \\ &= 27_{ten} \quad (\text{ A }) \end{aligned}$$

2010/1 NABTEB (Nov)

Convert 112_{three} to base 10

A 14 B13 C 12 D 10

Solution

Converting from other bases to base ten

$$\begin{aligned} 112_{three} &= 1 \times 3^2 + 1 \times 3^1 + 2 \times 3^0 \\ &= 9 + 3 + 2 \\ &= 14_{ten} \quad (\text{ A }) \end{aligned}$$

2010/26 NABTEB (Nov)

Express 556_{seven} as a denary number

A 245_{ten} B 286_{ten} C 450_{ten} D 550_{ten}

Solution

Converting from other bases to base ten (denary), expand in the powers of the old base

$$\begin{aligned} 556_{seven} &= 5 \times 7^2 + 5 \times 7^1 + 6 \times 7^0 \\ &= 5 \times 49 + 35 + 6 \\ &= 245 + 35 + 6 \\ &= 286_{ten} \end{aligned}$$

Conversion from base ten numbers plus decimal to others

2006/5 Neco

Express 12.625_{ten} in base two

A 101.101 B 101.110 C 1100.011
D 1100.101 E 1100.110

Solution

We solve in two parts

First, convert whole number 12_{ten} to base two

$$\begin{array}{c|l} 2 & 12 \\ \hline 2 & 6 + 0 \\ 2 & 3 + 0 \\ 2 & 1 + 1 \\ & 0 + 1 \end{array} \uparrow$$

$12_{ten} = 1100_{two}$

Convert fraction part $\cdot 625_{ten}$ to base two

Multiplication by 2 three times to show 3 places of decimal

$$\begin{array}{l} 0 \cdot 625 \times 2 \\ \hline 1 \cdot 25 \times 2 \Rightarrow 2 \times 0.25 \\ 0 \cdot 50 \times 2 \Rightarrow 2 \times 0.50 \\ 1 \cdot 0 \end{array} \downarrow$$

$0.625_{ten} = 101_{two}$

Thus $12.625_{ten} = 1100.101_{two}$ (D)

Example CTD2

Convert 87.65_{10} to a number in base 8

Solution

We solve in two parts

First convert whole number part 87_{ten} to base eight.

$$\begin{array}{c|l} 8 & 87 \\ \hline 8 & 10 + 7 \\ 8 & 1 + 2 \\ & 0 + 1 \end{array}\uparrow$$

$87_{ten} = 127_{eight}$

Next, we convert $0{\cdot}65_{10}$ to base eight

Multiplication by 8 two times to show 2 places of decimal

$$\begin{array}{l} 0 \cdot 65 \times 8 \\ \hline 5 \cdot 2 \times 8 \\ 1 \cdot 6 \end{array}\downarrow$$

$0.65_{ten} = 0{\cdot}51_{eight}$

Thus $87.65_{10} = 127.51_{\ eight}$

Example CTD3

Convert 0.54_{10} to a number in base three correct to 5 decimal places

Solution

We continue multiplying by 3 till we set five decimal places

$$\begin{array}{l} 0 \cdot 53 \times 3 \\ \hline 1 \cdot 59 \times 3 \Rightarrow 0{\cdot}59 \times 3 \\ 1 \cdot 77 \times 3 \Rightarrow 0{\cdot}77 \times 3 \\ 2 \cdot 31 \times 3 \Rightarrow 0{\cdot}31 \times 3 \\ 0 \cdot 93 \times 3 \Rightarrow 0{\cdot}93 \times 3 \\ 2 \cdot 79 \text{ stop} \end{array}\downarrow$$

$0{\cdot}54_{10} = 0{\cdot}11202_{three}$

2009/13 Neco (Nov)

Convert $\left(\frac{3}{8}\right)_{ten}$ to binary

A $11_{\ two}$ B 0.111_{two} C 0.11_{two}
D 0.011_{two} E 0.0011_{two}

Solution

Convert $\left(\frac{3}{8}\right)_{ten}$ to binary implies 0.375_{ten} to base 2

We multiply by 2 three times to show three places of decimal

$$\begin{array}{l} 0 \cdot 375 \times 2 \\ \hline 0 \cdot 75 \times 2 \Rightarrow 2 \times 0{\cdot}75 \\ 1 \cdot 50 \times 2 \Rightarrow 2 \times 0{\cdot}50 \\ 1 \cdot 0 \end{array}\downarrow$$

$0 \cdot 375_{ten} = 0{\cdot}011_{two}$ (D)

Exercise 7.4

Convert 0.37_{10} to base eight; leaves your answer in 3d.p

Exercise 7.5

Convert 127.35_{10} to base three

Exercise 7.6

Convert 0.41_{ten} to base two; leaves your answer in 3d.p

Exercise 7.7

Convert 144.41_{ten} to base four; leave your answer in 2d.p

Conversion of a given base number with decimal to base 10

2011/8 Neco

Express 101.01_{two} to denary

A 3.150 B 5.250 C 7.125 D 7.150 E 7.250

Solution

$101.01_{two} = 1\times2^2 + 0\times2^1 + 1\times2^0 + 0\times2^{-1} + 1\times2^{-2}$

$= 4 + 0 + 1 + 0 + \frac{1}{4}$

$= 5\frac{1}{4}$ ten i.e 5.25 (B)

1975/2a

Express the binary number 101.101 as a number in base ten

Solution

$101.101_{two} = 1\times2^2 + 0\times2^1 + 1\times2^0 + 1\times2^{-1} + 0\times2^{-2} + 1\times2^{-3}$

$= 4 + 0 + 1 + \frac{1}{2} + 0 + \frac{1}{8}$

$= 4 + 1 + 0.5 + 0 + 0.125$

$= 5.625_{ten}$

Example CBD3

Convert $11011.11_{\ 2}$ to a denary number

Solution

$11011.11_{\ 2} = 1\times2^4 + 1\times2^3 + 0\times2^2 + 1\times2^1 + 1\times2^0 + 1\times2^{-1} + 1\times2^{-2}$

$= 1\times16 + 1\times8 + 0 + 1\times2 + 1\times1 + 1\times\frac{1}{2} + 1\times\frac{1}{4}$

$= 16 + 8 + 0 + 2 + 1 + \frac{1}{2} + \frac{1}{4}$

$= 27 + \frac{2+1}{4} = 27\ \frac{3}{4}_{\ ten}$ or $27.75_{\ ten}$

Exercise 7.8

Convert 302.21_3 to base ten

Conversion from a non base ten to another non base ten number

2008/6

Convert 42_5 to a base three numeral

A 201_3 B 210_3 C 211_3 D 222_3

Solution

First, we convert 42_5 to base ten

$42_5 = 4 \times 5^1 + 2 \times 5^0$

$= 20 + 2$

$= 22_{\ ten}$

Next, we convert 22_{ten} to base 3

$$\begin{array}{c|l} 3 & 22 \\ \hline 3 & 7 + 1 \\ 3 & 2 + 1 \\ & 0 + 2 \end{array}\uparrow$$

$22_{\ ten} = 211_{\ 3}$ (C)

VTR – 11/7 NTI TCII

Express 132_{six} as a number in base FIVE

A.211 B.210 C.201 D.112 E.102

Solution

First; we convert to base ten

$132_{six} = 1 \times 6^2 + 3 \times 6^1 + 2 \times 6^0$

$= 6^2 + 3 \times 6 + 2 \times 1$

$= 36 + 18 + 2$

$= 56_{\ ten}$

2ndly; we convert 56_{ten} to base five

$$\begin{array}{r|l} 5 & 56 \\ \hline 5 & 11 + 1 \\ 5 & 2 + 1 \\ & 0 + 2 \end{array} \quad \text{i.e } 211_{five}$$

Thus $132_{six} = 211_{five}$ (A)

2003/3b Neco

Convert 223_{four} to a number in base three.

Solution

First; we convert to base ten

$223_{four} = 2 \times 4^2 + 2 \times 4^1 + 3 \times 4^0$

$= 2 \times 16 + 2 \times 4 + 3 \times 1$

$= 32 + 8 + 3$

$= 43_{ten}$

2ndly; we convert 43_{ten} to base three

$$\begin{array}{r|l} 3 & 43 \\ \hline 3 & 14 + 1 \\ 3 & 4 + 2 \\ 3 & 1 + 1 \\ & 0 + 1 \end{array} \quad \text{i.e } 1121_{three}$$

Thus, $223_{four} = 1121_{three}$

Exercise 7.9

Convert 241 in base 5 to base 8

A. 71_8 B. 107_8 C. 176_8 D. 241_8

Addition in bases

2014/22 Neco

Find the missing numbers, if the following addition is in base six

$$\begin{array}{ccc} 3 & 2 & 4 \\ * & * & * \\ \hline 10 & 0 & 1 \\ \hline \end{array}$$

A 534 B 432 C 323 D 234 E 233

Solution

In actual sense, the presentation is:

$$\begin{array}{rcccl} & 1 & 0 & 0 & 1_{six} \\ - & & 3 & 2 & 4_{six} \\ \hline & & 2 & 3 & 3_{six} \\ \hline \end{array} \quad (E)$$

Analysis

(Note any borrowed number is six)

After the carry – carry operations the new look is:

$$\begin{array}{cccc} 0 & 5 & 5 & 7_{six} \\ & 3 & 2 & 4_{six} \\ \hline & 2 & 3 & 3_{six} \\ \hline \end{array}$$

2014/31 Neco

Express $213_{four} + 202_{four}$ in denary

A 72_{ten} B 73_{ten} C 78_{ten} D 102_{ten} E 415_{ten}

Solution

First, we do the addition in base four

$$\begin{array}{rccl} & 2 & 1 & 3 \\ + & 2 & 0 & 2 \\ \hline & 10 & 2 & 1_{four} \\ \hline \end{array}$$

Next, we convert 1021_{four} to base ten

$1021_{four} = 1 \times 4^3 + 0 \times 4^2 + 2 \times 4^1 + 1 \times 4^0$

$= 64 + 0 + 8 + 1$

$= 73_{ten}$ (B)

2009/49 (Nov)

Find the missing number in the following addition of numbers in base two

$$\begin{array}{rcccccl} & & 1 & 0 & 1 & 1 & 1_{two} \\ + & & * & * & * & * & * \\ \hline & 1 & 0 & 0 & 0 & 0 & 0_{two} \end{array}$$

A 1001_{two} B 1100_{two} C 1101_{two} D 1011_{two}

Solution

By simple algebra $2 + x = 5$

Implies $x = 5 – 2$

So in actual sense, we are subtracting as:

$$\begin{array}{rcccccl} & 1 & 0 & 0 & 0 & 0 & 0_{two} \\ - & & 1 & 0 & 1 & 1 & 1_{two} \\ \hline & 0 & 0 & 1 & 0 & 0 & 1_{two} \\ \hline \end{array} \quad (A)$$

Explanation

The carry – carry operation results to

$$\begin{array}{rcccccl} & 0 & 1 & 1 & 1 & 1 & 2_{two} \\ - & & 1 & 0 & 1 & 1 & 1_{two} \\ \hline & 0 & 0 & 1 & 0 & 0 & 1_{two} \\ \hline \end{array}$$

2013/3

In what number base is the addition

465 + 24 + 225 = 1050?

A ten B nine C eight D seven

Solution

$$\begin{array}{cccc} & 4 & 6 & 5 \\ & & 2 & 4 \\ & 2 & 2 & 5 \\ \hline 1 & 0 & 5 & 0_{seven} \\ \hline \end{array} \quad (D)$$

Explanation

5 + 4 +5 is 14 i.e 20

Write 0 carry 2

10 + 2 is 12 i.e 15 in base seven

Write 5 carry1

4 + 2 is 6 plus 1 i.e 7 which is 10 in base seven written out in full

2009/4 NABTEB (Nov)

If 345 + 72 + 124 = 552, what is the base used?

A 9 B 8 C 7 D 6

Solution

First, we arrange the numbers in the old fashion arithmetic style

$$\begin{array}{rccc} & 3 & 4 & 5 \\ & & 7 & 2 \\ + & 1 & 2 & 4 \\ \hline & 5 & 5 & 2 \\ \hline \end{array}$$

What base was 5 + 2 + 4 = 11 in base ten becomes 2 or 12

Base 9 is suspect i.e write 2 carry 1

Confirm from next line

4 + (7 + 2) plus carried 1 in base 9 is 15 write 5; carry 1

Last line follows: 3 + 1 plus carried 1 is 5

2005/3 Exercise 7.10

In what base is 234 + 141 = 405?

A 5 B 6 C 7 D 8

2014/3 (Nov) Exercise 7.10b

Find the sum of 303_5 and 104_5

A 412_5 B 402_5 C 244_5 D 144_5

2015/2 Exercise 7.10c

The sum of 11011_2, 11111_2 and 1000_2 is $10m10n0_2$

Find the values of m and n

A m = 0, n = 0 B m= 1, n = 0 C m= 0, n = 1 D m= 1, n = 1

Subtraction in bases

2004/1a NABTEB

(ii) Evaluate: $11013_6 - 2534_6$, in base 6

Solution

$$\begin{array}{rrrrrl} & 1 & 1 & 0 & 1 & 3_{six} \\ - & & 2 & 5 & 3 & 4_{six} \\ \hline & & 4 & 0 & 3 & 5_{six} \\ \hline \end{array}$$

Explanation

The carry – carry operations result is

$$\begin{array}{rrrrrl} & 0 & 6 & 5 & 6 & 9 \\ - & & 2 & 5 & 3 & 4 \\ \hline & & 4 & 0 & 3 & 5_{six} \\ \hline \end{array}$$

2003/3b (i) NABTEB

Given P = 242_{five} and Q = 14_{five}

Calculate (i) P – Q

Solution

P – Q is the same as

$$\begin{array}{rrrl} & 2 & 4 & 2_{five} \\ - & & 1 & 4_{five} \\ \hline & 2 & 2 & 3_{five} \\ \hline \end{array}$$

Explanation

The carried or borrow operation result is

$$\begin{array}{rrrl} & 2 & 3 & 7_{five} \\ - & & 1 & 4_{five} \\ \hline & 2 & 2 & 3_{five} \\ \hline \end{array}$$

2006/2

Simplify: $11011_{two} - 1101_{two}$

A 101000_{two} B 1100_{two} C 1110_{two} D 1011_{two}

Solution

$$\begin{array}{rrrrrl} & 1 & 1 & 0 & 1 & 1_{two} \\ - & & 1 & 1 & 0 & 1_{two} \\ \hline & & 1 & 1 & 1 & 0_{two} \\ \hline \end{array} \quad (C)$$

Explanation

The carried – carry operation result is

$$\begin{array}{rrrrrl} & 0 & 2 & 2 & 1 & 1 \\ - & & 1 & 1 & 0 & 1 \\ \hline & & 1 & 1 & 1 & 0_{two} \\ \hline \end{array}$$

2014/34 Neco (Nov)

Subtract 2255_{six} from 20035_{six}

A 1334_{six} B 13340_{six} C 22334_{six}

D 23340_{six} E 31340_{six}

Solution

$$\begin{array}{rrrrrl} & 2 & 0 & 0 & 3 & 5_{six} \\ - & & 2 & 2 & 5 & 5_{six} \\ \hline & 1 & 3 & 3 & 4 & 0_{six} \\ \hline \end{array}$$

Explanation

5 – 5 = 0

3 – 5 cannot,

Carry one from zero, can't, move to 2, carry 1 (6) drop at zero. Carry 1 there and leave 5 behind drop 1 at zero to get six. Carry 1 (6) from there and leave 5 behind. Thus the new look is

$$\begin{array}{rrrrrl} & 1 & 5 & 5 & 9 & 5_{six} \\ - & & 2 & 2 & 5 & 5_{six} \\ \hline & 1 & 3 & 3 & 4 & 0_{six} \\ \end{array} \quad (B)$$

Note that this analysis step is for explanation only and not part of the working

2008/2 NABTEB (Nov)

Find the difference between 42_{nine} and 111_{five}

A 7 B 9 C 10 D 12

Solution

Since they are not in the same base, we convert both to base ten to reflect the options given, otherwise we would have converted only 42_{nine} to base five and subtract.

$$\begin{aligned} 42_{nine} &= 4 \times 9^1 + 2 \times 9^0 \\ &= 36 + 2 \\ &= 38_{ten} \end{aligned}$$

$$\begin{aligned} 111_{five} &= 1 \times 5^2 + 1 \times 5^1 + 1 \times 5^0 \\ &= 25 + 5 + 1 \\ &= 31_{ten} \end{aligned}$$

Thus their difference: 38 – 31 = 7 (A)

2008/1 NABTEB (Nov)

The difference between 10010_{two} and 1101_{two} is

A 11 B 101 C 110 D 111

Solution

Since they are in the same base, we can subtract directly

$$\begin{array}{rrrrr} 1 & 0 & 0 & 1 & 0 \\ & 1 & 1 & 0 & 1 \\ \hline \varnothing & \varnothing & 1 & 0 & 1 \\ \hline \end{array} \quad (B)$$

Analysis

0 – 1 cannot, carry 1 to zero makes it 2 & 2 – 1 = 1.

2^{nd} column becomes zero since we carried 1 from there 0 – 0 = 0

3rd column: 0 – 1 cannot, carry 1 from forth column; none there. Carry 1 from the 5^{th} column place at 4^{th} column 2 carry 1 from there to 3^{rd} column i.e 2,

2 – 1 = 1.

At 4^{th} column: 1 left there – 1 = 0

At 5^{th} column: 0 left there

2002/19 (Nov) Exercise 7.11

$$\text{If} \quad \begin{array}{rrrrl} & 5 & 1 & 2 & m \\ - & 3 & 5 & 4 & m \\ \hline & 1 & 2 & 5 & m\ , \\ \hline \end{array}$$

find the number base

A six B seven C eight E nine

2010/43 Exercise 7.12

The subtraction below is in base seven.

Find the missing number

$$\begin{array}{rrrrr} & 5 & 1 & 6 & 2 \\ - & 2 & 6 & 4 & 4 \\ \hline & 2 & * & 1 & 5 \\ \hline \end{array}$$

A 2 B 3 C 4 D 5

Multiplication in bases

1975/2c

Calculate $(212)_{three} \times (201)_{three}$ giving your answer as a number in base three

Solution

$$\begin{array}{rrrrrl} & & 2 & 1 & 2 & \\ & \times & 2 & 0 & 1 & \\ \hline & & 2 & 1 & 2 & \\ & 0 & 0 & 0 & * & \\ 1 & 2 & 0 & 1 & * & \\ \hline 1 & 2 & 1 & 0 & 1 & 2_{three} \\ \hline \end{array}$$

2007/1

Evaluate $(111_{two})^2$ and leave your answer in base 2

A 111001_{two} B 110001_{two} C 101001_{two} D 10010_{two}

Solution

$(111_{two})^2$ is same as:

$$\begin{array}{rrrrrr} & & & 1 & 1 & 1_{two} \\ & & \times & 1 & 1 & 1_{two} \\ \hline & & & 1 & 1 & 1 \\ & & 1 & 1 & 1 & * \\ & 1 & 1 & 1 & * & \\ \hline 1 & 1 & 0 & 0 & 0 & 1_{two} \\ \hline \end{array}$$

Note: 2 is 10, write zero carry one

4 is 20 write zero carry two

3 is 11 written out in full; it is the last operation

2005/1 NABTEB

Find the value of 1111×1001 in base two

A 1100 B 10101 C 101011 D 10000111

Solution

$$\begin{array}{rrrrrrrr} & & & & 1 & 1 & 1 & 1 \\ & & & \times & 1 & 0 & 0 & 1 \\ \hline & & & & 1 & 1 & 1 & 1 \\ & & & 0 & 0 & 0 & 0 & * \\ & & 0 & 0 & 0 & 0 & * & \\ & 1 & 1 & 1 & 1 & * & & \\ \hline 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1_{two} \\ \hline \end{array}$$ (D)

2003/3b (ii) NABTEB

Given $P = 242_{five}$ and $Q = 14_{five}$

Calculate (ii) PQ

Solution

PQ is

$$\begin{array}{rrrrr} & & 2 & 4 & 2_{five} \\ & \times & & 1 & 4_{five} \\ \hline & 2 & 1 & 2 & 3_{five} \\ & 2 & 4 & 2 & * \\ \hline 1 & 0 & 0 & 4 & 3_{five} \\ \hline \end{array}$$

Note: 11 is 21 in base five

Explanation

$4 \times 2 = 8$ i.e 13 in base five

Write 3 carry 1

4×4 is 16 plus carried 1 is 17 i.e 32,

Write 2 carry 3

2004 /1 NABTEB

Evaluate $(203_{four})^2$ in base four

A 2030 B 10012 C 12002 D 103021

Solution

$$\begin{array}{rrrrrr} & & & 2 & 0 & 3_{four} \\ & & \times & 2 & 0 & 3_{four} \\ \hline & & 1 & 2 & 2 & 1 \\ & & 0 & 0 & 0 & * \\ 1 & 0 & 1 & 2 & * & \\ \hline 1 & 0 & 3 & 0 & 2 & 1_{four} \\ \hline \end{array}$$ (D)

Explanation

$3 \times 3 = 9$ i.e 21 in base four (two four + 1)

Write 1 carry 2

$3 \times 0 = 0$ + carried 2 i.e 2

$3 \times 2 = 6$ i.e 12 written in full

Here 2×2 is 4 i.e. 10

Also $2 + 2$ is 4 i.e. 10

2004/1a NABTEB

(i) Evaluate: $213_6 \times 24_6$

Solution

$$\begin{array}{rrrrr} & & 2 & 1 & 3_{six} \\ & \times & & 2 & 4_{six} \\ \hline & 1 & 3 & 0 & 0 \\ & 4 & 3 & 0 & * \\ \hline 1 & 0 & 0 & 0 & 0_{six} \\ \hline \end{array}$$

Explanation

$4 \times 3 = 12$ i.e 20 in base six

Write 0 carry 2

$4 \times 1 = 4$ plus 2 i.e 6, which is 10 in base six

Write 0 carried 1

$4 \times 2 = 8$ i.e 12 plus 1 carried 13 write in full

$2 \times 3 = 6$ i.e 10 in base six

Write 0, carry 1

$3 + 3 = 6$ i.e. 10 in base six.

2007/2a NABTEB

Find the product of 324_6 and 15_6

Solution

Since they are in the same base we multiply directly bearing in mind that we are working in base 6

$$\begin{array}{rrrr} & 3 & 2 & 4 \\ & \times & 1 & 5 \\ \hline 2 & 5 & 1 & 2 \\ +3 & 2 & 4 & * \\ \hline 10 & 1 & 5 & 2 \\ \hline \end{array}$$

Explanation

5×4 is 32, write 2, carry 3

(it is $5 \times 4 = 20$, how many 6 in 20 is 3, balance 2)

5×2 is 10 + 3 carried is 13 i.e 21 in base six language write 1, carry 2

5×3 is 15 + 2 carried is 17 i.e 25 in base six language write it in full

$1 \times 4 = 4$ in base six

$1 \times 2 = 2$ in base six

$1 \times 3 = 3$ in base six

Note: $5 + 2 = 7$ i.e 11 in base six;

$2 + 3 + 1 = 6$ i.e 10 base six

2014/42 NABTEB (Nov)

Find $(101_2)^2$, expressing the answer in base 2

A 11001 B 10010 C 11101 D 10101

Solution

$(101_2)^2 = 101_2 \times 101_2$

First, we convert 101_2 to base ten

$$101_2 = 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$
$$= 4 + 0 + 1$$
$$= 5_{\text{ten}}$$

But $(101_2)^2 = 5 \times 5 = 25_{\text{ten}}$

Converting 25_{ten} to base 2, by division

$$\begin{array}{r|l} 2 & 25 \\ \hline 2 & 12 + 1 \\ 2 & 6 + 0 \\ 2 & 3 + 0 \\ 2 & 1 + 1 \\ & 0 + 1 \end{array}\uparrow$$

$(101_2)^2 = 11001_{\text{two}}$ A

2009/35 NABTEB (Nov) Exercise 7.13

Evaluate $1001_{\text{two}} \times 101_{\text{two}}$

A 100111_{two} B 101101_{two} C 11001_{two}
D 111001_{two}

Division in bases

1998/35 (Nov)

Evaluate $11110_{\text{two}} \div 110_{\text{two}}$

A 11_{two} B 100_{two} C 101_{two}
D 110_{two} E 111_{two}

Solution

First, convert all numbers to base ten

$$11110_{\text{two}} = 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$$
$$= 16 + 8 + 4 + 2 + 0$$
$$= 30_{\text{ten}}$$

$$110_{\text{two}} = 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$$
$$= 4 + 2 + 0$$
$$= 6_{\text{ten}}$$

Performing the division: $\frac{30}{6} = 5$

Next, we convert 5_{ten} to base 2

$$\begin{array}{r|l} 2 & 5 \\ \hline 2 & 2 + 1 \\ 2 & 1 + 0 \\ & 0 + 1 \end{array}\uparrow$$

$5_{\text{ten}} = 101_{\text{two}}$ (C)

2010/15 Neco

Evaluate $(110100)_{\text{two}} \div (100)_{\text{two}}$

A 10011_{two} B 1010_{two} C 1101_{two}
D 1100_{two} E 101_{two}

Solution

First, convert all numbers to base ten

$$110100_{\text{two}} = 1\times2^5+1\times2^4+0\times2^3+1\times2^2+0\times2^1+0\times2^0$$
$$= 32 + 16 + 0 + 4 + 0 + 0$$
$$= 52_{\text{ten}}$$

$$100_{\text{two}} = 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0$$
$$= 4 + 0 + 0$$
$$= 4_{\text{ten}}$$

Performing the division: $\frac{52}{4} = 13$

Next, we convert 13_{ten} to base two

$$\begin{array}{r|l} 2 & 13 \\ \hline 2 & 6 + 1 \\ 2 & 3 + 0 \\ 2 & 1 + 1 \\ & 0 + 1 \end{array}\uparrow$$

$13_{\text{ten}} = 1101_{\text{two}}$ (C)

Problems of bases in equation form

1975/2b

If $(23)_n = (1111)_{\text{two}}$, find n

Solution

We convert both sides to base ten

$23_n = 1111_{\text{two}}$ *becomes*

$$2 \times n^1 + 3 \times n^0 = 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$
$$2n + 3 = 8 + 4 + 2 + 1$$
$$2n + 3 = 15$$
$$2n = 15 - 3$$
$$2n = 12$$
$$n = 6$$

2005/2b NABTEB(Nov)

If $34_5 = 23_x$, find x

Solution

We convert both sides to base ten

$34_5 = 23_x$ becomes

$$3 \times 5^1 + 4 \times 5^0 = 2 \times x^1 + 3 \times x^0$$
$$15 + 4 = 2x + 3$$
$$19 - 3 = 2x$$
$$16 = 2x$$

Thus $x = 8$

2005/1 NABTEB (Nov)

Find x if $21_x = 7_{10}$

A 2 B 3 C 5 D 11

Solution

We convert the LHS to base ten

$21_x = 7_{10}$ becomes

$$2 \times x^1 + 1 \times x^0 = 7$$
$$2x + 1 = 7$$
$$2x = 7 - 1$$
$$2x = 6$$
$$x = 3 \quad (\text{ B })$$

2014/5a Neco

If $244_n = 1022_{four}$, find n

Solution

We convert both sides to base ten

$$2\times n^2 + 4\times n^1 + 4\times n^0 = 1\times 4^3 + 0\times 4^2 + 2\times 4^1 + 2\times 4^0$$
$$2n^2 + 4n + 4 = 64 + 0 + 8 + 2$$
$$2n^2 + 4n + 4 = 74$$

Divide through by 2

$$n^2 + 2n + 2 = 37$$
$$n^2 + 2n - 35 = 0$$

Factorizing the quadratic equation

$$n^2 + 7n - 5n - 35 = 0$$
$$n(n + 7) - 5(n + 7) = 0$$
$$(n - 5)(n + 7) = 0$$
$$n - 5 = 0 \text{ or } n + 7 = 0$$
$$n = 5 \text{ or } -7$$

We accept n = 5 since we are dealing with base.

2006/3 Neco (Nov)

Find the value of x, if $10001_{two} = 101_x$

A x = 1 B x = 2 C x = 3 D x = 4 E x = 5

Solution

We convert both sides to base ten

$10001_{two} = 101_x$ becomes

$$1\times 2^4 + 0\times 2^3 + 0\times 2^2 + 0\times 2^1 + 1\times 2^0 = 1\times x^2 + 0\times x^1 + 1\times x^0$$
$$16 + 0 + 0 + 0 + 1 = x^2 + 0 + 1$$
$$16 + 1 = x^2 + 1$$
$$16 = x^2$$

It follows that $x = \pm\sqrt{16}$

$= \pm 4$

We take x = + 4 (D)

2006/2b Neco

Find n, if five times 43_n is equal to 311_n

Solution

$$5(43_n) = 311_n$$

We convert both sides to base ten

$$5(4 \times n^1 + 3 \times n^0) = 3 \times n^2 + 1 \times n^1 + 1 \times n^0$$
$$5(4n + 3) = 3n^2 + n + 1$$
$$20n + 15 = 3n^2 + n + 1$$
$$3n^2 + n - 20n + 1 - 15 = 0$$
$$3n^2 - 19n - 14 = 0$$

Factorizing the quadratic equation

$$3n^2 - 21n + 2n - 14 = 0$$
$$3n(n - 7) + 2(n - 7) = 0$$
$$(3n + 2)(n - 7) = 0$$
$$3n + 2 = 0 \text{ or } n - 7 = 0$$
$$n = 7 \text{ accepted}$$

2014/2

If $23_x = 32_5$, find the value of x

A 7 B 6 C 5 D 4

Solution

We convert both sides to base ten

$23_x = 32_5$ becomes

$$2 \times x^1 + 3 \times x^0 = 3 \times 5^1 + 2 \times 5^0$$
$$2x + 3 = 15 + 2$$
$$2x = 17 - 3$$
$$2x = 14$$
$$x = 7 \quad (A)$$

2014/2b

If $124_n = 232_{five}$, find n

Solution

We convert both sides to base ten

$124_n = 232_{five}$ becomes

$$1 \times n^2 + 2 \times n^1 + 4 \times n^0 = 2 \times 5^2 + 3 \times 5^1 + 2 \times 5^0$$
$$n^2 + 2n + 4 = 50 + 15 + 2$$
$$n^2 + 2n + 4 - 67 = 0$$
$$n^2 + 2n - 63 = 0$$

Factorizing the quadratic equation

$$n^2 + 9n - 7n - 63 = 0$$
$$n(n + 9) - 7(n + 9) = 0$$
$$(n - 7)(n + 9) = 0$$
$$n - 7 = 0 \text{ or } n + 9 = 0$$
$$n = 7 \text{ accepted}$$

2014 /9a (Nov)

Find the value of n, if

$$44_n + 55_n = 121_n$$

Solution

$(44_n + 55_n)$ operation cannot be done since the base n is unknown

We convert both sides to base ten.

$44_n + 55_n = 121_n$ becomes

$$4\times n^1 + 4\times n^0 + 5\times n^1 + 5\times n^0 = 1\times n^2 + 2\times n^1 + 1\times n^0$$
$$4n + 4 + 5n + 5 = n^2 + 2n + 1$$
$$9n + 9 = n^2 + 2n + 1$$
$$n^2 + 2n - 9n + 1 - 9 = 0$$
$$n^2 - 7n - 8 = 0$$

Factorizing the quadratic equation

$$n^2 - 8n + n - 8 = 0$$
$$n(n - 8) + 1(n - 8) = 0$$
$$(n - 8)(n + 1) = 0$$
$$n - 8 = 0 \text{ or } n + 1 = 0$$
$$n = 8 \text{ accepted}$$

1978/1b

If $52_x - 24_x = 25_x$ calculate the value of x

Solution

$$52_x - 24_x = 25_x$$

$(52_x - 24_x)$ operation cannot be done since the base x is unknown

We convert both sides to base ten.

$52_x - 24_x = 25_x$ becomes

$$5 \times x^1 + 2 \times x^0 - (2 \times x^1 + 4 \times x^0) = 2 \times x^1 + 5 \times x^0$$
$$5x + 2 - (2x + 4) = 2x + 5$$
$$5x + 2 - 2x - 4 = 2x + 5$$
$$5x - 2x - 2x = 5 + 4 - 2$$
$$x = 7$$

2014/2b NABTEB (Nov) Exercise 7.14

Find the value of n in the equation $173_n = 81$

2008/1b NABTEB (Nov) Exercise 7.15

If $(23)_x = (1111)_{two}$, find x

2011/38 Exercise 7.16

Given that $124_x = 7(14_x)$, find the value of x

A 12 B 11 C 9 D 8

2006/5 (Nov)

If $243X_5 = 366$, find x

A 0 B 1 C 2 D 3

Solution

Convert the LHS to base ten

$243X_5 = 366$ becomes

$2 \times 5^3 + 4 \times 5^2 + 3 \times 5^1 + x \times 5^0 = 366$

$250 + 100 + 15 + x = 366$

$365 + x = 366$

$x = 366 - 365$

$x = 1$ (B)

1999/6 (Nov)

Given that $X = 111101_{two}$, find x

A 29 B 61 C 62 D 63

Solution

We convert the RHS to base ten

$111101_{two} = 1\times2^5 + 1\times2^4 + 1\times2^3 + 1\times2^2 + 0\times2^1 + 1\times2^0$

$= 32 + 16 + 8 + 4 + 0 + 1$

$= 61$ (B)

2007/5a Neco

If $(y_{two})^2 = 2211_{three} - 220_{four}$, find the value of y.

Solution

$(y_{two})^2 = 2211_{three} - 220_{four}$,

The RHS cannot be done since they are of different bases

Convert both sides to base ten.

$(y_{two})^2 = 2211_{three} - 220_{four}$ becomes

$(y\times2^0)^2 = 2\times3^3+2\times3^2+1\times3^1+1\times3^0 - (2\times4^2+2\times4^1+0\times4^0)$

$(y\times1)^2 = 54 + 18 + 3 + 1 - (32 + 8 + 0)$

$y^2 = 76 - 40$

$y^2 = 36$

$y^2 = 6^2$

$y = 6_{ten}$

We may convert 6_{ten} to base 2

$$\begin{array}{c|l} 2 & 6 \\ \hline 2 & 3 + 0 \\ 2 & 1 + 1 \\ & 0 + 1 \end{array}\uparrow$$

$6_{ten} = 110_{two}$

1979/2b

Solve for x in the following base two equation.

Leaving your answer in base two

$$10x - \frac{11(1+x)}{10} = 101$$

Solution

First, we convert all in terms from base two to base ten

$10x_{two} = [1 \times 2^1 + 0 \times 2^0]x = 2x_{ten}$

$11_{two} = 1 \times 2^1 + 1 \times 2^0 = 3_{ten}$

$10_{two} = 1 \times 2^1 + 0 \times 2^0 = 2_{ten}$

$101_{two} = 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 5_{ten}$

$$10x - \frac{11(1+x)}{10} = 101 \quad \textit{becomes} \quad 2x - \frac{3(1+x)}{2} = 5$$

Multiply the resultant equation through by 2 to clear fraction

$4x - 3(1 + x) = 10$

$4x - 3 - 3x = 10$

$x = 10 + 3$ i.e $x = 13$

Next, we convert 13_{ten} to base two

$$\begin{array}{c|l} 2 & 13 \\ \hline 2 & 6 + 1 \\ 2 & 3 + 0 \\ 2 & 1 + 1 \\ & 0 + 1 \end{array}\uparrow$$

$13_{ten} = 1101_{two}$

2008/1b Exercise 7.17

If all number in the equation $\frac{y}{y+101} = \frac{11}{10010}$ are in base two, solve for y

Problems of bases in equation form that are *simultaneous linear equations* in nature

2006/2 NABTEB (Nov)

Find the value of x and y in the following equations:

$32_x + 51_y = 10_{10}$

$23_x + 42_y = 7_{10}$

Solution

First, we convert the LHS of both equations to base 10, since their RHS is in base 10 already

$32_x + 51_y = 10_{10}$ becomes

$[3 \times x^1 + 2 \times x^0] + [5 \times y^1 + 1 \times y^0] = 10$

$3x + 2 + 5y + 1 = 10$

$3x + 5y = 7$ ------- (1)

Also $23_x + 42_y = 7_{10}$ becomes

$[2 \times x^1 + 3 \times x^0] + [4 \times y^1 + 2 \times y^0] = 7$

$2x + 3 + 4y + 2 = 7$

$2x + 4y = 7 - 5$

$2x + 4y = 2$ ------ (2)

Solving the resulting simultaneous linear equations; from 2

$2x = 2 - 4y$

$x = 1 - 2y$

Substitute this into (1)

$3x + 5y = 7$ becomes

$3(1 - 2y) + 5y = 7$

$3 - 6y + 5y = 7$

$3 - y = 7$

$3 - 7 = y$

$-4 = y$

Substitute y value into (2)

$2x + 4x = 2$ becomes

$2x + 4(-4) = 2$

$2x - 16 = 2$

$2x = 18$

$x = 9$

2009/6b Neco

Solve the simultaneous equations:

$11_x - 10_y = 101_{two}$ ------ (1)

$10_x + 11_y = 1001_{two}$ ----- (2)

Solution

First, we convert both sides of the two equations to base 10

$11x - 10y = 101_{two}$ becomes

$[1 \times x^1 + 1 \times x^0] - [1 \times y^1 + 0 \times y^0] =$

$1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$

$x + 1 - y = 5$

$x - y = 5 - 1$

$x - y = 4$ ----- (1)

Also $10x + 11y = 1001$ becomes

$[1 \times x^1 + 0 \times x^0] + [1 \times y^1 + 1 \times y^0] =$

$1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$

$x + 0 + y + 1 = 8 + 0 + 0 + 1$

$x + y + 1 = 9$

$x + y = 8$ ----- (2)

Solving the resulting simultaneous linear equations

$$\begin{array}{r} x - y = 4 \\ +(\ x + y = 8\ ) \\ \hline 2x \quad = 12 \\ x = 6 \end{array}$$

Substitute $x = 6$ into (1)

$x - y = 4$ becomes

$6 - y = 4$

$6 - 4 = y$ thus $y = 2$

2014/10b Neco (Nov)

Find x and y in the simultaneous equation:

$35_x + 50_y = 58_{10}$

$51_x + 34_y = 56_{10}$

Solution

First, we convert the LHS of both equations to base 10, since their RHS is in base 10 already

$35x + 50y = 58_{10}$ becomes

$[3 \times x^1 + 5 \times x^0] + [5 \times y^1 + 0 \times y^0] = 58$

$3x + 5 + 5y + 0 = 58$

$3x + 5y = 58 - 5$

$3x + 5y = 53$ ------- (1)

Also $51x + 34y = 56_{10}$ becomes

$[5 \times x^1 + 1 \times x^0] + [3 \times y^1 + 4 \times y^0] = 56$

$5x + 1 + 3y + 4 = 56$

$5x + 3y = 56 - 5$

$5x + 3y = 51$ ------ (2)

Solving the resulting simultaneous linear equations multiply (1) × 5 and (2) × 3 and subtract

$$\begin{array}{r} 15x + 25y = 265 \\ -(\ 15x + 9y = 153\ ) \\ \hline 16y = 112 \\ y = 7 \end{array}$$

Substitute this into (1)

$3x + 5y = 53$ becomes

$3x + 5(7) = 53$

$3x + 35 = 53$

$3x = 53 - 35$

$3x = 18$

$x = {}^{18}/_3$

$x = 6$

2005/10a Neco

The following simultaneous equations are given in base two:

$11_x + 10_y = 1100$

$101_x - y = 111$

Find the value of x and y, leaving your answers in base two

Solution

First, we convert both sides of the two equations to base 10

$11_x + 10_y = 1100$ becomes

$[1 \times x^1 + 1 \times x^0] + [1 \times y^1 + 0 \times y^0] =$

$1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0$

$x + 1 + y + 0 = 8 + 4 + 0 + 0$

$x + y = 11$ ----- (1)

Also $101x - y = 111$ becomes

$[1 \times x^2 + 0 \times x^1 + 1 \times x^0] - y = 1\times 2^2 + 1\times 2^1 + 1\times 2^0$

$x^2 + 0 + 1 - y = 4 + 2 + 1$

$x^2 - y = 7 - 1$

$x^2 - y = 6$ ------ (2)

From (1) $y = 11 - x$

Substitute it into (2)

$x^2 - (11 - x) = 6$

$x^2 - 11 + x = 6$

i.e $x^2 + x - 17 = 0$

Readers to complete the solving

Exercise 7.18

Find the number base x and y in the simultaneous equations

$25_x - 23_y = 6_{10}$

$34_x + 32_y = 36_{10}$

2014/10b Neco (Nov) Exercise 7.19

Find x and y in the simultaneous equations:

$35_x + 50_y = 58_{10}$

$51_x + 34_y = 56_{10}$

Miscellaneous cases

2006/1 NABTEB (Nov)

Which of the following is the greatest?

A 27_{nine} B 65_{seven} C 121_{eight} D 431_{five}

Solution

First, we convert all the options to base ten

$27_{nine} = 2 \times 9^1 + 7 \times 9^0$

$= 18 + 7$

$= 25_{ten}$

$65_{seven} = 6 \times 7^1 + 5 \times 7^0$

$= 42 + 5$

$= 47_{ten}$

$121_{eight} = 1 \times 8^2 + 2 \times 8^1 + 1 \times 8^0$

$= 64 + 16 + 1$

$= 81_{ten}$

$431_{five} = 4 \times 5^2 + 3 \times 5^1 + 1 \times 5^0$

$= 100 + 15 + 1$

$= 116_{ten}$

The greatest is 431_{five} (D)

2005/4 (Nov)

Arrange 39_{10}; 67_8; 201_3 in ascending order of magnitude

A 39_{10}; 67_8; 201_3 B 67_8; 39_{10}; 201_3

C 39_{10}; 201_3; 67_8 D 201_3; 39_{10}; 67_8;

Solution

First, we convert all numbers to base ten

39_{10} is already in base ten as 39_{ten}

$$67_8 = 6 \times 8^1 + 7 \times 8^0$$
$$= 48 + 7 \text{ i.e } 55_{ten}$$

$$201_3 = 2 \times 3^2 + 0 \times 3^1 + 1 \times 3^0$$
$$= 18 + 0 + 1 \text{ i.e } 19_{ten}$$

Ascending order of magnitude: 19, 39, 55

201_3; 39_{10}; 67_8; (D)

1999/5 (Nov)

The sum of three numbers in base two is 11101. If the first two numbers are 1011 and 1101, find the third number.

A 11_{two} B 101_{two} C 1011_{two} D 1111_{two}

Solution

We are simply asked to solve 11101 – (1011 + 1101)

First,
$$\begin{array}{rccccc} & & 1 & 0 & 1 & 1 \\ + & & 1 & 1 & 0 & 1 \\ \hline & 1 & 1 & 0 & 0 & 0_{two} \\ \hline \end{array}$$

Next,
$$\begin{array}{rccccc} & 1 & 1 & 1 & 0 & 1 \\ - & 1 & 1 & 0 & 0 & 0 \\ \hline & 0 & 0 & 1 & 0 & 1_{two} \\ \hline \end{array}$$

2009/6a (Nov)

By how much is 11000_2 greater than or less than $111_2 \times 11_2$? (Leaves your answer in base 2)

Solution

First, we multiply $111_2 \times 11_2$
$$\begin{array}{rccccc} & & & 1 & 1 & 1_2 \\ \times & & & & 1 & 1_2 \\ \hline & & & 1 & 1 & 1 \\ & & 1 & 1 & 1 & * \\ \hline & 1 & 0 & 1 & 0 & 1_{two} \\ \hline \end{array}$$

Next, we do the subtraction;

Note the order as 11000 – 10101 i.e
$$\begin{array}{ccccc} 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ \hline 0 & 0 & 0 & 1 & 1_{two} \\ \hline \end{array}$$

2007/7 Neco

Solve $11101_{two} + 11011_{two} - 11001_{two}$

A 11001_{two} B 11011_{two} C 11100_{two}

D 11110_{two} E 11111_{two}

Solution

First is addition
$$\begin{array}{rcccccc} & & 1 & 1 & 1 & 0 & 1 \\ + & & 1 & 1 & 0 & 1 & 1 \\ \hline & 1 & 1 & 1 & 0 & 0 & 0_{two} \\ \hline \end{array}$$

Next: subtraction
$$\begin{array}{rcccccc} & 1 & 1 & 1 & 0 & 0 & 0 \\ - & & 1 & 1 & 0 & 0 & 1 \\ \hline & & 1 & 1 & 1 & 1 & 1_{two} \\ \hline \end{array} \quad (E)$$

Explanation

After the carry – carry operation the new look is:
$$\begin{array}{cccccc} 0 & 2 & 2 & 1 & 1 & 2 \\ & 1 & 1 & 0 & 0 & 1 \\ \hline & 1 & 1 & 1 & 1 & 1_{two} \\ \hline \end{array}$$

2005/22 Neco

If $101_{two} \times 11_{two} + P_{two} = 10101_{two}$, find P

A 1101_{two} B 1011_{two} C 111_{two} D 110_{two} E 101_{two}

Solution

First, we perform the multiplication by BODMAS

$101_{two} \times 11_{two}$ is
$$\begin{array}{rcccc} & & 1 & 0 & 1 \\ & & \times & 1 & 1 \\ \hline & & 1 & 0 & 1 \\ & 1 & 0 & 1 & * \\ \hline & 1 & 1 & 1 & 1_{two} \\ \hline \end{array}$$

$101_{two} \times 11_{two} + P_{two} = 10101_{two}$ *becomes*

$P_{two} = 10101_{two} - (101_{two} \times 11_{two})$

$= 10101 - 1111$ *written out as*
$$\begin{array}{rccccc} & 1 & 0 & 1 & 0 & 1 \\ - & & 1 & 1 & 1 & 1 \\ \hline & & 0 & 1 & 1 & 0 \\ \hline \end{array} \quad (D)$$

2014/1 NABTEB Exercise 7.20

What is the place value of 2 in 243 base five?

A 2 B 10 C 50 D 200

2002/1a NABTEB Exercise 7.21

Copy and complete the table in base 5

x	2	3	4
2	4		
3			22
4		22	

2003/1 NABTEB Exercise 7.22

The place value of 3 in 5324_{six}

A 18 B 108 C 180 D 1204

Exercise 7.23

The total of four numbers is 1214_{five}. What is their average expressed in base five?

A. 303.10 B. 242.4 C. 141 D. 114

2009/4 Exercise 7.24

Arrange the following number in descending order of magnitude: 22_{three}, 34_{five}, 21_{six}

A 21_{six}, 22_{three}, 34_{five} B 21_{six}, 34_{five}, 22_{three}

C 22_{three}, 34_{five}, 21_{six} D 34_{five}, 21_{six}, 22_{three}

SURD

Irrational or non rational numbers are numbers whose value have no ending

Examples are $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$ e.t.c the square root of all non perfect squares are irrational likewise their multiples and fractions. Also π is an example of irrational numbers. we refer to the square root of non-perfect square as SURD.

Surd has it own basic rule of operation with addition, subtraction, division and multiplication.

Addition of surds

Example

Simplify: $\sqrt{2} + \sqrt{3}$

Solution

$\sqrt{2} + \sqrt{3} = \sqrt{2} + \sqrt{3}$

But $\sqrt{2} + \sqrt{3} \neq \sqrt{2+3}$ or $\sqrt{5}$

Generally

$\sqrt{a} + \sqrt{b} \neq \sqrt{a+b}$

Also $\sqrt{a} - \sqrt{b} \neq \sqrt{a-b}$

Multiplication of surds

$\sqrt{a} \times \sqrt{b} = \sqrt{a \times b}$

This true for all surds

Examples

a. simplify $\sqrt{2} \times \sqrt{6}$

Solution

$$\sqrt{2} \times \sqrt{6} = \sqrt{2\times6}$$
$$= \sqrt{12}$$
$$= \sqrt{4\times3} = \sqrt{4} \times \sqrt{3} = 2\sqrt{3}$$

b.Simplify $\sqrt{60}$

Solution

$$\sqrt{60} = \sqrt{4\times15}$$
$$= 2\sqrt{15}$$

c.Simplify $\sqrt{3} \times \sqrt{3}$

Solution

$$\sqrt{3} \times \sqrt{3} = \sqrt{3 \times 3}$$
$$= \sqrt{9}$$
$$= 3$$

d. Simplify $\sqrt{80} + \sqrt{20} - \sqrt{45}$

Solution

We bring out the perfect square factors in the given surd as:

$$\sqrt{80} + \sqrt{20} - \sqrt{45} = \sqrt{16\times5} + \sqrt{4\times5} - \sqrt{9\times5}$$
$$= \sqrt{16} \times \sqrt{5} + \sqrt{4} \times \sqrt{5} - \sqrt{9} \times \sqrt{5}$$
$$= 4\sqrt{5} + 2\sqrt{5} - 3\sqrt{5}$$
$$= (4 + 2 - 3)\sqrt{5}$$
$$= 3\sqrt{5}$$

Generally, $\sqrt{a} \times \sqrt{a}$ **= a**

2009/1a

Given that $(\sqrt{3} - 5\sqrt{2})(\sqrt{3} + \sqrt{2}) = a + b\sqrt{6}$ find a and b

Solution

We simply open up the bracket as:

$$(\sqrt{3} - 5\sqrt{2})(\sqrt{3} + \sqrt{2}) = \sqrt{3}\left(\sqrt{3} + \sqrt{2}\right) - 5\sqrt{2}\left(\sqrt{3} + \sqrt{2}\right)$$
$$= \sqrt{3\times3} + \sqrt{3\times2} - 5\sqrt{2\times3} - 5\sqrt{2\times2}$$
$$= 3 + \sqrt{6} - 5\sqrt{6} - (5 \times 2)$$
$$= 3 + \sqrt{6} - 5\sqrt{6} - 10$$
$$= -7 - 4\sqrt{6}$$
$$= -(7 + 4\sqrt{6})$$

2014/6

Simplify: $\sqrt{12}\left(\sqrt{48} - \sqrt{3}\right)$

A 18 B 16 C 14 D 12

Solution

First, we open the bracket using multiplication rule in surds

$$\sqrt{12}\left(\sqrt{48} - \sqrt{3}\right) = \sqrt{12 \times 48} - \sqrt{12 \times 3}$$
$$= \sqrt{12 \times 12 \times 4} - \sqrt{4 \times 3 \times 3}$$
$$= \sqrt{12^2 \times 2^2} - \sqrt{2^2 \times 3^2}$$
$$= 12 \times 2 - 2 \times 3$$
$$= 24 - 6$$
$$= 18 \quad (A)$$

2006/ 12a (Nov) Exercise 7.25

Simplify $\sqrt{3}(2\sqrt{2} + \sqrt{3}) + \sqrt{2}(\sqrt{3} - \sqrt{2})$, leaving the answer in surd form

Division in surds

$$\sqrt{\frac{m}{n}} = \frac{\sqrt{m}}{\sqrt{n}}$$

This true for all surds

Examples

1. Simplify $\dfrac{\sqrt{35}}{\sqrt{5}}$

Solution

$$\frac{\sqrt{35}}{\sqrt{5}} = \sqrt{\frac{35}{5}}$$
$$= \sqrt{7}$$

2. Simplify $\sqrt{\dfrac{17}{4}}$

Solution

$$\sqrt{\frac{17}{4}} = \frac{\sqrt{17}}{\sqrt{4}} = \frac{\sqrt{17}}{2}$$

Rationalization of surds

Examples

1. Rationalize $\sqrt{\frac{1}{3}}$

Solution

$$\sqrt{\frac{1}{3}} = \frac{1}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$= \frac{1 \times \sqrt{3}}{\sqrt{3} \times \sqrt{3}} = \frac{\sqrt{3}}{3}$$

Recall that $\sqrt{a} \times \sqrt{a} = a$

To rationalize, when the denominator is a single surd we multiply the numerator and denominator by the surd in the denominator.

2. Simplify $\frac{10\sqrt{2}}{\sqrt{5}}$

Solution

$$\frac{10\sqrt{2}}{\sqrt{5}} = \frac{10\sqrt{2}}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}}$$

$$= 10\ \frac{\sqrt{2\times5}}{5} \quad = 2\sqrt{10}$$

3. Simplify $2\sqrt{3} - \frac{6}{\sqrt{3}} + \frac{3}{\sqrt{27}}$

Solution

$$2\sqrt{3} - \frac{6}{\sqrt{3}} + \frac{3}{\sqrt{27}} = 2\sqrt{3} - \frac{6\sqrt{3}}{\sqrt{3}\times\sqrt{3}} + \frac{3\sqrt{27}}{\sqrt{27}\times\sqrt{27}}$$

$$= 2\sqrt{3} - \frac{6\sqrt{3}}{3} + \frac{3\sqrt{27}}{27}$$

$$= 2\sqrt{3} - 2\sqrt{3} + \frac{\sqrt{9\times3}}{9}$$

$$= \frac{3\sqrt{3}}{9} = \tfrac{1}{3}\sqrt{3}$$

2006/ 1b

By rationalizing the denominator, simplify: $\frac{7\sqrt{5}}{\sqrt{7}}$

(leave your answer in surd form)

Solution

$$\frac{7\sqrt{5}}{\sqrt{7}} = \frac{7\sqrt{5}\times\sqrt{7}}{\sqrt{7}\times\sqrt{7}}$$

$$= \frac{7\sqrt{35}}{7}$$

$$= \sqrt{35}$$

2009/ 2 (Nov) Exercise 7.26

Simplify $\frac{6}{\sqrt{3}}$

A $\frac{2\sqrt{3}}{3}$ B $\sqrt{3}$ C $2\sqrt{3}$ D $6\sqrt{3}$

2006/ 38(Nov) Exercise 7.27

Rationalize $\frac{5}{\sqrt{3}}$ and leave your answer in surd form

A $\frac{5\sqrt{3}}{3}$ B $\frac{3\sqrt{3}}{5}$ C $\frac{5\sqrt{3}}{9}$ D $\frac{9\sqrt{3}}{5}$

Rationalization of surd by Conjugate

Surd	Conjugate	their product
$2 + \sqrt{3}$	$2 - \sqrt{3}$	$2^2 - 3$
$2 - \sqrt{3}$	$2 + \sqrt{3}$	$2^2 - 3$
Generally		
$a + \sqrt{b}$	$a - \sqrt{b}$	$a^2 - b$

The last term is a general expression for Surd and their conjugates. Also from the table above we have seen that the product of a surd and its conjugate is not a surd since

$$(a + \sqrt{b})(a - \sqrt{b}) = a(a - \sqrt{b}) + \sqrt{b}(a - \sqrt{b})$$

$$= a^2 - a\sqrt{b} + a\sqrt{b} - \sqrt{b} \times \sqrt{b}$$

$$= a^2 - b$$

2006/4b

Simplify: $\frac{1+\sqrt{2}}{2-\sqrt{3}}$

Solution

We rationalize as:

$$\frac{1+\sqrt{2}}{2-\sqrt{3}} = \frac{(1+\sqrt{3})(2+\sqrt{3})}{(2-\sqrt{3})(2+\sqrt{3})}$$

$$= \frac{1(2+\sqrt{3})+\sqrt{3}(2+\sqrt{3})}{2^2-3}$$

$$= \frac{2+\sqrt{3}+2\sqrt{3}+3}{4-3}$$

$$= 5 + 3\sqrt{3}$$

2008/9

Rationalize the expression: $\frac{7-\sqrt{3}}{13-\sqrt{3}}$

A $\frac{44+3\sqrt{3}}{83}$ B $\frac{44+2\sqrt{3}}{83}$ C $\frac{44-3\sqrt{3}}{83}$

D $\frac{44-4\sqrt{3}}{83}$ E $\frac{44-2\sqrt{3}}{83}$

Solution

$$\frac{7-\sqrt{3}}{13-\sqrt{3}} = \frac{(7-\sqrt{3})(13+\sqrt{3})}{(13-\sqrt{3})(13+\sqrt{3})}$$

$$= \frac{7(13+\sqrt{3})-\sqrt{3}(13+\sqrt{3})}{13^2-3}$$

$$= \frac{91+7\sqrt{3} - 13\sqrt{3}-3}{169-3}$$

$$= \frac{88-6\sqrt{3}}{166}$$

Divide through by 2

$$= \frac{44-3\sqrt{3}}{83} \quad (C)$$

2010/8b Neco

Without using tables, find the value of $\frac{1}{\sqrt{11}-2} - \frac{1}{\sqrt{11}+2}$

Solution

Applying LCM to the denominators

$$\frac{1}{\sqrt{11}-2} - \frac{1}{\sqrt{11}+2} = \frac{\sqrt{11}+2-(\sqrt{11}-2)}{(\sqrt{11}-2)(\sqrt{11}+2)}$$

Applying the conjugation of surd to the denominator

$$= \frac{\sqrt{11}+2-\sqrt{11}+2}{11+2\sqrt{11}-2\sqrt{11}-2^2}$$

$$= \frac{4}{11-4} \quad \text{i.e} \quad \frac{4}{7}$$

2013/4 Neco

Rationalizing $\frac{1}{3-\sqrt{6}}$

A $\frac{3\sqrt{6}}{6}$ B $\frac{3+\sqrt{6}}{3}$ C $\sqrt{6}$ D $\sqrt{12}$ E 6

Solution

$$\frac{1}{3-\sqrt{6}} = \frac{1(3+\sqrt{6})}{(3-\sqrt{6})(3+\sqrt{6})}$$

$$= \frac{3+\sqrt{6}}{3^2-6}$$

$$= \frac{3+\sqrt{6}}{3} \quad (\text{B})$$

2006/5b NABTEB

Rationalize $\frac{2}{4+3\sqrt{2}}$

Solution

Applying conjugate rule

$$\frac{2}{4+3\sqrt{2}} = \frac{2(4-3\sqrt{2})}{(4+3\sqrt{2})(4-3\sqrt{2})}$$

$$= \left[\frac{8-6\sqrt{2}}{4^2-3\times3(2)}\right]$$

$$= \frac{8-6\sqrt{2}}{16-18}$$

$$= \frac{8-6\sqrt{2}}{-2}$$

$$= -4+3\sqrt{2}$$

2012/2 Neco Exercise 7.28

Which of the following represents the conjugate of $\sqrt{3}+\sqrt{2}$?

A $\sqrt{2}-\sqrt{3}$ B $\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$ C $\sqrt{3}-\sqrt{2}$

D $\sqrt{3}+\sqrt{2}$ E $\frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}}$

Exercise 7.29

Simplify : $\frac{3-2\sqrt{2}}{3+2\sqrt{2}}$

A. 1 B. $17-\sqrt{2}$ C. $17+12\sqrt{2}$ D. $17-12\sqrt{2}$

Simplification in surds

Generally, in solving a given question, sometimes we employ one or more of the principles treated so far.

2012/2

Simplify $\frac{3\sqrt{5}\times4\sqrt{6}}{2\sqrt{2}\times3\sqrt{3}}$

A $\sqrt{2}$ B $\sqrt{5}$ C $2\sqrt{2}$ D $2\sqrt{5}$

Solution

Applying multiplication rule in surd

$$\frac{3\sqrt{5}\times4\sqrt{6}}{2\sqrt{2}\times3\sqrt{3}} = \frac{3\times4\sqrt{5\times6}}{2\times3\sqrt{2\times3}}$$

$$= \frac{12\sqrt{30}}{6\sqrt{6}} \quad \text{i.e}$$

Applying division rule in surd

$$= 2\sqrt{\frac{30}{6}}$$

$$= 2\sqrt{5} \quad (\text{D})$$

2008/ 26 Neco (Dec)

Simplify $\left(\frac{\sqrt{2}-\sqrt{3}}{\sqrt{2}+\sqrt{3}}\right)^2$

A $-19-20\sqrt{6}$ B $-19+20\sqrt{6}$ C $19-20\sqrt{6}$

D $19+20\sqrt{6}$ E $49-20\sqrt{6}$

Solution

$$\left(\frac{\sqrt{2}-\sqrt{3}}{\sqrt{2}+\sqrt{3}}\right)^2 = \frac{(\sqrt{2}-\sqrt{3})^2}{(\sqrt{2}+\sqrt{3})^2} \text{ by law of indices}$$

$$= \frac{(\sqrt{2}-\sqrt{3})(\sqrt{2}-\sqrt{3})}{(\sqrt{2}+\sqrt{3})(\sqrt{2}+\sqrt{3})}$$

Opening up the brackets

$$= \frac{\sqrt{2}(\sqrt{2}-\sqrt{3}) - \sqrt{3}(\sqrt{2}-\sqrt{3})}{\sqrt{2}(\sqrt{2}+\sqrt{3}) + \sqrt{3}(\sqrt{2}+\sqrt{3})}$$

$$= \frac{2-\sqrt{6}-\sqrt{6}+3}{2+\sqrt{6}+\sqrt{6}+3}$$

$$= \frac{5-2\sqrt{6}}{5+2\sqrt{6}}$$

Rationalizing

$$= \frac{(5-2\sqrt{6})(5-2\sqrt{6})}{(5+2\sqrt{6})(5-2\sqrt{6})}$$

$$= \frac{25-10\sqrt{6}-10\sqrt{6}+(4\times6)}{25-10\sqrt{6}+10\sqrt{6}-(4\times6)}$$

$$= \frac{25-20\sqrt{6}+24}{25-24}$$

$= 49-20\sqrt{6}$ (E)

2011/32

If $\sqrt{72}+\sqrt{32}-3\sqrt{18}=x\sqrt{8}$, find the value of x

A 1 B $^3/_4$ C $^1/_2$ D $^1/_4$

Solution

Simplifying the surds

$\sqrt{72}+\sqrt{32}-3\sqrt{18}=x\sqrt{8}$ becomes

$$\sqrt{9\times8}+\sqrt{4\times8}-3\sqrt{9\times2}=x\sqrt{8}$$

$$3\sqrt{8}+2\sqrt{8}-3\times3\sqrt{2}=x\sqrt{8}$$

$$5\sqrt{8}-9\sqrt{2}=x\sqrt{8}$$

We simplify both sides to the lowest surd of $\sqrt{2}$

$$5\sqrt{4\times2}-9\sqrt{2}=x\sqrt{4\times2}$$

$$5\times2\sqrt{2}-9\sqrt{2}=2x\sqrt{2}$$

$$10\sqrt{2}-9\sqrt{2}=2x\sqrt{2}$$

$$\sqrt{2}=2x\sqrt{2}$$

Divide both sides by coefficient of x

$$\frac{\sqrt{2}}{2\sqrt{2}}=\frac{2x\sqrt{2}}{2\sqrt{2}}$$

$^1/_2$ = x (C)

2014/2a

Simplify $3\sqrt{75}-\sqrt{12}+\sqrt{108}$, leaving the answer in surd form (radicals)

Solution

$$3\sqrt{75}-\sqrt{12}+\sqrt{108}=3\sqrt{3\times25}-\sqrt{4\times3}+\sqrt{4\times3\times9}$$

$$=3\sqrt{3\times5^2}-\sqrt{3\times2^2}+\sqrt{2^2\times3\times3^2}$$

$$=3\times5\sqrt{3}-2\sqrt{3}+2\times3\sqrt{3}$$

$$=15\sqrt{3}-2\sqrt{3}+6\sqrt{3}$$

$$=19\sqrt{3}$$

2001/2a (Nov)

Without using mathematical tables or calculator, simplify $2\sqrt{3}(3\sqrt{3}-2)$ given that $\sqrt{3}=1.732$

Solution

Expanding the bracket

$$2\sqrt{3}(3\sqrt{3}-2)=6\times3-4\sqrt{3}$$

$$=18-4\sqrt{3}$$

Substituting for $\sqrt{3}$

= 18 – 4(1.732)
= 18 – 6.928
= 11.072

2005/6a (Nov)

Simplify $\sqrt{5}+\frac{1}{\sqrt{5}}-\frac{1}{\sqrt{125}}$

Solution

$$\sqrt{5}+\frac{1}{\sqrt{5}}-\frac{1}{\sqrt{125}}=\sqrt{5}+\frac{1}{\sqrt{5}}-\frac{1}{\sqrt{25\times5}}$$

$$=\sqrt{5}+\frac{1}{\sqrt{5}}-\frac{1}{5\sqrt{5}}$$

Rationalizing $=\sqrt{5}+\frac{\sqrt{5}}{5}-\frac{5\sqrt{5}}{25\times5}$

$$=\sqrt{5}+\frac{\sqrt{5}}{5}-\frac{\sqrt{5}}{25}$$

Applying LCM to the denominators

$$=\frac{25\sqrt{5}+5\sqrt{5}-\sqrt{5}}{25}$$

$$=\frac{29\sqrt{5}}{25}$$

2005/6b

Simplify: $\frac{8-4\sqrt{18}}{\sqrt{50}}$ leaving your answer in the form a + b$\sqrt{n}$ where a and b are rational numbers and n is an integer.

Solution

First, we simplify the surds to their lowest terms

$$\frac{8-4\sqrt{18}}{\sqrt{50}}=\frac{8-4\sqrt{9\times2}}{\sqrt{25\times2}}$$

$$=\frac{8-4\times3\sqrt{2}}{5\sqrt{2}}=\frac{8-12\sqrt{2}}{5\sqrt{2}}$$

Rationalizing

$$=\frac{(8-12\sqrt{2})(5\sqrt{2})}{5\sqrt{2}\times5\sqrt{2}}$$

$$=\frac{40\sqrt{2}-60\times2}{25\times2}$$

$$=\frac{40\sqrt{2}-120}{50}$$

Keeping to the given instruction $=-\frac{120}{50}+\frac{40}{50}\sqrt{2}$

$$=-\frac{12}{5}+\frac{4}{5}\sqrt{2}$$

2005/32

Simplify $\frac{6}{\sqrt{48}}-\sqrt{3}$

A $\frac{-\sqrt{3}}{2}$ B $\frac{2\sqrt{3}}{3}$ C $\frac{3\sqrt{3}}{2}$ D $2\sqrt{3}$

Solution

Using the lowest surd $\sqrt{3}$ as standard

$$\frac{6}{\sqrt{48}}-\sqrt{3}=\frac{6}{\sqrt{16\times3}}-\sqrt{3}$$

$$=\frac{6}{4\sqrt{3}}-\sqrt{3}$$

$$=\frac{3}{2\sqrt{3}}-\sqrt{3}$$

Rationalizing $=\frac{3\times2\sqrt{3}}{2\sqrt{3}\times2\sqrt{3}}-\sqrt{3}$

$$=\frac{6\sqrt{3}}{4\times3}-\sqrt{3}$$

$$=\frac{\sqrt{3}}{2}-\sqrt{3}=\frac{-\sqrt{3}}{2}\quad\text{(A)}$$

2005/29(Nov)

Simplify: $\frac{\sqrt{63}-\sqrt{28}}{\sqrt{7}}$

A $\sqrt{7}$ B $2\sqrt{7}$ C 1 D 5

Solution

$$\frac{\sqrt{63}-\sqrt{28}}{\sqrt{7}}=\frac{\sqrt{9\times7}-\sqrt{4\times7}}{\sqrt{7}}$$

$$=\frac{3\sqrt{7}-2\sqrt{7}}{\sqrt{7}}$$

$$=\frac{\sqrt{7}}{\sqrt{7}}=1\quad\text{(C)}$$

2006/49

Simplify: $\dfrac{20}{5\sqrt{28}-2\sqrt{63}}$

A $\dfrac{5\sqrt{7}}{7}$ B $\dfrac{5\sqrt{5}}{7}$ C $\dfrac{7\sqrt{5}}{5}$ D $\dfrac{7\sqrt{7}}{5}$

Solution

$$\frac{20}{5\sqrt{28}-2\sqrt{63}} = \frac{20}{5\sqrt{4\times7}-2\sqrt{9\times7}}$$

$$= \frac{20}{5\times2\sqrt{7} - 2\times3\sqrt{7}}$$

$$= \frac{20}{10\sqrt{7}-6\sqrt{7}} = \frac{20}{4\sqrt{7}} = \frac{5}{\sqrt{7}}$$

Rationalizing $= \dfrac{5\sqrt{7}}{\sqrt{7}\times\sqrt{7}}$

$$= \frac{5\sqrt{7}}{7} \quad (\text{A})$$

2006/3b Neco (Dec)

Simplify: $\dfrac{1}{\sqrt{2}} + 3\sqrt{2} - \dfrac{1}{3\sqrt{2}}$

Solution

First, we rationalizing the fractional surds

$$\frac{1}{\sqrt{2}} + 3\sqrt{2} - \frac{1}{3\sqrt{2}}$$

$$= \frac{1\times\sqrt{2}}{\sqrt{2}\times\sqrt{2}} + 3\sqrt{2} - \frac{1\times3\sqrt{2}}{3\sqrt{2}\times3\sqrt{2}}$$

$$= \frac{\sqrt{2}}{2} + 3\sqrt{2} - \frac{3\sqrt{2}}{9\times2}$$

$$= \frac{\sqrt{2}}{2} + 3\sqrt{2} - \frac{\sqrt{2}}{6}$$

Applying LCM to the denominators

$$= \frac{3\sqrt{2}+18\sqrt{2}-\sqrt{2}}{6} = \frac{20\sqrt{2}}{6} = \frac{10\sqrt{2}}{3}$$

2007/5

Simplify $3\sqrt{45} - 12\sqrt{5} + 16\sqrt{20}$, leaving the answer in surd form

A $29\sqrt{5}$ B $14\sqrt{15}$ C $12\sqrt{15}$ D $11\sqrt{5}$

Solution

Using the lowest surd $\sqrt{5}$ as standard

$$3\sqrt{45} - 12\sqrt{5} + 16\sqrt{20} = 3\sqrt{9\times5}-12\sqrt{5}+16\sqrt{4\times5}$$

$$= 3\times3\sqrt{5}-12\sqrt{5}+16\times2\sqrt{5}$$

$$= 9\sqrt{5}-12\sqrt{5}+32\sqrt{5}$$

$$= 29\sqrt{5} \quad (\text{A})$$

2009/ 44

Simplify: $\frac{1}{2}\sqrt{32}-\sqrt{18}+\sqrt{2}$

A 0 B $\sqrt{2}$ C $2\sqrt{2}$ D $4\sqrt{2}$

Solution

Using the lowest surd $\sqrt{2}$ as standard

$$\frac{1}{2}\sqrt{32}-\sqrt{18}+\sqrt{2} = \frac{1}{2}\sqrt{16\times2}-\sqrt{9\times2}+\sqrt{2}$$

$$= \frac{4}{2}\sqrt{2}-3\sqrt{2}+\sqrt{2}$$

$$= 2\sqrt{2}-3\sqrt{2}+\sqrt{2}$$

$$= 0 \quad (\text{A})$$

2009/1b (Nov)

Simplify without using tables or calculator $2+\sqrt{96}-4(\sqrt{6}-1)^2$ and express your answer in the form $m + n\sqrt{6}$ where m and n are real numbers.

Solution

First we treat the bracket items

$$2+\sqrt{96}-4(\sqrt{6}-1)^2 = 2+\sqrt{16\times6}-4(\sqrt{6}-1)(\sqrt{6}-1)$$

$$= 2+4\sqrt{6}-4(6-\sqrt{6}-\sqrt{6}+1)$$

$$= 2+4\sqrt{6}-4(7-2\sqrt{6})$$

$$= 2+4\sqrt{6}-28+8\sqrt{6}$$

$$= -26+12\sqrt{6}$$

2010/2a Neco

Express $\sqrt{243}+\dfrac{12}{\sqrt{3}}$ as a single surd.

Hence, find the value of $\dfrac{13}{\sqrt{3}}\left(\sqrt{243}+\dfrac{12}{\sqrt{3}}\right)$

Solution

Applying LCM to denominator

$$\sqrt{243}+\frac{12}{\sqrt{3}} = \frac{\sqrt{243\times3}+12}{\sqrt{3}}$$

$$= \frac{\sqrt{81\times3\times3}+12}{\sqrt{3}}$$

$$= \frac{(9\times3)+12}{\sqrt{3}} = \frac{27+12}{\sqrt{3}} \text{ i.e } \frac{39}{\sqrt{3}}$$

Thus $\dfrac{13}{\sqrt{3}}\left(\sqrt{243}+\dfrac{12}{\sqrt{3}}\right) = \dfrac{13}{\sqrt{3}}\left(\dfrac{39}{\sqrt{3}}\right) = \dfrac{507}{3} = 169$

2010/20 Exercise 7.30

Simplify: $2\sqrt{3} - \dfrac{6}{\sqrt{3}} + \dfrac{3}{\sqrt{27}}$

A 1 B $\frac{1}{3}\sqrt{3}$ C $2\sqrt{3}-5\frac{2}{3}$ D $6\sqrt{3}-17$

2008/4 Exercise 7.31

Simplify: $\sqrt{50}+\dfrac{10}{\sqrt{2}}$

A 10 B $10\sqrt{2}$ C 20 D $20\sqrt{2}$

2014/4 (Nov) Exercise 7.32

If $2\sqrt{2}+\sqrt{125}-\sqrt{45}+4 = a+b\sqrt{c}$,
evaluate (2a – b)

A. 8 B. 4 C. 2 D. 0

2015/32 Exercise 7.33

If $\dfrac{\sqrt{2}+\sqrt{3}}{\sqrt{3}}$ is simplified as $m+n\sqrt{6}$,
find the value of (m + n)

A. $^1/_3$ B. $^2/_3$ C. $1^2/_3$ D. $1^1/_3$

CHAPTER EIGHT

Statistics I (ungrouped data)

Statistics is the science of collecting, organizing and analyzing data for any given purpose. It helps us to reduce large and scattered data to an understandable level, thereby enabling us to make decisions in the face of uncertainty. At one time or the other in our day to day activities, we must have engaged statistical values in taking decision.

For instance, Zino's parent wants to send Zino to a college in Garki – Abuja. You will agree with them that there are so many schools to choose from. Through enquiry about various schools' qualities such as :

(i) School performance in JSCE & SSCE
(ii) Qualification of teaching staff
(iii) School location
(iv) Fees charged,

They will make a good decision on the best school that suit them. By the above act, they have engaged in the statistical activities of collecting, organizing and analysis of data for the purpose of sending their child to school.

Population

This is the totality of observation with which we are concerned. The **whole** Secondary Schools in Garki-Abuja forms the population of the above example.

Sample

It is basically the subset of population. You will agree with the author that it will not be Naira (Penny) wise for Zino's parents to check on **all** the schools in Garki – Abuja. Rather they will check on a few, say about 3 to 8 schools (sample) among the over 50 secondary schools

Raw data

They are collected data which have not been organized numerically, i.e they have not been arranged either in ascending or descending order.

E.g The scores of 10 students who sat for a mathematics test, are: 3, 5, 10, 0, 1,8, 10, 4, 2, 6.

ARRAY

This is the arrangement of raw numerical data in *ascending* or *descending* order of magnitude. From the data supplied above:

Ascending order: 0, 1, 2, 3, 4, 5, 6, 8, 10, 10
Descending order: 10, 10, 8, 6, 5, 4, 3, 2, 1,0.

Frequency & Tally

Frequency: this is the number of occurrences of any value in a given data while tally is the stroke representation of frequency where:

I	represents	1
II	represents	2
III	represents	3
IIII	represents	4
~~IIII~~	represents	5
~~IIII~~ I	represents	6
~~IIII~~ II	represents	7 and so on.

E.g 3 The ages of 25 students in JSS I of Ibru college Agbarha – Otor is given as follows;

7	8	10	15	8	9	7
8	8	9	10	12	7	13
14	8	8	9	10	15	8
7	10	11	11			

arrange the above data in a frequency table.

Solution

Arranging the above data in a frequency table.

Ages	Tally	No of students (freq.)
7	IIII	4
8	~~IIII~~ II	7
9	III	3
10	IIII	4
11	II	2
12	I	1
13	I	1
14	I	1
15	II	2
		Total = 25

The *centrality* and *spread* of any given data are of great importance to statisticians. These two mentioned concepts enable us to make appropriate decision.

Measure of central tendency (measure of location)

Measure of central tendency can be thought of as a measure which gives the location of the "center" of the data. The most commonly used ones are the mean, median and mode.

MEAN

The mean (arithmetic mean) denoted by $\bar{x}$ (x bar) of a set of n numbers $x_1, x_2, \dots x_n$ is

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

$$\bar{x} = \frac{\sum x}{n} \text{ (where } \sum \text{ is summation)}$$

2005/28

Find the mean of the numbers 1, 3, 4, 8, 8, 4 and 7

A 4 B 5 C 6 D 7

Solution

$$\text{Mean} = \frac{\sum x}{n}$$

$$= \frac{1+3+4+8+8+4+7}{7}$$

$$= \frac{35}{7} \text{ i.e } 5 \quad (\text{ B })$$

2014/28 NABTEB

Calculate the mean of 1, 0, 1, 2, 3, 2, 0, 2, 3, 2

A 1.6 B 1.7 C 1.8 D 2.0

Solution

$$\text{Mean} = \frac{\sum x}{n}$$

$$= \frac{1+0+1+2+3+2+0+2+3+2}{10}$$

$$= \frac{16}{10} \text{ i.e } 1.6 \quad (\text{ A })$$

2009/ 37 NABTEB (Nov)
The weight of 5 girls in kg are 48, x, 52, 50 and (2x – 5). If their average weight is 47 kg. Find the weight of the girl that has the heaviest weight
A 85kg B 65kg C 55kg D 50 kg
Solution

Average weight = $\frac{sum\ of\ weight}{No\ of\ girls}$

$$47 = \frac{48 + x + 52 + 50 + (2x - 5)}{5}$$

$47 \times 5 = 150 + x + 2x - 5$
$235 = 145 + 3x$
$235 - 145 = 3x$
$90 = 3x$
$30 = x$

Next, we list the weight: 48, 30, 52, 50, 55
Thus the heaviest weight is 55kg (C)

2004/15a NABTEB
The average cost of 20 articles is ₦ 213.00; the average cost of the first 9 is ₦ 115.00.Find the average cost of the remaining articles

Solution

Mean = $\frac{\sum x}{n}$

$213 = \frac{\sum x}{20}$

$\Sigma x = 20 \times 213$
$= 4260$

Also

$115 = \frac{\sum x}{9}$

$\Sigma x = 9 \times 115$
$= 1035$

Average cost of (20 – 9 article)

$$= \frac{4260 - 1035}{20 - 9}$$

$$= \frac{3225}{11} \text{ i.e } 293.18$$

2006/53 Neco (Nov)
The mean age of 10 students is 10 years 6 months. If the age of two of the students, 12 years 10 months and 13 years 6 months are subtracted from the ages of the ten students, what will be the mean age of the remaining 8 students?
A 4 years 1 months B 9years 6 months
C 9 years 10 months D 12years 3 months
E 13 years 2months

Solution

Mean = $\frac{\sum x}{n}$

$10.5 = \frac{\sum x}{10}$

$10.5 \times 10 = \Sigma x$ i.e 105
Sum ages of two students = $12^{10}/_{12} + 13\,^{6}/_{12}$
$= 25 + (0.83 + 0.5)$
$= 26.33$years
Then 26.33 years is subtracted from 105 year and 2 is subtracted from 10 students

Mean (8students) = $\frac{105 - 26.33}{10 - 2}$

$= \frac{78.67}{8}$

$= 9.83$ i.e 9 years 10 months (C)

2008/59 Neco
The ages to Abu, Segun, Kofi and Funmi are 17years, (2x – 13) years, 14 years and 16 years respectively. What is the value of x if their mean age is 17.5 years?
A 18 B 23 C 25 D 36 E 70

Solution

Mean = $\frac{\sum x}{n}$

$$17.5 = \frac{17 + (2x - 13) + 14 + 16}{4}$$

$17.5 \times 4 = 47 + 2x - 13$
$70 = 34 + 2x$
$70 - 34 = 2x$
$36 = 2x$
$18 = x$ (A)

2008/53 Neco
Twenty girls and y boys sat for an examination. The mean marks obtained by the girls and boys were 62 and 57 respectively. If the total scores for both girls and boys was 2950, find y
A 51 B 48 C 30 D 25 E 18

Solution

Let the twenty girls total score be x
Already, we are given y boys (not total score of boys)

From Mean = $\frac{\sum x}{n}$

Girls

$62 = \frac{x}{20}$

$62 \times 20 = x$

$1240 = x$

Boys

$57 = \frac{2950 - x}{y}$

substituting for x = 1240

$57 = \frac{2950 - 1240}{y}$

$57y = 1710$
$y = 30$ (C)

2011/54 Neco
The average of 10 boys was 12 years. A boy of 15 years was replaced with that of 5 years. Find the new average age of the boys.
A 7 years B 8 years C 9 years
D 10 years E 11 years

Solution

Let the sum of the 10 boys' ages be y, then

Mean = $\frac{\sum x}{n}$ here Σx is y

i.e $12 = \frac{y}{10}$

$12 \times 10 = y$
$y = 120$

But If 15 years boy was replaced by 5 years
Then the new $y = 120 - 15 + 5$
$= 110$
Our n i.e numbers of boys remain 10

New average = $\frac{110}{10}$ = 11 years (E)

2007/52 Neco
The mean of a set of numbers 68, 65, x, 69, 77, 48, 64 is 67 find x
A 78 B 68 C 67 D 66 E 62

Solution

$$\text{Mean} = \frac{\sum x}{n}$$

$$67 = \frac{68+65+x+69+77+48+64}{7}$$

$67 \times 7 = 391 + x$

$469 = 391 + x$

$469 - 391 = x$

$78 = x$ (A)

2013/58 Neco Exercise 8.1

If the mean of 6, 8, 9, x, 5, 7 is 8.
What is the value of x?

A 6 B 8 C 9 D 10 E 13

2007/30 Exercise 8.2

Divide the sum of 8, 6, 7, 2, 0, 4, 7, 2, 3 by their mean

A 9 B 8 C 7 D 6

2008/39 NABTEB (Nov) Exercise 8.3

Calculate the mean of the distribution: 4, 3, 6, 8, 9, 3, 9

A 6 B 7 C 8 D 9

Mean of frequency distribution

For any given data say $x_1, x_2, \ldots, x_k$ that occur $f_1, f_2, \ldots, f_k$ times respectively, then the mean

$$\bar{x} = \frac{f_1x_1 + f_2x_2 + \ldots + f_k x_k}{f_1 + f_2 + \ldots + f_k}$$

$$= \frac{\sum fx}{\sum f}$$

2008/12c NABTEB (Nov)

Calculate the mean of the distribution of the test scores below:

Scores	30	35	40	45	50
No of student	3	21	46	18	12

Solution

$$\text{Mean} = \frac{\sum fx}{\sum f}$$

Table for mean showing x, f and fx

Scores(x)	Frequency(f)	fx
30	3	90
35	21	735
40	46	1840
45	18	810
50	12	600
	$\Sigma f = 100$	$\Sigma fx = 4075$

Substituting

$$\text{Mean} = \frac{4075}{100} \text{ i.e } 40.75$$

2005/49 Neco

Find the mean of the distribution

Scores	1	2	3	4	5	6	7	8
Frequency	3	4	3	6	5	4	3	2

A 3 B 3.5 C 4.3 D 4.5 E 5

Solution

$$\text{Mean} = \frac{\sum fx}{\sum f}$$

Table for mean showing x, f and fx

Scores(x)	Frequency (f)	fx
1	3	3
2	4	8
3	3	9
4	6	24
5	5	25
6	4	24
7	3	21
8	2	16
	$\Sigma f = 30$	$\Sigma fx = 130$

Substituting

$$\text{Mean} = \frac{130}{30} \text{ i.e } 4.3 \quad (\text{ C })$$

2011/59 Neco

The table below shows the scores of certain number of students in a class.

Scores	2	4	5	6	7
No of students	5	3	6	4	2

59. Find the mean scores

A 4.8 B 4.5 C 4.0 D 3.8 E 3.5

Solution

$$\text{Mean} = \frac{\sum fx}{\sum f}$$

Table for mean showing x, f and fx

Scores (x)	Frequency (f)	fx
2	5	10
4	3	12
5	6	30
6	4	24
7	2	14
	$\Sigma f = 20$	$\Sigma fx = 90$

Substituting

$$\text{Mean} = \frac{90}{20} \text{ i.e } 4.5 \quad (\text{ B })$$

2005/44 (Nov)

The mean of the frequency distribution shown is 1.5. Find the value of p

x - value	0	p	3
Frequency	2	3	1

Solution

$$\text{Mean} = \frac{\sum fx}{\sum f}$$

Table for mean showing x, f and fx

x - value	frequency(f)	fx
0	2	0
p	3	3p
3	1	3
	$\Sigma f = 6$	$\Sigma fx = 3 + 3p$

Substituting

$$1.5 = \frac{3+3p}{6}$$

$$1.5 \times 6 = 3 + 3p$$
$$9 = 3 + 3p$$
$$9 - 3 = 3p$$
$$6 = 3p$$
Thus, $p = 2$ (D)

2010/58 Neco

The table shows the distribution of scores of students in a test. If the mean score is 3.5, find the value of x

Scores	1	2	3	4	5	6
Frequency	1	4	x	6	2	2

A 1 B 2 C 3 D 4 E 5

Solution

$$\text{Mean} = \frac{\sum fx}{\sum f}$$

Table for mean showing x, f and fx

Scores (x)	Frequency (f)	fx
1	1	1
2	4	8
3	x	3x
4	6	24
5	2	10
6	2	12
	$\Sigma f = 15 + x$	$\Sigma fx = 55 + 3x$

Substituting

$$3.5 = \frac{55+3x}{15+x}$$
$$3.5(15 + x) = 55 + 3x$$
$$52.5 + 3.5x = 55 + 3x$$
$$3.5x - 3x = 55 - 52.5$$
$$0.5x = 2.5$$
Thus, $x = 5$ (E)

2008/57 Neco Exercise 8.4

Scores	0	1	2	3	4	5
No of students	y	12	10	6	5	8

The table above shows the test scores of students in a class. If the average score is 2.2, find the value of y.

A 13 B 11 C 10 D 9 E 7

2013/57 Neco Exercise 8.5

The table below shows the distribution of some scores. Find the mean

Scores	1	2	3	4	5	6
Frequency	3	3	0	4	5	5

A 1.0 B 1.1 C 1.5 D 4.0 E 5.5

Median

The median of a set of numbers arranged in an **array** is the middle number (if their total number is odd) or the arithmetic mean of the two middle numbers (if the number of the given values is even). Median divides the data into two equal parts.

E.g 1 Find the median of the following

(a) 6, 1, 0, 3, 10, 2, 5

(b) 5, 7, 11, 9, 15, 12, 18, 6

Solution

Arranging the value in an array, we have:

(a) 0, 1, 2, 3, 5, 6, 10

The given values is odd, thus median = 3 (middle)

(b) 5, 6, 7, 9, 11, 12, 15, 18

The given values is even, thus

$$\text{Median} = \frac{9 + 11}{2} \left\{ \begin{matrix}\text{Arithmetic mean of the two} \\ \text{middle values}\end{matrix} \right\}$$
$$= 10$$

1995/48

Find the median of the following numbers:
2.64, 2.50, 2.72, 2.91 and 2.35

A2.91 B 2.71 C 2.64 D 2.50 E 2.35

Solution

First, we arrange the given data in ascending order: 2.35, 2.50, 2.64, 2.72, and 2.91

Since there are 5 numbers (odd)

Median = 2.64 (middle number)

2011/60 Neco

Calculate the median of the following set of numbers:
3, 2, 6, 8, 9, 6, 8, 12, 11, 12

A 9 B 8 C 7 D 6 E 3

Solution

First, we arrange the given data in ascending order:
2, 3, 6, 6, 8, 8, 9, 11, 12, 12

Since there are 10 numbers (even)

$$\text{Median} = \frac{8+8}{2} \text{ (Arithmetic mean of the two middle values)}$$
$$= 8$$

2009/29

What is the median of the following scores
22, 41, 35, 63, 82, 74?

A 82 B 52 C 49 D 22

Solution

First, we arrange the given data in ascending order:
22, 35, 41, 63, 74, 82

Since there are 6 scores (even)

$$\text{Median} = \frac{41+63}{2} \text{ (Arithmetic mean of the two middle values)}$$
$$= 52 \quad (B)$$

2007/53 Neco Exercise 8.6

Find the median of 16, 20, 8, 14, 12, 16, 10, 12, 7, 16

A 12 B 13 C 14 D 15 E 16

2014/28 Neco (Nov) Exercise 8.7

Calculate the median of the following set of numbers:
6, 8, 3, 2, 10, 6, 11, 9, 12, 4

A 3 B 6 C 7 D 8 E 9

Median (frequency distribution)

To get median for frequency data, we apply the idea of **odd** and **even** numbers to get the *median class* or *column* among other columns or classes. $\sum f$ is another important term for us to know, which is " Summation of frequency". The two ways of getting our median class or column are:

(a) Median class (column) $= \frac{\sum f}{2}$ if $\sum f$ is even

(b) Median class (column) $= \frac{\sum f + 1}{2}$ if $\sum f$ is odd

After we have gotten our median class, then keep adding the frequency of the given data till we get to the class or column where our number falls in – that position is our median. Examples on this will help us better

2005/12 (Nov)

A number of families were asked how many children they had. The results are as follows:

No of children	1	2	3	4	5	6
Frequency	10	15	8	6	4	7

(a) Calculate the: (ii) median

Solution

$\sum f = 10 + 15 + 8 + 6 + 4 + 7$

$= 50$ is *even*

Median class $= \frac{\sum f}{2}$

$= \frac{50}{2}$ i.e 25th class

Adding frequency cumulatively till we get to 25^{th} class:

10 + 15 i.e 2^{nd} column

Thus median = 2

2011/11

The table shows the scores obtained when a fair die was thrown a number of times

Scores	1	2	3	4	5	6
Frequency	2	5	x	11	9	10

If the probability of obtaining a 3 is 0.26,

(a) find the median

Solution

First, we find x by principle of probability

$$\text{Prob of 3} = \frac{freq(3)}{\sum f}$$

$$0.26 = \frac{x}{2+5+x+11+9+10}$$

$$0.26 = \frac{x}{37+x}$$

$$0.26(37 + x) = x$$

$$9.62 + 0.26x = x$$

$$9.62 = 1x - 0.26x$$

$$9.62 = 0.74x$$

$$\frac{9.62}{0.74} = x$$

$$x = 13$$

(a) Median working

$\sum f$ is 2 + 5 +13 +11 + 9 +10 i.e 50 which is even

Median class $= \frac{\sum f}{2}$

$= \frac{50}{2}$ i.e 25^{th} class

Adding frequency cumulatively till we get to 25^{th} class:

2 + 5 +13 +11 i.e 4^{th} column,

Thus, median = 4

2007/12a NABTEB

The distribution of the daily wages in ₦ 100.00 of some workers on a farm is as given below:

Wages	2	3	4	5	6	8	10
No of workers	2	4	10	11	15	10	3

(a) How many workers are on the farm?

(b) Calculate the (ii) median wage

Solution

(a) $\sum f = 2 + 4 + 10 + 11 + 15 + 10 + 3$

$= 55$

(ii) ***Median wage working***

$\sum f$ is 55 which is **odd**

Median class $= \frac{\sum f + 1}{2}$

$= \frac{55+1}{2}$

$= \frac{56}{2}$ i.e 28^{th} class

Adding frequency cumulatively to 28^{th} class:

2 + 4 +10 +11 +15 i.e 5^{th} column

Thus median = 6

= ₦ 600.00 since wages is in ₦ 100.00

Mode

The mode of a set of numbers is that value which occurs with greatest frequency. i.e. it is the most common value. The mode may not exist or it may not be unique (when we have more than one mode)

2009/29 NABTEB (Nov)
What is the mode of the scores below:
1, 2, 3, 4, 4, 5, 5, 5, 4, 2, 2, 3, 4, 5, 5, 6
A 1 B 3 C 4 D 5

Solution

Mode = 5 it occurred most (D)

2009/54 Neco (Nov)
The data below are the ages of a set of pupils
7, 9, 6, 10, 8, 8, 9, 11, 8, 7, 9, 6, 8, 9, 10, 7, 8, 7, 9, 8.
What is the mode?
A 10 B 9 C 8 D 7 E 6

Solution

Mode = number that occurred most
= 8 (C)

2009 /28 NABTEB (Nov) Exercise 8.8
A type of average that shows the most popular item is
A mean B median C mode D variances

Mode of frequency distribution

2009/16 NABTEB (Nov)
Determine the mode of the distribution below

x	2	6	10
f	5	10	9

A 5 B 6 C 9 D 10

Solution

Mode = number that occurred most
= 6 (B) it has highest frequency of 10

Measures of Spread (variation/dispersion)

This is the degree to which data are scattered (dispersed) about its average value. The several measures of spread are the range, mean deviation, inter quartile range, variance and standard deviation

2014/30 Neco (Nov)
Which of the following is **not** a measure of dispersion?
A interquartile range B mean C mean deviation
D range E standard deviation

Solution

Mean is **not** a measure of dispersion (B)

2002/26 (Nov) Exercise 8.9
Which of the following is / are not the measure(s) of dispersion?
I Range
II Variance
III mode
A 1 only B. II only C. III only D. 1 and II only.

Range

This is the simplest measure of dispersion and it is defined as the difference between the largest and smallest numbers in the set. Please **Note that** range is a poor measure of spread as it uses only two extreme values.

2008/50 Neco
The ages of 10 students in a class are:
15, 16, 15.5, 17, 14.9, 14.5, 14.1, 15.1, 15.2, 14.8
Find the range of their ages.
A 6.1 B 4.8 C 2.9 D 2.1 E 1.9

Solution

Range = biggest number – smallest number
= 17 – 14.1
= 2.9 (C)

2013/54 Neco
A group of numbers are written in ascending order of magnitude as $(x - 2)$, 8, $(5 + x)$, 12, $(x + 14)$. Find the range of the number.
A 12 B 14 C 16 D 18 E 20

Solution

Range = biggest number – smallest number
$= (x + 14) - (x - 2)$
In ascending order, the first number is the smallest while the last one is the biggest
$= x + 14 - x + 2$
= 16 (C)

2015/60 Neco
The table below shows the frequencies (f) of some children of age x years.

x	1	2	3	5	6	7	8
f	3	4	5	7	6	5	4

Find the range of their ages
A 1 B 3 C 4 D 7 E 8
Range = biggest number – smallest number
= 8 – 1
= 7 (D)

Interquartile and semi-interquartile range

Quartiles

This is the division of the cumulative frequency Into **4** portions.

(Lower quartiles) $Q_1 = \frac{1}{4}$ of cum freq,

(Middle quartile) $Q_2 = {}^2/_4$ of cum freq

(Upper quartile) $Q_3 = \frac{3}{4}$ of cum. freq.

Inter quartile range = $Q_3 - Q_1$

Semi – inter quartile range = $\dfrac{Q_3 - Q_1}{2}$

2007/55 Neco

The marks of a certain number of students in a class are 6, 7, 8, 12, 15, 8, 9, 5, 28, 15, 17, 21.

What is the semi – interquartile range?

A 4 B 5 C 6.5 D 8 E 15

Solution

First, we arrange the data in the table form (group)

Marks	5	6	7	8	9	12	15	17	21	28
Freq	1	1	1	2	1	1	2	1	1	1
Cum freq	1	2	3	5	6	7	9	10	11	12

Semi – interquartile range = $\dfrac{Q_3 - Q_1}{2}$

$= \dfrac{\frac{3}{4} \times 12 - \frac{1}{4} \times 12}{2}$

$= \dfrac{9th\ position - 3rd\ position}{2}$

$= \dfrac{15 - 7}{2}$

$= \dfrac{8}{2}$ i.e 4 (A)

2013/51 Neco

The table below gives the shoe sizes of 12 students in a class

Size	4	6	7	9	14
No of students	3	2	4	2	1

Find the semi – interquartile range

A 0.7 B 1.3 C 1.5 D 1.8 E 2.8

Solution

Semi – interquartile range = $\dfrac{Q_3 - Q_1}{2}$

$= \dfrac{\frac{3}{4}\ of\ cummulative\ freq - \frac{1}{4}\ of\ cum.freq}{2}$

Size	No of students (f)	Cumulative freq
4	3	3
6	2	3 + 2 = 5
7	4	5 + 4 = 9
9	2	9 + 2 = 11
14	1	11 + 1 = 12

Semi – interquartile range = $\dfrac{\frac{3}{4} \times 12 - \frac{1}{4} \times 12}{2}$

$= \dfrac{9th\ position - 3rd\ position}{2}$

$= \dfrac{7 - 4}{2} = {}^3/_2$ i.e 1.5 (C)

2014/11b (Nov) Exercise 8.10

The scores of students were recorded as follows

4	2	1	6	5
3	5	6	1	2
1	5	5	6	3
4	3	5	1	5

(i) Construct a frequency distribution table

(ii) Calculate the inter-quartile range

2015/27 Exercise 8.11

The scores of twenty students in a test are as follows: 44, 47, 48, 49, 50, 51, 52, 53, 53, 54, 58, 59, 60, 61, 63, 65, 67, 70, 73, 75. Find the third quartile.

A.62 B.63 C.64 D.65

Mean deviation (M.D)

The mean deviation of a set of n numbers $x_1, x_2, \ldots x_n$ is defined as:

M.D = $\dfrac{\sum|x - \bar{x}|}{n}$

Where $\bar{x}$ is the mean of the numbers and $|x - \bar{x}|$ is the absolute value of the deviation of x from the mean ($\bar{x}$) and n is the number of digits given

2010/57 Neco

Find the mean deviation of 2, 4, 6, 5 and 3

A 1.2 B 2.3 C 4.0 D 5.2 E 10

Solution

Mean deviation = $\dfrac{\sum|x - \bar{x}|}{n}$ But $\bar{x}$ is mean

and here mean = $\dfrac{2+4+6+5+3}{5}$ i.e 4

Mean deviation table

x	$x - \bar{x}$	$\lvert x - \bar{x} \rvert$
2	– 2	2
4	0	0
6	2	2
5	1	1
3	– 1	1
		$\Sigma \lvert x - \bar{x} \rvert = 6$

M.D = $\dfrac{6}{5}$ i.e 1.2 (A)

2002/ 38 (Nov)

Calculate the mean deviation of the numbers

0, –1, –3, 4, 5, 1

A 0 B 2 C ${}^7/_3$ D $2\,{}^4/_5$

Solution

Mean deviation = $\dfrac{\sum|x - \bar{x}|}{n}$ But $\bar{x}$ is mean

and here mean = $\dfrac{0+(-1)+(-3)+4+5+1}{6} = \dfrac{6}{6}$ i.e 1

Mean deviation table

x	$x-\bar{x}$	$\lvert x-\bar{x}\rvert$
0	– 1	1
– 1	– 2	2
– 3	– 4	4
4	3	3
5	4	4
1	0	0
		$\Sigma \lvert x-\bar{x}\rvert = 14$

$$M.D = \frac{14}{6} \text{ i.e } 2.33$$

2008/52 Neco Exercise 8.12
Find the mean deviation of 20, 25, 21, 27, 28, 29, to the nearest whole numbers
A 2 B 3 C 4 D 5 E 6

1999/41 (Nov) Exercise 8.13
Calculate the mean deviation of the numbers 6, 7, 8, 9, 10
A 0 B 1.2 C 1.5 D 4

2013/56 Neco Exercise 8.14
Find the mean deviation of the numbers 2, 3, 5, 6
A 1.0 B 1.2 C 1.4 D 1.5 E 1.6

Mean Deviation (frequency Distribution)

If $x_1, x_2, \ldots x_k$ occur with frequency $f_1, f_2, \ldots f_k$ then the mean deviation is given by

$$\text{Mean deviation} = \frac{\sum f\lvert x-\bar{x}\rvert}{\sum f} \quad \text{where } \bar{x} \text{ is mean}$$

2014/11

Scores	1	2	3	4	5	6
Frequency	2	5	13	11	9	10

The table shows the distribution of outcomes when a die is thrown 50 times. Calculate the: (a) mean deviation

Solution

$$\text{Mean deviation} = \frac{\sum f\lvert x-\bar{x}\rvert}{\sum f} \quad \text{where } \bar{x} \text{ is mean}$$

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{1\times2+2\times5+3\times13+4\times11+5\times9+6\times10}{50}$$

$$= \frac{200}{50} \text{ i.e } 4$$

Mean deviation table

x	$x-\bar{x}$	$\lvert x-\bar{x}\rvert$	f	$f\lvert x-\bar{x}\rvert$
1	– 3	3	2	6
2	– 2	2	5	10
3	– 1	1	13	13
4	0	0	11	0
5	1	1	9	9
6	2	2	10	20
				$\Sigma f\lvert x-\bar{x}\rvert = 58$

$$M.D = \frac{58}{50} \text{ i.e } 1.16$$

Variance & Standard deviation

The variance of a set of numbers $x_1, x_2 \ldots x_k$ is given as

$$V = \frac{\sum (x-\bar{x})^2}{n}$$

Where V is the variance and $(x-\bar{x})^2$ is the square of the deviation from the mean while **standard deviation** is the **square root** of variance.

$$\textbf{S.D} = \sqrt{\frac{\sum (x-\bar{x})^2}{n}}$$

If the numbers $x_1, x_2 \ldots x_k$ occur with frequency $f_1, f_2 \ldots, f_k$ respectively, then,

$$\text{Variance } V = \frac{\sum f(x-\bar{x})^2}{\sum f}$$

$$S.D = \sqrt{\frac{\sum f(x-\bar{x})^2}{\sum f}}$$

Eg1. Find the variance and standard deviation of the set of numbers, 2, 5, 6, 3 and 4

Solution

$$\text{Variance} = \frac{\sum (x-\bar{x})^2}{n}$$

$$\text{But mean} = \frac{20}{5} = 4$$

x	$x-\bar{x}$	$(x-\bar{x})^2$
2	-2	4
5	1	1
6	2	4
3	-1	1
4	0	0
		$\sum (x-\bar{x})^2 = 10$

$$\text{Var} = \frac{10}{5} = 2$$

$$S.D = \sqrt{\frac{\sum (x-\bar{x})^2}{n}} = \sqrt{2} = 1.4$$

Eg 2. Calculate the variance and standard deviation of the frequency distribution below.

x	1	2	3	4	5
f	2	1	2	1	2

Solution

$$\text{Variance} = \frac{\sum f(x-\bar{x})^2}{\sum f}$$

$$\text{But mean} = \frac{2\times1+1\times2+2\times3+1\times4+2\times5}{2+1+2+1+2}$$

$$= \frac{24}{8} = 3$$

x	$x-\bar{x}$	$(x-\bar{x})^2$	f	$f(x-\bar{x})^2$
1	-2	4	2	8
2	-1	1	1	1
3	0	0	2	0
4	1	1	1	1
5	2	4	2	8
			8	18

$$V = \frac{\sum f(x-x)^2}{\sum f}$$

$= {}^{18}/_{8}$

$= 2.25$

and S.D is square root of variance $\sqrt{2.25} = 1.5$

2002/3
The variance of given distribution is 25. what is the standard deviation?
A. 125 B. 75 C. 25 D . 5

Solution

Standard deviation is the square root of variance.

Thus SD = $\sqrt{variance}$

SD = $\sqrt{25}$ = 5 (D)

2013/55 Neco
Calculate the variance of the following set of numbers below: 5, 11, 13, 14, 17
A 5.3 B 7.3 C 8.0 D 10.3 E 16.0

Solution

Variance $= \frac{\sum\left(x - \bar{x}\right)^2}{n}$ But $\bar{x}$ is mean

here mean $\bar{x} = \frac{5+11+13+14+17}{5}$ i.e 12

Table for variance

x	$x - \bar{x}$	$\left(x - \bar{x}\right)^2$
5	– 7	49
11	– 1	1
13	1	1
14	2	4
17	5	25
		$\Sigma\left(x - \bar{x}\right)^2 = 80$

Variance = $\frac{80}{5}$ i.e 16.0 (E)

2012/27
Given that the mean of the scores 15, 21, 17, 26, 18 and 29 is 21, calculate the standard deviation of the scores
A $\sqrt{10}$ B 4 C 5 D $\sqrt{30}$

Solution

SD = $\sqrt{\frac{\sum\left(x - \bar{x}\right)^2}{n}}$ But $\bar{x}$ is mean i.e 21 given

Table for SD

x	$x - \bar{x}$	$\left(x - \bar{x}\right)^2$
15	– 6	36
21	0	0
17	– 4	16
26	5	25
18	– 3	9
29	8	64
		$\Sigma\left(x - \bar{x}\right)^2 = 150$

SD = $\sqrt{\frac{150}{6}}$

$= \sqrt{25}$ = 5 (C)

2014/19 NABTEB (Nov)
The standard deviation of the numbers 6, 0, 4, 3 and 2 is
A 2 B 1 C 1.5 D 2.2

Solution

SD = $\sqrt{\frac{\sum\left(x - \bar{x}\right)^2}{n}}$ But $\bar{x}$ is mean

and here mean $\bar{x} = \frac{6+0+4+3+2}{5}$ i.e 3

Table for SD

x	$x - \bar{x}$	$\left(x - \bar{x}\right)^2$
6	3	9
0	– 3	9
4	1	1
3	0	0
2	–1	1
		$\Sigma\left(x - \bar{x}\right)^2$ =20

SD = $\sqrt{\frac{20}{5}}$ = 2 (A)

2013/50 Neco
Calculate the standard deviation of the following set of numbers: 2, 3, 4, 4, 5, 6
A 1.29 B 1.35 C 2.04 D 2.50 E 3.56

Solution

SD = $\sqrt{\frac{\sum\left(x - \bar{x}\right)^2}{n}}$ But $\bar{x}$ is mean

here mean $\bar{x} = \frac{2+3+4+4+5+6}{6}$ i.e 4

Table for SD

x	$x - \bar{x}$	$\left(x - \bar{x}\right)^2$
2	– 2	4
3	– 1	1
4	0	0
4	0	0
5	1	1
6	2	4
		$\Sigma\left(x - \bar{x}\right)^2 = 10$

SD = $\sqrt{\frac{10}{6}}$ = 1.29 (A)

2011/11b
The table shows the scores obtained when a fair die was thrown a number of times

Scores	1	2	3	4	5	6
Frequency	2	5	x	11	9	10

If the probability of obtaining a 3 is 0.26,
(b) standard deviation of the distribution

Solution

First, we find x by principle of probability

Prob of 3 $= \frac{freq(3)}{\sum f}$

$0.26 = \frac{x}{2+5+x+11+9+10}$

$0.26 = \frac{x}{37+x}$

$$0.26(37 + x) = x$$
$$9.62 + 0.26x = x$$
$$9.62 = 1x - 0.26x$$
$$9.62 = 0.74x$$
$$\frac{9.62}{0.74} = x$$
$$x = 13$$

(b) $\text{S.D} = \sqrt{\frac{\sum f\left(x-\bar{x}\right)^2}{\sum f}}$

But $\bar{x} = \frac{1\times2+2\times5+3\times13+4\times11+5\times9+6\times10}{2+5+13+11+9+10}$

$= \frac{2+10+39+44+45+60}{50} = \frac{200}{50}$ i.e 4

Table for SD

x	$x-\bar{x}$	$\left(x-\bar{x}\right)^2$	f	$f\left(x-\bar{x}\right)^2$
1	– 3	9	2	18
2	– 2	4	5	20
3	– 1	1	13	13
4	0	0	11	0
5	1	1	9	9
6	2	4	10	40
			$\Sigma f =$ 50	$\Sigma f(x-\bar{x})^2$ =100

$\text{SD} = \sqrt{\frac{100}{50}}$

$= \sqrt{2} = 1.4$

2005/44 Neco Exercise 8.15
Calculate the standard deviation of 5, 5, 6, 6, 8
A 8.0 B 6.0 C 3.3 D 1.2 E 1.1

2006/51 Neco Exercise 8.16
Find the standard deviation of 3, 4, 5, 7, 6
A 1.41 B 2.00 C 4.00 D 4.47 E 5.00

2007/54 Neco Exercise 8.17
Find the standard deviation of 2, 5, 9, 2, 7
(correct to 2 d.p)
A 5.80 B 3.41 C 2.76 D 1.80 E 1.34

2009/56 Neco (Nov) Exercise 8.18
Calculate the standard deviation of the following set of numbers: 4, 5, 6, 7, 8
A 1.4 B 2.0 C 2.5 D 2.9 E 6.0

Miscellaneous cases involving measures of spread & location

2009/55 Neco (Nov)
If the mean and median of 170, 230, y, 215 and 235 are 210 and x respectively, find the values of x and y
A 215 and 215 B 215 and 210 C 215 and 200
D 210 and 200 E 210 and 215

Solution

$\text{Mean} = \frac{\sum x}{n}$

$210 = \frac{170+230+y+215+235}{5}$

$210 \times 5 = 850 + y$
$1050 = 850 + y$
$1050 - 850 = y$
$200 = y$

Median's working
Arranging the data in ascending order:
170, 200, 215, 230, 235 (odd)
Median = 215 i.e x

Thus, x and y are 215 and 200 (C)

2009/53 Neco (Nov)
If the mean of 2, 1, 2, 3, 1, 3, 4, 4, 5, 5 subtracted from the range the result is
A 1 B 2 C 3 D 4 E 5

Solution

$\text{Mean} = \frac{\sum x}{n}$

$= \frac{2+1+2+3+1+3+4+4+5+5}{10} = \frac{30}{10}$ i.e 3

Range = 5 – 1 i.e 4

Thus, range – mean = 4 – 3
= 1 (A)

2014/32 to 34 Neco (Nov)
The percentage scores of ten students in an examination are 20, 60, 50, 80, 70, 90, 20, 20, 60, and 30
Use the information to answer questions **32** to **34**

32. Find the mean score
A 30 B 50 C 60 D 70 E 80

Solution

$\text{Mean} = \frac{\sum x}{n}$

$= \frac{20+60+50+80+70+90+20+20+60+30}{10}$

$= \frac{500}{10}$ i.e 50 (B)

33. What is the median score?
A 50 B 55 C 60 D 70 E 80

Solution

First, we arrange the given data in ascending order:
20, 20, 20, 30, 50, 60, 60, 70, 80, 90
Since there are 10 scores (even)

Median = $\frac{50+60}{2}$ (Arithmetic mean of the two middle values)

$= \frac{110}{2}$ i.e 55 (B)

34. What is the modal score?

A 20 B 30 C 35 D 50 E 55

Solution

Mode = 20 (A) the number that occurred most

2008/51 Neco

Marks	1	2	3	4	5	6
Frequency	8	11	13	14	9	5

What is the sum of the median and mode of the above distribution?

A 5 B 7 C 8 D 9 E 10

Solution

<u>Median's working for frequency data</u>

Σf is 8 + 11 + 13 + 14 + 9 + 5 i.e 60

Median class = $\frac{\Sigma f}{2}$

= $\frac{60}{2}$ i.e 30^{th} class

Adding frequency from the beginning till we get to the 30^{th} class: 8 + 11 + 13 i.e 3^{rd} column

Thus median = 3 mark

<u>Mode's working</u>

Mode = 4 (has highest frequency)

Thus, sum of median and mode = 3 + 4

= 7 (B)

2014/55 to 57

Use the set of data below to answer questions **55** to **57**.

22, 27, 34, 20, 42

55. Find the mean of the data

A 22 B 24 C 27 D 29 E 36

Solution

Mean = $\frac{\sum x}{n}$

= $\frac{22+27+34+20+42}{5}$

= $\frac{145}{5}$ i.e 29 (D)

56. Calculate the mean deviation

A 4.6 B 6.0 C 7.2 D 9.0 E 11.0

Solution

Mean deviation = $\frac{\sum\left|x-\bar{x}\right|}{n}$ But $\bar{x}$ is mean

The mean is gotten as *29* from question **55**

Table for mean deviation

x	$x-\bar{x}$	$\left\|x-\bar{x}\right\|$
22	– 7	7
27	– 2	2
34	5	5
20	– 9	9
42	13	13
		$\Sigma\left\|x-\bar{x}\right\| = 36$

Mean deviation = $\frac{36}{5}$ i.e 7.2 (C)

57. Calculate the standard deviation of the data, correct to one decimal place.

A 2.7 B 3.8 C 5.4 D 7.2 E 8.1

Solution

SD = $\sqrt{\frac{\Sigma\left(x-\bar{x}\right)^2}{n}}$ But $\bar{x}$ is mean

The mean is gotten as *29* from question **55**

Table for standard deviation

x	$x-\bar{x}$	$\left(x-\bar{x}\right)^2$
22	– 7	49
27	– 2	4
34	5	25
20	– 9	81
42	13	169
		$\Sigma\left(x-\bar{x}\right)^2 = 328$

SD = $\sqrt{\frac{328}{5}} = \sqrt{65.6} \approx 8.1$ to 1 d.p (E)

2011/55 – 56 Neco

Given the following scores: 2, 3, 4, 5, 6, use the information to answer questions 55 and 56

55. Find the standard deviation

A 2.0 B 1.4 C 1.3 D 1.2 E 1.1

Solution

SD = $\sqrt{\frac{\Sigma\left(x-\bar{x}\right)^2}{n}}$ But $\bar{x} = \frac{2+3+4+5+6}{5} = 4$

Table for standard deviation

x	$x-\bar{x}$	$\left(x-\bar{x}\right)^2$
2	– 2	4
3	– 1	1
4	0	0
5	1	1
6	2	4
		$\Sigma\left(x-\bar{x}\right)^2 = 10$

SD = $\sqrt{\frac{10}{5}} = \sqrt{2}$ i.e 1.4 (B)

56. Find the variance

A 1.20 B 1.44 C 1.69 D 1.96 E 2.00

Solution

SD = $\sqrt{Variance}$

Thus, Variance = (standard deviation)^2

= $(\sqrt{2})^2$ = 2 (E)

2012/2 Neco

A survey was carried out to investigate the number of eggs in some bird's nests. The table below shows the findings:

No of eggs	frequency
2	11
3	12
4	25
5	30
6	12
7	10

(a) Calculate the mean number of eggs in the nests

(b) Calculate the mean deviation

(a) **Solution**

Mean = $\frac{\sum fx}{\sum f}$

Table for mean showing x, f and fx

No of eggs (x)	Frequency (f)	fx
2	11	22
3	12	36
4	25	100
5	30	150
6	12	72
7	10	70
	$\sum f$ =100	$\sum fx = 450$

Mean = $\frac{450}{100}$ i.e 4.5

b. **Solution**

Mean deviation = $\frac{\sum f\left|x - \bar{x}\right|}{\sum f}$ where $\bar{x}$ is mean 4.5

Mean deviation table

x	$x - \bar{x}$	$\left\|x-\bar{x}\right\|$	f	f$\left\|x-\bar{x}\right\|$
2	– 2.5	2.5	11	27.5
3	– 1.5	1.5	12	18
4	– 0.5	0.5	25	12.5
5	0.5	0.5	30	15
6	1.5	1.5	12	18
7	2.5	2.5	10	25
			Σf = 100	Σf$\left\|x-\bar{x}\right\|$ = 116

Mean deviation = $\frac{116}{100}$ i.e 1.16

2010/59 Neco

The table gives the scores of a group of students in a mathematics test. Use the information for question 59

Scores	2	3	4	5	6	8
No of students	3	4	2	7	2	2

59. What is the sum of the mode and the median?

A 5 B 7 C 10 D 12 E 20

Solution

Mode = **5** (it has the highest frequency)

Median's working

Σf is 20 even

Median class = $\frac{\sum f}{2}$

= $\frac{20}{2}$ i.e 10^{th} class

Adding frequency from the beginning till we get to 10^{th} class: 3 + 4 + 2 + 7 i.e 4^{th} column

Thus, median = **5**

Sum of mode and median = 5 + 5

= 10 (C)

2014/ 9b NABTEB (Nov)

The figure below shows the number of goals scored by a group of players in a match

No of goals	0	1	2	3	4	5	6
Frequency	26	29	14	15	5	6	3

Calculate the: (i) mean (ii) mode

(iii) standard deviation of the number of goals

Solution

(i) Mean = $\frac{\sum fx}{\sum f}$

Table for mean showing x, f and fx

No of goals (x)	Frequency (f)	fx
0	26	0
1	29	29
2	14	28
3	15	45
4	5	20
5	6	30
6	3	18
	$\sum f = 98$	$\sum fx = 170$

Substituting

Mean = $\frac{170}{98}$ = 1.73 ≈ 2 goals

(*There is nothing like 1.73 goals*)

(ii) Mode = 1

(iii) S.D = $\sqrt{\frac{\sum f\left(x - \bar{x}\right)^2}{\sum f}}$

Table for SD

x	$x - \bar{x}$	$\left(x - \bar{x}\right)^2$	f	f$\left(x - \bar{x}\right)^2$
0	– 2	4	26	104
1	– 1	1	29	29
2	0	0	14	0
3	1	1	15	15
4	2	4	5	20
5	3	9	6	54
6	4	16	3	48
			Σf = 98	Σf$(x-\bar{x})^2$ = 270

SD = $\sqrt{\frac{270}{98}}$ = $\sqrt{2.755}$ = 1.66

We left it in decimals since it is a deviation of the number of goals and not number of goals.

2005/12 (Nov)

A number of families were asked how many children they had. The results are as follows:

No of children	1	2	3	4	5	6
Frequency	10	15	8	6	4	7

(a) Calculate the: (i) mean

(b) Find the standard deviation of the distribution correct to two decimal places

Solution

(**i**) Mean = $\dfrac{\sum fx}{\sum f}$

Table for mean showing x, f and fx

No of children (x)	f	fx
1	10	10
2	15	30
3	8	24
4	6	24
5	4	20
6	7	42
	$\sum f = 50$	$\sum fx = 150$

Mean = $\dfrac{150}{50}$ i.e 3

(**b**) SD = $\sqrt{\dfrac{\sum f\left(x-\bar{x}\right)^2}{\sum f}}$ but $\bar{x} = 3$

Table for SD

x	$x-\bar{x}$	$\left(x-\bar{x}\right)^2$	f	$f\left(x-\bar{x}\right)^2$
1	– 2	4	10	40
2	– 1	1	15	15
3	0	0	8	0
4	1	1	6	6
5	2	4	4	16
6	3	9	7	63
			Σf = 50	$\Sigma f(x-\bar{x})^2 =$ 140

SD = $\sqrt{\dfrac{140}{50}}$

= $\sqrt{2.8}$ ≈ 1.67 to 2dp

2014/38 – 40 NABTEB (Nov)

The data below shows the frequency distribution of marks scored by a group of students in a class test. Use the information to answer questions 38 – 40

Score	2	3	4	5	6
Frequency	2	4	5	3	1

38. How many students took the test?

A 13 B 14 C 15 D 18

Solution

Σf = 2 + 4 + 5 + 3 + 1

= 15 (C)

39. What is the modal mark?

A 2 B 3 C 4 D 5

Solution

Mode = 4 (C)

40. Find the mean mark

A 2 B 1 C 3.8 D 1.3

Solution

Mean = $\dfrac{\sum fx}{\sum f}$

$= \dfrac{2\times2+3\times4+4\times5+5\times3+6\times1}{2+4+5+3+1}$

$= \dfrac{4+12+20+15+6}{15} = \dfrac{57}{15}$ i.e 3.8 (C)

2010/5 Neco

The table below shows the distribution of scores in percentage obtained by 20 students in a class test.

Scores	20	25	30	35	40	45	50
No of students	5	3	1	6	2	1	2

b. (i) Find the mode (ii) median of the distribution

c. Calculate the mean score

Solution

b. (i) Mode = 35 (it has the highest frequency)

(ii) Median's working

Σf is 20 which is **even**

Median class = $\dfrac{\sum f}{2}$

= $\dfrac{20}{2}$ i.e 10th class

Adding frequency cumulatively till we get to 10th class:

5 + 3 + 1 + 6 i.e 4th column

Thus median = 35

C. Mean's working

Mean = $\dfrac{\sum fx}{\sum f}$

Table for mean showing x, f and fx

Score (x)	Frequency (f)	fx
20	5	100
25	3	75
30	1	30
35	6	210
40	2	80
45	1	45
50	2	100
	Σf = 20	Σfx = 640

Substituting

Mean = $\dfrac{640}{20}$ i.e 32

2006/10

The following table shows the distribution of test scores in a class

Scores	1	2	3	4	5	7	8	9	10
No of pupils	1	1	5	3	k^2+1	6	2	3	4

(a) If the mean score of the class is 6, find the

(i) value of k (ii) median score

Solution

(i) Mean = $\dfrac{\sum fx}{\sum f}$

$6 = \dfrac{1\times1+2\times1+3\times5+4\times3+5(k^2+1)+7\times6+8\times2+9\times3+10\times4}{1+1+5+3+(k^2+1)+6+2+3+4}$

$6 = \dfrac{1+2+15+12+5k^2+5+42+16+27+40}{25+k^2+1}$

$6 = \dfrac{160+5k^2}{26+k^2}$

$6(26 + k^2) = 160 + 5k^2$

$156 + 6k^2 = 160 + 5k^2$

$6k^2 – 5k^2 = 160 – 156$

$k^2 = 4$

$k = 2$

Thus frequency $k^2 + 1$ **is** $2^2 + 1 =$ **5**

(ii) Median's score working

Σf is 30 which is **even**

Median class = $\frac{\sum f}{2}$

$= \frac{30}{2}$ i.e 15th class

Adding frequency cumulatively till we get to 10th class:

1 + 1 + 5 + 3 + 5 i.e 5th column

Thus median = 5

2006/5a Neco (Nov)

Age	11	12	13	14	15
Frequency	5	2	x – 1	x + 1	3

The table above gives the frequency distribution ages of students in a class. If the mean age is 13 years, find the: (i) value of x (ii)median age and (iii)modal age

Solution

(i) Mean = $\frac{\sum fx}{\sum f}$

$$13 = \frac{11\times5+12\times2+13(x-1)+14(x+1)+15\times3}{5+2+(x-1)+(x+1)+3}$$

$$13 = \frac{55+24+13x-13+14x+14+45}{10+x-1+x+1}$$

$$13 = \frac{125+27x}{10+2x}$$

13(10 + 2x) = 125 + 27x

130 + 26x = 125 + 27x

130 – 125 = 27x – 26x

5 = x

Thus x – 1 is 4 and x + 1 is 6

The new table

Age	11	12	13	14	15
Frequency	5	2	4	6	3

(ii) Median age

Σf is 5 + 2 + 4 + 6 + 3 i.e 20 **even**

Median class = $\frac{\sum f}{2}$

$= \frac{20}{2}$ i.e 10th class

Adding frequency from the beginning till we get to 10th class: 5 + 2 + 4 i.e 3rd column

Thus median = 13 age

(iii) Modal age is 14

2006/57 – 58 Neco (Nov)

Mass (kg)	55	56	57	58	59	60
No of students	4	7	12	10	8	9

The table above shows the masses to the nearest kilogram of 50 students in a class

57. What is the mean mass to the nearest kilogram?

A 56 B 57 C 58 60

Solution

Mean = $\frac{\sum fx}{\sum f}$

Table for mean showing x, f and fx

Mass(kg) x	No of students (f)	fx
55	4	220
56	7	392
57	12	684
58	10	580
59	8	472
60	9	540
	$\Sigma f = 50$	$\Sigma fx = 2888$

Mean = $\frac{2888}{50}$

= 57.76 ≈ 58 to the nearest kg (C)

58. The median of the distribution is

A 59kg B 58kg C 57kg D 56kg E 55 kg

Solution

Σf is 50 which is **even**

Median class = $\frac{\sum f}{2}$

$= \frac{50}{2}$ i.e 25th class

Adding frequency from the beginning till we get to 25th class: 4 + 7 + 12 +10 i.e 4th column

Thus, median = 58kg (B)

2005/48 Neco

The table below shows the scores of a set of students in a mathematics test. If x is the mode and y is the median, find (x, y)

Marks	4	5	6	7	8
Frequency	5	9	8	6	2

A (5, 5) B (6, 5) C (5, 6) D (8, 9) E (9, 8)

Solution

Mode = 5 (it has the highest frequency)

Median's working

Σf is 5 + 9 + 8 + 6 + 2 is 30 which is **even**

Median class = $\frac{\sum f}{2}$

$= \frac{30}{2}$ i.e 15th class

Adding frequency cumulatively to the 15th class:

5 + 9 + 8 i.e 3rd column

Thus median = 6

(x, y) is mode, median i.e (5, 6) C.

2007/12a NABTEB

The distribution of the daily wages in ₦ 100.00 of some workers on a farm is as given below:

Wages	2	3	4	5	6	8	10
No of workers	2	4	10	11	15	10	3

(a) How many workers are on the farm?

(b) Calculate the (i) mean wage

(iii) Modal wage.

Solution

(a) Σf = 2 + 4 + 10 + 11 + 15 + 10 + 3

= 55

(b) Mean = $\dfrac{\sum fx}{\sum f}$

$= \dfrac{2\times2+3\times4+4\times10+5\times11+6\times15+8\times10+10\times3}{2+4+10+11+15+10+3}$

$= \dfrac{4+12+40+55+90+80+30}{55}$

$= \dfrac{311}{55}$

= 5.65 = ₦ 565.00 (*wages in ₦ 100.00*)

(iii) Modal wage is 6 = ₦600(*wages in ₦ 100.00*)

2008/53 Neco (Nov)
Considering this set of numbers 3, 3, 4, 5, 6, 6, 6, 7. The modes, median and mean are respectively
A 5, 5.5, 6   B 5.5, 6, 5   C 6, 5, 5.5
D 5.5, 5, 6   E 6, 5.5, 5

Solution

Mode = 6

Median's working
The given data is already in ascending order:
3, 3, 4, 5, 6, 6, 6, 7 (8 of them even)

Median = $\dfrac{5+6}{2}$ (arithmetic mean of the two middle values)
= 5.5

Mean's working

Mean = $\dfrac{\sum x}{n}$

$= \dfrac{3+3+4+5+6+6+6+7}{8} = \dfrac{40}{8}$ i.e 5

Mode, median and mean: 6, 5.5, 5   (E)

2008/35 NABTEB (Nov)
The following are scores of 20 students in a mathematics test: 10, 2, 6, 5, 4, 4, 5, 6, 6, 4, 6, 1, 8, 9, 7, 7, 6, 6, 5, 6. Calculate the median and the mode of this distribution
A (6, 6)   B (6, 5)   C (6, 7)   D (5, 6)

Solution

Median's working
First, we arrange the given data in ascending order
1, 2, 4, 4, 4, 5, 5, 5, 6, 6, 6, 6, 6, 6, 6, 7, 7, 8, 9, 10
Since there are 20 numbers (even)

Median = $\dfrac{6+6}{2}$ (arithmetic mean of the two middle values)
= 6

Mode = 6

Thus median, mode is (6, 6)   A.

2002/12a NABTEB Exercise 8.19
The profits made by a hawker in 5 consecutive days of selling are: ₦ 8, ₦15, ₦20, ₦12, ₦15.
(i) What is the range of the profits?
(ii) Calculate, the mean, the median and the standard deviation of the profits

2005/47 Neco Exercise 8.20
Find the sum of the median and range of the following numbers 16, 13, 10, 11, 26, 9, 8, 38 and 14.
A 49   B 43   C 30   D 17   E 13

2005/8 Exercise 8.21
In a family survey, the numbers of children in some family are 4, 3, 1, 2, 4, 5, 1, 4, 3. By how much does the mode exceed the mean?
A 0   B 1   C 2   D 5

2006/12a NABTEB (Nov) Exercise 8.22
Below are amounts of money given to 15 students as gifts in a school, in naira?

2 3 7 5 3 9 5 6,
4 5 6 6 7 5 9

Calculate the (i) mean (ii) mode and
(iii) median, to the nearest ten kobo

2006/54 -56 Neco (Nov) Exercise 8.23
For a class of 30 students, the scores a mathematics test out of 10 marks are as follows:

4	5	7	2	3	6	5	5	8	9
5	4	2	3	7	9	8	7	7	7
3	4	5	5	2	3	6	7	7	2

54. What is the mode of the scores?
A 3   B 4   C 5   D 6   E 7

55. What is the median of the score?
A 3   B 4   C 5   D 6   E

56. What is the range of the distribution?
A 2   B 7   C 8   D 9   E 10

2015/16 Neco Exercise 8.24
Find the product of the median and the mode in the table below:

Score	0	1	2	3	4	5
Frequency	2	4	4	9	6	8

A.6   B.8   C.9   D.12   E.15

2015/20 to 22 Neco Exercise 8.25
Use the table below to answer questions **20** to **22**

Score	1	2	3	6	8	9
Frequency	4	3	5	7	2	1

20. Find the mean score
A 3.8   B 4.0   C 4.2   D 4.4   E 4.6

21. Find the median score
A 2   B 3   C 7   D 8   E 9

21. What is the modal score?
A 5   B 6   C 7   D 8   E 9

Moving average

This is a tool in statistics used to predict or forecast future results using past records. Any forecast is the average of the preceding **n** months' actual records

Example MA1

The number of books in thousands sold by OPL Sales Representative in the year 2014 is given in the table below:

Months	Jan	Feb	Mar	Apr	May	Jun
Books sold	450	440	460	410	380	400

July	Aug	Sep	Oct	Nov	Dec
370	360	410	450	470	490

Calculate three months moving averages

Solution

(i) **April** prediction = $\dfrac{Jan\ sale + Feb\ sales + Mar\ sales}{3}$

$= \dfrac{450+440+460}{3}$

$= \dfrac{1350}{3} = 450$

May prediction = $\dfrac{Feb\ sales + Mar\ sales + Apr\ sales}{3}$

$= \dfrac{440+460+410}{3}$

$= \dfrac{1310}{3} = 437$

June prediction = $\dfrac{Mar\ sales + Apr\ sales + May\ sales}{3}$

$= \dfrac{460+410+380}{3}$

$= \dfrac{1250}{3} = 417$

July prediction = $\dfrac{Apr\ sales + May\ sales + Jun\ sales}{3}$

$= \dfrac{410+380+400}{3}$

$= \dfrac{1190}{3} = 397$

August prediction = $\dfrac{May\ sales + Jun\ sales + July\ sales}{3}$

$= \dfrac{380+400+370}{3}$

$= \dfrac{1150}{3} = 383$

September prediction = $\dfrac{Jun\ sales + July\ sales + Aug\ sales}{3}$

$= \dfrac{400+370+360}{3}$

$= \dfrac{1130}{3} = 377$

October prediction = $\dfrac{July\ sales + Aug\ sales + Sept\ sales}{3}$

$= \dfrac{370+360+410}{3}$

$= \dfrac{1140}{3} = 380$

November prediction = $\dfrac{Aug\ sales + Sept\ sales + Oct\ sales}{3}$

$= \dfrac{360+410+450}{3}$

$= \dfrac{1220}{3} = 407$

December prediction = $\dfrac{Sept\ sales + Oct\ sales + Nov\ sales}{3}$

$= \dfrac{410+450+470}{3}$

$= \dfrac{1330}{3} = 443$

Example MA2

The number of books in thousands sold by OPL sales representative in the year 2014 is given in the table below:

Months	Jan	Feb	Mar	Apr	May	Jun
Books sold	450	440	460	410	380	400

July	Aug	Sep	Oct	Nov	Dec
370	360	410	450	470	490

Calculate the six monthly - moving averages

Solution

July prediction

$= \dfrac{Jan\ sale + Feb\ sales + Mar\ sales + Apil\ sales + May\ sales + Jun\ sales}{6}$

$= \dfrac{450+440+460+410+380+400}{6}$

$= \dfrac{2540}{6} = 423$

August prediction

$= \dfrac{Feb\ sales + Mar\ sales + Apil\ sales + May\ sales + Jun\ sales + July\ sales}{6}$

$= \dfrac{440+460+410+380+400+370}{6}$

$= \dfrac{2460}{6} = 410$

September prediction

$= \dfrac{Mar\ sales + Apil\ sales + May\ sales + Jun\ sales + July\ sale + Aug\ sales}{6} =$

$\dfrac{460+410+380+400+370+360}{6}$

$= \dfrac{2380}{6} = 397$

October prediction

$=$

$\dfrac{April\ sales + May\ sales + Jun\ sales + July\ sales + Aug\ sales + Sept\ sales}{6}$

$$= \frac{410+380+400+370+360+410}{6}$$

$$= \frac{2330}{6} = 388$$

November prediction =

$$\frac{May\,sales + Jun\,sales + July\,sales + Aug\,sales + Sept\,sale + Oct\,sales}{6}$$

$$= \frac{380+400+370+360+410+450}{6}$$

$$= \frac{2370}{6} = 395$$

December prediction =

$$\frac{Jun\,sales + July\,sales + Aug\,sales + Sept\,sale + Oct\,sales + Nov\,sales}{6}$$

$$= \frac{400+370+360+410+450+470}{6}$$

$$= \frac{2460}{6} = 410$$

2005/13 NABTEB

The prices of kerosene per litre on the first week of each of the 12 months of the year are as given in the table below

Months	Jan	Feb	Mar	Apr	May	Jun
Prices(₦)	18	21	25	30	40	52

July	Aug	Sep	Oct	Nov	Dec
48	50	55	43	26	18

Calculate the three – monthly moving averages for the period

Solution

April prediction = $\frac{Jan + Feb + March}{3}$

$$= \frac{18+21+25}{3}$$

$$= \frac{64}{3} = 21$$

May prediction = $\frac{Feb + March + Apirl}{3}$

$$= \frac{21+25+30}{3}$$

$$= \frac{76}{3} = 25$$

June prediction = $\frac{March + April + May}{3}$

$$= \frac{25+30+40}{3}$$

$$= \frac{95}{3} = 32$$

July prediction = $\frac{Apirl + May + June}{3}$

$$= \frac{30+40+52}{3}$$

$$= \frac{122}{3} = 41$$

August prediction = $\frac{May + June + July}{3}$

$$= \frac{40+52+48}{3}$$

$$= \frac{140}{3} = 47$$

September prediction = $\frac{June + July + August}{3}$

$$= \frac{52+48+50}{3}$$

$$= \frac{150}{3} = 50$$

October prediction = $\frac{July + August + Septeber}{3}$

$$= \frac{48+50+55}{3}$$

$$= \frac{153}{3} = 51$$

November prediction = $\frac{August + Sept + Oct}{3}$

$$= \frac{50+55+43}{3}$$

$$= \frac{148}{3} = 49$$

December prediction = $\frac{Sept\ sales + Oct\ sales + Nov\ sales}{3}$

$$= \frac{55+43+26}{3}$$

$$= \frac{124}{3} = 41$$

2008/12b NABTEB (Nov)

Calculation the 4 – point moving averages for the distribution: 17, 14, 11, 18, 15, 14, 13, 14

Solution

Point **15** prediction = $\frac{17+14+11+18}{4} = \frac{60}{4} = 15$

Point **14** prediction = $\frac{14+11+18+15}{4} = \frac{58}{4} = 14.5$

Point **13** prediction = $\frac{11+18+15+14}{4} = \frac{58}{4} = 14.5$

Point **14** prediction = $\frac{18+15+14+13}{4} = \frac{60}{4} = 15$

2002/15b NABTEB Exercise 8.26

Calculate the 4 – point moving average for the distribution: 25, 30, 32, 28, 27, 20, 16, 18 to two decimals

2004/12b NABTEB (Nov) Exercise 8.27

Calculate the 5 – point moving average for the distribution: 32, 31, 27, 28, 20, 18 and 16

CHAPTER NINE

SETS

Set is the collection of objects. (Objects mean numbers, things, places etc). Generally speaking, the above definition is acceptable but for precision;
* set is the collection of well defined objects.

Eg Suppose a new teacher is posted to your school to teach mathematics in SSII; he goes to the library to get some textbooks. Below is the Librarian response to his question. Mathematics teacher: Sir, where can I get SSII maths texts?

Librarian response; (I) You will get some in the mathematics shelves
(II) You will get some in shelf 2 of the mathematics shelves.

Of course, both responses are correct but statement II is precise. We usually denote sets by Capital letters while their members called **ELEMENTS** by lower case letters. Elements(members) of any set are usually enclosed in a curly bracket and frequently we may write for short.

{x : x satisfies a certain condition } or
{ x/x satisfies a certain condition}

Where the signs / or : means "such that"

Universal sets

This is the mother set from which other subsets emerge or relate and it is denoted by ξ or *U*

E.g. 1 ξ = {All SSIII students of Esi College, Warri}
These students include:
A = {All science students}
B = {All social science students}
C = {All arts students}
E.g. 2 ξ ={x : x is the set of all workers Living in Abuja}
Then the subsets will include:
S = {All civil servants}
T = {All taxi drivers}
V = {All private sector workers}

Singleton Sets

These are single element (member) sets
E.g. 1. A = {x : x is the principal of Ibru college Agbarha-Otor}

E.g. 2 B = {x : x is the president of Nigeria}

You will agree with the assertion that there is only one principal in a college and one president in a country at a given time.

Null or empty set

This type of set contains no element. It is the subset of every set and is denoted by $\varnothing$ or { }
E.g. W = {y : y is the set of a JSS one student who is a principal}
= $\varnothing$

Note : The set J = {0} is not an empty set because zero is an element.

Subsets

This is the child or children of a given universal set
E.g. 1 ξ = {a, b, c, d, e}
X = {a, b, c}
Y = {a, k, c, d}
X is a subset of ξ denoted by $\subset$ or $\subseteq$ i.e. $x \subseteq \xi$
Also, we say that the universal set ξ contains x i.e. $\xi \supseteq x$. While $y \not\subseteq \xi$ and $\xi \not\supseteq y$ means y is not a subset of ξ and ξ does not contain y respectively.

Proper and improper subsets

Any universal set is an improper subset of itself.
Also the empty set is an improper subset of all sets but every other subset originating from the universal set is a proper subset.

Power sets

This is the collection of all the subsets of a given set say s, denoted by 2^s
E.g list all the subsets of
A = { 1, 2, 3, }
The power sets = 2^s
= 2^3
= 8 (eight subsets)
They are: Ø, { 1, 2, 3}, { 1}, { 2 } , {3 }, {1,2 }, {1, 3 }, { 2, 3}

Equal sets

Two or more sets are equal if they contain the same elements, order of arrangement immaterial
E.g If X = {a , b, c, d}
Y = {b, d, a, c, d}
Z = {first four letters of the alphabet}
K = {a, b, c, d, e, f }
We say that X = Y i.e. They contain the same elements when counted once;letter d repetition in Y is immaterial.
Also X = Z but X ≠ K, since k contains e and f which are not elements of X

Equivalent sets

Two or more sets are equivalent when they contain equal elements denoted by ≡**.**
E.g Which of the sets listed below are equivalent.
A = {1, 2, 3, 4}
B = {a, c, d, e}
C = {a, b, b, e, c, d}
A ≡ B since one element of A can pair with only one element of B without remainder. But *A ànd* B *are not equal*. Neither A nor B is equivalent to C. This is so because C has a sixth element, which can not be paired. Though B is equal to C when the elements of C are counted once.

Other set Notations

"Element of'' is denoted by $\in$
E.g 1. Y = {x : x is between 10 and 15}
Then $11 \in Y$, $12 \in Y$ and $13 \in Y$ but any number outside the range *is not an element of Y*. Denoted by $\notin$. For instance $30 \notin Y$ means that 30 is not an element of Y.

Cardinality of set: refers to the number of elements in a given set - Say X and it is denoted by n(X).

E.g 1 If $X = \{i, j, k, l\}$

Then the Cardinality of X is $n(X) = 4$

E.g 2 If $B = \{a, b, c, d, e, f\}$ then find n(B)

Solution

$n(B) = 6$

2010/15

Given that $T = \{x : -2 < x \leq 9\}$ where x is an integer. What is n(T)?

A 9 B 10 C 11 D 12

Solution

n(T) Implies number of elements in set T

First, we list the members of set T

$T = \{-1, 0, 1, 2, 3, 4, 5, 6, 8, 9\}$

– 2 was not included because of the strict inequality there

9 was included because of the weak inequality there

$n(T) = 11$ C.

Union

The union of two or more sets is the <u>combination of elements</u> of the sets <u>without repetition.</u> It is denoted by $\cup$ and not U.

Mathematically if A and B are two sets then,

$A \cup B = \{x / x \in A$ <u>or</u> $x \in B\}$

E.g 1 If $A = \{1, 2, 3, 4, 5, 6\}$

$B = \{5, 2, 2, 7, 1, 3, 2\}$

then find $A \cup B$.

Solution

$A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$

E.g 2 List the elements of $X \cup Y$

If $X = \{a, b, c, d\}$ and $Y = \{a, d, e, a, f\}$

Solution

$X \cup Y = \{a, b, c, d, e, f\}$

Intersection

The intersection of two or more sets has a result of a set that contains only the <u>common elements</u> in them. It is denoted by $\cap$ and not n.

If A and B are two subsets, mathematically

$A \cap B = \{x: x \in A$ <u>and</u> $x \in B\}$

E.g 1. The sets of two groups of students who took part in the finals of shell and chevron Scholarship test are given as.

S = {Ada, Olu, Akpos}

C = {Ejiro, Akpos, Iluwa}

Find $S \cap C$

Solution

$S \cap C$ = {Akpos}

Complement of set

The complement of any set say K is the set that has <u>every element</u> in the universal set <u>except</u> those of K.

it is denoted by K^1 or K^c

i.e. $U - K = K^c$

2008/13 Neco (Dec)

Given a universal set $U = \{2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and the subsets $A = \{3, 5, 6, 7, 10\}$, $B = \{2, 5, 8, 9, 10\}$

Find $A^1 \cap B^1$

A {4} B {5} C {3, 4} D {4, 5} E {5, 10}

Solution

$A^1 = \{2, 4, 8, 9\}$ and $B^1 = \{3, 4, 6, 7\}$

$A^1 \cap B^1 = \{4\}$ A.

Finite set

These are <u>countable</u> sets.

E.g. 1 T = {All the State Governors in Nigeria}

E.g. 2 X = {All the students who registered for NECO exams in 2002}

Even a primary school pupil can easily count all the governors in the 36 states of Nigeria. The example 2 is more detailed because NECO headquarters has the total figure of students who sat for the exam in the year 2002.

Infinite set

These are <u>uncountable</u> sets

E.g.1 $D = \{x : x \in \mathbb{N}\}$

={All natural or counting number}

E.g 2. $F = \{x: x \in \mathbb{Z}\}$

= {Set of all integers}

$= \{\ldots, -2, -1, 0, 1, 2, 3, \ldots\}$

2014/14 Neco (Dec) Exercise 9.0

Which of the following is an example of an infinite set?

A { Even numbers less than 20 } B { Multiples of three }

C { Number of lecturers in Nigerian Universities }

D { Number of Local Governments in Nigeria}

E { Prime numbers greater than two but less than 97 }

Disjoint set

Two sets A and B are said to be disjoint when they have no element in common i.e. $A \cap B = \varnothing$

Non disjoint set

Two sets C and D are non-disjoint when they have at least one element in common

i.e. $C \cap D \neq \varnothing$

Examples

Given that X = {a, b, c, d}, Y = {a, b} and Z = {c, d}

Then find the pair of sets that are:

(a) Non – disjoint (b) disjoint.

Solution

(a) To establish non-disjoint. We find intersection.

$X \cap Y = \{a, b\}$

Thus X and Y are non-disjoint

(b) $Y \cap Z = \varnothing$ thus,

Y and Z are disjoint.

2009/7 Exercise 9.1

Two sets are disjoint if

A They are both empty

B Their union is an empty set

C Their intersection is an empty set

D one of them is a subset of the other

Range Sets

Here elements of inequality are employed to define the given sets. They are strict inequalities < (less than) and > (greater than) & the weak inequalities $\leq$ (less than or equal to) and $\geq$ (greater than or equal to).
Any set defined by strict inequalities does not accept numbers at the boundaries (limits). Similarly, any set defined by weak inequalities accepts numbers at the boundaries (limits).

Examples

List the elements of the sets below:

(i) $G = \{x : 1 < x < 5\}$

(ii) $L = \{y : 16 < y \leq 19\}$

(iii) $M = \{a : 25 \leq a < 30\}$

(iv) $N = \{b : 3 \leq b \leq 6\}$

Solution

(i) $G = \{2, 3, 4\}$
Reasons: Both boundaries are strict inequalities hence 1 and 5 are not included in the set G.

(ii) $L = \{17, 18, 19\}$
Reason : 16 is not included because the left half inequality is strict, while 19 is included because the right half inequality is weak.

(iii) $M =\{25, 26, 27, 28, 29\}$
Reason: 25 is included because the left half inequality is weak while 30 is not because the right half inequality is strict.

(iv) $N = \{3, 4, 5, 6, \}$
Reason: 3 and 6 are included because of the weak inequalities at both ends.

Number sets

This type of set deals with properties of the numbers such as *prime, odd, even, multiples, odd- prime and divisibility.* For the purpose of emphasis the author will briefly define and give examples of the above named properties.

Prime numbers

A number is said to be prime if it has only two factors; one and itself.

Example:

List the elements of the set
$A = \{z : z \text{ is a prime number from } 1 \text{ to } 10\}$
$A = \{2, 3, 5, 7\}$

Note: 1 is not a prime number because it has only one factor.

Even numbers

These are numbers divisible by 2 without remainder.

Example:

Given that $W = \{x: x \text{ is even number from 5 to 15}\}$: list the elements of W

Solution

$W = \{6, 8, 10 , 12, 14\}$

Odd numbers

Odd numbers are numbers not divisible by 2.

Example:

List the elements of the set
$R = \{k : k \text{ is an odd number between 2 and 11}\}$

Solution

The word "between'' here means 2 and 11 are not inclusive.
$R = \{3, 5, 7, 9\}$
Note: 1 is an odd number

Multiples

The Multiple of any number is the set, which contains the product of the given number and 1, 2, 3, … except when limits are given. Alternatively the multiples of any given number, say 4 are those numbers that can be divided by 4 without remainder.

Example: List the elements of the set
$J = \{t: 1 \leq t \leq 10 \text{ and t is the multiple of } 3\}$

Solution

The range $\{1 \leq t \leq 10\} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
The multiples of 3 here are $\{3, 6, 9,\}$
Therefore $J = \{3, 6, 9\}$

<u>Odd-prime Numbers</u>

These are numbers that have the odd numbers properties and are further subjected to the prime number properties.

Example:

If $U = \{x: 1 \leq x \leq 10\}$ and
$Y = \{x: x \text{ is odd} - \text{prime}\}$
Then list the elements of Y
$Y_{odd} = \{1, 3, 5, 7, 9 \}$
$Y_{odd - prime} = \{3, 5, 7\}$

<u>DIVISIBILITY</u>

This is the other term for multiple. Also not divisible and not a multiple are the same.

LAWS OF SET

1. **IDEMPOTENT LAW**
   $A \cup A = A; A \cap A = A$
2. **ASSOCIATIVE LAWS**
   $A\cup (B\cup C) = (A\cup B) \cup C$
   $A\cap (B\cap C) = (A\cap B) \cap C$
3. **COMMUTATIVE LAW**
   $A\cup B = B\cup A; A\cap B = B\cap A$
4. **DISTRIBUTIVE LAW**
   $A \cap (B\cup C) = (A\cap B) \cup (A\cap C)$
   $A \cup (B\cap C) = (A \cup B) \cap (A\cup C)$
5. **DE-MORGAN'S LAW**
   $(A\cup B)^1 = A^1\cap B^1$
   $(A\cap B)^1 = A^1\cup B^1$
6. **COMPLEMENT LAWS**
   $(A^1)^1 = A ; U^1 = \varnothing$
   $A\cup A^1 = \xi ; A\cap A^1 = \varnothing ; \varnothing^1 = \xi$
7. **IDENTITY LAWS**
   $A\cup\varnothing = A ; A\cap \xi = A$
   $A\cup\xi = \xi ; A\cap\varnothing = \varnothing$

Problems on number sets

1999/ 1 (Nov)

If $\xi = \{1, 2, 3, 4, 5\}$, $P = \{2, 3, 5\}$, $Q = \{1, 2, 3\}$

Find the complement of $P \cap Q$

A $\{2, 3, 5\}$ B $\{2, 4, 5\}$ C $\{1, 4, 5\}$ D $\{1, 3, 5\}$

Solution

$P \cap Q = \{2, 3\}$

$(P \cap Q)^1 = \{1, 4, 5\}$ C.

1998/ 13 (Nov)

If U = {Positive numbers less than 20}

P = {multiples of 4}, Q = {multiples of 6},find $P \cap Q$.

A $\{4, 6, 8, 12, 16, 18\}$ B $\{6, 12, 18\}$

C $\{4, 6\}$ D $\{4, 6, 12\}$ E $\{12\}$

Solution

$U = \{1, 2, 3, 4, 5, 6, \ldots 19\}$

$P = \{4, 8, 12, 16\}$ and $Q = \{6, 12, 18\}$

$P \cap Q = \{12\}$ E.

2007/ 12 Neco

Let the universal set U be the set of integers

$U = \{ x : 0 < x \le 10\}$. What is the complement of the Set $B = \{ x : x \in U,\ x \text{ is not divisible by } 3 \}$?

A $\{4\}$ B $\{4, 8\}$ C $\{9, 6, 3\}$

D $\{4, 8, 12, 16\}$ E $\{1, 2, 3, 5, 6, 7, 9\}$

Solution

Listing their elements

$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

$B = \{1, 2, 4, 5, 7, 8, 10\}$

$B^1 = \{3, 6, 9\}$ C.

1999/ 2 (Nov)

Given that $X = \{e, f, g\}$, $Y = \{e, f, h\}$ and $Z = \{g, h, k\}$.

How many elements are there in $X \cup Y \cup Z$?

A3 B 5 C 7 D 9

Solution

We are to find $n(X \cup Y \cup Z)$

Firstly, $X \cup Y \cup Z = \{e, f, g, h, k\}$

$n(X \cup Y \cup Z) = 5$

2003/8a

$A = \{1, 2, 5, 7\}$ and $B = \{1, 3, 6, 7\}$ are subsets of the universal set $U = \{1, 2, 3, 4, \ldots, 10\}$.

Find (i) A^1 (ii) $(A \cap B)^1$ (iii) $(A \cup B)^1$

(iv) the subsets of B each of which has three elements

Solution

$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

$A = \{1, 2, 5, 7\}$

(i) $A^1 = \{3, 4, 6, 8, 9, 10\}$

(ii) $A \cap B = \{1, 7\}$

$(A \cap B)^1 = \{2, 3, 4, 5, 6, 8, 9, 10\}$

(iii) $A \cup B = \{1, 2, 3, 5, 6, 7\}$

$(A \cup B)^1 = \{4, 8, 9, 10\}$

(iv) $\{1, 3, 6\}$, $\{1, 3, 7\}$, $\{3, 6, 7\}$, $\{1, 6, 7\}$

2004/ 25(Nov)

Given that $\xi = \{-1, 0, 1, 2, 3, 4, 5, 6\}$, where P = {Prime numbers} and Q = {Prime factors of 6} are subsets of ξ. Find $(P \cap Q)^1$.

A $\{2, 3\}$ B $\{-1, 0, 1, 2, 3\}$ C $\{-1, 0, 1, 4, 5, 6\}$

D $\{-1, 1, 2, 3\}$

Solution

$P = \{2, 3, 5\}$

$Q = \{2, 3\}$

Note: Factors of 6 are 1, 2, 3, 6

Prime factors of 6 are 2, 3

$P \cap Q = \{2, 3\}$

$(P \cap Q)^1 = \{-1, 0, 1, 4, 5, 6\}$ C

2004/ 26

Given that $\xi = \{1, 2, 3, \ldots, 10\}$, $P = \{x : x \text{ is prime}\}$ and $Q = \{ y : y \text{ is odd}\}$, find $P^1 \cap Q$

A $\{2\}$ B $\{1, 9\}$ C$\{ 3, 5, 7\}$ D$\{4, 6, 8, 10\}$

Solution

$\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

Applying the definition for prime numbers between 1 and 10

$P = \{2, 3, 5, 7\}$

$P^1 = \{1, 4, 6, 8, 9, 10\}$

Applying the definition of odd numbers between 1 and 10

$Q = \{1, 3, 5, 7, 9\}$

$P^1 \cap Q = \{1, 9\}$ B.

2005/ 34

$P = \{3, 9, 11, 13\}$ and $Q = \{3, 7, 9, 15\}$ are subsets of the universal set $\xi = \{1, 3, 7, 9, 11, 13, 15\}$. Find $P^1 \cap Q^1$

A $\{3, 9\}$ B $\{5, 7, 15\}$ C $\{1\}$ D $\{1, 11\}$

Solution

$P^1 = \{1, 7, 15\}$

$Q^1 = \{1, 11, 13\}$

$P^1 \cap Q^1 = \{1\}$ C.

2009/7 (Nov)

The sets $P = \{2, 5, 8\}$ and $Q = \{2, 3, 5, 7\}$ are subsets of the universal set $\mu = (1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Find $P^1 \cap Q$

A $\{3\}$ B $\{2, 5\}$ C $\{3, 7\}$ D $\{2, 3, 5, 7\}$

Solution

$P^1 = \{1, 3, 4, 5, 6, 7, 9, 10\}$

$Q = \{2, 3, 5, 7\}$

$P^1 \cap Q = \{3, 7\}$ C

2005/35

$P = \{3, 9, 11, 13\}$ and $Q = \{3, 7, 9, 15\}$ are subsets of the universal set $\xi = \{1, 3, 7, 9, 11, 13, 15\}$.Find $P^1 \cap Q^1$

A $\{3, 9\}$ B $\{5, 7, 15\}$ C $\{1\}$ D $\{1, 11\}$

Solution

$P^1 = \{1, 7, 15\}$ $Q^1 = \{1, 11, 13\}$

$P^1 \cap Q^1 = \{1\}$ C.

2006/42

If $P = \{1, 3, 5, 7, 9\}$ and $Q = \{2, 4, 6, 8, 10\}$ are subset of a universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, what are the elements of $P^1 \cap Q^1$?

A $\{1, 3, 5, 7, 9\}$ B $\{2, 4, 6, 8, 10\}$ C ϕ

D $\{1, 2, 6, 8, 10\}$

Solution

$P^1 = \{2, 4, 6, 8, 10\}$

$Q^1 = (1, 3, 5, 7, 9\}$

$P^1 \cap Q^1 = \phi$ C.

2007/ 10a NABTEB

If $U = \{1, 2, 3, \ldots, 10\}$ and $A = \{4, 6, 8, 10\}$

$B = \{1, 4, 5, 11\}$ and $C = \{4, 5, 11, 12\}$.

Find $C^1 \cup (A \cap B)$

Solution

Starting with the bracket items

$A \cap B = \{4\}$

$C^1 = \{1, 2, 3, 6, 7, 8, 9, 10\}$

$C^1 \cup (A \cap B) = \{1, 2, 3, 4, 6, 7, 8, 9, 10\}$

2008/3

Let P, Q, R, S be subsets of R defined by the following sets:

$P = \{x : (x – 2)(x + 3)(x – 4)(x + 5) = 0\}$

$Q = \{x : (x + 1)(x – 1)(x – 2) = 0\}$

$R = \{x : (x – 5)(x + 2)(x + 1) = 0\}$

$S = \{x : (x – 1)(x + 2)(x – 3) = 0\}$

Find: (i) $P \cup Q$ (ii) $P \cup Q \cup R$ (iii) $R \cap S$

Solution

First, we list the elements of each set.

Applying the principle of solving quadratic equations

Set P

$(x – 2)(x + 3)(x – 4)(x + 5) = 0$

$x – 2 = 0$ or $x + 3 = 0$ or $x – 4 = 0$ or $x + 5 = 0$

$x = 2$ or $– 3$ or 4 or $–5$

Listing the elements in order of magnitude

$P = \{– 5, – 3, 2, 4\}$

Set Q

$(x + 1)(x – 1)(x – 2) = 0$

$x + 1 = 0$ or $x – 1 = 0$ or $x – 2 = 0$

$x = –1$ or 1 or 2

Listing the elements in order of magnitude

$Q = \{– 1, 1, 2\}$

Set R

$(x – 5)(x + 2)(x + 1) = 0$

$x – 5 = 0$ or $x + 2 = 0$ or $x + 1= 0$

$x = 5$ or $–2$ or $–1$

Listing the elements in order of magnitude

$R = \{–2, –1, 5\}$

Set S

$(x – 1)(x + 2)(x – 3) = 0$

$x – 1 = 0$ or $x + 2 = 0$ or $x – 3 = 0$

$x = 1$ or $–2$ or 3

Listing the elements in order of magnitude

$S = \{– 2, 1, 3\}$

(i) $P \cup Q = \{–5, –3, –1, 1, 2, 4\}$

(ii) $P \cup Q \cup R = \{–5, –3, –2, –1, 1, 2, 4, 5\}$

(iii) $R \cap S = \{– 2\}$

2009/ 16 Neco (Dec)

Given that U is the universal set defined as

$U = \{x : 1 \le x \le 12 \}$, P and Q are subsets such that

$P = \{x : x < 12\}$ and $Q = \{x : x$ is an even number$\}$.

Find $P^1 \cap Q$ (where P^1 = complement of P)

A { } B {12} C {1, 2} D {2, 4, 6, 8, 12}

E {2, 3, 5, 7, 9, 11}

Solution

Listing the elements of the universal set and the subsets

$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

1 and 12 were included because of the weak inequality there.

$P = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$

12 was NOT included because of the strict inequality there.

$Q = \{2, 4, 6, 8, 10, 12\}$

$P^1 = \{12\}$

Thus, $P^1 \cap Q = \{12\}$ B.

2010/4b Neco

Let $U = \{3, 6, 9, \dots , 45\}$

$A = \{y : y < 22\}$ $B = \{y : y$ is a factor of $45\}$

Where $y \in U$. Find : (i) n(U) (ii) $A \cup B$, (iii) $(A \cap B)$

Solution

(i) n(U); here we apply nth term formula for AP, since elements of U follows a simple AP.

$T_n = a + (n – 1)d$

$45 = 3 + (n –1)3$

$45 – 3 = (n –1)3$

$42 = (n –1)3$

$42/3 = n – 1$

$14 + 1 = n$

$n = 15$

Thus $n(U) = 15$

Alternatively

(i) By listing

$U = \{3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, 39, 42, 45\}$

Thus, $n(U) = 15$

(ii) $A \cup B = ?$

First, we list elements of A, B

$A = \{3, 6, 9, 12, 15, 18, 21\}$

$B = \{1, 3, 5, 9, 15, 45\}$ Applying simple definition of factors

$A \cup B = \{1, 3, 5, 6, 9, 12, 15, 18, 21, 45\}$.

$A \cap B = \{3, 9, 15\}$

2010/13

Given that $P = \{x : 1 \le x \le 6\}$ and $Q = \{x : 2 < x < 10\}$ where x is an integer. Find $n(P \cap Q)$

A 4 B 6 C 8 D 10

Solution

$n(P \cap Q)$ implies number of elements in $P \cap Q$

First, we list the elements of set P, set Q then $P \cap Q$.

$P = \{1, 2, 3, 4, 5, 6\}$

1 and 6 were included because of the weak inequalities there

$Q = \{3, 4, 5, 6, 8, 9\}$

2 and 10 were NOT included because of the strict inequalities there.

$P \cap Q = \{3, 4, 5, 6\}$

Thus, $n(P \cap Q) = 4$ A.

2011 1b Neco

If $U = \{x :$ the solution set of the inequalities $2x – 3 < 27$ and $2x – 1 \le 2 + 3x\}$

P is a subset of U such that P = {Prime numbers}

List the elements in : (i) U (ii) P

Solution

(i) We get elements of U by solving the two inequalities

$2x – 3 < 27$	$2x – 1 \le 2 + 3x$
$2x < 27 + 3$	$2x – 3x \le 2 + 1$
$2x < 30$	$– 1x \le 3$
$\frac{2x}{2} < \frac{30}{2}$	$\frac{-1x}{-1} \ge \frac{3}{-1}$
$x < 15$	$x \ge –3$

$U = \{–3, –2, –1, 0, 1, 2, 3, 4, 5, \dots, 13, 14\}$

(ii) $P = \{2, 3, 5, 7, 11, 13\}$

2012 / 13 – 14 Neco

Given that U = $\{x : –3 < x \leq 15\}$

A = $\{x : 3 \leq x < 15\}$, B = $\{x : –1 < x < 8\}$

13. Find $(A \cup B)^1$

A {– 2, –1, 0, 15} B {–2, –1, 1, 15} C {– 2, –1, 15}

D {0, 1, 2, 15} E {1, 2, 15}

Solution

Listing elements

U = $(x : – 3 < x \leq 15\}$

= {–2, –1, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}

–3 was omitted because of the strict inequalities there

15 was included because of the weak inequalities there

A = $\{x : 3 \leq x < 15\}$

= {3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14}

3 was included because of the weak inequalities there

15 was NOT included because of the strict inequalities

B = $\{x : –1 < x < 8\}$

= {0, 1, 2, 3, 4, 5, 6, 7}

–1and 8 were NOT included because of the strict inequality there

$A \cup B$ = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14}

$(A \cup B)^1$ = {– 2, –1, 15} C.

14. Find $A^1 \cap B$

A {–1, 0, 1, 15} B {–2, –1, 0, 1, 2} C {2, 4, 6}

D {0, 1, 2} E {0, 1, 2, 15}

Solution

A = {3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14}

A^1 = {–2, –1, 0, 1, 2, 15} and

B = {0, 1, 2, 3, 4, 5, 6, 7}

$A^1 \cap B$ = {0, 1, 2} D.

2012/37

If P = {Prime factors of 210} and

Q = {Prime numbers less than 10}

Find $P \cap Q$

A = {1, 2, 3,} B = {2, 3, 5} C = {1, 3, 5, 7}

D = {2, 3, 5, 7}

Solution

Using the elementary way of listing elements of prime factors

2	210
3	105
5	35
7	7
	1

Thus P = {2, 3, 5, 7} and

Q = {2, 3, 5, 7}

$P \cap Q$ = {2, 3, 5, 7, } D.

2014/ 12

Find the truth set of the equation $x^2 = 3(2x + 9)$

A {x : x = 3, x = 9} B {x : x = –3, x = –9}

C {x : x = 3, x = –9} D {x : x = –3, x = 9}

Solution

We simply solve the resulting quadratic equation

$x^2 = 3(2x + 9)$

$x^2 = 6x + 27$

$x^2 – 6x – 27 = 0$

Factorizing

$x^2 – 9x + 3x – 27 = 0$

$x(x – 9) + 3(x – 9) = 0$

$(x + 3)(x – 9) = 0$

$x + 3 = 0$ or $x – 9 = 0$

$x = – 3$ or 9 D.

i.e {x : x = – 3, x = 9)

2009/18 NATEB Exercise 9.2

If ξ = {1, 2, 3, …15, 16} and T = {3, 6, 9, 12, 15}

S = {1, 4, 9, 16}. Then $(T \cup S)^1$ is

A {1, 3, 4, 6, 9, 10, 15, 16} B {2, 3, 4, … 15, 16}

C {2, 5, 7, 8, 10, 11, 13, 14} D {3, 4, 6, 9, 10, 15, 16}

2010/1 Exercise 9.3

A = {2, 4, 6, 8}, B = {2, 3, 7, 9} and

C = $\{x : 3 < x < 9\}$ are subsets of the universal set

U = {2, 3, 4, 5, 6, 7, 8, 9} Find (a) $A \cap (B^1 \cap C^1)$;

(b) $(A \cup B) \cap (B \cup C)$

2014/ 11 Exercise 9.4

If X = {0, 2, 4, 6}, Y = {1, 2, 3, 4} and Z = {1, 3}

are subsets of U = $\{x : 0 \leq x \leq 6\}$, find $x \cap (Y^1 \cup Z)$

A {0, 2, 6} B {1, 3} C {0, 6} D { }

2008/4 Neco Exercise 9.5

Given that : P = { x : (x – 2)(x + 3)(x – 4)(x + 5) = 0 }.

The elements in P are

A (– 5, –2, 3, 4} B {–5, –3, 2, 4}

C {–5, –3, –2, 4} D{– 4, –3, – 2, 5} E {2, 3, 4, 5}

2006/17 Neco Exercise 9.6

If μ = {K, S, A, P, M, E, C}, P = {S, P, M, E}

Q = {K, A, P, C}, R = {A, P, M, C}

Find $Q^1 \cup (P \cap R)$

A {k, a, m, e} B {k, s, m, c} C {s, a, p, m

D {s, a, e, c} E {s, p, m, e}

2008/2b NATEB (Dec) Exercise 9.7

If U = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} A = {2, 3, 5, 8},

B = {1, 3, 7, 8, 9}. Write down the sets

(i) $A \cap B$ (ii) $A^1 \cup B^1$

2014/4b (Nov) Exercise 9.8

Given that X = $\{x : 10 \leq x < 15\}$ and

Y = { even numbers < 18} are subsets of

U = { 10, 11, 12, …, 20 }, find : (i) $X \cap Y$ (ii) $n(X^1 \cap Y)$

2009/45 NATEB Exercise 9.9

Given that ξ = {1, 2, 3, 4 …10}, A = {a : a is even) and

B = {b : b is prime}, then $A^1 \cap B$ is

A {5} B {2, 8} C {3, 5, 7} D ϕ

2012/ 6 Neco Exercise 9.10

If P and Q are subsets of a universal set U,

evaluate $P^1 \cap (Q \cup P)$

A ϕ B P C Q D $P^1 \cap Q$ E $P^1 \cup Q$

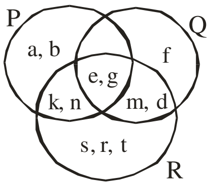
Venn diagram

This is a representation of set in shapes whereby subsets are represented by circles while rectangle enclosing the circle(s) represents the universal set.

Number Venn diagram

This is a representation of a given set of numbers or set's solution in Venn diagram.

2013/5



The diagram above consists of the sets P, Q and R.
Find $R\cap Q$

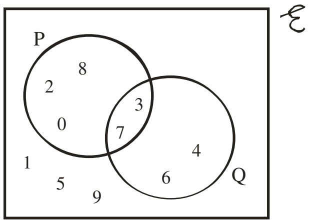
A {m, d} B {d, e, n} C {d, e, g, m}
D {e, g, k, m} E {k, n, e, g}

Solution

$R\cap Q = \{d, e, g, m\}$ C.

On the other hand, if we are asked to
find $R\cap Q\cap P^1$ = $R\cap Q$ only
= {d, m}

2000/ 3b



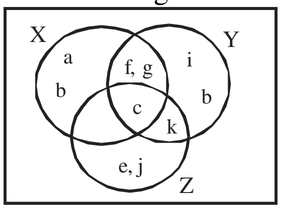
The Venn diagram above represents a universal set ξ of integers and its subsets P and Q.
List the elements of the following sets
(i) $P\cup Q$; (ii) $P\cap Q$; (iii) $(P\cup Q)^1$ (iv) $(P\cap Q)^1$

Solution

(i) $P\cup Q = \{0, 2, 8, 7, 3, 6, 4\}$
(ii) $P\cap Q = \{3, 7\}$
(iii) $(P\cup Q)^1 = \{1, 5, 9\}$
(iv) $(P\cap Q)^1 = \{0, 2, 8, 6, 4, 1, 5, 9\}$

2005/ 27 – 28 (Nov)

Use the Venn diagram to answer questions 27 and 28



27. What is $X^1\cap Y^1\cap Z$?
A {a, d} B {b, i} C {e, j} D { f, g }

Solution

$X^1\cap Y^1\cap Z$ is elements of Z only = {e, j} C.

28. What is $X\cap Y\cap Z^1$?
A {a, b) B {e, j} C {b, i} D {f, g}

Solution

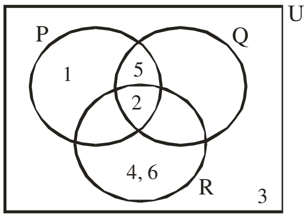
$X\cap Y\cap Z^1 = X\cap Y$ only
= {f, g} D.

2005/13b

Three sets P, Q and R are defined as follows:
P = {1, 2, 5}, Q = {2, 5}, R = {2, 4, 6} and the universal set μ = {1, 2, 3, 4, 5, 6}
(i) Draw a Venn diagram to represent the information
(ii) List the elements of P^1, $(P\cup R)^1\cap(P\cup Q)^1$ and $(P\cap Q)\cup(Q\cap R)$

Solution

(i) $P\cap Q\cap R = \{2\}$, $P\cap Q\cap R^1 = \{5\}$,
$P\cap Q^1\cap R = \{\ \}$, $P^1\cap Q\cap R = \{\ \}$



(ii) $P^1 = \{3, 4, 6\}$
$(P\cup R)^1\cap(P\cup Q)^1$ starting from the brackets

$P\cup R = \{1, 2, 4, 5, 6\}$	$P\cup Q = \{1, 2, 5\}$
$(P\cup R)^1 = \{3\}$	$(P\cup Q)^1 = \{3, 4, 6\}$

Thus $(P\cup R)^1\cap(P\cup Q)^1 = \{3\}$

$(P\cap Q)\cup(Q\cap R)$ starting from the brackets
$P\cap Q = \{2, 5\}$ $Q\cap R = \{2\}$
Thus $(P\cap Q)\cup(Q\cap R) = \{2, 5\}$

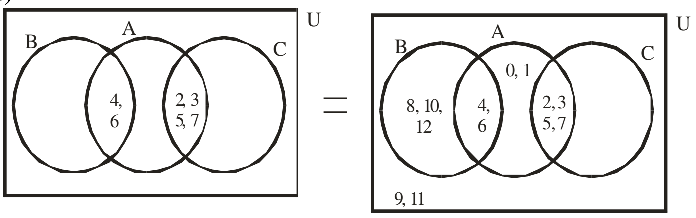
1998/1

A, B and C are subsets of the universal set μ such that:
μ = {0, 1, 2, 3, … , 12};
A = {x : 0 ≤ x ≤ 7}; B = {4, 6, 8, 10, 12};
C = {1< y < 8} where y is a prime number;
(a) Draw a Venn diagram to illustrate the information given above.
(b) Find: (i) $(B\cup C)^1$; (ii) $A^1\cap B\cap C$

Solution

μ = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12}
A = {0, 1, 2, 3, 4, 5, 6, 7}
B = {4, 6, 8, 10, 12}
C = {2, 3, 5, 7}
$A^1\cap B\cap C = \{\ \}$, $A\cap B\cap C^1 = \{4, 6\}$
$A\cap B^1\cap C = \{2, 3, 5, 7\}$

(a)

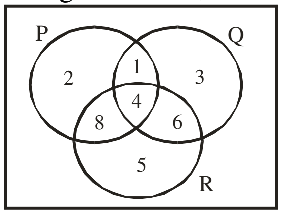


First filling of intersection elements

(b) (i) $B\cup C = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 12\}$
$(B\cup C)^1 = \{0, 1, 9, 11\}$
(ii) $A^1\cap B\cap C = \{\ \}$

2009/17 Neco (Dec) Exercise 9.11

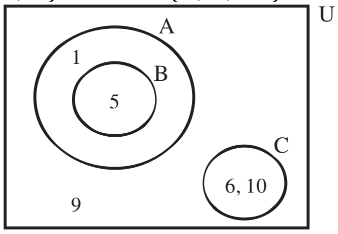
In the Venn diagram below, what is (Q∩R)?



A {2} B {4, 6} C {2, 3, 5, 8}
D {1, 2, 3, 5, 8} E {1, 2, 3, 4, 5, 8}

2005/ 1b Exercise 9.12

μ = {1, 2, 3, 4, 5, …,10}, A = {1, 2, 3, 4, 5}
B = {2, 3, 5} and C = {6, 8, 10}

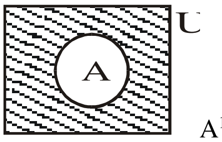


(i) Given that the Venn diagram represents the sets above, copy and till in the elements
(ii) Find A∩C (iii) Find $A∩B^1$

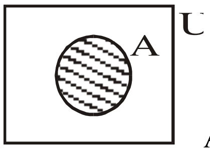
Venn diagram shading

This is the representation of set relationships such as intersections, union and complements by appropriate shading in Venn diagrams

One subset



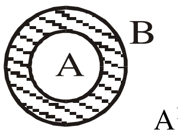
A^1 shaded



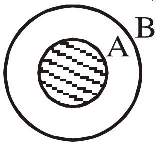
A shaded

Two subsets

(i) **When A ⊆ B (A is a subset of B)**



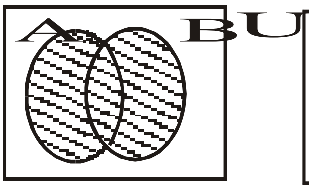
A^1 Shaded



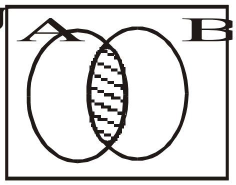
A∩B shaded

A∪B shaded

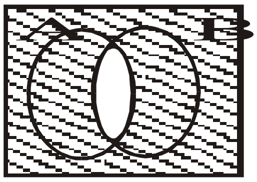
(ii) Independent subsets



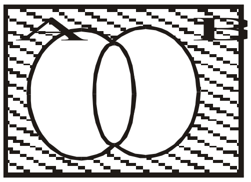
A∪B shaded



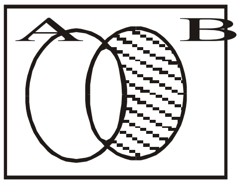
A∩B shaded



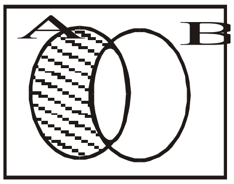
$(A∩B)^1$ shaded



$(A∪B)^1$ shaded

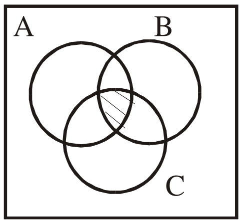


$A^1∩B$ shaded

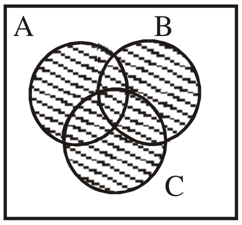


$A∩B^1$ shaded

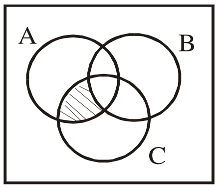
(iii) Three subsets



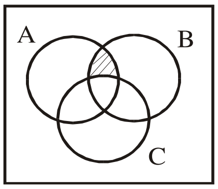
A∩B∩C shaded



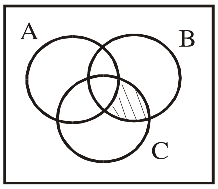
A ∪B∪C shaded



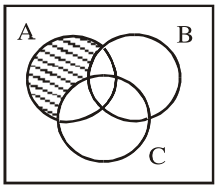
$A∩B^1∩C$ shaded



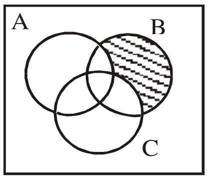
$A∩B∩C^1$ shaded



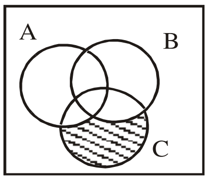
$A^1∩B∩C$ shaded



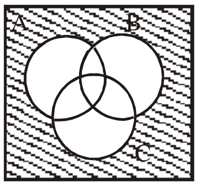
$A∩B^1∩C^1$ shaded



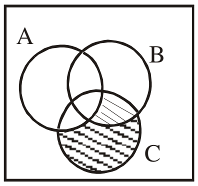
$A^1∩B∩C^1$ shaded



$A^1∩B^1∩C$ shaded

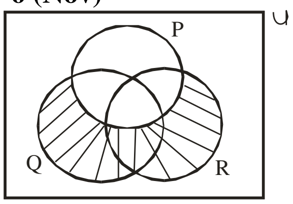


$(A ∪B∪C)^1$ shaded



$(B∩C) ∪A^1∩C$ shaded

2009/ 6 (Nov)



P, Q and R are three intersecting sets in the universal set μ. Which of the following represents the shaded portion?

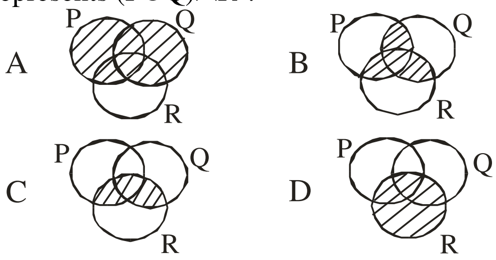
A $(P\cup Q)\cap R$ B $(P\cup Q)^1\cap R$ C $(R\cup Q)\cap P^1$
D $(P\cup Q)^1\cap Q$

Solution

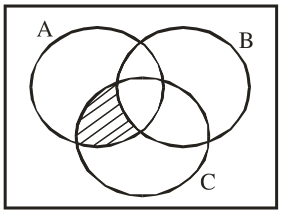
The shaded portion is $(R\cup Q)\cap P^1$

2009/48 Exercise 9.13

Which shaded region in the following diagrams represents $(P\cup Q)\cap R$?



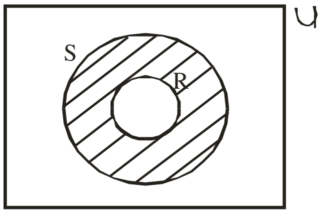
2005/18 Neco Exercise 9.14



In the Venn diagram below, the area shaded represents

A $A^1\cup(B\cup C)$ B $A\cap(B^1\cap C)$ C $A\cup(B\cup C^1)$
D $A^1\cap(B\cap C)$ E $A\cup(B^1\cup C)$

2015/4 (Nov) Exercise 9.15

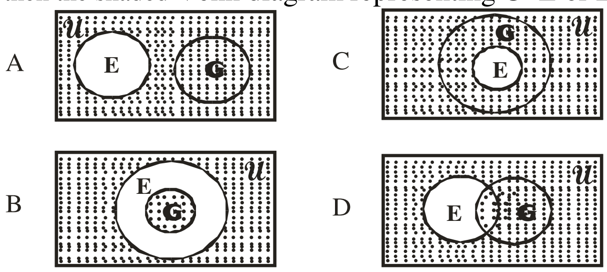


In the Venn diagram, R is a subset of S. Find the set which represents the shaded portion

A R^1 B S C $R^1\cup S$ D $R^1\cap S$

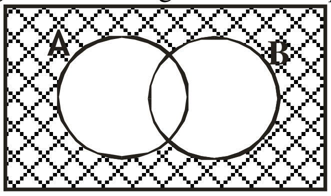
2006/46 UME Exercise 9.16

If $E\subseteq G\subseteq U$, where U is the universal set, then the set, then the shaded Venn diagram representing U–E or E^C is



2014/36 Neco (Dec) Exercise 9.17

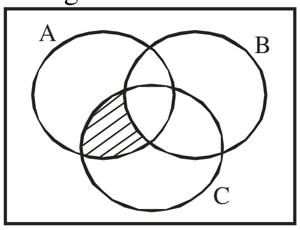
The shaded region in the diagram below represents



A $A\cap B$ B $A^1\cap B^1$ C $(A\cup B)^1$
D $A\cup B$ E $A^1\cup B$

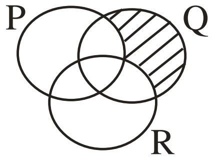
2015/6 Neco Exercise 9.18

Which of the following is illustrated by the shaded portion in the Venn diagram below?



A $(A\cap C)\cap B^1$ B $(A\cup C)\cap B^1$ C $(A\cap C)\cup B^1$
D $(A\cup C)\cup B^1$ E $(A\cup C)\cap(B\cup C)$

2015/42 Exercise 9.19



Describe the shaded portion in the diagram.

A $P^1\cap Q\cap R^1$ B $(P\cap R)^1\cap Q$ C $P^1\cap Q\cap R$ D $(P\cup Q)^1\cap R$

Two subsets problems

For any two subsets A and B not disjoint i.e $A\cap B \neq \varnothing$

$n(A\cup B) = n(A) + n(B) - n(A\cap B)$

1998/12 (Nov)

M and N are two sets such that n(M) = 10, n(N) = 7 and $n(M\cup N) = 13$. Find $n(M\cap N)$

A 16 B 6 C 4 D 3 E 1

Solution

Applying two subsets formula

$n(M\cup N) = n(M) + n(N) - n(M\cap N)$

$13 = 10 + 7 - n(M\cap N)$

$13 = 17 - n(M\cap N)$

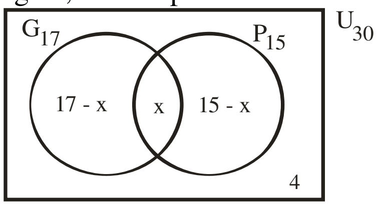
$n(M\cap N) = 17 - 13$ i.e 4 C.

2009/4a

Out of 30 Candidates applying for a post, 17 have degrees, 15 diplomas and 4 neither degree nor diploma. How many of them have both?

Solution

G – degree, P – diploma and x – both



$17 - x + x + 15 - x + 4 = 30$

$36 - x = 30$

$36 - 30 = x$ thus $x = 6$

2007/ 5b Neco

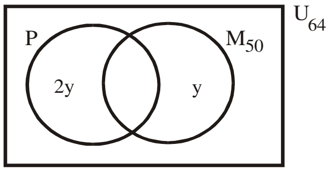
In class of 64 students, each student offers either Physics or Mathematics or both. If 50 students offer Mathematics and the number of students offering Mathematics only is twice the number of students offering Physics only, how many students offer both subjects?

Solution

M – Mathematic $\quad n(M) = 50$

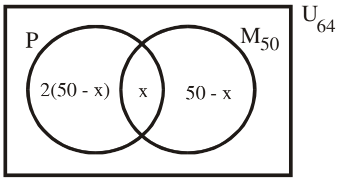
P – Physics $\quad n(P) =$

$n(U) = 64$



Let $n(P \cap M) = x$ then $y = 50 - x$

Re - labeling our diagram, we have



$$2(50 - x) + x + 50 - x = 64$$
$$100 - 2x + 50 = 64$$
$$150 - 64 = 2x$$
$$86 = 2x$$
$$x = 43$$

2006/ 4a

In a class of 45 students, 32 offered Physics (P), 28 offered Government (G) and12 did not offer any of the two subjects.

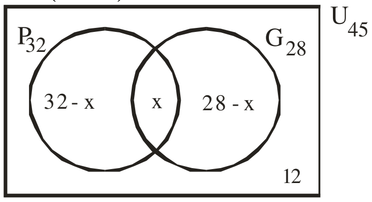
(i) Draw the Venn diagram to represent the information

(ii) How many students offered both Physics and Government?

(iii) What is $n(P \cup G)$?

Solution

(i) Let $n(P \cap G) = x$



(ii) $32 - x + x + 28 - x + 12 = 45$

$$72 - x = 45$$
$$72 - 45 = x$$
$$x = 27$$

(iii) $n(P \cup G) = 32 - x + x + 28 - x$

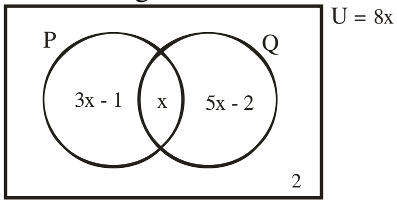
$= 60 - x$

$= 60 - 27$

$= 33$

2012/ 7 Neco

From the Venn diagram below, find the value of x



A 1 B 2 C 3 D 4 E 5

Solution

We equate everything in the box to the universal set

$$3x - 1 + x + 5x - 2 + 2 = 8x$$
$$9x - 1 = 8x$$
$$9x - 8x = 1$$
$$x = 1 \text{ (A)}$$

2012/ 6b

A number of tourists were interviewed on their choice of means of travel. Two – thirds said that they traveled by road, $\frac{13}{30}$ by air and $\frac{4}{15}$ by both air and road. If 20 tourists did not travel by either air or road,

(i) represent the information on a Venn diagram;

(ii) how many tourists

(a) were interviewed;

(b) Travelled by air only?

Solution

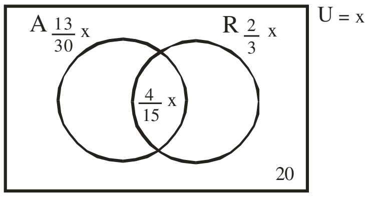
R – Road

A – Air

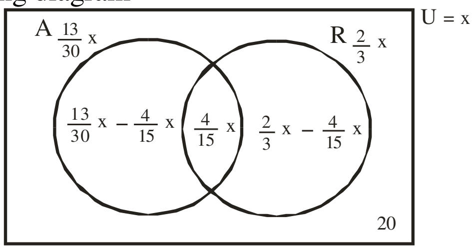
Let the number of tourists be x

$n(A \cap R) = \frac{4}{15}x$, $\quad n(R) = \frac{2}{3}x$, $\quad n(A) = \frac{13}{30}x$

(i)



Resulting diagram



(ii) a.

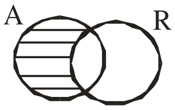
$$\left(\frac{13}{30}x - \frac{4}{15}x\right) + \frac{4}{15}x + \left(\frac{2}{3}x - \frac{4}{15}x\right) + 20 = x$$

$$\frac{5}{30}x + \frac{4}{15}x + \frac{6}{15}x + 20 = x$$

Multiply through by LCM of 15 and 30 to clear fractions

$$5x + 8x + 12x + 600 = 30x$$
$$30x - 25x = 600$$
$$5x = 600$$
$$x = {}^{600}/_{5}$$
$$x = 120 \text{ tourists}$$

(b) n (A) only



$$= \left(\frac{13}{30}x - \frac{4}{15}x\right)$$

$$= \frac{5}{30}x$$

$$= \frac{5}{30} \times 120 = 20 \text{ tourists.}$$

2008/33

Every staff in an office owns either a Mercedes and /or a Toyota car. 20 own Mercedes, 15 own Toyota and 5 own both. How many staffers are there in the office?

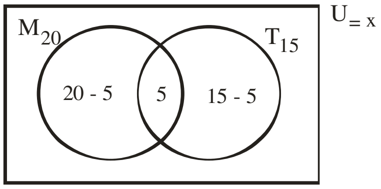
A 25 B 30 C 35 D 45

Solution

M – Mercedes

T – Toyota

$n(M) = 20, \quad n(T) = 15, \quad n(M \cap T) = 5, \quad n(U) = x$



$20 - 5 + 5 + 15 - 5 = x$

$30 = x$ (B)

2014/6a (Nov) Exercise 9.20

In a class of 52 students, 16 are Science students. If $\frac{1}{3}$ of the boys and $\frac{1}{4}$ of the girls are Science students, how many boys are in the class?

2011/ 2 Neco Exercise 9.21

Among 40 applicants for a sales manager's post, 26 have B.Sc degree and 18 have NCE, 6 applicants have no certificate. Use a Venn diagram to illustrate the information. How many applicants have:

(i) both B.Sc and NCE; (ii) B.Sc only

(iii) NCE only?

2009/ 5 Neco (Dec) Exercise 9.22

In a class of 25 pupils, 12 offer Physics and 18 offer Chemistry. If every pupil offers at least one of the two, how many offer both subjects?

A 5 B 6 C 7 D 12 E 13

2012/10 UTME Exercise 9.23

In a class of 46 students, 22 play football and 26 play volleyball. If 3 students play both games, how many play neither?

A 1 B 2 C 3 D 4

2009/11 PCE Exercise 9.24

In a class of 40 students, 30 students take Economics and 20 take Accounting. If 8 students take neither Economics nor Accounting, find the number of students who take Economics but not Accounting.

A 18 B 12 C 8 D 2

Three subsets problems

For any three subsets A, B and C not disjoint i.e $A \cap B \cap C \neq \varnothing$

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$

If the given three-subset problem is disjoint apply only the Venn diagram approach. Otherwise both methods can be used.

2013/ 4 Neco

In a class of 55 students, 21 study Physics, 24 study Geography and 23 study Economics, 6 study both Physics and Geography, 8 study both Geography and Economics and 5 study both Economics and Physics. If x study all the 3 subjects and 2x study none of the three subjects. Find

(i) the value of x,

(ii) the number of students that study physics only

(iii) the number of students that study only two subjects.

Solution

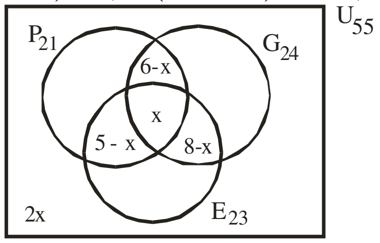
P – Physics $n(P) = 21$

G – Geography $n(G) = 24$

E – Economics $n(E) = 23$

$n(P \cap G) = 6, \quad n(G \cap E) = 8, \quad n(E \cap P) = 5$

$n(P \cap G \cap E) = x, \quad n(P \cup G \cup E)^1 = 2x, \quad n(U) = 55$



Here $n(P \cup G \cup E) = n(U) - n(P \cup G \cup E)^1$

$$n(U) - n(P \cup G \cup E)^1 = n(P) + n(G) + n(E) - n(P \cap G) - n(G \cap E) - n(E \cap P) + n(P \cap G \cap E)$$

$$55 - 2x = 21 + 24 + 23 - 6 - 8 - 5 + x$$
$$55 - 2x = 68 - 19 + x$$
$$55 - 49 = 2x + x$$
$$6 = 3x$$
Thus, $x = 2$

(ii) $n(P)$ only $= 21 - (6 - x + x + 5 - x)$

$= 21 - (11 - x)$

Substituting for the value of x

$= 21 - (11 - 2)$

$= 12$

(iii) Two subjects only $= 6 - x + 5 - x + 8 - x$

$= 19 - 3x$

$= 19 - 6$

$= 13$

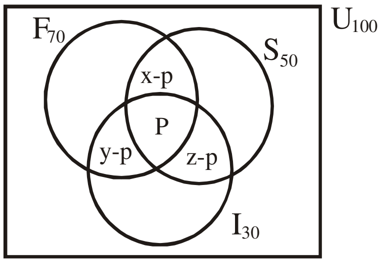
Example TSS 2

In a survey, out of 100 out–patients who reported at a clinic in a week it was found out that 70 complained of fever, 50 had stomach ache and 30 had injuries. All the 100 patients had at least one of the complaints and 44 had exactly two of the complaints. How many patients had all the three complaints?

Solution

This questions calls for application of our knowledge of solving three subsets formula/Venn diagram problems let the first letters represents each complaint

$n(F\cap S) = x \quad n(F\cap I) = y \quad n(S\cap I) = z$
and $n(F\cap S\cap I) = p$



But $x - p + y - p + z - p = x + y + z - 3p$
and $x + y + z$ is 44
Thus $x - p + y - p + z - p = 44 - 3p$

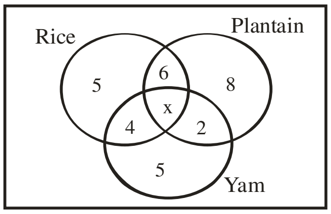
From the formula method

$n(F\cup S\cup I) = n(F) + n(S) + n(I) - n(F\cap I) - n(F\cap S) - n(S\cap I) + n(F\cap S\cap I)$

$= n(F) + n(S) + n(I) - [n(F\cap I) + n(F\cap S) + n(S\cap I)] + n(F\cap S\cap I)$

$100 = 70 + 50 + 30 - [44 + 3P] + P$
$100 = 150 - 44 - 3P + P$
$100 = 106 - 2P$
$2P = 106 - 100$
$2P = 6$ Thus $P = {}^6/_2$ i.e 3

2007/ 38 – 39



The Venn diagram shows the choice of food of a number of visitors to a canteen. Use the information to answer questions 38 and 39

38. If there were 35 visitors in all, find the value of x
A 5 B 4 C 3 D 2

Solution

$5 + 6 + x + 4 + 5 + 2 + 8 = 35$
$30 + x = 35$
$x = 35 - 30$ i.e 5 (A)

39. How many people took at-least two kinds of food
A 10 B 12 C 15 D 17

Solution

At-least two kinds of food implies
two kinds of food *plus* three kind of food.
$= 4 + 6 + 2 + x$
$= 12 + 5$
$= 17$ (D)

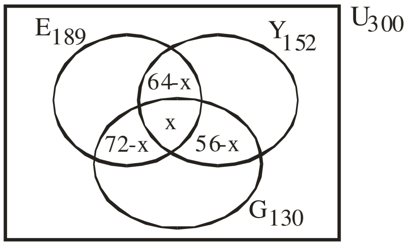
2010/9 Neco

In a survey of 300 viewers who watched 2009 National Festival of Art and Culture, it was found that 189 liked Efik dances, 152 liked Yoruba dances and 130 liked Gwape dances and each liked at least one of the three dances. If 64 liked Efik and Yoruba dances, 72 liked Ekik and Gwape dances and 56 liked Yoruba and Gwape dances.
(a) Draw a Venn diagram to illustrate the information
(b) How many viewers liked
(i) all the three dances?
(ii) exactly two of the dances?
(iii) exactly one of the dances?
(iv) Gwape dance only?

Solution

E – Efik $n(E) = 189$
Y – Yoruba $n(Y) = 152$
G – Gwape $n(G) = 130$
$n(E\cap Y) = 64$, $n(E\cap G) = 72$, $n(Y\cap G) = 56$
$n(E \cap Y\cap G) = x$,
$n(E\cup Y\cup G) = n(U)$ *since each viewer like at least one of the three dances.*

(a)



(b) i. Applying the formula for three subsets problem
$n(E\cup Y\cup G) = n(E) + n(Y) + n(G) - n(E\cap Y) - n(E\cap G) - n(Y\cap G) + n(E\cap Y\cap G)$

Substituting
$300 = 189 + 152 + 130 - 64 - 72 - 56 + x$
$300 = 471 - 192 + x$
$300 = 279 + x$
$300 - 279 = x$
$x = 21$

(ii) Exactly two of the dances
$= 64 - x + 72 - x + 56 - x$
$= 64 - 21 + 72 - 21 + 56 - 21$
$= 43 + 51 + 35$
$= 129$

(iii) Exactly one of the dances
$= 189 - (64 - x + x + 72 - x) + 130 - (72 - x + x + 56 - x) + 152 - (64 - x + x + 56 - x)$
$= 189 - (136 - 21) + 130 - (128 - 21) + 152 - (120 - 21)$
$= 189 - 115 + 130 - 107 + 152 - 99$
$= 74 + 23 + 53$
$= 150$

(iv) Gwape dance only
$= 130 - (72 - x + x + 56 - x)$
$= 130 - (128 - 21)$
$= 130 - 107$
$= 23$

2011/ 6

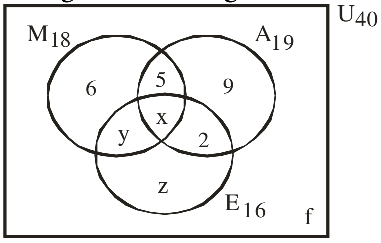
In a class of 40 students, 18 passed Mathematics, 19 passed Accounts, 16 passed Economics, 5 Mathematics and Accounts only, 6 Mathematics only, 9 Accounts only , 2 Accounts and Economics only. If each student offered **at least** one of the subjects,
(a) how many students failed in all the subjects?
(b) find the percentage number who failed in **at least** one of Economics and Mathematics;
(c) calculate the probability that a student selected at random **failed** in accounts

Solution

$n(M\cap A\cap E) = x$, $n(M\cap E)$ only $= y$, $n(E)$ only $= z$

$n(M\cup A\cup E)^1 = f$

Drawing its Venn diagram



Solving for the unknowns

$5 + 9 + 2 + x = 19$ *subset* Accounts

$x = 19 - 16$

$x = 3$

$6 + 5 + x + y = 18$ *subset* Mathematics

Substituting for x value

$6 + 5 + 3 + y = 18$

$y = 18 - 14$

$= 4$

$y + x + 2 + z = 16$ *subset* Economics

Substituting for x and y values

$4 + 3 + 2 + z = 16$

$z = 16 - 9$

$= 7$

(a) $6 + 5 + x + y + z + 2 + 9 + f = 40$

Substituting for x, y and z values

$6 + 5 + 3 + 4 + 7 + 2 + 9 + f = 40$

$f = 40 - 36$

$= 4$ students failed

(b) 100 – (%pass in at least one of Econs and Maths)

$= 100 - \left(\frac{y+x}{40} \times 100\right)$

$= 100 - \left(\frac{7}{40} \times 100\right)$

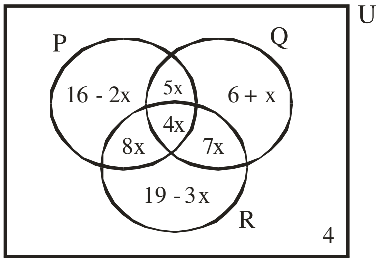
$= 100 - 17.5$ i.e 82.5

(C) Pr(failed Accts) = 1 – Pr(Pass Accts)

$= 1 - \frac{19}{40}$

$= \frac{21}{40}$

2014/ 9a



In the Venn diagram, P, Q and R are subsets of the universal set U. If n(U) = 125, find:

(i) the value of x ; (ii) $n(P\cup Q\cap R^1)$

Solution

n(U) = 125 implies everything inside the box equals 125

$16 - 2x + 5x + 4x + 8x + 19 - 3x + 7x + 6 + x + \mathbf{4} = 125$

4 is bolden because it is not in any circle but in the universal sets.

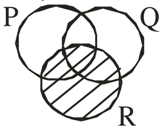
$45 + 25x - 5x = 125$

$20x = 125 - 45$

$20x = 80$

$x = 80/20$ i.e 4

(ii) $n(P\cup Q\cap R^1)$ refers to P and Q circle without R



$n(P\cup Q\cap R^1) = 16 - 2x + 5x + 6 + x$

Substituting for x = 4

$= 16 - 2(4) + 5(4) + 6 + 4$

$= 16 - 8 + 20 + 6 + 4$

$= 38$

1991/3

In a certain class, 22 pupils take one or more of chemistry, Economics and Government. 12 take Economics (E), 8 take Government (G) and 7 take Chemistry (C) nobody takes Economics and Chemistry and 4 pupils take Economics and Government.

(a), (i) Using set notation and the letters indicated above, write down two statements in the last sentence

(ii) Draw a Venn diagram to illustrate the information

(b). How many pupils take

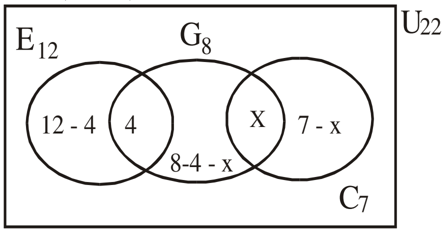
i.) Both Chemistry and Government

ii.) Government only?

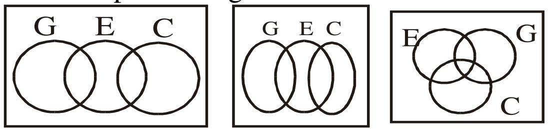
Solution

a (i) $n(E\cap C) = 0$ or $E\cap C = \varnothing$

$n(E\cap G) = 4$



The acceptable diagram is as shown above and **NOT**



Because econs and chem. are disjoint

b(i.) $12 - 4 + 4 + 8 - 4 - x + x + 7 - x = 22$

$23 - x = 22$

$23 - 22 = x$

$x = 1$

$n(C\cap G) = 1$

(ii) $n(G)$only $= 8 - 4 - x$

$= 4 - 1$ i.e 3

2008/8 NABTEB

In a Technical College, 115 students sat for Federal Craft Certificate Examination (FCCE), 69 of them passed Physics, 70 passed Technical Drawing and 80 passed Mathematics. Of these, 44 passed both Physics and Mathematics and 45 passed Technical Drawing and Mathematics. Given that 14 of them passed all the three subjects and 5 failed the three subjects, find the

(i) number of students who passed only Physics?

(ii) number of students who passed only one of the three subjects ?

(iii) number of students who passed only two of the three subjects?

(iv) probability of the number that passed only mathematics?

Solution

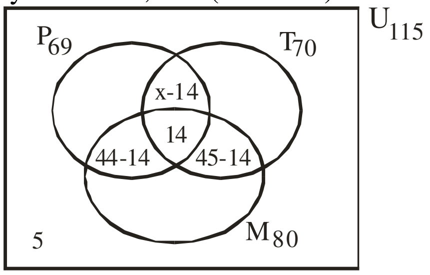
P – Physics $n(P) = 69$

T – Technical drawing $n(T) = 70$

M – Mathematics $n(M) = 80$

$n(P\cap M) = 44$, $n(T\cap M) = 45$, $n(P\cap M\cap T) = 14$

$n(P\cap T)$ only $= x - 14$, $n(P\cup M\cup T)^1 = 5$



Since $n(P\cup M\cup T) = n(U) - n(P\cup M\cup T)^1$

$$n(P\cup M\cup T) = n(P) + n(M) + n(T) - n(P\cap M) - n(P\cap T) - n(T\cap M) + n(P\cap M\cap T)$$

$$115 - 5 = 69 + 80 + 70 - 44 - x - 45 + 14$$

$$110 = 144 - x$$

$$x = 144 - 110$$

$$x = 34$$

(i) $n(P)\text{only} = 69 - [\,44 - 14 + 14 + (x - 14)\,]$

$= 69 - [\,30 + 14 + (34 - 14)]$

$= 69 - [\,30 + 14 + 20\,]$

$= 5$

(ii) $n(P)\text{only} + n(T)\text{only} + n(M)\text{only}$

$n(T)\text{only} = 70 - [\,45 - 14 + 14 + (x - 14)\,]$

$= 70 - [\,45 - 14 + 14 + 20\,]$

$= 5$

Also $n(M)\text{only} = 80 - [\,44 - 14 + 14 + 45 - 14\,]$

$= 5$

Thus $n(P)\text{only} + n(T)\text{only} + n(M)\text{only} = 5+5+5$ i.e 15

(iii) $n(P\cap T)\text{only} + n(T\cap M)\text{only} + n(P\cap M)\text{only}$

$= 44 - 14 + 45 - 14 + x - 14$

$= 30 + 31 + 20$

$= 81$

(iv) $Pr(\text{Maths})\text{only} = \dfrac{5}{115}$ i.e $\dfrac{1}{23}$

Exercise 9.25

In a Science Class, 18 students read Physics, 25 read Mathematics, 23 read Chemistry, 9 read Physics and Mathematics, 10 read Mathematics and Chemistry and 6 read Physics and Chemistry. If there are 50 students altogether and 5 students did not read any of the three subjects. How many students read

(i) all three subjects (ii) only Mathematics

(iii) Chemistry but not Mathematics

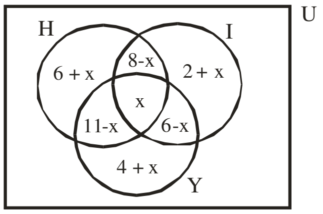
(iv) Physics and Chemistry but not Mathematics

Exercise 9.26

A sample poll of 128 voters revealed the following information in favour of three candidates A, B and C of a certain party in Agbarha – Otor. 25 were in favour of A alone. 66 favoured B, 18 favoured C alone. 9 were sitting on the fence. 39 were in favour of exactly two candidates. 72 were in favour of exactly one candidate. 30 were in favour of candidate A and atleast one other candidate. How many people were in favour of

(i) both A and C (ii) B and C only (iii) A or B

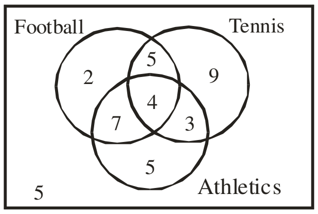
2007/10 PCE Exercise 9.27



From the diagram above if the universal set is 40, find x.

A 1 B 2 C 3 D4

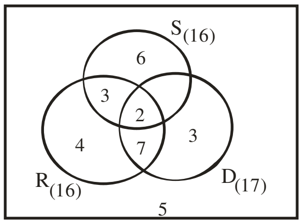
2005/23 UME Exercise 9.28



The Venn diagram above shows a class of 40 students with the games they play. How many of the students play two games only?

A 16 B 4 C 19 D 15

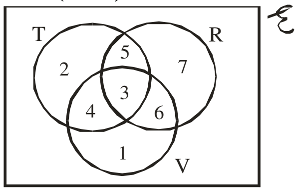
2011/ 44 Exercise 9.29



The Venn diagram shows the number of students in a class who like reading (R), dancing (D) and swinging (S). How many students like dancing and swimming?

A 7 B 9 C 11 D 13

2001/ 4 (Nov) Exercise 9.30



The Venn diagram shows the number of trades that sell radio (R), television (T) and video (V) in a market

(a) How many of the traders sell:

(i) both radio and television? (ii) at least two of the items?

2005/10 Neco Exercise 9.31

In a quiz competition of 60 competitors, 24drank Coca-cola, 25 drank Fanta while 22 drank Maltina, 6 drank Coca- cola and Maltina, 8drank Coca-cola and Fanta and 5 drank Fanta and Maltina. If x number of competitors drank the three types of soft drinks and 5 did not drink any of them,

(a) Draw a Venn-diagram to illustrate this piece of information

(b) How many drank:

(i) the three types of soft drink? (ii) Coca-cola only

(iii) Fanta only? (vi) Maltina only?

2014/ 10a Neco (Nov) Exercise 9.32

In a science class of 120 students, 56 passed Physics, 61 passed Chemistry, 60 passed biology, 20 passed both Physics and Chemistry, 24 passed both Physics and Biology, 26 passed both Chemistry and Biology and 5 passed none of the three subjects. Draw a Venn diagram to illustrate the information.

Find the number of students who passed:

(i) all the three subjects, (ii) Physics only,

(iii) Chemistry only.

CHAPTER TEN

Probability

This is a branch of mathematical statistics basically concerned with drawing conclusions or inferences from occurrences or experiments involving uncertainties. For any conclusion or inference made to be reasonably accurate an understanding of some probability theories is essential.

What does statements such as "Nigeria will probably win the football match against Norway", You will probably pass SSCE", There is a 30% chance that this job will not be finished on time".

In each case we are expressing the outcome of which we are not certain but because of part information or from understanding of the structure or the experiment or event bring about some degree of confidence in the validity of statement. *Chances, possibly, likely, Success, failure* etc are examples of words used in probability statements.

Definitions of basic terms used in probability

Sample space

A set whose elements represent all possible outcomes of an experiment usually denoted by **S**

Eg 1 A single throw of a die
S = {1, 2, 3, 4, 6}

Eg 2 The tossing of a coin once
S = {H, T}

Eg 3 The tossing of two coins
S = {HH, HT, HT, TT}

Eg 4 The launching of a missile from a Submarine. S = {Success, failure}

Event

Any Subset of the Sample space

Eg 5 Consider the experiment of tossing two coins
S = {HH, HT, TH, TT}

Then the event (E) that head appears on the first coin is
E = {HH, HT}

Outcome

It refers to all possible distinct results of a trial. Considering the examples given so far under basic definitions.

Examples	Outcomes
1	6
2	2
3	4
4	2

The concept of outcomes is similar to cardinality in set i.e n (A) or n (B) depending on the given set.

Experiment

This is the process through which outcomes are generated. The acts of throwing a die, tossing a coin, e.t .c are examples of experiment. Experiment can also be called **trial.**

…At most,…
The given number or less. For example; If Tosin asked his classmate to lend him at most ₦ 200.00. The classmate can as well give him ₦5, or ₦ 10. 00, ₦ 190. 00; as any of these amounts will solve his problem provided the money is not above ₦ 200.00.

…, At least,…
The given number or more. If the "at most" is replaced by "at least" in the previous example then the mentioned amount of ₦ 10 .00 or ₦ 190.00 or ₦ 190.99 will not take care of Tosin's demand except ₦ 200. 00 and above.

General formula

The probability of an event happening can be given a numerical value x ; where

$$x = \frac{\text{number of required outcomes}}{\text{number of possible outcomes}}$$

(a) When x = 1
If the probability of any event is1, then it is <u>sure</u> or <u>certain</u> to occur.

(b) When x = 0
If the value of x is 0 then the event is impossible. It will never occur.

(c) The range of solution to probability problem is between 0 and 1 i.e $0 \le Pr(x) \le 1$

The implication is that we do not have negative answers nor values greater than 1

(d) For any event, the probability of success plus failure:
Pr (Success) + Pr (failure) = 1
$Pr(x) + Pr(\bar{x}) = 1$

(e) The probability of **failure or non - occurrence or not happening :**
$Pr(\bar{x}) = 1 - Pr(x)$

General formula problems in probability

2009/15NATEB (Nov)
A box contains 9 blue, 6 red and 10 white beads. If a bead is taken at random, from the box, find the probability that it is white

A $^1/_5$ B $^2/_5$ C $^3/_5$ D $^4/_5$

Solution

$$Pr(\text{white bead}) = \frac{\text{Number of white beads}}{\text{Total number of beads}}$$

$$= \frac{10}{25} \text{ i.e } {}^2/_5 \quad (B)$$

2006/50 Neco
A bag contains 3 black, 4 yellow and 7 red balls. A ball is picked from the bag at random. What is the probability that it is white?

A 1 B $^7/_{14}$ C $^4/_{14}$ D $^3/_4$ E 0

Solution

$$Pr(\text{white}) = \frac{\text{number of white balls}}{\text{Total balls}}$$

$$= \frac{0}{14} \text{ i.e } 0 \quad (E)$$

it is zero since there is no white ball in the bag.

2009/39 NATEB (Nov)
A letter is chosen at random from the letters NIGERIA. What is the probability that it is an 'I'
A $^1/_7$ B $^2/_7$ C $^3/_7$ D $^4/_7$

Solution

$$\text{Pr(letter 'I')} = \frac{\text{Number of letter 'I'}}{\text{Total number of letters in Nigeria}} = \frac{2}{7} \quad (\text{B})$$

2012/ 59 Neco
A box contains the following set of balls numbered 3, 5, 7, 9, 11, 13, 3, 2, 3, 5, 8, 10. If a ball is picked at random, what is the probability that it is numbered 3?
A $^1/_{12}$ B $^1/_6$ C $^1/_4$ D $^2/_3$ E $^5/_{12}$

Solution

$$\Pr(3) = \frac{f(3)}{\Sigma f} = \frac{3}{12} \text{ i.e } \frac{1}{4} \quad (\text{C})$$

2013/60 Neco
What is the probability of winning the first prize by Dele who purchased 20 tickets in a lottery when 1000 tickets were sold?
A $^1/_{100}$ B $^1/_{50}$ C $^1/_{25}$ D $^1/_8$ E $^1/_2$

Solution

$$\text{Pr(winning)} = \frac{20}{1000} = {^1/_{50}} \quad (\text{B})$$

Theoretical probability

Theoretical probability is the probability calculated without the experiment being performed, i.e using only information that is known about the physical situation

EXAMPLES

1. Two numbers are chosen at random from 1, 3, 6. Find the probability that the sum of the two is not odd.

Solution

Arranging the given values in a tabular form, we have

+	1	3	6.
1	2	4	7
3	4	6	9
6	7	9	12

Pr (not odd) = 1 – Pr (odd)

$$= 1 - \frac{4}{9} \text{ i.e } {^5/_9}$$

2. Find the probability that a number selected at random from 41 to 56 is a multiple of 9.

Solution

Listing the numbers: 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56.
Total numbers = 16 i.e. (56 – 41) + 1
The multiples of 9 are two i.e. 45 & 54
Hence Pr (multiple of 9) = $^2/_{16}$ i.e $^1/_8$

2006/7
A fair coin is tossed three times. Find the probability of getting two heads and one tail
A $^1/_2$ B $^3/_8$ C $^1/_4$ D $^1/_8$

Solution

Recall the formula for power set 2^s
Here element of the sample space is $2^3 = 8$
{HHH, **HHT**, **HTH**, **THH**, TTH, THT, HTT, TTT}

$$\text{Pr(Two heads and one tail)} = \frac{3}{8} \quad (\text{B})$$

2006/5
Two fair dice are tossed together once.
(a) Draw the sample space for the possible outcomes
(b) Find the probability of getting a total;
(i) of 7 or 8 (ii) less than 4

Solution

(a)

Total	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

(b) i. $\text{Pr(Total 7 or 8)} = \frac{11}{36}$

ii. $\text{Pr(Total less than 4)} = \frac{3}{36} \text{ i.e } \frac{1}{12}$

2008/54 Neco (Dec)
A number is chosen at random from the set {1, 2, …,10}. What is the probability that it is a prime number?
A 0.3 B 0.4 C 0.5 D 0.6 E 0.7

Solution

First, we list out the members of each set
μ = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} ten of them
Prime numbers = {2, 3, 5, 7} four of them

$$\text{Pr(Prime number)} = \frac{4}{10} \text{ i.e } 0.4 \quad (\text{B})$$

2008/54 Neco (Dec) ***Adjusted***
A number is chosen at random from the set {1, 2, …,10}. What is the probability that it is an odd prime number?
A 0.3 B 0.4 C 0.5 D 0.6 E 0.7

Solution

First, we list out the members of each set
μ = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} ten of them
Odd - Prime numbers = {3, 5, 7} three of them

$$\text{Pr(Prime number)} = \frac{3}{10} \text{ i.e } 0.3 \quad (\text{A})$$

2008/60 Neco (Dec)
A die and a coin were thrown once, what is the probability of getting a head and six?
A $^1/_{12}$ B $^1/_6$ C $^1/_2$ D $^2/_3$ E 1

Solution

The sample space is as shown below:

		Die					
		1	2	3	4	5	6
	H	H1	H2	H3	H4	H5	H6
coin	T	T1	T2	T3	T4	T5	T6

Total sample space : 12

Head and six: 1

$$Pr\ (H6) = \frac{1}{12} \quad (A)$$

Alternatively: Applying multiplication rule

Coin	**Die**
Sample space {H, T}	{1, 2, 3, 4, 5, 6}
$Pr(H) = \frac{1}{2}$	$Pr(\text{a six}) = \frac{1}{6}$

$$\text{Thus } Pr(\text{a head and six}) = \frac{1}{2} \times \frac{1}{6}$$

$$= \frac{1}{12} \quad (A)$$

Empirical (experimental) probability

Empirical (Experimental) probability is calculated using the results of an experiment that has been performed a number of times.

In Experimental probability

$$Pr\ (E) = \frac{f\ (E)}{\sum f}$$

Where f(E) is the number of times the event E occurred

$\sum f$ is the total frequency

EXAMPLES

1. Some families were measured by the number of children they have; to give the following frequency distribution

No of children	0	1	2	3	4	5 or more
No of families	12	28	22	8	2	2

Use this information to calculate the probability that another family of this type will have

(a) 2 (b) 3 or more (c) less than 2 children

Solution

Here, $\sum f = 74$

(a) $Pr\ (2\ \text{children}) = \frac{f\ (2)\ \text{children}}{\sum f}$

$$= \frac{22}{74} = \frac{11}{37}$$

(b) $Pr\ (3\ \text{or more}) = \frac{f\ (3) + f\ (4) + f\ (5)}{\sum f}$

$$Pr\ (3\ \text{or more}) = \frac{8 + 2 + 2}{74}$$

$$= \frac{12}{74} = \frac{6}{37}$$

(c) $Pr\ (\text{less than 2 children}) = \frac{f\ (1) + f\ (0)}{\sum f}$

$$= \frac{28 + 12}{74}$$

$$= \frac{40}{74} = \frac{20}{37}$$

2. The table below shows the amount of money, which some students possess in their wallet at a given time.

Amount in ₦	1	2	3	4	5	6
No of students	1	3	2	5	1	4

If a student is chosen at random from the group, what is the probability that the student has

(a) ₦ 4 (b) ₦ 2

Solution

(a) $Pr\ (₦\ 4) = \frac{f\ (₦\ 4)}{\sum f}$

Here $\sum f$ is the total number of students

$$= \frac{5}{16}$$

(a) $Pr\ (₦2) = \frac{f\ (₦\ 2)}{\sum f}$

$$= \frac{3}{16}$$

2006/60 Neco (Dec)

Mass (kg)	55	56	57	58	59	60
No of students	4	7	12	10	8	9

The table above shows the masses to the nearest kilogram of 50 students in a class. If a student is chosen at random, what is the probability that he has mass of at least 58kg?

A $^1/_5$ B $^{17}/_{50}$ C $^{23}/_{50}$ D $^{27}/_{50}$ E $^{33}/_{50}$

Solution

$$Pr(\text{at least 58kg}) = \frac{f(58kg) + f(59kg) + f(60kg)}{\sum f}$$

$$= \frac{10 + 8 + 9}{50}$$

$$= {}^{27}/_{50} \quad (D)$$

2009/3a(iii)

The table shows the number of children per family in a community.

No. of children	0	1	2	3	4	5
No. of families	3	5	7	4	3	2

Find the probability that a family has **at least** 2 children

Solution

Pr(at least 2 children)

= Pr(2 children) + Pr(3 children) + Pr(4 children) + Pr(5 children)

$$= \frac{f(2)}{\Sigma f} + \frac{f(3)}{\Sigma f} + \frac{f(4)}{\Sigma f} + \frac{f(5)}{\Sigma f}$$

$$= \frac{7}{24} + \frac{4}{24} + \frac{3}{24} + \frac{2}{24}$$

$$= \frac{16}{24}$$

$$= \frac{2}{3}$$

2010/19

Number of pets	0	1	2	3	4
Number of students	8	4	5	10	3

The table shows the number of pets kept by 30 students in a class. If a student is picked at random from the class, what is the probability that he/she kept **more than** one pets?

A $^1/_5$ B $^2/_5$ C $^3/_5$ D $^4/_5$

Solution

$$Pr\ (\text{more than one pet}) = \frac{f(2) + f(3) + f(4)}{\Sigma f}$$

$$= \frac{5 + 10 + 3}{30}$$

$$= \frac{18}{30} \quad i.e \quad \frac{3}{5} \quad (C)$$

Addition Rule

Under this subheading *addition rule* we have twin concepts namely **mutually Exclusive (M.E)** and **non-mutually Exclusive Events.** The words *or* & *either* are used in both cases.

(a) Mutually Exclusive events (M.E)

Two events A and B are M.E if the occurrence of any one of them excludes the other.

Mathematically

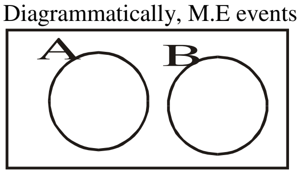
$Pr(A \text{ or } B) = Pr(A \cup B)$ recall that

$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$

But $Pr(A \cap B) = \varnothing$ Since nothing is common to both

Hence $Pr(A \cup B) = Pr(A) + Pr(B)$ M. E

Diagrammatically, M.E events A & B are represented below.



Examples of M.E Events

(a) The events of going to School by foot and by car are M. E; This is so because no single student can go to school by foot and at the same time by car . Impossible!

(b) The events of cooking a particular pot of soup using stove and gas cooker at the same time are M. E

(c) The events of the current principal of the Government college Ughelli to be serving and be retired at the same time are M.E

E.g. 1. A haulage contractor has 3 type A, 2 type B and 7 type C lorries available for deliveries, all of which are used frequently. What is the probability that a lorry delivering a load will be of type A or C?

Solution

If the contractor decides to use lorry A for the supply, automatically he cannot use lorry type C; thus, M. E

$$Pr(A \text{ or } C) = Pr(A) + Pr(C)$$
$$= \frac{3}{12} + \frac{7}{12}$$
$$= \frac{10}{12} = \frac{5}{6}$$

2009/59 NATEB (Nov)

A fair die is rolled once. What is the probability of obtaining either a 2 or a 5?

A $^1/_{36}$ B $^1/_6$ C $^1/_3$ D $^1/_2$ E $^2/_3$

Solution

Die sample space = {1, 2, 3, 4, 5, 6,}

Pr(either a 2 or a 5) = Pr(2) + Pr(5)

Nothing is common

$$= {}^1/_6 + {}^1/_6$$
$$= {}^2/_6 \text{ i.e. } {}^1/_3 \quad (C)$$

2012/50 Neco

A fair die is rolled once. What is the probability of obtaining 2 or 6?

A $^1/_{36}$ B $^1/_6$ C $^1/_3$ D $^1/_2$ E $^2/_3$

Solution

Die sample space = {1, 2, 3, 4, 5, 6} *i.e 6 elements*

Pr(2 or 6) = Pr(2) + Pr(6)

$$= \frac{1}{6} + \frac{1}{6}$$
$$= {}^2/_6 \text{ i.e } {}^1/_3 \quad (C)$$

2003/37 NABTEB

Two dice are thrown at the same time. What is the probability that the sum will be 7 or 11 ?

A. $^1/_9$ B. $^2/_9$ C. $^1/_3$ D. $^4/_9$

Solution

The sample space is as shown below :

sum	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Pr(sum 7 or 11) = Pr (sum 7) + Pr(sum 11)

$$= {}^6/_{36} + {}^2/_{36}$$
$$= {}^8/_{36}$$
$$= {}^2/_9 \text{ (B)}$$

2003/40 Neco

In a basket of fruits there are 6 grapes, 11 bananas and 13 oranges. If one fruit is chosen at random, what is the probability that the fruit is either a grape or banana ?

A. $^{17}/_{30}$ B. $^{11}/_{30}$ C. $^6/_{30}$ D. $^5/_{30}$ E $^{11}/_{150}$

Solution

Total fruit 6 + 11 + 13 i.e 30

Pr(a grape or banana) = Pr (grape) + Pr (banana)

$$= {}^6/_{30} + {}^{11}/_{30}$$
$$= {}^{17}/_{30} \quad (A)$$

2004/35

The table below gives the marks scored by a group of students in a test. Use the table to answer question 35

35. What is the probability of selecting a student from the group that scored 2 or 3 ?

Mark	0	1	2	3	4	5
Frequency	1	2	7	5	4	3

A. $^1/_{11}$ B. $^5/_{22}$ C. $^7/_{22}$ D. $^6/_{11}$

Solution

Pr(mark 2 or 3) = Pr(mark 2) + Pr(mark 3)

$$= \frac{f(2)}{\sum f} + \frac{f(3)}{\sum f}$$
$$= \frac{7}{22} + \frac{5}{22}$$
$$= \frac{12}{22} = \frac{6}{11} \quad (D)$$

VTR – 12/ 6a

A number is chosen at random from the set 15, 16, 17,… 32. What is the probability that the chosen number is

(i) a multiple of 7?

(ii) a prime number?
(iii) a prime or a multiple of 7?
Solution
Sample space : {15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31 32} 18 of them

Multiple of 7 : { 21, 28} 2 of them

Prime number: {17, 19, 23, 29, 31} 5 of them

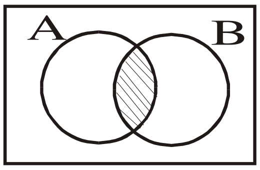
(i) Pr (multiple of 7) = $^2/_{18}$ i.e $^1/_9$

(ii) Pr (prime number) = $^5/_{18}$

(iii) Pr (prime or multiple of 7) = $^2/_{18} + ^5/_{18}$
$= ^7/_{18}$

Non – mutually Exclusive events

Two events A and B are non – mutually exclusive, if both events share something in common i.e. $A \cap B \neq \varnothing$
Mathematically, Pr (A or B) = Pr $(A \cup B)$,
$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$
Diagrammatically, non – M. E events A and B are represented below.



E.g If an event A implies obtaining prime number and event B implies obtaining an odd number in a single throw of a die. Find the probability of events A or B

Solution
S = {1,2, 3, 4, 5, 6}
A = {2, 3, 5} and B = {1, 3, 5}, also $A \cap B$ = {3, 5}
Pr (A) = $^3/_6$ Pr (B) = $^3/_6$ Pr $(A \cap B)$ = $^2/_6$

Thus Pr (A or B) = Pr (A) + Pr (B) - Pr $(A \cap B)$
$= ^3/_6 + ^3/_6 - ^2/_6$
$= \frac{3 + 3 - 2}{6}$
$= ^4/_6 = ^2/_3$

Note
When any of the word *or* & *either* is used, students should not jump into solving for addition. Rather, they should *first* identify the class of the problem (i.e. M. E or non – M. E).

2008/ 54 Neco
Bello chooses a number randomly from 1 to 10.
What is the probability that it is either odd or prime?
A $^1/_{10}$ B $^2/_5$ C $^1/_2$ D $^3/_5$ E $^9/_{10}$

Solution
The word **or** here tells us of addition but first we check for common terms
μ = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} Ten of them
Odd = {1, 3, 5, 7, 9} five of them
Prime = {2, 3, 5, 7} four of them
Odd and Prime = {3, 5, 7} three of them
Pr(either odd or prime) = Pr(odd) + Pr(prime) – Pr (odd and prime)

$= \frac{5}{10} + \frac{4}{10} - \frac{3}{10}$

$= \frac{5 + 4 - 3}{10}$

$= ^6/_{10}$ i.e $^3/_5$ (D)

2011/6
If a number is chosen at random from the set{x : 4 ≤ x ≤15}, find the probability that it is a multiple of 3 or a multiple of 4
A $^1/_{12}$ B $^5/_{12}$ C $^1/_2$ D $^{11}/_{12}$

Solution
First, we list the mother set and their subsets
X = {4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}
Multiples of 3 = {6, 9, 12, 15} *four of them*
Multiples of 4 = {4, 8, 12} *three of them*
$A \cap B$ = {12} *one item*
Pr(multiple of 3 **or** a multiple of 4)
= Pr(Multiple 3) + Pr(Multiple 4) – Pr(common item)
$= \frac{4}{12} + \frac{3}{12} - \frac{1}{12}$
$= \frac{6}{12}$ *i.e* $\frac{1}{2}$ (C)

2008/ 60
A class of 15 students offers either Physics or Chemistry or both. If 11 offer Physics and 9 offer Chemistry. What is the probability that a student chosen at random offers both Physics and Chemistry?
A $^1/_4$ B $^1/_3$ C $^3/_5$ D $^3/_4$ E 1

Solution
Apply set formula for two subsets
$n(P \cup C) = n(P) + n(C) - n(P \cap C)$
$15 = 11 + 9 - n(P \cap C)$
$n(P \cap C) = 20 - 15$
$= 5$
Thus Pr $(P \cap C) = \frac{5}{15}$ i.e $\frac{1}{3}$ (B)

VTR- 12 /45
There are twelve cards numbered 1 to 12. A card is selected at random. What is the probability that it is either even or a perfect square ?
A.$^1/_8$ B.$^1/_4$ C.$^7/_{12}$ D.$^2/_3$ E.$^3/_4$
Solution
Sample space: {1, 2, 3, 4, 5, 6. 7, 8, 9, 10, 11, 12}
12 of them
Even cards : {2, 4, 6, 8, 10, 12,} 6 of them

Perfect square cards: {4, 9,} 2 of them

$E \cap P$ = {4} 1 of them
Pr(even or perfect square) = Pr(E) + Pr(P) – Pr$(E \cap P)$
$= ^6/_{12} + ^2/_{12} - ^1/_{12}$
$= \frac{6 + 2 - 1}{12} = ^7/_{12}$ (C)

Multiplication rule

Under multiplication rule there is also twin concepts of **independent** and **dependent** events. The words "*and*" or "*both*" is used in both cases

Independent events

Two events A and B are said to be independent if the occurrence or non – occurrence of event A does not affect the probability of event B

Mathematically.

$$Pr(A \text{ and } B) = Pr(A) \times Pr(B)$$

i.e. $Pr(A \cap B) = Pr(A) \times Pr(B)$

Examples

1. Three balls are drawn successively from a box containing 8 red balls, 6 white balls and 4 black balls. Find the probability that they are drawn in order red, white and black if each ball is:

(i) Replaced.

Solution

You will agree with the author that once a ball is removed and replaced it does not affect the 2nd draw. Hence this is an independent event.

Pr (RWB) = Pr (R) x Pr (W) x Pr (B)

$$= \frac{8}{18} \times \frac{6}{18} \times \frac{4}{18}$$

$$= \frac{8}{243}$$

2008/ 38 NATEB (Nov) counter e.g

A president and a secretary are to be chosen for a club from a group of 4 girls and 6 boys. What is the probability that both are girls?

A $^4/_{25}$ B $^2/_{15}$ C $^1/_3$ D $^2/_5$

Solution

Pr(President and Secretary) = Pr(girl president) × Pr(girl secretary)

$$= \frac{4}{10} \times \frac{3}{9}$$

$$= \frac{2}{15} \quad \text{(B) dependent event}$$

2008/ 36 NATEB (Nov)

A bag contains 8 red balls and 12 white balls of the same size. If a ball is picked at random from the bag and replaced and a second picked. What is the probability that they are of the same colour?

A $^4/_{25}$ B $^9/_{25}$ C $^{13}/_{25}$ D $^{17}/_{25}$

Solution

Pr(same colour) = Pr(RR) or Pr(WW)

$$= \frac{8}{20} \times \frac{8}{20} + \frac{12}{20} \times \frac{12}{20}$$

Replaced here implies independent event hence the first ball picking did not affect the 2nd pick

$$= \frac{13}{25} \quad \text{(C)}$$

VTR – 11/42 NTI TCII

A pair of fair dice with each face numbered 1 to 6 are thrown once. Find the probability of scoring a 2 on one die and a 5 on the other

A. $^1/_{36}$ B. $^1/_{18}$ C. $^7/_{36}$ D. $^1/_3$ E. $^2/_3$

Solution

"and" shows multiplication rule. The probability of scoring any number on one die is independent of scoring a number on the second die. Thus

First die 2nd die

Pr(2) = $^1/_6$ Pr(5) = $^1/_6$

Pr (2 and 5) = Pr(2) x Pr(5)

$= {}^1/_6 \times {}^1/_6$

$= {}^1/_{36}$ (A)

2003/11a NABTEB

A bag contains 3 black balls and 2 red balls. A ball is picked at random from the bag and then replaced. A second ball is chosen at random, What is the probability that

(i) they are both black ?

(ii) one is black and the other is red ?

Solution

The words "both" in (i) "and" in (ii) shows multiplication rule.

(i) Pr(BB) = Pr (B) x Pr (B)

$= {}^3/_5 \times {}^3/_5$

$= {}^9/_{25}$

(ii) Pr (B and R) = Pr(B) xPr (R)

$= {}^3/_5 \times {}^2/_5$

$= {}^6/_{25}$

2004/5(a) (Nov)

A class consists of 30 boys and 20 girls. 60% of the boys and 40% of the girls can swim. A boy and a girl are chosen at random, find the probability that both of them can swim

Solution

The word "Both" reminds us of multiplication rule ; while the boy's ability to swim is independent of the girl's ability.

Thus, Pr (B and G) = Pr (B) xPr(G)

$$Pr(\text{swim}) = \frac{\text{Number of swim able pupils}}{\text{Total number of pupils}}$$

$$\text{Boys (swim)} = \frac{60}{100} \times 30 = 18$$

Total boys = 30

$$Pr(\text{swim able boys}) = \frac{18}{30}$$

$$\text{Girls (swim)} = \frac{40}{100} \times 20 = 8$$

Total girls = 20

$$Pr(\text{swim able girls}) = \frac{8}{20}$$

$$\text{Therefore, } Pr(B \text{ and } G) = \frac{18}{30} \times \frac{8}{20} = \frac{6}{25}$$

Alternatively

The ability of the boy to swim is independent of the girls ability.

Pr(Boys and girls swimming) = Pr (B) x Pr (G)

$$= \frac{60}{100} \times \frac{40}{100}$$

$$= \frac{3}{5} \times \frac{2}{5} = \frac{6}{25}$$

Dependent events

Two events A and B are said to be dependent, if the occurrence of event A affect the occurrence of B. then

Pr (A and B) = Pr (A) x Pr (B / A)

i.e Pr ($A \cap B$) = Pr (A) x Pr (B / A)

The Pr (B / A) reads probability of B given that A has occurred. Though there are cases where the candidate will represent the above by Pr ($A \cap B$) = Pr (A) x Pr (B) and still apply the sense of dependence in the working. It is acceptable at O/L.

Examples

a. From e. g 1 under independent events lets impose a second condition i.e

(ii) not replaced.

Solution

If each ball drawn is not replaced then the number left inside the box must be affected; hence this case must be dependent.

Pr (RWB) = Pr (R) x Pr (W/R) x Pr (B/RW)

$$= \left(\frac{8}{18}\right) \times \left(\frac{6}{17}\right) \times \left(\frac{4}{16}\right)$$

8 + 6 + 4 7 + 6 + 4 7 +5 + 4

$$= \frac{2}{51}$$

Analysis

In the first draw, the total balls and the number of red balls is not affected. In the 2nd draw for white, the total balls has reduced as a result of the non – replacement of the first drawn ball but the number of white balls is not affected.In the 3rd draw, the total is affected because of the 1st and 2nd drawn balls that are not replaced, though the number of black balls is not affected.

b. A science class consisting of 2 boys and 8 girls all participated in an end of year quiz. Find the probability of choosing both slot A and B by girls from the group to represent the college at the local government level.

Solution

The word "both" reminds us of multiplication rule.

Let G1 – Girls choose the first slot

Let G2 – Girls choose the 2nd slot

Then

Pr (G1 and G2) = Pr (G1) x Pr (G2/G1)

$$= \frac{8}{10} \times \frac{7}{9} = 0.62$$

Once a girl has chosen the first slot, there are now 7 girls left and the total number of the group reduces to 9.

2012/56 Neco

If 2 balls are drawn from a bag containing 5 blue and 10 red balls without replacement, find the probability that the balls drawn are blue.

A $^2/_{21}$ B $^1/_9$ C $^8/_{21}$ D $^{13}/_{21}$ E $^2/_3$

Solution

$Pr(B_1B_2) = Pr(B_1) \times Pr(B_2 / B_1)$

$$= \frac{5}{15} \times \frac{4}{14}$$

After the first draw without replacement; the next draw for blue will have 4 blue balls and the total reduced to14

$$= \frac{2}{21} \quad (A)$$

2012/ 28

A bag contains 4 red and 6 black balls of the same size. If the balls are shuffle briskly and two balls are drawn one after the other **without** replacement, find the probability of picking balls of different colours.

A $^8/_{15}$ B $^{13}/_{25}$ C $^{11}/_{15}$ D $^{13}/_{15}$

Solution

Drawing without replacement is a dependent event

Pr(Different colours) = Pr(R) Pr(B) or Pr(B) Pr(R)

$$= \left[\frac{4}{10} \times \frac{6}{9}\right] + \left[\frac{6}{10} \times \frac{4}{9}\right]$$

Once the first ball is picked, it will affect the total number of balls in the 2nd draw

$$= \frac{4}{15} + \frac{4}{15}$$

$$= {}^8/_{15} \quad (A)$$

2003/38 Neco

A packet contains 4 red, 5 blue and 6 black identical biros. Two biros are picked at random without replacement; find the probability of picking a red and a black biro.

A. $^{73}/_{105}$ B. $^2/_3$ C. $^4/_{35}$ D. $^8/_{35}$ E. $^8/_{125}$

Solution

Total biros: 4 + 5 + 6 i.e 15

Pr (R and B) = Pr(R) x Pr (B/R)

$= {}^4/_{15} \times {}^6/_{14}$

The second draw is affected in the total because of the NO replacement

$= {}^4/_5 \times {}^1/_7$

$= {}^4/_{35}$ (C)

General problems on probability

2000/11a (Nov)

Chinedu and Kareen take part in a test. The probability that Chinedu pass the test is $^1/_3$ and the probability that kareen pass is $^4/_5$. Calculate the probability that Chinedu or Kareen passes the test.

Solution

Pr(Chinedu or Kareen pass) =

Pr(Chinedu pass)Pr(Kareen fail) or Pr(Chinedu fail) Pr(Kareen pass)

$$= \frac{1}{3}\times\left(1-\frac{4}{5}\right) + \left(1-\frac{1}{3}\right)\times\frac{4}{5}$$

$$= \frac{1}{3}\times\frac{1}{5} + \frac{2}{3}\times\frac{4}{5}$$

$$= \frac{1}{15} + \frac{8}{15}$$

$$= \frac{9}{15} \text{ or } \frac{3}{5}$$

2003/ 24

Out of 60 members of an Association, 15 are Doctors and 9 Lawyers. If a member is selected at random from the Association, what is the probability that the member is neither a Doctor nor a Lawyer?

A $^3/_5$ B $^9/_{10}$ C $^3/_{20}$ D $^1/_4$

Solution

Pr(neither Doctor nor Lawyer) = 1 – Pr(Doctor or Lawyer)

$$= 1-\left(\frac{15}{60}+\frac{9}{60}\right)$$

$$= 1 - \frac{24}{60}$$

$$= \frac{36}{60} \text{ i.e } \frac{3}{5} \quad (A)$$

2016/29

Find the probability of picking the letter T from the word "STUDENT"

A $^1/_7$ B $^1/_6$ C $^2/_7$ D $^1/_3$

Solution

$$Pr(T) = \frac{\textit{Numbers of letter T}}{\textit{Total numbers of letters in STUDENT}}$$

$$= \frac{2}{7} \ (C)$$

2005/7a Neco ***good technical example***

The probability that Audu and Tunde will pass a mathematics test is 0.6, 0.8 respectively and the probability that Nnah will not pass the same test is 0.25. Find the probability that:

(a) all of them passed;
(b) none of them passed;
(c) one of them failed;

Solution

(a) Pr(all pass) = Pr(Audu pass) Pr(Tunde pass) Pr(Nnah pass)

$= 0.6 \times 0.8 \times (1 - 0.25)$
$= 0.6 \times 0.8 \times 0.75$
$= 0.36$

(b) Pr(all fail) = Pr(Audu fail) Pr(Tunde fail) Pr(Nnah fail)

$= (1 - 0.6)(1 - 0.8) \times 0.25$
$= 0.4 \times 0.2 \times 0.25$
$= 0.02$

(c) Pr(One of them failed) = Pr(A) Pr(T) Pr($\overline{N}$) **or**

Pr(A) Pr($\overline{T}$) Pr(N) **or** Pr($\overline{A}$) Pr(T) Pr(N)

$= 0.6 \times 0.8 \times 0.25 + 0.6 \times (1 - 0.8)\ 0.75 + (1 - 0.6)0.8 \times 0.75$
$= 0.12 + 0.09 + 0.24$
$= 0.45$

2005/42 Neco

A bag contains 5 yellow, 6 red and 4 blue balls. If a ball is picked from the bag at random, find the probability that it is NEITHER red NOR yellow

A $^{11}/_{15}$ B $^3/_5$ C $^2/_5$ D $^1/_3$ E $^4/_{15}$

Solution

We have 15 balls in all

Pr(neither red nor yellow) = 1 – Pr(red or yellow)

$$= 1 - \left(\frac{6}{15} + \frac{5}{15}\right)$$

$$= 1 - \frac{11}{15}$$

$$= \frac{4}{15} \quad (E)$$

2005/44

A fair die numbered 1 to 6 is rolled once. What is the probability of obtaining 3 or 5?

A $^1/_6$ B $^1/_4$ C $^1/_3$ D $^1/_2$

Solution

A die sample space is {1, 2, 3, 4, 5, 6}

Pr(3 or 5) = Pr(3) + Pr(5)

This is so since nothing is common to both

$$= \frac{1}{6} + \frac{1}{6}$$

$$= \frac{2}{6} \text{ i.e } \frac{1}{3} \quad (C)$$

2005/50

A bag contains four red, three white and five green balls. If one ball is picked at random, what is the probability that it is **not** green?

A $^7/_{12}$ B $^5/_{12}$ C $^1/_7$ D $^1/_{12}$

Solution

Total ball is 12

Pr(not green) = 1 – Pr(green)

$$= 1 - \frac{5}{12}$$

$$= \frac{7}{12} \quad (A)$$

2005/26 (Nov)

The probability that a member of a club owns a car is $^1/_3$. If two members are selected at random, what is the probability that only one of the two owns a car?

A $^1/_9$ B $^2/_9$ C $^4/_9$ D $^5/_9$

Solution

Let the two members be X and Y

Pr(only one owns car) = Pr(X) Pr($\overline{Y}$) **or** Pr($\overline{X}$) Pr(Y)

$$= \frac{1}{3}\left(1 - \frac{1}{3}\right) + \left(1 - \frac{1}{3}\right)\frac{1}{3}$$

$$= \frac{1}{3} \times \frac{2}{3} + \frac{2}{3} \times \frac{1}{3}$$

$$= \frac{2}{9} + \frac{2}{9}$$

$$= \frac{4}{9} \quad (C)$$

2005/ 5a (Nov)

A bag contains 5 white and 3 red identical balls. Two balls are drawn at random, one after the other without replacement. Find the probability of drawing balls of the same colour.

Solution

We have total of 8 balls

Pr(same colour) = Pr(WW) **or** Pr(RR)

Note: without replacement implies 1st draw will affect 2nd draw i.e dependent event

$$= \frac{5}{8} \times \frac{4}{7} + \frac{3}{8} \times \frac{2}{7}$$

$$= \frac{5}{14} + \frac{3}{28}$$

$$= \frac{10 + 3}{28} = \frac{13}{28}$$

2006/52 Neco (Dec) ***Prob of NOT counter eg***

The probability that Usman, Femi and Emeka fail a test are $^1/_4$, $^1/_5$ and $^1/_6$ respectively. Find the probability that all the boys pass the test.

A $^{37}/_{60}$ B $^1/_2$ C $^1/_3$ D $^7/_{30}$ E $^1/_6$

Solution

$$\Pr(\text{all pass}) = \Pr(U)\Pr(F)\Pr(E)$$
$$= (1 - {}^1/_4)(1 - {}^1/_5)(1 - {}^1/_6)$$
$$= \frac{3}{4} \times \frac{4}{5} \times \frac{5}{6}$$
$$= {}^1/_2 \ (B)$$

2006/8a NATEB (Nov)

Two perfect dice are thrown together. Calculate the probability that the sum is (i) 9 or 10 (ii) at most 5

Solution

Sum	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

(i) $\Pr(9 \text{ or } 10) = \frac{7}{36}$

(ii) At most 5 implies total 5 and other lowest numbers

$\Pr(\text{at most } 5) = \frac{10}{36}$ i.e $\frac{5}{18}$

2006/27 (Nov)

A class has 22 boys and 18 girls while another has 20 boys and 20 girls. A student is chosen at random from each class. Find the probability that both are boys.

A $^{31}/_{40}$ B $^{11}/_{20}$ C $^{11}/_{40}$ D $^9/_{40}$

Solution

Pr(Both boys) = Pr(Boy**1st** class) Pr(Boy in **2nd** class)

$$= \frac{22}{40} \times \frac{20}{40}$$
$$= \frac{11}{40} \ (C)$$

2006/26 (Nov)

A bag contains 3 red, 2 blue and 5 green marbles. One marble is drawn from the bag at random. What is the probability that it is **neither** red **nor** green?

A $^2/_3$ B $^3/_5$ C $^1/_2$ D $^1/_5$

Solution

Total marbles is ten

$$\Pr(\text{neither red nor green}) = 1 - \Pr(\text{red or green})$$
$$= 1 - \left(\frac{3}{10} + \frac{5}{10}\right)$$
$$= 1 - \frac{8}{10}$$
$$= \frac{2}{10} \text{ i.e } \frac{1}{5} \ (D)$$

2007/51 Neco

The probability of Abdul passing any examination is $^2/_3$. If Abdul takes three examinations, what is the probability that he will not pass any of them?

A $^2/_3$ B $^4/_9$ C $^1/_3$ D $^8/_{27}$ E $^1/_{27}$

Solution

Let the three exams be A, B and C

$$\Pr(\text{He will not pass any of them}) = \Pr(\text{fail all})$$
$$= \Pr(\overline{A})\Pr(\overline{B})\Pr(\overline{C})$$
$$= (1 - {}^2/_3)(1 - {}^2/_3)(1 - {}^2/_3)$$
$$= \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3}$$
$$= \frac{1}{27} \ (E)$$

2007/5

Out of the 24 apples in a box, 6 are bad. If three apples are taken from the box at random, with replacement, find the probability that:

(a) the first two are good and the third is bad

(b) all the three are bad

(c) all the three are good.

Solution

Let the 3 apples be X, Y and Z

Bad apples are **6** implies good apples are 24 – 6 i.e **18**

Replacement implies **independent** events

(a) Pr(First two good and third bad)

$$= \Pr(X)\Pr(Y)\Pr(\overline{Z}) + \Pr(X)\Pr(Z)\Pr(\overline{Y}) + \Pr(Y)\Pr(Z)\Pr(\overline{X})$$
$$= \frac{18}{24} \times \frac{18}{24} \times \frac{6}{24} + \frac{18}{24} \times \frac{18}{24} \times \frac{6}{24} + \frac{18}{24} \times \frac{18}{24} \times \frac{6}{24}$$

First take does not affect the second one, this is so because they are independent events

$$= \frac{3}{4} \times \frac{3}{4} \times \frac{1}{4} + \frac{3}{4} \times \frac{3}{4} \times \frac{1}{4} + \frac{3}{4} \times \frac{3}{4} \times \frac{1}{4}$$
$$= 3\left(\frac{3}{4} \times \frac{3}{4} \times \frac{1}{4}\right)$$
$$= \frac{27}{64}$$

(b) $\Pr(\text{all three bad}) = \Pr\ \Pr(\overline{X})\Pr(\overline{Y})\Pr(\overline{Z})$

$$= \frac{6}{24} \times \frac{6}{24} \times \frac{6}{24}$$

With replacement, no take affects the other

$$= \frac{1}{4} \times \frac{1}{4} \times \frac{1}{4}$$
$$= \frac{1}{64}$$

(c) $\Pr(\text{all three good}) = \Pr(X)\Pr(Y)\Pr(Z)$

$$= \frac{18}{24} \times \frac{18}{24} \times \frac{18}{24}$$

With replacement, no take affects the other

$$= \frac{3}{4} \times \frac{3}{4} \times \frac{3}{4}$$
$$= \frac{27}{64}$$

2008/ 55 Neco

Two fair dice are tossed together once. What is the probability of getting a total of at most 7 from the outcomes?

A $^7/_{12}$ B $^5/_{12}$ C $^1/_4$ D $^7/_{36}$ E $^1/_6$

Solution

A total of **at most** 7 implies total of 7 or less

Solution

(a)

Total	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Pr(at most 7) = $\frac{21}{36}$ i.e $\frac{7}{12}$ (A)

2008/56

The chances of Usman and Dele passing a Mathematics test are $^3/_5$ and $^1/_3$ respectively. What is the probability that neither of them passes the test?

A $^3/_5$ B $^2/_5$ C $^4/_{15}$ D $^1/_5$ E $^2/_{15}$

Solution

Pr(Neither pass) = Pr(both failing)

= Pr(U fail) Pr(D fail)

= $(1 - ^3/_5)(1 - ^1/_3)$

= $\frac{2}{5} \times \frac{2}{3}$

= $\frac{4}{15}$ (C)

2008/40 NATEB (Nov)

A die is thrown twice. What is the probability that each throw results in a 2?

A $^1/_{36}$ B $^1/_9$ C $^1/_3$ D $^2/_5$

Solution

a, a	1	2	3	4	5	6
1	1,1	1,2	1,3	1,4	1,5	1,6
2	2,1	2,2	2,3	2,4	2,5	2,6
3	3,1	3,2	3,3	3,4	3,5	3,6
4	4,1	4,2	4,3	4,4	4,5	4,6
5	5,1	5,2	5,3	5,4	5,5	5,6
6	6,1	6,2	6,3	6,4	6,5	6,6

Pr(each throw results in a 2) = $\frac{1}{36}$ (A)

2008/ 47 – 49

Given the sets A = {2, 4, 6, 8} and B = {2, 3, 5, 9}

Use the information to answer the questions 47 – 49.

47. If a number is selected at random from set B, what is the probability that the number is prime?

A 1 B $^3/_4$ C $^1/_2$ D $^1/_4$

Solution

B = {2, 3, 5, 9} four of them

Prime at B = {2, 3, 5} three of them

Thus Pr(prime at B) = $\frac{3}{4}$

48. If a number is picked at random from each of the two sets, what is the probability that their product is odd?

A 1 B $^3/_4$ C $^1/_4$ D 0

Solution

A

	×	2	4	6	8
	2	4	8	12	16
B	3	6	12	18	24
	5	10	20	30	40
	9	18	36	54	72

Pr(product odd) = 0 (D)

All their products are even, no odd number among them

49. If a number is picked at random from each of the two sets, what is the probability that their difference is 6 or 7?

A $^1/_{256}$ B $^1/_{16}$ C $^1/_8$ D $^1/_2$

Solution

We take absolute values of difference, since the question is not restricted to any order in which the number be picked

A

	–	2	4	6	8
	2	0	2	4	6
B	3	1	1	3	5
	5	3	1	1	3
	9	7	5	3	1

Pr(Difference 6 of 7) = Pr(6) + Pr(7)

Nothing is common to both

= $\frac{1}{16} + \frac{1}{16}$

= $^2/_{16}$ = $^1/_8$ (C)

2009/41

A box contains black, white and red identical balls. The probability of picking a black ball at random from the box is $^3/_{10}$ and the probability of picking a white ball at random is $^2/_5$. If there are 30 balls in the box, how many of them are red?

A 3 B 7 C 9 D 12

Solution

Numbers of black balls

Pr(black) = $\frac{\text{number of black ball } (\mathbf{b})}{\text{Total balls}}$

$\frac{3}{10} = \frac{b}{30}$

$\frac{3 \times 30}{10} = b$

b = 9 black balls

Numbers of white balls

Pr(white) = $\frac{\text{number of white ball (w)}}{\text{Total balls}}$

$\frac{2}{5} = \frac{w}{30}$

$\frac{2 \times 30}{5} = w$

w = 12 white balls

Thus red balls = 30 – (9 +12)

= 9 C.

2009/ 31 (Nov)

The probability that Shehu passes his examination is $^3/_5$ and the probability that he gets admitted to a University is $^2/_3$. What is the probability that he passes his examination but does not get admitted?

A $^1/_5$ B $^2/_{15}$ C $^4/_{15}$ D $^2/_5$

Solution

Pr(passes exam but not admitted)

$$= \Pr(\text{passes exam}) \Pr(\text{not admitted})$$
$$= \frac{3}{5} \times \left(1 - \frac{2}{3}\right)$$
$$= \frac{3}{5} \times \frac{1}{3}$$
$$= {}^{1}/_{5} \quad (\text{A})$$

2009/3 (Nov)

The probability that a malaria patient (M) survives when administered with a newly discovered drug is 0.27 and the probability that a typhoid patient (T) survives when injected with another newly discovered drug is 0.85. What is the probability that

(a) **either** of the two patients survives?

(b) **neither** of the two patients survives?

(c) **at least** one of the two patients survives?

Give your answers correct to 2 significant figures.

Solution

(a) Pr(either M or T survives)

M survives T dies **or** M dies T survives

$= \Pr(M)\ \Pr(\overline{T})$ **or** $\Pr(\overline{M})\ \Pr(T)$

$= 0.27 \times (1 - 0.85) + (1 - 0.27) \times 0.85$

$= 0.27 \times 0.15 + 0.73 \times 0.85$

$= 0.0405 + 0.6205$

$= 0.661$

≈ 0.66 to 2 s.f

M dies T dies

(b) Pr(Neither survives) $= \Pr(\overline{M})\ \Pr(\overline{T})$

$= (1 - 0.27)(1 - 0.85)$

$= 0.73 \times 0.15$

$= 0.1095$

≈ 0.11 to 2 s.f

(C) Pr(**at least** one survives)

M survives T dies **or** M dies T survives **or** M survives T survives

$= \Pr(M)\Pr(\overline{T})$ **or** $\Pr(\overline{M})\Pr(T)$ **or** $\Pr(M)\Pr(T)$

$= 0.27 \times (1 - 0.85) + (1 - 0.27)(0.85) + (0.27)(0.85)$

$= 0.27 \times 0.15 + 0.73 \times 0.85 + 0.27 \times 0.85$

$= 0.0405 + 0.6205 + 0.2295$

$= 0.8905$

≈ 0.89 to 2 s.f

2010/ 55 Neco

A number is selected randomly from the set of integers 1 to 20 inclusive. Find the probability that the number is less or equal to 9

A $^4/_9$ B $^4/_{10}$ C $^9/_{20}$ D $^{11}/_{20}$ E $^6/_{10}$

Solution

Set of integers = {1, 2, 3, … 20}

20 of them since 1 and 20 are inclusive

Subset of numbers less or equal to 9 ={1, 2, 3,…9} 9 of them

Pr(number less or equal to 9) $= \frac{9}{20}$ (C)

2010/25 NABTEB (Nov)

A bag contains three red balls, four blue balls, five white balls and six black balls. A ball is picked at random. What is the probability that it is either red or blue?

A $^5/_{18}$ B $^7/_{15}$ C $^4/_5$ D $^7/_{18}$

Solution

Pr(R or B) = Pr(R) + Pr(B)

Mutually exclusive events i.e nothing is common

$$= \frac{3}{18} + \frac{4}{18}$$
$$= {}^{7}/_{18} \quad (\text{D})$$

2010/24 NABTEB (Nov)

A coin is tossed and a die is thrown. What is the probability of getting a head and perfect square?

A $^1/_{12}$ B $^1/_4$ C $^1/_6$ D $^1/_3$

Solution

The sample space is as shown below:

Die

	1	2	3	4	5	6
H	H1	H2	H3	H4	H5	H6
T	T1	T2	T3	T4	T5	T6

Total sample space : 12

Head and a perfect square is 1

Pr(head and a perfect square) $= \frac{1}{12}$

2010/ 5

Two fair dice are thrown.

M is the event described by "the sum of the scores is 10" and N is the event described by "the difference between the scores is 3".

(a) Write out the element of M and N

(b) Find the probability of M and N

(c) Are M and N mutually exclusive? Give reasons

Solution

We generate two different tables

(a) Table for M

Sum	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Table for N

Difference	1	2	3	4	5	6
1	0	1	2	3	4	5
2	1	0	1	2	3	4
3	2	1	0	1	2	3
4	3	2	1	0	1	2
5	4	3	2	1	0	1
6	5	4	3	2	1	0

We take absolute values of difference, two fair dice are identical and when tossed, we subtract the smaller value from bigger value

Pr(M and N) = Pr(M) × Pr(N)

$$= \frac{3}{36} \times \frac{6}{36} = \frac{1}{72}$$

(c) M and N are mutually exclusive event .

Event M sum occurring does excludes event N of difference. In other words we can't take sum and difference at the same time on the values generated by the throw of the two dice.

2010/ 47

In an athletics competition, the probability that an athlete wins a 100m race is $^1/_8$ and the probability that he wins in high jump is $^1/_4$. What is the probability that he wins **only one** of the events?

A $^2/_{32}$ B $^3/_{16}$ C $^7/_{32}$ D $^5/_{16}$

Solution

Pr(wins only one) =

Pr(win race) Pr(fail jump) **or** Pr(fail race) Pr(win jump)

$$= \frac{1}{8} \times \left(1 - \frac{1}{4}\right) + \left(1 - \frac{1}{8}\right) \times \frac{1}{4}$$

$$= \frac{1}{8} \times \frac{3}{4} + \frac{7}{8} \times \frac{1}{4}$$

$$= \frac{3}{32} + \frac{7}{32}$$

$$= \frac{10}{32} \text{ i.e } \frac{5}{16} \quad (\text{ D })$$

2011/ 3a Neco

A fair die is rolled twice. Find the probability of getting:

(i) the same number

(ii) at least a total of 7

Solution

The *two problems* will carry different tables for clarity

(i) Table for the same number

a, a	1	2	3	4	5	6
1	1,1	1,2	1,3	1,4	1,5	1,6
2	2,1	2,2	2,3	2,4	2,5	2,6
3	3,1	3,2	3,3	3,4	3,5	3,6
4	4,1	4,2	4,3	4,4	4,5	4,6
5	5,1	5,2	5,3	5,4	5,5	5,6
6	6,1	6,2	6,3	6,4	6,5	6,6

$$\text{Pr(same number)} = \frac{6}{36} \; i.e \; \frac{1}{6}$$

(ii) Table for total 7 and above

Sum	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

$$\text{Pr(at least total 7)} = \frac{21}{36} \; i.e \; \frac{7}{12}$$

2011/57 Neco

The probability that Musa passes English, Mathematics and Geography are $^1/_3$, $^2/_5$ and $^3/_4$ respectively. What is the probability of passing all?

A $^3/_4$ B $^8/_{15}$ C $^2/_{15}$ D $^1/_{10}$ E $^1/_{12}$

Solution

Pr(passing all) = Pr(E) Pr(M) Pr(G)

$$= \frac{1}{3} \times \frac{2}{5} \times \frac{3}{4} = \frac{1}{10} \quad (\text{ D })$$

2011/58 Neco

In a bag containing 12 oranges and 18 apples, what is the probability of not picking apple?

A $^2/_3$ B $^3/_5$ C $^2/_5$ D $^1/_3$ E $^1/_6$

Solution

Pr(not picking apple) = 1 – Pr(picking apple)

$$= 1 - \frac{18}{30}$$

$= {}^{12}/_{30}$ i.e $^2/_5$ (C)

2012/52 Neco

The probability that Ahmed and Ade will fail Physics are $^1/_5$ and $^2/_3$ respectively. What is the probability that at least one of them will pass?

A $^1/_{15}$ B $^2/_{15}$ C $^8/_{15}$ D $^3/_5$ E $^{13}/_{15}$

Solution

Note that: *at least* is that number or more

Pr(at least one pass) = Pr(Ade pass) Pr(Ahmed fail) **or** Pr(Ahmed pass) Pr(Ade fail) **or** Pr(Ahmed pass)Pr(Ade pass)

Pr(Ahmed pass) is 1 – $^1/_5$ i.e $^4/_5$

Pr(Ade pass) is 1 – $^2/_3$ i.e $^1/_3$

$$\text{Pr(at least one pass)} = \frac{1}{3} \times \frac{1}{5} + \frac{4}{5} \times \frac{2}{3} + \frac{4}{5} \times \frac{1}{3}$$

$$= \frac{1}{15} + \frac{8}{15} + \frac{4}{15}$$

$= {}^{13}/_{15}$ (E)

2012/12 Neco

The probabilities that Ahmed, Abu and Ola score 70% in a Mathematic examination are $^2/_7$, $^1/_6$ and $^3/_5$ respectively. If the three of them sit for the mathematics examination, what is the probability that:

(a) all of them score below 70%;

(b) at most two of them score 70%;

(c) all of them score 70% ?

Solution

Let A – Ahmed, B – Abu, O – Ola

(a) Pr(all score below 70%) = Pr(all fail)

$= \Pr(\overline{A}) \Pr(\overline{B}) \Pr(\overline{O})$

$= (1 - {}^2/_7)(1 - {}^1/_6)(1 - {}^3/_5)$

$= ({}^5/_7)({}^5/_6)({}^2/_5)$

$= {}^5/_{21}$

(b) Pr(at most two of them score 70%)

= Pr(two pass) + Pr(one pass) + Pr(none pass)

$= \Pr(A)\Pr(B)\Pr(\overline{O})$ **or** $\Pr(A)\Pr(\overline{B})\Pr(O)$ **or** $\Pr(\overline{A})\Pr(B)\Pr(O)$

$+ \Pr(A)\Pr(\overline{B})\Pr(\overline{O})$ **or** $\Pr(\overline{A})\Pr(B)\Pr(\overline{O})$ **or** $\Pr(\overline{A})\Pr(\overline{B})\Pr(O)$

$+ \Pr(\overline{A})\Pr(\overline{B})\Pr(\overline{O})$

$$= \frac{2}{7} \times \frac{1}{6} \times \frac{2}{5} + \frac{2}{7} \times \frac{5}{6} \times \frac{3}{5} + \frac{5}{7} \times \frac{1}{6} \times \frac{3}{5}$$

$$+ \frac{2}{7} \times \frac{5}{6} \times \frac{2}{5} + \frac{5}{7} \times \frac{1}{6} \times \frac{2}{5} + \frac{5}{7} \times \frac{5}{6} \times \frac{3}{5}$$

$$+ \frac{5}{7} \times \frac{5}{6} \times \frac{2}{5}$$

$= \frac{2}{105} + \frac{1}{7} + \frac{1}{14} + \frac{2}{21} + \frac{1}{21} + \frac{5}{14} + \frac{5}{21}$

Applying the LCM of the dominators 105, 21, 14 and 7

$= \frac{4 + 30 + 15 + 20 + 10 + 75 + 50}{210}$

$= \frac{204}{210}$ i.e $\frac{102}{105}$

2012/ 54 Neco

Two dice are thrown once. What is the probability of getting a sum which is divisible by both 2 and 3?

A $^2/_3$ B $^1/_2$ C $^1/_3$ D $^1/_4$ E $^1/_6$

Solution

We mark multiple of 2 and 3 in the sample space

+	1	2	3	4	5	6
1	2	3*	4	5	6*	7
2	3*	4	5	6*	7	8
3	4	5	6*	7	8	9*
4	5	6*	7	8	9*	10
5	6*	7	8	9*	10	11
6	7	8	9*	10	11	12*

⬭ represents divisible by 2, * represent divisible by 3

Pr(sum divisible by both 2 and 3)

= Pr(sum divisible by 2) × Pr(sum divisible by 3)

$= \frac{18}{36} \times \frac{12}{36}$

$= {}^1/_6$ (E)

Note that some elements are circled and starred.
Let us treat a case where this apply

2012/ 54 Neco Adjusted

Two dice are thrown once. What is the probability of getting a sum which is divisible by 2 or 3?

A $^2/_3$ B $^1/_2$ C $^1/_3$ D $^1/_4$ E $^1/_6$

Solution

We mark multiple of 2 and 3 in the sample space

+	1	2	3	4	5	6
1	2	3*	4	5	6*	7
2	3*	4	5	6*	7	8
3	4	5	6*	7	8	9*
4	5	6*	7	8	9*	10
5	6*	7	8	9*	10	11
6	7	8	9*	10	11	12*

⬭ represents divisible by 2, * represent divisible by 3

Pr(sum divisible by 2 or 3)

= Pr(sum divisible by 2) + Pr(sum divisible by 3) – Pr(sum divisible by both 2 or 3)

$= \frac{18}{36} + \frac{12}{36} - \frac{6}{36}$

$= \frac{24}{36}$ i.e $^2/_3$ (A)

2012/ 60 Neco

Two dice were thrown once. What is the probability of having a difference of absolute value of 2?

A $^1/_9$ B $^2/_9$ C $^3/_9$ D $^4/_9$ E $^5/_9$

Solution

Difference of absolute value of 2 implies subtraction but all negative results must be written in positive form. The resulting table is

/– /	1	2	3	4	5	6
1	0	1	2	3	4	5
2	1	0	1	2	3	4
3	2	1	0	1	2	3
4	3	2	1	0	1	2
5	4	3	2	1	0	1
6	5	4	3	2	1	0

Pr(difference of absolute 2) $= \frac{8}{36}$

$= {}^2/_9$ (B)

2012/4a

A box contains 40 identical discs which are either red or white. If the probability of picking a red disc is $^1/_4$, calculate the number of:

(i) white discs;

(ii) red discs that should be added such that the probability of picking a red disc will be $^1/_3$

Solution

First, we get number of red discs i.e x

Pr(red disc) $= \frac{1}{4}$

$\frac{1}{4} = \frac{Number\ of\ red\ disc}{Total\ discs}$

$\frac{1}{4} = \frac{x}{40}$

$\frac{40}{4} = x$ i.e $x = 10$

(i) Number of white disc = 40 – 10
(since we have either red or white) = 30

(ii) Presently number of red is 10
Let r be the red discs to be added
Then from the statement in (ii)

$\frac{1}{3} = \frac{10 + r}{40 + r}$

$40 + r = 3(10 + r)$

$40 + r = 30 + 3r$

$40 - 30 = 3r - r$

$10 = 2r$

$r = 5$

2013/ 53 Neco

A basket contains 50 eggs, 15 of them are bad. If an egg is taken out, find the probability that it is good.

A $^3/_{20}$ B $^1/_5$ C $^3/_{10}$ D $^7/_{10}$ E $^{17}/_{20}$

Solution

Pr(good eggs) $= \frac{50 - 15}{50}$ *since we have 15 bad eggs*

$= \frac{35}{50}$ i.e $^7/_{10}$ (D)

2013/ 59 Neco

What is the probability that an integer x : 1≤ x ≤ 30 is divisible by both 3 and 5?

A $^1/_{15}$ B $^2/_{15}$ C $^8/_{15}$ D $^{17}/_{30}$ E $^{19}/_{30}$

Solution

x : 1≤ x ≤ 30 = {1, 2, 3, … 30} i.e μ

n(μ) = 30

Subset divisible by 3 = {3, 6, 9, 12, 15, 18, 21, 24, 27, 30}

Subset divisible by 5 = {5, 10, 15, 20, 25, 30}

Intersection: divisible by both 3 and 5 = {15, 30}

n(intersection) = 2

$$\text{Pr(divisible by both 3 and 5)} = \frac{2}{30}$$

$= {}^1/_{15}$ (A)

2014/ 46

The probability of an event P happening is $^1/_5$ and that of event Q is $^1/_4$. If the events are independent, what is the probability that neither of them happens?

A $^4/_5$ B $^3/_4$ C $^3/_5$ D $^1/_{20}$

Solution

Pr(P not happening) Pr(Q not happening)

$= (1 - {}^1/_5)(1 - {}^1/_4)$

$= {}^4/_5 \times {}^3/_4$

$= {}^3/_5$ (C)

2014/5

A building contractor tendered for two independent contracts, X and Y. The probability that he will win contract X is 0.5 and **not** win contract Y is 0.3. What is the probability that he will win:

(a) **both** contracts;

(b) **exactly** one of the contracts;

(c) **neither** of the contracts?

Solution

(a) Pr(X and Y win) = Pr(win X) Pr(win Y)

= (0.5)(1 – 0.3) *since not win y is 0.3, win is 1 – 0.3*

= (0.5) (0.7)

= 0.35

(b) Pr(win X)Pr(not win Y) or Pr(not win X)Pr(winY)

= (0.5)(0.3) + (1 – 0.5)(0.7)

since win x is 0.5 not win x is 1 – 0.5

= (0.5)(0.3) + (0.5)(0.7)

= 0.15 + 0.35

= 0.5

(c) Pr(not win X) Pr(not win Y) = (0.5) (0.3)

= 0.15

2014/ 43

If a number is selected at random from **each** of the sets P = {1, 2, 3} and Q = {2, 3, 5}, find the probability that the sum of the number is prime.

A $^5/_9$ B $^4/_9$ C $^1/_3$ D $^2/_9$

Solution

Sum	1	2	3
2	3	4	5
3	4	5	6
5	6	7	8

Pr(sum prime) = $^4/_9$ (B)

2014/11b

Score	1	2	3	4	5	6
frequency	2	5	13	11	9	10

The table shows the distribution of outcomes when a die is thrown 50 times. Calculate the probability that a score selected at random is **at least** a 4.

Solution

$$\text{Pr(score at least 4)} = \frac{f(4) + f(5) + f(6)}{\Sigma f}$$

$$= \frac{11 + 9 + 10}{50}$$

$= {}^{30}/_{50}$ i.e $^3/_5$

2009/3 Neco

During a school's visiting day, a mother visited her daughter. She carried a bag containing 12 oranges, 16 apples and 8 mangoes, all of the same size. She draws 3 fruits for her daughter's friend without replacement. Find the probability that: (i) the three are oranges

(ii) one is apple, the other two mangoes:

(iii) one is orange, one apple and a mango in that order.

Solution

O – Oranges, A – apples and M – mangoes

Note: "without replacement" implies one draw will affect the other i.e dependent event.

(i) $\text{Pr(three oranges)} = \frac{12}{36} \times \frac{11}{35} \times \frac{10}{34}$

This is so, as after the first draw, the number of oranges reduced by 1 likewise total fruits. This is also the case after the 2nd draw.

$$= \frac{11}{357}$$

(ii) $\text{Pr(Apple, two mangoes)} = \frac{16}{36} \times \frac{12}{35} \times \frac{11}{34}$

After the first draw, the total fruit is reduced by 1. Now the 3rd draw being mango is affected by the 2nd draw both in mango number and total fruits.

$$= \frac{88}{1785}$$

(iii) $\text{Pr(Orange, apple and mangoes)} = \frac{12}{36} \times \frac{16}{35} \times \frac{8}{34}$

Here only the total fruits reduces after each draw

$$= \frac{64}{1785}$$

1992/5

(a) In a game, a fair die is rolled once and two unbiased coins are tossed once. What is the probability of obtaining 3 and a tail?

Solution

Die	Coins
S = {1, 2, 3, 4, 5, 6}	S = {HH, HT, TH, TT}
E (3) = {3}	E (T) = {HT, TH}
Pr (3) = $^1/_6$	Pr (a tail) = $^2/_4$

Thus Pr (3 and tail) = $^1/_6 \times {}^2/_4$

$= {}^2/_{24} = {}^1/_{12}$

Alternatively

Tabulate the values as shown below

	1	2	3	4	5	6
HH	HH1	HH2	HH3	HH4	HH5	HH6
HT	HT1	HT2	HT3	HT4	HT5	HT6
TH	TH1	TH2	TH3	TH4	TH5	TH6
TT	TT1	TT2	TT3	TT4	TT5	TT6

Pr(3 and a tail) = $^2/_{24} = ^1/_{12}$

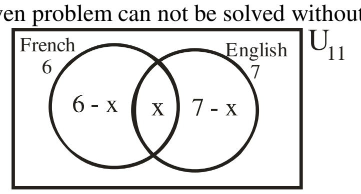
1999/14

A group of eleven people can speak either English or French or both. Seven can speak English and six can speak French. What is the probability that a person chosen at random can speak both English and French?

A. $\frac{2}{11}$ B. $\frac{4}{11}$ C $\frac{5}{11}$ D. $\frac{11}{13}$

Solution

The given problem can not be solved without applying set theory.



$6 – x + x + 7 – x = 11$

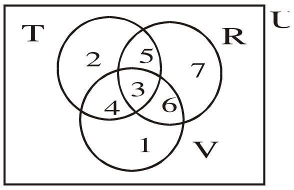
$13 – x = 11$

$13 – 11 = x$

$x = 2$ people speak both languages

Thus, Pr (speak both) = $\frac{2}{11}$ (A)

2001/4 (b) & (c) Nov



The venn diagram shows the number of traders that sell radio (R) television (T) and video (V) in a market.

(b) A man goes to the market to buy a set of radio and television. What is the probability that the first trader he meets:

(i) sells one of the three items only?

(ii) Sells two of the three items only?

(c) What is the probability that the man in (b) above, buys the radio and television in a video shop.

Solution

A good knowledge of set as treated in this book will give candidate an upper hand in answering this question.

b.(i.) Pr (one of three) = $\frac{f(E)}{\sum f}$

$= \frac{2 + 7 + 1}{28}$

$= \frac{10}{28} = \frac{5}{14}$

(ii) Pr (two of three items) = $\frac{f(E)}{\sum f}$

$= \frac{5 + 4 + 6}{28}$

$= \frac{15}{28}$

C. Pr (R & TV) = $\frac{3}{28}$

2007/27 Exercise 10.1

The probability of an event occurring is x, what is the probability of the event not occurring?

A $1 – x$ B $x – 1$ C 0 D $^1/_x$

2005/43 Neco Exercise 10.2

When two dice are thrown once, what is the probability that sum of the out come is less than 4?

A $^1/_{12}$ B $^1/_9$ C $^5/_{36}$ D $^1/_6$ E $^7/_{36}$

2005/4a Exercise 10.3

Two fair dice are thrown once. Find the probability of getting: (i) the same digit (ii) a total score greater than 5

2015/5 (Nov) Exercise 10.4

A fair die is thrown **two** times.

(a) Construct a table of the outcomes.

(b) Calculate the probability that the:

(i) sum of the outcomes is 8

(ii) product of the outcomes is less than 10

(iii) outcomes contain **at least** a 3

2015/29 (Nov) Exercise 10.5

Out of the 20 girls in a class, 12 like music and 15 like movies. If a girl is selected at random from the class, what is the probability that she likes **both** hobbies?

A $^3/_4$ B $^3/_5$ C $^7/_{20}$ D $^3/_{10}$

2015/1b Exercise 10.6

A number is selected at random from each of the sets { 2, 3, 4 } and { 1, 3, 5 }. Find the probability that the sum of the two numbers is **greater than** 3 and **less than** 7

2015/1b Exercise 10.7

A letter is selected from the letters of the English alphabet. What is the probability that the letter selected is from the word **MATHEMATICS** ?

A $^9/_{13}$ B $^{11}/_{26}$ C $^4/_{13}$ D $^1/_{26}$

2015/4a Neco Exercise 10.8

A basket contains 80 mangoes and 60 oranges. If two fruits are picked one after the other without replacement, what is the probability that

(i)one of each fruit is picked? (ii) one type of fruit is picked?

2015/58 Neco Exercise 10.9

Five cards are numbered one to five and then mixed up. Two of the cards are then chosen at random without replacement. Find the probability that at least one of them is a prime number

A $^3/_5$ B $^{18}/_{25}$ C $^3/_4$ D $^{19}/_{25}$ E $^9/_{10}$

2015/59Neco Exercise 10.10

A number is chosen at random from the set {1, 2, 3, …,19, 20}. What is the probability that the number is prime and even ?

A $^1/_{20}$ B $^7/_{20}$ C $^2/_5$ D $^{13}/_{20}$ E $^{19}/_{20}$

2014/5 Neco (Nov) Exercise 10.11

Two fair dice, each of which has its face numbered 1, 2, 3, 4, 5, 6 are tossed at the same time.

(a) Construct a table for the sample space of the products of the numbers shown

(b) Find the probability of the products which are perfect cube

(c) What is the probability of obtaining the products which are divisible by or a multiple of 3 ?

2014/58 to 60 Neco (Nov) Exercise 10.12

The table below shows the marks scored by a number of students in geography test.

Marks(%)	12	16	27	30	36	49	64
No. of Cand.	12	6	3	10	10	14	5

Use the information above to answer questions **58** to **60**

58. What is the probability that a student picked at random did not score a mark that is a perfect cube?
A $^{2}/_{15}$ B $^{5}/_{12}$ C $^{13}/_{15}$ D $^{11}/_{12}$ E $^{19}/_{20}$

59. If 50 is the pass mark, what is the probability that a student failed the exam?
A $^{1}/_{12}$ B $^{7}/_{20}$ C $^{29}/_{60}$ D $^{31}/_{60}$ E $^{11}/_{12}$

60. What is the probability that a student picked at random scored a mark that is a perfect cube?
A $^{7}/_{20}$ B $^{5}/_{12}$ C $^{29}/_{60}$ D $^{1}/_{2}$ E $^{7}/_{12}$

2014/6 NABTEB (Nov) Exercise 10.13

A die numbered 1 to 6 is rolled once. What is the probability of obtaining 8 or 10 ?
A 0 B $^{1}/_{2}$ C $^{1}/_{3}$ D $^{1}/_{4}$

2014/8a NABTEB (Nov) Exercise 10.14

Two dice are thrown together once. Find the probability that the following are obtained:
(i) a 5 (ii) a multiple of 3 (iii) equal likely numbers

2014/38 NABTEB Exercise 10.15

A bag contains 4 blue biros, 5 red biros and 6 black biros. If a student picks one biro at random. What is the probability that it is **not** black ?
A $^{2}/_{25}$ B $^{4}/_{25}$ C $^{6}/_{25}$ D $^{9}/_{25}$

2013/49 UTME Exercise 10.16

What is the probability that an integer x ($1 \leq x \leq 25$) chosen at random is divisible by both 2 and 3?
A $^{4}/_{25}$ B $^{3}/_{4}$ C $^{1}/_{25}$ D $^{3}/_{5}$

2014/49 UTME Exercise 10.17

Numbers	1	2	3	4	5	6
Frequency	18	32	20	16	10	14

The table above represents the outcome of throwing a die 100 times. What is the probability of obtaining at least a 4 ?
A $^{2}/_{5}$ B $^{3}/_{4}$ C $^{1}/_{5}$ D $^{1}/_{2}$

2014/50 UTME Exercise 10.18

A number is chosen at random from 10 to 30 inclusive, what is the probability that the number is divisible by 3?
A $^{1}/_{3}$ B $^{3}/_{5}$ C $^{2}/_{15}$ D $^{1}/_{10}$

2006/37 PCE Exercise 10.19

A number is chosen at random from 1 to 24 inclusive. What is the probability that the number is divisible by 5?
A $^{1}/_{5}$ B $^{1}/_{6}$ C $^{1}/_{12}$ D $^{1}/_{24}$

2008/50 PCE Exercise 10.20

If a die is tossed twice, what is the probability that the sum of the outcome is 6?
A $^{1}/_{18}$ B $^{1}/_{12}$ C $^{1}/_{9}$ D $^{5}/_{36}$

CHAPTER ELEVEN

Sequence (AP & GP) and series.

When numbers appear or are presented one after the other in an orderly manner, we then refer to the set of numbers as sequence.

Examples;
1, 2, 3, 4,…
5, 10, 20, 40,…
-3, -1, 1, 3,…

Series is the summation (addition) of the terms of a sequence. For example, the series equivalent of the above sequence are;
1 + 2 + 3 + 4 +…
5 + 10 + 20 + 40 +…
(-3) + (-1) + 1 + 3 +…

Arithmetic Progression (AP)

Arithmetic progression is a form of sequence in which each term is gotten by the <u>addition</u> of a <u>common difference</u> to the preceding one. Alternatively, we can say that APs or **linear sequence** are sequence that follows simple addition rule.

Examples; let us examine the pattern of the APs below;

(1) 1, 2, 3, 4, 5,…
We observe that;

Items	pattern
1	
2	= 1 + 1
3	= 2 + 1
4	= 3 + 1
5	= 4 + 1

Conclusion; 1 is the common difference and first term is 1.

(**2**) 12, 19, 26, 33,…
We observe that

Terms	pattern
12	
19	= 12 + 7
26	= 19 + 7
33	= 26 + 7

Conclusion; 7 is the common difference and first term is 12

nth term of an AP

Conventionally, we denote the first term by 'a' and the common difference by 'd'. Let n represent the nth term, we observe that the examples given above follow a simple pattern:

a, a + d, a + 2d, a + 3d
1st 2nd 3rd 4th

<u>For (1)</u>

$T_1 = 1$
$T_2 = 2 = 1 + (2 – 1) 1$
$= 1 + 1$

$T_3 = 3 = 1 + (3 – 1) 1$
$= 1 + 2$

$T_4 = 4 = 1 + (4 – 1) 1$
$= 1 + 3$

Then $T_n = 1 + (n – 1) 1$
$= 1 + n – 1$
$= n$

As for (2)

$T_1 = 12$

$T_2 = 19 = 12 + (2 - 1)7$
$= 12 + 7$

$T_3 = 26 = 12 + (3 - 1)7$
$= 12 + 14$

$T_4 = 33 = 12 + (4 - 1)7$
$= 12 + 21$

Then $T_n = 12 + (n - 1)7$
$= 12 + 7n - 7$
$= 5 + 7n$

Generally if we are given any A.P: a, a + d, a + 2d,…
Then

$$\mathbf{T_n = a + (n - 1)d}$$

This is the formula for finding the nth term of an AP

2006/35 (Nov)

What is the 23rd term of the Arithmetic Progression
$-7, -3, 1, \ldots$?

A 65 B 77 C81 D90

Solution

$a = -7$, $T_{23} = ?$ $d = -3 - (-7) = -3 + 7$ i.e 4

$T_n = a + (n - 1)d$

$T_{23} = -7 + (23 - 1)4$
$= -7 + (22) \times 4$
$= -7 + 88$
$= 81$ (C)

2014/48

Find the number of terms in the Arithmetic Progression (A.P) 2, –9, –20, … , –141

A 11 B12 C 13 D 14

Solution

$a = 2$, $d = -9 - 2$ i.e -11, $n = ?$ But $T_n = -141$

Applying $T_n = a + (n - 1)d$

Substituting

$-141 = 2 + (n - 1)(-11)$
$-141 = 2 - 11n + 11$
$-141 = 13 - 11n$
$11n = 13 + 141$
$11n = 154$
$n = {}^{154}/_{11}$ i.e 14 (D)

2006/2 Neco

The 12th term of an AP is – 41. If the first term is 3, find the 20th term.

A – 77 B – 73 C 77 D 79 E 83

Solution

$n = 12$, $T_{12} = -41$, a is 3, $d = ?$

$T_n = a + (n - 1)d$

Substituting

$T_{12} \Rightarrow -41 = 3 + (12 - 1)d$
$-41 = 3 + 11d$
$-41 - 3 = 11d$
$-44 = 11d$
${}^{-44}/_{11} = d$ Thus $d = -4$

20th term: $T_{20} = 3 + (20 - 1)(-4)$
$= 3 + (19)(-4)$
$= 3 - 76$
$= -73$ (B)

2010/10 Neco

Find the common difference in an Arithmetic Progression given that their first and 28th terms are –14 and – 5 respectively.

A – 3 B ${}^{-1}/_3$ C ${}^1/_3$ D $1^1/_4$ E 3

Solution

$a = -14$, $T_{28} = -5$, $d = ?$

Applying $T_n = a + (n - 1)d$

$T_{28} \Rightarrow -5 = -14 + (28 - 1)d$
$-5 = -14 + 27d$
$14 - 5 = 27d$
$9 = 27d$

Thus ${}^9/_{27} = d$ i.e ${}^1/_3$ (C)

2006/9 Neco (Nov)

The 12th term of an AP is 32. If the 1st term is $1^1/_2$, what is its common difference? (Correct to 1 decimal place)

A 2.7 B 2.8 C 5.4 D 5.5 E 5.6

Solution

$d = ?$, $n = 12$, $T_{12} = 32$, $a = 1^1/_2$ i.e ${}^3/_2$

$T_n = a + (n - 1)d$

Substituting

$$T_{12} \Rightarrow 32 = \frac{3}{2} + (12 - 1)d$$

$$32 = \frac{3}{2} + 11d$$

Multiply through by 2 to clear fraction

$64 = 3 + 22d$
$64 - 3 = 22d$
$61 = 22d$
$d = {}^{61}/_{22}$
$= 2.773$
≈ 2.8 to 1 dp (B)

2014/10 NABTEB (Nov) counter eg

The first term of an arithmetic progression is 8. If the tenth term is double the second term, the common difference is

A ${}^4/_3$ B ${}^8/_7$ C ${}^7/_8$ D 1

Solution

$a = 8$, $T_{10} = 2 \times T_2$, $d = ?$

Applying $T_n = a + (n - 1)d$

From the statement

$T_{10} = 2 \times T_2$
$8 + (10 - 1)d = 2 \times [8 + (2 - 1)d]$
$8 + 9d = 2 \times [8 + d]$
$8 + 9d = 16 + 2d$
$9d - 2d = 16 - 8$
$7d = 8$ thus $d = {}^8/_7$ (B)

2006/11b

The 2nd, 3rd, and 4th terms of an AP are x – 2, 5 and x + 2 respectively. Calculate the value of x.

Solution

In AP common difference rule is

3rd term – 2nd term = 4th term – 3rd term

$5 - (x - 2) = (x + 2) - 5$
$5 - x + 2 = x + 2 - 5$
$7 - x = x - 3$
$7 + 3 = x + x$
$10 = 2x$
${}^{10}/_2 = x$ thus $x = 5$

2007/ 2b

If $\frac{1}{2}, \frac{1}{x}, \frac{1}{3}$ are successive terms of an arithmetic progression (A.P), show that $\frac{2-x}{x-3} = \frac{2}{3}$

Solution

In AP, common difference rule is

2nd term – 1st term = 3rd term – 2nd term

$$\frac{1}{x} - \frac{1}{2} = \frac{1}{3} - \frac{1}{x}$$

Solving both sides separately by LCM

$$\frac{2-x}{2x} = \frac{x-3}{3x}$$

Having our target in mind

$$\frac{3x(2-x)}{2x} = x-3$$

Divide through by 2 – x

$$\frac{3x}{2x} = \frac{x-3}{2-x}$$

$$\frac{3}{2} = \frac{x-3}{2-x}$$

Taking the reciprocal of both sides

$$\frac{2}{3} = \frac{2-x}{x-3} \quad \text{QED}$$

2012/6b Neco

If 8, x, y, z, 32 is an AP, find the value of x, y, and z

Solution

$T_5 = 32$, n = 5, a = 8, d = ?

Applying $T_n = a + (n-1)d$

Substituting

$T_5 \Rightarrow 32 = 8 + (5-1)d$

$32 = 8 + 4d$

$32 - 8 = 4d$

$24 = 4d$

$^{24}/_4 = d$ thus d = 6

	x	y	z	
AP : 8,	**14**,	**20**,	**26**,	32, …

2005/4 Neco

If 2, a, b, c, d, 22 are terms of an Arithmetic Progression, find bd – ac

A 4 B 16 C 38 D 96 E 108

Solution

If we use common difference principle

a – 2 = 22 – d it leads no where

Applying nth term formula

$T_n = a + (n-1)d$

With n = 6

$T_6 = 22$ and a = 2, d = ?

$T_6 \Rightarrow 22 = 2 + (6-1)d$

$22 = 2 + 5d$

$22 - 2 = 5d$

$20 = 5d$

d = $^{20}/_5$ i.e 4

	a	b	c	d	
AP : 2,	**6,**	**10,**	**14,**	**18,**	22

Thus bd – ac = 10 × 18 – 6 × 14

= 180 – 84

= 96 (D)

2006/12b (Nov) Exercise 11.1

The numbers 11, p, q, $21^1/_2$, … form an arithmetic progression (A.P). Find the value of p and q

2006/4b NABTEB (Nov) Exercise 11.2

If – 8, x, y, 19 are a sequence in Arithmetic Progression (A.P), find the value of x and y

2008/13a Exercise 11.3

If 3, x, y, 18 are in Arithmetic progression (AP), find the value of x and y.

2015/47 (Nov) Exercise 11.4

The first four terms of a linear sequence (AP) are 5, 9, 13, 17. What is the 10th of the sequence ?

A 29 B 30 C 41 D 45

2015/9a Exercise 11.5

The first term of an Arithmetic Progression (AP) is – 8 If the ratio of the 7th term to 9th term is 5 : 8, find the common difference of the AP

Term's formula for particular AP

It has been noted earlier that:

$T_n = a + (n-1)d$

This is the general formula for finding the nth term of any AP. But, every A.P has its own particular term formula which is a short-cut to deriving the other members of the progression

E.g 1 Find the nth term of the AP 10, 13, 16, …

Analysis and solution

Applying the general formula

Tn = a + (n – 1) d

n = n, a = 10, d = 3

Substituting

Tn = 10 + (n – 1) 3

= 10 + 3n – 3

= 7 + 3n or = 3n + 7

2009/9 Neco

Find the nth term of an AP whose first three elements are $^3/_4$, 2, $3^1/_4$

A ½(4 – 3n) B $^1/_3$(4n – 1) C $^1/_4$(5n – 2)

D $^1/_5$(3 – 4n) E $^1/_6$(4 – 5n)

Solution

a = $^3/_4$, d = 2 – $^3/_4$ i.e $^5/_4$, n = ?

$T_n = a + (n-1)d$

$T_n = {^3/_4} + (n-1){^5/_4}$

$$= \frac{3}{4} + \frac{5}{4}n - \frac{5}{4}$$

$$= \frac{3}{4} - \frac{5}{4} + \frac{5}{4}n$$

$$= \frac{-2}{4} + \frac{5}{4}n \quad \text{i.e} \quad \frac{5}{4}n - \frac{2}{4} = \frac{1}{4}(5n-2) \qquad (\text{C})$$

2005/1b NABTEB (Nov)

$U_n = 3n + 2$ is the nth term of an AP.

List the first three terms of the sequence

Solution

1st term i.e n = 1

$U_1 = 3 \times 1 + 2$

$= 3 + 2$

$= 5$

2nd term i.e n = 2

$$U_2 = 3 \times 2 + 2 = 6 + 2 = 8$$

3rd term i.e n = 3

$$U_3 = 3 \times 3 + 2 = 9 + 2 = 11$$

1995/7 Exercise 11. 6

Find the nth term Un, of the A. P. 11, 4, -3…

A. Un = 19 + 7n B. Un = 19 – 7n C. Un = 18 – 7n
D. Un = 18 + 7n E. Un = 17 – 7n

2002/17 Exercise 11. 7

Find the nth term of the sequence 4, 10, 16 …

A. 2 (3n –1) B. $2(2 + 3^{n-1})$ C. $2^n + 2$
D. 2 (3n + 2)

1993/25 PCE Exercise 11.8

Given the progression 5, 8, 11, 14,…

Find the expression for the nth term of the progression

A. n(n + 1) B. 5 + 3n C. 3n + 2 D. 5 + 2n

Unknown 'a' and 'd' (resulting to simultaneous Linear equations)

There are cases in A. P where two terms of the progression are given and we are asked to find the first term "**a**" and common difference "**d** " ; the problem that follows leads us to simultaneous equations that can be solved either by substitution or elimination method.

Eg1 In an A.P the 10^{th} term is 68 while its 4^{th} term is 26
Find (a) common difference (b) first term

Solution

$T_{10} = 68 = a + (10 – 1) d$

$68 = a + 9d$ ----(1)

Also $T_4 \Rightarrow 26 = a + (4 – 1) d$

$26 = a + 3d$ ------ (2)

Solving equations (1) and (2)

$$\begin{array}{r} a + 9d = 68 \\ \underline{-\ (a + 3d = 26\ )} \\ 6d = 42 \end{array}$$

$$d = \frac{42}{6} = 7$$

(b) Substituting d value into (2)

a + 3d = 26 will become

$a + 3 \times 7 = 26$

a + 21 = 26

a = 26 – 21 = 5

2014/11b NABTEB (Nov)

The 4th term of an Arithmetic Sequence is 13 and the second term is 3. Find the common difference and the fifth term

Solution

$T_4 = 13,\ T_2 = 3,\ d = ?$

$T_n = a + (n – 1)d$

$T_4 \Rightarrow 13 = a + (4 – 1)d$

$13 = a + 3d$ ------ (1)

Also $T_2 \Rightarrow 3 = a + (2 – 1)d$

$3 = a + d$ ----- (2)

Solving the resulting simultaneous linear equations

$$\begin{array}{r} a + 3d = 13 \\ \underline{-\ (\ a + d\ =\ 3\ )} \\ 2d = 10 \end{array}$$

$$d = {}^{10}/_2 \text{ i.e } 5$$

Substituting d value into (2)

a + d = 3 becomes

a + 5 = 3

a = 3 – 5 i.e –2

$T_5 = –2 + (5 – 1)5$

$= –2 + (4 \times 5)$

$= –2 + 20$ i.e 18

1994/ 11a

The fourth term of an A. P is 37 and the 6th term is 12 more than the fourth term. Find the first and seventh terms

Solution

Applying the formula Tn = a + (n – 1) d, we have,

Fourth term i.e. $T_4 = 37 = a + (4 - 1) d$

$37 = a + 3d$ ----- (1)

Sixth term $T_6 = a + (6 – 1) d$

$T_6 = a + 5d$

'…the 6^{th} term is 12 more than the fourth term.'

$T_6 = T_4 + 12$

i.e $T_6 = 37 + 12$

i.e. 37 + 12 = a + 5d

49 = a + 5d ------- (2)

Solving equations (1) and (2),

$$\begin{array}{r} a + 5d = 49 \\ \underline{-\ (a + 3d = 37)} \\ 2d = 12 \end{array}$$

$$d = {}^{12}/_2 = 6$$

Substitute d value into (1)

a + 3d = 37 will become

a + 3(6) = 37

a = 37 – 18

a = 19, first term

(b) 7th term

$T_7 = 19 + (7- 1) 6$

$= 19 + (6) 6$

$= 19 + 36 = 55$

2000/ 12a

The first term of an A. P is – 8. The ratio of the 7^{th} term to the 9th term is 5 : 8 . Calculate the common difference of the progression.

Solution

Applying the formula $T_n = a + (n – 1) d$, we have

$T_7 = a + (7 – 1)d$

$= a + 6d$

Also $T_9 = a + 8d$

But the question specified that

$$\frac{T_7}{T_9} = \frac{5}{8}$$

$$\frac{a + 6d}{a + 8d} = \frac{5}{8}$$

Cross multiplying,

8 (a + 6d) = 5 (a + 8d)

8a + 48d = 5a + 40d

8a – 5a = 40d – 48d

3a = – 8d But a = – 8, thus,

3 (– 8) = – 8d

– 24 = – 8d

$$\frac{-24}{-8} = d$$, d = 3, common difference.

2014/ 19 UTME Exercise 11.9
The fourth term of an A.P is 13 while the 10th term is 31. Find the 24 term.
A 73 B 69 C 89 D 75

2010/22 UTME Exercise 11.10
The 3rd term of an A. P is –9 and the 7th term is –29. Find the 10th term of the progression.
A 44 B –165 C –44 D 165

2004/7b Neco (Nov) Exercise 11.11
The third and seventh terms of an AP are 27 and 91 respectively Find:
(i) the first term and common difference
(ii) the 22nd term of the Arithmetic progression.

Sum of an A. P

The sum of terms of an A. P refers to the addition of all the terms in the given A. P. It is usually denoted by S_n; where n is the number of terms in the progression.
Given any A. P. where $T_1 = a$ 1st term

$T_2 = a + d$ (2nd term)
$T_3 = a + 2d$ (3rd term)
$T_n = a + (n-1)d$ (nth term).

Then, the sum from the 1st term to the nth term
$S_n = a + (a + d) + (a + 2d) + \ldots + [a + (n-1)d]$ - -1
Also the sum from the nth term to the 1st term
$S_n = [a + (n-1)d] + [a + (n-2)d] + \ldots + (a + d) + a$ - -2
Adding (1) and (2) termly as they occur
$2S_n = [2a+(n-1)d] + [2a+(n-1)d] + \ldots + [2a + (n-1)d]$
We observe that $[2a + (n-1)d]$ is reoccurring n times in the right hand side of the equation
Thus, we can say
$2S_n = n \times \{2a + (n-1)d\}$

$\therefore$ $\mathbf{S_n = \frac{n}{2}\{2a + (n-1)d\}}$

A case where the first term (**a**) and last term (**L**) are given is:

$\mathbf{S_n = \frac{n}{2}(a + L)}$

2002/8c NABTEB
The first and last terms of an A.P are 79 and – 5.
If the sum is 814, find:
(i) the number of terms in the AP and
(ii) the common difference

Solution

The question wants us to find n from the formula for the 1st and last term sum i.e.

$S_n = \frac{n}{2}(a + L)$

$814 = \frac{n}{2}\{79 + (-5)\}$

$814 \times 2 = n(79 - 5)$

$814 \times 2 = 74n$

$\frac{814 \times 2}{74} = n$

$22 = n$

(ii) find d from $T_n = a + (n-1)d$
Since $n = 22$, $T_{22} = -5$, $a = 79$, $d = ?$

$T_{22} \Rightarrow -5 = 79 + (22-1)d$
$-5 = 79 + 21d$
$-5 - 79 = 21d$
$-84 = 21d$
$-\frac{84}{21} = d$
$d = -4$

2006/10 NABTEB (Nov)
Find the sum of the first 10 terms of the AP:
$2, 2\frac{1}{2}, 3, \ldots$
A $6\frac{1}{2}$ B $7\frac{1}{2}$ C $32\frac{1}{2}$ D $42\frac{1}{2}$

Solution

$S_{10} = ?$ Applying $S_n = \frac{n}{2}\{2a + (n-1)d\}$
$n = 10$, $a = 2$, $d = 2\frac{1}{2} - 2$ i.e $\frac{1}{2}$
Substituting

$S_{10} = \frac{10}{2}\left\{2 \times 2 + (10-1)\frac{1}{2}\right\}$

$= 5\{4 + (9)\frac{1}{2}\}$

$= 5\{4 + \frac{9}{2}\} = 5(\frac{17}{2}) = \frac{85}{2}$ i.e $42\frac{1}{2}$

2005/7 NABTEB (Nov)
Find the sum of the first 10terms of an AP whose first term and common difference are –1 and 2 respectively.
A 80 B 85 C 90 D 95

Solution

$S_{10} = ?$ Applying $S_n = \frac{n}{2}\{2a + (n-1)d\}$

$S_n = \frac{10}{2}\{2(-1) + (10-1)2\}$
$= 5\{-2 + (9)2\}$
$= 5\{-2 + 18\}$
$= 5(16)$ i.e 80 (A)

2008/6 NABTEB (Nov)
Find the sum of the first 12 terms of the A.P 2, 5, 8, 11, …
A 26 B52 C 222 D 444

Solution

$S_{12} = ?$ $n = 12$, $a = 2$, $d = 5 - 2$ i.e 3

Applying $S_n = \frac{n}{2}\{2a + (n-1)d\}$

$S_{12} = \frac{12}{2}\{2 \times 2 + (12-1)3\}$
$= 6\{4 + (11)3\}$
$= 6(4 + 33)$
$= 6(37)$
$= 222$ (C)

2009/1a Neco (Nov)
$1, \frac{4}{3}, \frac{5}{3}, \ldots$ is an AP. Find the:
(i) 10th term (ii) sum up to the 16th terms

Solution

(i) 10^{th} terms is T_{10}, Applying $T_n = a + (n-1)d$

$n = 10$, $a = 1$, $d = {}^4/_3 - 1$ i.e ${}^1/_3$

Substituting

$$T_{10} = 1 + (10-1) \times \frac{1}{3}$$

$$= 1 + 9 \times \frac{1}{3}$$

$$= 1 + 3 \quad \text{i.e } 4$$

(ii) $S_{16} = ?$ Applying $S_n = \frac{n}{2}\{2a+(n-1)d\}$

Here $n = 16$, $a = 1$, $d = {}^1/_3$

$$S_{16} = \frac{16}{2}\left\{2\times1 + (16-1)\frac{1}{3}\right\}$$

$$= 8(2 + 15 \times \frac{1}{3})$$

$$= 8(2 + 5)$$

$$= 8(7)$$

$$= 56$$

2007/13 Neco ***counter eg on sum***

In an Arithmetic Progression (A.P), the first term is 3, and the sum of the first and the sixth terms is 20. What is the 8th term?

A 22.6 B 25.6 C 28.6 D31.6 E 34.6

Solution

$a = 3$, $T_1 + T_6 = 20$, $T_8 = ?$

Applying $T_n = a + (n-1)\,d$

$$T_1 + T_6 = 20$$

$$3 + [3 + (6-1)d] = 20$$

$$3 + 3 + 5d = 20$$

$$5d = 20 - 6$$

$$5d = 14$$

$$d = \frac{14}{5}$$

Thus, $T_8 = 3 + (8-1) \times \frac{14}{5}$

$$= 3 + 7 \times \frac{14}{5}$$

$$= 3 + \frac{98}{5}$$

$$= \frac{113}{5} \text{ i.e } 22.6 \quad (\text{A})$$

2005/7a NABTEB (Nov)

The first term of an AP is 7 and the fourth term is 32, find the sum of the first 10 terms

Solution

$S_{10} = ?$ $a = 7$, $T_4 = 32$

To find S_{10}, we need a and d

a is given but we get d from $T_4 = 32$

Applying $T_n = a + (n-1)d$

$$T_4 \Rightarrow 32 = 7 + (4-1)d$$

$$32 = 7 + 3d$$

$$32 - 7 = 3d$$

$$25 = 3d$$

$${}^{25}/_3 = d$$

Applying $S_n = \frac{n}{2}\{2a+(n-1)d\}$

Thus, $S_{10} = \frac{10}{2}\left\{2\times7+(10-1)\frac{25}{3}\right\}$

$$= 5\{ 14 + (9)\ \frac{25}{3} \}$$

$$= 5\{14 + 75\}$$

$$= 5(89)$$

$$= 445$$

2014/9 Neco (Nov)

The sum of 6 terms of an A.P is 6 and the sum of 18 terms is 72. Find the common difference

A $-{}^1/_2$ B $-{}^1/_4$ C ${}^1/_4$ D ${}^1/_2$ E ${}^3/_4$

Solution

$S_6 = 6$ and $S_{18} = 72$, $d = ?$

Applying $S_n = \frac{n}{2}\{2a+(n-1)d\}$

$$S_6 \Rightarrow 6 = \frac{6}{2}\{2a+(6-1)d\}$$

$$6 = 3(2a + 5d)$$

Divide both sides by 3

$$2 = 2a + 5d \text{ ------- (1)}$$

$$\text{Also } S_{18} \Rightarrow 72 = \frac{18}{2}\{2a+(18-1)d\}$$

$$72 = 9(2a + 17d)$$

Divide both sides by 9

$$8 = 2a + 17d \text{ ------ (2)}$$

Solving the resultant simultaneous linear equations

$$\begin{array}{r} 2a + 17d = 8 \\ -\ (\ 2a + 5d\ =\ 2\ ) \\ \hline 12d\ = 6 \end{array}$$

$$d = {}^6/_{12} \text{ i.e } {}^1/_2 \quad (\text{D})$$

2013/6 Neco

The sum of the first twelve terms of an A.P is – 96 and the sum of the first eighteen terms is – 252. Find the:

(a) common difference (b) first terms

(c) product of 15^{th} and 4^{th} terms

Solution

$S_{12} = -96$, $S_{18} = -252$, $d = ?$, $a = ?$

Applying $S_n = \frac{n}{2}\{2a+(n-1)d\}$

$$S_{16} \Rightarrow -96 = \frac{12}{2}\{2a+(12-1)d\}$$

$$-96 = 6(2a + 11d)$$

Divide both sides by 6

$$-16 = 2a + 11d \text{ ------ (1)}$$

$$\text{Also, } S_{18} \Rightarrow -252 = \frac{18}{2}\{2a+(18-1)d\}$$

$$-252 = 9(2a + 17d)$$

Divide both sides by 9

$$-28 = 2a + 17d \text{ ------- (2)}$$

Solving the resultant simultaneous linear equations

$$\begin{array}{r} 2a + 17d\ =\ -28 \\ -\ (2a +\ 11d\ = -16\ ) \\ \hline 6d\ = -12 \end{array}$$

$$d = {}^{-12}/_6 \text{ i.e } -2$$

Substituting d value into (1)

$2a + 11d = -16$ becomes

$2a + 11(-2) = -16$
$2a - 22 = -16$
$2a = -16 + 22$
$2a = 6$
$a = {}^6/_2$ i.e 3

(c) $T_{15} \times T_4 = ?$
$T_{15} = 3 + (15 - 1)(-2)$
$= 3 + 14(-2)$
$= 3 - 28$ i.e -25
$T_4 = 3 + (4 - 1)(-2)$
$= 3 + 3(-2)$
$= 3 - 6$ i.e -3
Thus $T_{15} \times T_4 = -25 \times (-3)$
$= 75$

2009/4 Neco
Twelve iron poles are to be used for pillar and the lengths of the poles form an Arithmetic Progression (A.P). If the third pole is 4m and the seventh is 10m, find the:
(a) lengths of the poles in order of the A.P;
(b) cost of all the poles if a meter costs ₦ 45.00

Solution
$T_3 = 4m$, $T_7 = 10m$
$T_n = a + (n - 1)d$
$T_3 \Rightarrow 4 = a + (3 - 1)d$
$4 = a + 2d$ ------- (1)
$T_7 \Rightarrow 10 = a + (7 - 1)\,d$
$10 = a + 6d$ ------ (2)
Solving the resultant simultaneous linear equations

$$\begin{array}{r} a + 6d = 10 \\ \underline{-(a + 2d = 4)} \\ 4d = 6 \\ d = {}^6/_4 \text{ i.e } {}^3/_2 \end{array}$$

Substituting d value into (1)
$a + 2d = 4$ will become
$a + 2({}^3/_2) = 4$
$a + 3 = 4$
$a = 4 - 3$
$a = 1$
(a) Lengths of poles in AP (12 of them)
$1m, 1 + {}^3/_2, (1 + {}^3/_2) + {}^3/_2, \ldots$
$1m, 2^1/_2, 4m, 5^1/_2m, 7m, 8^1/_2m, 10m, 11^1/_2m, 13m, 14^1/_2m, 16m, 17^1/_2m$
For any A.P the first three terms are Ok except otherwise directed by the question as in this case.

(b) First we find sum of all the 12 poles length

$$S_n = \frac{n}{2}\{2a + (n-1)d\}$$

$$S_{12} = \frac{12}{2}\left\{2\times 1 + (12-1)\frac{3}{2}\right\}$$

$$= 6(2 + 11 \times \frac{3}{2})$$

$$= 6(2 + \frac{33}{2})$$

$$= 6 \times \frac{37}{2} \text{ i.e } 111m$$

Thus cost of the 12 poles at ₦ 45.00 per meter
$= 111 \times 45$
$=$ ₦ 4,995.00

2003/5 NABTEB ***good case***
The first term of an AP is 3 and the fifth term is 9. Find the number of terms in the progression if the sum is 81

Solution
$S_n = 81$, find n $a = 3$, $T_5 = 9$
To find n from $S_n = 81$, we need a and d
a is given as 3, next we get d from $T_5 = 9$
Applying $T_n = a + (n - 1)d$
$T_5 \Rightarrow 9 = 3 + (5 - 1)d$
$9 = 3 + 4d$
$9 - 3 = 4d$
${}^6/_4 = d$ i.e $d = {}^3/_2$

$$\text{Thus } S_n \Rightarrow 81 = \frac{n}{2}\left\{2\times 3 + (n-1)\frac{3}{2}\right\}$$

$$81 = \frac{n}{2}\left(6 + \frac{3}{2}n - \frac{3}{2}\right)$$

$$81 \times 2 = n\left\{6 - \frac{3}{2} + \frac{3}{2}n\right\}$$

$$162 = n\left\{\frac{9}{2} + \frac{3}{2}n\right\}$$

$$162 = \frac{9}{2}n + \frac{3}{2}n^2$$

Multiply through by 2 to clear fractions
$324 = 9n + 3n^2$
$3n^2 + 9n - 324 = 0$
Divide through by 3
$n^2 + 3n - 108 = 0$
Factorizing
$n^2 + 12n - 9n - 108 = 0$
$n(n + 12) - 9(n + 12) = 0$
$(n + 12)(n - 9) = 0$
$n = -12$ or 9
We accept $n = 9$

2013/9 Neco Exercise 11.12
The sum of the first three terms of an AP is 12. If its 4th term is 8, find its common difference
A 2 B 4 C 6 D 8 E 12

2011/5a Neco Exercise 11.13
Find the sum of the first 9 terms of the AP:
$-24, -18, -12, \ldots$

2006/8b Neco Exercise 11.14
The fourth and ninth terms of an arithmetic progression are -3 and 12 respectively. Find the
(i) Common difference (ii) fifth term,
(iii) number of terms which will give a sum of 135

2009/18 UME Exercise 11.15
The sum of the first n terms of the arithmetic progression 5, 11, 17, 23, 29, 35, … is
A $n(3n - 0.5)$ B $n(3n + 2)$ C $n(3n + 2.5)$ D $n(3n + 5)$

2006/36 UME Exercise 11.16
The sum of the first n positive integers is
A $n(n - 1)$ B $n(n + 1)$ C $\frac{1}{2}n(n + 1)$ D $\frac{1}{2}n(n - 1)$

Geometric Progression (GP)

A geometric progression (GP) is a form of sequence, which has a common ratio between any of the terms and its preceding term.

Examples

1.) 2, 4, 8, …

We observe that common ratio(r) i.e. $^4/_2 = ^8/_4 = 2$

2.) 14, 98, 686, …

Common ratio (r) = $\frac{98}{14} = \frac{686}{98} = 7$

nth term of a G. P.

Conventionally in a GP, we denote the first term by '**a**' and the common ratio by '**r**'

Analyzing the given examples, we have

	T_1	T_2	T_3	T_n
Eg 1	2	2×2^1	$2 \times 2^2 = 8$	$2 \times 2^{n-1}$ $= \mathbf{ar^{n-1}}$
Eg 2	14	$14 \times 7^1 = 98$	$14 \times 7^2 = 686$	$14 \times 7^{n-1}$ $= \mathbf{ar^{n-1}}$

In general, for any given G. P a, ar, ar^2 ... ar^n

$$\mathbf{T_n = ar^{n-1}}$$

2013/8 Neco

Find the 7^{th} term of the geometric progression 4, 12, 36, …

A 365 B 729 C 1458 D 2916 E 4374

Solution

$a = 4$, $r = ^{12}/_4$ i.e 3, $T_7 = ?$

Applying $T_n = ar^{n-1}$

$T_7 = 4(3)^{7-1}$

$= 4(3^6)$

$= 4(729) = 2916$ (D)

2009/9 NABTEB (Nov)

The first term of a geometric progression is 6 and its common ratio is 3. Find the sixth term

A 18^5 B 3×9^5 C 3×6^5 D 6×3^5

Solution

$a = 6$, $r = 3$, $T_6 = ?$

Applying $T_n = ar^{n-1}$

$T_6 = 6(3)^{6-1}$

$= 6(3^5)$ i.e 6×3^5 (D)

2007/2 Neco

If y + 2, y + 6 and y + 14 are consecutive terms of a geometric progression (GP), find the:

(i) value of y and hence the GP

(ii) 42nd term of the GP leaving your answer in index form.

Solution

(i) For the G.P: y + 2, y + 6, y + 14, …

By common ratio principle

$$\frac{y+6}{y+2} = \frac{y+14}{y+6}$$

Cross multiply

$$(y + 6)(y + 6) = (y + 14)(y + 2)$$
$$y(y + 6) + 6(y + 6) = y(y + 14) + 2(y + 14)$$
$$y^2 + 6y + 6y + 36 = y^2 + 14y + 2y + 28$$
$$y^2 + 12y + 36 = y^2 + 16y + 28$$
$$36 - 28 = y^2 - y^2 + 16y - 12y$$
$$8 = 4y$$
$$^8/_4 = y \text{ i.e } y = 2$$

GP: 4, 8, 16, …

(ii) $T_{42} = ?$ $a = 4$, $r = ^8/_4$ i.e 2

Applying $T_n = ar^{n-1}$

$T_{42} = 4(2)^{42-1}$

$= 4(2^{41})$

$= 2^2 \times 2^{41}$

$= 2^{43}$ in index form

2006/11 NABTEB (Nov)

The first and third terms of a GP are 1 and 9 respectively. What is the second term?

A 4 B 3 C 2 D $^1/_3$

Solution

$a = 1$, $T_3 = 9$, $T_2 = ?$

Applying $T_n = ar^{n-1}$

$T_3 \Rightarrow 9 = 1 \times r^{3-1}$

$9 = r^2$

Raise LHS in power 2

$3^2 = r^2$ Thus $3 = r$

$T_2 = 1 \times 3^{2-1}$

$= 1 \times 3^1$

$= 3$ (B)

Alternatively

Applying common ratio principle:

For a GP: a, ar, ar^2

a T_2 T_3

Common ratio r : $\frac{T_3}{T_2} = \frac{T_2}{a}$

$a(T_3) = (T_2)^2$

$1 \times 9 = (T_2)^2$

$3^2 = (T_2)^2$

$3 = T_2$ (B)

2014/9 Neco

The first, second and last term of a GP are 3, 6 and 1536 respectively. Calculate the number of terms in the G.P

A 8 B 9 C 10 D12 E 14

Solution

$a = 3$, $ar = 6$, $T_n = 1536$, $n = ?$

We can get n from $T_n = ar^{n-1}$

First, we get r from : $ar = 6$ *and a = 3*

$(3)r = 6$

$r = ^6/_3$ i.e 2

Then from $T_n = 1536$

$T_n \Rightarrow 1536 = 3(2)^{n-1}$

Divide both sides by 3

$$\frac{1536}{3} = 2^{n-1}$$

$512 = 2^{n-1}$

Next, we express LHS in 2 and its power

$2^9 = 2^{n-1}$

Both sides are now in same base 2, we equate powers

$9 = n - 1$

$9 + 1 = n$

$10 = n$ (C)

2015/4 Neco Exercise 11.17

Find the 12th term of the G.P –3, 6, –12, …

A –12288 B – 6144 C – 2048 D 2048 E 6144

2005/2a Exercise 11.18

The 6th term of a GP is 1215. If the common ratio is 3, find its 3rd term

2012/4 f/maths Exercise 11.19

Given that $\sqrt{6}$, $3\sqrt{2}$, $3\sqrt{6}$, $9\sqrt{2}$, … are the first four terms of an exponential sequence (GP). Find in its simplest form, the 8th term.

A $27\sqrt{2}$ B $27\sqrt{6}$ C $81\sqrt{2}$ D $81\sqrt{6}$

2014/20 UTME Exercise 11.20

What is the common ratio of the G.P

$\left(\sqrt{10} + \sqrt{5}\right) + \left(\sqrt{10} + 2\sqrt{5}\right) + \ldots?$

A 3 B 5 C $\sqrt{2}$ D $\sqrt{5}$

Unknown 'a' and ' r ' leading to equation

There are cases where two different terms of a GP that are not close to each other will be given and we are asked to find the first term *a* and common ratio *r*.

2005/8b NABTEB

The 2nd and 4th terms of a G.P are 10 and 40 respectively. Find the

(i) Common ratio (ii) first term

(iii) 8th term of the series

Solution

(i) $T_2 = 10$, $T_4 = 40$, $r = ?$ a is not given

Applying $T_n = ar^{n-1}$

$T_2 \Rightarrow 10 = ar^{2-1}$

$10 = ar$ ----- (1)

$T_4 \Rightarrow 40 = ar^{4-1}$

$40 = ar^3$ ----- (2)

Divide (2) by (1)

$$\frac{40}{10} = \frac{ar^3}{ar}$$

$$4 = r^2$$

Raise LHS to a number in power 2

$$2^2 = r^2$$

Thus $2 = r$

(ii) Substitute r value into (1)

$ar = 10$ becomes

$a(2) = 10$

$a = {}^{10}/_2$ i.e 5

(iii) $T_8 = 5 \times 2^{8-1}$

$= 5 \times 2^7$

$= 640$

2011/6 Neco

The fifth term of a GP is 8 times the 2nd term. Find its common ratio

A – 4 B – 2 C ${}^1/_2$ D 2 E 4

Solution

$T_5 = 8(T_2)$, $r = ?$, a is not given

Applying $T_n = ar^{n-1}$

$T_5 = 8(T_2)$ becomes

$ar^{5-1} = 8(ar^{2-1})$

$ar^4 = 8ar$

Divide both sides by ar

$$\frac{ar^4}{ar} = 8$$

$$r^3 = 8$$

Next, we raise RHS to power 3

$$r^3 = 2^3$$

Thus, $r = 2$ (D)

2005/5 Neco

The 4th term of a GP is 384 and the third term is 96. Find the first term.

A 2 B 4 C 6 D 24 E 288

Solution

$T_4 = 384$, $T_3 = 96$, $a = ?$, r is not given

Applying $T_n = ar^{n-1}$

$T_4 \Rightarrow 384 = ar^{4-1}$

$384 = ar^3$ ------- (1)

Also $T_3 \Rightarrow 96 = ar^{3-1}$

$96 = ar^2$ ------- (2)

Divide (1) by 2

$$\frac{ar^3}{ar^2} = \frac{384}{96}$$

$$r = 4$$

Substitute r value into (2)

$96 = a(4)^2$ $[4^2 = 16]$

$\frac{96}{16} = a$ Thus $6 = a$ (C)

2010/13a

The third term of a Geometric Progression (G.P) is 24 and its seventh term is $4\frac{20}{27}$. Find its first term.

Solution

$T_3 = 24$, $T_7 = 4\frac{20}{27}$, $a = ?$ r is not given

Applying $T_n = ar^{n-1}$

$T_3 \Rightarrow 24 = ar^{3-1}$

$24 = ar^2$ ------- (1)

$T_7 \Rightarrow \frac{128}{27} = ar^{7-1}$

$\frac{128}{27} = ar^6$ ------- (2)

(2) ÷ (1)

$$\frac{ar^6}{ar^2} = \frac{128}{27} \div 24$$

$$r^4 = \frac{128}{27} \times \frac{1}{24}$$

$$r^4 = \frac{16}{27 \times 3}$$

Raise the RHS to number in power four

$$r^4 = \frac{2^4}{3^4} \quad i.e \left(\frac{2}{3}\right)^4$$

Thus, $r = {}^2/_3$

Next, we substitute r value into (1)

$ar^2 = 24$ becomes

$a({}^2/_3)^2 = 24$

$a \times \frac{4}{9} = 24$

$$a = \frac{24 \times 9}{4} = 54$$

2011/19 f/maths Exercise 11.21
The fourth term of a geometric sequence is 2 and sixth term is 8.Find the common ratio.
A ±1 B ±2 C ±3 D ±4

2013/12 f/maths Exercise 11.22
The fourth term of an exponential sequence is 192 and its ninth term is 6. Find the common ratio of the sequence
A $^1/_3$ B $^1/_2$ C 2 D 3

2007/19 PCE Exercise 11.23
The 5th term of a G.P is 9 times the 3rd term. What is the positive value of the common ratio?
A 5 B 4 C 3 D 2

Sum of n terms of a GP.

The sum, S_n of n terms of a GP is :
$S_n = a + ar + ar^2 + .. + ar^{n-2} + ar^{n-1}$ -----(1)
$rS_n = ar + ar^2 + ar^3 + \ldots + ar^{n-1} + ar^n$ -----(2)
Subtract (2) from (1)
On the RHS, there will be elimination of all the terms except a & ar^n
$Sn - rS_n = a - ar^n$
$Sn (1 - r) = a (1 - r^n)$

$$\mathbf{Sn = \frac{a(1 - r^n)}{1 - r}}$$ - (formula one)

This formula holds when **r < 1**
Also, subtracting (1) from (2)
$rSn - Sn = ar^n - a$
$Sn (r - 1) = a (r^n - 1)$

$$Sn = \mathbf{\frac{a(r^n - 1)}{r - 1}}$$ - (formula two)

This formula is used when **r >1**.
Candidates should note that the usage of either formula one or formula two depends only on the common ratio (r) of the Geometric progression given.

2006/35 Neco
Find the sum of the first 4 terms of a G.P, whose first term is 5 and the common ratio is $^1/_2$
A $^{75}/_{32}$ B $^{75}/_{16}$ C $^{75}/_8$ D $^{150}/_8$ E $^{35}/_2$

Solution

First, we check the common ratio $^1/_2$ is less than 1

Hence $S_n = \dfrac{a(1-r^n)}{1-r}$

$$S_4 = \frac{5\left[1 - \left(\frac{1}{2}\right)^4\right]}{1 - \frac{1}{2}}$$
$$= 5\,[1 - {}^1/_{16}] \div {}^1/_2$$
$$= 5\,[\tfrac{15}{16}] \times \tfrac{2}{1}$$
$$= 5\,[{}^{15}/_8] \text{ i.e } {}^{75}/_8 \quad (C)$$

2005/10 NABTEB
In a geometric series a = 2 and r = $^1/_2$, find the sum of the first 5 terms
A $^1/_8$ B $3\,^3/_4$ C $3\,^7/_8$ D 4

Solution

S_5 = ? Since r = $^1/_2$ < 1

Hence $S_n = \dfrac{a(1-r^n)}{1-r}$

$$S_5 = \frac{2\left[1 - \left(\frac{1}{2}\right)^5\right]}{1 - \frac{1}{2}}$$
$$= 2\,[1 - \tfrac{1}{32}] \div \tfrac{1}{2}$$
$$= 2\,[\tfrac{31}{32}] \times \tfrac{2}{1}$$
$$= {}^{31}/_8 \text{ i.e } 3\,^7/_8 \quad (C)$$

2012/7b Neco
$^{10}/_3, {}^5/_3, {}^5/_6, \ldots$ is a G.P. Find the
(i) 8th term (ii) sum of the first 10 terms?

Solution

(i) T_8 = ? Applying $T_n = ar^{n-1}$

$$a = \frac{10}{3}, \quad r = \frac{5}{3} \div \frac{10}{3} \quad i.e\ \frac{5}{3} \times \frac{3}{10} = \frac{1}{2}$$
$$T_8 = \frac{10}{3}\left(\frac{1}{2}\right)^{8-1}$$
$$= \frac{10}{3}\left(\frac{1}{2}\right)^7$$
$$= \frac{10}{3} \times \frac{1}{128}$$
$$= \frac{5}{3\times 64} \quad i.e\ \frac{5}{192}$$

(ii) S_{10} = ? First we check the r i.e $^1/_2$, it is less than 1

Thus, $S_n = \dfrac{a(1-r^n)}{1-r}$

$$S_{10} = \frac{\frac{10}{3}\left[1 - \left(\frac{1}{2}\right)^{10}\right]}{1 - \frac{1}{2}}$$
$$= \frac{10}{3}\left[1 - \frac{1}{1024}\right] \div \frac{1}{2}$$
$$= \frac{10}{3}\left[\frac{1023}{1024}\right] \times \frac{2}{1}$$
$$= \frac{10}{3}\left(\frac{1023}{512}\right) \quad i.e\ \frac{1705}{256}$$

2008/12b
(i) The sum of the second and third terms of a geometric progression is six times the fourth term. Find the two possible values of the common ratio
(ii) If the second term is 8 and the common ratio is positive, find the first six terms.

Solution

(i) $T_2 + T_3 = 6 \times T_4$ Applying $T_n = ar^{n-1}$
$ar^{2-1} + ar^{3-1} = 6 \times ar^{4-1}$
$ar + ar^2 = 6ar^3$
Rearranging
$6ar^3 - ar^2 - ar = 0$

Factor out common term

$ar(6r^2 - r - 1) = 0$

$ar = 0$ or $6r^2 - r - 1 = 0$

Solving the resultant quadratic equation for r

$6r^2 - r - 1 = 0$ becomes

$6r^2 - 3r + 2r - 1 = 0$

$3r(2r - 1) + 1(2r - 1) = 0$

$(2r - 1)(3r + 1) = 0$

$2r - 1 = 0$ or $3r + 1 = 0$

$2r = 1$ or $3r = -1$

$r = {}^1/_2$ or ${}^{-1}/_3$

(ii) $T_2 = 8$, r is positive i.e ${}^1/_2$ here instead of $-{}^1/_3$
we are asked to list a, ar, ar^2, ar^3, ar^4, ar^5, …

$T_2 = 8$

$ar^{2-1} = 8$

$ar = 8$

But r is ${}^1/_2$

$a(\frac{1}{2}) = 8$

$a = 8 \times 2$ i.e $a = 16$

GP are 16, $16 \times {}^1/_2$, $16 \times {}^1/_4$, $16 \times {}^1/_8$, $16 \times {}^1/_{16}$, …
16, 8, 4, 2, 1, ${}^1/_2$, …

2007/7a NABTEB

Find the sum of the first three terms of the GP whose third term is 27 and whose 6th term is 8

Solution

$S_3 = ?$ $T_3 = 27$ and $T_6 = 8$

We know that to find S_n, we need **a** and **r**.
These we can get from T_3 and T_6 that are given

Applying $T_n = ar^{n-1}$

$T_3 \Rightarrow 27 = ar^{3-1}$

$27 = ar^2$ ------ (1)

$T_6 \Rightarrow 8 = ar^{6-1}$

$8 = ar^5$ ------- (2)

To get r, we divide (2) by (1)

$$\frac{ar^5}{ar^2} = \frac{8}{27}$$

$r^3 = {}^8/_{27}$

We raise the RHS to numbers in power 3

$$r^3 = \frac{2^3}{3^3} \quad i.e \quad \left(\frac{2}{3}\right)^3$$

Thus, $r = {}^2/_3$

Substitute r value into (2)

$ar^5 = 8$ becomes

$a({}^2/_3)^5 = 8$

$a(\frac{32}{243}) = 8$

$32a = 8 \times 243$

$$a = \frac{8 \times 243}{32} \quad i.e \quad \frac{243}{4}$$

$S_3 = ?$ Since r is ${}^2/_3 < 1$

$$S_3 = \frac{a(1 - r^n)}{1 - r}$$

$$S_3 = \frac{243}{4}\left[1 - \left(\tfrac{2}{3}\right)^3\right] \div \left[1 - \frac{2}{3}\right]$$

$$= \frac{243}{4}\left[1 - \frac{8}{27}\right] \div \frac{1}{3}$$

$$= \frac{243}{4}\left[\frac{19}{27}\right] \times \frac{3}{1}$$

$$= \frac{27 \times 19}{4} = \frac{513}{4} \quad \text{i.e } 128\tfrac{1}{4}$$

2014/3 Neco Exercise 11.24

In a G.P, the 5th term exceeds the 4th term by 24 and the 4th term exceeds the 3rd by 8, find the:
(i) Common ratio (ii) first term,
(iii) Sum of the first five terms of the G.P

2015/1b Neco Exercise 11.25

The 5th term of a geometric progression (G.P) is $\frac{2}{81}$.
If the first term is 2, Find the : (i) common ratio
(ii) sum of the first five terms of the G.P

2014/6b (Nov) Exercise 11.26

The sum of the first third terms of a Geometric Progression (GP) is 40 while the fourth and the sixth terms are in the ratio 1 : 4.
Find the : (i) common ratio (ii) fifth term

Sum To Infinity (S_∞)

A GP whose common ratio is between – 1 and + 1 say $-{}^1/_2$, ${}^1/_2$, ${}^{-1}/_4$, ${}^1/_4$ e.t.c. has a sum which approaches a finite value as n approaches infinity. It is given as

$$S_\infty = \frac{a}{1 - r}$$

E.g. The sum to infinity of the GP 4, 2, 1, … is ?

Solution

$$S_\infty = \frac{a}{1 - r}$$

$a = 4$, $r = {}^1/_2$

$$S_\infty = \frac{4}{1 - {}^1/_2}$$

$= 4 \div {}^1/_2$

$= 8.$

2008/9 Neco (Nov)

The sum to infinity of a G.P is 80. If the first term of the series is 20, find the second term.

A 15 B $11^1/_4$ C 5 D $1^1/_4$ E ¾

Solution

$S_\infty = 80$, $a = 20$, $T_2 = ?$ We can get T_2 from $T_n = ar^{n-1}$,
First, we get r from $S_\infty = 80$

$$S_\infty = \frac{a}{1 - r}$$

$$S_\infty \Rightarrow 80 = \frac{20}{1 - r}$$

$$80(1 - r) = 20$$
$$80 - 80r = 20$$
$$80 - 20 = 80r$$
$$60 = 80r$$
$$\frac{60}{80} = r \quad \text{i.e } r = \frac{3}{4}$$

Thus, $T_2 = 20 \times \left(\frac{3}{4}\right)^{2-1}$

$= 20 \times \frac{3}{4}$

$= 15$ (A)

2007/26 UME Exercise 11.27
Find the sum to infinity of the series
$2 + \frac{3}{2} + \frac{9}{8} + \frac{27}{32} + \ldots$
A 1 B 2 C 8 D 4

2006/13 f/maths Exercise 11.28
Find the sum of the exponential series 96 + 24 + 6 +…
A 144 B 128 C 72 D 64

2009/18 UME Exercise 11.29
Find the sum to infinity, of the sequence
$1, \frac{9}{10}, \left(\frac{9}{10}\right)^2, \left(\frac{9}{10}\right)^3, \ldots$
A.10, B. 9 C. $^{10}/_9$ D. $^{9}/_{10}$

2012/20 UTME Exercise 11.30
The sum to infinity of a geometric progression is $-\frac{1}{10}$ and the first term is $-\frac{1}{8}$. find the common ratio of the progression
A $-{}^1/_5$ B $-{}^1/_4$ C $-{}^1/_3$ D $-{}^1/_2$

Problems on Sequence

The Author deliberately left problems on sequence for the later part of this chapter since the term 'sequence' could mean AP, GP or any other particular pattern of numbers. Once a sequence or a progression is mentioned without specifying whether it is an AP or a GP, or any other patterned numbers, it is the candidates responsibility to identify the type of sequence (progression) and deal with it accordingly.

2013/5
If $U_n = n(n^2 + 1)$, evaluate $U_5 - U_4$
A 18 B 56 C 62 D 80

Solution

$U_n = n(n^2 + 1)$
$U_5 = 5(5^2 + 1)$
$= 5(25 + 1)$
$= 5(26)$ i.e 130

Also
$U_4 = 4(4^2 + 1)$
$= 4(16 + 1)$
$= 4(17)$ i.e 68

Thus $U_5 - U_4 = 130 - 68$
$= 62$ (C)

2005/25
A sequence is given by $2^1/_2, 5, 7^1/_2, \ldots$
If the nth term is 25, find n
A 9 B 10 C 12 D 15

Solution

The sequence is 2.5, 5, 7.5, …

Test for AP: has **a** 2.5 and **d** as 2.5 (i.e 5 – 2.5 or 7.5 – 5)
$Tn = a + (n - 1)d$
$25 = 2.5 + (n - 1)2.5$
$25 = 2.5 + 2.5n - 2.5$
$25 = 2.5n$
$\frac{25}{2.5} = \frac{2.5n}{2.5}$
$10 = n$ (B)

2010/30
The nth term of the sequence: –2, 4, –8, 16, …is given by
A $T_n = 2^n$ B $T_n = (-2)^n$ C $T_n = (-2n)$ D $T_n = n^2$

Solution

Sequence –2, 4, –8, 16, …
Test for AP: a is – 2, and d = 4 – (– 2) or – 8 – 4
= 6 or – 12 impossible
Test for GP: a is – 2, $r = \frac{4}{-2}$ or $\frac{-8}{4}$ or $\frac{16}{-8}$
= – 2 or – 2 or – 2 true

Applying $T_n = ar^{n-1}$
$= -2(-2)^{n-1}$
$= \frac{-2(-2)^n}{-2}$
$= (-2)^n$ (B)

2012/5
The nth term of a sequence is $Tn = 5 + (n - 1)^2$,
Evaluate $T_4 - T_6$
A 30 B 16 C – 16 D – 30

Solution

$T_n = 5 + (n - 1)^2$
$T_4 = 5 + (4 - 1)^2$
$= 5 + 3^2$
$= 5 + 9$ i.e 14
Also $T_6 = 5 + (6 - 1)^2$
$= 5 + 5^2$
$= 5 + 25$ i.e 30
Thus $T_4 - T_6 = 14 - 30$
$= -16$ (C)

2005/6 NABTEB (Nov)
What is the 13th term of the series – 4 + 1 + 6 + 11 + …
A 51 B 56 C 61 D 66

Solution

Series: – 4 + 1 + 6 + 11 + …
A.P Test : a is – 4 and **d** = 1 – (– 4) or 6 – 1 or 11 – 6
= 5 or 5 or 5 true
$T_{13} = ?$ Applying $T_n = a + (n - 1)d$
$T_{13} = -4 + (13 - 1)5$
$= -4 + (12)5$
$= -4 + 60$
$= 56$ (B)

2008/11 NABTEB (Nov)
Given the sequence 1, 4, 9, 16, …
The (n + 1)th term of the sequence is
A 25 B n^2 C $(n + 1)^2$ D $n^2 + 1$

Solution

A.P Test: **a** is 1 and **d** = 4 – 1 or 9 – 4 or 16 – 9
= 3 or 5 or 7 not true

GP Test: **a** is 1 and **r** = $\frac{4}{1}$ or $\frac{9}{4}$ or $\frac{16}{4}$
= 4 or $^9/_4$ or 4 not true

But from the AP test a pattern is already generated, Difference between the number increases by simple addition of 2

1 4 9 16, …
3 5 7

This will not help us

A second look shows 2^2 is 4
3^2 is 9
4^2 is 16
Thus 1^2 is 1

The nth term is n^2
and (n + 1)th term is $(n + 1)^2$ C.

2012/3 Neco

The nth term of a sequence is 2^{3n-2}, which term of the sequence is 2^{13}?

A 6th B 5th C 4th D 3rd E 2nd

Solution

$T_n = 2^{3n-2}$
$2^{13} = 2^{3n-2}$

Since both sides are in the same base, we equate powers

13 = 3n – 2
13 + 2 = 3n
15 = 3n
$\frac{15}{3} = \frac{3n}{3}$
5 = n (B)

2014/2 Neco

The nth term of a sequence is given by $(-1)^{n-2} \bullet 2^{n-1}$
Find the sum of the second and the third terms

A – 2 B 1 C 2 D 6 E 12

Solution

$T_2 + T_3$ = ? From $T_n = (-1)^{n-1} \bullet 2^{n-1}$
$T_2 = (-1)^{2-2} \bullet 2^{2-1}$
$= (-1)^0 \bullet 2^1$
$= 1 \times 2$
$= 2$

Also $T_3 = (-1)^{3-2} \bullet 2^{3-1}$
$= (-1)^1 \bullet 2^2$
$= -1 \times 4$
$= -4$

Thus $T_2 + T_3 = 2 + (-4)$
$= 2 - 4$
$= -2$ (A)

2011/5 Neco

Find the sum of the first 12 terms of the sequence
18, 11, 4, –3, …

A 678 B – 38 C – 41 D – 241 E – 246

Solution

AP Test: **a** is 18 and **d** = 11 – 18 or 4 – 11
= – 7 or – 7 true

S_{12} = ? Applying $S_n = \frac{n}{2}\{2a + (n-1)d\}$
$= \frac{12}{2}\{2 \times 18 + (12-1)(-7)\}$
$= 6\{36 + (11)(-7)\}$
$= 6(36 - 77)$
$= 6(-41)$
$= -246$ (E)

2014/8 Neco (Nov)

Find the sum of the first 28 terms of the sequence
8, x, 18, 23, …

A 2002 B 2114 C 4004 D 4228 E 8008

Solution

Assuming the sequence to be A.P
a = 8, d = 23 – 18 i.e 5
To confirm **23** is T_4 and $T_4 = 8 + (4 - 1)5$
$= 8 + (3)5$
$= 23$
18 is T_3 and $T_3 = 8 + (3 - 1)5$
$= 8 + (2)5 = 18$

We conveniently say the sequence is an A.P
Thus x = 8 + 5
= 13

and S_{28} = ? Applying $S_n = \frac{n}{2}\{2a + (n-1)d\}$
$S_{28} = \frac{28}{2}\{2 \times 8 + (28-1)5\}$
$= 14\{16 + (27)5\}$
$= 14(16 + 135)$
$= 14(151)$
$= 2114$ (B)

2008/8 Neco (Nov)

Find the 32[nd] term of the sequence 482, 480, 478, 476,…

A – 420 B 420 C 450 D 514 E 544

Solution

Sequence is : 482, 480, 478, 476,…

AP Test: **a** is 482 and **d** = 480 – 482 or 478 – 480
= – 2 or – 2 true

T_{32} = ? Applying $T_n = a + (n - 1)d$
$T_{32} = 482 + (32 - 1)(-2)$
$= 482 + (31)(-2)$
$= 482 - 62$
$= 420$ (B)

2009/15 Neco (Nov)

The nth term of a sequence is given by $(3)2^{n-1}$. Write down the first three terms of the sequence given that $n \geq 0$

A $^3/_2$, 3, 6 B ¾, $^3/_2$, 3 C $^3/_2$, 3, $^1/_3$
D $^3/_2$, $^3/_4$, 3 E $^3/_4$, 0, 6

Solution

$T_n = (3)2^{n-1}$
For $n \geq 0$, first three terms are n = 0, n = 1, n = 2

<u>n = 0</u>
$(3)2^{0-1} = (3)2^{-1}$
$= \frac{3}{2}$

<u>n = 1</u>
$(3)2^{1-1} = (3)2^0$ i.e 3×1
$= 3$

<u>n = 2</u>
$(3)2^{2-1} = (3)2^1$
$= 6$ Thus, sequence is : $^3/_2$, 3, 6 (A)

2014/17 NABTEB (Nov)
Find the 28th term of the sequence 3, 8, 13, 18, 23,…
A 114 B 133 C 135 D138

Solution

A.P test: **a** is 3 and **d** = 8 – 3 or 13 – 8 or 18 – 13
= 5 or 5 or 5

T_{28} = ? Applying $T_n = a + (n - 1)d$

$$T_{28} = 3 + (28 - 1)5$$
$$= 3 + (27)5$$
$$= 3 + 135$$
$$= 138 \quad (D)$$

2014/9 NABTEB
Find the sum of the first 90 positive integers
A90 B 2045 C 4095 D 5050

Solution

Pattern of positive integers: 1, 2, 3, 4, 5, …
A.P Test: **a** is 1 and **d** = 2 – 1 or 3 – 2 or 4 – 3
= 1 or 1 or 1 true

S_{90} = ? Applying $S_n = \frac{n}{2}\{2a + (n-1)d\}$

$$S_{90} = \frac{90}{2}\{2\times 1 + (90-1)1\}$$
$$= 45\{2 + (89)1\}$$
$$= 45(2 + 89)$$
$$= 45(91) = 4095 \quad (C)$$

2007/8b NABTEB
Simplify without using mathematical tables the sum of the first 20 terms of the series 3 + 6 + 9 + 12 +…

Solution

A.P Test: **a** is 3 and **d** = 6 – 3 or 9 – 6 or 12 – 9
= 3 or 3 or 3 true

S_{20} = ? Applying $S_n = \frac{n}{2}\{2a + (n-1)d\}$

$$S_{20} = \frac{20}{2}\{2\times 3 + (20-1)3\}$$
$$= 10\{6 + (19)3\}$$
$$= 10(6 + 57)$$
$$= 10(63) = 630$$

2006/1Neco (Nov) Exercise 11.31
The nth term of a series is given by the formula $2a + 2^{n-1}$. If a = 6
(i) what is the value of the 9th term?
(ii) find the number of terms if the last term is:
(a) 76, (b) 524

2015/5 Exercise 11.32
Find the 7th term of the sequence : 2, 5, 10, 17, 26,…
A 37 B 48 C 50 D 63

2014/6 (Nov) Exercise 11.33
The nth term of the sequence 5, 8, 11, … is 383. Find n
A 125 B 126 C 127 D 194

2015/3 Neco Exercise 11.34
The nth term of the sequence –1, 5, –7, 17,… is given by
A 2^n B $-1 + (-2)^n$ C $-1 - 2^n$
D $1 + (-2)^n$ E $1 - 2^n$

2005/28 Exercise 11.35
What is the nth term of the sequence 3, 8, 13…?
A 5n + 3 B 5n – 2 C 5n – 3 D 5n – 5

CHAPTER TWELVE
Variation

Variation is the term that refers to the relationship between two or more quantities. A change in one of the quantities results in a change in the other (s). The symbol $\propto$ is used to indicate variation.
We shall consider four types of variations namely:
Direct variation
Inverse variation
Joint variation
Partial variation

Direct variation

$$y \propto x$$

reads: y varies directly as x *or* y is directly proportional to x.
To introduce equality sign, we must introduce a constant of variation i.e

$$y = kx$$

Where k is the constant of variation or constant of proportionality and $k = \frac{y}{x}$

2006/ 15 Neco (Nov)
If x α y, when x = 5, y = 15, what is y when x = 1.5?
A 50 B 45 C 5 D 4.5 E 2.25

Solution

$$x \; \alpha \; y$$

Introducing a constant k to enable us replace α by =

$$x = ky \text{ ----------- } **$$
$$\frac{x}{y} = k$$

When x = 5, y = 15 *then*

$$\frac{5}{15} = k$$
$$\frac{1}{3} = k$$

Substitute k value into ** to get the relation

$$x = \frac{1}{3}y$$

Next, when x = 1.5, y = ?

$$1.5 = \frac{y}{3}$$
$$1.5 \times 3 = y$$
$$4.5 = y \quad (D)$$

2006/ 43 Neco
The time of oscillation of a pendulum varies as the square root of its length. If the length of the pendulum which oscillates for 35 seconds is 49cm, find the time of oscillation of a pendulum with length 121cm
A 77sec B 55sec C 18sec D 11sec E 7 sec

Solution

$$T \; \alpha \; \sqrt{L}$$

Introducing a constant K to enable in replace α by =

$$T = K\sqrt{L} \text{ --------} **$$
$$\frac{T}{\sqrt{L}} = K$$

When T = 35s, L = 49cm *then*

$$\frac{35}{\sqrt{49}} = K \text{ i.e } \frac{35}{7} = K$$

$$5 = K$$

Substitute K value into ** to get the relation

$$T = 5\sqrt{L}$$

Next, when L = 121cm, T = ?

$$T = 5\sqrt{121}$$
$$= 5 \times 11$$
$$= 55 \text{ sec} \quad (B)$$

2011/ 33

G varies directly as the square of H. If G is 4 when H is 3, find H when G = 100

A 15 B 25 C 75 D 225

Solution

$$G \; \alpha \; H^2$$

Introducing a constant k to enable us replace α by =

$$G = k H^2 \text{ ---------- **}$$

$$\frac{G}{H^2} = k$$

When G = 4, H = 3

$$\frac{4}{3^2} = k$$

$$\frac{4}{9} = k$$

Substitute k value into ** to get the relation

$$G = \frac{4H^2}{9}$$

When G = 100, H = ?

$$100 = \frac{4H^2}{9}$$

$$100 \times 9 = 4H^2$$

$$\frac{100 \times 9}{4} = H^2$$

$$25 \times 9 = H^2$$

To get H under square we take the square root of both sides

$$\sqrt{25 \times 9} = H$$
$$5 \times 3 = H$$
$$15 = H \quad (A)$$

2014/11a NABTEB (Nov)

The electrical resistance, R ohm, of a piece of wire is proportional to its length, Lm. A piece of wire of length 12m has resistance 25 ohms

(i) Write the equation connecting R and L

(ii) Find the length of wire for a resistance of 40 ohms

Solution

$$R \, \alpha \, L$$

Introducing a constant k to enable us replace α by =

$$R = KL \text{ -------- **}$$

$$\text{Thus } \frac{R}{L} = K$$

When L = 12m, R = 25ohms *then*

$$\frac{25}{12} = K$$

Substitute K value into ** to get the relation

$$R = \frac{25}{12}L$$

Next when R = 40 ohms, L = ?

$$40 = \frac{25}{12}L$$

$$40 \times 12 = 25L$$

$$\frac{40 \times 12}{25} = L \quad \text{i.e } L = 19.2\text{m}$$

2006/ 18 – 19 ***Counter example***

The table below satisfies the relation $y = k\sqrt{x}$

Where k is a positive constant. Use it to answer questions 18 – 19

x	1	4	p
y	0.5	1	2.5

18. Find the value of k

A 0.5 B 1 C 1.5 D 2

Solution

We can pick any suitable set of values of x and y from the table, say x = 4 and y = 1

$$y = k\sqrt{x} \quad \text{becomes}$$

$$1 = k\sqrt{4}$$
$$1 = 2k$$
$$\tfrac{1}{2} = k \quad \text{i.e } k = 0.5 \quad (A)$$

19. Find the value of p

A 2 B 4 C 10 D 25

Solution

The relation now is

$$y = \tfrac{1}{2}\sqrt{x}$$

When y = 2.5, **x** i.e p = ?

$$2.5 = \tfrac{1}{2}\sqrt{x}$$

$$2.5 \times 2 = \sqrt{x}$$

$$5 = \sqrt{x}$$

To remove square root from x, we take the square of both sides

$$5^2 = x \quad \text{i.e } 25 = x \quad (D)$$

2006/40(Nov)

Q varies directly as the cube of P. when P = 1, Q = 3. Find P when Q = 24

A $\frac{1}{3}$ B $\frac{1}{2}$ C 2 D 3

Solution

$$Q \; \alpha \; P^3$$

Introducing a constant k to enable us replace α by =

$$Q = k P^3 \text{ --------- **}$$

$$\frac{Q}{P^3} = k$$

When P = 1, Q = 3 *then*

$$\frac{3}{1^3} = k \quad \text{i.e } k = 3$$

Substitute k value into ** to get the relation

$$Q = 3P^3$$

Next, when Q = 24, P = ?

$$24 = 3P^3$$

$$\frac{24}{3} = P^3$$

$$8 = P^3$$

To get P under cube, we take the cube root of both sides

$$\sqrt[3]{8} = P \quad \text{i.e } P = 2 \quad (C)$$

2014/7 Neco (Nov)
If P varies directly as the cube root of Q and P = 3 and Q = 125, find Q when P = $^{18}/_{5}$
A 169 B 216 C 343 D 450 E 512

Solution

$P \propto \sqrt[3]{Q}$

Introducing a constant k to enable us replace α by =

$P = K\sqrt[3]{Q}$ ---------- **

$$\frac{P}{\sqrt[3]{Q}} = K$$

When P = 3, Q = 125 *then*

$$\frac{3}{\sqrt[3]{125}} = K$$

$$\frac{3}{5} = K$$

Substitute K value into ** to get the relation

$$P = \frac{3\sqrt[3]{Q}}{5}$$

Next, when $P = \frac{18}{5}$, Q = ?

$$\frac{18}{5} = \frac{3\sqrt[3]{Q}}{5}$$

$$\frac{18}{5} \times 5 = 3\sqrt[3]{Q}$$

$$\frac{18}{5} \times \frac{5}{3} = \sqrt[3]{Q}$$

$$6 = \sqrt[3]{Q} \quad \text{i.e } 6 = Q^{\frac{1}{3}}$$

To remove cube root from Q, we take cube of both sides

$6^3 = Q$

$216 = Q$ (B)

2007/3b NABTEB
If S is directly proportional to T and T = 120, when S = 30, find the value of T when S = 136

Solution

$S \propto T$

Introducing a constant k to enable us replace α by =

S = kT ----------- **

$$\frac{S}{T} = k$$

T = 120, when S = 30 *then*

$$\frac{30}{120} = k$$

$$\frac{1}{4} = k$$

Substitute k value into ** to get the relation

$$S = \frac{1}{4}T$$

Next, when S = 136, T = ?

$$136 = \frac{T}{4}$$

$136 \times 4 = T$

$544 = T$

2013/24 Neco
If a is directly proportional to b and a = –1 while b = –4, the formula connecting a and b is
A $a = \frac{1}{4}b$ B $a = \frac{1}{2}b$ C a = b
D a = 2b E a = 4b

Solution

$a \propto b$

Introducing a constant k to enable us replace α by =

a = kb ---------- **

$$\frac{a}{b} = k$$

When a = – 1, b = – 4 *then*

$$\frac{-1}{-4} = k \quad \text{i.e} \quad \frac{1}{4} = k$$

Substitute k value into ** to get the relation or connecting formula

$a = \frac{1}{4}b$ (A)

1975/5
y varies directly as an unknown power of x. Express y as a function of x, given that when x = 2, y = 0.4 and when x = 4, y = 6.4 (Take the unknown power of x to be n)

Solution

$y \propto x^n$

Introducing the constant of variation

$y = kx^n$ ------------**

When x = 2, y = 0.4 into **

$0.4 = k\,2^n$

$$k = \frac{0.4}{2^n} \quad \ldots\ldots\ldots\ldots(1)$$

When x = 4, y = 6.4 into **

$6.4 = k\,4^n$

$$k = \frac{6.4}{4^n} \quad \ldots\ldots\ldots\ldots(2)$$

From (1) and (2) we equate k values

$$k \Rightarrow \frac{0.4}{2^n} = \frac{6.4}{4^n}$$

$$4^n \times 0.4 = 2^n \times 6.4$$

$$\frac{4^n}{2^n} = \frac{6.4}{0.4}$$

$$\frac{2^{2n}}{2^n} = 16 \quad (\text{ Note } 2^{2n-n} = 2^n)$$

$2^n = 16$

$2^n = 2^4$

n = 4

Next, we substitute n = 4 into (1)

$$k = \frac{0.4}{2^4} = \frac{0.4}{16} \text{ i.e } \frac{1}{40}$$

The relation from $y = kx^n$ is

$$y = \frac{1}{40}x^4$$

2012/7a Exercise 12.1
The electrical resistance R ohms of a wire varies directly as the length Lcm. If R = 0.6 ohms when L = 7.5cm, find R when L = 35.5cm

2008/23 Neco (Nov) Exercise 12.2

P varies directly as the cube root of Q. When P = 20, Q = 125. Find the value of P when Q = 27

A 12 B 15 C 24 D 25 E 30

2005/24 Neco Exercise 12.3

If A varies directly as the square root of B and A = 35, when B = 25. Find B when A = 14

A 14 B 7 C 6 D 5 E 4

2014/7 Neco (Nov) Exercise 12.4

If p varies directly as the cube root of q and p = 3 when q = 125, find q when p = $^{18}/_{5}$

A 169 B 216 C 343 D 450 E 512

2015/40 Exercise 12.5

The rate of consumption of petrol by a vehicle varies directly as the square of the distance covered. If 4 litres of petrol is consumed on a distance of 15 km, how far would the vehicle go on 9 litres of petrol?

A $22^1/_2$ km B 30km C $33^3/_4$ km D 45km

Inverse variation

If y varies inversely as x, then mathematically, we write

$$y \propto \frac{1}{x}$$

Introducing a constant of variation,

$$y = \frac{k}{x}$$

2009/5 (Nov)

If y is inversely proportional to x and x = 3 when y = 4, find y when x = 2

A 1 B 3 C 6 D 9

$$y \ \alpha \ \frac{1}{x}$$

Introducing a constant k to enable us replace α by =

$$y = \frac{k}{x} \quad \text{------------**}$$

$$k = xy$$

When x = 3, y = 4 *then*

$$k = 3 \times 4$$
$$= 12$$

Substitute k value into ** to get the relation

$$y = \frac{12}{x}$$

When x = 2, y = ?

$$y = \frac{12}{2}$$
$$y = 6 \qquad (\text{C})$$

2014/ 26 Neco

y varies inversely as x^2. If y = 3 when x = 4, find y when x = 2

A 8 B 9 C 10 D 11 E 12

Solution

$$y \ \alpha \ \frac{1}{x^2}$$

Introducing a constant of variation k

$$y = \frac{k}{x^2} \quad \text{-----------**}$$

Thus $yx^2 = k$

Substituting for the first set of y and x values

$$k = 3 \times (4)^2$$
$$= 48$$

Substituting k value into ** to get the relation

The relation is $y = \dfrac{48}{x^2}$

When x = 2, find y

$$y = \frac{48}{2^2}$$
$$= \frac{48}{4} \text{ i.e } 12 \qquad (\text{E})$$

2008/10

T varies inversely as the square root of F, when T = 7, F = $2^1/_4$. Find T when F = $2\,^7/_9$

A 1.5 B 6.3 C 10.5 D 12.6 E 21

Solution

$$T \ \alpha \ \frac{1}{\sqrt{F}}$$

Introducing a constant of variation k to replace α by =

$$T = \frac{k}{\sqrt{F}} \quad \text{-----------**}$$

When T = 7, F = $2^1/_4$. *then*

$$7 = \frac{k}{\sqrt{\frac{9}{4}}}$$

$$k = 7 \times \sqrt{\frac{9}{4}}$$
$$= 7 \times {}^3/_2$$
$$= {}^{21}/_2$$

Substitute k value into ** to get the relation

$$T = \frac{21}{2\sqrt{F}}$$

When F = $2^7/_9$, T is

$$T = \frac{21}{2\sqrt{2\frac{7}{9}}} \quad \text{i.e} \quad \frac{21}{2\sqrt{\frac{25}{9}}}$$
$$= \frac{21}{2 \times \frac{5}{3}}$$
$$= 21 \div {}^{10}/_3$$
$$= 21 \times {}^3/_{10}$$
$$= {}^{63}/_{10}$$
$$= 6.3 \qquad (\text{B})$$

2006/10b Neco

The weight, w, of a body varies inversely as the square of its distance, d, from the centre of the earth. A body weighs 80 Newtons on the earth surface, if the radius of the earth is 6400km,

i. Obtain an expression connecting the weight of a body and its distance from the centre of the earth;
ii. Calculate the distance of a body of 20 Newtons from the centre of the earth;
iii. Calculate the weight of a body which is 1,600 km above the surface of the earth.

Solution

(i) $w \ \alpha \ \frac{1}{d^2}$

Introducing a constant of variation k to replace α by =

$$w = \frac{k}{d^2} \quad \text{-----------**}$$

When w = 80N, d = 6400km *then*

$$80 = \frac{k}{6400^2}$$

$$k = 80 \times (6400)^2$$

$$= 3.2768 \times 10^9$$

Substitute k value into ** to get the relation

$$w = \frac{3.2768 \times 10^9}{d^2}$$

(ii) w = 20N, find d

Substituting into the relation

$$20 = \frac{3.2768 \times 10^9}{d^2}$$

$$d^2 = \frac{3.2768 \times 10^9}{20}$$

$$d^2 = 163840000$$

$$d = \sqrt{163840000} = 12800\text{km}$$

(iii) From the centre of the earth taken as reference

Here d = (1600 + 6400) = 8000km ,

w = ?

Substituting into the relation $w = \frac{k}{d^2}$

$$w = \frac{3.2768 \times 10^9}{(8000)^2}$$

$$= \frac{3276800000}{64000000} = 51.2\text{N}$$

2012/ 43

If x and y are variables and k is a constant, which of the following describe an inverse relationship between x and y?

A $y = kx$ B $y = \frac{k}{x}$ C $y = k\sqrt{x}$ D $y = x + k$

Solution

$$y \ \alpha \ \frac{1}{x}$$

Introducing a constant k to enable us replace α by =

$$y = \frac{k}{x} \qquad (B)$$

2009/26 Neco (Dec)

If (y + 2) varies inversely as x and y = 3 when x = 2. Find y when x = 5.

A 0 B 2 C 4 D 7 E 11

Solution

$$(y + 2) \ \alpha \ \frac{1}{x}$$

Introducing a constant of variation k

$$(y + 2) = \frac{k}{x} \quad \text{-----------**}$$

$$x(y + 2) = k$$

When y = 3, x = 2 *then*

$$k = 2(3 + 2)$$

$$= 2(5)$$

$$= 10$$

Substitute k value into ** to get the relation

The relation is $(y + 2) = \frac{10}{x}$

When x = 5, then find y

$$(y + 2) = \frac{10}{5}$$

$$y + 2 = 2$$

$$y = 2 - 2$$

$$y = 0 \qquad (A)$$

2007/24 Neco

n varies inversely as the square root of m. If n = 5 when m = 9. State the relationship between n and m

A n = 15m B m = 15n C $m = \frac{k}{15n}$

D $n = \frac{405}{\sqrt{m}}$ E $n = \frac{15}{\sqrt{m}}$

Solution

$$n \ \alpha \ \frac{1}{\sqrt{m}}$$

Introducing a constant k to enable us replace α by =

$$n = \frac{k}{\sqrt{m}} \quad \text{---------- **}$$

$$k = n\sqrt{m}$$

When n = 5, m = 9 *then*

$$k = 5\sqrt{9}$$

$$= 5 \times 3$$

$$= 15$$

Substitute k value into ** to get the relation

$$n = \frac{15}{\sqrt{m}} \qquad (E)$$

2006/13 Neco

Y varies inversely as the cube root of x.

If Y = 4 when x = 0.125, find y when x = 8

A 5 B 4 C 3 D 2 E 1

Solution

$$Y \ \alpha \ \frac{1}{\sqrt[3]{x}}$$

Introducing a constant k to enable us replace α by =

$$Y = \frac{k}{\sqrt[3]{x}} \quad \text{---------- **}$$

$$k = y\sqrt[3]{x}$$

When y = 4, x = 0.125

$$k = 4\sqrt[3]{0.125}$$

$$= 2$$

Substitute k value into ** to get the relation

$$Y = \frac{2}{\sqrt[3]{x}}$$

Next, when x = 8, y = ?

$$Y = \frac{2}{\sqrt[3]{8}}$$

$$= \frac{2}{2} \text{ i.e } 1 \qquad (E)$$

2005/48 Exercise 12.6

If $y \alpha \frac{1}{x^2}$ and $y = 1\frac{1}{4}$ when $x = 4$, find the value of y when $x = \frac{1}{2}$

A $2\frac{1}{2}$ B 5 C 10 D 80

2014/43 Neco (Nov) Exercise 12.7

The resistance of an electric wire varies inversely as the square of the potential difference. When the resistance is 0.7Ω , the potential difference is 0.4v. Find the potential difference when the resistance is 44.8 Ω

A 0.05V B 0.13V C 0.50V D0.52V E 5.18V

2005/36 Exercise 12.8

If p varies inversely as the square of q and p = 8 when q = 2, find p when q = 4

A 2 B 4 C 8 D 16

2013/28 Neco Exercise 12.9

Given that y is inversely proportional to x and y = 20 when $x = \frac{1}{4}$, find x when y = 30

A $\frac{1}{6}$ B $\frac{1}{5}$ C $\frac{1}{4}$ D $\frac{1}{3}$ E $\frac{1}{2}$

Joint variation

Joint variation involves more than two variables. In most cases it is a combination of direct and inverse variation

Example

If p varies directly as the square of q and inversely as r

Then $p \propto \frac{q^2}{r}$

Introducing a constant of variation

$$p = \frac{kq^2}{r}$$

2007/23 Neco

r varies directly as the cube of p and inversely as the square of q. If k is a constant which of the following is correct?

A $q^2 = kp^3r$ B $q^2 = \frac{kr}{p^3}$ C $q^3 = kp^2r$

D $q^2 = \frac{kp^3}{r}$ E $q^3 = \frac{kp^2}{r}$

Solution

$$r \alpha \frac{p^3}{q^2}$$

Introducing a constant of variation

$$r = \frac{kp^3}{q^2}$$

From the options, we are to make q^2 the subject of formula

$$q^2r = kp^3$$

Thus, $q^2 = \frac{kp^3}{r}$ (D)

2008/11 Neco

x varies directly as y and inversely as z. when x = 5, y = 2 and z = 1. What is the value of x when y = 5 and z = 2 ?

A 2.5 B 5.0 C 6.25 D 6.52 E 7.5

Solution

$$x \alpha \frac{y}{z}$$

Introducing a constant of variation

$$x = \frac{ky}{z} \text{ ----------**}$$

When x = 5, y = 2 and z = 1 *then*

$$5 = \frac{k2}{1}$$

$$5 = 2k$$

$$\frac{5}{2} = k$$

Substitute k value into ** to get the relation

$$x = \frac{5y}{2z}$$

When y = 5, z = 2 x = ?

$$x = \frac{5 \times 5}{2 \times 2}$$

$$= \frac{25}{4} \text{ i.e } 6.25 \quad (C)$$

2012/15 Neco

The cost of material for making a drum varies as the cube of the radius and inversely as the surface area. When the cost is ₦150.00, the area is $2250cm^2$ and the radius is 15cm. Find the cost of the materials when the radius is 18cm and the area is $2700cm^2$

A ₦270.00 B ₦216.00 C ₦196.00

D ₦180.00 E ₦168.00

Solution

$$C \alpha \frac{r^3}{A}$$

Introducing a constant of variation

$$C = \frac{kr^3}{A} \text{ ---------**}$$

When C = ₦150.00, A = $2250cm^2$, r = 15cm *then*

$$150 = \frac{k \times 15^3}{2250}$$

$$\frac{150 \times 2250}{15 \times 15 \times 15} = k$$

$$100 = k$$

Substitute k value into ** to get the relation

$$C = \frac{100r^3}{A}$$

When r = 18cm, Area = $2700cm^2$ C = ?

$$C = \frac{100 \times 18^3}{2700}$$

$$= \frac{18 \times 18 \times 18}{27}$$

= ₦216.00 (B)

2008/22 Neco (Nov)

The energy E possessed by a moving object varies directly as its mass m and the square of its velocity v. when m = 8kg, $v = 5ms^{-1}$ and E = 100 joules. Find E when m = 6kg and $v = 2ms^{-1}$

A 6J B 12J C 24J D 48J E 96J

Solution

$E \propto mv^2$

Introducing a constant of variation

$E = kmv^2$ ----------**

When m = 8kg, v = $5ms^{-1}$ and E = 100 joules *then*

$100 = k \times 8 \times 5^2$

$\frac{100}{8 \times 25} = k$

$\frac{1}{2} = k$

Substitute k value into ** to get the relation

$$E = \frac{1}{2}mv^2$$

When m = 6kg, v = $2ms^{-1}$, E = ?

$$E = \frac{6 \times 2^2}{2}$$

$= 12j$ (B)

2010/28 Neco

$F \propto \frac{Q}{T^2}$, when Q = 32, T = 4 and F = 20.

Find F when Q = 49 and T = 7

A 7 B 10 C 14 D 49 E 160

Solution

$F \propto \frac{Q}{T^2}$

Introducing a constant of variation

$F = \frac{kQ}{T^2}$ ---------**

When Q = 32, T = 4 and F = 20 *then*

$20 = \frac{k \times 32}{4^2}$ i.e $\frac{k \times 32}{4 \times 4}$

$20 = k \times 2$

$\frac{20}{2} = k$ i.e k = 10

Substitute k value into ** to get the relation

$$F = \frac{10Q}{T^2}$$

When Q = 49, T = 7 F = ?

$F = \frac{10 \times 49}{7^2}$ i.e $\frac{10 \times 49}{49}$

F = 10 (B)

2012/19 Neco

Given that $x \propto by$ when x = 2, y = 3 and b = 2, what is the value of b when x = 3 and y = 1?

A $^1/_3$ B 3 C 6 D 9 E 12

Solution

$x \propto by$

Introducing a constant of variation

$x = kby$ ----------**

When x = 2, y = 3 and b = 2 *then*

$2 = k \times 2 \times 3$

$\frac{2}{2 \times 3} = k$

$^1/_3 = k$

Substitute k value into ** to get the relation

$$x = \, ^1/_3 \, by$$

When x = 3, y = 1, b = ?

$3 = \frac{b \times 1}{3}$

$3 \times 3 = b$

$9 = b$ (D)

2008/6b NABTEB (Nov)

$A \propto \frac{B}{C^2}$ and A = 20 when B = 5 and C = $^1/_3$

Find the (i) relation between A, B and C

(ii) value of A when B = 4 and C = $^1/_6$

Solution

$A \propto \frac{B}{C^2}$

Introducing a constant of variation

$A = \frac{kB}{C^2}$ ------------**

If A = 20, B = 5, C = $^1/_3$ *then*

$20 = \frac{k5}{(\frac{1}{3})^2}$

$20 \times \, ^1/_9 = 5k$

$20 \times \, ^1/_9 \times \, ^1/_5 = k$

$^4/_9 = k$

Substitute k value into ** to get the relation

$$A = \frac{4B}{9C^2}$$

(ii) When B = 4, C = $^1/_6$, A = ?

$A = \frac{4 \times 4}{9 \times (\frac{1}{6})^2}$

$= 4 \times 4 \div \, ^1/_4$

$= 4 \times 4 \times 4$

$= 64$

2005/ 12b Neco

The energy (E) varies directly as the resistance (R) and inversely as the square of the distance (D). If E = $^{32}/_{25}$, when R = 16 and D = 10, calculate the value of R when E = 32 and D = 7

Solution

$E \propto \frac{R}{D^2}$

Introducing a constant of variation

$E = \frac{KR}{D^2}$ ---------- **

When E = $^{32}/_{25}$, R = 16 and D = 10, *then*

$\frac{32}{25} = \frac{k16}{10^2}$

$\frac{32}{25} \times 10^2 = 16k$

$\frac{32}{25} \times 100 \times \frac{1}{16} = k$

$8 = k$

Substitute k value into ** to get the relation

$$E = \frac{8R}{D^2}$$

When E = 32, D = 7, R =?

$32 = \frac{8 \times R}{7^2}$

$32 \times 7^2 = 8R$

$$\frac{32 \times 49}{8} = R$$
$$196 = R$$

2014/7 (Nov)

A quantity z varies directly as the square root of x and inversely as the cube of s. If z = 8 when x = 4 and s = $^1/_2$, express z in terms of x and s.

A $Z = \frac{2\sqrt{x}}{S^3}$ B $Z = \frac{\sqrt{x}}{S^3}$ C $Z = \frac{2S^3}{\sqrt{x}}$ D $Z = \frac{\sqrt{x}}{2S^3}$

Solution

$$Z \ \alpha \ \frac{\sqrt{x}}{s^3}$$

Introducing a constant of variation

$$Z = \frac{k\sqrt{x}}{s^3} \quad \text{------------}**$$

When z = 8, x = 4 and s = $^1/_2$, *then*

$$8 = \frac{k\sqrt{4}}{(\frac{1}{2})^3}$$
$$8 = \sqrt{4}\ k \div \left(\tfrac{1}{2}\right)^3$$
$$8 = 2k \div {}^1/_8$$
$$8 = 2k \times {}^8/_1$$
$$8 = 16k$$
$$^8/_{16} = k \text{ i.e } ^1/_2 = k$$

Substitute k value into ** to get the relation

$$Z = \frac{\sqrt{x}}{2S^3} \quad (\text{ D })$$

2014/16

If x varies inversely as y and y varies directly as z, what is the relationship between x and z?

A $x \alpha z$ B $x \ \alpha \ \frac{1}{z}$ C $x \ \alpha z^2$ D $x \ \alpha \ \frac{1}{z^2}$

Solution

$$x \ \alpha \ \frac{1}{y} \text{ and } y \ \alpha \ z$$

By simple reasoning, the second statement simply means take y as z

Thus the relationship between x and y is the same as between x and z

$$\therefore \ x \ \alpha \ \frac{1}{z}$$

2011/13a Exercise 12.10

P varies directly as Q and inversely as the square of R. If P = 1 when Q = 8 and R = 2, find the value of Q when P = 3 and R = 5.

2008/ 6b Exercise 12.11

P, Q and R are related in such a way that $P \ \alpha \ \frac{Q^2}{R}$.

When P = 36, Q = 3 and R = 4, Calculate Q when P = 200 and R = 2

2005/6c Exercise 12.12

x, y and z are related such that x varies directly as the cube of y and inversely as the square of z. if x = 108 when y = 3 and z = 4, find z when x = 4000 and y = 10

2008/7 NABTEB (Nov) Exercise 12.13

If $x \ \alpha \ y^2z$ when y = 5 and z = 6, x = 45.
Find x when y = 3 and z = 7

A 189 B 181 C 18.9 D 1.89

2008/6b NABTEB (Nov) Exercise 12.14

$A \ \alpha \ \frac{B}{C^2}$ and A = 20 when B = 5 and C = $\frac{1}{3}$

Find the (i) relation between A, B and C

(ii) value of A when B = 4 and C = $\frac{1}{6}$

2005/36 Neco Exercise 12.15

The mass of a spherical ball varies as the cube of its radius and inversely as its surface area, when the radius is 4cm, its area is $16cm^2$ and the mass is $20kg /cm^3$. Find its mass when the area is $54cm^2$ and the radius is 3cm

A $20kg/cm^3$ B $18kg/cm^3$ C $6^2/_3\ kg/cm^3$
D $5^1/_3 kg/cm^3$ E $2^1/_2\ kg/cm^3$

1976/8b Exercise 12.16

The electrical conductivity in a wire of circular cross-section varies directly as the square of the radius and inversely as the length. One wire is twice as long as the other but the radius of the cross-section of the shorter is $\frac{3}{2}$ times that of the longer. Find the ratio of their conductivity

1976/6 (Nov) Exercise 12.17

Given that the energy E varies directly as the resistance R and inversely as the square of the distance d, get an equation connecting E, R and d. If E = $\frac{32}{25}$ when R = 8 and d = 5, calculate:

(a) the value of R when E = 16 and d = 3
(b) the value of d when R = 5 and E = $\frac{5}{6}$
(c) the percentage increase in R when E and d each increases by 3%

Partial variation

When a variation is expressed as a sum of terms, we refer to that type of variation as *partial.* For instance,

y = k + ax

Reads: "y is partly constant and partly varies as x". The letter 'a' is a constant of variation. If we introduce same to K say b, then we will have ' bk' but constant multiply by a constant is also a constant. Thus, we do not apply another constant to a constant as we do to variables.

The statement "t varies partly as r and partly as inverse of s" is represented as

$$t = ar + b\frac{1}{s}$$

2013/7b Neco

The cost of sewing a garment is partly constant and partly varies with the amount of time it takes to sew it. If it takes 6 hours to sew the garment, the cost is ₦2800.00 and if it takes 10 hours to sew it, the cost is ₦3600.00. Find the cost if it takes 4 hours to sew the garment.

Solution

c = k + t

Introducing constant b to variable t

c = k + bt

When t = 6hours, c = ₦2800.00

2800 = k + 6b ------ (1)

When t = 10hours, c = ₦3600.00

$$3600 = k + 10b \text{ ----- (2)}$$

Solving equations (1) and (2)

$$\begin{array}{r} 3600 = k + 10b \\ -\ (2800 = k + 6b) \\ \hline 800 = 4b \end{array}$$

$$\frac{800}{4} = b \quad \text{i.e } b = 200$$

Substitute b value into (2)

$3600 = k + 10b$ will become

$$3600 = k + 10(200)$$

$$3600 = k + 2000$$

$$3600 - 2000 = k$$

$$1600 = k$$

The relationship is **c = 1600 + 200t**

Next, when t = 4hrs, c = ?

$$c = 1600 + 200(4)$$

$$= 1600 + 800$$

$$= ₦2400.00$$

2014/ 56 Neco

y varies partly as the square of x and partly as the inverse of the square root of x. Write down the expression for y if

y = 2 when x = 1and y = 6 when x = 4

A $y = x^2 + \frac{1}{x}$ B $y = x^2 + \frac{1}{\sqrt{x}}$

C $y = \frac{x^2}{31} + \frac{1}{31\sqrt{x}}$ D $y = \frac{10}{31}\left(x^2 + \frac{1}{\sqrt{x}}\right)$

E $y = \frac{10x^2}{31} + \frac{52}{31\sqrt{x}}$

Solution

$$y = x^2 + \frac{1}{\sqrt{x}}$$

Introducing constants a and b

$$y = ax^2 + b\frac{1}{\sqrt{x}}$$

If y = 2, when x = 1

$$2 = a(1^2) + \frac{b}{\sqrt{1}}$$

$$2 = a + b \text{ ------ (1)}$$

y = 6 when x = 4

$$6 = a(4^2) + \frac{b}{\sqrt{4}}$$

$$6 = 16a + \frac{b}{2}$$

Multiply through by 2 to clear fraction

$$12 = 32a + b \text{ ------- (2)}$$

Solving equations (1) and (2)

$$\begin{array}{r} 12 = 32a + b \\ -\ (2 = a + b) \\ \hline 10 = 31a \end{array}$$

$$a = {}^{10}/_{31}$$

Substitute a value into (1)

$2 = a + b$ will become

$$2 = \frac{10}{31} + b$$

$$2 - \frac{10}{31} = b \quad \text{thus } b = \frac{52}{31}$$

The relationship is $y = \frac{10x^2}{31} + \frac{52}{31\sqrt{x}}$ (E)

2008/1a Neco

A varies partly as B and partly as C. When A = 4, B = 2 and C = – 2 and when A = 3, B = 3 and C = 1.5.

What is the value of C when A = 0.5 and B = 2.7?

Solution

$$A = B + C$$

Introducing constant k and d

$$A = kB + dC$$

When A = 4, B = 2 and C = – 2

$$4 = k(2) + d(-2)$$

$$4 = 2k - 2d \text{ ------ (1)}$$

Next, when A = 3, B = 3 and C = 1.5

$$3 = 3k + 1.5d \text{ ------- (2)}$$

(1) × 3 and (2) × 2 then subtract

$$\begin{array}{r} 12 = 6k - 6d \\ -\ (6 = 6k + 3d) \\ \hline 6 = \quad -9d \end{array}$$

$$-{}^{6}/_{9} = d \quad \text{i.e } d = -{}^{2}/_{3}$$

Substitute d value into (1)

$4 = 2k - 2d$ will become

$$4 = 2k - 2(-{}^{2}/_{3})$$

$$4 = 2k + {}^{4}/_{3}$$

$$4 - {}^{4}/_{3} = 2k$$

$${}^{8}/_{3} = 2k$$

Multiply through by ${}^{1}/_{2}$

$${}^{8}/_{3} \times {}^{1}/_{2} = k$$

$${}^{4}/_{3} = k$$

The relationship is $A = \frac{4}{3}B - \frac{2}{3}C$

When A = 0.5, B = 2.7, C = ?

Substituting into the relation, we have

$$0.5 = \frac{4}{3} \times 2.7 - \frac{2}{3}C$$

$$0.5 = \frac{10.8}{3} - \frac{2c}{3}$$

$$0.5 = \frac{10.8 - 2c}{3}$$

$$0.5 \times 3 = 10.8 - 2c$$

$$1.5 = 10.8 - 2c$$

$$2c = 10.8 - 1.5$$

$$2c = 9.3$$

$$c = {}^{9.3}/_{2} = 4.65$$

2007/25 Neco

z is partly constant and partly varies as y, when y = 1, z = 2 and when y = 2, z = 1. State the equation connecting y and z

A z = 2 – y B z = 3 – 2y C z = 3 – y

D z = 2 + y E z = 2 + 3y

Solution

$$z = k + y$$

Introducing constant b to the variable y

$$z = k + by$$

When y = 1, z = 2, *then*

$$2 = k + b \text{ -------- (1)}$$

Next, when y = 2, z = 1

$$1 = k + 2b \text{ -------- (2)}$$

Solving equations (1) and (2)

$$\begin{array}{rl} 2 = & k + b \\ -(\ 1 = & k + 2b\ ) \\ \hline 1 = & -b \end{array}$$

Divide both sides by –1

$$\frac{1}{-1} = \frac{-b}{-1}$$

$$-1 = b$$

Substitute b value into (1)

$2 = k + b$ will become

$$2 = k - 1$$

$$2 + 1 = k \text{ i.e } k = 3$$

The relationship (connecting equation) is

$$z = 3 - y \qquad (\text{C})$$

2011/11a Neco

The resistance R to the motion of a car is partly constant and partly varies as the square of the speed V. At 80kmhr^{-1}, the resistance is 1060N and at 120kmhr^{-1}, its resistance is 1460N. What will be the resistance at 140km hr^{-1}?

Solution

$$R = k + V^2$$

Introducing constant to the variable V

$$R = k + bV^2$$

When $V = 80\text{km hr}^{-1}$, R = 1060N,

$$1060 = k + b(80^2)$$

$$1060 = k + 6400b \text{ ------- (1)}$$

Next, when $V = 120\text{km hr}^{-1}$, R = 1460N

$$1460 = k + b(120^2)$$

$$1460 = k + 14400b \text{ ------- (2)}$$

Solving equations (1) and (2)

$$\begin{array}{rl} 1460 = & k + 14400b \\ -(\ 1060 = & k + 6400b\ ) \\ \hline 400 = & 8000b \end{array}$$

$$\frac{400}{8000} = b \text{ i.e } b = \frac{1}{20}$$

Substitute b value into (1)

$1060 = k + 6400b$ will become

$$1060 = k + 6400 \times \frac{1}{20}$$

$$1060 = k + 320$$

$$1060 - 320 = k$$

$$740 = k$$

The relationship is $R = 740 + \frac{V^2}{20}$

Next, when $V = 140\text{kmhr}^{-1}$, R = ?

$$R = 740 + \frac{(140)^2}{20}$$

$$= 740 + \frac{140 \times 140}{20}$$

$$= 740 + 980 = 1720 \text{ N}$$

2006/4 Neco (Nov)

The quantity y is partly constant and partly varies inversely as the cube of x

(i) Given that x = 1, y = 5 and x = 2, y = – 1, find the relationship between x and y

(ii) Hence, find the value of y when x = 3

Solution

$$y = k + \frac{1}{x^3}$$

Introducing constant to the variable x

$$y = k + \frac{b}{x^3}$$

(i) When x = 1, y = 5,

$$5 = k + \frac{b}{1^3}$$

$$5 = k + b \text{ -------- (1)}$$

Next, when x = 2, y = – 1

$$-1 = k + \frac{b}{2^3}$$

$$-1 = k + \frac{b}{8}$$

Multiply through by 8 to clear fraction

$$-8 = 8k + b \text{ ----------(2)}$$

Solving equations (1) and (2)

$$\begin{array}{rl} 5 = & k + b \\ -(\ -8 = & 8k + b\ ) \\ \hline 13 = & -7k \end{array}$$

$$\frac{-13}{7} = k$$

Substitute k value into (1)

$5 = k + b$ will become

$$5 = \frac{-13}{7} + b$$

$$5 + \frac{13}{7} = b$$

$$\frac{48}{7} = b$$

The relationship is $y = \frac{-13}{7} + \frac{48}{7x^3}$

When x = 3, y = ?

$$y = \frac{-13}{7} + \frac{48}{7(3^3)}$$

$$= \frac{-13}{7} + \frac{48}{189}$$

$$= \frac{-351 + 48}{189} = \frac{-303}{189} \text{ i.e } -1.603$$

2005/37 Neco ***Counter example***

x varies partly as y and partly as the square of y. When y = 2, x = 5 and when y = 5, x = 57.5, find x when y = 4

A 57.5 B 50.0 C 40.0 D 34.0 E 25.0

Solution

$$x = y + y^2$$

Introducing constant to the variable y

$$x = ay + by^2$$

When y = 2, x = 5

$$5 = 2a + 4b \text{ ------- (1)}$$

When y = 5, x = 57.5

$$57.5 = 5a + 25b$$

Multiply through by 2 to clear decimal

$$115 = 10a + 50b \text{ -------- (2)}$$

Solving the equations: (1) × 5 and subtract

$$\begin{array}{rl} 115 = & 10a + 50b \\ -(\ 25 = & 10a + 20b\ ) \\ \hline 90 = & 30b \end{array}$$

$$\frac{90}{30} = \frac{30b}{30}$$

$3 = b$

Substitute b value into (1)

$5 = 2a + 4b$ will become

$5 = 2a + 4(3)$

$5 = 2a + 12$

$5 - 12 = 2a$

$-7 = 2a$

$-{}^7/_2 = a$

The relationship is $x = \frac{-7}{2}y + 3y^2$

When y = 4, x = ?

$$x = \frac{-7}{2} \times 4 + 3\times 4^2$$

$= -14 + 48$

$= 34$ (D)

2008/20 NABTEB (Nov) Exercise 12.18

If a and b are constant and P varies partly as Q and partly as the reciprocal of the square root of Q, which of these statements is correct ?

A $P = a + \frac{b}{Q}$ B $P = a + \frac{b}{Q^2}$

C $P = aQ + \frac{b}{\sqrt{Q}}$ D $P = aQ + \frac{b}{Q^2}$

2008/31 Exercise 12.19

If $x \ \alpha \left(45 + \frac{1}{2}y\right)$ which of the following is true?

A x varies directly as y

B x varies inversely as y

C x is partly constant and partly varies as y

D x varies jointly as 45 and directly as y

2009/4 Neco (Dec) Exercise 12.20

y is partly constant and partly varies as the square of x. when y = 16, x = 2 and when y = 24, x = 3. Find :

(i) The relationship between x and y

(ii) y when x = 4

2006/44 Neco Exercise 12.21

y is partly constant and partly varies as x. when x = 1, y = 7 and when x = 2, y = 11. Find the equation connecting x and y

A y = 3x + 2 B y = 2x – 3 C y = 4x + 3

D y = 2x + 4 E y = 3x + 4

2015/9a Neco Exercise 12.22

W is partly constant and partly varies inversely as the square of t. When W = 24, t = 4 and When W = 18, t = 2

(i) Determine the law connecting W and t

(ii) Find t when W = – 46

2015/6b Exercise 12.23

The cost (c) of producing n bricks is the sum of a fixed amount, h and variable amount y, where y varies directly as n. If it costs GH₡ 950.00 to produce 600 bricks and GH₡ 1,030.00 to produce 1000 bricks,

(i) find the relationship between c, h and n;

(ii) calculate the cost of producing 500 bricks

1975/7b (Nov) Exercise 12.24

The cost of feeding a number of students during a certain period is partly constant and partly varies directly as the number of students. The cost of feeding 75 students during a certain period is Le875.00 and the cost of feeding 100 students during the same period of time is Le1000.00. Find the cost of feeding 220 students over the same period of time

1977/7b Exercise 12.25

The resistance R to the motion of a car is partly constant and partly proportional to the square of the speed v. When the speed is 30km/h, the resistance is 190 newtons and When the speed is 50km/h, the resistance is 350 newtons . Find for what speed the resistance is 302.5 newtons

CHAPTER THIRTEEN

Trigonometry

Trigonometry is simply the study of triangles in relation to their sides and angles.

For any right angled triangle of the form below.

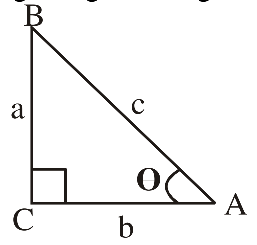


Fig I

Where θ is facing, is opposite (BC)

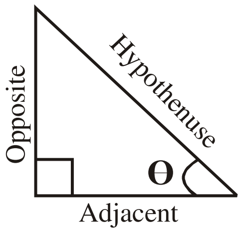
Where right angle is facing, is the **Hypotenuse** (AB)

While the remaining side is the **Adjacent** (AC)

Also, where the capital letter A is facing is the small letter a; same for B and C

Diagrammatically, we have

Fig II



The three basic trig. ratios are Sine, Cosine and Tangent written in short form Sin, Cos and Tan respectively.

We will apply SOH CAH TOA in Fig I above to define them as :

Applying **SOH** in defining Sin θ

Here **S** – sine, **O** – opposite, **H** – hypotenuse

$$\text{Sin}\ \theta = \frac{\text{Opposite}}{\text{Hypotenuse}} = \frac{a}{c}$$

Applying **CAH** in defining Cos θ

Here **C** – cosine, **A** – adjacent, **H** – hypotenuse

$$\text{Cos}\ \theta = \frac{\text{Adjacent}}{\text{Hypotenuse}} = \frac{b}{c}$$

Similarly we apply **TOA** to define Tan θ

Here **T** – tangent , **O** – opposite, **A** – adjacent,

$$\text{Tan}\ \theta = \frac{\text{Opposite}}{\text{Adjacent}} = \frac{a}{b}$$

The three basic trig ratios have reciprocals namely

Cosec (cosecant) $= \dfrac{1}{\text{Sine}}$

Sec (secant) $= \dfrac{1}{\text{Cosine}}$

Cot (cotangent) $= \dfrac{1}{\text{Tangent}}$

Deductively, we have that

$$\text{Cosec } \theta = \frac{1}{\text{Sine } \theta} = \frac{\text{Hypo}}{\text{Opp}} = \frac{c}{a}$$

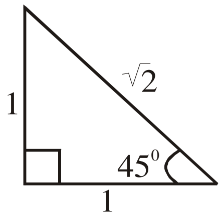
$$\text{Sec } \theta = \frac{1}{\text{Cos}\theta} = \frac{\text{Hypo}}{\text{Adj}} = \frac{c}{b}$$

and $\text{Cot } \theta = \dfrac{1}{\text{Tan}\theta} = \dfrac{\text{Adj}}{\text{Opp}} = \dfrac{b}{a}$

Special angles trigonometric ratios

Angle of 45^0

To achieve the desired goal here; candidates must allow for the mathematical assumption that follows. A right – angled triangle whose θ is 45^0 with opposite and adjacent sides of unit 1. Then by Pythagoras theorem the hypotenuse is $\sqrt{2}$



Applying SOH CAH TOA

$$\text{Sin}45^0 = \frac{\text{Opp}}{\text{Hyp}} = \frac{1}{\sqrt{2}} \quad \text{rationalizing, we have}$$

$$= \frac{\sqrt{2}}{2}$$

$$\text{Cos } 45^0 = \frac{\text{Adj}}{\text{Hyp}} = \frac{1}{\sqrt{2}} \quad \text{rationalizing, we have}$$

$$= \frac{\sqrt{2}}{2}$$

$$\text{Tan } 45^0 = \frac{\text{Opp}}{\text{Adj}} = \frac{1}{1} = 1$$

Angles of 30^0 and 60^0

Here the mathematical assumption is that of an equilateral triangle of sides 2 unit. Then on bisecting the triangle from the top angle to the base. We have 30^0 each at the top and 1 unit each at the base of the resulting two right – angled triangles.

Then by Pythagoras theorem the bisector (height) is $\sqrt{3}$ represented below by **Fig. 1** and **Fig. II**

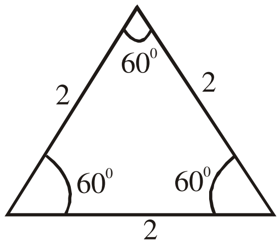
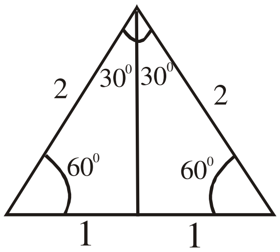


Fig I



FigII

In practice, we adopt one part of Fig II i.e.

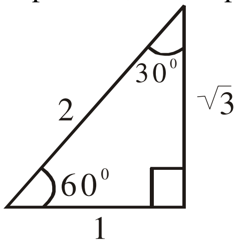


Fig III

Applying SOH CAH TOA to fig III

$\text{Sin } 30^0 = {}^1/_2$

$$\text{Cos } 30^0 = \frac{\sqrt{3}}{2}$$

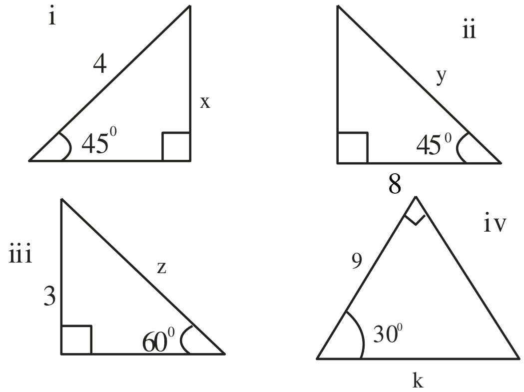
$$\text{Tan } 30^0 = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3} \quad \text{by rationalization.}$$

Similarly,

$$\text{Sin } 60^0 = \frac{\sqrt{3}}{2} \qquad \text{Cos } 60^0 = \frac{1}{2}$$

$$\text{Tan } 60^0 = \frac{\sqrt{3}}{1} = \sqrt{3}$$

Eg 1 Find the value of the unknown sides in the diagrams below:



Solution

(i) x – Opp; 4 – Hypo (OH $\Rightarrow$ SOH)

$$\therefore \text{Sin } 45^0 = \frac{\text{Opp}}{\text{Hypo}} = \frac{x}{4}$$

Substituting for Sin 45^0 from special trig. ratio

$$\frac{1}{\sqrt{2}} = \frac{x}{4}$$

$$\frac{4}{\sqrt{2}} = x \quad \text{Rationalizing, } \frac{4}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = x$$

$$\frac{4\sqrt{2}}{2} = x \quad \text{i.e } x = 2\sqrt{2}$$

(ii) 8 – Adj; y – Hypo (AH $\Rightarrow$ CAH)

$\text{Cos } 45^0 = \frac{\text{Adj}}{\text{Hyp}} = \frac{8}{y}$

Substituting for Cos45⁰ from special trig. ratio

$\frac{1}{\sqrt{2}} = \frac{8}{y}$

$y = 8\sqrt{2}$

(iii) 3 – opp; z – Hypo (OH $\Rightarrow$ SOH)

$\text{Sin } 60^0 = \frac{\text{opp}}{\text{Hypo}} = \frac{3}{z}$

Substituting for Sin 60^0 from special trig. ratio

$\frac{\sqrt{3}}{2} = \frac{3}{z}$

$z\sqrt{3} = 6$

$z = \frac{6}{\sqrt{3}}$ rationalising , we have $= \frac{6 \times \sqrt{3}}{\sqrt{3} \times \sqrt{3}}$

$= \frac{6\sqrt{3}}{3} = 2\sqrt{3}$

(iv) 9 – Adj; k – Hypo (AH $\Rightarrow$ CAH)

$\text{Cos } 30^0 = \frac{\text{Adj}}{\text{Hypo}} = \frac{9}{k}$

Substituting for Cos30^0 from special trig. ratio

$\frac{\sqrt{3}}{2} = \frac{9}{k}$

$k\sqrt{3} = 18$

$k = \frac{18}{\sqrt{3}}$ Rationalising, $k = \frac{18\sqrt{3}}{3} = 6\sqrt{3}$

E.g 2. Without using tables evaluate

$$\frac{1 + \tan 60^0 \tan 30^0}{\tan 60^0 + \tan 30^0}$$

Solution

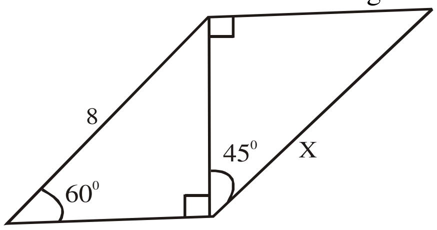
Substituting for the various special trig. ratio

$= \frac{1 + \sqrt{3}\,(1/\sqrt{3})}{\sqrt{3} + 1/\sqrt{3}} = \frac{2}{\sqrt{3} + 1/\sqrt{3}}$

We simplify as in ordinary fractions:

$= \frac{\mathbf{2}}{\frac{\sqrt{3} \times \sqrt{3} + 1}{\sqrt{3}}} = \frac{\mathbf{2}}{\frac{3+1}{\sqrt{3}}} = \frac{\mathbf{2}}{\frac{4}{\sqrt{3}}} = \frac{2\sqrt{3}}{4} = \frac{\sqrt{3}}{2}$

E.g. 3 Find the value of x in the diagram below:



Solution

Candidates can only find x if side of the sub-triangle of 45^0 is known. The other triangle of 60^0 can be used if and only if we find the value of the side opposite 60^0 using the available data.

In the right - angled triangle 60^0

Let the Opp be y ; Hyp – 8 (OH $\Rightarrow$ SOH) Thus,

$\text{Sin } 60^0 = \frac{y}{8}$

Substituting for Sin 60^0 from special trig. ratio

$\frac{\sqrt{3}}{2} = \frac{y}{8}$

$y = \frac{8\sqrt{3}}{2} = 4\sqrt{3}$

Then in right – angled Δ 45^0

$y = 4\sqrt{3}$ – Adj; x – Hypo (AH $\Rightarrow$ CAH) Thus,

$\text{Cos } 45^0 = \frac{4\sqrt{3}}{x}$

Substituting for Cos45^0 from special trig. ratio

$\frac{1}{\sqrt{2}} = \frac{4\sqrt{3}}{x}$

$x = 4\sqrt{3} \times \sqrt{2} = 4\sqrt{6}$

Special angles trig ratios substitution Problems

2009/ 17 NABTEB (Nov)

Determine, without the use of table, tan 60^0

A $\frac{1}{\sqrt{3}}$ B $\frac{2}{\sqrt{3}}$ C $\sqrt{3}$ D $\frac{\sqrt{2}}{3}$

Solution

By special trig ratios

$\tan 60^0 = \sqrt{3}$ (C)

2012/46 Neco

Express $\frac{1+\cos 30^0}{1-\sin 30^0}$ in surd form

A $\sqrt{3}-2$ B $2+\sqrt{3}$ C $2\sqrt{2}+\sqrt{3}$
D $2\sqrt{3}+\sqrt{2}$ E $2\sqrt{3}+2\sqrt{2}$

Solution

Substituting for the values of the special trig ratios

$$\frac{1+\cos 30^0}{1-\sin 30^0} = \left(1+\frac{\sqrt{3}}{2}\right) \div \left(1-\frac{1}{2}\right)$$

$$= \left(\frac{2+\sqrt{3}}{2}\right) \div \left(\frac{1}{2}\right)$$

$$= \frac{2+\sqrt{3}}{2} \times \frac{2}{1} = 2+\sqrt{3} \quad \text{(B)}$$

2008/49 Neco (Dec)

Evaluate without using table $\frac{1 - \sin 45^0}{1 + \cos 30^0}$

A $4-2\sqrt{3}-4\sqrt{2}-\sqrt{6}$ B $4+2\sqrt{3}-4\sqrt{2}+\sqrt{6}$

C $4+2\sqrt{3}-3\sqrt{2}+\sqrt{6}$ D $4-2\sqrt{3}-2\sqrt{2}-\sqrt{6}$

E $4+2\sqrt{3}-5\sqrt{2}+\sqrt{6}$

Solution

Substituting for the value of the special trig ratio

$$\frac{1 - \sin 45^0}{1 + \cos 30^0} = \left(1-\frac{1}{\sqrt{2}}\right)\div\left(1+\frac{\sqrt{3}}{2}\right)$$

$$= \left(\frac{\sqrt{2}-1}{\sqrt{2}}\right)\div\left(\frac{2+\sqrt{3}}{2}\right)$$

Changing from division to multiplication

$$= \left(\frac{\sqrt{2}-1}{\sqrt{2}}\right)\times\left(\frac{2}{2+\sqrt{3}}\right)$$

$$= \frac{2\ (\sqrt{2}-1)}{\sqrt{2}\ (2+\sqrt{3})}$$

$$= \frac{2\sqrt{2}-2}{2\sqrt{2}+\sqrt{6}}$$

Rationalizing by conjugate of the denominator surd

$$= \frac{(2\sqrt{2}-2)(2\sqrt{2}-\sqrt{6})}{2\sqrt{2}+\sqrt{6}(2\sqrt{2}-\sqrt{6})}$$

$$= \frac{2\sqrt{2}\ (2\sqrt{2}-2)-\sqrt{6}(2\sqrt{2}-2)}{2\sqrt{2}\ (2\sqrt{2}+\sqrt{6})-\sqrt{6}\ (2\sqrt{2}+\sqrt{6})}$$

$$= \frac{4\times 2 - 4\sqrt{2} - 2\sqrt{12} + 2\sqrt{6}}{4\times 2 + 2\sqrt{12} - 2\sqrt{12} - 6}$$

Note that $\sqrt{12} = \sqrt{4}\times\sqrt{3}$ i.e $2\sqrt{3}$

$$= \frac{8-4\sqrt{2}-2\times 2\sqrt{3}+2\sqrt{6}}{8-6}$$

$$= \frac{8-4\sqrt{2}-4\sqrt{3}+2\sqrt{6}}{2}$$

$$= 4-2\sqrt{2}-2\sqrt{3}+\sqrt{6}$$

2005/1b

Without using mathematical tables or calculator, evaluate, leaving your answer in surds

$$\frac{\sin 45^0 + \cos 60^0}{\tan 30^0}$$

Solution

Substituting for the values of the special trig ratios

$$\frac{\sin 45^0 + \cos 60^0}{\tan 30^0} = \left(\frac{1}{\sqrt{2}}+\frac{1}{2}\right)\div\frac{1}{\sqrt{3}}$$

$$= \left(\frac{2+\sqrt{2}}{2\sqrt{2}}\right)\times\frac{\sqrt{3}}{1}$$

$$= \frac{\sqrt{3}(2+\sqrt{2})}{2\sqrt{2}}$$

$$= \frac{2\sqrt{3}+\sqrt{6}}{2\sqrt{2}}$$

Rationalizing : $= \frac{(2\sqrt{3}+\sqrt{6})}{2\sqrt{2}}\times\frac{2\sqrt{2}}{2\sqrt{2}}$

$$= \frac{4\sqrt{6}+2\sqrt{12}}{8}$$

$$= \frac{4\sqrt{6}+2\times 2\sqrt{3}}{8}$$

$$= \frac{4\sqrt{6}+4\sqrt{3}}{8} = \frac{1}{2}(\sqrt{6}+\sqrt{3})$$

2009 /25

Given that $\sin 60^0 = \frac{\sqrt{3}}{2}$ and $\cos 60^0 = \frac{1}{2}$

Evaluate: $\frac{1-\sin 60^0}{1+\cos 60^0}$

A $\frac{2+\sqrt{3}}{3}$ B $\frac{1-\sqrt{3}}{3}$ C $\frac{1+\sqrt{3}}{3}$ D $\frac{2-\sqrt{3}}{3}$

Solution

$$\frac{1-\sin 60^0}{1+\cos 60^0} = \left(1-\frac{\sqrt{3}}{2}\right)\div\left(1+\frac{1}{2}\right)$$

$$= \left(\frac{2-\sqrt{3}}{2}\right)\div\left(\frac{2+1}{2}\right)$$

$$= \left(\frac{2-\sqrt{3}}{2}\right)\div\left(\frac{3}{2}\right)$$

$$= \left(\frac{2-\sqrt{3}}{2}\right)\times\frac{2}{3}$$

$$= \frac{2-\sqrt{3}}{3} \quad \text{(D)}$$

2005/4b

Given that $x = \cos 30^0$ and $y = \sin 30^0$, evaluate without using a mathematical table or calculator

$$\frac{x^2+y^2}{y^2-x^2}$$

Solution

$$\frac{x^2+y^2}{y^2-x^2} = \frac{(\cos 30^0)^2 + (\sin 30^0)^2}{(\sin 30^0)^2 - (\cos 30^0)^2}$$

Substituting for the values of the special trig ratios

$$= \frac{\left(\frac{\sqrt{3}}{2}\right)^2+\left(\frac{1}{2}\right)^2}{\left(\frac{1}{2}\right)^2-\left(\frac{\sqrt{3}}{2}\right)^2}$$

$$= \left(\frac{3}{4}+\frac{1}{4}\right)\div\left(\frac{1}{4}-\frac{3}{4}\right)$$

$$= \frac{3+1}{4} \div \frac{1-3}{4}$$

$$= \frac{4}{4} \div \left(\frac{-2}{4}\right)$$

$$= 1\div\left(\frac{-1}{2}\right)$$

$$= 1\times\left(\frac{-2}{1}\right)$$

$$= -2$$

2013/45 Neco Exercise 13.1

What is the value of $\sin 45^0 \sin 30^0 - \cos 45^0 \cos 30^0$ leaving the answer in surd form?

A $\frac{\sqrt{3}-\sqrt{2}}{4}$ B $\frac{\sqrt{2}-1}{2}$ C $\frac{\sqrt{2}+\sqrt{6}}{4}$

D $\frac{\sqrt{2}-\sqrt{6}}{4}$ E $\frac{\sqrt{6}-\sqrt{2}}{4}$

2010/33 NABTEB (Nov) Exercise 13.2

The value of sin 45^0 + cos 45^0 in surd form without rationalizing is

A $\sqrt{2}$ B $\frac{1}{\sqrt{2}}$ C 2 D $\frac{2}{\sqrt{2}}$

2014/54 Neco (Nov) Exercise 13.3

If $x = 30^0$ and $y = 60^0$, evaluate without using tables or calculator $\frac{\sin x \ + \ \cos x}{\sin y \ + \ \cos y}$

A 3 B 2 C 1 D –1 E – 2

2015/10a Exercise 13.4

Without using Mathematical tables or calculators, simplify:

$$\frac{2\tan 60^0 \ + \ \cos 30^0}{\sin 60^0}$$

2007/49 UME Exercise 13.5

Find the value of $\frac{\tan 60^0 \ - \ \tan 30^0}{\tan 60^0 \ + \ \tan 30^0}$

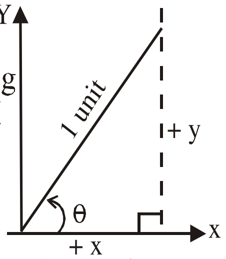
A $\frac{4}{\sqrt{3}}$ B $\frac{2}{\sqrt{3}}$ C 1 D $\frac{1}{2}$

Trigonometric ratios of angles between 0^0 and 360^0

These type of trig ratios does not carry the condition "acute angle" or $0^0 \leq \theta \leq 90^0$. Apart from our knowledge of special angles of 45^0, 30^0 and 60^0 ; we are also required to know the required quadrant we are dealing with as discussed below:

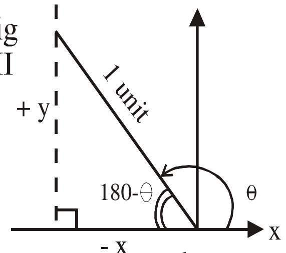
Diagrams of trigonometric ratio of angles between 0^0 and 360^0.

Fig I

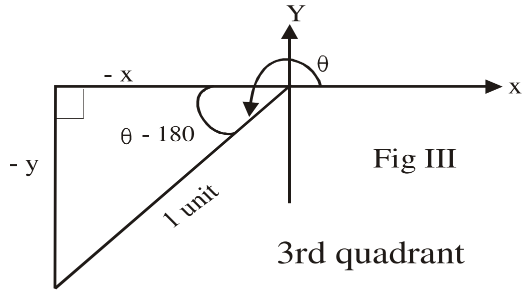


1^{st} quadrant

Fig II

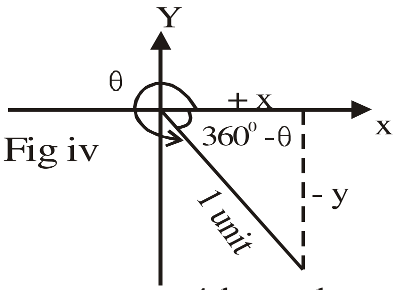


2^{nd} quadrant



3rd quadrant

Fig iv



4th quadrant

The x and y axes divide the plane into four quadrants as shown in Fig. I to Fig.IV. Any angle measured positively in an anticlockwise direction will be located in any of these quadrants.

Fig. I (1st quadrant between 0^0 and 90^0)

The resulting right – angled triangle has both x and y positive, the hypotenuse that is 1 unit plays little or no role. Here Sin θ, Cos θ and Tan θ are all positive.

Fig. II (2nd Quadrant between 90^0 and 180^0).

The right – angled triangle here has negative x, which is the adjacent side and a positive y - which is the opposite side. Hence, only

Sin θ = Sin (180^0 - θ)

$= \frac{+y}{1} = +y$ is positive

Others are negative in 2nd Quadrant i.e Cos θ and Tan θ

Fig III (3rd Quadrant between 180^0 and 270^0).

The right - angled triangle formed here has both x and y negative.

Only Tan θ = Tan (θ - 180^0)

$= \frac{-y}{-x}$

$= \frac{y}{x}$ is positive

Others are negative 3rd quadrant i.e Cos θ and Sin θ

Fig. IV (4th Quadrant between 270^0 & 360^0)

Our right – angled triangle here has negative y which is the opposite side but a positive x which is the adjacent side. Hence only

Cos θ = Cos (360 - θ)

$= \frac{+x}{1}$

= x is positive

Others are negative in 4th quadrant i.e Tan θ and Sin θ

Summarily

We use "Acts" to recall the mentioned properties of trig.ratio between θ ^{0} and 360^0 where

A is for **all** positive

C – **Cosine** only positive

T – **Tangent** only positive

S – **Sine** only positive

Written

S 2nd Quad	A 1st Quad
T 3rd Quad	C 4th Quad

1993/39 UME

If $\sin\theta = \cos\theta$, find θ between 0^0 and 360^0

A. $45^0, 225^0$ B. $125^0 \cdot 315^0$

C. $45^0 , 225^0$ D. $135^0, 225^0$

Solution

$\sin\theta = \cos\theta$

Then, divide both sides by $\cos\theta$

$\frac{\sin\theta}{\cos\theta} = \frac{\cos\theta}{\cos\theta}$

$\tan\theta = 1 = \frac{1}{1}$ and $\tan\theta$ is positive

Since we have $\frac{1}{1}$, then it must be 45^0 special trig. ratios

If $\tan\theta = {}^1/_1$ then $\theta = 45^0$ Thus 45^0 in the first quadrant

Also tan is positive in the 3^{rd} now with formula $\theta - 180^0$, then

$\theta - 180^0 = 45$

$\theta = 45 + 180^0$

$= 225^0$

$\theta = 45^0, 225^0$ (A)

2002/34 PCE

The tangent of 135^0 has the value

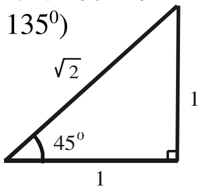
A. – 1 B. – $^1/_2$ C. $^1/_2$ D. 1

Solution

Tan 135^0; tan is negative in the second quadrant where 135 falls into. The formula here is $180 - \theta$

Thus, $\tan 135^0 = \tan(180 - 135^0)$

$= -\tan 45^0$



From special angles 45^0

$\tan 45^0 = \frac{1}{1}$ i.e 1

Hence $\tan 135^0 = -\tan 45^0$

$= -1$ (A)

1990/37 PCE

Evaluate $\frac{\cos 60^0 + \sin 30^0}{\sin 150^0}$

A. 2 B. $^1/_2$ C. – 2 D. – $^1/_2$

Solution

Ordinarily, on seeing $\cos 60^0$ and $\sin 30^0$ the author would have treated this question under acute angles trig ratios but for the $\sin 150^0$ which is in a quadrant beyond 90^0.

S	A
T	C

Sin 150^0 is the 2nd quadrant where sine is positive with the general formula $180^0 - \theta$. Thus

$\sin 150^0 = +\sin(180^0 - 150^0)$

$= \sin 30^0$

We can now conveniently say;

$\frac{\cos 60^0 + \sin 30^0}{\sin 150} = \frac{\cos 60^0 + \sin 30^0}{\sin 30^0}$

$= \frac{\frac{1}{2} + \frac{1}{2}}{\frac{1}{2}}$

$= 1 \div \frac{1}{2} = 1 \times \frac{2}{1}$

$= 2$ (A)

1994/3 (Nov) fm

If $\sin\theta = -\frac{1}{2}$ and $0^0 < \theta < 270^0$ find θ

A 30^0 B 60^0 C 120^0 D 150^0 E 210^0

Solution

$\sin\theta = -0.5$

$\theta = \sin^{-1}(-0.5)$

But $\sin^{-1} 0.5$ is 30^0 and $\sin\theta$ is negative in 3^{rd} and 4^{th} quadrants but our restriction here is 3^{rd} with formula $\theta - 180^0$

Thus, $\theta - 180^0 = 30^0$

$\theta = 30 + 180$

$= 210^0$ (E)

1995/7 (Nov) fm

Find the values of x, for which $2\cos x + 1 = 0$, $0^0 \le x \le 360^0$

A $0^0, 180^0$ B $90^0, 270^0$ C $-60^0, 60^0$

D $60^0, 300^0$ E $120^0, 240^0$

Solution

$2\cos x + 1 = 0$

$2\cos x = -1$

$\cos x = -\frac{1}{2}$

$x = \cos^{-1}(-0.5)$

But $\cos^{-1}(0.5)$ is 60^0 and that cosine is negative in the 2^{nd} and 3^{rd} quadrants with formula

$180 - \theta$ and $\theta - 180$

Thus, $x = 180 - 60$ or $\theta - 180 = 60$

$= 120^0$ or 240^0 (E)

2005/18 fm

If $\sin\Theta = \frac{1}{2}$, $0^0 < \Theta \le 450^0$, find the possible values of Θ

A $30^0, 210^0, 390^0$ B $60^0, 330^0, 390^0$

C $60^0, 210^0, 450^0$ D $30^0, 150^0, 390^0$

Solution

$\sin\Theta = {}^1/_2$

$\Theta = \sin^{-1}(0.5)$

$= 30^0$ or $(180 - 30^0)$ Sine is +ve in 2^{nd} quad.

$= 30^0$ or 150^0

$= 30^0$ or 150^0 or $30 + 360^0$ i.e Plus complete cycle

$= 30^0$ or 150^0 or 390^0 (D)

2012 Neco

Without using tables, find the value of $\sin 120^0$

A $\frac{\sqrt{3}}{4}$ B $\frac{\sqrt{3}}{2}$ C $\frac{1}{2}$ D $\sqrt{3}$ E 1

Solution

Sin 120^0 is in the 2^{nd} quadrant with the formula $180 - \theta$ i.e $180 - 120 = 60^0$ and sine is positive there.

Thus $\sin 120^0 = \sin 60^0$

By special trig ratio

$= \frac{\sqrt{3}}{2}$ (B)

2014/34 NABTEB Exercise 13.6

Which of these is equivalent to $\tan 300^0$?

A $\tan 60^0$ B $-\tan 60^0$ C $\sin 60^0$ D $-\cos 60^0$

1993/43 PCE Exercise 13.7

Evaluate $\frac{\tan 45^0 + \cos 60^0}{\sin 150^0}$

A. – 3 B. 3 C. $-(\sqrt{2} + \sqrt{1})$ D $\sqrt{2} + 1$

2005/15 UME Exercise 13.8

If $\sin\theta = -\frac{1}{2}$ for $0^0<\theta < 360^0$, the value of θ is

A 210^0 and 330^0 B 150^0 and 330^0
C 150^0 and 210^0 D 30^0 and 150^0

2012/35 UTME Exercise 13.9

If angle θ is 135^0, evaluate $\cos\theta$

A $\frac{1}{2}$ B $-\frac{\sqrt{2}}{2}$ C $\frac{\sqrt{2}}{2}$ D $-\frac{1}{2}$

2004/34 PCE Exercise 13.10

The value of $\tan 315^0$ is

A -1 B 0 C 1 D 2

Trigonometry of Negative angles (i.e $-\theta$)

They are treated by adding 360^0 to it or its multiples as the case demands. We use the resulting angle to identify the quadrant it belongs.

1994/38 UME

What is the value of $\sin(-690^0)$

A. $\frac{\sqrt{3}}{2}$ B. $-\frac{\sqrt{3}}{2}$ C. $-\frac{1}{2}$ D. $\frac{1}{2}$

Solution

$-690^0 + 360^0 = -330^0$

So $\sin(-690^0) = \sin(-330^0)$

and 330^0 falls into the 4th quadrant where sin is negative with the general formula $360^0 - \theta$. Thus

$\sin(-690^0) = \sin(-330^0)$

$= -\sin(360^0 - 330^0)$

$= -\sin 30^0$ From special trig ratios

$= -\frac{1}{2}$ (C)

2001/35 PCE

What is the value of $\cos(-840^0)$

A. $-\frac{\sqrt{3}}{2}$ B. $-\frac{1}{2}$ C. $\frac{1}{2}$ D. $\frac{\sqrt{3}}{2}$

Solution

$-840^0 + 720^0 = -120^0$ (720 is 2×360^0)

So $\cos(-840^0) = \cos(-120^0)$

and 120 falls into 2nd quadrant where cosine is negative with quadrant formula$180 - \theta$. thus

$\cos(-840^0) = \cos(-120^0)$

$= -\cos(180^0 - 120^0)$

$= -\cos 60^0$

$= -\frac{1}{2}$ (B)

2012/5 f/maths

Given that $\sin x = -\frac{\sqrt{3}}{2}$ and $\cos x > 0$, find the value of x

A 300^0 B 240^0 C 120^0 D 60^0

Solution

$$\sin x = -\frac{\sqrt{3}}{2}$$

$$x = -\sin^{-1}\frac{\sqrt{3}}{2}$$

$$x = -60^0$$

Sine is negative in the 3rd and 4th quadrants

But the condition $\cos x > 0$ restricts us to 4th quadrant

With formula $360^0 - \theta$

$x = 360^0 - 60^0$

$= 300^0$ (A)

2005/45 Neco (Dec) Exercise 13.11

If $\cos 30^0 = 0.866$, find the value of the angles whose cosine is -0.866

A 120^0 and 330^0 B 120^0 and 210^0 C 150^0 and 210^0
D 150^0 and 330^0 E 210^0 and 330^0

1978/44 fm Exercise 13.12

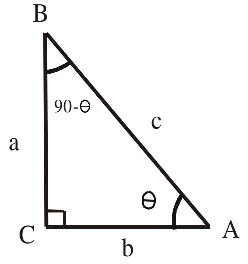
Evaluate without using tables $\sin(-1290^0)$

A. $-\frac{\sqrt{3}}{2}$ B. $\frac{\sqrt{3}}{2}$ C. $\frac{\sqrt{2}}{2}$ D. 1 E. $-½$

Identities

Complementary angles

If θ is an acute angle, then the angles θ and $90^0 - \theta$ are complementary which appear together in the right angled triangle below;



1 $\sin\theta = {}^a/_c = \cos(90 - \theta)$
$\cos\theta = {}^b/_c = \sin(90 - \theta)$
$\tan\theta = {}^a/_b = \cot(90 - \theta)$

2. $\tan\theta = {}^a/_b = \frac{\sin\theta}{\cos\theta}$

Reason $\frac{\sin\theta}{\cos\theta} = \frac{a}{c} \div \frac{b}{c}$

$= \frac{a}{c} \times \frac{c}{b} = \frac{a}{b}$

3. $\sin^2\theta + \cos^2\theta = 1$

<u>Proof</u>

By Pythagoras rule

$a^2 + b^2 = c^2$

Divide through by c^2

$$\frac{a^2}{c^2} + \frac{b^2}{c^2} = \frac{c^2}{c^2}$$

But ${}^a/_c = \sin\theta$ hence $a^2/c^2 = \sin^2\theta$

Similarly $b^2/c^2 = \cos^2\theta$ but $c^2/c^2 = 1$

Then substituting

$\sin^2\theta + \cos^2\theta = 1$ or

$\cos^2\theta + \sin^2\theta = 1$

Since addition is commutative i.e. 2 + 3 = 3 + 2

From identity 3 we can derive others

i.e. $\cos^2\theta + \sin^2\theta = 1$

Divide through by $\cos^2\theta$

$$\frac{\cos^2\theta}{\cos^2\theta} + \frac{\sin^2\theta}{\cos^2\theta} = \frac{1}{\cos^2\theta}$$

4. $1 + \tan^2\theta = \sec^2\theta$

Similarly dividing identity 3 by $\sin^2\theta$

$$\frac{\cos^2\theta}{\sin^2\theta} + \frac{\sin^2\theta}{\sin^2\theta} = \frac{1}{\sin^2\theta}$$

5. $\cot^2\theta + 1 = \operatorname{cosec}^2\theta$

Complementary angles problems

2002/19 (Nov) fm

Given that $\sin(90^0 - 5\theta) = \cos(180^0 - \theta)$ find the value of θ.

A 15^0 B 22.5^0 C 30^0 D 45^0

Solution

$\sin(90 - 5\theta) = \cos(180 - \theta)$

Under complementary angle

$\sin(90 - 5\theta) = \sin[90 - (180 - \theta)]$

$\Rightarrow 90 - 5\theta = 90 - 180 + \theta$

$90 - 5\theta = \theta - 90$

$90 + 90 = \theta + 5\theta$

$180 = 6\theta$ Thus, $\theta = 30^0$ (C)

2005/3a Neco fm

If $\sin(x - \alpha) = \cos(x + \alpha)$, prove that $\tan x = 1$

Solution

Under complementary angle

$\sin(x - \alpha) = \cos(x + \alpha)$, implies

$\sin(x - \alpha) = \sin[90 - (x + \alpha)]$

Thus $x - \alpha = 90 - x - \alpha$

$x + x - \alpha + \alpha = 90$

$2x = 90$

$x = 45^0$

Hence $\tan x \Rightarrow \tan 45^0 = {}^1/_1 = 1$ QED

1981/6 UME

If $\sin x^0 = \frac{a}{b}$, what is $\sin(90 - x)^0$?

A. $\frac{a}{b}$ B. $1 - \frac{a}{b}$ C. $\frac{\sqrt{b^2 - a^2}}{b}$

D. $\frac{\sqrt{a^2 - b^2}}{b}$ E. $\sqrt{b^2 - a^2}$

Solution

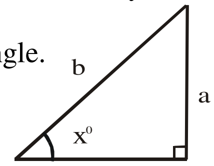
Recall that, under complementary angles

$\cos\theta = \sin(90 - \theta)$. So that question is indirectly asking us to find $\cos x$.

First, let us complete the trig ratio triangle.

Applying SOH

$\sin x^0 = \frac{a}{b}$ implies



By Pythagoras rule the adjacent is equal to $\sqrt{b^2 - a^2}$

Hence $\sin(90 - x)^0 \Rightarrow \cos x^0 = \frac{\sqrt{b^2 - a^2}}{b}$ (C)

2010/9 fm Exercise 13.13

If $\sin x = -\sin 70^0$, $0^0 < x < 360^0$, determine the two possible values of x

A $110^0, 250^0$ B $110^0, 290^0$ C $200^0, 250^0$ D $250^0, 290^0$

2000/7a fm Exercise 13.14

Solve the equation $\cos 7\theta = \sin 5\theta$ Where $0 < \theta < 90^0$

2007/46 Neco Exercise 13.15

If $\sin(x + 20^0) = \cos(x - 10^0)$, find the value of x

A 90^0 B 60^0 C 45^0 D 40^0 E 30^0

Problems on identities I

1999/1b (Nov) fm

Simplify $\sqrt{\frac{1 - \cos^2\theta}{\sec^2\theta - 1}}$

Solution

By trig identities

$$\sqrt{\frac{1 - \cos^2\theta}{\sec^2\theta - 1}} = \sqrt{\frac{\sin^2\theta}{\tan^2\theta}}$$

$$= \frac{\sin\theta}{\tan\theta}$$

$$= \sin\theta \div \frac{\sin\theta}{\cos\theta}$$

$$= \sin\theta \times \frac{\cos\theta}{\sin\theta}$$

$$= \cos\theta$$

2003/20 fm

Given that $\sin\alpha = \frac{a - b}{a + b}$ what is $\sqrt{1 - \cos^2\alpha}$

A $\frac{a - b}{a + b}$ B $\sqrt{\frac{a - b}{a + b}}$ C $\sqrt{\frac{a + b}{a - b}}$ D $\left(\frac{a - b}{a + b}\right)^2$

E $\left(\frac{a + b}{a - b}\right)^2$

Solution

By trig identities $\sin^2\alpha + \cos^2\alpha = 1$

$\sin^2\alpha = 1 - \cos^2\alpha$

Taking the square root of both sides

$\sin\alpha = \sqrt{1 - \cos^2\alpha}$

Thus, $\frac{a - b}{a + b} = \sqrt{1 - \cos^2\alpha}$ (A)

1982/20 UME

If $\sin\theta = \frac{m - n}{m + n}$, find the value of $1 + \tan^2\theta$.

A. $\frac{2(m^2 + n^2)}{m + n}$ B. $\frac{2(m^2 + n^2)}{m + n}$

C. $\frac{m^2 + n^2 + 2mn}{2mn}$ D. $\frac{m^2 + n^2 + 2mn}{4mn}$ E. $\frac{(m + n)^2}{4mn}$

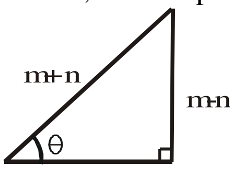
Solution

Recall that $1 + \tan^2\theta = \sec^2\theta$

and $\sec\theta = \frac{1}{\cos\theta}$

Then we are to find $\sec^2\theta$ but first, lets complete our trig ratio triangle.

$\sin\theta = \frac{m - n}{m + n}$ implies



Applying SOH

By Pythagoras rule, the adjacent side

$= \sqrt{(m + n)^2 - (m - n)^2}$

Such terms are not in our options so we simplify further.

$= \sqrt{[m + n - (m - n)][m + n + (m - n)]}$

$= \sqrt{[m + n - m + n][m + n + m - n]}$

$= \sqrt{(2n)(2m)}$

$= \sqrt{4mn}$

i.e Adj $= 2\sqrt{mn}$

Hence $\sec^2\theta = (\sec\theta)^2$

$$= \left(\frac{Hypo}{Adj}\right)^2 = \left(\frac{m+n}{2\sqrt{mn}}\right)^2$$

$$= \frac{(m+n)^2}{(2\sqrt{mn})^2}$$

$$= \frac{(m+n)(m+n)}{4mn}$$

$$= \frac{m^2+n^2+2mn}{4mn} \quad (D)$$

1985/34 UME

Without using tables, calculate the value of $1 + \sec^2 30^0$

A. $2\,^1/_3$ B. 2 C. $1\,^1/_3$ D. ¾ E. $^3/_2$

Solution

Recall that $1 + \tan^2\theta = \sec^2\theta$

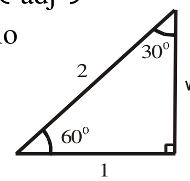
$\tan^2\theta = \sec^2\theta - 1$

$-\tan^2\theta = 1 - \sec^2\theta$

The only way out here is direct substitution

$$\sec^2 30^0 = \frac{1}{\cos^2 30^0} = \left(\frac{hyp}{adj}\right)^2$$

From special angles 30^0 trig ratio



$\cos 30^0 = \frac{\sqrt{3}}{2}$

From above deduction

$$\sec^2 30^0 = \left(\frac{2}{\sqrt{3}}\right)^2$$

$$= \frac{4}{3}$$

Hence $1 + \sec^2 30^0 = 1 + \frac{4}{3} = \frac{7}{3}$ i.e $2\,^1/_3$ (A)

1985/38 UME

If $\cos\theta = \frac{\sqrt{3}}{2}$ and θ is less than 90^0, calculate

$$\frac{\cot(90-\theta)}{\sin^2\theta}$$

A. $\frac{4\sqrt{3}}{3}$ B. $4\sqrt{3}$ C. $\frac{3}{2}$ D. $\frac{1}{\sqrt{3}}$ E. $\frac{2}{\sqrt{3}}$

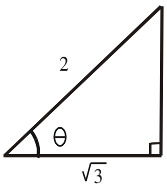
Solution

Recall that $\cot(90-\theta) = \tan\theta$ and $\sin^2\theta = 1 - \cos^2\theta$

But the latter is longer to solve for

Hence $\frac{\cot(90-\theta)}{\sin^2\theta} = \frac{\tan\theta}{\sin^2\theta}$

Let's now complete the trig ratio triangle



$\cos\theta = \frac{\sqrt{3}}{2} \Rightarrow$ applying CAH

By Pythagoras rule, the

opposite $= \sqrt{2^2 - (\sqrt{3})^2}$

$= \sqrt{4-3}$

$= \sqrt{1}$ i.e 1

So, $\tan\theta = \frac{1}{\sqrt{3}}$ and $\sin\theta = \frac{1}{2}$

Hence $\frac{\tan\theta}{\sin^2\theta} = \frac{1}{\sqrt{3}} \div \left(\frac{1}{2}\right)^2$

$$= \frac{1}{\sqrt{3}} \times \frac{4}{1}$$

$$= \frac{4}{\sqrt{3}} \text{ rationalizing } \frac{4\sqrt{3}}{3} \quad (A)$$

2008/2 fm

If $\sin A = \frac{3}{5}$ and $\cos B = \frac{15}{17}$, where A is obtuse and B is acute, find the value of $\cos(A+B)$

Solution

$\cos(A+B) = \cos A \cos B - \sin A \sin B$

To proceed we first find cosA and sinB

By trig identities

If $\sin A = \frac{3}{5}$ it follows that $\cos A = \sqrt{1-\sin^2 A}$

$$\cos A = \sqrt{1-\frac{9}{25}} = \pm\frac{4}{5} \text{ i.e } -\frac{4}{5} \text{ (since A is Obtuse)}$$

If $\cos B = \frac{15}{17}$ it follows that $\sin B = \sqrt{1-\cos^2 B}$

$$\sin B = \sqrt{1-\frac{225}{289}} = \pm\frac{8}{17} \text{ i.e } +\frac{8}{17} \text{ (since B is Acute)}$$

$\cos(A+B) = \cos A \cos B - \sin A \sin B$

$$= -\frac{4}{5} \times \frac{15}{17} - \frac{3}{5} \times \frac{8}{17}$$

$$= -\frac{12}{17} - \frac{24}{85}$$

$$= -\frac{84}{85}$$

2014/2 fm

Simplify $(1-\sin\theta)(1+\sin\theta)$

A $\sin^2\theta$ B $\sec^2\theta$ C $\tan^2\theta$ D $\cos^2\theta$

Solution

Applying difference of two squares principles

$(1-\sin\theta)(1+\sin\theta) = 1^2 - (\sin\theta)^2$

$= 1 - \sin^2\theta$

By trig identities $= \cos^2\theta$ (D)

2005/1b (Nov)

Given that $\sin(A+B) = \sin A\cos B + \cos A\sin B$, without using mathematical table or calculator evaluate $\sin 105^0$, leaving your answer in surd form [Hint $105^0 = 60^0 + 45^0$]

Solution

Using the given hint and the quoted formula

$\sin 105^0 \Rightarrow \sin(60^0 + 45^0) = \sin 60^0 \cos 45^0 + \cos 60^0 \sin 45^0$

Substituting for the values of the special trig ratios

$$= \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}} + \frac{1}{2} \times \frac{1}{\sqrt{2}}$$

$$= \frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}}$$

$$= \frac{\sqrt{3}+1}{2\sqrt{2}}$$

Rationalizing: $= \dfrac{(\sqrt{3}+1)\,2\sqrt{2}}{2\sqrt{2} \times 2\sqrt{2}}$

$$= \frac{(\sqrt{3}+1)\,2\sqrt{2}}{4 \times 2}$$

$$= \frac{2\sqrt{6} + 2\sqrt{2}}{8} = \frac{1}{4}(\sqrt{6}+\sqrt{2})$$

1990/39 UME Exercise 13.16

If $\cos\theta = \dfrac{12}{13}$, find $1 + \cot^2\theta$

A. $\dfrac{169}{25}$ B. $\dfrac{25}{169}$ C $\dfrac{169}{144}$ D $\dfrac{144}{169}$

2002/31 UME Exercise 13.17

If $\tan\theta = \dfrac{4}{3}$, calculate $\sin^2\theta - \cos^2\theta$

A. $\dfrac{16}{25}$ B $\dfrac{24}{25}$ C. $\dfrac{7}{25}$ D. $\dfrac{9}{25}$

2006/9 UME Exercise 13.18

If $\tan\theta = \dfrac{5}{4}$, find $\sin^2\theta - \cos^2\theta$

A. $\dfrac{41}{9}$ B. $\dfrac{9}{41}$ C. 1 D. $\dfrac{5}{4}$

1986/32 UME Exercise 13.19

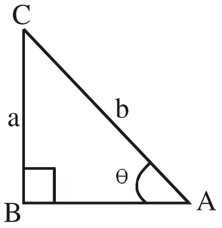
If $\cos\theta = \dfrac{a}{b}$, find $1 + \tan^2\theta$

A. $\dfrac{b^2}{a^2}$ B. $\dfrac{a^2}{b^2}$ C. $\dfrac{a^2+b^2}{b^2-a^2}$ D. $\dfrac{2a^2+b^2}{a^2+b^2}$

DERIVED TRIG RATIOS

Given a trigonometric ratio such as $\sin\theta = {}^a/_b$, where θ is acute; we can obtain the trig ratios for $\cos\theta$ and $\tan\theta$ and their reciprocals as follows:

Draw a right –angled triangle, which contains θ as shown below. The side opposite θ is **a** unit and the hypotenuse (i.e. the side facing the right angle) is **b** unit



Applying Pythagoras rule, we have

$AB^2 = AC^2 - BC^2$

$= b^2 - a^2$

$AB = (b^2 - a^2)^{1/2}$

$\cos\theta = \dfrac{Adj}{Hypo} = \dfrac{AB}{AC} = \dfrac{(b^2-a^2)^{1/2}}{b}$

$\tan\theta = \dfrac{opp}{Adj} = \dfrac{BC}{AB} = \dfrac{a}{(b^2-a^2)^{1/2}}$

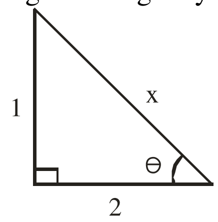
2012/48 Neco

If $\tan\theta = \dfrac{1}{2}$ and θ is an acute angle, what is $\cos\theta$?

A $\dfrac{\sqrt{2}}{5}$ B $\dfrac{2\sqrt{5}}{5}$ C $\dfrac{\sqrt{5}}{2}$ D $\sqrt{\tfrac{5}{2}}$ E $\sqrt{5}$

Solution

We find the unknown side of the resulting right – angled triangle by applying Pythagoras rule



$x^2 = 1^2 + 2^2$

$x^2 = 1 + 4$

$x^2 = 5$

$x = \sqrt{5}$

Thus $\cos\theta = \dfrac{2}{\sqrt{5}}$

Rationalizing: $= \dfrac{2}{\sqrt{5}} \times \dfrac{\sqrt{5}}{\sqrt{5}}$

$= \dfrac{2\sqrt{5}}{5}$ (B)

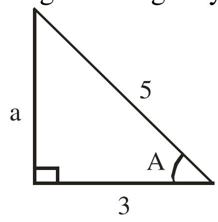
2010/43 NABTEB (Nov)

Find the value of sinA + cosA if $\cos A = \dfrac{3}{5}$

A 7.0 B 5.0 C 1.5 D 1.4

Solution

We find the unknown side of the resulting right – angled triangle by applying Pythagoras rule



$5^2 = a^2 + 3^2$

$5^2 - 3^2 = a^2$

$25 - 9 = a^2$

$16 = a^2$

$\sqrt{16} = a$ i.e $a = 4$

Thus, $\sin A + \cos A = \dfrac{4}{5} + \dfrac{3}{5}$

$= \dfrac{7}{5}$ i.e 1.4 (D)

2006/8 Neco

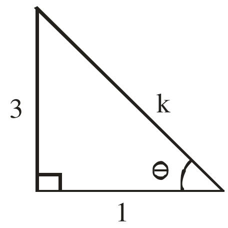
If $\tan\theta = 3$, find the value of $\sin\theta \bullet \cos(90^0 - \theta)$

A $\dfrac{9}{10}$ B $\dfrac{7}{10}$ C $\dfrac{3}{5}$ D $\dfrac{3}{10}$ E $\dfrac{1}{9}$

Solution

We complete the trig- ratio triangle, applying TOA

$\tan\theta = \frac{3}{1}$ implies



By Pythagoras rule: $k^2 = 3^2 + 1^2$

$k = \sqrt{10}$

From complementary angles

$\cos(90^0 - \theta) = \sin\theta$

$\sin\theta \bullet \cos(90^0 - \theta) = \sin\theta \sin\theta$

$= \frac{3}{\sqrt{10}} \times \frac{3}{\sqrt{10}}$

$= \frac{9}{10}$ (A)

2011/ 8a

Given that $\sin x = 0.6$ and $0^0 \le x \le 90^0$,

evaluate $2\cos x + 3\sin x$, leaving your answer in the form $\frac{m}{n}$ where m and n are integers

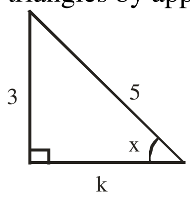
Solution

Though our problem is presented in decimals, we are directed to work in fractions

$\sin x = 0.6$

$\sin x = \frac{6}{10}$ i.e $\sin x = \frac{3}{5}$

Next, we find the unknown side of the resulting right – angled triangles by applying Pythagoras rule



$5^2 = 3^2 + k^2$

$5^2 - 3^2 = k^2$

$25 - 9 = k^2$

$16 = k^2$

$\sqrt{16} = k$ i.e $k = 4$

Thus, $2\cos x + 3\sin x = 2\times\frac{4}{5} + 3\times\frac{3}{5}$

$= \frac{8}{5} + \frac{9}{5} = \frac{17}{5}$

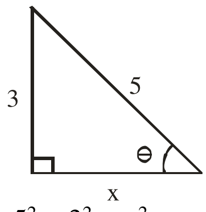
2008/28 NABTEB (Nov)

If $\sin\theta = \frac{3}{5}$ and θ is acute, find the value of $2\tan\theta$

A $\frac{5}{8}$ B $\frac{4}{5}$ C $1\frac{1}{5}$ D $1\frac{1}{2}$

Solution

We find the unknown side of the resulting right – angled triangle by applying Pythagoras rule



$5^2 = 3^2 + x^2$

$5^2 - 3^2 = x^2$

$25 - 9 = x^2$

$16 = x^2$

$\sqrt{16} = x$ i.e $x = 4$

Thus, $2\tan\theta = 2\times\frac{3}{4}$

$= \frac{3}{2}$ i.e $1^1/_2$ (D)

2008/27

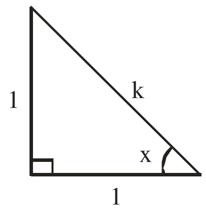
If $\tan x = 1$, evaluate $\sin x + \cos x$, leaving your answer in the surd form

A $2\sqrt{2}$ B $\frac{1}{2}\sqrt{2}$ C $\sqrt{2}$ D 2

Solution

We find the unknown side of the resulting right – angled triangle by applying Pythagoras rule

Note $\tan x = 1$ *implies* $\tan x = \frac{1}{1}$



$k^2 = 1^2 + 1^2$

$k^2 = 2$

$k = \sqrt{2}$

Thus, $\sin x + \cos x = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}$

$= \frac{2}{\sqrt{2}}$

Rationalizing $= \frac{2}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$

$= \frac{2\sqrt{2}}{2} = \sqrt{2}$ (C)

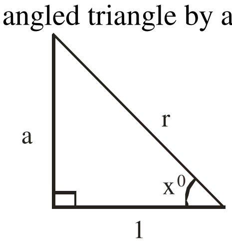
2012/3

Given that $\cos x^0 = \frac{1}{r}$, express $\tan x^0$ in terms of r

A $\frac{1}{\sqrt{r}}$ B $\sqrt{r}$ C $\sqrt{r^2+1}$ D $\sqrt{r^2-1}$

Solution

We find the unknown side of the resulting right – angled triangle by applying Pythagoras rule



$r^2 = 1^2 + a^2$

$$r^2 - 1^2 = a^2$$
$$r^2 - 1 = a^2$$
$$\sqrt{r^2 - 1} = a$$
$$\text{Thus } \tan x^0 = \frac{\sqrt{r^2 - 1}}{1}$$
$$= \sqrt{r^2 - 1} \qquad (\text{ D })$$

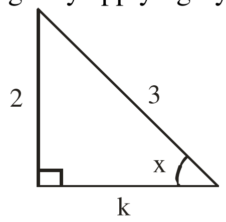
2012/1b

Given that $\sin x = \frac{2}{3}$, evaluate, leaving your answer in surd form and without using table or calculator

$$\tan x - \cos x$$

Solution

We find the unknown side of the resulting right – angled triangle by applying Pythagoras rule



$$3^2 = 2^2 + k^2$$
$$3^2 - 2^2 = k^2$$
$$9 - 4 = k^2$$
$$5 = k^2$$
$$\sqrt{5} = k$$

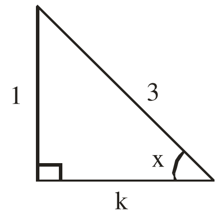
$$\text{Thus, } \tan x - \cos x = \frac{2}{\sqrt{5}} - \frac{\sqrt{5}}{3}$$
$$= \frac{6-5}{3\sqrt{5}}$$
$$= \frac{1}{3\sqrt{5}}$$

2014/22 (Nov)

If $\sin x = \frac{1}{3}$, $0 < x < 90^0$, calculate the value of cos x

A $\frac{1}{8}$ B $\frac{2}{5}$ C $\frac{\sqrt{2}}{3}$ D $\frac{2\sqrt{2}}{3}$

Solution



$$3^2 = 1^2 + k^2$$
$$3^2 - 1^2 = k^2$$
$$9 - 1 = k^2$$
$$8 = k^2$$
$$\sqrt{8} = k \quad \text{i.e} \quad 2\sqrt{2} = k$$

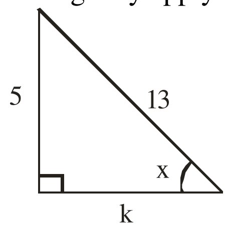
$$\text{Thus, } \cos x = \frac{Adj}{Hyp} = \frac{2\sqrt{2}}{3} \qquad (\text{ D })$$

2014/ 3b (Nov)

If $\sin x = \frac{5}{13}$, $0^0 \leq x \leq 90^0$, evaluate, without table or calculator $\frac{\cos x - 2\sin x}{2\tan x}$

Solution

We find the unknown side of the resulting right – angled triangle by applying Pythagoras rule



$$13^2 = 5^2 + k^2$$
$$13^2 - 5^2 = k^2$$
$$169 - 25 = k^2$$
$$144 = k^2$$
$$\sqrt{144} = k \quad \text{i.e} \quad k = 12$$

$$\text{Thus, } \frac{\cos x - 2\sin x}{2\tan x} = \left(\frac{12}{13} - 2\times\frac{5}{13}\right) \div \left(2\times\frac{5}{12}\right)$$
$$= \left(\frac{12}{13} - \frac{10}{13}\right) \div \frac{5}{6}$$
$$= \left(\frac{12-10}{13}\right) \times \frac{6}{5}$$
$$= \frac{2}{13} \times \frac{6}{5} = \frac{12}{65}$$

2008/48 Neco

If $\tan \alpha = \frac{1}{2}$ and $\tan \beta = \frac{1}{3}$ and both α and β are acute,

find $\tan(\alpha + \beta)$

A 0.015 B 1.00 C 10.00 D 39.81 E 45.00

Solution

By trigonometric identities:

$$\tan(\alpha + \beta) = \frac{\tan \propto + \tan \beta}{1 - \tan \propto \tan \beta}$$
$$= \left(\frac{1}{2} + \frac{1}{3}\right) \div \left(1 - \frac{1}{2} \times \frac{1}{3}\right)$$
$$= \left(\frac{3+2}{6}\right) \div \left(1 - \frac{1}{6}\right)$$
$$= \frac{5}{6} \div \frac{5}{6}$$
$$= \frac{5}{6} \times \frac{6}{5}$$
$$= 1 \qquad (\text{ B })$$

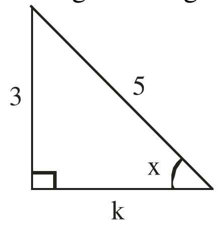
2008/46 Neco (Dec)

If $\sin x = \frac{3}{5}$ and $\tan x = \frac{3}{4}$ which of the following is correct?

A $0^0 < x < 90^0$ B $90^0 < x < 180^0$
C $180^0 < x < 270^0$ C $270^0 < x < 360^0$
E The value of x cannot be determined from the given information

Solution

Using $\sin x = \frac{3}{5}$, let us complete the Pythagorean triple, by right – angled triangle



$5^2 = 3^2 + k^2$
$5^2 - 3^2 = k^2$
$25 - 9 = k^2$
$16 = k^2$
$\sqrt{16} = k$ i.e $k = 4$

Thus, $\tan x = \frac{Opp}{Adj} = \frac{3}{4}$ which same as given above

We can conclude that $0^0 < x < 90^0$

2008/44 Neco (Dec)

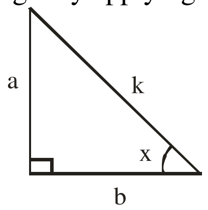
If $\tan x = \frac{a}{b}$, $0 < x < 90^0$, find $1 + \cos^2 x$

A $\frac{a^2 - b^2}{a^2 + b^2}$ B $\frac{2a^2 + b^2}{a^2 + b^2}$ C $\frac{a^2 + 2b^2}{a^2 + b^2}$

D $\frac{a^2}{a^2 + b^2}$ E $\frac{b^2}{a^2 + b^2}$

Solution

We find the unknown side of the resulting right – angled triangle by applying Pythagoras rule



$k^2 = a^2 + b^2$
$k = \sqrt{a^2 + b^2}$

Thus, $1 + \cos^2 x = 1 + \left(\frac{b}{\sqrt{a^2 + b^2}}\right)^2$

$= 1 + \frac{b^2}{a^2 + b^2}$

Applying LCM in fraction method

$= \frac{a^2 + b^2 + b^2}{a^2 + b^2}$

$= \frac{a^2 + 2b^2}{a^2 + b^2}$ (C)

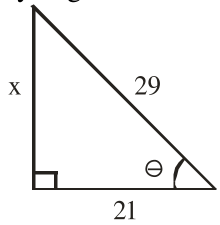
2009/50 Neco (Dec)

Given that $\cos \theta = \frac{21}{29}$, where θ is acute. Find cosec θ

A $\frac{20}{29}$ B $\frac{21}{29}$ C $\frac{20}{21}$ D $\frac{29}{21}$ E $\frac{29}{20}$

Solution

We find the unknown side of the resulting right – angled triangle by applying Pythagoras rule



$29^2 = 21^2 + x^2$
$29^2 - 21^2 = x^2$
$841 - 441 = x^2$
$400 = x^2$
$\sqrt{400} = x$ i.e $20 = x$

Thus, $\text{cosec } \theta = \frac{1}{\sin\theta} = \frac{hypothenuse}{opposite} = \frac{29}{20}$ (E)

2005/7

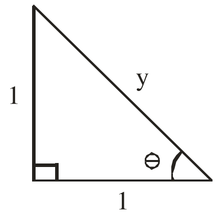
If $\tan \theta = 1$, evaluate $\cos\theta$. $(0 \le \theta \le 90^0)$

A -1 B $\frac{1}{\sqrt{2}}$ C $-\frac{1}{\sqrt{2}}$ D $\frac{1}{2}$

Solution

We find the unknown side of the resulting right – angled triangle by applying Pythagoras rule

Note that $\tan \theta = 1$ *implies* $\tan \theta = \frac{1}{1}$



$y^2 = 1^2 + 1^2$
$y^2 = 2$
$y = \sqrt{2}$ Thus, $\cos\theta = \frac{Adj}{Hyp} = \frac{1}{\sqrt{2}}$ (B)

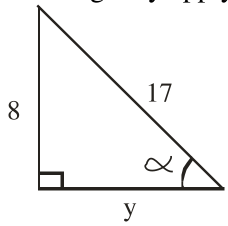
2011/44 Neco

Given that $\sin \alpha = \frac{8}{17}$, $0<\alpha<90^0$, find the value of $\cos \alpha$

A $\frac{14}{15}$ B $\frac{15}{17}$ C $\frac{14}{17}$ D $\frac{8}{15}$ E $\frac{8}{17}$

Solution

We find the unknown side of the resulting right – angled triangle by applying Pythagoras rule



$17^2 = 8^2 + y^2$
$17^2 - 8^2 = y^2$
$289 - 64 = y^2$
$225 = y^2$
$\sqrt{225} = y$
$15 = y$ Thus, $\cos \alpha = \frac{Adj}{Hyp} = \frac{15}{17}$ (B)

2013/46 Neco Exercise 13.20

Given that $\sin\theta = \frac{3}{5}$ and $0^0 \le \theta \le 90^0$

find $\tan\theta + \cos\theta$

A $\frac{1}{20}$ B $\frac{20}{31}$ C $\frac{3}{4}$ D $\frac{4}{5}$ E $\frac{31}{20}$

2014/9c Neco (Nov) Exercise 13.21

If $\tan\theta = \frac{5}{12}$ and θ is acute, determine $\cos\theta$

2014/25 Neco (Nov) Exercise 13.22

Given that $\cos\theta = \frac{\sqrt{3}}{2}$ and $0 \le \theta \le 90^0$,

find the value of $\tan^2\theta$

A $^1/_3$ B $^3/_4$ C 3 D 4 E 5

2014/53 Neco (Nov) Exercise 13.23

If $\sin\alpha = \frac{4}{5}$ and α is an acute angle,

evaluate $15(\tan\alpha - \cos\alpha)$

A 29 B 11 C $1^{14}/_{15}$ D $1^4/_{11}$ E $^{11}/_{15}$

2015/23 Exercise 13.24

Given that $\tan x = \frac{2}{3}$, where $0^0 \le x \le 90^0$,

find the value of $2\sin x$

A $\frac{2\sqrt{13}}{13}$ B $\frac{3\sqrt{13}}{13}$ C $\frac{4\sqrt{13}}{13}$ D $\frac{6\sqrt{13}}{13}$

2015/4a (Nov) Exercise 13.25

If $\sin x = \frac{3}{5}$, where $0^0 \le x \le 90^0$, without using tables

or calculator, calculate $\frac{\cos x + \tan x}{\sin x}$

2010/38 UTME Exercise 13.26

If $\cot\theta = \frac{8}{15}$, where θ is acute, find $\sin\theta$,

A $^{13}/_{15}$ B $^{15}/_{17}$ C $^8/_{17}$ D $^{16}/_{17}$

INVERSE or ARC TRIG. RATIOS

These are the inverse of the various trig ratios we already discussed

Trig. ratios	Inverse
$\cos\theta$	$\cos^{-1}\theta$ or $\arccos\theta$
$\sin\theta$	$\sin^{-1}\theta$ or $\arcsin\theta$
$\tan\theta$	$\tan^{-1}\theta$ or $\arctan\theta$

2008/46 Neco

Find y if $\sin y = \cos 48^0$

A 21^0 B 24^0 C 42^0 D 48^0 E 102^0

Solution

$\sin y = \cos 48^0$

$\sin y = 0.6691$

$y = \sin^{-1} 0.6691$

$= 42^0$ (C)

2006/31 (Nov)

Find x if $\cos x = \sin 40^0$

A 40^0 B 45^0 C 50^0 D 90^0

Solution

$\cos x = \sin 40^0$

$\cos x = 0.64\ 28$

$x = \cos^{-1} 0.6428$

$= 50^0$ (C)

2006/10

If $\tan y = 0.404$, where y is acute, find $\cos 2y$

A 0.035 B 0.719 C 0.808 D 0.927

Solution

$\tan y = 0.404$

$y = \tan^{-1} 0.404$

$= 21.999$

Since y is acute, we are restricted to the 1st quadrant only

$\cos 2y = \cos 2(21.999)$

$= \cos 43.998$

$= 0.719$ (B)

2008/28

If $\cos (x + 25)^0 = \sin 45^0$, find the value of x

A 20 B 30 C 45 D 60

Solution

$\cos (x + 25)^0 = \sin 45^0$

$\cos (x + 25)^0 = \frac{1}{\sqrt{2}}$ from special trig ratios

$(x + 25)^0 = \cos^{-1} \frac{1}{\sqrt{2}}$

By special trig ratio triangle $\frac{1}{\sqrt{2}}$ appears in 45^0

Thus $(x + 25)^0 = 45^0$

$x^0 = 45^0 - 25^0$

$= 20$ (A)

2012/49 Neco

What is the value of α if $\sin 2\alpha = \cos 52^0$

A 18^0 B 19^0 C 38^0 D 64^0 E 72^0

Solution

$\sin 2\alpha = \cos 52^0$

$\sin 2\alpha = 0.6157$

$2\alpha = \sin^{-1} 0.6157$

$2\alpha = 38^0$

$\alpha = \frac{38}{2}$

$= 19^0$ (B)

Inverse trig. ratios and equations

2012/24

If $\cos (x + 40)^0 = 0.0872$, what is the value of x

A 85^0 B 75^0 C 65^0 D 45^0

Solution

$\cos (x + 40)^0 = 0.0872$

$x + 40 = \cos^{-1} 0.0872$

$x + 40 = 85^0$

$x = 85^0 - 40$

$= 45^0$ (D)

2008/ 26 NABTEB (Nov)

If $5\cos\theta - 3 = 0$, find the value of θ to 1 decimal place, $0^0 \le \theta \le 360^0$

A 126.9, 306.9 B 126.9, 233.1 C 53.1, 233.1

D 53.1, 306.9

Solution

$5\cos\theta - 3 = 0$

$5\cos\theta = 3$

$\cos\theta = {}^3/_5$ i.e 0.6

$\theta = \cos^{-1} 0.6$

$= 53.1^0$

Within the range of $0^0 \le \theta \le 360^0$ cosine is also positive in 4^{th} quadrant with formula $360 - \theta$

$\theta = 360 - 53.1$

$= 306.9^0$

Thus $\theta = 53.1^0, 306.9^0$ (D)

2006/46 Neco (Dec)

Solve the equation $3\cos\theta + 1 = 0$

A 360^0 B 270^0 C 109.47^0 D 70.53^0 E 0^0

Solution

$3\cos\theta + 1 = 0$

$3\cos\theta = -1$

$\cos\theta = \frac{-1}{3}$

$\cos\theta = -0.3333$

$\theta = -\cos^{-1} 0.3333$

$= 70.53$

Cosine is negative in 2^{nd} and 3^{rd} quadrants with formula $180 - \theta$ and $\theta - 180$

$180 - 70.53$ applies here

$= 109.47^0$ (C)

2009/2a

If $9\cos x^0 - 7 = 1$ and $0^0 \le \theta \le 90^0$, find x

Solution

$9\cos x^0 - 7 = 1$

$9\cos x = 1 + 7$

$9\cos x^0 = 8$

$\cos x^0 = \frac{8}{9}$

$x^0 = \cos^{-1} \frac{8}{9}$

$= \cos^{-1} 0.8889$

$= 27.26^0$

2009/52 Neco (Dec)

Solve the equation $3\sin\theta - 4\cos\theta = 0$, if θ is acute

A 53.12^0 B 46.48^0 C 46.43^0 D 41.66^0 E 36.38^0

Solution

$3\sin\theta - 4\cos\theta = 0$

$3\sin\theta = 4\cos\theta$

$\sin\theta = \frac{4}{3}\cos\theta$

Divide both side by $\cos\theta$

$\frac{\sin\theta}{\cos\theta} = \frac{4}{3}$

$\tan\theta = {}^4/_3$

$\theta = \tan^{-1} 1.333$

$= 53.12^0$

2014/52 Neco (Nov) Exercise 13.27

If $0^0 \le \alpha \le 90^0$, find the value of α in the equation

$\sin(2\alpha - 4) = \cos(5\alpha + 3)$

A 12^0 B 13^0 C 14^0 D 30^0 E 45^0

2014/11a Neco (Nov) Exercise 13.28

Given that $\cos\beta = \frac{\sqrt{2}\sin\beta + x^2}{4x}$, find the possible values of x if $\beta = 45^0$

Exercise 13.29

Solve: $2\cos x - 1 = 0$ for $0^0 \le x \le 360^0$

2015/37 fm Exercise 13.30

If $\sin^2\theta = 1 + \cos\theta$, $0^0 \le \theta \le 90^0$, find the value of θ

A 90^0 B 60^0 C 45^0 D 30^0

Mathematical tables trig ratio problems

2010/22 (Nov)

Evaluate $\cos 116^0$ using tables

A 0.5354 B 0.4384 C – 0.3384 D – 0.4384

Solution

$\cos 116^0$ is in the 2^{nd} quadrant with formula $180 - \theta$ and cosine is negative there.

Thus, $\cos 116^0 = -\cos(180 - 116)^0$

$= -\cos 64$

Check up 64^0 under zero in cosine of natural angles

$= -0.4384$ (D)

1990/40

Use mathematical tables to evaluate ($\cos 40^0 - \sin 30^0$)

A – 0. 2660 B. – 0. 0266 C. 0. 0266

D. 0. 2660 E. 1. 5320

Solution

Simply check both cos 40 and sin 30 in the table of cosines of angles and sine of angles respectively.

(0. 7660 – 0. 5000) = 0. 2660 (D)

1987/4b GCE

Evaluate ,correct to three significant figures, the expression.

$\frac{\tan 53.2^0 \sin 42.3^0}{\cos 28.2^0}$

Solution

$\frac{\tan 53.2^0 \sin 42.3^0}{\cos 28.2^0} = \frac{1.337 \times 0.6730}{0.8813}$

$= 1.02099$

≈ 1.02 is 3sf

Using logarithm of angles

No	log	
Tan 53. 2^0	0.1260	
Sin 42. 3^0	$\bar{1}.8280$	
	$\bar{1}.9540$	$\bar{1}.9540$
Cos 28.2^0	$\bar{1}.9451$	$\bar{1}.9451$
		0.0089

Check up antilog values 0.00 under 8 difference 9

0.0089 antilog is 1019

$\quad$ + 2

i.e 1021

1.021 = 1.02 to 3sf

1995/6a

Using mathematical tables,
Find (i) $2 \sin 63.35^0$ (ii) $\log \cos 44.74^0$

Solution

i. check up 63 under 0. 3 difference of 0. 05 i.e. $0.05^0 \times 60 = 3^1$ = difference of 3^1 which is 4

Hence sin 63. 35 = 0. 8938

Thus $2 \sin 63.35 = 2 \times 0.8938 = 1.7876$

ii. Check up log cos 44. 74 in the log of cosines. We look up 44^0 under 0.7^0 difference of 0.04^0 i.e. $0.04 \times 60 = 2^1$ difference which is 2. but **note** *"subtract difference"*

Hence $\log \cos 44.74 = \bar{1}.8517$

$$\begin{array}{r} \bar{1}.8517 \\ -2 \\ \hline \bar{1}.8515 \end{array}$$

2000/2a Exercise 13.31

Given that $\cos x = 0.7431$, $0^0 < x < 90^0$,

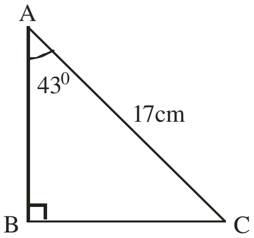
use tables to find the values of (i) $2\sin x$ (ii) $\tan \frac{x}{2}$

Trig. ratios triangular problems

2005/10b NABTEB

The hypotenuse of a right angle triangle is 17cm and one of the angles is 43^0, find the
(i) third angle (ii) side opposite the smallest angle

Solution



(i) The third angle C is

$43 + 90 + C = 180$ (sum of angle in Δ)

$C = 180 - 133$

$= 47^0$

(ii) Side opposite the smallest angle is BC

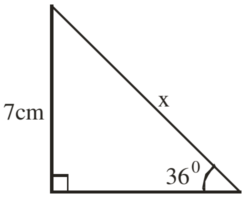
Using 43^0, the sides relevant to it are Opp and Hypo (SOH)

$$\sin 43 = \frac{BC}{17}$$

$$17 \sin 43 = BC$$

$$11.59\text{cm} = BC$$

2014/36 NABTEB



Calculate the value of x in the figure above

A 10.7cm B 11.1cm C 12.2 cm D 13.7cm

Solution

The sides relevant to 36^0 are Opp and Hypo (SOH)

$$\sin 36^0 = \frac{7}{x}$$

$$x \sin 36^0 = 7$$

$$x = \frac{7}{\sin 36}$$

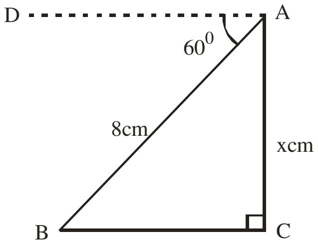
$$= \frac{7}{0.5878}$$

$$= 11.91 \text{ cm}$$

$$\approx 11.9\text{cm}$$

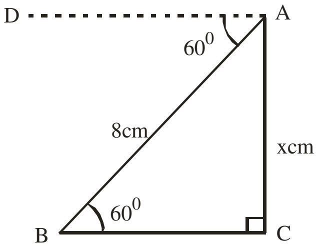
2006/3 NABTEB (Nov)

Given that $D\hat{A}B = 60^0$, $A\hat{C}B = 90^0$, $\overline{AB} = 8$cm with $\overline{AD} \,//\, \overline{BC}$, calculate x



A $\frac{16\sqrt{3}}{3}$ cm B $\frac{8\sqrt{3}}{3}$ cm C $4\sqrt{3}$ cm D 4cm

Solution



$\angle DAB = \angle ABC = 60^0$ (alternate angles)

In ΔABC, the sides relevant to 60^0 are Opp and Hypo (SOH)

$$\sin 60^0 = \frac{x}{8}$$

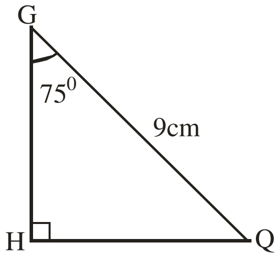
$$8 \sin 60^0 = x$$

$$8 \times \frac{\sqrt{3}}{2} = x$$

$$4\sqrt{3} \text{ cm} = x \quad (C)$$

2010/36 NABTEB (Nov)

In the diagram below, find the value of the length /HQ/ given that $G\hat{H}Q = 90^0$, $H\hat{G}Q = 75^0$ and ΔGHQ is right – angled



A 8.70cm B 3.00cm C 2.50cm D 2.33cm

Solution

The sides relevant to 75^0 are Opp and Hypo (SOH)

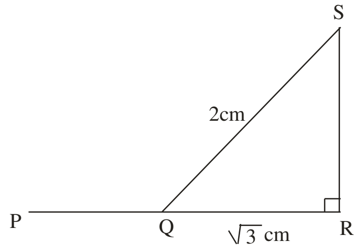
$$\sin 75^0 = \frac{HQ}{9}$$

$$9 \sin 75^0 = HQ$$

$$8.693 \text{ cm} = HQ$$

$$8.7 \text{ cm} \approx HQ \quad (A)$$

2008/4b



In the diagram, PQR is a straight line $\overline{QR} = \sqrt{3}$ cm and

$\overline{SQ}$ = 2cm. Calculate, correct to one decimal place $\angle$PQS

Solution

Let $\angle$ SQR = θ,

Sides of relevance here are Adj and Hypo (CAH)

$$\cos\theta = \frac{\sqrt{3}}{2}$$

$$\theta = \cos^{-1}\frac{\sqrt{3}}{2}$$

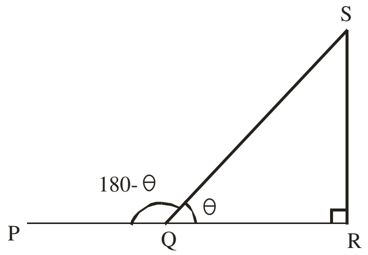
By special trig ratio

$\theta = 30^0$

$\angle$ PQS = $180^0 - \theta$ (sum of angles on a straight line)

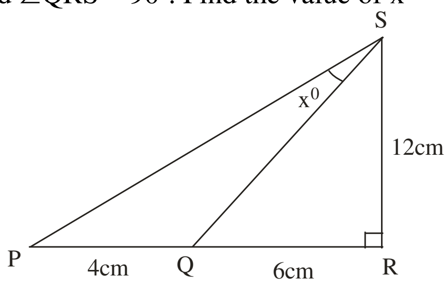
= 180 – 30

= 150^0



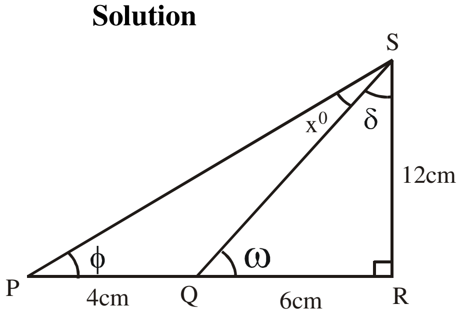
2007/32 *good case*

In the diagram /PQ/ = 4cm, /QR / = 6cm, /RS / = 12cm and $\angle$QRS = 90^0. Find the value of x



A 27 B 26 C 18 D 13

Solution



To find x^0, first, we find ω, ϕ and δ.

In ΔQRS, the relevant sides to ω are Opp and Adj (TOA)

$$\tan\omega = \frac{12}{6} = 2$$

$\omega = \tan^{-1}2$

$= 63.43^0$

Also $90^0 + \omega + \delta = 180^0$ (sum of angles in Δ)

$90 + 63.43 + \delta = 180$

$\delta = 180 - 153.43$

$= 26.57^0$

In ΔPRS, the relevant sides to ϕ are Opp and Adj (TOA)

$$\tan\phi = \frac{12}{4+6} = 1.2$$

$\phi = \tan^{-1}1.2$

$= 50.19^0$

Also $\phi + 90 + (x^0 + \delta) = 180^0$ (sum of angle in Δ)

$50.19 + 90 + x^0 + 26.57 = 180$

$x^0 = 180 - 166.76$

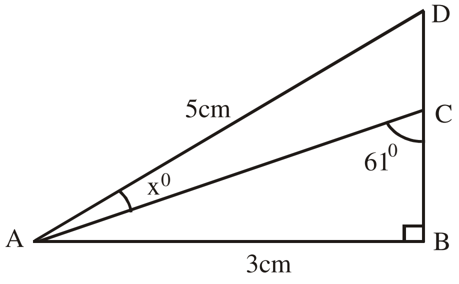
$= 13.24$

$\approx 13^0$ (D)

2005/12a

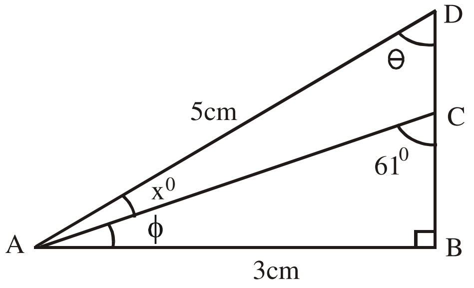
In the diagram, ΔABD is right – angle at B.

/AB/ = 3cm, /AD/ = 5cm, $A\hat{C}B = 61^0$ and $D\hat{A}C = x^0$



Calculate, correct to one decimal place, the value of x.

Solution



To find x^0, first we find ϕ and θ

In ΔABC $\phi + 61^0 + 90^0 = 180^0$ (sum of angles in Δ)

$\phi = 180 - 151$

$= 29^0$

In ΔABD,

the relevant sides to θ are Opp and Hypo (SOH)

$$\sin\theta = \frac{3}{5}$$

$\theta = \sin^{-1}0.6$

$= 36.87^0$

Also in Δ ABD

$\theta + 90 + (\phi + x^0) = 180^0$

$36.87 + 90 + 29 + x^0 = 180$

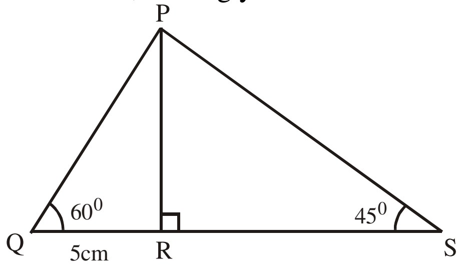
$x^0 = 180 - 155.87$

$= 24.13$

$\approx 24.1^0$ to 1 d.p

2006/13

In the diagram, /QR/ = 5cm, $P\hat{Q}R = 60^0$ and $P\hat{S}R = 45^0$. Find /PS/, leaving your answer in surd form.



A $4\sqrt{5}$ cm B $3\sqrt{7}$ cm C $4\sqrt{6}$ cm D $5\sqrt{6}$ cm

Solution

To find PS, first we solve for PR in ΔPQR

In ΔPQR, the sides relevant to 60^0 are Opp and Adj (TOA)

$$\tan 60^0 = \frac{PR}{QR}$$

$$\tan 60^0 = \frac{PR}{5}$$

$$5 \tan 60^0 = PR$$

Instruction is leaving your answer in surd form

Thus by special trig ratio

$5 \times \sqrt{3}$ cm = PR i.e PR = $5\sqrt{3}$ cm

Next, in ΔPRS,

the side relevant to 45^0 are Opp and Hypo (SOH)

$$\sin 45^0 = \frac{PR}{PS}$$

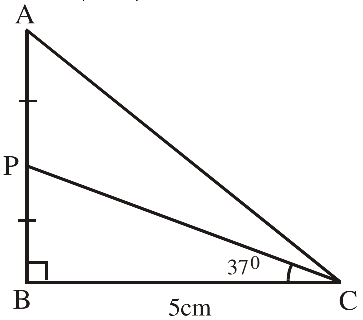
$$PS \sin 45^0 = PR$$

$$PS = \frac{PR}{\sin 45^0}$$

$$PS = 5\sqrt{3} \div \frac{1}{\sqrt{2}}$$

$$= 5\sqrt{3} \times \sqrt{2} = 5\sqrt{6} \text{ cm} \quad (D)$$

2006/7 (Nov)



In the diagram, ΔABC is right - angled at B,

P is the mid – point of AB, $P\hat{C}B = 37^0$ and /BC/ = 5cm

(a) Calculate, correct to three significant figures:

(i) /AB/; (ii) /AC/.

(b) Calculate $P\hat{C}A$ correct to the nearest degree

Solution

(i) To find /AB/, first we solve for BP in ΔBCP

In ΔBCP, the relevant sides to 37^0 are Opp and Adj (TOA)

$$\tan 37^0 = \frac{BP}{5}$$

$$5 \tan 37^0 = BP$$

$$3.7678\text{cm} = BP$$

But AB = AP + BP i.e 2BP (AP = BP)

$= 2 \times 3.7678$

$= 7.5356$ cm

≈ 7.54cm to 3 s.f

(ii) By Pythagoras rule in ΔABC

$AC^2 = AB^2 + BC^2$

$AC^2 = 7.54^2 + 5^2$

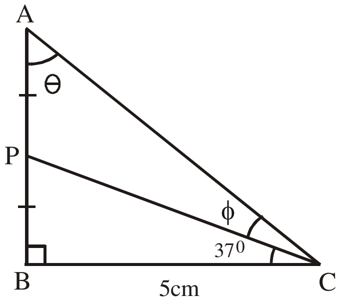
$= 56.8516 + 25$

$AC^2 = 81.8516$

$AC = \sqrt{81.8516}$

$= 9.047$cm ≈ 9.05cm to 3 s.f

(b)



To find $P\hat{C}A$ i.e ϕ, first we solve for θ in ΔABC,

the sides relevant to θ are Opp and Adj (TOA)

$$\tan\theta = \frac{5}{7.54} = 0.6631$$

$$\theta = \tan^{-1} 0.6631$$

$$= 33.55^0$$

Also $\theta + 90 + (37 + \phi) = 180^0$ (sum of angles in Δ)

$33.55 + 90 + 37 + \phi = 180$

$\phi = 180 - 160.55$

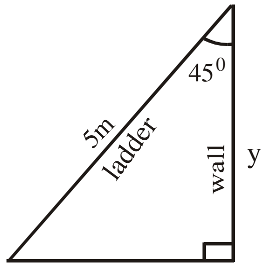
$= 19.45 \approx 19^0$ to nearest degree

2010/ 31 NABTEB (Nov)

A ladder leans on a wall of a building at an angle of 45^0. If the length of the ladder is 5m, find the height of the wall at which the ladder touches it (leave your answer in surd form)

A $\frac{5}{2}\sqrt{2}$ m B $\sqrt{2}$ m C $\frac{1}{\sqrt{2}}$ m D $\sqrt{5}$ m

Solution



The statement "ladder leans on a wall of building at an angle of 45^0" shows that 45^0 is fitted as shown above.

The relevant sides to 45^0 are Adj and Hypo (CAH)

$$\cos 45^0 = \frac{y}{5}$$

$$5 \cos 45^0 = y$$

$$5 \times \frac{1}{\sqrt{2}} = y$$

$$y = \frac{5}{\sqrt{2}}$$

Rationalizing $y = \frac{5}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{5}{2}\sqrt{2}$ m (A)

2005/49 (Nov) Exercise 13.32

A ladder leans against vertical wall at an angle 60^0 to the wall. If the foot of the ladder is 7cm away from the wall, calculate the length of the ladder

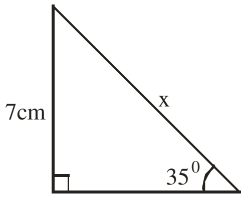
A $\frac{7\sqrt{3}}{3}$ m B 7m C $\frac{14\sqrt{3}}{3}$ m D $7\sqrt{3}$ m

2005/12 Neco Exercise 13.33

A ladder 25m long rest against a vertical wall. If ladder makes an angle of 60^0 with a wall, find the distance between the foot of the ladder and the wall, correct to two place of decimal.

A 25.00m B 21.65m C 20.00m D 19.65m E 12.50m

2014/36 NABTEB Exercise 13.34

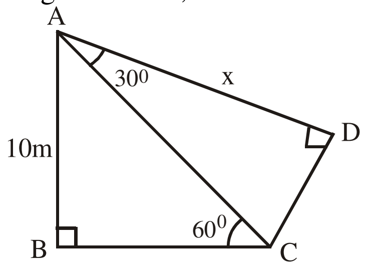


Calculate the value of x in the figure above

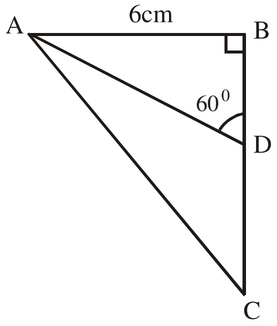
A 10.7cm B 11.1cm C 12.2 cm D 13.7cm

2005/2b Neco Exercise 13.35

In the diagram below, find the value of x



2015/24 Neco Exercise 13.36



In the above figure, D is such that BD : CD = 1 : 2. Find /BC/.

A $\sqrt{3}$ B $2\sqrt{3}$ C $4\sqrt{3}$ D $6\sqrt{3}$ E $8\sqrt{3}$

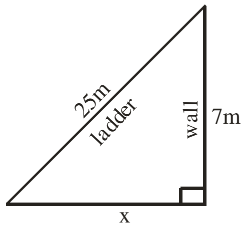
Right – angled triangular problems (Pythagoras rule)

2009/36 NABTEB (Nov)

A ladder 25m long leans against a wall. If the top of the ladder is 7m from the base of the wall, how far is the foot of the ladder from the base of the wall?

A 12m B 24m C 49m D 54m

Solution

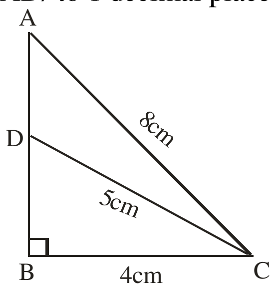


Applying Pythagoras rule:

$$25^2 = 7^2 + x^2$$
$$25^2 - 7^2 = x^2$$
$$625 - 49 = x^2$$
$$576 = x^2$$
$$\sqrt{576} = x \quad \text{thus } x = 24\text{m} \quad (B)$$

2003/7b

Use the dimensions given in the diagram below to calculate /AD/ to 1 decimal place



Solution

First, in ΔBCD, by Pythagoras rule

$$DC^2 = BD^2 + BC^2$$
$$5^2 = BD^2 + 4^2$$
$$5^2 - 4^2 = BD^2$$
$$25 - 16 = BD^2$$
$$9 = BD^2$$
$$\sqrt{9} = BD \quad \text{i.e } BD = 3$$

In ΔABC, by Pythagoras rule

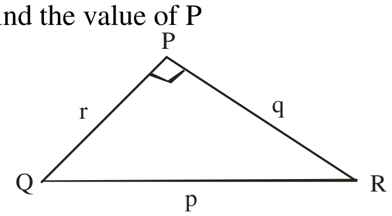
$$AC^2 = AB^2 + BC^2$$
$$8^2 = AB^2 + 4^2$$
$$8^2 - 4^2 = AB^2$$
$$64 - 16 = AB^2$$
$$48 = AB^2$$
$$\sqrt{48} = AB \quad \text{i.e } AB = 6.928\text{cm}$$

Thus, **/AD/** = AB – BD

= 6.928 – 3 = 3.9cm to 1d.p

2008/16

In the diagram, $\angle QPR = 90^0$. if $q^2 = 25 - r^2$, find the value of P



A 3 B 4 C 5 D 6

Solution

By Pythagoras rule

$p^2 = r^2 + q^2$

Substituting of q^2

$p^2 = r^2 + 25 - r^2$

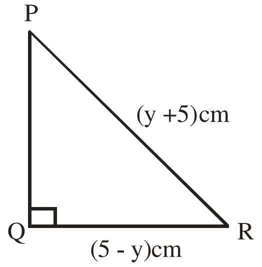
$p^2 = 25$

$p = \sqrt{25}$

$p = 5$ (C)

2005/12 (Nov)

In the diagram, /PR/ = (y + 5)cm, /QR/ = (5 – y) cm and $\angle PQR = 90^0$, find /PQ/



A $4\sqrt{5y}$ cm B $2\sqrt{5y}$ cm C $\sqrt{y^2 - 50}$ cm

D $\sqrt{y^2 + 50}$ cm

Solution

Since ΔPQR is a right – angled; by Pythagoras rule

$(y + 5)^2 = (5 - y)^2 + PQ^2$

$(y + 5)^2 - (5 - y)^2 = PQ^2$

By difference of two squares rule

$[(y + 5) + (5 - y)]\,[(y + 5) - (5 - y)] = PQ^2$

$(y + 5 + 5 - y)\,(y + 5 - 5 + y) = PQ^2$

$10 \times 2y = PQ^2$

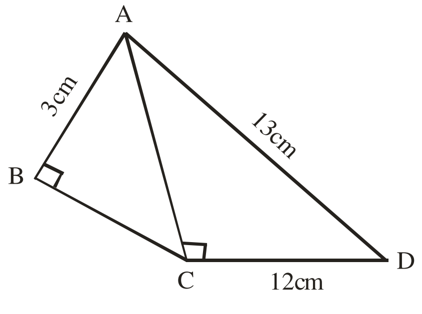
$20y = PQ^2$

$\sqrt{20y} = PQ$

$\sqrt{4 \times 5y} = PQ$

$2\sqrt{5y}$ cm $= PQ$ (B)

2005/12 NABTEB



ABCD is a quadrilateral such that AD = 13cm, CD = 12cm, AB = 3cm, $\angle ABC = 90^0$, $\angle ACD = 90^0$

12. What is the length of the side BC?

A 4cm B 5cm C 10cm D 12cm

Solution

First, in ΔACD, by Pythagoras rule

$AD^2 = AC^2 + CD^2$

$13^2 = AC^2 + 12^2$

$13^2 - 12^2 = AC^2$

$169 - 144 = AC^2$

$25 = AC^2$

$\sqrt{25} = AC$ i.e AC = 5cm

Next, in ΔABC, by Pythagoras rule

$AC^2 = AB^2 + BC^2$

$5^2 = 3^2 + BC^2$

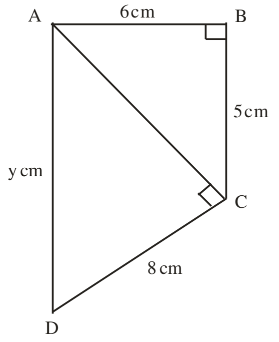
$5^2 - 3^2 = BC^2$

$25 - 9 = BC^2$

$16 = BC^2$

$\sqrt{16} = BC$ thus BC = 4cm

2014/ 17 Neco



Find the value of y in the figure above

A 6.00cm B 7.81cm C 8.00cm

D 9.00cm E 11.18cm

Solution

First, in ΔABC, by Pythagoras rule

$AC^2 = 6^2 + 5^2$

$AC^2 = 36 + 25$

$AC^2 = 61$

$AC = \sqrt{61}$ cm

Thus in ΔACD, by Pythagoras rule

$AD^2 \Rightarrow y^2 = AC^2 + CD^2$

$y^2 = (\sqrt{61})^2 + 8^2$

$y^2 = 61 + 64$

$y^2 = 125$

$y = \sqrt{125} = 11.18$cm (E)

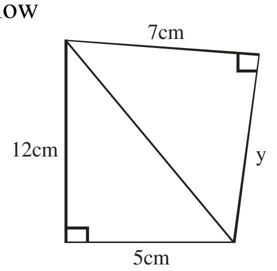
2005/19 NABTEB (Nov) Exercise 13.37

A ladder is placed on top of a vertical wall with its bottom 5m away from the wall on the level ground. If the ladder is 13m in length, how high up is the wall?

A 8m B 12m C 14m D 18m

2010/36 Neco Exercise 13.38

Calculate the value of y to 3 significant figures in the diagram below

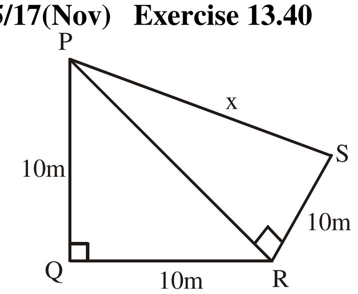


A 13.00 cm B 11.00cm C 10.95cm

D 10.9cm E 9.35cm

2014/22 Neco (Nov) Exercise 13.39
A ladder 23m long rests against a vertical wall so that the foot of the ladder is 11m from the wall. Find the angle that the ladder makes with the wall to the nearest degree.
A 26^0 B 29^0 C 61^0 D 64^0 E 66^0

2015/17(Nov) Exercise 13.40



In the diagram, /PQ/ = /QR/ = /RS/ = 10cm, $\angle PQR = \angle PRS = 90^0$.
Calculate the perimeter of quadrilateral PQRS
A $(10 + \sqrt{3})$cm B $10\sqrt{3}$ cm C $10(2 + \sqrt{3})$cm
D $10(3 + \sqrt{3})$cm

CHAPTER FOURTEEN

Angles Of Elevation & Depression

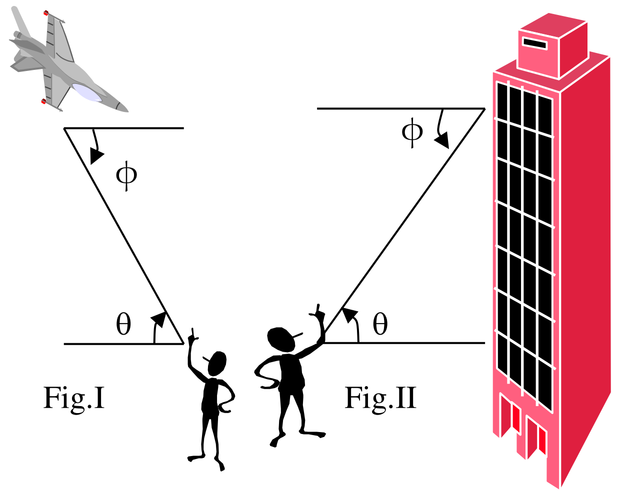
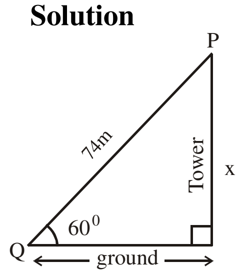


Fig.I shows a boy standing on the ground viewing a plane with an angle of elevation θ while the pilot views the boy at an angle of depression Φ.
Fig II shows a man viewing his daughter in her office at an angle of elevation θ and the daughter views her father at an angle of depression Φ.
Deductively in both fig I and fig II
θ = Φ (Alternate angles).
The angles of elevation and depression can be used in calculating among others
(a) the height of an object above the ground
(b) the distance of an object from a given point of observation.

Single-triangle cases involving angle of elevation

2008/47 Neco (Nov)
The angle of elevation of a point P on a tower from a point Q on the horizontal ground is 60^0. If /PQ/ = 74m, how high is P above the ground?
A $\frac{37\sqrt{3}}{2}$ m B $\frac{74\sqrt{3}}{3}$ m C $37\sqrt{3}$ m D $74\sqrt{3}$ m E 37m

Solution

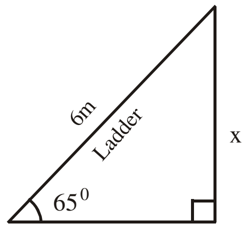


The relevant sides to 60^0 are Opp and Hypo (SOH)

$$\sin 60^0 = \frac{x}{74}$$
$$74 \sin 60 = x$$
$$74 \times \frac{\sqrt{3}}{2} = x$$
$$37\sqrt{3} \text{ m} = x \quad (C)$$

2003/4b NABTEB
A ladder 6m long leans against a wall so that it makes an angle of 65^0 with the horizontal ground. Calculate how far up the wall the ladder reaches, to 3 significant figures

Solution

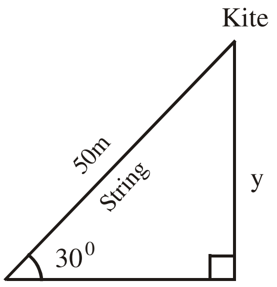


The relevant sides to 65^0 are Opp and Hypo (SOH)

$$\sin 65^0 = \frac{x}{6}$$
$$6 \sin 65^0 = x$$
$$5.438\text{m} = x$$
$$x \approx 5.44\text{m to 3 s.f}$$

2005/33 NABTEB (Nov)
A boy flies a kite with a string of length 50m. If the string makes an angle of 30^0 with the ground, what is the height of the kite above the ground?
A 25m B 45m C 56m D 100m

Solution



The relevant sides to 30^0 are Opp and Hypo (SOH)

$$\sin 30^0 = \frac{y}{50}$$

$$50 \sin 30 = y$$

$$50 \times \frac{1}{2} = y$$

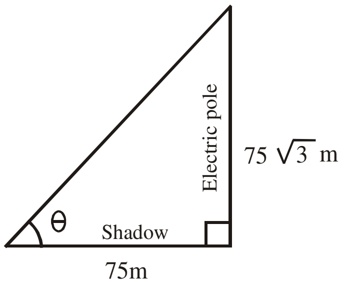
$$25m = y \quad (A)$$

2014/27 Neco (Dec)

The shadow of an electric pole $75\sqrt{3}$ m high is 75m long. Determine the angle of elevation of the sun

A 30^0 B 60^0 C 90^0 D 120^0 E 150^0

Solution



The relevant sides to θ are Opp and Adj (TOA)

$$\tan\theta = \frac{75\sqrt{3}}{75}$$

$$\tan\theta = \sqrt{3}$$

$$\theta = \tan^{-1}\sqrt{3}$$

By special angle trig ratios

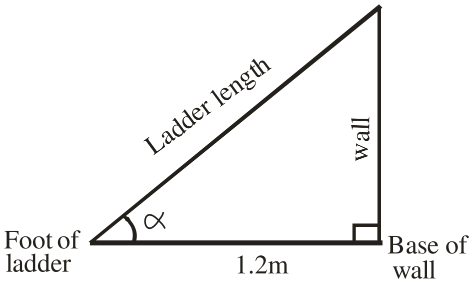
$$\theta = 60^0 \quad (B)$$

2006/20 Neco *good case*

A ladder leans against a vertical wall making an angle whose cosine is 0.6 with the ground. The distance between the foot of the ladder and the base of the wall is 1.2m. Calculate the length of the ladder

A 2.50m B 2.00m C 1.50m D 0.96m E 0.72m

Solution



We are informed that

$$\cos\alpha = \frac{1.2}{ladder\ length}$$

$$0.6 = \frac{1.2}{ladder\ length}$$

$$0.6 \times \text{ladder length} = 1.2$$

$$\text{Ladder length} = \frac{1.2}{0.6}$$

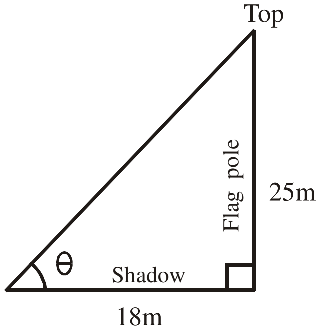
$$= 2.00m \quad (B)$$

2006/19 Neco

The shadow of a flagpole 25m long is 18m. What is the angle of elevation of the top of the flagpole from the shadow, correct to 1 decimal place?

A 35.8^0 B 43.9^0 C 46.1^0 D 53.4^0 E 54.2^0

Solution



The relevant sides to 60^0 are Opp and Adj (TOA)

$$\tan\theta = \frac{25}{18} = 1.3889$$

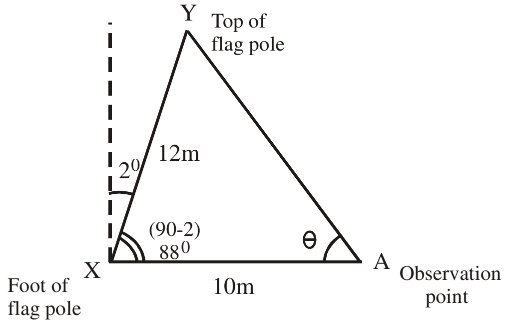
$$\theta = \tan^{-1} 1.3889$$

$$= 54.2^0 \text{ to 1 decimal place}$$

2008/7a Neco ***counter example***

A flagpole XY of length 12m tilts towards an observation point A at an angle of 2^0 to the vertical. If A is on the same level as the foot X of the flagpole and /AX/ = 10m, find the angle of elevation of the top of the flagpole from A to the nearest degree

Solution



Here no special trig ratio angles of 30^0, 60^0 and 45^0.

The resulting triangle has two sides, the third side /YA/ = x can be gotten by cosine rule

$$x^2 = 12^2 + 10^2 - 2(10)(12)\cos 88^0$$

$$= 144 + 100 - 240\cos 88$$

$$= 244 - 240\cos 88$$

$$x^2 = 235.62$$

$$x = \sqrt{235.62} = 15.35m$$

Next, we find θ using sine rule

$$\frac{12}{\sin\theta} = \frac{15.35}{\sin 88}$$

$$\frac{12\sin 88}{15.35} = \sin\theta$$

$$\frac{11.99}{15.35} = \sin\theta$$

$$\sin\theta = 0.7811$$

$$\theta = \sin^{-1} 0.7811$$

$$= 51.36^0$$

$$\approx 51^0 \text{ to nearest degree}$$

2005/ 43 Exercise 14.1

The angle of elevation of the top of a cliff 15metres high from a landmark is 60^0. How far is the landmark from the foot of the cliff? Leave your answer in surd form.

A $15\sqrt{3}$ m B $15\sqrt{2}$ m C $10\sqrt{3}$ m D $5\sqrt{3}$ m

2006/50 Exercise 14.2

A ladder 16m long leans against an electric pole. If the ladder makes an angle of 65^0 with the ground, how far up the electric pole does its top reach?

A 6.8m B 14.5m C 17.7m D 34.3m

2005/10 Neco Exercise 14.3

A man stands on the ground 12m away from a building which is 16m high. Find the angle of elevation of the top of the building from the man's feet.

A 53.13^0 B 48.59^0 C 41.41^0 D 36.87^0 E 30.96^0

2006/49 Neco (Nov) Exercise 14.4

If point A is 20m away from the foot of an electric pole of height 15m, calculate the angle of elevation of the top of the pole from point A.

A 18.43^0 B 36.87^0 C 51.34^0 D 73.74^0 E 75.25^0

Further cases on single triangle

2013/13

When one end of a ladder, LM, is placed against a vertical wall at a point 5 metres above the ground, the ladder makes an angle of 37^0 with the horizontal ground. (a) Represent this information in a diagram

(b) Calculate, correct to 3 significant figures, the length of the ladder

(c) If the foot of the ladder is pushed towards the wall by 2 metres, calculate correct to the nearest degree, the angle which the ladder now makes with the ground

Solution

(a)

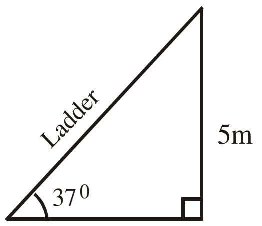


Fig I

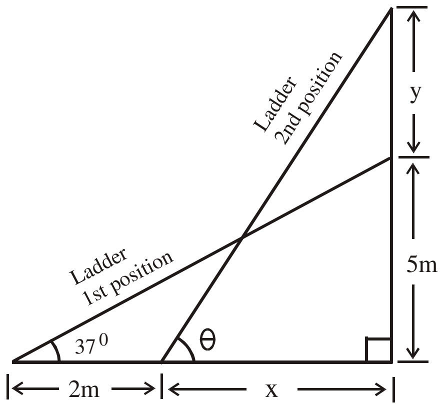
(b) The relevant sides to 37^0 are Opp and Hypo (SOH)

$$\sin 37^0 = \frac{5}{ladder\ length}$$

$$\text{Ladder length} \times \sin 37^0 = 5$$

$$\text{Ladder length} = \frac{5}{\sin 37^0}$$

$$= 8.3082\text{m} \approx 8.31\text{m to 3 s.f}$$



(c)

Since ladder length remains unchanged in both triangles, we find x from triangle with 37^0, thereby getting two sides in triangle with θ.

In triangle with 37^0, relevant sides are Opp and Adj (TOA)

$$\tan 37^0 = \frac{5}{2+x}$$

$$(2 + x) \tan 37^0 = 5$$

$$2 + x = \frac{5}{\tan 37^0}$$

$$2 + x = 6.6352$$

$$x = 6.6352 - 2$$

$$= 4.6352\text{m}$$

Next, in triangle with θ,

the relevant sides are Adj and Hypo (CAH)

$$\cos\theta = \frac{x}{ladder\ length}$$

$$\cos\theta = \frac{4.6352}{8.3082}$$

$$\cos\theta = 0.5579$$

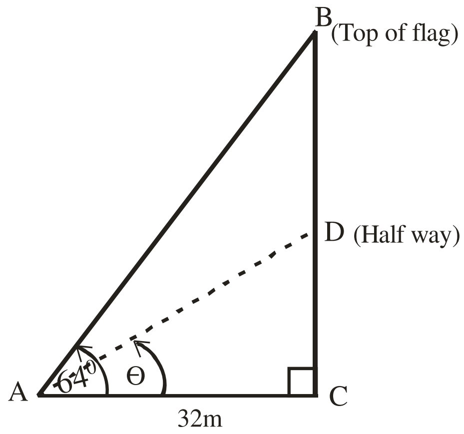
$$\theta = \cos^{-1} 0.5579$$

$$= 56.089 \approx 56^0 \text{ to nearest degree}$$

2006/3b Neco ***counter example***

The angle of elevation of the top of a flagpole is 64^0 from a point 32m away from the foot of the flagpole. Find the angle of elevation of a flag half way up the flagpole from that point correct to the nearest degree

Solution



Let's get AC from ΔABC

$$\tan 64^0 = \frac{AC}{32}$$

$$AC = 32 \tan 64^0$$

It was stated that Length CD = $^1/_2$ AC = $16 \tan 64^0$

$$\text{From } \Delta ADC \quad \tan\theta = \frac{CD}{BC}$$

$$= \frac{16 \tan 64^0}{32}$$

$$\tan\theta = {}^1/_2 (\tan 64^0)$$

$$\theta = \tan^{-1} 1.02515$$

$$= 45.71^0 \approx 46^0 \text{ to nearest degree}$$

2013/5

A boy 1.2m tall stands 6m away from the foot of a vertical lamp pole 4.2m long. If the lamp is at the tip of the pole.

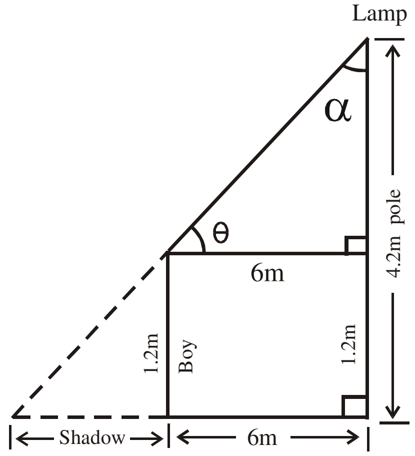
(a) represent this information in a diagram

(b) Calculate

(i) length of the shadow of the boy cast by lamp

(ii) angle of elevation of the lamp from the boy correct to the nearest degree

Solution



First, we calculate θ, before shadow length.
In the Δ with angle θ,
the relevant sides are Opp and Adj (TOA)

$$\tan\theta = \frac{4.2 - 1.2}{6}$$

$$\tan\theta = \frac{3}{6}$$

$$\theta = \tan^{-1} 0.5$$

$$= 26.57^0 \approx 27^0$$

$$\theta + \alpha + 90^0 = 180^0$$

$$27 + \alpha + 90^0 = 180^0$$

$$\alpha = 180 - 117 = 63^0$$

In the whole triangle (bigger) triangle with α ,
The relevant sides are Opp and Adj (TOA)

$$\tan\ \alpha = \frac{shadow + 6}{4.2}$$

$$\tan 63^0 = \frac{shadow + 6}{4.2}$$

$$4.2 \tan 63^0 = \text{shadow} + 6$$

$$8.2430 = \text{shadow} + 6$$

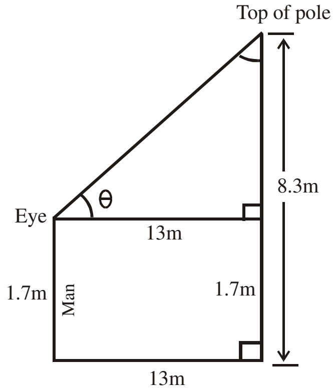
$$8.2430 - 6 = \text{shadow length}$$

$$2.2430\text{m} = \text{shadow length}$$

shadow length = 2m to the nearest whole number

2014/38
A man's eye level is 1.7m above the horizontal ground and 13m from a vertical pole. If the pole is 8.3m high, calculate, correct to the nearest degree, the angle of elevation of the top of the pole from his eyes.
A 33^0 B 32^0 C 27^0 D 26^0

Solution



In the right – angled triangle formed
The relevant sides to θ are Opp and Adj (TOA)

$$\tan\theta = \frac{8.3 - 1.7}{13}$$

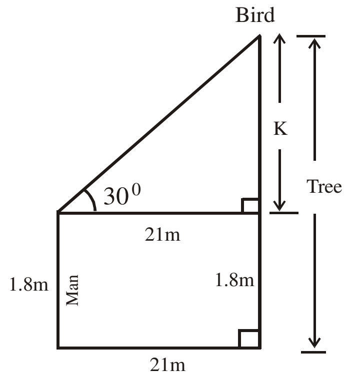
$$\tan\theta = \frac{6.6}{13} = 0.5077$$

$$\theta = \tan^{-1} 0.5077$$

$\theta = 26.92^0 \approx 27^0$ to the nearest degree

2013/44 Neco
A 1.8m tall man observes a bird on top of a tree. If the man is 21m away from the tree and his angle of sighting the bird is 30^0. Calculate the height of the tree
A 10.50m B 11.48m C 12.13m D 13.92m E 18.19m

Solution



In the resulting right – angle triangle,
the sides relevant to 30^0 are Opp and Adj (TOA)

$$\tan 30 = \frac{k}{21}$$

$$21 \tan 30 = k$$

$$12.12\text{m} = k$$

Thus height of tree = k + 1.8m
= 12.12m + 1.8m = 13.92m (D)

2005/14 Neco Exercise 14.5
A man, 1.5m tall, observes a bird at the top of a tree, 4.5m high. If the bird is 8m away from the man, calculate the distance between the feet of the man and the base of the tree
A $\sqrt{55}$ m B $\sqrt{61}$ m C $\sqrt{73}$ m D 11m E 14m

2014/26 (Nov) Exercise 14.6
The angle of elevation of a bird in the air from a hunter standing on the ground is 35^0. If the hunter is 1.4m tall and the bird is 81.7m away along the hunter's line of sight, how high, correct to1 decimal place, is the bird from the ground?
A 48.3m B 58.8m C 68.3m D 71.1m

2005/20(Nov) Exercise 14.7
The angle of elevation of the top of a fence from a boy 1.2m tall is 57^0, if the boy is 24m away from the fence, find, correct to two decimal place, the height of the fence.
A 13.07m B 20.13m C 36.96m D 38.16m

2015/25 Exercise 14.8
The angle of elevation of an aircraft from a point K on the horizontal ground is 30^0. If the aircraft is 800m above the ground, how far is it from K?
A 400.00m B 692.82m C 923.76m D 1,600.00m

2015/57 Neco Exercise14.9
The angle of elevation of the top of a communication mast from a point A is 60^0. If the mast is 120m high, how far is the point A from the foot of the mast, correct to one decimal place?
A 60.0m B 69.3m C 103.9m D 120.0m E207.8m

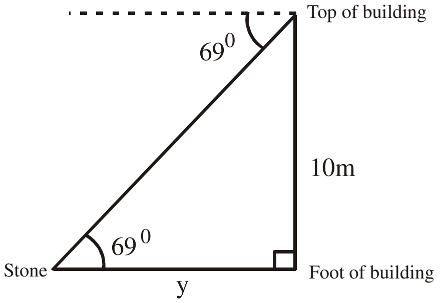
Single-triangle cases involving angle of depression

2014/30 NABTEB (Nov)

From the top of a building 10m high, the angle of depression of a stone lying on the horizontal ground is 69^0. Calculate, correct to 1 decimal place, the distance of the stone from the foot of the building

A 3.6m B 3.8m C 26.1m D 9.3m

Solution



The sides relevant to 69^0 are Opp and Adj (TOA)

$$\tan 69^0 = \frac{10}{y}$$

$$y \tan 69^0 = 10$$

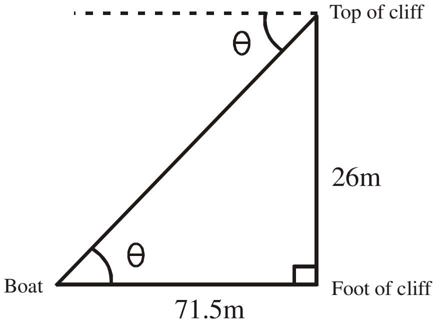
$$y = \frac{10}{\tan 69^0}$$

$$= 3.8386\text{m} \approx 3.8\text{m to 1 d.p}$$

2014/4a Neco (Dec)

A boat can be sighted at the sea 71.5m from the foot of a cliff which is 26m high. Calculate the angle of depression of the boat from the top of the cliff, correct to 2 sig. figures.

Solution



The sides relevant to θ are Opp and Adj (TOA)

$$\tan\theta = \frac{26}{71.5} = 0.3636$$

$$\theta = \tan^{-1} 0.3636$$

$$= 19.98$$

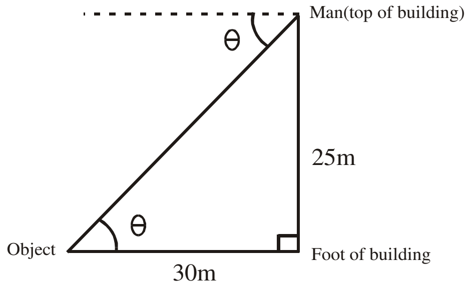
$$\approx 20^0 \text{ to 2 s.f}$$

2006/50 Neco (Nov)

A man at the top of a building 25m high looks down at an object 30m away from the foot of the building. Calculate the angle of depression of the object from the man

A 19.9^0 B 39.8^0 C 50.2^0 D 70.1^0 E 79.6^0

Solution



Angle of depression is equal to angle of elevation.

The side relevant to θ are opp and adj (TOA)

$$\tan\theta = \frac{25}{30} = 0.8333$$

$$\theta = \tan^{-1} 0.8333$$

$$\theta = 39.804$$

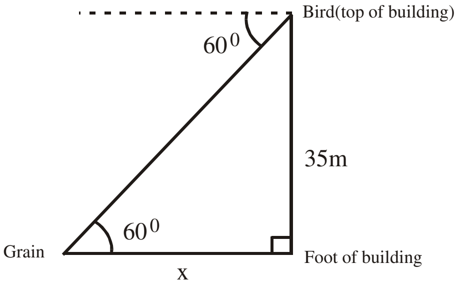
$$\approx 39.8^0 \quad (\text{B})$$

2013/49 Neco

A bird on top of a building roof edge 35m high sighted a grain on a level ground at angle 60^0. How far is the grain to the foot of the building?

A 17.50m B 20.21m C 40.42m D 70.00m E 159.79m

Solution



The sides relevant to 60^0 are Opp and Adj (TOA)

$$\tan 60^0 = \frac{35}{x}$$

$$x \tan 60^0 = 35$$

$$x = \frac{35}{\tan 60}$$

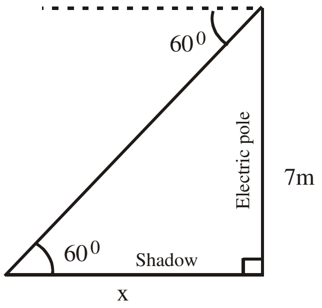
$$= 20.21\text{m} \quad (\text{B})$$

2010/50 Neco

An erected electric pole is 7m high from the ground. It casts a shadow on the horizontal ground when the altitude of the sun is 60^0. Calculate the length of the shadow.

A $\frac{7\sqrt{3}}{3}$ m B 7m C $7\sqrt{3}$ m D 21m E $21\sqrt{3}$ m

Solution



The sides relevant to 60^0 are Opp and Adj (TOA)

$$\tan 60^0 = \frac{7}{x}$$

$$x \tan 60^0 = 7$$

$$x = 7 \div \tan 60$$

$= 7 \div \sqrt{3}$

$= 7 \times \frac{1}{\sqrt{3}}$ i.e $\frac{7}{\sqrt{3}}$

Rationalizing

$= \frac{7}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{7\sqrt{3}}{3}$ m (A)

2010/2a

The angle of depression of a boat from the mid – point of a vertical cliff is 35^0. If the boat is 120m from the foot of the cliff, calculate the height of the cliff

Solution

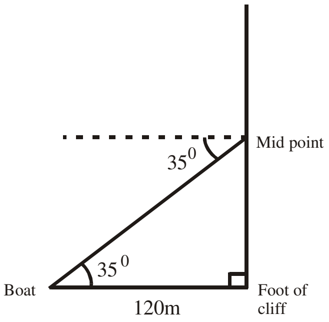


Fig I

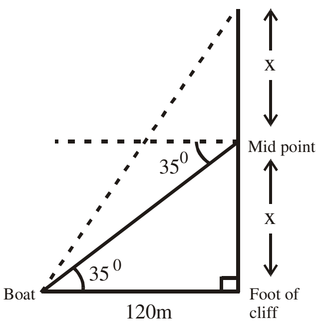


Fig II

From fig II, in Δ with 35^0,
the relevant sides to 35^0 are Opp and Adj (TOA)

$\tan 35^0 = \frac{x}{120}$

$120 \tan 35^0 = x$

$84.02\text{m} = x$

Height of cliff = x + x

= 84.02 + 84.02

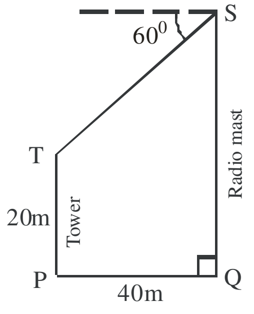
= 168.04m

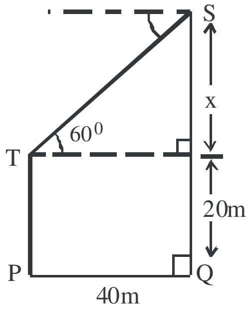
2008/49 Neco

From the top S of a radio mast QS, the angle of depression of the top T of a 20 metres tower PT is 60^0. If P and Q are on level ground and /PQ/ = 40m, find the height of the mast to the nearest metres.

A 40m B 43m C 69m D 89m E 120m

Solution





Height of mast is 20m + x
In triangle with 60^0,
The relevant sides are Opp and Adj (TOA)

$\tan 60^0 = \frac{x}{40}$

$40 \tan 60^0 = x$

$69.28\text{m} = x$

Height of mast = 20 + x

= 20 + 69.28

= 89. 28m ≈ 89m (D)

2013/5b Neco Exercise 14.10

A boat is on the same horizontal level as the foot of a cliff and the angle of depression of the boat from the top of the cliff is 60^0. If the boat is 150m away from the foot of the cliff, find the height of the cliff correct to three significant figures.

2006/32 (Nov) Exercise 14.11

A man on top of a cliff 100m high observes that the angle of depression of a boat at sea is 15^0. How far is the boat from the foot of the cliff?

A 268m B 373m C 425m D 500m

2013/25 Exercise 14.12

An object is 6m away from the base of a mast. The angle of depression of the object from the top of the mast is 50^0. Find, correct to 2 decimal places, the height of the mast

A 8.60m B 7.51m C 7.15m D 1.19m

2011/ 49 Exercise 14.13

A boy looks through a window of a building and sees a mango fruit on the ground 50m away from the foot of the building. If the window is 9m from the ground, calculate, correct to the nearest degree, the angle of depression of the mango from the window

A 9^0 B 10^0 C 11^0 D 12^0

2005/13 Exercise 14.14

The angle of depression of a boat at sea from the top of a cliff is 72^0. What is the angle of elevation of the top of the cliff from the boat?

A 18^0 B 36^0 C 72^0 D 90^0

2015/19 Neco Exercise 14.15

The angle of depression of a fowl on the ground from the top of a pole is 30^0. If the distance from the fowl to the foot of the pole is 63m, calculate the height of the pole and leave your answer in surd form.

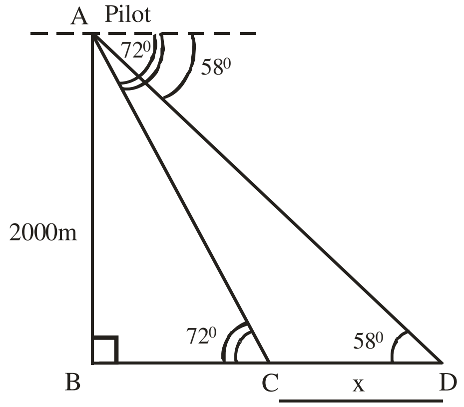
A $7\sqrt{3}$ m B 21m C $21\sqrt{3}$ m D 63m E $63\sqrt{3}$ m

Double triangles cases

2009/13b (Nov)

The pilot of an air craft 2,000 metres above the sea level observes at an instance that the angles of depression of two boats which are in direct straight line are 58^0 and 72^0. Find, correct to the nearest metres, the distance between the two boats.

Solution



The distance between the two boats is x = BD – BC

In ΔABC, $\tan 72^0 = \frac{2000}{BC}$

$BC = \frac{2000}{\tan 72^0}$

In ΔABD, $\tan 58^0 = \frac{2000}{BD}$

$BD = \frac{2000}{\tan 58^0}$

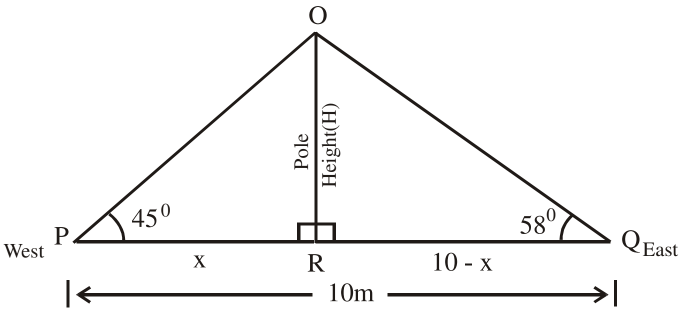
$$x = BD - BC$$
$$x = \frac{2000}{\tan 58^0} - \frac{2000}{\tan 72^0}$$
$$= 1249.74 - 649.84$$
$$= 599.9\text{m}$$
$$\approx 600\text{m to nearest metres}$$

2014/10b (Nov)

From a point P on a level ground and directly west of a pole, the angle of elevation of the top of the pole is 45^0 and from point Q east of the pole, the angle of elevation of the top of the pole is 58^0. If /PQ/ = 10m, calculate, correct to 2 significant figures, the:

(i) distance from P to the pole; (ii) height of the pole

Solution



In ΔOPR, $\tan 45^0 = \frac{H}{x}$

$H = x \tan 45^0$ -------- (i)

Also in Δ OQR, $\tan 58^0 = \frac{H}{10 - x}$

$H = \tan 58(10 - x)$ --------- (2)

Equating (1) to (2)

$H \Rightarrow x \tan 45^0 = \tan 58(10 - x)$

$x \times 1 = 1.6\ (10 - x)$

$x = 16 - 1.6x$

$x + 1.6x = 16$

$2.6x = 16$

$x = \frac{16}{2.6}$ i.e 6.15m

$x = 6.15\text{m} \approx 6.2\text{m}$ to 2 s.f

(ii) Substitute x value into (1)

$H = x \tan 45^0$ becomes

$= 6.15 \times 1$

$\approx 6.2\text{m}$ to 2 s.f

Alternatively

Substitute x value into (2)

$H = \tan 58^0\ (10 - x)$ becomes

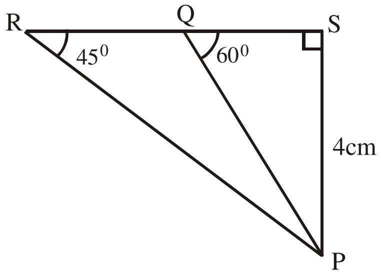
$H = \tan 58^0\ (10 - 6.15)$

Don't use approximated values for substitutions

$= \tan 58^0\ (3.85)$

$= 6.16\text{m} \approx 6.2\text{m}$ to 2 s.f

2005/11 Neco



In the figure above, the angles of depression of P from Q and R are 60^0 and 45^0 respectively. If /PS/ = 4m, find the length of RQ

A 4.0m B 3.2m C 2.3m D 1.7m E 0.8m

Solution

$RQ = RS - QS$

In ΔPRS, $\tan 45^0 = \frac{4}{RS}$

$RS \tan 45^0 = 4$

$RS = \frac{4}{\tan 45} = 4\text{m}$

In ΔPQS, $\tan 60^0 = \frac{4}{QS}$

$QS \tan 60^0 = 4$

$QS = \frac{4}{\tan 60^0} = 2.309\text{m}$

Thus, $RQ = RS - QS$ RQ

$= 4 - 2.309$

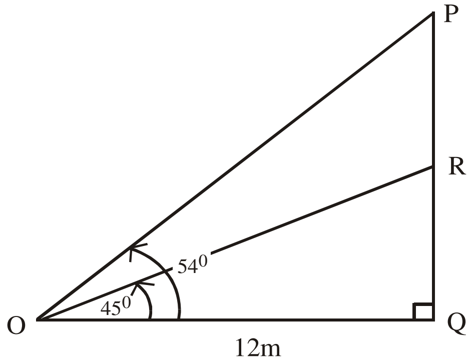
$= 1.691\text{m}$

$= 1.7\text{m}$ (D)

2005/6a NABTEB (Nov)

A flag – pole is placed on the top of a tower. The angles of elevation of the bottom and top of the flag – pole from a point on the ground 12m away are 45^0 and 54^0 respectively. Find the length of the flag – pole to 2 decimal places

Solution



To find PR = PQ – QR

In ΔOQR, $\tan 45^0 = \frac{QR}{12}$

$12 \tan 45^0 = QR$

$12\text{m} = QR$

In ΔOPQ, $\tan 54^0 = \frac{PQ}{12}$

$12 \tan 54^0 = PQ$

$16.5166\text{m} = PQ$

Thus $PR = PQ - QR$

$= 16.5166 - 12$

$= 4.5166\text{m}$

$\approx 4.52\text{m}$ to 2 d.p

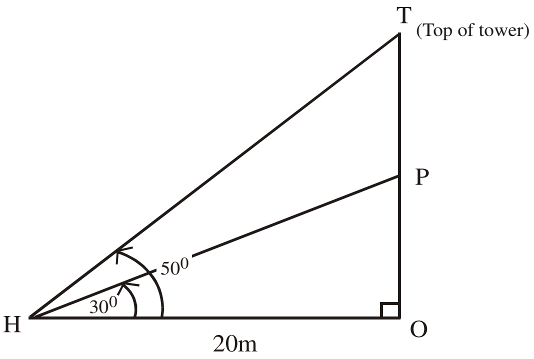
2012/8 *different case*

A point H is 20m away from the foot of a tower on the same horizontal ground. From the point H, the angle of elevation of the point (P) on the tower and the top (T) of the tower are 30^0 and 50^0 respectively. Calculate correct to 3 significant figures (a) /PT/

(b) The distance between H and the top of the tower;

(c) The position of H if the angle of depression of H from the top of the tower is to be 40^0

Solution



(a) To find PT, we solve for OP and OT and subtract

$$\text{In } \Delta OPH,\ \tan 30^0 = \frac{OP}{20}$$

$$20 \tan 30^0 = OP$$

$$11.547m = OP$$

$$\text{Also in } \Delta OTH,\ \tan 50^0 = \frac{OT}{20}$$

$$20 \tan 50^0 = OT$$

$$23.835m = OT$$

Thus PT = OT – OP

= 23.835 – 11.547

= 12.288m ≈ 12.3m to 3sf

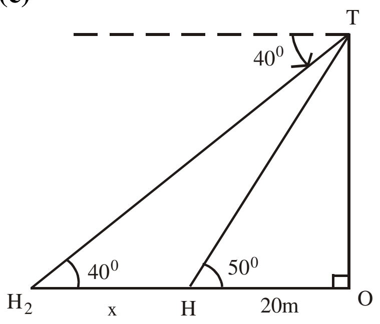
(b) $\text{In } \Delta OTH,\ \cos 50^0 = \dfrac{20}{HT}$

$$HT \cos 50^0 = 20$$

$$HT = \frac{20}{\cos 50}$$

$$= 31.114m \quad \approx 31.1m \text{ to 3s.f}$$

(c)



$$\text{In } \Delta OTH_2,\ \tan 40^0 = \frac{OT}{x+20}$$

$$\tan 40^0(x + 20) = 23.835$$

$$0.8391x + 16.782 = 23.835$$

$$0.8391x = 23.835 - 16.782$$

$$0.8392x = 7.053$$

$$x = \frac{7.053}{0.8391} = 8.405 \approx 8.41m \text{ to 3s.f m}$$

The position of **H** would be (8.41 + 20)m i.e 28.41m away from the foot of the tower

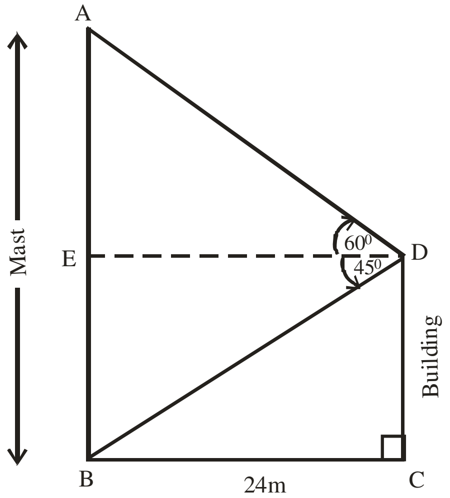
2009/7b Neco

A vertical mast is erected 24m away from a building. From the top of the building, the angle of elevation and depression of the top and foot of the mast are 60^0 and 45^0 respectively. Calculate the:

(i) height of the mast

(ii) distance between the top of the building and the top of the mast (correct all answers to the nearest metres)

Solution



(i) Height of the mast AB = AE + BE

$$\text{In } \Delta ADE,\ \tan 60^0 = \frac{AE}{24}$$

$$AE = 24 \tan 60^0$$

$$= 24 \times 1.732 = 41.57m$$

$$\text{In } \Delta ABE, \quad \tan 45^0 = \frac{BE}{24}$$

$$BE = 24 \tan 45^0$$

$$= 24 \times 1.0 = 24m$$

Thus AB = AE + BE

= 41.57 + 24 = 65.57m ≈ 66m to nearest metres

(ii) AD = ?

$$\text{In } \Delta ADE, \quad \cos 60^0 = \frac{24}{AD}$$

$$AD = \frac{24}{\cos 60^0} = \frac{24}{0.5}$$

$$= 48m$$

2005/13a

Two towers XA and YB on the same level are 200m apart. If their heights are 150m and 85m respectively, calculate the

(i) angle of elevation of the top of XA from the top of YB;

(ii) distance between the tops of the two towers, correct to the nearest metres

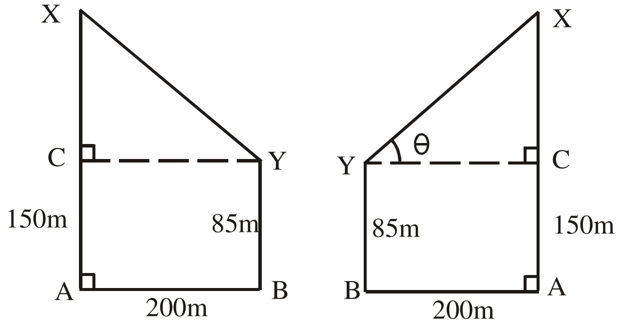


Fig I **fig II**

(i) Fig I will give us headache to find θ

From Fig II in ΔCXY,
the sides relevant to θ are Opp and Adj (TOA)

$$\tan\theta = \frac{XC}{YC} \quad (XC = 150 - 85) \text{ and } (YC = AB)$$

$$\tan\theta = \frac{65}{200} \text{ i.e } 0.325$$

$$\theta = \tan^{-1} 0.325$$
$$= 18^0$$

(ii) XY can be gotten from ΔCXY by Pythagoras rule

$$XY^2 = XC^2 + YC^2$$
$$= 65^2 + 200^2$$
$$XY^2 = 4225 + 40000$$
$$XY = \sqrt{44225}$$
$$= 210.297\text{m}$$
$$\approx 210\text{m to the nearest metres}$$

2015/10b Exercise 14.16
From an aeroplane in the air and at a horizontal distance of 1050m, the angles of depression of the top and base of a control tower at an instance are 36^0 and 41^0 respectively. Calculate, correct to the nearest metre, the
(i) height of the control tower;
(ii) shortest distance between the aeroplane and the base of the control tower

2007/11b Exercise 14.17
From two points on opposite sides of a pole 33m high, the angles of elevation of the top of the poles are 53^0 and 67^0. If the two points and the base of the pole are on the same horizontal level, calculate, correct to three significant figures, the distance between the two points.

2008/47 Neco Exercise 14.18
Two ladders of length 5m and 7m lean against a pole and make angles 45^0 and 60^0 with the ground respectively. What is their distance apart on the pole correct to two decimal places?
A 9.60m B 6.06m C 2.54m D 2.53m E 2.00m

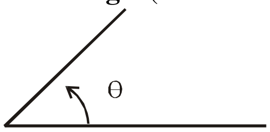
CHAPTER FIFTEEN

Bearing & Elements Of Plane Geometry

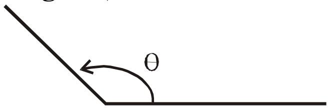
Elements of plane geometry

Angles
The units of measuring angles are degrees and minutes.
1 degree = 60 minutes
i.e. $1^0 = 60^1$
Sometimes, smaller units like seconds are used;
1 minute = 60 seconds
i.e. $1^1 = 60^{11}$

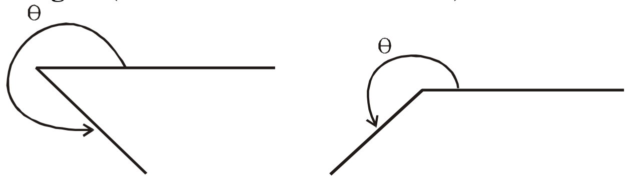
Acute angle (i.e. θ between 0^0 and 90^0)



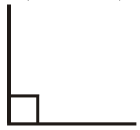
Obtuse angle (i.e. θ between 90^0 and 180^0)



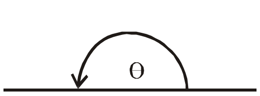
Reflex angle (i.e. θ between 180^0 and 360^0)



Others
Right angle (i. e. 90^0)



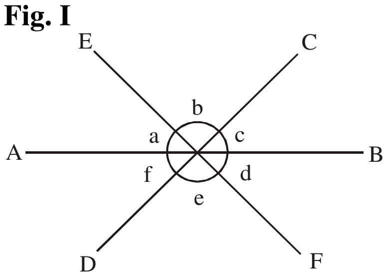
Straight line angles is 180^0



SOME BASIC FACTS ON THE FORMATION OF ANGLES BETWEEN LINES

STRAIGHT LINES

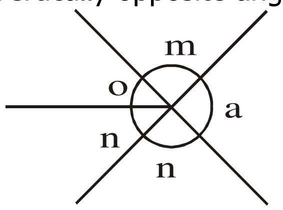
(i) Sum of angles on a straight line is 180^0
Fig. I



From Fig. I, the following holds :
$a + b + c = 180^0$ on line AB
$b + c + d = 180^0$ on line EF
$f + e + d = 180^0$ on line AB
$a + f + e = 180^0$ on line EF

(ii) Sum of angles at a point is 360^0
From Fig. I
$a + b + c + d + e + f = 360^0$

(iii) Vertically opposite angles are equal



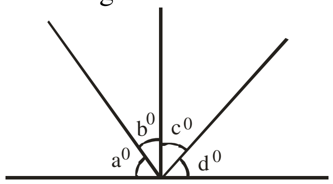
m = n (vertically opposite angles)
o + n = a (vertically opposite angles)

Problems on angles formed between straight lines

2014/51 Neco (Dec)

From the diagram below, $b = \frac{1}{2}d$, $c = \frac{1}{3}a$ and $d = 32^0$.

Find the angle marked a^0



A 9 B 33 C 48 D 99 E 132

Solution

If $b = \frac{1}{2}d$ then $2b = d$

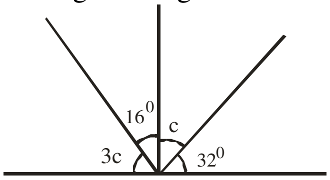
If $c = \frac{1}{3}a$ then $3c = a$

and $d = 32^0$

$2b = 32^0$

$b = 16^0$

Re- labeling our diagram



Thus, the marked angle $a^0 = 3c$

$3c + 16 + c + 32^0 = 180^0$ (sum of angles on a straight line)

$4c + 48 = 180$

$4c = 180 - 48$

$4c = 132$

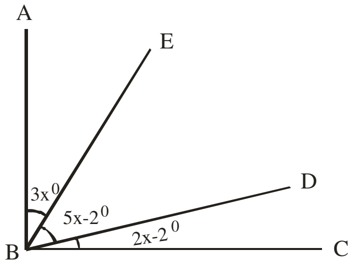
$c = {}^{132}/_4$ i.e 33

and $3c = 33 \times 3$

$= 99^0$ D.

2014/50 Neco

In the figure below, $\angle ABC$ is a right angle. Find the value of $\angle EBD$ correct to one decimal place



A 21.6^0 B 26.4^0 C 42.0^0 D 65.6^0 E 68.4^0

Solution

We are to find $5x - 2$

$3x + 5x - 2 + 2x - 2 = 90^0$ (sum of angles of a right–angle)

$10x - 4 = 90$

$10x = 90 + 4$

$10x = 94$

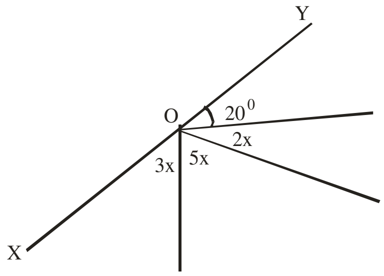
$x = 9.4^0$

Thus, $5x - 2 = 5(9.4) - 2$

$= 47 - 2$

$= 45^0$

2011/38 Neco



If XOY is a straight line, find the value of x

A 10^0 B 16^0 C 18^0 D 19^0 E 25^0

Solution

$20 + 2x + 5x + 3x = 180^0$ (sum of angles on straight line)

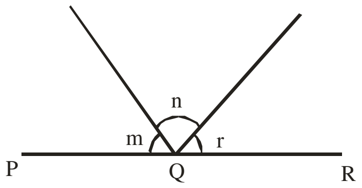
$20 + 10x = 180$

$10x = 180 - 20$

$10x = 160$

$x = {}^{160}/_{10}$ i.e 16^0 B.

2014/20



In the diagram, PQR is a straight line, $(m + n) = 120^0$ and $(n + r) = 100^0$. Find $(m + r)$

A 110^0 B 120^0 C 140^0 D 160^0

Solution

$m + n + r = 180^0$ (sum of $\angle$s on a straight line)

But $m + n = 120^0$

$120 + r = 180^0$

$r = 180 - 120$

$r = 60^0$

Substituting for r value into

$n + r = 100$

$n + 60 = 100$

$n = 100 - 60$

$n = 40^0$

Substituting for n value into

$m + n = 120$

$m + 40 = 120$

$m = 120 - 40$

$m = 80^0$

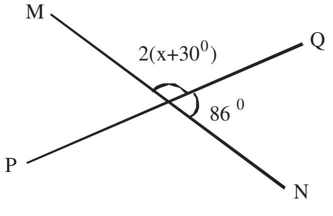
Thus $m + r = 80 + 60$

$= 140^0$ C.

2005/16

In the diagram, PQ and MN are straight lines.

Find the value of x



A 13^0 B 17^0 C 28^0 D 30^0

Solution

$2(x + 30) + 86 = 180^0$ (sum of angles on a straight line)

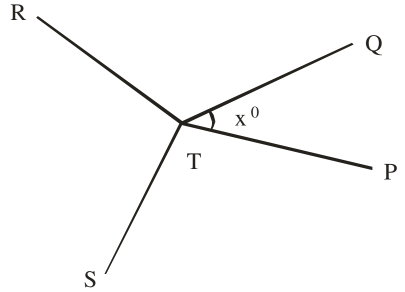
$2x + 60 + 86 = 180$

$2x = 180 – 146$

$2x = 34$

$x = {}^{34}/_2 = 17^0$ B.

2005/4

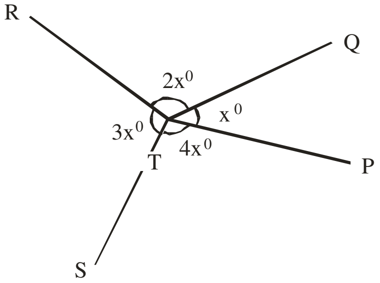


In the diagram, $\angle PTQ = x^0$, $\angle QTR$ is twice as big as $\angle PTQ$, $\angle RTS$ is three times as big as $\angle PTQ$ and $\angle PTS$ is four times as big as $\angle PTQ$. Find $\angle RTS$

A 54^0 B 72^0 C 108^0 D 144^0

Solution

Let us label the diagram as instructed



We are to find 3x

$x + 2x + 3x + 4x = 360^0$ (sum of angles at a point)

$10x = 360$

$x = {}^{360}/_{10}$ i.e 36

Thus $3x = 3 \times 36$ i.e 108^0 C.

2006/44

The value of three angles at a point are $3y – 45^0$, $y + 25^0$ and y^0. Find the value of y.

A 40^0 B 58^0 C 68^0 D 76^0

Solution

$3y – 45 + y + 25 + y = 360^0$ (sum of angles at a point)

$5y – 20 = 360$

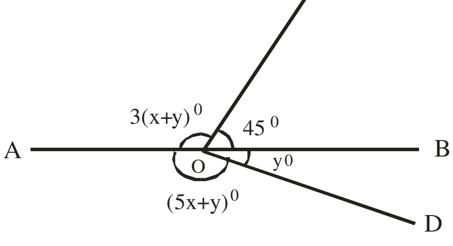
$5y = 360 + 20$

$5y = 380$

$y = {}^{380}/_5$

$y = 76^0$ D.

2007/11a



In the diagram, AOB is a straight line, $\angle AOC = 3(x + y)^0$, $\angle COB = 45^0$, $\angle AOD = (5x + y)^0$ and $\angle DOB = y^0$. Find the value of x and y

Solution

$3(x + y) + 45 = 180^0$ ----- (1) (sum of angles on line AB)

$(5x + y) + y = 180^0$ ----- (2) (sum of angles on line AB)

From (1) $3x + 3y + 45 = 180$

$3x + 3y = 180 – 45$

$3x + 3y = 135$

Divide through by 3

$x + y = 45$ -------- (a)

Also from (2) $5x + y + y = 180$

$5x + 2y = 180$ -------- (b)

(a) × 2 and subtract

$$\begin{array}{r} 5x + 2y = 180 \\ -\ (2x + 2y = 90) \\ \hline 3x \qquad = 90 \end{array}$$

$x = {}^{90}/_3$

$x = 30^0$

Substituting x value into (a)

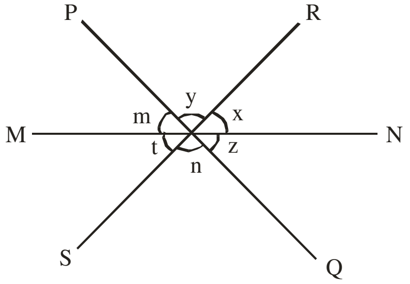
$x + y = 45$ becomes

$30 + y = 45$

$y = 45 – 30$

$y = 15^0$

2012/23



In the diagram, MN, PQ and RS are three intersecting straight lines.Which of the following statement(s) is/are **true**

I $t = y$ **II** $x + y + z + m = 180^0$

III $x + m + n = 180^0$ **IV** $x + n = m + z$

A I and IV only B II only C III only D IV only

Solution

Option I : t can only be equal to x (Vertically opp. ∠s) and not $t = y$

Option II: $x + y + z = 180^0$ (sum of ∠s on line PQ) and not $x + y + z + m = 180^0$

we can accept $x + y + m = 180^0$ and others as $z = m$ (vertically opp ∠s)

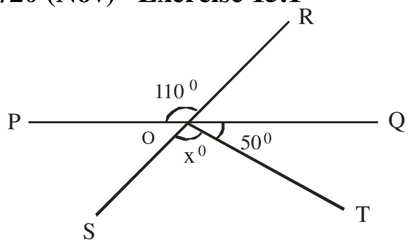
Option III: $x + m + n = 180^0$ (sum of angles on line MN) True as $x + m + y = 180^0$

Here $y = n$ (vertically opp angles)

Option IV : $x + n = m + z$ (not true)

$x + n = m + t$ or $z + t$ (vertically opp angles)

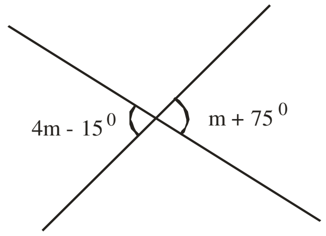
2009/20 (Nov) Exercise 15.1



In the diagram, POQ, ROS and OT are straight lines. $\angle POR = 110^0$ and $\angle QOT = 50^0$. Calculate the value of x

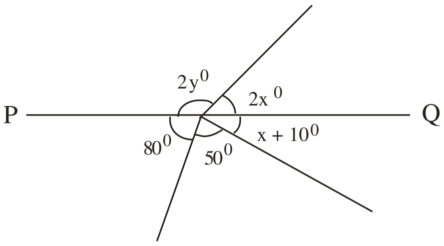
A 70^0 B 60^0 C 55^0 D 44^0

2010/10 Exercise 15.2



What is the value of m in the diagram?
A 20^0 B 30^0 C 40^0 D 50^0

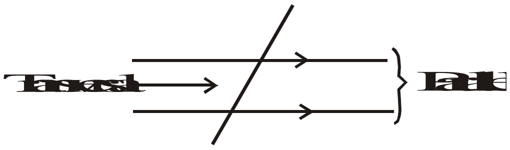
2013/39 Exercise 15.3



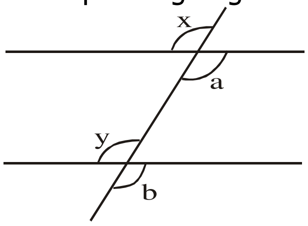
In the diagram, PQ is a straight line. Calculate the value of the angle labeled $2y^0$
A 130^0 120^0 C 110^0 D 100^0

PARALLEL LINES

Two lines are parallel; if they *do not* and *can never* meet (intersect) when their length is increased. Any line cutting across a pair of parallel lines is called transversal.



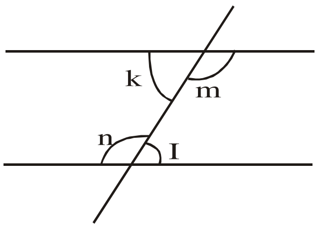
(i) Corresponding angles are equal



x = y (corresponding angles)
a = b (corresponding angles)

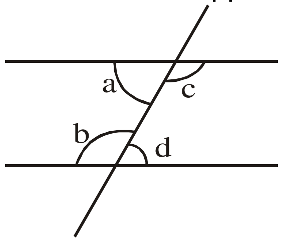
In corresponding angles, one angle is inside while the other angle is outside and both angles are on the same side.

(ii) Alternate angles are equal



k = I (alternate angles)
m = n (alternate angles)
Under alternate angles both angles are within the parallel lines but on different sides.

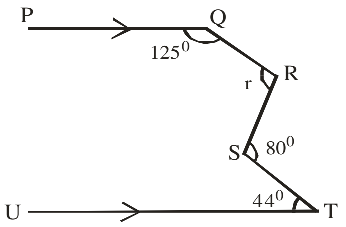
(iii) Sum of interior opposite angles is 180^0



$a + b = 180^0$ (sum of interior opposite angles)
$c + d = 180^0$ (sum of interior opposite angles)

Problems on angles formed between parallel lines

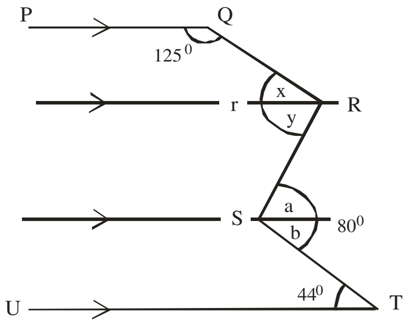
2010/ 3a



In the diagram, $\angle PQR = 125^0$, $\angle QRS = r$, $\angle RST = 80^0$ and $\angle STU = 44^0$. Calculate the value of r

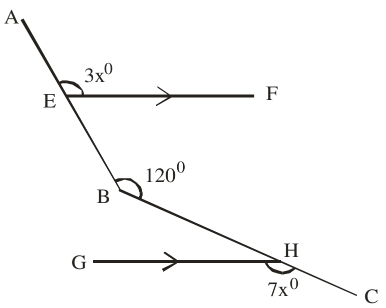
Solution

Produce parallel lines at R and S and label as:



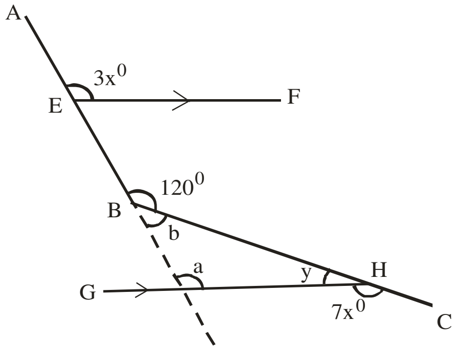
$b = 44^0$ (alternate angles)
$a + b = 80^0$
$a + 44 = 80$
$a = 80 - 44$
$a = 36^0$
$y = a$ i.e 36^0 (Alternate angles)
$125 + x = 180^0$ (sum of interior opp $\angle$s)
$x = 180 - 125 = 55^0$
Thus $r = x + y$
$= 55^0 + 36^0$
$= 91^0$

2014/1b



In the diagram, $\overline{EF}$ is parallel to $\overline{GH}$ .If $\angle AEF = 3x^0$, $\angle ABC = 120^0$ and $\angle CHG = 7x^0$, find the value of $\angle GHB$

Solution



$a = 3x^0$ (corresponding angles)

$b + 120 = 180^0$ (sum of angles on a straight line)

$b = 180 - 120 = 60^0$

$y + 7x = 180^0$ (sum of angles on a straight line)

$y = 180 - 7x$

$a + b + y = 180^0$ (sum of angles in a Δ)

$3x + 60 + (180 - 7x) = 180$

$-4x + 240 = 180$

$4x = 240 - 180$

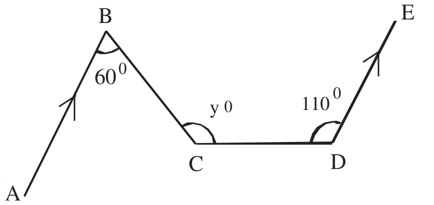
$x = {}^{60}/_4 = 15^0$

Next, we substitute x = 15 into $y = 180 - 7x$

$y = 180 - 7(15)$

$= 180 - 105 = 75^0$

2006/36 Neco

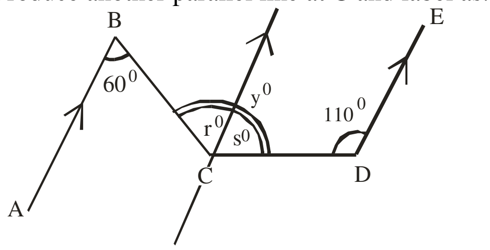


Find the value of y in the figure above in which AB//DE, $\angle ABC = 60^0$ and $\angle CDE = 110^0$

A 60^0 B 70^0 C 120^0 D 130^0 E 170^0

Solution

Produce another parallel line at C and label as:



$y^0 = r^0 + s^0$

$r^0 = 60^0$ (Alternate angles)

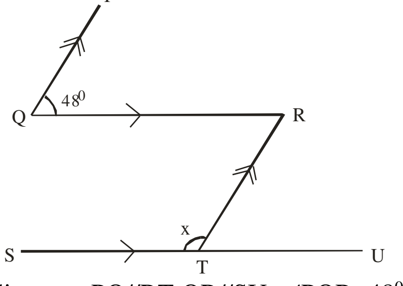
$s^0 + 110^0 = 180^0$ (sum of interior opp $\angle$s)

$s^0 = 180 - 110^0 = 70^0$

Thus $y^0 = r + s$

$= 60 + 70$ i.e 130^0 (D)

2015/18

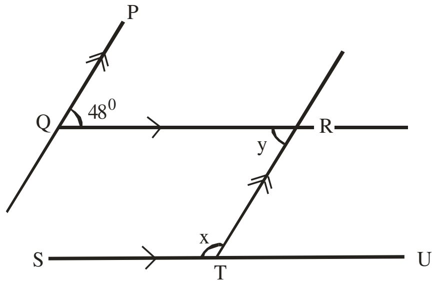


In the diagram, PQ//RT,QR//SU, $\angle PQR = 48^0$ and $\angle RTS = x$. find the value of x

A 134^0 B 132^0 C 96^0 D 48^0

Solution

To get a clear picture, produce the parallel lines a little longer and label as shown below:



$y = 48^0$ (alternate angles)

$y + x = 180^0$ (sum of interior opp. $\angle$s)

$48 + x = 180$

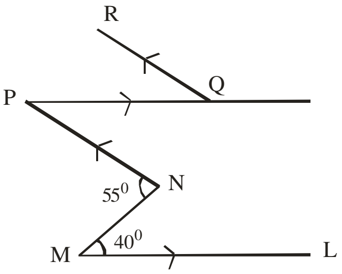
$x = 180 - 48$

$= 132^0$ (B)

2014/37 (Nov) NABTEB

In the diagram below, ML//PQ and NP//QR.

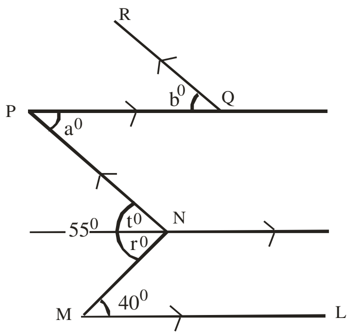
If $\angle LMN = 40^0$ and $\angle MNP = 55^0$, find $\angle PQR$



A 15^0 B 25^0 C 35^0 D 40^0

Solution

Produce another parallel line at N and label as shown below:



We are to find $\angle PQR$ i.e b^0

$r^0 = 40^0$ (Alternate angles)

$t^0 + r^0 = 55^0$

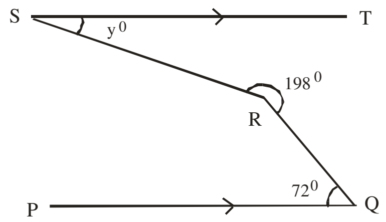
$t = 55^0 - 40^0$

$= 15^0$

$a^0 = t^0$ i.e 15^0 (Alternate angles)

$b^0 = a^0$ i.e 15^0 (Alternate angles) A.

2013/43

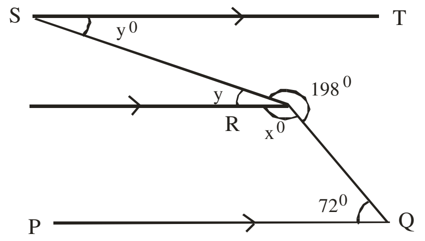


In the diagram, $\overline{ST} // \overline{PQ}$ reflex angle SRQ = 198^0 and $\angle RQP = 72^0$. Find the value of y^0

A 18^0 B 54^0 C 92^0 D 108^0

Solution

Produce another parallel line at R and label as shown below



$72 + x = 180^0$ (sum of interior opp $\angle$s)

$x = 180 - 72$

$= 108^0$

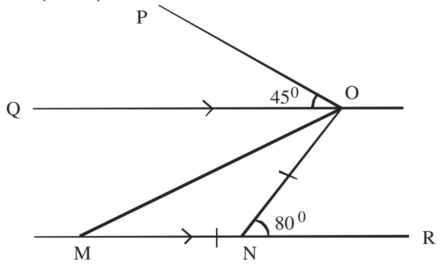
$x^0 + y + 198^0 = 360^0$ (sum of $\angle$s at a point)

$108 + y + 198 = 360$

$y = 360 - 306$ i.e 54^0

$y^0 = y$ i.e 54^0 (alternate angles) B.

2009/19 (Nov)

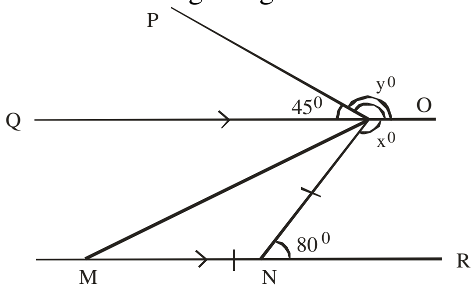


In the diagram, MR//QO, /MN/ = /NO/, $\angle ONR = 80^0$ and $\angle QOP = 45^0$. Find the size of the reflex $\angle PON$

A 135^0 B 235^0 C 280^0 D 315^0

Solution

Let us label our target angles as:



We are to find $x^0 + y^0$

$80 + x^0 = 180^0$ (sum of interior opp. $\angle$s)

$x^0 = 180 - 80$ i.e 100^0

$45^0 + y^0 = 180^0$ (sum of $\angle$s on straight line)

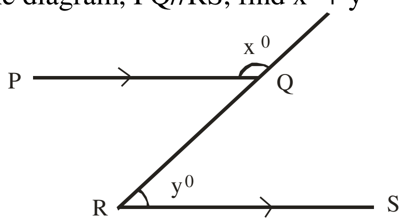
$y^0 = 180 - 45^0$ i.e 135^0

Thus, $x^0 + y^0 = 100 + 135$

$= 235^0$ (B)

2006/15

In the diagram, PQ//RS, find $x^0 + y^0$



A 360^0 B 300^0 C 270^0 D 180^0

Solution

$\angle PQR = y^0$ (alternate angle)

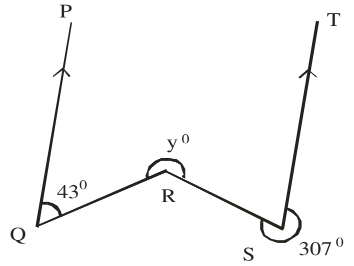
$x^0 + \angle PQR = 180^0$ (sum of angles on a straight line)

$x^0 + y^0 = 180^0$ D.

2006/3a Neco

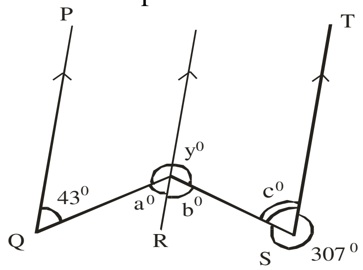
Find the value of y^0 in the diagram below.

If $\overline{QP}$ is parallel to $\overline{ST}$



Solution

Produce another parallel line at R and label as :



$c^0 + 307^0 = 360^0$ (sum of $\angle$s at a point)

$c^0 = 360^0 - 307^0$

$= 53^0$

$b^0 = c^0$ i.e 53^0 (alternate angles)

$a^0 = 43^0$ (alternate angles)

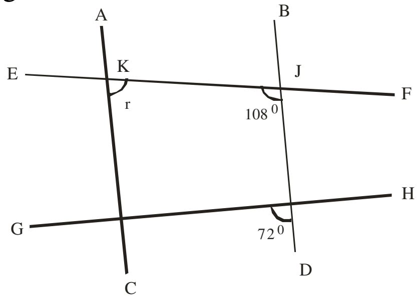
$a^0 + b^0 + y^0 = 360^0$ (sum of $\angle$s at a point)

$43 + 53 + y^0 = 360$

Thus, $y = 360 - 96$ i.e 264^0

2008/43 Neco

In the figure below, what is the value of the angle marked r?



A 36^0 B 72^0 C 108^0 D 144^0 E 160^0

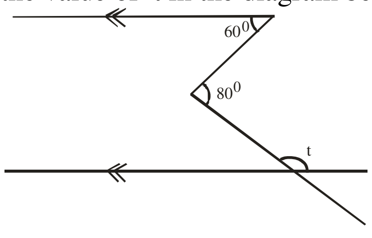
Solution

$r + 108^0 = 180^0$ (sum of interior opp. $\angle$s)

$r = 180 - 108 = 72^0$

2010/37 Neco

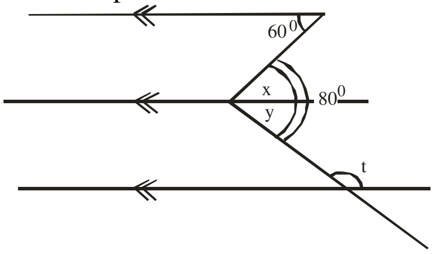
Find the value of t in the diagram below:



A 20^0 B 60^0 C 80^0 D 100^0 E 160^0

Solution

Produce another parallel line and label as shown below:



$x = 60^0$ (alternate angles)

$x + y = 80^0$

$y = 80 - 60$

$= 20^0$

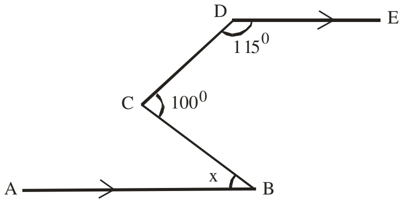
$y + t = 180^0$ (sum of interior opp. $\angle$s)

$t = 180 - 20$

$= 160^0$ E.

2006/33 Neco

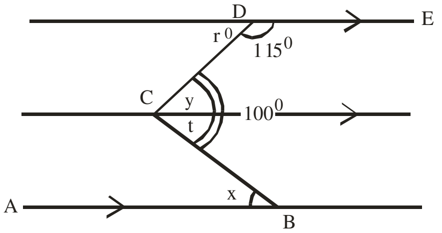
Find the value of x in the diagram below with AB//DE, $\angle CDE = 115^0$ and $\angle BCD = 100^0$



A 15^0 B 35^0 C 65^0 D 80^0 E 100^0

Solution

Produce another parallel line at C and extend DE and AB as:



$r + 115^0 = 180^0$ (sum of $\angle$s on a straight line)

$r = 180 - 115^0$

$= 65^0$

$y = r^0$ (alternate angles)

But $y + t = 100^0$

$65 + t = 100$

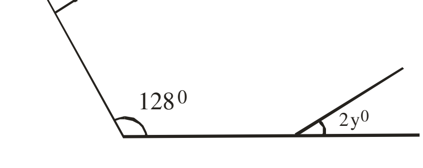
$t = 100 - 65$ i.e 35^0

$x = t$ (alternate angles)

$x = 35^0$ B.

2008/42 Neco

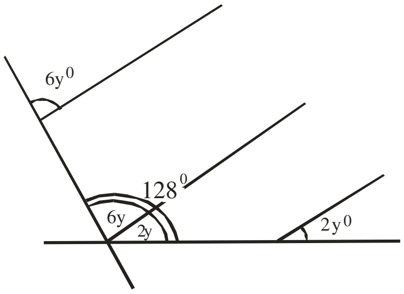
Find y in the figure below



A 16 B 21 C 32 D 96 E 120

Solution

Produce another parallel line at 128^0 as:



$6y^0 = 6y$ (corresponding angles)

$2y^0 = 2y$ (corresponding angles)

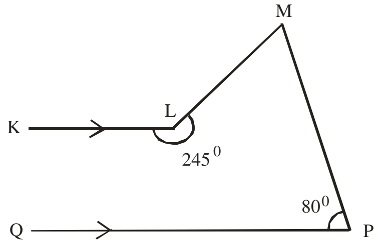
$6y + 2y = 128^0$

$8y = 128$

$y = \frac{128}{8}$

$y = 16^0$ (A)

2005/52 Neco

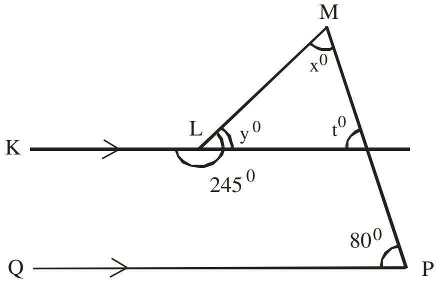


In the figure above, $\overline{LK} // \overline{PQ}$. Reflex angle KLM = 245^0 and angle QPM = 80^0. What is the value of angle LMP?

A 35^0 B 65^0 C 80^0 D 100^0 E 115^0

Solution

Produce KL to N and label as:



$t^0 = 80^0$ (corresponding angles)

$\angle KLM + 245^0 = 360^0$ (sum of angles at a point)

$\angle KLM = 360 - 245$

$= 115^0$

$\angle KLM + y^0 = 180^0$ (sum of angles on a straight line)

$115 + y^0 = 180$

$y = 180 - 115$ i.e 65^0

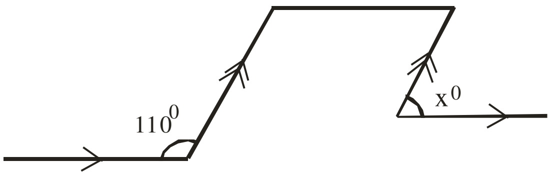
$\angle LMP(x^0) + y^0 + t^0 = 180^0$ (sum of $\angle$s in Δ)

$x^0 + 65 + 80^0 = 180^0$

$x^0 = 180 - 145$

$x^0 = 35^0$ A.

2007/19

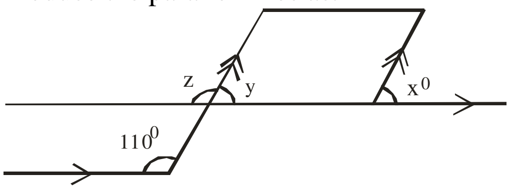


From the diagram, find the value of x

A 80^0 B 70^0 C 55^0 D 35^0

Solution

Produce the parallel lines as:



$z = 110^0$ (corresponding angles)

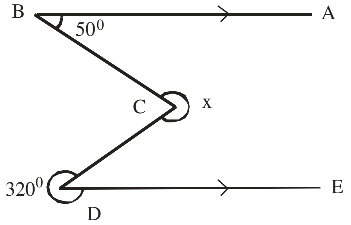
$z + y = 180^0$ (sum of angles on a straight line)

$110 + y = 180$

$y = 180 - 110 = 70^0$

$x^0 = y$ i.e 70^0 (corresponding angle) B.

2005/53 Neco Exercise 15.4

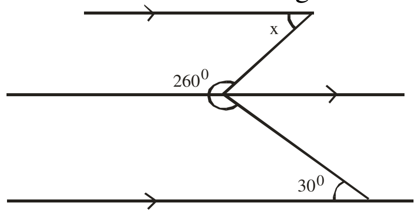


In the above figure, $\overline{BA} // \overline{DE}$. Reflex $\angle CDE = 320^0$, $\angle CBA = 50^0$. Find the value of x

A 40^0 B 130^0 C 140^0 D 270^0 E 320^0

2009/46 Exercise 15.5

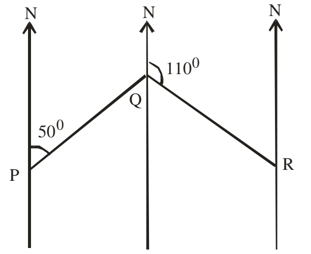
Find the value of x in the diagram



A 50^0 B 60^0 C 70^0 D 80^0

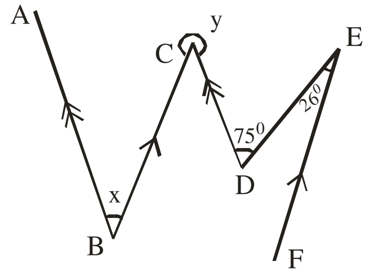
2006/27 Exercise 15.6

In the diagram, P, Q and R as three points in a plane such that the bearing of R from Q is 110^0 and the bearing of Q from P is 050^0. Find angle PQR



A 60^0 B 70^0 C 120^0 D 160^0

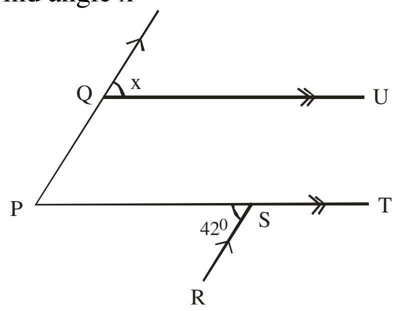
2008/11a Exercise 15.7



In the diagram, AB//CD and BC//FE. $C\hat{D}E = 75^0$ and $D\hat{E}F = 26^0$. Find the angle market x and y

2008/22 Exercise 15.8

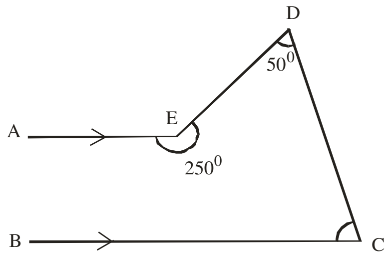
In the diagram, PQ//RS, QU//PT and $\angle PSR = 42^0$. Find angle x



A 84^0 B 48^0 C 42^0 D 32^0

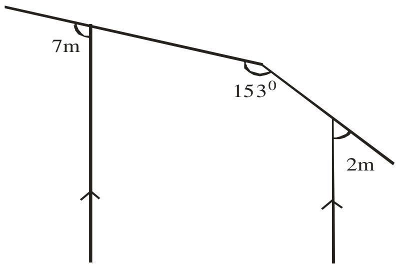
2009/44 Neco (Dec) Exercise 15.9

In the figure below, AE //BC. Reflex angle AED = 250^0 and angle EDC = 50^0. What is the value of angle BCD?



A 20^0 B 50^0 C 60^0 D 70^0 E 110^0

2013/22 Exercise 15.10

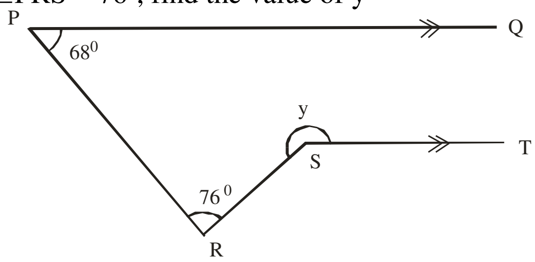


Find the value of m in the diagram

A 34^0 B 27^0 C 23^0 D 17^0

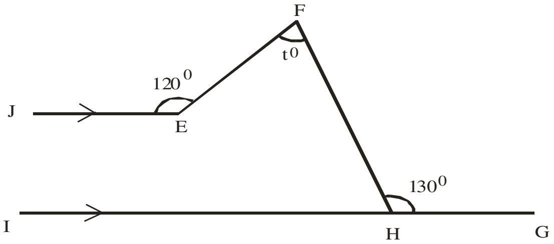
2007/39 Neco Exercise 15.11

In the figure below, PQ//ST, $\angle QPR = 68^0$ and $\angle PRS = 76^0$, find the value of y



A 252^0 B 224^0 C 216^0 D 192^0 E 144^0

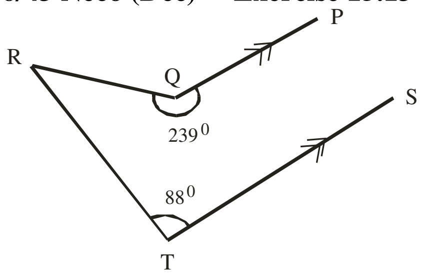
2009/21 Exercise 15.12



In the diagram, IG is parallel to JE, $J\hat{E}F = 120^0$ and $F\hat{H}G = 130^0$. Find the angle marked t.

A 40^0 B 70^0 C 80^0 D 100^0

2006/43 Neco (Dec) Exercise 15.13

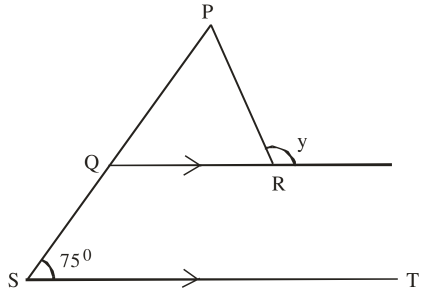


In the diagram above, QP//TS, reflex angle PQR = 239^0 and $\angle STR = 88^0$. What is the value of $\angle QRT$?

A 33^0 B 59^0 C 88^0 D 121^0 E 147^0

Problems on angles formed between lines in Triangles, Quadrilaterals

2010/11



In the diagram, QR//ST, /PQ/ = /PR/ and $\angle PST = 75^0$. Find the value of y

A 105^0 B 110^0 C 130^0 D 150^0

Solution

$\angle PQR = 75^0$ (corresponding angles)

Also $\angle PRQ = 75^0$ (base angle of isosceles ΔQPR)

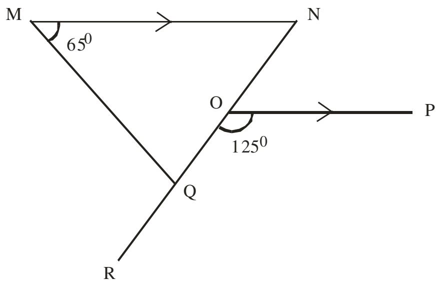
$\angle PRQ + y = 180^0$ (sum of angles on a straight line)

$75 + y = 180$

$y = 180 - 75$

$= 105^0$ A.

2011/23



In the diagram, MN//OP, $\angle NMQ = 65^0$ and $\angle QOP = 125^0$. What is the size of $\angle MQR$?

A 110^0 B 120^0 C 130^0 D 160^0

Solution

$\angle NOP + 125^0 = 180^0$ (sum of angles in straight line)

$\angle NOP = 180 - 125$

$= 55^0$

$\angle MNQ = \angle NOP$ (alternate angles)

$65^0 + \angle MNQ + \angle MQN = 180^0$ (sum of $\angle$s in Δ QMN)

$65 + 55 + \angle MQN = 180$

$\angle MQN = 180 - 120$

$= 60^0$

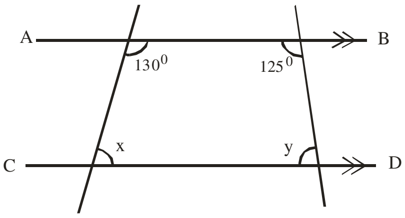
$\angle MQN + \angle MQR = 180^0$ (sum of $\angle$s on a straight line)

$60^0 + \angle MQR = 180^0$

$\angle MQR = 180 - 60$

$= 120^0$ B.

2012/4a Neco



If AB//CD in the above figure, find x and y

Solution

$x + 130^0 = 180^0$ (sum of interior opp. $\angle$s)

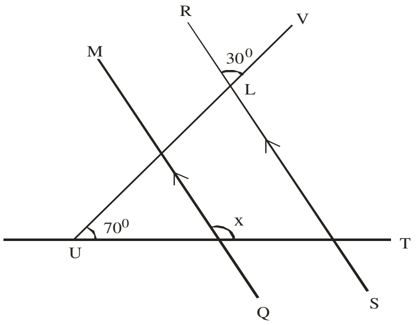
$x = 180 - 130$

$= 50^0$

$y + 125 = 180^0$ (sum of interior opp. $\angle$s)

$y = 180 - 125 = 55^0$

2012/22

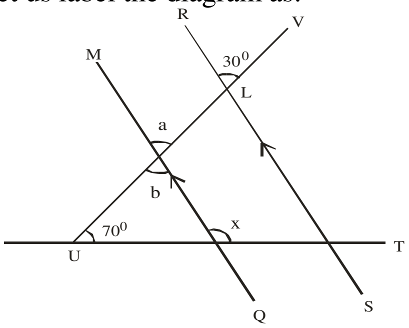


In the diagram, MQ//RS, $\angle TUV = 70^0$ and $\angle RLV = 30^0$. Find the value of x

A 150^0 B 110^0 C 100^0 D 95^0

Solution

Let us label the diagram as:



$a = 30^0$ (corresponding angles)

$b = a$ (vertically opp. angles)

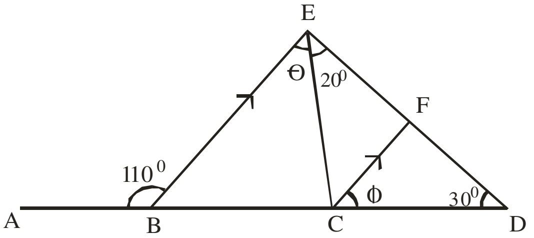
$70 + b = x$ (exterior angle equal to interior opp. $\angle$s)

$70 + 30 = x$

$100^0 = x$ C.

2012/38 Neco

In the diagram below BE//CF, determine the value of $\theta + \phi$



A 130^0 B 100^0 C 70^0 D 60^0 E 50^0

Solution

$110^0 + \angle EBD = 180^0$ (sum of angles on a straight line)

$\angle EBD = 180 - 110$

$= 70^0$

$\angle EBD + (\theta + 20^0) + 30^0 = 180^0$ (sum of $\angle$s in Δ EBD)

$70 + \theta + 20^0 + 30 = 180$

$\theta = 180 - 120$

$\theta = 60^0$

$\angle EBD + \theta + \angle ECB = 180^0$ (sum of $\angle$s in Δ BEC)

$70 + 60 + \angle ECB = 180^0$

$\angle ECB = 180 - 130$

$= 50^0$

$\angle ECF = \theta$ i.e 60^0 (alternate angles)

$\angle ECB + \angle ECF + \phi = 180^0$ (sum of $\angle$s on a straight line)

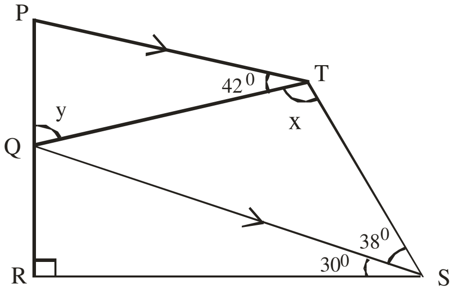
$50 + 60 + \phi = 180$

$\phi = 180 - 110$ i.e 70^0

Thus, $\theta + \phi = 60 + 70$

$= 130^0$ A.

2011/4a



In the diagram, PQRST is a quadrilateral. PT//QS, $\angle PTQ = 42^0$, $\angle TSQ = 38^0$ and $\angle QSR = 30^0$. If $\angle QTS$ = x and $\angle PQT$ = y, find : (i) x (ii) y

Solution

(i) $\angle TQS = 42^0$ (alternate angles)

$\angle TQS + x + 38^0 = 180^0$ (sum of $\angle$s in Δ QTS)

$42 + x + 38 = 180$

$x = 180 - 80$

$x = 100^0$

(ii) $\angle RQS + 90^0 + 30^0 = 180^0$ (sum of angle in Δ RQS)

$\angle RQS = 180 - 120$

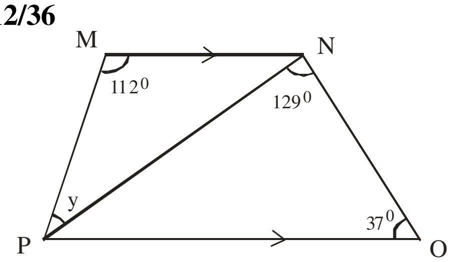
$= 60^0$

$y + \angle TQS + \angle RQS = 180^0$ (sum of $\angle$s on a straight line)

$y + 42 + 60 = 180$

$y = 180 - 102 = 78^0$

2012/36



In the diagram, MN//PO, $\angle PMN = 112^0$, $\angle PNO = 129^0$, $\angle NOP = 37^0$ and $\angle MPN$ = y. Find the value of y.

A 51^0 B 54^0 C 56^0 D 68^0

Solution

$(\angle MNP + 129^0) + 37^0 = 180^0$ (Sum of interior opp $\angle$s)

$\angle MNP + 129 + 37 = 180$

$\angle MNP = 180 - 166$

$= 14^0$

$y + 112^0 + \angle MNP = 180^0$ (Sum of $\angle$s in ΔPMN)

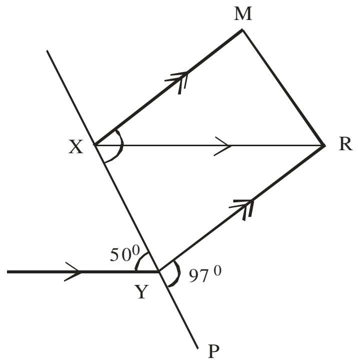
$y + 112 + 14 = 180^0$

$y = 180 - 126$

$= 54^0$ B.

2014/11 Neco

Find the value of $\angle XRY$ in the figure below



A 47^0 B 50^0 C 73^0 D 83^0 E 147^0

Solution

$\angle RYX + 97^0 = 180^0$ (Sum of angle on straight line)

$\angle RYX = 180 - 97^0$

$= 83^0$

$\angle YXR = 50^0$ (alternate angles)

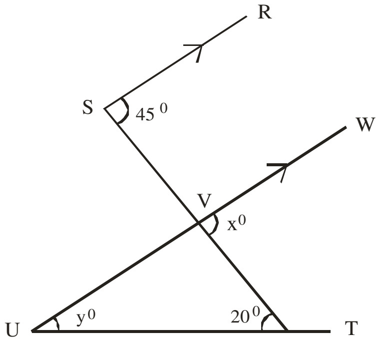
$\angle XRY + \angle YXR + \angle RYX = 180^0$ (Sum of $\angle$s in ΔYXR)

$\angle XRY + 50 + 83 = 180^0$

$\angle XRY = 180 - 133$

$= 47^0$ A.

2014 / 21 – 22



In the diagram, $\overline{SR}$ is parallel to $\overline{UW}$, $\angle WVT = x^0$, $\angle VUT = y^0$, $\angle RSV = 45^0$ and $\angle VTU = 20^0$ use this diagram to answer questions 21 and 22

21. Find the value of x

A 20 B 45 C 65 D 135

Solution

$x = 45^0$ (corresponding angles)

22. Calculate the value of y

A 20 B 25 C 45 D 65

Solution

$\angle UVT + x^0 = 180^0$ (sum of angles on a straight line)

$\angle UVT + 45 = 180$

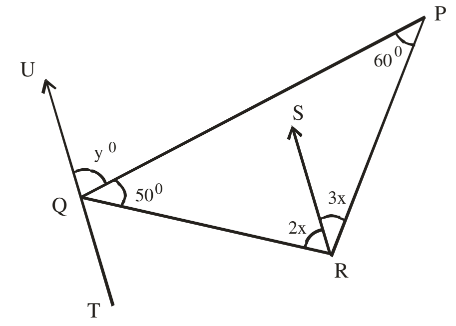
$\angle UVT = 180 - 45$

$= 135^0$

Next, $\angle UVT + 20 + y^0 = 180^0$ (sum of angles in Δ)

$135 + 20 + y^0 = 180$
$y^0 = 180 - 155$
$y^0 = 25$ B.

2008/45



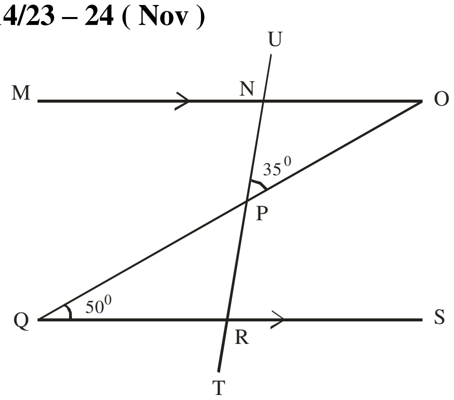
In the diagram, $\angle QPR = 60^0$, $\angle PQR = 50^0$, $\angle QRS = 2x^0$, $\angle SRP = 3x^0$, $\angle UQP = y^0$ and RS//TU. Calculate y.

A 102^0 B 78^0 C 70^0 D 60^0

Solution

$50^0 + 60^0 + (2x + 3x) = 180^0$(sum of angles in Δ)
$50 + 60 + 5x = 180$
$5x = 180 - 110$
$5x = 70$
$x = {}^{70}/_{5}$ i.e 14^0
$\angle TQR = 2x$ (alternate angles)
$\angle TQR = 2 \times 14$ i.e 28^0
$\angle TQR + 50 + y^0 = 180^0$ (sum of $\angle$s on a straight line)
$28 + 50 + y = 180$
$y = 180 - 78$
$y = 102^0$ A.

2014/23 – 24 (Nov)



In the diagram, TU and OQ are straight lines, MO//QS, angle OPN = 35^0 and angle RQP = 50^0.
Use the diagram to answer questions 23 and 24

23. Calculate the value of angle ONU

A 35^0 B 50^0 C 85^0 D 95^0

Solution

$\angle QOM = 50^0$ (Alternate angles)
$\angle ONP + 35 + 50 = 180^0$ (sum of angles in Δ PNO)
$\angle ONP = 180 - 85$
$= 95^0$
$\angle ONP + \angle ONU = 180^0$ (sum of $\angle$s on a straight line)
$95 + \angle ONU = 180^0$
$\angle ONU = 180 - 95$
$= 85^0$ C.

24. Find the size of angle SRT

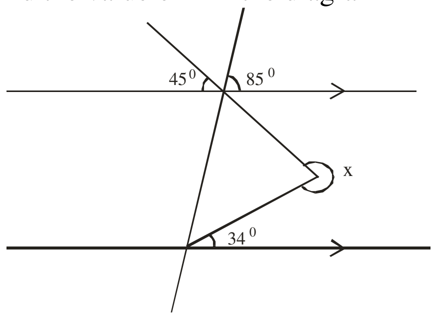
A 85^0 B 95^0 C 130^0 D 145^0

Solution

$\angle PRS = \angle ONU$ (corresponding angles)
$\angle PRS = 85^0$
$\angle PRS + \angle SRT = 180$ (sum of angles on a straight line)
$85 + \angle SRT = 180$
$\angle SRT = 180 - 85$
$= 95^0$ B.

2007/20

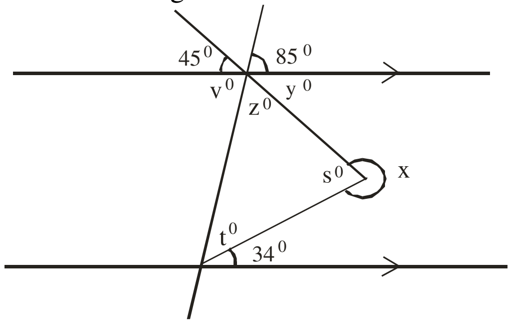
Find the value of x in the diagram



A 281^0 B 269^0 C 201^0 D 179^0

Solution

Let us label the diagram as:

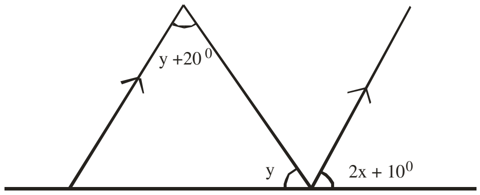


$y = 45^0$ (vertically opp. angles)
$85 + y + z = 180^0$ (sum of angles on a straight line)
$85 + 45 + z = 180$
$z = 180 - 130$
$= 50^0$
$v = 85^0$ (vertically opp. angles)
$t + 34^0 = 85$ (v^0) (alternate angles)
$t = 85 - 34$
$t = 51^0$
$t + z + s = 180^0$ (sum of angles in Δ)
$51 + 50 + s = 180$
$s = 180 - 101$
$s = 79^0$
$s + x = 360^0$ (sum of $\angle$s at a point)
$79 + x = 360$
$x = 360 - 79$
$= 281^0$ A.

2005/30 (Nov)

From the diagram, find an expression for y in terms of x

A $y = 75^0 + 2x$ B $y = 75^0 - x$ C $y = 75^0 - 2x$
D $y = 75^0 + x$



Solution

The vacant angle on the straight line where we have y and $2x + 10^0$ is $y + 20^0$. (Alternate angles)

$y + (y + 20) + 2x + 10 = 180^0$ (sum of ∠s on a straight line)

$y + y + 20 + 2x + 10 = 180$

$2y + 2x = 180 – 30$

$2y + 2x = 150$

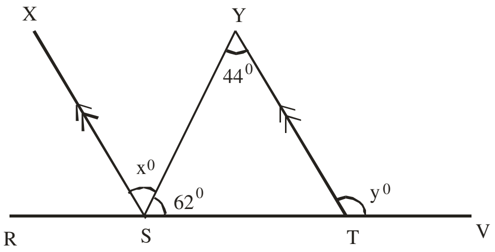
$2y = 150 – 2x$

Divide through by 2

$y = 75 – x$ B.

2006 /45 Neco (Dec)

In the diagram below, RSTV is a straight line, XS//YT, $\angle SYT = 44^0$, $\angle YST = 62^0$. Find $x^0 + y^0$



A 150^0 B 106^0 C 74^0 D 44^0 E 30^0

Solution

$y^0 = 44 + 62$ (exterior ∠ = sum of two interior opp ∠s)

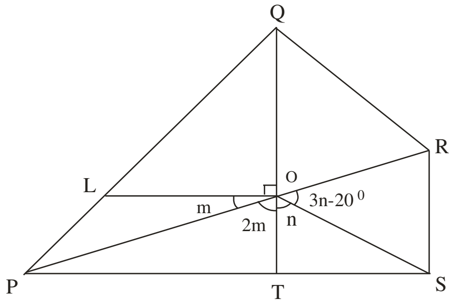
$y = 106^0$

$x = 44^0$ (alternate angles)

Thus, $x + y = 44 + 106$

$= 150^0$ A.

2015/33



In the diagram, $\overline{QT}$ and $\overline{PR}$ are straight lines, $\angle ROS = (3n – 20^0)$, $\angle POT = 2m$, $\angle SOT = n$, $\angle POL = m$ and $\angle QOL$ is a right angle. Find the value of n.

A 35^0 B 40^0 C 55^0 D 60^0

Solution

$90^0 + m + 2m = 180^0$ (sum of angles on a straight line)

$3m = 180 – 90$

$3m = 90$

$m = {}^{90}/_{3} = 30^0$

$2m + n + (3n – 20^0) = 180^0$ (sum of angles on a straight line)

$2m + n + 3n – 20 = 180^0$

$2(30) + 4n – 20 = 180$

$60 – 20 + 4n = 180$

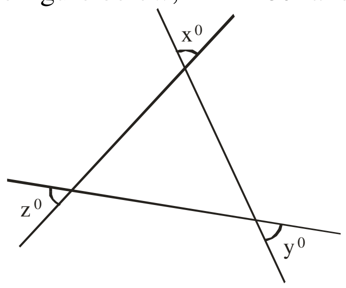
$4n = 180 – 40$

$4n = 140$

$n = {}^{140}/_{4} = 35^0$ A.

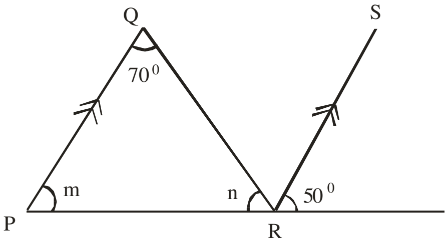
2010/43 Neco Exercise 15.14

In the figure below, if $x = 80^0$ and $z = 36^0$, find y



A 32^0 B 36^0 C 58^0 D 64^0 E 80^0

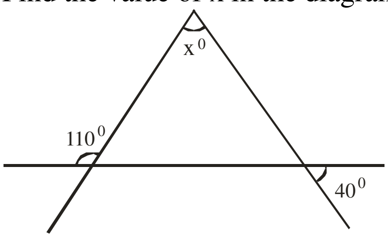
2012/5a Neco Exercise 15.15



In the above figure, PQR is a triangle and PQ//RS, find m and n

2006/19 (Nov) Exercise 15.16

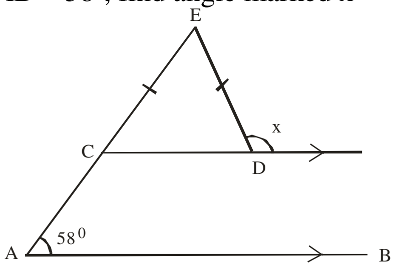
Find the value of x in the diagram



A 40^0 B 60^0 C 70^0 D 150^0

2010/44 Neco Exercise 15.17

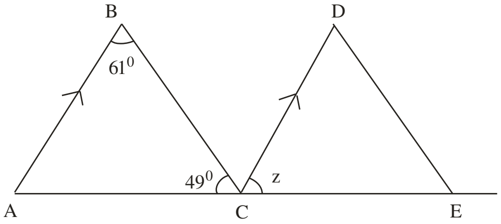
In the diagram below, $\overline{CE} = \overline{ED}$, $\overline{CD} // \overline{AB}$ and $\angle EAB = 58^0$, find angle marked x



A 32^0 B 58^0 C 61^0 D 64^0 E 122^0

2014/49 Neco (Dec) Exercise 15.18

Consider the diagram below. ACE is a straight line, AB//CD, $\angle BCA = 49^0$ and $\angle ABC = 61^0$. Find the angle marked z.



A 49^0 B 61^0 C 70^0 D 110^0 E 131^0

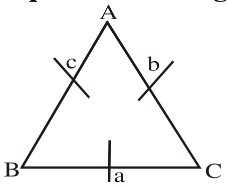
BEARING

This topic deals with measurement of points on the earth surface based on some principles of which part has been discussed. The resulting shapes are mostly triangles whose lengths and angles are used to solve for the unknown ones. This task (i.e. of solving for unknown angles and lengths) need some prerequisite knowledge. Among these are;

- Element of plane geometry as treated in the preceding unit.
- Types of triangles and ways of finding their unknown angles and lengths.

TYPES OF TRIANGLES (REVIEW)

Equilateral triangle

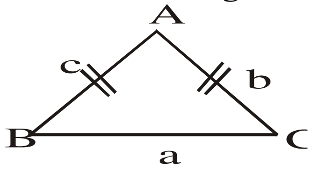


Properties

All sides are equal $a = b = c$

All angles are equal $\hat{A} = \hat{B} = \hat{C}$

Isosceles Triangle

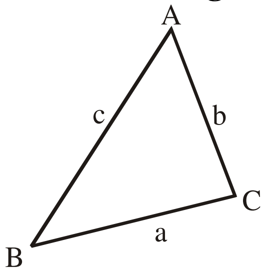


Properties

Two sides are equal $b = c$

Base angles are equal $\hat{B} = \hat{C}$

Scalene Triangle

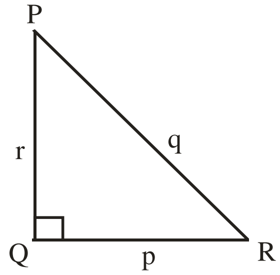


Properties

None of the sides are equal $a \neq b \neq c$

None of the angles are equal $\hat{A} \neq \hat{B} \neq \hat{C}$

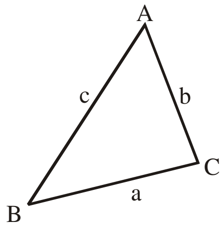
Right- angled Triangle



Properties

One of the angles is 90^0 i.e. $\hat{Q}$

Acute – angled Triangle

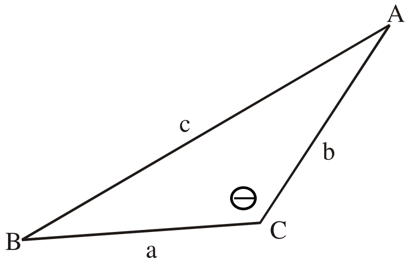


Properties

Three angles are less than 90^0

i. e. $\hat{A} < 90^0$, $\hat{B} < 90^0$, $\hat{C} < 90^0$

Obtuse – angled Triangle



Properties

One of the angles , θ is greater than 90^0 but less than 180^0

To find the unknown angles or lengths of the last two triangles (*acute and obtuse-angled triangles*) leads to *sine* and *cosine* rule.

The sine rule states that :

$$\frac{a}{\text{Sin } A} = \frac{b}{\text{Sin } B} = \frac{c}{\text{Sin } C}$$

The notations a, b & c and A, B & C are as applied to acute and obtuse angled triangles above.

Conditions for applying sine rule

(i) One length and all angles, or

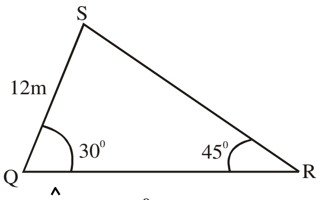
(ii) Two side and an angle opposite one of the given sides.

(iii) Two angles and one side opposite one of the given angles

Examples

(a) QRS is a triangle with QS = 12m, $\angle$ RQS = 30^0 and $\angle$QRS = 45^0, calculate the length RS

Solution



$30 + 45 + \hat{S} = 180^0$ (Sum of $\angle$ s in a Δ)

$\hat{S} = 180 - 75$

$= 105^0$

Two angles and one side opposite one of the given angles condition fulfilled hence sine rule will apply

$$\frac{RS}{\text{Sin } 30} = \frac{12}{\text{Sin } 45}$$

$RS = \dfrac{12 \sin 30}{\sin 45}$

$= \dfrac{12 \times {}^1/_2}{1/\sqrt{2}} \qquad = 6 \div \dfrac{1}{\sqrt{2}} \qquad = 6 \times \dfrac{\sqrt{2}}{1} \qquad = 6\sqrt{2}$

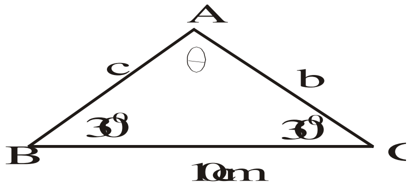
(b) Given that the two angles of a triangle are 30^0 each and the longest side is 10 cm. Calculate the length of each of the other sides.

Solution

The 3^{rd} angle should be 120^0 facing the longest side.
Since

$30 + 30 + \theta = 180$ (sum of $\angle$ s in Δ)

$\theta = 120^0$



All angles and one side hence sine rule applies

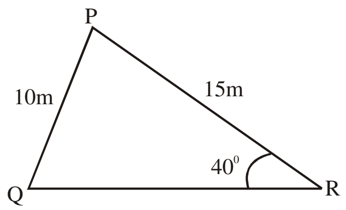
$\dfrac{10}{\sin 120} = \dfrac{b}{\sin 30}$

$b = \dfrac{10 \sin 30}{\sin 60}$ Since sine is positive in the 2^{nd} quad.

$= \dfrac{10 \times {}^1/_2}{\sqrt{3}\ /2} \quad = \dfrac{5 \times 2}{\sqrt{3}} \qquad = \dfrac{10\sqrt{3}}{3}$ cm

(c) In an acute-angled triangle PQR with PQ = 10m, PR = 15m , $\angle PRQ = 40^0$. Evaluate $\angle$ PQR

Solution



Two sides and one opposite angle, thus we can apply sine rule.

$\dfrac{10}{\sin 40} = \dfrac{15}{\sin Q}$

$\sin Q = \dfrac{15 \sin 40}{10}$

$= 1.5\ (0.6428)$

$\sin Q = 0.9642$

$Q = \sin^{-1} 0.9642$

$= 74.62^0$

Cosine Rule

Applying the usual notations in acute and obtuse angled triangles

Cosine rule states that:

$a^2 = b^2 + c^2 - 2bc \cos A$

$b^2 = a^2 + c^2 - 2ac \cos B$

$c^2 = a^2 + b^2 - 2ab \cos C$

Conditions for applying cosine Rule

(i) Given two sides and the included angle.

(ii) Given all the three sides of the triangle.
Here the above stated rule will have to be adjusted to make cos A or cos B or cos C the subject of the formula as the case may be i.e.

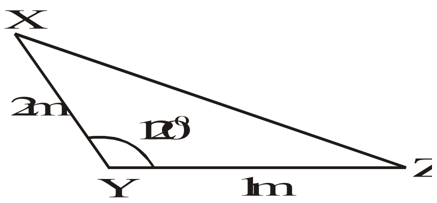
$\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$

$\cos B = \dfrac{a^2 + c^2 - b^2}{2ac}$

the others follow suit.

Examples

(i) Find /XZ/ in the triangle below



Solution

Two sides and the included angle; thus cosine rule applies here,

$/XZ/^2 = 2^2 + 1^2 - 2 \times 2 \times 1 \times \cos 120$

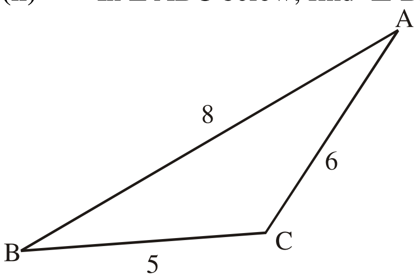
$= 5 + 4 \cos 60^0$ since Cosine is – ve in 2^{nd} quad

$= 5 + 4 \times {}^1/_2$

$/XZ/^2 = 7$

$/XZ/ = \sqrt{7}$

(ii) In Δ ABC below, find $\angle$ BAC



Solution

Three sides of Δ then ,the adjusted cosine rule will apply.

$\cos A = \dfrac{c^2 + b^2 - a^2}{2 \times c \times b}$

$\cos A = \dfrac{8^2 + 6^2 - 5^2}{2 \times 8 \times 6}$

$= \dfrac{64 + 36 - 25}{96}$

$\cos A = 0.7813$

$A = \cos^{-1} 0.7813$

$= 38.62^0$

$\approx 38.6^0$

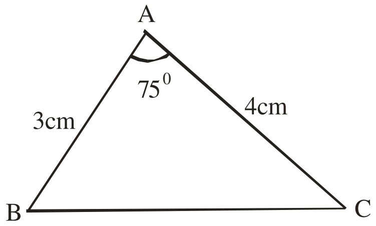
Problems on sine and cosine triangles

2014/17 Neco (Nov)

ABC is a triangle in which $\angle BAC = 75^0$, /AB/ = 3cm and /AC/ = 4cm. Calculate /BC/ correct to one decimal place

A 3.9cm B 4.0cm C 4.3cm D 5.0cm E 7.0cm

Solution



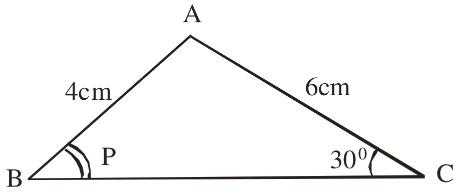
Applying cosine formula

since we have two sides and an included angle

$$BC^2 = 3^2 + 4^2 - 2 \times 3 \times 4 \cos 75^0$$
$$= 9 + 16 - 6.212$$
$$BC^2 = 18.788$$
$$BC = \sqrt{18.788}$$
$$= 4.3345$$
$$\approx 4.3\text{cm to 1 d.p (C)}$$

2006/ 45 NABTEB (Nov)

Calculate the angle marked P in the diagram below, given that $\overline{AB}$ = 4cm , $\overline{AC}$ = 6cm, $\angle ACB = 30^0$



A 48.6^0 B 40.0^0 C 30.0^0 D 24.0^0

Solution

Applying sine formula

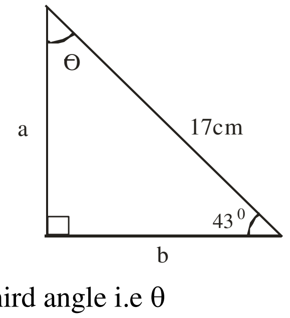
$$\frac{4}{\sin 30} = \frac{6}{\sin P}$$
$$4 \sin P = 6 \sin 30$$
$$\sin p = \frac{6 \sin 30}{4}$$
$$\sin p = 0.75$$
$$p = \sin^{-1} 0.75$$
$$= 48.59^0$$

2005/ 10b NABTEB ***Counter example***

The hypotenuse of a right – angled triangle is 17cm and one of the angles is 43^0, find the

(i) third angle (ii) side opposite the smallest angle

Solution



(**i**) Third angle i.e θ

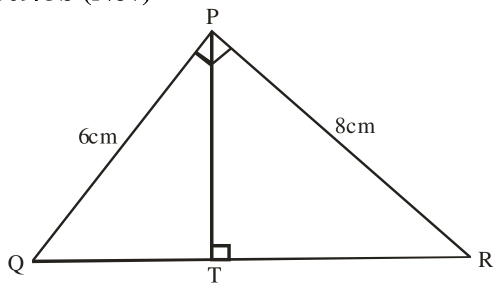
$$\theta + 43^0 + 90^0 = 180^0 \text{ (sum of } \angle\text{s in } \Delta)$$
$$\theta = 180 - 133$$
$$= 47^0$$

(**ii**) To find a

The sides relevant to 43^0 are opp and hypo (SOH)

$$\sin 43^0 = \frac{a}{17}$$
$$17 \sin 43^0 = a$$
$$11.59\text{cm} = a$$

2009/5b (Nov)



In the diagram $\angle QPR = \angle PTR = 90^0$, /PR/ = 8cm and /QP/ = 6cm. find /TR/

Solution

In ΔPQR, by Pythagoras rule

$$QR^2 = 6^2 + 8^2$$
$$QR^2 = 36 + 64$$
$$QR^2 = 100$$
$$QR = \sqrt{100} \text{ i.e } 10 \text{ cm}$$

Next, Applying sine rule in ΔPQR

$$\frac{10}{\sin 90} = \frac{6}{\sin R}$$
$$10 \sin R = 6 \sin 90$$
$$\sin R = \frac{6 \sin 90}{10}$$
$$\sin R = 0.6$$
$$R = \sin^{-1} 0.6$$
$$= 36.87^0$$

In right – angled ΔPTR,

the relevant sides to 36.87^0 are Adj and Hypo (CAH)

$$\text{Cos } 36.87^0 = \frac{TR}{8}$$
$$8 \times \cos 36.87^0 = TR$$
$$6.40\text{cm} = TR$$

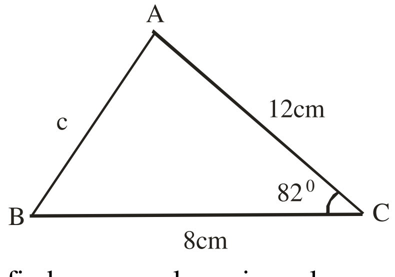
2006/47 Neco (Dec)

In ΔABC, a = 8cm, b = 12cm, C = 82^0. Find c

A 6.73cm B 13.45cm C 29.92cm

D 33.46cm E 50.92cm

Solution



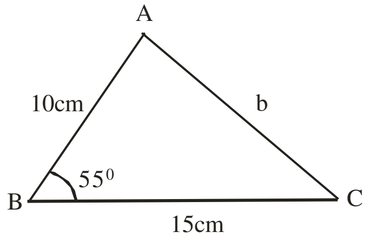
To find c, we apply cosine rule

since we have two sides and an included angle

$$c^2 = 12^2 + 8^2 - 2 \times 12 \times 8 \cos 82^0$$
$$c^2 = 144 + 64 - 26.72$$
$$c^2 = 181.28\text{cm}$$
$$c = \sqrt{181.28}$$
$$= 13.46 \text{ cm} \quad \text{B.}$$

2009/2a Neco (Dec)
Two sides of a triangle measuring 10cm and 15cm enclose an angle of 55^0.
(i) calculate the third side
(ii) find the two other angles

Solution



(i) Applying cosine formula to find b
since we have two sides and an included angle

$$b^2 = 10^0 + 15^2 - 2 \times 10 \times 15 \cos 55^0$$
$$= 100 + 225 - 172.07$$
$$b^2 = 152.93$$
$$b = \sqrt{152.93}$$
$$= 12.37\text{cm}$$

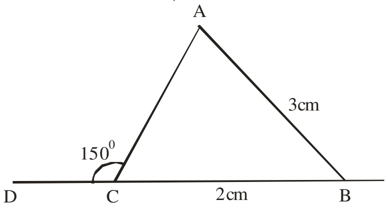
(ii) Applying sine rule to find A

$$\frac{12.37}{\sin 55} = \frac{15}{\sin A}$$
$$12.37 \times \sin A = 15 \sin 55$$
$$\sin A = \frac{15 \sin 55}{12.37}$$
$$\sin A = 0.9933$$
$$A = \sin^{-1} 0.9933$$
$$= 83.36^0$$

Applying sine rule to find C

$$\frac{12.37}{\sin 55} = \frac{10}{\sin C}$$
$$12.37 \sin C = 10 \sin 55$$
$$\sin C = \frac{10 \sin 55}{12.37}$$
$$\sin C = 0.6622$$
$$C = \sin^{-1} 0.6622$$
$$= 41.47^0$$

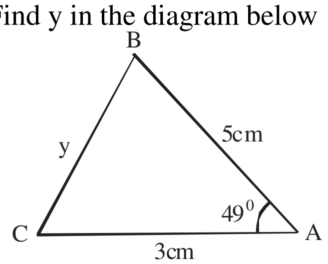
2010/47 Neco Exercise 15.19
In the figure below, /BC/ = 2cm, /AB/ = 3cm and $\angle ACD = 150^0$, what is the value of sin A?



A $\sqrt{3}$ B $\frac{\sqrt{3}}{2}$ C $\frac{2}{3}$ D $\frac{1}{2}$ E $\frac{1}{3}$

2014/42 Neco Exercise 15.20
A vehicle makes a journey that forms a triangular shape such that the distance CD = 40km, AD = 50km, $\angle ACD = 115^0$ and $\angle D = 25^0$. Find the third side of the triangular shape.
A 43.32km B 32.14km C 32.31km
D 20.20km E 18.65km

2012/31 Neco Exercise 15.21
Find y in the diagram below



A 4.01cm B 3.87cm C 3.78cm
D 2.87cm E 2.78cm

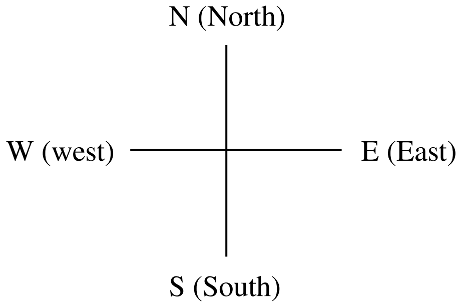
2009/9b (Nov) Exercise 15.22
In triangle XYZ, /XY/ = 9cm, /XZ/ = 10cm and $\angle YXZ = 75^0$. Find /YZ/

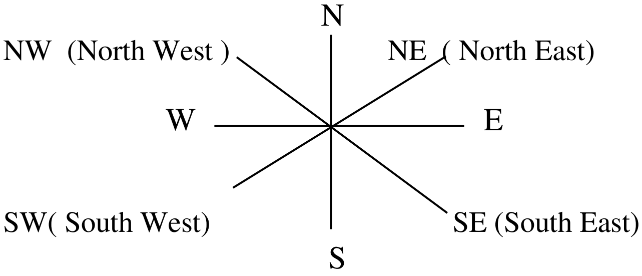
TYPES OF BEARING

Any given bearing angle is represented in two ways, namely
- Compass bearing (cardinal points)
- Three-digit bearing.

Compass bearing (cardinal points)

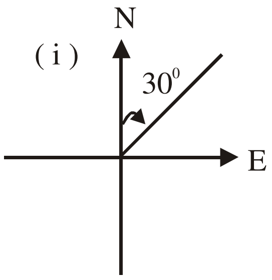
This concept is based on cardinal points as can be obtained in any standard compass such as shown below.

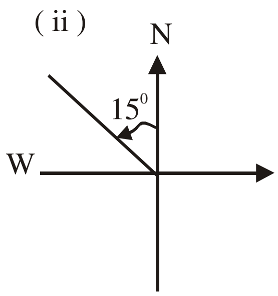


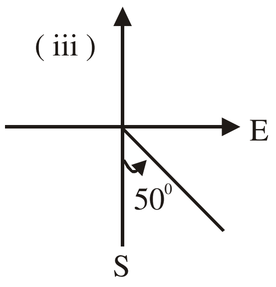


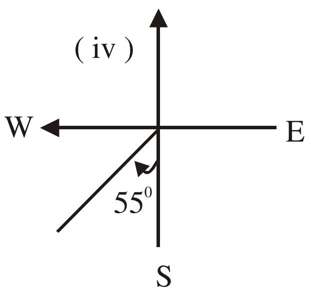
The starting point are based on **North** or **South** which are measured to either **East** or **West**. You will agree with the author that from North to East or North to west, we can only cover between 0^0 and 90^0. The same applies to South to East or South to West. Below are examples of compass bearing

(i) N 30^0 E (ii) N 15^0 W (iii) S 50^0E (iv) S 55^0 W









Three digit bearing

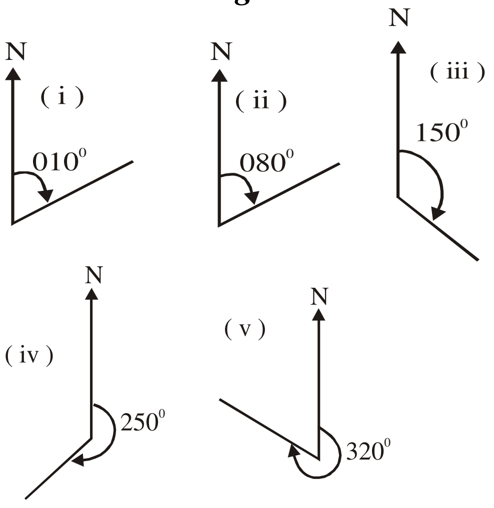
This type of bearing bases its starting point from the **North only;** then moves in a clockwise direction. The movement is between 0^0 and 360^0.
Below are examples of 3 - digit bearing

(i) 010^0 (ii) 080^0 (iii) 150^0 (iv) 250^0 (v) 320^0

The concept of quadrant is important here though to be numbered in clockwise direction instead of the usual anti – clockwise numbering thus we have the format as:

4^{th} Quadrant 270^0 and 360^0	1^{st} quadrant 0^0 and 90^0
3^{rd} quadrant 180^0 and 270^0	2^{nd} quadrant 90^0 and 180^0

Solution diagrams



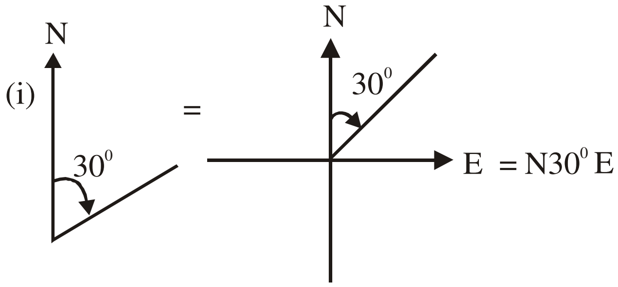
Relationship between three digits and compass bearing

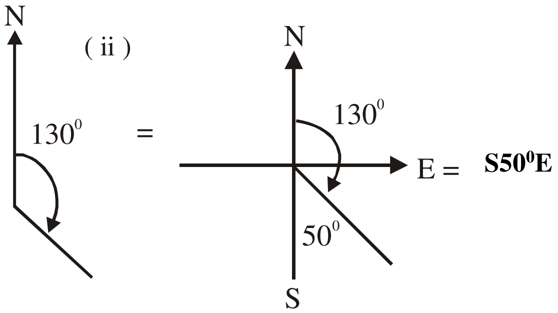
Any three – digit bearing can be converted to a compass bearing and vice versa.

Examples

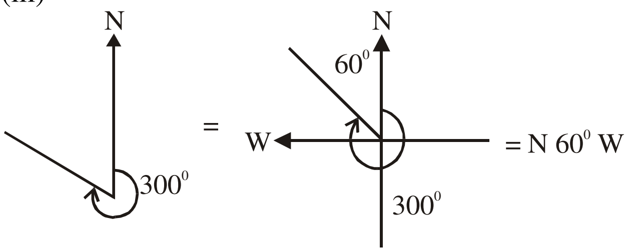
(a) Covert the following 3 digit bearing to compass bearing.
(i) 030^0 (ii) 130^0 (iii) 300^0

Solution





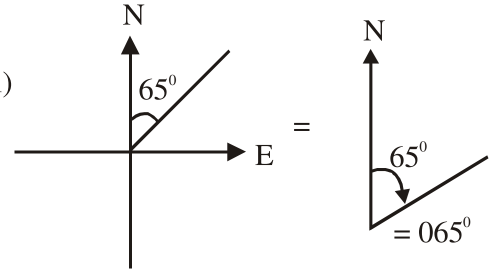
(iii)



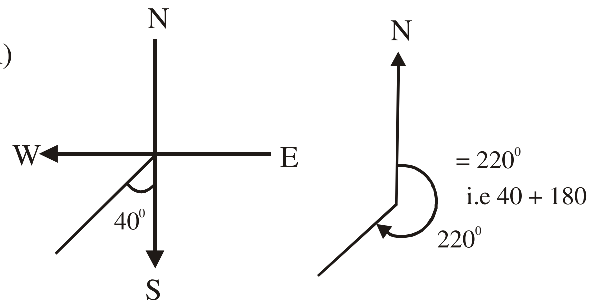
(b) Convert the following compass bearing to a 3 digit bearing
(i) N 65^0 E (ii) S 40^0 W

Solution

(i)

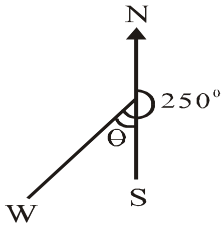


(ii)



1997/40
Express the true bearing of 250^0 as a compass bearing.
A. N 20^0 E B. S 20^0 E C. N 20^0 W
D. S70^0 W E. S 70^0 E

Solution



The required angle is θ
But θ = 250 – 180
= 70^0
Also θ is between South and West
Hence θ = S 70 W (D)

2001/26 Exercise 15.23
The bearing S 40^0 E is the same as
A. 040^0 B. 050^0 C. 130^0 D. 140^0

2002/32 (Nov) Exercise 15.24
The bearing S 50^0 W is the same as
A. 050^0 B. 130^0 C. 140^0 D. 230^0

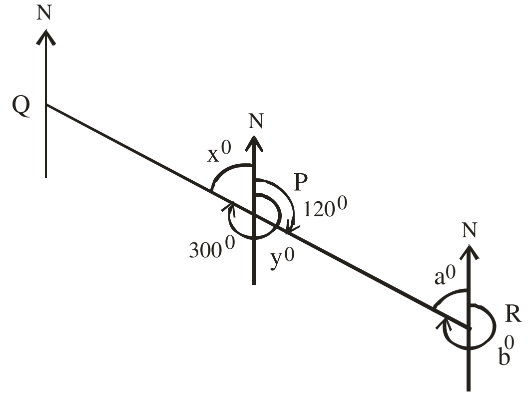
2004/48 Neco Exercise 15.25
The bearing S 40^0 W is the same as
A 040^0 B 050^0 C 140^0 D 220^0 E 230^0

REVERSED BEARING

If the bearing of a point A is described from another point B. Then, the bearing of B can be described from A based on the first description.

2012/47 Neco
If P, Q, R are points such that the bearing of Q from P is 300^0 and the bearing of R from P is 120^0, find the bearing of Q from R
A 060^0 B 120^0 C 150^0 D 180^0 E 300^0

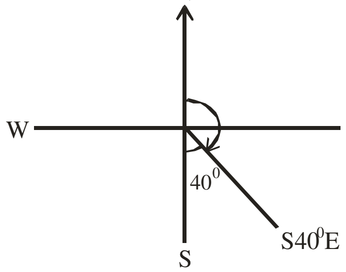
Solution



$x^0 + 300^0 = 360^0$ (sum of angles at a point)
$x^0 = 360 - 300 = 60^0$
$y^0 = x^0$ i.e 60^0 (vertically opposite angles)
Alternatively
$y^0 + 120^0 = 180^0$ (sum of angles on a straight line)
$y^0 = 180 - 120 = 60^0$
$a^0 = y^0$ i.e 60^0 (Alternate angles)
The bearing of Q from R is b^0
$a^0 + b^0 = 360^0$ (sum of angles at a point)
$b^0 = 360 - a^0$
$b = 360 - 60 = 300^0$

2013/48 Neco
The bearing S 40^0E is the same as
A 040^0 B 050^0 C 130^0 D 140^0 E 180^0

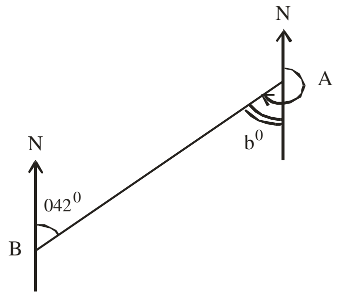
Solution



The compass bearing is $= 180^0 - 40$
$= 140^0$ (D)

2010/49 Neco
The bearing of a point A from a point B is 042^0. Calculate the bearing of B from A
A 228^0 B 222^0 C 138^0 D 48^0 E 42^0

Solution

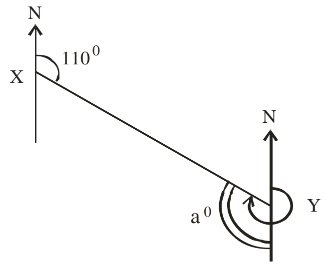


$b^o = 42^0$ (Alternate angles)

Bearing of B from A $= 180^0 + b^0$
$= 180 + 42^0$
$= 222^0$ B.

2008/33 Neco (Dec)
Find the bearing of a point X from Y. If the bearing of Y from X is 110^0.
A 070^0 B 110^0 C 250^0 D 290^0 E 310^0

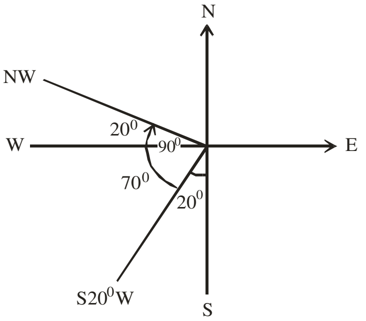
Solution



$a^0 = 110^0$ (Alternate angles)
Bearing of X from Y $= 180 + a^0$
$= 180 + 110$
$= 290^0$ D.

2011/12
Esther was facing S20^0W. She turned 90^0 in the clockwise direction. What direction is she facing?
A N 70^0 W B S 70^0 E C N 20^0 W D S 20^0 E

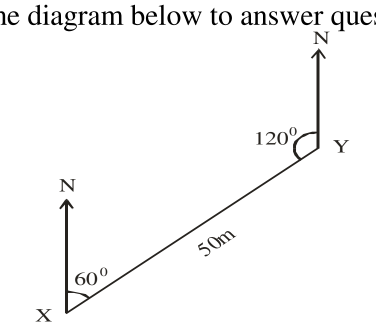
Solution



Her new position is NW and **not** WN and we can not say N 20^0 W here; it would have been correct if we had W20^0N.
From north to west is 90^0 ,
20^0 is out of it is remaining 70^0
The new position is N 70^0 W (A)

2009/ 26 – 27

Use the diagram below to answer questions 26 and 27



26. Find the bearing of X from Y

A 300^0 B 240^0 C 120^0 D 60^0

Solution

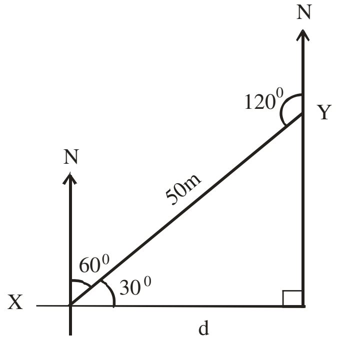
Bearing of X from Y = 360 – 120

= 240^0 B

27. If /XY/ = 50m, how far east of X is Y?

A 25.5m B 40.6m C 40.8m D 43.3m

Solution



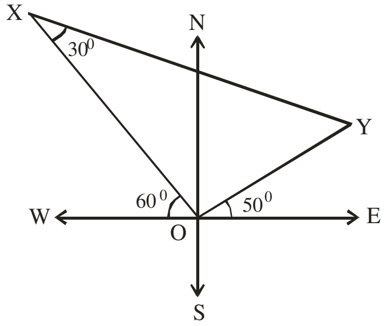
Using 30^0 in the resulting right – angled triangle the relevant sides are Adj and Hypo (CAH)

$$\text{Cos } 30^0 = \frac{d}{50}$$

$$50 \cos 30^0 = d$$

$$43.30\text{m} = d \quad (\text{ D })$$

2010/46

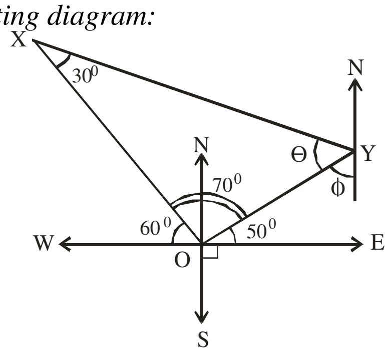


In the diagram, $\angle WOX = 60^0$, $\angle YOE = 50^0$ and $\angle OXY = 30^0$.What is the bearing of X from Y?

A 300^0 B 240^0 C 190^0 D 150^0

Solution

Resulting diagram:



Bearing of X from Y is $\theta + \phi + 180^0$

$70^0 + 30^0 + \theta = 180^0$ (sum of $\angle$s in Δ)

$\theta = 180 - 100$

$= 80^0$

$\phi + (50 + 90^0) = 180^0$ (sum of interior opp. $\angle$s)

$\phi = 180 - 140$

$= 40^0$

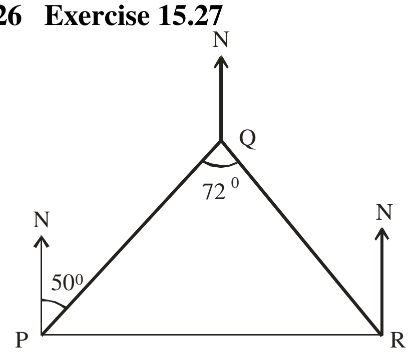
Thus bearing of X from Y is $80 + 40 + 180 = 300^0$ A.

2011/48 Neco Exercise 15.26

A from Q is 120km from a town B in the direction 050^0. What is the bearing of B from Q?

A 040^0 B 050^0 C 130^0 D 230^0 E 310^0

2012/26 Exercise 15.27



The position of three ships P, Q and R at sea are illustrated in the diagram. The arrows indicate the North direction. The bearing of Q from P is 050^0 and $\angle PQR = 72^0$. Calculate the bearing of R from Q

A 130^0 B 158^0 C 222^0 D 252^0

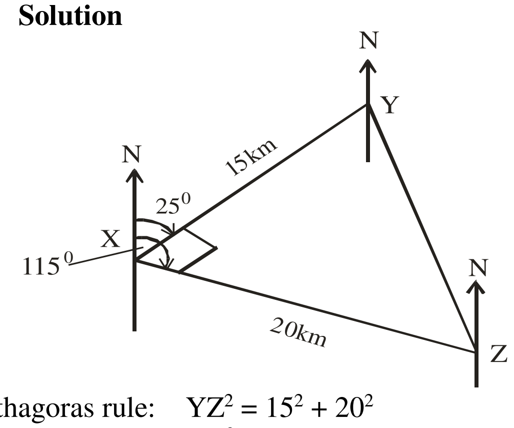
Problems on bearings I

2006/15 Neco

A village Y is 15km from a point X on a bearing of 025^0. Another village Z is 20km from X on a bearing 115^0. Calculate the distance YZ.

A 35km B 25km C 20km D 15km E 5km

Solution



By Pythagoras rule: $YZ^2 = 15^2 + 20^2$

$YZ^2 = 225 + 400$

$YZ^2 = 625$

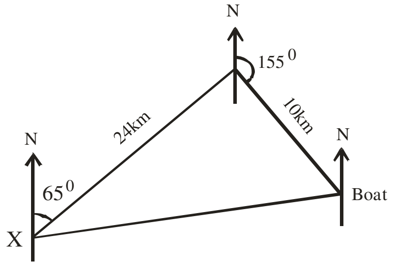
$YZ = \sqrt{625} = 25\text{km}$ (B)

2005/15 Neco

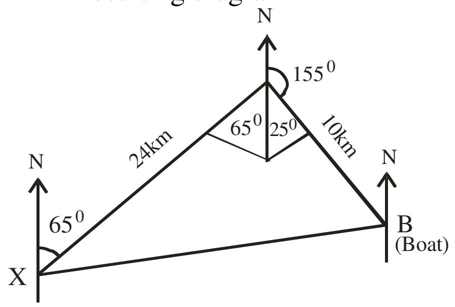
A boat sails 24km from a port X on a bearing of 065^0 and thereafter 10km on the bearing of 155^0. What is the distance of the boat from X?

A 34km B 30km C 26km D 24km E 14km

Solution



Resulting diagram



We are to find XB

By Pythagoras rule: $XB^2 = 24^2 + 10^2$

$XB^2 = 576 + 100$

$XB^2 = 676$

$XB = \sqrt{676}$

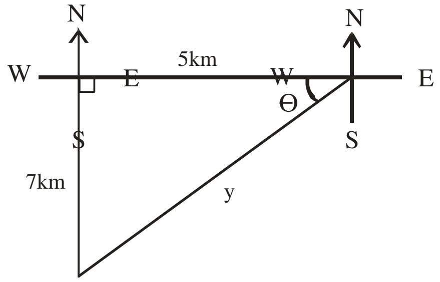
$= 26km$ (C)

2014/25 (Nov)

A ship sails 5km due west and then 7km due south. Find; correct to the nearest degree its bearing from the original position.

A 055^0 B 056^0 C 215^0 D 216^0

Solution



The bearing from the original position = $360 - (\theta + 90^0)$

By Pythagoras rule: $y^2 = 7^2 + 5^2$

$y^2 = 49 + 25$

$y^2 = 74$

$y = \sqrt{74} = 8.6km$

θ can be gotten by sine rule

$$\frac{8.6}{\sin 90} = \frac{7}{\sin\theta}$$

$8.6 \sin\theta = 7 \sin 90^0$

$$\sin\theta = \frac{7\sin 90}{8.6}$$

$\sin\theta = 0.8140$

$\theta = \sin^{-1} 0.8140$

$= 54.49^0$

Thus bearing from original position

$= 360 - (\theta + 90)$

$= 360 - (54.49 + 90)$

$= 360 - 144.49$

$= 215.51^0$

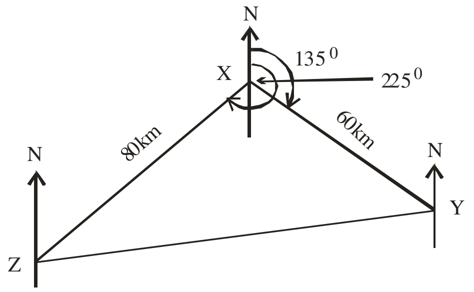
$\approx 216^0$ to nearest degree D.

2007/4

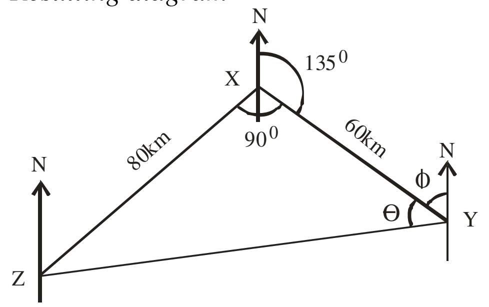
Y is 60km away from X on a bearing of 135^0. Z is 80km away from X on a bearing of 225^0. Find the:

(a) distance of Z fromY; (b) bearing of Z from Y.

Solution



Resulting diagram



(i) Applying cosine rule

since we have two sides and an included angle

$ZY^2 = 80^2 + 60^2 - 2 \times 80 \times 60 \cos 90$

$ZY^2 = 80^2 + 60^2$ (reduces to Pythagoras rule)

$= 6400 + 3600$

$ZY^2 = 10,000$

$ZY = \sqrt{10,000} = 100km$

(ii) The bearing of Z from y = $360 - (\phi + \theta)$

$135^0 + \phi = 180^0$ (sum of interior opp. $\angle$s)

$\phi = 180 - 135 = 45^0$

Applying sine rule to find θ

$$\frac{100}{\sin 90} = \frac{80}{\sin\theta}$$

$100 \sin\theta = 80 \sin 90$

$$\sin\theta = \frac{80\sin 90}{100}$$

$\sin\theta = 0.8$

$\theta = \sin^{-1} 0.8$

$= 53^0$

Thus the bearing of Z from Y = $360 - (53 + 45)$

$= 360 - 98 = 262^0$

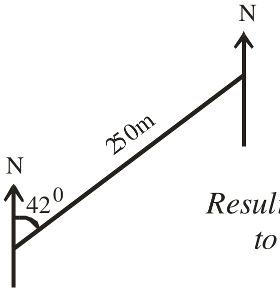
2006/9a Neco counter example

A hunter walked 250m on a bearing 042^0. Calculate, correct to the nearest kilometer, the

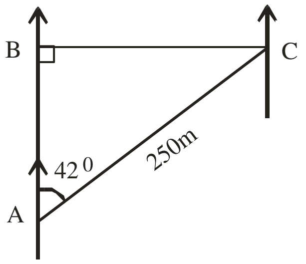
(i) vertical height through which he has moved;

(ii) horizontal distance covered

Solution



Resulting to



(i) We are to find AB;
the sides relevant to 42^0 are Adj and Hypo (CAH)

$$\cos 42^0 = \frac{AB}{250}$$

$$250 \cos 42^0 = AB$$

$$AB = 185.79\text{km}$$

$$\approx 186\text{km to nearest km}$$

(ii) To find BC;
The sides relevant to 42^0 are Opp and Hypo (SOH)

$$\sin 42^0 = \frac{BC}{250}$$

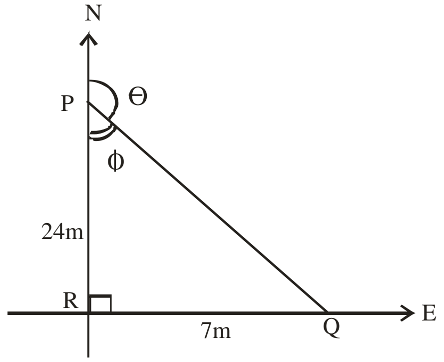
$$250 \sin 42^0 = BC$$

$$BC = 167.28\text{km}$$

$$\approx 167\text{km to nearest km}$$

2005/10 (Nov)
Points P and Q are respectively 24m north and 7m east of point R. What is the bearing of Q from P to the nearest degree?
A 016^0 B 050^0 C 074^0 D 164^0

Solution



We are to find θ with the aid of ϕ
The sides relevant to ϕ are opp and adj (TOA)

$$\tan \phi = \frac{7}{24}$$

$$\phi = \tan^{-1} 0.2917$$

$$\phi = 16.26^0$$

But $\theta + \phi = 180^0$ (sum of angles on a straight line)

$$\theta = 180 - 16.26$$

$$= 163.74^0$$

$$\approx 164^0$$ D.

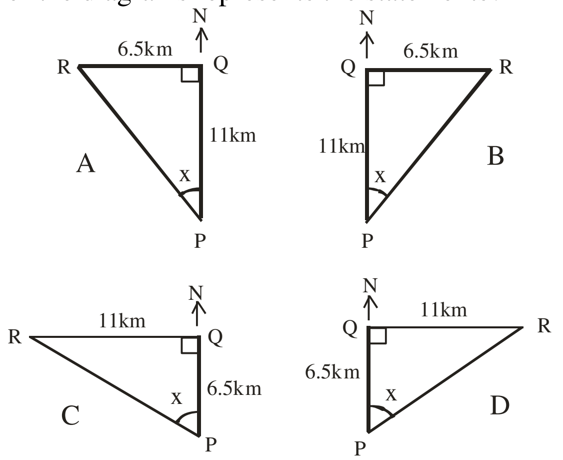
2009/49 Neco(Dec) Exercise 15.28
The bearing of two points Q and R from a point P are 030^0 and 120^0 respectively. If /PQ/ = 24m and /PR/ = 7m. Find the distance QR
A 7m B 15m C 19m D 21m E 25m

2014/6b NABTEB Exercise 15.29
A man starts at **A** and walks 2km on a bearing of 017^0. He then walks 3km on a bearing of 107^0 to **C**. What is the bearing of **C** from **A**?

2006/48 Neco (Dec) Exercise 15.30
A man who could not trace his route was directed on the following instructions. Walk 6km due East and 8km due North. Find his bearing from his initial position, correct to 1 decimal place.
A 18.55^0 B 26.55^0 C 36.9^0 D 53.1^0 E 126.9^0

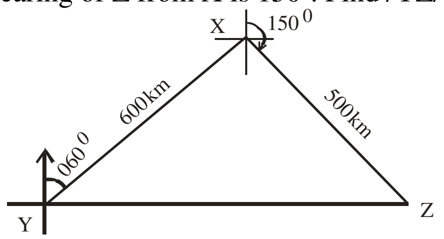
2014/26 Neco (Dec) Exercise 15.31
A boat sails 8km from a port Q on a bearing of 055^0 and then 15km on a bearing of 145^0. What is the distance of the boat from Q?
A 7km B 11km C 15km D 17km E 23km

2006/33 (Nov) Exercise 15.32
A man walk 11km due north from P to Q and then 6.5km due east to R. Given that x is the bearing of R from P, which of the diagrams represents the statements?



2014/18 Neco Exercise 15.33
Three towns P, Q and R are situated such that PQ = 15km and PR = 20km. The bearing of Q from P is 030^0 while the bearing of R from P is 300^0. Calculate the distance RQ
A 5km B 10km C 20km D 25km E 35km

2009/25 (Nov) Exercise 15.34
In the diagram, the bearing of X from Y is 060^0.
The bearing of Z from X is 150^0. Find /YZ/



A 7.81km B 78.10km C 237.00km D 781.02km

2007/48 Neco Exercise 15.35
The bearing of two points Y and Z from a point X are 060^0 and 150^0 respectively. If /XY/ = 12m and /XZ/ = 5m, find the distance /YZ/
A 13m B 11m C 9m D 7m E 5m

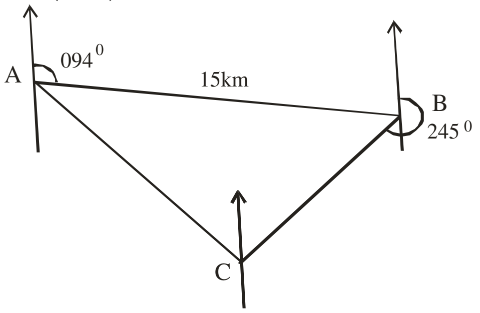
2005/18 Exercise 15.36
From a point R, 300m north of P, a man walks eastwards to a place Q which is 600m from P. Find the bearing of P from Q correct to the nearest degree
A 026^0 B 045^0 C 210^0 D 240^0

2004/22 NABTEB Exercise 15.37
Three observation points P, Q and R are such that Q is due East of P and R is due north of Q. If /PQ/ = 5km and /PR/ = 10km, find /QR/
A 5.0km B 7.5km C 7.6km D 8.7km

2007/31 Exercise 15.38
Q is 32km away from P on a bearing 042^0 and R is 25km from P on a bearing of 132^0. Calculate the bearing of R from Q
A 122^0 B 184^0 C 190^0 D 226^0

Problems on bearings II

2001/12a (Nov)



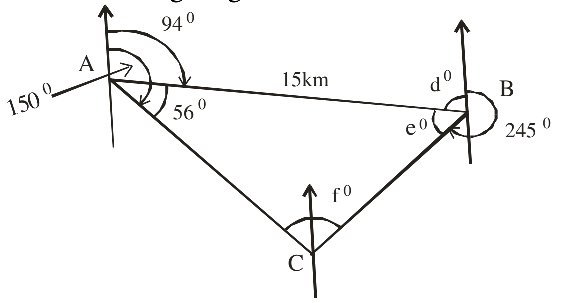
(a) In the diagram, A and B are two ports 15km apart. B is on bearing 094^0 from A and C is a ship on the sea. The ship is seen from A on bearing 150^0 and from B on bearing 245^0.

(i) Calculate, correct to one decimal place, the distance of the ship from A

(ii) How far east of A is C?

Solution

Resulting diagram



Since we are given only one distance 15km, it is obvious we shall be working with sine rule; so we target the various angles in the resulting triangle

(i) $\angle CAB = 150 - 94$ i.e 56^0

$d^0 + 94^0 = 180^0$ (sum of interior opp. $\angle$s)

$d^0 = 180 - 94$ i.e 86^0

$d^0 + e^0 + 245^0 = 360^0$ (sum of $\angle$s at a point)

$86 + e^0 + 245 = 360$

$e^0 = 360 - 331 = 29^0$

$56^0 + e^0 + f^0 = 180^0$ (sum of $\angle$s in Δ)

$56^0 + 29^0 + f^0 = 180^0$

$f^0 = 180 - 85 = 95^0$

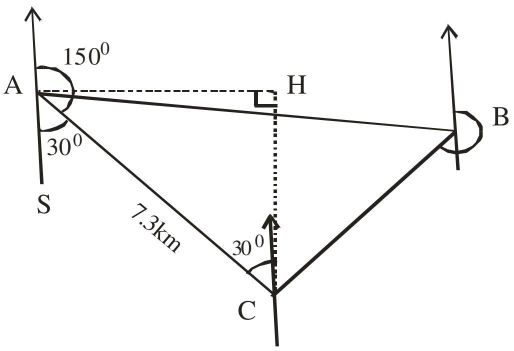
$$\frac{15}{\sin 95^0} = \frac{AC}{\sin 29}$$

$15 \sin 29 = AC \sin 95$

$$AC = \frac{15 \sin 29}{\sin 95^0}$$

$$= \frac{7.272}{0.9962} = 7.300 \text{ km to 1 d.p}$$

(ii)



We are to find AH :

Note that $\angle ACH = 30^0$ (alternate $\angle$s).

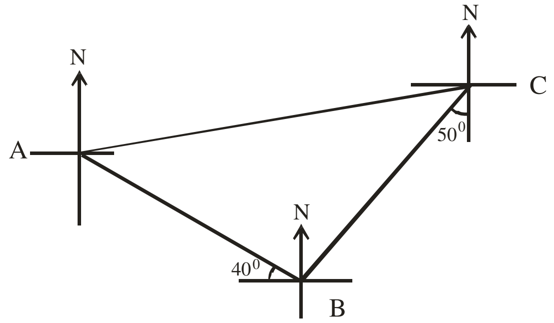
The sides relevant to 30^0 in ΔACH are opp and hypo (SOH)

$$\text{Sin } 30^0 = \frac{AH}{7.3}$$

$7.3 \sin 30 = AH$

$3.65 \text{ km} = AH$

2010/9



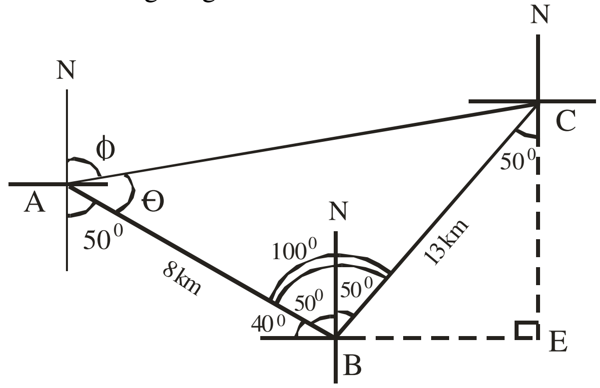
In the diagram, /AB/ = 8km, /BC/ = 13km, the bearing of A from B is 310^0 and the bearing of B from C is 230^0. Calculate, correct to 3 significant figures

(a) the distance AC; (b) the bearing of C from A

(c) how far east of B, C is

Solution

The resulting diagram is:



Breakdown:

$\angle ABN = 360 - 310$ i.e 50^0 OR $90 - 40 = 50^0$

$\angle CBN = 50^0$ (Alternate $\angle$s)

angle at A = 50^0 (Alternate $\angle$s)

(a) We can find AC by cosine rule since there are two sides and included angle

$AC^2 = 8^2 + 13^2 - 2 \times 8 \times 13 \cos 100$

$= 64 + 169 - [208 \times (-0.1736)]$

$= 233 + 36.11$

$AC^2 = 269.11$

$AC = \sqrt{269.11} = 16.40\text{km} \approx 16.4\text{km to 3 s.f}$

(b) The bearing of C from A

$\phi = 180 - (50 + \theta)$

we get θ by sine rule

$$\frac{16.40}{\sin 100} = \frac{13}{\sin\theta}$$

$16.40 \sin\theta = 13 \sin 100$

$$\sin\theta = \frac{13 \sin 100}{16.40}$$

$\sin\theta = 0.7806$

$\theta = \sin^{-1} 0.7806 = 51.32^0$

$\therefore \phi = 180 - (50 + 51.32^0) \approx 78.7^0$ to 3 s.f

(c) In ΔBCE, the sides relevant to 50^0 are opp and hypo (SOH)

$$\sin 50^0 = \frac{BE}{13}$$

$$13 \sin 50 = BE$$

$$BE = 9.959 \text{ km}$$

$$\approx 9.96 \text{ km to 3s.f}$$

2005/7 NABTEB

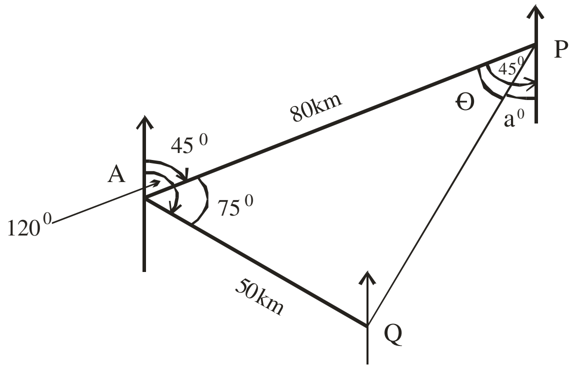
The bearings of point P and Q from A are 045^0 and 120^0 respectively. If the distance AP is 80km and AQ is 50km, calculate

(a) the distance between P and Q to 3 significant figures

(b) the bearing of Q from P to the nearest degree

(c) how far east of A is Q

Solution



(a) $PQ^2 = 80^2 + 50^2 - 2 \times 80 \times 50 \cos 75$

$PQ^2 = 6400 + 2500 - 2070.55$

$PQ^2 = 6829.45$

$PQ = \sqrt{6829.45} = 82.64\text{km} \approx 82.6\text{km to 3 s.f}$

(b) the bearing of Q from P

$$\frac{82.64}{\sin 75} = \frac{50}{\sin\theta}$$

$$\sin\theta = \frac{50\sin 75}{82.64}$$

$$\sin\theta = 0.5844$$

$$\theta = \sin^{-1} 0.5844$$

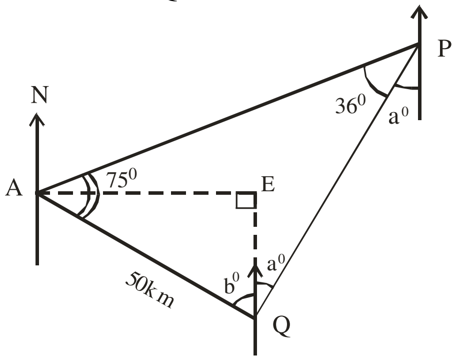
$$= 35.76^0$$

Note that : $a^0 = 45 - 35.76^0$

$= 9.24^0 \approx 9^0$

Bearing of Q from P $= 180 + 9^0$ i.e 189^0

(c) how far east of A is Q i.e **AE**



$(75 + 36) + b^0 + a^0 = 180$ (Sum of angles in triangle)

$180 - (75 + 36) = b^0 + a^0$

$69 = b^0 + a^0$ but $a^0 = 9^0$

$69 = b^0 + 9^0$

$69 - 9^0 = b^0$ i.e $b = 60^0$

The side relevant to b^0 in ΔAEQ are opp and hyp (SOH)

$$\sin 60^0 = \frac{AE}{50}$$

$$50 \sin 60^0 = AE$$

$$43.33\text{km} = AE$$

2005/6 Neco

Three points, P, Q and R are on a horizontal plane. Point Q is on a bearing 120^0, at a distance 80m from point P. Point R is on a bearing of 045^0 and a distance of 150m from point Q.

(a) Calculate, correct to 2 significant figures the,

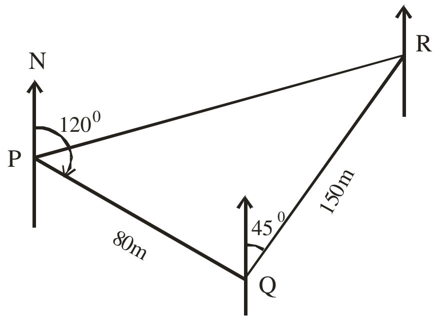
(i) distance between point P and point R;

(ii) bearing of point P from point R

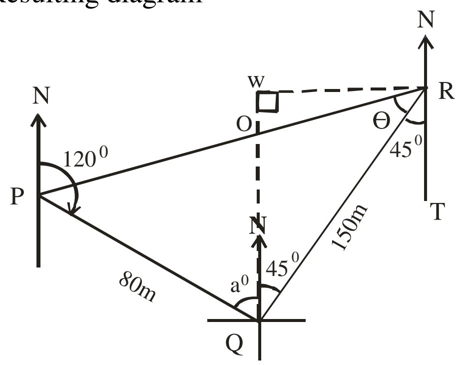
(b) How far west of R is Q to 2 decimal places?

(c) Find the perpendicular distance from Q to $\overline{PR}$ to 2 d.p

Solution



Resulting diagram



Breakdowns:

∠NQR = ∠QRT i.e 45^0 (Alternate angles)

$120^0 + a^0 = 180^0$ (sum of interior opp. ∠s)

$a^0 = 180 - 120$ i.e 60^0

Thus ∠PQR $= a^0 + 45^0$ i.e 105^0

a (i) PR can be gotten using cosine rule since we have two sides and an include angle

$PR^2 = 80^2 + 150^2 - 2 \times 80 \times 150 \cos 105^0$

$= 6400 + 22500 - 24000(-0.2588)$

$= 28900 + 6211.2$

$PR^2 = 35111.2$

$PR = \sqrt{35111.2} = 187.38\text{m} \approx 190\text{m to 2s.f}$

a (ii) The bearing of P from R is $= \theta + 45^0 + 180^0$

we find θ by sine rule

$$\frac{187.38}{\sin 105} = \frac{80}{\sin\theta}$$

$$187.38 \sin\theta = 80 \sin 105$$

$$\sin\theta = \frac{80\sin 105}{187.38}$$

$$\sin\theta = 0.4124$$

$$\theta = \sin^{-1} 0.4124$$

$$\theta = 24.36^0$$

Thus bearing $= 24.36 + 45^0 + 180$

$= 249.36^0 \approx 250^0$ to 2 s.f

(b) Our attention here will be on ΔQOR
and we are to find WR

The sides relevant to 45^0 are opp and hypo (SOH)

$$\text{Sin } 45^0 = \frac{WR}{150}$$

$150 \sin 45 = WR$

$106.066m = WR$

$WR \approx 106.07m$ to 2d.p

(c) We are to find QO in ΔQOR.

$\angle QOR = 180 - (45 + 24.36)$ *sum of ∠s in a Δ*

$= 110.64^0$

By sine rule : $\dfrac{QO}{\sin 24.36} = \dfrac{150}{\sin 110.64}$

$$QO = \frac{150 \sin 24.36}{\sin 110.64}$$

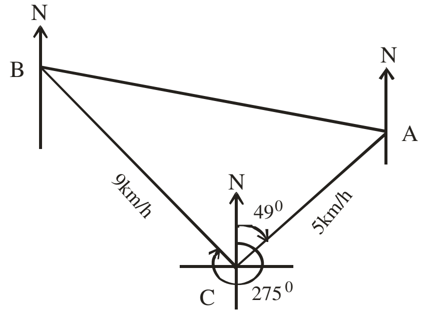
$QO \approx 66.11m$ to 2 d.p

2006/6 Neco (Dec)

Two cyclists A and B leave a town C at the same time. cyclist A speeds at the rate of 5 km/h on a bearing of 049^0 and B at the rate of 9km/h on a bearing of 275^0. After two hours, calculate:

(i) how far apart they are, correct to the nearest km,
(ii) the bearing of B from A to the nearest degree and
(iii) the speed at which A will get to B in 4 hours.

Solution



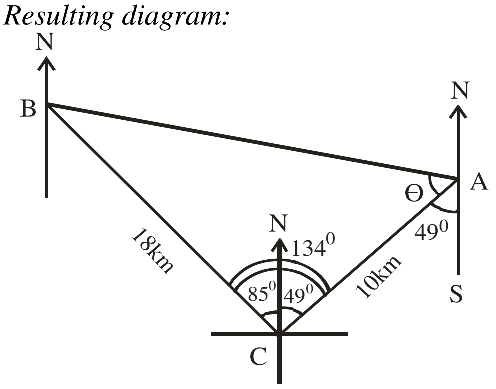
$$\text{Speed} = \frac{distance}{time}$$

Distance = speed × time

Distance BC = 9 × 2 i.e 18km

Distance AC = 5 × 2 i.e 10km

Resulting diagram:



Breakdown:

$\angle NCB = 360^0 - 275^0$ (∠s at a point)

$= 85^0$

$\angle NCA = \angle CAS$ (alternate ∠s)

(i) Distance AB can be gotten by cosine formula since we have two sides and an included angle

$AB^2 = 18^2 + 10^2 - 2 \times 18 \times 10 \cos 134^0$

$= 324 + 100 - 360(-0.6947)$

$= 424 + 250.092$

$AB^2 = 674.092$

$AB = \sqrt{674.092}$

$= 25.96$ km

≈ 26 km to the nearest km

(ii) Bearing of B from A = $\theta + 49^0 + 180^0$

But θ can be gotten by sine rule

$$\frac{25.96}{\sin 134} = \frac{18}{\sin\theta}$$

$25.96 \sin\theta = 18 \sin 134$

$$\sin\theta = \frac{18 \sin 134}{25.96}$$

$\sin\theta = 0.4988$

$\theta = \sin^{-1} 0.4988$

$= 29.92^0 \approx 30^0$ to the nearest degree

$\therefore$ Bearing of B from A = $30^0 + 49^0 + 180^0 = 259^0$

(iii) $\text{Speed} = \dfrac{distance\ AB}{time}$

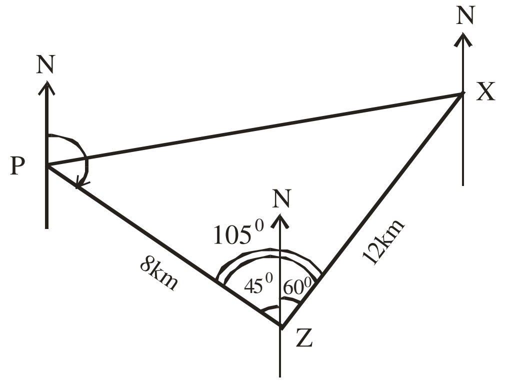
$= \dfrac{25.96}{4} = 6.49$ km/hr

2013/47 Neco

A cyclist on training cycles for 4 hours at an average speed of 2 km/hr from a point P on a bearing of 135^0 to another point Z, then to another point X, 12km on a bearing of 060^0. Calculate the distance /PX/ correct to two decimal places

A 12.14km B 15.45km C 15.46km
D 16.05km E 19.55km

Solution



Breakdown:

Distance PZ = speed × time

= 2 × 4 i.e 8km

$135^0 + \angle PZN = 180^0$ (sum of interior opp. ∠ s)

$\angle PZN = 180 - 135$ i.e 45^0

PX can be gotten by cosine formula since we two sides and an included angle

$PX^2 = 8^2 + 12^2 - 2 \times 8 \times 12 \cos 105^0$

$= 64 + 144 - 192(-0.2588)$

$= 208 + 49.6896$

$PX^2 = 257.6896$

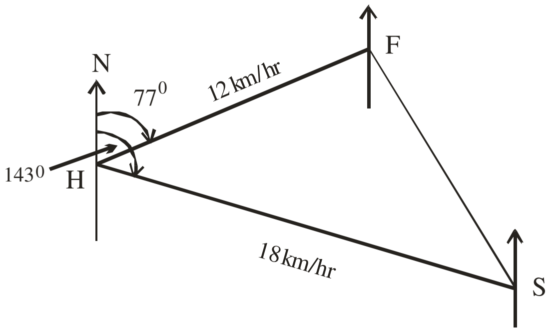
$PX = \sqrt{257.6896} \approx 16.05$km to 2 dp

2014/11b Neco (Dec)

Two boats take off from a harbour at the same time. The first travels at 12 km/h on a bearing 077^0 while the second travels at 18km/h on a bearing of 143^0. Calculate their distance apart after 3 hours

Solution

H for Harbour, **F** for first *and* **S** for second travel



Speed = $\frac{distance}{time}$

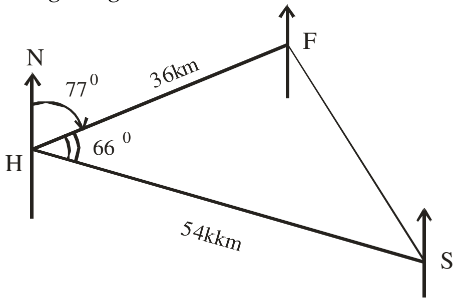
Distance = speed × time

Distance H F = 12 × 3 i.e 36km

Distance HS = 18 × 3 i.e 54km

$\angle$F HS = $143^0 - 77^0$ i.e 66^0

Resulting diagram:



Thus distance apart FS can be gotten by cosine rule since we have two sides and an included angle

$FS^2 = 36^2 + 54^2 - 2 \times 36 \times 54 \cos 66^0$

$= 1296 + 2916 - 1581.39$

$FS^2 = 2630.61$

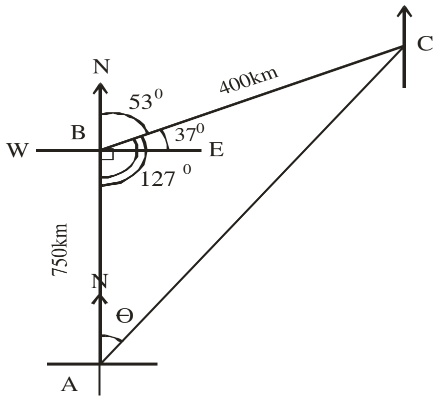
$FS = \sqrt{2630.61}$

$\approx$ 51.29km

2014/8 Neco (Dec)

An aeroplane leaves an international airport and flies due north for $1^1/_2$ hours at 500km/hr. It then flies for 30 minutes at 800km/hr on a bearing of N53^0E. Calculate its distance and bearing from the airport.

Solution



Breakdown:

Distance = speed × time

Distance AB = 500 × 1.5 i.e 750km

Distance BC = 800 × 0.5 i.e 400km

Note: 30 min is 0.5 hour

$\angle$CBE = 90 – 53 i.e 37^0

Distance AC can be gotten by cosine formula since we have two sides and included angle

$AC^2 = 750^2 + 400^2 - 2 \times 750 \times 400 \times \cos 127^0$

$= 562500 + 160000 - 600000\ (-0.6018)$

$AC^2 = 722500 + 361080$

$AC^2 = 1083580$

$AC = \sqrt{1083580}$

= 1040.95km

Bearing from the airport i.e bearing of C from A which is θ. It can be gotten by sine rule

$$\frac{1040.95}{\sin 127} = \frac{400}{\sin \theta}$$

$1040.95 \sin \theta = 400 \sin 127$

$\sin \theta = \frac{400 \sin 127}{1040.95}$

$\sin \theta = 0.3069$

$\theta = \sin^{-1} 0.3069$

$= 17.87^0$

2014/12b

The bearing of Q from P is 150^0 and the bearing of P from R is 015^0. If Q and R are 24 km and 32km respectively from P:

(i) Represent this information in a diagram;

(ii) Calculate the distance between Q and R,

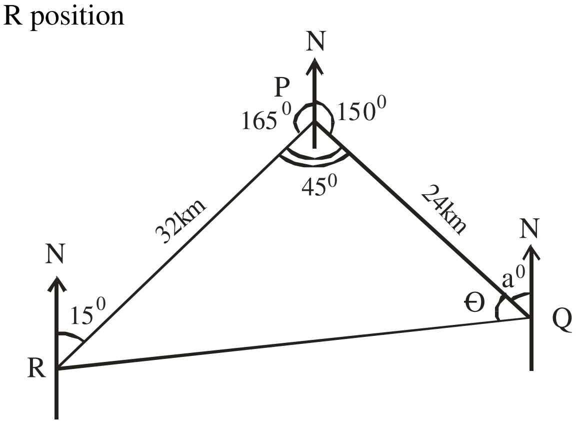
(iii) Find the bearing of R from Q, correct to the nearest degree

Solution

If the bearing of P from R is 015^0 then, the angle at P + 15^0 = 180^0.

Reason ¨sum of interior opp angles

Thus angle at P is 165^0 as shown which enabled us to locate R position



$\angle$RPQ is 360 – (165 + 150) = 45^0

$QR^2 = 32^2 + 24^2 - 2 \times 32 \times 24 \cos 45$

$QR^2 = 1024 + 576 - 1086.12$

$QR^2 = 513.88$

$QR = \sqrt{513.88}$ = 22.669km ≈ 22.67km to 2d.p

(iii) The bearing of R from Q = $360 - (\theta + a^0)$

$a^0 + 150^0 = 180^0$ (sum of interior opp. $\angle$s)

$a^0 = 180 - 150$ i.e 30^0

θ can be gotten by sine rule

$$\frac{22.67}{\sin 45^0} = \frac{32}{\sin \theta}$$

$22.67 \sin \theta = 32 \sin 45^0$

$\sin \theta = \frac{32 \sin 45^0}{22.67}$

$\sin \theta = 0.9981$

$\theta = \sin^{-1} 0.9981 = 86.47^0$

Thus bearing of R from Q = $360 - (86.47 + 30^0)$
$= 360 - 116.47$
$= 243.53^0$
$\approx 244^0$ to the nearest degree

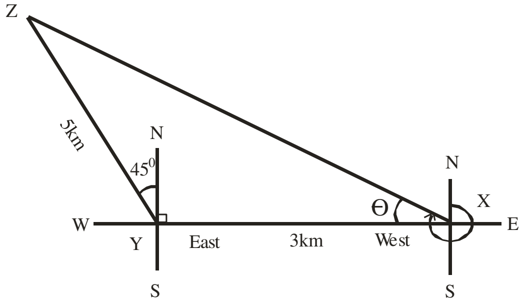
2011/10b

A cyclist starts from a point X and rides 3km due west to a point Y. At Y, he changes direction and rides 5km north – west to a point Z.

(i) How far is he from the starting point, correct to the nearest km?

(ii) Find the bearing of Z from X to the nearest degree

Solution



Note: North – west implies 45^0 as shown
$\angle ZYX = 45^0 + 90$ i.e 135^0

(i) ZX can be gotten by cosine formula since we have two sides and an included angle

$ZX^2 = 3^2 + 5^2 - 2 \times 3 \times 5 \cos 135$
$= 9 + 25 - 30(-0.7071)$
$= 34 + 21.213 = 55.213$

$ZX = \sqrt{55.213} = 7.43$km $\approx$ 7km to the nearest km

(ii) Bearing of Z from X = $270^0 + \theta$
we solve for θ using sine rule

$$\frac{7.43}{\sin 135} = \frac{5}{\sin\theta}$$

$7.43 \sin \theta = 5 \sin 135$

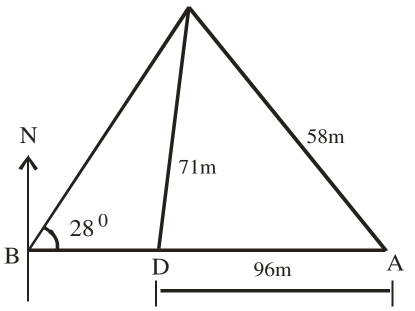
$$\sin \theta = \frac{5\sin 135}{7.43}$$

$\sin \theta = 0.4758$

$\theta = \sin^{-1} 0.4758 = 28.41^0$

Bearing of Z from X $= 270 + 28.41$
$= 298.41^0 \approx 298^0$

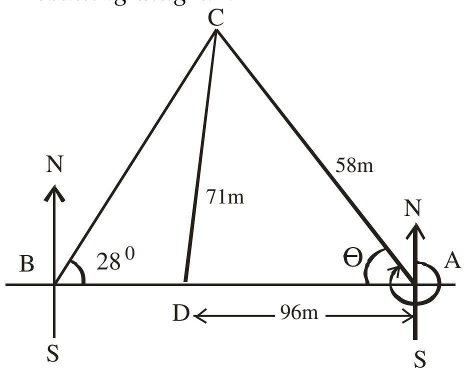
2005/7 (Nov)



In the diagram, **A** is a house which lies due east of a mosque **B**. Feeder roads run from **A** to **C** and from **C** to **B**. A foot path leads from **C** to the point **D** to **AB**. The distance between these places are measured as follows: /AC/= 58m, /CD/ = 71m, /AD/ = 96m and $A\hat{B}C = 28^0$, calculate: (a) The bearing of **C** from **A**, correct to the nearest degree (b) The distance **BD**, correct to the nearest metre.

Solution

Resulting diagram



The bearing of C from A = $\theta + 270^0$
But θ can be gotten from ΔACD by cosine rule since we have all sides without any angle

$$71^2 = 58^2 + 96^2 - 2 \times 58 \times 96 \cos \theta$$

$2 \times 58 \times 96 \cos \theta = 58^2 + 96^2 - 71^2$

$$\cos \theta = \frac{58^2 + 96^2 - 71^2}{2 \times 58 \times 96}$$

$$\cos \theta = \frac{3364 + 9216 - 5041}{11136}$$

$\cos \theta = 0.6770$

$\theta = \cos^{-1} 0.6770$

$\theta = 47.39^0$

Bearing of C from A $= 47.39^0 + 270^0$
$= 317.39^0$
$\approx 317^0$ to the nearest degree

(b) Distance BD can be gotten from AB in ΔABC as:

$\hat{C} + 28^0 + 47.39 = 180^0$ (sum of $\angle$s in Δ)

$\hat{C} = 180 - 75.39$
$= 104.61^0$

Applying sine rule

$$\frac{AB}{\sin 104.61} = \frac{58}{\sin 28^0}$$

$AB \sin 28^0 = 58 \sin 104.61^0$

$$AB = \frac{58\sin 104.61^0}{\sin 28^0}$$

$$AB = \frac{58 \times 0.9677}{0.4695}$$

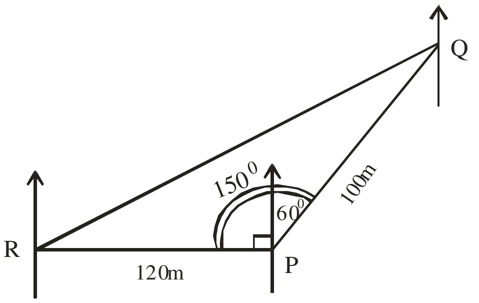
AB = 119.55m

Thus BD = 119.55 – 96
= 23.55m $\approx$ 24m to the nearest metre

2009/3a Neco (Dec)

Point Q is on a bearing 060^0 from P, point R is due west of P, If the distance PQ is 100m and PR is 120m, find /QR/

Solution



$QR^2 = 120^2 + 100^2 - 2 \times 120 \times 100 \cos 150$
$= 14400 + 10000 - [2 \times 120 \times 100 \times (-0.8660)]$

$QR^2 = 24400 + 20784$

$QR^2 = 45184$

$QR = \sqrt{45184}$

$= 212.57m$

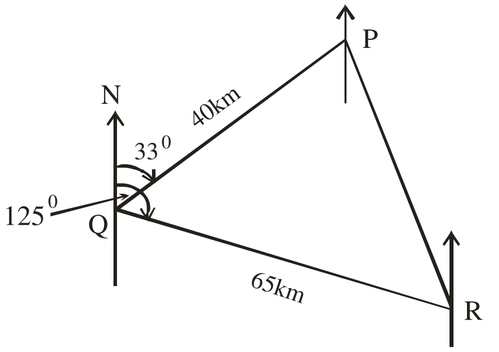
2006/13b (Nov) counter example

A point P is 40km away from point Q and on a bearing of 033^0. Another point R is 65km from Q and on a bearing of 125^0. Calculate, correct to the nearest whole number, the

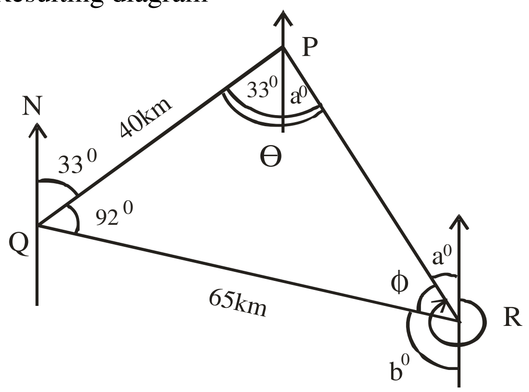
(i) distance between P and R;

(ii) bearing of P from R

Solution



Resulting diagram



(i) PR can be gotten by cosine rule since we have two sides and an included angle

$PR^2 = 40^2 + 65^2 - 2 \times 40 \times 65 \times \cos 92$

$PR^2 = 1600 + 4225 - 2 \times 40 \times 65 \times (-0.0349)$

$PR^2 = 5825 + 181.48$

$PR^2 = 6006.48$

$PR = \sqrt{6006.48} = 77.5km$

$\approx 78km$ to nearest whole number

(ii) The bearing of P from R is $\phi + b^0 + 180^0$

$b^0 = 125^0$ ($33^0 + 92^0$) *Alternate angles*

We find ϕ by sine rule

$$\frac{77.5}{\sin 92} = \frac{40}{\sin \phi}$$

$77.5 \sin\phi = 40 \sin 92$

$$\sin\phi = \frac{40 \sin 92}{77.5}$$

$\phi = \sin^{-1} 0.5158$

$\phi = 31.05^0$

The bearing of P from R is $31.05^0 + 125^0 + 180^0$

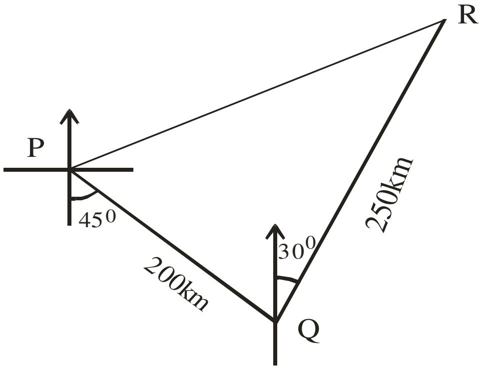
$= 336.05^0$

$\approx 336^0$ to nearest whole number

2009/5b Neco Exercise 15.39

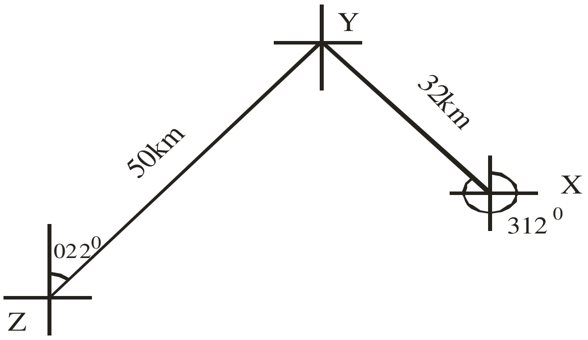
A man travels from station A on a bearing of 050^0 to station B, a distance of 12km. He then leaves to station C, 10km away from B on a bearing of 110^0. How far is he from his starting point? (Answer to 2d.p)

2010/8a Neco Exercise 15.40



The above diagram represents the flight of an aeroplane from city P on a bearing of $S45^0E$ to another city Q a distance of 200km. It then flew a distance of 250km on a bearing of $N30^0E$ from city Q to city R. If it flew back directly to city P, Calculate the (i) total distance covered, correct to there significant figures (ii) bearing of city R from city P, correct to the nearest degree.

2006/13b Exercise 15.41



The diagram shows the positions of three points X, Y and Z on a horizontal plane. The bearing of Y from X is 312^0 and that of Y from Z is 022^0. If /XY/ = 32km and /ZY/ = 50km. Calculate, correct to one decimal place:

(i) /XZ/ ; (ii) the bearing of Z from X

2014/8c NABTEB (Nov) Exercise 15.42

A ship steams 4km due North from a point and then 3km on a bearing of 040^0. How far is the ship from the point?

1978/5 Exercise 15.43

A man sets out to travel from A to C via B. From A he travels a distance of 8km on a bearing $N30^0E$ to B. From B he travels a further 6 km due east.

(a) Calculate how far C is

(i) north of A (ii) east of A

(b) Then, calculate the distance AC, correct to1 decimal place.

2007/3 Neco Exercise 15.44

A hunter walks from one forest X on a bearing of 080^0 to another forest Y which is 40km away. From there, he went another forest Z a distance of 50km on a bearing of 220^0 in search of an antelope which he shot. Calculate:

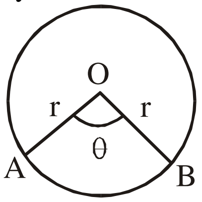
(i) distance of Z from X to the nearest kilometer

(ii) bearing of Z from X to the nearest degree

` # CHAPTER SIXTEEN

Latitude & Longitude

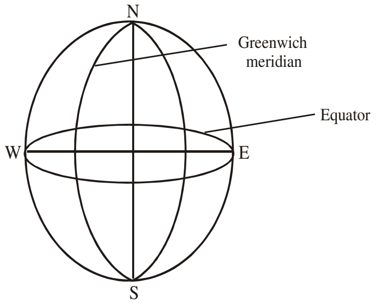
This is one of the theory based areas to be tested in certificate exams. The formula of the length of an *arc of a circle* plays a crucial role here.



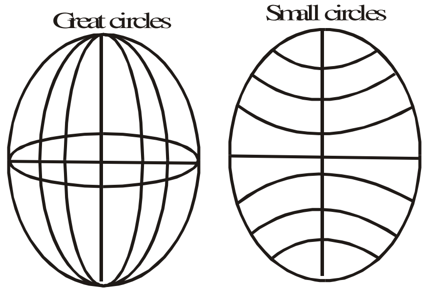
circle with centre O

Length of arc AB = $\frac{\theta}{360} \times 2\pi r$

The Earth is spherical in shape. Any cut or slice through it horizontally (Latitude) or vertically (Longitude) is circular in shape. To find the distance between any two points on such slice we have to use the above-mentioned formula

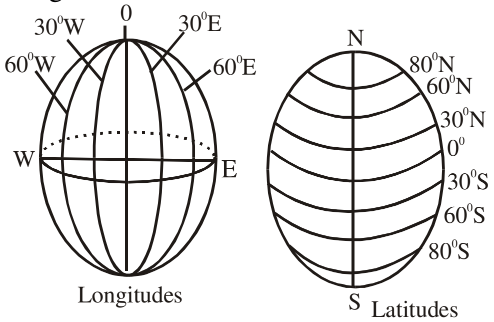


The measurement of any point on the latitudes either to the east or west must start from the Greenwich meridian; while that of longitude either to south or north must start from the equator - *please note the difference.* Any point on the earth surface is the intersection of (latitude, longitude) like in co-ordinates (x,y)



Units of Measurement

Degrees and seconds are the units of measurement. Latitude is from 0^0 to 90^0 south or North ; while Longitude is from 0^0 to 180^0 East or West

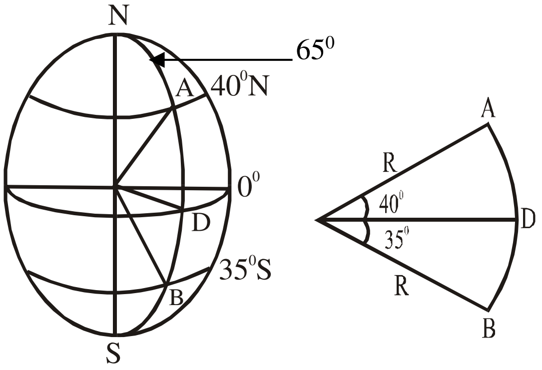


Longitudes   Latitudes

Example 1

Find the angular distance between A (40^0 N, 65^0E) and B (35^0 S, 65^0E)

Solution



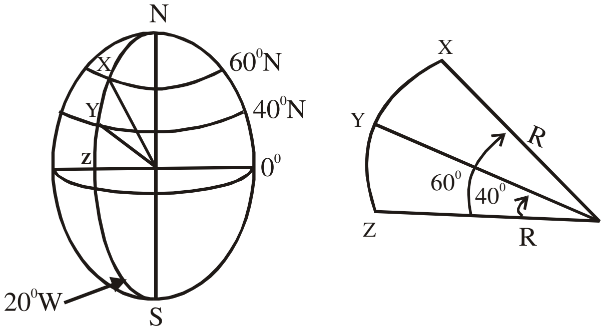
Angular distance = $40^0 + 35^0$
$= 75^0$

Note that we add when the points are in different hemispheres i.e South and North.

Example 2

Find the angular distance between X (60^0N, 20^0W) and Y (40^0 N, 20^0W)

Solution



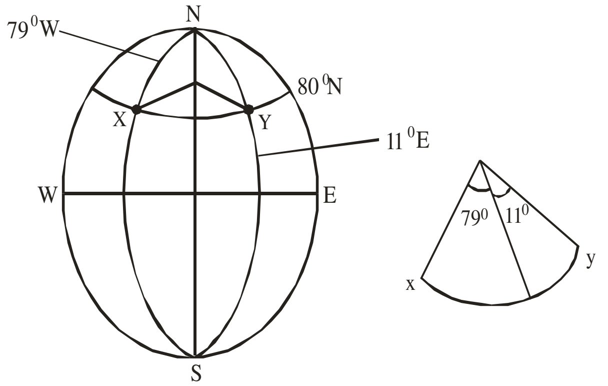
Angular distance = $60^0 - 40^0 = 20^0$

2010/ 32 Neco

X and Y are two places on the same circle of latitude 80^0N, X is on longitude 79^0W and Y is on longitude 11^0E. What is the angular difference between X and Y?

A 60^0 B 69^0 C 80^0 D 90^0 E 170^0

Solution



Angular difference between X and Y = 79 + 11
$= 90^0$ (D)

Reason X and Y are on different sides of the globe west & east

Distance along great circles

Distance along *any* of the *longitudes* and distance along the *equator only* among the latitudes are called distance along great circles. The formula used is

Lengths of arc = $\frac{\theta}{360}$ x $2\pi R$

Where R is the radius of the earth.

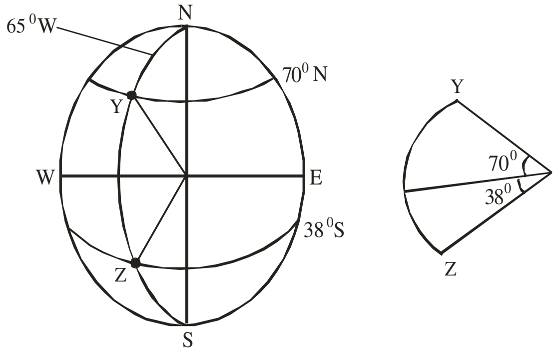
2005/36 – 37 NABTEB

Y and Z are two places on the surface of the earth along the meridian such that Y (lat 70°N, long 65°W) and Z (lat 38°S, long 65°W)

36. Calculate the sectorial angle of YZ

A 32° B 38° C 70° D 108°

Solution



Sectorial angle of YZ = 70°+38°

= 108° (D)

37. Calculate the distance YZ if the radius of the earth is 6400km

A 12800πkm B 8960πkm C 3840πkm D 1138πkm

Solution

Distance YZ ⇒ Distance along a great circle = $\frac{\theta}{360} \times 2\pi R$

$= \frac{108}{360} \times 2 \times \pi \times 6400$

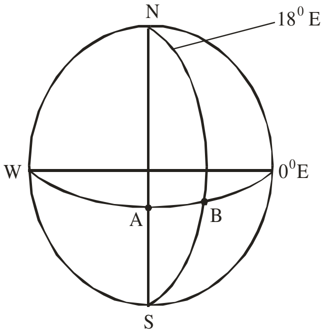
$= 3840\pi$km (C)

2011/33 Neco

An air craft flew from city A (lat 0^0, long 0^0) to another city B (lat 0^0, long 18^0) at an average speed of 1000km/hr. Calculate the distance travelled to the nearest km, (Take $\pi = {}^{22}/_7$, R = 6370km)

A 2000km B 2002km C 2100km D 2200km E 2220km

Solution



Distance travelled along lat 0^0 is the equator and that is distance along great circle

$= \frac{\theta}{360} \times 2 \times \pi R$

$= \frac{18}{360} \times 2 \times \frac{22}{7} \times 6370$

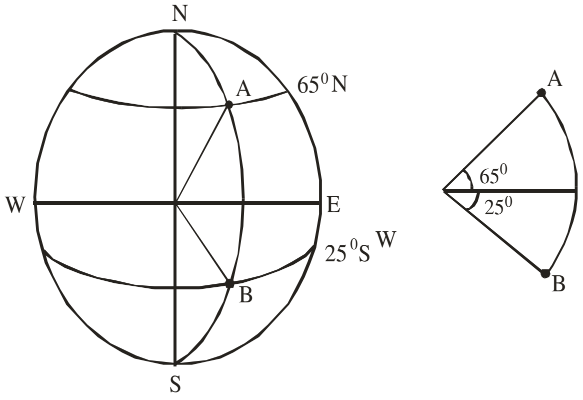
= 2002km (B)

2008/24 NABTEB (Nov)

Find the distance between A(65^0N, 3^0 E) and B(25^0S, 3^0E) along their common meridian on the earth's surface (Use $\pi = {}^{22}/_7$, R = 6370km)

A 10010Km B 12460Km C12590Km D 13085Km

Solution



Distance AB (Great circle) $= \frac{\theta}{360} \times 2 \times \pi R$

Here θ is 65° + 25° i.e 90^0 since they are N and S

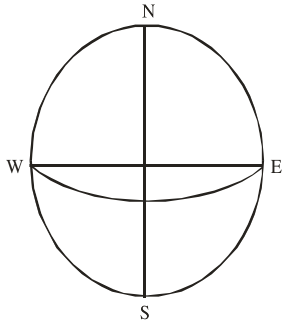
$= \frac{90}{360} \times 2 \times \frac{22}{7} \times 6370$

= 10010km (A)

2007/6c NABTEB

Two places on the equator are 7900km apart measured along the equator. Find the difference in their longitudes. (Take R = 6370km and $\pi = 3.14$)

Solution



Distance along the equator is in a great circle $= \frac{\theta}{360} \times 2\pi R$

$7900 = \frac{\theta}{360} \times 2 \times 3.14 \times 6370$

$\frac{7900 \times 360}{2 \times 3.14 \times 6370} = \theta$

$71.09^0 = \theta$

2015/39 Neco Exercise 16.1

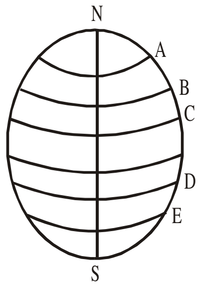
Two places on the same meridian have latitudes 10^0S and 53^0N.What is their distance apart measured along the meridian? (Take $\pi = {}^{22}/_7$, R = 6370km)

A 7000km B 7040km C 7100km

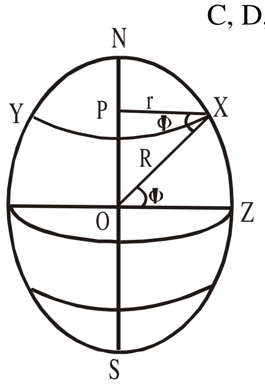
D 7300km E 7400km

Distance along small circles
(Distance along parallel of latitudes)

This is any distance along the latitudes *except the equator.*



Here distance between any two points on latitudes along A, B, C, D, E is derived as shown next



Assuming X to be a point on the earth surface with latitudes Φ (phi), of course XY is a small circle with radius r. The resulting Δ OPX has

$\cos \Phi = \frac{r}{R}$

Where R is the radius of the earth

r – small r

R – Big R

Thus to calculate the distance of any two points on the parallel of latitudes, we apply the formula

$\text{Distance} = \frac{\theta}{360} \times 2\pi r$ But $r = R\cos\Phi$

$= \frac{\theta}{360} \times 2\pi R \cos\Phi$

Where **θ** is the angular distance of the points on the longitudes to East or West.

Φ is the angle between the given parallel of latitude and the equator i.e latitude measurements is to North or South

Note: θ and Φ are dummy variables; they can be changed But always remember what is attached to them

2014/37 NABTEB

Calculate the radius of latitude 75^0N

A 1656km B 3911km C 4229km D 6182km

Solution

Radius $r = R\cos\phi$

$= 6400 \times \cos 75^0$

$= 6400 \times 0.2588$

$= 1656.32$km

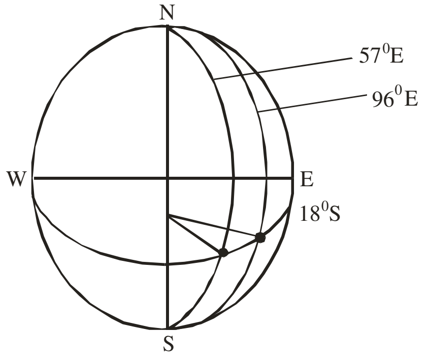
≈ 1656km (A)

2010/23 NABTEB

Find the distance measured along the parallel of latitude between two places with the same latitude, 18^0S and with longitudes 96^0E and 57^0E

A 4020km B 4120km C 4130km D 4220km

Solution



Distance along parallel of latitude = $\frac{\theta}{360} \times 2\pi R \cos\phi$

Here θ is 96 – 57 i.e 39 and ϕ is 18

$= \frac{39}{360} \times 2 \times \frac{22}{7} \times 6400 \times \cos 18$

$= \frac{39}{360} \times 2 \times \frac{22}{7} \times 6400 \times 0.9511$

$= 4144.98$km

≈ 4145km

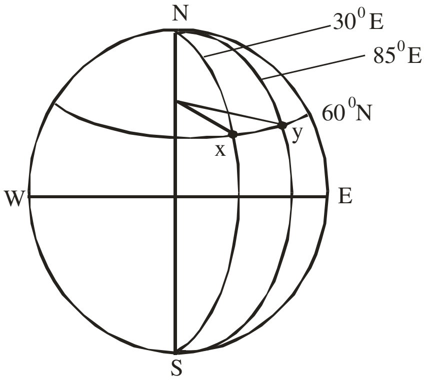
2005/ 34 – 35 NABTEB (Nov)

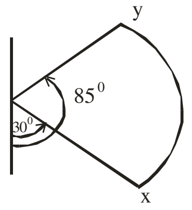
X and Y are two places on the earth surface such that X (lat 60^0N, long 30^0E) and Y (lat 60^0N, long 85^0E)

34. Find the sectorial angle XY measured along the parallel of latitude

A 5^0 B 55^0 C 60^0 D 115^0

Solution





Sectorial angle XY = 85 – 30

$= 55^0$ (B)

35. If the radius of the earth is 6400km, find the radius of the parallel of latitude

A 3.200km B 3670km C 3671km D 5243km

Solution

Distance along parallel of latitude = $\frac{\theta}{360} \times 2\pi r$

Where $r = R\cos\phi$

and ϕ is the latitude measure to N or S

Here we are simply asked to find r in

$R = r\cos\phi$

$R = 6400 \times \cos 60$

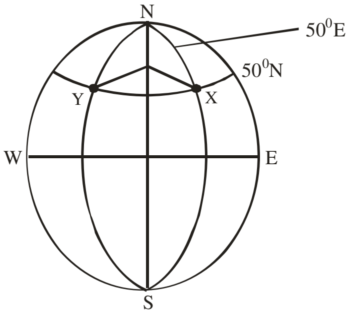
$= 3{,}200$km (A)

2009/32 Neco (Nov) ***counter example***

Two points X and Y are on latitude 50^0N and are directly opposite each other. If the longitude of X is 50^0E, what is the longitude of Y?

A 130^0W B 130^0E C 50^0W D 40^0W E 40^0E

Solution



Longitude of Y is 50^0W

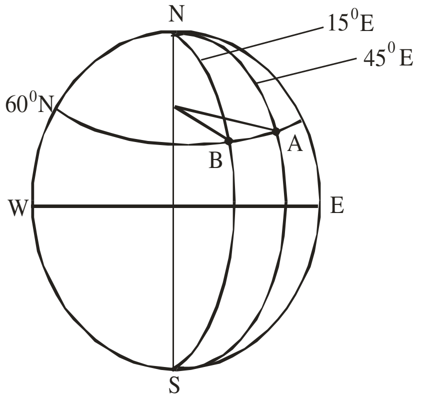
(Points X and Yare directly opposite each other on the same latitude).

2006/31 Neco (Nov)

A(latitude 60^0N, longitude 45^0E) and B(latitude 60^0N, longitude 15^0E) are two towns on the earth surface. Calculate, correct to 3 significant figures, the distance between A and B along the parallel of latitude (Take R = 6400km)

A 6700km B 3530km C 3350km
D 1860km E 1680km

Solution



Distance along parallel of latitude AB = $\frac{\theta}{360} \times 2\pi R \cos\phi$

Here θ is 45 – 15 i.e 30 and ϕ is 60

$= \frac{30}{360} \times 2 \times \frac{22}{7} \times 6400 \times \cos 60$

$= \frac{30}{360} \times 2 \times \frac{22}{7} \times 6400 \times 0.5$

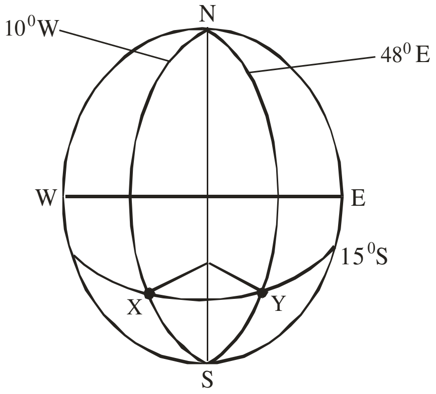
= 1676.19 km

≈ 1680km to 3 s.f (E)

2006/8b NABTEB (Nov)

An aeroplane flies at 650km per hour along the parallel of latitude from a point X(15^0S, 10^0W) to Y(15^0S, 48^0E). Calculate the time spent by the aeroplane to fly from X to Y to the nearest 1 hour (Take R = 6400km and π = 3.142)

Solution



Speed = $\frac{Distance\ XY}{time}$

Here speed is 650km/hr, time is?

Distance XY is

Distance along parallel of latitude = $\frac{\theta}{360} \times 2\pi R \cos \phi$

Here θ is 10 + 48 i.e 58 and ϕ is 15

$= \frac{58}{360} \times 2 \times 3.142 \times 6400 \times \cos 15$

$= \frac{58}{360} \times 2 \times 3.142 \times 6400 \times 0.9659$

= 6258.55 km

Thus, speed = $\frac{Distance\ XY}{time}$

$650 = \frac{6258.55}{time}$

Time = $\frac{6258.55}{650}$

= 9.629hrs

≈ 10hrs to the nearest 1 hour

2004/35 NABTEB (Dec) Exercise 16.2

What is the angle between two points P and Q whose longitudes are 102^0E and 38^0W lying on latitude 30^0S

A 140^0 B 94^0 C 64^0 D 52^0

2008/22 NABTEB (Nov) Exercise 16.3

Calculate the radius of the parallel of latitude 60^0N. (use R = 6400km)

A 110km B 3200km C 4600km D 6340km

2015/53 Neco Exercise 16.4

Two villages are on (15^0S, 107^0E) and (15^0S, 17^0E) respectively. What is their distance apart, along the line of latitude? (Leave your answer in terms of π and R, the radius of the earth)

A $2\pi R \cos 15^0$ B $\frac{\pi R}{4} \sin 15^0$ C $\frac{\pi R}{4} \cos 15^0$

D $\frac{\pi R}{2} \sin 15^0$ E $\frac{\pi R}{2} \cos 15^0$

Joint cases on small and great circles

2009/9a (Nov)

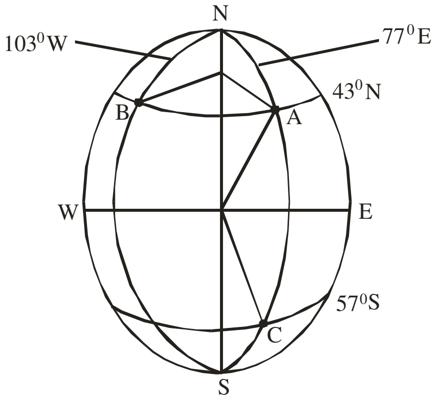
A(lat 43^0N, long 77^0E), B(lat 43^0N, long 103^0W) and C(lat 57^0S, long 77^0E) are three points on the surface of the earth. Find the distance from:

(i) A to B along latitude 43^0N

(ii) A to C along the great circle joining the two points

[Take $\pi = 3.142$ and R = 6400km].

Solution



(i) Distance AB (Small circle) = $\frac{\theta}{360} \times 2\pi R \cos\phi$

Here $\theta = 103 + 77 = 180^0$ and $\phi = 43^0$

(Addition: east to west)

$$\text{Distance AB} = \frac{180}{360} \times 2 \times 3.142 \times 6400 \times \cos 43^0$$
$$= 3.142 \times 6400 \times \cos 43^0$$
$$= 14706.64\text{km}$$

(ii) Distance AC (Great circle) = $\frac{\theta}{360} \times 2\pi R$

Here $\theta = 43 + 57$ (North to South) $= 100^0$

$$\text{Distance AC} = \frac{100}{360} \times 2 \times 3.142 \times 6400$$
$$= 11171.56\text{km}$$

2010/10 Neco

Two points P and Q on latitudes 40^0S are on longitude 15^0W and 45^0W respectively.

(a) Find the distance between P and Q along their parallel of latitude, correct to the nearest 100km.

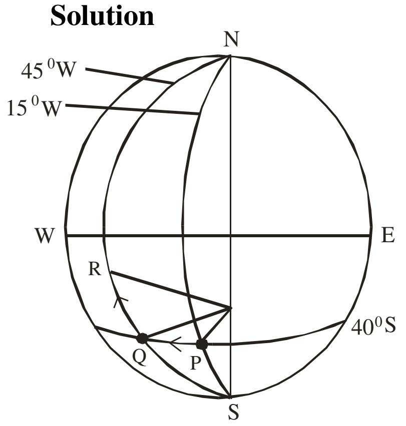
(b) A third point R is on the same longitude as Q but due north of it. If the distance between Q and R is 6700km,

(i) Find the total time taken by an aircraft which flies at a constant speed of 850km/hr from P to Q and then to R.

(ii) What is the latitude of R to the nearest degree?

(Take $\pi = {}^{22}/_{7}$ and radius of the earth as 6400km)

Solution



(a) Distance along parallel of latitude PQ = $\frac{\theta}{360} \times 2\pi R \cos\phi$

Here θ is 45 – 15 i.e 30 and ϕ is 40

$$= \frac{30}{360} \times 2 \times \frac{22}{7} \times 6400 \times \cos 40$$
$$= \frac{30}{360} \times 2 \times \frac{22}{7} \times 6400 \times 0.7660$$
$$= 2567.92\text{km}$$
$$\approx 2600 \text{ km to the nearest 100km}$$

(b) (i) speed = $\frac{Distance\ PQR}{time}$

$$850 = \frac{\overset{PQ}{2567.92} + \overset{QR}{6700}}{time}$$

$850 \times \text{time} = 9267.92$

$$\text{Time} = \frac{9267.92}{850}$$
$$= 10.9 \text{ hrs}$$

(ii) Latitude of R can be gotten from θ in:

Distance along longitude QR = $\frac{\theta}{360} \times 2\pi R$

$$6700 = \frac{\theta}{360} \times 2 \times \frac{22}{7} \times 6400$$
$$= \frac{6700 \times 360 \times 7}{2 \times 22 \times 6400} = \theta$$
$$59.96^0 = \theta$$

Thus latitude of R = 59.96 – 40

= 19.96 N

$\approx 20^0$ N to the nearest degree

As the plane flies from South to north, once θ exceeds 40^0, it has entered the Northern Hemisphere

2012/9a Neco

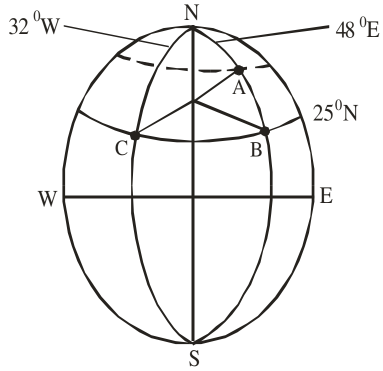
The longitude of town B in 48^0E and the longitude of another town C is 32^0W. Both towns B and C lie on latitude 25^0N. Calculate, correct to 3 significant figures, the:

(i) distance between town B and town C along the parallel of latitude;

(ii) Chord BC

(b) Town A is 4800km north of town B. If the latitude of town B is 25^0N, what is the latitude of town A correct to the nearest degree?

(Take $\pi = 3.142$ and radius of the earth = 6400km)



(i) Distance along parallel of latitude BC = $\frac{\theta}{360} \times 2\pi R \cos\phi$

Here θ is 32 + 48 i.e 80 and ϕ is 25

$= \frac{80}{360} \times 2 \times 3.142 \times 6400 \times \cos 25$

$= \frac{80}{360} \times 2 \times 3.142 \times 6400 \times 0.9063$

$= 8099.82$ km

≈ 8100km to 3 significant figures

(ii) Chord BC $= 2 r \sin \frac{\theta}{2}$

$= 2 R \cos \phi \sin \frac{\theta}{2}$

Substituting

$= 2 \times 6400 \times \cos 25 \sin 40$

$= 2 \times 6400 \times 0.9063 \times 0.6428$

$= 7456.89$km

≈ 7460km to 3 s.f

(b) Latitude of town A can be gotten from θ in:

Distance along longitude AB $= \frac{\theta}{360} \times 2\pi R$

$4800 = \frac{\theta}{360} \times 2 \times 3.142 \times 6400$

$\frac{4800 \times 360}{2 \times 3.142 \times 6400} = \theta$

$42.97^0 = \theta$

Latitude of A $= (25 + 42.97)^0$ N

$= 67.97^0$N

$\approx 68^0$N to the nearest degree

Reason: we added because latitude A is in upper part of lat B; higher latitude than B i.e increasing latitude so we cannot subtract.

2011/13b

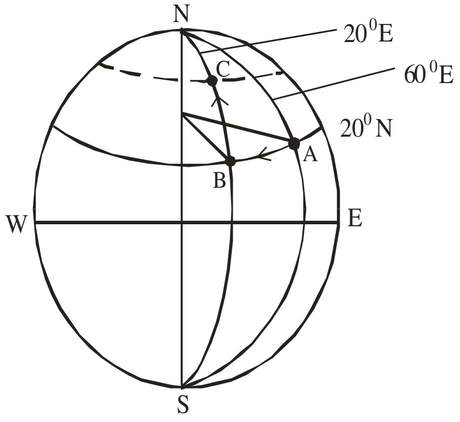
An aeroplane flies from town A(20^0N, 60^0E) to town B(20^0N, 20^0E).

(i) if the journey takes 6 hours, calculate, correct to 3 significant figures, the average speed of the aeroplane.

(ii) if it then flies due north from town B to town C, 420km away, calculate, correct to the nearest degree, the latitude of town C

[Take radius of the earth = 6400km and $\pi = 3.142$]

Solution



(i) Speed $= \frac{distance}{time}$

Here distance is

Distance along parallel of latitude AB $= \frac{\theta}{360} 2\pi R \cos \phi$

Here θ is 60 – 20 i.e 40^0 and ϕ is 20^0

$= \frac{40}{360} \times 2 \times 3.142 \times 6400 \times \cos 20$

$= \frac{40}{360} \times 2 \times 3.142 \times 6400 \times 0.9397$

$= 4199.16$ km

Thus speed $= \frac{4199.16}{6}$

$= 699.86$km/hr

≈ 700km/hr to 3 s.f

(ii) The latitude of C can be gotten from θ in:

Distance along longitude BC $= \frac{\theta}{360} \times 2\pi R$

$420 = \frac{\theta}{360} \times 2 \times 3.142 \times 6400$

$\frac{420 \times 360}{2 \times 3.142 \times 6400} = \theta$

$3.76^0 = \theta$

Thus latitude of C $= 20^0 + 3.76^0$ N

$= 23.76$ N

$\approx 24^0$ N to the nearest degree

Reason we added because latitude C is upper part of lat B; higher latitude that B: that is increasing in latitude so you cannot subtract

2014/7 Neco

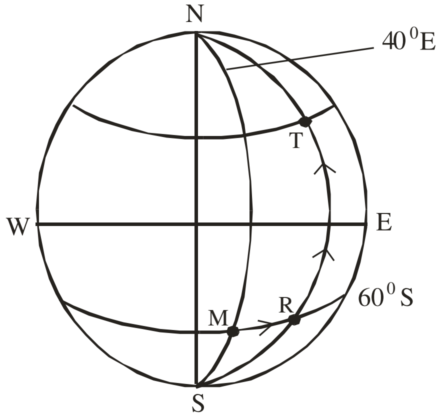
An aeroplane flies eastward from a town M(60^0S, 40E) for 4hours at an average speed of 300km/hr to another town R. it then flies due north from R to another town T a distance of 1800km. calculate the:

(a) Longitude of R;

(b) Latitude of T, correct to the nearest degree

(Take $\pi = {}^{22}/_{7}$, R = 6400km)

Solution



(a) Longitude of R can be gotten from:

Distance along parallel of latitude MR $= \frac{\theta}{360} \times 2\pi r$

where r is R cos ϕ

But Distance MR = speed from M to R × time

$= 300 \times 4$

$= 1200$km

Substituting

$1200 = \frac{\theta}{360} \times 2 \times \frac{22}{7} \times 6400 \times \cos 60^0$

$\theta = \frac{1200 \times 360 \times 7}{2 \times 22 \times 6400 \times 0.5}$

$= 21.48^0$

(which is the angular difference between M and R)

But longitude of R is $40^0 + \theta$

$= 40 + 21.48 = 61.48^0E$

$\approx 61^0E$ to the nearest degree

(b) The latitude can be gotten from :

Distance along longitude RT = $\frac{\theta}{360} \times 2\pi R$

$$1800 = \frac{\theta}{360} \times 2 \times \frac{22}{7} \times 6400$$

$$\theta = \frac{1800 \times 360 \times 7}{2 \times 22 \times 6400}$$

$$= 16.11^0$$

Latitude of T is $60 - 16.11 = 43.89^0$

$\approx 44^0$ to the nearest degree

We subtracted because the flight to T from R was due North

2005/8

A plane flies due East from A(lat 53^0N, long 25^0E) to a point B(lat 53^0N, long 85^0E) at an average speed of 400km/h. The plane then flies south from B to a point C 2000km away. Calculate, correct to the nearest whole number:

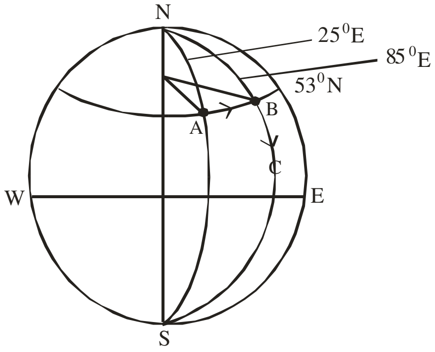
(a) the distance between A and B;

(b) the time the plane takes to reach point B;

(c) the latitude of C

[Take radius of the earth to be = 6400km and $\pi = {}^{22}/_{7}$]

Solution



(a) Distance between A and B is

Distance along parallel of latitude = $\frac{\theta}{360} \times 2\pi R \cos\phi$

Here θ is 85 – 25 i.e 60^0 (θ is in the same east, east), ϕ is 53^0

Substituting

$$= \frac{60}{360} \times 2 \times \frac{22}{7} \times 6400 \cos 53$$

$$= \frac{60}{360} \times 2 \times \frac{22}{7} \times 6400 \times 0.6018$$

$= 4034.93$km

≈ 4035km to the nearest km

(b) Speed = $\frac{distance}{time}$

$$400 = \frac{4034.93}{time}$$

$$\text{Time} = \frac{4034.93}{400} = 10.09\text{hrs}$$

≈ 10 hrs to the nearest whole number

(c) The latitude of C can be gotten from θ in

Distance along longitude BC = $\frac{\theta}{360} \times 2\pi R$

$$2000 = \frac{\theta}{360} \times 2 \times \frac{22}{7} \times 6400$$

$$\frac{2000 \times 360 \times 7}{2 \times 22 \times 6400} = \theta$$

$$17.90^0 = \theta$$

Since the plane flew south from B to C

Latitude of C = $53^0\text{N} - 17.90^0$

= 35.1N

$\approx 35^0$N to the nearest whole number

Point C did not enter the southern Hemisphere as shown on the sketch

2013/12 Exercise 16.5

An aeroplane flies due north from a town **T** on equator at a speed of 950km per hour for 4 hours to another town **P**. It then flies eastward to town **Q** on longitude 65^0E. If the longitude of T is 15^0E,

(a) Represent this information in a diagram

(b) Calculate the:

(i) Latitude of P, correct to the nearest degree;

(ii) Distance between P and Q, correct to 4 significant figures

[Take $\pi = {}^{22}/_{7}$, Radius of the earth = 6400km]

2009/11b Neco Exercise 16.6

P(55^0N, 16^0W), Q(55^0N, X^0E) and S(Y^0S, 16^0W) are three towns on the earth's surface. The distance between P and Q along the parallel of latitude 55^0N is 3800km and the distance between P and S along the meridian 16^0W is 5,000km. Calculate, correct to the nearest degree, the

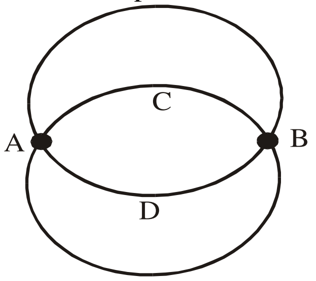
(i) value of x^0,

(ii) angle subtended by arc PS at the centre of the earth

(Take R = 6400km and $\pi = {}^{22}/_{7}$)

Shortest distance between any two points on the parallel of latitude i . e (Distance along the great circle of two points on the parallel of latitude).

Apart from the normal questions on distance along the great circle or along the parallel of latitudes, there are other questions on the shortest distance between any two points on the parallel of latitudes or distance along the great circle of any two points on the parallel of latitude. Below is a sketch of how the problem looks like:



This form of question are interwoven in the sense that the data given concerns mainly arc AB through D but candidates are required by the question to find out some facts on arc AB through C. This can be done in two ways namely:

Split and direct formula methods

Example 1

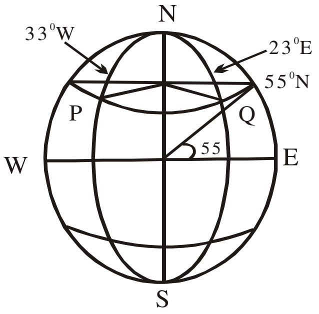
P and Q are two points on latitude 55^0N and their longitudes are 33^0 W and 23^0 E respectively. Calculate the distance between P and Q measured along:

(a) the parallel of latitude

(b) great circle

(Take $\pi = {}^{22}/_7$, Radius of the earth = 6400km)

Solution



(a) From the resulting diagram,

Distance PQ = $\dfrac{\theta}{360} \times 2\pi R \cos\phi$

Here $\theta = 33 + 23$ i.e 56^0 and $\Phi = 55^0$

PQ = $\dfrac{56 \times 2 \times 22 \times 6400 \times \cos 55}{360 \times 7}$

$= \dfrac{44 \times 1280 \times 0.5736}{9}$

= 3589 km

(b)

The distance between P and Q measured along the great circle is also the shortest distance between P and Q.

Method I (SPLIT)

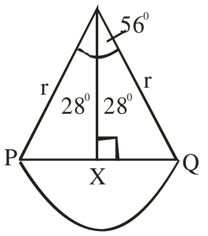
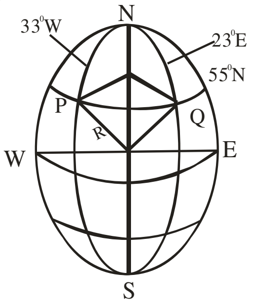


Fig I



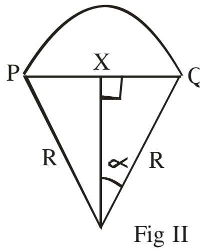


Fig II

From fig I

The length of chord PQ = 2 XQ

$= 2r \sin \dfrac{\theta}{2}$

$= 2 R \cos \Phi \sin \dfrac{\theta}{2}$

$= 2 \times 6400 \times \cos 55^0 \sin 28$

$= 2 \times 6400 \times 0.5736 \times 0.4695$

= 3447 km

From fig II

Arc PQ is the required distance $= \dfrac{2\alpha}{360} \times 2\pi R$

α is derived from fig. II as shown below:

$\sin\alpha = \dfrac{XQ}{R}$

$= \dfrac{\frac{1}{2}\ length\ of\ chord\ PQ}{R} = \dfrac{\frac{1}{2} \times 3447}{6400}$

$\sin\alpha = 0.2693$

$\alpha = \sin^{-1}(0.2691) = 15.6^0$

Thus, Arc PQ = $\dfrac{2 \times 15.6 \times 2 \times 22 \times 6400}{360 \times 7}$

$= 3486.47 \approx 3486$ km

Method II (Applied formula)

To avoid error due to students' judgment of *three-dimensional* shape in two dimensions, the above procedures can be summed up as follows;

Distance PQ along great circle $= \dfrac{2\sin^{-1}\left(\cos\phi \sin\dfrac{\theta}{2}\right) \times 2\pi R}{360}$

Where θ = angular distance or longitude difference.

Φ = is the given latitude to North or South

Substituting

Distance PQ = $\dfrac{2\sin^{-1}\left(\cos 55 \sin\dfrac{56}{2}\right) \times 2 \times 22 \times 6400}{360 \times 7}$

$= \dfrac{2\sin^{-1}(0.2691) \times 281600}{2520}$

$= \dfrac{31.20 \times 281600}{2520} = 3486$km

Note

θ and Φ are dummy variables, they can be changed to r and s, or a and b etc. But the values attached to them remain unchanged.

The Author applied, method I to solve the given problem for students to appreciate and be able to recall easily formula II which can be used without drawing much diagrams.

Candidates should note that the answer to the great circle distance is always <u>smaller</u> in value than that of the distance along parallel of latitude.

1975/11

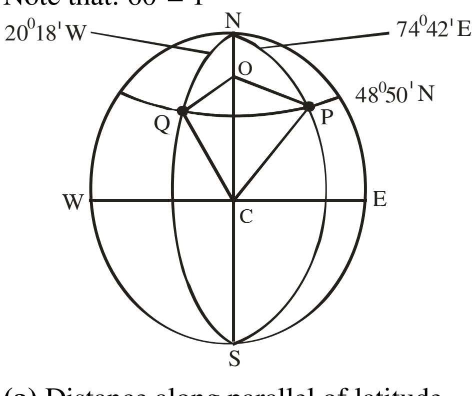
P is a town on longitude $74^0 42'$E and Q is another town on longitude $20^0 18'$W. Both towns are on latitude $48^0 50'$N. Calculate: (a)the distance between P and Q measured along their common latitude (b) the length of chord PQ

(c) the angle subtended by PQ at the centre of the Earth

[Take π to be 3.142, R = 6400km]

Solution

Note that: $60' = 1^0$



(a) Distance along parallel of latitude $= \dfrac{\theta}{360} \times 2\pi R \cos\phi$

Here θ is $20^0 18' + 74^0 42'$ i.e $94^0 60' = 95^0$, ϕ is $48^0\ 50'$ i.e 48.8^0

Substituting

$$= \frac{95}{360} \times 2 \times 3.142 \times 6400 \times \cos 48.8^0$$

$$= \frac{95}{360} \times 2 \times 3.142 \times 6400 \times 0.6587$$

$= 6990.77$km

(b) Length of chord PQ = $2\, r \sin \frac{\theta}{2}$

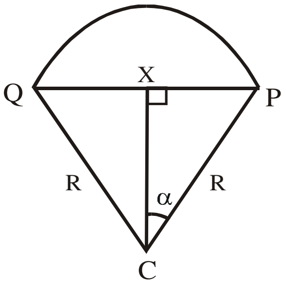
Where $r = R\cos\phi$

$$= 2 \times 6400 \times \cos 48.8^0 \times \sin \frac{95}{2}$$

$$= 2 \times 6400 \times 0.6587 \times 0.7373$$

$= 6216.44$km

(c) The angle subtended by PQ at the centre of the earth C is 2α in the diagram below derived from the main diagram.



In ΔCXP, $\sin\alpha = \frac{XP}{R}$

$$= \frac{\frac{1}{2}\ length\ of\ chord\ PQ}{R}$$

$$= \frac{\frac{1}{2} \times 6216.44}{6400} \quad \text{(6216.44 from b)}$$

$\sin\alpha = 0.4857$

$\alpha = \sin^{-1} 0.4857$

$= 29.06^0$

Thus $2\alpha = 2 \times 29.06$

$= 58.12^0$

2007/12 Neco

Two points X(32^0S, 86^0W) and Y(32^0 S, 14^0W) are on the earth's surface. Calculate correct to three significant figures the:

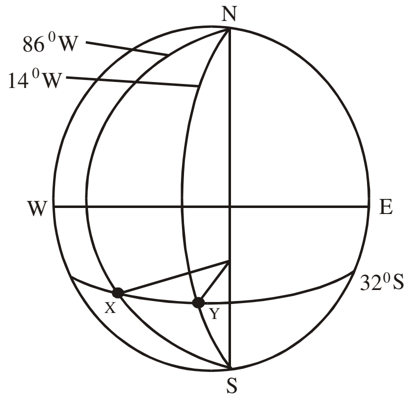
(a) Length of chord XY

(b) Distance between X and Y along the:

(i) parallel of latitude (ii) great circle

[Take R = 6400 and $\pi = 3.142$]

Solution



(a) Length of chord XY = $2r \sin\frac{\theta}{2}$

Here θ is $86 - 14 = 72^0$ and $r = R\cos\phi$

Where ϕ is the latitude angle 32^0

Thus length of chord XY = $2 \times 6400 \times \cos 32 \times \sin\frac{72}{2}$

$= 1280 \times 0.8480 \times 0.5878$

$= 6380.22$km

≈ 6380km to 3 s.f

(b)i Distance between X and Y along parallel of latitude

$$= \frac{\theta}{360} \times 2\pi R\cos\phi$$

$$= \frac{72}{360} \times 2 \times 3.142 \times 6400 \times \cos 32$$

$$= \frac{72}{360} \times 2 \times 3.142 \times 6400 \times 0.8480$$

$= 6820.90$km

≈ 6820km to 3 s.f

(b)ii Distance between X and Y along great circle

$$= \frac{2\sin^{-1}\left(\cos\phi \sin\frac{\theta}{2}\right) \times 2\pi R}{360}$$

$$= \frac{2\sin^{-1}\left(\cos 32 \sin\frac{72}{2}\right) \times 2 \times 3.142 \times 6400}{360}$$

$$= \frac{2\sin^{-1}(0.8480 \times 0.5878) \times 40217.6}{360}$$

$$= \frac{2\sin^{-1}(0.4985) \times 40217.6}{360}$$

$$= \frac{2 \times 29.9 \times 40217.6}{360}$$

$= 6680.59$km ≈ 6680km to 3 s.f

2006/9 Neco (Nov)

If two places P and Q are on the same parallel of latitude 30^0N and they differ in longitude by 45^0. Calculate the;

(i) distance between P and Q measured along the parallel of latitude

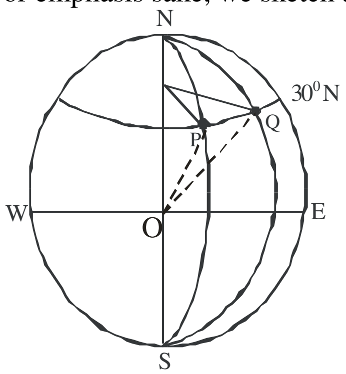
(ii) angle subtended at the centre of the earth by PQ and

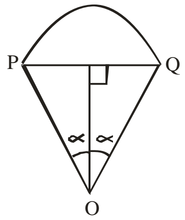
(iii) shortest distance between P and Q

(R = 6400km)

Solution

For emphasis sake, we sketch as:





(i) Distance PQ along parallel of latitude = $\frac{\theta}{360} \times 2\pi R\cos\phi$

$$= \frac{45}{360} \times 2 \times \frac{22}{7} \times 6400 \times \cos 30$$
$$= \frac{45}{360} \times 2 \times \frac{22}{7} \times 6400 \times 0.8660$$
$= 4354.74$km

(ii) Angle subtended at the centre of the earth by PQ $= \angle POQ$
$= 2\alpha$

But $2\alpha = 2\sin^{-1}\left(\cos\phi \sin\frac{\theta}{2}\right)$

Where ϕ is common latitude 30^0
θ is longitude difference 45^0

$$\therefore \angle POQ = 2\alpha = 2\sin^{-1}\left(\cos 30 \sin\frac{45}{2}\right)$$
$$= 2\sin^{-1}(0.8660 \times \sin 22.5)$$
$$= 38.7^0$$

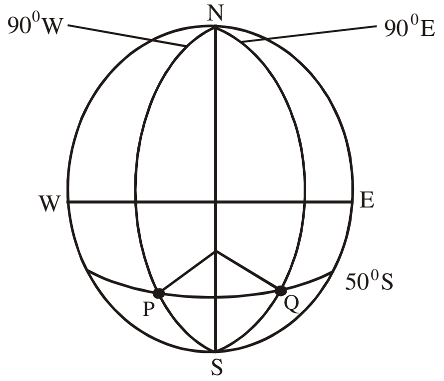
(iii) Shortest distance between P and Q

$$= \frac{2\sin^{-1}\left(\cos\phi\sin\frac{\theta}{2}\right)\times 2\pi R}{360}$$
$$= \frac{2\sin^{-1}\left(\cos 30\sin\frac{45}{2}\right)\times 2\times 22\times 6400}{360\times 7}$$
$$= \frac{2\sin^{-1}(0.8660\times 0.3827)\times 2\times 22\times 6400}{360\times 7}$$
$$= \frac{2\sin^{-1}(0.3314)\times 281600}{2520}$$
$$= \frac{38.70\times 281600}{2520}$$
$= 4324.57$km

2014/4 NABTEB (Nov)

Find the distance between P(50^0S, 90W) and Q(50^0S, 90^0E), along a
(i) circle of latitude (ii) great circle
(Take $\pi = ^{22}/_7$, R = 6400km)

Solution



(i) Distance PQ along circle of latitude $= \frac{\theta}{360} \times 2\pi R\cos\phi$

Here θ is 90 + 90 i.e 180, ϕ is 50
Substituting

$$= \frac{180}{360} \times 2 \times \frac{22}{7} \times 6400 \times \cos 50$$
$$= \frac{22}{7} \times 6400 \times 0.6428$$
$= 12929.46$km

(ii) Distance PQ along great circle

$$= \frac{2\sin^{-1}\left(\cos\phi\sin\frac{\theta}{2}\right)\times 2\pi R}{360}$$
$$= \frac{2\sin^{-1}\left(\cos 50\sin\frac{180}{2}\right)\times 2\times 22\times 6400}{360\times 7}$$
$$= \frac{2\sin^{-1}(0.6428\times 1)\times 2\times 22\times 6400}{2520}$$
$$= \frac{2\times 40\times 281600}{2520}$$
$= 8939.68$km

2015/7a Neco Exercise 16.7

A(Lat 4^0N, long15^0E) and B(Lat 4^0N, long 40^0W) are two points on the earth's surface. Calculate to the nearest km, the distance along
(i) the parallel of latitudes: (ii) a great circle.
(Take radius of the earth to be 6400km)

CHAPTER SEVENTEEN
Statistics II (Grouped data)

Frequency Distribution

When summarizing large masses of raw data, it is often useful to distribute the data into classes or categories and to determine the number of individuals belonging to each class called the class frequency. The tabular arrangement of data by classes together with their corresponding class frequency is called frequency distribution or frequency table.

Example

Mass (kg)	Number of Students
40 – 42	5
43 – 45	18
46 – 48	42
49 – 51	27
52 – 54	8

Total = 100

The table above shows a frequency distribution of masses (recorded to the nearest Kg) of 100 students attending SSCE lesson.

Any data organized and summarized as in the above frequency distribution are often called Group data.

Class interval & Class limits

A symbol defining a class such as 40 – 42, 43 – 45 in the above table is called a class interval – while the end numbers 40 and 42, 43 and 45 are called class limits; the smaller numbers 40, 43, ... are the lower class limits and the bigger numbers: 42, 45; ... are the upper class limits

Class boundaries

If masses are recorded to the nearest kg, the class interval 40 – 42 theoretically includes all measurements from 39.5 to 42.5 kgs which are called class boundaries or True class limits. The smaller number 39.5 is the lower class boundary while the bigger number 42. 5 is the upper class boundary. Displaying the given table to reflect class boundaries, we have

Class interval	**Class boundaries**	**Frequency**
40 – 42	39.5 – 42.5	5
43 – 45	42.5 – 45.5	18
46 – 48	45.5 – 48.5	42
49 – 51	48.5 – 51.5	27
52 – 54	51.5 – 54.5	8

In practice, the class boundaries are obtained by adding the upper limit of one class interval to the lower limit of the next higher class interval and dividing by two

i.e $\frac{43 + 42}{2}$, $\frac{46 + 45}{2}$, $\frac{49 + 48}{2}$

Class Size

This is the difference between the lower and upper class boundaries or the difference between two upper limits, or two lower limits. The class size is also called the class width or class strength. In this case 42.5 – 39.5 = 3 or 43 – 40 = 3 or 45 – 42 = 3

Class marks (Midpoints)

This is the mid point of the class interval and is obtained by adding the lower and upper class limits and dividing by two (2). Thus the class mark of the interval 40 – 42 is

$$\frac{40 + 42}{2} = 41$$

Measure of location for grouped data (Mean, median & mode)

Mean

Mean for grouped data can be calculated in two ways:

(i) Mean for problems without assumed mean

$$\bar{x} = \frac{\sum fx}{\sum f}$$

Where x is the class mark or class midpoint

(ii) Mean for problems with assumed mean

$$\bar{x} = A + \frac{\sum fd}{\sum f}$$

where A = Assumed mean ;

d = deviation from the mean (x – A)

The first method though easy to manipulate involves bigger values and if the question did not restrict you to any particular method i.e if the word "otherwise" is used in questions involving assumed mean; then students should stick to the easiest one for safety.

Example 1

The weights to the nearest kilogram, of a group of 50 students in a college of technology are given below:

65	70	60	46	51	55	59	63	68	53
47	53	72	58	67	62	64	70	57	56
73	56	48	51	58	63	65	62	49	64
53	59	63	50	48	72	67	56	61	64
66	52	49	62	71	58	53	69	63	59

(a) Prepare a grouped frequency table with class intervals 45 – 49, 50 – 54, 55 – 59 etc.

Solution

Class internal	Tally	Frequency
45 – 49	~~IIII~~ I	6
50 – 54	~~IIII~~ III	8
55 – 59	~~IIII~~ ~~IIII~~ I	11
60 – 64	~~IIII~~ ~~IIII~~ II	12
65 – 69	~~IIII~~ II	7
70 – 74	~~IIII~~ I	6

Applicable to all raw or scattered data problems

Note

The problem of grouping data when they are scattered or raw can be solved, if students use four figure table plus mathematical set or any opaque object to cover the data leaving one column or row at a time. Allocate the numbers there to their various class intervals with the aid of tally. Then move to the next row or column as shown below. Alternatively write out the data as they appear in the column or row; then allocate them to the various class intervals with the aid of tally. Repeat the process for the next column or row depending on your chosen pattern.

	Class interval	Tally
65	45 – 49	I
47	50 – 54	I
73	55 – 59	
53	60 – 64	
66	65 – 69	II
	70 – 74	I

Next, we attend to 2nd column

	Class interval	tally
70		
53	45 – 49	I
56	50 – 54	I **I I**
59	55 – 59	**I I**
52	60 – 64	
	65 – 69	II
	70 – 74	I **I**

Next, we attend to 3rd column

	Class interval	tally
60		
72	45 – 49	I I I
48	50 – 54	I **I I**
63	55 – 59	**I I**
49	60 – 64	I I
	65 – 69	II
	70 – 74	I **I** I

The pattern continues, each addition to the tally was differentiated for emphasis.

The lesser the figures the better. A student who chooses the row aspect of this data will likely have problems or slower compared to the person picking his data from column. The pattern to be followed depends on the way the data given are presented.

Column
65
47
73
53
66

Row: 66 52 49 62 71 58 53 69 63 59

2008/8d

The age in years of 50 teachers in a school are given below:

21	37	49	27	49	42	26	33	46	40
50	29	23	24	29	31	36	22	27	38
30	26	42	39	34	23	21	32	41	46
46	31	33	29	28	43	47	40	34	44
26	38	34	49	45	27	25	33	39	40

(a) Form a frequency distribution of the data using the internals 21 – 25, 26 – 30, 31 – 35, etc

(b) Calculate the mean age

Solution

$$\bar{x} = \frac{\sum fx}{\sum f}$$

We prepare a table to reflect x, f and fx items in the formula.

Class internal	x	tally	f	fx
21 – 25	23	~~IIII~~ II	7	161
26 – 30	28	~~IIII~~ ~~IIII~~ I	11	308
31 – 35	33	~~IIII~~ IIII	9	297
36 – 40	38	~~IIII~~ IIII	9	342
41 – 45	43	~~IIII~~ I	6	258
46 – 50	48	~~IIII~~ III	8	384
			Σf = 50	Σfx = 1750

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{1750}{50}$$

$$= 35$$

2009/8

The marks scored by 50 students in a Geography examination are as follows:

60	54	40	67	53	73	37	55	62	43
44	69	39	32	45	58	48	67	39	51
46	59	40	52	61	48	23	60	59	47
65	58	74	47	40	59	68	51	50	50
71	51	26	36	38	70	46	40	51	42

(a) Using class interval 21 – 30, 31 – 40, … prepare a frequency distribution table

(b) Calculate the mean mark of the distribution

(c) What percentage of the students scored **more** than 60%

Solution

$$\bar{x} = \frac{\sum fx}{\sum f}$$

We prepare a table to reflect x, f and fx formula items.

Class interval	x	tally	f	fx
21 – 30	25.5	II	2	51
31 – 40	35.5	~~IIII~~ ~~IIII~~	10	355
41 – 50	45.5	~~IIII~~ ~~IIII~~ II	12	546
51 – 60	55.5	~~IIII~~ ~~IIII~~ ~~IIII~~	15	832.5
61 – 70	65.5	~~IIII~~ III	8	524
71 – 80	75.5	III	3	226.5
			Σf = 50	Σfx = 2535

(b) Mean ($\bar{x}$) = $\frac{\sum fx}{\sum f}$

$$= \frac{2535}{50} = 50.7$$

(c) Percentage mark more than 60%

$$= \frac{\text{freq of more than 60\%}}{\text{Total freq}} \times 100$$

$$= \frac{8+3}{50} \times 100 = \frac{11}{50} \times 100 = 22\%$$

2014/7 f/maths

The table shows the distribution of the lengths of 20 iron rods measured in metres

Length (m)	1.0-1.1	1.2-1.3	1.4-1.5	1.6-1.7	1.8-1.9
frequency	2	3	8	5	2

Using an assumed mean of 1.45, calculate the mean of the distribution

Solution

Assumed mean formula for mean is $\bar{x} = A + \frac{\sum fd}{\sum f}$

We prepare be table to reflect x, f, d = x – 1.45, fd

Length (M)	x	d	f	fd
1.0 – 1.1	1.05	– 0.4	2	– 0.8
1.2 – 1.3	1.25	– 0.2	3	– 0.6
1.4 – 1.5	1.45	0.0	8	0.0
1.6 – 1.7	1.65	0.2	5	1.0
1.8 – 1.9	1.85	0.4	2	0.8
			Σf = 20	Σfd = 0.4

Mean $(\bar{x}) = A + \frac{\sum fd}{\sum f}$

$= 1.45 + \frac{0.4}{20}$

$= 1.45 + 0.02 \quad = 1.47$

2005/11a Neco

The marks obtained by students in a Mathematics test are recorded as follows:

Marks	1-10	11-20	21-30	31- 40	41-50	51-60	61-70	71-80
Number of students	10	20	50	90	110	70	40	10

What is the modal class?

Solution

Mode formula for grouped data is given as

Mode = $L_1 + \left[\frac{\Delta_1}{\Delta_1 + \Delta_2}\right] \times C$

Where : **L_1** = lower class boundary of the modal class

Δ_1 = Difference between the modal frequency and the frequency of the next lower class i.e class before it .

Δ_2 = Difference between the modal frequency and the frequency of the next higher class i.e class after it.

C = Size of the modal class

Modal table

Marks	class boundaries	f
1 – 10	0.5 – 10.5	10
11 – 20	10.5 – 20.5	20
21 – 30	20.5 – 30.5	50
31 – 40	30.5 – 40.5	90
41 – 50	40.5 – 50.5	110
51 – 60	50.5 – 60.5	70
61 – 70	60.5 – 70.5	40
71 – 80	70.5 – 80.5	10

First identify the modal class: the class with the highest frequency which is 41 – 50

$L_1 = 40.5$, $\Delta_1 = 110 - 90 = 20$, $\Delta_2 = 110 - 70 = 40$ and C = 10

Mode = $40.5 + \left[\frac{20}{20+40}\right] \times 10$

$= 40.5 + \frac{20}{60} \times 10 \quad = 43.83$

2014/12 Neco (Dec)

The scores of 50 students in a mid – term test were as shown below:

11	20	30	24	13	28	33	40	23	28
40	8	30	13	15	34	8	34	32	22
26	21	25	18	26	10	19	3	27	18
18	24	26	25	27	29	28	13	35	24
9	24	14	28	27	38	40	32	33	34

(a) Construct a frequency table using the class intervals 1 – 5, 6 – 10, 11 – 15, …

(b) Estimate the:

(i) mean score (ii) median score (iii) modal score

Solution

Class interval	x	tally	f	fx
1 – 5	3	I	1	3
6 – 10	8	IIII	4	32
11 – 15	13	~~IIII~~ I	6	78
16 – 20	18	~~IIII~~	5	90
21 – 25	23	~~IIII~~ IIII	9	207
26 – 30	28	~~IIII~~ ~~IIII~~ III	13	364
31 – 35	33	~~IIII~~ III	8	264
36 – 40	38	IIII	4	152
			Σf = 50	Σfx = 1190

(i) $\bar{x} = \frac{\sum fx}{\sum f} = \frac{1190}{50} = 23.8$

For median

Median formula for grouped data is given as

Median $= L_1 + \left[\frac{\frac{n}{2} - Cf_b}{f_m}\right] C$

Where

L_1 = Lower class boundary of the median class

n = Total frequency

Cf_b = Cumulative frequency before the median class

F_m = Frequency of median class

C = Size of the median class

we prepare a table to reflect class boundaries and cumulative frequencies

Marks	class boundaries	f	cf
1 – 5	0.5 – 5.5	1	1
6 – 10	5.5 – 10.5	4	5
11 – 15	10.5 – 15.5	6	11
16 – 20	15.5 – 20.5	5	16
21 – 25	20.5 – 25.5	9	25
26 – 30	25.5 – 30.5	13	38
31 – 35	30.5 – 35.5	8	46
36 – 40	35.5 – 40.5	4	50

(ii) Median, first, we find $\frac{n}{2}$ i.e $\frac{50}{2} = 25$ median position

Next, we find the class interval where the median lies with the aid of cumulative frequency:

At cumulative frequency 25 occurs in class interval 21 – 25

$L_1 = 20.5$, $Cf_b = 16$, C = 5 i.e 25 – 20 or 26 – 21

$F_{median} = 9$

Median = $20.5 + \left(\frac{25-16}{9}\right) \times 5$

$= 20.5 + \frac{9}{9} \times 5$

$= 20.5 + 5 = 25.5$

(iii) Mode:

$$\text{mode} = L_1 + \left[\frac{\Delta_1}{\Delta_1 + \Delta_2}\right]C$$

First, identify the modal class

Class with highest frequency which is 26 – 30

$L_1 = 25.5$, $\Delta_1 = 13 - 9$ i.e 4, $\Delta_2 = 13 - 8$ i.e 5 and C = 5

$$\text{Mode} = 25.5 + \left[\frac{4}{4+5}\right]\times 5$$
$$= 25.5 + 2.22$$
$$= 27.72$$

Measures of *spread (plus measure of location cases)*

Standard deviation and mean deviation

Standard deviation and mean deviation are not the only measures of spread. The subtopic measure of spread has been treated under *statistics I*.

2009/12 Neco

62	54	53	44	46	55	46	56	68	63
59	61	66	54	39	48	47	53	59	57
50	35	40	30	46	44	36	49	54	51
57	56	45	33	38	41	40	45	53	58
51	45	48	34	36	46	43	49	63	52

(a) Using class interval 30 – 34, 35 – 39, 40 – 44, … construct the frequency distribution table

(b) Calculate the:

(i) Mean; (ii) Standard deviation

(All answer correct to 2 decimal places)

Solution

$$\text{Mean } (\bar{x}) = \frac{\sum fx}{\sum f} \text{ and S.D} = \sqrt{\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2}$$

We prepare our table to reflect formula items x, x^2, f, fx, fx^2

Class interval	tally	f	x	fx	x^2	fx^2
30 – 34	III	3	32	96	1024	3072
35 – 39	~~IIII~~	5	37	185	1369	6845
40 – 44	~~IIII~~ I	6	42	252	1764	10584
45 – 49	~~IIII~~ ~~IIII~~ II	12	47	564	2209	26508
50 – 54	~~IIII~~ ~~IIII~~	10	52	520	2704	27040
55 – 59	~~IIII~~ III	8	57	456	3249	25992
60 – 64	IIII	4	62	248	3844	15376
65 – 69	II	2	67	134	4489	8978
		Σf = 50		Σfx = 2455		Σfx^2 = 124395

(b)i Mean $(\bar{x}) = \frac{\sum fx}{\sum f} = \frac{2455}{50} = 49.1$

(ii) $$\text{S.D} = \sqrt{\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2}$$
$$= \sqrt{\frac{124395}{50} - \left(\frac{2455}{50}\right)^2}$$
$$= \sqrt{2487.9 - (49.1)}$$
$$= \sqrt{2487.9 - 2410.81}$$
$$= \sqrt{77.09}$$
$$= 8.780$$
$$\approx 8.78 \text{ to 2 d.p}$$

2013/12 Neco

The marks of 100 students who passed Economics in a promotion examination are given below

Mark %	51– 60	61- 70	71- 80	81– 90	91-100
Frequency	11	23	39	17	10

Find the: (i) mean mark (ii) mean deviation and

(iii) Standard deviation correct to the nearest whole number

Solution

$$\text{Mean } (\bar{x}) = \frac{\sum fx}{\sum f}, \quad \text{M.D} = \frac{\sum f\,/x-\bar{x}/}{\sum f} \quad \text{and}$$

$$\text{S.D} = \sqrt{\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2}$$

Next we prepare a table to reflect the formula items x, f, fx, $/x-\bar{x}/$, f$/x-\bar{x}/$, x^2 and fx^2

Marks %	x	f	fx	$/x-\bar{x}/$	f$/x-\bar{x}/$	x^2	fx^2
51 – 60	55.5	11	610.5	19.5	214.5	3080.25	33882.75
61 – 70	65.5	23	1506.5	9.5	218.5	4290.25	98675.75
70 – 80	75.5	39	2944.5	0.5	19.5	5700.25	222309.75
81 – 90	85.5	17	1453.5	10.5	178.5	7310.25	124274.25
91 – 100	95.5	10	955	20.5	205	9120.25	91202.5
		Σf = 100	Σfx = 7470		Σf$/x-\bar{x}/$ = 836		Σfx^2 = 570345

(i) $$\text{Mean } (\bar{x}) = \frac{\sum fx}{\sum f}$$
$$= \frac{7470}{100} = 74.7 \approx 75 \text{ to nearest whole number}$$

(ii) $$\text{Mean deviation} = \frac{\sum f\,/x-\bar{x}/}{\sum f}$$
$$= \frac{836}{100}$$
$$= 8.36 \approx 8 \text{ to nearest whole number}$$

(iii) $$\text{S.D} = \sqrt{\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2}$$
$$= \sqrt{\frac{570345}{100} - \left(\frac{7470}{100}\right)^2}$$
$$= \sqrt{5703.45 - (74.7)^2}$$
$$= \sqrt{5703.45 - 5580.09}$$
$$= \sqrt{123.36}$$
$$= 11.107 \approx 11 \text{ to the nearest whole number}$$

2012/14 f/maths

The table gives the distribution of heights in metres of 100 students

Height	1.40-1.42	1.43-1.45	1.46-1.48
Freq	2	4	19

1.49-1.51	1.52-1.54	1.55-1.57	1.58-1.60	1.61-1.63
30	24	14	6	1

(a) Calculate the

(i) mean height (ii) mean deviation of the distribution.

(b) What is the probability that the height of a student selected at random is greater than the mean height of the distribution?

Solution

Mean ($\bar{x}$) = $\frac{\sum fx}{\sum f}$, mean deviation = $\frac{\sum f\,/\,x-\bar{x}\,/}{\sum f}$

Next, we prepare a table for the formula items:
x, f, fx , $/x-\bar{x}/$ and f $/x-\bar{x}/$

Height	x	f	fx	$/x-\bar{x}/$	f $/x-\bar{x}/$
1.40 – 1.42	1.41	2	2.82	0.1	0.2
1.43 – 1.45	1.44	4	5.76	0.07	0.28
1.46 – 1.48	1.47	19	27.93	0.04	0.76
1.49 – 1.51	1.50	30	45.00	0.01	0.30
1.52 – 1.54	1.53	24	36.72	0.02	0.48
1.55 – 1.57	1.56	14	21.84	0.05	0.70
1.58 – 1.60	1.59	6	9.54	0.08	0.48
1.61 – 1.63	1.62	1	1.62	0.11	0.11
		Σf =100	Σfx = 151.23		Σf $/x-\bar{x}/$ = 3.31

(i) Mean ($\bar{x}$) = $\frac{\sum fx}{\sum f}$

$= \frac{151.23}{100} = 1.5123$

≈ 1.51m to 2d.p (approximated base on the given data)

(ii) Mean deviation = $\frac{\sum f\,/\,x-\bar{x}\,/}{\sum f}$

$= \frac{3.31}{100} = 0.0331$

≈ 0.03 to 2d.p

(b) Prob (height > mean height)

$= \frac{\text{sum freq greater than 1.51}}{\text{Total freq}}$

$= \frac{24+14+6+1}{100}$

$= \frac{45}{100} = \frac{9}{20}$ = 0.45

2013/6 f/maths

The table shows the distribution of ages of 22 students in a school

Age (years)	12 –14	15 –17	18 –20	21 –23	24 –26
Frequency	6	10	3	2	1

Using an assumed mean of 19, calculate, correct to three significant figures, the:
(a) Mean age (b) standard deviation of the distribution

Solution

Assumed mean formula for mean and S.D

$\bar{x} = A + \frac{\sum fd}{\sum f}$, S.D = $\sqrt{\frac{\sum fd^2}{\sum f} - \left(\frac{\sum fd}{\sum f}\right)^2}$

We prepare a table to reflect the formula items
x, f, d = x – 19, fd, d^2, fd^2

Age(years)	x	d	f	fd	d^2	fd^2
12 – 14	13	–6	6	–36	36	216
15 – 17	16	–3	10	–30	9	90
18 – 20	19	0	3	0	0	0
21 – 23	22	3	2	6	9	18
24 – 26	25	6	1	6	36	36
			Σf =22	Σfd = –54		Σfd^2 =360

(a) Mean ($\bar{x}$) = A + $\frac{\sum fd}{\sum f}$

$= 19 + \left(\frac{-54}{22}\right) = 19 - 2.45$

= 16.55yrs ≈ 16.6 to 3sf

(b) S.D = $\sqrt{\frac{\sum fd^2}{\sum f} - \left(\frac{\sum fd}{\sum f}\right)^2} = \sqrt{\frac{360}{22} - \left(\frac{-54}{22}\right)^2}$

$= \sqrt{16.36 - 6.025} = \sqrt{10.335} = 3.215$ ≈ 3.22yrs to 3sf

2007/14 f/maths

The table shows the distribution of ages of workers in a company

Age (in yrs)	17 – 21	22 – 26	27 – 31	32 – 36	37 – 41	42 – 46	47 – 51	52 – 56
frequency	12	24	30	37	45	25	10	7

(a) Using an assumed mean of 39, calculate the
(i) Mean (ii) standard deviation of the distribution
(b) If a worker is selected at random from the company for an award, what is the probability that he is at most 36 years old?

Solution

Assumed mean formula for mean and S.D

$\bar{x} = A + \frac{\sum fd}{\sum f}$, S.D = $\sqrt{\frac{\sum fd^2}{\sum f} - \left(\frac{\sum fd}{\sum f}\right)^2}$

We prepare a table to reflect the formula items
x, f, d = x – 39, fd, d^2, fd^2

Age (yrs)	x	d	f	fd	d^2	fd^2
17 – 21	19	–20	12	–240	400	4800
22 – 26	24	–15	24	–360	225	5400
27 – 31	29	–10	30	–300	100	3000
32 – 36	34	–5	37	–185	25	925
37 – 41	39	0	45	0	0	0
42 – 46	44	5	25	125	25	625
47 – 51	49	10	10	100	100	1000
52 – 56	54	15	7	105	225	1575
			Σf = 190	Σfd = –755		Σfd^2 = 17325

(a) i Mean ($\bar{x}$) = A + $\frac{\sum fd}{\sum f}$

$= 39 + \left(\frac{-755}{190}\right) = 39 - 3.97 = 35.03$

(ii) S.D = $\sqrt{\frac{\sum fd^2}{\sum f} - \left(\frac{\sum fd}{\sum f}\right)^2} = \sqrt{\frac{17325}{190} - \left(\frac{-755}{190}\right)^2}$

$= \sqrt{91.184 - (3.97)^2} = \sqrt{91.184 - 15.7609} = \sqrt{75.4231}$

= 8.6846 ≈ 8.68

(b) Pr(at most 36) = Pr(≤ 36)

$= \frac{f(36)+f(31)+f(26)+f(21)}{Total\ \ freq}$

$= \frac{37+30+24+12}{190}$

$= \frac{103}{190}$ = 0.542

Discrete data & continuous data

Discrete data have distinct values without intermediate points. The number of student(s) in a class could be 0, 1, 2, 3, 4, 5,… **but not** 1.5 or 25.6 or 50.1 impossible ! Frequency can also be attached such as: the number of girls, number of boys and number of absentees in a class.

Where as if we are to consider the weight of students in a class; values such as 16kg, 16.4kg, 17.1kg, 19.3kg 20.0kg etc. the result will be both whole and fractional or intermediate values. This will require a range say 16–20. Hence **continuous data have** values over a range of whole number or fraction.

Note that when a continuous data is treated as fixed, it becomes discrete

2016/9

The weight(in kg) of 50 contestants at a competition is as follows:

65	66	67	66	64	66	65	63	65	68
64	62	66	64	67	65	64	66	65	67
65	67	66	64	65	64	66	65	64	65
66	65	64	65	63	63	67	65	63	64
66	64	68	65	63	65	64	67	66	64

(a) Construct a frequency table for the discrete data

(b) Calculate, correct to 2 decimal places, the:
 i. mean ii. standard deviation; of the data

Solution

Discrete data implies the given data are not continuous as it may appear to be

(b) i. Mean $\bar{x} = \frac{\sum fx}{\sum f}$

Table for mean

Weight(kg)	tally	frequency	fx
62	**I**	**1**	**62**
63	**~~IIII~~**	**5**	**315**
64	**~~IIII~~ ~~IIII~~ II**	**12**	**768**
65	**~~IIII~~ ~~IIII~~ IIII**	**14**	**910**
66	**~~IIII~~ ~~IIII~~**	**10**	**6600**
67	**~~IIII~~ I**	**6**	**402**
68	**II**	**2**	**136**
		Σf = 50	**Σfx = 9193**

Mean $\bar{x} = \frac{\sum fx}{\sum f} = \frac{9193}{50} = 183.36$

(ii) S.D $= \sqrt{\frac{\sum f\left(x-\bar{x}\right)^2}{\sum f}}$

Table for SD

x	$x-\bar{x}$	$\left(x-\bar{x}\right)^2$	f	$f\left(x-\bar{x}\right)^2$
62	–121.36	14728.250	1	14728.250
63	–120.36	14486.530	5	72432.650
64	–119.36	14246.810	12	170961.720
65	–118.36	14009.090	14	196127.260
66	–117.36	13773.370	10	137733.700
67	–116.36	13539.650	6	81237.900
68	–115.36	13307.930	2	26615.86
			Σf =50	$\Sigma f(x-\bar{x})^2 =$ 699,837.34

$$\text{SD} = \sqrt{\frac{699837.34}{50}}$$

$$= \sqrt{13996.7468} = 118.3078 \approx 118.31 \text{ to 2d.p}$$

2007/13 Adjusted Exercise 17.1

In a college, the number of absentees recorded over a period of 30 days was as shown in the frequency distribution table

Number of absentees	0 – 4	5 – 9	10 – 14	15– 19	20 – 24
Number of days	1	5	10	9	5

Calculate the: (a) Mean (b) standard deviation,
(c) mean deviation correct to two decimal places

2016/12 Neco Exercise 17.2

The scores of 40 students in a Geography test are given below:

62	64	67	68	55	61	62	65	67	69
69	64	61	68	70	70	69	64	63	61
63	70	61	62	69	61	64	68	65	68
50	51	53	54	55	57	52	56	58	60

(a) Using a class interval 50 – 54, 55 – 59,…, construct a frequency distribution table

(b) Find the: **i.** mean; **ii.** variance **iii.** Standard deviation

2012/10 Exercise 17.3

Class interval	Frequency
60 – 64	2
65 – 69	3
70 – 74	6
75 – 79	11
80 – 84	8
85 – 89	7
90 – 94	2
95 – 99	1

The table shows the distribution of marks scored by students in an examination, calculate, correct to 2 decimal places, the
(a) Mean (b) standard deviation of the distribution

2014/15a & c Neco f/maths Exercise 17.5

A cattle rearer recorded the weight in kilograms of 50 cows in his farm as follows:

28	45	46	52	56	83	26	48	28	53
72	50	32	77	41	43	53	50	80	45
69	64	72	66	43	65	62	51	52	67
54	31	32	52	65	36	42	60	53	75
60	53	46	88	64	83	56	88	26	64

(a) Construct a cumulative frequency table using the class intervals: 26 – 35, 36 – 45, …

(b) From your table, calculate the mode of the distribution.

CHAPTER EIGHTEEN

Graphs (Charts)

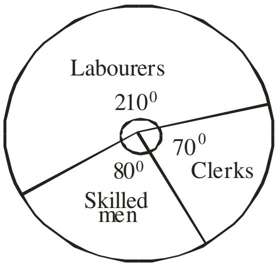
Statistical graphs (Charts)

PIE CHART

A pie chart is a circular diagram in which each sector of the circle represents a given frequency expressed in degrees.

2012/51 Neco

The pie chart below shows the division of the workforce of a factory



51. What fraction of the workforce is skilled men?

A $\frac{2}{9}$ B $\frac{3}{8}$ C $\frac{4}{5}$ D $\frac{5}{3}$ E $\frac{5}{3}$

Solution

Fraction of skilled men = $\frac{80}{360} = \frac{8}{36} = \frac{2}{9}$ (A)

2011/53

A father was to share to his children ₦12,000,000.00 and demonstrated the sharing on a pie chart. If the sectorial angle of his only son is 60^0, how much did the son take?

A ₦ 10,000,000.00 B ₦ 8,000,000.00

C ₦ 6,000,000.00 D ₦ 4,000,000.00

E ₦ 2,000,000.00

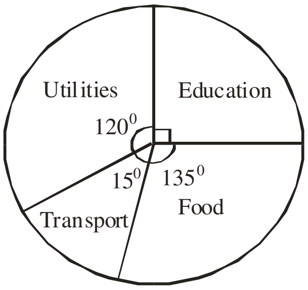
Solution

Son's share = $\frac{60}{360} \times 12{,}000{,}000$

= ₦ 2,000,000.00 (E)

2014/35 – 37 Neco

The pie chart below shows the expenditure on a man's income whose expenditure on education is ₦180,000.00. Use the information to answer questions **35** to **37**



35. What is the total income of the man?

A ₦180,000.00 B ₦270,000.00

C ₦300,000.00 D ₦ 400,000.00 E ₦720,000.00

Solution

The sector with useful information here is education

Expenditure on education is ₦ 180,000

Sector of Education is 90^0

$$\frac{90}{360} \times \text{Total income} = 180,000$$

$$90 \times \text{Total income} = 180{,}000 \times 360$$

$$\text{Total income} = \frac{180\,000 \times 360}{90}$$

= ₦ 720,000 (E)

36. Calculate the amount spent on transport

A ₦ 20,010.00 B ₦ 30,000.00

C ₦ 400,000.00 D ₦ 720,000.00 E ₦ 800,000.00

Solution

Amount spent on transport = $\frac{15}{360} \times 720{,}000$

= ₦ 30,000 (B)

37. What percentage of the income was spent on utilities?

A 33.3% B 32.1% C 30% D 25.5% E 20.2%

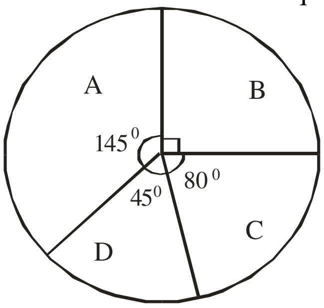
Solution

% on utilities = $\frac{120}{360} \times 100$

= 33.3% (A)

2006/30 and 31 NABTEB (Nov)

The pie chart below illustrates the age of student in a Technical college. The age brackets are A, B, C and D Use the information to answer questions **30** and **31**



30. If there are 45 students under B, how many students are in the college?

A 90 B 180 C 360 D 720

Solution

Sector under B

Age of B student is 45

Sector of B is 90^0

$$\frac{45}{\textit{Total students}} \times 360 = 90$$

$$45 \times 360 = 90 \times \text{Total students}$$

$$\frac{45 \times 360}{90} = \text{total student}$$

180 = total student (B)

31. How many students are under C?

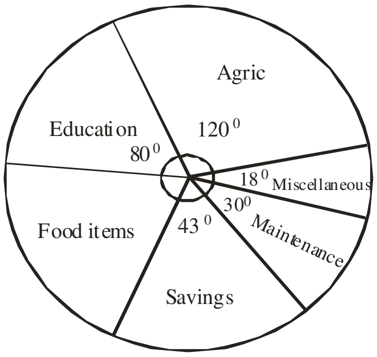
A 160 B 80 C 40 D 30

Solution

Students under C = $\frac{80}{360} \times 180$

= 40 (C)

2014/31 – 32 NABTEB (Nov)
The annual salary of Mr. Segun for 2001 was ₦12,000.00. He spent this on agricultural project, education of his children, food items, savings, maintenance and miscellaneous item as show in the pie chart



31. How much did he spend on food item?
A ₦ 9700.00 B ₦ 6700.00
C ₦ 3700.00 D ₦ 2300.00

Solution

Amount spent on food item = $\frac{food\ sector}{360} \times 12{,}000$

(Note the pattern here)

$= \frac{360-(80+120+18+30+43)}{360} \times 12{,}000$

$= \frac{360-291}{360} \times 12{,}000$

$= \frac{69}{360} \times 12{,}000$

= ₦ 2,300 (D)

32. How much did he spend on agriculture?
A ₦ 1200.00 B ₦ 1440.00
C ₦ 4000.00 D ₦ 2910.00

Solution

Amount spent on Agriculture = $\frac{120}{360} \times 12{,}000$

= ₦ 4000.00 (C)

2005/45 Neco
The distribution of students during lessons is as shown below:

Subject	Eng	Math's	Bio	Agric	Fine art
No of sutudents	9	11	6	5	5

Calculate the sum of the sectoral angles of Maths and Biology
A 360^0 B 170^0 C 100^0 D 70^0 E 50^0

Solution

Maths sectorial angle = $\frac{11}{9+11+6+5+5} \times 360$

$= \frac{11}{36} \times 360$

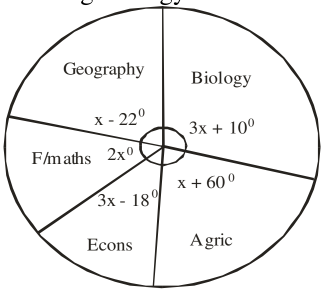
$= 110^0$

Biology sectorial angle = $\frac{6}{36} \times 360$

$= 60^0$

Sum of Maths and Biology sectorial angle = 110 + 60
$= 170^0$ (B)

2006/54 Neco
The pie chart below shows the distribution of students offering Further Mathematics, Geography, Economics, Biology, and Agricultural Science. What angle represents the proportion of students offering Biology?



A 33^0 B 43^0 C 90^0 D 109^0 E 120^0

Solution

First, we find value of x

$x - 22 + 3x + 10 + x + 60 + 3x - 18 + 2x = 360$

$10x + 30 = 360$

$10x = 360 - 30$

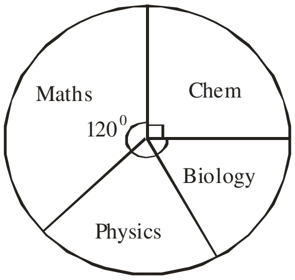
$10x = 330$

$x = \frac{330}{10} = 33$

Sector angle of Biology = 3x + 10
= 3(33) + 10
$= 99 + 10 = 109^0$ (D)

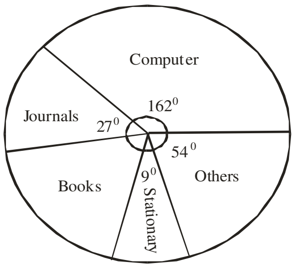
2014/29 Neco (Nov) Exercise 18.1
The following numbers are to be represented on a pie chart: 20, 25, 30, 35, 40. What is the sectorial angle representing 25?
A 25^0 B 48^0 C 60^0 D 72^0 E 84^0

2008/37 NABTEB (Nov) Exercise 18.2
Using the pie chart below, calculate the number of candidates that offer chemistry if the total number of candidate is 240



A 60 B 70 C 80 D 90

2005/20 – 21 Exercise 18.3



The pie chart shows the items purchased for a local library.
Use the information to answer question 20 and 21

20. If the cost of the computer is ₦ 90,000.
Calculate the cost of the books
A ₦15,000 B ₦30,000 C ₦50,000 D ₦60,000

21. What was the total amount spent for the library?
A ₦185,000 B ₦195,000 C ₦200,000 D ₦220,000

Live diagrams in pie chart

Live diagrams in pie chart is a circular diagram in which each sector of the circle represents a given frequency expressed in degrees; this is done with the aid of a protractor to mark out the various angles.

2008/3

The table below shows how a man spends his income in a month

Items	Amount spent
Food	₦ 4500
House rent	₦ 3000
Provisions	₦ 2500
Electricity	₦ 2000
Transportation	₦ 5000
Others	₦ 3000

(a) Represent the information on a pie chart

(b) What percentage of his income is spent on transportation?

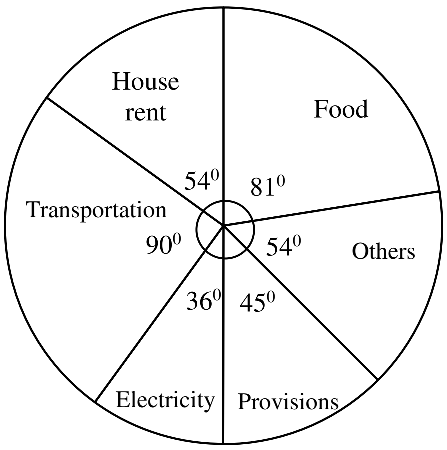
Solution

Total amount spent = ₦ 20,000.00

(4500 + 3000+ 25000 + 2000 + 5000 + 3000)

Items	**amount spent**	**sector angle (degrees)**
Food	₦ 4500	$\frac{4500}{20,000}\times 360 = 81^0$
House rent	₦ 3000	$\frac{3000}{20,000}\times 360 = 54^0$
Provisions	₦ 2500	$\frac{2500}{20,000}\times 360 = 45^0$
Electricity	₦ 2000	$\frac{2000}{20,000}\times 360 = 36^0$
Transportation	₦ 5000	$\frac{5000}{20,000}\times 360 = 90^0$
Others	₦ 3000	$\frac{3000}{20,000}\times 360 = 54^0$

(a) The required diagram is as shown below: it is drawn with the **aid of a protractor** to mark out the various angles



(b) Transportation % = $\frac{5000}{20000}\times 100$

= 25%

2014/1a Neco

In a certain year, the government of a certain state bought 480 cars. 180 were Peugeot, 108 were Datsun, 72 were Ford, 56 were Toyota and the rest were Volkswagen. Represent the above information on a pie chart.

Solution

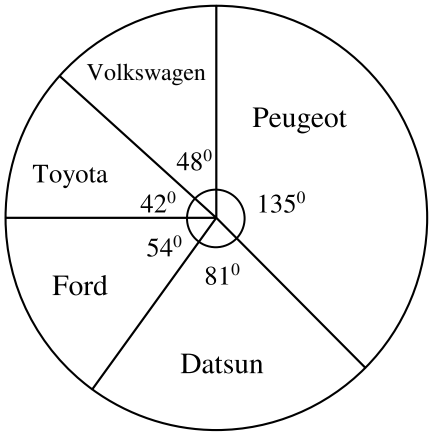
Total number of cars = 480 (given)

Portion for Volkswagen = 480 – (180 + 108 + 72 + 56)

= 64

Cars (brand)	**No of cars**	**sector angle (degrees)**
Peugeot	180	$\frac{180}{480}\times 360 = 135^0$
Datsun	108	$\frac{108}{480}\times 360 = 81^0$
Ford	72	$\frac{72}{480}\times 360 = 54^0$
Toyota	56	$\frac{56}{480}\times 360 = 42^0$
Volkswagen	64	$\frac{64}{480}\times 360 = 48^0$

The required diagram is as shown below: it is drawn with the **aid of a protractor** to mark out the various angles.



2006/1 Neco Exercise 18.4

The data below shows the number of students that represented a Government Secondary School in a Mathematics Competition

Class	No. of students
JS 1	12
JS 2	14
JS 3	24
SS 1	16
SS 2	13
SS 3	11

(a) Represent this information on a pie chart

(b) Find the probability that a student picked at random is in SS 1

2005/8b NABTEB (Nov) Exercise 18.5

Represent the following information on a pie chart using a radius of 6cm

Letter	A	B	C	D	E
Frequency	10	8	5	7	6

1979/3 Exercise 18.6

The result of an opinion poll among 900 students in a Grammar School regarding the country they would like to visit outside Africa is given below. Represent the information on a pie chart

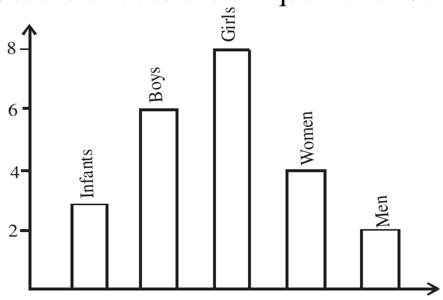
Country they wish to visit	No. of students
Japan	90
U.S.S.R.	225
Canada	155
U.K.	185
USA	135
China	110

BAR CHART

A bar chart is a chart in which rectangular shapes are used to represent frequency. The rectangles have equal width and unequal length. The lengths depend on the frequency of each class. The distinct bars are separated by a uniform gap all through. Students choose the scale used but it must be reasonable.

2014/26 - 27 NABTEB

Use the chart to answer questions **26** and **27**



26. What is the mode?

A 6 B 8 C Boys D Girls

Solution

Mode = girls (since girl has the highest frequency of 8)

27 How many people are there altogether?

A 25 B 23 C 20 D 17

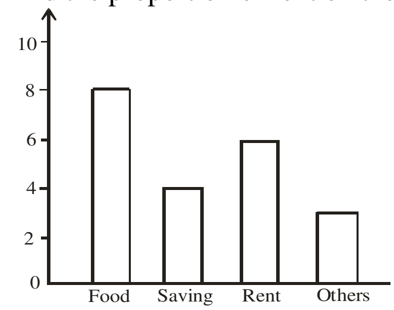
Solution

Total people = 3 + 6 + 8 + 4 +2

= 23 (B)

2005/46 Neco

The bar chart below shows how a family spends its income. Find the proportion of rent on the total income



A $^{8}/_{21}$ B $^{2}/_{7}$ C $^{5}/_{21}$ D $^{1}/_{7}$ E $^{2}/_{21}$

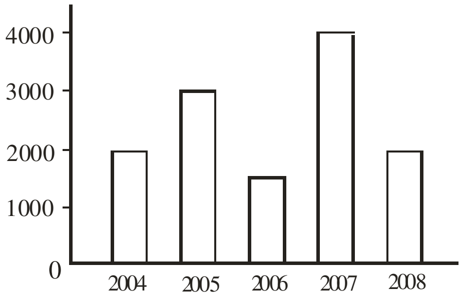
Solution

Proportion of rent on total income = Pr(rent)

$$= \frac{freq\ of\ rent}{Total\ freq} = \frac{6}{8+4+6+3} = \frac{6}{21} = \frac{2}{7} \quad (B)$$

2013/52 Neco

The bar chart below shows the number of items produced by cat bol ventures over a period of 5 years



52. What is the ratio of the least frequent production to that of the most frequent production?

A $^{3}/_{8}$ B $^{1}/_{3}$ C $^{2}/_{5}$ D $^{3}/_{7}$ E $^{3}/_{5}$

Solution

Ratio

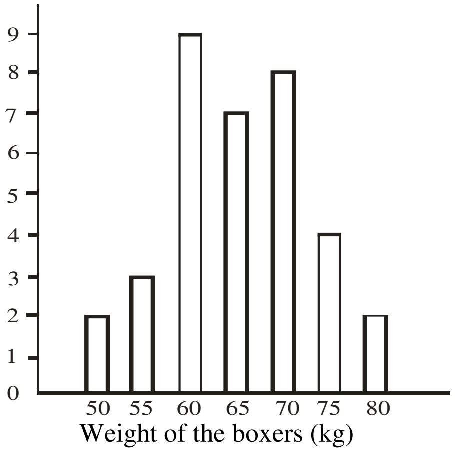
Least frequent : most frequent

1500 (2006) : 4000 (2007)

15 : 40

3 : 8 or $^{3}/_{8}$ (A)

2011/ 50 – 52 Neco



The bar chart above shows the distribution of boxers weight in a row for a boxing competition: use the chart to answer questions 50 to 52

50. How many boxers have weight of 65kg and below?

A 35 B 21 C 15 D 14 E 2

Solution

Freq freq freq freq

Weight of 65kg and below = 65 + 60 + 55 + 50

= 7 + 9 +3 + 2

= 21 B

51. How many boxers registered for the competition?

A 14 B 15 C 35 D 49 E 56

Solution

Total boxers = 2 + 3 + 9 + 7 + 8 + 4 + 2

= 35 C

52. If a boxer is chosen at random. What is the probability that his weight is less than or equal to 60kg?

A $^{1}/_{12}$ B $^{3}/_{16}$ C $^{2}/_{5}$ D $^{3}/_{4}$ E $^{11}/_{12}$

Solution

freq freq freq

$$Pr(\le 60kg) = \frac{9 + 3 + 2}{35}$$

$$= \frac{14}{35} = \frac{2}{5} \qquad C$$

Live diagrams in bar chart

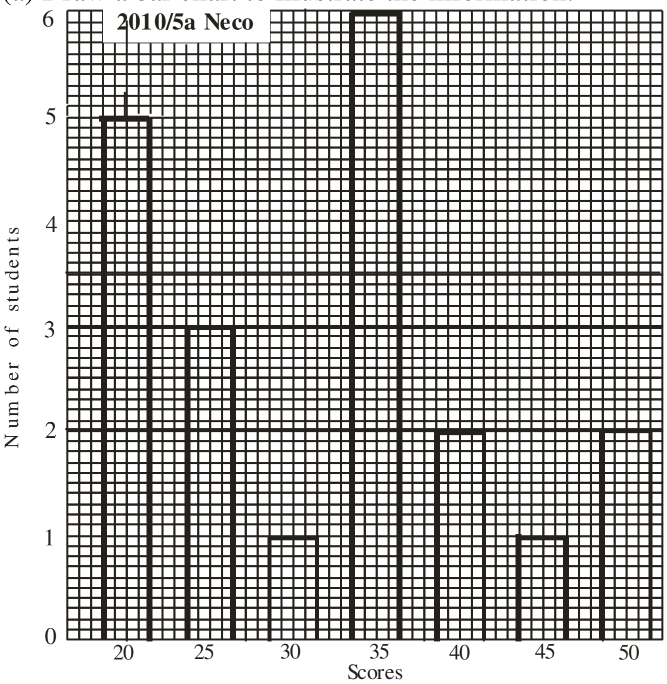
Here we are required to draw the graph in a graph sheet as shown below

2010/5a Neco

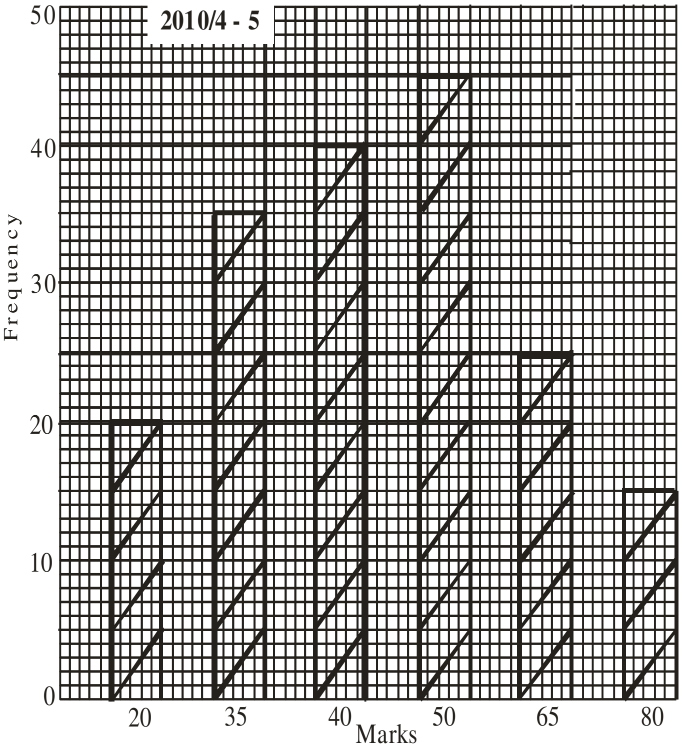
The table below shows the distribution of scores in percentage obtained by 20 students in a class test

Scores	20	25	30	35	40	45	50
No. of students	5	3	1	6	2	1	2

(a) Draw a bar chart to illustrate the information.



2010/ 4 and 5



The bar chart shows the marks distribution in an English test. Use it to answer questions 4 and 5

4. If 50% is the pass mark, how many students passed the test?

A 100 B 85 C 80 D 70 E 45

Solution

Number of students who passed at 50% pass mark

= freq of 50% + freq of 65% + freq of 80%

= 45 + 25 + 15

= 85 (B)

5. What percentage of the students has mark ranging from 35 to 50?

A $55\frac{1}{3}\%$ B 60% C 65% D $65\frac{2}{3}\%$

Solution

Percentage of marks between 35 to 50

$$= \frac{freq\,35 + freq\,40 + freq\,50}{Total\ freq} \times 100$$

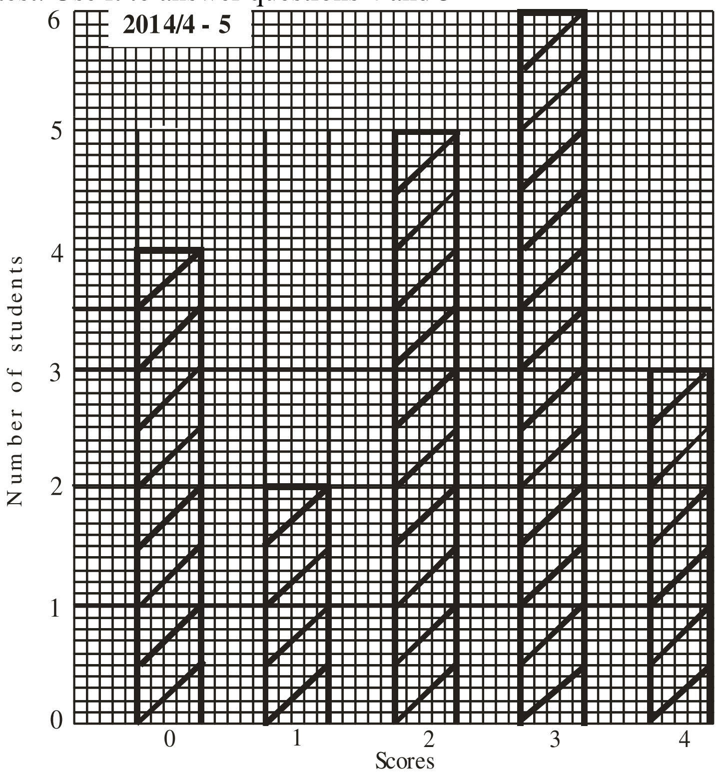
$$= \frac{35+40+45}{20+35+40+45+25+15} \times 100$$

$$= \frac{120}{180} \times 100$$

$$= 66\frac{2}{3}\% \quad (\text{D})$$

2014/ 4 and 5 Exercise 18.7

The bar chart below shows the scores of some students in a test. Use it to answer questions 4 and 5



4. How many students took the test?

A 18 B 19 C 20 D 22

5. If one student is selected at random find the probability that he/she scored **at most** 2 marks

A $^{11}/_{18}$ B $^{11}/_{20}$ C $^{7}/_{22}$ D $^{5}/_{19}$

2005/4 (old) Exercise 18.8

The table shows the frequency distribution of the scores obtained in a test by a group of 50 students

Score	1	2	3	4	5	6	7	8
Frequency	2	4	7	12	11	6	4	4

(a) Draw a bar chart for the distribution

HISTOGRAM

Histogram is almost the equivalent of bar chart. It is used for frequency distributions. Unlike the bar chart, histogram bars have no space between them. We use *class boundaries* to plot against the *frequencies*. At this level Histogram is used to **estimate mode.**

2009/ 7 (Nov)

The marks scored by 50 students in a Geography examination are as follow:

60	54	40	67	53	73	37	55	62	43
44	69	39	32	45	58	48	67	39	51
46	59	40	52	61	48	23	60	59	47
65	58	74	47	40	59	68	51	50	50
71	51	26	36	38	70	46	40	51	42

(a) Using a class interval of 21, 31, 40, …, prepare a frequency distribution table

(b) Draw a histogram to represent the distribution

(c) Use your histogram to estimate the modal mark

(d) If a student is selected at random, find the probability that he/she obtains a mark **greater** than 60

Solution

Class interval	Tally	frequency	class boundaries
21 – 30	II	2	20.5 – 30.5
31 – 40	~~IIII~~ ~~IIII~~	10	30.5 – 40.5
41 – 50	~~IIII~~ ~~IIII~~ II	12	40.5 – 50.5
51 – 60	~~IIII~~ ~~IIII~~ ~~IIII~~	15	50.5 – 60.5
61 – 70	~~IIII~~ III	8	60.5 – 70.5
71 – 80	III	3	70.5 – 80.5

(b) Histogram is shown on page 260

(c) 50.5 + 3 = 53.5

(d) Pr (> 60) = $\frac{8+3}{50} = \frac{11}{50}$

2001/5 (Nov)

An experimental census count gives the following data

Age (yrs)	10 – 14	15 – 19	20 – 24	25 – 29	30 – 34	35 – 39	40 – 44
Number	3	8	16	26	18	12	6

(a) (i) Using a scale of 2cm to 5 units on the vertical axis, draw the histogram

(ii) Estimate the modal age

(b) Identify the median class

(c) What is the maximum age limit of the group?

Solution

Age(year)	Class boundaries	frequency
10 – 14	9.5 – 14.5	3
15 – 19	14.5 – 19.5	8
20 – 24	19.5 – 24.5	16
25 – 29	24.5 – 29 .5	26
30 – 34	29.5 – 34.5	18
35 – 39	34.5 – 39.5	12
40 – 44	39.5 – 44.5	6

(a) i histogram is shown on page 260

ii 24.5 + 2.8 = 27.3 years

(b) Our total frequency is 89 and half of (89 + 1) = 90 is 45 and by cumulative frequency 3+8+16+26: 45 falls here

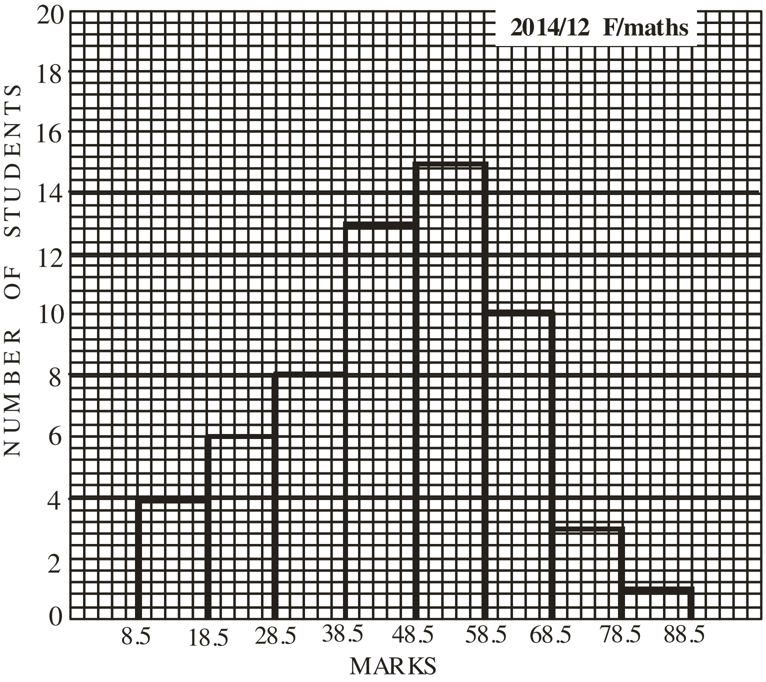
Thus median class 25–29.

(c) 44.5 years

Recall that the other name for class boundaries is true class limits

2014/12a F/math Adjusted

The histogram below represents the score of same candidate in an examination



(a) Using the histogram, construct a frequency distribution table indicating clearly the class intervals

Solution

Class interval	Class boundaries	f	cf
9.0 – 18.0	8.5 – 18.5	4	4
19.0 – 28.0	18.5 – 28.5	6	10
29.0 – 38.0	28.5 – 38.5	8	18
39.0 – 48.0	38.5 – 48.5	13	31
49.0 – 58.0	48.5 – 58.5	15	46
59.0 – 68.0	58.5 – 68.5	10	56
69.0 – 78.0	68.5 – 78.5	3	59
79.0 – 88.0	78.5 – 88.5	1	60

2006/3 (Nov) Exercise 18.9

The table shows the frequency distribution of marks obtained by 30 students in test

Marks	1 – 5	6 – 10	11 –15	16 –20	21 –25
Freq	4	6	11	8	1

(a) Draw a histogram of the distribution

(b) Use your histogram to estimate the mode

(c) If a student is selected at random, what is the probability that he scored at most 15 Mark?

2008/8 Exercise 18.10

The ages in years, of 50 teachers in a school are given below:

21	37	49	27	49	42	26	33	46	40
50	29	23	24	29	31	36	22	27	38
30	26	42	39	34	23	21	32	41	46
46	31	33	29	28	43	47	40	34	44
26	38	34	49	45	27	25	33	39	40

(a) Form a frequency distribution of the data using the intervals: 21 – 25, 26 – 30, 31 – 35 e.t.c

(b) Draw the histogram of the distribution

(c) Use your histogram to estimate the mode

2005/11 (New) Exercise 18.11

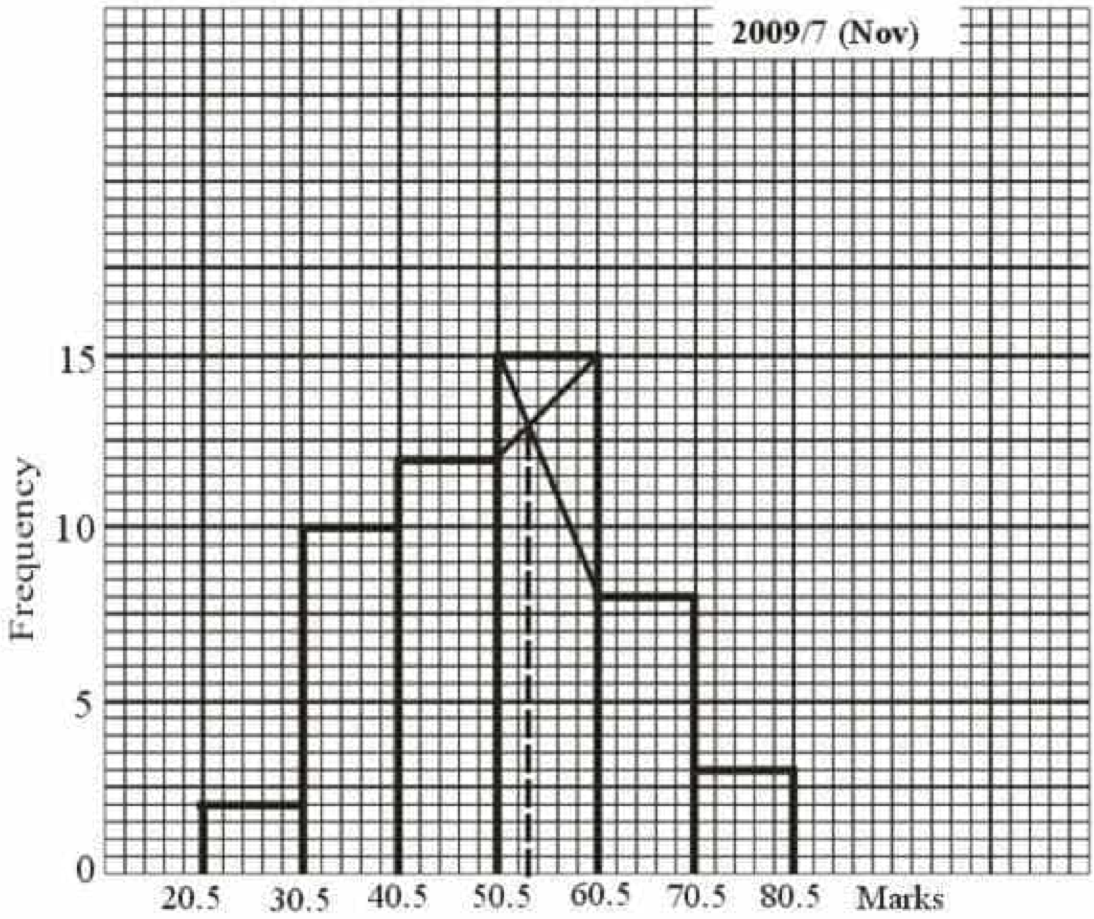
The table shows the age distributions of the members of a club

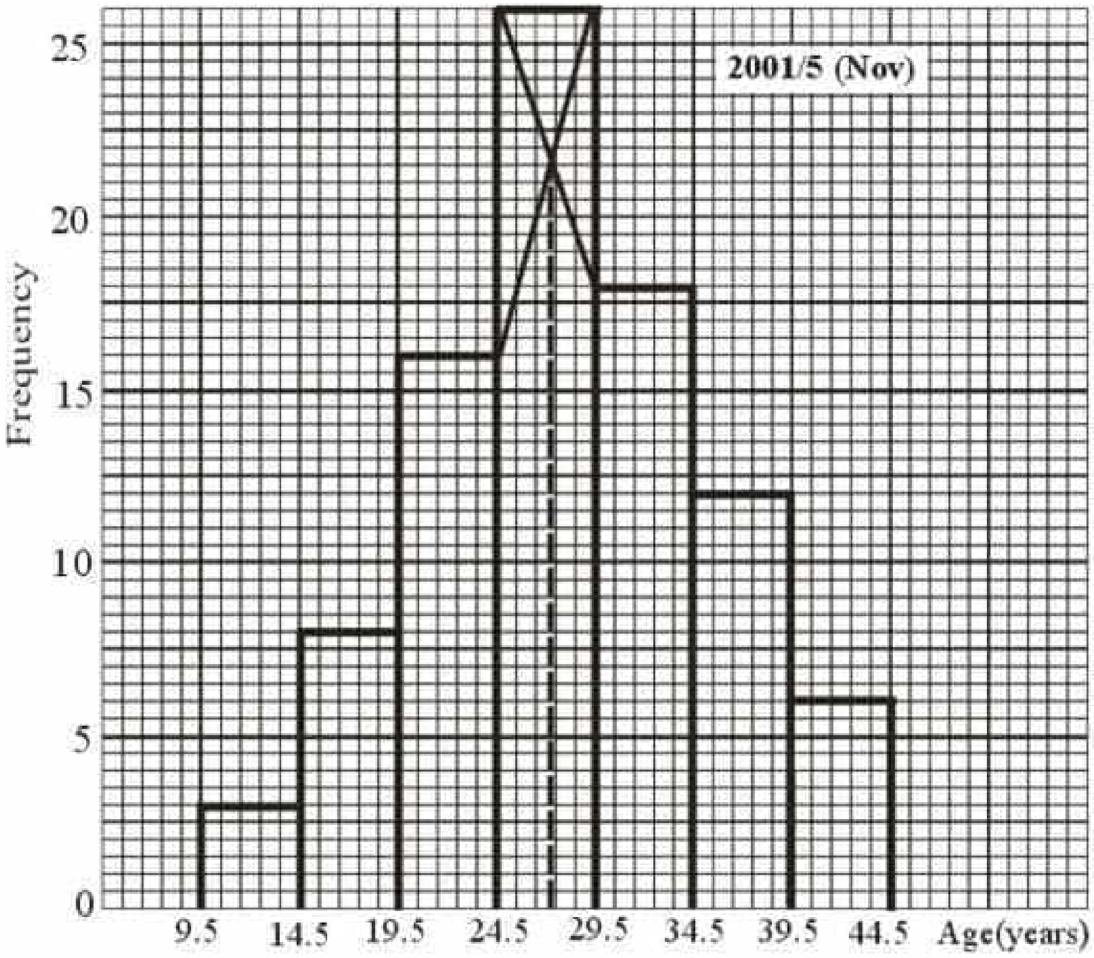
Age (year)	10– 14	15 –19	20 –24	25 –29	30 –34	35 –39
Frequency	7	18	25	17	9	4

b (i) Draw a histogram to illustrate the information

(ii) Use the histogram to estimate the modal age

c If a member is selected at random, what is the probability that he/she is in the modal class?


2009/7 (Nov)
Frequency
15
10
5
0
20.5
30.5
40.5
50.5
60.5
70.5
80.5
Marks


2001/5 (Nov)
Frequency
25
20
15
10
5
0
9.5
14.5
19.5
24.5
29.5
34.5
39.5
44.5
Age(years)

Cumulative Frequency Curve (O - give)

Cumulative frequency curve or O – give is a statistical graph gotten by plotting the upper class boundaries against cumulative frequencies. It is used to determine among others:

Median
Percentiles (100 divisions)
Deciles (10 divisions)
Quartiles (4 divisions)

Median in O – give

The median is simply half of the cumulative frequency traced to the curve then to the class boundaries to get the required value. Also the 50th percentile is a measure of median likewise the 5th Deciles (D_5) and 2nd Quartile (Q_2)

Percentile

This is the division of the cumulative frequency into 100 points. For instance

$$75\% = \frac{75}{100} \times \text{cumulative frequency}$$

$$20\% = \frac{20}{100} \times \text{cumulative frequency}$$

Then we trace the required values in the graph as discussed in median

Deciles

This is the division of the cumulative frequency into 10 portions.

For instance, $D_1 = \frac{1}{10} \times$ cumulative frequency

$D_5 = \frac{5}{10} \times$ cumulative frequency

Quartiles

This is the division of the cumulative frequency into 4 portions. Where:

(Lower quartiles) $Q_1 = \frac{1}{4}$ of cumulative frequency

(Middle quartile) $Q_2 = \frac{2}{4}$ of cumulative frequency

(Upper quartile) $Q_3 = \frac{3}{4}$ of cumulative frequency

Inter quartile range $= Q_3 - Q_1$

Semi – inter quartile range $= \dfrac{Q_3 - Q_1}{2}$

2006/10 (Nov)

The table shows the distribution of marks obtained by 200 candidates in a selection test for appointment

Marks	11–20	21–30	31–40	41–50	51–60	61–70	71–80	81–90	91–100
No of Cand.	9	25	43	49	29	22	10	8	5

(a) (i) Construct a cumulative frequency table for the distribution
(ii) Draw a cumulative frequency curve to represent the information
(b) Use your curve to find the:
(i) Inter-quartile range:
(ii) Minimum pass mark if 5% of the candidates are to be considered for appointment

Solution

Marks	Class boundaries	f	cf
11 – 20	10.5 – 20.5	9	9
21 – 30	20.5 – 30.5	25	34
31 – 40	30.5 – 40.5	43	77
41 – 50	40.5 – 50.5	49	126
51 – 60	50.5 – 60.5	29	155
61 – 70	60.5 – 70.5	22	177
71 – 80	70.5 – 80.5	10	187
81 – 90	80.5 – 90.5	8	195
91 – 100	90.5 – 100.5	5	200

(a) ii Graph is shown on **page 262**

b (i) inter-quartile range $= Q_3 - Q_1 = 58.5 - 34.5 = 24$

(ii) 5% of the candidates to be considered implies 100 – 5 = 95% of the candidate

$$= \frac{95}{100} \times 200 = 190$$

Next, we trace 190 at Cf – axis (y) to the o-give then to the mark – axis (x): 84.5

2007/11 Neco

The table below shows the marks scored by a group of students in a test

Marks	1–10	11–20	21–30	31–40	41–50	51–60	61–70	71–80	81–90	91–100
Freq	4	6	9	12	20	15	7	5	0	2

(a) Construct the cumulative frequency table (b) Draw the o-give
(c) From your o-give, find the: (i) Median (ii) Lower quartile
(d) A student was picked at random from the group, what is the probability that the student (using o-give)
(i) Obtained a distinction grade of 75% and above
(ii) failed the test if the pass mark is 40%

Solution

Marks	Class boundaries	f	cf
1 – 10	0.5 – 10.5	4	4
11 – 20	10.5 – 20.5	6	10
21 – 30	20.5 – 30.5	9	19
31– 40	30.5 – 40.5	12	31
41 – 50	40.5 – 50.5	20	51
51 – 60	50.5 – 60.5	15	66
61 – 70	60.5 – 70.5	7	73
71 – 80	70.5 – 80.5	5	78
81 – 90	80.5 – 90.5	0	78
91 – 100	90.5 – 100.5	2	80

C (i) Median mark: Half of the Cf traced to the mark axis: $\frac{1}{2}$ of 80 = 40 is traced to the mark axis = 45.5

(ii) Lower quartile $Q_1 = \frac{1}{4}$ of 80 = 20

Trace Cf = 20 to the curve then to mark axis: 31.5

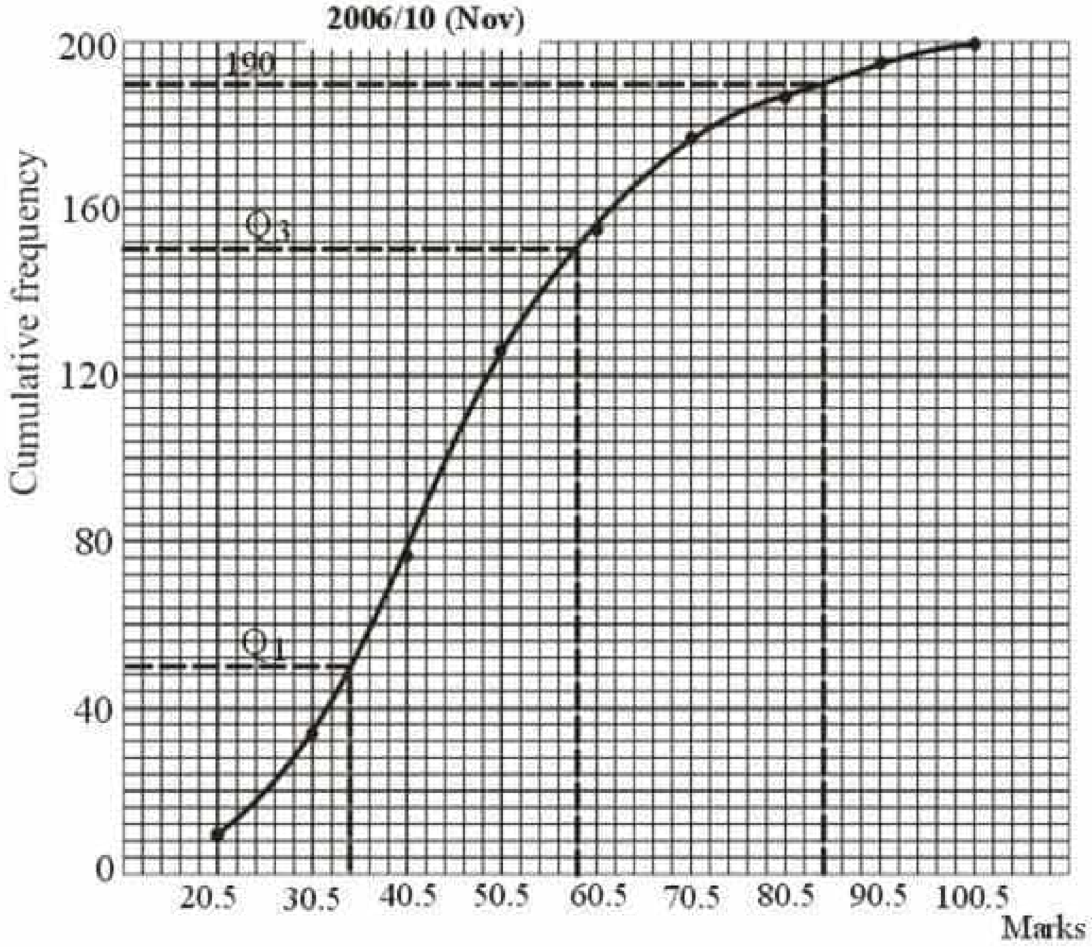
(d)(i) This is the number of students that scored 75% to 100% divided by 80

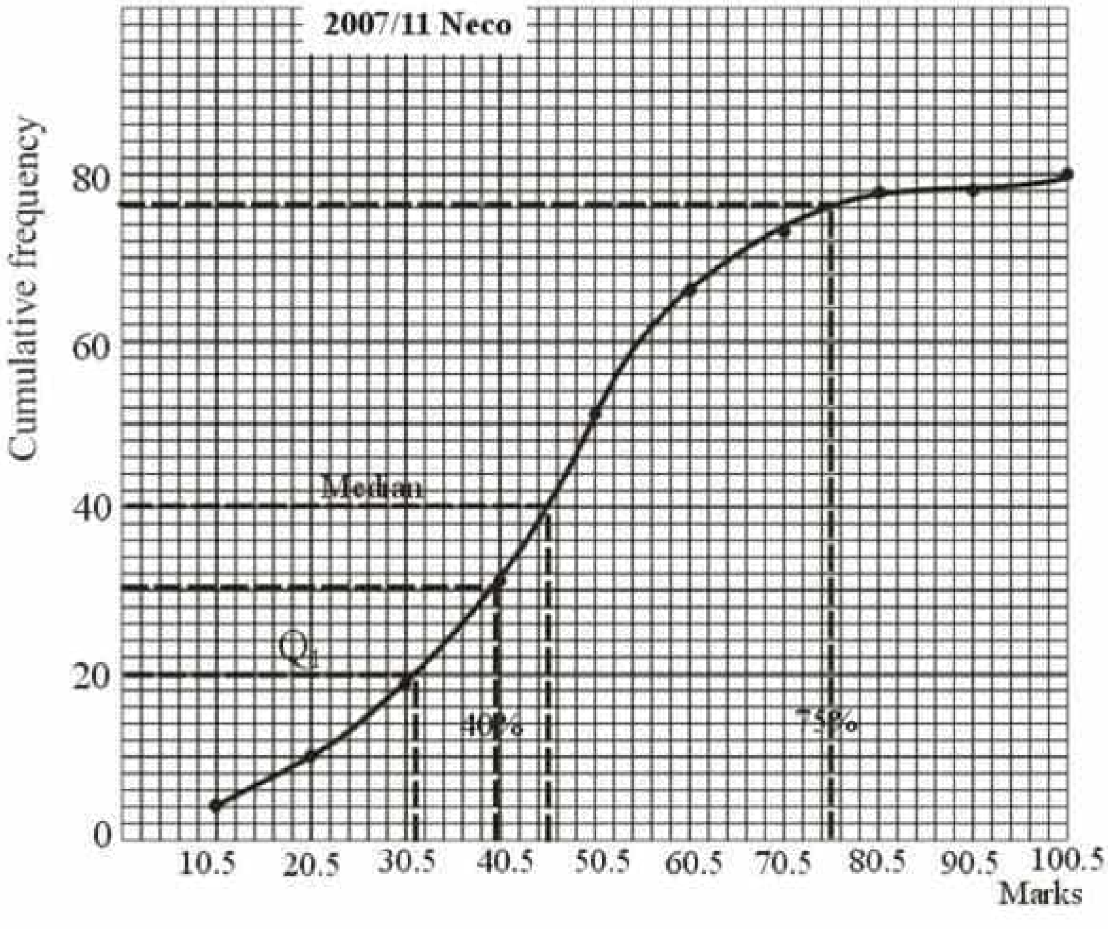
Trace 75 at x –axis to the curve then to the Cf – axis : 77

$$\text{Pr (75\% and above)} = \frac{77}{80}$$

(ii) Trace 40 at the mark – axis to the curve then to Cf – axis which will give 30

$$\text{Pr(failure at 40\% mark)} = \frac{30}{80} = \frac{3}{8}$$


2006/10 (Nov)
Cumulative frequency
200
160
120
80
40
0
190
Q_3
Q_1
20.5
30.5
40.5
50.5
60.5
70.5
80.5
90.5
100.5
Marks


2007/11 Neco
Cumulative frequency
80
60
40
20
0
Median
Q_1
40%
75%
10.5
20.5
30.5
40.5
50.5
60.5
70.5
80.5
90.5
100.5
Marks

2015/13

The table shows the marks scored by some candidates in an examination.

Marks (%)	0– 9	10–19	20–29	30–39	40–49
Frequency	7	11	17	20	29

50–59	60–69	70–79	80–89	90–99
34	30	25	21	6

(a) Construct a cumulative frequency table for the distribution and draw a cumulative frequency curve

(b)Use the curve to estimate, correct to one decimal place, the

(i) lowest mark for distribution if 5% of the candidates passed with distinction

(ii) probability of selecting a candidate who scored **at most** 45%

Solution

Marks (%)	Class boundaries	f	cf
0 – 9	0 – 9.5	7	7
10 – 19	9.5 – 19.5	11	18
20 – 29	19.5 – 29.5	17	35
30 – 39	29.5 – 39.5	20	55
40 – 49	39.5 – 49.5	29	84
50 – 59	49.5 – 59.5	34	118
60 – 69	59.5 – 69.5	30	148
70 – 79	69.5 – 79.5	25	173
80 – 89	79.5 – 89.5	21	194
90 – 99	89.5 – 99.5	6	200

Graph is shown on **page 264**

(b) i. Distribution if 5% of candidates passed implies $100 - 5 = 95\%$

Next, $\frac{95}{100} \times 200 = 190$

Trace 190 at cf to the o–give then to the Mark – axis = 85.5

ii. At most 45% mark is 45% or less

Trace 45 on the Mark – axis to the curve then to the cf – axis = 70

Pr (at most 45%) = $\frac{70}{200} = \frac{7}{20}$

2014/10 Neco

The table below shows the marks scored by 200 candidates in a certain examination

Marks	1– 10	11–20	21–30	31–40	41–50
No. of Cand.	6	9	12	19	28

51– 60	61– 70	71– 80	81– 90	91– 100
37	41	28	13	7

(a) Construct a

(i) Cumulative frequency table

(ii) Cumulative frequency curve (o-give) of the distribution

(b) From your o-give, estimate the:

(i) Median mark;

(ii) Pass mark if 80% of the candidates passed the examination

(iii) Minimum mark scored by the top 45% of the candidates

Solution

Marks	Class boundaries	f	cf
1 – 10	0.5 – 10.5	6	6
11 – 20	10.5 – 20.5	9	15
21 – 30	20.5 – 30.5	12	27
31 – 40	30.5 – 40.5	19	46
41 – 50	40.5 – 50.5	28	74
51 – 60	50.5 – 60.5	37	111
61 – 70	60.5 – 70.5	41	152
71 – 80	70.5 – 80.5	28	180
81 – 90	80.5 – 90.5	13	193
91 – 100	90.5 – 100.5	7	200

Graph is shown on **page 264**

(i) Median mark: Half of the Cf traced to the mark axis:

$\frac{1}{2}$ of 200 is 100 traced to the Mark axis = 58.0

(ii) 80% pass implies 20% fail

$\frac{20}{100} \times 200 = 40;$

Trace Cf = 40 to the o-give then to Mark axis = 37.5

(iii) It is the same as 100 – 45 = 55% of 200

$\frac{55}{100} \times 200 = 110$

Trace Cf =110 to the o-give then to the Mark axis= 60.5

2006/12 Neco Exercise 18.12

The following are the ages of 50 workers who worked in a factory years back:

71	79	46	35	25	28	56	82	52	68
68	64	93	95	78	43	58	72	57	60
50	98	62	63	70	73	53	44	86	79
72	68	88	51	32	59	72	73	46	85
40	55	52	61	96	67	82	72	48	59

(a) Using class intervals 21–30, 31– 40, 41–50, … draw a frequency table for the distribution

(b) Draw the cumulative frequency curve (o-give) and determine

(i) the median (ii) inter – quartile range

(c) If a worker is picked at random, what is the probability that he is more than 60 years?

2009/11 Neco (Dec) Exercise 18.13

The table below shows the height of maize plants in a given farm

Height of plants	No of plants
45 – 49	10
50 – 54	36
55 – 59	64
60 – 64	52
65 – 69	28
70 – 74	10

(i) Construct the cumulative frequency table

(ii) Draw a cumulative frequency curve (o-give) for the distribution using a scale of 2cm = 20 units on the y – axis and 2cm = 5 units on the x – axis

(iii) Use your graph to determine the inter-quartile range

2005/11 (old) Exercise 18.14

The table shows the examination marks for 120 students

Mark	30–39	40– 49	50–59	60– 69	70–79	80– 89
Freq	11	25	45	15	10	14

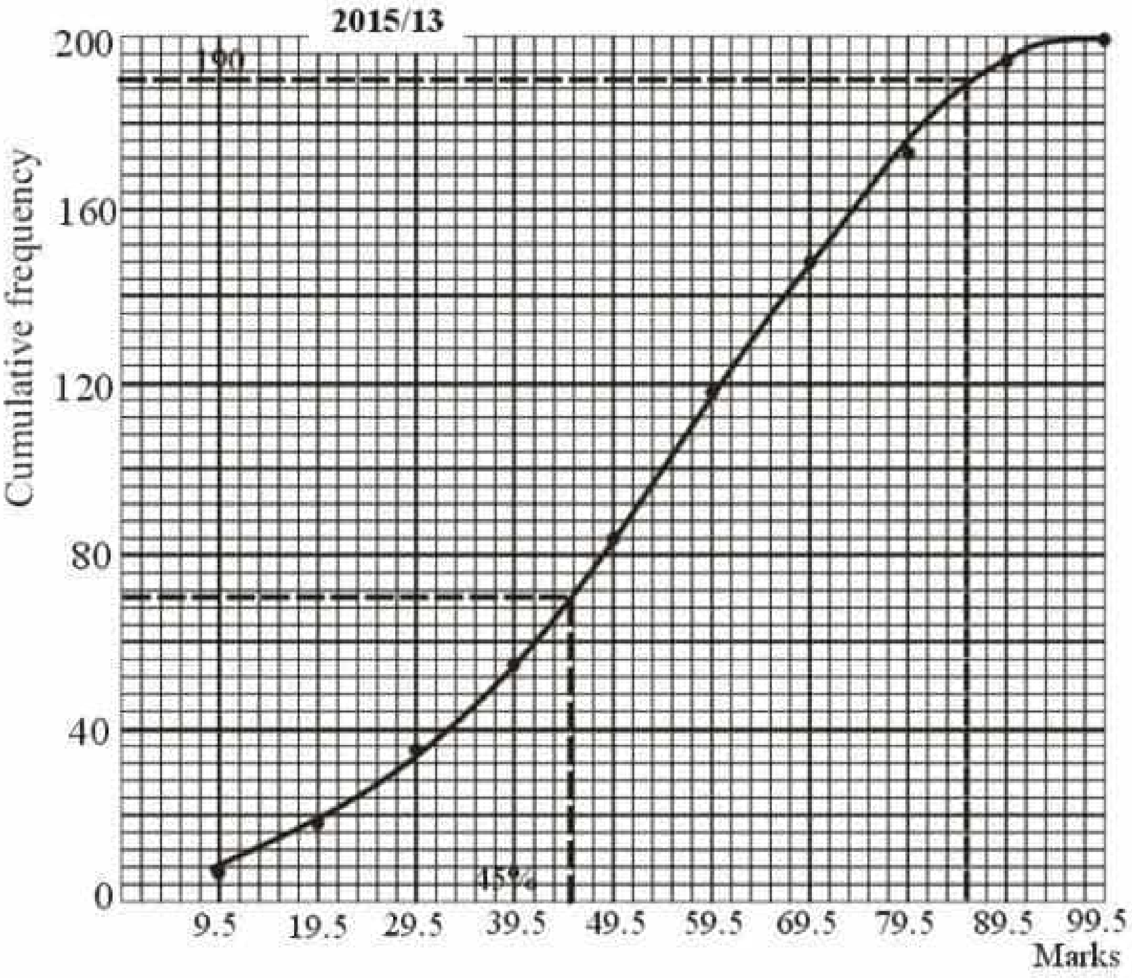
(a) i. Draw the cumulative frequency table for the distribution

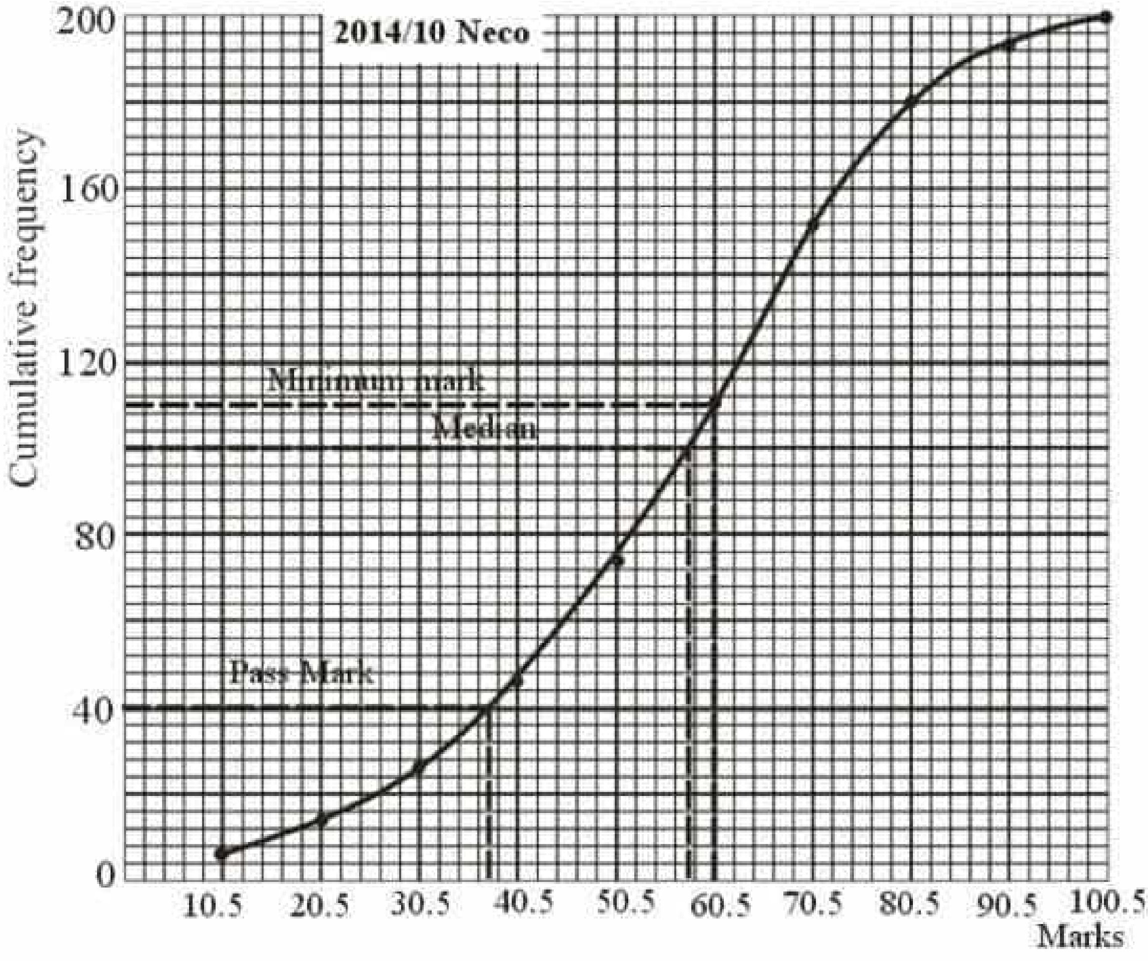
ii. Draw the cumulative frequency curve

(b) Use the cumulative frequency curve to estimate:

i. the median ii. the upper quartile

(c) What is the probability that a student chosen at random scored between 50 and 69 marks

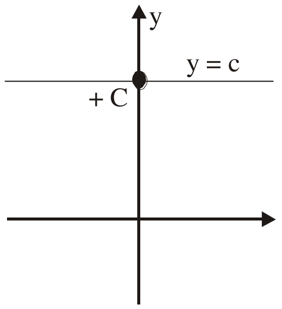

2015/13
190
45%
Cumulative frequency
200
160
120
80
40
0
9.5
19.5
29.5
39.5
49.5
59.5
69.5
79.5
89.5
99.5
Marks


2014/10 Neco
Minimum mark
Median
Pass Mark
Cumulative frequency
200
160
120
80
40
0
10.5
20.5
30.5
40.5
50.5
60.5
70.5
80.5
90.5
100.5
Marks

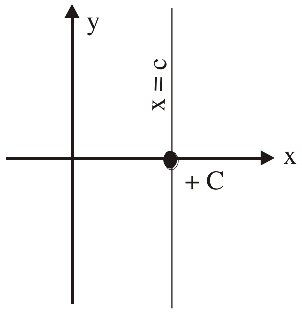
Linear equations

The general form of a linear equation is y = mx + c, where m is the gradient and x is the independent variable; c is the intercept at the y – axis.

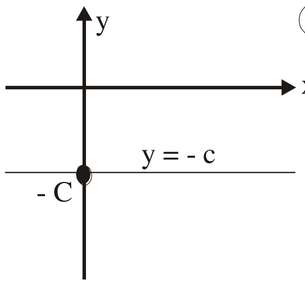
When there is **no gradient** we could have two major types of line such as:



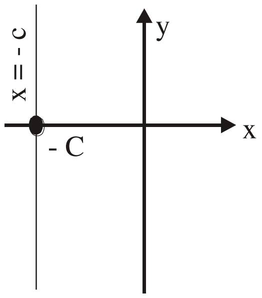
and



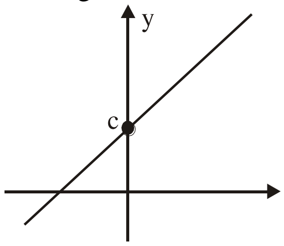
Alternatively, it may occur as:



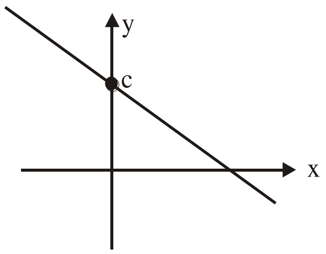
and



Where the **gradient exist** ; then it may be any of the forms given below:



OR



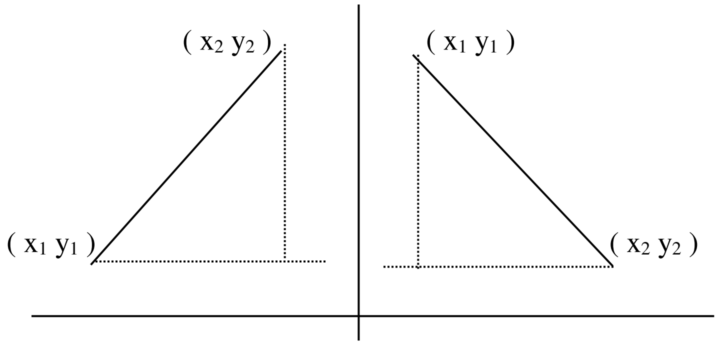
A straight line where we can form a triangle under it or on it.

There are basically two types of gradients namely:

Positive and **Negative** gradients

Gradient (slope) = $\frac{Change\ in\ y}{Change\ in\ x}$

Positive gradient	**Negative gradient**
$\frac{\text{Increase in y}}{\text{Increase in x}}$	$\frac{\text{Decrease in y}}{\text{Increase in x}}$
In physics it is called - acceleration since y increases as x increases	In physics it is called - deceleration since y decreases as x increases
y-axis starts at the down part and ends at the top	y-axis starts at the upper part and ends at the down part



Gradient $= \frac{y_2 - y_1}{x_2 - x_1}$

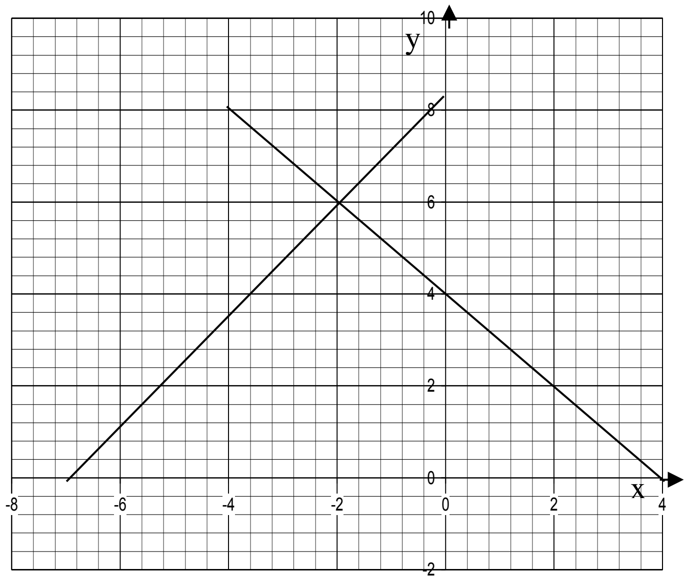
From the above table note that : y_2 , y_1 and x_1 , x_2 positions differ for positive and negative gradients

Simultaneous linear equations graphical solution

This is the case where two straight lines graphs are drawn on same graph sheet and their point of intersection traced to the x and y axes to give a coordinate value (x, y) which is the solution

2007/22 Neco

The graphical solution of simultaneous equations is represented on the graph below. What is the solution of the simultaneous equation?



A (–2, 6) B (6, –2) C (–2, –6) D (0, 6) E (4, 0)

Solution

The solution is the point of intersection of the two lines traced to the x – axis and y – axis

x = – 2 and y = 6 (–2, 6) A

2015/ 21 and 22

The table below shows some values of a linear graph.

x	0	$1\frac{1}{4}$	2	4
y	3	$5\frac{1}{2}$		

Use it to answer questions 21 and 22

21 Find the gradient of the line

A 1 B 2 C 3 D 4

Solution

Gradient = $\frac{y_2 - y_1}{x_2 - x_1}$

$$= \frac{5.5-3}{1.25-0} = \frac{2.5}{1.25} = 2 \quad (B)$$

22. What is the value of y when x = 2

A 5 B 7 C 9 D 11

Solution

For us to know the value of y when x = 2

First, we find the equation for the line from:

y = mx + c

Here m is the gradient which is gotten earlier as 2

Next we find C

When x = 0, y = 3, from the table

y = mx + c becomes

3 = 0 + c thus c = 3

Thus our line equation is y = 2x + 3

When x = 2

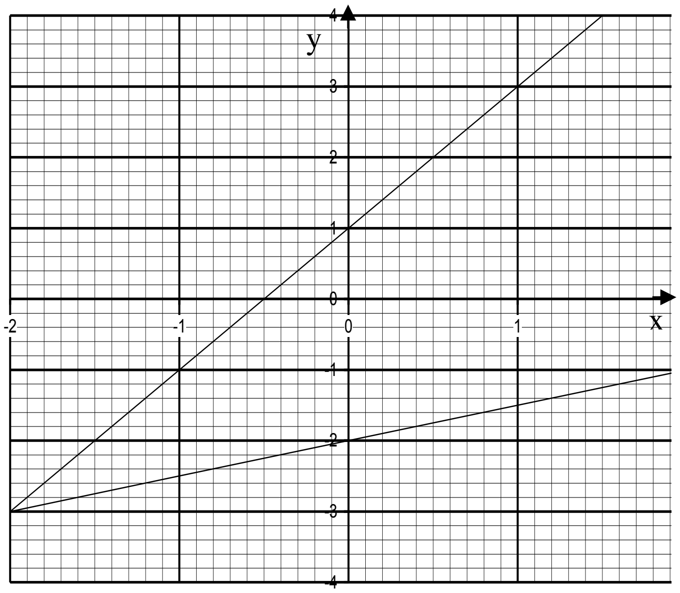
y = 2(2) + 3

= 4 + 3

= 7 (B)

2005/26 Neco

Find the solution of the simultaneous equations in the graph below



A (–2, 3) B (–2, –5) C (–2, –3) D (–3, –2) E (2, –3)

Solution

The solution is the point of intersection between the two lines: x = – 2 and y = – 3

(–2, –3) C

2005/8b (old) ***adjusted***

The table below gives the relationship between x and y

x	– 1	0	1	2	3
y	– 4.0	– 2.5	– 1.0	0.5	2.0

(i) Using a scale of 2cm to 1unit on y-axis and 2cm to 0.5unit on x-axis , use the table to draw the graph of the relationship between x and y

(ii) From your graph, find the value of y when x = 4

(iii) Use the graph to calculate the gradient of the line

(iv) What is the equation of the relation?

Solution

Graph is shown on page 267

(i) The straight line graph stop at x = 3 and y = 2.0

(ii) For us to find y when x = 4, we have to extend the straight line as shown by dotted line. A straight line can be extended anytime without knowing the equation of the line unlike curves

Thus when x = 4, y = 3.6

(iii) y = mx + c equation of straight line with Gradient (m)

$$= \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{2-(-3)}{3-(-0.3)} = \frac{2+3}{3+0.3} = \frac{5}{3.3} = 1.5$$

(iv) Where the line cut across the y – axis C = – 2.5

Substituting

y = 1.5x – 2.5

2y = 3x – 5

2010/39 and 40

x	0	2	4	6
y	1	2	3	4

The table is for the relation y = mx + c where m and c are constants. Use it to answer questions 39 and 40

39. What is the equation of the line described in the table?

A y = 2x B y = x + 1 C y = x D y = $\frac{1}{2}$x + 1

Solution

y = mx + c

$$m(\text{gradient}) = \frac{y_2 - y_1}{x_2 - x_1} \text{ (take any two points)}$$

$$= \frac{4-1}{6-0} = \frac{3}{6} = \frac{1}{2}$$

To find c, we take any coordinate

When x = 2, y = 2

y = mx + c becomes

2 = $\frac{1}{2}$(2) + c

2 = 1 + c

2 – 1 = c thus c = 1

The line is y = $\frac{1}{2}$x + 1 (D)

40. What is the value of x when y = 5?

A 8 B 9 C 10 D 11

Solution

When y = 5

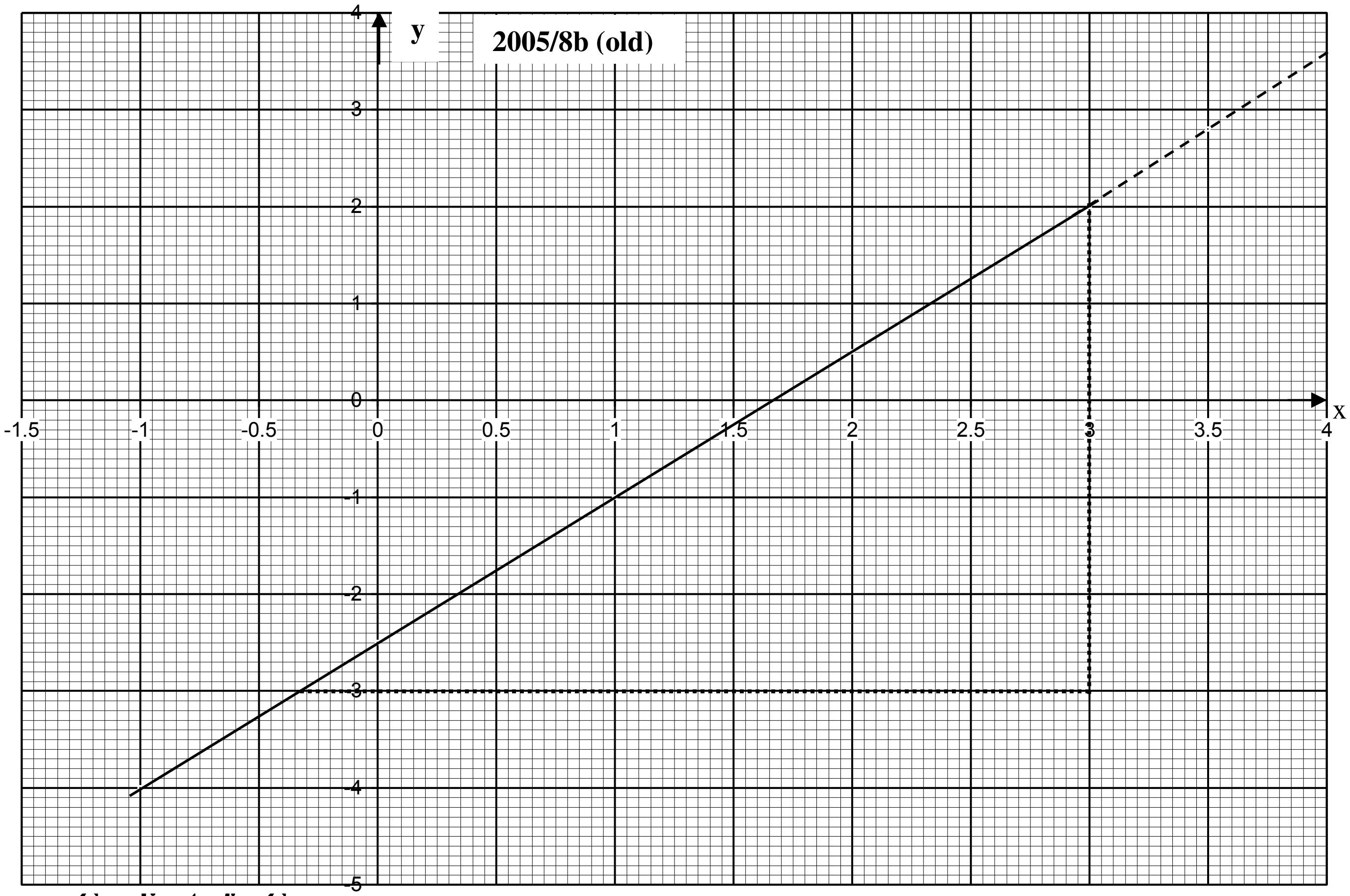
y = $\frac{1}{2}$x + 1 becomes

5 = $\frac{1}{2}$x + 1

5 – 1 = $\frac{1}{2}$x

4 = $\frac{1}{2}$x

4 × 2 = x Thus 8 = x (A)



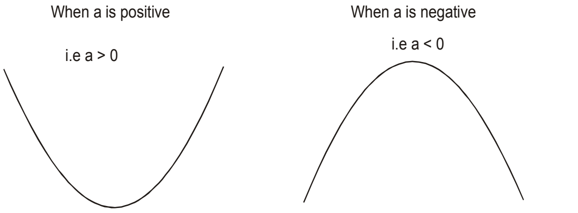
Gradient of a Curve

The gradient of any point on a curve is the gradient of the tangent to the curve at that point. The tangent must be produced equidistant to the point

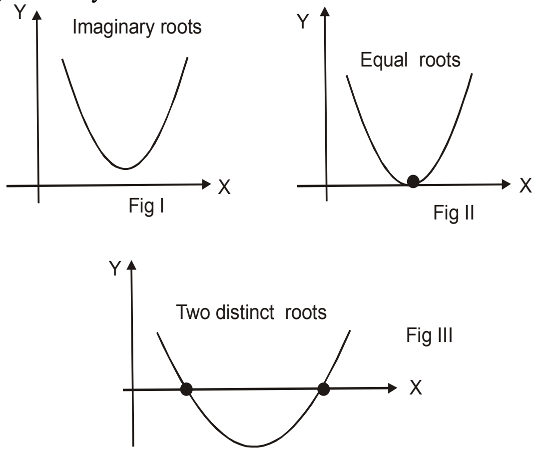
Quadratic Equations

Graphical Solution

There are two major types of quadratic curves from the general form: $ax^2 + bx + c$



The roots or solutions or zeros of any quadratic equation may take any of the forms listed below:



Sometimes, quadratic equation comes along with linear equation. The following examples will illustrate the required concepts

2006/8 (Nov)

(a) Copy and complete the table

x	– 3	– 2	–1	0	1	2	3	4	5
$y = 2x^2 – 5x – 4$		14		– 4		– 6		8	

(b) Using the scale 2cm = 1 unit on the x –axis and 2cm to 5 units on the y – axis, draw the graph of $y = 2x^2 – 5x – 4$ for $– 3 \le x \le 5$

(c)Use your graph to find the:

(i) roots of the equation $2x^2 – 5x – 4 = 0$;

(ii) Range of values of x for which $y \le 3$;

(iii) Minimum value of y

Solution

Table for $y = 2x^2 – 5x – 4$

x	– 3	– 2	– 1	0	1	2	3	4	5
$2x^2$	18	8	2	0	2	8	18	32	50
– 5x	15	10	5	0	– 5	– 10	– 15	– 20	–25
– 4	– 4	– 4	– 4	– 4	– 4	– 4	– 4	– 4	– 4
y	29	14	3	– 4	– 7	– 6	– 1	8	21

From the table, we note that:

x – axis

Has almost equal positive and negative sides – 3 and + 5

Thus rule close to the centre on the y – axis

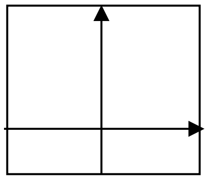
y – axis

Has very small negative side – 7

Has very big positive side 29

The ruling gives more space to positive side in x – axis

The graph can be ruled in the format below:



(b) **Graph is on page 269**

(c)(i) The roots of the equation $2x^2 - 5x - 4 = 0$ is the point where the curve cut the x – axis
– 0.65 and 3.15

(ii) Range of value of x for $y \le 3$ is gotten as:
Draw a line at y = 3 to touch the curve, then trace to x – axis – 1and 3.5,
in inequality form $-1 \le x \le 3.5$

(iii) The lowest position of the curve traced to y – axis as shown by dotted line – 7.

2006/10 NABTEB

(b) solve graphically, the simultaneous equation:
$y = x^2 - 7x + 10$ and $y = x + 3$ using the interval $0 \le x \le 8$ and a scale of 2cm to 1unit on the x – axis and 1cm to 2units on the y – axis

(c) Use your graph in (b) to find the roots of
(i) $x^2 - 7x + 10 = 0$
(ii) $x^2 - 7x + 5 = 0$

Solution

Table for $y = x^2 - 7x + 10$

x	0	1	2	3	4	5	6	7	8
x^2	0	1	4	9	16	25	36	49	64
– 7x	0	–7	–14	–21	–28	–35	–42	– 49	–56
+10	+10	+10	+10	+10	+10	+10	+10	+10	+10
y	10	4	0	–2	–2	0	4	10	18

From the table, it will be observed that
x – axis
It has only positive axis
We rule y – axis from the beginning of the graph
y– axis
Has very small negative side – 2
Has very big positive side 18
We rule the x – axis with much space for the positive side.
Thus the graph is to be ruled in the format below:

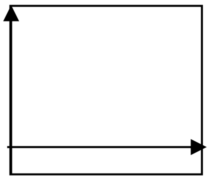


Table for y = x + 3

x	0	2	5
x	0	2	5
+3	+3	+3	+3
y	3	5	8

Graph is shown on page 269

C (i) 2 and 5 the two points of intersection traced to the x – axis

(ii) $x^2 - 7x + 5 = 0$

What must be added to both sides to make it the original equation $x^2 - 7x + 10$

$$x^2 - 7x + 5 + 5 = 5$$
$$x^2 - 7x + 10 = 5$$

Next, we rule a straight line at y = 5 where it touches the curve are traced to the x – axis as shown by dotted lines 0.8 and 6.2

2009/12 Neco (Dec)

(a) Copy and complete the following table of values for $y = 2x^2 - 9x - 1$

x	– 1	0	1	2	3	4	5	6
y		– 1	– 8	– 11				17

(b) Using a scale of 2cm to represent 1unit on the x– axis and 2cm to represent 5units on the y– axis , draw the graph of $y = 2x^2 - 9x - 1$ for $-1 \le x \le 6$.

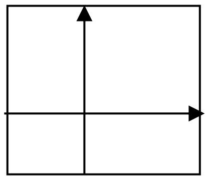
(c) Use your graph to find the
(i) roots of the equation $2x^2 - 9x = 4$, correct to one decimal place
(ii) gradient of the curve $y = 2x^2 - 9x - 1$ at x = 3

Solution

Table for $y = 2x^2 - 9x - 1$

x	– 1	0	1	2	3	4	5	6
$2x^2$	2	0	2	8	18	32	50	72
–9x	9	0	– 9	–18	–27	–36	–45	–54
– 1	– 1	– 1	– 1	– 1	– 1	– 1	– 1	– 1
y	10	– 1	– 8	–11	–10	– 5	4	17

From the table, we observed that
x – axis
Has very small negative value of –1and big positive + 6
We rule y – axis with more space for positive side
y – axis
Has positive side + 17 and negative side – 11
We rule the x – axis with little more space for the positive side.
Thus, we rule the graph as

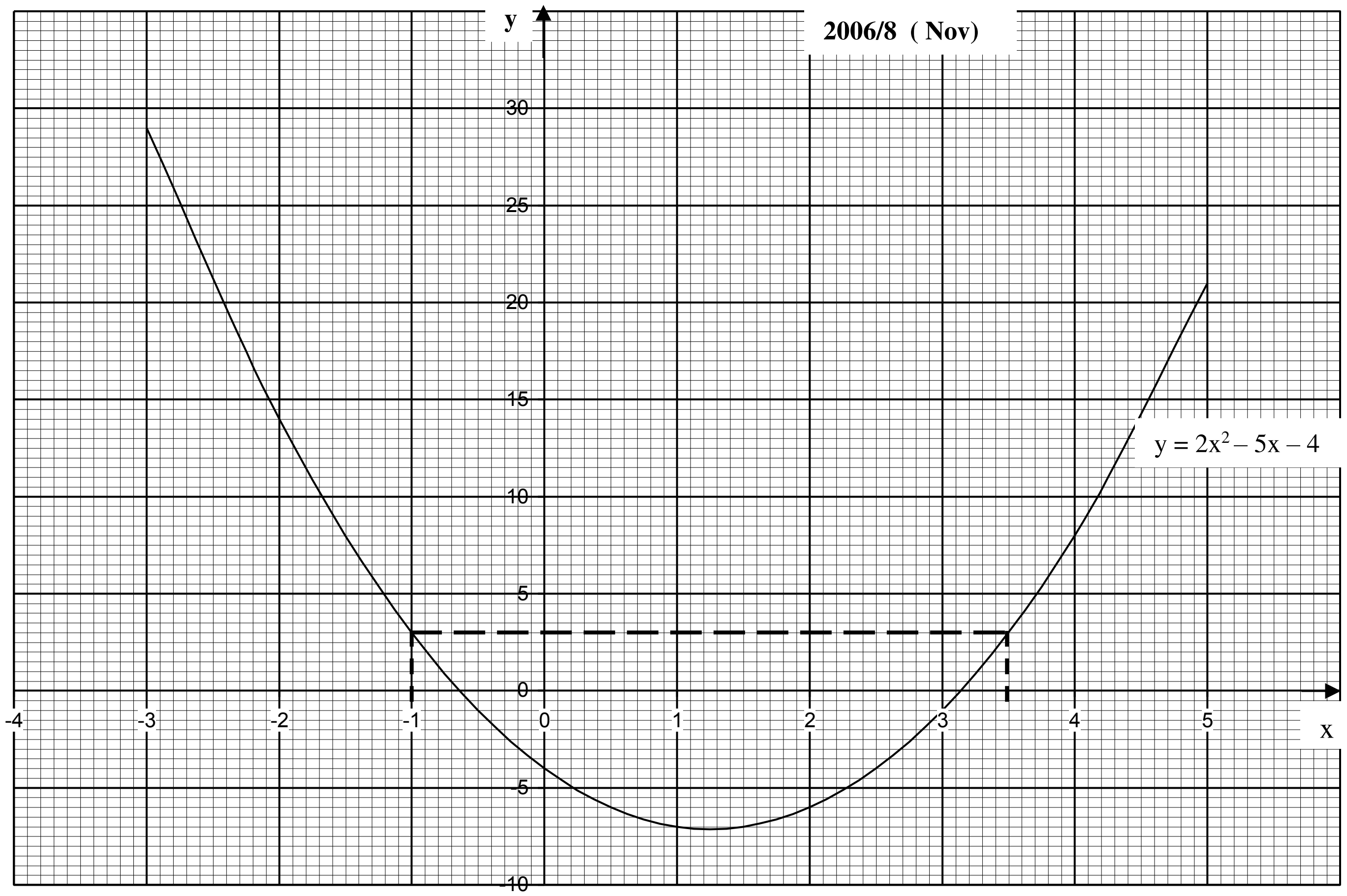


(b) The graph is shown on page 270

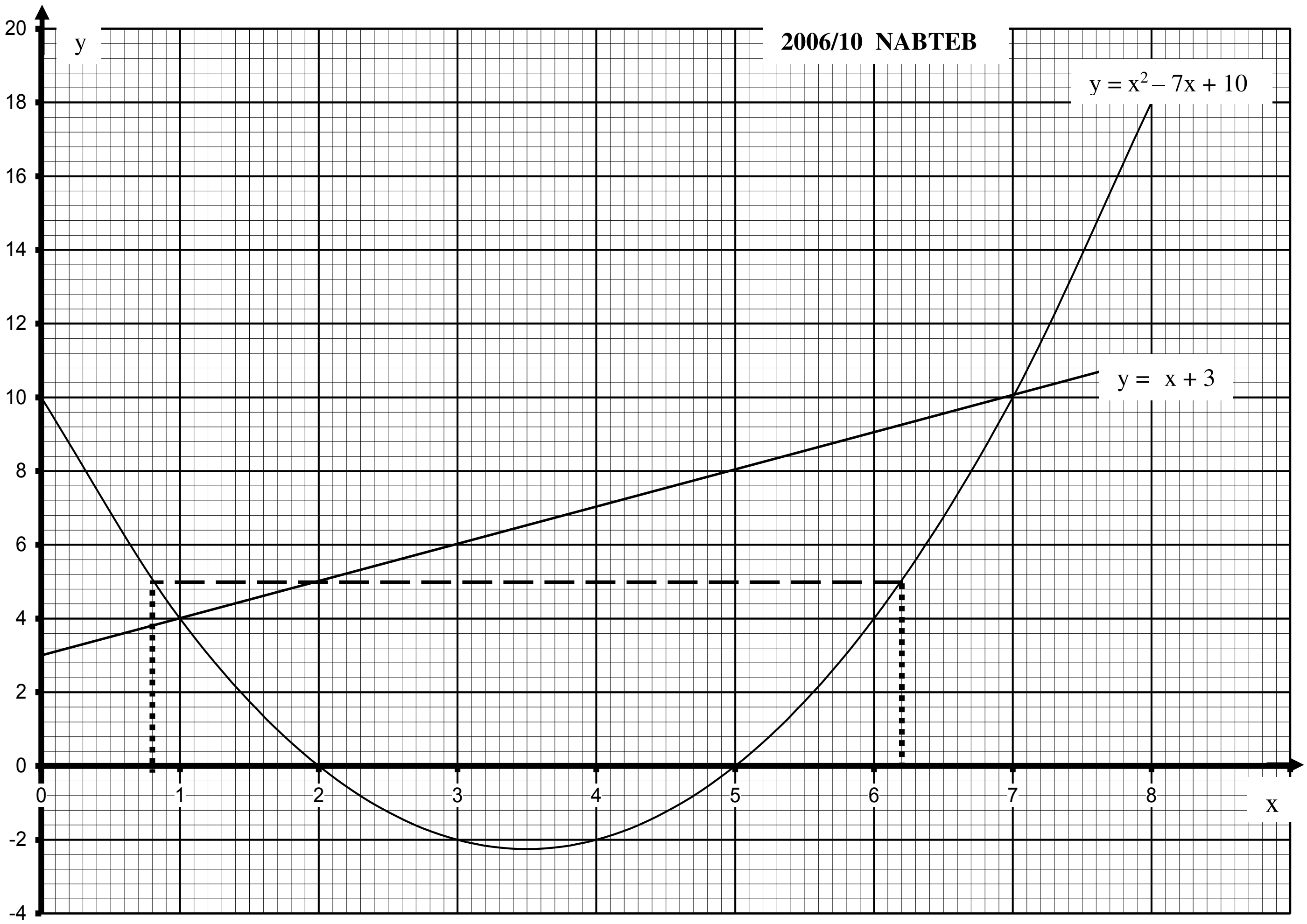
(c)i The roots of the equation $2x^2 - 9x = 4$ is gotten as:
Rearrange the equation
$$2x^2 - 9x = 4$$
$$2x^2 - 9x - 4 = 0$$
Add 3 to both sides to get our original equation
$$2x^2 - 9x - 4 + 3 = 3$$
$$2x^2 - 9x - 1 = 3$$
On the graph, draw a line at point y = 3 to cut the curve.
These (two) points traced to x – axis gives the solution
– 0.4 and 4.9

(c)ii At point x = 3 traced to the curve, we draw an equidistant line as shown, then take the slope

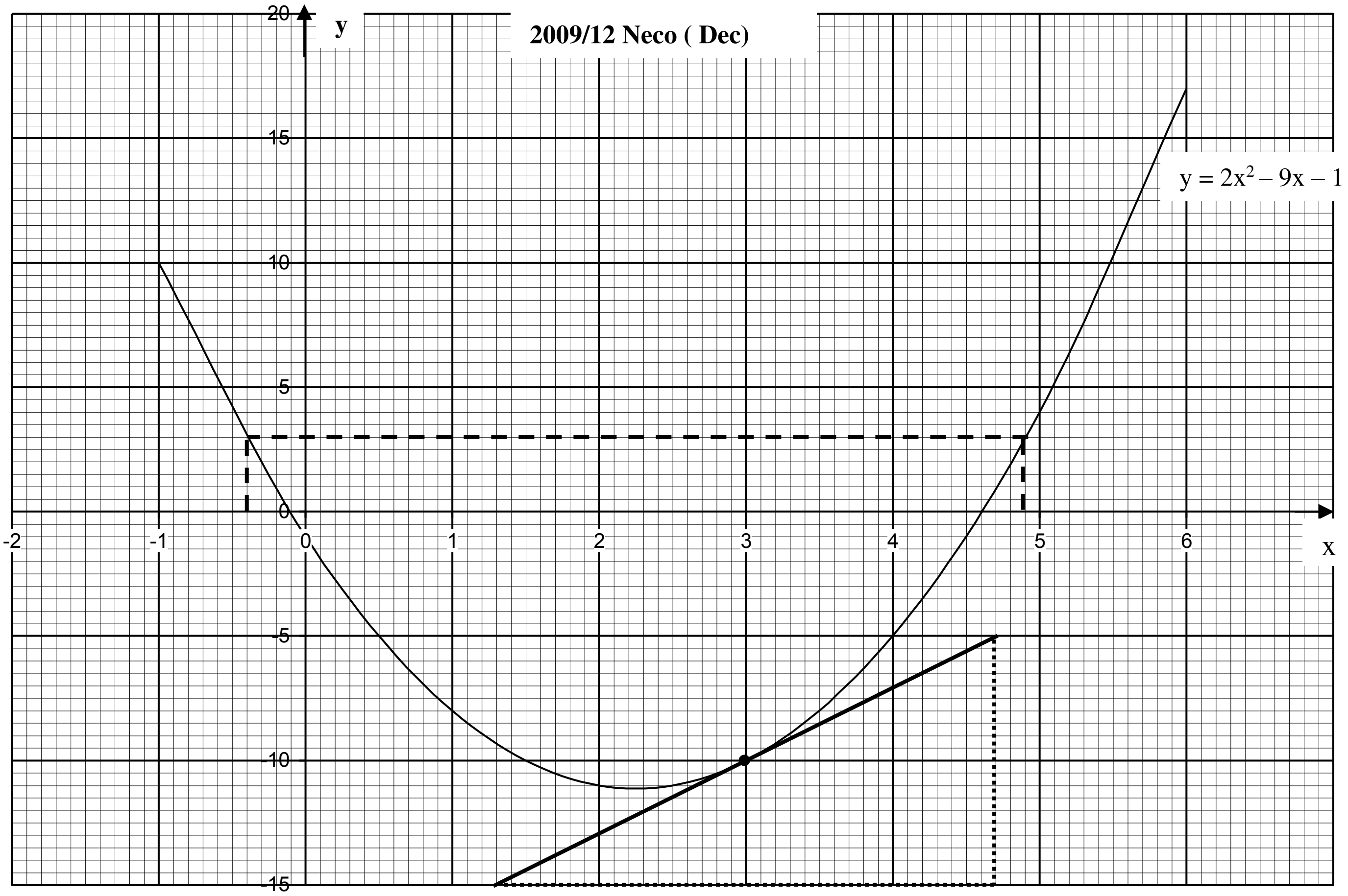
$$\text{Slope} = \frac{\text{Change in } y}{\text{Change in } x} = \frac{y_2 - y_1}{x_2 - x_1}$$
$$= \frac{-5-(-15)}{4.7-1.3}$$
$$= \frac{10}{3.4} = 2.94$$

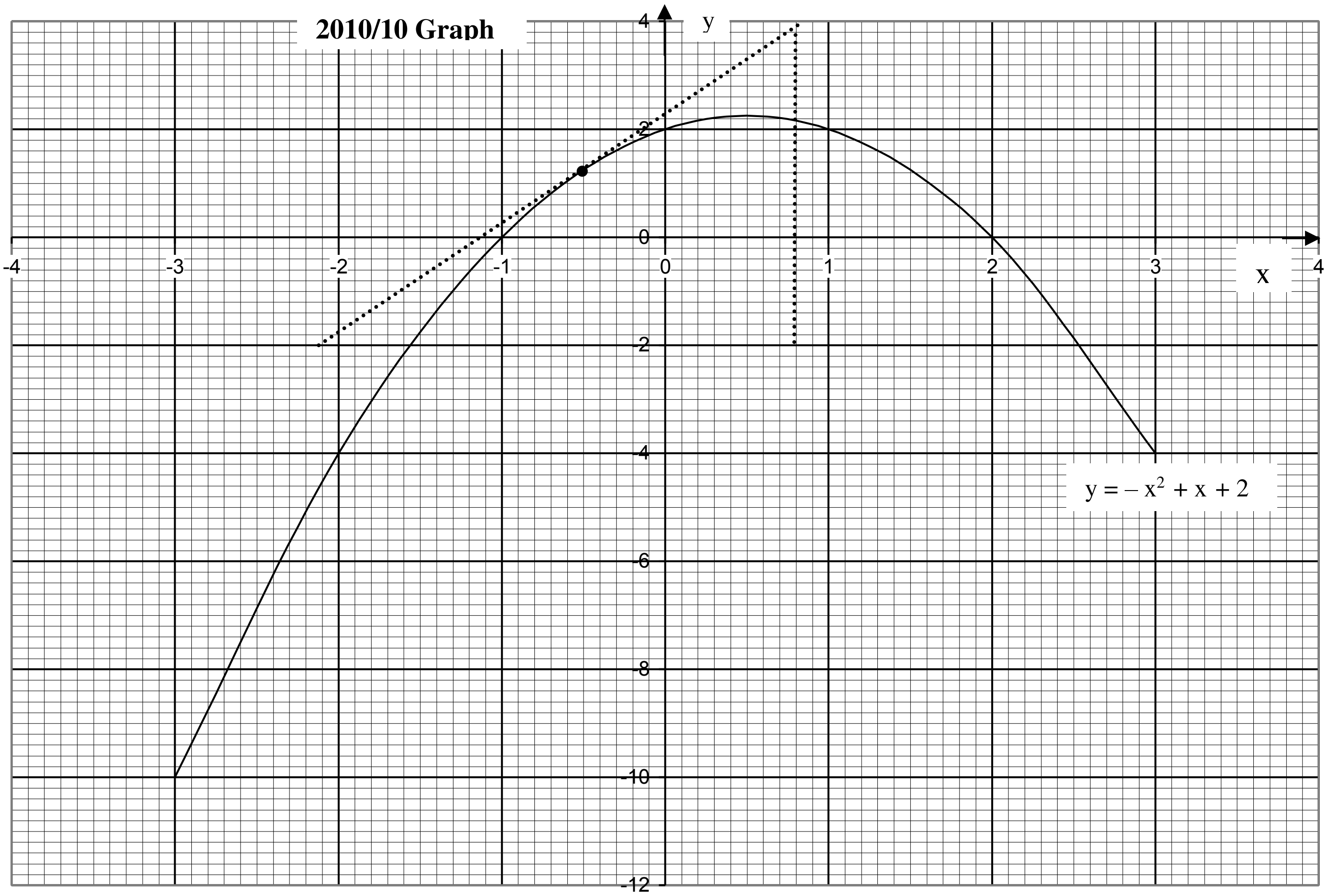


2006/8 (Nov)
y
x
$y = 2x^2 - 5x - 4$
30
25
20
15
10
5
0
-5
-10
-4
-3
-2
-1
0
1
2
3
4
5



2006/10 NABTEB
y
x
$y = x^2 - 7x + 10$
$y = x + 3$
20
18
16
14
12
10
8
6
4
2
0
-2
-4
0
1
2
3
4
5
6
7
8


2009/12 Neco (Dec)
y
x
y = 2x² – 9x – 1
20
15
10
5
0
-5
-10
-15
-2
-1
0
1
2
3
4
5
6


2010/10 Graph
y
x
y = – x² + x + 2
4
2
0
-2
-4
-6
-8
-10
-12
-4
-3
-2
-1
0
1
2
3
4

2010/10

(a) Copy and complete the table of values for the relation $y = -x^2 + x + 2$ for $-3 \le x \le 3$

x	-3	-2	-1	0	1	2	3
y	-4			2			-4

(b) Using scales of 2cm to 1unit on the x – axis and 2cm to 2 units on the y – axis,
draw a graph of the relation $y = -x^2 + x + 2$.

(c) From the graph, find the;

(i) minimum value of y;

(ii) root of the equation $x^2 - x - 2 = 0$

(iii) Gradient of curve at $x = -0.5$

Solution

Table for $y = -x^2 + x + 2$

x	-3	-2	-1	0	1	2	3
$-x^2$	-9	-4	-1	0	-1	-4	-9
$+x$	-3	-2	-1	0	1	2	3
$+2$	$+2$	$+2$	$+2$	$+2$	$+2$	$+2$	$+2$
y	-10	-4	0	2	2	0	-4

From the table, we observe that

x – axis

Has equal sides – 3 and + 3

We rule the y – axis at the centre

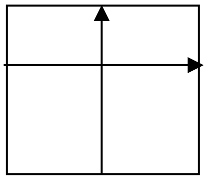
y – axis

Has very small positive side + 2

Has bigger negative side – 10

We rule the x – axis with much space for the negative side

Thus the graph should be ruled as:



(b) Graph is shown on page 270

(c) i. Typing error from the question :

This type of graph does not have maximum value it has maximum value 2.3

ii. – 1 and 2 where the curve cut the x – axis

iii. At $x = -0.5$, we trace it to the curve and draw an equidistant line as show by dotted line

$$\text{Gradient} = \frac{change\ in\ y}{change\ in\ x}$$

$$= \frac{4-(-2)}{0.8-(-2.1)} = \frac{4+2}{0.8+2.1} = \frac{6}{2.9} = 2.07$$

2009/ 11 (Nov)

(a) Copy and complete the table of values for $y = x^2 - 2$ for $-3 \le x \le 4$

x	-3	-2	-1	0	1	2	3	4
y		2		-2		2		

(b) Using a scale of 2cm to 1unit on the x – axis and a scale of 2cm to 2units on y–axis, draw the graph of $y = x^2 - 2$

(c) Use your graph to find the:

(i) roots of the equation $x^2 - 2 = 0$

(ii) values of x for which $x^2 - 3 = 0$

(iii) gradient of the curve at the point where $x = -1$

Solution

Table for $y = x^2 - 2$

x	-3	-2	-1	0	1	2	3	4
x^2	9	4	1	0	1	4	9	16
-2	-2	-2	-2	-2	-2	-2	-2	-2
y	7	2	-1	-2	-1	2	7	14

From the table, we observed that:

x – axis

Has almost equal sides – 3 and + 4

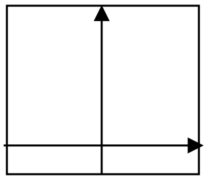
We rule the y – axis at the centre

y – axis

Has small negative side – 2 and very big positive side 14

We rule x – axis with much space for the positive side

Thus the graph is to be ruled in the format below:



(b) Graph to be drawn

(c) i. The point where the curve cuts the x – axis

– 1.4 and 1.4

ii. $x^2 - 3 = 0$

What can be added to both sides to get $y = x^2 - 2$

$x^2 - 3 + 1 = +1$

$x^2 - 2 = 1$

Next, we rule a straight line at y = 1 on the graph, where it cuts the curve is traced to x – axis with dotted line as shown – 1.7 and 1.7

iii. Trace the point $x = -1$ to the curve and draw an equidistant line there

$$\text{Gradient} = \frac{change\ in\ y}{change\ in\ x} = \frac{y_2 - y_1}{x_2 - x_1},$$

$$= \frac{-4-2}{0.5-(-2.6)} = \frac{-6}{3.1} = -1.94$$

2015/7

The table is for the relation $y = px^2 - 5x + q$

x	-3	-2	-1	0	1	2	3	4	5
y	21	6		-12				0	13

(a) (i) use the table to find the values of p and q

(ii) Copy and complete the table

(b) Using scales of 2cm to 1unit on the x – axis and 2cm to 5 units on the y – axis, draw the graph of the relation for $-3 \le x \le 5$

(c) Use the graph to find

(i) y when x =1.8

(ii) x when y = – 8

Solution

a (i) substituting any 2 sets of values from the table

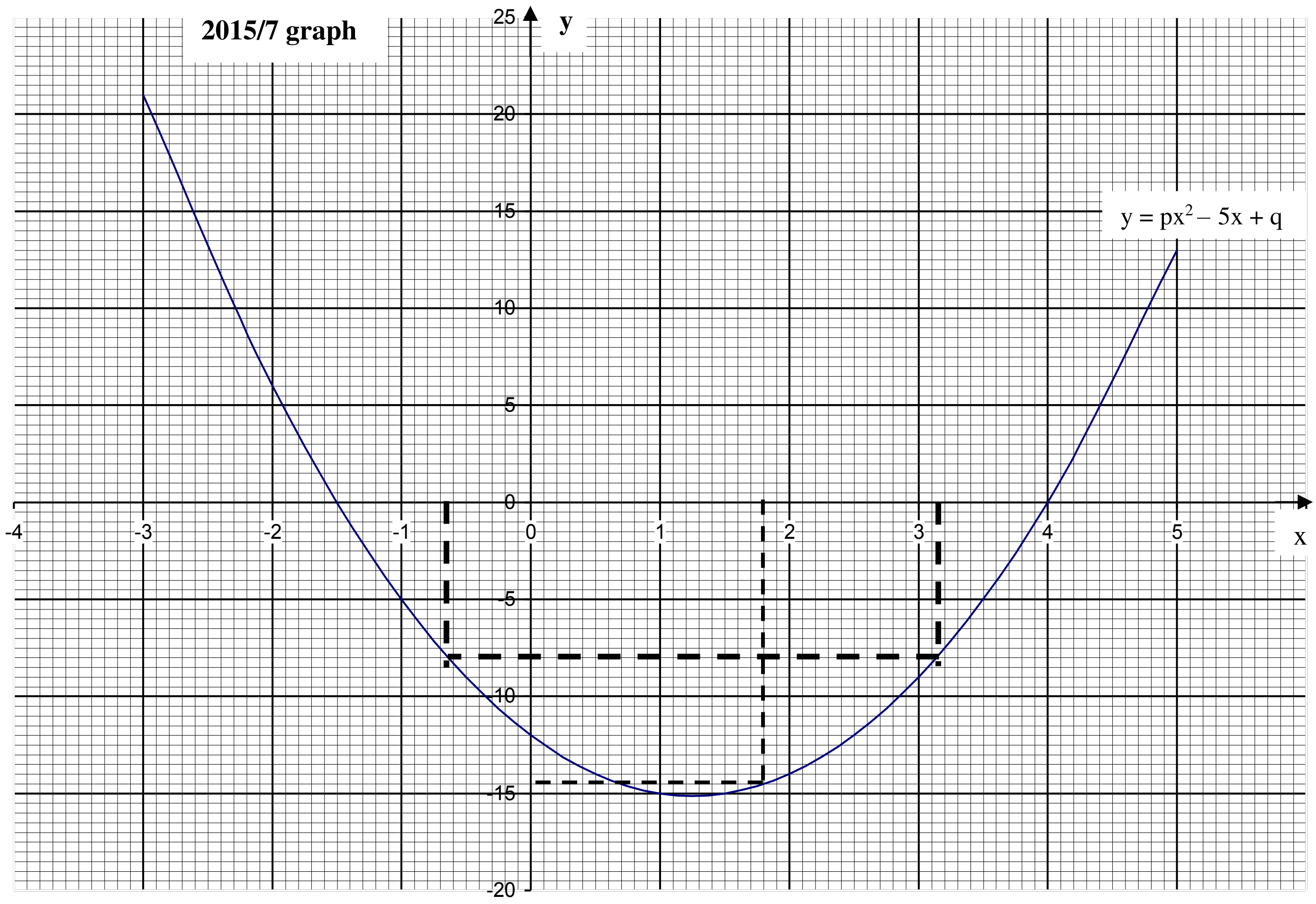
x	-3	-2
y	21	6

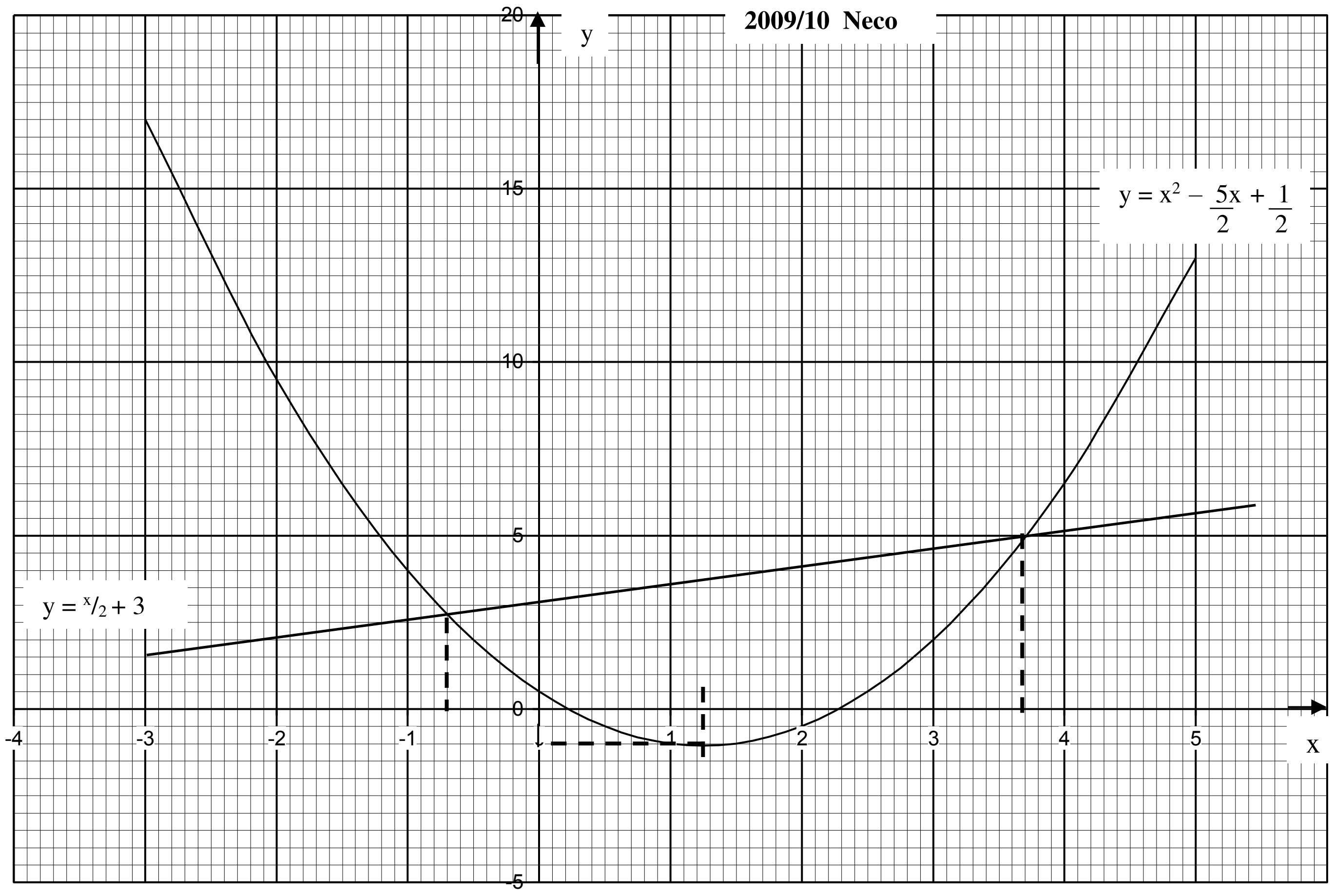
into $y = px^2 - 5x + q$

$21 = p(-3)^2 - 5(-3) + q$

$21 = 9p + 15 + q$

$6 = 9p + q$ ----- (1)


2015/7 graph
y
x
$y = px^2 - 5x + q$


2009/10 Neco
y
x
$y = x^2 - \frac{5x}{2} + \frac{1}{2}$
$y = {}^x/_2 + 3$

Also

$6 = p(-2)^2 - 5(-2) + q$

$6 = 4p + 10 + q$

$-4 = 4p + q$ ---- (2)

Solving the resulting simultaneous liner equations

$$\begin{array}{r} 6 = 9p + q \\ -(-4 = 4p + q) \\ \hline 10 = 5p \end{array}$$

$p = 2$

Substitute p value into (1)

$6 = 9p + q$ becomes

$6 = 9(2) + q$

$6 = 18 + q$

$6 - 18 = q, \Rightarrow q = -12$

Thus the equation is $y = 2x^2 - 5x - 12$

a (ii)Table for $y = 2x^2 - 5x - 12$

x	– 3	– 2	– 1	0	1	2	3	4	5
$2x^2$	18	8	2	0	2	8	18	32	50
$-5x$	15	10	5	0	–5	–10	–15	–20	–25
-12	–12	–12	–12	–12	–12	–12	–12	–12	–12
y	21	6	– 5	–12	–15	–14	– 9	0	13

From the table, it will be noted that

x – axis

Has a little higher positive side +5 than negative side of –3

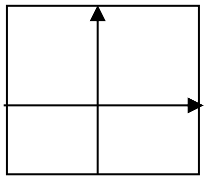
We rule the y – axis with little more space for positive side

y – axis

Has a little higher positive side +21than negative side of -15

We rule the x – axis with little more space for positive side

Thus the graph is to be ruled in the format below



(b) Graph is shown on page 272

(c) i. At point x = 1.8, a dotted line is traced to the curve then traced to the y – axis – 14.5

ii. A straight line is ruled at y = – 8 to touch the curve and then traced to x – axis. – 0.65 and 3.15

2009/10 Neco

(a) Copy and complete the following table for the relation $y = x^2 - \frac{5x}{2} + \frac{1}{2}$

x	– 3	– 2	– 1	0	1	2	3	4	5
y			4	0.5	– 1	0.5			

(b) Using a scale of 2cm to 1unit on the x – axis and 2cm to 5units on the y – axis, draw the graph of

$$y = x^2 - \frac{5x}{2} + \frac{1}{2}$$

(c) Using the same scale and axes, draw the graph of

$$y = \frac{x}{2} + 3$$

(d) From your graph, determine the:

i. least value of y and the corresponding values of x:

ii. Solution of the equations. $x^2 - \frac{5x}{2} + \frac{1}{2} = \frac{x}{2} + 3$

Solution

Table for $y = x^2 - \frac{5x}{2} + \frac{1}{2}$

x	– 3	– 2	– 1	0	1	2	3	4	5
x^2	9	4	1	0	1	4	9	16	25
$-^{5x}/_2$	7.5	5	2.5	0	–2.5	– 5	–7.5	–10	–12.5
$+ ^1/_2$	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
y	17	9.5	4	0.5	– 1	–0.5	2	6.5	13

From the table, we observed that:

x – axis

Has a little more positive side of +5

Has a small negative side of – 3

We rule the y –axis with little more space to positive side

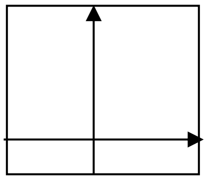
y – axis

Has a very small negative side – 1

Has a very big positive side 17

We rule the x – axis with much space for positive side

The graph should be ruled in the format below:



(b) Graph is shown on page 272

(c) Table for $y = \frac{x}{2} + 3$

x	– 2	0	2
$^x/_2$	– 1	0	1
+ 3	+ 3	+ 3	+ 3
y	2	3	4

The straight line is shown on the graph

(d) i. lowest point of the curve: x = 1.25 and y = 1

ii. – 0.7 and 3.7 are the points of intersection of the curve and straight line traced to x – axis

2014/9 (Nov)

An object is thrown vertically upwards from the top of a cliff and its height y metres, above sea level after t second is given by $y = -16t^2 + 64t + 5$

(a) Copy and complete the table of values for $y = -16t^2 + 64t + 5$; $0 \le t \le 4.0$

t	0.0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0
y	5			65			53		

(b) Using scales of 2cm to 0.5second on the t – axis and 2cm to 10m on the y – axis draw the graph of $y = -16t^2 + 64t + 5$ for $0 \le t \le 4.0$

(c) Use the graph to find the:

i. height reached when t = 1.75 seconds

ii. times the object was at a height of 50m

iii. maximum height reached.

Solution

(a) Table for $y = -16t^2 + 64t + 5$

t	0.0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0
$-16t^2$	0	– 4	–16	–36	–64	-100	– 144	–196	–256
64t	0	32	64	96	128	160	192	224	256
+ 5	+5	+5	+5	+5	+5	+5	+5	+5	+5
y	5	33	53	65	69	65	53	33	5

From the table, it will be noted that

x – axis (**t**)

Has only positive side from 0.0 to 4.0

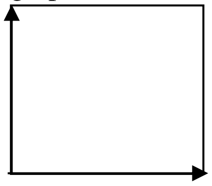
we rule the y –axis at the beginning part

<u>y – axis</u>

Has only positive side from 5 to 69

we rule the x –axis at the beginning part

Thus the graph is to be ruled in the format below



(b) The graph is shown on page 275

(c) i. 68m is indicated by dotted line

t = 1.75 traced to the curve then to y – axis

ii. 0.9 and 3.08 seconds is indicated by dotted line

y = 50 traced to the curve then to t – axis

iii. 69m is the highest point of the curve traced to the y – axis

2014/12 Neco

(a) Draw the graph of $y = 4 + 5x - 3x^2$ for which $-2 \le x \le 3$. Use a scale of 2cm to represent 1unit on the x – axis and 1cm to represent 2units on the y – axis

(b) Find the range of values of x for which $4 + 5x - 3x^2$ is greater than zero

(c) Using the same scale and axes, draw the graph of $y = 4x - 2$

(d) Write down the solution of the equations which are given by the point of intersection of the two graphs

(e) Find the maximum value of $y = 4 + 5x - 3x^2$

Solution

Table for $y = 4 + 5x - 3x^2$

x	– 2	– 1	0	1	2	3
4	4	4	4	4	4	4
+5x	– 10	–5	0	5	10	15
$-3x^2$	– 12	–3	0	–3	–12	–27
y	–18	–4	4	6	2	–8

From the table, it will be noted that

<u>x – axis</u>

Has almost equal sides of – 2 and + 3

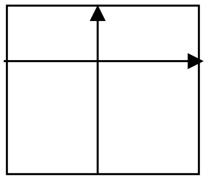
We rule the y – axis almost at the centre

<u>y – axis</u>

Has higher negative axis –18 than positive side of + 6

We rule the x – axis with more space for the negative side

Thus the graph can be ruled in the format below



(a) The graph is shown on page 275

(b) This is simply the points where the curve cuts the x – axis i.e the upper portion of the curve above the x –axis; it is also the roots presented in inequality form

$-0.6 < x < 2.25$

(c) Table for y = 4x – 2

x	– 2	0	3
4x	– 8	0	12
– 2	– 2	– 2	– 2
y	– 10	– 2	10

The straight line graph is as plotted on the graph

(d) it is the point of intersection traced to the x – axis as shown by dotted lines – 1.24 and 1.6

(e) The highest point of the curve traced to y – axis 6

2005/13 (New)

(a) Copy and complete the table $y = x^2 - 2x - 2$ for $-4 \le x \le 4$

x	– 4	– 3	– 2	–1	0	1	2	3	4
y	22				– 2			1	6

(b) Using a scale of 2cm to 1unit on the x – axis and 2cm to 5units on y– axis, draw the graph of

$y = x^2 - 2x - 2$

(c) Use your graph to find:

(i) the roots of the equation $x^2 - 2x - 2 = 0$

(ii) the values of x for which $x^2 - 2x - 4^1/_2 = 0$

(iii) the equation of the line of symmetry of the curve

Solution

Table for $y = x^2 - 2x - 2$

x	– 4	– 3	– 2	–1	0	1	2	3	4
x^2	16	9	4	1	0	1	4	9	16
–2x	8	6	4	2	0	– 2	– 4	– 6	– 8
–2	–2	– 2	– 2	– 2	– 2	– 2	– 2	– 2	– 2
y	22	13	6	1	– 2	– 3	– 2	1	6

From the table we observe that

<u>x – axis</u>

Has equal sides – 4 and +4

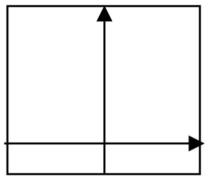
We rule the y– axis at the centre

<u>y – axis</u>

Has small negative sides –3 and very big positive side 22

We rule the x – axis with much space for the positive side.

Thus the graph can be ruled in the format below



(b) The graph is shown on page 276

(c) i. Roots of $x^2 - 2x - 2 = 0$ is the point where the curve cuts the x – axis – 0.75 and 3.3

ii. value of x for which $x^2 - 2x - 4^1/_2 = 0$

What must be added to the equation to get the original one

$x^2 - 2x - 4.5 + 2.5 = 2.5$

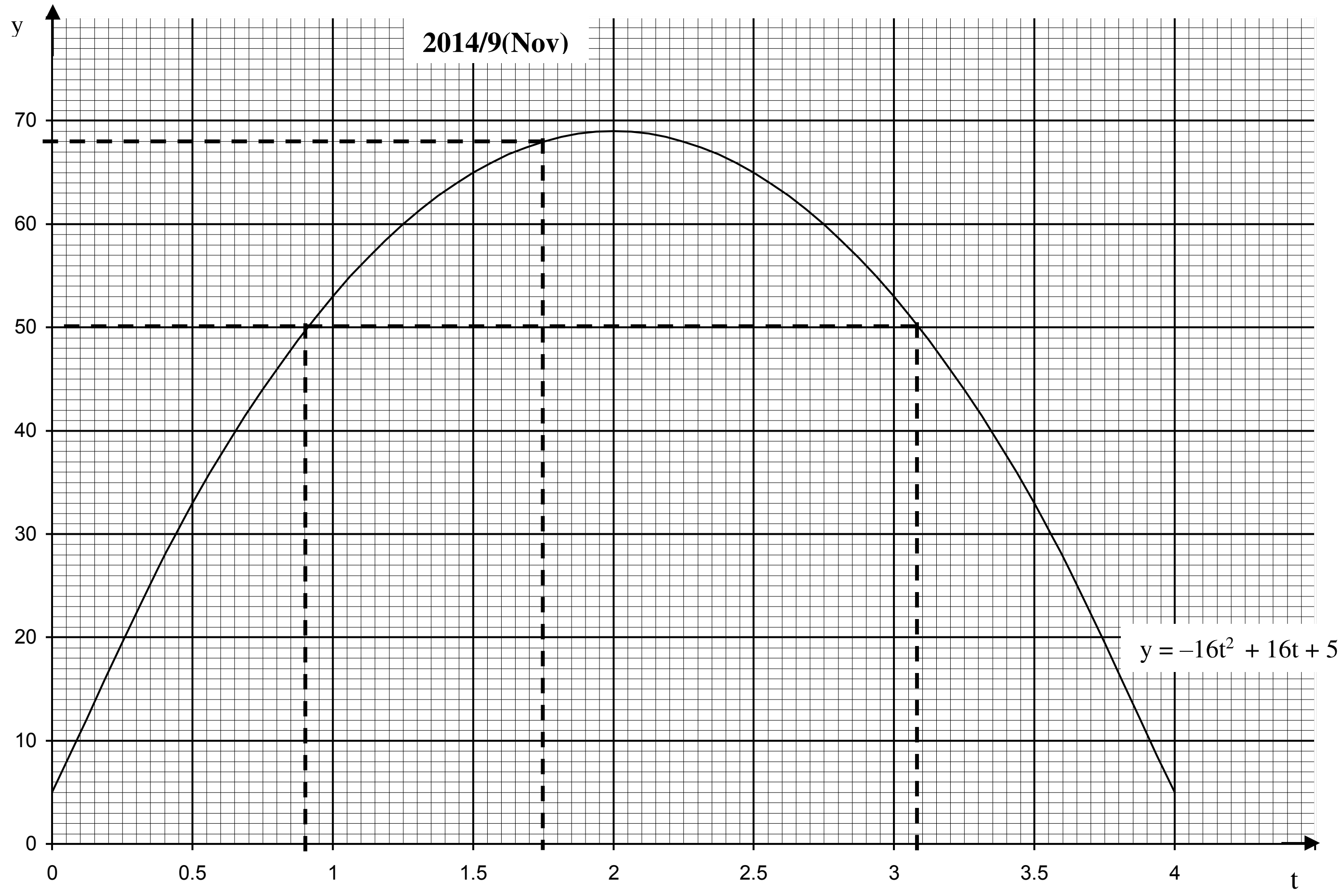
we just added 2.5 to both sides

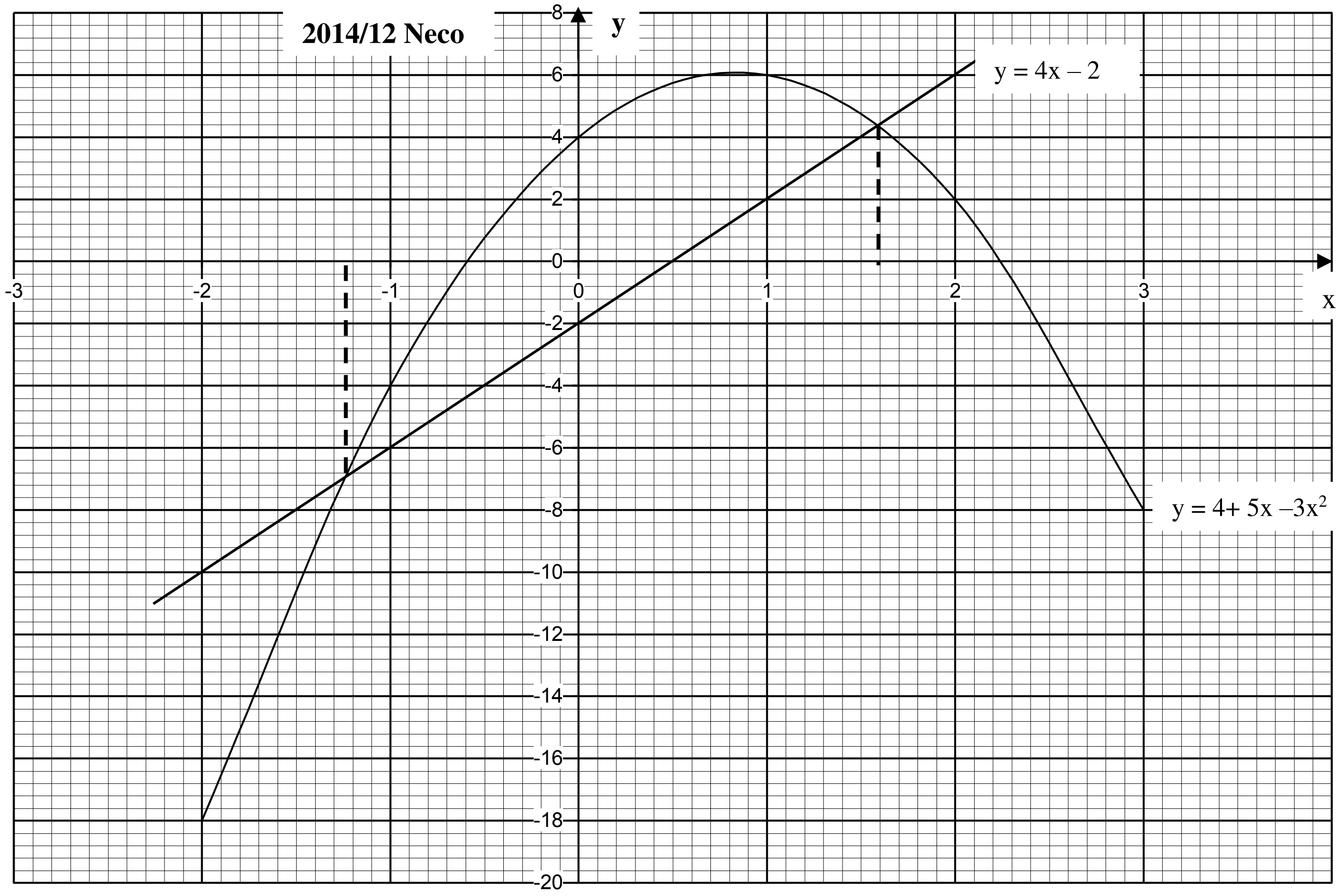
$x^2 - 2x - 2 = 2.5$

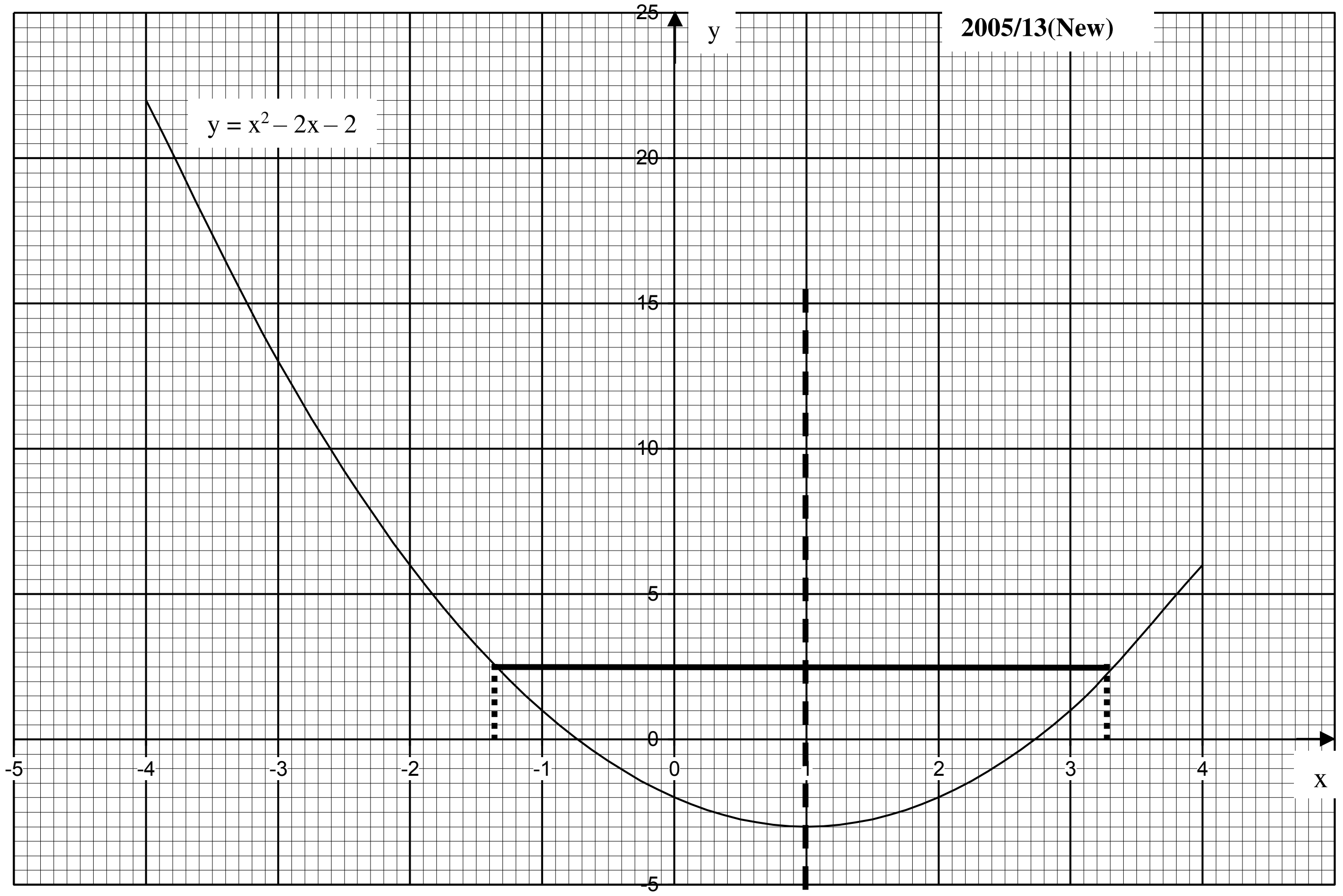
Next, we rule a straight line at y = 2.5 in the graph where it touches the curve is traced to the x – axis as shown by dotted line – 1.35 and 3.3

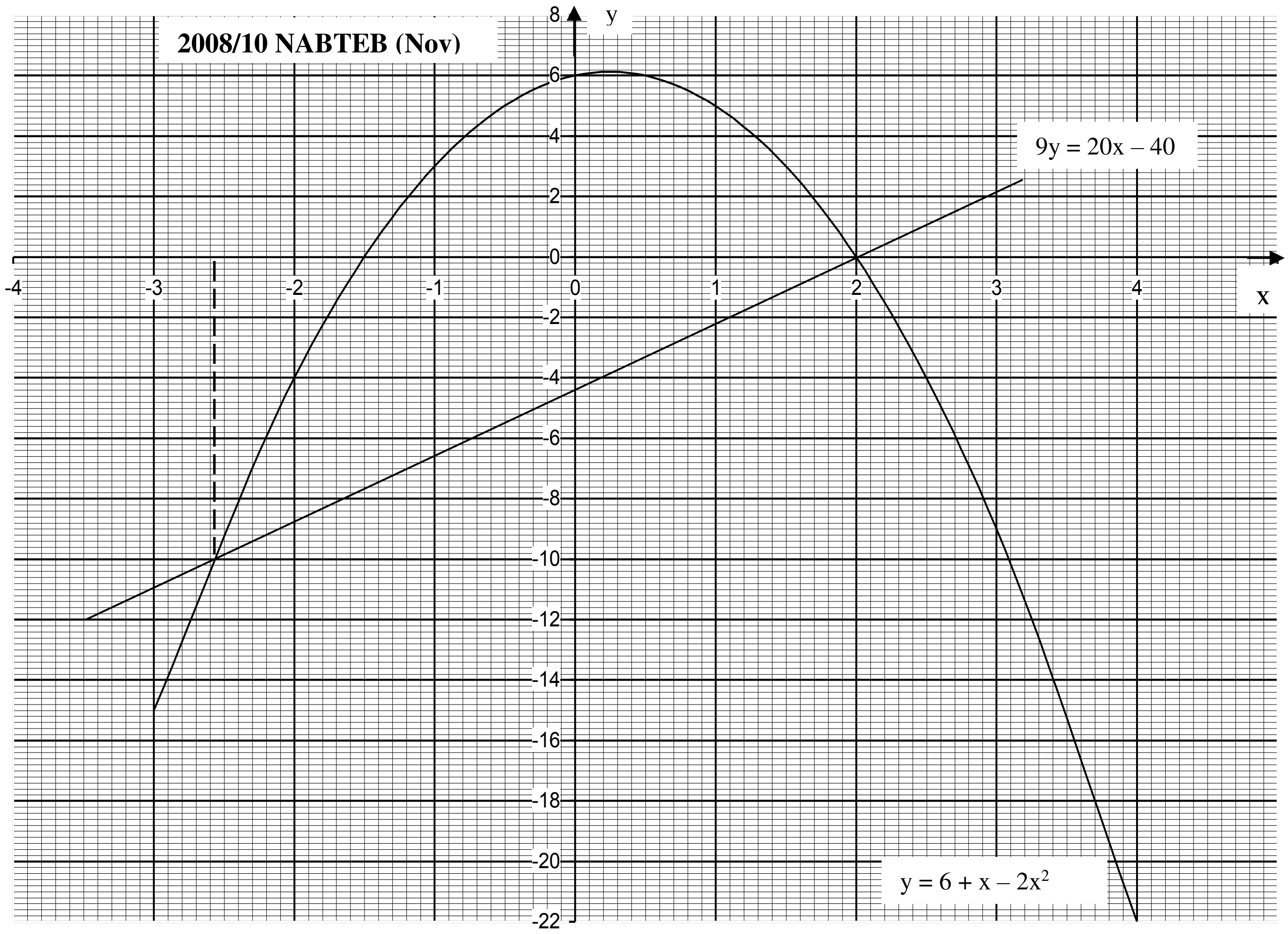
iii. Line of symmetry is the straight line that divides the graph into two equal halves as shown by dotted line

x = 1


2014/9(Nov)
y
70
60
50
40
30
20
10
0
0
0.5
1
1.5
2
2.5
3
3.5
4
t
y = –16t² + 16t + 5


2014/12 Neco
y
8
6
4
2
0
-2
-4
-6
-8
-10
-12
-14
-16
-18
-20
-3
-2
-1
0
1
2
3
x
y = 4x – 2
y = 4+ 5x –3x²


2005/13(New)
y = x² – 2x – 2
y
x


2008/10 NABTEB (Nov)
9y = 20x – 40
y = 6 + x – 2x²
y
x

2008/10 NABTEB (Nov)

(a) Copy and complete the following table for values of $y = 6 + x - 2x^2$

x	− 3	− 2	− 1	0	1	2	3	4
y	− 15			6		0		− 22

(i) Draw the graph of $y = 6 + x - 2x^2$ for values of x from –3 to +4 taking 2cm to represent 1unit on x–axis and 2cm to represent 2units on y – axis

(ii) On the same axes and with same scale draw the graph of $9y = 20x - 40$

(iii) From your graphs, determine the value of x for which $6 + x - 2x^2 = \frac{1}{9}(20x - 40)$

(iv) Find the maximum value of y

Solution

(a) Table for $y = 6 + x - 2x^2$

x	− 3	− 2	− 1	0	1	2	3	4
6	6	6	6	6	6	6	6	6
+ x	− 3	− 2	− 1	0	1	2	3	4
$-2x^2$	− 18	− 8	− 2	0	–2	− 8	− 18	− 32
y	- 15	- 4	3	6	5	0	- 9	-22

From the table, it should be noted that:

x – axis

Has almost equal sides – 3 and + 4

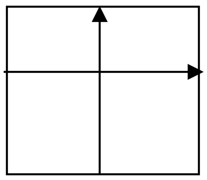
we rule y – axis line almost at the centre

y – axis

Has small positive side +6 and very big negative side – 22

We rule the x – axis line with much space for the negative side

Thus the graph is to be ruled in the format below:



(i) Graph is shown on page 276

(ii) Table for $y = \frac{20x}{9} - \frac{40}{9}$

x	− 3	0	3
$\frac{20x}{9}$	− 6.67	0	6.67
$-{}^{40}/_{9}$	− 4.44	− 4.44	− 4.44
y	− 11.11	− 4.44	2.23

The straight is as shown on the graph

(iii) This is the points of intersection of the straight line and the curve as shown by the dotted lines – 2.58 and 2

(iv) It is the highest point on the curve 6.2

2013/10 Neco

(a) The table below shows the relation $y = 2x^2 - 5x - 7$, copy and complete the table

x	− 2	− 1	0	1	2	3	4
y	11		− 7			− 4	

(b) Using a scale of 2cm to 1unit on x – axis and 2cm to 2 units on y –axis, draw the graph of the relation.

(c) From your graph, find the least point of the relation $y = 2x^2 - 5x - 7$

(d) Using the same scale and axes, draw the graph of $y = x - 2$

(e) Use your graph to solve the equation $2x^2 - 6x - 5 = 0$

Solution

(a) Table for $y = 2x^2 - 5x - 7$

x	− 2	− 1	0	1	2	3	4
$2x^2$	8	2	0	2	8	18	32
− 5x	10	5	0	− 5	− 10	− 15	− 20
− 7	− 7	− 7	− 7	− 7	− 7	− 7	− 7
y	11	0	− 7	− 10	− 9	− 4	5

From the table, we observed that

x – axis

Has small negative side – 2 and times two positive side 4

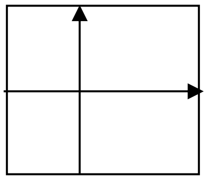
We rule the y – axis line in that order

y – axis

Have almost equal sides of –10 and 11

We rule the x – axis line at the centre

This graph is ruled in the format below



(b) Graph is shown on page 278

(c) The lowest point of curve traced to the y – axis as shown by dotted line y = – 10.2

(d) Table for $y = x - 2$

x	− 2	0	2
x	− 2	0	2
− 2	− 2	− 2	− 2
y	− 4	− 2	0

The straight line is as plotted on the graph

(e) $2x^2 - 6x - 5 = 0$ is a combination of $y = 2x^2 - 5x - 7$ and $y = x - 2$ as:

$$2x^2 - 5x - 7 = x - 2$$

Collect like terms together

$$2x^2 - 5x - x - 7 + 2 = 0$$

$$2x^2 - 6x - 5 = 0$$

Thus the solution to $2x^2 - 6x - 5 = 0$ is the point of intersection of the curve and the straight line traced to the x – axis as shown by dotted lines – 0.7 and 3.65

2010/12 Neco

(a) Copy and complete the table of values for the relation $y = 2x^2 - 3$ for $-3 \le x \le 3$

x	− 3	− 2	− 1	0	1	2	3
y		5					15

(b) Draw the graph of $y = 2x^2 - 3$ using 2cm to 1unit on x – axis and 2cm to 5units on y – axis

(c) From the graph, find the

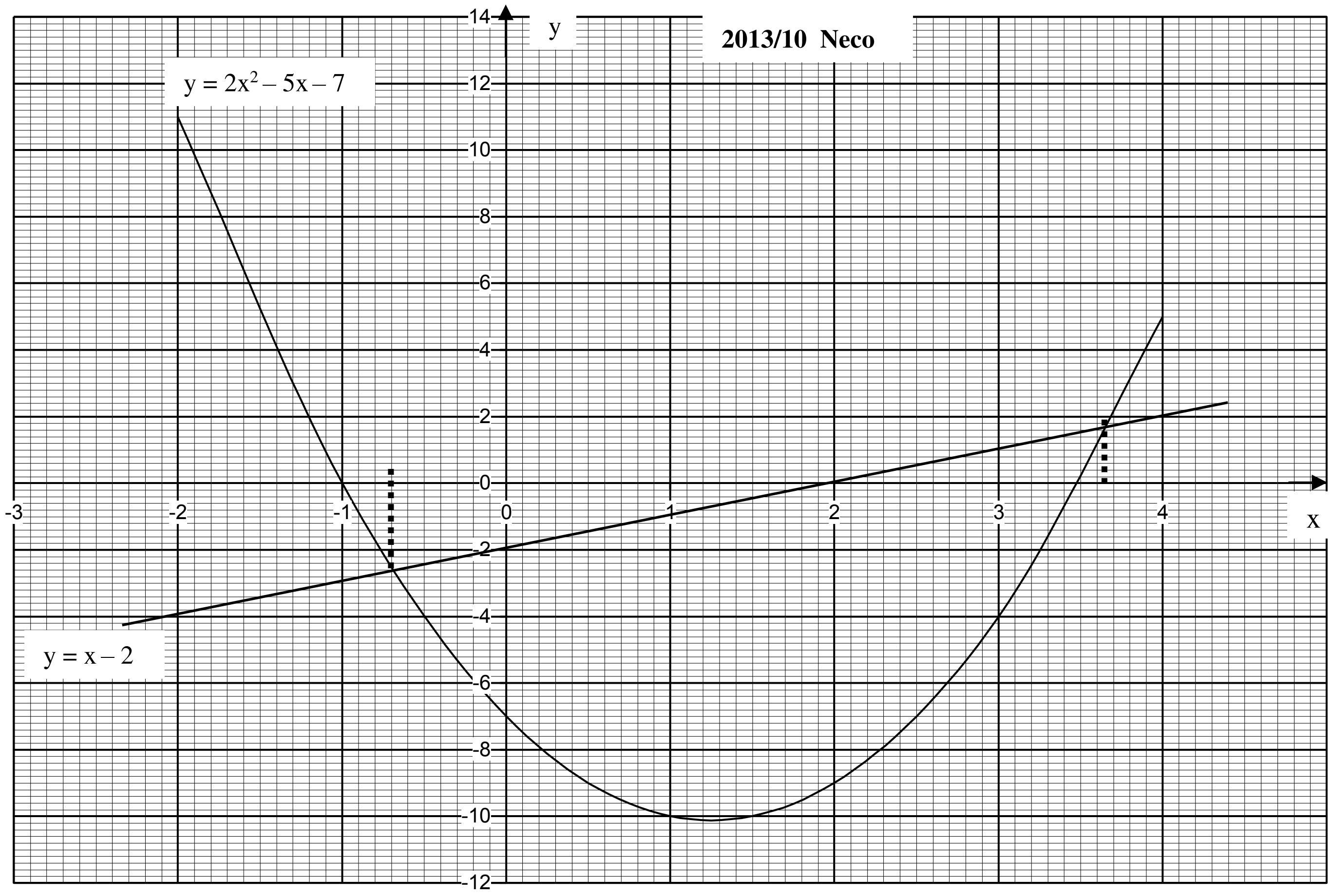
(i) roots of $2x^2 - 3 = 0$

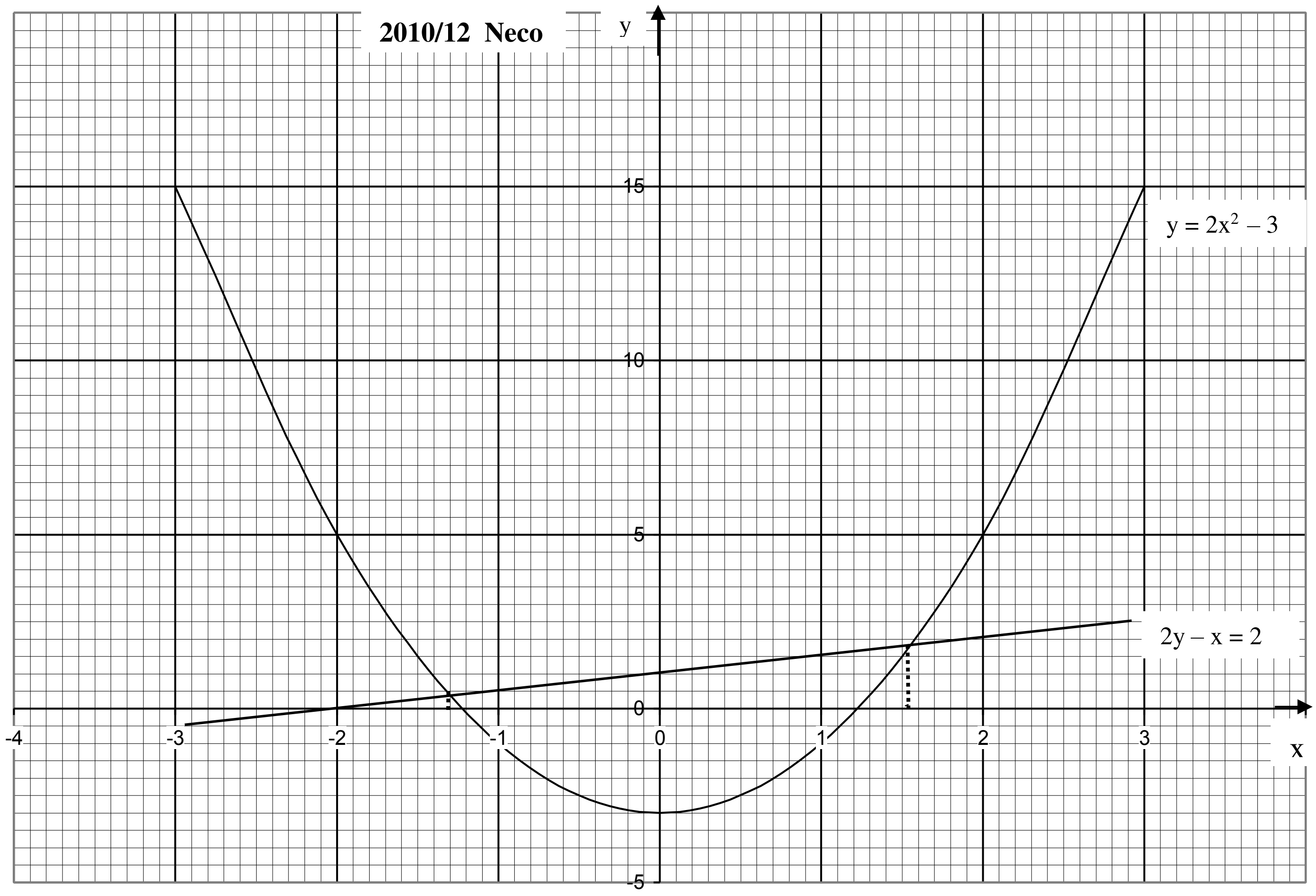
(ii) least value of y and the corresponding value of x.

(d) Using the same axes, draw the graph of $2y - x = 2$

(e) Use your graphs to find the values of x for which

$$2x^2 - 3 = \frac{1}{2}(x + 2)$$


2013/10 Neco
y
x
$y = 2x^2 - 5x - 7$
$y = x - 2$
14
12
10
8
6
4
2
0
-2
-4
-6
-8
-10
-12
-3
-2
-1
0
1
2
3
4


2010/12 Neco
y
x
$y = 2x^2 - 3$
$2y - x = 2$
15
10
5
0
-5
-4
-3
-2
-1
0
1
2
3

Solution

Table for $y = 2x^2 - 3$

x	– 3	– 2	– 1	0	1	2	3
$2x^2$	18	8	2	0	2	8	18
– 3	– 3	– 3	– 3	– 3	– 3	– 3	– 3
y	15	5	– 1	– 3	– 1	5	15

From the table, it will be noted that:

x – axis

Has equal positive and negative values –3 and + 3

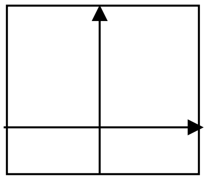
We rule the y – axis line at the centre

y – axis

Has small negative value of –3 bigger positive value of 15

We rule the x – axis line with more space for positive side

Thus the graph is to be ruled in the format below



(b) Graph is shown on page 278

(c) i. Roots of $2x^2 - 3 = 0$ are – 1.20 and 1.20 where the curve cuts the x – axis

ii. The lowest point of the curve traced to y and x – axis x = 0 and y = – 3

(d) Table for $y = \frac{x}{2} + 1$ (When we make y subject formula)

x	– 2	0	2
$x/2$	– 1	0	1
+1	1	+1	+1
y	0	1	2

(e) – 1.3 and 1.55

It is the point of intersection of the curve and the line traced to x – axis as shown by dotted lines

2006/11 NABTEB (Nov)

(a) Construct a table of values for $-1 \le x \le 5$ for the function $y = 2x^2 - 7x - 4$

(b) Using your table of values, plot the graph of $y = 2x^2 - 7x - 4$ taking 2cm to represent 1unit and 4units on the x – axis and y – axis respectively.

(c) On the same axes and with the same scale, draw the graph of y = 3x + 1

(d) Use your graph to find the:

(i) least value of $y = 2x^2 - 7x - 4$ and the corresponding value(s) of x

(ii) roots of the equation $2x^2 - 10x - 5 = 0$

Solution

(a) Table for $y = 2x^2 - 7x - 4$

x	– 1	0	1	2	3	4	5
$2x^2$	2	0	2	8	18	32	50
–7x	7	0	– 7	–14	–21	–28	–35
– 4	– 4	– 4	– 4	– 4	– 4	– 4	– 4
y	5	– 4	– 9	–10	–7	0	11

From the table, it can be noted that :

x – axis

Has very small negative side –1 and very big positive side + 5

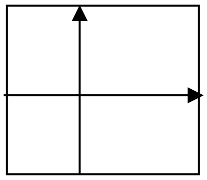
We rule the y – axis line giving much space for positive side

y – axis

Have almost equal sides – 10 and 11

We rule the x – axis line almost at the centre

Thus the graph can be ruled in the format below:



(b) Graph is shown on page 280

(c) Table for y = 3x + 1

x	– 1	0	3
3x	– 3	0	9
+1	+1	+1	+1
y	– 2	1	10

The straight line is shown on the graph

(d) i. the lowest point of curve, y = – 10.1 and x = 1.75

ii. This is the points of intersection of the straight line and the curve traced to the x – axis.

Here the two only meet at one point – 0.45

Candidates don't need to bother themselves when such cases arise in exam. Work strictly by the question – No addition No subtraction; any part of the question not clear – leave it alone.

For Practice only

If we extend the range of values from –1 to 5 to –1 to 6 $(-1 \le x \le 6)$.

Table for $y = 2x^2 - 7x - 4$

x	–1	0	1	2	3	4	5	6
$2x^2$	2	0	2	8	18	32	50	72
– 7x	7	0	– 7	– 14	– 21	– 28	– 35	– 42
– 4	– 4	– 4	– 4	– 4	– 4	– 4	– 4	– 4
y	5	– 4	– 9	– 10	– 7	0	11	26

We don't need to extend the table for the straight line, as straight can be extended to any given length with only two points.

The resulting graph **is shown on page 280** will give us a better result for **(d) ii** – 0.45 and 5.4

2007/9 Neco

Copy and complete the table for $y = 2x^2 - 4x - 5$ and $-2 \le x \le 4$

x	– 2	– 1	0	1	2	3	4
$y = 2x^2 - 4x - 5$				–7		1	

(a) Use the table to draw the graph of $y = 2x^2 - 4x - 5$

(b) On the same axes draw the graph of y = 2x + 1

(c) Use the graph to solve

(i) $2x^2 - 4x - 5 = 0$

(ii) $2x^2 - 4x - 5 = 4$

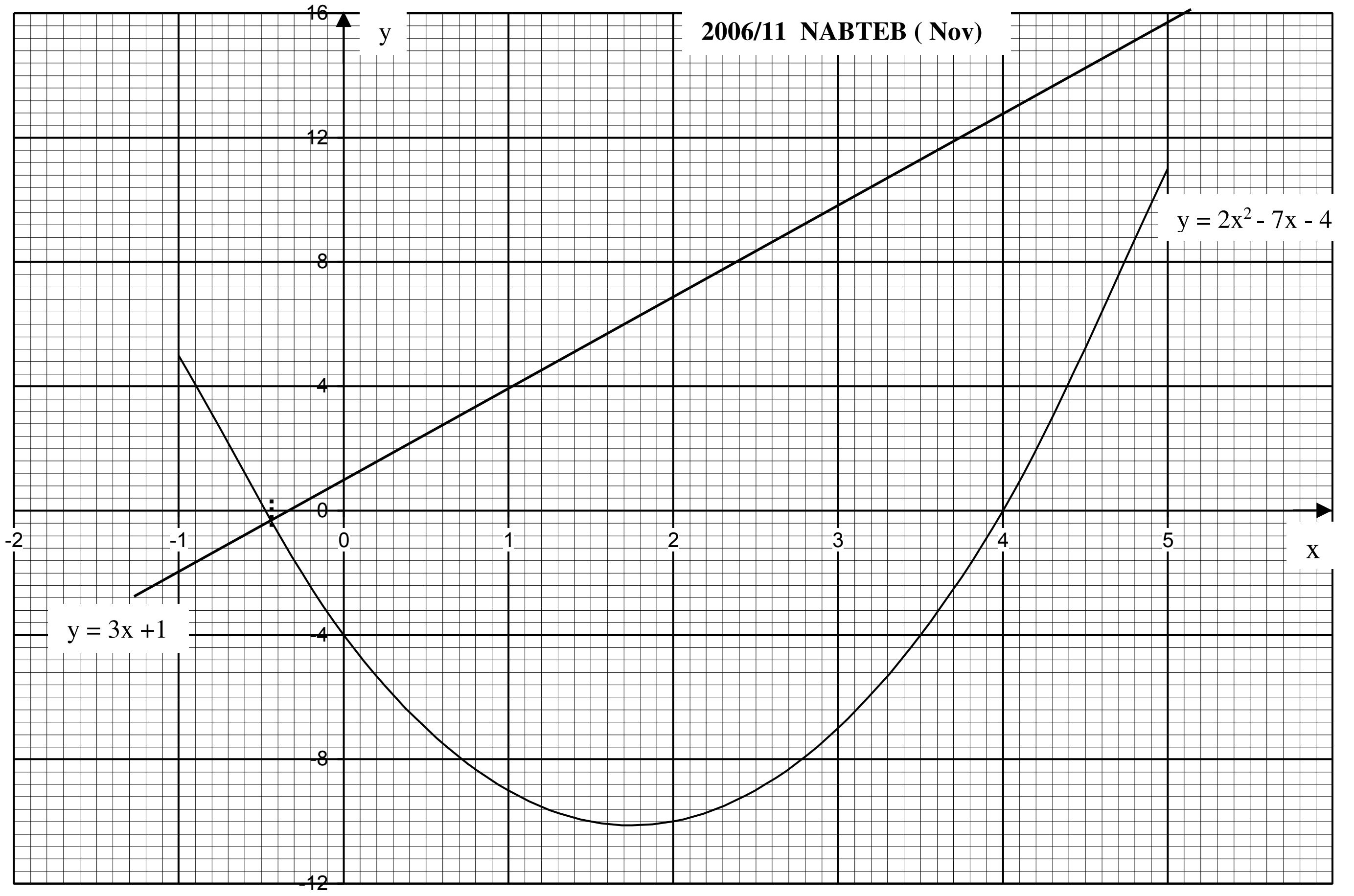
(d) Use the graph to solve $x^2 - 3x - 3 = 0$

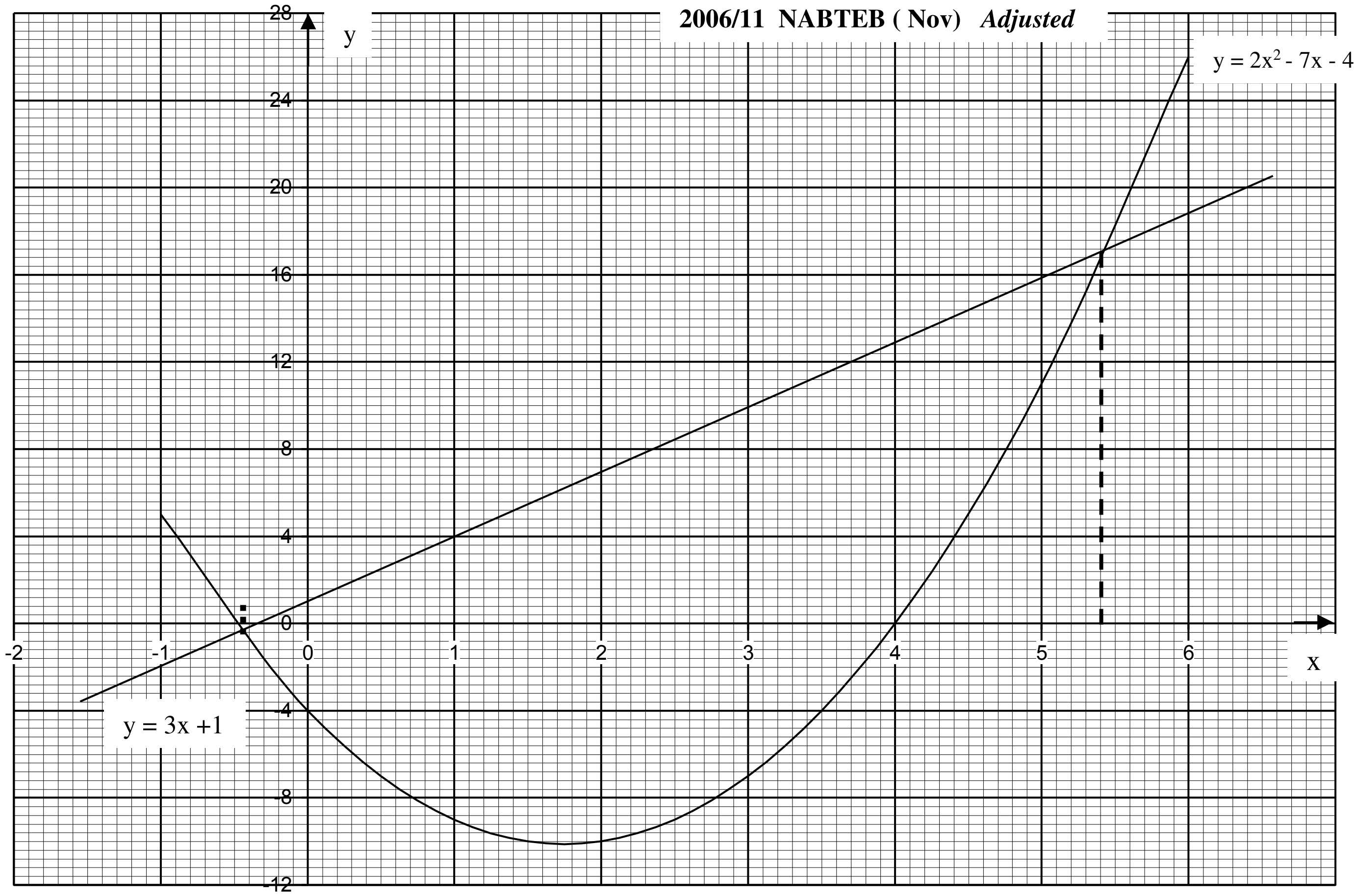
(e) What is the minimum value of y and the corresponding value of x?

Solution

Table for $y = 2x^2 - 4x - 5$

x	– 2	– 1	0	1	2	3	4
$2x^2$	8	2	0	2	8	18	32
– 4x	8	4	0	– 4	– 8	– 12	– 16
– 5	– 5	– 5	– 5	– 5	– 5	– 5	– 5
y	11	1	– 5	– 7	– 5	1	11


2006/11 NABTEB (Nov)
y
x
$y = 2x^2 - 7x - 4$
$y = 3x + 1$
16
12
8
4
0
-4
-8
-12
-2
-1
0
1
2
3
4
5


2006/11 NABTEB (Nov) *Adjusted*
y
x
$y = 2x^2 - 7x - 4$
$y = 3x + 1$
28
24
20
16
12
8
4
0
-4
-8
-12
-2
-1
0
1
2
3
4
5
6

From the table, we observed that :

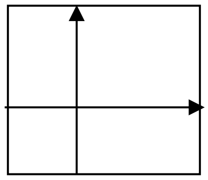
x – axis

Has small negative side – 2 and bigger positive side + 4
Rule the y – axis line with twice space for the positive side

y – axis

Has small negative side – 7 and big positive side 11
We rule the x – axis line with little more space for the positive side.

The graph should be ruled in the format:



The graph is shown on **page 282** though no scale was given

(b) Table for $y = 2x + 1$

x	– 2	0	3
2x +1	– 4 +1	0 +1	6 +1
y	– 3	1	7

The straight line is drawn on the graph

C (i) it is the points where the curve cuts the x – axis
– 0.88 and 2.88

(ii) a line is drawn at y = 4 to touch the curve then traced to x – axis – 1.3 and 3.3

(d) It is off target as a result of typing error or examiner's omission.

(e) y = 7 and x = 1 (lowest point of the curve)

2011/7b

The diagram **on page 282** shows the graph of $y = ax^2 + bx + c$ and $y = mx + k$ where a, b, c, m and k are constants.

Use the graph(s) to:

(i) find the root of the equation $ax^2 + bx + c = mx + k$;

(ii) determine the values of a, b and c using the coordinate of points **L**, **M** and **N** and hence write the equation of the curve;

(iii) determine the line of symmetry of the curve $y = ax^2 + bx + c$

Solution

(i) This is the points of intersection of the curve and the straight line traced to the x – axis i.e **P** and **H**

P = – 1.45 and H = 1.48

(ii) $y = ax^2 + bx + c$ -----**

y – Intercepts

At **M** x = 0, y = 2 substituting into **

$2 = a(0) + b(0) + c$

Implies c = 2

At **L** (– 1.0, 0) and C is 2 substituting into **

$0 = a(-1)^2 + b(-1) + 2$

$a - b = -2$ ----- (1)

At **N**(2.0, 0) and C is 2 substituting into **

$0 = a(2)^2 + b(2) + 2$

$4a + 2b = -2$ ------ (2)

Solving the resulting simultaneous linear equations

From (1) $a = b - 2$

Substituting into (2)

$4(b - 2) + 2b = -2$

$4b - 8 + 2b = -2$

$4b + 2b = 8 - 2$

$6b = 6$

$b = 1$

Thus $a = 1 - 2$

$a = -1$

a = – 1, b = 1 and c = 2

Equation of the curve: $y = -x^2 + x + 2$

Or $y = 2 + x - x^2$

(iii) Line which divide the curve into equal halves

x = 0.4

2009/9b Exercise 18.16

The graph of the equation $y = Ax^2 + Bx + C$ passes through the points (0, 0), (1, 4) and (2, 10). Find the

i. value of C
ii. values of A and B
iii. co-ordinates of the other point where the graph cuts the x – axis

2013/20 Neco Exercise 18.17

A student plots a graph and gets the shape U, the equation may be of the form

A $y = ax^2$ B $y = -bx^3$ C $y = -ax^2$ E $y = ax^4$

2015/12 Exercise 18.18

Ada draws the graph of $y = x^2 - x - 2$ and $y = 2x - 1$ on the same axes. Which of these equations is she solving?

A $x^2 - x - 3 = 0$ B $x^2 - 3x - 1 = 0$
C $x^2 - 3x - 3 = 0$ D $x^2 + 3x - 1 = 0$

1978/9 Exercise 18.19

Copy and complete the following table of values for the function $y = 2x^2 - 4x + 3$, from x = –2 to x = 4

x	– 2	– 1	0	1	2	3	4
y	19		3	1			

Draw the graph of $y = 2x^2 - 4x + 3$, using a scale of 2cm to 1unit on the x – axis and 1cm to 2units on the y –axis.

From your graph, find:

(a) the equation of the line of symmetry of the curve
(b) the gradient of the curve at x = 3
(c) the values of x for which $2x^2 - 4x - 3 = 0$
(d) the range of values of x for which $2x^2 - 4x + 1 < 0$

1975/10 Exercise 18.20

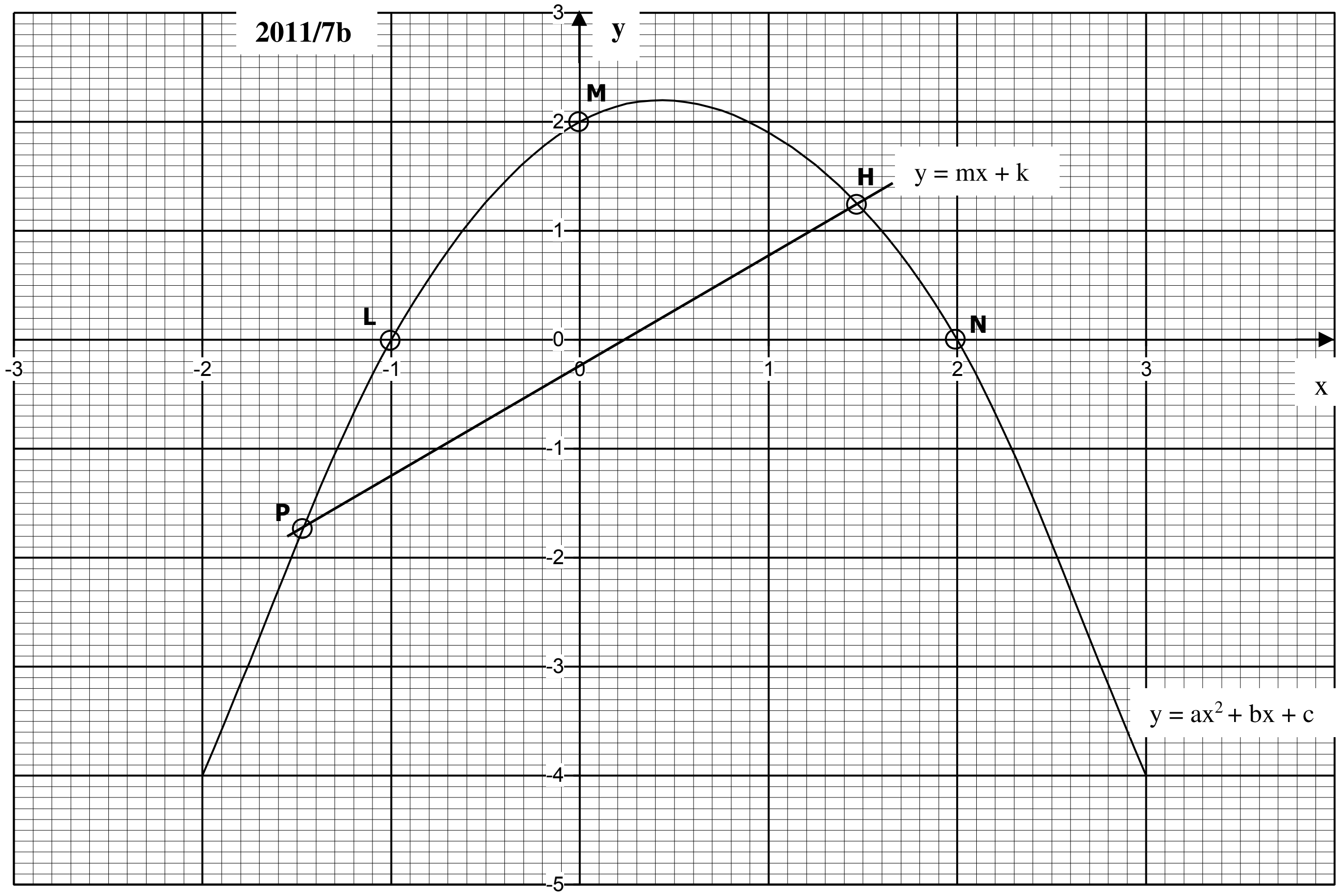
A ball is projected vertically upwards from a point O on the ground. The height of the ball after t seconds have passed is y metres, where $y = 16 + 60t - 16t^2$

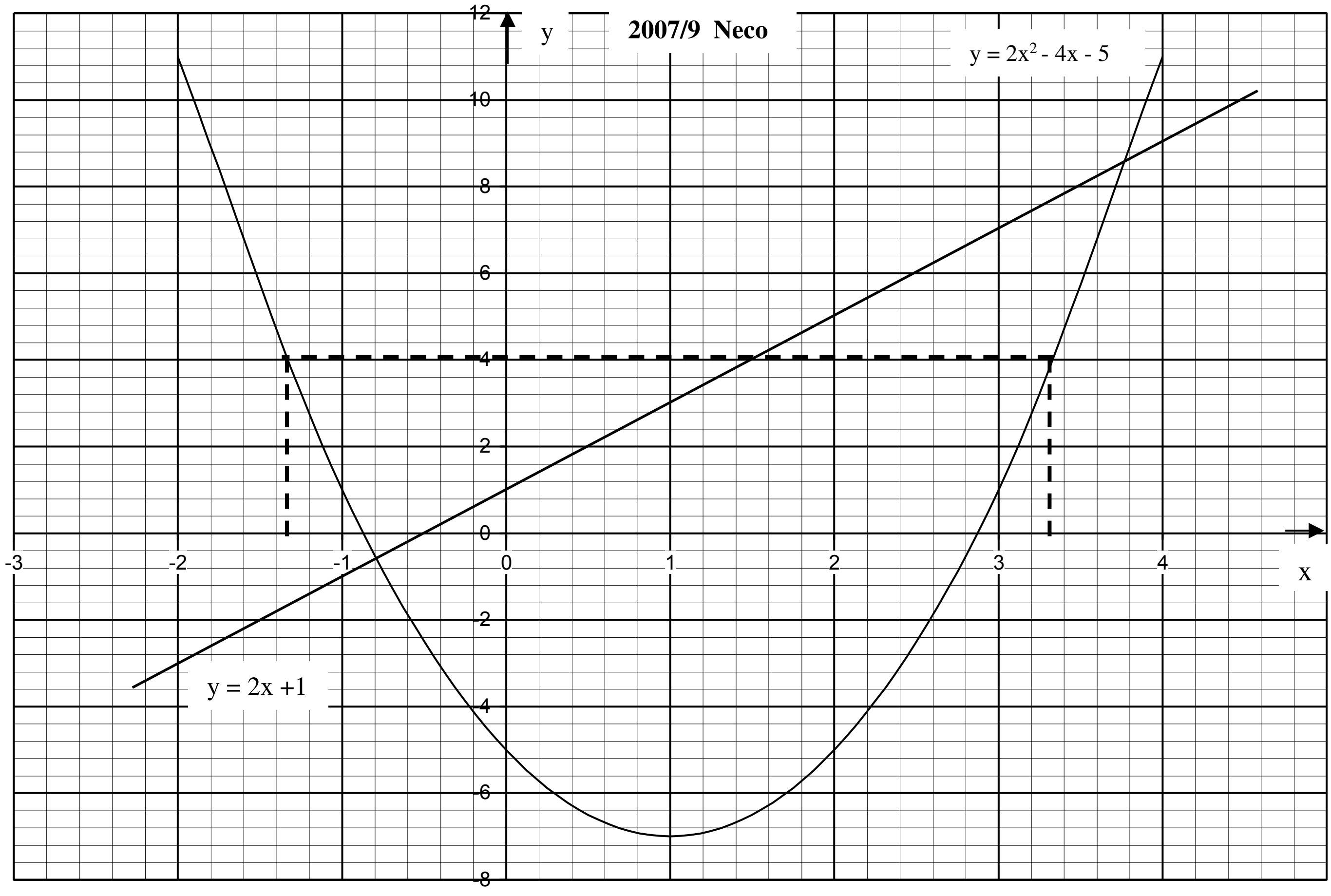
Draw the graph of y for t = 0 to t = 4, using a scale of 2cm to 1second along the x – axis and 1cm to 10meters along the y – axis.

Make use of intermediate values of t, i.e $t = {}^1/_2, {}^3/_2, {}^5/_2$ and ${}^7/_2$

From your graph, find:

(a) The greatest height above the ground which the ball reaches.
(b) The time that passes before the ball hits the ground.
(c) The velocity of the ball at t = 3
(Note: this is the same as the gradient of the curve at the time t = 3)


2011/7b
y
x
M
H
L
N
P
y = mx + k
y = ax² + bx + c


2007/9 Neco
y
x
y = 2x² - 4x - 5
y = 2x +1

1975/11 (Nov) Exercise 18.21

Draw the graph of $y = 3x(4 - x)$ for valves of x ranging from -2 to $+6$. On the same graph draw the line $y = 5(x - 2)$. Use a scale of 1cm to represent 1unit along the x – axis and 5units along the y – axis.

From your graph:

(a) Find the roots of the equation $3x(4 - x) = 0$

(b) Write down the maximum value of $y = 3x(4 - x)$ and its value of x

(c) Write down the equation of the axis of symmetry of the curve $y = 3x(4 - x)$

(d) Deduce the roots of the equation $3x(4 - x) = 5(x - 2)$

2005/13 (Nov) Exercise 18.22

(a) Copy and complete the following table of values for the relation $y = 2x - x^2$ for $-2 \le x \le 4$

x	–2	–1.5	– 1	– 0.5	0	0.5	1	1.5	2	2.5	3	3.5	4
y	– 8		–3		0				0	–1.25		–5.25	

(b) Using 2cm to 1unit on the x – axis and 2cm to 2units on the y – axis, draw the graph of the relation

(c) On the same axis, draw the graph of $y = x - 4$

(d) Use your graph to find the:

(i) roots of the equation $4 + x - x^2 = 0$;

(ii) line of symmetry of the curve

(iii) value of x for which $y \ge 0$

1976/11 (Nov) Exercise 18.23

Copy and complete the table below and use it to draw the graph of $y = 3x^2 - 5x + 4$. Use a scale of 1cm to 1unit along the x – axis and 10units along the y – axis

x	– 3	– 2	– 1	0	1	2	3	4
$3x^2$		12	3	0	3	12	27	
$-5x + 4$	19		9		–1		–11	–16
y	46	26		4				32

Use your graph:

(a) to solve the equation $3x^2 - 5x + 4 = 0$

(b) to solve the equation $3x^2 - 5x = 2$

(c) to determine the range of value of x for which $3x^2 - 5x < 20$

2012/8 Neco Exercise 18.24

(a) Copy and complete the table of the values for $y = 3x^2 - 5x - 1$

x	– 3	– 2	– 1	0	1	2	3	4
y	41			– 1			11	

(b) Using the scale 2cm = 1unit on the x –axis and 1cm = 5units on the y – axis, plot the graph of the equation $y = 3x^2 - 5x - 1$

(c) (i) On the graph, find x when y = 0

(ii) Draw the line of symmetry of the equation $y = 3x^2 - 5x - 1$ and determine its equation

2009/11 f/maths Exercise 18.25

(a) On the same axes, draw the graphs of $y = 15 - x - 6x^2$ and $y = 2x - 1$ for $-3 \le x \le 3$, using a scale of 2cm to 1 unit on the x – axis and 2cm to 5 units on the y – axis .

(b) Using your graph

i. find the roots of the equation $16 - 3x - 6x^2 = 0$

ii. state the equation of the line of symmetry

iii. find the range of x for which $16 - 3x - 6x^2 \le 0$

(c) Find the gradient of $y = 15 - x - 6x^2$ at the point $x = -2$

Trigonometry (2)

Trigonometric functions like others have their graphical representations, which is of great importance to scientist. Thus, it is one of the basic knowledge required in mathematics. The sine, cosine and tangent can be represented graphically in either degrees or radians as units of measurements, though degree is often used.

Sine Graph

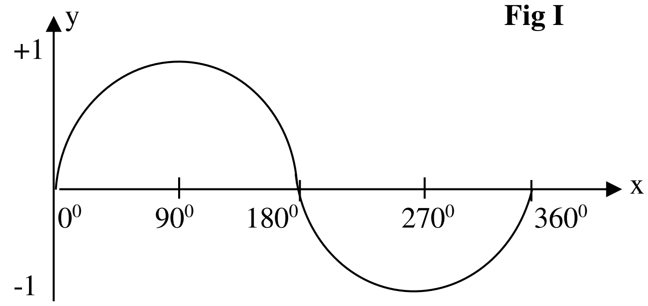


Fig I

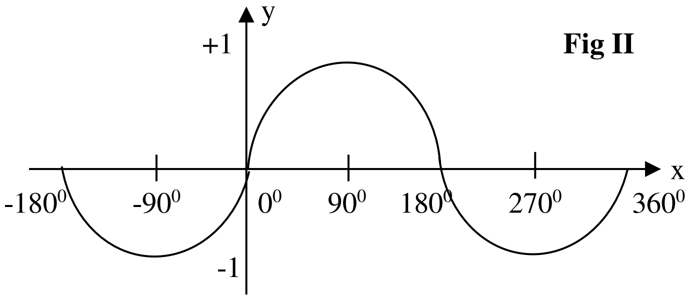


Fig II

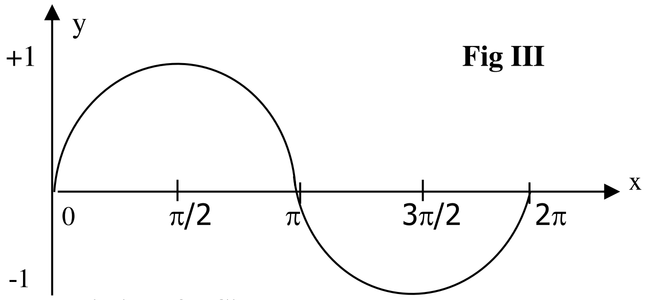


Fig III

Characteristics of a Sine curve

When plotting or sketching a sine curve on the positive x – axis only (fig 1), it starts from the origin $(0^0, 0)$ and reach its peak of +1 at 90^0 $(\pi/2)$ then falls to 0 at $x = 180^0$ (π) from this point, it descends downward to a lowest point of -1 at $x = 270^0$ $(3\pi/2)$. It then attains 0 at $x = 360^0$. The cycle continues in that order; other forms of sine curves are multiples of this basic one.

Sine graph for AMPLITUDE multiples

When y = sin θ

Actually the amplitude here is the coefficient of sinθ which is 1. Thus, the highest and lowest points of the graph are + 1 and –1 respectively

when y = 2 sinθ.

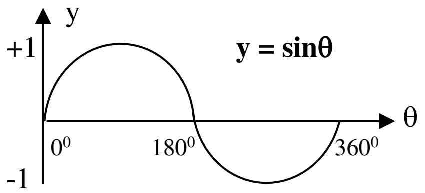
The amplitude (coefficient of sin θ) is 2

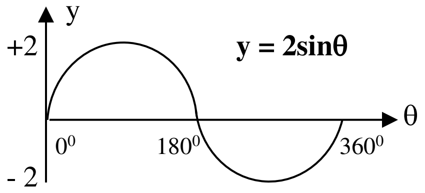
Thus, the highest and lowest points are +2 and – 2 respectively.

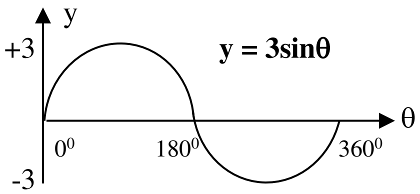
Similarly, when y = 3 sinθ.

The amplitude (coefficient of sin θ) is 3

Thus, the highest and lowest points are +3 and – 3 respectively.







Sine graph for PERIODICITY Multiples

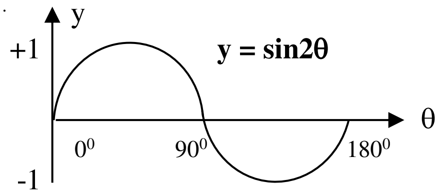
The periodicity of trigonometry graph is a term for the repetition of the graph pattern at intervals; or completing of the graph cycle. Usually within 360^0

When y = sinθ

Its periodicity is 360^0 . This is the basic sine graph shown earlier.

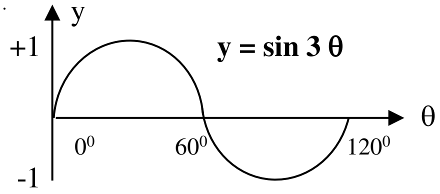
When y = sin2θ

Its periodicity is 180^0 – since two multiplies 180^0 to give 360^0. The graph pattern is shown below:

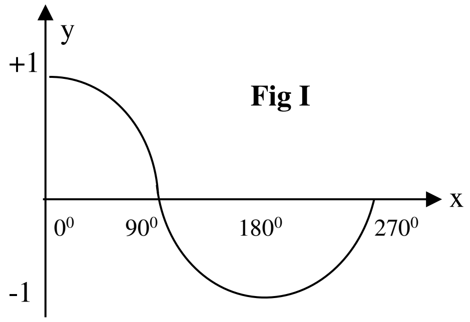


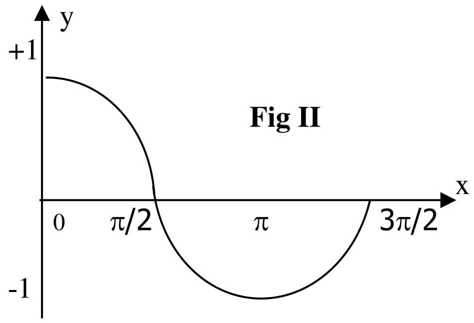
When y = sin 3 θ

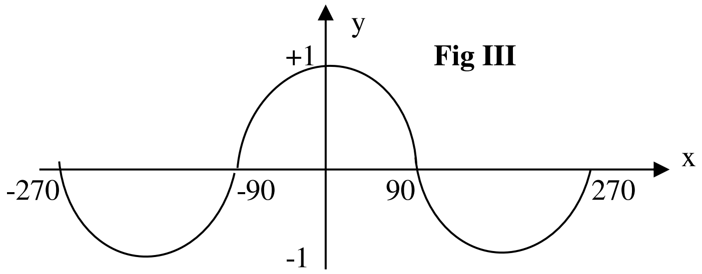
The periodicity is 120^0 – since three multiplies 120^0 to give 360^0. The graph pattern is shown below.



Cosine Graph







Characteristic of a cosine curve

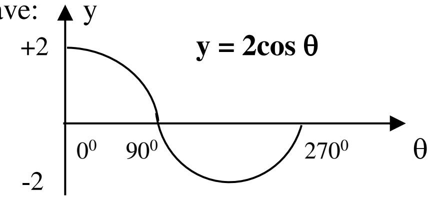
Sketching cosine curve on the positive x – axis as shown in fig, I and II. It starts from its peak of +1 (0^0,1) and then falls to zero at x = 90^0 (π/2). From this point it continues to descend till it get to its lowest point of –1 at x = 180^0 (π) . It then starts climbing to zero at x = 270^0 (3π/2).

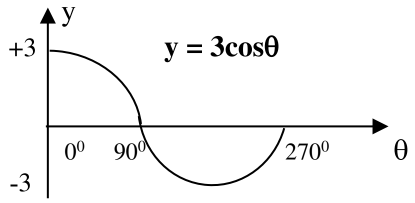
The cycle continues in this pattern.

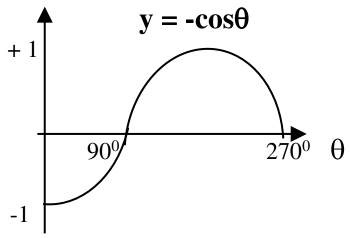
Other forms of cosine graph are either multiples of its amplitude or periodicity.

Cosine graph for AMPLITUDE multiples

The principles applied in sine graph multiples are same here. Thus, we have:

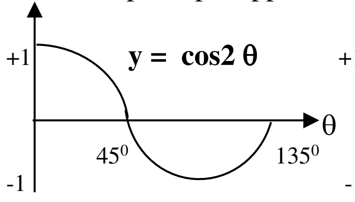


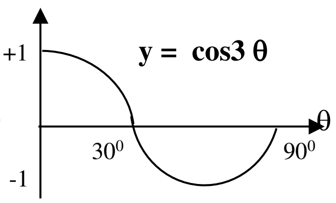




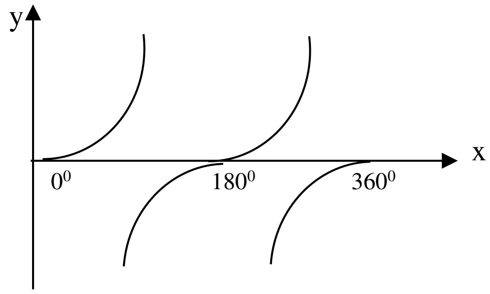
Cosine Graph for PERIODICITY Multiples

The same principle applies as in the case of sine function.





Tangent graph



The curve of tangent function is not a continuous one like that of sine or cosine. It appears in three distinct curves as shown above

GRAPH PLOTTING IN TRIGONOMETRY

The methods applied in table of trigonometry curves are similar to that of quadratic graph.

2008/12

(a) Copy and complete the table of values for

$y = 3\sin x + 2\cos x$ for $0^0 \le x \le 360^0$

x	0^0	60^0	120^0	180^0	240^0	300^0	360^0
y	2.00						2.00

(b) Using a scale of 2cm to 60^0 on x – axis and 2cm to 1unit on y – axis, draw the graph of

$y = 3\sin x + 2\cos x$ for $0^0 \le x \le 360^0$

(c) Use your graph to solve the equation:

$3\sin x + 2\cos x = 1.5$

(d) Find the range of values of x for which

$3\sin x + 2\cos x < -1$

Solution

(a) Table for $y = 3\sin x + 2\cos x$

x	0^0	60^0	120^0	180^0	240^0	300^0	360^0
3sinx	0	2.60	2.60	0	–2.60	–2.60	0
+ 2cosx	2.00	1.00	- 1.00	-2.00	- 1.00	1.00	2.00
y	2.00	3.60	1.60	-2.00	- 3.60	- 1.60	2.00

In completing the table we stick to the decimal place given by $x = 0^0$ *and* $y = 2.00$

(b) Graph is shown on page 286

(c) Rule a line at y = 1.5 to cut the curve, then trace the points to the x – axis 122^0 and 352^0

(d) Rule a line at y = –1, the points where it cuts the curve, we require the lower part of the curve in range form $162^0 < x < 312^0$

2006/12

(a) Copy and complete the table of the relation

$y = 2\sin x - \cos 2x$

x	0^0	30^0	60^0	90^0	120^0	150^0	180^0
y		0.5					–1.0

Using a scale of 2 cm to 30^0 on the x – axis and 2cm to 0.5 unit on the y – axis, draw the graph of

$y = 2\sin x - \cos 2x$ for $0^0 \le x \le 180^0$

(b) Using the same axes, draw the graph of y = 1.25

(c) Use your graph to find the:

(i) value of x for which $2\sin x - \cos 2x = 0$

(ii) roots of the equations $2\sin x - \cos 2x = 1.25$

Solution

(a) Table for $y = 2\sin x - \cos 2x$

x	0^0	30^0	60^0	90^0	120^0	150^0	180^0
2sinx	0	1.0	1.7	2.0	1.7	1.0	0
- cos2x	- 1.0	- 0.5	0.5	1.0	0.5	- 0.5	- 1.0
y	- 1.0	0.5	2.2	3.0	2.2	0.5	- 1.0

In completing table we stick to the decimal place (one d.p) given from $x = 30^0$ and y = 0.5

Graph is shown on page 286

(b) A straight line at y = 1.25 is drawn to cut the curve as shown on the graph.

(c) **i.** the points where the curve cuts the x – axis

20^0 and 160^0

ii. The points of intersection between the curve and the straight line traced to the x – axis as shown by dotted lines 43^0 and 137^0

2011/12 Neco

(a) Copy and complete the table of value for

$y = 2\sin x + \cos x$ for $0^0 \le x \le 360^0$

x	0^0	60^0	120^0	180^0	240^0	300^0	360^0
y			1.23				1

(b) Using a scale of 2cm to 60^0 on x – axis and 2cm to y – axis, draw the graph of $y = 2\sin x + \cos x$ for $0^0 \le x \le 360^0$

(c) Use your graph to solve:

(i) $2\sin x = -\cos x$

(ii) $2\sin x + \cos x = 0.5$

Solution

(a) Table for $y = 2\sin x + \cos x$

x	0^0	60^0	120^0	180^0	240^0	300^0	360^0
2sinx	0.0	1.73	1.73	0.0	–1.73	–1.73	0.0
+cosx	1.0	0.5	- 0.5	- 1.0	- 0.5	0.5	1.0
y	1.0	2.23	1.23	- 1.0	- 2.23	- 1.23	1.0

(b) Graph is shown on page 287

(c) i. Using the graph to solve for $2\sin x = -\cos x$ *is same as*

$2\sin x + \cos x = 0$

These are the points where the curve cuts the x – axis:

153^0 and 335^0

ii. Rule a straight line at y = 0.5

The point where it touches the curve traced to x – axis as shown by dotted line: 141^0 and 348^0

2009/12

(a) Copy and complete the table of value for

$y = \sin x + 2\cos x$ correct to one decimal place.

x	0^0	30^0	60^0	90^0	120^0	150^0	180^0	210^0	240^0
y		2.2				- 1.2	- 2.0		- 1.9

(b) Using a scale of 2cm to 30^0 on the x – axis and 2cm to 0.5unit on the y – axis, draw the graph of

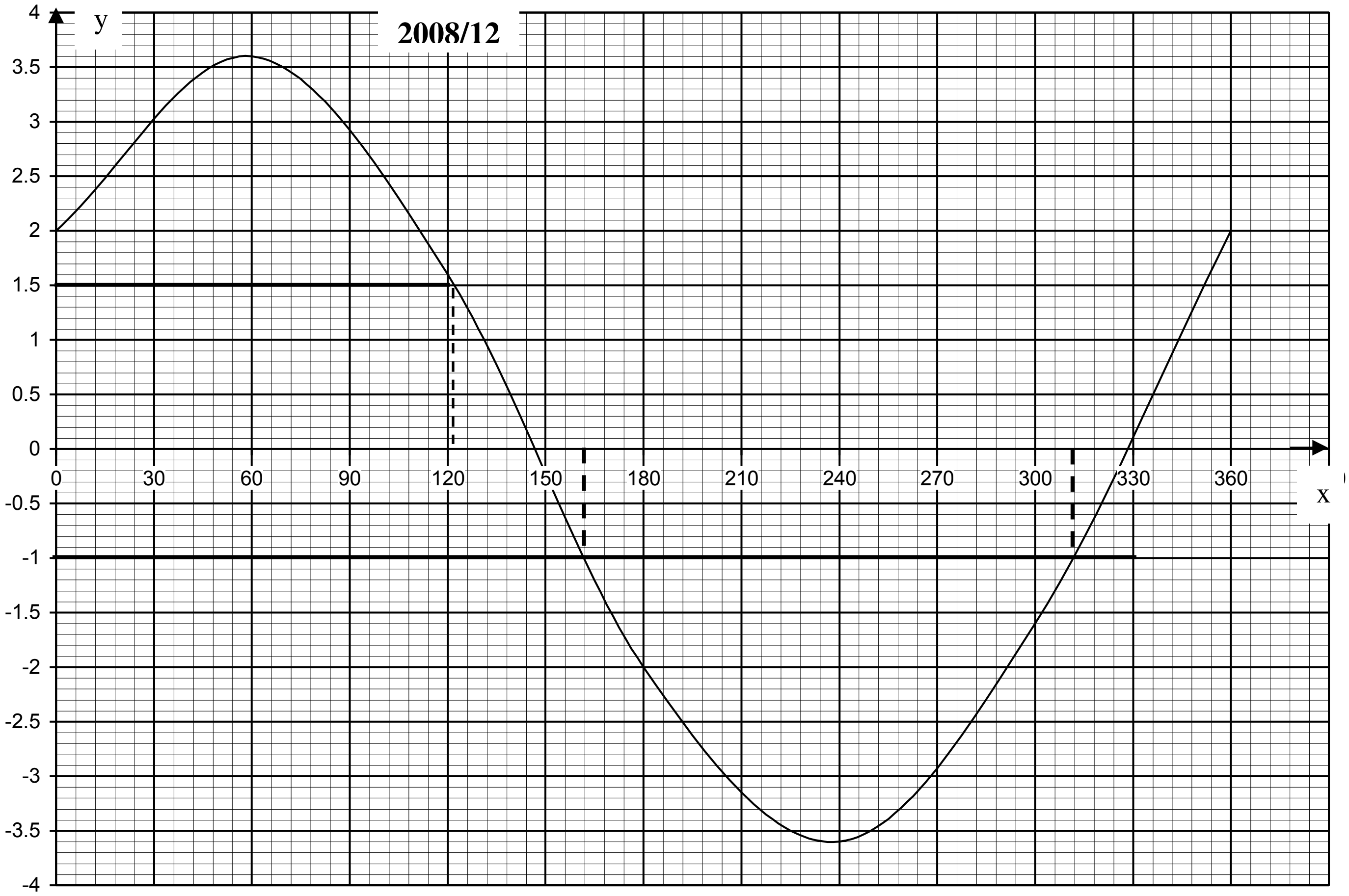
$y = \sin x + 2\cos x$, for $0^0 \le x \le 240^0$

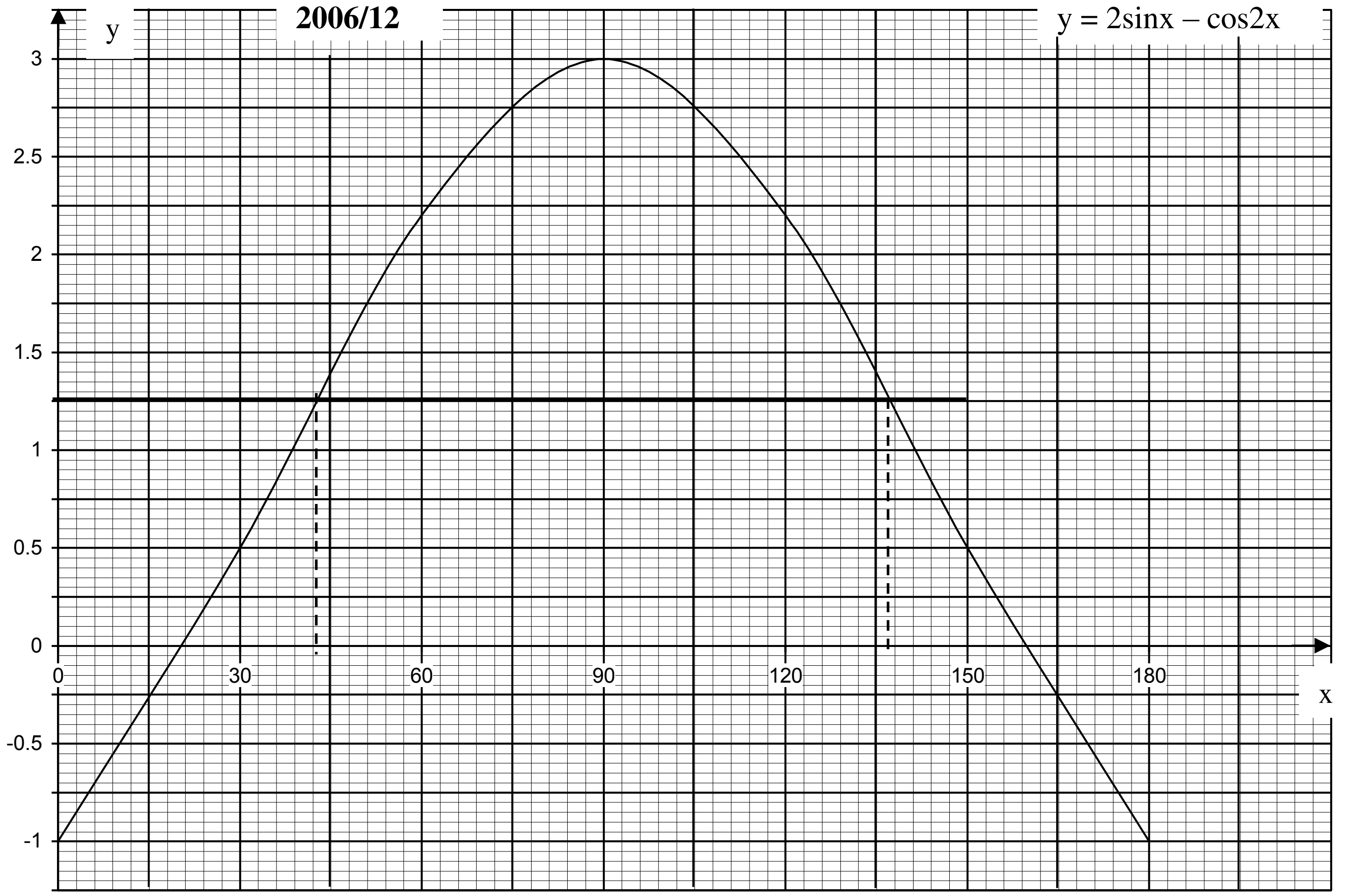
(c) Use your graph to solve the equation:

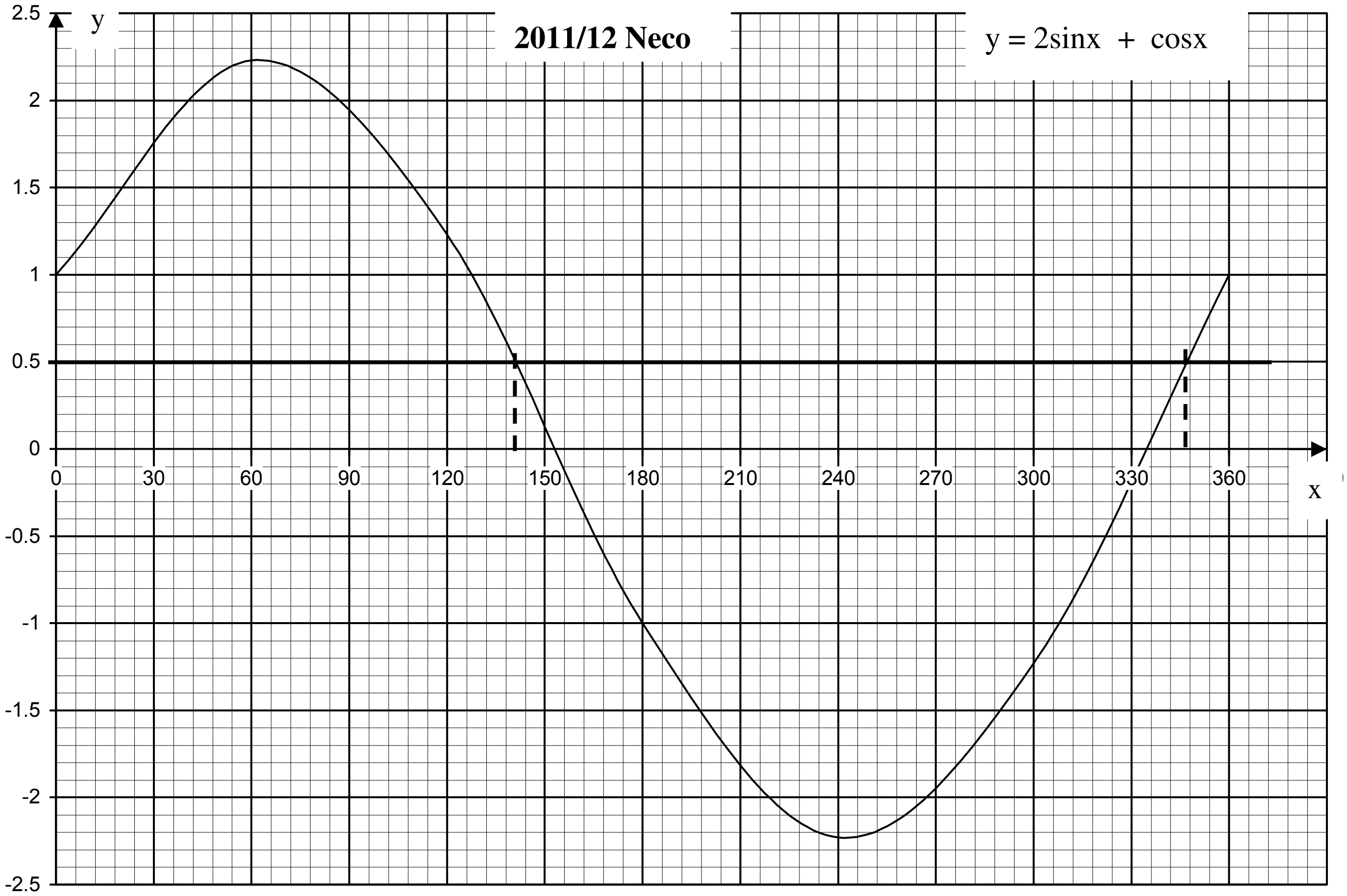
(i) $\sin x + 2\cos x = 0$

(ii) $\sin x = 2.1 - 2\cos x$

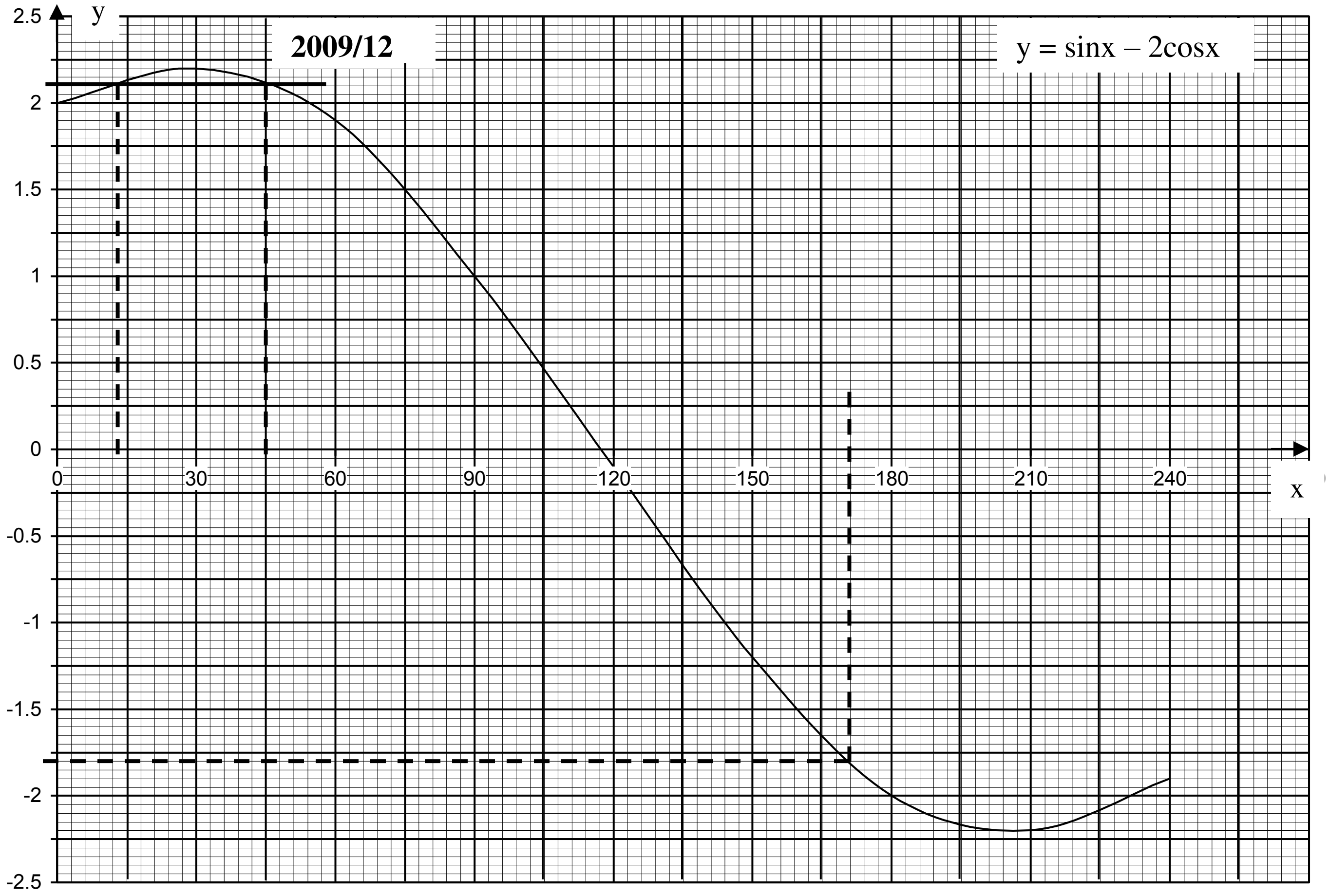
(d) From the graph, find y when $x = 171^0$


2008/12
y
x
4
3.5
3
2.5
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
-2.5
-3
-3.5
-4
0
30
60
90
120
150
180
210
240
270
300
330
360


2006/12
y = 2sinx – cos2x
y
x
3
2.5
2
1.5
1
0.5
0
-0.5
-1
0
30
60
90
120
150
180



2011/12 Neco
y = 2sinx + cosx
y
x
2.5
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
-2.5
0
30
60
90
120
150
180
210
240
270
300
330
360



2009/12
y = sinx – 2cosx
y
x
2.5
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
-2.5
0
30
60
90
120
150
180
210
240

Solution

(a) Table for y = sin x + 2 cos x

x	0^0	30^0	60^0	90^0	120^0	150^0	180^0	210^0	240^0
sin x	0	0.5	0.9	1.0	0.9	0.5	0	–0.5	–0.9
2cos x	2.0	1.7	1.0	0	–1.0	–1.7	–2.0	–1.7	–1.0
y	2.0	2.2	1.9	1.0	–0.1	–1.2	–2.0	–2.2	–1.9

(b) Graph is show on page 287

(c) i. 117^0 This is the point where the curve cuts the x – axis

ii. Rearranging to fit our original equation

sinx = 2.1 – 2cos x becomes

sinx + 2cos x = 2.1

Next, we draw a straight line at y = 2.1 to cut the curve, trace the points to the x – axis as shown by dotted lines

12^0 and 45^0

(d) At x – axis mark 171^0 and trace to the curve then to y – axis as shown by dotted lines

y = – 1.8

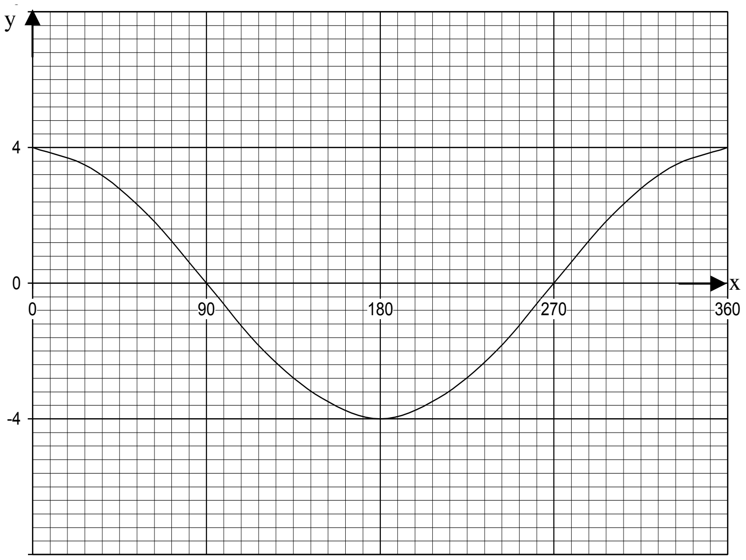
45. What is $\cos^{-1}(-0.8)$?

A 144^0 or 221^0 B 147^0 or 216^0 C 138^0 or 221^0

D 138^0 or 216^0 E 132^0 or 210^0

Solution

At the negative y – axis, trace – 0.8 to cut the curve then to x – axis: 144^0 or 216^0



2007/49 Neco Exercise 18.26

Which of the following represent the graph above?

A $y = \sin\frac{1}{4}x$ B y = cos 4x C y = 4 sin x

D y = 4 cos x E $y = \cos\frac{1}{4}x$

2008/44 and 45 Neco

The graph below is the cosine curve. Use it to answer questions 44 and 45

44. Find $\cos 138^0$

A 0.75 B 0.70 C – 0.75 D – 0.70 E – 0.068

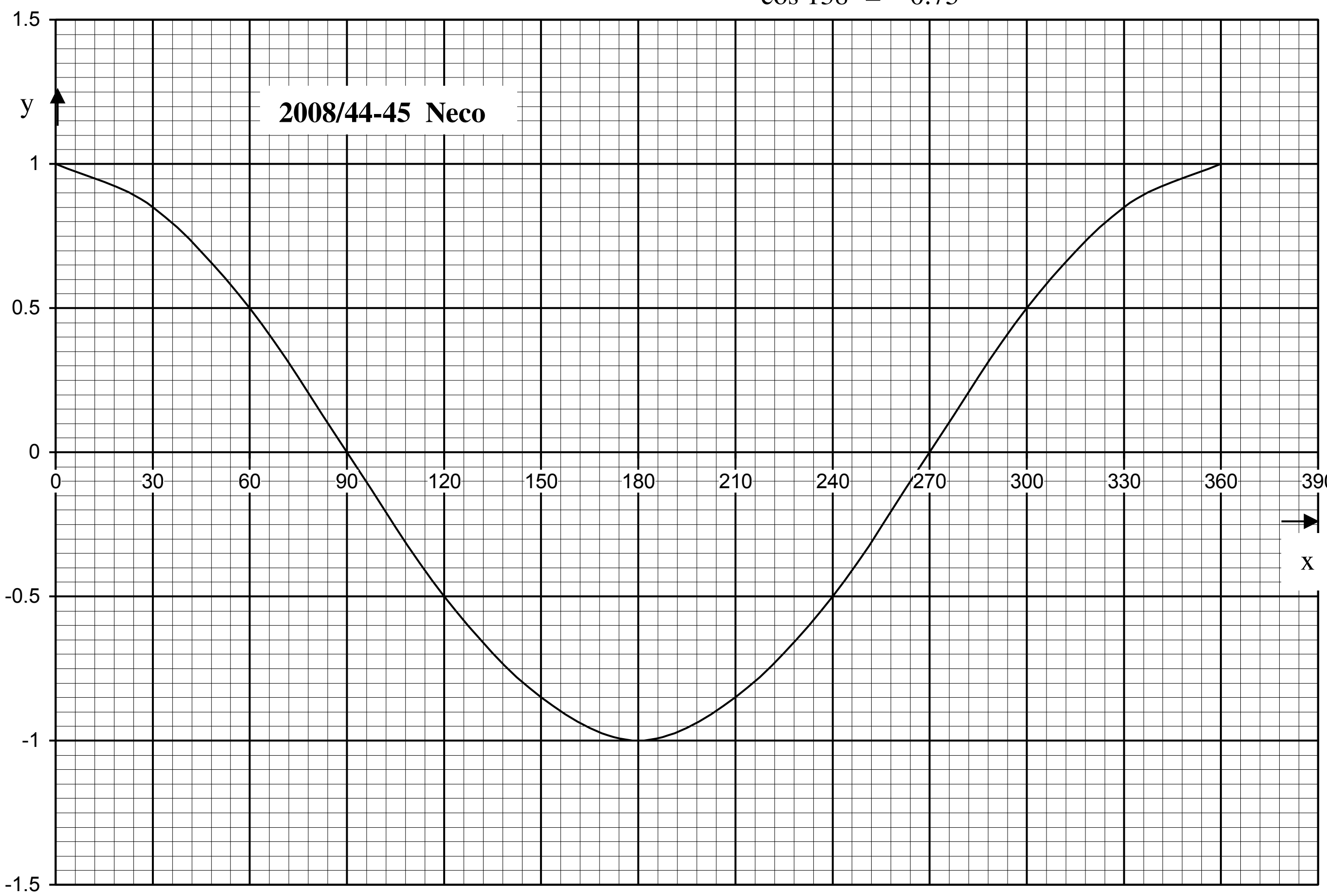
Solution

Mark out 138^0 in the x – axis by 138 – 120 = 18^0

Thus three boxes after 120^0 gives 138^0 (since each box is 6^0)

trace it to the curve then to y – axis

$\cos 138^0 = -0.75$



2014/7 Exercise 18.27

Copy and complete the table of value for the relation

$y = 2\sin x + 1$

x	0^0	30^0	60^0	90^0	120^0	150^0	180^0	210^0	240^0	270^0
y	1.0				2.7			0.0	-0.7	

(b) Using a scale of 2cm to 30^0 on the x – axis and 2cm to 1unit on the y – axis, draw the graph of $y = 2\sin x + 1$ for $0^0 \le x \le 270^0$

(c) Use the graph to find the value of x for which $\sin x = \frac{1}{4}$

1995/8 Exercise 18.28

Copy and complete the table of value for the relation

$y = 2\cos 2x - 1$

x	0^0	30^0	60^0	90^0	120^0	150^0	180^0
$y = 2\cos 2x - 1$	1.0	0					1.0

(b) Using a scale of 2cm = 30^0 on the x – axis and 2cm = 1unit on the y – axis, draw the graph of $y = 2\cos 2x - 1$ for $0^0 \le x \le 180^0$

(c) On the same axes draw the graph of

$$y = \frac{1}{180}(x - 360)$$

(d) Use your graph to find the :

i. values of x for which $2\cos 2x + \frac{1}{2} = 0$

ii. roots of the equation $2\cos 2x - \frac{x}{180} + 1 = 0$

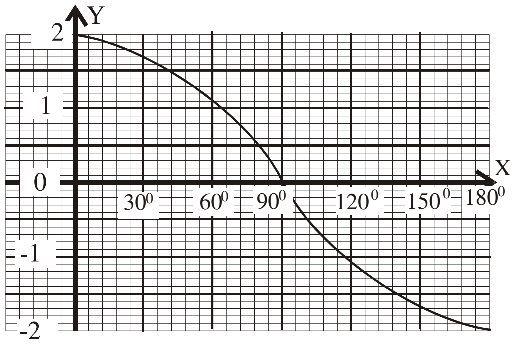
2013/10 f/maths Exercise 18.29

(a) Using a scale of 2cm to 30^0 on the x – axis and 2cm to 0.2 unit on the y – axis, on the same graph sheet, draw the graphs of $y = \sin 2x$ and $y = \cos x$ for $0^0 \le x \le 210^0$ at intervals of 30^0.

(b) Using the graphs in 10(a), find the truth set of:

i. $\sin 2x = 0$ ii. $\sin 2x - \cos x = 0$

2004/26 f/maths Exercise 18.30



The sketch above represents the graph of

A $y = 2\sin x$ B $y = 2\cos x$ C $y = \cos 2x$ D $y = \sin 2x$

Linear inequality II

Linear inequalities in two variables.

This is another phase of inequalities, which is related to linear equation in two variables. We shall concern ourselves mainly with the graphical solution to inequality in two variables.

Here broken lines represents strict inequalities (< or >) while thick straight line shows weak inequality (≤ or ≥). We always use the linear equation equivalent of equality to get the two points necessary to plot the straight line.

Examples of inequality graphs are shown below

(1) Solve the inequality : $y < 4x + 2$

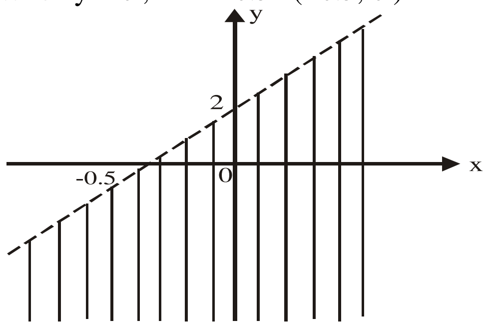
Solution

We consider the line equivalent

$y = 4x + 2$

When x = 0 , y = 2 (0, 2)

When y = 0 , x = - 0.5 (-0.5, 0)



The shaded region satisfies the inequality $y < 4x + 2$ but points on the broken line are not included since we are dealing with strict inequality.

2. Show graphically the region represented by $x + y > 5$

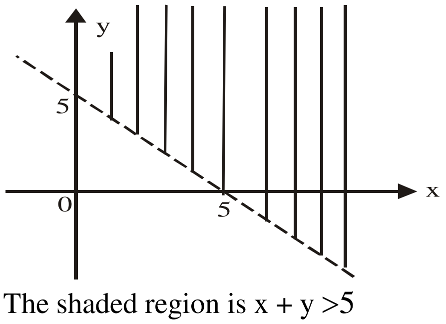
Solution

First we consider the line equivalent

$x + y = 5$

When x = 0, y = 5 (0, 5)

When y = 0, x = 5 (5, 0)



The shaded region is $x + y > 5$

3 . Show graphically the region represented by $2x + y \le 8$

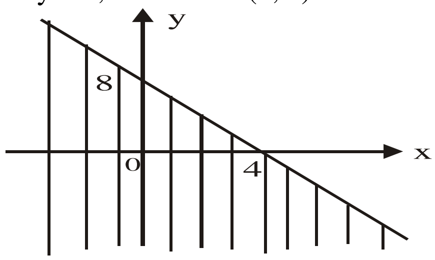
Solution

Considering the line equivalent,

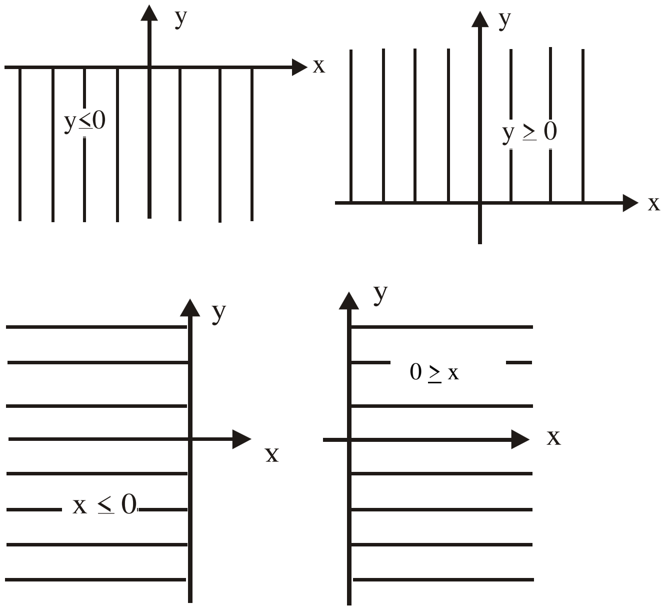
$2x + y = 8$

When x = 0, y = 8 (0, 8)

When y = 0, x = 4 (4, 0)

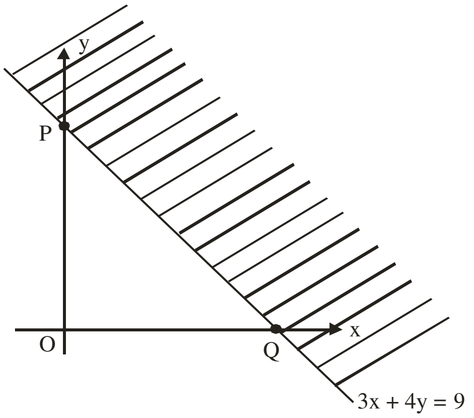


Some inequality regions are shown below (shaded)



2015/30 and 31 Neco

Use the inequality graph below to answer questions **30** and **31**



30. Find the coordinates of point P

A $(0, 2^1/_4)$ B $(0, 3)$ C $(2^1/_4, 0)$

D $(3, 0)$ E $(3, 4)$

Solution

At point p

$x = 0$, then $3x + 4y = 9$ becomes

$3(0) + 4y = 9$

$4y = 9$

$y = {}^9/_4 = 2^1/_4$

Coordinates of point P $(0, 2^1/_4)$ A.

31. The shaded portion shows the outer boundary of the half plane defined by the linear inequality

A $3x + 4y = 9$ B $3x + 4y < 9$ C $3x + 4y > 9$

D $3x + 4y \leq 9$ E $3x + 4y \geq 9$

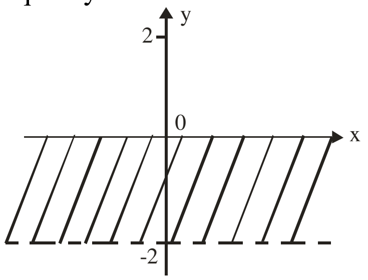
Solution

Since the outer boundary is the region referred to in this case and we have a thick line (weak inequality)

$3x + 4y \geq 9$ (E)

2000/18 Neco

The shaded region on the Cartesian plane below represents the inequality:



A $-2 < x < 0$ B $-2 \leq x < 0$ C $-2 < y < 0$

D $-2 \leq y < 0$ E $-2 < y \leq 0$

Solution

The straight line at –2

$y = -2$

The shaded part is the upper part i.e > or ≥

The line is broken i.e strict inequality of >

Thus $y > -2$

The line at $x = 0$

The shaded part is the lower part and it is a thick line

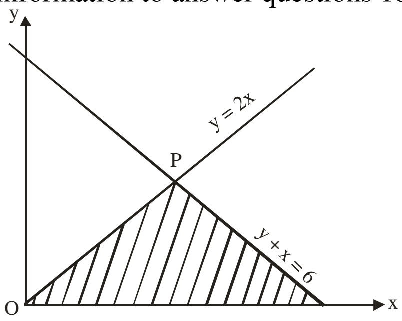
$y \leq 0$

Combining the two inequalities

$-2 < y \leq 0$ (E)

2005/18 and 19 (Nov)

The diagram below shows the graph of lines $y = 2x$ and $y + x = 6$. P is the point of intersection of the two lines. Use the information to answer questions 18 and 19



18. Find the coordinates of point P

A $(2, 3)$ B $(3, 4)$ C $(3, 6)$ D $(2, 4)$

Solution

The coordinates of the point P is gotten by equating the two equations and solving

$y = 2x$ and $y + x = 6$ i.e $y = 6 - x$

$2x = 6 - x$

$2x + x = 6$

$3x = 6$

$x = {}^6/_3 = 2$

Substitute x value into $y = 2x$

$y = 2(2)$

$y = 4$

Thus P $(2, 4)$ D.

19. Which set of inequalities satisfies the shaded region ?

A $y \geq 0,\ y \leq 2x,\ y + x \geq 6$ B $y \geq 0,\ y \leq 2x,\ y + x \leq 6$

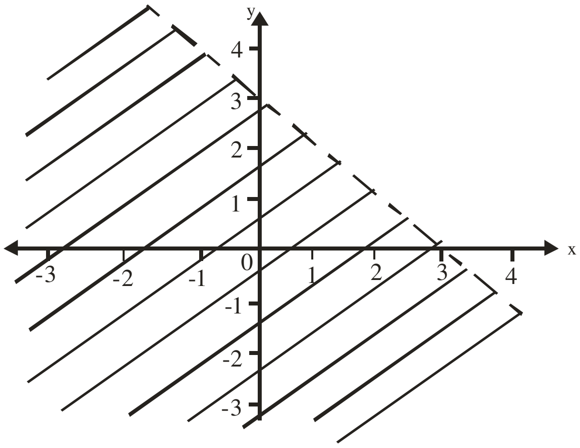
C $y \geq 0,\ y \geq 2x,\ y + x \leq 6$ D $y \leq 0,\ y \leq 2x,\ y + x \leq 6$

Solution

All the lines are thick ones-no broken line. The result is weak inequality in all the values here.

The shaded region is:
the positive side of y i.e $y \geq 0$
the lower part of line $y = 2x$ i.e $y \leq 2x$
the lower part of line $y + x = 6$ i.e $y + x \leq 6$
Thus the set of inequalities is $y \geq 0$, $y \leq 2x$, $y + x \leq 6$ (B)

2010/33



The shaded portion in the diagram is the solution of
A $x + y \leq 3$ B $x + y < 3$ C $x + y > 3$
D $x + y \geq 3$

Solution

Applying the formula for straight line $y = mx + c$,
Here c = 3 i.e where the line cuts the y – axis

$$m(\text{Slope}) = \frac{y_2 - y_1}{x_2 - x_1}$$

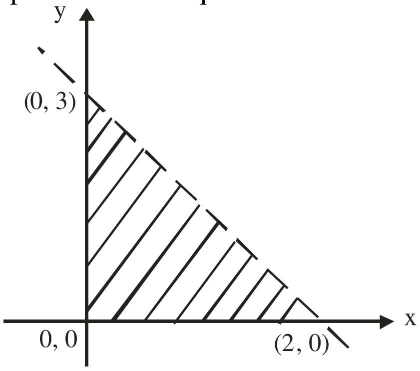
Note it is negative slope where y_2 is the lower part

$$= \frac{0 - 3}{3 - 0} = \frac{-3}{3} = -1$$

Substituting $y = -1x + 3$
$y + x = 3$
Broken line implies > or <
Lower part of the line implies <
Thus $y + x < 3$ (B)

2014/58 Neco

The shaded region on the Cartesian plane below represents the inequalities:



A $x\geq 0$, $y\geq 0$, $2y + 3x < 6$ B $x\geq 0$, $y< 0$, $2y + 3x < 6$
C $x> 0$, $y< 0$, $2y + 3x < 6$ D $x\geq 0$, $y> 0$, $2y + 3x \leq 6$
E $x> 0$, $y\geq 0$, $2y + 3x \geq 6$

Solution

Applying the formula for straight line $y = mx + c$,
Here c = 3 i.e where the line cuts the y – axis

$$m(\text{Slope}) = \frac{y_2 - y_1}{x_2 - x_1}$$

Note it is negative slope where y_2 is the lower part

$$= \frac{0 - 3}{2 - 0} = \frac{-3}{2}$$

Substituting $y = -\frac{3}{2}x + 3$
Multiply through by 2 to clear fraction
$2y = -3x + 6$
$2y + 3x = 6$
Broken line implies > or <
Lower part of the line implies <
Thus $2y + 3x < 6$
The shaded region is:
The positive side of y - axis $x \geq 0$
The positive side of x - axis $y \geq 0$
Option A

System of inequalities (more than one equation)

The solution of a system of linear inequalities in two variables is the region of intersection of all the equations involved.

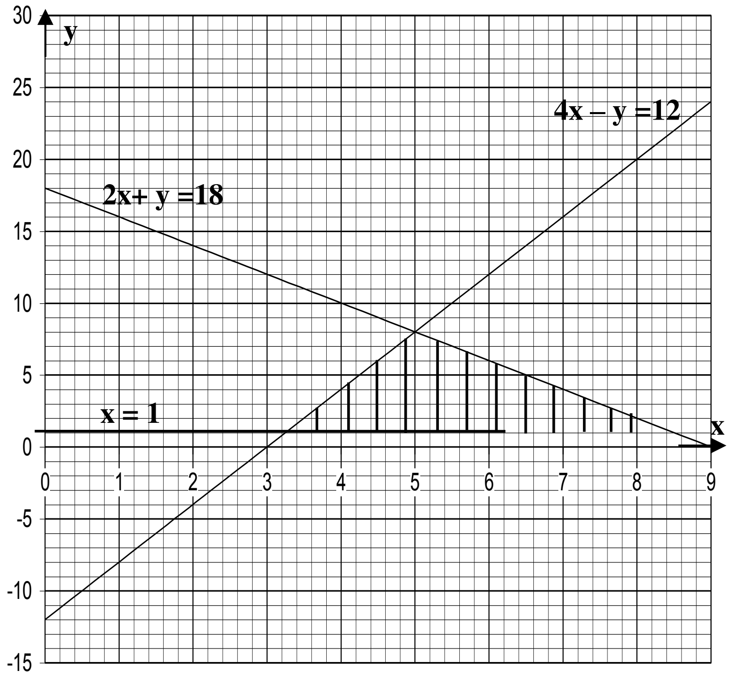
2007/8a Neco

Show the region on the graph which satisfies the inequalities :

$$4x - y \leq 12$$
$$2x + y \leq 18$$
$$x = 1$$
$$y \geq 0$$

Solution

Line $4x - y = 12$ (Thick line and shade the back part)
When y = 0 , $x = {}^{12}/_{4} = 3$ (3, 0)
When x = 0 , y = –12 (0, –12)
Line $2x + y = 18$ (Thick line and shade the back part)
When y = 0 , $x = {}^{18}/_{2} = 9$ (9, 0)
When x = 0 , y = 18 (0, 18)
Line x = 1 (ordinary line)
Line y = 0 (Thick line and shade the positive side)



2012/7

(i) Using a scale of 2cm to 1unit on both axes, on the same graph sheet, draw the graphs of $y - \frac{3x}{4} = 3$ and $y + 2x = 6$

(ii) From your graph, find the coordinates of the point of intersection of the two graphs

(iii) Show on the graph sheet, the region satisfied by the inequality $y - \frac{3x}{4} \geq 3$

Solution

Line $y - \frac{3x}{4} = 3$

When $x = 0$, $y = 3$ (0, 3)

$y = 0$, $-3x = 12$ i.e $x = -4$ (– 4, 0)

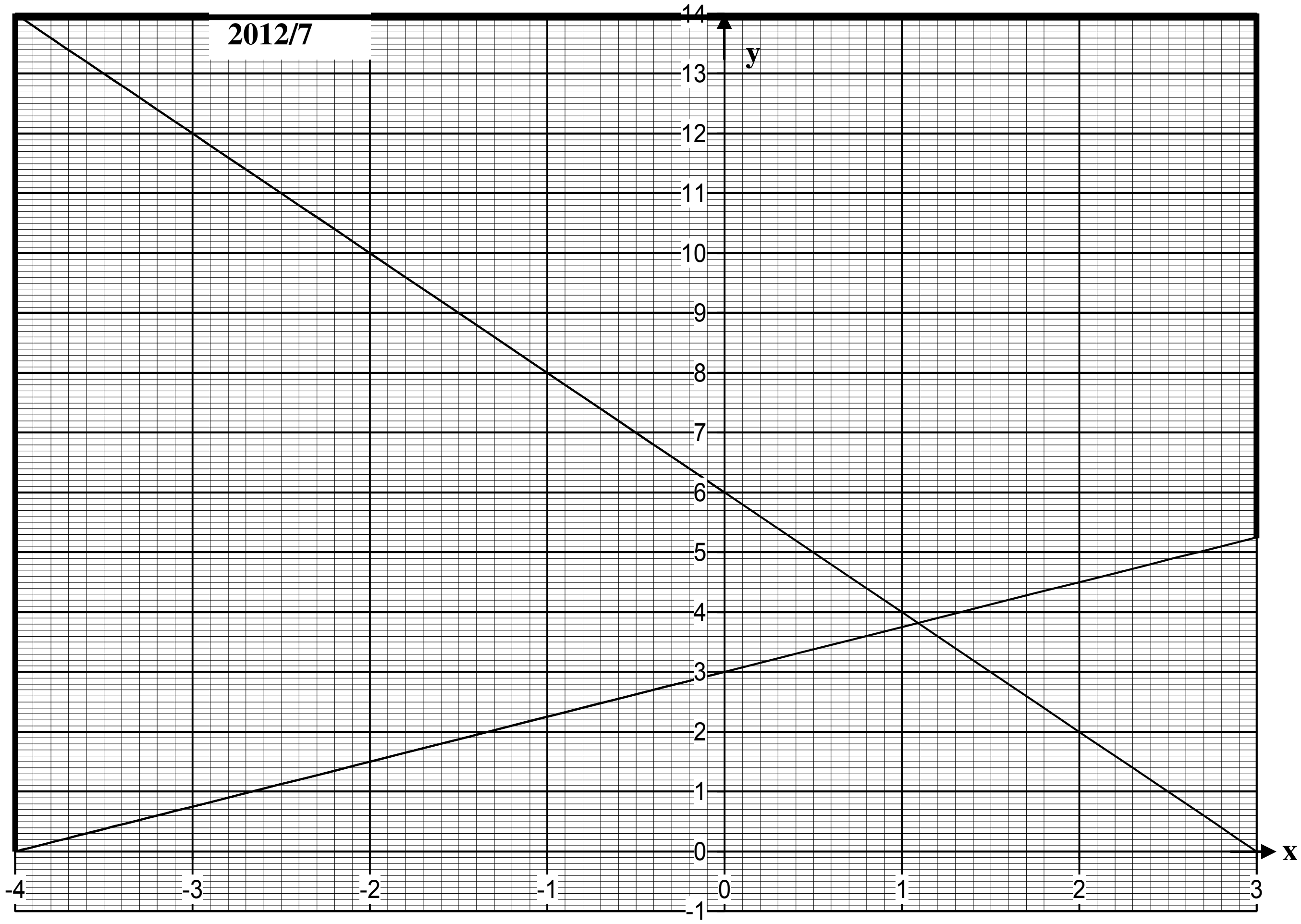
Line $y + 2x = 6$

When $x = 0$, $y = 6$ (0, 6)

$y = 0$, $2x = 6$ i.e $x = 3$ (3, 0)

(ii) Coordinates of the point of intersection (1.1, 3.8)

(iii) Region is shown by area bounded by the **thick line** on the graph below (we avoided shading for clearer graph)



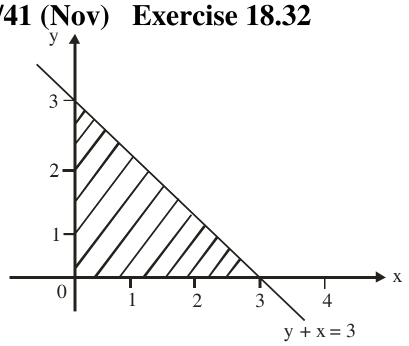
2005/12a Neco Exercise 18.31

Show graphically the region R which satisfies the following inequalities

$3x + 4y \leq 6$

$x + 4y \leq 10$, $y \geq 0$, $x \geq 0$

2009/41 (Nov) Exercise 18.32



Which of the following inequalities represent the shaded portion?

A $y + x \leq 3$, $x \geq 0$, $y \geq 0$ B $y + x \geq 3$, $x \geq 0$, $y \geq 0$
C $y + x \leq 3$, $x \geq 0$, $y \leq 0$ D $y + x \geq 3$, $x \leq 0$, $y \geq 0$

Exercise 18.33

(i) Using a scale of 1cm to 1unit on both axes, on the same graph sheet, draw the graphs of

$2x + 4y \leq 16$

$x - 2y \geq 10$, $y \geq 0$, $x \geq 0$

Show on the graph sheet, the region satisfied by the inequalities

CHAPTER NINETEEN

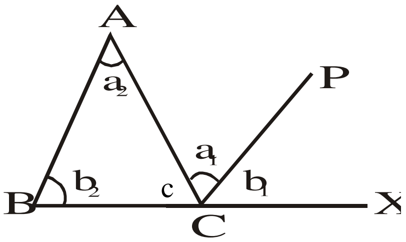
Plane and Circle Geometry

Plane Geometry

Angles between straight and parallel lines, types of triangles have been discussed under 'elements of plane Geometry' (**Bearing).** Readers and students to take note.

Triangles

Theorem: Sum of angles in a triangle is 180^0 (2 right angles)



Given: any Δ ABC

To prove: $\hat{A} + \hat{B} + \hat{C} = 180^0$

Construction: Produce $\overline{BC}$ to point X. Draw CP//BA

Proof:

$a_1 = a_2$ (alternative $\angle$s)

$b_1 = b_2$ (corresponding $\angle$s)

$c + a_1 + b_1 = 180$ ($\angle$s on straight line)

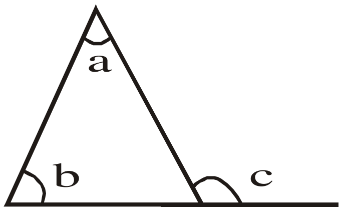
$\therefore\ c + a_2 + b_2 = 180^0$

$\therefore\ A\hat{C}B + \hat{A} + \hat{B} = 180^0$

$\therefore\ \hat{A} + \hat{B} + \hat{C} = 180^0$

Note: *The above theorem and its proof is to be mastered by students as the theorem is among the required ones to be proven in exams:*

The exterior angle of a triangle is equal to the sum of the two interior opposite angles.

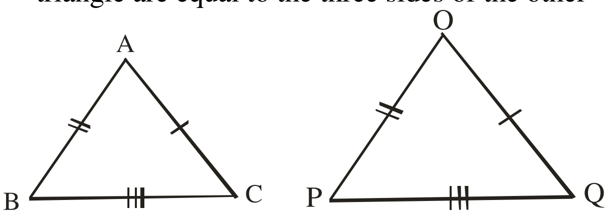


C = a + b

Congruent triangles

Two triangles are congruent ($\equiv$) if they pass **one** of the **four** tests listed below:

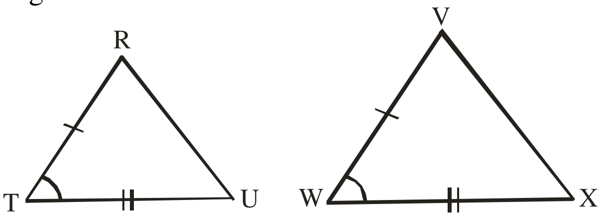
I . SSS : **Side, side, side** are equal i.e three sides of one triangle are equal to the three sides of the other



Here AB = OP, BC = PQ and AC = OQ

Thus $\Delta ABC \equiv \Delta OPQ$ (SSS)

II. SAS : **Side, angle, side** two sides and the included angle of one triangle are equal to the two sides and the included angle of the other



Here TR = WV, TU = WX, $\angle T = \angle W$

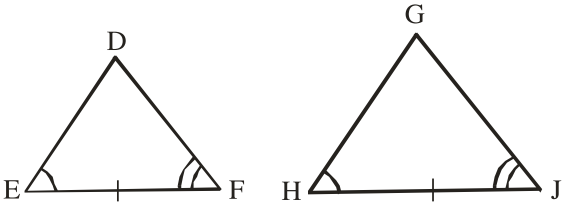
Thus $\Delta RTU = \Delta VWX$ (SAS)

III. ASA : **angle, side, angle**

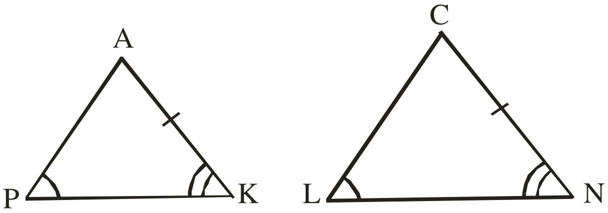
AAS : **angle, angle, side**

Two angles and a side of one triangle are respectively equal to two angles and a side of the other

ASA

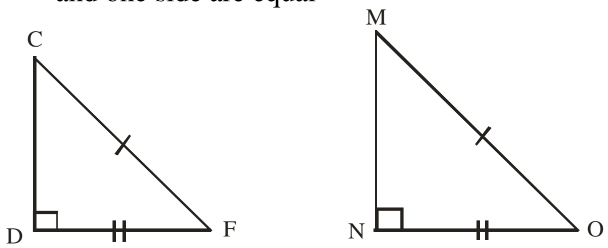


AAS



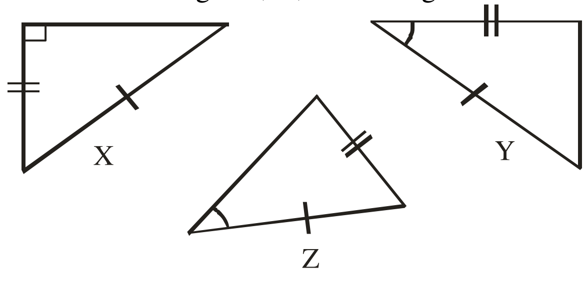
IV. RHS: **right – angled, hypotenuse, side**

Both triangles are right – angled, the hypotenuse and one side are equal



2006/56 Neco

Which of the triangle X, Y, Z are congruent?



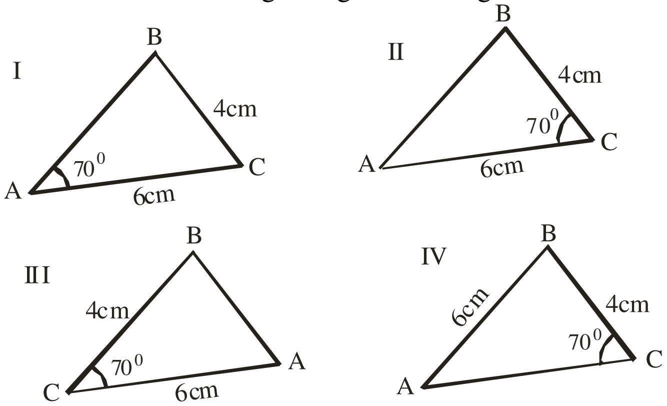
A X and Y only B X and Z only C Y and Z only

D All the three E None of them

Solution

X = Y (RHS) A.

2008/35 Neco (Dec)
Which of the following triangles are congruent?



A I and III only B II and IV only C I and II only
D II and III only E I and IV only

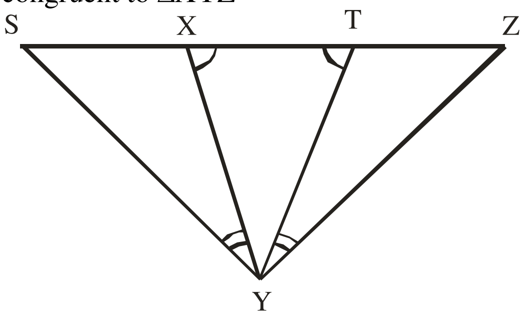
Solution

II and III only (SAS) D.

2005/55 Neco Exercise19.1
Which of the following conditions is NOT always true about congruent triangles P and Q?
A The area of P is equal to the area of Q
B Three angles of P are equal to three angles of Q
C Three sides of P are equal to three sides of Q
D Two sides and the included angle of P are equal to two sides and the included angle of Q
E Two angles and one side of P are equal to two angles and one side of Q

2008/40 Neco Exercise 19.2
In proving the congruence of two triangles, which of the following is NOT important?
A Two sides and the included angles.
B Two angles and a side
C Three sides
D Three angles
E Right – angle, hypotenuse and another side

2003/48 NABTEB Exercise 19.3
In the diagram below, name the triangle which is congruent to ΔXYZ



A ΔTYS B ΔTYZ C ΔXYS D ΔXYZ

Similar triangles

For two triangles to be similar it must meet (fulfill) one of the three conditions below:

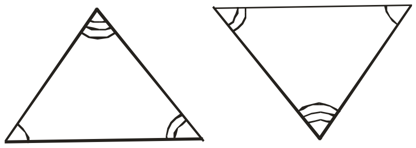
Two triangle are similar if

I. "Two angles in one triangle are equal to two angles in the other" of course the third angle must follow suit. It can be restated as "Two triangles are similar if three angles of one are respectively equal to the three angles of the other"

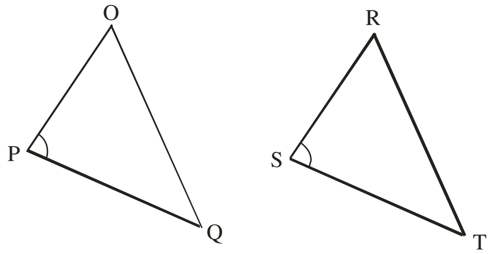
II. Two triangles are similar if two pairs of sides are in the same ratio and their included angles are the same

III. Two triangles are similar if the ratios of the corresponding sides are equal

Condition I

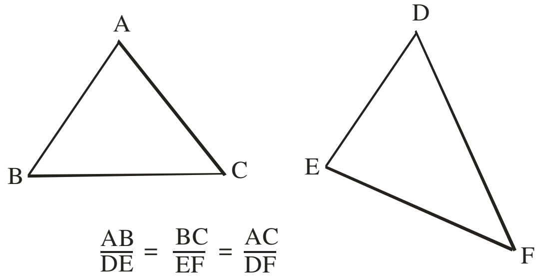


Condition II



$\angle OPQ = \angle RST$, $\frac{OP}{RS} = \frac{PQ}{ST}$

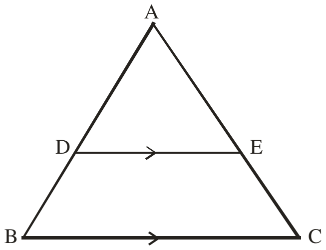
Condition III



$\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$

Intercept Theorem

If a straight line is drawn parallel to one side of a triangle, the straight line divides the other two sides in the same ratio

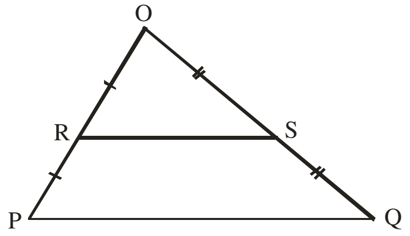


Here DE Divides side AB and AC in equal ratios

$$\frac{AD}{DB} = \frac{AE}{EC}$$

Midpoint theorem

The straight line joining the midpoints of two sides of a triangle is parallel to the third side and equal to half of it

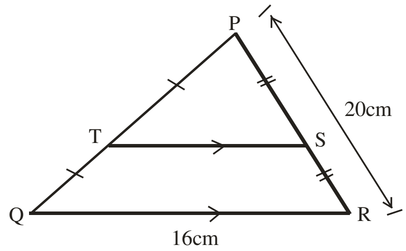


RS//PQ and RS = $\frac{1}{2}$ PQ

2006/47 NABTEB (Nov)

In ΔPQR, $\overline{PT} = \overline{TQ}$, $\overline{PS} = \overline{SR}$ and $\overline{TS} // \overline{QR}$.

If $\overline{QR}$ = 16cm and $\overline{PR}$ = 20cm, find $\overline{TS}$



A 8cm B 10cm C 32cm D 40cm

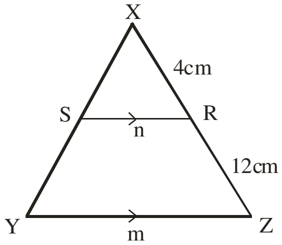
Solution

TS = $\frac{1}{2}$ QR

= $\frac{1}{2} \times 16$ = 8cm (A)

2007/49

In the diagram, /XR/ = 4cm, /RZ/ = 12cm, /SR/ = n, /YZ/ = m and $\overline{SR} // \overline{YZ}$. Find M in terms of n.

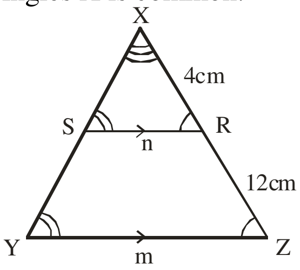


A m = 2n B m = 3n C m = 4n D m = 5n

Solution

Applying corresponding angles rule in parallel lines and transversals to indicate angles.

Angles X is common.



$$\frac{XR}{XZ} = \frac{SR}{YZ}$$

$$\frac{4}{4+12} = \frac{n}{m}$$

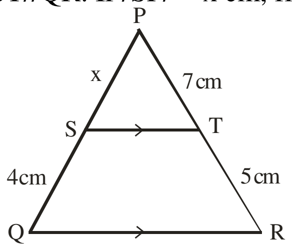
4m = n(4 + 12)

4m = 16n

m = 4n (C)

2009/24

In the diagram, /SQ/ = 4cm, /PT/ = 7cm, /TR/ = 5cm and ST//QR. If /SP/ = x cm, find the value of x.

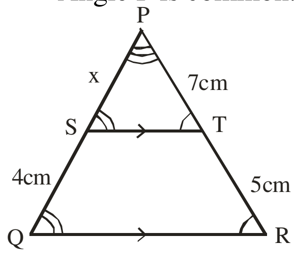


A 5.6 B 6.5 C 6.6 D 6.8

Solution

We apply corresponding angles rule in parallel lines and transversal to indicate the angles.

Angle P is common.



$$\frac{PT}{PR} = \frac{PS}{PQ}$$

$$\frac{7}{7+5} = \frac{x}{x+4}$$

$$\frac{7}{12} = \frac{x}{x+4}$$

7(x + 4) = 12x

7x + 28 = 12x

28 = 12x – 7x

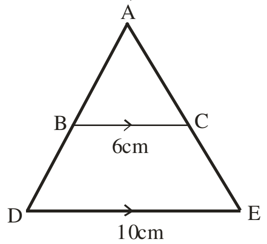
28 = 5x

x = 5.6 (A)

2005/48 NABTEB

In the figure below, BC is parallel to DE.

If /AE/ = 15cm, find the length of CE.

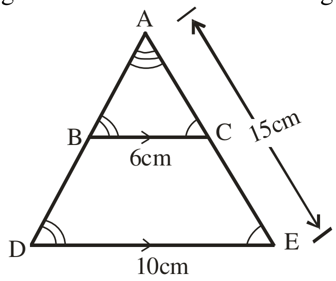


A 4cm B 6cm C 9cm D 10cm

Solution

We are dealing with similar angles. Applying corresponding angles format in parallel lines and transversal to indicate the angles.

Angle A is common to both triangles



CE = AE – AC

But $\frac{BC}{DE} = \frac{AC}{AE}$

$\frac{6}{10} = \frac{AC}{15}$

$6 \times 15 = 10 \times AC$

$\frac{6 \times 15}{10} = AC$

AC = 9cm

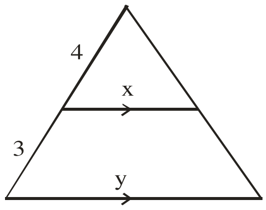
Thus CE = 15 – 9

= 6cm (B)

2005/47 NABTAB (Nov)

What is the ratio x : y in the figure below?

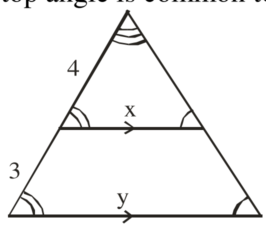
A 3 : 4 B 4 : 3 C 3 : 7 D 4 : 7



Solution

We are dealing with similar triangles. Applying corresponding angles format in parallel lines and transversal to indicate the angles.

The top angle is common to both triangles



$\frac{4}{4+3} = \frac{x}{y}$

$\frac{4}{7} = \frac{x}{y}$

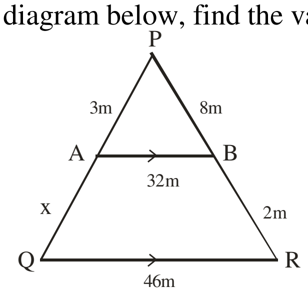
In ratio form

x : y

4 : 7 (D)

2013/38 Neco

In the diagram below, find the value of x



A 0.44m B 0.75m C O.82m D 1.83m E 2.74m

Solution

$\frac{8}{8+2} = \frac{3}{3+x}$

8(3 + x) = 3 × 10

24 + 8x = 30

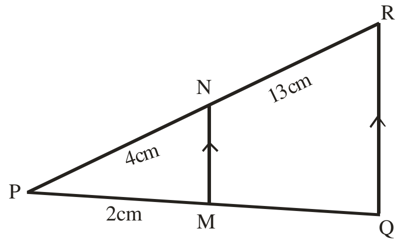
8x = 30 – 24

8x = 6

x = $^6/_8$ = 0.75m (B)

2004/45 NABTEB

Find /PQ/ in the diagram, given that /PM/ = 2cm, /PN/ = 4cm and /NR/ = 13cm



A $^3/_4$cm B $2^3/_4$cm C $5^1/_3$cm D $7^1/_3$cm

Solution

Note that PQ = 2 + MQ

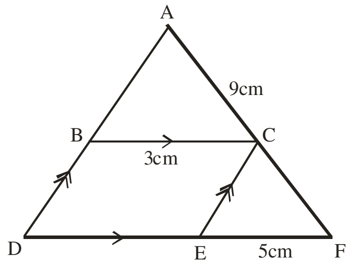
$\frac{4}{4+13} = \frac{2}{2+MQ}$

4(2 + MQ) = 2 × 17

2 + MQ = $\frac{34}{4} = \frac{17}{2}$ = $8^1/_2$

2010/40 Neco ***counter example***

In the diagram below, BC//DF, BD//CE, /AC/ = 9cm, /BC/ = 3cm and /EF/ = 5cm. Find the value of $\overline{CF}$



A 3cm B 15cm C 24cm D 45cm E 72cm

Solution

ΔABC and ΔADF are similar Δs

$\frac{BC}{DF} = \frac{AC}{AF}$

But DF = DE + EF and DE = 3cm (opposite side of ‖gm)

and AF = 9 + CF

$\frac{3}{3+5} = \frac{9}{9+CF}$

3(9 + CF) = 9 × 8

27 + 3CF = 72

3CF = 72 – 27

3CF = 45 thus CF = 15cm (B)

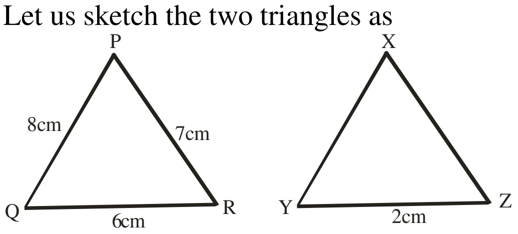
2014/19 (Nov) ***counter example***

Triangles PQR and XYZ are similar. If the sides of triangle PQR are 6cm, 7cm and 8cm and the shortest side of the triangle XYZ is 2cm, find the length of the longest side of triangle XYZ.

A 4cm B $3^1/_2$cm C $2^2/_3$cm D $2^2/_7$ cm

Solution

Let us sketch the two triangles as



Since they are similar the ratio of their corresponding sides are equal

Shortest side here is QR and YZ.

Longest side here is PQ and XY.

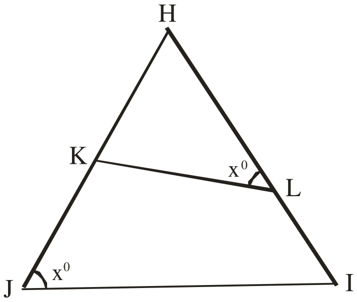
$$\frac{6}{2} = \frac{8}{XY}$$

$$6\,XY = 2 \times 8$$

$$XY = \frac{2 \times 8}{6}$$

$$= {}^{8}/_{3}\text{ cm} \quad = 2^{2}/_{3}\text{cm} \quad (\text{ C })$$

2010/21



In the diagram, triangles HKL and HIJ are similar.

Which of the following ratios is equal to $\frac{LH}{JH}$?

A $\frac{KL}{JI}$ B $\frac{HK}{JK}$ C $\frac{JI}{KL}$ D $\frac{HK}{LK}$

Solution

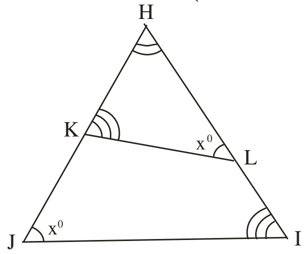
Here the task is to identify the corresponding sides, since it is their ratio that is of concern.

In Δs HJI and HKL

$\angle J = \angle L = x^0$ (given)

$\angle H$ is common to both triangles

It follows that $\angle K = \angle I$ (3rd $\angle$ of Δ)



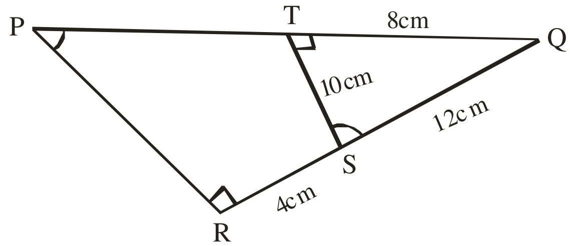
$$\frac{LH}{JH} = \frac{HK}{HI} = \frac{KL}{JI}$$

In this case $\frac{LH}{JH}$,are originating side from a given angle x, we can't take angle H, it is common and not distinct, so we are left with $\angle$ K and $\angle$ I and the originating line there are KL and IJ though written as JI, the ratio is

$\frac{KL}{JI}$ Option A

2004/48 and 49 NABTEB

Use the information in the figure below to answer questions 48 and 49



The marked angles are equal, $\angle QTS = \angle QRP = 90^0$, /TQ/ = 8cm, /TS/ = 10cm, /RS/ = 4cm and /QS/ = 12cm

48. Which of the following statement is not correct about QTPRS?

A /TS/ is proportional to /RP/

B /QT/ is proportional to /QR/

C /QS/ is proportional to /SR/

D /QS/ is proportional to /QP/

Solution option D

49. Calculate the length of PR

A 14cm B 16cm C 20cm D 24cm

Solution

Ratio: side facing ordinary angle: side facing common angle at Q

$$\frac{8}{4+12} = \frac{10}{PR}$$

$$8PR = 10 \times 16$$

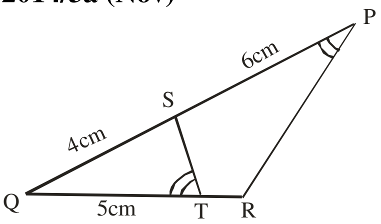
$$PR = \frac{10 \times 16}{8}$$

$$= 20\text{cm} \quad C$$

Note the ratio of side facing right angled triangle here is $\frac{12}{8+TP}$ and not $\frac{12}{12+4}$

Since the right – angle in the big triangle is not facing 12 + 4

2014/3a (Nov)



In the diagram, PQ and QR are straight lines, /PS/ = 6cm, /QS/ = 4cm, /QT/ = 5cm and $\angle QTS = \angle RPQ$.

Calculate /TR/

Solution

Taking ratio of side facing marked angles **:** no common angle at Q

$$\frac{4}{5+TR} = \frac{5}{4+6}$$

$$4 \times 10 = 5(5 + TR)$$

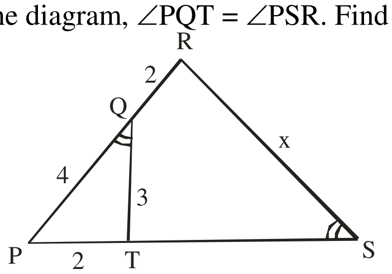
$$40 = 25 + 5TR$$

$$40 - 25 = 5TR$$

$$15 = 5TR \quad \text{thus } 3\text{cm} = TR$$

2005/47 (Nov)

In the diagram, $\angle PQT = \angle PSR$. Find the value of x



A 5 B 6 C 8 D 9

Solution

The ratio of sides facing the marked angles: common angle P

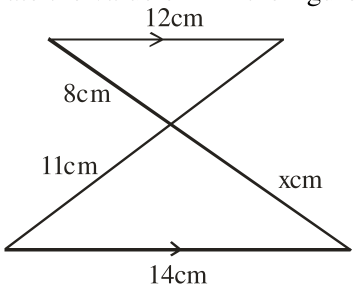
$$\frac{2}{2+4} = \frac{3}{x}$$

$$2x = 3 \times 6$$

$$x = \frac{3 \times 6}{2} = 9 \quad (\text{ D })$$

2008/41 Neco

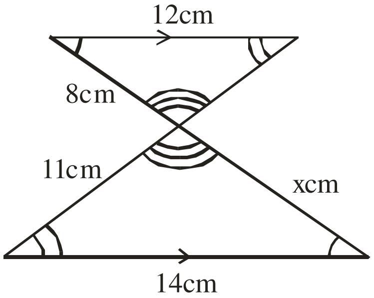
Calculate the value of x in the figure below



A $7^{1}/_{2}$ B $9^{1}/_{3}$ C $9^{3}/_{7}$ D 10 E $10^{2}/_{3}$

Solution

First we identity the equal angles,
applying alternate angles and vertically opposite angles



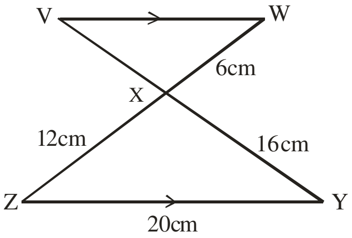
$$\frac{12}{14} = \frac{8}{x}$$

$$12x = 8 \times 14$$

$$x = \frac{8 \times 14}{12}$$

$$= \frac{28}{3} = 9^{1}/_{3}\text{cm} \quad (\text{B})$$

2015/30

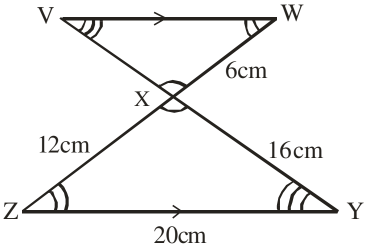


In the diagram, VW//YZ, /WX/ = 6cm, /XY/ = 16cm, /YZ/ = 20cm and /ZX/ = 12cm. Calculate /VX/

A 3cm B 4cm C 6cm D 8cm

Solution

First, we identify the equal angles
applying alternate and vertically opposite angles rule



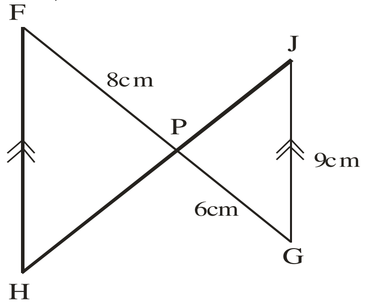
$$\frac{VX}{16} = \frac{6}{12}$$

$$VX = \frac{6 \times 16}{12}$$

$$= 8\text{cm} \quad (\text{D})$$

2007/40 Neco Exercise 19.4

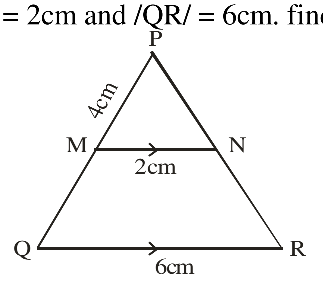
Two lines FG and HJ intersect at P. If FH//JG, /PG/ = 6cm, /FP/ = 8cm and /GJ/ = 9cm find /FH/



A 16cm B 15cm C 13cm D 12cm E 10cm

2004/47 NABTEB (Dec) Exercise 19.5

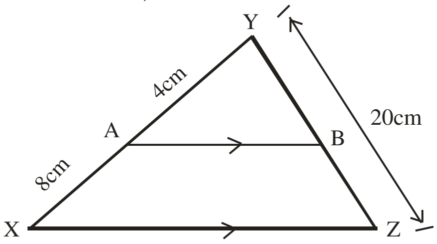
In the figure below MN is parallel to QR. /PM/ = 4cm, /MN/ = 2cm and /QR/ = 6cm. find the length of MQ



A 4cm B 8cm C 12cm D 16cm

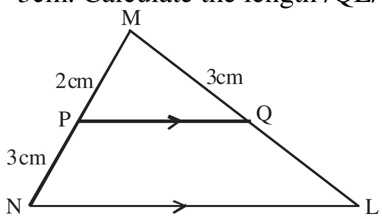
2003/4a NABTEB Exercise 19.6

Calculate the value of /BZ/ in ΔXYZ, if /XA/ = 8cm, /AY/ = 4cm, /YZ/ = 20cm and $\overline{AB} // \overline{XZ}$



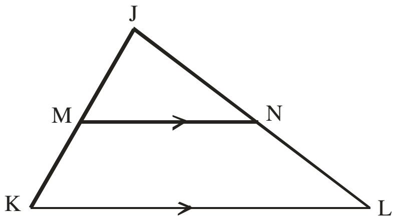
2003/42 NABTEB Exercise 19.7

In the figure below, /MP/ = 2cm, /MQ/ = 3cm, /PN/ = 3cm. Calculate the length /QL/



A 3.5cm B 4.0cm C 4.5cm D 5.0cm

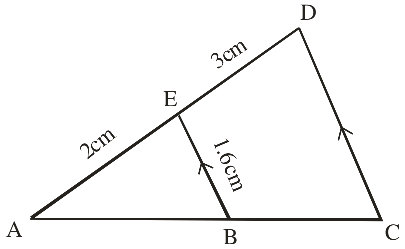
2005/54 Neco Exercise 19.8



In the figure above, $\overline{MN}$ is parallel to $\overline{KL}$, $\overline{JM}$ = 6cm, $\overline{MK}$ = 8cm and $\overline{JL}$ = 21cm. Find $\overline{JN}$

A 6cm B 9cm C 10cm D 12cm E 15cm

2005/26 Exercise 19.9

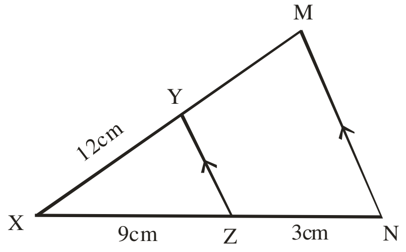


In the diagram, BE//CD, /AE/ = 2cm, /BE/ = 1.6cm and /ED/ = 3cm. Calculate /CD/.

A 2.4cm B 2.67cm C 3.25cm D 4.0cm

2006/46 Exercise 19.10

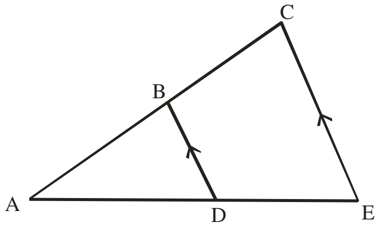
In the diagram, /XY/ = 12cm, /XZ/ = 9cm, /ZN/ = 3cm and ZY// NM. Calculate /MY/



A 3cm B 4cm C 5cm D 6 cm

2011/36 Neco Exercise 19.11

Which of the following is correct about the figure shown below?



A ΔADB is similar to ΔADC
B ΔABD is similar to ΔACE
C AB : AD is similar to BC : DE
D BDEC is similar to ΔABD

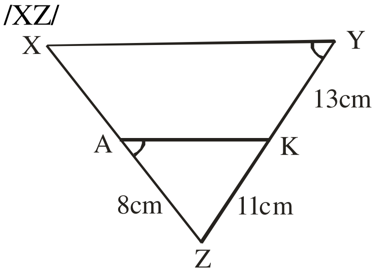
2004/48 NABTEB (Nov) Exercise 19.12

Which of the following statements is always true of two similar triangles? Their

A areas are equal B corresponding sides are equal
C corresponding angles are equal
D shapes are not the same.

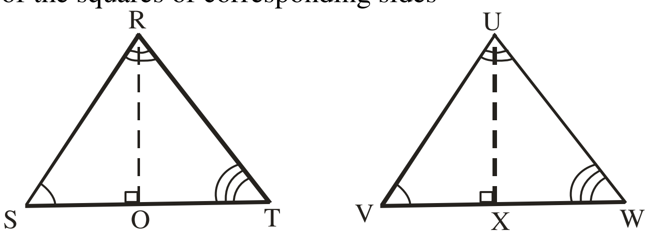
2014/5b Neco Exercise 19.13

In the diagram below, $X\hat{Y}Z = K\hat{A}Z$, /YZ/ = 24cm. /KY/ = 13cm, /ZK/ = 11cm and /AZ/ = 8cm, calculate /XZ/



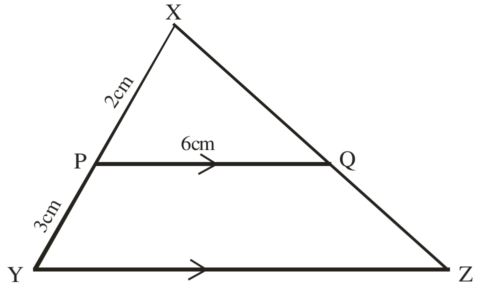
Area of similar triangles

The ratio of the areas of similar triangles is equal to the ratio of the squares of corresponding sides



$$\frac{Area\ of\ \Delta\ RST}{Area\ of\ \Delta\ UVW} = \frac{RS^2}{UV^2} = \frac{RT^2}{UW^2} = \frac{ST^2}{VW^2} = \frac{RO^2}{UX^2}$$

2007/12b



In the diagram, $\overline{PQ}\ //\ \overline{YZ}$, /XP/ = 2cm, /PY/ = 3cm, /PQ/ = 6cm and the area of ΔXPQ = 24cm^2. Calculate the area of the trapezium PQZY

Solution

Area of trapezium PQZY = area of ΔXYZ – XPQ

But $\frac{Area\ of\ \Delta\ XPQ}{Area\ of\ \Delta\ XYZ} = \frac{XP^2}{XY^2}$ (Similar triangles)

$$\frac{24}{Area\ of\ \Delta\ XYZ} = \frac{2^2}{(2+3)^2}$$

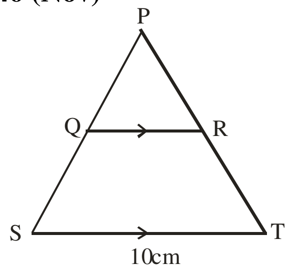
$24 \times 5^2 = 2^2 \times$ Area of Δ XYZ

$\frac{24 \times 5^2}{2^2}$ = Area of Δ XYZ

150cm^2 = area of Δ XYZ

Thus area of trapezium PQZY = 150 – 24
= 126cm^2

2009/46 (Nov)



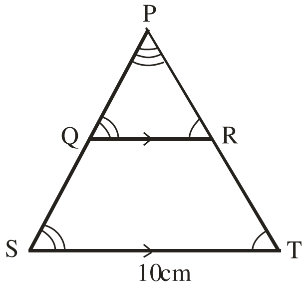
In the diagram, QR is parallel to ST and /ST/ = 10cm. The areas of ΔPQR and quadrilateral QRTS are 9cm^2 and 16cm^2 respectively. Calculate /QR/.

A 5.6cm B 6.0cm C 7.5cm D 8.0cm.

Solution

We are dealing with similar Δs. Applying corresponding angles rule in parallel lines and transversals to indicate the angles.

Here P is common angle.



$$\frac{Area\ \ of\ \ \Delta PQR}{Area\ \ of\ \ \Delta PST} = \frac{QR^2}{ST^2}$$

But Area of ΔPST = Area of ΔPQR + Area of quad QRTS
$= 9 + 16$
$= 25cm^2$

$$\text{Thus } \frac{9}{25} = \frac{QR^2}{10^2}$$

$$9 \times 10^2 = 25 \times QR^2$$

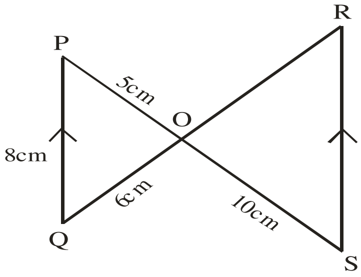
$$\frac{9 \times 100}{25} = QR^2$$

$$QR = \sqrt{36}$$

$= 6cm$ (B)

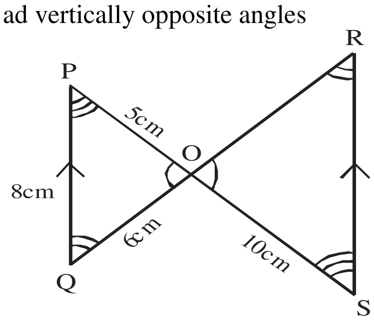
2005/6c (Nov)

In the diagram, QP is parallel to SR and /PQ/ = 8cm, /QO/ = 6cm, /PO/ = 5cm and /SO/ = 10cm.
If the area of ΔPOQ = $40cm^2$, Calculate the area of ΔROS



Solution

Ratio of the areas of similar triangles is equal to the ratio of the squares of corresponding sides, To identify the corresponding sides we mark out equal angles by applying alternate ad vertically opposite angles



$$\frac{Area\ \ of\ \Delta\ POQ}{Area\ of\ \Delta\ ROS} = \frac{5^2}{10^2}$$

Note: We maintain the order of the Δs, Taking the smaller one first in area and sides at numeration

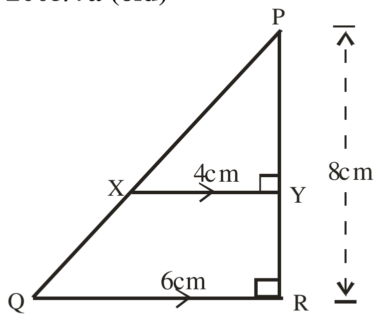
$$\frac{40}{area\ \ of\ \ \Delta\ ROS} = \frac{25}{100}$$

$40 \times 100 = 25$ area of ΔROS

$$\frac{40 \times 100}{25} = \text{Area of } \Delta ROS$$

$160cm^2$ = Area of ΔROS

2005/7a (old)



In the diagram, XY//QR, /XY/ = 4cm, /QR/ = 6cm and /PR/ = 8cm. Calculate the area of ΔPXY

Solution

Two triangles are similar (RHS)

$$\frac{Area\ \ of\ \Delta\ PXY}{Area\ of\ \Delta\ PQR} = \frac{XY^2}{QR^2}$$

Note that, we maintain order; small Δ at numerator on both sides

But Area of Δ PQR = $\frac{1}{2} \times 6 \times 8 \sin 90^0$
$= 24cm^2$

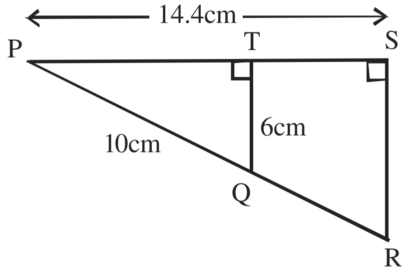
Substituting

$$\frac{Area\ \ of\ \Delta\ PXY}{24} = \frac{4^2}{6^2}$$

$$\text{Area of } \Delta\ PXY = \frac{16 \times 24}{6 \times 6}$$

$= 10.67cm^2$

2010/11a



In the diagram, $\angle PTQ = \angle PSR = 90^0$, /PQ/ = 10cm, /PS/ = 14.4cm and /TQ/ = 6cm. Calculate the area of quadrilateral QRST

Solution

Area of quad QRST = Area of Δ PRS – area of Δ PQT
The two triangles here are similar (RHS)

$$\frac{Area\ \ of\ \Delta\ PQT}{Area\ of\ \Delta\ PRS} = \frac{PT^2}{14.4^2}$$

But by Pythagoras rule in ΔPQT

$$10^2 = 6^2 + PT^2$$

$$PT = \sqrt{64} = 8cm$$

Thus area of ΔPQT = $\frac{1}{2} \times 6 \times 8 \times \sin 90^0$ (sin 90 = 1)
$= 24cm^2$

Substituting

$$\frac{24}{Area\ of\ \Delta\ PRS} = \frac{8^2}{14.4^2}$$

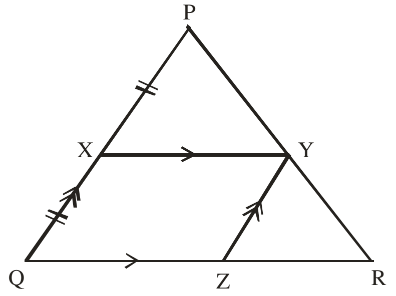
$24 \times 14.4^2 = 8^2 \times$ area of ΔPRS

$$\frac{24 \times 14.4^2}{8^2} = \text{Area of } \Delta PRS$$

$77.76cm^2$ = Area of ΔPRS

Area of quad QRST = 77.76 – 24
$= 53.76cm^2$

2008/24 ***counter example***
In the figure /PX/ = /XQ/, PQ//YZ and XY//QR.
What is the ratio of the area of XYZQ to the area of ΔYZR?



A 1 : 2 B 2 : 1 C 1 : 3 D 3 : 1

Solution

XYZQ is a parallelogram and YZR is a triangle both in same base QR and between same parallel XY and QR. Thus

$\frac{1}{2}$ Area parallelogram = area of Δ

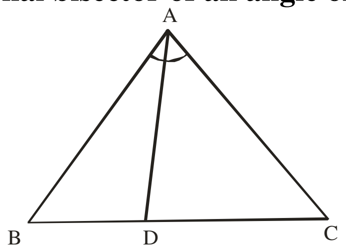
Multiply through by 2

Area of parallelogram = 2 Area of Δ

1 × YZQ : 2 YZR

1 : 2 (A)

Internal bisector of an angle of a triangle

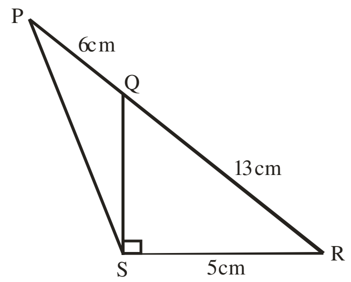


Angle A is bisected by line AD

The internal bisector of an angle of a triangle divides the opposite side in the ratio of the sides containing the angle

$$\frac{BD}{DC} = \frac{AB}{AC}$$

2012/13(a)



In the diagram, /PQ/ = 6cm, /QR/ = 13cm, /RS/ = 5cm and ∠RSQ is a right angle. Calculate, correct to one decimal place, /PS/

Solution

By the theorem of internal bisector of an angle of a triangle

$$\frac{QR}{PQ} = \frac{SR}{PS}$$

$$\frac{13}{6} = \frac{5}{PS}$$

13 PS = 6 × 5

PS = $\frac{30}{13}$

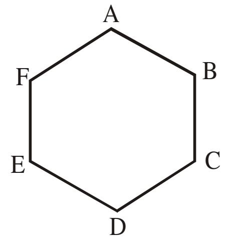
= 2.3077cm ≈ 2.3cm to 1d.p

Polygons

A closed figure bounded by three or more straight sides is called a polygon. A polygon is regular; if it has all its sides and angles equal, otherwise it is irregular.

Sum of interior and exterior angles

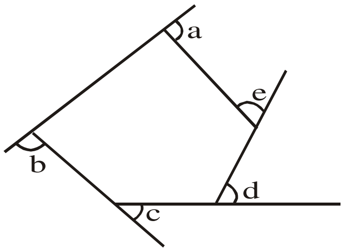
The sum of interior angles of an n – sided polygon is (2n – 4) x 90^0



A + B + C + … = (2n – 4) x 90^0 and

Each interior angles of n – sides = $\frac{(2n-4) \times 90^0}{n}$

While the sum of the exterior angles of an n – sided polygon is 4 × 90^0



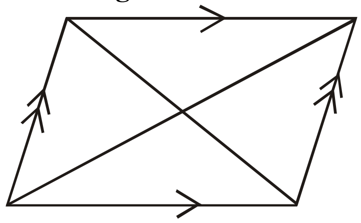
a + b + c + … = 360^0 and

each exterior angle of n sides = $\frac{4 \times 90^0}{n}$

Quadrilaterals

Quadrilaterals are four sided figures. Quadrilaterals whose *diagonals* bisect each other at right angles are square, kite and rhombus. Below is a list of some quadrilaterals and their properties:

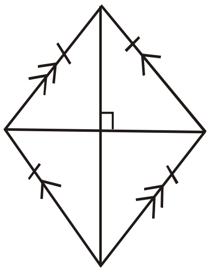
Parallelogram



Opposite sides are equal and parallel.
Opposite angles are equal and the diagonals bisect each other. *Note:* other parallelograms among the quadrilaterals are rhombus, rectangle and square; NOT kite & trapezium

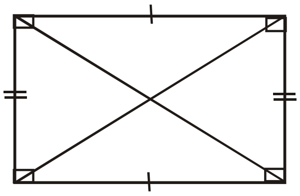
Rhombus

All sides are equal, opposite angles are equal and the diagonals bisect each other at right angles.



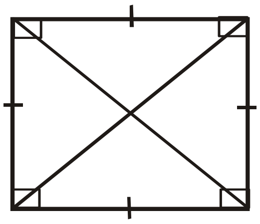
Rectangle

Opposite sides are equal and the angles are 90^0 .
The diagonal bisect each other and are of equal length.



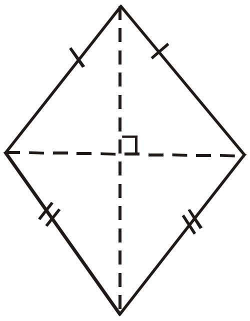
Square

All sides are equal and the angles are 90^0. The diagonals bisect each other at right angles and are of equal length



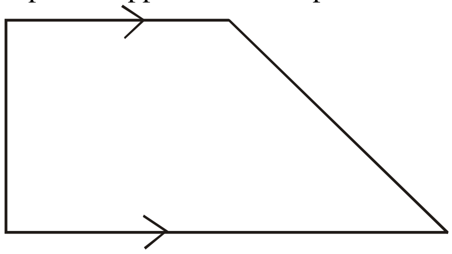
Kite

Adjacent sides are equal and the diagonals bisect each other at right angles



Trapezium

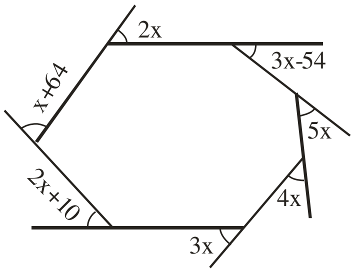
One pair of opposite sides is parallel



Polygons table

Name	No of sides
Triangle	3
Quadrilateral	4
Pentagon	5
Hexagon	6
Heptagon	7
Octagon	8
Nonagon	9
Decagon	10

2009/23



Find the value of x in the diagram

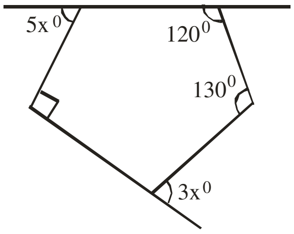
A 50^0 B 30^0 C 22^0 D 17^0

Solution

Sum of exterior angles of a polygon = 360^0

$$2x + x + 64 + 2x + 10 + 3x + 4x + 5x + 3x – 54 = 360$$
$$20x + 74 – 54 = 360$$
$$20x + 20 = 360$$
$$20x = 360 – 20$$
$$20x = 340$$
$$x = {}^{340}/_{20}$$
$$= 17^0 \quad (D)$$

2007/27

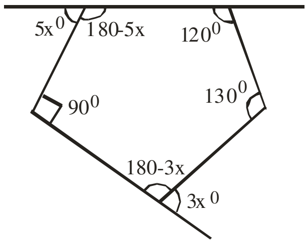


From the diagram, find the value of x

A 40 B 30 C 25 D 20

Solution

We re-label the diagram to reflect all the interior angles

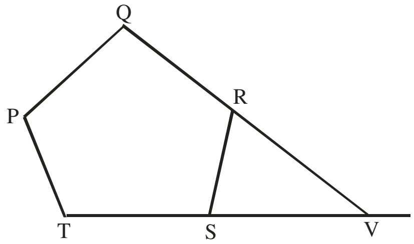


It is five sided polygon

Sum of interior angle of polygon = (n – 2)180

$$180 – 5x + 90 + 180 – 3x + 130 + 120 = (5 – 2)180$$
$$700 – 8x = 540$$
$$700 – 540 = 8x$$
$$160 = 8x$$
$$20 = x \quad (D)$$

2013/37



In the diagram, PQRST is a regular polygon with side QR and TS produced to meet at V. Find the size of $\angle RVS$

A 36^0 B 54^0 C 60^0 D 72^0

Solution

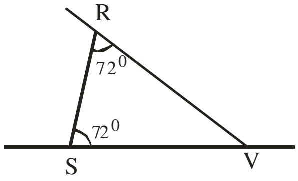
The polygon PQRST is five sided. In this case all the sides and interior angles are equal.

Sum of interior angle of polygon = (n – 2)180
= (5 – 2)180
= 540

Each of the interior angle = $^{540}/_5$
= 108^0

Exterior angle at R is same as at S

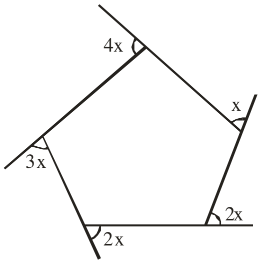
But exterior angle at R = 180 – 108
= 72^0



Thus $\angle RVS + 72 + 72 = 180^0$ (sum of $\angle$s in Δ).

$\angle RVS = 180 - 144$
$= 36^0$ (A)

2012/47



The diagram is a polygon. Find the largest of its interior angles

A 30^0 B 100^0 C 120^0 D 150^0

Solution

The figure is a five sided polygon with given exterior angles

The largest interior angle is gotten from the smallest exterior angle x

Sum of exterior angles of polygon = 360

4x + 3x + 2x + 2x + x = 360

12x = 360

x = $^{360}/_{12}$

x = 30

x + interior angle = 180 (sum of angles in straight line)

30 + interior angle = 180

Interior angle = 180 – 30
= 150^0 (D)

2012/40

The sum of the interior angle of a regular polygon is 1800^0. How many sides has the polygon?

A 16 B 12 C 10 D 8

Solution

Sum of the interior angles of a polygon = (n – 2)180

1800 = (n – 2)180

$$\frac{1800}{180} = \frac{(n-2)180}{180}$$

10 = n – 2

10 + 2 = n

12 = n (B)

2005/5a Neco

The interior angle of a heptagon are x^0, $(2x - 15)^0$, $(x + 10)^0$, $(3x + 5)^0$, $(2x - 34)^0$, $(x + 5)^0$ and $(2x + 5)^0$. Find the value of x

Solution

Heptagon is a 7 sided polygon

Sum of interior angles of a polygon = (n – 2)180

x + (2x –15)+ (x + 10) + (3x + 5)+ (2x –34)+(x + 5)+(2x + 5) = (7– 2)180

x + 2x – 15 + x + 10 + 3x + 5 + 2x – 34 + x + 5 + 2x + 5 = 5 ×180

12x + 25 – 49 = 900

12x – 24 = 900

12x = 900 + 24

12x = 924

x = $^{924}/_{12}$

= 77^0

2009/6b Neco (Dec)

The interior angles of a pentagon x^0, $(x + 10)^0$, $(2x - 25)^0$, $(3x + 5)^0$ and $(x - 10)^0$,

(i) Find x

(ii) What is the value of the biggest interior angle?

(iii) Determine the exterior angle at the vertex where the interior angle is $(2x - 25)^0$

Solution

Pentagon is 5 sided polygon

Sum of interior angle of polygon = (n – 2)180

x + (x + 10) + (2x – 25) + (3x + 5) + (x – 10) = (5 – 2)180

x + x + 10 +2x – 25 + 3x +5 + x – 10 = 3 ×180

8x + 15 – 35 = 540

8x – 20 = 540

8x = 540 + 20

$$x = \frac{560}{8} = 70^0$$

(ii) Biggest angle is $(3x + 5)^0 = 3 \times 70 + 5$
$= 210 + 5 = 215^0$

(iii) Exterior angle + interior angle = 180^0
(sum of $\angle$s on straight line)

Exterior angle + (2x – 25) = 180

Exterior angle = 180 – (2x – 25)
= 180 – (2 × 70 – 25)
= 180 – (140 – 25)
= 180 – 115
= 65^0

2008/23

The angles of a quadrilateral are $(x + 10)^0$, $2y^0$, 90^0 and $(100 - y)^0$. Find y in terms of x.

A y = 160 + x B y = 100 + x C y = 160 – x

D y = x – 100

Solution

A quadrilateral is a four sided polygon

Sum of interior angle of polygon = (n – 2)180

$x + 10 + 2y + 90 + 100 - y = (4 - 2)180$

$x + y + 200 = 360$

$y = 360 - 200 - x$

$= 160 - x$ (C)

2007/36 Neco

If the exterior angles of a quadrilateral are $2y^0$, $(3y - 5)^0$, $(3y - 15)^0$ and $(5y - 10)^0$, find y

A 61.43^0 B 60^0 C 52.86^0 D 51.43^0 E 30^0

Solution

Quadrilateral is a 4 sided figure

Sum of exterior angles of polygon = 360

$2y + (3y - 5) + (3y - 15) + (5y - 10) = 360$

$2y + 3y - 5 + 3y - 15 + 5y - 10 = 360$

$13y - 30 = 360$

$13y = 360 + 30$

$13y = 390$

$y = {}^{390}/_{13}$

$= 30^0$ (E)

2008/4a

A pentagon is such that one of its exterior angles is 60^0. Two other are $(90 - m)^0$ each while the remaining angles are $(30 + 2m)^0$ each. Find the value of m

Solution

Sum of exterior angles of polygon = 4 × 90

Here pentagon has 5 sides (angles)

$60 + 2 \times (90 - m) + 2(30 + 3m) = 4 \times 90$

So after one, two angles, the remaining must be two angles as well

$60 + 180 - 2m + 60 + 4m = 360$

$4m - 2m + 300 = 360$

$2m = 60$

$m = {}^{60}/_{2} = 30^0$

2005/7a ***counter example***

A regular polygon of n sides is such that each interior angle is 120^0 greater than the exterior angle.

Find : (i) the value of n

(ii) the sum of all the interior angles

Solution

Interior angle = exterior + 120^0

Recall that:

interior angle + exterior angle = 180(sum of angles on straight line)

It will be easier to work with exterior angle here. Thus

exterior angle + 120 + exterior angle = 180

2exterior angle + 120 = 180

2exterior angle = 60

Exterior angle = ${}^{60}/_{2} = 30^0$

But each exterior angle of a polygon = $\frac{360}{n}$

$30 = \frac{360}{n}$

$30n = 360$

$n = \frac{360}{30}$

$= 12$

(ii) Sum of interior angles = $(2n - 4)90$

$= (2 \times 12 - 4)90$

$= (24 - 4)90$

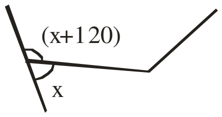
$= 20 \times 90 = 1800$

Alternatively (by algebraic representation)

Let each exterior angle be x

Each interior angle will be x + 120,

We sketch as :



$x + 120 + x = 180$ (sum of angles in straight line)

$2x + 120 = 180$

$2x = 180 - 120$

$2x = 60$

$x = {}^{60}/_{2} = 30^0$

Other part follows as solved above

2013/42

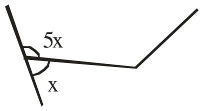
An interior angle of a regular polygon is 5 times each exterior angle. How many sides has the polygon?

A 15 B 12 C 9 D 6

Solution

Let each exterior angle be x

Then each interior angle will be 5x



$5x + x = 180$(Sum of ∠s on straight line)

$6x = 180$

$\frac{6x}{6} = \frac{180}{6}$

$x = 30^0$

But each exterior angle of polygon = $\frac{360}{n}$

$30 = \frac{360}{n}$

$30n = 360$

$n = \frac{360}{30}$

$= 12$ (B)

2011/7b Neco

The interior angle of a regular polygon is four times the exterior angle. Calculate the:

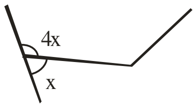
(i) exterior angle;

(ii) number of sides of the polygon

Solution

Let each exterior angle be x

Then each interior angle will be 4x



$4x + x = 180$ (sum of ∠s on straight line)

$5x = 180$

$\frac{5x}{5} = \frac{180}{5}$

$x = 36$

(ii) Each exterior angle of polygon = $\frac{360}{n}$

$$36 = \frac{360}{n}$$
$$36n = 360$$
$$n = \frac{360}{36} = 10 \text{ sides}$$

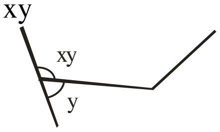
2013/4a Neco ***counter example***

If the interior angle of a regular polygon is x times the exterior angle, express the number of sides of the polygon in terms of x

Solution

Let each exterior angle be y

Then each interior angle will be xy



$xy + y = 180^0$ (sum of $\angle$s on straight line)

$y(x + 1) = 180$

$$y = \frac{180}{x+1}$$

Each exterior angle of polygon = $\frac{360}{n}$

$$\frac{180}{x+1} = \frac{360}{n}$$

Make n subject of formula

$$180n = 360(x + 1)$$
$$n = \frac{360(x+1)}{180}$$
$$n = 2(x + 1)$$

2006/6b Neco ***counter example***

A polygon has three of its interior angles to be 160^0 each and others are 120^0 each. Calculate the number of sides of the polygon

Solution

Sum of interior angles in polygon = $(n - 2) \times 180$

The sum of its three interior angles = $160 \times 3 = 480$

sum of the remaining interior angles = $(n - 3) \times 120$

Recall that sum of interior angles in polygon = $(n - 2) \times 180$

$$480 + (n - 3)120 = (n - 2)180$$
$$480 + 120n - 360 = 180n - 360$$
$$480 - 360 + 360 = 180n - 120n$$
$$480 = 60n$$
$$\frac{480}{60} = \frac{60n}{60}$$
$$8 = n$$

2012/40 Neco

Two of interior angles of a polygon are 63^0 each and the remaining interior angles are 166^0 each. How many sides has the polygon?

A 7 B 9 C 10 D 11 E 13

Solution

Sum of interior angles in polygon = $(n - 2)180$

The sum of its two interior angles = $63 \times 2 = 126^0$

Sum of the remaining interior angles = $(n - 2) \times 166$

Recall that: Sum of interior angles in polygon = $(n - 2)180$

$126 + (n - 2)166 = (n - 2)180$

$126 + 166n - 332 = 180n - 360$

$126 - 332 + 360 = 180n - 166n$

$$154 = 14n$$
$$\frac{154}{14} = \frac{14n}{14}$$
$$11 = n \quad (D)$$

2014/2a Neco (Dec) ***interior angles***

One interior angle of a hexagon is 140^0

Calculate, to the nearest degree, each of the remaining angles, given that they are all equal to each other.

Solution

We are dealing with 6 sided polygon

Sum of interior angle of polygon = $(n - 2)180$

Out of the six angles, one is 140

Let x be each of the five angles left which are equal.

That is 5x

$$140 + 5x = (6 - 2)\,180$$
$$140 + 5x = 4 \times 180$$
$$5x = 720 - 140$$
$$5x = 580$$
$$x = \frac{580}{5}$$
$$= 116^0$$

2013/40 Neco ***interior angles***

The sum of six angles of an eleven sided polygon is 1000^0. The other five angles are equal. Find the size of each of the equal angles

A 47^0 B 77^0 C 92^0 D 124^0 E 154^0

Solution

We are dealing with 11 sided polygon

Since the angle is not specified to be exterior,

it must be interior angle

Sum of interior angles of polygon = $(n - 2)180$

Out of the eleven angles; six are 1000

Let x be each of the five angles left which are equal.

That is 5x

$$1000 + 5x = (11 - 2)180$$
$$1000 + 5x = 9 \times 180$$
$$5x = 1620 - 1000$$
$$5x = 620$$
$$x = \frac{620}{5} = 124^0 \quad (D)$$

2005/7b Neco ***interior angles***

Three of the angles of a hexagon are 122^0, 71^0 and 95^0 and the other three angles are in the ratio 2 : 3 : 3. Calculate the size of each of the unknown angles.

Solution

Hexagon is a six sided polygon

Since the angle is not specified to be exterior

It must be interior angle

Sum of interior angles of polygon = $(n - 2)180$

Three of the six angles are given

Let x be the value of the angles which are in ratio 2:3:3

$$2x : 3x : 3x$$

Resulting to 8x

Thus $122 + 71 + 95 + 8x = (6 - 2)180$

$$288 + 8x = 4 \times 180$$
$$8x = 720 - 288$$

$$x = \frac{432}{8}$$

$$x = 54^0$$

$\therefore$ $2x = 2 \times 54 = 108^0$

$3x = 3 \times 54 = 162^0$

$3x = 3 \times 54 = 162^0$

2010/27

The sum of the exterior angle of an n – sided convex polygon is half the sum of its interior angles. Find n

A 6 B 8 C 9 D 12

Solution

Sum of exterior angle = $\frac{1}{2}$ sum interior angle

$$4 \times 90 = \tfrac{1}{2}(2n - 4) \times 90$$

90 will cancel out

$$4 = n - 2$$

$$4 + 2 = n$$

$$6 = n \quad (A)$$

2011/11

A regular polygon of n sides has each exterior angle equal to 45^0. Find the value of n

A 6 B 8 C 12 D 12

Solution

Each exterior angle of polygon = $\frac{360}{n}$

$$45 = \frac{360}{n}$$

$$45n = 360$$

$$n = \frac{360}{45}$$

$$= 8 \quad (B)$$

2009/38 (Nov)

A regular polygon has 15 sides. Find the size of each of the exterior angle

A 42^0 B 30^0 C 24^0 D 18^0

Solution

Each exterior angle of polygon = $\frac{360}{n}$

$$= \frac{360}{15}$$

$$= 24^0 \quad (C)$$

2009/39

Each of the interior angles of a regular polygon is 140^0. Calculate the sum of all the interior angles of the polygon

A 1080 B 1260 C 1800 D 2160

Solution

Each interior angle of polygon = $\frac{(n-2)180}{n}$

$$140 = \frac{(n-2)180}{n}$$

$$140n = (n - 2)\ 180$$

$$140n = 180n - 360$$

$$360 = 180n - 140n$$

$$360 = 40n$$

$$\frac{360}{40} = n \quad \text{i.e} \quad n = 9$$

Sum of interior angles of polygon = $(n - n)180$

$= (9 - 2)180$

$= 7 \times 180$

$= 1260^0$ (B)

2007/21 Exercise 19.14

Each interior angle of a regular polygon is 162^0. How many sides has the polygon?

A 8 B 12 C 16 D 20

2007/ 28 Neco Exercise 19.15

The sum of the interior angles of a polygon is 26 right angles. How many sides does the polygon have?

A 45 B 35 C 25 D15 E 5

2011/39 Neco Exercise 19.16

How many sides has a regular polygon if each of the interior angles is 120^0?

A 8 B 6 C 5 D 4 E 3

2014/49 Neco Exercise 19.17

The sum of the interior angles of a regular polygon is 3060^0. How many sides has the polygon?

A 9 B 15 C 17 D 19 E 21

2006/37 Neco (Dec) Exercise 19.18

What is the name of a regular polygon in which one of its interior angles is 135^0?

A Decagon B Heptagon C Hexagon
D Nonagon E Octagon

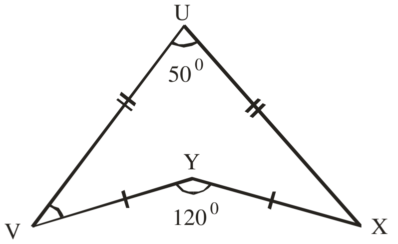
2014/2b Neco (Dec) Exercise 19.19

A polygon has the following interior angles. k^0, $(k + 15)^0$, $(k + 30)^0$, $(k + 35)^0$, $(k + 45)^0$ and $(k + 55)^0$. Find the value of k

Other cases

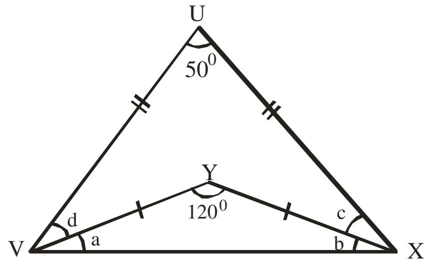
2010/39 Neco

In the diagram below, /UV/ = /UX/ and /VY/ = /YX/, $\angle VUX = 50^0$ and $\angle VYX = 120^0$. Find $\angle UVY$



A 170^0 B 70^0 C 65^0 D 35^0 E 30^0

Solution



In ΔVXY,

$a = b$ (base $\angle$s in IsoscelesΔ)

$120 + a + a = 180^0$ (sum of $\angle$s in Δ)

$2a = 180 - 120$

$a = {}^{60}/_2 = 30^0$

In ΔUVX,

$a + d = b + c$ (Base $\angle$s of isoscelesΔ)

$30 + d = 30 + c$

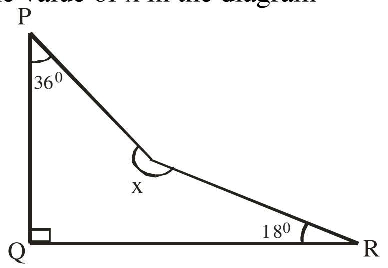
$50 + 30 + d + 30 + d = 180^0$ (sum of $\angle$s in Δ)

$2d = 180 - 110$

$d = {}^{70}/_2 = 35^0$ (D)

2009/22 (Nov)

Find the value of x in the diagram



A 184^0 B 196^0 C 204^0 D 216^0

Solution

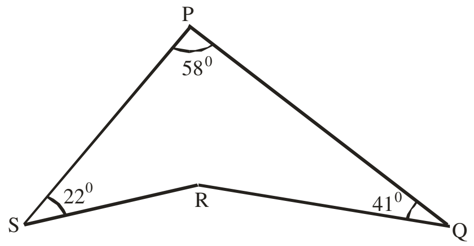
$36 + 90 + 18 + x = 360^0$ (sum of $\angle$s in Quad)

$144 + x = 360$

$x = 360 - 144$

$= 216^0$ (D)

2010/3



In the diagram, $\angle PSR = 22^0$, $\angle SPQ = 58^0$ and $\angle PQR = 41^0$. Calculate the obtuse angle QRS

A 99^0 B 100^0 C 121^0 D 165^0

Solution

Obtuse angles are between 90^0 and 180^0

$58 + 22 + \text{reflex angle } QRS + 41 = 360^0$ (sum of $\angle$s in Quad)

$121 + \text{reflex } \angle QRS = 360$

$\text{Reflex } \angle QRS = 360 - 121$

$= 239$

$\text{Reflex } \angle QRS + \text{obtuse } \angle QRS = 360^0$ (sum of $\angle$s at a point)

$\text{Obtuse } \angle QRS = 360 - 239$

$= 121^0$ (C)

2012/33 Neco

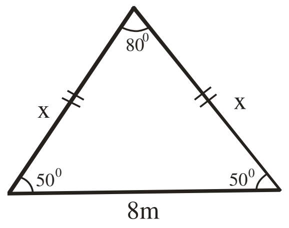
The vertex angle of an isosceles triangle is 80^0. If the base is 8cm long, what is the length of each of the equal sides?

A 6.2cm B 8.0cm C 8.8cm D 9.0cm E 12.4cm

Solution

The isosceles triangle's other two equal angles are

$= \frac{180-80}{2} = 50^0$ each



Applying sine rule: $\frac{8}{\sin 80} = \frac{x}{\sin 50}$

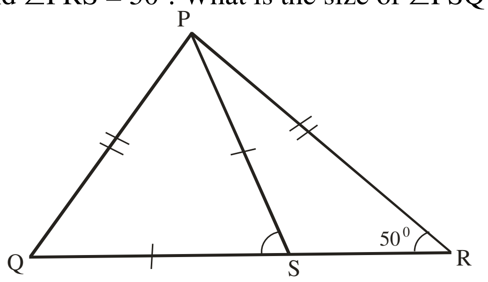
$\frac{8\sin 50}{\sin 80} = x$

$\frac{8 \times 0.7660}{0.9848} = x$

$6.2\text{cm} = x$ (A)

2006/33 Neco (Dec)

Below is ΔPQR such that /PQ/ = /PR/, /SQ/ = /PS/ and $\angle PRS = 50^0$. What is the size of $\angle PSQ$?



A 25^0 B 50^0 C 80^0 D100^0 E 110^0

Solution

$\angle PQS = 50^0$ (base angles of Isosceles ΔPQR)

$\angle QPS = 50^0$ (Base angles of Isosceles Δ PSQ)

In ΔPSQ

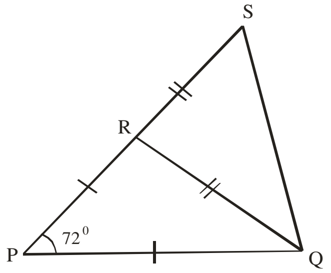
$\angle PQS + \angle QPS + \angle PSQ = 180^0$ (sum of angles in Δ)

$50 + 50 + \angle PSQ = 180$

$\angle PSQ = 180 - 100$

$= 80^0$ (C)

2006/52 Neco



In the Figure above, $\overline{PQ} = \overline{PR}$, $\overline{RQ} = \overline{RS}$ and PRS is a straight line. If $\angle QPR = 72^0$, find $\angle QSR$

A 72^0 B 63^0 C 54^0 D 36^0 E 27^0

Solution

In ΔQPR,

Let $x = \angle PRQ = \angle PQR$ (Base angles of isosceles ΔQPR)

$72 + x + x = 180^0$ (sum of $\angle$s in Δ)

$2x = 180 - 72$

$x = {}^{108}/_2 = 54^0$

Next, $\angle PRQ + \angle QRS = 180^0$ (sum of $\angle$s on straight line)

$54 + \angle QRS = 180$

$\angle QRS = 180 - 54$

$= 126^0$

In ΔQRS

Let $y = \angle QSR = \angle RQS$ (Base angles of isosceles ΔQRS)

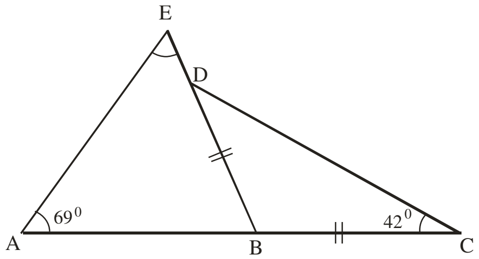
$126 + y + y = 180^0$ (sum of $\angle$s in ΔQRS)

$2y = 180 - 126$

$y = {}^{54}/_2$

$= 27^0$ (E)

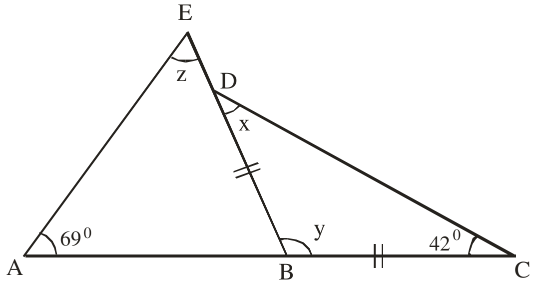
2006/59 Neco



In the diagram above, AEB and BCD are two triangles on the straight line AC. If $\overline{BC} = \overline{BD}$ and $\angle EAB = 69^0$ and $\angle BCD = 42^0$, find $\angle AEB$

A 96^0 B 84^0 C 69^0 D 48^0 E 27^0

Solution



$x = 42^0$ (Base angle of isosceles ΔDBC)

$x + y + 42^0 = 180^0$ (sum of $\angle$s in Δ)

$42 + y + 42 = 180$

$y = 180 - 84$

$= 96$

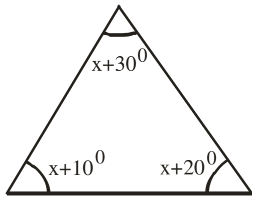
Next, $69 + z = 96$ (exterior $\angle$s and interior opposite $\angle$s in Δ)

$z = 96 - 69$

$= 27^0$ (E)

2012/43 Neco

Find the value of x in the triangle below



A 10^0 B 20^0 C 30^0 D 40^0 E 50^0

Solution

$(x + 30^0) + (x + 10^0) + (x + 20^0) = 180^0$ (sum of $\angle$s in Δ)

$x + 30 + x + 10 + x + 20 = 180$

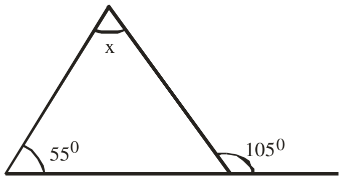
$3x + 60 = 180$

$3x = 180 - 60$

$3x = 120$

$x = {}^{120}/_3 = 40^0$ (D)

2012/36



Find the value of x in the above diagram

A 50^0 B 55^0 C 100^0 D 105^0 E 110^0

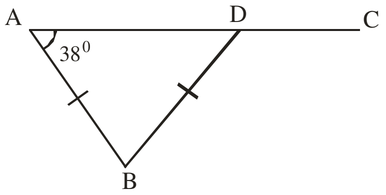
Solution

$x + 55 = 105^0$ (exterior$\angle$s and interior opposite $\angle$s in Δ)

$x = 105 - 55$

$= 50^0$ (A)

2013/41 Neco



In the diagram above /AB/ = /BD/ and $\angle DAB = 38^0$. Find $\angle BDC$

A 38^0 B 71^0 C 104^0 D 109^0 E 142^0

Solution

$\angle ADB = 38^0$ (Base angle of isosceles Δ)

$\angle ADB + \angle BDC = 180^0$ (sum of $\angle$s on straight line)

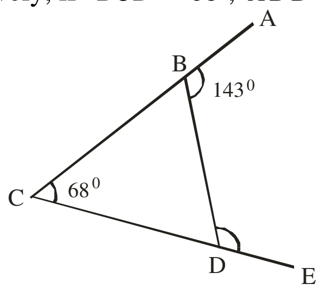
$38 + \angle BDC = 180$

$\angle BDC = 180 - 38$

$= 142^0$ (E)

2009/40 Neco (Dec)

Given triangle BCD with $\overline{CB}$ and $\overline{CD}$ produced to A and E respectively, if $B\hat{C}D = 68^0$, $A\hat{B}D = 143^0$. Find $B\hat{D}E$



A 37^0 B 68^0 C 75^0 D 105^0 E 112^0

Solution

$\angle CBD + 143^0 = 180^0$ (sum of angles on straight line)

$\angle CBD = 180 - 143$

$= 37^0$

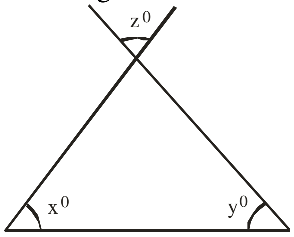
$68 + \angle CBD = \angle BDE$ (exterior $\angle$s and interior opp $\angle$s in Δ)

$68 + 37 = \angle BDE$

$105^0 = \angle BDE$ (D)

2006/43

In the diagram, what is x + y in terms of z?



A $180^0 - z$ B $180^0 + z$ C $z - 180^0$ D z

Solution

Exterior z = interior z (vertically opp. $\angle$s)

$x + y + z = 180^0$ (sum of $\angle$s in Δ)

$x + y = 180 - z$ (A)

2005/17 (old)

The angles of a triangle are in the ratio 5:3:2.
What is the size of the smallest angles?

A 90^0 B 72^0 C 54^0 D 36^0

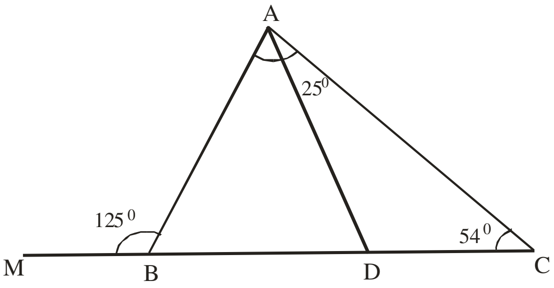
Solution

Sum of angles in Δ is 180^0,
sum of ratio is 5 + 3 + 2 = 10

Smallest angle = $\frac{2}{10} \times 180$

$= 36^0$

2005/57 Neco



In the diagram above, $\angle DCA = 54^0$, $\angle DAC = 25^0$ and $\angle ABM = 125^0$. Calculate $\angle BAD$

A 134^0 B 101^0 C 79^0 D 55^0 E 46^0

Solution

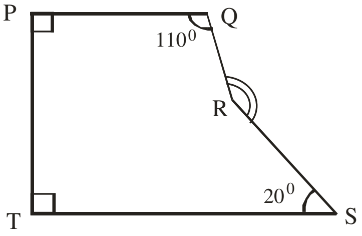
$125^0 = (\angle BAD + 25) + 54^0$ *(Exterior angle equal sum of opposite interior angles)*

$125 = \angle BAD + 25 + 54$

$125 - 79 = \angle BAD$

$46^0 = \angle BAD$ (E)

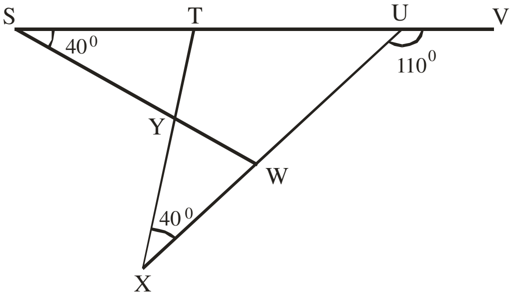
2014/15 Exercise 19.20



In the diagram, $\angle QPT = \angle PTS = 90^0$, $\angle PQR = 110^0$ and $\angle TSR = 20^0$. Find the size of the obtuse angle QRS

A 140^0 B 130^0 C 120^0 D 110^0

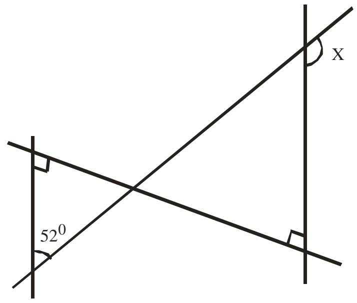
2011/37 Exercise 19.21



In the diagram, STUV is a straight line. $\angle UXY = 40^0$ and $\angle VUW = 110^0$. Calculate $\angle$ TYW

A 150^0 B 140^0 C 130^0 D 120^0

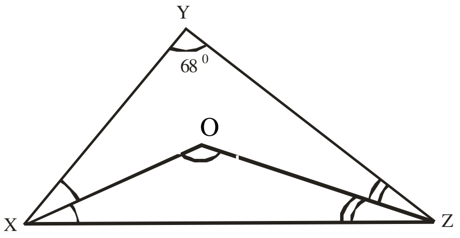
2011/10 Exercise 19.22



Find the size of the angle marked x in the diagram

A 108^0 B 112^0 C 128^0 D 142^0

2015/50 Exercise 19.23

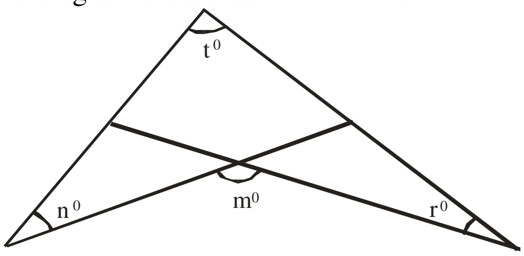


In the diagram, $\overline{OX}$ bisects $\angle YXZ$ and $\overline{OZ}$ bisect $\angle YZX$. If $\angle XYZ = 68^0$, Calculate the value of $\angle XOZ$

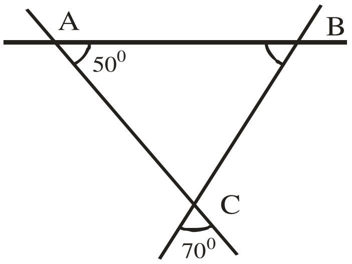
A 68^0 B 72^0 C 112^0 D 124^0

2005/9a (Nov) Exercise 19.24

In the diagram show that r = m – n – t

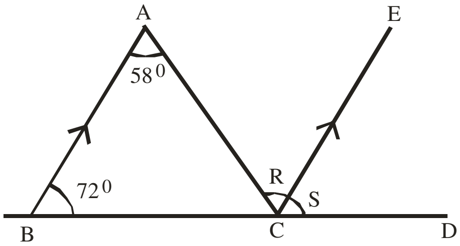


2011/40 Neco Exercise 19.25
In the diagram below, what is the value of $\angle$ABC ?



A 130^0 B 120^0 C 70^0 D 60^0 E 50^0

2013/35 and 36 Neco Exercise 19.26

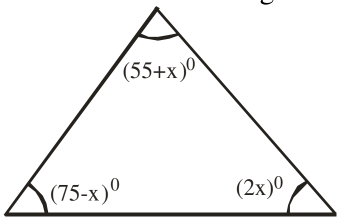


Use the figure above to answer questions 35 and 36

35. What is the value of R?
A 14^0 B 58^0 C 72^0 D 122^0 E 130^0

36. The $\angle$ACD equals
A 14^0 B 58^0 C 72^0 D 122^0 E 130^0

2006/41 Neco (Dec) Exercise 19.27
Find the value of x in the diagram below

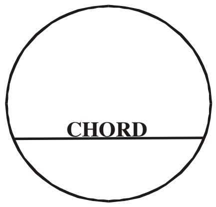


A 50^0 B 25^0 C 20^0 D 18^0 E 12^0

2007/25 Exercise 19.28
In a ΔPQR, $\angle PQR = \angle PRQ = 45^0$. Which of the following statements is/are correct?
I. ΔPQR is an equilateral triangle
II. ΔPQR is an isosceles triangle
III. ΔPQR is a right – angled triangle
A II only B I and II only C II and III only
D I and II only

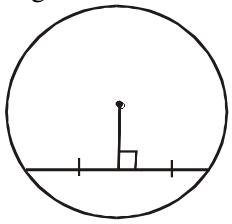
2005/16 (Old) Exercise 19.29
The angles of a triangle are $(2x-30)^0$, $(2x-60)^0$ and $(x-30)^0$. What is the value of x?
A 105 B 60 C 40 D 30

CIRCLE GEOMETRY



Properties of Chords

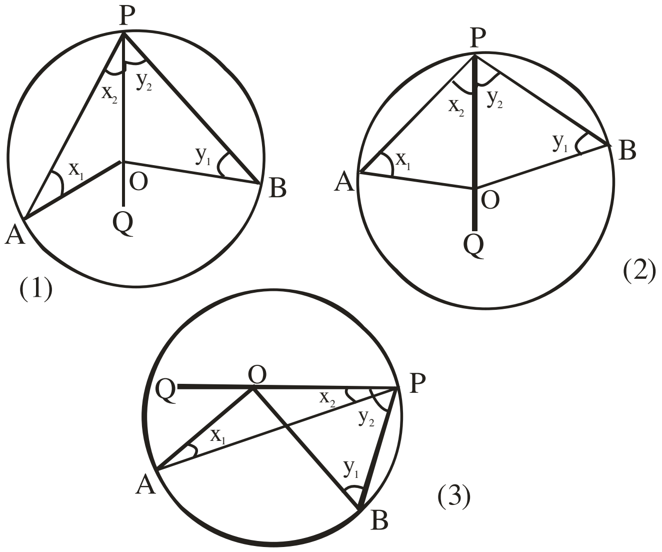
(a)The line joining the center to the mid-point of a chord is perpendicular to the chord as indicated in the diagram below



(b) Equal chords of a circle are equidistant from the centre and conversely

(1) THEOREM : T

The angle which an arc of a circle subtends at the centre is twice that which it subtends at any point on the remaining part of the circumference.



Given: a circle APB with centre O
To prove: $A\hat{O}B = 2 \times A\hat{P}B$
Construction: Join PO and produce it to any point Q
Proof:
/OA/ = /OB/ (radii)
$\therefore x_1 = x_2$ (Base $\angle$s of iso. Δ)
$\therefore A\hat{O}Q = x_1 + x_2$ (Ext. $\angle$ of Δ AOP)
$A\hat{O}Q = 2x_2$ $(x_1 = x_2)$
Similarly , $B\hat{O}Q = 2y_2$

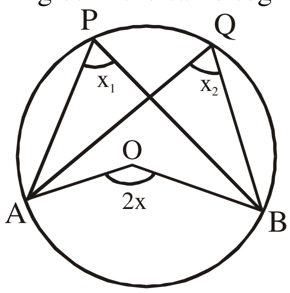
$$\left.\begin{array}{l}(1)\ A\hat{O}B \\ (2)\ \text{reflex } A\hat{O}B\end{array}\right\} = A\hat{O}Q + B\hat{O}Q$$
$$= 2x_2 + 2y_2$$
$$= 2(x_2 + y_2)$$
$$= 2 \times A\hat{P}B$$

From (3) $A\hat{O}B = B\hat{O}Q - A\hat{O}Q$
$$= 2y_2 - 2x_2$$
$$= 2(y_2 - x_2)$$
$$= 2 \times A\hat{P}B$$

In all cases $A\hat{O}B = 2 \times A\hat{P}B$

(2) Theorem :

Angles in the same segment of a circle are equal



Given :Point P,Q on the circumference of a circle APQB.

To prove : $A\hat{P}B = A\hat{Q}B$

Construction: Join A and B to O, the center of the circle.

Proof:

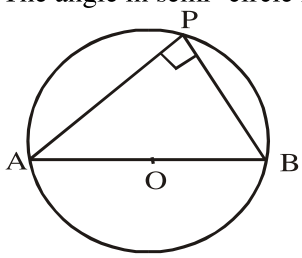
$A\hat{O}B = 2x_1$ ($\angle$s at center = 2 × $\angle$ at circum)

$= 2x_2$

$\therefore\ x_1 = x_2$

(3) Theorem:

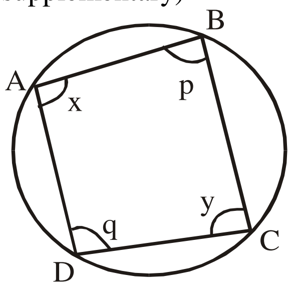
The angle in semi- circle is 90^0



$\hat{P} = 90^0$

(4) Theorem:

Angles in opposite segments are supplementary.
(opposite angles of a cyclic quadrilateral are supplementary)

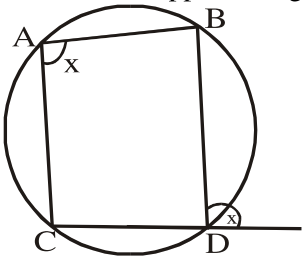


$x + y = 180^0$

$p + q = 180^0$

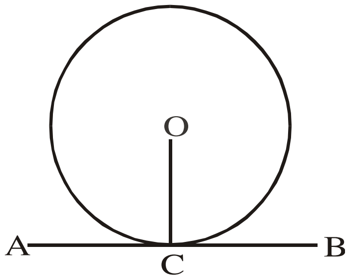
(5) Theorem:

The exterior angle of a cyclic quadrilateral is equal to the interior - opposite angle.



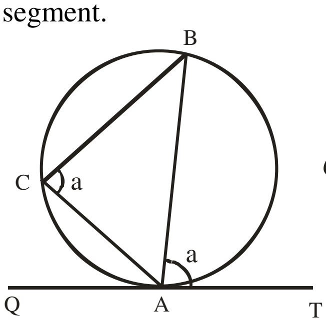
(6) Theorem

A tangent is perpendicular to the radius at the point of contact

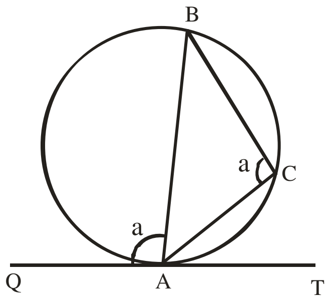


(7)Theorem

The angle between a tangent to a circle and a chord through the point of contact is equal to the angle in the alternate segment.

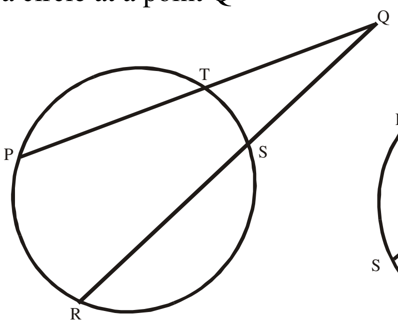


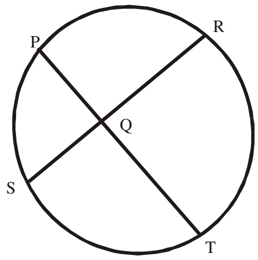
OR



(8)Theorem

Two chords PT and RS of a circle intersect outside or inside a circle at a point Q

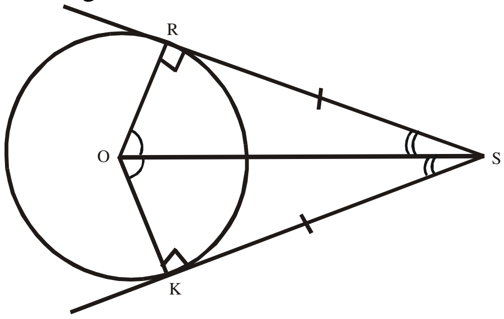




$PQ \cdot QT = QR \cdot QS$

(9)Theorem

Two tangents SR and SK to a circle with cenre O are equal in length



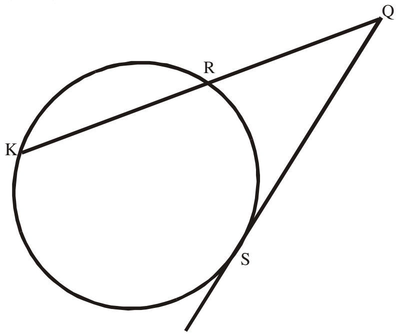
SR = SK,

$\angle OSR = \angle OSK$

$\angle SOR = \angle SOK$

$\hat{R} = \hat{K} = 90^o$

(10)Theorem

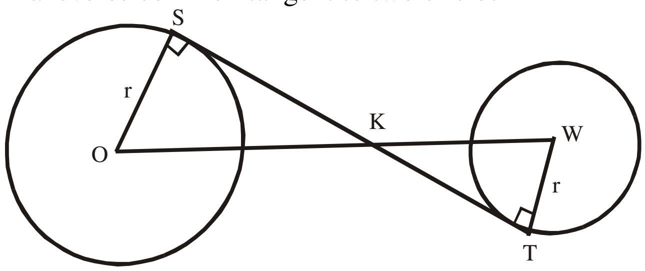


If QK is a secant and QS is a tangent to a circle, then

$SQ^2 = QR \cdot QK$

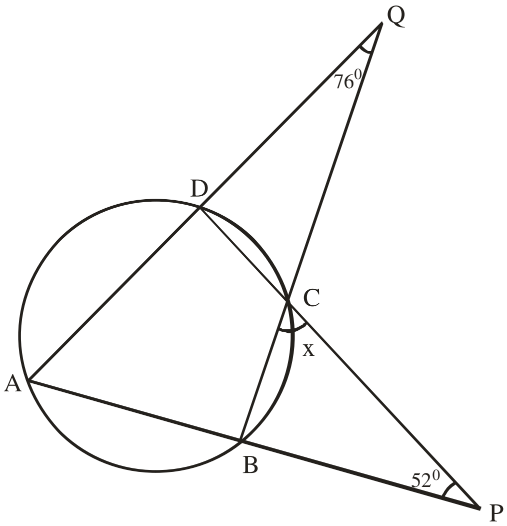
(11)Theorem

Transverse common tangent to two circles



$\frac{SK}{TK} = \frac{OK}{WK} = \frac{r}{r}$

2015/9a



In the diagram, ABCD is a cyclic quadrilateral. AB and DC are produced to meet at P, AD and BC are produced to meet at Q. if $\angle DQC = 76^0$, $\angle BPC = 52^0$ and $\angle BCP = x$, calculate the value of x.

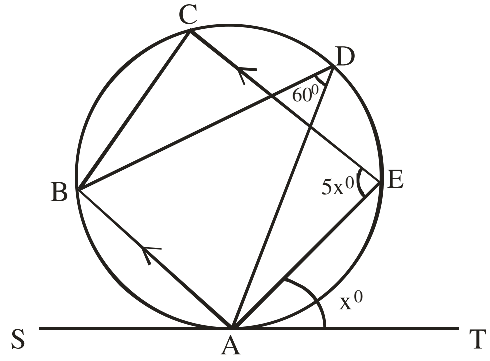
Solution

$\angle DAB = x$ (Exterior and interior $\angle$s of cyclic quad ABCD)
$\angle DCQ = x$ (Vertically opp. $\angle$s)
$\angle DCB = 180 - x$ (angles on straight line)
Thus $\angle QCP = x + x + 180 - x$
$= x + 180$
$\angle QAP + 76 + 52 + x + 180 = 360$(sum of $\angle$s quad AQCP)
$x + 76 + 52 + x + 180 = 360$
$2x = 360 - 308$
$x = {}^{52}/_{2} = 26^0$

2015/9a Alternative solution

$\angle ABC = 52 + x$ (Exterior and interior $\angle$s in ΔCBP)
$\angle DCQ = x$ (vertically opposite $\angle$s)
Also, $\angle ADC = 76 + x$ (Exterior and interior $\angle$s in ΔDCQ)
$\angle ABC + \angle ADC = 180$ ($\angle$s in opp. Segment in Quad ABCD)
$52 + x + 76 + x = 180$
$2x + 128 = 180$
$2x = 180 - 128$
$2x = 52$
$x = 26^0$

2010/3b

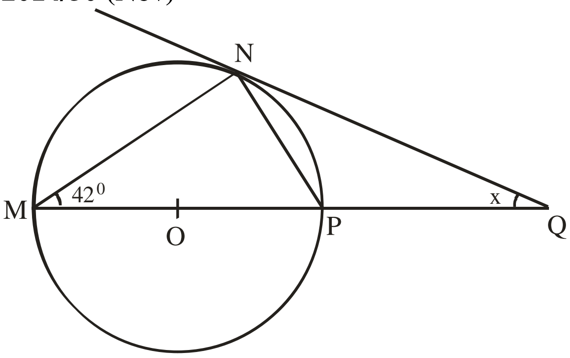


In the diagram, TS is a tangent to the circle at A. AB//CE, $\angle AEC = 5x^0$, $\angle ADB = 60^0$ and $\angle TAE = x$. Find the value of x

Solution

$\angle BAS = 60^0$ (Alternate segment $\angle$s)
$\angle BAE + 5x = 180^0$ (Sum of interior opp. $\angle$s)
$\angle BAE = 180 - 5x$
But $\angle BAS + \angle BAE + x = 180^0$ (Sum of $\angle$s on line SAT)
$60 + 180 - 5x + x = 180$
$240 - 4x = 180$
$240 - 180 = 4x$
$60 = 4x$
$\frac{60}{4} = x$ thus $x = 15^0$

2014/50 (Nov)



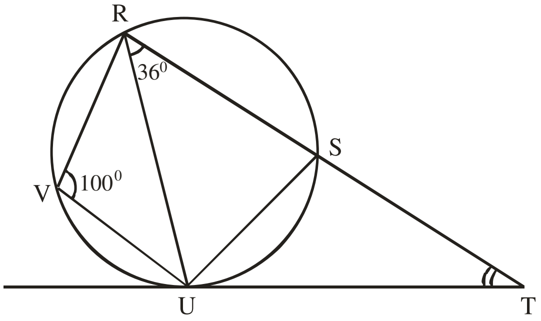
In the diagram, $\overline{MP}$ is a diameter of the circle, MQ is a straight line, $\angle NMP = 42^0$, $\angle NQP = x$ and $\overline{NQ}$ is a tangent to the circle at N. Calculate the value of x.

A 48^0 B 48^0 C 42^0 D 6^0

Solution

$\angle QNP = 42^0$ (Tangent and chord, alternate segment)
$\angle NMQ + \angle MNQ + x = 180$(sum of $\angle$s in ΔMNQ)
$42 + (90 + 42) + x = 180$ (sum of $\angle$s in ΔMNQ)
$174 + x = 180$
$x = 180 - 174$
$x = 6^0$ (D)

2012/3a



In the diagram, $\overline{TU}$ is a tangent to the circle. $\angle RVU = 100^0$ and $\angle URS = 36^0$, calculate the value of angle STU

Solution

$100 + \angle USR = 180^0$ (opp. segment $\angle$s)

$\angle USR = 180 - 100$

$= 80^0$

$\angle RVU = \angle UST = 100^0$ (Exterior & interior opp. $\angle$s)

$\angle SUT = 36^0$ (Alternate segment $\angle$s)

In Δ STU,

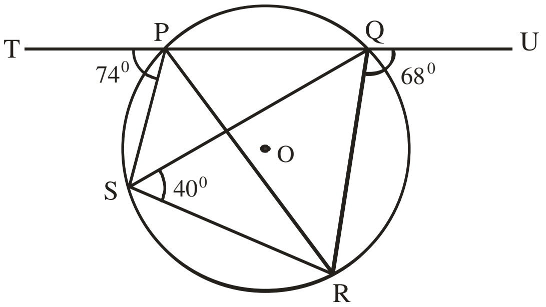
$\angle SUT + \angle UST + \angle STU = 180$ (Sum of $\angle$s in Δ)

$36 + 100 + \angle STU = 180$

$\angle STU = 180 - 136$

$= 44^0$

2015/3b



The diagram shows a circle PQRS with centre O, $\angle UQR = 68^0$, $\angle TPS = 74^0$ and $\angle QSR = 40^0$. Calculate the value of $\angle PRS$

Solution

$\angle PSR = 68^0$ (Exterior and interior opp $\angle$s)

$\angle SRQ = 74^0$ (Exterior and interior opp $\angle$s)

$\angle SPQ + \angle SRQ = 180^0$ (Opp $\angle$s in cyclic quad)

$\angle SPQ + 74 = 180$

$\angle SPQ = 180 - 74$

$= 106^0$

$\angle PSR + \angle PQR = 180^0$ (Opp $\angle$s in cyclic quad)

$68 + \angle PQR = 180$

$\angle PQR = 180 - 68$

$= 112^0$

$\angle PSQ = \angle PSR - 40^0$

$= 68 - 40$

$= 28^0$

In ΔSPQ, $\angle S + \angle P + \angle Q = 180^0$ (Sum of $\angle$s in Δ)

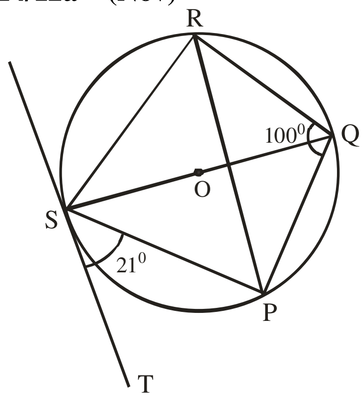
$28 + 106 + \angle Q = 180$

$\angle Q = 180 - 134$

$= 46^0$

Thus $\angle PRS = \angle PQS = 46^0$ (same segment $\angle$s)

2014/12a (Nov)



In the diagram, $\overline{TS}$ is a tangent to the circle at S. If O is the center of the circle, $\angle TSP = 21^0$ and $\angle RQP = 100^0$, find, with reasons.

(i) $\angle SPR$; (ii) $\angle QSR$

Solution

$\angle SRP = 21^0$ (Alternate segment $\angle$s)

$\angle RSP + 100^0 = 180^0$ (Opposite segment $\angle$s)

$\angle RSP = 180 - 100$

$= 80$

$\angle SRP + \angle RSP + \angle SPR = 180$ (Sum of $\angle$s in Δ)

$21 + 80 + \angle SPR = 180$

$\angle SPR = 180 - 101$

$= 79^0$

(ii) $\angle SPQ = 90^0$ (Semi – circle angle)

$\angle RSP + \angle RQP = 180^0$ (Opp. segment$\angle$s)

$\angle RSP + 100 = 180$

$\angle RSP = 180 - 100$

$= 80$

$\angle QST = 90^0$ (Radius and tangent at contact)

$\angle QSP = 90 - 21$

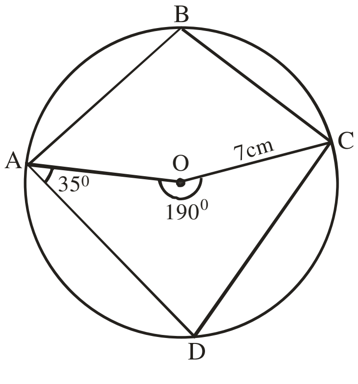
$= 69^0$

Thus $\angle QSR = \angle RSP - \angle QSP$

$= 80 - 69$

$= 11^0$

2008/11 b & c



(b) The diagram shows a circle ABCD with center O and radius 7cm. The reflex angle AOC = 190^0 and angle DAO = 35^0. Find (i) $\angle ABC$ (ii) $\angle ADC$

(c) Using the diagram in 11(b) above, calculate, correct to 3 significant figure, the length of:

(i) arc ABC (ii) the chord AD (Take $\pi = 3.142$)

Solution

b. $\angle AOC = 360 - 190$ (Sum of $\angle$s at a point)

$= 170$

$\angle ADC = \frac{170}{2}$ (center & circumference $\angle$s)

$= 85^0$

$\angle ABC + \angle ADC = 180^0$ (Opp. $\angle$s in cyclic quad)

$\angle ABC + 85 = 180$

$\angle ABC = 180 - 85 = 95^0$

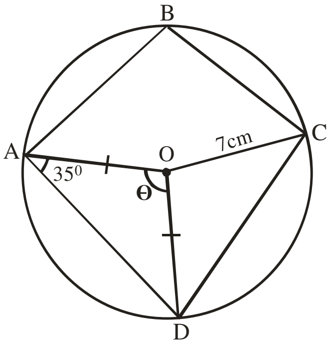
C (i) Length of arc ABC = $\frac{\theta}{360} \times 2\pi r$

$= \frac{170}{360} \times 2 \times 3.142 \times 7$

$= 20.77$cm

≈ 20.8cm to 3s.f

(ii) Chord AD = $2r \sin\frac{\theta}{2}$



In ΔAOD : $\angle ADO = 35^0$ (Base $\angle$s of isosceles Δ)

$35 + 35 + \theta = 180$ (Sum of $\angle$s in Δ)

$\theta = 180 - 70$

$= 110^0$

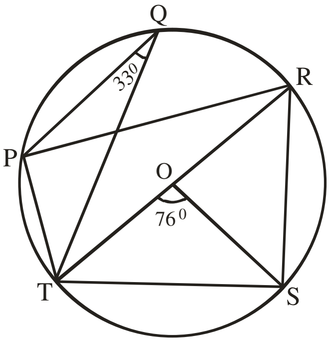
Chord AD = $2 \times 7 \times \sin\frac{110}{2}$

$= 14 \times \sin 55$

$= 14 \times 0.8192$

$= 11.47$cm $\approx$ 11.5cm to 3s.f

2015/19 and 20



In the diagram, O is the centre of the circle, $\overline{RT}$ is a diameter, $\angle PQT = 33^0$ and $\angle TOS = 76^0$. Use the diagram to answer questions **19** and **20**

19. Calculate the value of angle PTR

A 73^0 B 67^0 C 57^0 D 37^0

Solution

$\angle PRT = 33^0$ (Same segment $\angle$s)

In ΔTOS : $\angle OTS = \angle OST$ (Base $\angle$s of isoscelesΔ)

$\angle OTS + \angle OST + \angle TOS = 180^0$ (Sum of $\angle$s in Δ)

$2\angle OTS + 76 = 180$

$2\angle OTS = 180 - 76$

$2\angle OTS = 104$

$\angle OTS = \frac{104}{2} = 52$

$\angle TSR = 90^0$ (Semi – circle angle)

In ΔTSR:

$\angle RTS + \angle TSR + \angle TRS = 180$(Sum of $\angle$s in Δ)

$52 + 90 + \angle TRS = 180$

$\angle TRS = 180 - 142$

$= 38^0$

$\angle PTS + \angle PRS = 180^0$ (Opp. $\angle$s in cyclic quad)

$(\angle PTR + 52) + (33 + 38) = 180^0$

$= 180 - 123 = 57^0$ (C)

20. Find the size of angle PRS

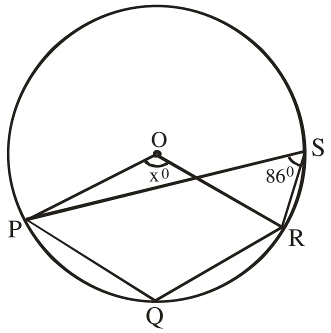
A 76^0 B 71^0 C 38^0 D 33^0

Solution

$\angle PRS = 33 + 38$

$= 71^0$ (B)

2012/20



In the diagram, O is the center of the circle PQRS and $\angle PSR = 86^0$. If $\angle POR = x^0$, find x

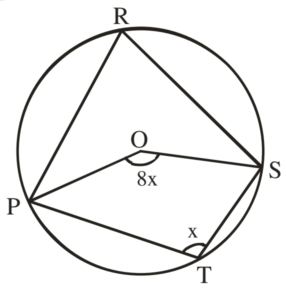
A 274 B 172 C 129 D 86

Solution

$x = 2 \times 86$ (Centre and circumference $\angle$s)

$= 172^0$

2012/41



The diagram is a circle with center O. PRST are points on the circle. Find the value of $\angle PRS$

A 144^0 B 72^0 C 40^0 D 36^0

Solution

$\angle PRS = \frac{8x}{2}$ (Centre and circumference $\angle$s)

$= 4x$

$\angle PRS + \angle PTS = 180^0$ (Opp. $\angle$s in cyclic quad)

$4x + x = 180$

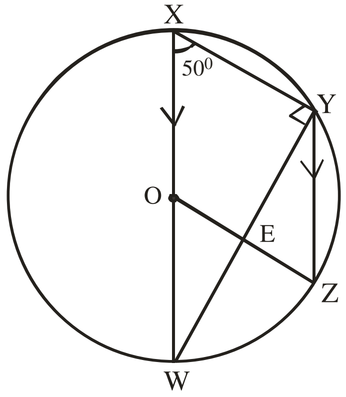
$5x = 180$

$x = \frac{180}{5} = 36$

Thus $\angle PRS = 4x$

$= 4 \times 36 = 144^0$ (A)

2014/9b



In the diagram, O is the centre of the circle.
If WX is parallel to YZ and $\angle WXY = 50^0$, find the value of: (i) $\angle WYZ$; (ii) $\angle YEZ$

Solution

In ΔWXY

$\angle XWY + 50 + 90^0 = 180^0$ (Sum of $\angle$s in Δ)

$\angle XWY = 180 - 140$

$= 40^0$

$\angle WYZ = \angle XWY = 40^0$ (Alternate $\angle$s between parallel)

(ii) $\angle WOZ = 2\angle WYZ$ (Centre $\angle$ = 2 circumference$\angle$)

$\angle WOZ = 2 \times 40^0$

$\angle WOZ = 80^0$

$\angle WOZ = \angle YZO$ (Alternate $\angle$s)

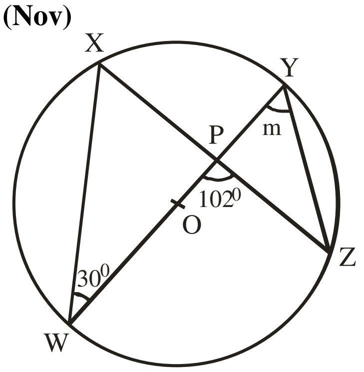
$\angle YEZ + \angle YZO + \angle WYZ = 180^0$ (Sum of $\angle$s in Δ)

$\boldsymbol{\angle YEZ} = 180 - (80 + 40)$

$\angle YEZ = 180 - 120$

$= 60^0$

2014/21 (Nov)



In the diagram, O is the center of the circle ,
WY and XZ are straight lines, $\angle WPZ = 102^0$,
$\angle XWY = 30^0$ and $\angle PYZ = m$. Find the value of m.

A 78^0 B 72^0 C 51^0 D 36^0

Solution

$\angle XPW + 102 = 180^0$ (Sum of $\angle$s on line XPZ)

$\angle XPW = 180 - 102$

$= 78^0$

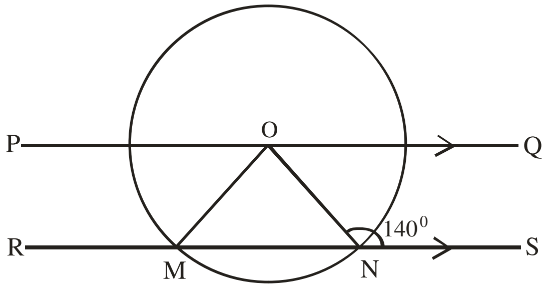
In ΔXPW : $30 + 78^0 + \angle PXW = 180^0$ (Sum $\angle$s in Δ)

$\angle PXW = 180 - 108$

$= 72^0$

$m = \angle PXW = 72^0$ (Same segment $\angle$s)

2015/16



In the diagram, O is the centre of the circle. If PQ//RS and $\angle ONS = 140^0$, find the size of $\angle POM$.

A 40^0 B 50^0 C 60^0 D 80^0

Solution

ΔMON is isosceles (OM = ON Radii)

$\angle ONM + 140^0 = 180^0$ (Sum of $\angle$s on line MNS)

$\angle ONM = 180 - 140$

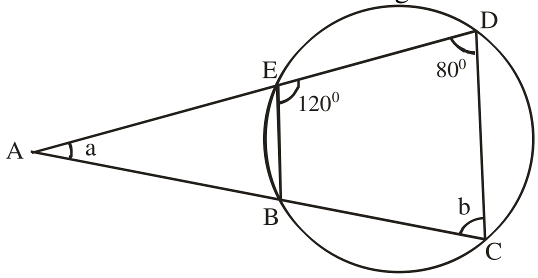
$= 40^0$

$\angle OMN = \angle ONM = 40^0$ (Base $\angle$s of isoscelesΔ)

$\angle POM = \angle OMN = 40^0$ (Alternate $\angle$s between parallel lines)

2005/6a NABTEB

Find the value of a and b in the figure below



Solution

$120 + b = 180^0$ (Opp. segment $\angle$s)

$b = 180 - 120$

$= 60^0$

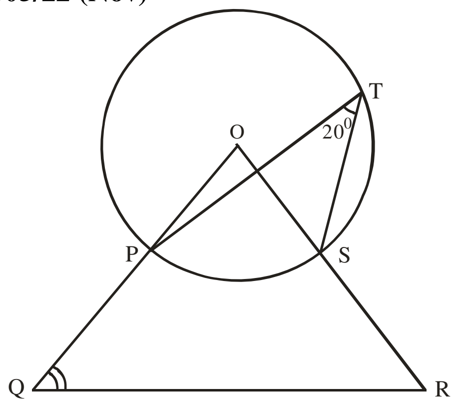
$a + 80 + b = 180^0$ (sum of $\angle$s in Δ)

$a + 80 + 60 = 180$

$a = 180 - 140$

$= 40^0$

2005/22 (Nov)



In the diagram, O is the centre of the circle, $\angle PTS = 20^0$ and /OQ/ = /OR/. Calculate $\angle OQR$

A 140^0 B 80^0 C 70^0 D 40^0

Solution

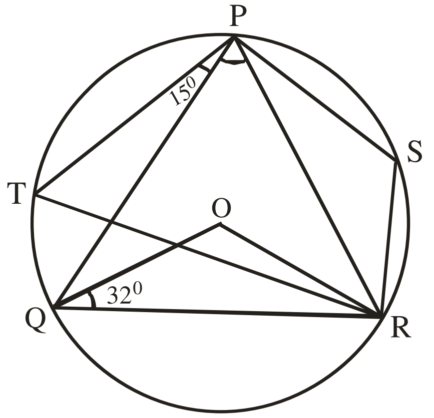
$\angle POS = 2 \times 20^0$ (centre and circumference $\angle$s)

$= 40^0$

In ΔQOR : $\hat{Q} = \hat{R}$ (Base $\angle$s of isoscelesΔ)

$$\hat{O} + \hat{Q} + \hat{R} = 180^0 \text{ (Sum of } \angle\text{s in}\Delta)$$
$$\hat{O} + 2\hat{Q} = 180$$
$$40 + 2\hat{Q} = 180$$
$$2\hat{Q} = 180 - 40 = 140$$
$$\hat{Q} = \frac{140}{2} = 70^0 \text{ (C)}$$

2009/7b



In the diagram, O is the centre of the circle, $\angle OQR = 32^0$ and $\angle TPQ = 15^0$.
Calculate : (i) $\angle QPR$; (ii) $\angle TQO$

Solution

In ΔQOR:

$\angle QRO = 32^0$ (Base $\angle$s of isoscelesΔ)

$32 + 32 + \angle QOR = 180^0$ (Sum of $\angle$s in Δ)

$\angle QOR = 180 - 64$

$= 116^0$

$\angle QPR = \frac{1}{2}\angle QOR$ (Center & circumference $\angle$s)

$= \frac{116}{2} = 58^0$

(ii) $\angle QTR = 58^0$ (Same segment $\angle$s on line QR)

$\angle TRQ = 15^0$ (Same segment $\angle$s on line TQ)

In ΔTQR : $58 + 15 + \angle TQR = 180^0$ (Sum of $\angle$s in Δ)

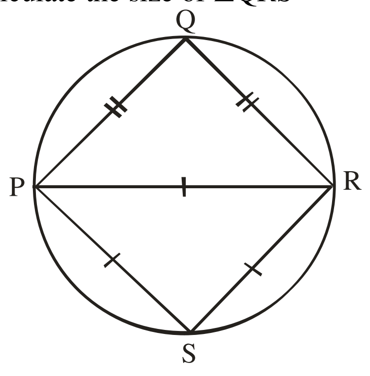
$\angle TQR = 180 - 73$

$= 107$

$\therefore \angle TQO = 107 - 32 = 75^0$

2008/20

In the diagram, /PQ/ = /QR/ and /PR/ = /RS/ = /SP/.
Calculate the size of $\angle QRS$



A 150^0 B 120^0 C 90^0 D 60^0

Solution

In ΔPRS : $\angle P = \angle S = \angle R = 60^0$ (Equilateral Δ)

$\angle PSR + \angle PQR = 180^0$ (Opp $\angle$s in cyclic quad)

$60 + \angle PQR = 180$

$\angle PQR = 180 - 60$

$= 120^0$

In ΔPQR: $\angle P + \angle Q + \angle R = 180^0$ (Sum of $\angle$s in Δ)

But $\angle P = \angle R$ (Base $\angle$s of isosceles Δ)

Thus, $2\angle R + 120 = 180$

$2\angle R = 180 - 120$

$2\angle R = 60$

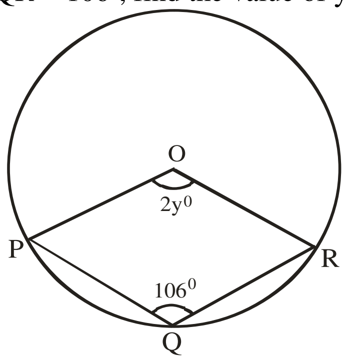
$\angle R = 30^0$

$\therefore \angle QRS = 30 + 60$ i.e $\angle QRP + \angle PRS$

$= 90^0$ (C)

2007/24

In the diagram, O is the centre of the circle and $\angle PQR = 106^0$, find the value of y.



A 16 B 37 C 74 D 127

Solution

Reflex $\angle POR = 106 \times 2$ (Centre and circumference $\angle$s)

$\angle POR = 212$

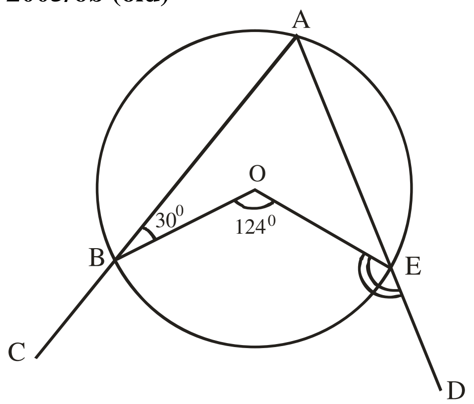
$212 + 2y = 360^0$ (Sum of $\angle$s at a point)

$2y = 360 - 212$

$2y = 148$

$y = \frac{148}{2} = 74^0$ (C)

2005/6b (old)



In the diagram, O is the centre of the circle, $\angle ABO = 30^0$ and $\angle BOE = 124^0$. Calculate $\angle OED$

Solution

$\angle BAE = \frac{124}{2}$ (Centre and circumference $\angle$s)

$= 62$

Reflex $\angle BOE = 360 - 124^0$ (Sum of $\angle$s at a point)

$= 236^0$

$236^0 + 30^0 + 62^0 + \angle OEA = 360^0$ (Sum of $\angle$s in Quad)

$\angle OEA = 360 - 328$

$= 32^0$

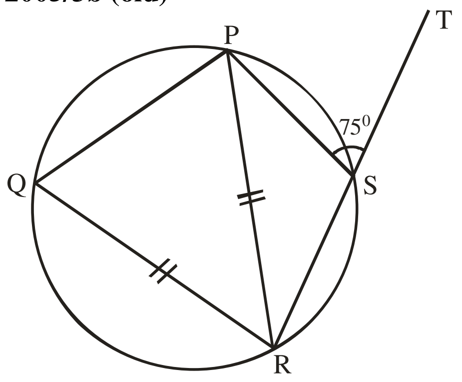
$\angle OEA + \angle OED = 180^0$ (Sum of $\angle$s on a line)

$32 + \angle OED = 180^0$

$\angle OED = 180 - 32^0$

$= 148^0$

2005/3b (old)



In the diagram, PQRS is a cyclic quadrilateral with /PR/ = /QR/ and $P\hat{S}T = 75^0$, calculate $\angle QRP$

Solution

$\angle PQR = 75^0$ (Exterior angle = interior opp angle)

$\angle QPR = \angle PQR = 75^0$ (Base $\angle$s of isoscelesΔ)

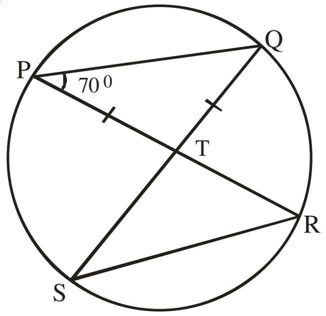
In ΔQPR : $75 + 75 + \angle QRP = 180^0$ (Sum of $\angle$s in Δ)

$\angle QRP = 180 - 150$

$= 30^0$

2005/42

In the diagram PQRS is a circle, /PT/ = /QT/ and $\angle QPT = 70^0$. What is the size of $\angle PRS$



A 40^0 B 70^0 C 80^0 D 140^0

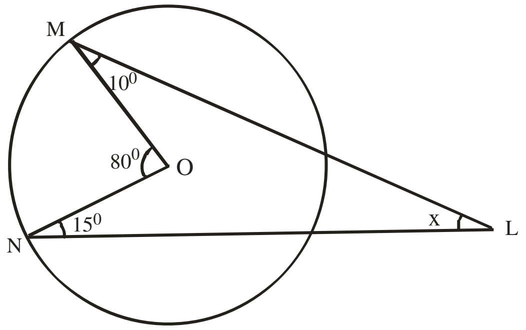
Solution

$\angle PQT = 70^0$ (Base $\angle$s of isosceles Δ)

$\angle PRS = \angle SQP = 70^0$ (Same segment $\angle$s) B.

2005/23

In the diagram, O is the centre of the circle, $\angle MON = 80^0$, $\angle LMO = 10^0$ and $\angle LNO = 15^0$. Calculate the value of x



A 40^0 B 50^0 C 55^0 D 75^0

Solution

Reflex $\angle MON = 360 - 80$ (Sum of $\angle$s at a point)

$= 280$

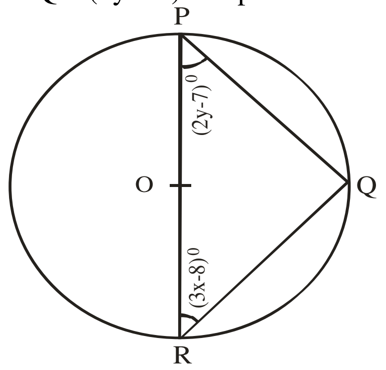
$10^0 + 280 + 15^0 + x = 360^0$ (Sum of $\angle$s in Quad)

$x = 360 - 305$

$= 55^0$ (C)

2005/10

In the diagram, PR is a diameter, $\angle PRQ = (3x - 8)^0$ and $\angle RPQ = (2y - 7)^0$. Express x in terms of y



A $x = \dfrac{75 - 2y}{3}$ B $x = \dfrac{105 - 3y}{2}$

C $x = \dfrac{105 - 2y}{3}$ D $x = \dfrac{75 - 3y}{2}$

Solution

$\angle PQR = 90^0$ (Semi – circle $\angle$s)

$(2y - 7)^0 + (3x - 8)^0 + 90^0 = 180^0$ (Sum of $\angle$s in Δ)

$2y - 7 + 3x - 8 = 180 - 90$

$2y + 3x - 15 = 90$

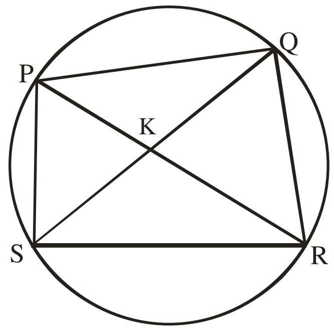
$2y + 3x = 90 + 15$

$2y + 3x = 105$

$3x = 105 - 2y$

$x = \dfrac{105 - 2y}{3}$ (C)

2011/4



The diagram shows a cyclic quadrilateral PQRS with its diagonals intersecting at K. Which of the following triangles is similar to triangle QKR?

A ΔPQK B ΔPKS C ΔSKR D ΔPSR

Solution

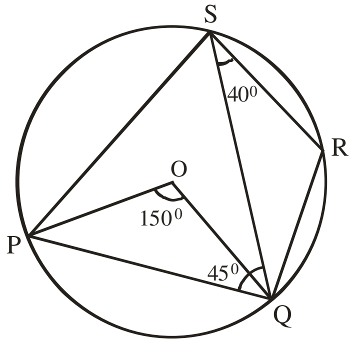
$\angle SPR = \angle SQR$ (Angles on same segment)

Line SR is same base to ΔSPR and ΔSQR

Thus ΔSPR is similar to ΔSQR

$\Rightarrow \Delta QKR$ is similar to ΔPKS (B)

2011/4b



In the diagram, PQRS is a circle centre O.
If $\angle POQ = 150^0$, $\angle QSR = 40^0$ and $\angle SQP = 45^0$, calculate $\angle RQS$

Solution

$\angle PSQ = \frac{150}{2}$ (centre & circumference $\angle$s)

$= 75^0$

$\angle PQR + \angle PSR = 180^0$ (Opp. $\angle$s in cyclic Quad)

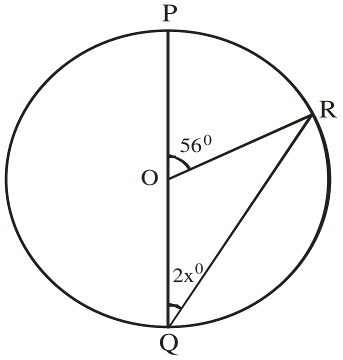
$(45^0 + \angle RQS) + (75 + 40^0) = 180$

$45 + \angle RQS + 115 = 180$

$\angle RQS = 180 - 160 = 20^0$

2010/44

In the diagram, O is the centre of the circle.
Find the value of x



A 34 B 29 C 17 D 14

Solution

$56 + \angle ROQ = 180^0$ (sum of $\angle$s on line POQ)

$\angle ROQ = 180 - 56 = 124^0$

$2 \times (2x) + 124 = 180^0$ (Sum of $\angle$s in isosceles Δ)

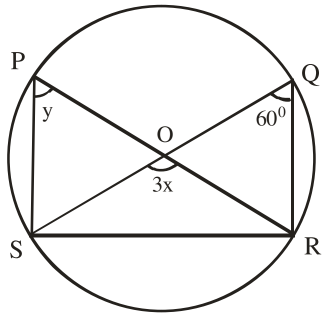
$4x + 124 = 180$

$4x = 180 - 124$

$4x = 56$

$x = \frac{56}{4} = 14^0$ (D)

2010/31



In the diagram, O is the centre of the circle $\angle SQR = 60^0$, $\angle SPR = y$ and $\angle SOR = 3x$. Find the value of $(x + y)$

A 110^0 B 100^0 C 80^0 D 70^0

Solution

$y = 60^0$ (same segment $\angle$s)

$3x = 2 \times 60^0$ (centre & circumference $\angle$s)

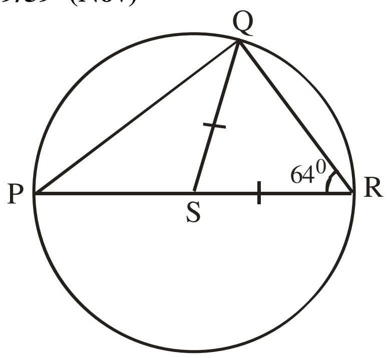
$3x = 120$

$x = \frac{120}{3} = 40^0$

Thus $x + y = 40 + 60$

$= 100^0$ (B)

2009/39 (Nov)



In the diagram, PR is a diameter, /QS/ = /SR/ and $\angle SRQ = 64^0$. Find $\angle PQS$

A 26^0 B 32^0 C 45^0 D 64^0

Solution

$\angle SQR = 64^0$ (Base $\angle$s in isosceles Δ)

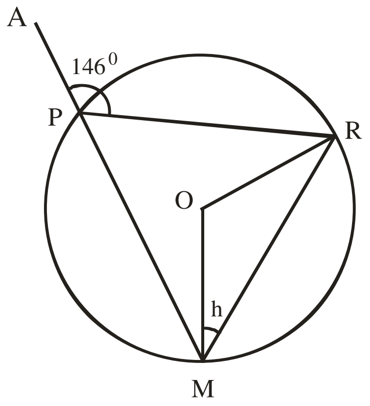
$\angle PQR = 90^0$ (Semi – circle angle)

$\angle PQS = \angle PQR - \angle SQR$

$= 90 - 64$

$= 26^0$ (A)

2014/46 Neco



In the figure above, O is the centre of the circle and $\angle APR = 146^0$. Find the value of h.

A 34^0 B 56^0 C 68^0 D 73^0 E 146^0

Solution

$146 + \angle RPM = 180^0$ (Sum of $\angle$s on line APM)

$\angle RPM = 180 - 146$

$= 34^0$

$\angle ROM = 2 \times 34^0$ (centre and circumference $\angle$s)

$= 68^0$

$2h + 68 = 180^0$ (Sum of $\angle$s in isosceles Δ)

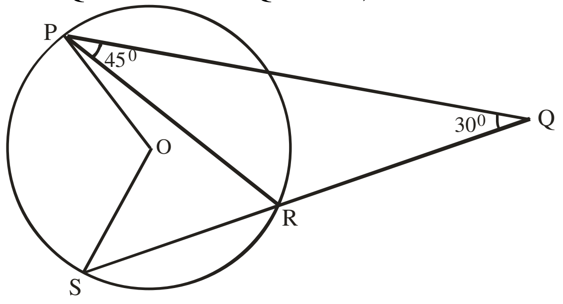
$2h = 180 - 68$

$2h = 112$

$h = \frac{112}{2} = 56^0$ (B)

2005/59 Neco

In the figure below, O is the centre of the circle. If $\angle RPQ = 45^0$ and $\angle PQR = 30^0$, find the value of $\angle POS$



A 45^0 B 60^0 C 75^0 D 135^0 E 150^0

Solution

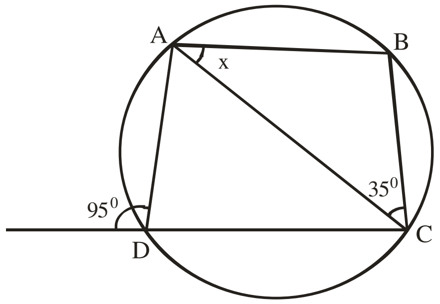
$\angle PRS = 45 + 30^0$ (Exterior & interior opp. $\angle$s in Δ)

$= 75$

$\angle POS = 2 \times 75^0$ (Centre and circumference$\angle$)

$= 150^0$ E.

2005/58 Neco



In the above diagram, what is the size of x?

A 130^0 B 120^0 C 85^0 D 60^0 E 50^0

Solution

$\angle ABC = 95^0$ (Exterior angle = interior opp $\angle$s)

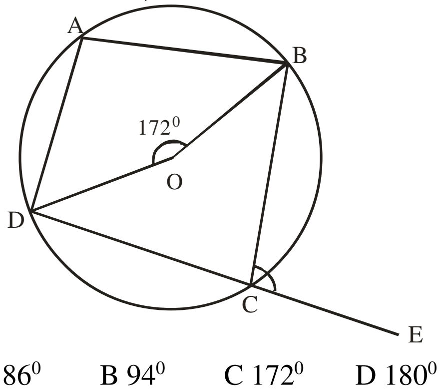
In ΔABC : $x + \angle ABC + 35^0 = 180$ (sum of $\angle$s in Δ)

$x + 95 + 35 = 180$

$x = 180 - 130 = 50^0$ (E)

2006/60 Neco

Figure ABCD is a cyclic quadrilateral with $\overline{DC}$ produced to E. If O is the centre of the circle and $\angle BOD = 172^0$, Calculate $\angle BCE$



A 86^0 B 94^0 C 172^0 D 180^0 E 188^0

Solution

$\angle BCD = \dfrac{172^0}{2}$ (Centre and circumference $\angle$s)

$= 86^0$

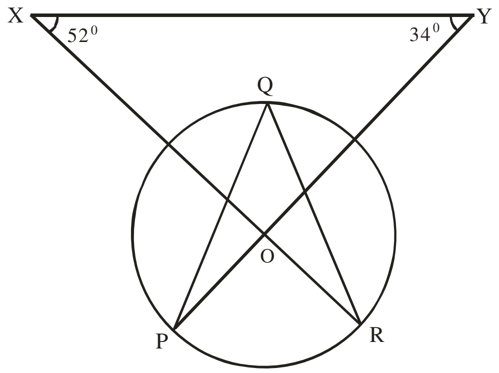
$\angle BCD + \angle BCE = 180$ (Sum of $\angle$s on line DCE)

$86 +$ BCE $= 180$

$\angle BCE = 180 - 86$

$= 94^0$ (B)

2006/32 Neco



In the figure above, P, Q, R are points on the circumference of a circle with centre O; if $\angle OXY = 52^0$ and $\angle OYX = 34^0$, find $\angle PQR$

A 94^0 B 86^0 C 62^0 D 47^0 E 43^0

Solution

In ΔOXY : $52^0 + 34^0 + \angle XOY = 180$ (Sum of $\angle$s Δ)

$\angle XOY = 180 - 86$

$= 94^0$

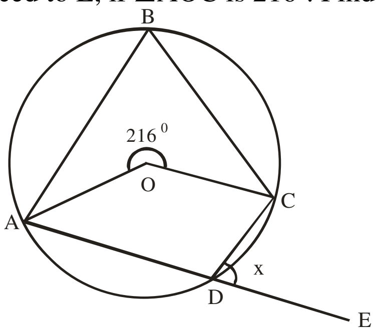
$\angle POR = 94^0$ (Vertically opp. $\angle$s)

$\angle PQR = \dfrac{94}{2}$ (Centre and circumference $\angle$s)

$= 47^0$ (D)

2008/43 Neco (Dec)

Given a cyclic quadrilateral ABCD centre O with $\overline{AD}$ produced to E, if $\angle AOC$ is 216^0. Find the value of $\angle CDE$



A 72^0 B 108^0 C 144^0 D 180^0 E 216^0

Solution

Obtuse $\angle AOC = 360 - 216$ ($\angle$s at appoint 360^0)

$\angle AOC = 144^0$

$\angle ABC = \dfrac{144}{2}$ (centre and circumference $\angle$s)

$= 72^0$

$\angle ABC + \angle ADC = 180^0$ (Opposite segment $\angle$s)

$72^0 + \angle ADC = 180^0$

$\angle ADC = 180^0 - 72^0$

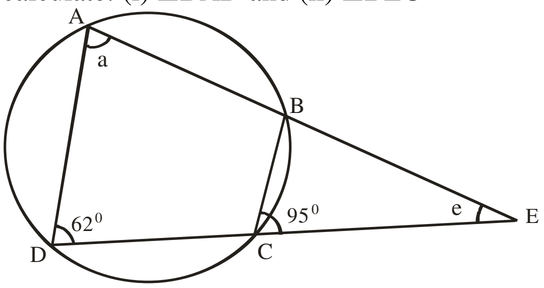
$= 108^0$

$\angle ADC + x = 180^0$ (sum of angles on line ADE)

$x = 180 - 108^0$

$= 72^0$ (A)

2010/7b Neco
In the diagram below, ABCD is a cyclic quadrilateral and lines AB and DC produced meet at E. If $\angle BCE = 95^0$ and $\angle ADC = 62^0$,
calculate: (i) $\angle BAD$ and (ii) $\angle BEC$



Solution

$a = 95^0$ (Exterior and interior opp. $\angle$s in cyclic quad)

(ii) $\angle ABC + 62^0 = 180^0$ (opp. $\angle$s in cyclic quad)

$\angle ABC = 180 - 62$

$= 118^0$

$\angle EBC = 180 - \angle ABC$ (Sum of $\angle$s on line ABC)

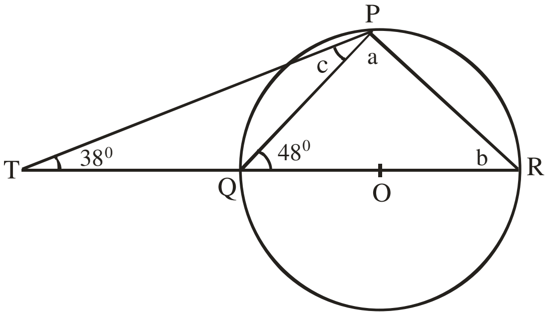
$= 180 - 118$

$= 62^0$

In ΔBEC : $95 + 62^0 + \angle BEC = 180^0$ (Sum of $\angle$s Δ)

$\angle BEC = 180 - 157 = 23^0$

2011/11b Neco



In the diagram above, O is the centre of the circle PQR, what are the values of a, b and c?

Solution

$a = 90^0$ (Semi – circle angle)

$\angle TQP = 180^0 - 48^0$ ($\angle$s on line TQR)

$= 132^0$

In ΔTPQ : $38^0 + 132^0 + c = 180^0$ (Sum of $\angle$s in Δ)

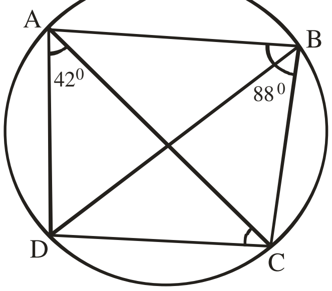
$c = 180 - 170 = 10^0$

$a + b = \angle TQP = 132^0$ (Exterior $\angle$ & interior opp $\angle$s in Δ)

$90 + b = 132$

$b = 132 - 90 = 42^0$

2014/ 45 Neco
In the diagram below, ABCD is a cyclic quadrilateral, $\angle ABC = 88^0$ and $\angle DAC = 42^0$. Calculate $\angle ACD$



A 42^0 B 46^0 C 48^0 D 52^0 E 88^0

Solution

$\angle ADC + \angle ABC = 180^0$ (opp. $\angle$s cyclic quad)

$\angle ADC + 88 = 180$

$\angle ADC = 180 - 88 = 92^0$

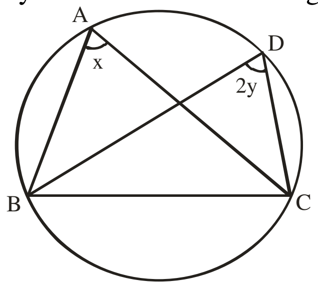
In ΔADC

$42^0 + \angle ADC + \angle ACD = 180^0$ (sum of $\angle$s in Δ)

$42^0 + 92^0 + \angle ACD = 180^0$

$\angle ACD = 180 - 134 = 46^0$ (B)

2012/39 Neco
Find y in terms of x in the diagram below



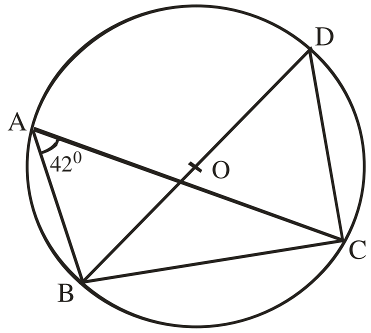
A $y = x$ B $y = 2x$ C $y = \frac{x}{2}$ D $y = \sqrt{x}$ E $y = x^2$

Solution

$2y = x$ (same – segment $\angle$s)

$y = \frac{x}{2}$ C.

2013/37 Neco
In the diagram below, O is the centre of the circle, BOD is a diameter and $\angle BAC$ is 42^0. What is $\angle CBD$?



A 42^0 B 48^0 C 77^0 D 90^0 E 138^0

Solution

$\angle BDC = 42^0$ (Same segment $\angle$s)

$\angle BCD = 90^0$ (Semi – circle$\angle$)

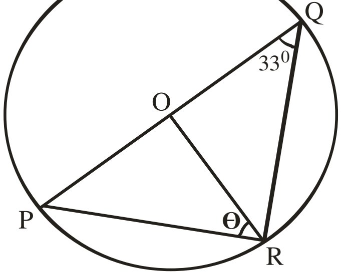
In ΔCBD ,

$\angle BDC + \angle BCD + \angle CBD = 180^0$ (Sum of $\angle$s in Δ)

$42^0 + 90^0 + \angle CBD = 180$

$\angle CBD = 180 - 132 = 48^0$ (B)

2006/39 Neco



In the above diagram, O is the center of the circle.
Calculate the value θ

A 90^0 B 66^0 C 57^0 D 33^0 E 24^0

Solution

$\angle PRQ = 90^0$ (semi – circle $\angle$s)

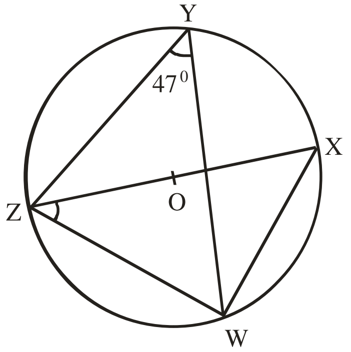
$\angle ORQ = 33^0$ (Base $\angle$s in isosceles Δ)

$\theta = \angle PRQ - \angle ORQ$

$= 90 - 33 = 57^0$

2008/42 Neco (Dec)

In the diagram below, O is the centre of the circle.
If ZOX is a diameter and $\angle WYZ = 47^0$, find $\angle WZX$



A 137^0 B 117^0 C 55^0 D 47^0 E 43^0

Solution

$\angle ZXW = 47^0$ (Same segment $\angle$s)

$\angle ZWX = 90^0$ (Semi – circle$\angle$)

In ΔWZX,

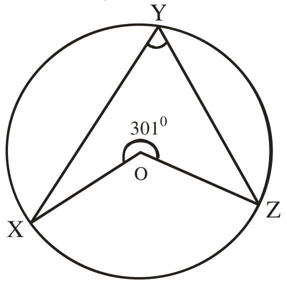
$\angle ZXW + \angle ZWX + \angle WZX = 180^0$ (Sum of $\angle$s in Δ)

$47 + 90 + \angle WZX = 180$

$\angle WZX = 180 - 137 = 43^0$ (E)

2006/39 Neco (Dec)

In the diagram below, O is the centre of the circle XYZ.
If $\angle XOZ = 301^0$, find $\angle XYZ$



A 150.5^0 B 118^0 C 59.5^0 D 59^0 E 29.5^0

Solution

Obtuse $\angle XOZ = 360 - 301$ (Sum of $\angle$s at a point)

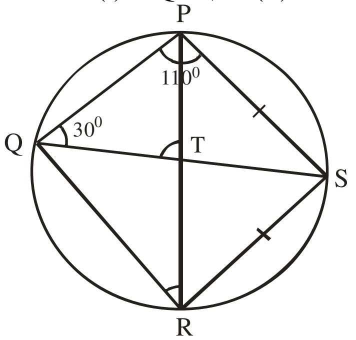
$\angle XOZ = 59^0$

$\angle XYZ = \dfrac{59}{2}$ (Centre & circumference $\angle$s)

$= 29.5^0$ E.

2009/8a Neco

PQRS is a cyclic quadrilateral $\angle QPS = 110^0$, $\angle PQS = 30^0$, $\overline{PS} = \overline{SR}$ and $\overline{PS}$ intersects QS at T.
Calculate : (i) $\angle QTP$; (ii) $\angle PRQ$



Solution

$\angle PQS = \angle PRS = 30^0$ (Same segment $\angle$s)

$\angle RPS = \angle PRS = 30^0$ (Base $\angle$s of isosceles Δ)

$\angle QPR = 110 - \angle RPS$

$= 110 - 30$

$= 80^0$

In ΔQTP: $\angle PQT + \angle QPT + \angle QTP = 180^0$ (Sum of $\angle$s in Δ)

$30 + 80 + \angle QTP = 180$

$\angle QTP = 180 - 110$

$= 70^0$

(ii) $\angle QPS + \angle QRS = 180^0$ (Opp $\angle$s in cyclic quad)

$110 + \angle QRS = 180$

$\angle QRS = 180 - 110$

$= 70^0$

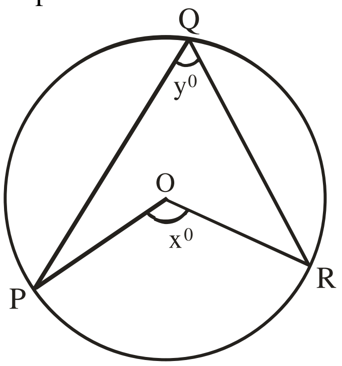
$\angle PRQ = \angle QRS - \angle PRS$

$= 70 - 30$

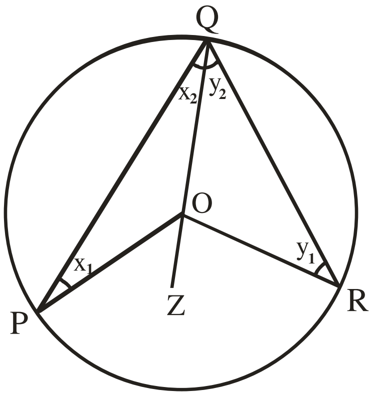
$= 40^0$

2009/10a Neco (Dec) PROOF

In the diagram below, O is the centre of the circle.
P, Q, R are points on the circumference, show that $x = 2y$



Solution



Given: a circle PQR with centre O

To prove: $P\hat{O}R = 2 \times P\hat{Q}R$

Construction: Join QO and produce it to any point Z

Proof:

/OP/ = /OR/ (radii)

$\therefore x_1 = x_2$ (Base $\angle$s of iso. Δ)

$\therefore P\hat{O}Z = x_1 + x_2$ (Ext. $\angle$ of Δ POQ)

$P\hat{O}Z = 2x_2$ $(x_1 = x_2)$

Similarly , $R\hat{O}Z = 2y_2$

$$\left.\begin{matrix}(1)\ P\hat{O}R \\ (2)\ \text{Obtuse } P\hat{O}R\end{matrix}\right\} = P\hat{O}Z + R\hat{O}Z$$

$= 2x_2 + 2y_2$

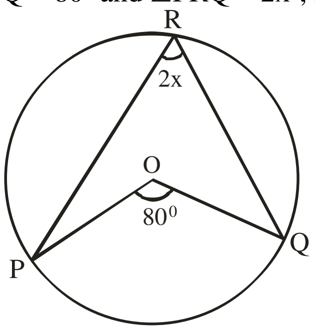
$= 2(x_2 + y_2)$

$= 2 \times P\hat{Q}R$

$\Rightarrow x = 2y$

2010/45 Neco

In the diagram below, O is the centre of circle PQR. If $\angle POQ = 80^0$ and $\angle PRQ = 2x^0$, find the value of x



A 10 B 20 C 40 D 80 E 90

Solution

$80 = 2(2x^0)$ [Centre and circumference $\angle$s]

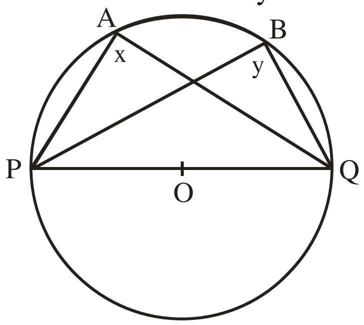
$80 = 4x$

$\frac{80}{4} = x$

$20 = x$ B.

2011/35 Neco

In the diagram below, O is the circle.
Find the value of x + y



A 90^0 B 120^0 C 180^0 D 270^0 E 360^0

Solution

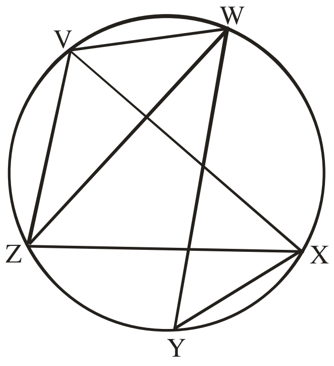
$x = 90^0$ (angle in Semi – circle)

$y = 90^0$ (angle in semi – circle)

$x + y = 180^0$ C.

2012/37 Neco Exercise 19.30

Which of the angle(s) is/are equal to $\angle WZX$ in the diagram below?



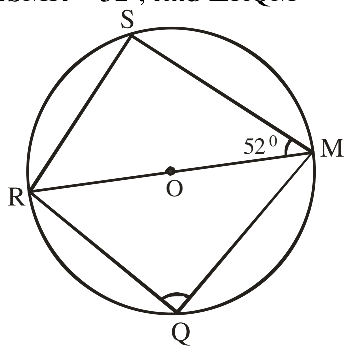
I $\angle WVX$ II $\angle WYX$ III $\angle WVZ$

A I only B II only C III only

D I & II only E II & III only

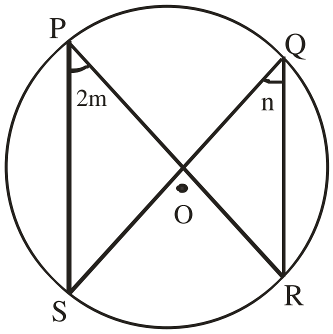
204/47 Neco Exercise 19.31

In the figure below, O is the centre of the circle MQRS and $\angle SMR = 52^0$, find $\angle RQM$



A 26^0 B 38^0 C 88^0 D 90^0 E 104^0

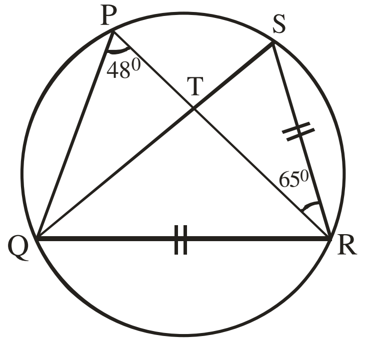
2012/21 Exercise 19.32



The diagram is a circle centre O. If $\angle SPR = 2m$ and $\angle SQR = n$, express m, in terms of n

A $m = \frac{n}{2}$ B $m = 2n$ C $m = n - 2$ D $m = n + 2$

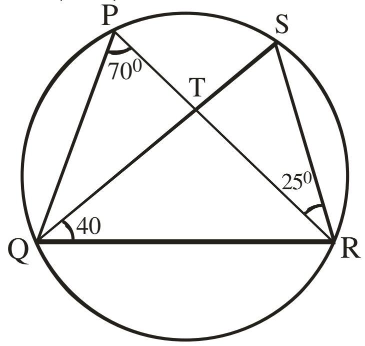
2012/44 Exercise 19.33



In the diagram, /SR/ = /QR/, $\angle SRP = 65^0$ and $\angle RPQ = 48^0$, find $\angle PRQ$

A 65^0 B 45^0 C 25^0 D 19^0

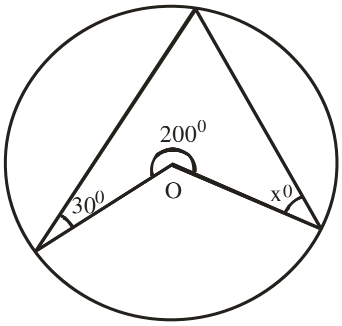
2009/21 (Nov) Exercise 19.34



In the diagram, $\angle QPR = 70^0$, $\angle SQR = 40^0$ and $\angle SRT = 25^0$. Find $\angle PRQ$

A 45^0 B 50^0 C 55^0 D 60^0

2009/38 Exercise 19.35

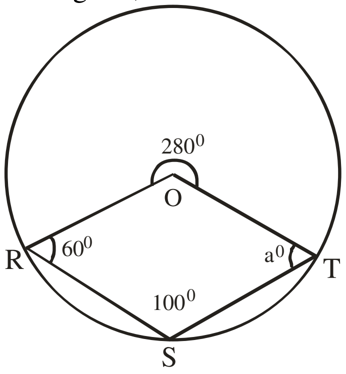


In the diagram, angle 200^0 is subtended at the centre of the circle. Find the value x.

A 30^0 B 50^0 C 80^0 D 100^0

2008/46 Exercise 19.36

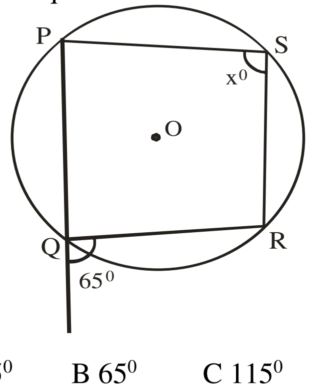
In the diagram, find the size of the angle marked a^0



A 60^0 B 80^0 C 120^0 D 160^0

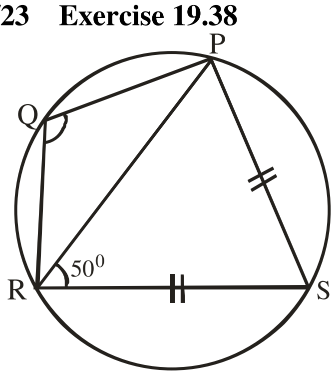
2008/25 Exercise 19.37

In the diagram, O is the centre of the circle and PQRS is a cyclic quadrilateral. Find the value of x



A 25^0 B 65^0 C 115^0 D 130^0

2007/23 Exercise 19.38



In the diagram, /SP/ = /SR/ and $\angle PRS = 50^0$.

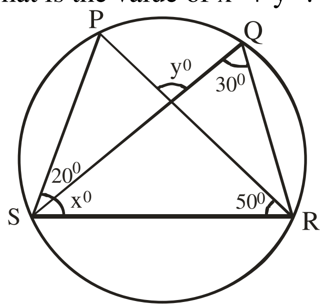
Calculate $\angle PQR$.

A 120^0 B 110^0 C 100^0 D 80^0

2006/21 Exercise 19.39

In the diagram P, Q, R, S are points on the circle, $\angle RQS = 30^0$, $\angle PRS = 50^0$ and $\angle PSQ = 20^0$.

What is the value of $x^0 + y^{-0}$?

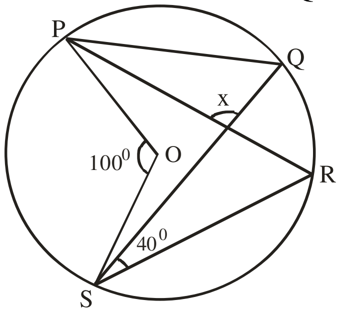


A 260^0 B 130^0 C 100^0 D 80^0

2006/37 Neco Exercise 19.40

In the figure below, PQRS is a circle with centre O.

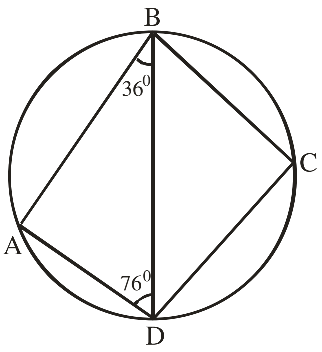
If $\angle POS = 100^0$ and $\angle RSQ = 40^0$, find the marked angle x



A 40^0 B 50^0 C 90^0 D 100^0 E 140^0

2009/45 Neco (Dec) Exercise 19.41

ABCD is a cyclic quadrilateral. Given that $A\hat{D}B = 76^0$ and $A\hat{B}D = 36^0$, find $B\hat{C}D$

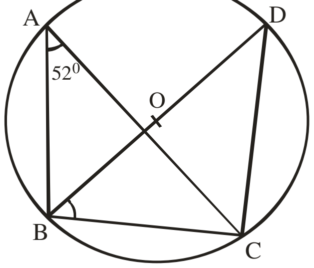


A 56^0 B 68^0 C 76^0 D 112^0 E 248^0

2012/14 Neco Exercise 19.42

In the diagram below, O is the centre of the circle.

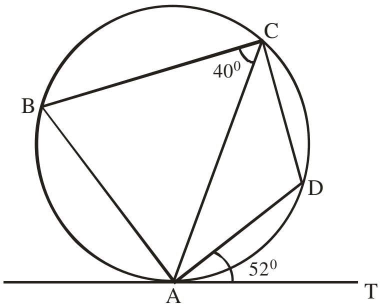
BOD is diameter and angle BAC is 52^0. Find $\angle DBC$



A 14^0 B 38^0 C 52^0 D 128^0 E 142^0

Circle geometry II

2006/3b



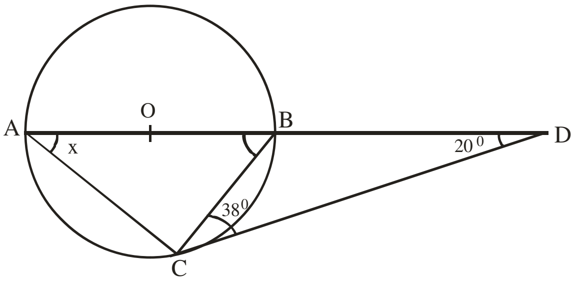
In the diagram, TA is a tangent to the circle at A.
If $\angle BCA = 40^0$ and $\angle DAT = 52^0$, find $\angle BAD$

Solution

$\angle DCA = 52^0$ (Alternate segment $\angle$s)
$\angle BAD + \angle BCD = 180$ (Opp. segment $\angle$s)
$\angle BAD + (40 + 52) = 180$
$\angle BAD = 180 - 92 = 88^0$

2010/38 Neco

Find the value of x in the diagram below where O is the centre of the circle, given that $\angle BCD = 38^0$ and $\angle BDC = 20^0$



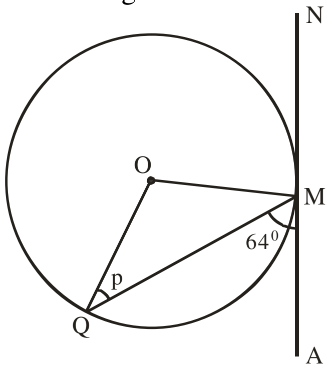
A 142^0 B 128^0 C 110^0 D 58^0 E 32^0

Solution

$\angle CBD + 20^0 + 38^0 = 180^0$ (Sum of $\angle$s in ΔCBD)
$\angle CBD = 180 - 58$
$= 122$
$\angle ABC + \angle CBD = 180^0$ (Sum of $\angle$s on a straight line)
$\angle ABC = 180 - 122$
$= 58$
$x + \angle ACB + 58^0 = 180^0$ (sum of $\angle$s in ΔABC)
$x + 90^0 + 58 = 180$
($\angle ACB$ is 90^0, angle in semi – circle)
$x = 180 - 148$
$= 32^0$ E.

2011/37 Neco

In the diagram below, O is the centre of the circle.
If /AN/ is a tangent to M and $\angle AMQ = 64^0$. Calculate the value of angle P

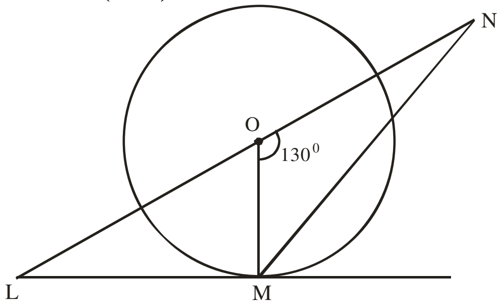


A128^0 B 64^0 C 52^0 D 32^0 E 26^0

Solution

$\angle OMA = 90^0$ (Radius and tangent $\angle$s)
$\angle QMO = 90^0 - 64^0$
$= 26^0$
$p = \angle QMO = 26^0$ (Base angles of isosceles Δ)

2014/35 (Nov)



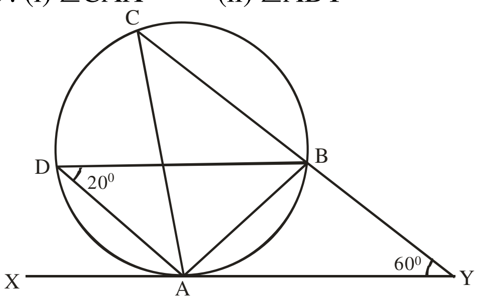
In the diagram, O is the centre of the circle, LM is a tangent and angle MON is 130^0. Find the size of angle OLM
A 65^0 B 50^0 C 45^0 D 40^0

Solution

$\angle OML = 90^0$ (Radius and tangent at contact)
$\angle MOL = 180 - 130^0$ ($\angle$s on line LON)
$= 50^0$
In ΔOLM : $\angle OLM + 50 + 90 = 180^0$ (sum of $\angle$s in Δ)
$\angle OLM = 180 - 140$
$= 40^0$ D.

2005/5a

In the diagram, A, B, C and D are points on the circumference of a circle. XY is a tangent at A.
Find : (i) $\angle CAX$ (ii) $\angle ABY$

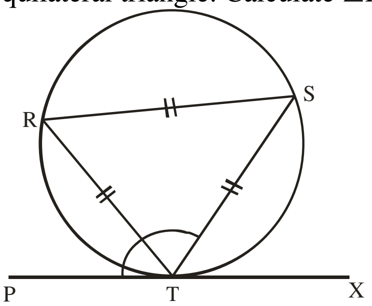


Solution

$\angle ACB = 20^0$ (angle in same segment)
In ΔACY : $\hat{A} + \hat{C} + 60^0 = 180^0$ (sum of $\angle$s in Δ)
$\hat{A} = 180 - (60 + 20)$
$= 100^0$
(i) $\angle CAX + \angle CAY = 180^0$ (sum of $\angle$s on line XAY)
$\angle CAX + 100 = 180$
$\angle CAX = 180 - 100$
$= 80^0$
(ii) $\angle BAY = 20^0$ (Alternate segment)
In ΔABY: $\angle ABY + 20 + 60 = 180^0$ (sum of $\angle$s in Δ)
$\angle ABY = 180 - 80$
$= 100^0$

2008/44

In the diagram, PX is a tangent to the circle and RST is an equilateral triangle. Calculate $\angle PTS$.



A 60^0 B 90^0 C 120^0 D 150^0

Solution

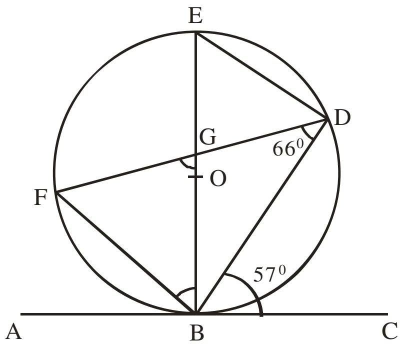
In ΔRTS : $\angle R = \angle T = \angle S = 60^0$ (equilateral Δ)

$\angle PTR = \angle RST = 60^0$ (Alternate segment)

$\angle PTS = \angle PTR + \angle RTS$

$= 60^0 + 60^0 = 120^0$ C.

2007/8b



In the diagram, O is the centre of the circle and ABC is a tangent at B. If $\angle BDF = 66^0$ and $\angle DBC = 57^0$, calculate (i) $\angle EBF$ and (ii) $\angle BGF$

Solution

$\angle BFD = 57^0$ (Alternate segment)

$\angle BED = \angle BFD = 57^0$ (Angles in same segment)

$\angle OBC = 90^0$ (Radius and tangent at contact)

Thus $\angle OBD = 90 - 57 = 33^0$

In ΔBDF: $\angle BFD + 66 + \angle FBD = 180^0$(Sum of $\angle$s in Δ)

$57 + 66 + \angle FBD = 180$

$\angle FBD = 180 - 123 = 57^0$

$\therefore \angle EBF = 57 - 33 = 24^0$

(ii) In ΔBGF

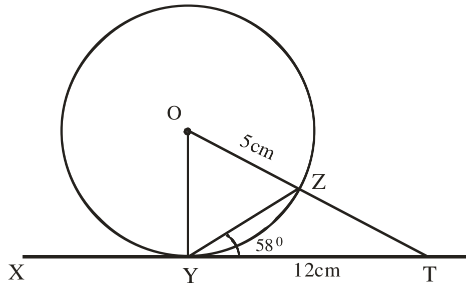
$57 + 24 + \angle BGF = 180$

$\angle BGF = 180 - 81 = 99^0$

2005/49 and 50 NABTEB (Nov)

Use the diagram below to answer question **49** and **50**

XYT is a tangent to the circle centre O and radius 5cm. /YT/ is 12cm and $\angle ZYT$ is 58^0



49. What is the length of OT?

A 5cm B 7cm C 13cm D 17cm

Solution

$\angle OYT = 90^0$ (Radius and tangent at contact)

$OY = OZ = 5cm$ (Radius)

By Pythagoras rule in ΔOYT

$OT^2 = 5^2 + 12^2$

$= 25 + 144$

$OT^2 = 169$

$OT = \sqrt{169} = 13cm$ C.

50. Find the size of $\angle OZY$

A 26^0 B 32^0 C 58^0 D 64^0

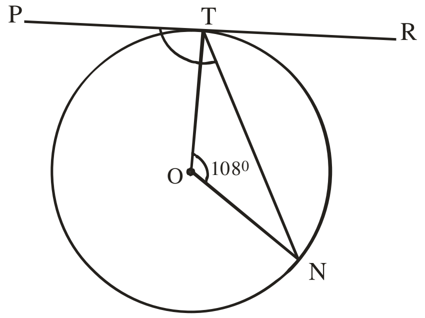
Solution

$\angle OYZ = 90 - 58$

$= 32^0$

$\angle OZY = \angle OYZ = 32^0$ (Base $\angle$s of isosceles Δ)

2015/17



In the diagram, PTR is a tangent to the circle centre O. If angle $TON = 108^0$, calculate the size of angle PTN

A 132^0 B 126^0 C 108^0 D 102^0

Solution

$\angle ONT = \angle OTN$ (Base angles of isosceles Δ)

$108 + \angle ONT + \angle OTN = 180^0$ (Sum of $\angle$s in Δ)

$108 + 2\angle OTN = 180$

$2\angle OTN = 180 - 108$

$\angle OTN = 72 \div 2$

$= 36^0$

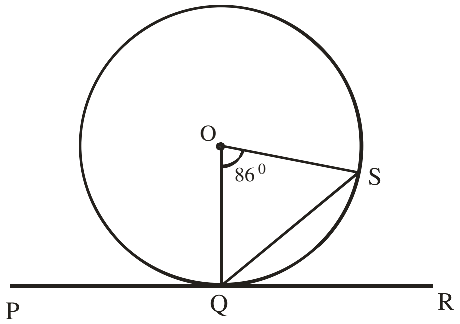
$\angle OTP = 90^0$ (Radius and tangent)

But $\angle PTN = \angle OTP + \angle OTN$

$= 90 + 36$

$= 126^0$ B.

2014/44



In the diagram, O is the centre of the circle $\overline{PR}$ is a tangent to the circle at Q and $\angle SOQ = 86^0$.

Calculate the value of $\angle SQR$

A 43^0 B 47^0 C 54^0 D 86^0

Solution

$\angle OQS = \angle OSQ$ (Base $\angle$s of isosceles Δ)

$86 + \angle OQS + \angle OSQ = 180^0$ (Sum of $\angle$s in Δ)

$2\angle OQS = 180 - 86$

$2\angle OQS = 94$

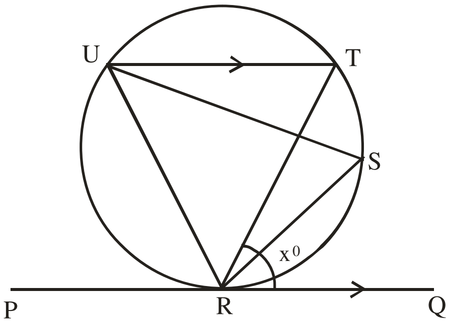
$\angle OQS = 47^0$

$\angle OQR = 90^0$ (tangent and radius at a point of contact)

Thus, $\angle SQR = \angle OQR - \angle OQS$

$= 90 - 47 = 43^0$ A.

2014/32



In the figure, PQ is a tangent to the circle at R and UT is parallel to PQ. If $\angle TRQ = x^0$, find $\angle URT$ in terms of x

A $2x^0$ B $(90 - x)^0$ C $(90 + x)^0$ D $(180 - 2x)^0$

Solution

$\angle RUT = x^0$ (Alternate segment Angles)

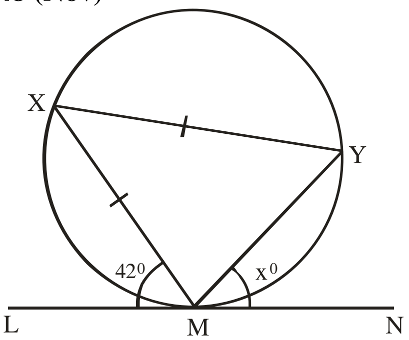
$\angle UTR = \angle TRQ = x^0$ (Alternate $\angle$s between parallel lines)

$\angle URT + \angle RUT + \angle UTR = 180^0$ (Sum of $\angle$s in ΔURT)

$\angle URT + x^0 + x^0 = 180$

$\angle URT = (180 - 2x)^0$ D.

2009/48 (Nov)



In the diagram, XMY is a triangle inscribed in a circle. LMN is a tangent to the circle at M, /XY/ = /XM/ and $\angle XML = 42^0$. Find the value of x

A 42 B 84 C 96 D 138

Solution

$\angle XYM = 42^0$ (Alternate segment)

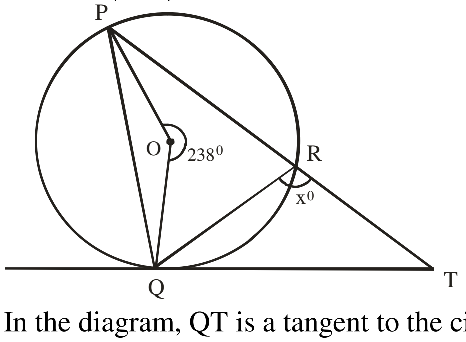
$\angle XMY = 42^0$ (Base $\angle$s of isoscelesΔ)

$\angle LMX + \angle XMY + x^0 = 180^0$ (Sum of $\angle$s on line LMN)

$42 + 42 + x = 180$

$x = 180 - 84 = 96^0$ C.

2009/37 (Nov)



In the diagram, QT is a tangent to the circle PQR centre O. If PT is a straight line, reflex $\angle POQ = 238^0$ and $\angle QRT = x^0$, find the value of x

A 100^0 B 119^0 C 122^0 D 130^0

Solution

Reflex $\angle QOP = 360 - 238$ (angle at a point)

$= 122^0$

$2 \times QRP = 122^0$ (circumference and centre $\angle$s)

$QRP = 122 \div 2$

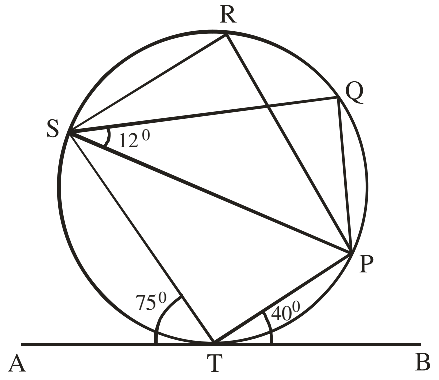
$= 61$

$x + 61^0 = 180^0$ (Sum of $\angle$s on line TRP)

$x = 180 - 61$

$= 119^0$ B.

2013/7a Neco



In the above diagram, ATB is a tangent to the circle at T.

If $A\hat{T}S = 75^0$, $B\hat{T}P = 40^0$ and $P\hat{S}Q = 12^0$, calculate the size of : (i) $\angle SRP$ (ii) $\angle SQP$ (iii) $\angle SPQ$

Solution

$75 + \angle STP + 40^0 = 180^0$ (Sum of $\angle$s on line ATB)

$\angle STP = 180 - 115 = 65^0$

$\angle SRP + \angle STP = 180^0$ ($\angle$s in opp. segment)

$\angle SRP = 180 - 65$

$= 115^0$

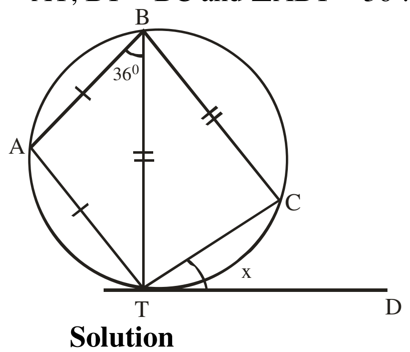
$\angle SQP = \angle SRP = 115^0$ ($\angle$s in same segment)

In ΔSPQ : $12 + \angle SPQ + 115^0 = 180^0$ (Sum of $\angle$s in Δ)

Thus $\angle SPQ = 180 - 127 = 53^0$

2012/11b Neco

In the figure below, TD is a tangent to the circle ABCT, AB = AT, BT = BC and $\angle ABT = 36^0$. Calculate $\angle CTD$



Solution

In ΔABT : $\angle T = 36^0$ (Base angles of isosceles Δ)

$\angle A + \angle B + \angle T = 180^0$ (Sum of $\angle$s in Δ)

$\angle A + 36 + 36 = 180^0$

$\angle A = 180 - 72 = 108^0$

$x = \angle CBT$ (Alternate segment $\angle$s)

$\angle BAT + \angle BCT = 180^0$ ($\angle$s in opposite segment)

$\angle BCT = 180 - 108 = 72^0$

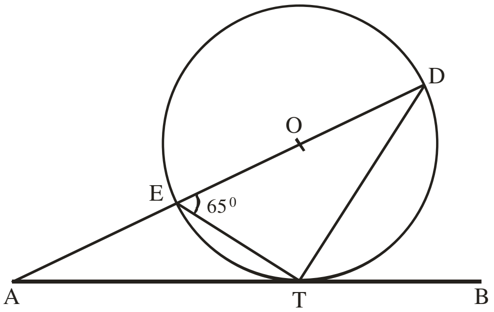
$\angle BCT = \angle BTC\ 72^0$ (Base $\angle$s of ΔBTC)

$\angle CBT + 2\angle BTC = 180^0$ (sum of $\angle$s in Δ)

$\angle CBT = 180 - 144 = 36^0$

$x = \angle CBT = 36^0$ (Alternate segment $\angle$s)

2005/60 Neco



IN the diagram above, $\overline{AB}$ is a tangent to a circle with centre O at T. Given that $\angle TED = 65^0$, find $\angle EAT$

A 25^0 B 40^0 C 65^0 D 70^0 E 115^0

Solution

$\angle ETD = 90^0$ (Angle in semi – circle)

In ΔETD : $65 + 90 + \angle D = 180^0$ (Sum of $\angle$s in Δ)

$\angle D = 180 - 155$

$= 25^0$

$\angle ATE = 25^0$ [$\angle D$] (Alternate segment, chord and tangent)

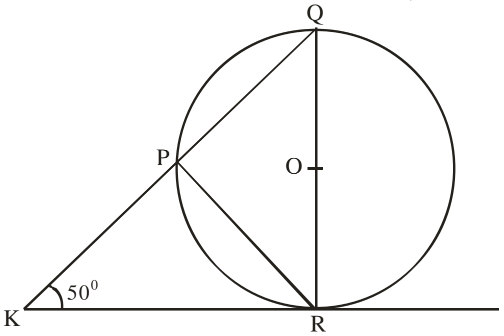
In ΔDAT : $\angle EAT + (25 + 90) + 25 = 180^0$ (Sum of $\angle$s in Δ)

$\angle EAT = 180 - 140$

$= 40^0$ B.

2005/56 Neco

In the diagram below, $\overline{KR}$ is a tangent to the circle with centre O at R. If $\angle RKP = 50^0$, find $\angle RQP$



A 140^0 B 130^0 C 90^0 D 50^0 E 40^0

Solution

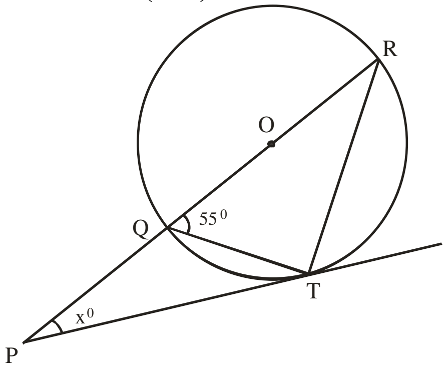
$\angle QRK = 90^0$ (tangent is perpendicular to the radius at contact point)

In ΔKRQ : $50 + 90 + \angle Q = 180^0$ (Sum of $\angle$s in Δ)

$\angle Q = 180 - 140$

$= 40^0$

2006/36 Neco (Dec)



In the diagram above, O is the centre of the circle QRT and PT is a tangent to the circle at T. Calculate angle x

A 145^0 B 125^0 C 55^0 D 35^0 E 20^0

Solution

$\angle QTR = 90^0$ (angles in semi circle)

In ΔQTR : $55 + \angle T + \angle R = 180^0$ (sum of $\angle$s in Δ)

$55 + 90 + \angle R = 180$

$\angle R = 180 - 145 = 35^0$

$\angle PTQ = \angle R = 35^0$ (Angle between tangent and chord through point of contact equal to angle in alternate segment)

In Δ PTR : $x + \angle T + \angle R = 180^0$ (sum of $\angle$s in Δ)

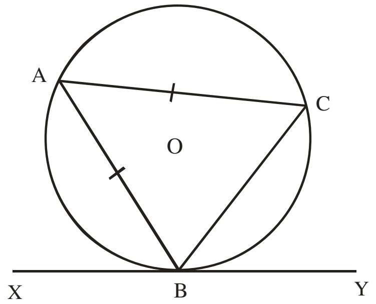
$x + (35 + 90) + 35 = 180^0$

Note that: $\angle T$ in Δ PTR is $\angle PTQ + \angle QTP$

$x = 180 - 160 = 20^0$ E.

2006/38 Neco (Dec)

In the diagram below, XBY is a tangent to the circle ABC of centre O at B. If $\angle CBY = 54^0$ and /AB/ = /AC/, find $\angle XBA$



A126^0 B117^0 C 108^0 D 63^0 E 54^0

Solution

$\angle BAC = \angle CBY = 54^0$ (Angle between tangent and chord through point of contact equal to angle in alternate segment)

ΔABC is isosceles ($AB = AC \Rightarrow \angle B = \angle C$)

Thus $\angle A(54) + \angle B + \angle C = 180^0$ (Sum of $\angle$s in Δ)

$54 + 2\angle B = 180^0$

$2\angle B = 180 - 54$

$2\angle B = 126$

$\angle B = 63$

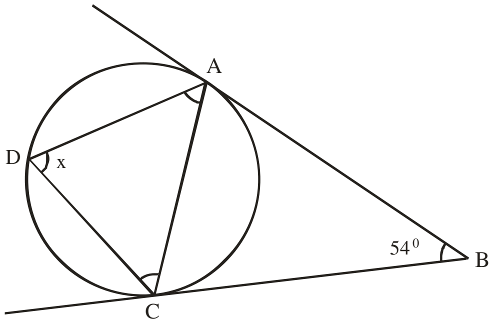
$\angle XBA + \angle ABC + \angle CBY = 180^0$ (Sum of $\angle$s on line XBY)

$\angle XBA + 63 + 54 = 180$

$\angle XBA = 180 - 117 = 63^0$

2009/43 Neco (Dec)

In the diagram below, $\overline{AB}$ and $\overline{CB}$ are tangent to the circle. Given that $C\hat{B}A = 54^0$, calculate $A\hat{D}C$



A 54^0 B 58^0 C 63^0 D 108^0 E 126^0

Solution

ΔABC is isosceles (Tangents BA and BC are equal in lengths)

Thus in ΔABC, $\angle A = \angle C$

$\angle A + \angle C + 54^0 = 180^0$ (Sum of $\angle$s in Δ)

$2\angle A + 54 = 180$

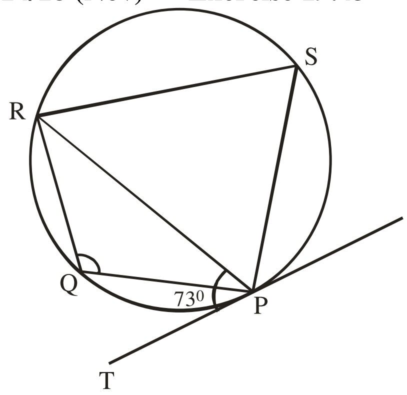
$2\angle A = 180 - 54$

$2\angle A = 126$

$\angle A = 63^0$

$\angle A = x = 63^0$ (Angle between tangent and chord through point of contact equal to angle in alternate segment)

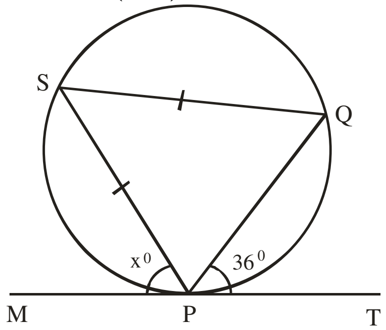
2014/18 (Nov) Exercise 19.43



In the diagram, $\overline{TP}$ is a tangent to the circle PQRS and $\angle RPT = 73^0$. Find $\angle PQR$

A 146^0 B 134^0 C 113^0 D 107^0

2006/57 Neco (Dec) Exercise 19.44

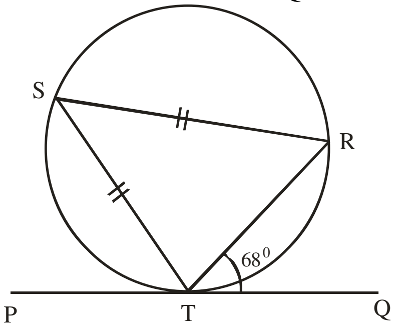


MT is a tangent to a circle SQP at P. If $\angle QPT = 36^0$, find the angle marked x^0.

A 18^0 B 36^0 C 54^0 D 72^0 E 144^0

2014/48 Neco Exercise 19.45

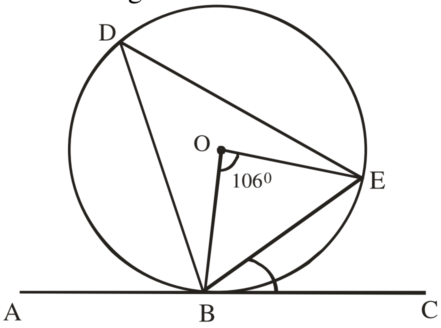
In the diagram below, PQ is the tangent to the circle RST at T. /ST/ = /SR/ and $\angle RTQ = 68^0$. Find $\angle PTS$.



A 34^0 B 56^0 C 68^0 D 72^0 E 112^0

2013/39 Neco Exercise 19.46

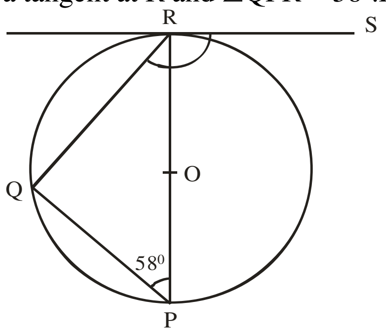
In the diagram below, BDE is a circle with centre O. ABC is a tangent at B and $\angle BOE = 106^0$. Find $\angle CBE$



A 37^0 B 45^0 C 53^0 D 74^0 E 90^0

2006/17 Exercise 19.47

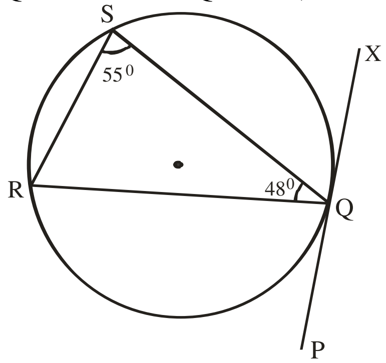
In the diagram, PR is a diameter of a circle centre O. RS is a tangent at R and $\angle QPR = 58^0$.Find $\angle QRS$



A 112^0 B 116^0 C 122^0 D 148^0

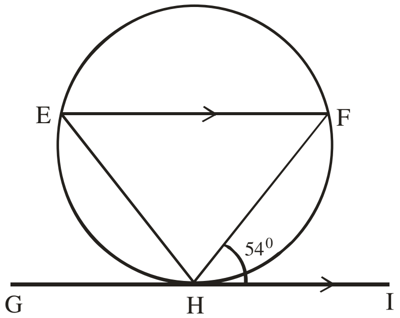
2005 /48 (old) Exercise 19.48

In the diagram PQX is a tangent to the circle at Q, $\angle RSQ = 55^0$ and $\angle RQS = 48^0$, calculate $\angle SQX$



A 42^0 B 48^0 C 55^0 D 77^0

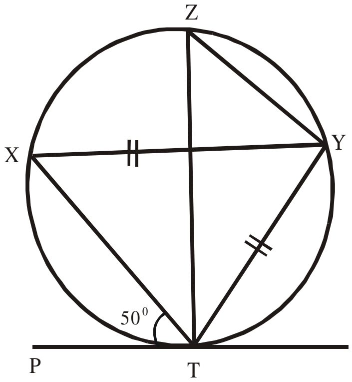
2010/35 Exercise 19.49



In the diagram, GI is a tangent to the circle at H.
If EF//GI, calculate the size of $\angle EHF$

A 126^0 B 72^0 C 54^0 D 28^0

1995/27 UME Exercise 19.50

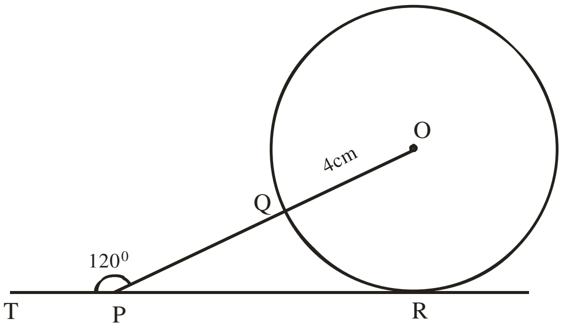


PT is a tangent to the circle TYZX, YT = YX and $\angle PTX = 50^o$. Calculate $\angle TZY$.

A. 50^o B. 65^o C. 85^o D.130^o

Circle geometry III

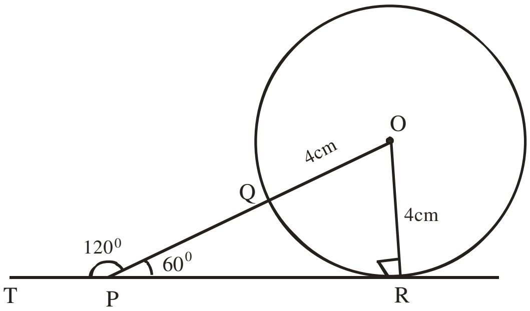
2012/42



The diagram is a circle of radius |OQ| = 4cm. $\overline{TR}$ is a tangent to the circle at R. If $\angle$TPO = 120^0, find |PQ|

A 2.32cm B 1.84cm C 0.62cm D 0.26cm

Solution



$\angle$OPR = 60^0 (120 + $\angle$OPR = 180^0 straight line$\angle$)
$\angle$ORP = 90^0 (Radius and tangent to a circle)
The resulting triangle is right – angled
The sides relevant to 60^0 here are :
Hypotenuse PO and opposite OR i.e SOH

$$\text{Sin } 60^0 = \frac{4}{PO}$$

$$0.8660 = \frac{4}{PO}$$

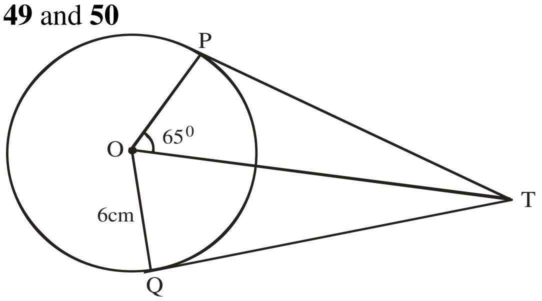
$$PO = \frac{4}{0.8660}$$

= 4.6189cm

But PQ = PO – QO
= 4.6189 – 4
= 0.6189cm
≈ 0.62cm . C

2005/49 and 50 NABTEB

Use the information given in the figure below to answer questions **49** and **50**



PT and QT are tangents to the circle centre O and radius 6cm | PT | = 12cm and $\angle$ POT = 65^0

49. Calculate the length of OT

A 6cm B 8cm C 10cm D 14cm

Solution

Applying two tangents to a circle theorem
$\angle$OPT = $\angle$ OQT = 90^0
OP = OQ = 6cm
Thus by Pythagoras rule
$OT^2 = PT^2 + OP^2$
$= 12^2 + 6^2$
$OT^2 = 144 + 36$
$OT = \sqrt{180}$
= 13.416cm

50. Find $\angle$ PTQ

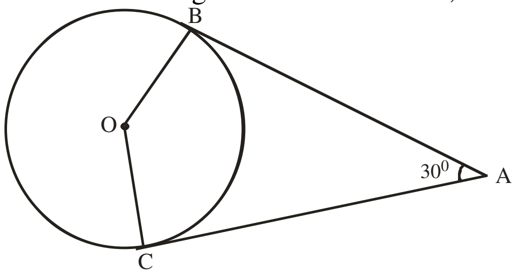
A 50^0 B 65^0 C 115^0 D 130^0

Solution

In ΔOPT : 90 + 65 + $\angle$PTO = 180^0 (Sum of $\angle$s in Δ)
$\angle$ PTO = 180 – 155
= 25^0
Similarly $\angle$QTO = 25^0
Thus $\angle$PTQ = 25 + 25
= 50^0 A.

2006/50 NABTEB (Nov)

In the diagram below, O is the centre of the circle with $\overline{AB}$ and $\overline{AC}$ as tangents. If $\angle$BAC = 30^0, find $\angle$BOC

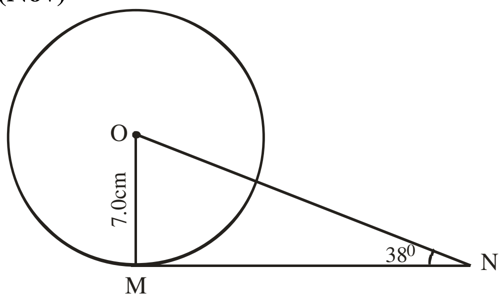


A 270^0 B 165^0 C 150^0 D 75^0

Solution

Applying two tangents to a circle theorem
$\angle$ABO = $\angle$ACO = 90^0
90 + 30 + 90 + $\angle$BOC = 360^0 (Sum of $\angle$s in Quad ABOC)
$\angle$BOC = 360 – 210
= 150^0 C.

2009/26 (Nov)



In the diagram, O is the centre of the circle and MN is a tangent to the circle at M. If the radius of the circle is 7.0cm and $\angle$MNO = 38^0, calculate |MN|

A 10.70cm B 8.96cm C 7.86cm D 6.42cm

Solution

$\angle$OMN = 90^0 (Radius and tangent to circle)
The resulting triangle is right – angled
The sides relevant to 38^0 here are:
Adjacent MN and opposite MO i.e TOA

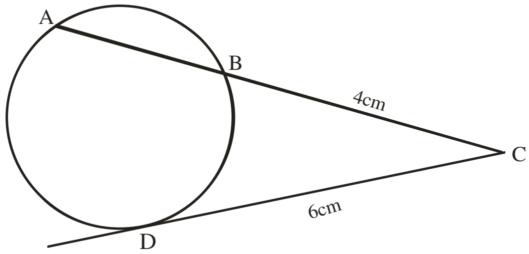
$\text{Tan } 38^0 = \dfrac{7}{MN}$

$0.7813 = \dfrac{7}{MN}$

$MN = \dfrac{7}{0.7813}$

$= 8.9594\text{cm} \approx 8.96\text{cm}$ B.

2008/39 Neco



Find |AB| in the above figure if |BC| = 4cm and |DC| = 6cm

A 4 B 5 C 6 D 9 E 14

Solution

Applying tangent and secant theorem

$DC^2 = BC \cdot AC$

$6^2 = 4 \cdot AC$

$\dfrac{36}{4} = AC$

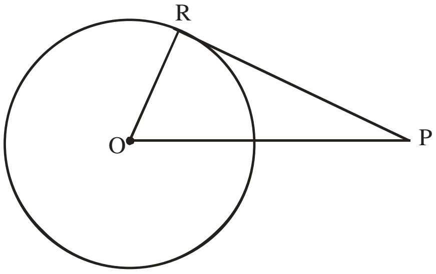
$9\text{cm} = AC$

But $AB = AC - BC$

$= 9\text{cm} - 4\text{cm} = 5\text{cm}$ B.

2004/50 NABTEB

In the diagram PR is a tangent at a point R on circle centre O and radius OR = 8cm



If angle OPR is 60^0, calculate the length of OP to the nearest whole number

A 4cm B 7cm C 9cm D 16cm

Solution

$\angle ORP = 90^0$ (Radius and tangent to circle)

Applying special angle trig ratio

The side relevant to 60^0 here are:

Hypotenuse OP and opposite OR i.e SOH

$\text{Sin } 60^0 = \dfrac{OR}{OP}$

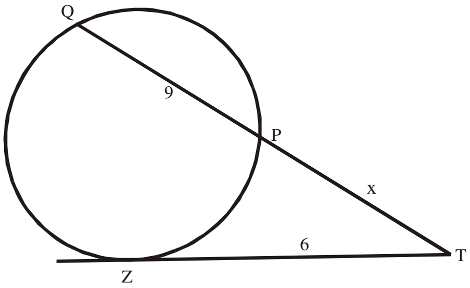
$\text{Sin } 60 = \dfrac{8}{OP}$

$\dfrac{\sqrt{3}}{2} = \dfrac{8}{OP}$

$OP = \dfrac{2 \times 8}{\sqrt{3}} = \dfrac{16 \times \sqrt{3}}{3}$

$= 9.238\text{cm} \approx$ 9cm to whole number

1999/31 UME



In the figure above, TZ is tangent to the circle QPZ.

Find x if TZ = 6 units and PQ = 9units

A. 3 B. 4 C. 5 D. 6

Solution

Applying tangent and secant theorem

$ZT^2 = PT \cdot QT$

$ZT^2 = x(PQ + x)$

$6^2 = x(9 + x)$

$36 = 9x + x^2$

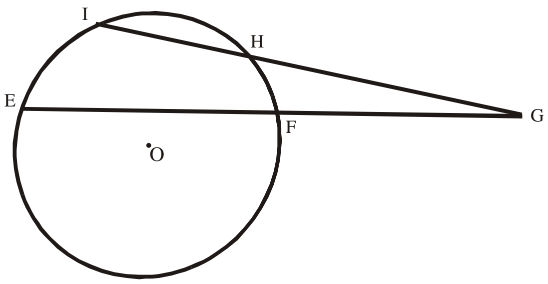
$x + 9x - 36 = 0$

factorizing: $(x + 12)(x - 3) = 0$

$x = 3$ or -12

$x = 3$ (A)

1991/33 PCE



Two chords EF and IH intersect at G when produced,

If EG = 8cm, GF = 3cm, and IG = 6cm, find GH.

A. 8cm B. 6cm C. 4cm D. 2cm

Solution

Applying two chords of circle intersecting outside theorem

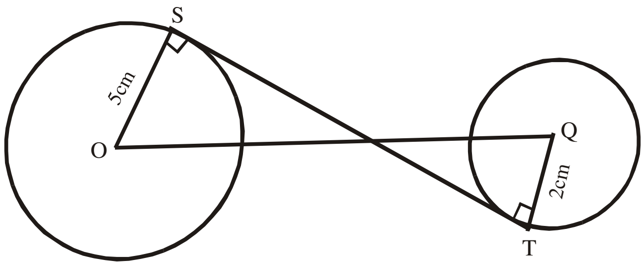
IG.GH = EG.GF

$6 \times GH = 8 \times 3$

$GH = \dfrac{8 \times 3}{6}$

GH = 4cm (C)

1993/34 UME

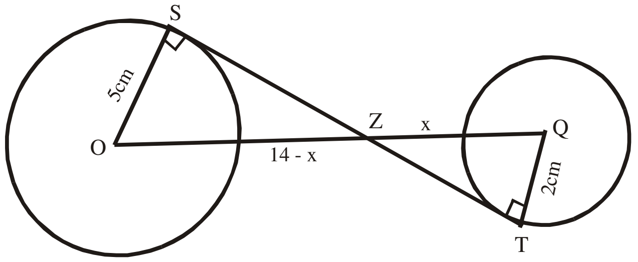


In the figure above, the line segment ST is a tangent to the two circles at S and T. O and Q are the centers of the circles with OS = 5cm, QT = 2cm and OQ = 14cm. Find ST

A. $7\sqrt{3}$ cm B. 12cm C. $\sqrt{87}$ cm D. 7cm

Solution

Let the point of intersection of the two lines be Z. Then let us re-label applying transverse common tangent theorem



$$\frac{OZ}{QZ} = \frac{5}{2}$$

$$\frac{14 - x}{x} = \frac{5}{2}$$

$2(14 - x) = 5x$

$28 - 2x = 5x$

$28 = 5x + 2x$

$28 = 7x$

Thus, $x = 4$

We can only get ST by Pythagoras rule in each of the right – angled triangles.

In Δ OSZ

$(14 - x)^2 = 5^2 + SZ^2$

$(14 - 4)^2 = 5^2 + SZ^2$

$10^2 = 5^2 + SZ^2$

$10^2 - 5^2 = SZ^2$

$(10 - 5)(10 + 5) = SZ^2$

$5 \times 15 = SZ^2$

$\sqrt{5 \times 5 \times 3} = SZ$

$5\sqrt{3} = SZ$

Also in Δ QTZ

$x^2 = 2^2 + TZ^2$

$4^2 = 2^2 + TZ^2$

$4^2 - 2^2 = TZ^2$

$(4 - 2)(4 + 2) = TZ^2$

$2 \times 6 = TZ^2$

$\sqrt{2 \times 2 \times 3} = TZ$

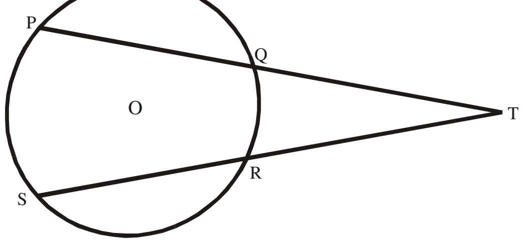
$2\sqrt{3} = TZ$

Therefore; ST = SZ + TZ

$= 5\sqrt{3} + 2\sqrt{3} = 7\sqrt{3}$ cm (A)

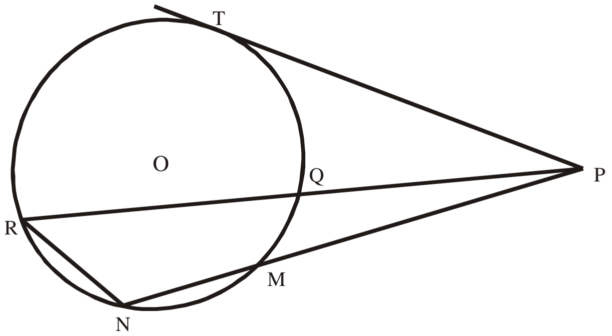
1986/33 UME Exercise 19.51

In the diagram below, PQ and RS are chords of a circle center O which meet at T outside -the circle. If TP = 24cm, TQ = 8cm and TS = 12cm, find TR



A.16cm B.14cm C.12cm D.8cm

1991/35 and 36 UME Exercise 19.52



PMN and PQR are two secants of the circle MQTRN and PT is a tangent.

35. If PM = 5cm, PN = 12cm and PQ = 4.8cm ,calculate the respective lengths of PR and PT in centimeters.

A. 7.3, 5.9 B. 7.7, 12.5 C. 12.5, 7.7 D. 5.9, 7.3

36. If $\angle PNR = 110^0$ and $\angle PMQ = 55^0$, find $\angle MPQ$

A 40^0 B 30^0 C 25^0 D 15^0

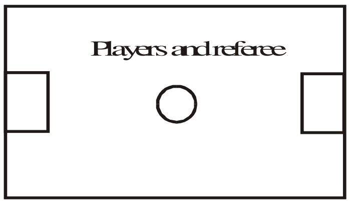
CHAPTER TWENTY

Plane mensuration

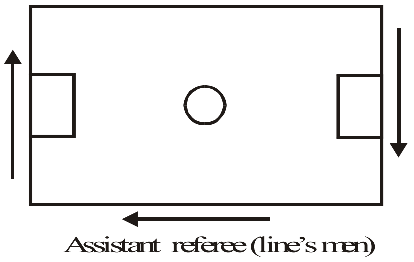
Area & perimeter of plane shapes

Area and perimeter are terms used in mensuration. Students tend to confuse the two terms thus, interchanging the formulae to be used.

To give a clear difference between them, let us consider an example of a football field that is rectangular in shape. The 11 players of both teams play within the pitch area and the referee officiates within the pitch (area)

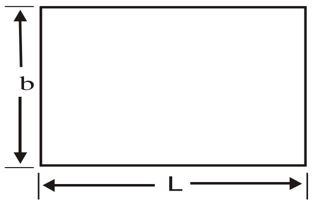


As for the assistant referee (lines' men) they work within the lines outside the field (perimeter). They run round the length and breadth of the field but <u>not within</u> the field as the players



Thus, area is the content of any plane shape while the perimeter is the distance round the shape.

Rectangle



Area of rectangle $= L \times b$

Perimeter of rectangle $= 2L + 2b$

$= 2 (L + b)$

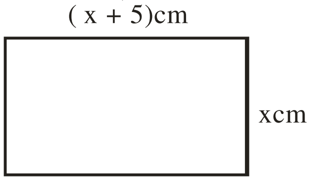
2006/6b (Nov)

The length of a rectangle is 5cm longer than the width and the area is $36cm^2$. Find the:

(i) width (ii) perimeter of the rectangle

Solution

Let the width be x, then we sketch as:



Area of rectangle $= L \times B$

$36 = (x + 5) \times x$

$36 = x^2 + 5x$

Rearranging

$x^2 + 5x - 36 = 0$

Factorizing 36; 2×18, $2 \times 2 \times 9$ i.e 4 and 9

$x^2 + 9x - 4x - 36 = 0$

$x(x + 9) - 4(x + 9) = 0$

$(x + 9)(x - 4) = 0$

$x + 9 = 0$ or $x - 4 = 0$

$x = -9$ or 4

We accept x = 4cm as width

(ii) Perimeter $= 2(L + b)$

$= 2 [(x + 5) + x]$

$= 2 [(4 + 5) + 4]$

$= 2 (13)$

$= 26cm$

2014/46 (Nov)

Find the dimensions of a rectangle whose perimeter and area are 46cm and $112cm^2$ respectively

A 16cm by 7cm B 17cm by 6cm C 14cm by 9cm D 12cm by 6cm

Solution

Perimeter of rectangle $= 2(L + b)$

$46 = 2 (L+ b)$

$23 = L + b$

$23 - L = b$

Also area of rectangle $= L \times b$

$112 = L \times b$

Substituting for b

$112 = L (23 - L)$

$112 = 23L - L^2$

$L^2 - 23L + 112 = 0$

Factorising

$L^2 - 16L - 7L + 112 = 0$

$L(L - 16) - 7 (L - 16) = 0$

$(L - 16) (L - 7) = 0$

$L - 16 = 0$ or $L - 7 = 0$

L = 16cm or 7cm

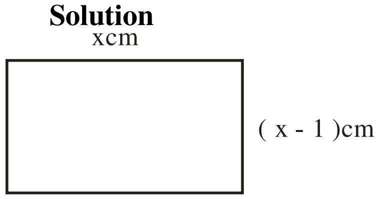
Thus, dimensions of rectangle = 16cm by 7cm

2010/24

A rectangle has length x cm and width (x – 1) cm. if the perimeter is 16cm, find the value of x

A $3^1/_2$ cm B 4cm C $4^1/_2$ cm D 5cm

Solution



Perimeter of rectangle $= 2(L + b)$

$16 = 2 [x + (x - 1)]$

$8 = x + x - 1$

$8 = 2x - 1$

$8 + 1 = 2x$

$9 = 2x$

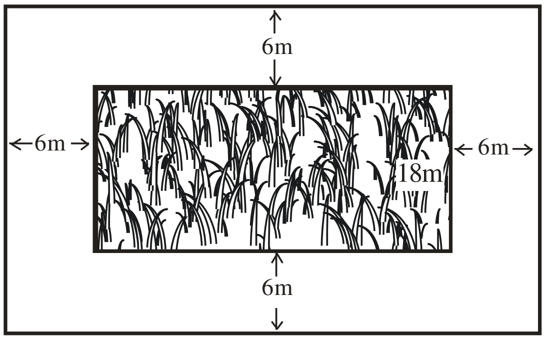
$^9/_2 = x,$

$x = 4^1/_2$ cm (C)

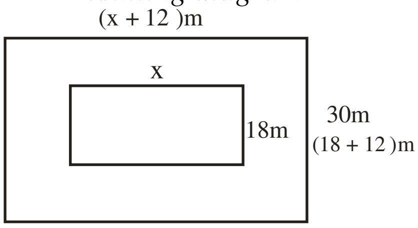
2010/6b

A rectangular playing field is 18m wide. It is surrounded by a path 6m wide such that its area is equal to the area of the path. Calculate the length of the field

Solution



Resulting diagram



Area of rectangular field = area of path

Area of rect. field = area of big rect. – area of small rectangle

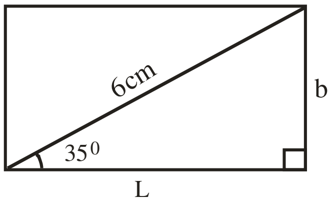
$$x \times 18 = (x + 12) \times 30 - x \times 18$$
$$18x = 30x + 360 - 18x$$
$$18x = 12x + 360$$
$$18x - 12x = 360$$
$$6x = 360$$
$$x = 60m$$

2009/15 (Nov)

The diagonal of a rectangle is 6cm long and makes an angle of 35^0 with one of the sides. Find, correct to 1 decimal place, the perimeter of the rectangle.

A 6.9cm B 9.8cm C 16.7cm D 16.9cm

Solution



Perimeter of rectangle = 2 (L × b)

We solve for L and b in the resulting right–angled Δ by trig ratio

$$\text{Sin } 35^0 = \frac{b}{6}$$
$$6 \sin 35^0 = b$$
$$3.44cm = b$$

$$\text{Also } \cos 35^0 = \frac{L}{6}$$
$$6 \cos 35 = L$$
$$4.91cm = L$$

Thus, perimeter of rectangle = 2(4.91 + 3.44)
= 2(8.35)
= 16.7cm to 1 dp

2009/8 (Nov)

The area of a rectangular football field is $7200m^2$ while its perimeter is 360m. Calculate the:

(a) dimensions of the field

(b) cost of clearing the field at ₦ 6.50per square metre, leaving a margin of 2m wide along the longer sides;

(c) Percentage of the part **not** cleared

Solution

Area of rectangular field = 7200

$$L \times b = 7200 \quad \text{----- (1)}$$

Perimeter of rectangular field = 360

$$2(L + b) = 360 \text{ - --- (2)}$$
$$L + b = 180$$
$$L = 180 - b$$

Substitute L = 180– b into (1)

$$(180 - b)b = 7200$$
$$180b - b^2 = 7200$$

Rearranging

$$b^2 - 180b + 7200 = 0$$

Factorizing (– 120 and –60)

$$b^2 - 120b - 60b + 7200 = 0$$
$$b(b - 120) - 60(b - 120) = 0$$
$$(b - 120)(b - 60) = 0$$
$$b - 120 = 0 \text{ or } b - 60 = 0$$
$$b = 120m \text{ or } 60m$$

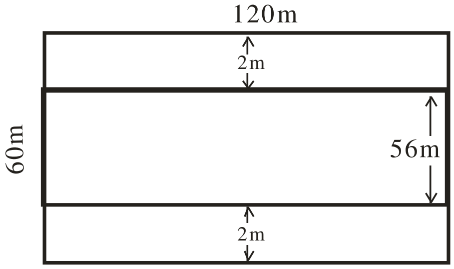
Alternative step to factorisation

When to factorise look difficult like we have here, we may apply the general formula as

$$b = \frac{-(-180) \pm \sqrt{(-180)^2 - 4 \times 1 \times 7200}}{2 \times 1}$$
$$= \frac{180 \pm \sqrt{32400 - 28800}}{2}$$
$$= \frac{180 \pm \sqrt{3600}}{2}$$
$$= \frac{180 \pm 60}{2} = 90 \pm 30$$
$$= 120 \text{ or } 60$$

Without solving further L is 120m and b is 60m since both satisfies (1) and (2)

(b)



Here the length remains the same as 120m the breadth is reduced by 2m + 2m i.e 4m to become 56m

Thus cost of clearing the field = per square metre
= 120 × 56 × 6.5
= ₦ 43,680.00

C. percentage of part **not** cleared

$$= \frac{\text{Area of part not cleared}}{\text{Total area of rectangle}} \times 100$$

Area not cleared
= area of total rectangle – area of smaller rectangle
= 120 × 60 – 120 × 56
= 7200 – 6720
= $480m^2$

$$\text{Thus, \% of part not cleared} = \frac{480}{7200} \times 100$$
$$= 6.67\%$$

2009/11b

A rope 60cm long is made to form a rectangle. If the length is 4 times its breadth, calculate, correct to one decimal place the (i) length (ii) diagonal of the rectangle

Solution

The first sentence implies

Perimeter of rectangle = 60cm

$2(L + b) = 60$

$L + b = 30$ ----- (1)

Next, we are given that

$L = 4b$ ------- (2)

Substitute (2) into (1)

$L + b = 30$ becomes

$4b + b = 30$

$5b = 30$

$b = 6$cm

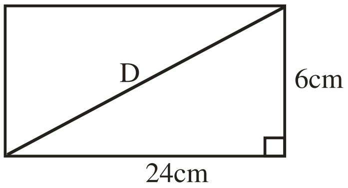
Substitute b value into (2) to get L

$L = 4b$ becomes

$L = 4 \times 6$

$= 24$cm

(ii)



By Pythagoras rule

$D^2 = 24^2 + 6^2$

$D^2 = 576 + 36$

$D^2 = 612$

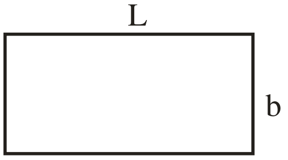
$D = \sqrt{612}$

$= 24.7386$cm ≈ 24.7cm to 1d.p

2008/5a

A rectangle field is l metre long and b metre wide. Its perimeter is 280metres. If the length is two and a half times its breadth, find the value of L and b

Solution



Perimeter of rectangle = $2(L + b)$

$2(L + b) = 280$ ----- (1)

and $L = 2^1/_2 \times b$ ---- (2)

From (1)

$2(L + b) = 280$

$L + b = 140$ ----- (a)

From (2)

$L = 2^1/_2 \times b$

$L = \frac{5b}{2}$ --- (c)

Substitute (c) into (a)

$\frac{5b}{2} + b = 140$

Multiply through by 2 to clear fraction

$5b + 2b = 280$

$7b = 280$

$b = 40$m

Next, we substitute b value into (a)

$L + 40 = 140$

$L = 140 – 40 = 100$m

2006/45

A rectangle carpet 2.5m long and 2.4m wide covers 5% of a rectangular floor. Calculate the area of the floor

A $30m^2$ B $57m^2$ C $120m^2$ D $225m^2$

Solution

Let the area of the rectangular floor be x

5% of x = area of rectangular carpet

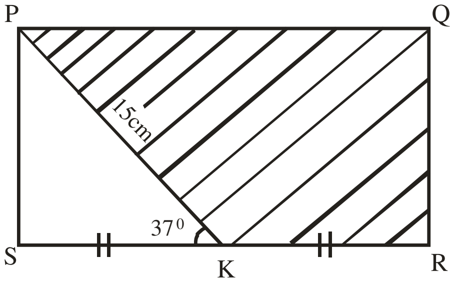
$\frac{5}{100}x = 2.5 \times 2.4$

$\frac{5}{100}x = 6$

$5x = 6 \times 100$

$x = \frac{6 \times 100}{5} = 120m^2$ (C)

2007/10b



In the diagram, PQRS is a rectangle. /PK/ = 15cm, /SK/ = /KR/ and $P\hat{K}S = 37^0$. Calculate, correct to three significant figures:

(i) /PS/; (ii) /SK/ and (iii) the area of the shaded portion

Solution

(i) Applying trig ratio to ΔPKS

$\text{Sin } 37 = \frac{PS}{15}$

$15 \text{ Sin } 37 = PS$

$15 \times 0.6018 = PS$

$9.027\text{cm} = PS$

$PS \approx 9.03$cm to sf

(ii) Also by trig ratio in ΔPKS

$\text{Cos } 37 = \frac{SK}{15}$

$15 \cos 37 = SK$

$15 \times 0.7986 = SK$

$11.979\text{cm} = SK$

$SK \approx 11.98$cm to 3sf

(iii) Area of shaded portion = area of rectangle – area of Δ

$= /PS/ \times /SR/ - \frac{1}{2} /SK/ \times /PK/ \text{ Sin } 37^0$

$= 9.03 \times (2 \times SK) - \frac{1}{2} \times 11.98 \times 15 \times 0.6018$

$= 9.03 \times (2 \times 11.98) – 54.07173$

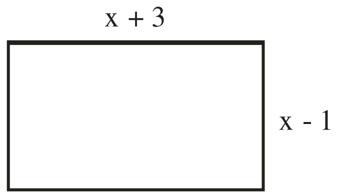
$= 216.3588 – 54.07173$

$= 162.28707\ cm^2$

$\approx 162cm^2$ to 3 s.f

2005/25 (old)
The sides of a rectangle are (x – 1) and (x + 3). If the area of the rectangle is 45, find the positive value of x.
A 4 B 5 C 6 D 8

Solution



Length is bigger, of course x + 3 is bigger
Area of rectangle = length × breadth

$45 = (x + 3)(x - 1)$
$45 = x^2 - x + 3x - 3$
$45 = x^2 + 2x - 3$
$x^2 + 2x - 48 = 0$

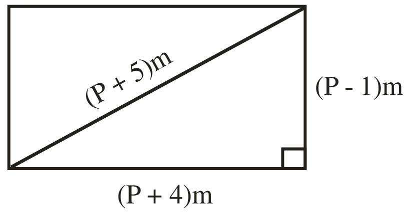
Factorizing
48, 2× 24, 2× 2 ×12, 2 × 2× 2 × 6 i.e 8 × 6

$x^2 + 8x - 6x - 48 = 0$
$x(x + 8) - 6(x + 8) = 0$
$(x + 8)(x - 6) = 0$
$x + 8 = 0$ or $x - 6 = 0$
$x = -8$ or 6

We accept x = 6 C.

2014/9a Neco (Dec)
A rectangle lawn of length (p + 4) m is (p – 1)m wide. If the diagonal is (p + 5) m, find the
(i) value of P correct to 2 decimal places;
(ii) area of the lawn correct to 1 decimal place

Solution



(i) Applying Pythagoras rule to the resulting right – angled Δ

$(p + 5)^2 = (p + 4)^2 + (p - 1)^2$
$p^2 + 10p + 25 = p^2 + 8p + 16 + p^2 - 2p + 1$
$p^2 + 10p + 25 = 2p^2 + 6p + 17$
$2p^2 - p^2 + 6p - 10p + 17 - 25 = 0$
$p^2 - 4p - 8 = 0$

The resulting quadratic equation is not factorisable
Apply the general formula

$$P = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$p = \frac{-(-4) \pm \sqrt{(-4)^2 - 4 \times 1 \times (-8)}}{2 \times 1}$$

$$= \frac{4 \pm \sqrt{16 + 32}}{2}$$

$$= \frac{4 \pm \sqrt{48}}{2} = \frac{4 \pm 6.928}{2} = \frac{4 + 6.928}{2} \text{ or } \frac{4 - 6.928}{2}$$

$$= \frac{10.928}{2} \text{ or } \frac{-2.928}{2}$$

We accept the positive value since we are dealing with length, breadth and diagonal

$= 5.464$m
≈ 5.46m to 2dp

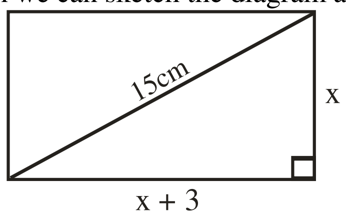
(ii) Area of rectangle lawn = length × breadth

$= (p + 4) \times (p - 1)$
$= (5.46 + 4) \times (5.46 - 1)$
$= 9.46 \times 4.46$
$= 42.1916\text{m}^2$
$\approx 42.2\text{m}^2$ to 1 d.p

2006/5a NABTEB
A diagonal of a rectangle is 15cm. If the length is 3cm greater than the breadth, find the perimeter of the rectangle.

Solution

Let the breadth be x
then we can sketch the diagram as:



By Pythagoras rule

$15^2 = (x + 3)^2 + x^2$
$225 = x^2 + 6x + 9 + x^2$
$225 = 2x^2 + 6x + 9$

Rearranging
$2x^2 + 6x - 216 = 0$
Reducing further
$x^2 + 3x - 108 = 0$
Factorizing [108: 2 × 54; 4 × 27; 4 × 3 × 9 i.e 12 × 9]

$x^2 + 12x - 9x - 108 = 0$
$x(x + 12) - 9(x + 12) = 0$
$(x + 12)(x - 9) = 0$
$x + 12 = 0$ or $x - 9 = 0$
$x = -12$ or 9

Accept x = 9
Thus perimeter of the rectangle = 2 (L + b)

$= 2[x + (x + 3)]$
$= 2(9 + 9 + 3)$
$= 2(21)$
$= 42$cm

2001/21 (Nov)
The perimeter of a rectangle is 35cm. the ratio of the length of the rectangle to its width is 3 : 2. Calculate the dimension of the rectangle.

Solution

Perimeter of rectangle = 35cm

2 (L + b) = 35 ----- (1)

Next, length : breadth

L : b
3 : 2

i.e $\frac{L}{b} = \frac{3}{2}$

$L = \frac{3b}{2}$

Substitute L into (1)
2 (L + b) = 35 is rearranged as

$L + b = \frac{35}{2}$

$\frac{3b}{2} + b = \frac{35}{2}$

Multiply through by 2 to clear fractions

$3b + 2b = 35$

$5b = 35$

$b = 7cm$

Substitute b value into (1)

$2(L + 7) = 35$

$L + 7 = 17.5$

$L = 17.5 - 7 = 10.5cm$

2010/28 Exercise 20.1

What is the length of a rectangular garden whose perimeter is 32cm and area $39cm^2$?

A 25cm B 18cm C 13cm D 9cm

2011/41 Exercise 20.2

A rectangular garden measures 18.6m by 12.5m Calculate, correct to three significant figures, the area of the garden

A $230m^2$ B $231m^2$ C $232m^2$ D $233m^2$

2008/30 Neco (Dec) Exercise 20.3

The area of a rectangle is $30cm^2$ and its perimeter is 22cm. state the dimensions of the rectangle

A 10cm × 3cm B 6cm × 5cm C 15cm × 2cm D 12cm × 2.5cm E 25cm × 1.2cm

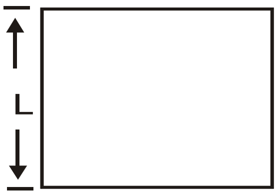
2005/9c (old) Exercise 20.4

EFGH is the rectangular floor of a room in which a carpet 5m by 3m is laid, leaving a uniform margin of x metres round it. If the total area of the margin is $20m^2$, find the value of x

1975/8a Exercise 20.4b

The area of a rectangular park is $8000m^2$. If it had been 10m shorter and also 10m wider, the area would have increased by $100m^2$. Calculate the original length and width of the park

Square



Area of square = $L \times L = L^2$

Perimeter of square = $4L$

2008/15 NABTEB (Nov)

Find the area of a square with one side 4.5cm

A $16.25cm^2$ B $18.75cm^2$ C $20.25cm^2$ D $21.50cm^2$

Solution

Area of square = side squared

$= 4.5^2$

$= 20.25cm^2$ (C)

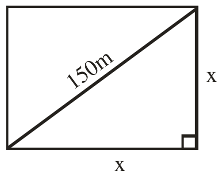
2009/21 NABTEB (Nov)

Find the area of a square field with each diagonal 150m long

A 32,500m B $32,500m^2$ C $22,500m^2$ D 22,500m

Solution

Square has all sides equal



By Pythagoras rule : $150^2 = x^2 + x^2$

$22500 = 2x^2$

$11250 = x^2$

Since area of square $= x \times x$

$= x^2$

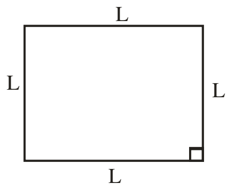
$= 11250m^2$

2007/14

The area of a square field is $110.25m^2$. Find the cost of fencing it round at ₦ 75.00per metre.

A ₦ 1,575.00 B ₦ 3,150.00 C ₦ 4,734.30 D ₦ 8,268.75

Solution



Perimeter of a shape = distance round it

$= L + L + L + L$

$= 4L$

First, we find L from the given area

Area of square = L^2

$110.25 = L^2$

$\sqrt{110.25} = L$

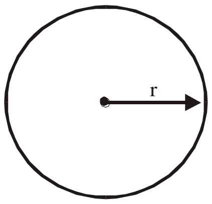
Perimeter = 4×10.5

$= 42m$

Cost of fencing = 42×75

= ₦ 3,150.00 (B)

Circle



Area of circle = πr^2

Perimeter of circle (circumference) = $2\pi r$

2008/16 NABTEB (Nov)

The area of a circle is $81\pi cm^2$. Find the diameter of the circle.

A 7cm B 9cm C 13cm D 18cm

Solution

Note that diameter is radius r times 2

Area of circle = πr^2

$81\pi = \pi r^2$

$81 = r^2$

$\sqrt{81} = r$

$9cm = r$

Thus, diameter = $2 \times r$

$= 2 \times 9cm$

$= 18cm$ (D)

2008/17 NABTEB (Nov)

Calculate the circumference of a circle, centre O and radius 3.5cm? (Take $\pi = {}^{22}/_{7}$)

A 11cm B 15cm C 19cm D 22cm

Solution

Circle circumference = $2\pi r$

$= 2 \times \frac{22}{7} \times 3.5$

$= 22cm$ (D)

2005/22 (old)

If the circumference of a circle is 88cm, find in m^2, the area of the circle (Take $\pi = 3.143$)

A 0.00616 B 0.0616 C 0.616 D 61.6

Solution

Area of circle = πr^2

We can get r from the given circumference

Circumference of circle = $2\pi r$

$88 = 2 \times 3.143 \times r$

$$\frac{88}{2 \times 3.143} = r$$

$13.999cm = r$

Thus, area of circle = πr^2

First, we convert r in 13.999cm to metres $\Rightarrow$ 13.999/100

$$= 3.143 \times \left(\frac{13.999}{100}\right)^2$$

$= 0.0616m^2$ B.

2005/20

What is the diameter of a circle of area $77cm^2$? (Take $\pi = {}^{22}/_{7}$)

A $\frac{\sqrt{2}}{7}$ cm B $3^1/_2$ cm C 7cm D $7\sqrt{2}$ cm

Solution

Diameter is twice of radius and radius can be gotten from

Area of circle = πr^2

$$77 = \frac{22}{7} \times r^2$$

$77 \times 7 = 22r^2$

$$\frac{77 \times 7}{22} = r^2$$

$$\frac{49}{2} = r^2$$

$$\sqrt{\frac{49}{2}} = r$$

$$\frac{7}{\sqrt{2}} cm = r$$

Thus ,diameter = 2r

$$= 2 \times \frac{7}{\sqrt{2}}$$

Rationalizing the surd = $\frac{2 \times 7 \times \sqrt{2}}{2}$

$= 7\sqrt{2}$ cm (D)

2005/39 Neco

Find the circumference of a circle whose area is $1386cm^2$ (Take $\pi = {}^{22}/_{7}$)

A 21cm B 132 cm C 264 cm

D 441 cm E 2772cm

Solution

Circumference of circle (perimeter) = $2\pi r$

But r can be gotten from the given area of circle

Area of circle = πr^2

$$1386 = \frac{22}{7} \times r^2$$

$1386 \times 7 = 22r^2$

$$\frac{1386 \times 7}{22} = r^2$$

$441 = r^2$

$\sqrt{441} = r$

r = 21 cm

Thus, circumference of circle = $2 \times \frac{22}{7} \times 21$

= 132cm (B)

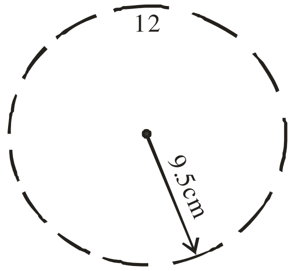
2012/ 42 Neco

The minute – hand of a wall clock is 9.5m long. What distance does its tip cover in 72 hours?

A 59.7m B 59.0m C 58.0m

D 45.0m E 43.0m

Solution



The minute - hand makes a complete cycle of 360^0 in 60min to cover 1 hour

$\therefore$ 72 hours would require 72 complete cycles of the minute - hand

Distance covered in 72hrs = $72 \times$ perimeter of circle

$= 72 \times 2\pi r$

$= 72 \times 2 \times \frac{22}{7} \times 9.5\text{cm}$

$= 4{,}299.43\text{cm}$

Converting to metres $= 4{,}299.43 \times 10^{-2}$ m

$= 42.99\text{m}$

$\approx 43.0\text{m}$ (E)

2009/17 (Nov)

X and Yare circle such that the radius of X is twice the radius of Y. if the radius of Y is 5cm, find in terms of π, the area of x

A 6.25π B 10π C 20π D 100π

Solution

Circle		Circle
X	and	Y
2r (radius)		r
$(2 \times 5\text{cm})$		5cm

Area of circle x = πr^2

$= \pi (2\times5)^2$

$= \pi\, 10^2$

$= 100\pi$ (D)

2010/32

How many times, correct to the nearest whole number, will a man run round a circular track of diameter 100m to cover a distance of 1000m?

A 3 B 4 C 5 D 6

Solution

Once round the track = circumference of circular track

$= 2\pi r$

$= 2 \times \frac{22}{7} \times \frac{100}{2}$

$= 314.2857$

Number of times it will cover 1000m

$= \frac{1000}{314.2857}$

$= 3.18$

≈ 3 times (A)

2014/16 (Nov)

A bicycle wheel covers 100cm in one revolution, find in terms of π, the radius of the wheel.

A $\frac{50}{\pi}$cm B $\frac{100}{\pi}$cm C 50πcm D 100πcm

Solution

One revolution = circumference of bicycle wheel.

$100 = 2\pi r$

In terms of π, radius r is

$\frac{100}{2\pi} = r$

$\frac{50}{\pi}\text{cm} = r$ (A)

2007/30 Neco

A bicycle wheel of radius 21cm is rolled over a distance of 33m. How many revolutions does it make?

(Take $\pi = {}^{22}/_{7}$)

A 2.5 B 5 C 25 D 50 E 55

Solution

One revolution = perimeter of bicycle wheel.

$= 2\pi r$

$= 2 \times \frac{22}{7} \times 21$

$= 132\text{cm}$

A distance of 33m is $33 \times 100\text{cm} \Rightarrow 3300\text{cm}$

Thus number of revolutions is $\frac{3300}{132} = 25$ (C)

2006/41 counter example

The wheel of a tractor has diameter 1.4m. What distance does it cover in 100 complete revolutions?

(Take $\pi = {}^{22}/_{7}$)

A 140m B 220m C 280m D 440m

Solution

One revolution is the perimeter of the tractor wheel.

$= 2\pi r$

Here radius is diameter divided by two

$= 2 \times \frac{22}{7} \times \frac{1.4}{2}$

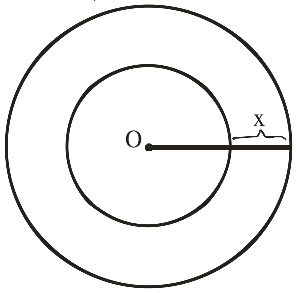
$= 4.4\text{m}$

Distance covered in 100 revolutions = $100 \times 4.4\text{m}$

$= 440\text{m}$ (D)

2006/33

In the diagram, the two circles have a common centre O. If the area of the lager circle is 100π and that of the smaller circle is 49π, find x



A 2 B 3 C 4 D 6

Solution

We are asked to find **R – r** in

Larger circle area is $\pi R^2 = 100\pi$

$R^2 = 100$

$R = 10$

Similarly, smaller circle area is $\pi r^2 = 49\pi$

$r^2 = 49$

$r = 7$

Thus $R - r = 10 - 7$

$= 3$ (B)

2009/46 Neco (Dec)

What is the area between two concentric circles of diameters 21cm and 18cm respectively

A 91.93 cm^2 B 254.57 cm^2 C 346.50cm^2

D 367.70cm^2 E 601.07cm^2

Solution

Area of two concentric circle = $\pi R^2 - \pi r^2$

$= \pi(R^2 - r^2)$

Here big circle **R** is $^{21}/_2$cm and small circle **r** is $^{18}/_2$cm

$$= \frac{22}{7}(10.5^2 - 9^2)$$

$$= \frac{22}{7}(110.25 - 81)$$

$$= \frac{22}{7}(29.25)$$

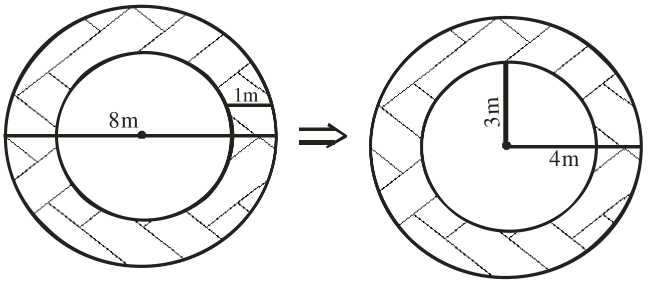
$= 91.93\text{cm}^2$ (A)

2014/15 NABTEB

Calculate the area of the path round a circular lake of diameter 8m, if the path is 1m wide

A πcm^2 B 7πcm^2 C 8πcm^2 D 9πcm^2

Solution



The shaded area is the path

Area of path round the lake = $\pi R^2 - \pi r^2$

$= \pi(4^2 - 3^2)$

$= \pi(4 - 3)(4 + 3)$

$= \pi(1)(7)$

$= 7\pi\text{ cm}^2$ B.

2006/47 (Nov)

The sum of the areas of two circles each of diameter 64cm is equal to the area of radius rcm. Find the value of r.

A 38.40cm B 45.25cm C 64.00cm D 90.50cm

Solution

Since the two circles have same diameter

Area of 2 circles = $2\pi R^2$

The other circle with radius r has area

$= \pi r^2$

Next, we are given that : $2\pi R^2 = \pi r^2$

$2R^2 = r^2$

$$2 \times \left(\frac{64}{2}\right)^2 = r^2$$

$2 \times 32^2 = r^2$

$2048 = r^2$

$r = \sqrt{2048} = 45.25\text{cm}$ (B)

2007/47

Two circles have radii 16cm and 23cm. What is the difference between their circumferences (Take $\pi = {}^{22}/_7$)

A 244.92cm B 149.92cm C 44.00cm D 43.96cm

Solution

Let R for big circle and r for small circle

Difference in circumference = $2\pi R - 2\pi r$

$= 2\pi(R - r)$

$$= 2 \times \frac{22}{7}(23 - 16)$$

$$= 2 \times \frac{22}{7} \times 7 = 44\text{cm} \quad (C)$$

2006/17 (Nov) Exercise 20.5

Find the area between two concentric circles of radii 9cm and 2cm. ($\pi = {}^{22}/_7$)

A 254. 57cm^2 B 242.00cm^2 C 12.57cm^2 D 9.00cm^2

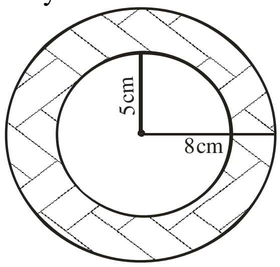
2009/16 Exercise 20.6

The diameter of a bicycle wheel is 42cm. If the wheel makes 16 complete revolutions, what will be the total distance covered by the wheel? (Take $\pi = {}^{22}/_7$)

A 672cm B 1056cm C 2112cm D 4224cm

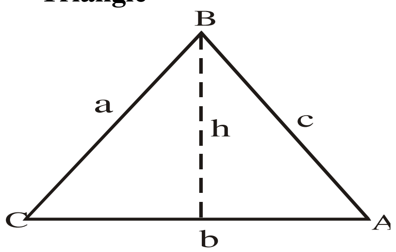
2008/18 NABTEB (Nov) Exercise 20.7

The inner and outer radii of the ring below are 5cm and 8cm respectively. Calculate the area of the shaded part



A $28^2/_7$cm^2 B $122^4/_7$cm^2 C $345^6/_7$cm^2 D 616cm^2

Triangle



Area of triangle = $\frac{1}{2}$ b x h ;

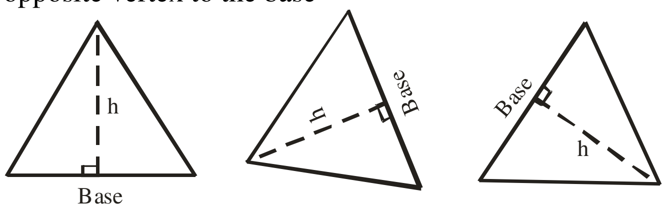
Area of triangle = $\frac{1}{2}$ base x height

$= \frac{1}{2}$ ab sin C

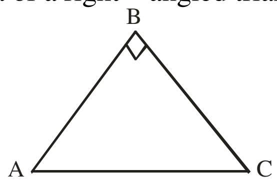
or = ½ ac sin B

or = ½ bc sin A

In triangles; any one of the three sides may by taken as base, while the height is the perpendicular length from the opposite vertex to the base



Area of a right – angled triangle



The perpendicular length from A to BC is AB

Thus area of triangle ABC = $\frac{1}{2}$ BC × AB

If all the sides of a triangle are given: use Hero's formula

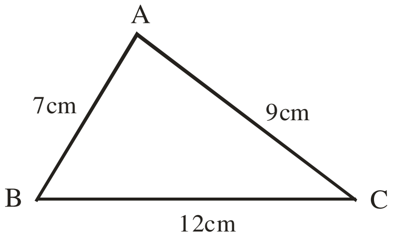
Area of triangle = $\sqrt{s(s-a)(s-b)(s-c)}$

Where s = ½ (a + b + c)

2009/9a Neco (Dec)

Calculate the smallest angle of triangle ABC in which /AB/ = 7cm, /BC/ = 12cm and /AC/ = 9cm. Hence or otherwise, find the area of the triangle (correct to 2 decimal places)

Solution



The smallest angle is the angle facing the smallest side 7cm i.e angleC

Applying cosine rule

$7^2 = 12^2 + 9^2 - 2 \times 12 \times 9 \cos C$

$49 = 144 + 81 - 216 \cos C$

$49 = 225 - 216 \cos C$

$216 \cos C = 225 - 49$

$\cos C = \frac{176}{216} = 0.8148$

$C = \cos^{-1} 0.8148 = 35.43^0$

(ii) Area of Δ ABC = $\frac{1}{2}$ ab sin c

$= \frac{1}{2} \times 12 \times 9 \times \sin 35.43$

$= 6 \times 9 \times 0.5797 = 31.30\text{cm}^2$ to 2d.p

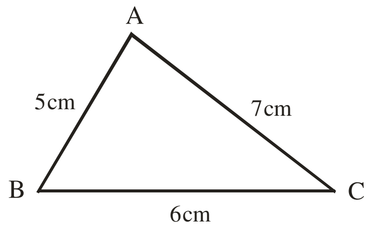
2011/30 Neco

Calculate, correct to 2 decimal places, the area of a triangle whose sides are 5cm, 6cm and 7cm

A 12.89cm^2 B 14.70cm^2 C 22.06cm^2

D 38.24cm^2 E 60.28cm^2

Solution



Area of Δ = $\frac{1}{2}$ a b sin C

We solve for angle C using cosine rule

$5^2 = 6^2 + 7^2 - 2 \times 6 \times 7 \cos C$

$25 = 36 + 49 - 84 \cos C$

$25 = 85 - 84 \cos C$

$84 \cos C = 85 - 25$

$\cos C = \frac{60}{84} = 0.7143$

$C = \cos^{-1} 0.7143$

$C = 44.41^0$

Substituting

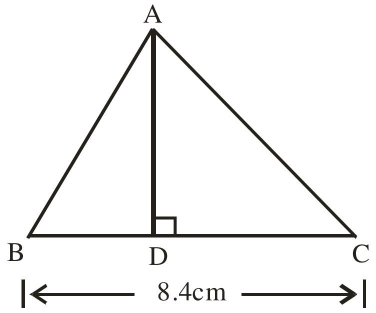
Area of Δ = $\frac{1}{2} \times 6 \times 7 \times \sin 44.41$

$= 14.696\text{cm}^2$

$\approx 14.70\text{cm}^2$ to 2d.p

2005/2b NABTEB

ABC is a triangle with $\overline{BC}$ = 8.4cm, $A\hat{D}C = 90^0$ and area 40.16cm^2. Find $\overline{AD}$



Solution

Area of triangle = $\frac{1}{2}$ base × height

$40.16 = \frac{1}{2} \times 8.4 \times AD$

$40.16 \times 2 = 8.4 \times AD$

$\frac{40.16 \times 2}{8.4} = AD$

9.56cm = AD

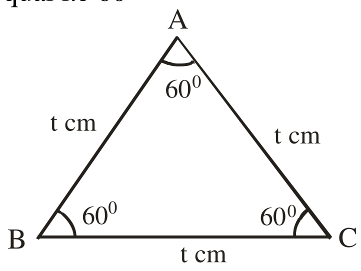
2011/42 Neco

Calculate the area of an equilateral triangle of side t cm, leaving your answer in surd from

A $\frac{t^2}{4}$ B $\frac{\sqrt{3}}{4}t^2$ C $\frac{t^2}{2}$ D $\frac{\sqrt{3}}{2}t^2$ E t^2

Solution

Equilateral Δ implies all sides are equal and the three angles are equal i.e 60^0



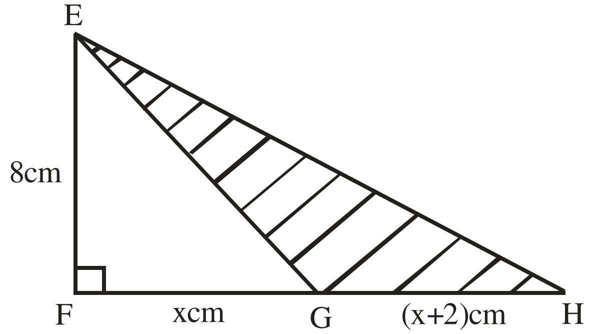
Area of triangle = $\frac{1}{2}$ a × b × sin c

$= \frac{1}{2} \times t \times t \times \sin 60^0$

$= \frac{1}{2} t^2 \times \frac{\sqrt{3}}{2}$

$= \frac{t^2}{4}\sqrt{3}\,cm^2$ or $\frac{\sqrt{3}}{4}t^2\,cm^2$ (B)

2010/34



In the diagram, /EF / = 8cm, /FG/ = xcm /GH/ = (x + 2) cm, $\angle EFG = 90^0$. If the area of the shaded portion is $40cm^2$, find the area of ΔEFG.

A $128cm^2$ B $72cm^2$ C $64cm^2$ D $32cm^2$

Solution

Area of Δ EFG = area of Δ EFH – area of Δ EGH

Area of triangle = $\frac{1}{2}$ a b sin C

In Δ EFH

Area of triangle = $\frac{1}{2} \times 8 \times [x + (x + 2)] \times \sin 90^0$

$= 4 \times (x + x + 2) \times 1$

$= 4\times (2x + 2)$

$= (8x + 8)\ cm^2$

Also in Δ EFG

Area of triangle = $\frac{1}{2} \times 8 \times x \times \sin 90^0$

$= 4x\ cm^2$

But Area of ΔEFH – ΔEFG = ΔEGH

$(8x + 8) – 4x = 40$

$8x + 8 – 4x = 40$

$8x – 4x = 40 – 8$

$4x = 32\ cm^2$ (D)

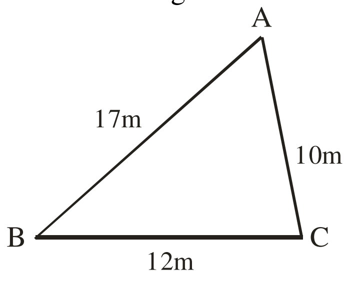
2008/9a

The triangle ABC has sides AB = 17m, BC = 12m and AC = 10m. Calculate the:

(i) largest angle of the triangle; (ii) area of the triangle

Solution

Sketch the diagram as:



(i) The largest angle of the triangle is the angle facing the longest side i.e C

By cosine rule

$17^2 = 10^2 + 12^2 – 2 \times 10 \times 12 \cos \theta$ (Here cosθ is C)

$289 = 100 + 144 – 240 \cos \theta$

$240 \cos \theta = 244 – 289$

$\text{Cos } \theta = \frac{-45}{240}$

$\text{Cos } \theta = – 0.1875$

$\theta = \cos^{-1}(– 0.1875) = 100.81^0$

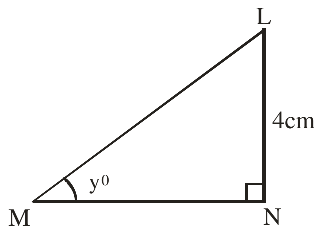
(ii) Area of Δ ABC = $\frac{1}{2}$ a b sin C

$= \frac{1}{2} \times 12 \times 10 \times \sin 100.81$

$= 60 \times 0.9823 = 58.938m^2$

2007/17

In the diagram, /LN/ = 4cm, $\angle$ LNM = 90^0 and tan y = $\frac{2}{3}$, what is the area of the ΔLMN?



A $24cm^2$ B $12cm^2$ C $10cm^2$ D $6cm^2$

Solution

Area of Δ LMN = $\frac{1}{2}$ /MN/ $\times$ /LN/ $\times$ sin N

/MN/ can be gotten from tan y = $\frac{2}{3}$

tan y = $\frac{Opp}{Adj}$

In the Δ opposite side is 4

But the given value in tan y is 2 which implies 2×2 to give 4 and what we do to the numerator at tan y

We do same to denominator

Thus the actual value of 3 is $3 \times 2 = 6$ i.e Adj : MN

Substituting

Area of Δ LMN = $\frac{1}{2} \times 6 \times 4 \times \sin 90$

$= 12cm^2$ (B)

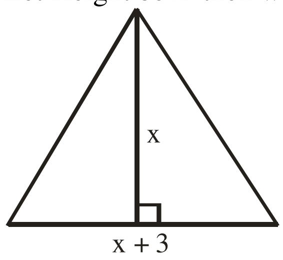
2012/29 Neco

The base of a triangle is 3cm longer than height. If the area is $44cm^2$, find the length of its base.

A 8cm B 11cm C 12cm D 20cm

Solution

Let Height be x then we sketch as:



Area of triangle = $\frac{1}{2}$ base $\times$ height

$$44 = \frac{1}{2}(x + 3) \times x$$

$$44 \times 2 = x(x + 3)$$

$$88 = x^2 + 3x$$

$$x^2 + 3x - 88 = 0$$

Factorizing

$$x^2 + 11x - 8x - 88 = 0$$

$$x(x + 11) - 8(x + 11) = 0$$

$$(x - 8)(x + 11) = 0$$

$$x - 8 = 0 \text{ or } x + 11 = 0$$

$$x = 8 \text{ or } -11$$

We accept x = 8cm

Thus base $x + 3 = 8 + 3$

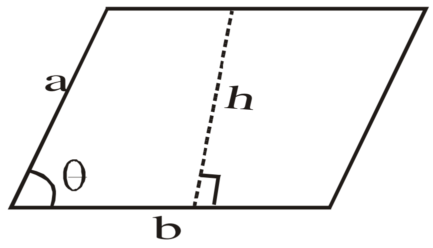
$= 11\text{cm}$ (B)

2014/12 NABTEB (Nov) Exercise 20.8

Calculate the area of an equilateral triangle of side 8cm

A $8\sqrt{3}$ cm^2 B 16cm^2 C $4\sqrt{3}$ cm^2 D $16\sqrt{3}$ cm^2

Parallelogram



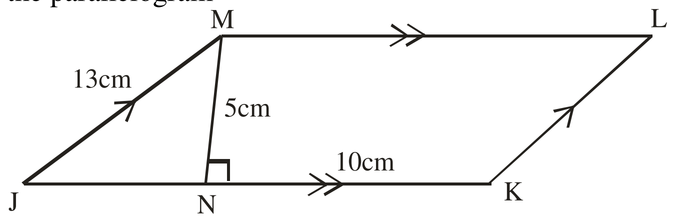
Area of parallelogram = b x h

= base x height

= ab sin θ

2007/34 Neco

JKLM is a parallelogram and MN⊥ JK. If /JM/ = 13cm, /MN/ = 5cm and /NK/ = 10cm, find the area of the parallelogram



A 50cm^2 B 65cm^2 C 80cm^2 D 110cm^2 E 130cm^2

Solution

Area of parallelogram = Base × height

The base here is JK, NK is given, JN can be gotten by Pythagoras rule in the right – angled Δ

$$13^2 = 5^2 + JN^2$$

$$169 = 25 + JN^2$$

$$169 - 25 = JN^2$$

$$144 = JN^2$$

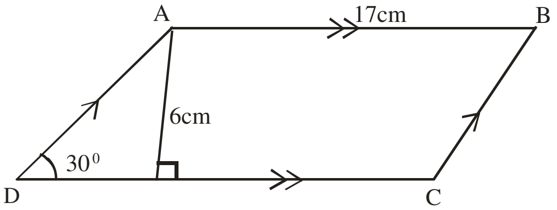
$$\sqrt{144} = JN\text{ ; } JN = 12\text{cm}$$

Thus, area of parallelogram = $(12 + 10) \times 5$

$= 22 \times 5$

$= 110\text{cm}^2$ D.

2014/13 NABTEB



ABCD is a parallelogram of height 6cm, $A\hat{D}C = 30^0$ and /AB / = 17cm. Calculate the perimeter

A 29cm B 34cm C 46cm D 58cm

Solution

Perimeter = distance round a particular shape

Here AB = DC = 17cm (opposite sides of llgm)

BC = AD

We solve AD in the right – angled Δ by trig – ratio

$$\text{Sin } 30 = \frac{6}{AD}$$

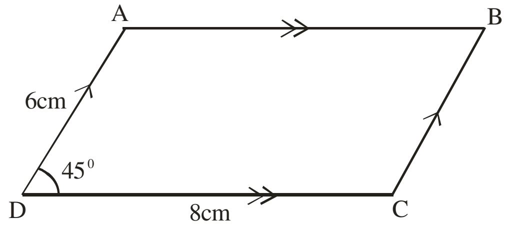
$$0.5\,AD = 6$$

$$AD = 12\text{cm}$$

Thus perimeter of llgm = 12 + 17 + 12 + 17

= 58cm (D)

2011/47 Neco



In the diagram above, determine the area of the parallelogram.

A $12\sqrt{2}$ cm^2 B 24cm^2 C $24\sqrt{2}$ cm^2

D 48cm^2 E $48\sqrt{2}$ cm^2

Solution

Area of parallelogram = a b sin θ

$$= 6 \times 8 \times \sin 45$$

$$= 6 \times 8 \times \frac{1}{\sqrt{2}}$$

Rationalizing the surd

$$= 6 \times 8 \times \frac{\sqrt{2}}{2}$$

$$= 24\sqrt{2}\text{ cm}^2$$

2008/9 NABTEB (Nov)

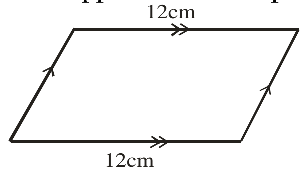
The perimeter of a parallelogram is 38cm. if one of the sides is 12cm, find the other side

A 7cm B 9cm C 13cm D 26cm

Solution

Perimeter = distance round a shape

Note that: opposite sides of parallelogram are equal

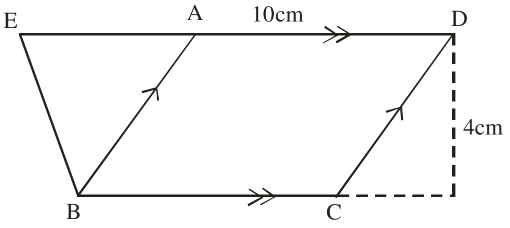


The other side = $\dfrac{38-(12+12)}{2}$

$= \dfrac{14}{2} = 7\text{cm}$ (A)

2006/34 Neco (Dec) counter example

In the diagram below, EAD is a straight line. ABCD is a parallelogram with base BC and height 4cm. If /AD/ = 10cm and the area of ΔBAE is 30cm^2, find the area of trapezium BCDE



A 100cm^2 B 50cm^2 C 20cm^2 D 15cm^2 E 10cm^2

Solution

Here area of trapezium =

area of ΔBAE + area of llgm ABCD + area of right– angled Δ

Area of llgm ABCD = base × height

= 10 × 4

= 40cm^2

Note that Δ between same base and same parallel have equal areas.

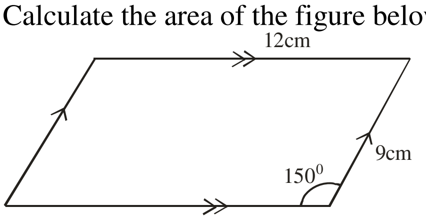
Area of ΔBAE = 30cm^2 same for right– angled Δ

Thus, area of trapezium = 30 + 40 + 30

= 100cm^2 (A)

2013/29 Neco Exercise 20.9

Calculate the area of the figure below

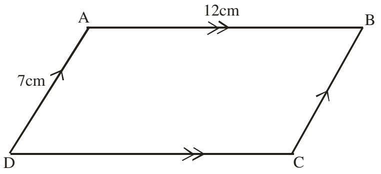


A 1.5cm^2 B 10.5cm^2 C 27cm^2

D 54cm^2 E 104.3cm^2

2009/ 30 Neco(Dec) Exercise 20.10

The area of parallelogram ABCD is 425cm^2, /AB / = 12cm and / AD / = 7cm. Find the perpendicular height of the parallelogram to 2 decimal place.

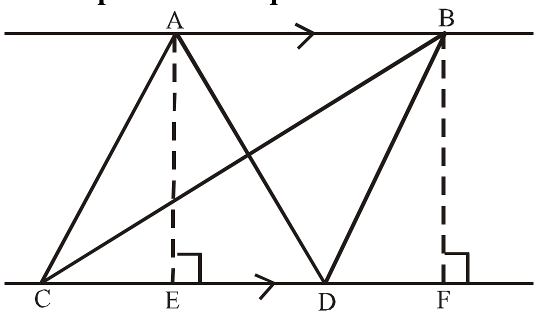


A 17.74cm B 31.07cm C 35.42cm

D 62.14cm E 84.00cm

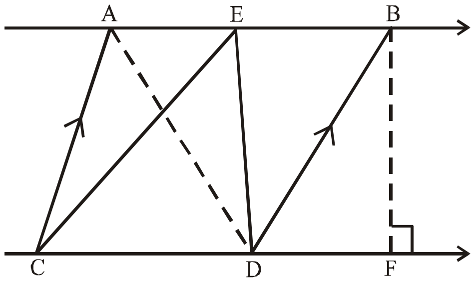
Properties of Triangles & Parallelograms on the same base and between the same parallels.

1. **Triangles on the same base and between the same parallel are equal in area.**



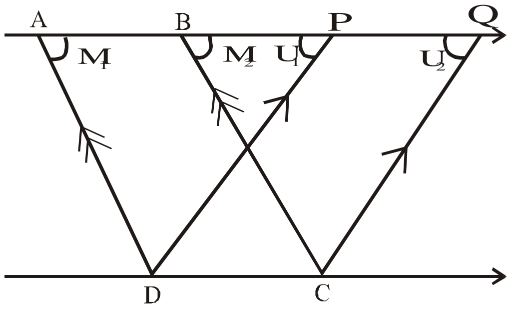
Area of ΔACD = Area of ΔBCD

2. **If a triangle and a parallelogram are on the same base and between same parallels, the area of the triangle is equal to half of that of the parallelogram**



Area of ΔECD = ½ area of parallelogram ABCD

3. **Parallelogram on the same base and between the same parallels are equal in area.**



Given: Parallelograms ABCD, PQCD on the same base DC and between the same parallels AQ and DC.

To prove: ABCD = PQCD

Proof:

In Δ s APD and BQC

$M_1 = M_2$ (Corr. ∠s)

$U_1 = U_2$ (Corr. ∠ s)

AD = BC (Opp. Sides of llgm)

Therefore Δ APD = Δ BQC (AAS)

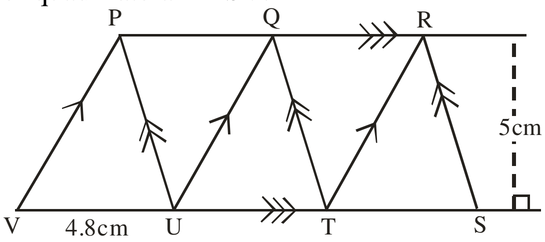
Thus, Quad. AQCD - ΔAPD is equal quad.

AQCD - ΔBQC is equal quad.

Hence PQCD = ABCD

2006/11

In the diagram, PQUV, PQTU, QRTU and QRST are parallelograms. /UV/ = 4.8cm and the perpendicular distance between PR and VS is 5cm. Calculate the area of quadrilateral PRSV.



A $96cm^2$ B $72cm^2$ C $60cm^2$ D $24cm^2$

Solution

Area of quadrilateral PRSV

= area of llgm PQUV + area of llgm QRUT + area of ΔRTS

Recall that parallelogram between same base and between same parallel are equal in area and half area of llgm is equal to area of triangle between same base and same parallel.

Area of llgm = base × height

$= 4.8 \times 5 = 24cm^2$

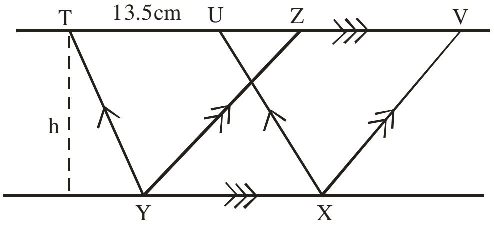
Area of $\Delta = \frac{1}{2}$ area of llgm

$= \frac{1}{2} \times 24 = 12cm^2$

Thus, area of quadrilateral PRSV = 24 + 24 + 12

$= 60cm^2$ (C)

2006/35 Neco (Dec)



In the diagram above, TUXY and ZVXY are parallelograms. If /TU/ = 13.5cm and the area of parallelogram ZVXY is $67.5cm^2$, Calculate its height.

A 54cm B 27cm C 8.5cm D 5cm E 2.7cm

Solution

Parallelogram between same base and same parallel are equal in area

Thus ZVXY = TUXY

and area of llgm TUXY = base × height

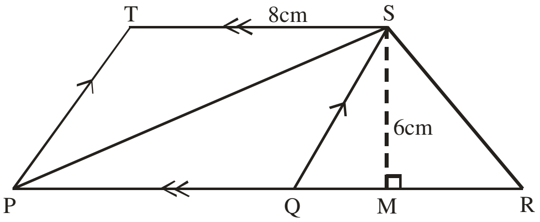
Note YX is same as TU (opp side of llgm)

$67.5 = 13.5 \times h$

$\frac{67.5}{13.5} = h$

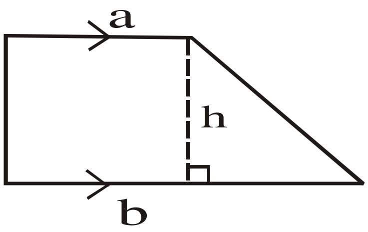
5cm = h (D)

2016/5a Exercise 20.10b



In the diagram, PQST is a parallelogram, PR is a straight line, |TS| = 8cm, |SM| = 6cm and area of triangle PSR = $36cm^2$. Find the value of |QR|

Trapezium



Area of trapezium = $\frac{1}{2}(a + b) \times h$

$= \frac{1}{2}$ (sum of two parallel sides) x height

2013/30 Neco

Calculate the area of a trapezium ABCD with /AB/ = 20cm, /DC/ = 45cm and height = 8cm

A $245cm^2$ B $260cm^2$ C $287cm^2$

D $344cm^2$ E $480cm^2$

Solution

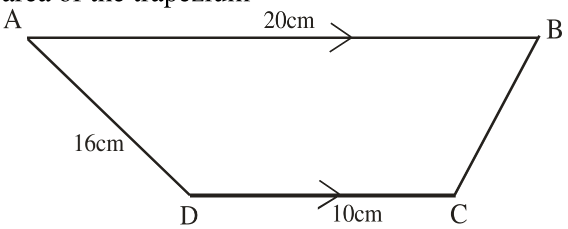
Area of trapezium $= \frac{1}{2}(20 + 45) \times 8$

$= \frac{1}{2} \times 65 \times 8$

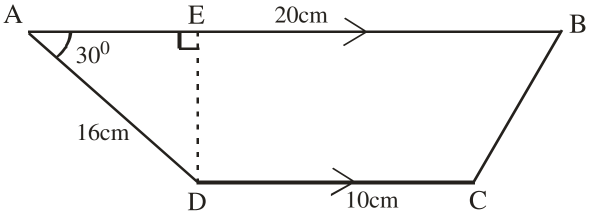
$= 260cm^2$ (B)

2005/3a NABTEB (Nov)

ABCD is a trapezium in which AB//CD, $\overline{AB} = 20cm$, $\overline{DC} = 10cm$, $\overline{AD} = 16cm$ and $B\hat{A}D = 30^0$, calculate the area of the trapezium



Solution



Area of trapezium $= \frac{1}{2}(10 + 20) \times DE$

We solve for DE in ΔADE using trig ratio

$\text{Sin } 30^0 = \frac{DE}{16}$

16 sin 30 = DE

16 × 0.5 = DE

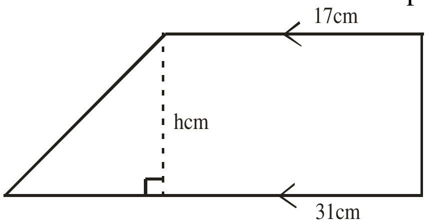
8cm = DE

Thus area of trapezium $= \frac{1}{2}(10 + 20) \times 8$

$= 30 \times 4$

$= 120cm^2$

2009/49 NABTEB (Nov)
The trapezium below has an area of $456cm^2$?
Calculate the distance between its parallel sides



A 29cm B 27cm C 19cm D 17cm

Solution

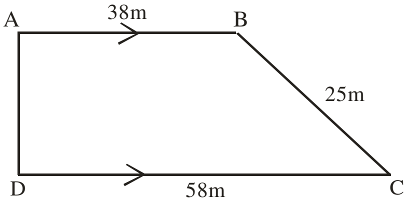
Area of trapezium = $\frac{1}{2}(17 + 31) \times h$

$456 = \frac{1}{2}(48) \times h$

$456 = 24h$

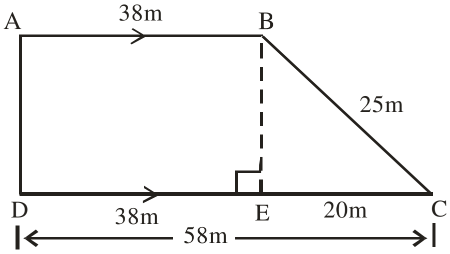
$\frac{456}{24} = h$ thus, 19cm = h (C)

2005/38 Neco



In the trapezium above, if $\overline{AB} = 38m$
$\overline{DC} = 58m$, $\overline{BC} = 25m$ and $\angle BAD = \angle ADC = 90^0$.
Find the length AD
A 38m B 33m C 25m D 20m E 15m

Solution



AD = BE
Appling Pythagoras rule to the resulting right – angled Δ

$25^2 = 20^2 + BE^2$

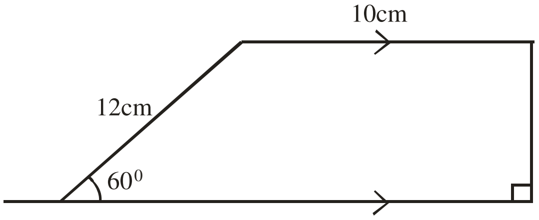
$625 = 400 + BE^2$

$625 - 400 = BE^2$

$225 = BE^2$

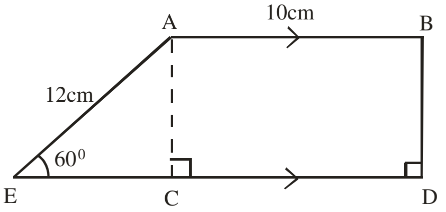
$\sqrt{225} = BE$ thus,15m = BE (E)

2006/30 Neco



The figure above is a trapezium. Calculate its area
A $156\sqrt{3}\ cm^2$ B $78\sqrt{3}\ cm^2$ C $120cm^2$
D $39\sqrt{3}\ cm^2$ E $60cm^2$

Solution



It is common knowledge that AB = CD = 10cm;
length of rectangle ABCD
Applying Pythagoras rule to the resulting right – angled Δ ACE

$\text{Sin } 60^0 = \frac{AC}{12}$

$12 \sin 60 = AC$

$12\frac{\sqrt{3}}{2}$ cm = AC Thus AC = $6\sqrt{3}$ cm

Also $\cos 60^0 = \frac{EC}{12}$

$12 \cos 60^0 = EC$

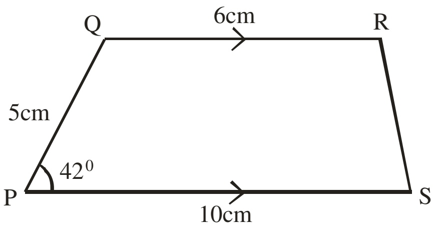
$12 \times \frac{1}{2} = EC$; EC = 6cm

Thus, area of trapezium = $\frac{1}{2}(AB + ED) \times AC$

$= \frac{1}{2}(10 + 16) \times 6\sqrt{3}$

$= \frac{1}{2} \times 26 \times 6\sqrt{3} = 78\sqrt{3}\ cm^2$ (B)

2009/17 and 18 Exercise 20.11

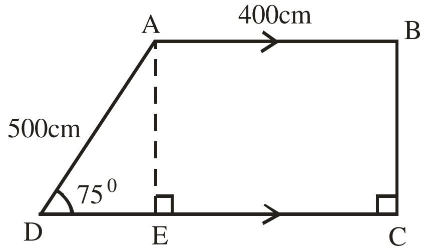


PQRS is a trapezium. QR//PS, /PQ/ = 5cm,
/QR/ = 6cm, /PS/ = 10cm and angle QPS = 42^0
Use the information to answer questions 17 and 18

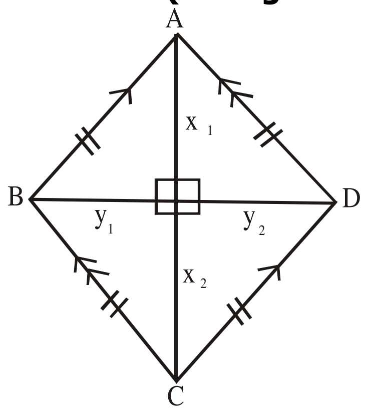
17. Calculate the perpendicular distance between the parallel sides
A 3.35cm B 3.72cm C 4.50cm D 4.62cm

18. Calculate, correct to the nearest cm^2, the area of the trapezium
A $27cm^2$ B $30cm^2$ C $36cm^2$ D $37cm^2$

1976/4 Exercise 20.11b
In the accompanying figure which is not drawn to scale, ABCD is a piece of land in the form of trapezium with AB parallel to DC, AB = 400cm, AD = 500cm, $\angle ADC = 75^0$, $\angle BCD = 90^0$. Calculate the distance between the parallel sides and hence calculate the area of the piece of land, giving your answer in standard notation



Rhombus (with given diagonal)



Diagonal BD = y **Where y = $y_1 + y_2$**
and diagonal AC = x **Where x = $x_1 + x_2$**
Area of rhombus = ½ **x y** **or** = ½ $x_1 y_1$ x 4
Where x and y are the two stated diagonals

2010/10 NABTEB (Nov)
Calculate the area of a rhombus whose diagonals are 10cm and 6cm long
A $32cm^2$ B $30cm^2$ C $28cm^2$ D $20cm^2$

Solution

Area of rhombus = $\frac{1}{2}$ xy (where x and y are diagonals)

$= \frac{1}{2} \times 10 \times 6$

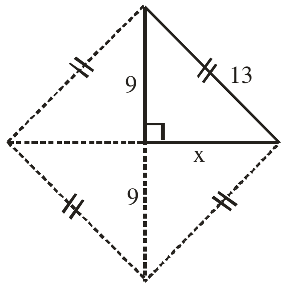
$= 30cm^2$ (B)

2005/5b Neco
A Rhombus of side 13cm has a diagonal of 18cm. Find the area of the rhombus correct to 2 decimal places

Solution

Area of rhombus = $\frac{1}{2}$ x y (where x and y are diagonals)

one diagonal is given, we find the other one from the given data :



By Pythagoras rule : $13^2 = 9^2 + x^2$

$169 = 81 + x^2$

$169 - 81 = x^2$

$88 = x^2$

$\sqrt{88} = x$

The second diagonal is $2 \times 9.381 = 18.762cm$

Thus, area of rhombus = $\frac{1}{2} \times 18 \times 18.762$

$= 168.858cm^2$

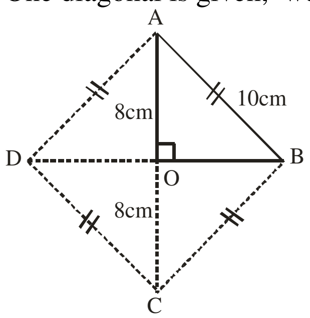
$\approx 168.86cm^2$ to 2dp

2008/14 NABTEB (Nov)
The side of a rhombus is 10cm. if one of the diagonal is 16cm, find the area
A $48cm^2$ B $96cm^2$ C $160cm^2$ D $180cm^2$

Solution

Area of rhombus = $\frac{1}{2}$ x y(where x and y are diagonals)

One diagonal is given, we sketch as:



The second diagonal is DB but DB = DO + OB
and OB can be gotten by Pythagoras rule in ΔAOB

$10^2 = 8^2 + OB^2$

$100 = 64 + OB^2$

$100 - 64 = OB^2$

$36 = OB^2$

$OB = 6cm$

$\Rightarrow DB = 2 \times 6cm$

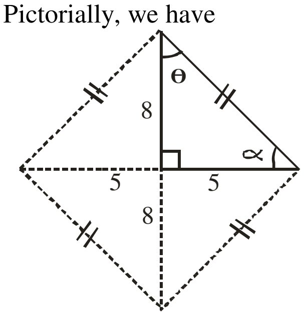
$= 12cm$

Thus area of rhombus = $\frac{1}{2} \times 16 \times 12 = 96cm^2$

2006/8a Neco
Calculate the angles of a rhombus whose diagonals are of lengths 10cm and 16cm

Solution

Pictorially, we have



One set of the required angles is 2θ
But by trig – ratio

$\tan\theta = \frac{5}{8}$

$\tan\theta = 0.625$

$\theta = \tan^{-1} 0.625$

$= 32^0$ Thus $2\theta = 2 \times 32^0 = \mathbf{64^0}$

Also another set of the required angles is 2α
In the right – angled triangle :

$\theta + 90 + \alpha = 180^0$ (sum of ∠s in Δ)

$32 + 90 + \alpha = 180^0$

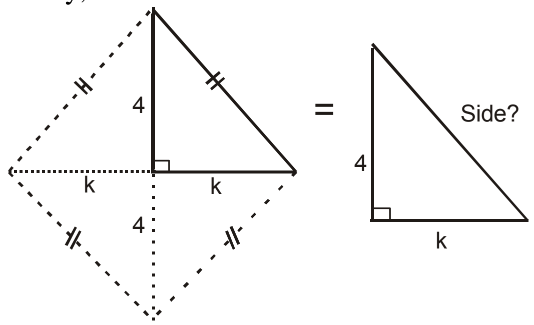
$\alpha = 180 - 122 = 58^0$ Thus $2\alpha = 2 \times 58^0 = \mathbf{116^0}$

Thus the angles of the rhombus are 64^0, 64^0, 116^0 and 116^0

2010/32 NABTEB (Nov)
The area of a rhombus is $24cm^2$ and one of its diagonals is 8cm. Find the length of a side
A 3cm B 4cm C 5cm D 6cm.

Solution

Pictorially, we have



For us to know the rhombus side;
We first find K i.e half the 2nd diagonal of rhombus

Area of rhombus $= \frac{1}{2}$ x y

$$24 = \frac{1}{2} \times 8 \times y$$

$$\frac{2 \times 24}{8} = y$$

$$6\text{cm} = y$$

Thus, $K = \frac{6}{2} = 3\text{cm}$

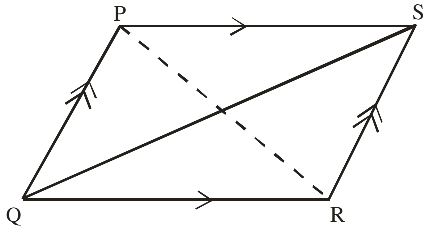
By Pythagoras rule

Rhombus side $= \sqrt{4^2 + 3^2}$

$= \sqrt{16+9}$

$= \sqrt{25}$ i.e 5cm C.

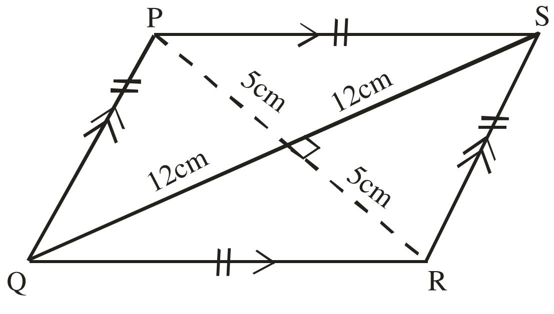
2008/19



In the diagram, PQRS is a rhombus. /PR/ = 10cm and /QS/ = 24cm. calculate the perimeter of the rhombus

A 34cm B 52cm C 56cm D96cm

Solution



Perimeter of rhombus = PS + SR + RQ + PQ

All sides of rhombus are equal

By Pythagoras rule: $SR^2 = 12^2 + 5^2$

$SR^2 = 144 + 25$

$SR^2 = 169$

$SR = \sqrt{169}$

$= 13\text{cm}$

Thus, perimeter of rhombus = 13 + 13 + 13 + 13

= 52cm (B)

2012/49 Exercise 20.12

A side and a diagonal of a rhombus are 10cm and 12cm respectively. Find its area

A 20cm^2 B 24cm^2 C 48cm^2 D 96cm^2

2006/5a (Nov) Exercise 20.13

In a quadrilateral ABCD, the diagonals $\overline{AC}$ and $\overline{BD}$ bisect each at right angles, /AC/ = 16cm and /BD/ = 30cm

(i) What type of quadrilateral is ABCD?

(ii) Find /AB/

(iii) Calculate the area of the quadrilateral

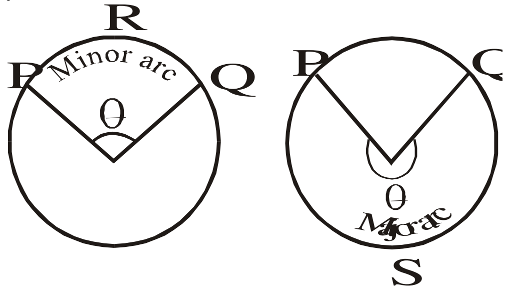
2015/36 Exercise 20.14

The area of a rhombus is 110cm^2. If the diagonal are 20cm and (2x + 1)cm long , find the value of x.

A 5.0 B 4.0 C 3.0 D 2.5

Length of an arc

For any circle, with arc PRQ (minor) or arc PSQ (major)



Subtending an angle θ at the center of the circle,

length of the arc $= \frac{\theta}{360} \times 2\pi r$

2007/42 Neco

An arc of a circle of radius 8cm subtends an angle of 85^0 at the centre of the circle. Find the length of the major arc (Take $\pi = {}^{22}/_{7}$)

A 11.87cm B 38.41cm C 272.45cm
D 289.62cm E 474.87cm

Solution

$$\text{Length of arc} = \frac{\theta}{360} \times 2\pi r$$

Here the minor arc subtends 85^0

But the major arc subtend $360 - 85 = 275^0$

Thus length of major arc $= \frac{275}{360} \times 2 \times \frac{22}{7} \times 8$

= 38.41cm B

2010/33 Neco

What is the length of an arc which subtends an angle of 60^0 at the circumference of a circle of radius 21cm? (Take $\pi = {}^{22}/_{7}$)

A 141cm B 132cm C 44cm D 22cm E 11cm

Solution

$$\text{Length of arc} = \frac{\theta}{360} \times 2\pi r$$

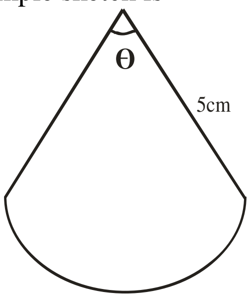
$$= \frac{60}{360} \times 2 \times \frac{22}{7} \times 21$$

= 22cm (D)

2005/23 (Nov)
The pendulum of a clock is 5cm long and swings through an arc of 8cm. Through what angle, in degrees, does the pendulum swing? (Take $\pi = {}^{22}/_7$)
A 36^0 B 46^0 C 72^0 D 92^0

Solution

The simple sketch is



Length of arc = $\frac{\theta}{360} \times 2\pi r$

$$8 = \frac{\theta}{360} \times 2 \times \frac{22}{7} \times 5$$

$$8 = \frac{220\theta}{2520}$$

$$\frac{8 \times 2520}{220} = \theta$$

$91.62^0 = \theta$

$\theta \approx 92^0$ (D)

2011/24 Counter example
A circle is divided into two sectors in the ratio 3: 7.
If the radius of the circle is 7cm, calculate the length of the minor arc of the circle
A 18.85cm B 13.20cm C 12.30cm D 11.30cm

Solution

Length of arc = $\frac{\theta}{360} \times 2\pi r$

Here $\frac{\theta}{360}$ is replaced by ratio 3:7 their sum is 10

and for the minor arc $\frac{\theta}{360}$ is $\frac{3}{10}$

Length of minor arc = $\frac{3}{10} \times 2 \times \frac{22}{7} \times 7$

= 13.2cm B.

2009/16 (Nov)
In a circle, an arc of length 110cm subtends an angle of 210^0 at the centre. Find the radius of the circle
(Take $\pi = {}^{22}/_7$)
A 75cm B 60cm C 55cm D 30cm

Solution

Length of arc = $\frac{\theta}{360} \times 2\pi r$

$$110 = \frac{210}{360} \times 2 \times \frac{22}{7} \times r$$

$$\frac{110 \times 360 \times 7}{210 \times 2 \times 22} = r$$

30cm = r (D)

2008/35
An arc of a circle, radius 14cm, is 18.33cm long. Calculate, correct to the nearest degree the angle which the arc subtends at the centre of the circle (Take $\pi = {}^{22}/_7$)
A 11^0 B 20^0 C 22^0 D 75^0

Solution

Length of arc = $\frac{\theta}{360} \times 2\pi r$

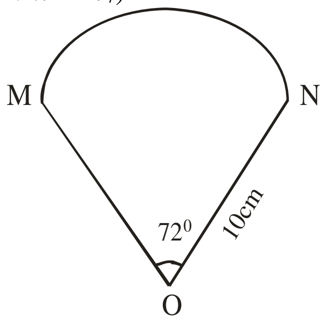
$$18.33 = \frac{\theta}{360} \times 2 \times \frac{22}{7} \times 14$$

$$\frac{18.33 \times 360 \times 7}{2 \times 22 \times 14} = \theta$$

$74.986^0 = \theta$

$\theta \approx 75^0$ to the nearest degree

2006/40
The diagram shows an arc MN of a circle centre O, with radius 10cm. If $\angle MON = 72^0$, calculate the length of the arc, correct to three significant figures
(Take $\pi = {}^{22}/_7$)



A 6.28cm B 12.6cm C 22.6cm D 62.8cm

Solution

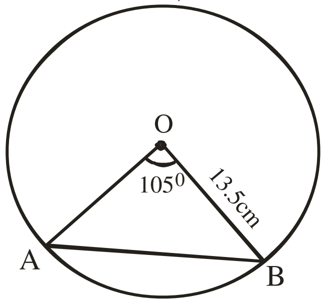
Length of arc = $\frac{\theta}{360} \times 2\pi r$

$$= \frac{72}{360} \times 2 \times \frac{22}{7} \times 10$$

= 12.57cm

$\approx$ 12.6cm to 3s.f

2008/5 NABTEB (Nov)
An arc of a circle of radius 13.5cm subtends an angle of 105^0 at the centre. Calculate, correct to three significant figures, the:



(i) chord length AB
(ii) arc length AB (Take $\pi = 3.142$)

Solution

(i) Length of chord = $2\, r \sin \frac{\theta}{2}$

$$= 2 \times 13.5 \times \sin \frac{105}{2}$$

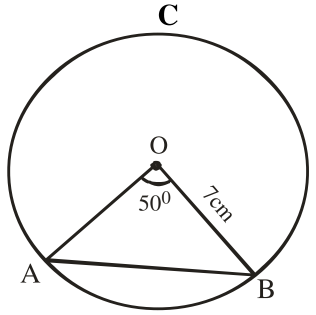
$= 27 \times \sin 52.5$
$= 27 \times 0.7934$
$= 21.42\text{cm}$
$\approx 21.4\text{cm}$ to 3s.f

(ii) Length of arc $= \dfrac{\theta}{360} \times 2\pi r$

$= \dfrac{105}{360} \times 2 \times \dfrac{22}{7} \times 13.5$

$= 24.75\text{cm} \approx 24.8\text{cm}$ to 3s.f

2011/31 Neco



Calculate the length of the major arc ACB correct to 2 decimal places

A 132.61cm B 37.89cm C 21.39cm
D 6.11cm E 5.42cm

Solution

Length of arc $= \dfrac{\theta}{360} \times 2\pi r$

Here the minor arc is 50^0

But the major arc θ is 360 – 50 i.e 310^0

Thus, length of major arc $= \dfrac{310}{360} \times 2 \times \dfrac{22}{7} \times 7$

$= 37.888\text{cm} \approx 37.89\text{cm}$ to 2d.p

2008/34 Neco (Dec)

Calculate, in terms of π, the length of an arc of a circle of radius 12cm which subtends angle 240^0 at the centre of the circle.

A 16πcm B 24πcm C 32πcm D 48πcm E 69πcm

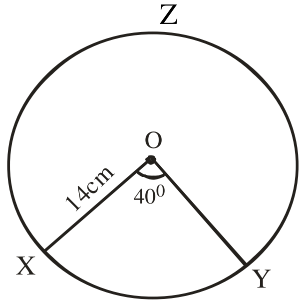
Solution

Length of arc $= \dfrac{\theta}{360} \times 2\pi r$

$= \dfrac{240}{360} \times 2 \times \pi \times 12 = 16\pi\text{cm}$ (A)

2008/40 and 41 Neco (Dec)

In the diagram below, O is the centre of the circle of radius 14cm, $\angle XOY = 40^0$. Use this information to answer questions 40 and 41 (Take $\pi = {}^{22}/_7$)



40. What is the length of the major arc XZY? (Correct to 2 d.p)

A 4.89cm B 9.76cm C 9.78cm
D 19.54cm E 78.22cm

Solution

Length of arc $= \dfrac{\theta}{360} \times 2\pi r$

Major arc here θ is 360 – 40 i.e 320^0

$= \dfrac{320}{360} \times 2 \times \dfrac{22}{7} \times 14$

$= 78.22\text{cm}$

41. What is the area of the minor sector XOY? (Correct to 2d.p)

A 547.56cm^2 B 273.77cm^2 C 78.22cm^2
D 68.44cm^2 E 58.66cm^2

Solution

Area of sector $= \dfrac{\theta}{360} \times \pi r^2$

Minor sector has θ as 40^0

$= \dfrac{40}{360} \times \dfrac{22}{7} \times 14^2$

$= 68.44\text{cm}^2$

2012/16

The lengths of the minor and major arcs of a circle are 54cm and 126cm respectively. Calculate the angle of the major sector

A 306^0 B 252^0 C 246^0 D 234^0

Length of minor arc $= \dfrac{\theta}{360} \times 2\pi r$

Length of major arc $= \dfrac{360 - \theta}{360} \times 2\pi r$

$54 = \dfrac{\theta}{360} \times 2\pi r$ and $126 = \dfrac{360 - \theta}{360} \times 2\pi r$

$54 \times 360 = \theta(2\pi r)$ and $126 \times 360 = (360 - \theta)2\pi r$

$19440 = \theta(2\pi r)$ ---(1) and $45360 = (360 - \theta)2\pi r$ ---(2)

Equation (2) ÷ (1)

$$\frac{45360}{19440} = \frac{(360-\theta)2\pi r}{\theta(2\pi r)}$$

$$2.333 = \frac{360 - \theta}{\theta}$$

$2.333\theta = 360 - \theta$

$2.333\theta + \theta = 360 - \theta$

$3.333\theta = 360$

$\theta = \dfrac{360}{3.333}$

$\theta = 108^0$

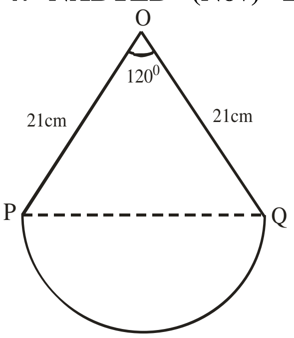
Angle of the major sec tor $= 360 - \theta$
$= 360 - 108$
$= 252^0$ (B)

2008/12 NABTEB (Nov) Exercise 20.15

Find the length of arc of a circle of radius 15.4cm which subtends an angle of 60^0 at the centre of the circle (Take $\pi = {}^{22}/_7$)

A 16.13cm B 15.83cm C 8.26cm D 7.67cm

2010/47 NABTEB (Nov) Exercise 20.16



Find the length of arc /PQ/

A 44cm B 21cm C 22cm D 33cm

2009/34 Exercise 20.17

An arc of a circle subtends an angle of 60^0 at the centre. If the radius of the circle is 3cm, find, in terms of π, the length of the arc

A πcm B 2πcm C 3πcm D 6πcm

2009/40 (Nov) Exercise 20.18

A chord EF subtends an angle 60^0 at the centre of a circle. If the radius of the circle is 7cm, calculate the length of the major arc formed by the chord

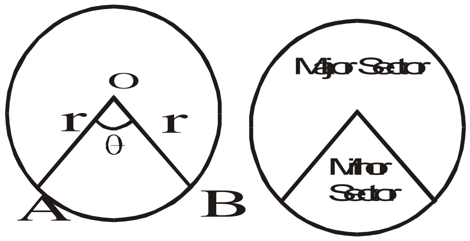
A 7.3cm B 14.7cm C 29.3cm D 36.7cm

2009/45 (Nov) Exercise 20.19

An arc of a circle of radius 3.5cm is 6.6cm long. What angle does the arc subtend at the centre of the circle? (Take $\pi = {}^{22}/_7$)

A 54^0 B 72^0 C 90^0 D 108^0

Perimeter of a Sector



For any circle with center O and radius r from which a sector AOB is formed. The perimeter of the sector is the sum of the two radii and the length of the arc AB since perimeter is the distance round a given shape.

Perimeter of a sector = 2r + L of arc

Where L of arc = $\frac{\theta}{360} \times 2\pi r$

2007/35 Neco

The angle of a sector of a circle of radius 5.5cm is 60^0. What is the perimeter of the sector? (Take $\pi = {}^{22}/_7$)

A 5.76cm B 11.26cm C 16.76cm D 30.25cm E 36.01cm

Solution

Perimeter of a sector = 2r + Length of arc

$$= 2 \times 5.5 + \frac{60}{360} \times 2 \times \frac{22}{7} \times 5.5$$

$$= 11\text{cm} + 5.76\text{cm}$$

$$= 16.76\text{cm} \quad (\text{C})$$

2012/17

A sector of a circle which subtends 172^0 and the centre of the circle has a perimeter of 600cm. find, correct to the nearest cm, the radius of the circle. (Take $\pi = {}^{22}/_7$)

A 120cm B 116cm C 107cm D 100cm

Solution

Perimeter of a sector = 2r + Length of arc

$$600 = 2r + \frac{172}{360} \times 2 \times \frac{22}{7} \times r$$

$$600 = 2r + \frac{7568r}{2520}$$

Applying LCM principle to the RHS

$$600 = \frac{5040r + 7568r}{2520}$$

$$600 = \frac{12608r}{2520}$$

Cross multiply to clear fraction

$$1512000 = 12608\ r$$

$$\frac{1512000}{12608} = \frac{12608r}{12608}$$

$$119.92\text{cm} = r$$

$$r \approx 120\text{cm to nearest cm}$$

2015/14

A sector of a circle with radius 6cm subtends an angle of 60^0 at the centre. Calculate its perimeter in terms of π.

A $2(\pi + 6)$ cm B $2(\pi + 3)$ cm C $2(\pi + 2)$ cm D $(\pi + 12)$ cm

Solution

Perimeter of a sector = 2r + Length of arc

$$= 2 \times 6 + \frac{60}{360} \times 2 \times \pi \times 6$$

$$= 12 + 2\pi$$

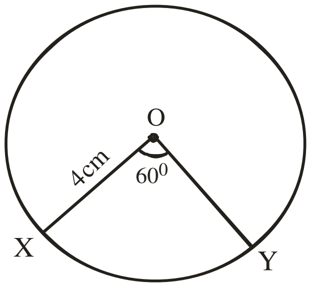
$$= 2(6 + \pi)\text{ cm} \quad \text{A.}$$

2005/42 (old)

OX and OY are radii of a circle centre O. if /OX/ is 4cm and $\angle XOY$ is 60^0, calculate correct to the nearest whole number, the perimeter of sector XOY of the circle (Take $\pi = {}^{22}/_7$)

A 2cm B 4cm C 8cm D 12cm

Solution



Perimeter of a sector = 2r + Length of arc

$$= 2 \times 4 + \frac{60}{360} \times 2 \times \frac{22}{7} \times 4$$

$$= 8 + 4.19\text{cm}$$

$$= 12.19\text{cm}$$

$$\approx 12\text{cm to nearest whole number}$$

2009/38 Neco (Dec)

A sector is bounded by radii 6cm long and an arc subtending an angle of 50^0. Find, correct to 2 decimal places, its perimeter.

A 5.24cm B 11.24cm C 15.24cm
D 17.24cm E 27.24cm

Solution

Perimeter of a sector = 2r + Length of arc

$= 2 \times 6 + \frac{50}{360} \times 2 \times \frac{22}{7} \times 6$

= 12cm + 5.238cm

= 17.238cm

≈ 17.24cm to 2d.p (D)

2014/44 Neco

Find the radius of a sector whose perimeter is 32cm and its arc length is 25cm.

A 14.0cm B 7.0cm C 5.2cm D 4.0cm E 3.5cm

Solution

Perimeter of a sector = 2r + Length of arc

Substituting for the given values

$32 = 2r + 25$

$32 - 25 = 2r$

$7 = 2r$

$\frac{7}{2} = r$ thus, r = 3.5cm (E)

2014/39

A chord subtends an angle of 120^0 at the centre of a circle of radius 3.5cm. Find the perimeter of the minor sector containing the chord (Take $\pi = {}^{22}/_{7}$)

A $14\,{}^{1}/_{3}$cm B $12\,{}^{5}/_{6}$cm C $8\,{}^{1}/_{7}$cm D $7\,{}^{1}/_{3}$cm

Solution

Perimeter of a sector = 2r + Length of arc

$= 2 \times 3.5 + \frac{120}{360} \times 2 \times \frac{22}{7} \times 3.5$

$= 7 + \frac{1}{3} \times 2 \times \frac{22}{7} \times \frac{7}{2}$

$= 7 + \frac{22}{3}$

$= \frac{43}{3}$ = $14\,{}^{1}/_{3}$cm (A)

2011/32 Neco Exercise 20.20

Calculate the perimeter of a sector of a circle of radius 15cm, where the angle of the sector is 60^0

(Take $\pi = {}^{22}/_{7}$)

A 90.24cm B 45.71cm C 22.82cm
D 15.71cm E 14.29cm

2013/5a Neco Exercise 20.21

The angle of a sector of a circle of a radius 9cm is 120^0. Calculate the perimeter of the sector, correct to three significant figures. (Take $\pi = {}^{22}/_{7}$)

2012/32 Neco Exercise 20.22

A sector of a circle of radius 14cm subtends an angle of 135^0 at the centre of the circle. What is the perimeter of the sector? (Take $\pi = {}^{22}/_{7}$)

A 47cm B 51cm C 61cm D 88cm E 231cm

2005/30 Exercise 20.23

An arc of a circle of radius 14cm subtends angle 300^0 at the centre. Find the perimeter of the sector formed by the arc. (Take $\pi = {}^{22}/_{7}$)

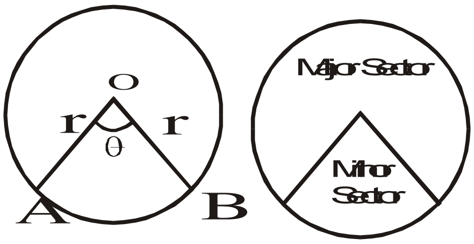
A 14.67cm B 42.67cm C 101.33cm D 513.33cm

2006/3 Neco Exercise 20.24

Find the perimeter of a sector of a circle of radius 14cm which subtends an angle of 135^0 at the centre of the circle. (Take $\pi = {}^{22}/_{7}$)

A 33cm B 47cm C 61cm D 75cm E 91cm

Area of a sector



The area of a sector AOB of a circle radius r of center O is given as

Area of sector = $\frac{\theta}{360}$ x area of circle

$= \frac{\theta}{360} \times \pi r^2$

2006/5a NABTEB

Calculate the area of the major sector of a circle which subtends an angle of 130^0 at the centre and having radius 14cm (Take $\pi = 3.14$)

Solution

Area of sector = $\frac{\theta}{360} \times \pi r^2$

$= \frac{130}{360} \times 3.14 \times 14^2$

= $222.24cm^2$

2008/34 Neco

A sector of angle 120^0 is cut out from circle of radius 13.5cm. What area of the circle is remaining? (Take $\pi = {}^{22}/_{7}$)

A $14.1cm^2$ B $95.5cm^2$ C $190.9cm^2$
D $381.9cm^2$ E $763.7cm^2$

Solution

Area of sector = $\frac{\theta}{360} \times \pi r^2$

Here θ is 360 – 120 i.e 240

$= \frac{240}{360} \times \frac{22}{7} \times 13.5^2$

= $381.857cm^2$

≈ $381.9cm^2$ (D)

2005/45
XOY is a sector of a circle centre O of radius 3.5cm which subtends an angle of 144^0 at centre. Calculate, in terms of π, the area of the sector
A $1.4\pi cm^2$ B $2.8\pi cm^2$ C $4.9\pi cm^2$ D $9.8\pi cm^2$

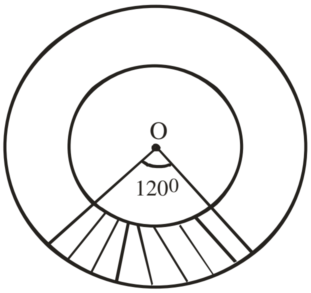
Solution

Area of sector = $\frac{\theta}{360} \times \pi r^2$

$= \frac{144}{360} \times \pi \times 3.5^2$

$= 4.9\pi cm^2$

2009/4c NABTEB (Nov)
In the diagram below, O is the centre of concentric circle of radii 13cm and 10cm respectively. Find the area of the shaded portion in the sector with an angle of 120^0 at the centre



A $^1/_3\ \pi cm^2$ B πcm^2 C $3\pi cm^2$ D $23\pi cm^2$

Solution

Area of shaded portion
= area of big circle sector minus area of small circle sector

$= \frac{120}{360} \times \pi \times 13^2 - \frac{120}{360} \times \pi \times 10^2$

$= \frac{120}{360} \pi\ (13^2 - 10^2)$

$= \frac{1}{3} \pi\ (13-10)(13+10)$

$= \frac{1}{3} \pi\ (3)(23)$

$= 23\pi cm^2$ D.

2010/3 Neco
An arc of a circle of radius 12cm subtends an angle of x^0 at the centre. If the perimeter of the sector formed by the arc is 46cm, calculate the:
(i) value of x
(ii) area of the sector (Take $\pi = ^{22}/_7$)
Solution
Perimeter of a sector = 2r + Length of arc

$$46 = 2 \times 12 + \frac{\theta}{360} \times 2 \times \frac{22}{7} \times 12$$

Here θ is x^0

$$46 = 24 + \frac{528\theta}{2520}$$

$$46 - 24 = \frac{528\theta}{2520}$$

$$\frac{22 \times 2520}{528} = \theta$$

$105^0 = \theta$ i.e x^0

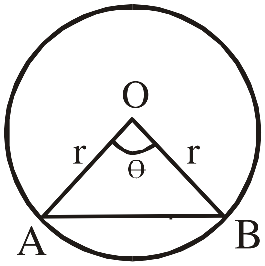
(ii) Area of sector = $\frac{\theta}{360} \times \pi r^2$

$= \frac{105}{360} \times \frac{22}{7} \times 12^2 = 132cm^2$

2008/17 Exercise 20.25
POR is a sector of a circle centre O, radius 4cm.
If $P\hat{O}R = 30^0$, find, correct to 3 significant figures, the area of sector POR (Take $\pi = ^{22}/_7$)
A $4.19cm^2$ B $8.38cm^2$ C $10.5cm^2$ D $20.9cm^2$

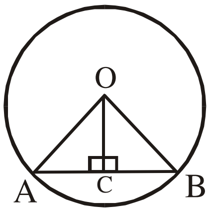
2014/26 NABTEB (Nov) Exercise 20.26
The angle of a sector of a circle of diameter 8cm is 135^0, find the area of the sector (Take $\pi = ^{22}/_7$)
A $9^3/_7 cm^2$ B $12\ ^4/_7 cm^2$ C $18^6/_7 cm^2$ D $25^1/_7 cm^2$

Length of a chord



Length of chord AB = 2r sin $\frac{\theta}{2}$

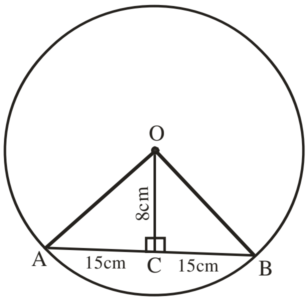
On the other hand if θ is not given but a shape like the one below is given; we apply Pythagoras theorem to solve for the length of chord.



2006/29 Neco (Dec)
A chord of length 30cm is 8cm away from the centre of the circle. What is the radius of the circle?
A 17cm B 18cm C 19cm D 20cm E 21cm

Solution

The sketch is as:



AB = 30 cm i.e AC + AB

By Pythagoras rule in ΔOAC

$OA^2 = AC^2 + OC^2$

$OA^2 = 15^2 + 8^2$

$OA^2 = 225 + 64$

$OA^2 = 289$

$0A = \sqrt{289} = 17cm$ (A)

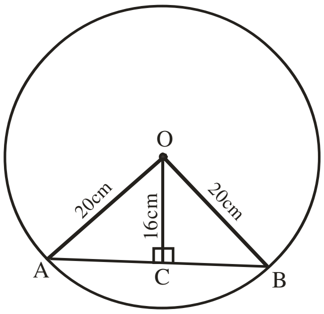
2006/40 Neco (Dec)

The distance of a chord of a circle of radius 20cm from the centre of the circle is 16cm.Calculate the length of the chord

A 36cm B 24cm C 12cm D 6cm E 4cm

Solution

The sketch is as:



Required length is AB

But AB = AC + BC and AC = BC

In ΔOAC: By Pythagoras rule

$20^2 = 16^2 + AC^2$

$AC^2 = 20^2 - 16^2$

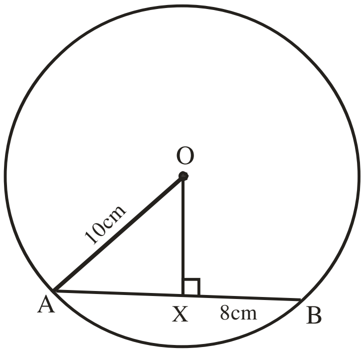
$AC^2 = 400 - 256$

$AC^2 = 144$

$AC = \sqrt{144} = 12cm$

Thus $AB = AC + BC \Rightarrow 12 + 12 = 24cm$ (B)

2011/43 Neco



O is the centre of the above circle and AXB is a chord. Find /OX /

A 9cm B 8cm C 7cm D 6cm E 4cm

Solution

By Pythagoras rule : $OA^2 = AX^2 + OX^2$

AX = BX (The line joining the centre of a circle to the mid – point of a chord is perpendicular to the chord)

$$10^2 = 8^2 + OX^2$$

$$10^2 - 8^2 = OX^2$$

$$(10 - 8)(10 + 8) = OX^2$$

$$2(18) = OX^2$$

$$36 = OX^2$$

$$\sqrt{36} = OX$$

$$OX = 6 \text{ cm} \quad (\text{ D })$$

2007/29 Neco

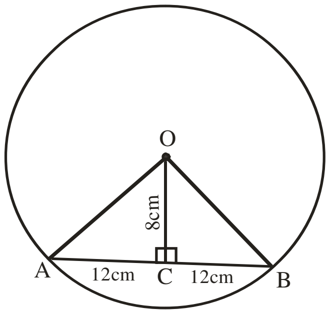
The length of a chord is 24cm. if the distance of the chord is 8cm from the centre of the circle, find the radius of the circle (correct to 2 d.p)

A 12.65cm B 14.42cm C 20.50cm

E 22.63cm E 25.30cm

Solution

The sketch is as:



AB = 24cm i.e AC + CB

By Pythagoras rule in ΔOAC

$OA^2 = AC^2 + OC^2$

$OA^2 = 12^2 + 8^2$

$OA^2 = 144 + 64$

$OA^2 = 208$

$0A = \sqrt{208} = 14.422cm \approx 14.42$ cm to 2d.p

2014/1 Neco (Dec)

An arc PQ of a circle of radius 14cm subtends an angle of 74^0 at the centre of the circle. Calculate, correct to 2 decimal places, the:

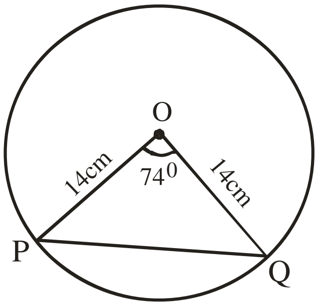
(a) length of the chord PQ;

(b) distance of the chord PQ from the centre of the circle;

(c) Area of Δ POQ

Solution

Sketch of the circle with centre O



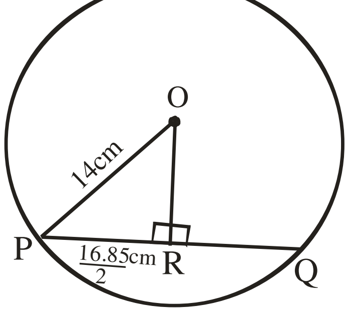
(a) Length of the chord = $2r \sin \frac{\theta}{2}$

$= 2 \times 14 \times \sin \frac{74}{2}$

$= 28 \times \sin 37$

$= 28 \times 0.6018 = 16.8504cm \approx 16.85cm$ to 2d.p

(b)



The required distance is OR

By Pythagoras rule in ΔOPR

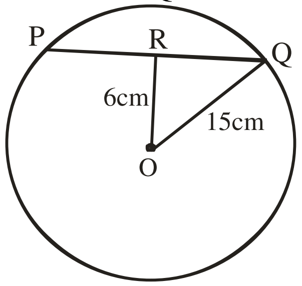
$$14^2 = OR^2 + PR^2$$
$$14^2 = OR^2 + 8.425^2$$
$$196 = OR^2 + 70.981$$
$$196 - 70.981 = OR^2$$
$$125.019 = OR^2$$
$$\sqrt{125.019} = OR$$
$$11.181cm = OR$$
$$OR \approx 11.18cm \text{ to 2d.p}$$

(c) Area of $\Delta POQ = {}^1/_2\, r^2 \sin\theta$

$$= {}^1/_2 \times 14^2 \times \sin 74$$
$$= {}^1/_2 \times 14^2 \times 0.9613$$
$$= 94.2074cm^2$$
$$\approx 94.21cm^2 \text{ to 2d.p}$$

2014/6 Neco Counter example

In the diagram below, O is the centre of the circle, PQ is a chord where /OQ/ = 15cm and /OR/ = 6cm



Calculate the:

(i) length of the chord, correct to 1 decimal place

(ii) angle subtended by POQ, correct to the nearest degree;

(iii) perimeter of the minor segment, correct to 3 significant figures (Take $\pi = {}^{22}/_7$)

Solution

Length of chord is $2 \times RQ$

By Pythagoras rule in ΔROQ

$$15^2 = 6^2 + RQ^2$$
$$225 - 36 = RQ^2$$
$$189 = RQ^2$$
$$\sqrt{189} = RQ$$
$$13.75cm = RQ$$

Thus length of chord $= 2 \times 13.75cm$

$= 27.5cm$ also approximated to 3s.f

(ii) Angle POQ is $2 \times ROQ$

In ΔROQ, by trig ration, let $\angle ROQ$ be θ

$$\text{Cos } \theta = \frac{6}{15}$$
$$\text{Cos } \theta = 0.4$$
$$\theta = \text{Cos}^{-1} 0.4$$
$$= 66.422^0$$

Thus, $POQ = 2 \times 66.422^0$

$$= 132.844^0$$
$$\approx 133^0 \text{ to the nearest degree}$$

(iii) Perimeter of minor segment

= length of arc + length of chord

$$= \frac{\theta}{360} \times 2\pi r + 27.5cm$$
$$= \frac{132.8}{360} \times 2 \times \frac{22}{7} \times 15 + 27.5$$
$$= 34.78 + 27.5$$
$$= 62.28cm$$
$$\approx 62.3cm \text{ to 3s.f}$$

2008/30 Neco

A chord XY of a circle with centre O and radius 5.32cm has $\angle XOY = 140^0$. What is the length of the chord to the nearest centimeter?

A 10cm B 6cm C 5cm D 3cm E 1cm

Solution

Length of chord $= 2r \sin \frac{\theta}{2}$

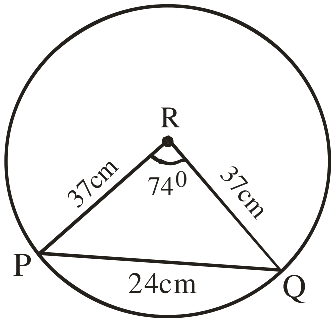
$$= 2 \times 5.32 \times \sin \frac{140}{2}$$
$$= 10.64 \times \sin 70$$
$$= 9.998cm$$
$$\approx 10cm \text{ to nearest cm} \quad (A)$$

2013/3a Neco Counter example

A chord $\overline{PQ}$ of length 24cm is drawn in a circle of radius 37cm. If point R is the centre of the circle, find the area of ΔPRQ

Solution

We sketch as:



Since all the sides of ΔPRQ is given, we apply

Area of Δ PRQ $= \sqrt{S(s-a)(s-b)(s-c)}$

Where $S = \frac{1}{2}(37 + 37 + 24)$

$$= \frac{1}{2}(98) = 49$$

And a is 37, b is 37 and c is 24

Area Δ PRQ $= \sqrt{49(49-37)(49-37)(49-24)}$

$$= \sqrt{49(12)(12)(25)}$$
$$= \sqrt{7^2 \times 12^2 \times 5^2}$$
$$= 7 \times 12 \times 5$$
$$= 420cm^2$$

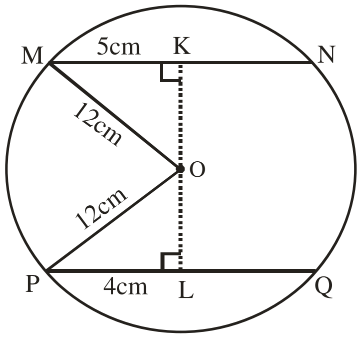
2014/8b NABTEB (Nov)

There are two chords /MN/ and /PQ/ in a circle /MN/ = 10cm and /PQ/ = 8cm and the radius of the circle is 12cm. What is the distance of each chord from the centre of the circle?

Solution

We sketch the diagram as:

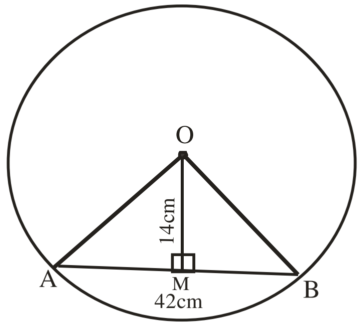
MN is 10cm $\Rightarrow$ MK is 5cm



PQ is 8cm $\Rightarrow$ PL is 4cm

We are asked to find OL in ΔOPL and OK in ΔOMK, readers to complete it.

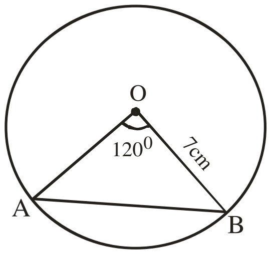
2014/23 NABTEB Exercise 20.27



In the figure, O is the centre of the circle, /AB/ = 42cm and /OM/ = 14cm. if $O\hat{M}A = 90^0$, Calculate /OA/

A 24cm B 25cm C 63cm D 73cm

2014/46 NABTEB Exercise 20.28



Calculate the length of chord /AB/

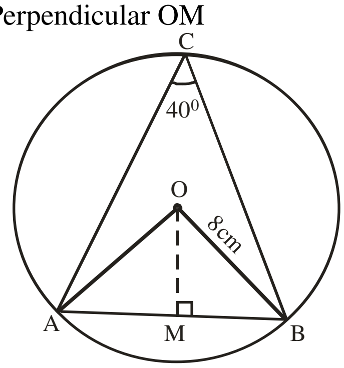
A $2\sqrt{3}$ cm B $3\sqrt{3}$ cm C $5\sqrt{3}$ cm D $7\sqrt{3}$ cm

2005/5 NABTEB Exercise 20.29

The centre of the circle ABC is O. If its radius is 8cm and $\angle ACB = 40^0$, calculate the length of the

(a) Chord AB

(b) Perpendicular OM

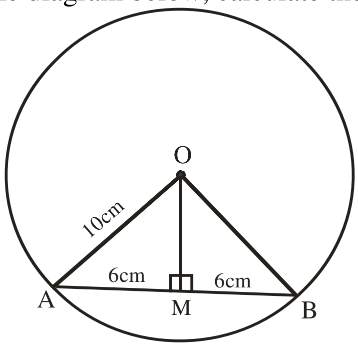


2006/11a NABTEB Exercise 20.30

In a circle of radius 6cm, calculate the distance from the centre to a chord which is 8.5cm long

2009/23 NABTEB (Nov) Exercise 20.31

In the diagram below, calculate the value of /OM /



A 8cm B 15cm C 20cm D 25cm

2006/48 (Nov) Exercise 20.32

A chord subtends an angle of 120^0 at the centre of a circle of radius 10cm. Find the length of the chord.

A 15.00cm B 17.32cm C 18.42cm D 19.32cm

2006/15 (Nov) Exercise 20.33

A Chord of length 8.50cm is drawn in a circle of radius 8.50cm. Calculate, correct to two decimal places, the distance of the chord from the centre of the circle.

A 7.36cm B 7.63cm C 8.00cm D 9.00cm

2014/24 Exercise 20.34

A chord, 7cm long, is drawn in a circle with radius 3.7cm. Calculate the distance of the chord from the centre of the circle

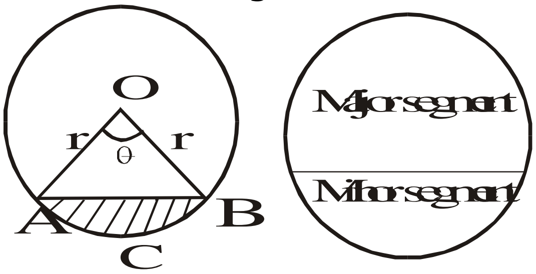
A 0.7cm B 1.2cm C 2.0cm D 2.5cm

2007/15 Exercise 20.35

A chord of length 6cm is drawn in a circle of radius 5cm. Find the distance of the chord from the centre of the circle.

A 2.5cm B 3.0cm C 3.5cm D 4.0cm

Perimeter of Segment of a circle



The perimeter of a segment

= length of arc ACB + length of chord AB

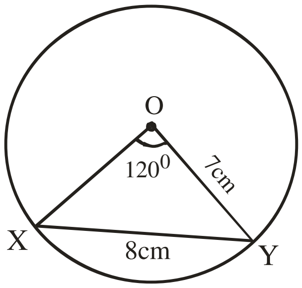
$$= \frac{\theta}{360} 2\pi r + 2r \sin\frac{\theta}{2}$$

2008 /37

XY is a chord of circle centre O and radius 7cm. The chord XY is 8cm long subtends an angle of 120^0 at the centre of the circle. Calculate the perimeter of the minor segment (Take $\pi = {}^{22}/_{7}$)

A 14.67cm B 22.67cm C 29.33cm D 37.33cm

Solution



Perimeter of a segment

= length of arc XY + length of chord XY

Of course minor segment θ is 120^0

$$= \frac{\theta}{360}2\pi r + 8$$

$$= \frac{120}{360}\times 2\times\frac{22}{7}\times 7 + 8$$

$$= 14.667 + 8$$

$$= 22.67\text{cm (B)}$$

2014/43 Neco

Calculate the perimeter of a minor segment of a circle of radius 14cm. subtending an angle 120^0 at the centre. ($\pi = {}^{22}/_{7}$)

A 23.33cm B 24.25cm C 38.92cm
D 53.58 E 126.92cm

Solution

Perimeter of segment = length of arc + length of chord

$$= \frac{\theta}{360}2\pi r + 2r\sin\frac{\theta}{2}$$

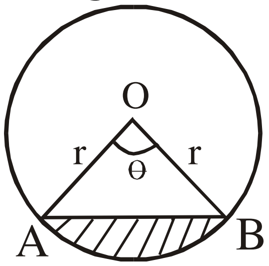
$$= \frac{120}{360}\times 2\times\frac{22}{7}\times 14 + 2\times 14\times\sin\frac{120}{2}$$

$$= 29.333 + 24.249$$

$$= 53.582\text{cm}$$

$$\approx 53.58\text{cm} \quad (\text{D})$$

Area of segment of a circle

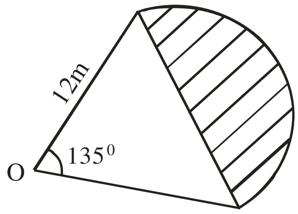


Area of segment = area of sector – area of triangle

$$= \frac{\theta}{360}\times\pi r^2 - \frac{1}{2}r^2\sin\theta$$

$$= \frac{r^2}{2}\left[\frac{\pi\theta}{180} - \sin\theta\right]$$

2005/16 Neco



The figure above shows a sector of a circle radius 12m which subtends angle 135^0 at the centre O. Calculate the area of the shaded segment

A $(84\pi - 48\sqrt{2})$ m² B $(66\pi - 42\sqrt{2})$ m²
C $(54\pi - 36\sqrt{2})$ m² D $(42\pi - 32\sqrt{2})$ m² E $(36\pi - 28\sqrt{2})$ m²

Solution

Area of segment = area of sector – area of triangle

$$= \frac{\theta}{360}\times\pi r^2 - \frac{1}{2}r^2\sin\theta$$

$$= \frac{135}{360}\times\pi\times 12^2 - \frac{1}{2}\times 12^2\times\sin 135$$

But sin 135 is same as sin (180 – 135) i.e sin 45^0

$$= \frac{135}{360}\times\pi\times 12\times 12 - \frac{1}{2}\times 12\times 12\times\sin 45$$

$$= 54\pi - 6\times 12\times\frac{1}{\sqrt{2}}$$

Rationalizing $\frac{1}{\sqrt{2}}$ $= 54\pi - 6\times 12\times\frac{\sqrt{2}}{2}$

$$= (54\pi - 36\sqrt{2})\text{ m}^2 \quad \text{C.}$$

2005/3a Neco

A segment of a circle of diameter 24cm subtends angle of 90^0 at the centre. Calculate the area of the segment to the nearest cm² (Take $\pi = {}^{22}/_{7}$)

Solution

Area of segment = area of sector – area of triangle

$$= \frac{\theta}{360}\times\pi r^2 - \frac{1}{2}r^2\sin\theta$$

Here r is diameter ÷ 2 i.e ${}^{24}/_{2}$ is 12cm

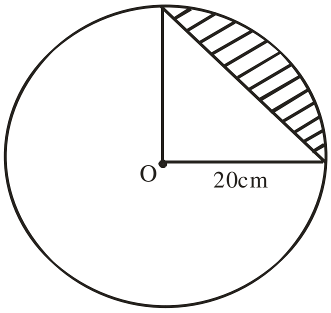
$$= \frac{90}{360}\times\frac{22}{7}\times 12^2 - \frac{1}{2}\times 12^2\times\sin 90$$

$$= 113.14 - 72$$

$$= 41.14\text{cm}^2$$

$$\approx 41\text{cm}^2 \text{ to nearest cm}^2$$

2009/39 Neco (Dec)



Find the area of the shaded portion in the diagram above if O is the centre of the circle. (Correct to 1 decimal place)

A 114.3cm^2 B 125.7cm^2 C 157.1cm^2
D 314.3cm^2 E 514.3cm^2

Solution

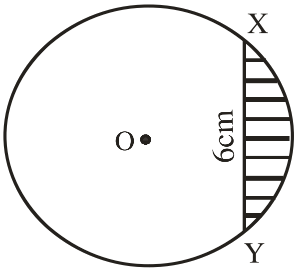
Area of segment = area of sector – area of triangle

$$= \frac{\theta}{360} \times \pi r^2 - \frac{1}{2} r^2 \sin \theta$$

$$= \frac{90}{360} \times \frac{22}{7} \times 20^2 - \frac{1}{2} \times 20^2 \times \sin 90$$

= 314.29 – 200
= 114.29cm^2
≈ 114.3cm^2 to 1 d.p

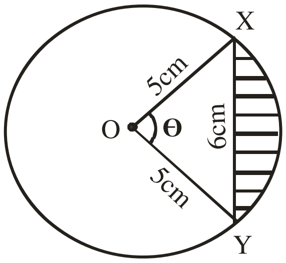
2012/5



In the diagram, O is the centre of the circle and XY is a chord. If the radius is 5cm and /XY/ = 6cm, Calculate, correct to 2 decimal places, the:
(a) angle which XY subtends at the centre O
(b) area of the shaded portion

Solution

We sketch as



Applying cosine rule to find θ

$6^2 = 5^2 + 5^2 - 2 \times 5 \times 5 \cos \theta$

$36 = 25 + 25 - 50 \cos \theta$

$50 \cos \theta = 50 - 36$

$\text{Cos } \theta = \frac{14}{50}$

$\theta = \cos^{-1} 0.28$
$= 73.739^0$
$\approx 73.74^0$

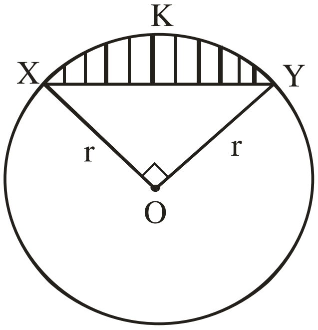
(b) Area of shaded portion = area of segment
= area of sector – area of Δ

$$= \frac{\theta}{360} \times \pi r^2 - \frac{1}{2} r^2 \sin \theta$$

$$= \frac{73.74}{360} \times \frac{22}{7} \times 5^2 - \frac{1}{2} \times 5^2 \times \sin 73.74$$

= 16.094 – 12.000
= 4.094cm^2
≈ 4.09cm^2 to 2 d.p

2014/4a



(NOT DRAWN TO SCALE)

In the diagram, O is the centre of the circle radius r cm and ∠XOY = 90^0, if the area of the shaded part is 504cm^2 , calculate the value of r. (Take π = $^{22}/_7$)

Solution

Area of segment = area of sector – area of triangle

$$= \frac{\theta}{360} \times \pi r^2 - \frac{1}{2} r^2 \sin \theta$$

$$\text{Area of segment} = \frac{r^2}{2}\left[\frac{\pi\theta}{180} - \sin\theta\right]$$

$$504 = \frac{r^2}{2}\left[\frac{22 \times 90}{180 \times 7} - \sin 90\right]$$

$$504 = \frac{r^2}{2}[1.571 - 1]$$

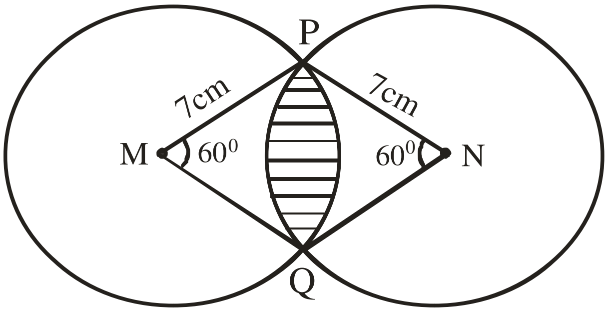
$504 \times 2 = 0.571r^2$

$$\frac{504 \times 2}{0.571} = r^2$$

$1765.32 = r^2$

Thus, $r = \sqrt{1765.32}$
$= 42.02$cm

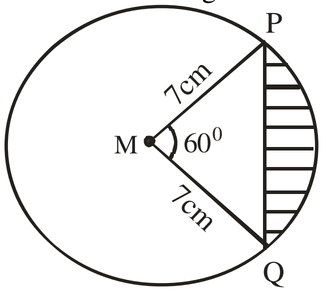
2014/10b



In the diagram, M and N are the centres of two circle of equal radii 7cm. The circle intercept at P and Q.
if ∠PMQ = ∠PNQ = 60^0, calculate, correct to the nearest whole number, the area of the shaded portion (Take π = $^{22}/_7$)

Solution

When you draw a straight line from P to Q it become a case of area of minor segment times two



Area of minor segment = area of sector – area of Δ

$$= \frac{\theta}{360} \times \pi r^2 - \frac{1}{2} r^2 \sin \theta$$

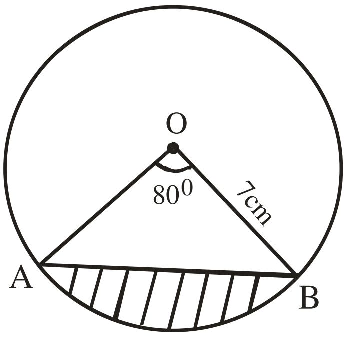
$$= \frac{60}{360} \times \frac{22}{7} \times 7^2 - \frac{1}{2} \times 7^2 \times \sin 60^0$$

$= 25.667 - {}^1/_2 \times 49 \times 0.8660$
$= 25.667 - 21.217$
$= 4.45$ cm^2

Thus, area of shaded portion = 2×4.45cm^2
= 8.9cm^2
= 9cm^2 to the nearest whole number

2005/4 NABTEB (Nov) Exercise 20.36

In the figure below, O is the centre of the circle of radius 7cm and $A\hat{O}B = 80^0$

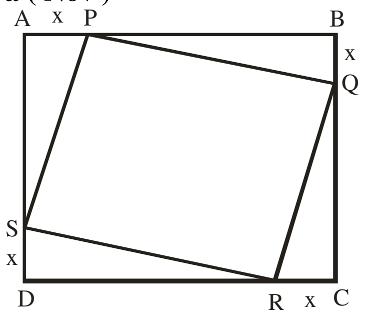


Calculate the (a) length of the minor arc AB
(b) Area of the shaded portion
(Take $\pi = {}^{22}/_7$)

Irregular plane - shapes
(Area & Perimeter)

Irregular plane - shapes are figures that are combinations of two or more plane shapes . Methods of solving these types of problems do not have a particular formula. We apply our experience in plane shapes mensuration to solve them in parts or as a whole.

2015/ 7a (Nov)



In the diagram, ABCD is a square of side 5m, P, Q, R and S are points on the square such that |AP| = |BQ| = |CR| = |DS| = x.
(i) Show that the area of the square PQRS is $(2x^2 - 10x + 25)$m^2
(ii) Find, correct to two decimal places, the values of x for which the area of PQRS is $\frac{3}{5}$ of that of ABCD

Solution

In right-angled ΔQRC, $\angle C = 90^0$, /QC/ = 5 – x and /RC/ = x . QR can be gotten by Pythagoras rule as:

$QR^2 = x^2 + (5 - x)^2$
$= x^2 + 5(5 - x) - x(5 - x)$
$= x^2 + 25x - 5x - 5x + x^2$
$QR^2 = 2x^2 - 10x + 25$

Coincidentally, area of square PQRS = L^2
$= QR^2$
$= (2x^2 - 10x + 25)$m^2

(ii)

Area of square ABCD = 5^2 i.e 25 m^2

Next Area of square PQRS = $\frac{3}{5}$ of ABCD

$$2x^2 - 10x + 25 = \frac{3}{5} \times 25$$

$2x^2 - 10x + 25 = 15$
$2x^2 - 10x + 10 = 0$
$x^2 - 5x + 5 = 0$

Applying the general quadratic formula

$$x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4 \times 1 \times 5}}{2 \times 1}$$

$$= \frac{5 \pm \sqrt{25 - 20}}{2}$$

$$= \frac{5 \pm \sqrt{5}}{2}$$

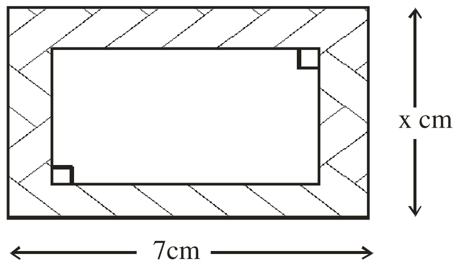
$$= \frac{5 \pm 2.236}{2}$$

$$= \frac{5 + 2.236}{2} \text{ or } \frac{5 - 2.236}{2}$$

$$= \frac{7.236}{2} \text{ or } \frac{2.764}{2}$$

$= 3.618$ m^2 or 1.382 m^2
≈ 3.62 m^2 or 1.38 m^2 to 2 d.p

2005/27 (old)



In the diagram, the shaded portion is 1cm wide. Calculate, in terms of x, the area of the shaded portion

A $(2x + 5)\ cm^2$ B $(2x + 10)\ cm^2$ C $(5x – 10)\ cm^2$ D $(7x + 2)\ cm^2$

Solution

Area of shaded portion

= area of big rectangle – area of small rectangle.

$= 7 \times x – [(7 – 2)(x – 2)]$

Note that in the small rectangle, the 1cm width of it is subtracted twice from the length and breadth

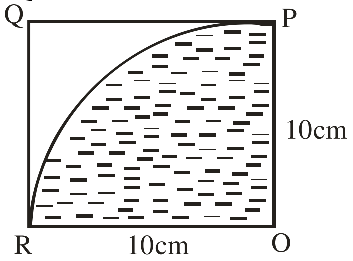
$= 7x – 5(x – 2)$

$= 7x – 5x + 10$

$= (2x +10)\ cm^2$ B.

2005/31 (old)

In the diagram, OPQR is a square of sides 10cm. The shaded portion is a quadrant of a circle centre O, radius OP. Calculate, leaving your answer in terms of π, the ratio of the area of the shaded potion to that of the square OPQR



A $\pi : 2$ B $\pi : 3$ C $\pi : 4$ D $\pi : 5$

Solution

The shaded portion is a sector with centre at O (90^0) or quadrant of a circle implies $\frac{1}{4}$ of 360^0 i.e 90^0

Thus the ratio of shaded portion to area square OPQR

Area of sector ORP : area of square OPQR

$\frac{\theta}{360}\pi r^2 \quad : \quad L \times L$

$\frac{90}{360} \times \pi \times 10^2 \quad : \quad 10 \times 10$

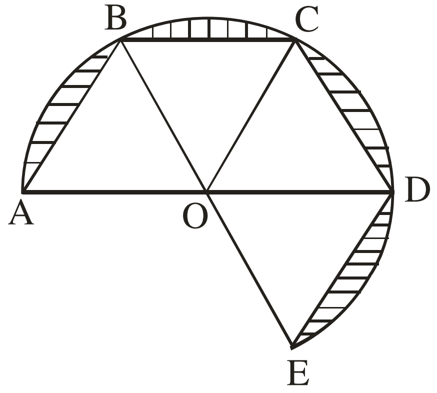
Ten squared will cancel out

$\frac{1}{4}\pi \quad : \quad 1$

Multiply through by 4

$\pi \quad : \quad 4$ (C)

2006/8



In the diagram, ABCDEO is two – thirds of a circle centre O. The radius AO is 7cm and /AB/ = /BC/ = /CD/ = /DE/. Calculate, correct to the nearest whole number, the area of the shaded portion (Take $\pi = {}^{22}/_{7}$)

Solution

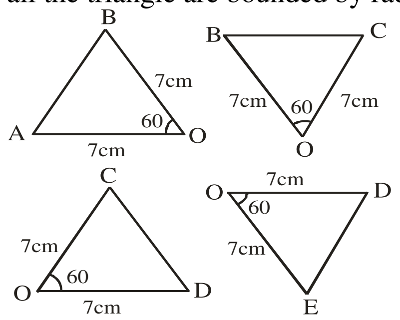
Angle on AD is 180^0

$\angle AOB + \angle BOC + \angle COD = 180^0$

And since they are equal

$\angle AOB = \frac{180}{3} = 60^0$

Since all the triangle are bounded by radius twice



Area of shaded portion

$= \text{area of } \frac{2}{3}(\text{circle}) – 4 \times (\text{area triangle})$

$= \frac{2}{3} \times \pi r^2 - 4 \times \frac{1}{2} r^2 \times \sin\theta$

$= \frac{2}{3} \times \frac{22}{7} \times 7^2 - 4 \times \frac{1}{2} \times 7^2 \times \sin 60$

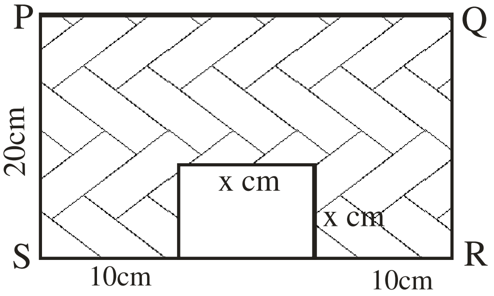
$= 102.667 – 98 \times 0.8660$

$= 102.667 – 84.868$

$= 17.799\ cm^2$

$= 18cm^2$ to nearest whole number

2015/2b



The diagram shows a rectangle PQRS from which a square of side x cm has been cut. If the area of the shaded portion is $484cm^2$, find the value of x

Solution

Area of rectangle = length × breadth

Here the breadth is 20cm and length is (10 + x + 10) and **not** (10 + x + x + 10)

Area of shaded portion = area of rectangle PQRS – area of square

$484 = 20 \times (10 + x + 10) - x \times x$

$484 = 400 + 20x - x^2$

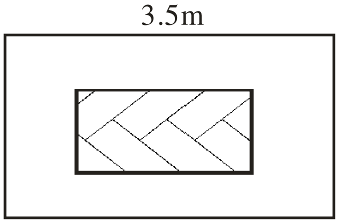
$x^2 - 20x + 84 = 0$

factorising

$(x - 6)(x - 14) = 0$

$x = 6$cm or 14cm

2015/45



In the diagram, the shaded part is a carpet laid in a room with dimensions 3.5m by 2.2m leaving a margin of 0.5m round it. Find the area of the margin

A 4.7m^2 B 4.9m^2 C 5.7m^2 D 5.9m^2

Solution

Area of the margin

= area of big rectangle – area of small rectangle

= 3.5× 2.2 – (3.5 – 1.0) (2.2 – 1.0)

Note that the margin is subtracted from the length and breadth twice

= 7.7 – (2.5) (1.2)

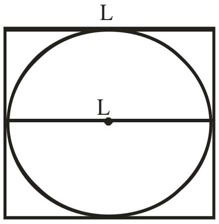
= 7.7 – 3.0

= 4.7m^2 A.

2009/11a

A circle is inscribed in a square. If the sum of the perimeter of the square and the circumference of the circle is 100cm, calculate the radius of the circle (Take $\pi = {}^{22}/_{7}$)

Solution



Perimeter of square is 4L

(Circumference) perimeter of circle is $2\pi r$

Here **r** is **L÷ 2**

From the question's data, we have

$4L + 2\pi r = 100$ becomes

$4L + 2\pi(\frac{L}{2}) = 100$

$4L + 2 \times \frac{22}{7} \times \frac{L}{2} = 100$

$4L + \frac{44l}{14} = 100$

$\frac{56l + 44l}{14} = 100$

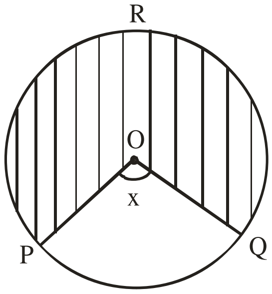
$56L + 44L = 1400$

$100L = 1400$

$L = 14$cm

But **r** is $\frac{L}{2}$; thus r = 14 ÷ 2 is 7cm

2010/6c



The diagram shows a circle centre O. if ∠POQ = x^0, the diameter of the circle is 7cm and the area of the shaded portion is 27.5cm^2, find, correct to the nearest degree, the value of x. (Take $\pi = {}^{22}/_{7}$)

Solution

The shaded part is the major sector of the circle

Area of sector = $\frac{\theta}{360}\pi r^2$

Here θ is the reflex angle 360 – x and not x and radius is diameter ÷ 2 i.e $^7/_2$

Substituting

$27.5 = \frac{360 - x}{360} \times \frac{22}{7} \times 3.5^2$

$27.5 = \frac{(360 - x)269.5}{2520}$

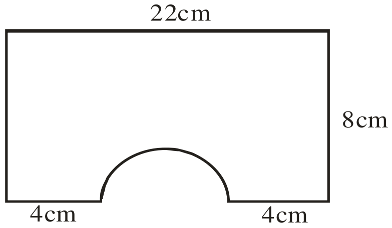
$\frac{27.5 \times 2520}{269.5} = 360 - x$

$257.14 = 360 - x$

$x = 360 - 257.14$

$= 102.86^0 \approx 103^0$ to nearest degree

2010/42



The diagram shows a rectangular cardboard from which a semi – circle is cut off. Calculate the remaining part.

A 44cm^2 B 99cm^2 C 154cm^2 D 165cm^2

Solution

Area of remaining part

= area of rectangle – area of semi – circle

But diameter of semi – circle is 22 – (4 + 4) i.e 14cm

It implies that its radius is 7cm

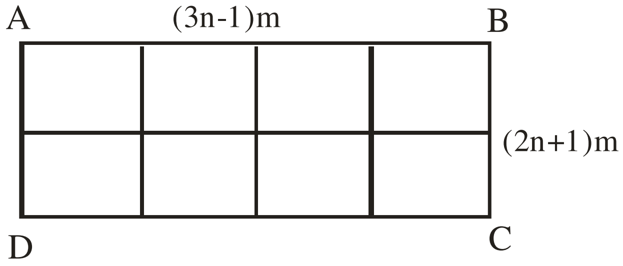
$= L \times b - \frac{1}{2}(\pi r^2)$

$= 22 \times 8 - \frac{1}{2} \times \frac{22}{7} \times 7^2$

$= 176 - 77$

$= 99\text{cm}^2$

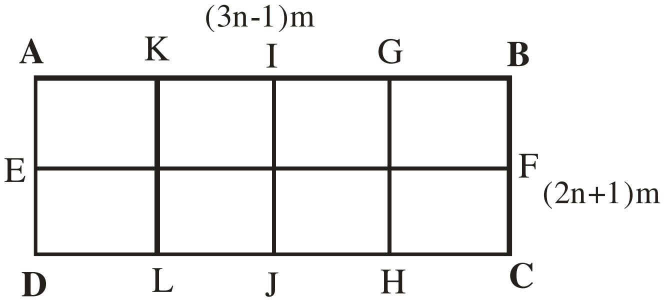
2010/11a



In the diagram, ABCD is a rectangular garden (3n – 1)m long and (2n + 1)m wide. A wire – mesh 135m long is used to mark its boundary and to divide it into 8 equal plots. Find the value of n

Solution

We will apply the principle of perimeter here in relabeling the diagram



Our concern now is the total length of the wire mesh 135m long

At breadth: BC + GH + IJ + KL + AD = **5** × (2n + 1) m

Since they are equal

At length: DC + EF + AB = **3** × (3n – 1) m

Since they are equal

Perimeter of wire – mesh = 135

$5(2n + 1) + 3(3n – 1) = 135$

$10n + 5 + 9n – 3 = 135$

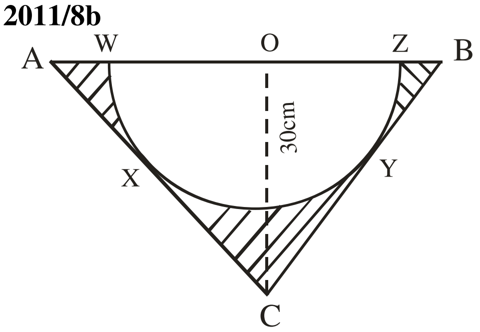
$19n = 135 – 2$

$19n = 133$

$$\frac{19n}{19} = \frac{133}{19}$$

$n = 7$

2011/8b



In the diagram, a semi – circle WXYZ with centre O is inscribed in an isosceles triangle ABC. If / AC/ = /BC/, /OC/ = 30cm and $A\hat{C}B = 130^0$, calculate, correct to one decimal place, the

(i) radius of the circle

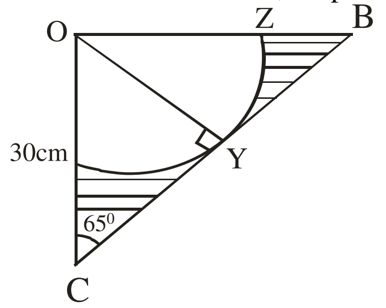
(ii) area of the shaded portion (Take $\pi = {}^{22}/_7$)

Solution

Radius is OW or OX or OZ or OY.

OY can be gotten from ΔOCB when it is bisected as shown below:

Note $\angle ACB = 130^0$, Implies $\angle OCB = 65^0$



By trig ratios $\sin 65^0 = \dfrac{OY}{OC}$

$\text{Sin } 65 = \dfrac{OY}{30}$

$30 \sin 65 = OY$

$30 \times 0.963 = OY$

$OY = 27.189$ cm

Radius OY ≈ 27.2cm to 1d.p

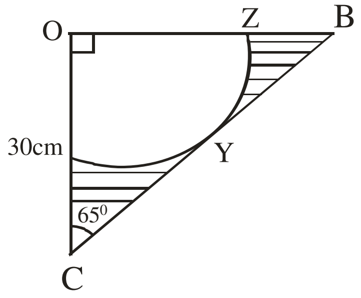
(ii) Area of shaded portion

= area of ΔABC – area of semi – circle

Area of ΔABC = $\frac{1}{2}$ base × height

Half base here is OA or OB

To find $\frac{1}{2}$ base : In Δ BCO, by trig ratio (Not bisected)



$\text{Tan } 65 = \dfrac{OB}{OC}$

$\text{Tan } 65 = \dfrac{OB}{30}$

$30 \tan 65 = OB$

Substituting

Area of ΔABC = 30 tan 65 × 30

= 3 0 × 2.145 × 30

= $1930.5cm^2$

Thus area of shaded portion = $1930.5 - \dfrac{\pi r^2}{2}$

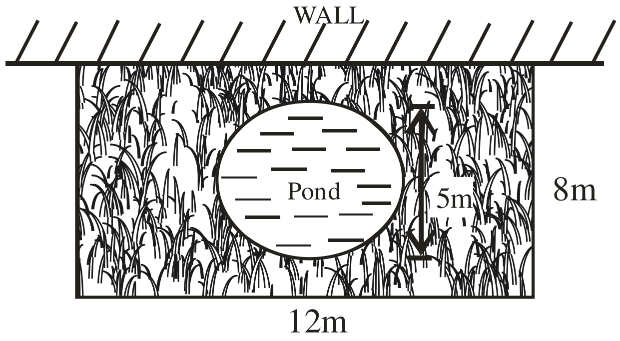
$= 1930.5 - \dfrac{22}{7} \times \dfrac{(27.2)^2}{2}$

$= 1930.5 – 1162.6$

$= 767.9cm^2$ to 1dp

2005/8 (Nov)

The diagram shows a rectangular lawn of length 12m and width 8m. Along one of the side of the lawn is a concrete wall and in the middle is a pond of diameter 5m and surrounded by grass.



(a) Calculate the cost of fencing the remaining sides of the lawn at the rate of D45 per metre

(b) Calculate, correct to three significant figures the :

(i) area covered by grass

(ii) Area covered by grass as a percentage of area of the whole lawn

(c) If the pond is filled with water to a depth of 1.2m, calculate the volume of water in litres.

(Take $\pi = 3.142$)

Solution

(a) The remaining part is 8cm + 12m +8m = 28m

cost of fencing at D45 perimeter = D45 × 28

= D1260

(b) (i) Area covered by grass

= Area of rectangle – area of circular pond

$= l \times b - \pi r^2$

$= 12 \times 8 - 3.142 \times (^5/_2)^2$

$= 96 - 3.142 \times 2.5^2$

$= 96 - 19.6375$

$= 76.3625 m^2$

(ii) $\frac{Area(i)}{Total\ rectangle\ area} \times 100 = \frac{76.3625}{12 \times 8} \times 100$

$= \frac{76.3625}{96} \times 100$

= 79.54%

(c) Volume of water in circular pond $= \pi r^2 h$

$= 3.142 \times 2.5^2 \times 1.2$

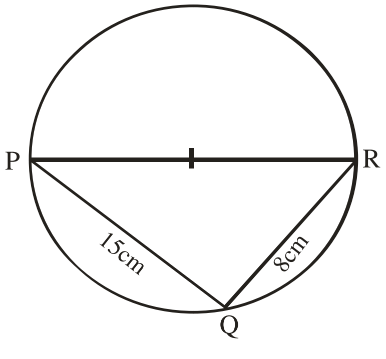
$= 23.565 m^3$

Volume in litres = 23.565 × 1000

= 23565 litres

2014/14 and 15 (Nov)

In the diagram, PR is a diameter /PQ/ = 15cm and /QR/ = 8cm. use the information to answer questions 14 and 15



14. Calculate the area of triangle PQR

A 23cm^2 B 60cm^2 C 68cm^2 D 120cm^2

Solution

Area of $\Delta PQP = \frac{1}{2} \times 15 \times 8 \sin Q$

Note that $\angle PQR$ is 90^0 (angle in semi – circle)

$= \frac{1}{2} \times 15 \times 8 \times \sin 90$

$= \frac{1}{2} \times 15 \times 8 \times 1$

$= 60 cm^2$ (B)

15. Calculate, correct to 1 decimal place, the perimeter of the semi – circle PQR. (Take $\pi = ^{22}/_7$)

A 106.9cm B 47.1cm C 43.7cm D 35.4cm

Solution

Perimeter of semi – circle PQR = $\frac{1}{2}(2\pi r)$

But r is half of PR and PR can be gotten by Pythagoras rule in PQR, since $\angle$ PQR is 90^0 (angle in semi – circle)

$PR^2 = 15^2 + 8^2$

$PR^2 = 225 + 64$

$PR^2 = 289$

$PR = \sqrt{289}$

= 17cm,

Radius = ½ of PR

= $^1/_2$ of 17 = 8.5cm

Thus perimeter of semi – circle PQR = $\frac{1}{2}(2\pi r)$

$= \pi r$

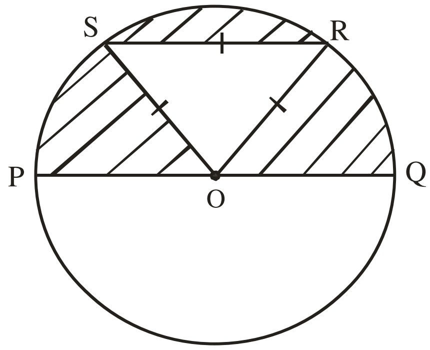
$= ^{22}/_7 \times 8.5$

= 26.71cm

≈ 26.7cm to one d.p

2007/50

In the diagram, O is the centre of the circle and PQ is a diameter. Triangle RSO is equilateral triangle of side 4cm. Find the area of the shaded region



A 43.36cm^2 B 32.07cm^2 C 18.21cm^2 D 6.93cm^2

Solution

Area of shaded region = area of semi-circle – area of ΔRSO

Since all the sides of ΔRSO are given

Area of $\Delta RSO = \sqrt{s(s-a)(s-b)(s-c)}$

Where $S = ^1/_2 (a + b + c)$

$S = ^1/_2 (4 + 4 + 4) = 6cm$

Area of Δ RSO = $\sqrt{6(6-4)(6-4)(6-4)}$

$= \sqrt{6(2)(2)(2)}$

$= \sqrt{48}$

$= 6.928 cm^2$

Area of semi – circle = $\frac{1}{2} \times \pi r^2$

$= \frac{1}{2} \times \frac{22}{7} \times 4^2$

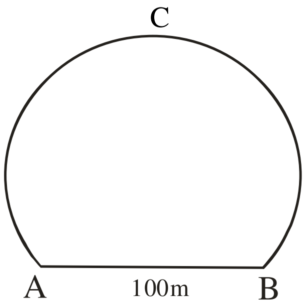
$= 25.143cm^2$

Thus area of shaded region = 25.143 – 6.928

= $18.215cm^2$

≈ $18cm^2$ to nearest whole number

2007/3



The diagram shows the cross – section of a railway tunnel. If /AB/ = 100m and the radius of the arc is 56m, calculate, correct to the nearest metres, the perimeter of the cross – section

Solution

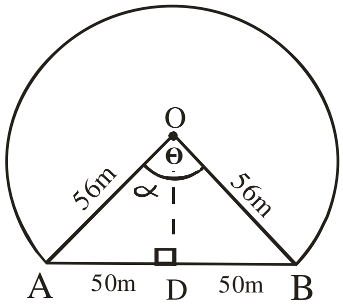
We sketch the diagram as:

The perimeter of the cross – section is perimeter of major segment

= length of arc ACB + length of chord AB

$= \frac{\theta}{360} 2\pi r + 100m$

We derive θ with the aid of the diagram below



In ΔAOD, $\sin \alpha = \frac{50}{56}$

$\sin \alpha = 0.8929$

$\alpha = \sin^{-1} 0.8929$

$= 63.24^0$

But θ = 2α i.e 2 × 63.24 = 126.48^0

This is θ in the minor segment of chord AB

θ in the major segment of chord AB = 360 – 126.48^0

= 233.52^0

Thus perimeter of cross section = $\frac{\theta}{360} \times 2\pi r + 100m$

$= \frac{233.52}{360} \times 2 \times \frac{22}{7} \times 56 + 100$

= 228.33 + 100

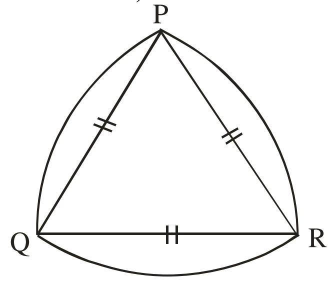
= 328.33m

≈ 328m to nearest metres

1979/9b

In the figure below, PQR is an equilateral triangle of side 20cm. Each arc is part of a circle whose centre is at the opposite vertex of the triangle. Calculate, correct to the nearest whole number, the area of the figure

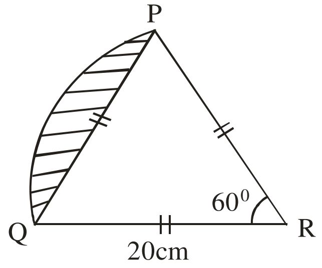
(Take $\pi = \frac{22}{7}$)



Solution

Note that we have three different circles

Since ΔPQR is equilateral, ∠R = 60^0 same as ∠Q and ∠P and PQ = QR = PR = 20cm. thus we sketch as:



Area of Δ PQR = $\frac{1}{2}$ a b sin C

$= \frac{1}{2} \times 20 \times 20 \times \sin 60$

= $173.2cm^2$

The arc PQ is part of the circle centre R

The sector PQR is part of the circle centre R

Area of sector PQR = $\frac{60}{360} \times \frac{22}{7} \times 20^2$

= $209.5cm^2$

Area of shaded portion = area of sector PQR – area of ΔPQR

= 209.5 – 173.2

= $36.3cm^2$

From our question's diagram, the other two circle segments are the same as the one we solved for.

Required area of figure

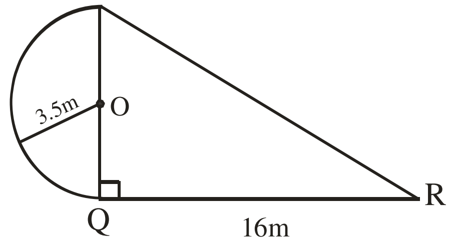
= area of ΔPQR + **3** × area of shaded portion

= $173.2cm^2$ + **3** × $36.3cm^2$

= $282.1cm^2$ ≈ $282cm^2$ to nearest whole number

2007/32 Neco

The composite figure below shows a semi-circle with centre O and triangle PQR, right-angled at Q. Calculate the area of the figure (Take $\pi = \frac{22}{7}$)



A $75\frac{1}{4}m^2$ B $87\frac{1}{2}m^2$ C $124m^2$

D 150 ½ m^2 E 175m^2

Solution

Area of figure = area of semi-circle + area of Δ PQR

$= \frac{1}{2}(\pi r^2) + \frac{1}{2}(PQ \times OR)\sin Q$

Note that : PQ is twice radius 3.5m

$= \frac{1}{2} \times \frac{22}{7} \times 3.5^2 + \frac{1}{2} \times 7 \times 16 \sin 90$

$= 19.25 + 56$

$= 75.25m^2$ i.e $75\ ^1/_4 m^2$

2005 /48 (Nov)

A square of area 64cm^2 has the same perimeter as a rectangle of length 10cm. What is the width of the rectangle?

A 6cm B 12cm C 16cm D 22cm

Solution

Area of square = perimeter of rectangle

$L^2 = (L + b)$

$64 = 2(10 + b)$

Divide through by 2

$32 = 10 + b$

$32 – 10 = b$

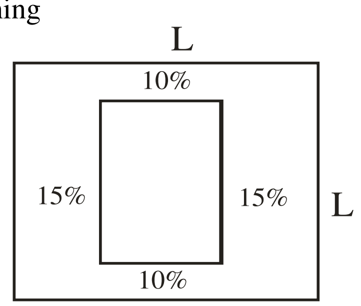
22cm = b (D)

2010/11b

Two opposite sides of a square are each decreased by 10% while the other two are each increased by 15% to form a rectangle. Find the ratio of the area of the rectangle to that of the square.

Solution

Sketching



Ratio of area of square : area of rectangle

L × L : L – 20% × L – 30%

Note that the decrease is subtracted twice from length and breadth.

100 L × 100 L : 80 L × 70 L

100 × 100 : 80 × 70

10 × 10 : 8 × 7

100 : 56

25 : 14

Reducing further; divide through by 5

Square 5 : 3 rectangle

Presenting properly as required 3 : 5

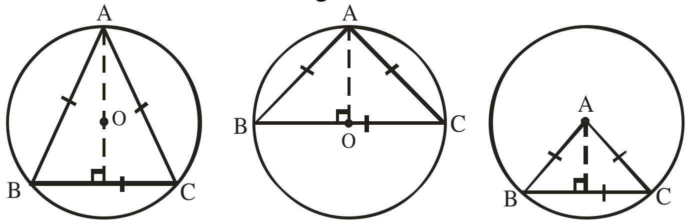
2006/16 (Nov)

An equilateral triangle of side 12cm is inscribed in a circle. Calculate the radius of the circle.

A $6\sqrt{3}$ cm B $4\sqrt{3}$ cm C $3\sqrt{3}$ cm D $2\sqrt{3}$ cm

Solution

We could sketch the diagram as:



It is only the center diagram that can help us.

Here radius of the circle is the height of the inscribed ΔABC

Area of ΔABC = $\frac{1}{2}$ base × height

Also area of ΔABC = $\frac{1}{2}$ a b sin C

Equilateral implies $a = b = c = 60^0$

$= \frac{1}{2} \times 12 \times 12 \times \sin 60$

$= 6 \times 12 \times \frac{\sqrt{3}}{2} = 36\sqrt{3}\ cm^2$

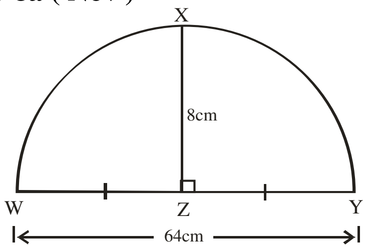
Substituting into area with height

Area of ΔABC = $\frac{1}{2}$ base × height

$36\sqrt{3} = \frac{1}{2} \times 12 \times AO$

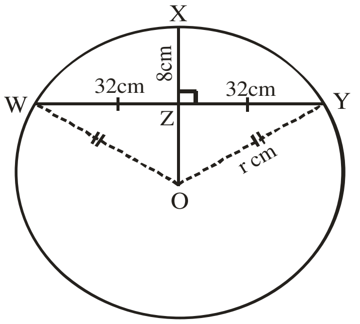
$36\sqrt{3} = 6AO$ thus , AO = $6\sqrt{3}$ cm = (A)

2015/ 8a (Nov)



In the diagram, WXYZ is a segment of a circle such that XZ is the perpendicular bisector of WY. If /XZ/ = 8cm and /WY/ = 64cm, calculate the radius of the circle.

Solution



Given that XZ⊥WY; /WY/ = 64cm, WZ = ZY = ½WY= 32cm

We draw a full circle centre O and radius r cm

Then OY = OX = r

and OZ = OX – ZX = r – 8

In right-angled ΔOZY : $OY^2 = OZ^2 + ZY^2$

$r^2 = (r – 8)^2 + 32^2$

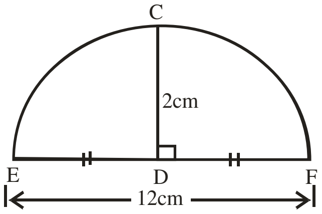
$r^2 = r^2 – 16r + 64 + 1024$

$16r = 1088$

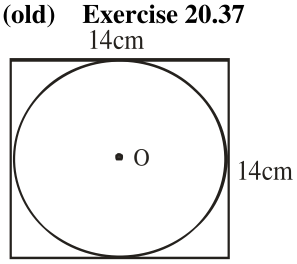
$r = 68cm$

1975/10b Exercise 20.36b

In the accompanying figure which is not drawn to scale, EF is a chord of the circle passing through E, C and F. CD bisects EF perpendicularly at D. If EF = 12cm and CD =2cm, calculate the length of the radius of the circle whose segment is shown.



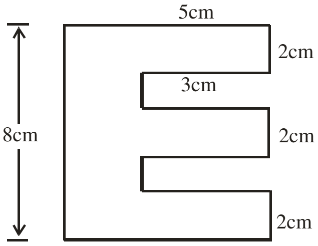
2005/29 (old) Exercise 20.37



The diagram shows a circle centre O inscribed inside a square of side 14cm. Calculate the area of the circle.

A 44cm^2 B 88cm^2 C 154cm^2 D 616cm^2

2014/16 NABTEB Exercise 20.38



Calculate the area of the figure

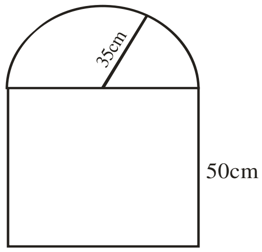
A 26cm^2 B 30cm^2 C 34cm^2 D 40cm^2

2009/42 Neco (Dec) Exercise 20.39

Find the area of a circular field enclosed inside a square field of side 21m such that each of the side of the square is tangent to the circle.

A 110.3m^2 B 173.3m^2 C 210.0m^2 D 346.5m^2 E 441.0m^2

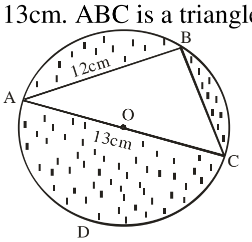
2005/9c Exercise 20.40



The diagram shows a window consisting of rectangular and semi-circular parts. The radius of the semi-circular part is 35cm and the height of the rectangular part is 50cm. Find the area of the window. (Take $\pi = {}^{22}/_{7}$).

2006/8a NABTEB Exercise 20.41

In the diagram, ABCD is a circle centre O with diameter 13cm. ABC is a triangle inscribed in the circle



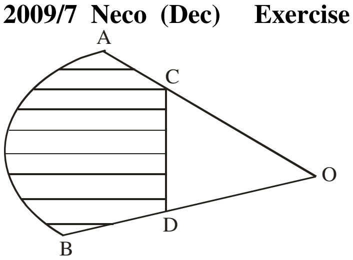
Find, correct to 3 significant figures, the

(i) area of the triangle ABC

(ii) total area of the shaded portion, and

(iii) perimeter of the shaded area ACD

2009/7 Neco (Dec) Exercise 20.42

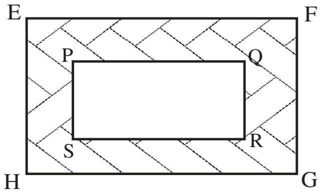


The diagram above shows a sector of a circle of radius 10cm which subtends an angle of 80^0 at the centre O, of the circle. If C and D are the mid point of $\overline{OA}$ and $\overline{OB}$ respectively, calculate, correct to the nearest whole number, the: (i) area and

(ii) perimeter of the shaded region (Take $\pi = {}^{22}/_{7}$)

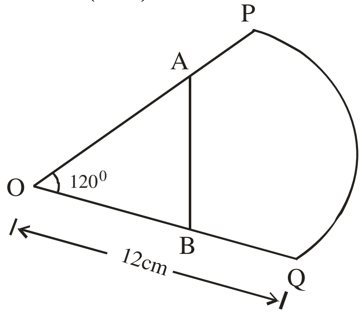
2005/8 (Nov) Exercise 20.43

In the diagram, the sides of rectangle EFGH and PQRS are in the ratio 3:1. Find the ratio of the area of shaded portion to the area of rectangle PQRS



A 3 : 1 B 6 : 1 C 8 : 1 D 9 : 1

2006/11 (Nov) Exercise 20.44



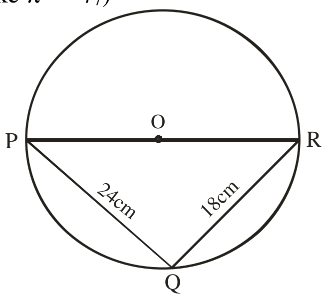
The diagram shows a piece of cardboard in the form of a sector of a circle. The radii OP and OQ are each equal to 12cm. A and B are the mid – points of OP and OQ respectively and $\angle POQ = 120^0$. Calculate, correct to one decimal place, the:

(a) area of ABQP; (b) perimeter of ABQP (Take $\pi = {}^{22}/_{7}$)

2005/25 (Nov) Exercise 20.45

In the diagram, O is the centre of the circle if /PQ/ = 24cm and /RQ/ = 18cm, Calculate, correct to the nearest whole number, the area of the circle.

(Take $\pi = {}^{22}/_{7}$)

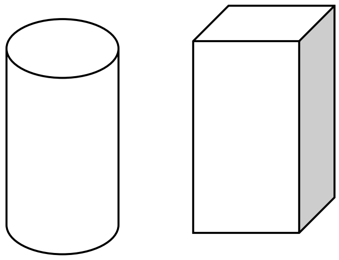


A 2829cm^2 B 1414cm^2 C 707cm^2 D 452cm^2

CHAPTER TWENTY ONE

Solid mensuration

A solid is a three dimensional shape. It is usually represented in two dimensions pictorially in books. Examples of solid are shown below



When we walk round our homes, it is observed that it has some shape which are the same on both sides – but on snapping a picture of the house, we can only see one plane view of it. That is another example of a three dimensional objects represented in two dimensions

Surface areas & volumes

A painter hired to paint our house can only paint the walls round the home (**surface area).** On the other hand, when we want to sit, eat, sleep or even bring in new piece of furniture, we occupy certain space in the house. We cannot sit on the wall, or lie on the wall – thus we occupy space (volume) in our homes. We can say that surface area in solids is the equivalent of perimeter in plane shapes whereas volume in solids is an equivalent of area in plane shapes.

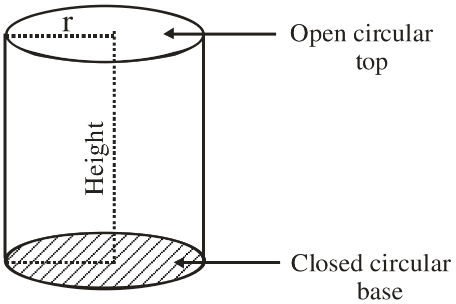
CYLINDER

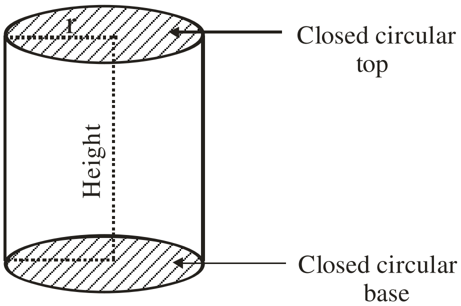
A cylinder is a solid with circular cross section. Examples are bournvita tin, tomatoes tin, milk tin, and open drum etc.

There are two types of cylinders namely:

Open top cylinder & Closed ends cylinder

Of the three properties: curved surface area, total surface area and volume of a cylinder to be measured, only the ***total surface area** that has different formulae for the two types of cylinders.*





Curved surface area & volume for one – open and closed end(s) cylinders

Curved surface area = base circumference × height

$= 2\pi r \times h$

$= 2\pi rh$ square unit.

Volume = base area × height

$= \pi r^2 \times h$

$= \pi r^2 \times h$ cubic units

Total Surface area for open top cylinder only

Total surface area = area of all the faces

= Area of base + curved surface area

$= \pi r^2 + 2\pi rh$

$= \pi r\,(r + 2h)$

Total Surface area for closed tops cylinder only

Total surface area = area of the faces

= Area of circular base + area of circular top + curved surface area.

$= \pi r^2 + \pi r^2 + 2\pi rh$

$= 2\pi r^2 + 2\pi rh$

$= 2\pi r\,(r + h)$ square units

2011/22

The height of a cylinder is equal to its radius. If the volume is $0.216\pi m^3$, calculate the radius

A 0.46m B 0.60m C 0.87m D 1.80m

Solution

Volume of cylinder $= \pi r^2 h$

*But here **r** is equal to **h***

$0.216\pi = \pi r^2 r$

$0.216 = r^3$

$\sqrt[3]{0.216} = r$

$0.6m = r$ (B)

2008/4b NABTEB (Nov)

Find the total surface area of a closed cylinder of diameter 10cm and height 21cm (Use $\pi = {}^{22}/_{7}$)

Solution

Total surface area of closed cylinder $= 2\pi r\,(r + h)$

Note: radius is diameter ÷ 2

$$= 2 \times \frac{22}{7} \times 5(5 + 21)$$

$$= 2 \times \frac{22}{7} \times 5 \times 26 = 817.14cm^2$$

2013/2b Neco

If the volume of a cylindrical container of base radius 4cm is found to be $352cm^3$, calculate the depth of the container. (Take $\pi = {}^{22}/_{7}$)

Solution

Here, we are to find height h in:

Volume of cylinder $= \pi r^2 h$

$$352 = \frac{22}{7} \times 4^2 \times h$$

$$\frac{352 \times 7}{22 \times 4^2} = h$$

$7cm = h$

2007/8a
A cylinder with radius 3.5cm has its two ends closed. If the total surface area is 209cm^2, calculate the height of the cylinder. (Take $\pi = {}^{22}/_{7}$)

Solution

Total surface area of closed ends cylinder $= 2\pi r\,(r + h)$

$$209 = 2\times\frac{22}{7}\times 3.5\,(3.5 + h)$$
$$\frac{209\times 7}{2\times 22\times 3.5} = (3.5 + h)$$
$$\frac{1463}{154} = 3.5 + h$$
$$9.5 = 3.5 + h$$
$$9.5 - 3.5 = h$$
$$6\text{cm} = h$$

2005/29
What is the total surface area of a closed cylinder of height 10cm and diameter 7cm? (Take $\pi = {}^{22}/_{7}$)
A 77cm^2 B 227cm^2 C 297cm^2 D 374cm^2

Solution

Total surface area of closed cylinder =
Area of circular base + area of circular top + curved surface area
$= \pi r^2 + \pi r^2 + 2\pi rh$
$= 2\pi r\,(r + h)$
$= 2\times\frac{22}{7}\times\frac{7}{2}\left(\frac{7}{2}+10\right)$ Note : radius is diameter ÷ 2
$= 22\left(\frac{27}{2}\right) = 297\text{cm}^2$ (B)

2014/46 Neco (Dec)
A tin of milk has a height of 10.3cm and base radius 6.8cm. Find the volume of liquid in litres that can fill the tin to capacity
A 1.153litres B 1.208litres C 1.411litres
D 1.497litres E 2.000litres

Solution

First, we find the volume in cm^3 then convert to litres.
Volume of cylinder $= \pi r^2 h$
$= \frac{22}{7}\times(6.8)^2\times 10.3$
$= 1496.85\text{cm}^3$

Converting to litres : $1496.85\text{cm}^3 = \frac{1496.85}{1000}$ litres
$= 1.4969$litres
≈ 1.497 litres (D)

2007/31 Neco
Find the capacity of a cylinder in litres whose diameter is 18cm and height 17cm (Take $\pi = {}^{22}/_{7}$)
A 4.328litres B 121.18litres C 43.28 litres
D 4327.71litres E 4328.8litres

Solution

First, we find the volume in cm^3 then convert to litres.
Note: radius is diameter ÷ 2
Volume of cylinder $= \pi r^2 h$
$= \frac{22}{7}\times 9^2\times 17$
$= 4327.71\text{cm}^3$

Converting to litres : $4327.71\text{cm}^3 = \frac{4327.71}{1000}$ litres
$= 4.3277$litres
≈ 4.328 litres (A)

2005/22 NABTEB (Nov)
A cylindrical tank with diameter 1 metre and height 2 metre is half filled with water. What is the volume of the water?
A $\frac{\pi}{4}m^3$ B $\frac{\pi}{2}m^3$ C πm^3 D 2πm^3

Solution

First, we find volume in πm^3 then take half of it
Volume of cylinder $= \pi r^2 h$
$= \pi\left(\frac{1}{2}\right)^2\times 2$
$= \frac{\pi}{2}m^3$

Thus, volume of half filled cylindrical tank
$= \frac{1}{2}\times\frac{\pi}{2}m^3 = \frac{\pi}{4}m^3$ (A)

2015/35 and 36 (Nov)
A cylindrical tin with base diameter 14cm and height 20cm is open at the top. (Take $\pi = {}^{22}/_{7}$)
Use this information to answer question 35 and 36

35. Find the total surface area of the tin.
A 976cm^2 B 1034cm^2 C 1188cm^2 D 2376cm^2

Solution

Total of surface area = area of base + curved surface area
$= \pi r^2 + 2\pi rh$
$= \pi r\,(r + 2h)$
Note: radius is diameter ÷ 2
$= \frac{22}{7}\times 7(7+2\times 20)$
$= 22(47)$
$= 1034\text{cm}^2$ (B)

36. Calculate the volume of water in the tin when it is full.
A 3080cm^3 B 4620cm^3 C 6160cm^3 D 12320cm^3

Solution

Volume of cylinder $= \pi r^2 h$
$= \frac{22}{7}\times 7^2\times 20$
$= 3080\text{cm}^3$ (A)

2006/28 Neco (Dec)
A solid cylindrical object of radius 7cm is 10cm high. Find its total surface area.
A 220cm^2 B 594cm^2 C 629cm^2
D 660cm^2 E 748cm^2

Solution

Total surface area of a **solid** cylindrical object is
total surface area of closed ends cylinder $= 2\pi\, r(r + h)$
$= 2\times\frac{22}{7}\times 7(7+10)$
$= 2\times\frac{22}{7}\times 7\times 17$
$= 748\text{cm}^2$ (E)

2013/31 Neco

A cylindrical tank of length 24cm has a volume of $14784cm^3$. Calculate its radius

A 9.38cm B 14cm C 14.8cm
D 15.0cm E 24.8cm

Solution

Volume of cylinder $= \pi r^2 h$

$$14784 = \frac{22}{7} \times r^2 \times 24$$

$$\frac{14784 \times 7}{22 \times 24} = r^2$$

$$196 = r^2$$

$$r = \sqrt{196} = 14cm \quad (B)$$

2008/33 Neco

The radii of two similar cylindrical jugs are in the ratio 3 : 7. Calculate the ratio of their volumes

A 1:4 B 3:7 C 9:49 D 1:64 E 27:343

Solution

Applying volume of cylinder formula

	$\pi r^2 h$	:	$\pi r^2 h$	
In volume terms	r^2	:	r^2	
In radius terms	$r = 3$	:	$r = 7$	
In volume terms	3^2	:	7^2	
	9	:	49	(C)

2007/48

The volume of a cylinder is $1200cm^3$ and the area of its base is $150cm^2$. Find the height of the cylinder

A 80.00cm B 8.00cm C 0.80cm D 0.08cm

Solution

Volume of cylinder $= \pi r^2 h$

$$1200 = \pi r^2 h \text{ ----- (1)}$$

Area of base of cylinder $= \pi r^2$

$$150 = \pi r^2 \text{ ----- (2)}$$

Replace πr^2 in (1) by 150 from (2)

$$1200 = 150h$$

$$\frac{1200}{150} = h$$

$$8 \text{ cm} = h \quad (B)$$

2012/15

The curved surface area of a cylindrical tin is $704cm^2$. If the radius of its base is 8cm, find the height

(Take $\pi = {}^{22}/_{7}$)

A 14cm B 9cm C 8cm D 7cm

Solution

Curved surface area of cylindrical tin $= 2\pi rh$

$$704 = 2 \times \frac{22}{7} \times 8 \times h$$

$$\frac{704 \times 7}{2 \times 22 \times 8} = h$$

$$14cm = h \quad (A)$$

2014/15 NABTEB (Nov)

The volume of a cylinder is $216cm^3$. A similar cylinder has a volume of $125cm^3$. If the height of the larger cylinder is 30cm, find the height of the smaller cylinder.

A 25cm B 20cm C 15cm D 12cm

Solution

The only item to show their similarity here is volume

Volume of larger cylinder : volume of small cylinder

216 : 125

2 : 1

Thus if the height of larger cylinder is 30cm

$\therefore$ Height of small cylinder is 15cm (C)

2014/38 (Nov)

The ratio of the area of the base of a cylinder to the curved surface area of the cylinder is 1: 4. If the radius of the cylinder is 4cm, find the height of the cylinder

A 1cm B 2cm C 4cm D 8cm

Solution

Base area of cylinder : curved surface area

$\pi r^2 \quad : \quad 2\pi rh$

Some terms will cancel out

$r \quad : \quad 2h$

$1 \quad : \quad 4$

When $r = 1$, $2h = 4$

Thus, if $r = 4$, $2h = 4 \times 4$

$2h = 16$

$h = {}^{16}/_{2} = 8cm$ (D)

2014/9a Neco

The total surface area of a solid cylinder is $88cm^2$, while the height is 7cm. Find, correct to three significant figure, the:

(i) diameter of its cross section

(ii)volume of the solid (Take $\pi = {}^{22}/_{7}$)

Solution

(i) Diameter is twice radius

Total surface area of a solid cylindrical object is total surface area of closed ends cylinder $= 2\pi r(r + h)$

$$88 = 2 \times \frac{22}{7} \times r(r+7)$$

$$\frac{88 \times 7}{2 \times 22} = r(r+7)$$

$$14 = r(r+7)$$

$$14 = r^2 + 7r$$

$$r^2 + 7r - 14 = 0$$

Applying quadratic formula $r = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$$r = \frac{-7 \pm \sqrt{7^2 - 4 \times 1 \times (-14)}}{2 \times 1}$$

$$= \frac{-7 \pm \sqrt{105}}{2}$$

$$= \frac{-7 \pm 10.25}{2}$$

$$= \frac{-7 - 10.25}{2} \text{ or } \frac{-7 + 10.25}{2}$$

$$= \frac{-17.25}{2} \text{ or } \frac{3.25}{2}$$

$r = 1.625cm$ *the positive value*

Thus, diameter of its cross section $= 2 \times r$

$= 2 \times 1.625cm$

$= 3.25cm$

(ii) Volume of solid $= \pi r^2 h$

$$= \frac{22}{7} \times (1.625)^2 \times 7$$

$$= 58.094cm^3$$

$$\approx 58.1cm^3 \text{ to 3s.f}$$

2011/34 Neco Exercise 21.1
A right circular cylinder of base radius 8cm has height of 14cm. Find the volume of the cylinder.
A 988cm^3 B 1, 408cm^3 C 2, 816cm^3
D 3, 423cm^3 E 5, 632cm^2

2009/3b Neco (Dec) Exercise 21.2
Find the volume of a cylinder whose height is 8cm and radius 5cm, correct to 3 significant figures (Take $\pi = {}^{22}/_7$)

2011/9 Exercise 21.3
A cylindrical container has a base radius of 14cm and height 18cm. How many litres, correct to the nearest litre of liquid can it hold? (Take $\pi = {}^{22}/_7$)
A 11 B 14 C 16 D 18

2007/18 Exercise 21.4
Find the volume of a solid cylinder with base radius 10cm and height 14cm (Take $\pi = {}^{22}/_7$)
A 220cm^3 B 880cm^3 C 1400cm^3 D 4400cm^3

2005/9a NABTEB (Nov) Exercise 21.5
Calculate the volume of a cylinder of base radius 5.4cm and height 16.2cm.

2014/9a Neco Adjusted Exercise 21.6
The total surface area of a solid cylinder is 88cm^2.
While the height is 5cm. Find the:
(i) diameter of its cross section
(ii) volume of the solid (Take $\pi = {}^{22}/_7$)

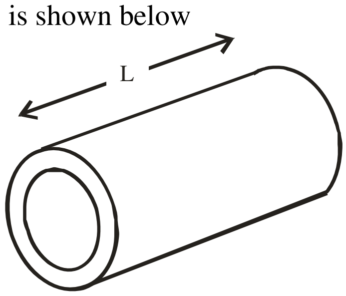
2014 /18 Neco (Dec) Exercise 21.7
Find the volume of a cylinder that is closed at both ends with a radius 5.7m and height 6.3m. Give your answer to the nearest m^3
A 18 B 113 C 205 D 600 E 643

2010/33 UTME Exercise 21.8
A cylindrical pipe 50m long with radius 7m has one end open. What is the total surface area of the pipe?
A 700πm^2 B 98πm^2 C 350πm^2 D 749πm^2

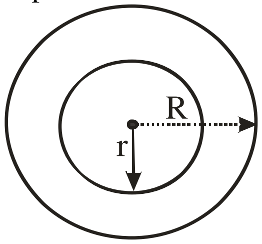
2016/3b Exercise 21.9
A cylindrical tin, 7cm high, is closed at one end. If its total surface area is 462 cm^2, calculate its radius
(Take $\pi = {}^{22}/_7$)

2016/16 Exercise 21.10
The curved surface area of a cylinder 5cm high is 110 cm^2. Find the radius of its base. (Take $\pi = {}^{22}/_7$)
A 2.6cm B 3.5cm C 3.6cm D7.0cm

HOLLOWED CYLINDER, PIPES & RINGS
This is a special type of cylinder, which is open at both ends but has very thick layer. Examples are culverts, ring pipes etc. Diagrammatic presentation of such shapes is shown below



Open end – side view



Area $= \pi R^2 - \pi r^2$
$= \pi (R^2 - r^2)$
Difference in areas between the big circle with radius R and small circle with radius r
Similarly,
Volume $= \pi R^2 L - \pi r^2 L$
$= \pi L(R^2 - r^2)$

2006/14 (Nov)
The internal and external radii of a water pipe are 9cm and 10cm respectively. If the pipe is 35cm long, find, cm^3, the volume of the material used in making it (Take $\pi = {}^{22}/_7$)
A 1, 100 B 1, 900 C 2,090 D 2,200

Solution

Here; volume of pipe $= \pi L (R^2 - r^2)$
$$= \frac{22}{7} \times 35 \ (10^2 - 9^2)$$
$$= 22 \times 5 \ (100 - 81)$$
$$= 22 \times 5 \times 19 = 2{,}090 cm^3 \ \text{(C)}$$

2005/2 (Nov)
A cylindrical pipe is 28metres long. Its internal radius is 3.5cm and its external radius is 5cm. Calculate:
(a) the volume of water, in litres, that the pipe can hold when full;
(b) the volume in cm^3 of metal used in making the pipe.
(Take $\pi = {}^{22}/_7$)

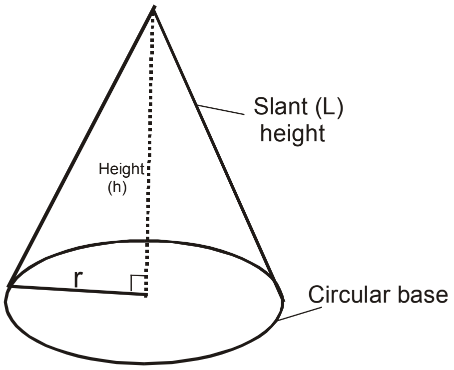
Solution

(a) First, we find the volume of water, it can hold in cm^3.
Volume of water is the same as:
Volume of cylinder using the internal radius given.
Volume of water in cylindrical pipe $= \pi r^2 h$
$$= \frac{22}{7} \times (3.5)^2 \times 28$$
$$= 1078 \ cm^3$$
Converting to litres $1078 cm^3 = \frac{1078}{1000}$ litres
$= 1.078$ litres
(b) Volume of metal used $= \pi L (R^2 - r^2)$
$$= \frac{22}{7} \times 28 (5^2 - 3.5^2)$$
$$= 22 \times 4 \ (25 - 12.25)$$
$$= 22 \times 4 \times 12.75$$
$$= 1122 cm^3$$

2002/14 (Nov) Exercise 21.10b
The internal and external radii of a cylindrical bronze pipe are 1.5cm and 2cm respectively. If the pipe is 10cm long, calculate the volume of the bronze used
(Take $\pi = {}^{22}/_7$)
A. $5^1/_2$ cm^3 B. 55cm^3 C. 196 $^2/_5 cm^3$ D 550cm^3

CONE

A cone is a solid with circular base and slant sides. The cone of concern to us is the right circular cone. *There is the ordinary right circular cone and cone formed from a sector of a circle*



Properties to be measured

Curved surface area = πrL

Total surface area

= Area of circular base + curved surface area

= $\pi r^2 + \pi rL$

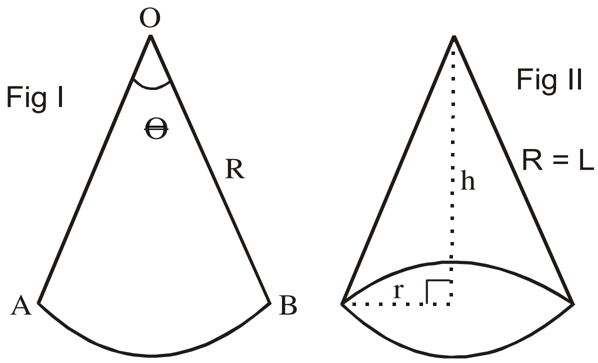
= $\pi r (r + L)$

Volume = $\frac{1}{3} \pi r^2 h$

Slant height (L) , radius (r) and height (h) are related by Pythagoras' theorem : $L^2 = h^2 + r^2$

Cones formed from a sector

The diagram 1 shown below is a sector of a circle radius R which subtends an angle θ in the center O. then FigII is the cone formed from the sector.



To measure the properties of a cone formed from a sector we need, L and h. This is gotten from the relationship between a sector and the cone it formed.

Relationship between a sector of a circle and the cone it forms

From fig. I and II

1. Length of arc = circumference of base of cone

$$\frac{\theta}{360} \times 2\pi R = 2\pi r$$

$$\frac{\theta R}{360} = r$$

Radius r of new cone formed

$$r = \frac{\theta R}{360}$$

2. R = L

Radius of the sector of a circle is equal to the slant height of the cone formed.

3. Recall $L^2 = h^2 + r^2$

$$h^2 = L^2 - r^2$$

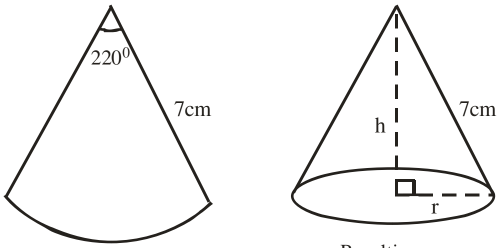
$$h = \sqrt{L^2 - \left(\frac{\theta R}{360}\right)^2}$$

Problems on cones formed from a sector

2007/7b NABTEB

A cone is formed by folding a major sector of a circle having angle 220^0 at the centre. Calculate the circumference of the base of the cone if the diameter of the circle is 14cm, correct to 1 decimal place.

Solution



sector Resulting cone

Cone base is circular, thus

Circumference of cone base = $2\pi r$

But $r = \frac{\theta R}{360}$ (circle radius)

$= \frac{220}{360} \times 7$

$= 4.278$cm

Thus circumference of cone base = $2 \times \frac{22}{7} \times 4.278$

= 26.89cm

≈ 26.9cm to 1d.p

2007/16

A sector of a circle of radius 14cm containing an angle 60^0 is folded to form a cone. Calculate the radius of the base of the cone

A $5\frac{1}{2}$cm B $4\frac{2}{3}$cm C $3\frac{1}{2}$cm D $2\frac{1}{3}$cm

Solution

Cone base radius $= \frac{\theta R}{360}$ (circle radius)

$= \frac{60 \times 14}{360}$

$= \frac{7}{3} = 2\frac{1}{3}$cm (D)

2013/50

An open cone with base radius 28cm and perpendicular height 96cm was stretched to form a sector of a circle. Calculate the area of the sector (Take $\pi = \frac{22}{7}$)

A $8800cm^2$ B $8448cm^2$ C $4400cm^2$ D $4224cm^2$

Solution

Area of sector = $\frac{\theta}{360} \times \pi r^2$

Cone base radius = $\frac{\theta R}{360}$ (radius of circle)

$28 = \frac{\theta \times R}{360}$

R = L and $L^2 = h^2 + r^2$

$L^2 = 96^2 + 28^2$

$= 9216 + 784$

$L^2 = 10,000$

$L = \sqrt{10,000} = 100$cm

Thus, Cone base radius = $\frac{\theta R}{360}$ (radius of circle)

$28 = \frac{\theta \times 100}{360}$

$\frac{28 \times 360}{100} = \theta$

$100.8^0 = \theta$

Area of sector = $\frac{100.8}{360} \times \frac{22}{7} \times 100^2$

$= 8800\text{cm}^2$ (A)

2009/4 (Nov)

A sector of angle 135^0 is removed from a thin circular metal sheet of radius 40cm. It is then folded with the straight edges coinciding, to form a right circular cone. Calculate the:

(a) base radius of the cone, correct to **two** decimal places:

(b) greatest volume of liquid which the cone can hold, leaving your answer correct to the nearest cm^3 (Take $\pi = {}^{22}/_7$)

Solution

(a) Cone base radius = $\frac{\theta R}{360}$ (circle radius)

$= \frac{135 \times 40}{360}$

= 15cm

(b) Volume of cone = $\frac{1}{3}\pi r^2 h$

But $L^2 = h^2 + r^2$

where L is radius of sector and r is cone base radius

$40^2 = h^2 + 15^2$

$40^2 - 15^2 = h^2$

$1600 - 225 = h^2$

$h^2 = 1375$

$h = \sqrt{1375}$

$= 37.08$cm

Volume of cone = $\frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \frac{22}{7} \times 15^2 \times 37.08$

$= 8740.29\text{cm}^3$

$= 8740\text{cm}^3$ to the nearest cm^3

2014/23

The area of a sector of a circle with diameter 12cm is 66cm^2. If the sector is folded to form a cone, calculate the radius of the base of the cone. (Take $\pi = {}^{22}/_7$)

A 3.0cm B 3.5cm C 7.0cm D 7.5cm

Solution

Cone base radius = $\frac{\theta R}{360}$ (circle radius)

Here radius is diameter divided by 2

But we find θ from: area of sector = $\frac{\theta}{360} \times \pi r^2$

$66 = \frac{\theta}{360} \times \frac{22}{7} \times 6^2$

$\frac{66 \times 360 \times 7}{22 \times 6^2} = \theta$

$210^0 = \theta$

Thus cone base radius r = $\frac{\theta R}{360}$ (circle radius)

$= \frac{210 \times 6}{360} = 3.5$cm (B)

2015/52 Neco Exercise 21.11

A sector which subtends an angle of 90^0 was cut from a circle of radius 8cm and folded to give a cone. Find the radius of the cone

A 1cm B 2cm C 3cm D 4cm E 5cm

2005/7b (old) Exercise 21.12

A sector of a circle of radius 12cm subtends an angle of 120^0 at the centre. If it is folded to form a cone, calculate:

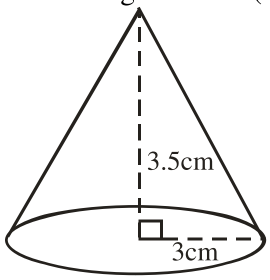
(i) the base radius of the cone

(ii) the height and volume of the cone, correct to three significant figures (Take $\pi = 3.142$)

Problems on cones

2014/33 NABTEB (Nov)

Calculate the volume of a cone with radius 3cm and perpendicular height 3.5cm (Take $\pi = {}^{22}/_7$)



A 3.3cm^3 B 33.0cm^3 C 65.0cm^3 D 66.0cm^3

Solution

Volume of cone = $\frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \frac{22}{7} \times 3^2 \times 3.5$

$= 33\text{cm}^3$

2009/2b Neco (Dec)

Find the volume of a cone whose height and base radius are 15cm and 6cm respectively

Solution

Volume of cone = $\frac{1}{3}\pi r^2 h$

$= \frac{1}{3}\times\frac{22}{7}\times 6^2\times 15$

$= 565.71cm^3$

2014/14 NABTEB

Find the total surface area of a cone radius 7cm and slant height 8cm.

A $56cm^2$ B $77cm^2$ C $110cm^2$ D $330cm^2$

Solution

Total surface area

= area of circular base + curved surface area

$= \pi r^2 + \pi rL$

$= \pi r\,(r + L)$

$= \frac{22}{7}\times 7\,(7+8)$

$= \frac{22}{7}\times 7\times 15 = 330cm^2$ (D)

2006/49 and 50 (Nov)

A hollow right circular cone has a base radius of 3.50cm and a slant height of 4.61cm.

Use the information to answer question 49 and 50 (Take $\pi = {}^{22}/_{7}$)

49. What is the curved surface area of the cone?

A $66.76cm^2$ B $50.71cm^2$ C $38.48cm^2$ D $32.94cm^2$

Solution

Curved surface area $= \pi rL$

$= \frac{22}{7}\times 3.5\times 4.61$

$= 50.71cm^2$ (B)

50. Find, correct to one decimal place, the volume of the cone

A $77.9cm^3$ B $38.5cm^3$ C $32.9cm^3$ D $11.5cm^3$

Solution

Volume of cone = $\frac{1}{3}\pi r^2 h$

But $L^2 = h^2 + r^2$

$(4.61)^2 = h^2 + (3.5)^2$

$21.25 - 12.25 = h^2$

$9 = h^2$

$h = 3cm$

Volume = $\frac{1}{3}\times\frac{22}{7}\times 3.5^2\times 3$

$= 38.5cm^3$ (B)

2009/48 NABTEB (Nov)

Find the total surface area of a solid cone of diameter 6cm and a slant height 5cm to the nearest whole number (Take $\pi = {}^{22}/_{7}$)

A $362cm^2$ B $350cm^2$ C $180cm^2$ D $75cm^2$

Solution

Total surface area

= area of circular base + curved surface area

$= \pi r^2 + \pi rL$

$= \pi r\,(r + L)$

$= \frac{22}{7}\times\frac{6}{2}\left(\frac{6}{2}+5\right)$

$= \frac{22}{7}\times 3\times 8$

$= 75.43cm^2 \approx 75cm^2$ to the nearest whole number

2014/47 NABTEB

Two similar cones have radii 3cm and 5cm. Find the ratio of their volumes.

A 2:1 B 3:5 C 9:25 D 27:125

Solution

Cone A	:	Cone B
$\frac{1}{3}\pi r^2 h$	:	$\frac{1}{3}\pi r^2 h$
r^2	:	r^2
3^2	:	5^2
9	:	25 (C)

2008/7b NABTEB (Nov)

A vertical cone has base diameter 16cm and slant height 17cm. Find the

(i) curved surface area

(ii) volume of the cone

Solution

(i) Curved surface cone = πrL

$= \frac{22}{7}\times\frac{16}{2}\times 17$

$= 427.43cm^2$

(ii) Volume of the cone = $\frac{1}{3}\pi r^2 h$

But $L^2 = h^2 + r^2$

h is not given: $L^2 = h^2 + r^2$

$17^2 = h^2 + 8^2$

$17^2 - 8^2 = h^2$

$225 = h^2$

$15cm = h$

Thus Volume of cone = $\frac{1}{3}\times\frac{22}{7}\times 8^2\times 15 = 1005.71cm^3$

2008/32 Neco

The ratio of the base area of a hollow cone to that of its curved surface is 1: 4. If its base radius is 7cm, calculate the slant height of the cone

A 7cm B 22cm C 28cm

D 49cm E 154cm

Solution

Base area of a cone is circular, πr^2

Base area	:	curved surface area
πr^2	:	πrL
1	:	4

Canceling out common items

r : L

Radius(r) 7cm = 4 × 7cm (L)

= 28cm (L) C .

2011/4b Neco

The base diameter of a cone is 18cm and height is 24cm. Find the volume and the curved surface area of the cone.(Correct to 2 decimal places)

Solution

Volume of cone = $\frac{1}{3}\pi r^2 h$

Radius is diameter divided by 2

$= \frac{1}{3} \times \frac{22}{7} \times 9^2 \times 24$

$= 2036.57cm^3$ to 2d.p

Curved surface area of cone = πrL

But $L^2 = h^2 + r^2$

$= 24^2 + 9^2$

$= 576 + 81$

$L^2 = 657$

$L = \sqrt{657} = 25.63cm$

Curved surface area = $\frac{22}{7} \times 9 \times 25.63$

$= 724.96cm^2$ to 2dp

2012/34 Neco

Find the radius of a cone with height 7cm and volume of $550cm^3$

A $3\sqrt{5}$ B $3\sqrt{2}$ C $5\sqrt{2}$ D $5\sqrt{3}$ E $5\sqrt{5}$

Solution

Volume of cone $= \frac{1}{3}\pi r^2 h$

$550 = \frac{1}{3} \times \frac{22}{7} \times r^2 \times 7$

$\frac{550 \times 3 \times 7}{22 \times 7} = r^2$

$75 = r^2$

$r = \sqrt{75}$

$= \sqrt{25} \times \sqrt{3}$

$= 5\sqrt{3}$ cm (D)

2012/30 Neco

The curved surface area of a cone of base radius 7cm is $550cm^2$. Find the vertical height of the cone. (Take $\pi = {}^{22}/_7$)

A 78.60cm B 25.96cm C 25.00cm
D 24.00cm E 10.70cm

Solution

$L^2 = h^2 + r^2$

h = ?, r = 7cm, L can be gotten from curved surface area.

Curved surface area of cone $= \pi rL$

$550 = \frac{22}{7} \times 7 \times L$

$\frac{550 \times 7}{22 \times 7} = L$

$25cm = L$

Thus $L^2 = h^2 + r^2$ becomes

$25^2 = h^2 + 7^2$

$25^2 - 49 = h^2$

$625 - 49 = h^2$

$576 = h^2$

$\sqrt{576} = h$

$24cm = h$ (D)

2013/33 Neco Exercise 21.13

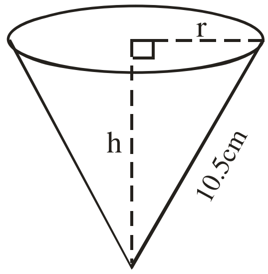
A cone has a base radius 7cm and height 9cm. Calculate its volume.

A $462cm^3$ B $624cm^3$ C $928cm^3$ D $987cm^3$ E $1,254cm^3$

2005/9b NABTEB (Nov) Exercise 21.14

A cone of height 12cm has base radius 5cm. Calculate the
(i) slant height,
(ii) curved surface area and (iii) volume (Take $\pi = {}^{22}/_7$)

2010/4 Exercise 21.14b



The diagram shows a cone with slant height 10.5cm. If the curved surface area of the cone is $115.5cm^2$, Calculate, correct to 3 significant figures, the:
(a) base radius r; (b) height, h; (c) volume of the cone (Take $\pi = {}^{22}/_7$)

2014/25 NABTEB Exercise 21.15

Calculate the volume of a cone base radius 3cm and slant height 5cm.

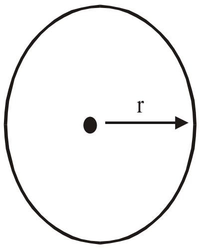
A $3\pi cm^3$ B $9\pi cm^3$ C $12\pi cm^3$ D $24\pi cm^3$

2013/47 Exercise 21.16

The slant height of a cone is 5cm and the radius of its base is 3cm. Find, correct to the nearest whole number, the volume of the cone (Take $\pi = {}^{22}/_7$)

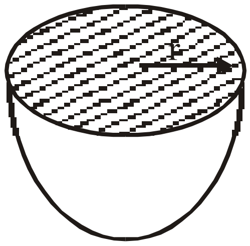
A $48cm^3$ B $47cm^3$ C $38cm^3$ D $13cm^3$

SPHERE



(I) Surface Area of Sphere = $4\pi r^2$
(II) Volume of sphere = ${}^4/_3 \pi r^3$

HEMISPHERE



i) Curved surface area = ½ $(4\pi r^2)$
(Half surface area of sphere) = $2\pi r^2$

ii) Total surface area

$= \text{Curved surface} + \text{circular surface area}$

$= 2\pi r^2 + \pi r^2$

$= 3\pi r^2$

iii) Volume of hemisphere $= \frac{1}{2}\left(\frac{4}{3}\pi r^3\right)$

$= \frac{2}{3}\pi r^3$ (half the volume of sphere)

2014/24 NABTEB

Calculate the total surface area of a hemisphere with radius 3cm

A $9\pi cm^2$ B $18\pi cm^2$ C $27\pi cm^2$ D $45\pi cm^2$

Solution

Total surface area of hemisphere

$= \text{curved surface area} + \text{circular surface area}$

$= 2\pi r^2 + \pi r^2$

$= 3\pi r^2$

$= 3 \times 3^2\pi = 27\pi cm^2$ (C)

2006/24 Neco

Find the radius of a sphere, if $^3/_4$ of its volume is $134.75cm^3$ (Take $\pi = {}^{22}/_7$)

A 176.67cm B 134.75cm C 101.06cm

D 42.87cm E 3.50cm

Solution

Volume of sphere $= \frac{4}{3}\pi r^3$

$\frac{3}{4}$ of $\frac{4}{3}\pi r^3 = 134.75$ (*find r*)

$\pi r^3 = 134.75$

$r^3 = 134.75 \times \frac{1}{\pi}$

$= 134.75 \times \frac{7}{22} = 42.875cm$

$r = \sqrt[3]{42.875} = 3.50cm$ (E)

2005/24 NABTEB

What is the capacity of a spherical tank whose diameter is 1.5m?

A $\frac{3}{4}\pi m^3$ B $\frac{9}{16}\pi m^3$ C $3\pi m^3$ D $4\frac{1}{2}\pi m^3$

Solution

Volume of sphere $= \frac{4}{3}\pi r^3$

$= \frac{4}{3}\times\pi\times\left(\frac{1.5}{2}\right)^3$

$= \frac{4}{3}\times\pi\times\left(\frac{3}{2}\times\frac{1}{2}\right)^3$

$= \frac{4}{3}\times\pi\times\left(\frac{3}{4}\right)^3 = \frac{9}{16}\pi\, m^3$ (B)

2006/7 Neco

Find the surface area of a sphere with radius 5cm (Take $\pi = {}^{22}/_7$)

A $78.6cm^2$ B $104.8cm^2$ C $220.0cm^2$

D $314.3cm^2$ E $550.0cm^2$

Solution

Surface area of sphere $= 4\pi r^2$

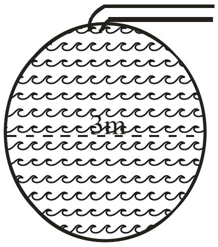
$= 4 \times \frac{22}{7} \times 5^2$

$= 314.29cm^2 \approx 314.3cm^2$ (D)

2014/7a (Nov)

A spherical tank of diameter 3m is filled with water from a pipe of radius 30cm at 0.2m per second. Calculate, correct to 3 significant figures, the time, in minutes, it takes to fill the tank (Take $\pi = {}^{22}/_7$)

Solution



Volume of spherical tank $= \frac{4}{3}\pi r^3$

$= \frac{4}{3}\times\frac{22}{7}\times\left(\frac{3}{2}\right)^3 = 14.14m^3$

Volume of water per second $= \pi r^2 h$

$= \frac{22}{7} \times (0.3)^2 \times 0.2$

$= 0.05657m^3/s$

Time taken in sec $= \frac{V\tan k}{V/\sec\ of\ \ pipe}$

$= \frac{14.14}{0.05657} = 249.96$ sec

Time taken in minutes $= \frac{249.96}{60} = 4.166$ minutes

≈ 4.17min to 3s.f

2015/8

A tap is leaking at the rate of $2cm^3$ per second into an empty container with capacity 45litres. How long will it take to fill the container?

A 8 hrs B 6 hrs 15 mins C 4 hrs 25 mins D 3 hrs

Solution

Volume of container in $cm^3 = 45 \times 1000$

$= 45000cm^3$

Time taken in sec $= \frac{Vol\ of\ container}{Rate\ (Vol)\ of\ leaking}$

$= \frac{45000}{2}$

$= 22500$ sec.

$= \frac{225000}{60}$ minutes $= 375$ minutes

$= \frac{375}{60}$ hours 6 hrs 15 mins (B)

2014/60 Neco Exercise 21.17

What will be the volume of an hemisphere of diameter 21cm?

A $231.0\ cm^3$ B $808.5\ cm^3$ C $1617.0\ cm^3$

D $2425.5\ cm^3$ E $7276.5\ cm^3$

2007/42 UME Exercise 21.18

The volume of a hemispherical bowl is $718^2/_3\ cm^3$. Find its radius ($\pi = {}^{22}/_7$)

A 4.0cm B 5.6cm C 7.0cm D 3.8cm

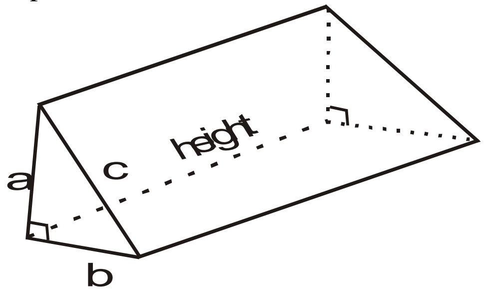
2009/29 UME Exercise 21.19

Find the radius of a sphere whose surface area is $154cm^2$ (Take $\pi = {}^{22}/_7$)

A 7.00cm B 3.50cm C 3.00cm D 1.75cm

TRIANGLE PRISM

A prism is a solid with uniform cross sectional area.

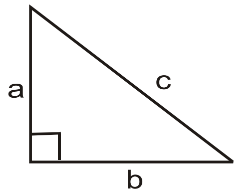


Volume = base area x height = $^1/_2$ ab ×h

Relationship between a, b and c is that of Pythagoras' rule

$c^2 = a^2 + b^2$

Cross sectional area = $^1/_2$ a b



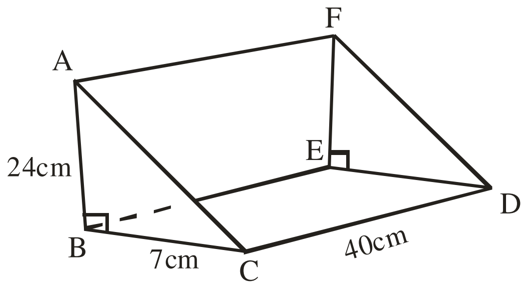
Note

C is not used; since the **area of a triangle**

= $^1/_2$ base × height (perpendicular height)

= $^1/_2$ a × b

2009/5



In the diagram, ABCDEF is a triangular prism. $\angle ABC = \angle DEF = 90^0$, $|AB| = 24$cm, $|BC| = 7$cm and $|CD| = 40$cm. Calculate :

(a) $|AC|$; (b) The total surface area of the prism.

Solution

(a) By Pythagoras rule in ΔABC

$AC^2 = 24^2 + 7^2$

$AC^2 = 576 + 49$

$AC = \sqrt{625}$

= 25cm

(b) Total surface area of prism

= Area of ΔABC + Area of ΔDEF + area of rectangle ACDF + Area of rect. BCDE + Area of rect. ABEF

Area of ΔABC = $^1/_2$base × height (perpendicular height)

Area of rectangle = length × breadth

Total surface area of prism = $\frac{1}{2} \times 7 \times 24 + \frac{1}{2} \times 7 \times 24 + 40 \times 25 + 40 \times 7 + 40 \times 24$

= 84 + 84 + 1000 + 280 + 960

= 2408cm^2

2014/40 (Nov)

The height of a triangular prism is 6cm. If the cross section of the prism is an equilateral triangle of side 8cm, find its volume

A $96\sqrt{3}$ cm^3 B $64\sqrt{3}$ cm^3 C $32\sqrt{3}$ cm^3 D $16\sqrt{3}$ cm^3

Solution

Volume of prism = base area × height

= cross sectional area × height

Cross sectional area = area of equilateral triangle 8cm

$= \sqrt{s(s-a)(s-b)(s-c)}$

Where s = $^1/_2$(a + b + c)

S = $^1/_2$(8 + 8 + 8)

= $^1/_2$(24) = 12

Cross sectional area = $\sqrt{12(12-8)(12-8)(12-8)}$

$= \sqrt{12(4)(4)(4)}$

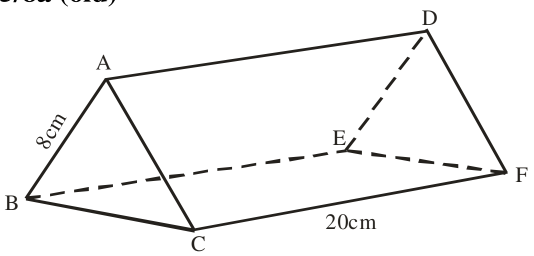
$= \sqrt{3 \times 4 \times 4 \times 4 \times 4}$

$= 16\sqrt{3}$ cm^3

Thus volume of triangular prism = $16\sqrt{3} \times 6$

$= 96\sqrt{3}$ cm^3 (A)

2005/8a (old)



The diagram shows a solid prism whose cross – section is an equilateral triangle of side 8cm. If the length of the prism is 20cm, Calculate its total surface area correct to three significant figures

Solution

Total surface area = area of ΔABC + area ΔDEF + Area of rect. ADCF + Area of rect.BECF + Area of rect.ABED

Since the two triangles are equal, and the rectangles are equal

Total surface area = 2(Area of Δ ABC) + 3(Area of rect.ADCF)

Area of triangle with all sides given = $\sqrt{s(s-a)(s-b)(s-c)}$

Where s = $^1/_2$(a + b + c)

s = $^1/_2$(8 + 8 + 8)

s = $^1/_2$(24) = 12

Area of Δ ABC = $\sqrt{12(12-8)(12-8)(12-8)}$

$= \sqrt{12(4)(4)(4)}$

$= \sqrt{768} = 27.71$cm^2

Area of rectangle ADCF = 20 × 8 = 160cm^2

∴ Total surface area = 2 (27.71) + 3(160)

= 55.42 + 480

= 535.42cm^2

2007/33 Neco

The cross – section of a prism is a right – angled triangle 13cm by 12cm by 5cm, calculate its volume if its height is 0.025m

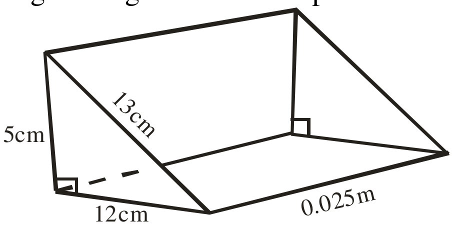
A $0.75cm^3$ B $1.95cm^3$ C $75cm^3$
D $195cm^3$ E $200cm^3$

Solution

Volume of prism = base area × height

$= \frac{1}{2}$ base × Δ height × height

By simple knowledge of Pythagoras triangle, we draw the right – angled side of the prism as



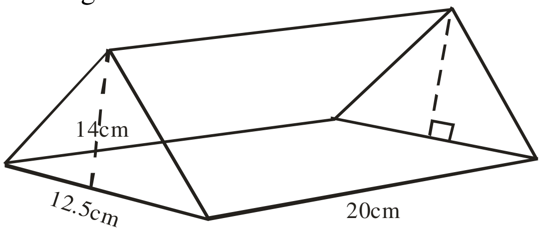
$= \frac{1}{2} \times 12 \times 5 \times (0.025 \times 100)$

Note 100cm is one metre

$= 75cm^3$ C.

2015/15 (Nov)

The dimensions of a triangular prism are as shown in the diagram. Find its volume



A $1750cm^3$ B $1770cm^3$ C $1870cm^3$ D $1970cm^3$

Solution

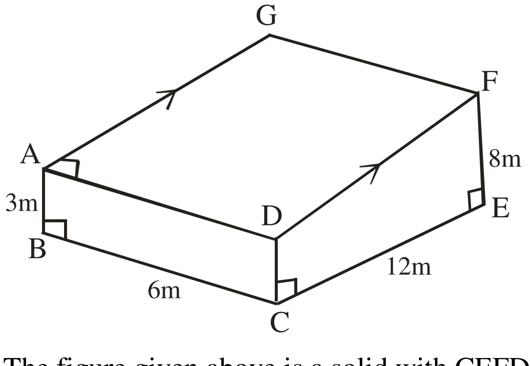
Volume of prism = base area × height

$= \frac{1}{2}$ base × Δ height × height

$= \frac{1}{2} \times 12.5 \times 14 \times 20$

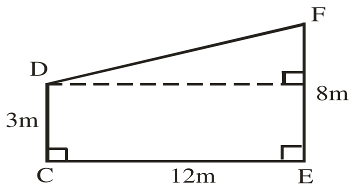
$= 1750cm^3$ (A)

2007/9b NABTEB



The figure given above is a solid with CEFD as the cross section, calculate the:
(i) area of CEFD and (ii) volume of the solid.

Solution



Area of CEFD = Area of trapezium

$= \frac{1}{2}(3 + 8) \times 12$

$= 66m^2$

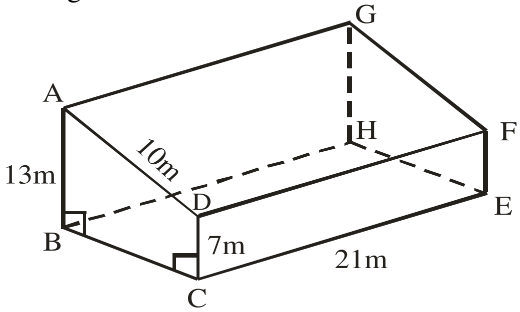
Volume of block = base area × height

$= \frac{1}{2}(3 + 8) \times 12 \times 6$

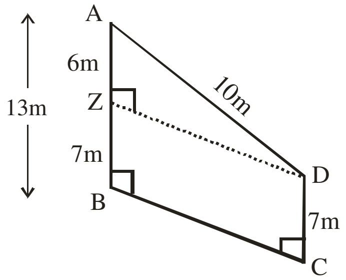
$= 396m^3$

Example P6

The dimensions of a prism with surface area ABCD is given in diagram below. Find its volume



Solution



Volume of prism = base area × height

= Area of trapezium ABCD × height

$= \frac{1}{2}(13 + 7) \times |ZD| \times 21$

By Pythagoras rule : $ZD^2 = 10^2 - 6^2$

$ZD = \sqrt{64} = 8m$

Volume of prism $= \frac{1}{2}(13 + 7) \times 8 \times 21$

$= 1680m^3$

2011/15 Exercise 21.20

The cross section of a uniform prism is a right – angled triangle with sides 3cm, 4cm and 5cm. If its length is 10cm, calculate the total surface area

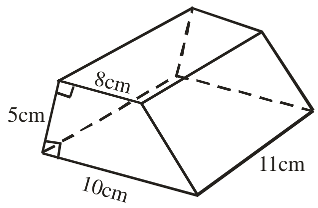
A $142cm^2$ B $132cm^2$ C $122cm^2$ D $112cm^2$

2008/31 NABTEB (Nov) Exercise 21.21

Calculate the volume of a triangular based prism with base area $24cm^2$ and height 10cm

A $12cm^3$ B $120cm^3$ C $240cm^3$ D $360cm^3$

1997/30 UME Exercise 21.22

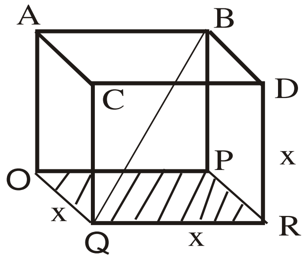


Find the volume of the prism above

A $990cm^3$ B $880cm^3$ C $550cm^3$ D $495cm^3$

Cube

A cube is a solid with six faces of equal dimension



Surface area (s) = 6 × area of one face

$= 6x^2$

x is the length of the side

Volume = base area × height

$= x^2 \times x$

$= x^3$

Length of diagonal QB $= \sqrt{x^2 + x^2 + x^2}$

$= \sqrt{3x^2}$

$= x\sqrt{3}$

Length of side diagonal $= \sqrt{x^2 + x^2}$

$= \sqrt{2x^2}$

$= x\sqrt{2}$

2009/41 Neco (Dec)

Find the volume of a cube of side 12cm

A $36cm^3$ B $144cm^3$ C $864cm^3$

D $1082cm^3$ E $1728cm^3$

Solution

Volume of cube $= x^3$

$= 12^3$

$= 1728cm^3$ (E)

2006/4 Neco

Calculate the cost of parting the surface area of a solid cube whose edge is 8m at ₦ 5.00 per m^2

A ₦ 960.00 B ₦ 1,280.00 C ₦ 1,920.00

D ₦ 2,560.00 E ₦ 3,200.00

Solution

Total surface area of cube $= 6x^2$

$= 6 \times 8^2$

$= 384^2$

Cost of painting at ₦ 5.00 per $m^2 = 384 \times 5$

= ₦ 1, 920.00 (C)

2010/48 NABTEB (Nov)

The volumes of two similar solid cubes are 729 cm^3 and $512cm^3$. Find the ratio of their lengths

A 4 : 3 B 9 : 7 C 8 : 7 D 9 : 8

Solution

Volume of cube $= x^3$

$\sqrt[3]{Cube\ volume}$ = x (length)

Ratio of lengths

$\sqrt[3]{729} : \sqrt[3]{512}$

9 : 8 (D)

2008/36

What is the length of a cube whose total surface area is Xcm^2 and whose volume is $\frac{x}{2}$ cm^3

A 3 B 6 C 9 D 12

Solution

Total surface area of cube $= 6x^2$

$X = 6x^2$

Let X be y thus $\mathbf{y = 6x^2}$

Volume of cube $= x^3$

$\frac{x}{2} = x^3$

Let X be y thus $\frac{y}{2} = \mathbf{x^3}$

Now we have $y = 6x^2$ ----- (1)

$\frac{y}{2} = x^3$ ------ (2)

From (2) $y = 2x^3$ ------ (4)

Equating y in (1) and (4)

$6x^2 = 2x^3$

Divide through by x^2

$\frac{6x^2}{x^2} = \frac{2x^3}{x^2}$

$6 = 2x$

$\frac{6}{2} = x$

$3 = x$ (A)

2009/18 (Nov)

Find the diagonal of a cube whose edge is 5cm long

A 8.66cm B 9.08cm C 10.00cm D 11.18cm

Solution

Length of diagonal of a cube $= x\sqrt{3}$

$= 5\sqrt{3}$ cm

= 8.66cm (A)

2008/18 Exercise 21.23

If the volume of a cube is $343cm^3$, find the length of its side

A 3cm B 6cm C 7cm D 8cm

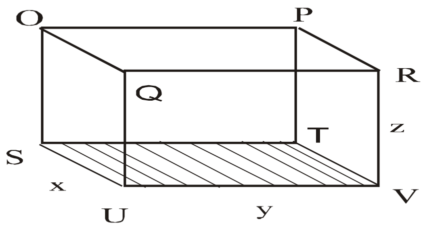
2005/6 (Old) Exercise 21.24

What is the total surface area of a cube of side 4 cm?

A $36cm^2$ B $64cm^2$ C $96cm^2$ D $144cm^2$

CUBOID

A cuboid is a solid shape with a rectangular base and sides.



It consists of 6 rectangles;. of which 2 opposite ones are the same in area. Thus ,

Surface area = 6 rectangles

$= 2xy + 2xz + 2yz$

Volume = Base area × height

$= xy \times z$

$= xyz$

Alternatively,

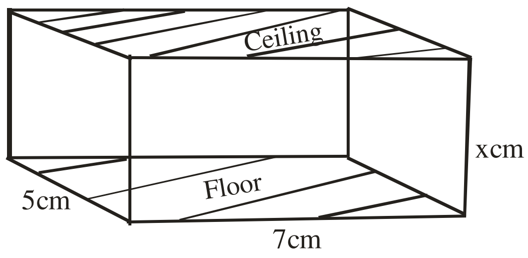
Volume = Length × breadth × Height

2005/39

The total surface area of the walls of a room, 7m long, 5m wide and xm high is $96m^2$. Find the value of x

A $^2/_3$ B 2 C 4 D 8

Solution



Total surface area = 4 rectangle of the rooms walls

$96 = 2(5 \times x) + 2(7 \times x)$

$96 = 10x + 14x$

$\frac{96}{24} = \frac{24x}{24}$

$4 = x$ (C)

2010/31 Neco

A square based box with a height of 8cm has a volume of $2048cm^3$. Find the length of the side of the base

A 4cm B 8cm C 16cm

D 64cm E 256cm

Solution

Volume of cuboid box = base area × height

= square base area × height

$2048 = x^2 \times 8$

$\frac{2048}{8} = x^2$

$256 = x^2$

$\sqrt{256} = x$

$16cm = x$ (C)

2012/48

The volume of a cuboid is $54cm^3$. If the length, width and height of the cuboid is in the ratio 2 : 1: 1 respectively, find its total surface area

A $108cm^2$ B $90cm^2$ C $80cm^2$ D $75cm^2$

Solution

Volume of cuboid = xyz

Applying the given ratio values

$54 = (2x) \times x \times x$

$54 = 2x^3$

Divide through by 2

$27 = x^3$

$\sqrt[3]{27} = x$

$3cm = x$

Thus the cuboid has length, width and height

$2 \times 3cm$, 3cm and 3cm

Total surface area of cuboid $= 2 \times 6 \times 3 + 2 \times 6 \times 3 + 2 \times 3 \times 3$

$= 36 + 36 + 18$

$= 90cm^2$ (B)

2009/33

The capacity of a water tank is 1,800litres. If the tank is in the form of a cuboid with base 600cm by 150cm, find the height of the tank.

A 2cm B 20cm C 200cm D 2000cm

Solution

First we convert capacity in litres to volume in cm^3

1,800litres = $1800 \times 1000cm^3$

= 1800, 000cm3

Volume of cuboid = base × height

$1800,000 = 600 \times 150 \times$ height

$\frac{1800000}{600 \times 150} =$ height

20cm = height (B)

2005/35 (Nov)

A closed box of matches is in the form of a cuboid with dimension 5.5cm by 3.5cm by 1.5cm. Find, correct to the nearest whole number, the total surface area of the box

A $66cm^2$ B $33cm^2$ C $29cm^2$ D $21cm^2$

Solution

Total surface area of cuboid = 2xy + 2xz + 2yz

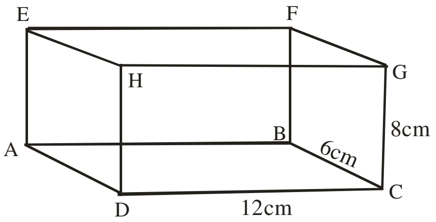
$= 2 \times 5.5 \times 3.5 + 2 \times 5.5 \times 1.5 + 2 \times 3.5 \times 1.5$

$= 38.5 + 16.5 + 10.5$

$= 65.5cm^2$

$\approx 66cm^2$ (A)

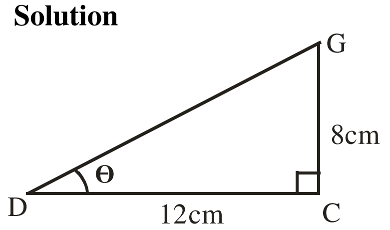
2008/38 Neco



In the above rectangular box, |DC| = 12cm, |BC| = 6cm and |FB| = 8cm. Calculate the angle between the planes AFGD and ABCD

A 56.3^0 B 41.8^0 C 36.9^0 D 33.7^0 E 26.6^0

Solution



Angle between two planes

$$\tan\theta = \frac{8}{12},$$

$$\theta = \tan^{-1}0.6667 = 33.7^0 \ (D)$$

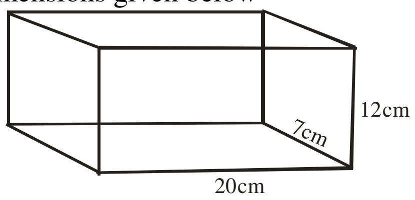
2005/21 NABTEB (Nov) Exercise 21.25

Find the total surface area of a cuboid
6cm by 5cm by 2cm

A $52cm^2$ B $60cm^2$ C $104cm^2$ D $120cm^2$

Exercise 21.26

Find the capacity in litres of the water tank with dimensions given below



2016/4 Neco Exercise 21.26b

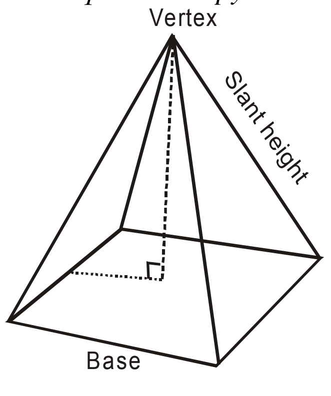
30,000 litres of fuel is transferred to a rectangular tank with dimensions $(7.5 \times 4.2 \times 1.2)$m.

a. Calculate, correct to 2d.p, the depth of fuel in the tank
b. How many more litres of fuel would be needed to fill the tank

Pyramid on a square base

Pyramid is a solid shape with triangular faces meeting at a point (vertex).The shape lies on a base, which may be square, triangle, rectangle e t c

Square base pyramid



Volume = $^1/_3$ area of base × height

$= {}^1/_3 \ L^2 \times h$

Since, the area of a square $= L \times L$

$= L^2$

2013/32 Neco

Find the volume of a right pyramid of square base 6cm and 42cm height

A $324cm^3$ B $385cm^3$ C $504cm^3$ D $572cm^3$ E $678cm^3$

Solution

Volume of pyramid $= \frac{1}{3}$ base area × height

Base area here is square(length^2)

$$= \frac{1}{3} \times 6^2 \times 42$$

$$= 504cm^3$$

2014/59 Neco

Find the perpendicular height of a right pyramid whose volume is $21cm^3$ and its base is a square of side 3cm

A 3cm B 4cm C 5cm D 6cm E 7cm

Solution

Volume of pyramid $= \frac{1}{3}$ base area × height

Base is square (length^2)

$$21 = \frac{1}{3} \times 3^2 \times \text{height}$$

$$21 = 3 \times \text{height}$$

$$\frac{21}{3} = \frac{3 \times height}{3}$$

$$7cm = \text{height} \ (E)$$

2010/38 NABTEB (Nov)

Find the area of the base of a square based pyramid whose volume and height are $2700cm^3$ and 9cm

A $900cm^2$ B $270cm^2$ C $210cm^2$ D $90cm^2$

Solution

Volume of pyramid $= \frac{1}{3}$ base area × height

Base area here is square (L^2)

$$2700 = \frac{1}{3} \times L^2 \times 9$$

$$2700 = 3L^2$$

$$\frac{2700}{3} = L^2$$

$900cm^2 = L^2$ (Base area) A.

2013/46

A pyramid has a rectangular base with dimensions 12m by 8m. If its height is 14m, calculate the volume

A $344m^3$ B $448m^3$ C $632m^3$ D $840m^3$

Solution

Volume of pyramid $= \frac{1}{3}$ base area × height

Base area here is rectangle (length × breadth)

$$= \frac{1}{3} \times 12 \times 8 \times 14$$

$$= 448m^3 \ (B)$$

2008/5b

The base of a pyramid is a 4.5m by 2.5m rectangle. The height of the pyramid is 4m. Calculate its volume

Solution

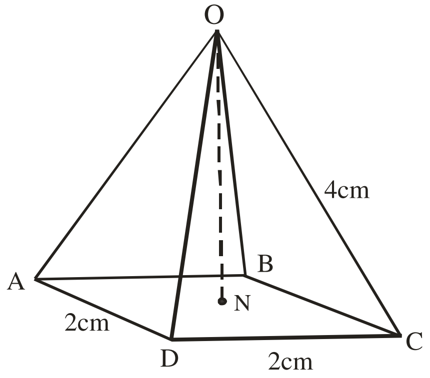
Volume of pyramid $= \frac{1}{3}$ base area × height

Base area here is rectangle (length × breadth)

$= \frac{1}{3} \times 4.5 \times 2.5 \times 4$

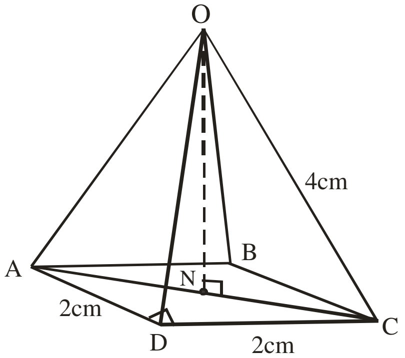
$= 15m^3$

2005/12b



In the diagram, OABCD is a pyramid with a square bases of side 2cm a slant height of 4cm. Calculate, correct to three significant figures;
(i) the vertical height of the pyramid
(ii) the volume of the pyramid

Solution



(i) To find ON, we need to know NC from AC

In ΔADC, $AC^2 = 2^2 + 2^2$ (Pythagoras rule)

$AC = \sqrt{8}$

Thus, $NC = \frac{1}{2}\sqrt{8}$

In ΔONC, $OC^2 = NC^2 + ON^2$

$4^2 = \left(\frac{\sqrt{8}}{2}\right)^2 + ON^2$

$16 = \frac{8}{4} + ON^2$

$16 - 2 = ON^2$

$14 = ON^2$

$ON = \sqrt{14}$ cm vertical height of the pyramid

≈ 3.74cm to 3s.f

(ii) Volume of pyramid $= \frac{1}{3}$ base area $\times$ height

Base area here is square

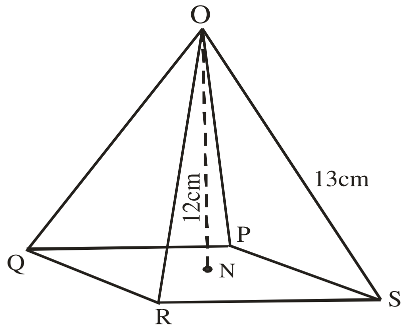
$= \frac{1}{3} \times 2^2 \times \sqrt{14}$

$= 4.989cm^3$

$\approx 4.99cm^3$ to 3 sf

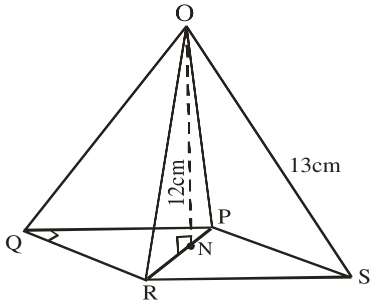
2005/64 (old)

In the diagram, OPQRS is a right pyramid on a square base OS = 13cm and ON = 12cm. Calculate the volume of the pyramid



A $100cm^3$ B $200cm^3$ C $310cm^3$ D $600cm^3$

Solution



Volume of pyramid $= \frac{1}{3}$ base area $\times$ height

But first, we find side QP

In ΔONR, $13^2 = 12^2 + NR^2$

$169 = 144 + NR^2$

$169 - 144 = NR^2$

$25 = NR^2$

$NR = 5cm$

In ΔQPR, $PR = 2 \times NR$ and $QP = QR$(Both lines are equal)

$PR^2 = QP^2 + QR^2$

$10^2 = QP^2 + QP^2$

$100 = 2QP^2$

$50 = QP^2$

$\sqrt{50}$ cm = QP

Base area here is square

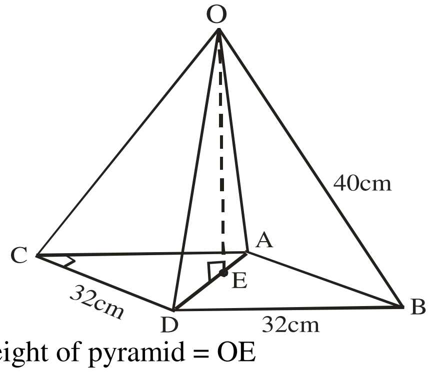
Volume of pyramid $= \frac{1}{3} \times \sqrt{50} \times \sqrt{50} \times 12$

$= {}^1/_3 \times 50 \times 12 = 200cm^3$ (B)

2013/9 Neco

The base ABCD of a right pyramid with vertex V, is a square of side 32cm and the length of a slant edge is 40cm. Calculate, correct to 1 decimal place the:
(a) height of the pyramid and
(b) total area of its surfaces.

Solution



Height of pyramid = OE

To find OE, we need ED which is half of AD

In right – angled ΔACD : $AD^2 = 32^2 + 32^2$

$AD^2 = 2048$

$$AD = \sqrt{2048} = 45.25\text{cm}$$

Thus, $ED = \frac{1}{2}(45.25) = 22.63\text{cm}$

In right – angled ΔOED : $OD^2 = ED^2 + OE^2$

$$40^2 = (22.63)^2 + OE^2$$

$$1600 - 512.12 = OE^2$$

$$\sqrt{1087.88} = OE$$

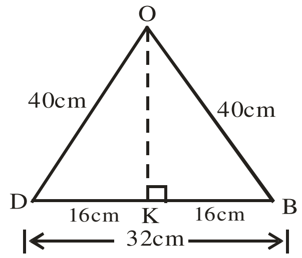
$$OE = 32.98\text{cm} \approx 33.0\text{cm to 1d.p}$$

(b) Total surface area

= square base area + Area of 4 triangles

It is so since the base is square and all the sides are equal

Area of $\Delta ODB = \frac{1}{2}\text{base} \times \text{height}$



$$OD^2 = DK^2 + OK^2$$

$$40^2 = 16^2 + OK^2$$

$$1600 - 256 = OK^2$$

$\sqrt{1344} = OK$ thus, $36.66\text{cm} = OK$

Total surface area $= 32^2 + 4(^1/_2 \times 32 \times 36.66)$

$= 1024 + 2346.24 = 3370.24\text{cm}^2$

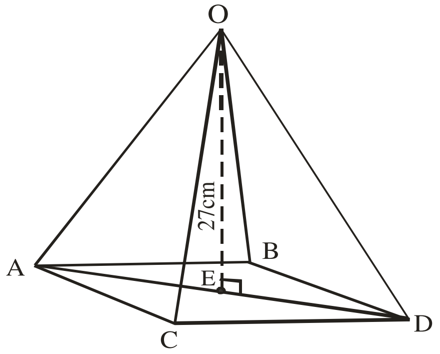
2015/8b (Nov)

A silo, in the form of a pyramid on a square base, has height 27cm and volume 2601cm^3. Calculate the:

(i) length of a side of the base of the silo, correct to 3 significant figures.

(ii) angle between a sloping face and the base of the silo,
correct to the nearest degree

Solution



(i) Volume of pyramid $= \frac{1}{3}\text{base area} \times \text{height}$

Base area is square (L^2)

$$2601 = \frac{1}{3} \times L^2 \times 27$$

$$2601 = 9L^2$$

$$\frac{2601}{9} = L^2$$

$\sqrt{289} = L$ Thus $17\text{cm} = L$

(ii) $\angle ODA = \angle ODE$

First, we find $ED = \frac{1}{2}AD$

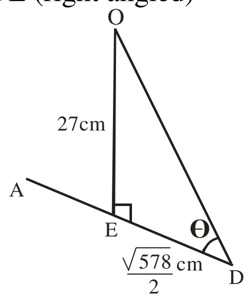
In ΔACD (right angled) : $AD^2 = AC^2 + CD^2$

$$AD^2 = 17^2 + 17^2$$

$$AD = \sqrt{578}\text{ cm}$$

Thus, $ED = \frac{1}{2}\sqrt{578}$

In ODE (right angled)



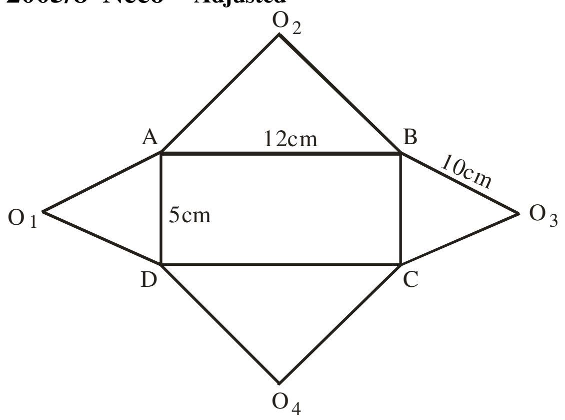
$$\tan\theta = 27 \div \frac{\sqrt{578}}{2}$$

$$\tan\theta = 27 \div 12.02$$

$$\tan\theta = 2.246$$

$$\theta = \tan^{-1} 2.246 = 66.0^0$$

2005/8 Neco Adjusted



The diagram above shows the net of a right rectangular pyramid in which ABCD is base and O_1, O_2, O_3, O_4 are brought together to form the vertex O. when the solid is formed, ABCD is the base $|AB| = 12\text{cm}$, $|AD| = 5\text{cm}$ and $|BO_3| = 10\text{cm}$

(a) Draw a sketch of the pyramid

(b) Calculate the:

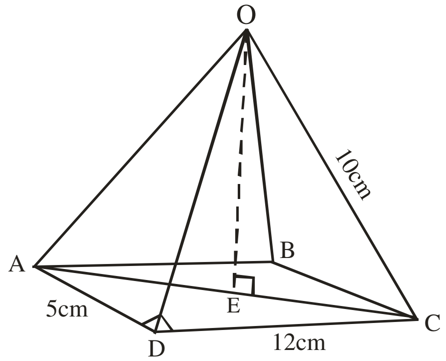
(i) height of the pyramid

(ii) volume of the pyramid

(iii) area of triangle AOB

(iv) inclination of triangle BOC to the base ABCD

Solution



(b) i. Height OE can be gotten from EC and EC from AC

In right – angled ΔADC:

$AC^2 = 12^2 + 5^2$

$AC = \sqrt{169} = 13cm$

$\therefore EC = \frac{13}{2}cm = 6.5cm$

In right – angled ΔOEC: $OC^2 = OE^2 + EC^2$

$10^2 = OE^2 + 6.5^2$

$100 - 42.25 = OE^2$

$\sqrt{57.75} = OE$

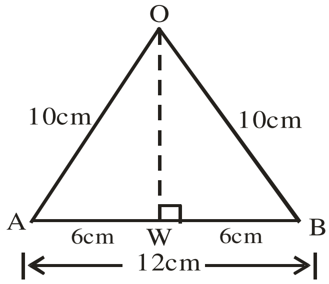
$7.6cm = OE$

(ii) Volume of pyramid = $\frac{1}{3}$ base area × height

Base area here is rectangle (length × breadth)

$= \frac{1}{3} \times 12 \times 5 \times 7.6 = 152cm^3$

(iii)



Area of $\Delta AOB = \frac{1}{2}$ base × height

But height OW can be gotten from

right – angled ΔOWB : $OB^2 = OW^2 + WB^2$

$10^2 = OW^2 + 6^2$

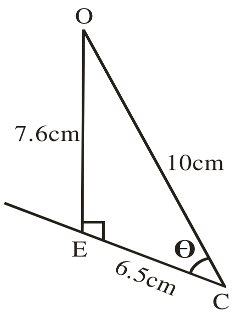
$100 - 36 = OW^2$

$\sqrt{64} = OW$

$8cm = OW$

$\therefore$ Area of $\Delta AOB = \frac{1}{2} \times 12 \times 8 = 48cm^2$

(iv)



From the sketch; $\tan \theta = \frac{7.6}{6.5}$

$\tan \theta = 1.169$

$\theta = \tan^{-1} 1.169 = 49.46^0$

2014/9b Neco (Dec) Exercise 21.27

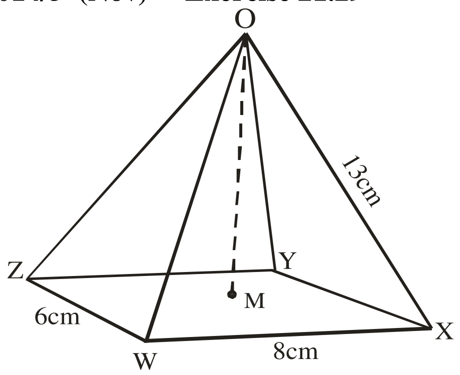
The base of a pyramid is 6.5m by 3.5m rectangle and its height is 7m. Calculate its volume

2009/33 Neco (Dec) Exercise 21.28

A pyramid 7cm high stands on a square base 12cm. Calculate the volume of the pyramid

A $246cm^3$ B $336cm^3$ C $433cm^3$
D $504cm^3$ E $528cm^3$

2014/5 (Nov) Exercise 21.29



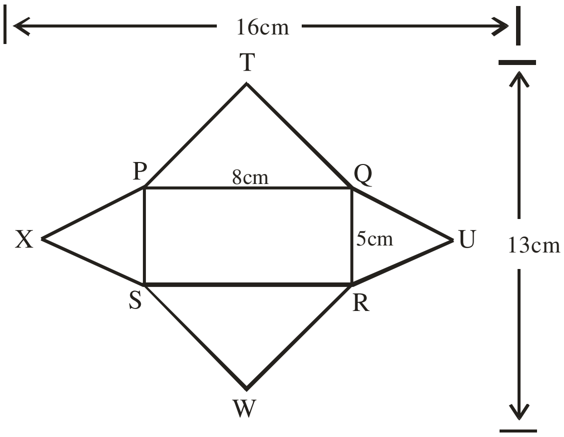
The diagram shows a right pyramid with a rectangular base WXYZ and vertex O. If |WX| = 8cm, |ZW| = 6cm and |OX| = 13cm, calculate the:

(a) height of the pyramid

(b) value of $\angle OXZ$, correct to the nearest degree

(c) volume of the pyramid

2010/41 Exercise 21.30

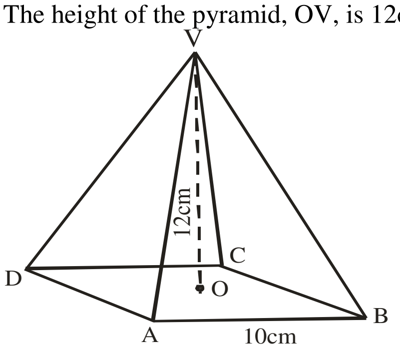


The diagram is a net of a right – rectangular pyramid. Calculate the total surface area

A $208cm^2$ B $112cm^2$ C $92cm^2$ D $76cm^2$

2006/7 NABTEB (Nov) Exercise 21.31

VABCD is a solid pyramid on a square base ABCD has vertex V. The height of the pyramid, OV, is 12cm and the length AB is 10cm



Calculate the: (a) total surface area and (b) volume of the pyramid

2005/14 (Nov) Exercise 21.32

The base of a solid pyramid is a square of side 6cm. If the height of the pyramid is 7cm, calculate the volume of the pyramid

A $84cm^3$ B $126cm^3$ C $198cm^3$ D $252cm^3$

2006/10 Neco (Dec) ***Adjusted*** **Exercise 21.33**

If a rectangular based pyramid VABCD is such that |AB| = 10cm, |BC| = 6cm and the slant height is 15cm, calculate the: (i) vertical height of the pyramid,

(ii) angle face VBC makes with the base and

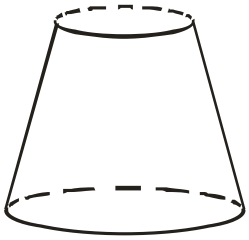
(iii) volume of the pyramid (iv) total surface area

Frustum

This is a shape obtained by cutting off the upper portion of a pyramid or a cone.
E.g A bucket is a frustum of a cone while some flower vase are frustums of a pyramid.

Comparing frustum of cone solving to frustum of pyramid

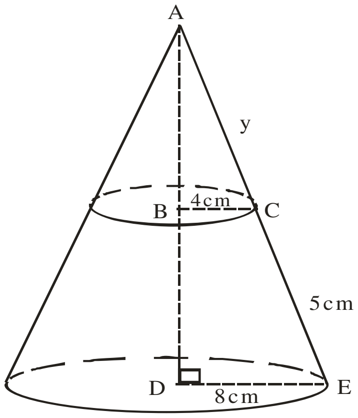
2014/44 NABTEB



The figure is a part of a cone with top radius 4cm and bottom radius 8cm. If the slant height is 5cm, calculate the vertical height

A 3cm B 4cm C 5 cm D 8cm

Solution



Triangles ABC and ADE are similar

$$\frac{BC}{DE} = \frac{AC}{AE}$$

$$\frac{4}{8} = \frac{y}{y+5}$$

$4(y + 5) = 8y$
$4y + 20 = 8y$
$20 = 8y - 4y$
$20 = 4y$ thus $5 = y$

By Pythagoras rule

Vertical height : $AD^2 = AE^2 - DE^2$
$= 10^2 - 8^2$
$= 100 - 64$
$= 36$

$AD = \sqrt{36} = 6cm$

By similar triangles

$$\frac{AB}{4} = \frac{AD}{8}$$

$$AB = \frac{6 \times 4}{8} = 3 \text{ cm}$$

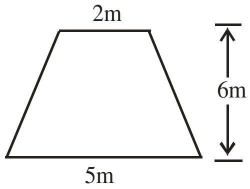
Vertical height = BD
= AD – AB
= 6cm – 3cm = 3cm (A)

2000/24 UME

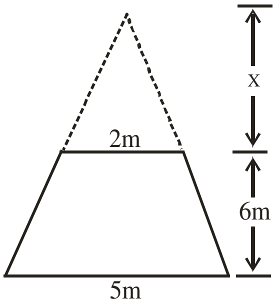
A frustum of a pyramid with square base has its upper and lower sections as squares of sizes 2m and 5m respectively and the distance between them 6m. Find the height of the pyramid from which the frustum was obtained.

A 8.0m B 8.4m C 9.0m D 10.0m

Solution



Problem diagram



Solution diagram

Height of the pyramid is x + 6m

By similar triangles

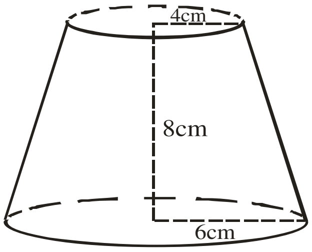
$$\frac{2}{5} = \frac{x}{x+6}$$

$2(x + 6) = 5x$
$2x + 12 = 5x$
$12 = 5x - 2x$
$12 = 3x$
$4m = x$

Thus, height of the pyramid is 4m + 6m = 10m (D)

2005/48 NABTEB (Nov) Exercise 21.34

Find the height of the cone cut off to leave the frustum below

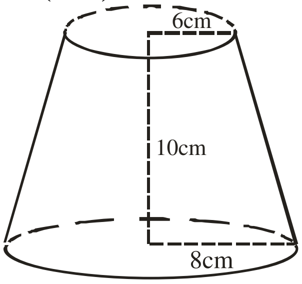


A 8cm B 14cm C 16cm D 18cm

Exercise 21.35

A frustum of a pyramid with square base has its upper and lower sections as squares of sizes 7m and 13m respectively and the distance between them 10m. Find the height of the pyramid from which the frustum was obtained.

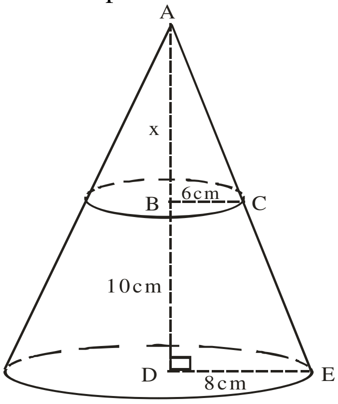
2005/9b (Nov)



The diagram shows a frustum which is part of a solid cone. The radii of the top and bottom are 6cm and 8cm respectively. Calculate the:
(i) height of the whole cone
(ii) volume of the frustum,correct to the nearest whole number

Solution

We draw the complete cone as shown below



Triangles ABC and ADE are similar

$$\frac{AB}{AD} = \frac{BC}{DE}$$

$$\frac{x}{10+x} = \frac{6}{8}$$

$8x = 6(10 + x)$
$8x = 6(10 + x)$
$8x = 60 + 6x$
$8x - 6x = 60$
$2x = 60$
$x = 30cm$

Thus height of whole cone = 10 + x
= 10 + 30 = 40cm

(ii) Volume of a frustum $= \frac{1}{3}\pi h\ (R^2 + Rr + r^2)$

Here h is10, r is 6 and R is 8. Substituting, we have

$$= \frac{1}{3} \times \frac{22}{7} \times 10(8^2 + 8\times 6 + 6^2)$$

$$= \frac{1}{3} \times \frac{22}{7} \times 10(64 + 48 + 36)$$

$$= \frac{1}{3} \times \frac{22}{7} \times 10 \times 148$$

$$= \frac{32560}{21}$$

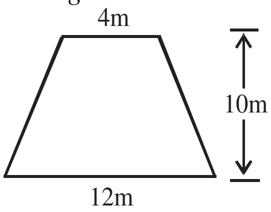
$= 1550.476cm^3$
$\approx 1550\ cm^3$ to the nearest whole number

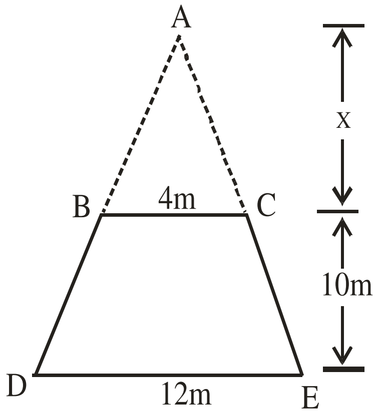
Example FSP2

A frustum of a squared base pyramid is 4m at the top and 12m at the bottom. If its height is 10m, calculate the :
(i) height of the whole pyramid
(ii) volume of the frustum, correct to the nearest whole number

Solution

Problem diagram





Solution diagram

(i) Height of the whole pyramid = x + 10m

By similar triangles

$$\frac{4}{12} = \frac{x}{x+10}$$

$4(x + 10) = 12x$
$4x + 40 = 12x$
$40 = 12x - 4x$
$40 = 8x$
$5m = x$

Thus, height of the whole pyramid is 5m + 10m = 15m

(ii) Volume of a frustum = vol of big – vol of small pyramid
= vol of ADE – vol of ABC

Recall that: volume of pyramid $= \frac{1}{3}$ base area × height

$$= \frac{1}{3} \times 12^2 \times 15 - \frac{1}{3} \times 4^2 \times 5$$

$= 720m^3 - 26.67m^3$
$= 693.33\ m^3$
$\approx 693\ cm^3$ to the nearest whole number

2005/10a NABTEB Exercise 21.36

A bucket is 28cm in diameter at the top, 18cm in diameter at the bottom and 20cm deep. Find the capacity, in litres, of the bucket. (Take $\pi = 3.142$)

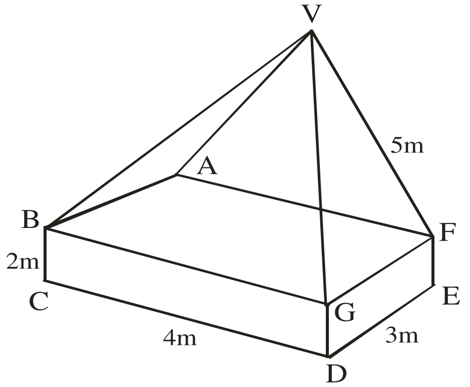
2015/6b Neco Exercise 21.37

A flower vase 8cm high is in the shape of the frustum of a square base pyramid of side 6cm at the bottom and 10cm at the top. What is the volume of water that will fill the vase when it is empty?

IRREGULAR SOLIDS

Irregular solids do not have any particular formula used; we apply the relevant solid formula as it affects a given shape. The use of common sense in interpreting diagrams is crucial so as to know the applicable formula

2006/11a



The diagram shows a pyramid standing on a cuboid. The dimensions of the cuboid are 4m by 3m by 2m and the slant edge of the pyramid is 5m. Calculate the volume of the shape

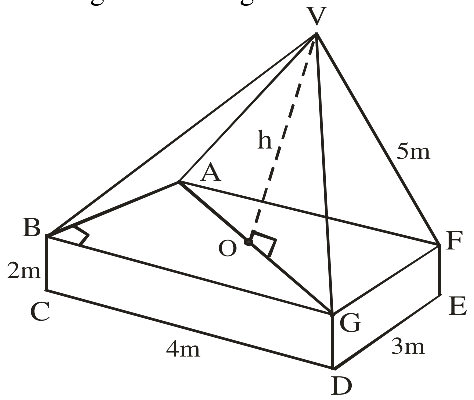
Solution

Volume of shape = volume of pyramid + volume of cuboid

Volume of pyramid = $\frac{1}{3}$base area × height

Here base area is rectangle (length × breadth)

Our height h can be gotten as:



H ⇒VO can be gotten from OG and OG from AG

In right – angled ΔABG : $AG^2 = BG^2 + AB^2$

$AG^2 = 4^2 + 3^2$

$AG = \sqrt{25} = 5m$

Thus, $OG = \frac{1}{2}AG = 2.5m$

In right – angled ΔVOG : $VG^2 = VO^2 + OG^2$

$5^2 = VO^2 + 2.5^2$

$25 - 6.25 = OV^2$

$\sqrt{18.75} = OV$

$4.33m = OV$ (Height)

Volume of shape

$= \frac{1}{3}$base area × height **+** length × breadth × height

$= \frac{1}{3} \times 4 \times 3 \times 4.33$ **+** $4 \times 3 \times 2$

$= 17.32 + 24$

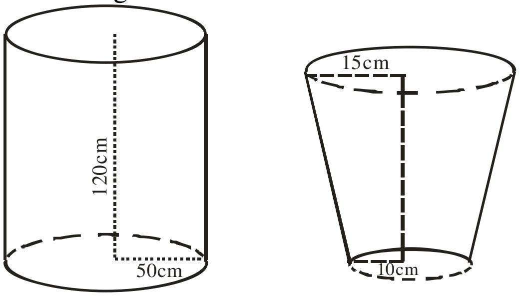
$= 41.32m^3$

2003/12 Neco (Dec)

A drum of radius 50cm and height of 1.2m is to be filled with water using a bucket of radius 15cm and 10cm for top and bottom respectively and height of 50cm. How many buckets filled with water will fill
the drum to the nearest whole bucket (Take $\pi = {}^{22}/_7$)

Solution

Drum height 1.2m = 120cm



Volume of drum(cylinder) = $\pi r^2 h$

$= \frac{22}{7} \times 50^2 \times 120$

$= 942857.14cm^3$

Volume of bucket (frustum)

$= \frac{1}{3}\pi h\ (R^2 + Rr + r^2)$

$= \frac{1}{3} \times \frac{22}{7} \times 50\ (15^2 + 15 \times 10 + 10^2)$

$= \frac{1100}{21}(225 + 150 + 100)$

$= \frac{1100}{21}\ (475)\ = 24880.95cm^3$

Number of buckets = $\frac{Volume\ of\ drum}{Volume\ of\ bucket}$

$= \frac{942857.14}{24880.95}$

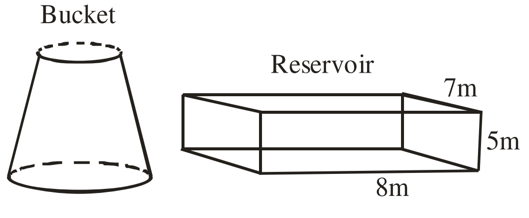
= 37.89 buckets ≈ 38 buckets

2009/40

A bucket holds 10 litres of water. How many buckets of water will fill a reservoir of size 8m × 7m × 5m.
(1 litre = $1000cm^3$)

A 28 B 280 C 2800 D 28000

Solution



Volume of bucket (frustum) = $10 \times 1000\ cm^3$

$= 10,000cm^3$

Volume of Reservoir (cuboid) = x×y×z

$= 800 \times 700 \times 500cm^3$

(Note: 100cm is 1m)

$= 280,000,000cm^3$

Number of buckets = $\frac{Volume\ of\ reservoir}{Volume\ of\ bucket}$

$= \frac{280,000,000}{10,000} = 28,000$ buckets

2014/17 (Nov)

A rectangular tank measuring 11m by 2m by 7m is equal in volume to a cylindrical tank of height 4m. Calculate the radius of the cylindrical tank

(Take $\pi = {}^{22}/_{7}$)

A 14.00m B 7.00m C 3.50m D 1.75m

Solution

Volume of rectangular tank (cuboid) = $11 \times 2 \times 7$

$= 154m^3$

Volume of cylinder = $\pi r^2 h$

$$154 = \frac{22}{7} \times r^2 \times 4$$

$$\frac{154 \times 7}{22 \times 4} = r^2$$

$$12.25 = r^2$$

$$\sqrt{12.25} = r \quad \text{thus, } 3.50m = r \quad (C)$$

1975/9 (Nov)

From a cylindrical object of diameter 70cm and height 84cm, a right solid cone having its base as one of the circular ends of the cylinder and height 84cm is removed. Calculate :

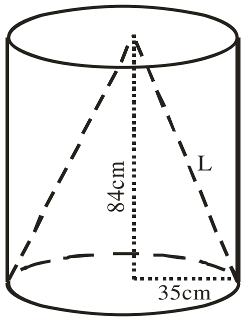
(a) the volume of the remaining solid object, expressing your answer in the form $a \times 10^n$ where $1 < a < 10$ and n is a positive integer

(b) the surface area of the remaining solid object

Solution

Height h is 84cm, radius is diameter ÷ 2 i.e 70÷ 2 = 35cm

We sketch as :



Volume of solid cylinder = $\pi r^2 h$

Volume of solid cone = ${}^1/_3\, \pi r^2 h$

Thus volume of remaining solid = $\pi r^2 h - \frac{1}{3}\pi r^2 h$

$$= \frac{2}{3}\pi r^2 h$$

$$= \frac{2}{3} \times \frac{22}{7} \times 35^2 \times 84 = 215600 \text{ cm}^3$$

$$= 2.156 \times 10^5 \text{ cm}^3$$

Surface area of the remaining solid object =

Inner surface area of cone shape left behind + outer surface area of a cylinder + area of circular outer top

$$= \pi rL + 2\pi rh + \pi r^2$$

$$= \pi r(L + 2h + r)$$

L can be gotten from the right- angled Δ in the diagram

$L^2 = 84^2 + 35^2$

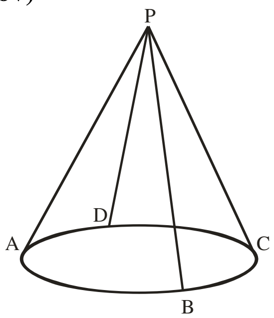
$L^2 = 7056 + 1225$

$L = \sqrt{8281} = 91$cm

Surface area of the remaining solid object = $\pi r(L + 2h + r)$

$$= \frac{22}{7} \times 35\,(91 + 2 \times 84 + 35) = 32340\text{cm}^2$$

1976/8(Nov)



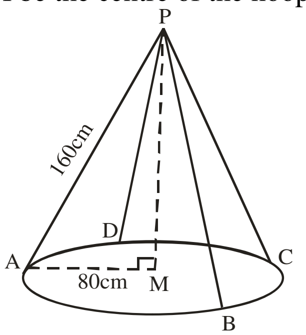
In the figure, a hoop of radius 80cm is suspended horizontally by four strings each 160cm long and each attached to a nail vertically above the hoop at P. The strings are attached to points A, B, C, D which are equally spaced on the hoop. Calculate:

(a) the angle which PA makes with the horizontal.

(b) the angle BPD

Solution

Let M be the centre of the hoop. Then we sketch as:



PA = PB = PC = PD and AB = BC = CD = DA(Given)

Thus, PM is perpendicular to the horizontal hoop

$\angle PMA = 90^0$.

In ΔPAM, $\cos A = \frac{AM}{PA} = \frac{80}{160}$

$$\cos A = \frac{1}{2}$$

$$A = \cos^{-1} 0.5$$

$$A = 60^0$$

PA makes an angle of 60^0 with the horizontal.

(b) In ΔBPD , BP = DP = 160cm

DB = diameter of hoop = 2× 80 = 160cm

Hence ΔBPD is equilateral.

∴ angle BPD = 60^0

2012/11b

A cylinder with base radius 14cm has the same volume as a cube of side 22cm. Calculate the ratio of the total surface area of the cylinder to that of the cube.

(Take $\pi = {}^{22}/_{7}$)

Solution

First, we get the height **h** of cylinder from its value of volume in cube

Volume of cube = x^3

$= 22^3$

$= 10648\text{cm}^3$

This is the same volume for the cylinder

Next, volume of cylinder = $\pi r^2 h$

$$10648 = \frac{22}{7} \times 14^2 \times h$$

$$\frac{10648 \times 7}{22 \times 14^2} = h$$

$$17.29\text{cm} = h$$

Total surface area of Cylinder **:** Total surface area of cube

$$2\pi r(r + h) \quad : \quad 6x^2$$

$$2 \times \frac{22}{7} \times 14\,(14 + 17.29) \quad : \quad 6 \times 22^2$$

$$2753.52 : 2904$$

$$2754 : 2904$$

$$1 : 1$$

2011/10a

The total surface areas of two spheres are in the ratio 9 : 49. If the radius of the smaller sphere is 12cm, find correct to the nearest cm^3 the volume of the bigger sphere

Solution

Surface area of sphere = $4\pi r^2$

Small sphere : bigger sphere

$4\pi r^2 = 9 \quad : \quad 4\pi R^2 = 49$

$$\frac{\textit{surface area of small sphere}}{\textit{surface area of bigger sphere}} = \frac{9}{49}$$

$$\frac{4\pi r^2}{4\pi R^2} = \frac{9}{49}$$

$$\frac{r^2}{R^2} = \frac{9}{49}$$

Substituting for r = 12cm,

$$\frac{12^2}{R^2} = \frac{9}{49}$$

$$\frac{12^2 \times 49}{9} = R^2$$

$$R = \sqrt{\frac{12^2 \times 7^2}{3^2}}$$

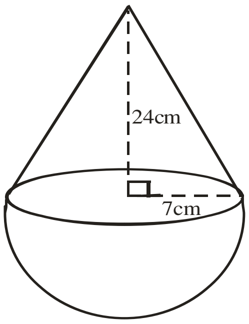
$$R = \frac{12 \times 7}{3} = 28 \text{ cm}$$

Thus volume of bigger sphere = $\frac{4}{3}\pi R^3$

$$= \frac{4}{3} \times \frac{22}{7} \times 28^3 = 91989.33\text{cm}^3$$

$$\approx 91989\text{cm}^3 \text{ to nearest cm}^3$$

2012/13b



The diagram shows a wooden structure in the form of a cone mounted on a hemispherical base. The vertical height of the cone is 24cm and the base radius 7cm. calculate , correct to 3 significant figures, the surface area of the structure

(Take $\pi = {}^{22}/_{7}$)

Solution

Surface area of solid

= surface area of cone + surface area of hemisphere

= **$\pi rL + 2\pi r^2$**

But slant height $(L)^2 = 7^2 + 24^2$

$= 49 + 576$

$L^2 = 625$

$L = \sqrt{625} = 25$cm

$$\text{Surface area of solid} = \frac{22}{7} \times 7 \times 25 + 2 \times \frac{22}{7} \times 7^2$$

$$= 550 + 308$$

$$= 858\text{cm}^2$$

2015/12

A water reservoir in the form of a cone mounted on a hemisphere is build such that the plane face of the hemisphere fits exactly to the base of the cone and the height of the cone is 6 times the radius of its base.

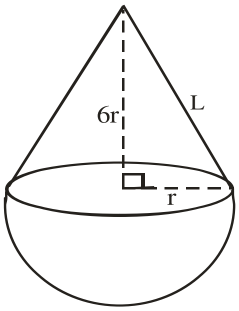
(a) illustrate this information in a diagram

(b) if the volume of the reservoir is $333^1/_3\pi\text{cm}^3$ calculate, correct to the nearest whole number, the:

(i) volume of the hemisphere

(ii) total surface area of the reservoir (Take $\pi = {}^{22}/_{7}$)

Solution



Volume of reservoir = volume of cone + volume of hemisphere

$$= \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3$$

$$333^1/_3\pi = \pi\left(\frac{1}{3}r^2 \times 6r + \frac{2}{3}r^3\right)$$

Pi will cancel out

$$\frac{1000}{3} = \frac{6}{3}r^3 + \frac{2}{3}r^3$$

$$\frac{1000}{3} = \frac{8}{3}r^3$$

$^1/_3$ will cancel out

$$1000 = 8r^3$$

$$\frac{1000}{8} = r^3$$

$$\sqrt[3]{\frac{1000}{8}} = r$$

$$\frac{10}{2} = r$$

$$r = 5\text{cm}$$

(i) Volume of hemisphere $= \frac{2}{3}\pi r^3$

$$= \frac{2}{3} \times \frac{22}{7} \times 5^3$$

$$= 261.90\text{cm}^3$$

(ii) Total surface area of reservoir

= surface area of cone + surface area of hemisphere

$= \pi rL + 2\pi r^2$

But $L^2 = (6r)^2 + r^2$

$L^2 = 30^2 + 5^2$

$L = \sqrt{925} = 30.41\text{cm}$

Total surface area of reservoir $= \frac{22}{7} \times 5 \times 30.41 + 2 \times \frac{22}{7} \times 5^2$

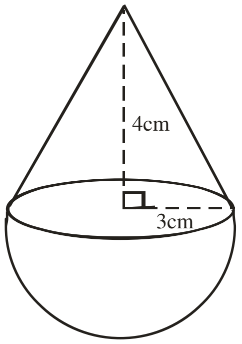
$= 477.87 + 157.14$

$= 320.73\text{cm}^2$

2005/3b Exercise 21.38

A cone and a right pyramid have equal heights and volume. If the area of the base of the pyramid is 154cm^2, find the base radius of the cone (Take $\pi = {}^{22}/_7$)

2014/41 Neco Exercise 21.39



The total surface area of the above shape in terms of π is

A $6\pi\text{cm}^2$ B $18\pi\text{cm}^2$ C $21\pi\text{cm}^2$ D $27\pi\text{cm}^2$ E $33\pi\text{cm}^2$

2015/15 Adjust ed Exercise 21.40

The dimensions of a rectangular tank are 2m by 10m by 11m. If its volume is equal to that of a cylindrical tank of height 7m. Calculate the base radius of the cylindrical tank (Take $\pi = {}^{22}/_7$)

A 14m B 10m C $3^1/_2$m D $1^3/_4$m

1976/9a (Nov) Exercise 21.41

A solid cube of side 8cm is dropped vertically into a cylindrical tank of radius 7cm. Calculate the rise in the water level if:

(i) the original depth of the water was 9cm

(ii) the original depth of the water was 5cm

2004/13b (Nov) Exercise 21.42

Two tanks, A and B contain 300 litres and 800 litres of water respectively. If 3x litres is drawn from B and x litres added to A, then B contains twice as much as A, Find: (i) the value of x

(ii) the quantity of water each tank now has

2000/10a (Nov) Exercise 21.43

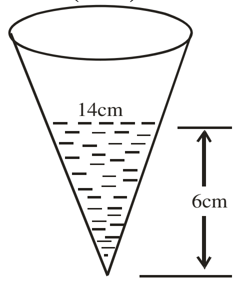
A Cylindrical tank of radius 35cm contains water 20cm high. The water is poured into a barrel of rectangular base 42cm by 30cm. Find, correct to two significant figures, the height of the water level in the barrel.

(Take $\pi = {}^{22}/_7$)

2000/8 Neco Exercise 21.44

A solid cone of height 56cm and base diameter 51cm is formed from a cylindrical drum equal in height and diameter with the cone. Find the volume of the remaining part of the drum

2004/12b (Nov) Exercise 21.45



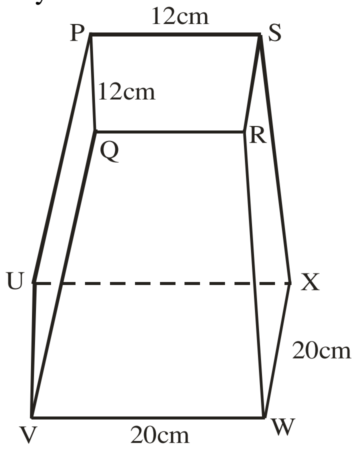
The diagram show a hollow cone dug out of a stone block. The diameter of the level of water inside the cone is 14cm and the height is 6cm. If 250cm^2 of water is added to the water inside the cone, calculate, correct to one decimal place, the rise in water level. (Take $\pi = {}^{22}/_7$)

2004/8 Neco Exercise 21.46

The volume of a frustum of a cone of height, hcm is given by the formula $\frac{h}{3}\pi(R^3 - r^3)$ where R and r are radii of its circular ends. If an open- ended metal pipe of external and internal diameters of 20.5cm and 10.5cm respectively and length 21cm are to be cast from a molten contents of the frustum of a cone of depth 1.5m and radii 0.5m and 0.3m. Calculate the number of pipes that will be cast to the nearest whole number (Take $\pi = {}^{22}/_7$)

1986/20 Exercise 21.47

The diagram represents a lampshade whose top PQRS and bottom UVWX are squares of sides 12cm and 20cm respectively.



If the vertical depth is 14cm, calculate, correct to the nearest whole number:

(a)the area of the material used in making the lampshade

(b) the volume of the pyramid of which the lampshade is a part

CHAPTER TWENTY TWO

Construction

Foundation facts on construction

To construct any given shape say triangle, we need A compass, protractor, sharp pencil, very neat ruler and a good eraser. All construction lines must not be thick. Do not erase any arc or guiding lines to your final result. Always confirm your lines or arc measurements with ruler or compass as the case may be

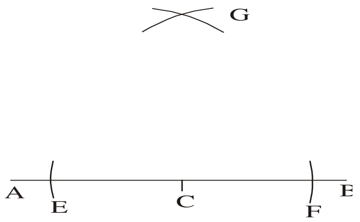
Construction of angle 90^0 at a given point C

Draw a line AB

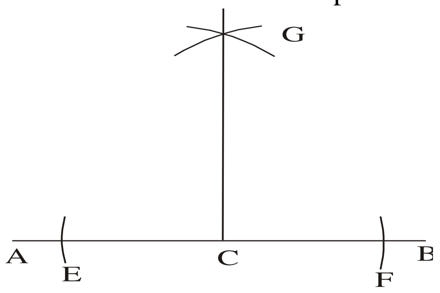
A ———————— B

With center at the point C, on line AB open your compass to a suitable radius, cut arc twice on line AB at points E & F

Changing or adjusting our compass to a bigger radius wider than CF or CE. Next, make E and F our centers, one at a time draw an arc to cut each other at top of C as shown below:

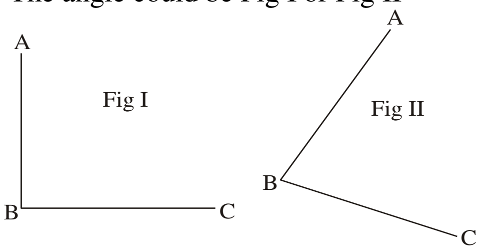


Join point of intersection G to line AB at C, we have $G\hat{C}B = 90^0$ and $G\hat{C}A = 90^0$ as shown below.
Confirm it with the use of a protractor.

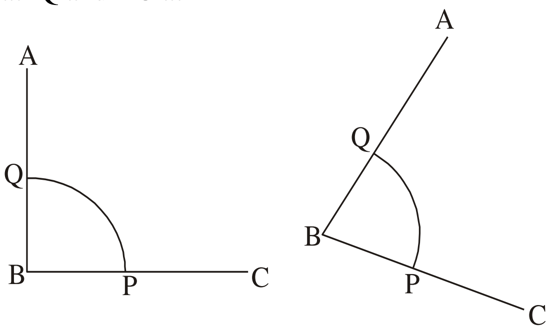


To bisect any given angle ABC

The angle could be Fig I or Fig II

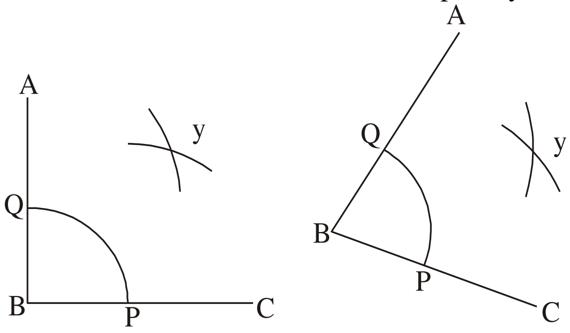


I: With center B and a suitable radius draw an arc to cut AB at Q and BC at P

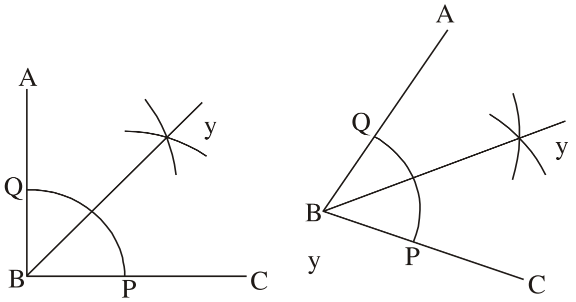


Where this arc already exist maintain it to step II

II: With centers at "P" and "Q" each at a time; using the same radius, draw arcs to intersect at point y.



III: Join B to Y. This is the Bisector

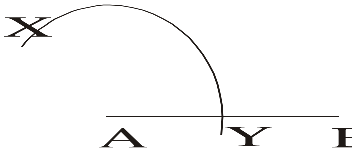


Construction of 60^0

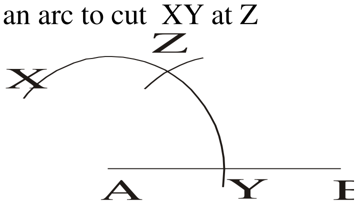
Step I : Draw line AB

A ———————— B

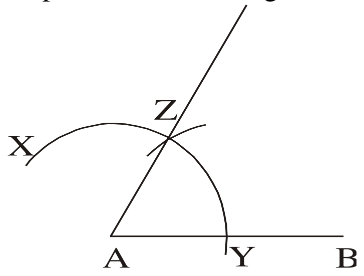
Step II: With center at A and a suitable radius draw an arc XY to cut AB at Y



Step III : With center at Y and same radius as before, draw an arc to cut XY at Z



Step IV: Join AZ . Angle ZAB = 60^0

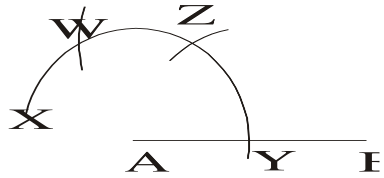


Construction of 120^0

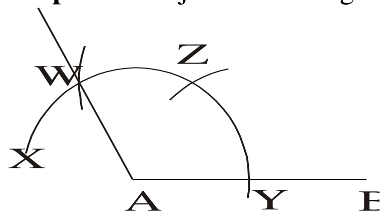
We use same steps as in 60^0, but for step IV

`Do not join A Z   Rather,`

Step IV: With same radius as before and center Z draw another arc to cut XY at W.



Step V: Then join AW. Angle WAB = 120^0



Construction of triangles with given length of the three sides (No included angles)

The steps we follow here are the same except for change in length based on the question.

Sketching is an important part of construction; so always remember to do so before the construction proper.

1997/5a Delta

Construct a triangle AB with /AB/ = 6cm, /BC/ = 8cm and /AC/ = 10 cm

Measure angle ABC

Diagram on page 391

Solution

(a) Draw line AB = 6 cm

(b) With center B and radius 8 cm, draw an arc

(c) With center A and radius 10 cm, draw an arc to intersect the previous one at C.

(d) Join AC and BC using a ruler

Angle ABC = 90^0 using a protractor to measure it.

1994/5 Delta

Construct a triangle XYZ in which XY =10 cm,YZ =6 cm and ZX = 11 cm. On XY mark a point D such that XD = 4.5 cm. Draw ZD, measure the length of ZD.

`Solution` **`Diagram on page 391`**

We have something to do with XY, So we let it lie on the floor . Steps:

Draw a line XY = 10 cm

(b) With center X and radius 11 cm draw an arc

(c) With center Y and radius 6 cm draw an arc to intersect the previous one at Z

(d) Join YZ and ZX using a ruler

Lastly, line ZD as instructed : with centre X and radius 4.5cm draw an arc to cut XY at D.Then join D to Z

To construct a triangle, given <u>two sides</u> and <u>one of its angles</u>

2002/5 Delta

Using a ruler and a pair of compasses only,

(a) Construct Δ PQR such that PQ = 6.5cm, QR = 8 cm and ∠ PQR = 45^0

(b) Measure ∠ QPR

Solution **Diagram on page 391**

Steps taken are:

(i) Draw QR = 8 cm

(ii) At "Q" construct angle Q = 45^0 i.e bisecting of 90^0 and measure out line QP = 6.5cm through it to P

(iii) With center Q and radius 6.5cm draw an arc to cut QP.

(iv) Join P to R

2000/5 Delta

Using a ruler and a pair of compasses, construct a ΔPQR such that PQ = 9 cm, ∠PQR = 60^0 and QR = 10 cm

Measure (i) PR (ii) ∠QPR

Solution ***Diagram on page 391***

For ease in measuring ∠ QPR and constructing ∠ PQR = 60^0 ; we let Q and P be on the floor.

Steps taken:

(a) Draw side PQ = 9 cm

(b) At point Q, construct ∠Q = 60^0

(c) Measure out QR = 10 cm

(d) Join PR

2004/1 FGC (JSS Neco) Exercise 22.1

Construct Δ ABC in which AB = 9cm, BC = 12cm and ABC = 60^0

(i) Construct the bisector of angle A and let it meet BC at D.

(ii) Measure DC

Construction of a triangle, given <u>two angles</u> and <u>one</u> side

1997/5b Delta

Construct a triangle PQR with /PQ/ = 10 cm ∠ RPQ = 30^0, ∠ PQR = 60^0. Measure PR.

Solution **Diagram on page 391**

Our Sketch must be reasonable enough to allow us construct the two given angles P and Q on the floor.

Steps taken are:

(a) Draw PQ = 10cm

(b) At P construct angle P = 30^0 i.e bisecting of 60^0

(c) At Q construct angle Q = 60^0

(d) The point of intersection of these lines is R.

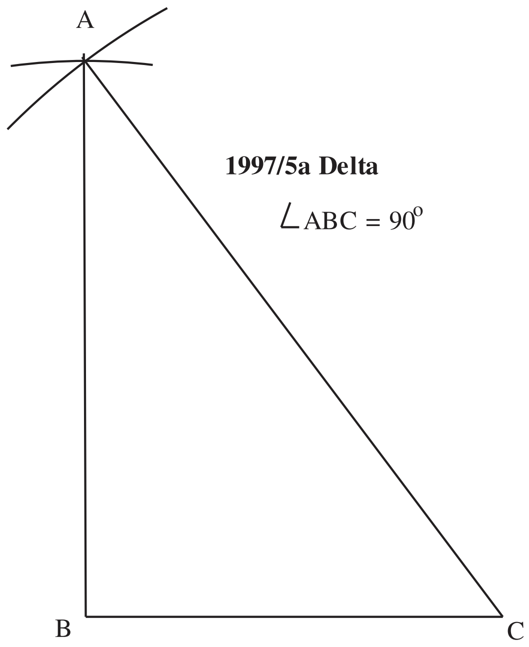
2004/5 Delta Exercise 22.2

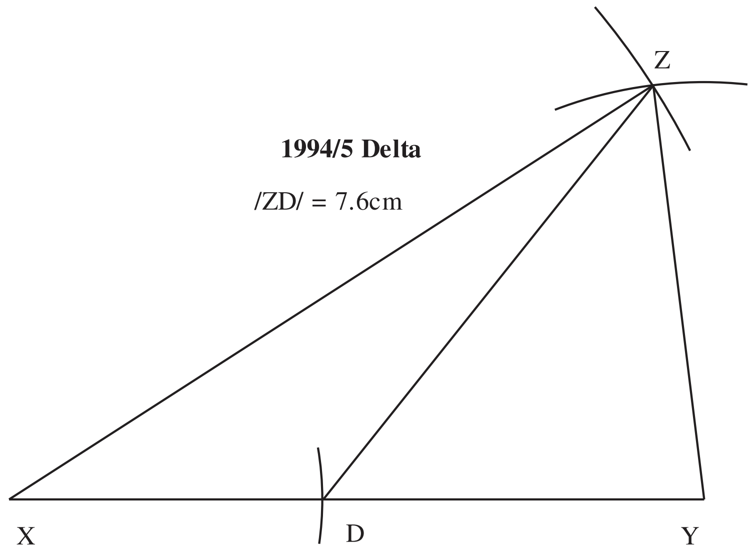
Construct ΔXYZ such that X = Z = 45^0 and XZ = 8 cm

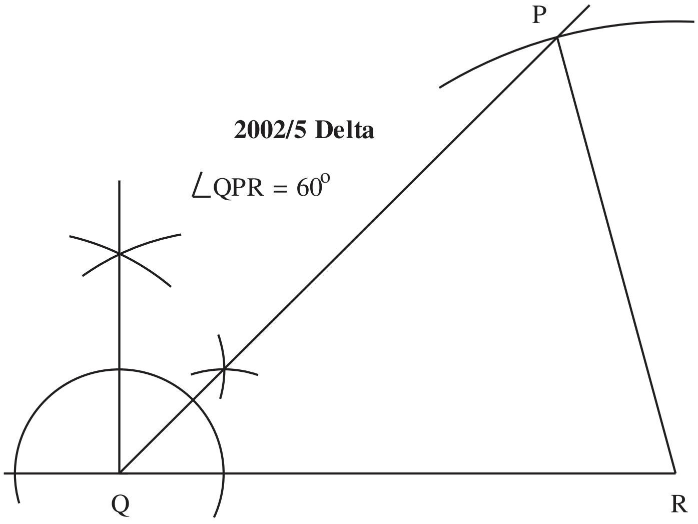
(b) Measure: (i) side XY and (ii) angle Y

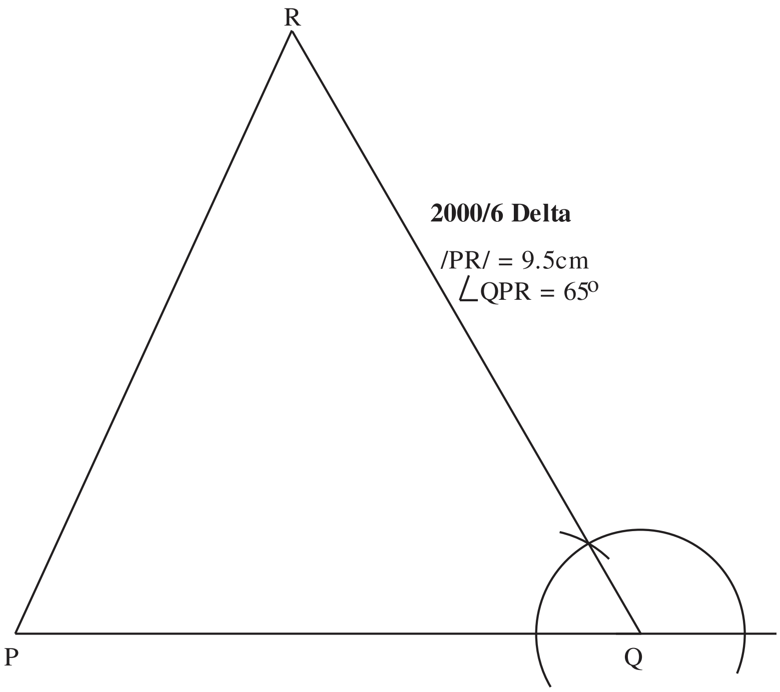
2004/3 Osun Exercise 22.3

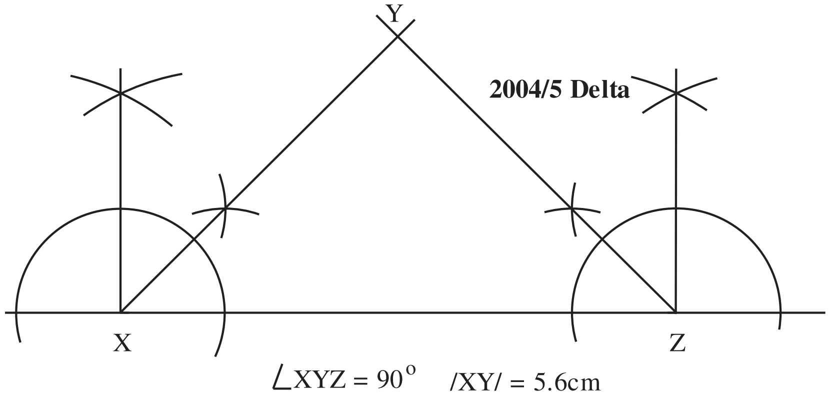
Using a pair of compasses and a ruler only,

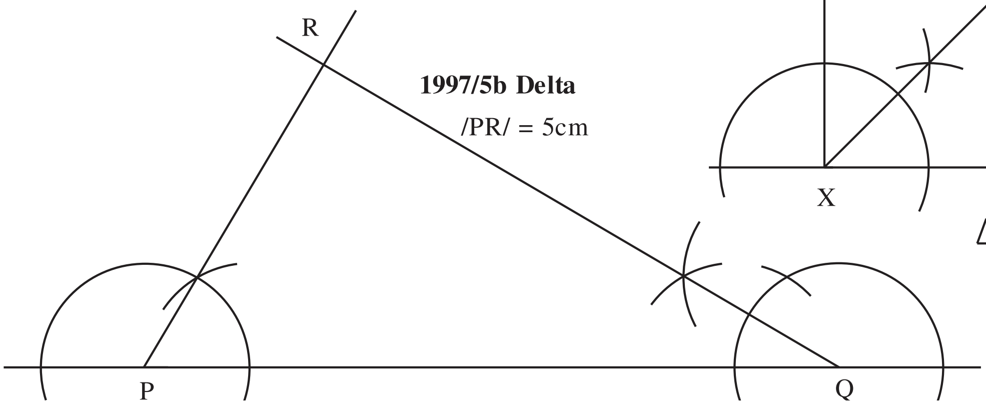

A
1997/5a Delta
∠ABC = 90°
B
C


Z
1994/5 Delta
/ZD/ = 7.6cm
X
D
Y


P
2002/5 Delta
∠QPR = 60°
Q
R


R
2000/6 Delta
/PR/ = 9.5cm
∠QPR = 65°
P
Q


Y
2004/5 Delta
X
Z
∠XYZ = 90°
/XY/ = 5.6cm


R
1997/5b Delta
/PR/ = 5cm
P
Q

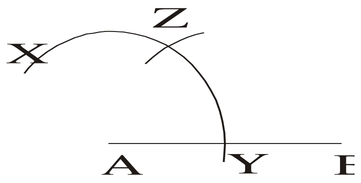
construct a triangle PQR in which /QR/ = 10cm , $\angle PQR = 60^0$ and $\angle PRQ = 30^0$
(i) measure /QP/ (ii) measure /PR/
(iii) measure $\angle QPR$
(iv) What type of triangle is PQR

SPECIAL ANGLES, AND LOCUS

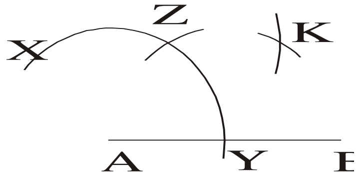
Under this sub topic we shall treat construction of 15^0, $75^0(60+15)$, $105^0(90+15)$, or $105^0(60+30+15)$, $135^0(120+15)$, or $135^0(90+45)$, $150^0(120+30)$.

Construction of 15^0

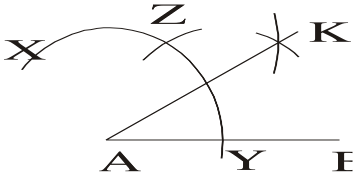
When we bisect angle 60^0, we get 30^0. Further bisecting of 30^0 gives 15^0
Steps taken :
1: Using the steps given earlier on the construction of angle 60^0, we have:



2:Using same radius and centers Z and Y bisect $\angle ZAY$ as shown below:

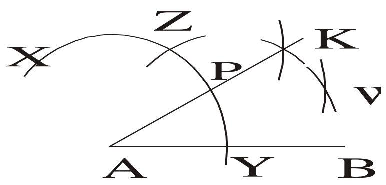


3 : Join AK. Angle BAK is 30^0 .Also angle ZAK is30^0

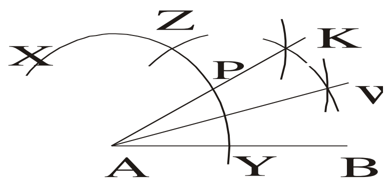


We are to either bisect angle BAK or ZAK
To make the work neat is always good to use an already made arc in the construction. The visible one here is arc YZ.If you have adjusted your compass before; ensure that you bring it back to the said arc YZ with centre at A on line AB.

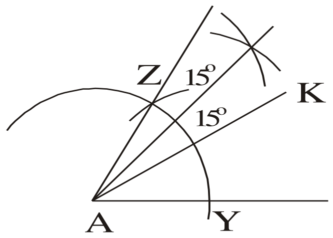
For emphasis sake we will note a point P on line AK i.e where the line touches the arc YZ
4 : Using same radius and with centers at P and Y bisect
$\angle BAK$ as shown below



5 : Join line AW and we have that $\angle BAW = 15^0$



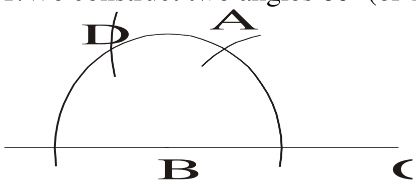
Note that $\angle KAW$ is also 15^0
Readers to bisect $\angle KAZ = 30^0$ at step5 to give 15^0 each as shown below



Most of the special angles is based on one or two manipulations of the above bisectings

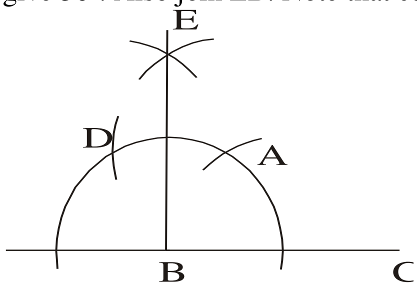
Construction of 75^0

Construction of 75^0 is basically construction of two 60^0 (120^0) and bisecting of one angle 60^0 to give 30^0 then further bisecting to give 15^0
Steps taken:
1:We construct two angles 60^0 (or 120^0 showing two arcs)

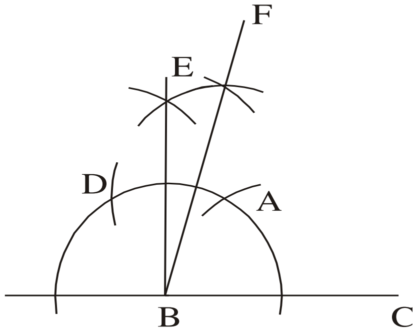


$\angle ABC = 60^0$ and $\angle ABD$ is the other 60^0 to be bisected

2 : Using same radius we bisect angle $ABD = 60^0$ at E to give 30^0. Also join EB. Note that centers here are D and A

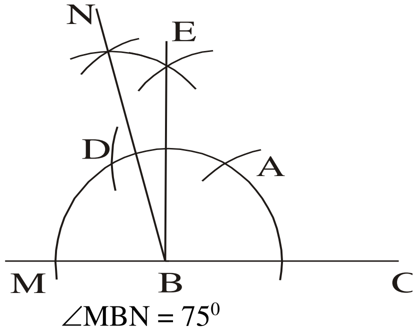


3 : Using same radius, we bisect angle $ABE = 30^0$ at F to give 15^0 . Centres here are A and the point of intersection between BE and arc AD



$\angle FBC = 75^0$

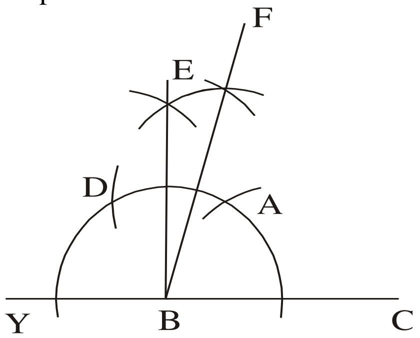
ALTERNATIVE STEP 3
From step2, using same radius we can bisect angle EBD = 30^0 to get 15^0(With centers at D and point of intersection between line EB and arc AD)



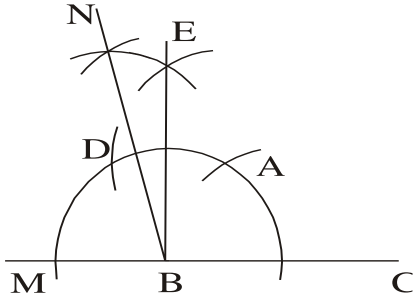
$\angle MBN = 75^0$

Construction of 105^0

Construction of two 60^0 (120^0) and bisecting of one angle 60^0 to give 30^0 each .The last of the 30^0 is bisected to give 15^0
so we have $60^0 + 30^0 + 15^0 = 105^0$
Steps taken are as shown above in construction of 75^0



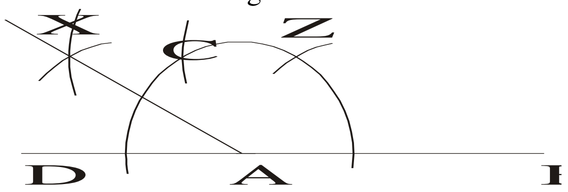
$\angle YBF = 105^0$ **OR** $\angle CBN = 105^0$ shown below:



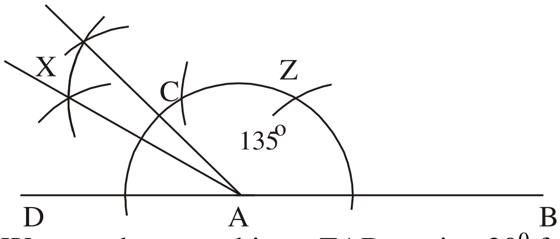
Construction of 135^0

$135^0(120 + 15)$, or $135^0(90 + 45)$

135^0(120 +15) case,
First we construct angle 120^0 then bisect the last part

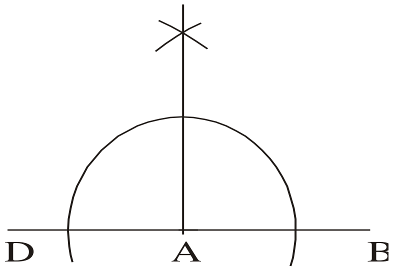


Secondly, we bisect $\angle XAC = 30^0$ to give 15^0

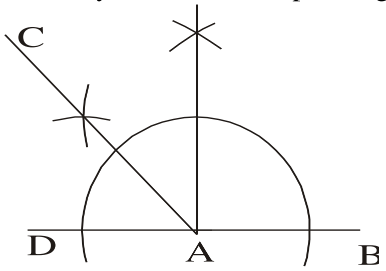


We can choose to bisect ZAB to give 30^0 further bisecting to get 15^0 . $\angle BAF = 135^0$

135^0(90 + 45)
First we construct angle 90^0at A



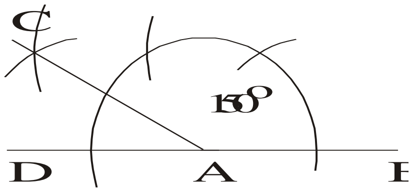
Secondly, we bisect one part to get 45^0



CAB = 135^0

Construction of 150^0

$150^0(120 + 30)$.
We construct angle 150^0 by bisecting the third part in 120^0



LOCUS

Locus has plural form loci. It can be defined as a set of points satisfying or obeying a given condition (rule)
There are basically three loci
1.LOCUS OF A POINT (IN A PLANE) WHICH IS EQUIDISTANT FROM A GIVEN POINT.
It is a circle at that point with radius equal to the fixed distance
2. LOCUS OF A POINT (IN A PLANE) WHICH IS EQUIDISTANT FROM TWO GIVEN POINTS.
It is the perpendicular bisector of the line between those two points
3. LOCUS OF A POINT (IN A PLANE) WHICH IS EQUIDISTANT FROM TWO INTERSECTING STRAIGHT LINES.
It is the bisector of the angle between them

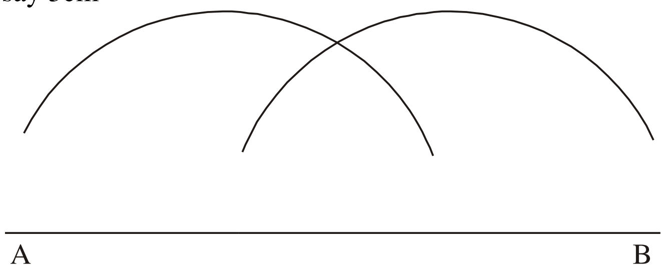
In subsequent examples we shall utilize the above locus as directed by the given questions

PARALLEL LINES

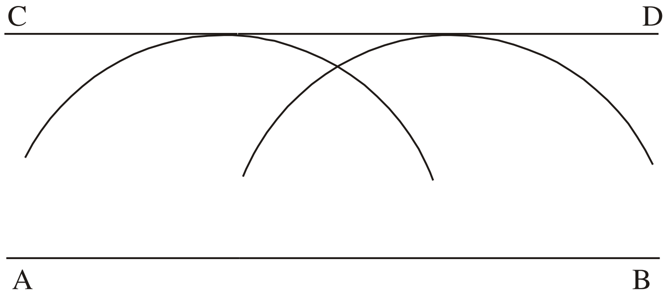
There are basically two ways of constructing parallel lines namely:

Case of parallel line to given line at a fixed distance apart.

Firstly we draw two arcs of almost the size of semi circle on the line. It's radius equal the given distance say 3cm

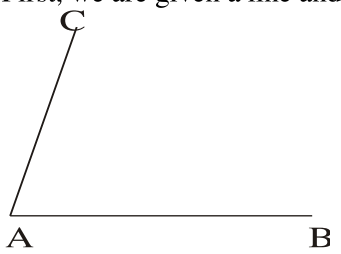


Secondly, we draw a straight line CD on the arcs as shown below:

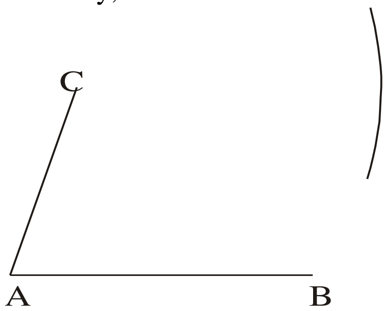


Case of parallel line to given line at a fixed point outside the line

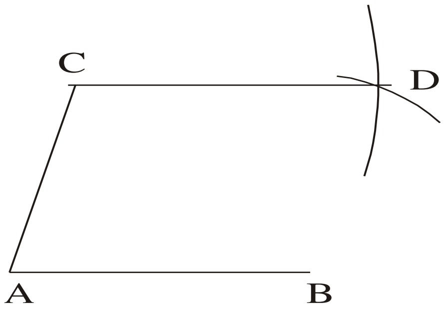
First, we are given a line and a point outside the line



Secondly, with radius AB but centre at **C** cut an arc as ;



Thirdly, with radius AC but centre at **B** draw an arc to cut the first one. Join CD; CD//AB as required



LINE RATIO CONSTRUCTION

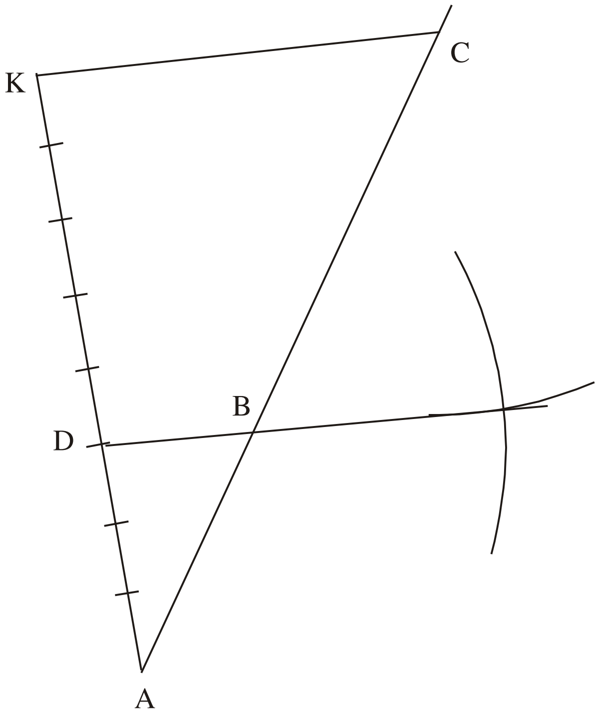
Find by a suitable construction, a point B on AC such that /AB/ : /BC/ = 3:5 .

Steps taken

1 : Draw a line AK within a reasonable angle AC

2 : Divide AK into 8 reasonable segments

3 : At D draw parallel line to KC using the steps for a parallel line at point outside a given line



CONSTRUCTION OF ORDINARY TRIANGLE I

2009/10 (Nov)

Using a ruler and a pair of compasses only. Construct a triangle ABC, such that $|AB| = 7.1$cm, $|AC| = 7$cm and $\angle BAC = 105^0$. Construct the bisector of $\angle BAC$ to meet $\overline{BC}$ at X and the bisector of $\overline{AC}$ to meet $\overline{AX}$ produce at Y. Measure: (a) $|XY|$; (b) $|BC|$

Analysis and solution ***Diagram on page 396***

Using free – hand, sketch the required diagram showing all the given data. Don't wipe out or erase your sketch.

Steps taken

1.Draw a line AC = 7cm with extensions as shown

2.At point A construct $\angle 105^0$ (i.e constructing 90^0 between 60^0 and 120^0 then bisecting it to give $15^0 + 90^0$)

3. Produce the straight line at 105^0 to 7.1cm long to get B then join B to C

4 . Bisect $\angle BAC$ to meet $\overline{BC}$ at X

5. Bisect line AC to meet $\overline{AX}$ at Y

6 . Measure (a) $|XY|$; (b) $|BC|$ with a ruler

2007/6 Neco

A triangular plot of land ABC is such that $|BC|$ is 105m, $\angle ABC = 45^0$, $\angle ACB = 75^0$

(a) With the aid of a ruler and a pair of compasses only construct the:

(i) triangular plot using a scale of 10m to 1cm

(ii) locus l_1 of points equidistant from B and C

(iii) locus l_2 of points equidistant from $\overline{AB}$ and $\overline{BC}$

(iv) locus l_3 of points 5cm equidistant from A

(b) (i) Locate T_1 and T_2 which are the points of intersection of l_1 and l_3

(ii) Measure $|T_1 T_2|$

Analysis and solution ***Diagram on page 396***

Using free – hand, sketch the required diagram showing all the given data. Don't wipe out or erase your sketch.

Steps taken

1. Draw a line BC = 10.5cm with extensions as shown
2. At point B construct $\angle 45^0$ (Bisecting of 90^0)
3. At point C construct $\angle 75^0$ (i.e 60^0, 120^0 then 90^0 in between, next bisect between 60^0 and 90^0)
4. Produce B and C to meet at A
5. Locus l_1 is the perpendicular bisector of line BC
6. Locus l_2 is the bisector of $\angle ABC$
7. Locus l_3 is a circle with radius 5cm and centre A
8. Measure $|T_1 T_2|$ with a ruler

2014/7b NABTEB(Nov) Exercise 22.4

Construct a triangle PQR for which $|PQ| = 8$cm, $|PR| = 9$cm and $\angle QPR = 60^0$. Measure $|QR|$

(i) construct the locus of points at a distance of 5cm from P

(ii) construct the locus of a point equidistant from P and R

(iii) The locus meet at X. Measure $|XQ|$

2007/9a NABTEB Exercise 22.5

With a pair of compasses and a ruler only, construct a triangle PQR in which $\angle RPQ = 120^0$, $\angle PQR = 45^0$ and $|PQ| = 6$cm

(i) Find a point M on RQ such that PM is perpendicular to RQ

(ii) Measure $|PM|$

1979/7 Exercise 22.6

Using a ruler and a pair of compasses only:

(a) Construct a triangle ABC such that AB = 6 cm, AC = 8.5 cm $\angle$ BAC = 120^0

(b) construct the locus l_1 of points equidistant from A and B

(c) construct the locus l_2 of points equidistant from AB and AC

(d) find the point of intersection P of l_1 and l_2 and measure PC

Construction of triangles with ratios

1995/9

(a)Using a ruler and a pair of compasses only, construct triangle ABC with /AB/ = 7.5cm, /BC/ = 8.1cm and $\angle ABC = 105^0$

(b) Locate a point D on BC such that /BD/ : /DC/ is 3:2

(c) Through D, construct a line L perpendicular to $\overline{BC}$.

(d) If the line L meets AC at P , measure /BP/ .

Analysis and solution ***Diagram on page 397***

Steps taken

1: Draw a line AB = 7.5cm with extensions as shown in the diagram

2 :At point B construct $\angle 105^0$ and produce a line 8.1cm through it to C

3: Join C to A to complete the triangle

4 : Draw a line CK and divide it into 5 reasonable equal segments; join K to B

5 : Mark the 3rd segment point d on CK

6 : At **d** construct a parallel line to BK using steps of a point outside a parallel line and join **d** to **D**

7 : At D construct a perpendicular to meet AC at P

8 : Join B to P using a dotted line and measure

1987/9(a) GCE Exercise 22.7

(a) Using a ruler, a protractor and a pair of compasses, construct a triangle ABC , given that /BC/ = 8.5 cm , /AB/ = 5.1 cm and $\angle BAC = 65^0$

(i) Measure /AC/.

(ii) Find by a suitable construction, a point D on AC such that /AD/ : /DC/ = 3 : 5 .

(iii) Measure angles ABD and CBD .

Author's hint: Use protractor to get $65^{0\wedge}$ and not by construction

Construction of triangles with inscribed circle

It is done by bisecting any two angles of the triangle. Their point of intersection is the centre of the circle touching all the sides of the circle.

2006/11 Neco

(a) Using a ruler and a pair of compasses only , construct :

(i) a triangle PQR such that $\overline{PQ}$ = 8.5cm , PR = 7cm and $\angle QPR = 75^0$.

(ii) the locus, L, of points equidistant from PQ and PR

(**iii)** ***an inscribed circle which lies inside the triangle and touches PQ , PR and QR.***

(b) Measure :

(i) $\overline{QR}$, (ii) the radius of the inscribed circle .

Analysis and solution ***Diagram on page 397***

Steps taken

1: Draw a line PR = 7cm with extensions as shown in the diagram

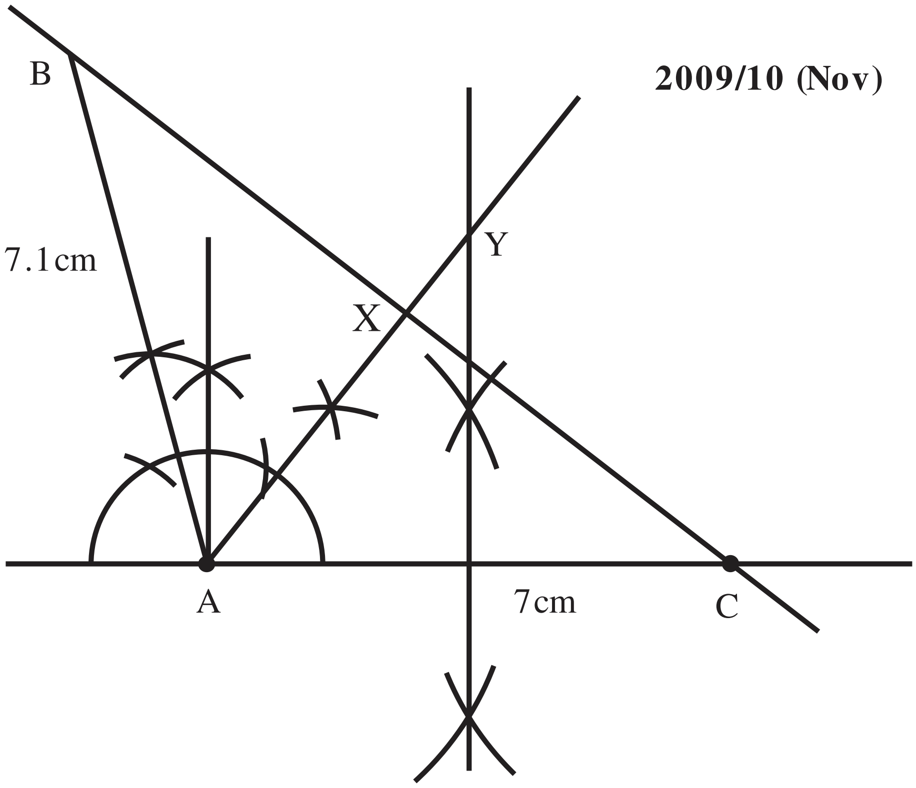
2 :At point P construct $\angle 75^0$ and produce a line 8.5cm through it to Q

3: Join Q to R to complete the triangle

4 : Construction of locus L at P is the bisecting of angle P

5 : For (iii) to be fulfilled; we must construct bisector of angle R or Q. Here we bisect R and the intersection K is the centre of the circle touching all the sides

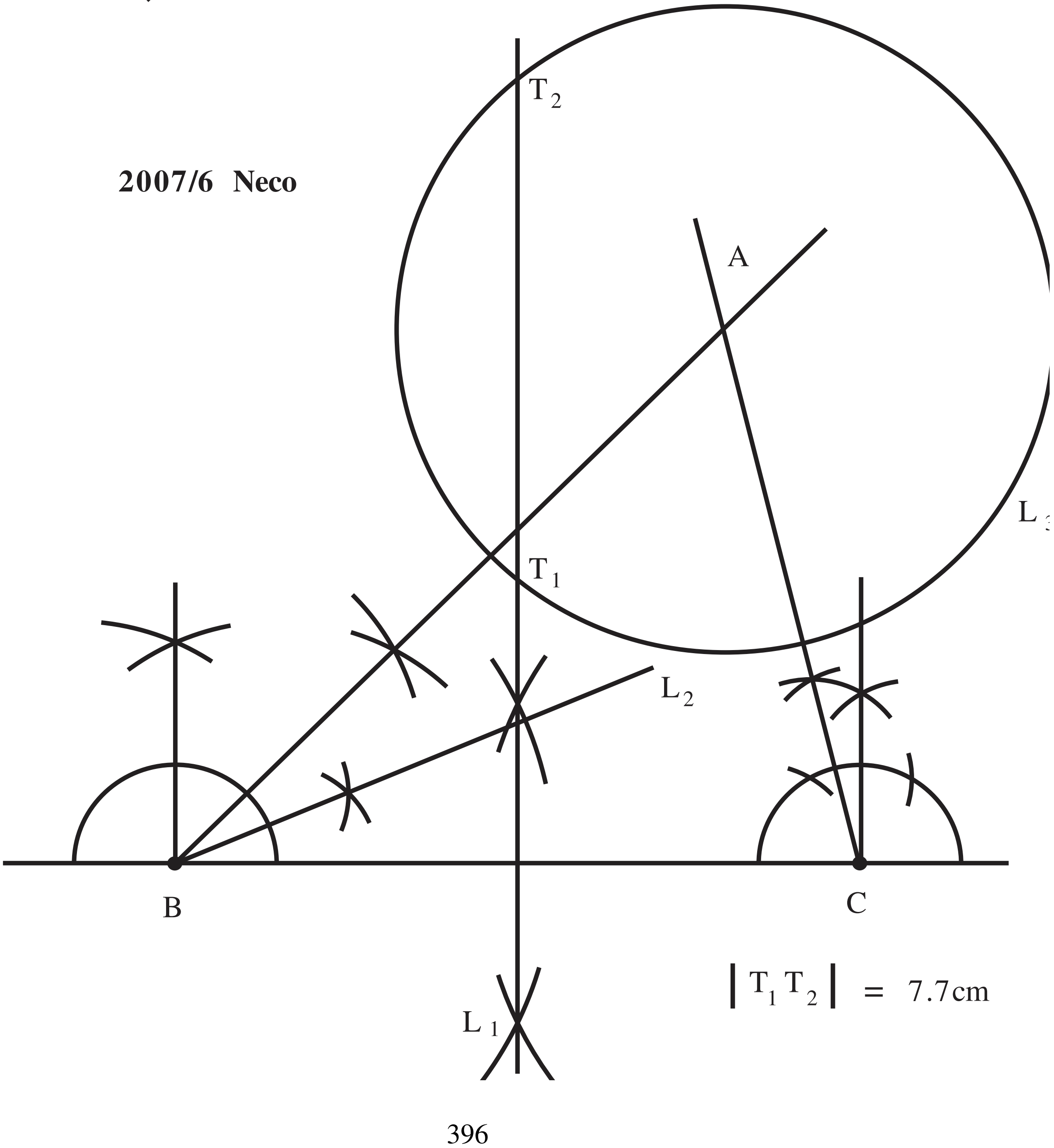
2009/10 (Nov)



XY = 1.3cm

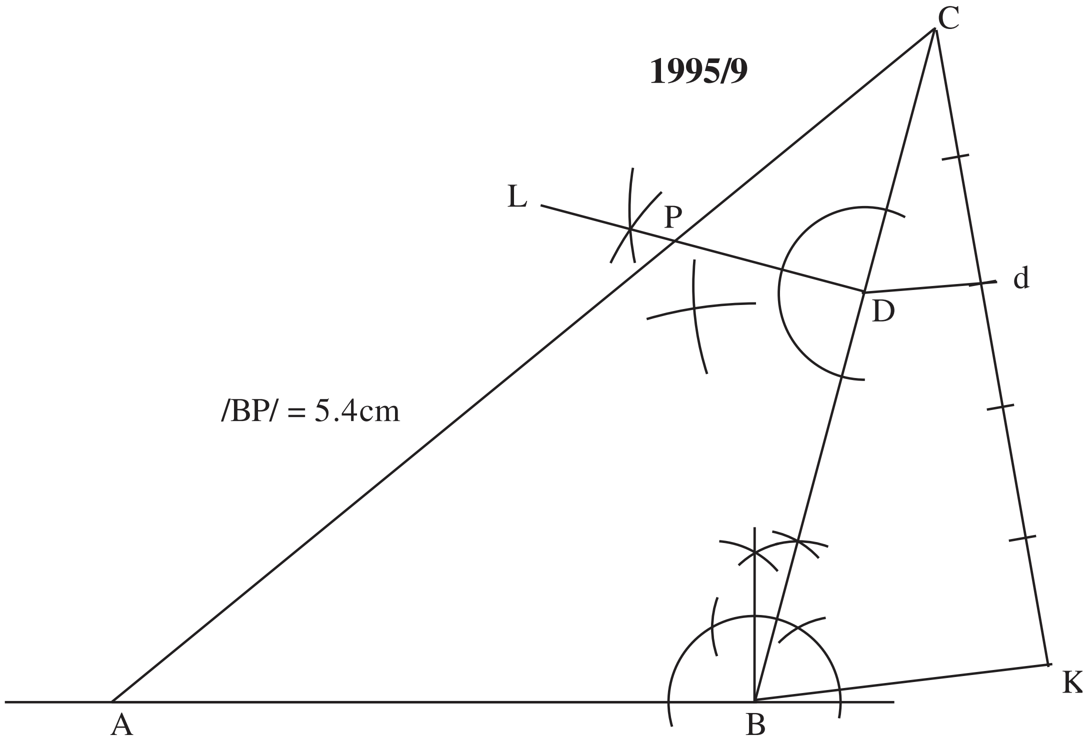
BC = 11cm

2007/6 Neco

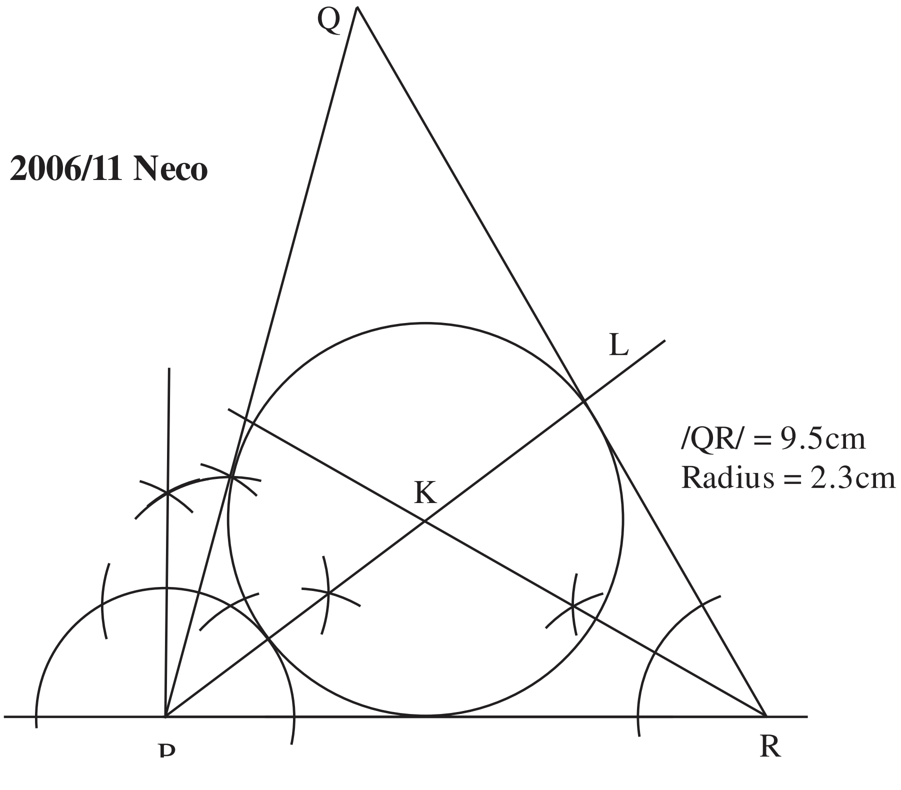


$|T_1 T_2| = 7.7cm$

1995/9



2006/11 Neco



2006/9(a) Exercise 22.8

(a) Using a ruler and a pair of compasses only :

(i) construct ΔXYZ such that /XY/ = 8cm and $\angle YXZ = \angle ZYX = 45^0$;

(ii) locate a point P inside the triangle equidistant from YX and YZ ;

(iii) construct a circle touching the three sides of the triangle .

(iv) measure the radius of the circle .

Construction of triangles with circumcircle

It is done by perpendicular bisector of any two sides of the triangle. Their point of intersection is the centre of the circle touching all the vertices of the triangle *and it is of equal distance from all the vertices*

2015/10 Neco

(a) Construct an equilateral triangle ABC with sides 8cm

(b) Construct the perpendicular bisectors of AB and AC to meet at point M

(c) Draw a circumcircle to pass through points A, B, C.

(d) Measure the radius of the circle

Analysis and solution ***Diagram on page 399***

Steps taken

1. Draw a line AB = 8cm with extensions as shown
2. With radius 8cm and centre at A draw an arc, next with centre at B and same radius draw another arc to cut the first one.
3. Join their point of intersection C to A and B
4. With centre at A, radius beyond centre AB draw an arc at top and bottom of AB
5. With same radius and centre B, draw an arc to cut the set of arc in 4.
6. Join their point of intersection
7. Repeat steps 4, 5 and 6 at AC
8. The point of intersection forms the centre of the circumcircle with radius A or B or C

2007/9

(a)Using a ruler and a pair of compasses only , Construct :

(i) A triangle PQR such that $|PQ| = 10cm$, $|QR| = 7cm$ and $\angle PQR = 90^0$.

(ii) the locus l_1 of points equidistant from Q and R;

(iii) the locus l_2 of points equidistant from P and Q

(b) Locate the point O equidistant from P, Q and R

(c) With O as centre, draw the circumcircle of the triangle PQR

(d) Measure the radius of the circumcircle

Analysis and solution ***Diagram on page 399***

Steps taken

1. Draw line QR = 7cm with extensions as shown
2. At Q, construct 90^0 with extension to P 10cm
3. join P to R
4. locus l_1 is the bisector of line QR
5. locus l_2 is the bisector of line PQ
6. Point O is the intersection of l_1 and l_2, also the centre of the circumcircle

2004/9 Exercise 22.9

Using a ruler and a pair of compasses only .

(a) Construct :

(i) ΔPQR such that /PQ/ = 8cm , /PR/ = 7cm and $\angle QPR = 105^0$.

(ii) locus l_1 of points equidistant from P and Q .

(iii) locus l_2 of points equidistant from Q and R

(b) (i) Label the point T where l_1 and l_2 intersect .

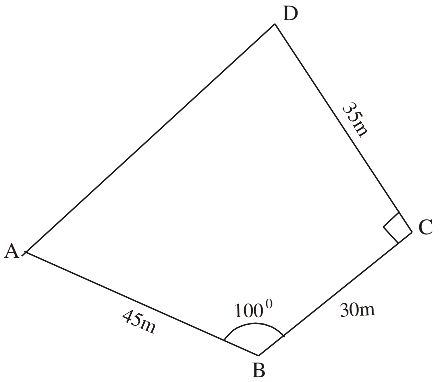
(ii) With centre T and radius /TQ/ , construct a circle l_3

(iii). Complete quadrilateral PQSR such that /RS/ = /QS/ and /TQ/ = /TS/ .

SCALED, COPIED DIAGRAM CONSTRUCTION

2002/11 (Nov)

The diagram below represents a lawn ABCD where AB = 45m, BC = 30m, CD = 35m, $\angle ABC = 100^0$ and $\angle BCD = 90^0$.



(a) Using a scale of 1cm : 5m draw an accurate diagram of the lawn .

(b) Construct the locus:

(i) l_1 of points equidistant from AD and DC ,

(ii) l_2 of points 25m from B

(c) Shade the region of all points in the lawn which are not more than 25m from B and nearer AD than DC .

Analysis and solution ***Diagram on page 400***

Steps taken

1 : Scale conversions- AB = 9cm from 45m, BC = 6cm from 30m, CD = 7cm from 35m

2 : Draw a line AB = 9cm with extension as shown in the diagram

3 : At B measure out 100^0 with your protractor.(Do not construct) and draw line BC = 6cm through it

4 : At C measure out 90^0 with protractor and draw line CD = 7cm through it

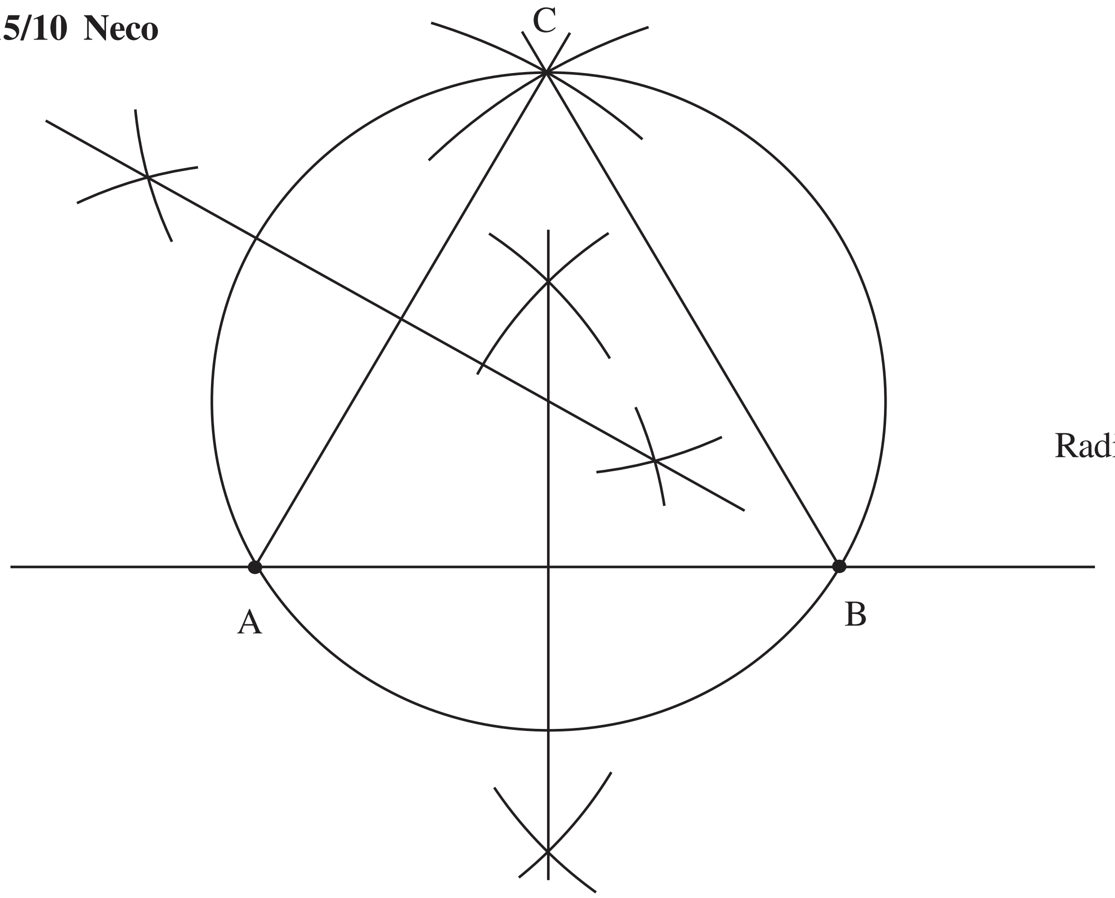
5 : Join D to A to complete diagram

6 : Construction of locus l_1 is the bisector of angle D with the aid of compass and ruler

7 : Construction of locus l_2 is a circle with radius 5cm(from 25m) and centre B

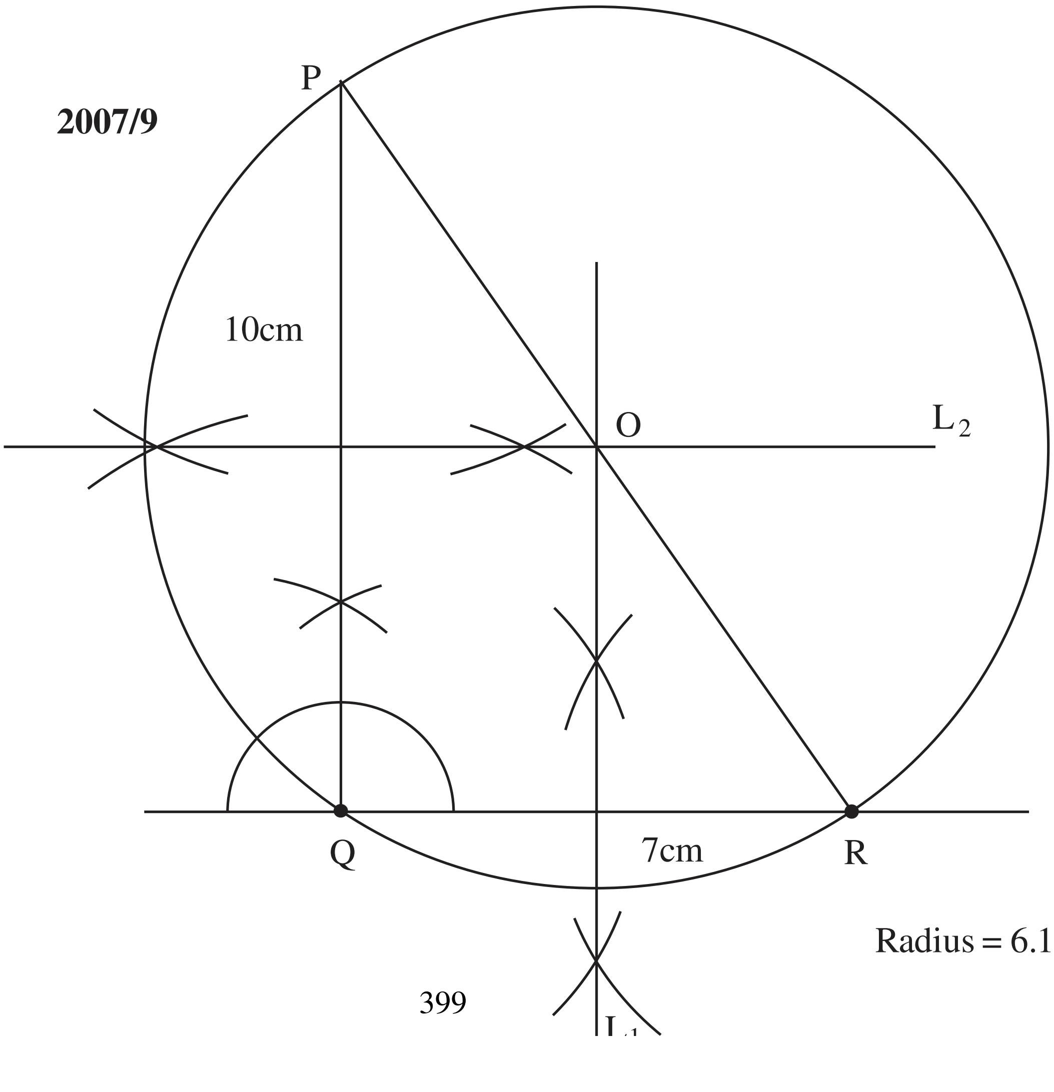
The region is shaded in the diagram

2015/10 Neco

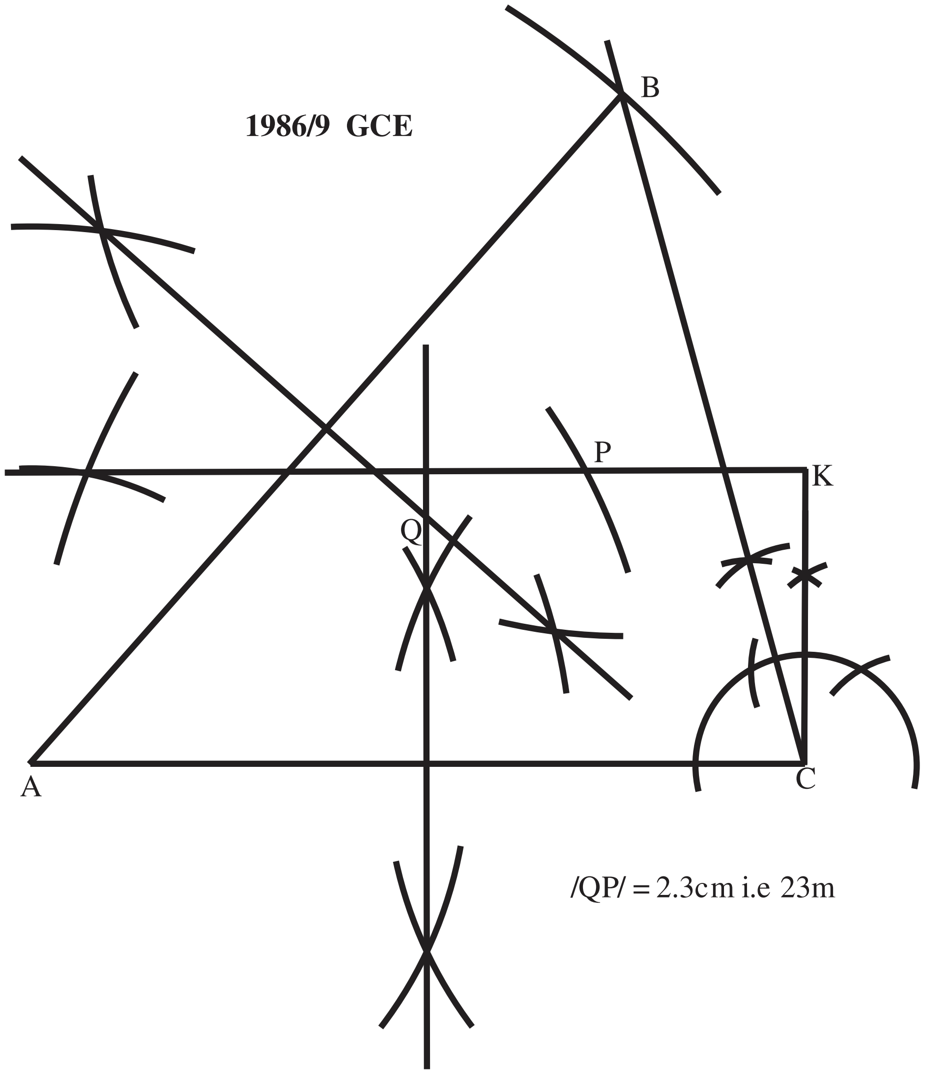


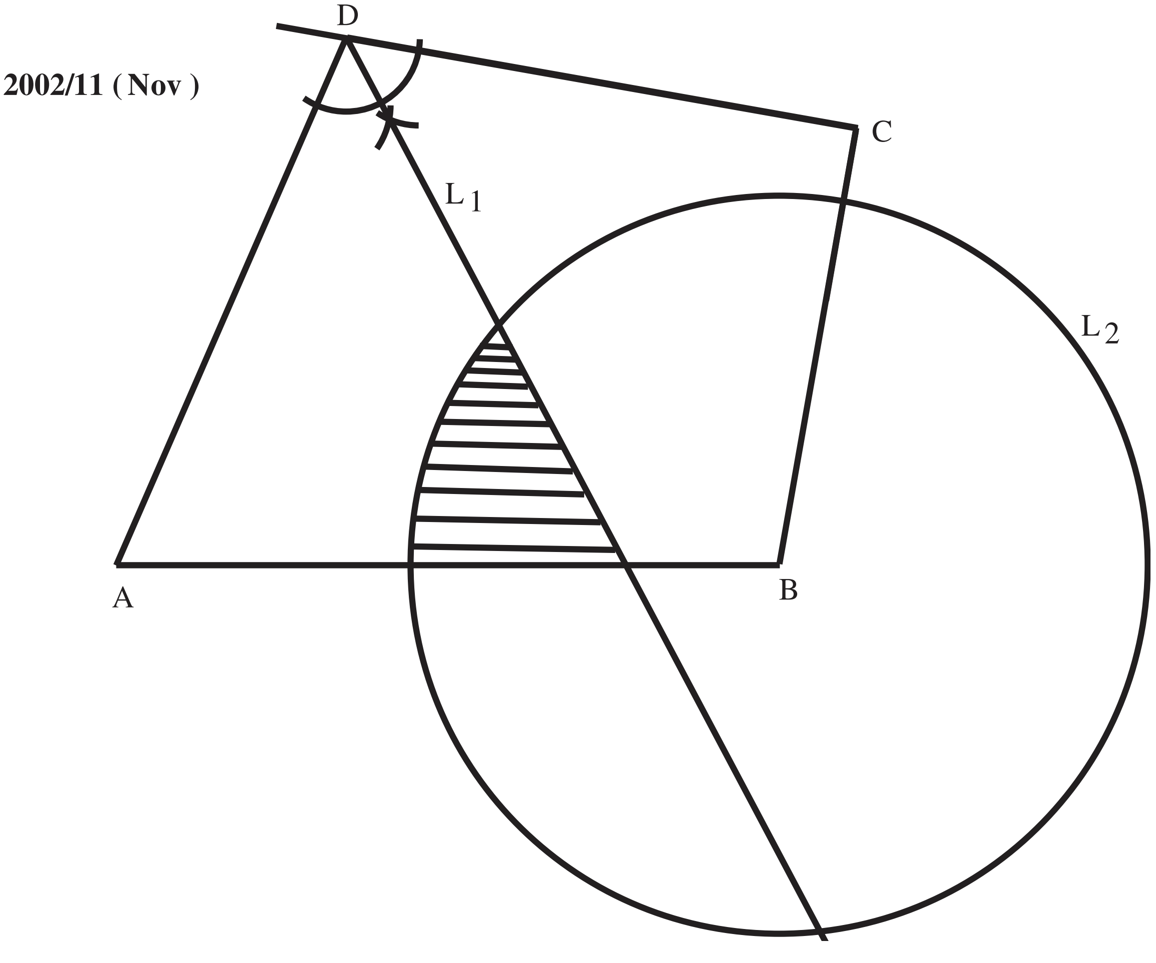
Radius 4.6cm

2007/9



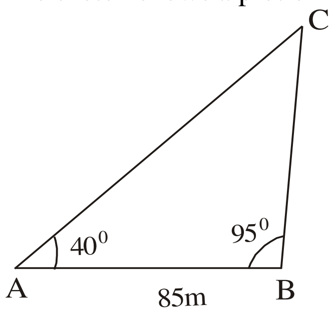
Radius = 6.1cm


1986/9 GCE
B
P
K
Q
A
C
/QP/ = 2.3cm i.e 23m


D
2002/11 (Nov)
C
L_1
L_2
A
B

2005/10 Exercise 22.10

The sketch shows a plot of land.



(a) Using a scale of 1cm to 10m draw an accurate diagram of the plot .

(b)Construct :

(ii) The locus l_1 of points equidistant from AC and BC .

(ii) The locus l_2 of points 60m from A.

(c) A tree T inside the plot, is on both $l_1$and l_2.

Locate T and find /TC/ in metres .

(d) A flag pole , P is to be placed such that it is nearer

AC than BC and more than 60m from A .

Shade the region where P can be located .

1986/9 GCE

A plot of land is in the form of a triangle ABC in which /AB/ = 121m , /AC/ = 105m and $\angle ACB = 75^0$.

Using a ruler and pair of compasses only, and a scale of 1cm to 10m

(a) construct :

(i) the triangle ABC ,

(ii) the position of a flagpole P within ΔABC which is 85m from A and 40m from AC ;

(iii) the position of another flagpole Q which is equidistant from A, B and C;

(b) find the distance between the flagpoles P and Q

Analysis and solution ***Diagram on page 400***

Steps taken

1 : Scale conversions- AB = 12.1cm from 121m, AC = 10.5cm from 105m

2 : Draw a line AC = 10.5cm with extension as shown

3 : At point C construct angle 75^0 and produce a line through it to a reasonable extent

4 : With centre at A and radius 12.1cm draw an arc to cut the line drawn through 75^0 at point B

5 : Join B to A to complete the triangle

6 : Construction of condition (ii)

First; Measure CK 4cm(40m) through 90^0 line as shown.

Secondly, at K construct a parallel line to AC using steps of point outside a parallel line

Thirdly, with centre at A and radius 8.5cm(85m) draw an arc to cut the parallel line at P

7 : Construction of condition (iii) means the point of intersection of constructed perpendicular bisectors of two sides of the triangle as shown in the diagram

Construction of quadrilaterals

The basic ideas about the properties of quadrilaterals are useful in their construction. We will break down this subtopic as:

*Triangles to quadrilateral construction

*Direct quadrilateral: whose name say trapezium, parallelogram, rhombus etc. is given in the beginning of the question

*Indirect quadrilateral; whose name is not mentioned. In this case just be guided in your sketch by the fact that it is a FOUR SIDED FIGURE . Don’t be bias to sketch in favour of any quadrilateral you know; let only the question’s conditions be your guide

Triangles to quadrilaterals construction

2016/10

Using a ruler and a pair of compasses only ,

(a) construct :

(i)ΔXYZ such that $|XY| = 10cm$, $\angle XYZ = 30^0$ and $\angle YXZ = 45^0$;

(ii) locus l_1 of points equidistant from Y and Z;

(iii) locus l_2 of points parallel to XY through Z

(b) locate point M, the point of intersection of l_1 and l_2

(c) Measure $\angle ZMY$

Analysis and Solution Diagram on page 402

Steps taken

1. Draw line XY = 10cm with extensions as shown
2. At Y construct angle 30^o (Bisecting of 60^0)
3. At X construct angle 45^o (Bisecting of 90^0)
4. Join the lines to meet at Z
5. locus l_1 is the perpendicular bisector of line YZ
6. locus l_2 is a parallel line to a given line at a fixed point outside the line
7. With radius XY and centre Z draw an arc
8. With radius YZ and centre X draw an arc to cut the 1^{st} one
9. Join Z to the point of intersection and produce to meet l_1 at M

1998/9

Using a ruler and a pair of compasses only , construct

(a) a triangle QRT with /QR/ = 8cm , /RT/ = 6cm and /QT/ = 4.5cm

(b) a quadrilateral QRSP which has a common base QR with ΔQRT such that QTP is a straight line . PQ//SR, /QP/ = 9cm and /RS/ = 4.5cm .

(i) Measure /PS/ .

(ii) Find the perpendicular distance between RS and PQ.

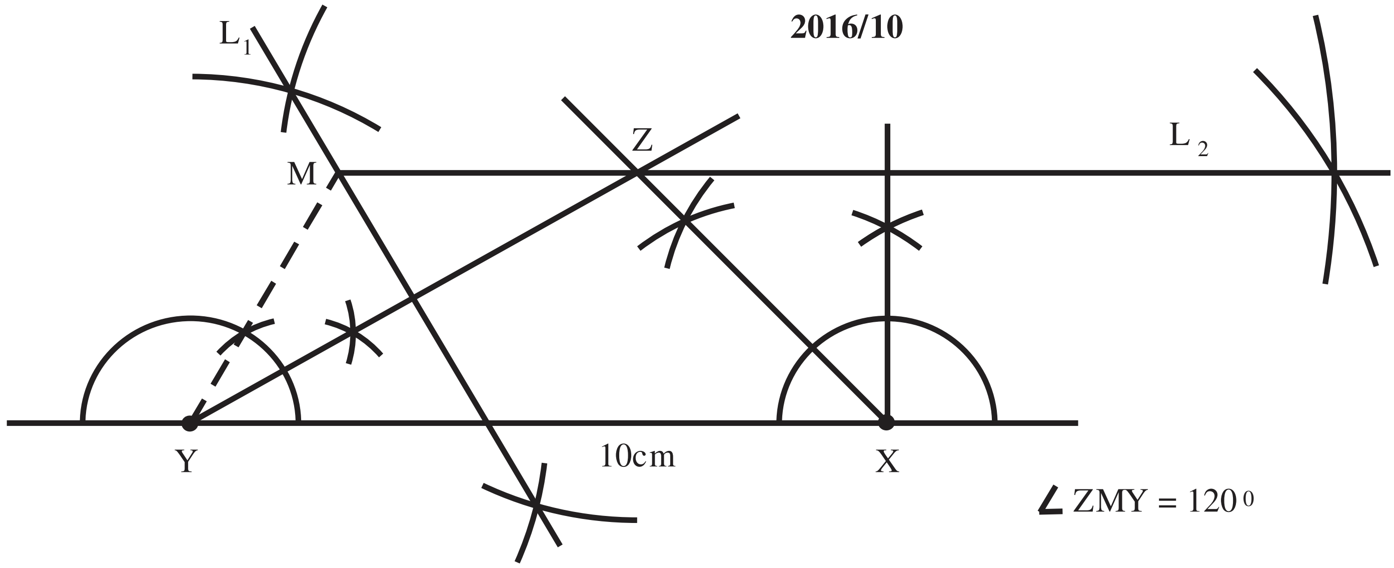
(iii) What is QRSP ?

Analysis and Solution Diagram on page *402*

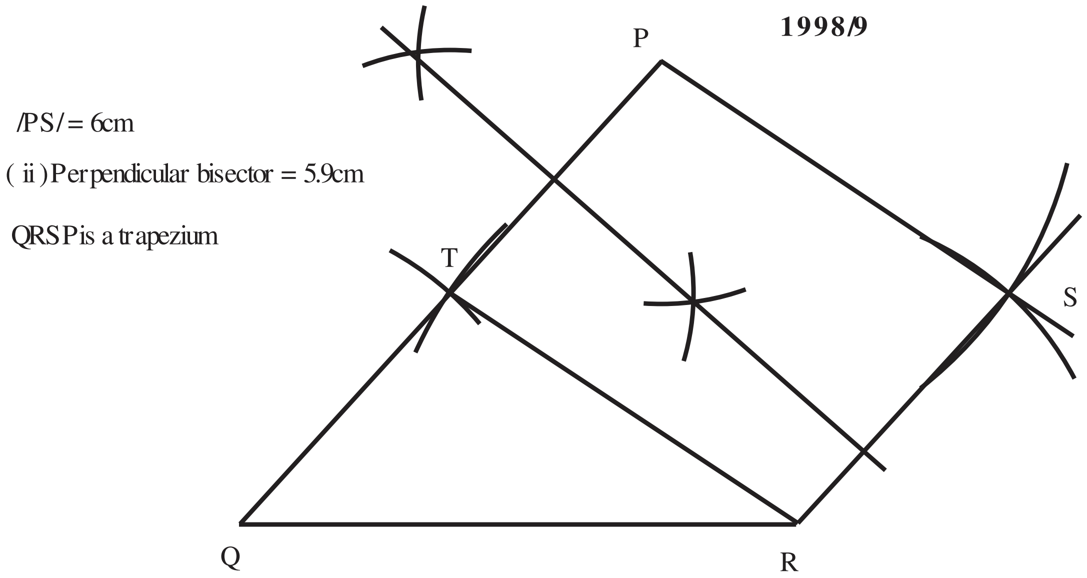
Steps taken:

1. Draw a line QR = 8cm with extensions as shown in the diagram.
2. With center Q and radius 4.5cm draw an arc.
3. With center R and radius 6cm draw another arc to cut the former one at T.
4. Join T to Q and T to R to complete the diagram.

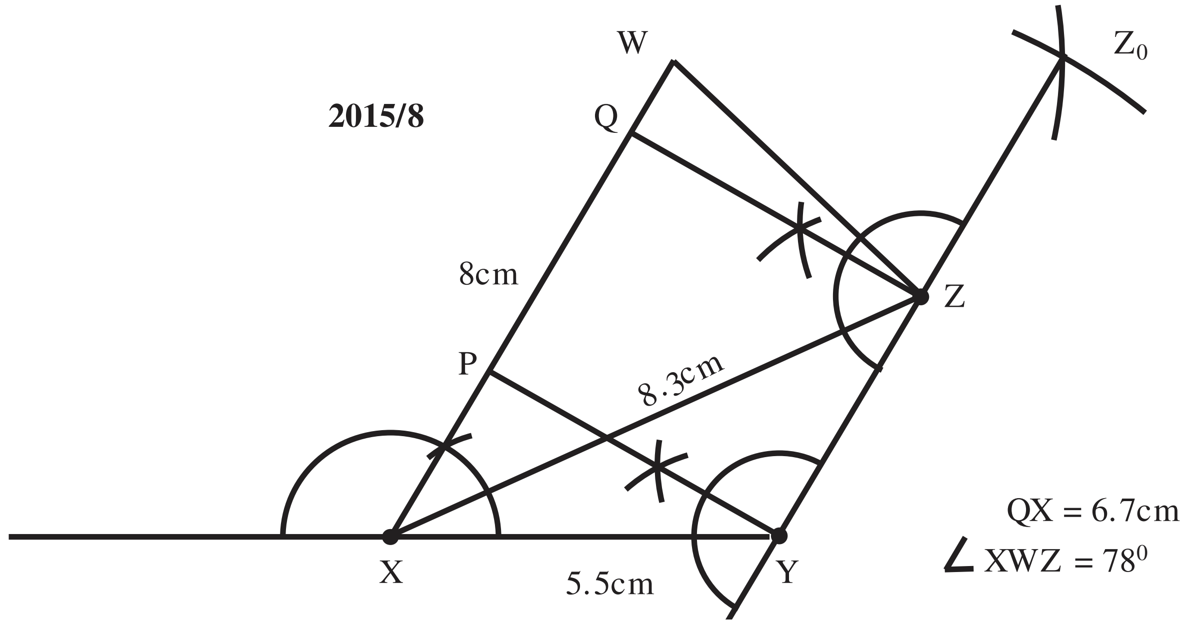
2016/10



1998/9



2015/8



5. Produce T to P 4.5cm long since QP = 9cm and QT = 4.5cm
6. At P construct a parallel line to TR using step of a point outside a parallel lines
7. Produce R 4.5cm long to meet the parallel line at S.
8. To fulfill condition (ii) we construct a perpendicular bisector on line TP and produce it to touch RS

1997/7 Exercise 22.11

(a) Using a ruler and a pair of compasses only, construct ΔABC in which /AB/ = 7cm , /BC/ = 5cm and $\angle ABC = 75^0$. Measure AC .

(b) In (a) above, locate by construction , a point D such that CD is parallel to AB and D is equidistant from points A and C . Measure $\angle BAD$

Direct quadrilateral construction

Trapezium

2015 /8

(a) Using a ruler and a pair of compasses only , construct a:

(i) trapezium WXYZ such that $|WX| = 8$cm, $|XY| = 5.5$cm, $|XZ| = 8.3$cm, $\angle WXY = 60^0$ and WX//ZY

(ii) rectangle PQYZ where P and Q are on $\overline{WX}$

(b) Measure : (i) $|QX|$; (ii) $\angle XWZ$

Analysis and Solution Diagram on page *404*

Steps taken

1. Draw a line WX = 8cm with extensions as shown
2. At X construct angle 60^o
3. Produce X to Y 5.5cm long
4. Construct a line ZY parallel to WX
5. With radius WX and centre Y, draw an arc
6. With radius XY and centre W, draw an arc to cut the 1st one
7. join Z_o to Y
8. Measure X to Z_o 8.3cm to give Z
9. At Y and Z, construct perpendicular bisectors to touch WX at P and Q

Alternative step 1 - 3

1. Draw a line XY = 5.5cm with extensions as shown
2. At X construct angle 60^o
3. Produce X to W 8cm long

step 4 – 9 are the same here

Alternative step 1 – 3 Diagram on page <u>402</u>

2012/10 Neco

(a) Using a ruler and a pair of compasses only , construct :

(i) a trapezium PQRS in which the parallel sides PQ and SR are 5cm apart, $\angle SPQ = 60^0$, $|PQ| = 9$cm and $|SR| = 6$cm

(ii) locus l_1 of points equidistant from P and Q

(iii) locus l_2 of points 7cm from P

(b) (i) Locate M_1 and M_2, points of intersection of l_1 and l_2

(ii) Measure $|M_1M_2|$

Analysis and Solution Diagram on page 404

Steps taken

1. Draw line PQ = 9cm with extension as shown
2. At P construct angle 60^o
3. To draw PQ//SR that is 5cm apart: with radius 5cm and two centres that are not too close on line PQ
4. Carefully draw a straight line SR = 6cm touching the tops of the two arcs and parallel to line PQ
5. Join R to Q.
6. locus l_1 is the perpendicular bisector of line PQ
7. locus l_2 is a circle with radius 7cm and centre P

1977/5 Exercise 22.12

(a) Using a ruler and a pair of compasses only , construct a trapezium ABCD, in which the parallel sides AB and DC are 2cm apart. $\angle DAB = 60^0$, AB = 4cm and BC = 2.5cm.
Measure the possible lengths of DC to the nearest cm

Parallelogram

2014/8 (Nov)

(a) Using a ruler and a pair of compasses only , construct

(i) parallelogram PQRS with RS as the base such that $|PQ| = 7.8$cm, $|QR| = 5.6$cm and angle QRS = 120^0;

(ii) rectangle ABRS, equal in area to parallelogram PQRS

(b) Measure : (i) $|AP|$; (ii) $|AS$

Analysis and Solution Diagram on page 405

Steps taken

1. Draw line RS with extensions as shown
2. At R construct angle 120^0
3. Produce R to Q 5.6cm long
4. Construct line QP parallel to RS
5. With radius QR centre S_o draw an arc
6. With radius RS_o centre Q draw another arc to cut the 1st
7. Join Q to P 7.8cm long through the intersection of the arcs
8. Construct another line PS parallel to QR
9. With radius QP and centre R draw an arc
10. With radius QR and centre P, draw an arc to cut the 1st one
11. Produce P to S
12. Draw a perpendicular at R and S to touch QP at A and B as specified rectangle AB**RS (i.e** RS as the base)

2013/11 Neco

(a) Using a ruler and a pair of compasses only: construct;

(i) a Parallelogram EFGH with $|EF| = 9$cm, $|EG| = 11.5$cm and $\angle EFG = 105^0$

(ii) a circle to touch angles E, F and G

(b) (i) measure $|FH|$ and

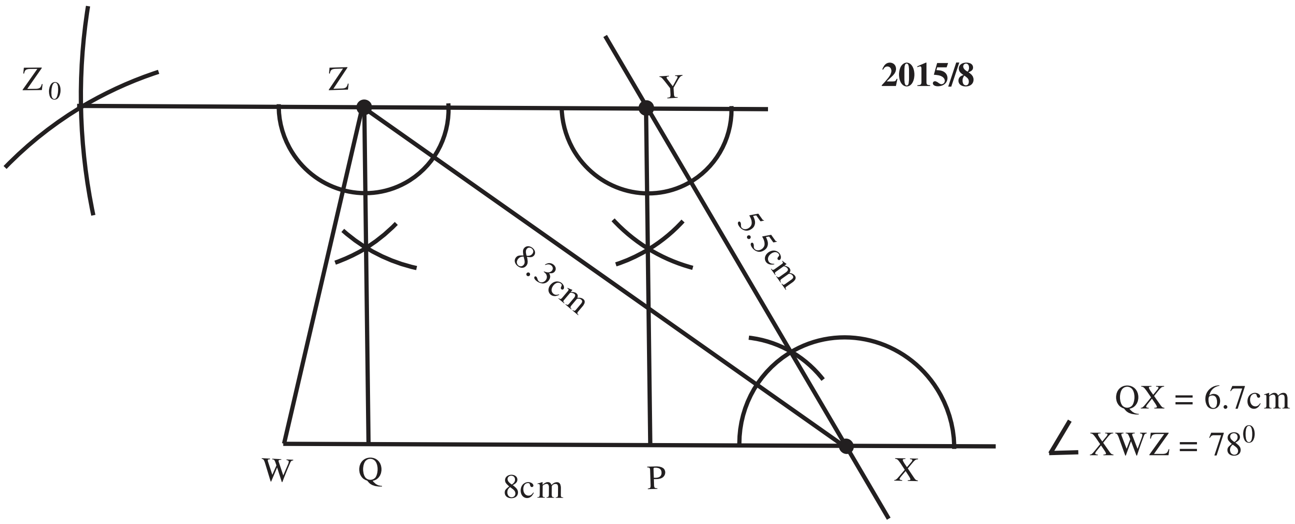
(ii) radius of the circle

(c) determine the area of the parallelogram

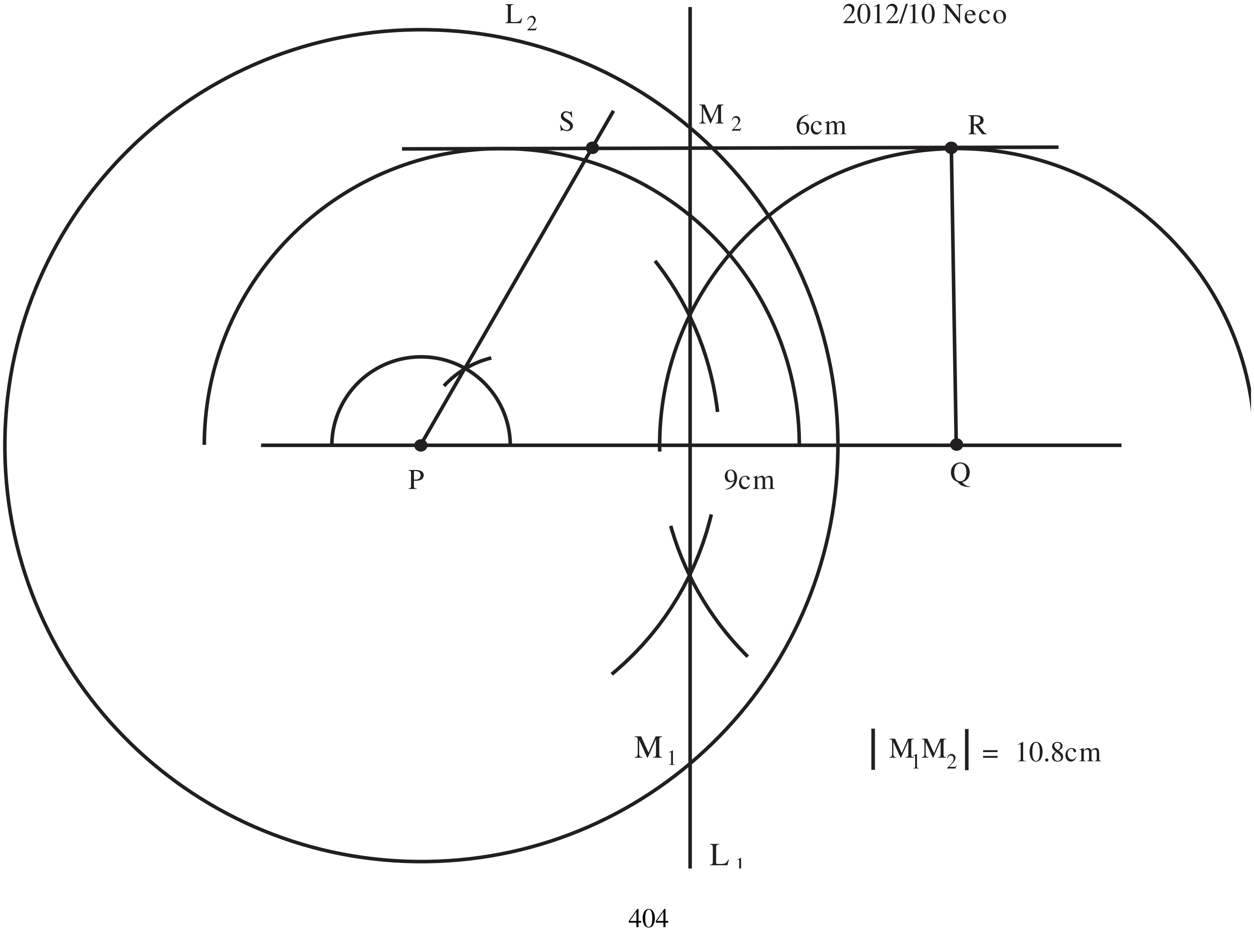
Analysis and Solution Diagram on page *405*

Steps taken

1. Draw line EF = 9cm with extensions as shown
2. At F construct angle 105^o (i.e constructing 90^0 between 60^0 and 120^0 then bisecting it to give $15^0 + 90^0$)
3. Join E to G at 11.5cm long

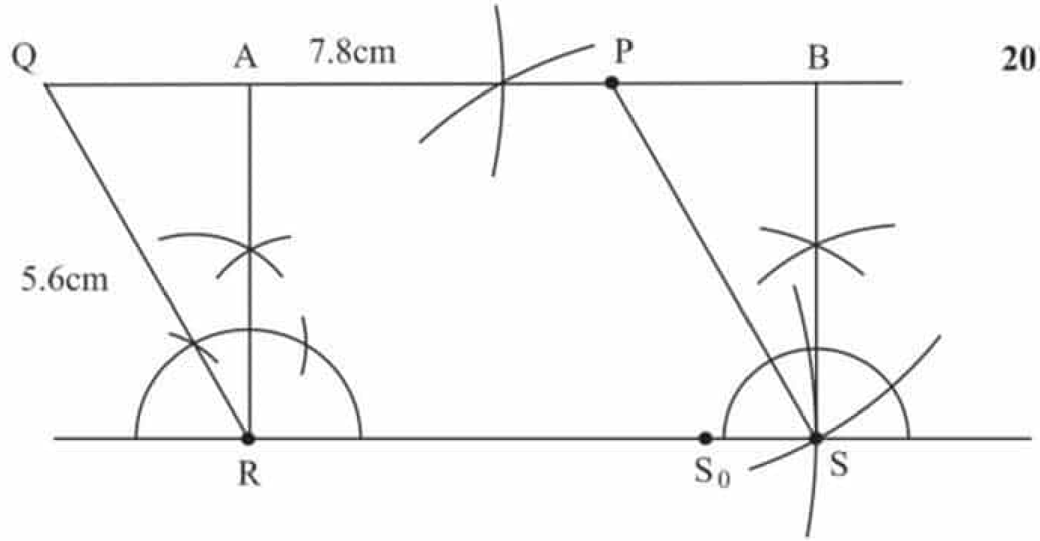


2015/8
Z0
Z
Y
5.5cm
8.3cm
W
Q
8cm
P
X
QX = 6.7cm
∠XWZ = 78⁰



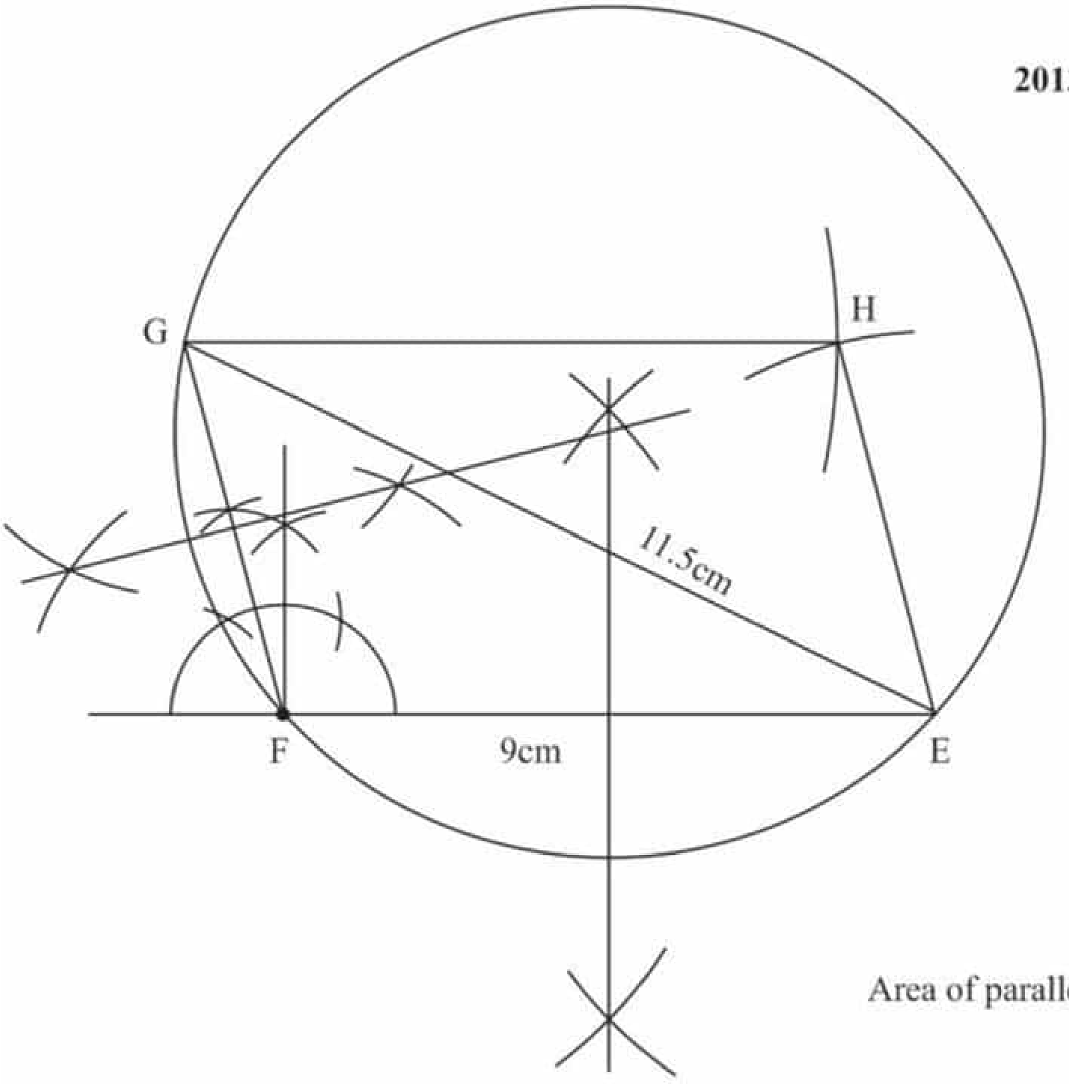
L2
2012/10 Neco
S
M2
6cm
R
P
9cm
Q
M1
|M1M2| = 10.8cm
L1

2014/8 (Nov)



AP = 5cm
AS = 9.2cm

2013/11 Neco



FH = 9cm
Radius = 6.0cm

Area of parallelogram = GF x EF sin 105^0
= 5.2 x 9 x 0.9659
= $45.20cm^2$

4. Construct a parallel line to EF
5. With radius EF and centre G draw an arc
6. With radius GF and centre E, draw another arc to cut the 1st one
7. Join G to H point of the arc intersection
8. Construct a circle to touch angles E, F and G is a circumcircle to ΔEFG
9. Bisect line EF and line GF, their point of intersection is the centre of the circle.

Rhombus

2011\9

Using a ruler and a pair of compasses only ,

(a) Construct a rhombus PQRS of side 7cm and $\angle PQR = 60^0$

(b) locate a point X , such that X lies on the locus of points equidistant from PQ and QR and also equidistant from Q and R :

(c) Measure $|XR|$

Analysis and Solution **Diagram on page *407***

Steps taken

1. Draw a straight at Q to a reasonable length
2. At Q construct 60^0 at the top and 60^0 at the bottom as shown
3. Extend Q to P 7cm long and Q to S 7cm long
4. Join S to the line at Q meeting at R 7cm long also join P to it 7cm long
5. locus l_1 of PQ and QR is bisector of angle PQR
6. locus l_2 of Q and R is the perpendicular bisector of line QR

2015\9b (Nov) **Exercise 22.13**

(i) Using a ruler and a pair of compasses only , Construct a rhombus MNOQ of length 10cm and $\angle MNO = 45^0$

(ii) Measure : (a) $|MO|$, (b) $|NQ|$

Indirect quadrilateral construction

2008/9 NABTEB (Nov)

(a) Using a ruler, a pair of compasses and a set squares only construct a quadrilateral PQRS such that $\angle PQR = 75^0$ $\angle QRS = 60^0$ PQ = 6 cm, QR = 8 cm and PS //QR.
Measure RS

(b) construct the

(i) locus l_1 of points equidistant from Q and R

(ii) locus l_2 of points equidistant from QR and RS

(c) Locate the point of intersection T of l_1 and l_2.
Measure PT

Analysis and Solution **Diagram on page 407**

Steps taken

1. Draw a line QR = 8cm with extensions as shown
2. At point Q construct angle 75^0 (120^0, then bisect between 60^0 and 60^0 to get 90^0, then bisect between 90^0 and 60^0 to get 75^0)
3. Produce line Q to P 6cm long
4. Construct line PS parallel to QR
5. With radius QR and centre P, draw an arc
6. With radius QP and centre R, draw another arc to cut the 1st one
7. Join P to the point of intersection S_o
8. At R construct 60^0
9. Produce R to meet PS_o at S
10. locus l_1 is the perpendicular bisector of line QR
11. locus l_2 is the bisector of angle QRS

2003/12 **Exercise 22.14**

Using a ruler and a pair of compasses only,

(a) Construct a quadrilateral PXYQ such that /PX/ = 9.9cm , /QX/ = 10.2cm , $\angle QPX = 75^0$, /QY/ = 10.4cm and PQ//XY.

(b) Construct the :

(i) locus l_1 of points equidistant from X and Y,

(ii) locus l_2 of points equidistant from QY and YX .

(c) Locate M, the point of intersection of l_1 and l_2 .

(d) Measure /PM/ .

2001/9 **Exercise 22.15**

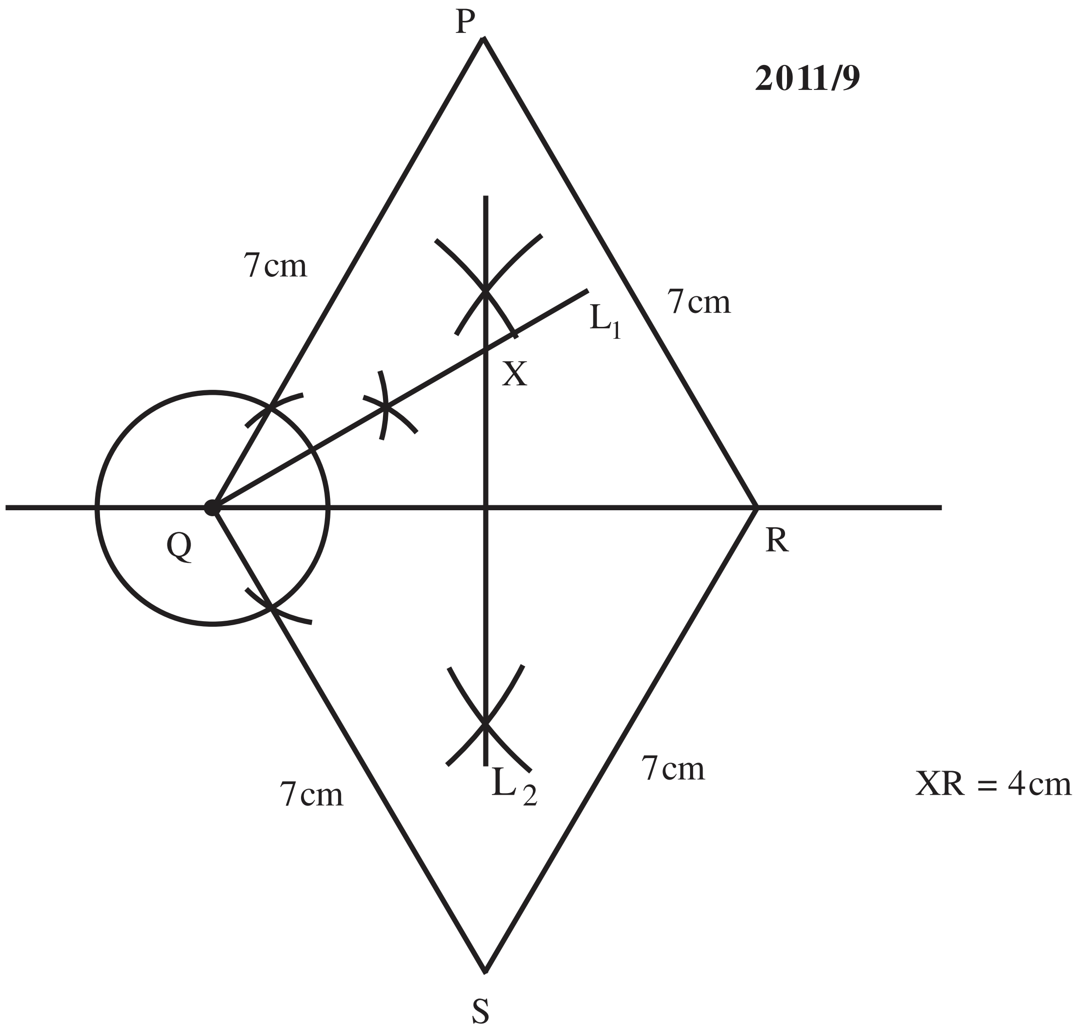
Using a ruler and a pair of compasses only ,

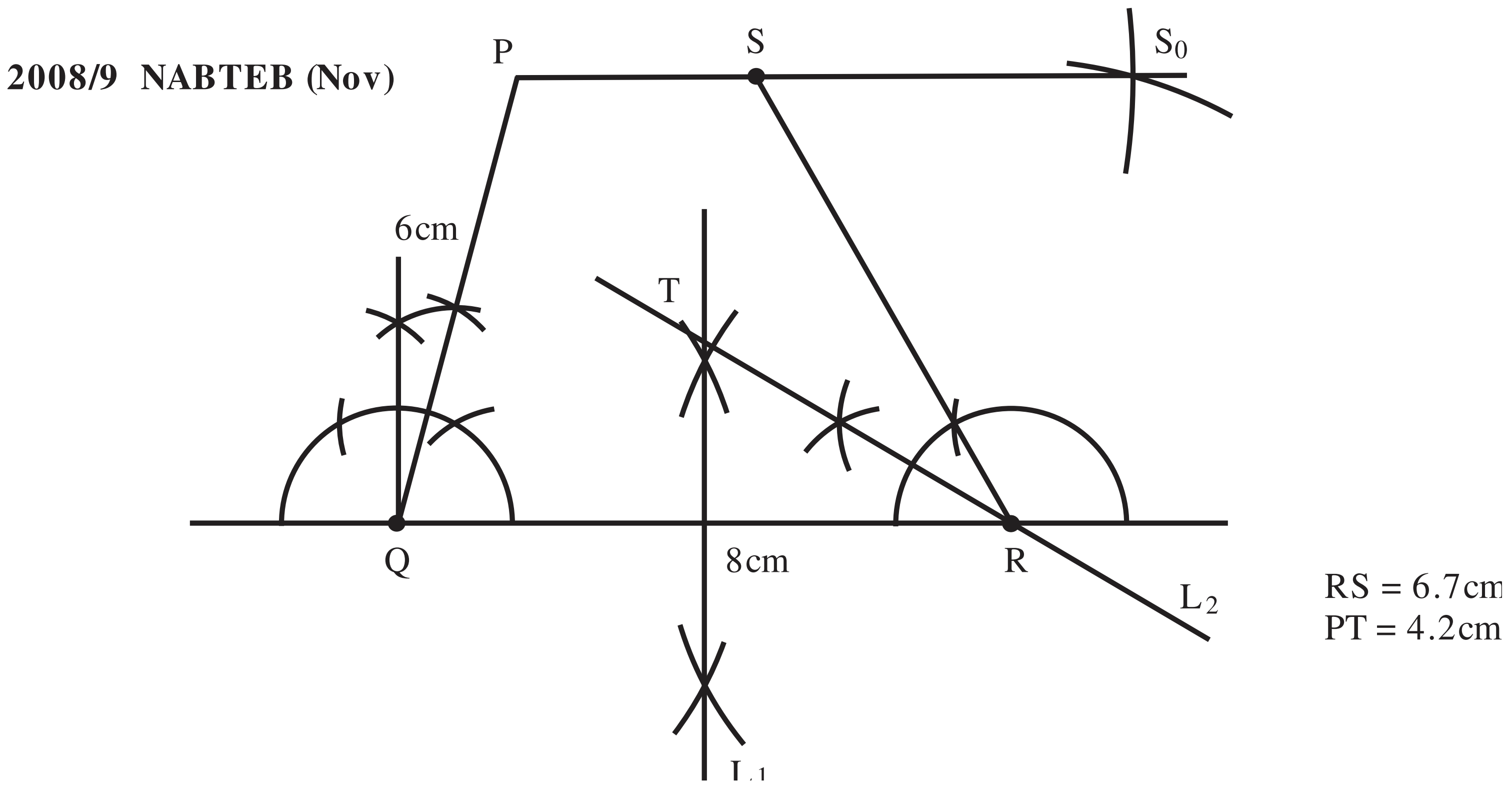
(a) Construct quadrilateral MNOP with /OP/ = 11cm, /PM/ = 6cm , /NO/ = 5cm , /MN/ = 8cm and /MO/ = 9cm .

(b) Measure $\angle MPO$.

(c) Locate the point Y which is equidistant from the straight lines ON and OP and also equidistant from points M and N.

(d) Draw $\overline{YP}$ and find the perimeter of ΔOYP.


P
2011/9
7cm
7cm
L1
X
Q
R
7cm
7cm
L2
XR = 4cm
S


P
S
S0
2008/9 NABTEB (Nov)
6cm
T
Q
8cm
R
L2
RS = 6.7cm
PT = 4.2cm

Chapter twenty three

Modulo arithmetic

This is the arithmetic of remainders when a number is divided by a lower number. From our counting numbers, each of them has its set of remainders (residues) :

4 is {0, 1, 2, 3}

5 is {0, 1, 2, 3, 4}

6 is {0, 1, 2, 3, 4, 5}

Generally, for any number **n, the residues are**

{0, 1, 2, 3, 4, …, n – 1 }

Some of the commonly used symbols are

≡ for congruent or equivalent

⊗ for multiplication and ⊕ for addition

2009/14a

(a) Copy and complete the multiplication table modulo 5 on the set {1, 2, 3, 4, }

∗	1	2	3	4
1	1		3	
2		4		
3	3			2
4		3		1

(b) From your table

i. Solve the expression 2n ∗ 4 ≡ 3

ii. Find the value of n for which

$2 * (3 * n) \equiv 2$

Solution

1 ∗ 2 = 2 to mod 5
= 2

1 ∗ 4 = 4 to mod 5
= 4

2 ∗ 4 = 13 to mod 5
= 3

3 ∗ 2 = 11 to mod 5
= 1

3 ∗ 3 = 14 to mod 5
= 4

4 ∗ 1 = 4 to mod 5
= 4

4 ∗ 3 = 22 to mod 5
= 2

Multiple table for {1, 2, 3, 4} mod 5

∗	1	2	3	4
1	1	2	3	4
2	2	4	1	3
3	3	1	4	2
4	4	3	2	1

b. (i) 2n ∗ 4 ≡ 3

From the table, which operation in column 2 gave RHS 3

i.e 2 ∗ 4 = 3

is only possible if 2 is unchanged

Thus, 2 × 1 ∗ 4 = 3

hence n = 1

(ii) 2 ∗ (3 ∗ n) ≡ 2

From the table, which operation in column 2 gave RHS 2

2 ∗ 1 = 2

Thus, 2 ∗ (3 ∗ n) ≡ 2 implies

3 ∗ n ≡ 1

From the table, which operation in column 3 gave RHS 1

3 ∗ 2 = 1

Hence n = 2

2016 / 13b

The operation Δ is defined on the set

T = {2, 3, 5, 7} by XΔY = (x + y + xy) mod 8

(i) Construct modulo 8 table for the operation Δ on the set T.

(ii) Use the table to find

i. 2 Δ (5 Δ 7)

ii. 2 Δ n = 5 Δ 7

		x			
	Δ	2	3	5	7
	2	0	3	1	7
y	3	3	7	7	7
	5	1	7	3	7
	7	7	7	7	7

2 Δ 2 = (2 + 2 + 2 × 2) to mod 8
= (4 + 4) = 8 *i.e 8 ÷ 8 = 1 remainder 0*
= **0**

2 Δ 3 = (2 + 3 + 2 × 2) to mod 8
= (5 + 6) = 11 *i.e 11 ÷ 8 = 1 remainder 3*
= **3**

2 Δ 5 = (2 + 5 + 2 × 5) to mod 8
= (7 + 10) = 17 *i.e 17 ÷ 8 = 2 remainder 1*
= **1**

2 Δ 7 = (2 + 7 + 2 × 7) to mod 8
= (9 + 14) = 23 *i.e 23 ÷ 8 = 2 remainder 7*
= **7**

3 Δ 2 = (3 + 2 + 3 × 2) to mod 8
= (5 + 6) = 11 *i.e 11 ÷ 8 = 1 remainder 3*
= **3**

3 Δ 3 = (3 + 3 + 3 × 3) to mod 8
= (6 + 9) = 15 and *15 ÷ 8 = 1 remainder 7*
= **7**

3 Δ 5 = (3 + 5 + 3 × 5) to mod 8
= (8 + 15) = 23 *i.e 23 ÷ 8 = 2 remainder 7*
= **7**

$3 \Delta 7 = (3 + 7 + 3 \times 7)$ to mod 8
$= (10 + 21) = 31$ *i.e 31 ÷ 8 = 3 remainder 7*
$= \mathbf{7}$

$5 \Delta 2 = (5 + 2 + 5 \times 2)$ to mod 8
$= (7 + 10) = 17$ *i.e 17 ÷ 8 = 2 remainder 1*
$= \mathbf{1}$

$5 \Delta 3 = (5 + 3 + 5 \times 3)$ to mod 8
$= (8 + 15) = 23$ *i.e 23 ÷ 8 = 2 remainder 7*
$= \mathbf{7}$

$5 \Delta 5 = (5 + 5 + 5 \times 5)$ to mod 8
$= (10 + 25) = 35$ *i.e 35 ÷ 8 = 4 remainder 3*
$= \mathbf{3}$

$5 \Delta 7 = (5 + 7 + 5 \times 7)$ to mod 8
$= (12 + 35) = 47$ *i.e 47 ÷ 8 = 5 remainder 7*
$= \mathbf{7}$

$7 \Delta 7 = (7 + 7 + 7 \times 7)$ to mod 8
$= (14 + 49) = 63$ *i.e 63 ÷ 8 = 7 remainder 7*
$= \mathbf{7}$

(ii) **i.** $2 \Delta (5 \Delta 7)$
From table $(5 \Delta 7) = 7$
Thus, $2 \Delta (5 \Delta 7) = 2 \Delta 7$
From table $2 \Delta 7 = 7$

ii. $2 \Delta n = 5 \Delta 7$
$2 \Delta n = 7$
From the table
$2 \Delta 7 = 7$
Thus n is 7

2015/ 11c

An operation $\otimes$ is defined on the set
$X = \{1, 3, 5, 6\}$ by $m \otimes n = m + n + 2 \pmod 7$,
where $m, n, \in X$.

i. Draw a table for the operation
ii. Using the table, find the truth set of:
 i. $3 \otimes n = 3$;
 ii. $n \otimes n = 3$

Solution

$m \otimes n = m + n + 2 \pmod 7$ for $\{1, 3, 5, 6\}$

$\otimes$	1	3	5	6
1	4	6	1	2
3	6	1	3	4
5	1	3	5	6
6	2	4	6	0

$1 \otimes 1 = 1 + 1 + 2$ (to mod 7)
$= 4$

$1 \otimes 3 = 1 + 3 + 2$ (to mod 7)
$= 6$

$1 \otimes 5 = 1 + 5 + 2$ (to mod7)
$= 8$ to mod 7 i.e *8 ÷ 7 = 1 remainder 1*
$= \mathbf{1}$

$1 \otimes 6 = 1 + 6 + 2$ (to mod 7)
$= 9$ to mod 7 i.e *9 ÷ 7 = 1 remainder 2*
$= \mathbf{2}$

$3 \otimes 3 = 3 + 3 + 2$ (to mod 7)
$= 8$ to mod 7 i.e *8 ÷ 7 = 1 remainder 1*
$= 1$

$3 \otimes 5 = 3 + 5 + 2$ (to mod 7)
$= 10$ to mod 7 i.e *10 ÷ 7 = 1 remainder 3*
$= 3$

$3 \otimes 6 = 3 + 6 + 2$ (to mod 7)
$= 11$ to mod 7 i.e *11 ÷ 7 = 1 remainder 4*
$= 4$

$5 \otimes 5 = 5 + 5 + 2$ (to mod 7)
$= 12$ to mod 7 i.e *12 ÷ 7 = 1 remainder 5*
$= 5$

$5 \otimes 6 = 5 + 6 + 2$ (to mod 7)
$= 13$ to mod 7 i.e *13 ÷ 7 = 1 remainder 6*
$= 6$

$6 \otimes 6 = 6 + 6 + 2$ (to mod 7)
$= 14$ to mod 7 i.e *14 ÷ 7 = 2 remainder 0*
$= 0$

(ii) i. $3 \otimes n = 3$
From the table, which number did 3 operate on to give 3
$3 \otimes 5 = 3$
Thus n = 5
ii. $n \otimes n = 3$
From the table, which number operate itself to give 3
It does Not exist.

2012 / 14b

Addition: Modulo 6

$\oplus$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	a	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

Multiplication: modulo 6

$\otimes$	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4	0	b	4
3	0	3	0	3	0	3
4	0	4	2	0	4	2
5	0	5	4	3	2	1

Using the addition $\oplus$ and multiplication $\otimes$tables for modulo 6,
i. find the value of a;
ii. determines the value of b;
iii. evaluate $4 \oplus (3 \otimes 2)$;
iv. find m if $5 + (m \otimes 3) = 2$

Solution

i. a is $3 \oplus 2 \pmod 6 = 5$ to mod 6
$= 5$
ii. b is $2 \otimes 4 \pmod 6 = 22$ to mod 6
$= 2$

iii. $4 \oplus (3 \otimes 2) = 4 \oplus 0$ [$3 \otimes 2 = 0$ from multip.table]

$= 4$ [$4 \oplus 0 = 4$ from addition table]

iv. $5 + (m \otimes 3) = 2$

From the addition table,

5 + what = 2

$5 + 3 = 2$

Thus, $m \otimes 3 = 3$

From multiplication table

value(s) that multiply 3 to give 3 are $m = 1, 3$ and 5

But **m = 3** is the accepted value

Based on our earlier assertion

$5 + 3 = 2$ from additional table.

2014/8a

Copy and complete the following table for multiplication modulo 11

⊗	1	5	9	10
1	1	5	9	10
5	5			
9	9			
10	10			

Use the table to:

i. Evaluate $(9 \otimes 5) \otimes (10 \otimes 10)$;

ii. Find the truth set of

i. $10 \otimes m = 2$,

ii. $n \otimes n = 4$

Solution

⊗mod 11 of { 1, 5, 9, 10}

⊗	1	5	9	10
1	1	5	9	10
5	5	3	1	6
9	9	1	4	2
10	10	6	2	1

$5 \otimes 5 = 5 \times 5 = 25$ to mod11 i.e *25 ÷ 11 = 2 remainder 3*

= 3

$5 \otimes 9 = 5 \times 9 = 45$ to mod11 i.e *45 ÷ 11 = 4 remainder 1*

= 1

$5 \otimes 10 = 5 \times 10 = 50$ to mod11 i.e *50 ÷ 11 = 4 remainder 6*

= 6

$9 \otimes 9 = 9 \times 9 = 81$ to mod11 i.e *81 ÷11 = 7 remainder 4*

= 4

$9 \otimes 10 = 9 \times 10 = 90$ to mod11 i.e *90 ÷ 11 = 8 remainder 2*

= 2

$10 \otimes 10 = 10 \times 10 = 100$ to mod11 i.e *100 ÷11 = 9 remainder 1*

= 1

(i) $(9 \otimes 5) \otimes (10 \otimes 10)$

From the table $(9 \otimes 5) = 1$ and $(10 \otimes 10) = 1$

Thus $(9 \otimes 5) \otimes (10 \otimes 10) = 1 \otimes 1$

From the table $1 \otimes 1 = 1$

(ii) i. $10 \otimes m = 2$

What did 10 multiply to give 2?

$10 \otimes 9 = 2$

Thus, $m = 9$

ii. $n \otimes n = 4$

Which number multiple itself to give 4?

$9 \otimes 9 = 4$

Thus, $n = 9$

2009 / 15 (Nov)

(a) Draw the

i. Addition ⊕

ii. Multiplication ⊗

tables for the set x = {1, 2, 3, 4, 5, 6} modulo 6

(b) From your tables, solve

i. $(4 \oplus n) \oplus n = 4$

ii. $t \oplus (t \otimes 3) = 2$

Solution

Mod 6 ⊕ {1, 2, 3, 4, 5, 6}

⊕	1	2	3	4	5	6
1	2	3	4	5	0	1
2	3	4	5	0	1	2
3	4	5	0	1	2	3
4	5	0	1	2	3	4
5	0	1	2	3	4	5
6	1	2	3	4	5	0

Mod 6 ⊗ {1, 2, 3, 4, 5, 6}

⊗	1	2	3	4	5	6
1	1	2	3	4	5	0
2	2	4	0	2	4	0
3	3	0	3	0	3	0
4	4	2	0	4	2	0
5	5	4	3	2	1	0
6	0	0	0	0	0	0

<u>Addition table</u>

1 + 1, 1 + 2, ... 1 + 4 result less than 6

So we write them down like that

$1+5 = 10$ to mod 6 [$1+5 = 6$ and $6 \div 6 = 1$ no remainder]

$= 0$

$1 + 6 = 11$ to mod 6

$= 1$

$2 + 6 = 12$ to mod 6 [$2 + 6 = 8$ and $8 \div 6 = 1$ remainder 2]

$= 2$

The pattern goes on

$6 + 6 = 20$ to mod 6 [$6 + 6 = 12$ and $12 \div 6 = 2$ remainder 0]

$= 0$

<u>Multiplication table</u>

1×1, 1×2, ... 1×5 result less than 6

So we write the result down unchanged

$1 \times 6 = 10$ to mod 6 [$1 \times 6 = 6$ and $6 \div 6 = 1$ remainder 0]

$= 0$

$2 \times 4 = 1\ 2$ to mod 6 [$2 \times 4 = 8$ and $8 \div 6 = 1$ remainder 2]

$= 2$

$3 \times 4 = 20$ to mod 6 [$3 \times 4 = 12$ and $12 \div 6 = 2$ remainder 0]

$= 0$

$3 \times 6 = 30$ to mod 6 [$3\times6 = 18$ and $18 \div 6 = 3$ remainder 0]
$= 0$

$6 \times 6 = 60$ to mod 6 [$6 \times 6 = 36$ and $36 \div 6 = 6$ remainder 0]
$= 0$

b (i) $(4 \oplus n) \oplus n = 4$
our working is in Row 4
Which number did 4 add to that produce 4
$4 \oplus 6 = 4$
Thus $n = 6$
Confirmation by working further
$(4 \oplus n) \oplus n = 4$ becomes
$4 \oplus 6 = 4$ same result

(ii) $t \oplus (t \otimes 3) = 2$
Here our attention will be on result 2 in the Addition tables
$1 + 1 = 2$
$2 + 6 = 2$
$3 + 5 = 2$
$4 + 4 = 2$
$5 + 3 = 2$
$6 + 2 = 2$

Our the results, the bracket item $t \otimes 3$ will force in to be concerned with multiples of 3.
Reason $t \oplus 1 = 2$ implies $1 + 1 = 2$
Then $t \otimes 3 = 1$
Is it true that $1 \otimes 3 = 1$ **<u>No</u>**
Hence we must restrict to multiples of 3
$2 \oplus 6 = 2$ Addition table
Then $2 \oplus (2 \otimes 3) = 2 \oplus 0$ dead end
$5 \oplus 3 = 2$
Then $5 \oplus (5 \otimes 3) = 5 \oplus 3$
$= 2$ **true**

2010/15c
If x is the positive integer, determine the least value of x for which $13 + 2x = 3 \pmod 8$

Solution
$13 + 2x = 3 \pmod 8$
Same as
$13 + 2x = 3 + \mathbf{8} \pmod 8$ it will not help us
Same as
$13 + 2x = 3 + \mathbf{16} \pmod 8$
Take away 13 from both sides
$2x = 6 \pmod 8$
Divide through by 2
$x = 3 \pmod 8$

2015/3 (Nov)
If x is a whole number such that $2x + 1 \equiv 4 \bmod 7$, find the **least** value of x
A. 2 B. 3 C. 4 D. 5

Solution
$2x + 1 = 4 \pmod 7$
Same as
$2x + 1 = 4 + 7 \pmod 7$
Take away 1 from both sides
$2x = 10 \pmod 7$
Since the RHS is already in the multiple of 2;
Divide through by 2
$x = 5$ D.

2003/14a
Find the least integral value of n such that
$4n + 3 \equiv 1 \pmod 6$

Solution
$4n + 3 = 1 \pmod 6$
Same as
$4n + 3 = 1 + 6 \pmod 6$
Take away 3 from both sides
$4n = 4 \pmod 6$
Divide through by 4
$n = 1$

2004/14a
Solve $2x + 4 = 0 \pmod 5$

Solution
$2x + 4 = 0 \pmod 5$
Same as
$2x + 4 = 0 + 5 \pmod 5$
Take away 4 from both sides
$2x = 1 \pmod 5$
We get the multiple of the coefficient of x i.e 2 at the RHS
$2x = 1 + 5 \pmod 5$
$2x = 6 \pmod 5$
Divide through by2
$x = 3$

2016/39
If 20(mod 9) is equivalent to y(mod 6), find y
A 1 B 2 C 3 D 4

Solution
20(mod 9) compared in terms of tens is 20(mod 10)
We say *two tens* i.e 20
Same applies to 20(mod 9)
two nines i.e 18
Next, convert 18 to mod 6
We divide 18 by 6 = 3 times
20(mod 9) = 30(mod 6)
Thus y = 3

2016/41 (Nov)
Evaluate 6 – 36(mod)9
A. 3 B. 4 C. 5 D. 6

Solution
36(mod)9 is *36 ÷ 9 = 4 remainder 0*
= 0
Thus 6 – 36(mod)9 = 6 – 0
= 6 (D)

2016/1a Neco Exercise 23.1
Copy and complete the multiplication table below in mod 6.

$\otimes$	1	2	3	4
1	1		3	
2	2	4		
3	3		3	
4	4			4

2002/14 Exercise 23.2
(a) Draw the :(i) addition $\oplus$ (ii) multiplication $\otimes$
Tables for the set { 2, 3, 5, 6 } modulo 7
(b) From your tables in (a)
(i) evaluate $(6 \otimes 5) \oplus (3 \otimes 2)$ (ii) find the truth set of $n \otimes 6 = 2$

2004/15 (Nov) Exercise 23.3
(a) Construct a multiplication $\otimes$ table modulo 7
On the set H = { 1, 2, 3, 5, 6}
(b) Using your table: (i) evaluate $6 \otimes 2 \otimes 5$ (ii) solve $n \otimes 6 = 2$
(iii) find the truth set of $n \otimes n = 1$

Chapter twenty four

logic

The basic problem of logic is the analysis of the methods of reasoning. Reasoning is that special kind of thinking called inference, in which conclusions are drawn from premises. As an example consider:

All dogs have four legs.
Bingo is a dog
Therefore, Bingo has four legs.
The premises of this argument are *all dogs have four legs* and *bingo is a dog*. The conclusion is Bingo has four legs.
The logician seeks the answer to this question. "If the premises are true, is one justified in saying that the conclusion is true?". If so, the reasoning that led to the conclusion is valid (logical); if not, the reasoning is invalid (illogical). The systematic formalization and cataloging of valid methods of reasoning is one of the major task of a logician. If his work uses abstract symbols, then it is called symbolic logic.

PROPOSITIONS

To avoid ambiguities, certain technical terms will now be defined before using them.

Proposition

A proposition is a declarative sentence, which is true or false but not both

Truth Value:

The truth value of a proposition is true if the proposition is true and false if the proposition is false.

Examples LOG 1

These sentences are propositions:

(A) Abuja is a town in Nigeria.
(B) France is one of the united states
(C) 3 + 2 = 5
(D) George is tall.

Assertion (a) is true since it corresponds to reality.
Assertion(b) is false since it does not correspond to reality
Assertion (c) is true.
Some logicians do not consider a sentence such as(d) to be a proposition because there are many individuals named George and because tall is relative term.
Here we shall consider that words such as George, I and they refer to specific persons and that arrangement on standards for words such as tall, short, and old has been reached. If these assumptions are made, (d) is a proposition whose truth value depends upon the particular George referred to and upon the standard of tallness agreed.

KINDS OF PROPOSITION

A Proposition is *simple* if it does not contain any other proposition as a component part. It is a *compound* proposition if it is composed of two or more proposition.

Examples LOG2

These are compound proposition:

(a) 1 + 1 = 2, and 4 is less than 9.
(b) Owls are nocturnal and cats are playful.
(c) If it rains, then I will not attend the game.

Connectives

The words and, or, not,… then, and if and only if are the basic words used to connect propositions. They are called logical connectives

Use of Symbols

The power of symbolic Logic lies in its use of symbols to represent proposition and connectives. We shall represent propositions by the lower case letters p, q, r,… usually, each letter will stand for a simple proposition and compound will be formed by linking two or more letters together by means of one or more of logical connectives and also give the name by which the compound is known

Connectives	Symbol	Names of compound
Not	$\sim$	Negation
And	$\wedge$	Conjunction
Or	$\vee$	Disjunction
If,…. then	$\rightarrow$	Conditional
If and only if	$\leftrightarrow$	Biconditional

1. **Example LOG3**

Given that:
p represents "The baby is crying"
q represents " The boys are singing" and
r represents" "The dog is barking"

Represent the following propositions symbolically.

(A) The dog is not barking
(B) The baby is crying and the birds are singing
(C) The dog is barking or the birds are singing
(D) The dog is barking and the birds are not singing
(E) If the dog is barking and the birds are not singing, then the baby is crying.
(F) The baby cries if and only if the dog barks.

Solution

(a) $\sim r$
(b) $p \wedge q$
(c) $r \vee q$
(d) $r \wedge \sim q$
(e) $(r \wedge \sim q) \rightarrow p$
(f) $p \leftrightarrow r$

Example LOG4

Let s and t stand for:
s : sally is smart.
t : tom is tall

Which sentence has the same meaning as:

(a) $\sim s$
(b) $s \wedge t$
(c) $(t \vee s) \rightarrow t$

Solution

(a) Sally is not smart
(b) Sally is smart and Tom is tall
(c) If Tom is Tall or Sally is smart then Tom is Tall

Examples LOG5

If p, q, r, stand for

p : Birds fly

q : The sky is blue

r : The grass is green

1.Write the sentence that has the same meaning as:

(a)	$\sim p$	(f)	$p \wedge \sim q \wedge \sim r$
(b)	$p \wedge q$	(g)	$\sim p \vee (\sim q)$
(c)	$p \vee q$	(h)	$(p \wedge q) \rightarrow r$
(d)	$\sim p \vee q$	(i)	$p \leftrightarrow q$
(e)	$p \wedge q \wedge r$		

2.Using the proposition p, q and r of the above problem, write symbolic representations of:

(a) The sky is blue and the grass is green.

(b) Birds fly or the sky is blue.

(c) Birds do not fly and the sky is not blue

(d) If the grass is green and the sky is not blue then the birds do not fly.

Solution

1. (a) Birds do not fly

(b) Birds fly and the sky is blue

(c) Birds fly or the sky blue

(d) Birds do not fly or the sky is blue

(e) Birds fly and the sky is blue and the grass is green

(f)Birds fly and the sky is not blue and the grass is not green

(g)Birds do not fly or the grass is not green

(h) If birds fly and the sky is blue then the grass is green.

(i) Birds fly if and only if the sky is blue

2. (a) $q \wedge r$

(b) $p\vee q$

(c) $\sim p \wedge \sim q$

(d) $(r \wedge \sim q) \rightarrow \sim p$

Truth values and truth tables.

In the preceding section, the five basic logical connectives were introduced. In this section, we consider the forms of compounds and the rules for assigning truth values to each of them.

Negation

The negation ~p is defined to be true if p is false and false if p true. It is customary to use symbol **T** to denote the truth-value of a true proposition and **F** to donate the truth value of a proposition that is false. Using these symbols, the relationship between the truth values of p and ~p are shown in the following truth table.

p	~p
T	F
F	T

Conjunction

The word "and" is defined to have the same meaning in logic as in common usage. The proposition $p\wedge q$ is true if p and q are **both true** and **false** in **other cases.** To know how many truth value are required. We use the formula $\mathbf{2^n}$, where n denotes the number of propositions in value. In the case of p, q two propositions are involved and hence the numbers of truth values required will be $2^2 = 4$. Our convention for arranging truth values in the first column of the truth tables is **TTFF,** and for those in second column is **TFTF.** The complete truth table for conjunction is:

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction The proposition $p\vee q$ is defined to be false if p and q are both false and true in all other cases.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Conditional

The proposition $p \rightarrow q$ is defined to be false if p is true and q is false, and true in all other cases

p	q	$p\rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Biconditional

The proposition $p \leftrightarrow q$ is defined to be true when p and q have the same truth values and false if they have different values.

p	q	$p\leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Constructing truth tables:

The tables that have been shown so far can be used in constructing truth tables for more complex propositions. The following examples will illustrate how this is done

Example LOG 6

Construct the truth table for $p\wedge\sim q$

Solution

We start the table with the standard four rows and two columns for compound involving the two propositions p and q. the proposition in question $p \wedge\sim q$ is the conjunction of p and ~q.

p	q	~q	$p\wedge\sim q$
T	T	F	F
T	F	T	T
F	T	F	F
F	F	T	F

After obtaining the truth values under ~q, we then use connective $\wedge$(and) to find the truth values under $p\wedge\sim q$

ExampleLOG7

Construct the truth for $\sim (p\vee q)$

p	q	$p\vee q$	$\sim (p\vee q)$
T	T	T	F
T	F	T	F
F	T	T	F
F	F	F	T

The values for the proposition $p\vee q$ are obtained and then negated to obtain the values for $\sim (p\vee q)$

ExampleLOG8

Construct the truth table for the following statement forms:

(a) $p \rightarrow \sim q$

(b) $p \leftrightarrow (q \rightarrow q)$

(c) $(p \wedge q) \rightarrow r$.

Solutions

(a)

p	q	~q	$p \rightarrow \sim q$
T	T	F	F
T	F	T	T
F	T	F	T
F	F	T	T

(b)

p	q	$q \rightarrow p$	$p \leftrightarrow (q \rightarrow p)$
T	T	T	T
T	F	T	T
F	T	F	T
F	F	T	F

(C) In this case, we have three proposition p, q and r so we shall have $2^3 = 8$ truth values and write them out as follows:

p	q	r	$p \wedge q$	$(p \wedge q) \rightarrow r$
T	T	T	T	T
T	T	F	T	F
T	T	F	T	T
T	F	T	F	T
F	T	T	F	T
F	T	F	F	T
F	F	T	F	T
F	F	F	F	T

Logically True propositions

A proposition is logically true if it is true for every possible arrangement of truth values of its component propositions, and it is false for every possible arrangement of truth values of its parts. **Logically true propositions** are often **called tautologies.**

Equivalence

Two proposition p and q are equivalent if the bi-conditional $p \leftrightarrow q$ is logically true i.e $p \leftrightarrow q$ is logically false, p and q are contradictions.

Examples LOG9

(a) $p \vee \sim p$ is a tautology

p	~p	$p \vee \sim p$
T	F	T
F	T	T

(b) $p \wedge \sim p$ is a contradiction

p	~p	$p \wedge \sim p$
T	F	F
F	T	F

(C) Show that $\sim p \vee q$ is equivalent to $p \rightarrow q$

Solution

p	q	~p	$\sim p \vee q$	$p \rightarrow q$	$(\sim p \vee q) \leftrightarrow (p \rightarrow q)$
T	T	F	T	T	T
T	F	F	F	F	T
F	T	T	T	T	T
F	F	T	T	T	T

From the table, $(\sim p \vee q) \leftrightarrow (p \rightarrow q)$ is a tautology .

Thus $\sim p \vee q$ and $p \rightarrow q$ are equivalent.

2010/8 NABTEB (Nov) Exercise 24.1

What is the negation of the sentence "John is older than me"

A john is not older than me

B john is neither older than me

C john is younger than me

D john is my age mate

2016/5 Exercise 24.2

Which of the following is a valid conclusion from the premise: "Nigerian footballers are good footballers"?

A Joseph plays football in Nigeria therefore he is a good footballer

B Joseph is a good footballer therefore he is a Nigerian footballer

C Joseph is a Nigerian footballer therefore he is a good footballer

D Joseph plays good football therefore he is a Nigerian footballer

2015/20 f/m Exercise 24.3

Given the statements:

p : the subject is difficult

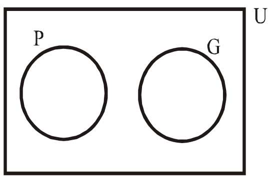
q : I will do my best

Which of the following is equivalent to "Although the subject is difficult, I will do my best"

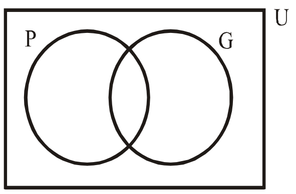
A $p \vee q$ B $\sim p \vee q$ C $p \wedge (\sim q)$ D $p \wedge q$

Logic and the use of Venn diagram

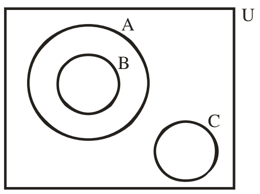
Here we apply basic Venn diagram in set such as listed below to interpret logic



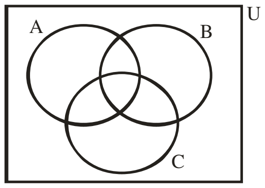
P and G have nothing in common



P and G have something in common



A contains B but A and B have nothing in common with C



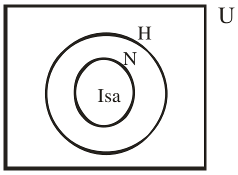
A, B and C have something in common

Example LOG10

Illustrate the statement below in a Venn diagram:
'All Northerners in Nigeria speak Hausa,
Isa is a northerner; Isa speak Hausa'
Is the conclusion valid?

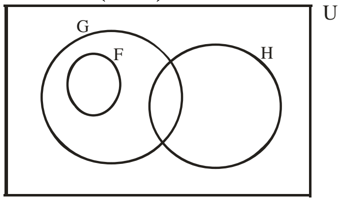
Solution

Universal set U: Nigeria
N for Northerner
H for Hausa
I for Isa



From the Venn diagram we conclude that:
The conclusion is valid

2014/33 (Nov)



In the Venn diagram, U = {students in a school}
G = {students in class 3G},
F = {students in the school's football team}
and H = {students in the school's hockey team}
Which of the following statements is **false**?

A No member of the hockey team is in the football team
B Only members of the class 3G are in the football team
C All of class 3G are members of the football team
D Some members of the hockey team are in class 3G.

Solution

Option C is false
as some members of the hockey team are in class 3G

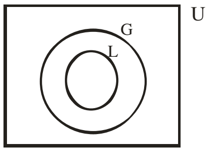
2015/6a

(i) illustrate the following statement in a Venn diagram;
All good Literature Students in a school are in the General Arts class.
(ii) Use the diagram to determine whether or not the following are valid conclusions from the given statement:

i Vivian is in the general Arts Class therefore she is a good Literature student;
ii Audu is not a good Literature student therefore he is not in the General Arts Class;
iii Kweku is not in the General Arts Class therefore he is not a good literature student.

Solution

First we break down the statement as:
Good literature student (L)
General arts (G)
School (S) i.e universal set
From the statement L is subset of G

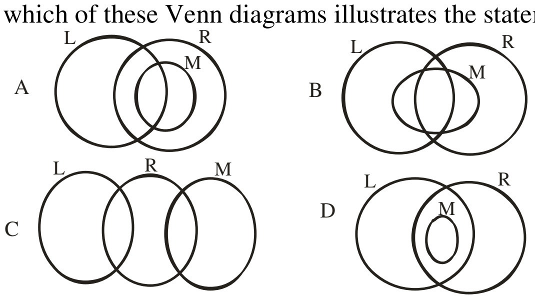


From the Venn diagram we conclude that:
I not valid ; II not valid ; III valid

2015/49

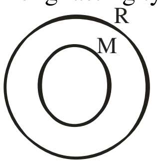
Consider the following statements:
X : locally manufactured types are attractive
Y : many locally manufactured types do not last long

Denoting locally manufactured types by **M**, attractive types by **R** and long lasting types by **L**, which of these Venn diagrams illustrates the statements?

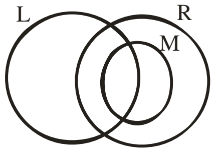


Solution

Local tyres **M**
Attractive tyres **R**
Long lasting tyres **L**



First impression
M is subset of R



Second impression
part of L is in M
Part of L is in R

2010/15

Consider the following statements:
X : all lazy students are careless
Y : most dull students are lazy

(a) Illustrate the information above on a Venn diagram.
(b) Using the Venn diagram or otherwise, determine whether or not **each** of the following statements is valid

(i) Stella is careless ⇒ Stella is lazy
(ii) Osei is lazy ⇒ Osei is careless
(iii) Kojo is dull ⇒ Kojo is lazy
(iv) Kwame is lazy ⇒ Kwame is dull
(v) Adwoa is lazy and dull ⇒ Adwoa is careless
(vi) Fiifi is careless and dull ⇒ Fiifi is lazy

Solution

Let Lazy students be **L**
,, Careless students be **C**
,, Dull student be **D**

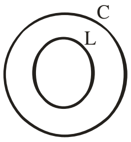


fig I
Statement X

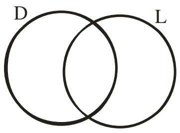


fig II
Statement y

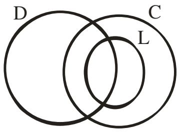
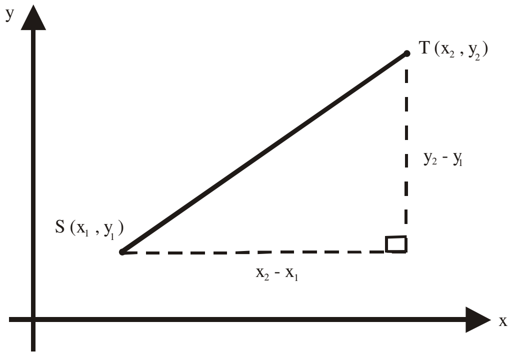


fig III
Result of X and Y

(i) Not valid, from **figI**, L is subset of C, other students are careless but not lazy
(ii) Valid, from **fig I**, L is inside C once you are lazy, you are also careless
(iii) Not valid, from **fig II,** dull and lazy only have intersection
(iv) Not valid from **fig II** same reason as in (iii)
(v) Valid, from **fig III**, portion for lazy and dull falls inside careless
(vi) Not valid; from **fig III**, there are portion left in the intersection between careless and dull that are not lazy

CHAPTER TWENTY FIVE
Coordinate geometry

Distance between two points



For any two points S and T on the Cartesian plane to be located, it will be through their x and y co-ordinates. i.e. (x, y). To find the distance between such points, we complete a hypothetical right angled triangle as shown above. By Pythagoras theorem

$(\text{Distance ST})^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$

It follows that

$$\text{Distance ST} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

2014/37 f/m

Find the distance between the points (2, 5) and (5, 9)

A. 4 units B. 5 units C. 12 units D. 14 units

Solution

Distance between two points $= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$= \sqrt{(5-2)^2 + (9-5)^2}$

$= \sqrt{3^2 + 4^2}$

$= \sqrt{9+16}$

$= \sqrt{25} = 5$ units B.

2006 / 12 PCE

Find the distance between points Y(7, 9) and Z(15, 11)

A $3\sqrt{17}$ B $2\sqrt{17}$ C $2\sqrt{34}$ D $4\sqrt{17}$

Solution

Distance between two points $= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$= \sqrt{(15-7)^2 + (11-9)^2}$

$= \sqrt{8^2 + 2^2}$

$= \sqrt{64 + 4}$

$= \sqrt{68}$

$= \sqrt{4 \times 17}$

$= 2\sqrt{17}$ B.

2008/35 PCE

If the distance between the point (–3, –2) and (1, y) is $2\sqrt{5}$ units, find the value of y

A. 4 B. 2 C. –2 D. – 4

Solution

Distance between two points $= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$2\sqrt{5} = \sqrt{[1-(-3)]^2 + [y-(-2)]^2}$

Take the square of both sides

$(2\sqrt{5})^2 = (1+3)^2 + (y+2)^2$

$4 \times 5 = 4^2 + (y+2)(y+2)$

$20 = 16 + y^2 + 4y + 4$

$y^2 + 4y = 0$

$y(y+4) = 0$

$y = 0$ or $y + 4 = 0$

$y = 0$ or -4

accept $y = -4$ D.

2010/35 UTME

Find the distance between the point $(\frac{1}{2}, \frac{1}{2})$ and $(-\frac{1}{2}, -\frac{1}{2})$

A. $\sqrt{2}$ B. 0 C. 1 D. $\sqrt{3}$

Solution

Distance between two points $= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$= \sqrt{\left(\frac{-1}{2} - \frac{1}{2}\right)^2 + \left(\frac{-1}{2} - \frac{1}{2}\right)^2}$

$= \sqrt{(-1)^2 + (-1)^2}$

$= \sqrt{1+1} = \sqrt{2}$ A.

2009 / 33 UME

What is the value of r, if the distance between the points (4, 2) and (1, r) is 3 units ?

A. 1 B, 2 C. 3 D. 4

Solution

Distance between two points $= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$3 = \sqrt{(1-4)^2 + (r-2)^2}$

$3^2 = (-3)^2 + (r-2)(r-2)$

$9 = 9 + r^2 - 4r + 4$

$r^2 - 4r + 4 = 0$

Factoring

$(r-2)(r-2) = 0$

$r - 2 = 0$ twice

$r = 2$ B.

2012/33 UTME

The distance between the point (4, 3) and the intersection of $y = 2x + 4$ and $y = 7 - x$ is

Solution

Distance b/w two points $= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

(x_1, y_1) is (4, 3)

(x_2, y_2) can be gotten from the intersection of $y = 2x + 4$ and $y = 7 - x$

Solving the simultaneous linear equations

$$\begin{array}{r} y - 2x = 4 \\ -(y + x = 7) \\ \hline -3x = -3 \\ x = 1 \end{array}$$

Substitute $x = 1$ into $y = 2x + 4$

$y = 2 + 4$

$= 6$

Thus (x_2, y_2) is $(1, 6)$

$$\text{Distance} = \sqrt{(1-4)^2 + (6-3)^2}$$
$$= \sqrt{(-3)^2 + (3)^2}$$
$$= \sqrt{9+9}$$
$$= \sqrt{18} = \sqrt{9 \times 2} = 3\sqrt{2}$$

2004/2 UME

Find the value of $\alpha^2 + \beta^2$ if $\alpha + \beta = 2$ and the distance between the points $(1, \alpha)$ and $(\beta, 1)$ is 3 units

A.14 B.3 C.5 D.11

Solution

Distance b/w two points $= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)}$

$3^2 = (\beta - 1)^2 + (1 - \alpha)^2$

$9 = \beta^2 - 2\beta + 1 + 1 - 2\alpha + \alpha^2$

$9 = \alpha^2 + \beta^2 - 2\alpha - 2\beta + 2$

$9 - 2 = \alpha^2 + \beta^2 - 2(\alpha + \beta)$

put $\alpha + \beta = 2$

$7 = \alpha^2 + \beta^2 - 2(2)$

$7 = \alpha^2 + \beta^2 - 4$

$7 + 4 = \alpha^2 + \beta^2$

$11 = \alpha^2 + \beta^2$ (D)

Coordinates of mid - point

is given as $\left[\frac{1}{2}(x_2 + x_1), \ \frac{1}{2}(y_2 + y_1) \right]$

2014/32 UTME

Find the mid point of $S(-5, 4)$ and $T(-3, -2)$

A –4, 1 B 4, –1 C –4, 2 D 4, –2

Solution

$$\text{Mid point coordinates} = \frac{1}{2}(x_2 + x_1), \ \frac{1}{2}(y_2 + y_1)$$
$$= \frac{1}{2}(-3 - 5), \ \frac{1}{2}(-2 + 4)$$
$$= -\frac{8}{2}, \ \frac{2}{2}$$
$$= -4, \ 1 \quad \text{(A)}$$

2009/33 PCE

Find the coordinates of the midpoint of the straight line joining $M(6, 10)$ and $N(4, 2)$

A (2, 8) B (10, 12) C (–5, –6) D (5, 6)

Solution

$$\text{Mid Point Coordinates} = \frac{x_2 + x_1}{2}, \ \frac{y_2 + y_1}{2}$$
$$= \frac{4 + 6}{2}, \ \frac{2 + 10}{2}$$
$$= \frac{10}{2}, \ \frac{12}{2}$$
$$= (5, 6) \quad \text{D.}$$

2013/31 UTME

If the mid point of the line PQ is (2, 3) and the point P is (–2, 1), find the coordinate of the point Q

A (8, 6) B (5, 6) C (0, 4) D (6, 5)

Solution

Let the coordinate of the point Q be (a, b)

$$\text{Mid Point Coordinates} = \frac{x_2 + x_1}{2}, \ \frac{y_2 + y_1}{2}$$
$$2, \ 3 = \frac{a - 2}{2}, \ \frac{b + 1}{2}$$

Equating the corresponding coordinates

$2 = \frac{a-2}{2}$ and $3 = \frac{b+1}{2}$

$4 = a - 2$ and $6 = b + 1$

$4 + 2 = a$ and $6 - 1 = b$

$6 = a$ and $5 = b$

Q(6, 5) D.

2008/33 PCE

If the mid point of the line joining points $P(m, n)$ and $Q(1, 3)$ is $R(2, 4)$, find the values of m and n respectively

A 4, 3 B 2, 7 C 4, 2 D 3, 5

Solution

Here (m, n) is (x, y)

$$\text{Mid Point Coordinates} = \frac{x_2 + x_1}{2}, \ \frac{y_2 + y_1}{2}$$
$$2, \ 4 = \frac{1 + m}{2}, \ \frac{3 + n}{2}$$

Equating the corresponding coordinates

$2 = \frac{1+m}{2}$ and $4 = \frac{3+n}{2}$

$4 = 1 + m$ and $8 = 3 + n$

$4 - 1 = m$ and $8 - 3 = n$

$3 = m$ and $5 = n$

P(3, 5) D.

2007/32 PCE

M $(3, -4)$ is the midpoint of $\overline{PQ}$. If P is $(6, 5)$, find the coordinates of Q.

A (0, –13) B (4.0, 0.5) C (3, 2.5) D (12, –3)

Solution

The coordinates of Q are (x_2, y_2)

$$\text{Mid point coordinates} = \frac{x_2 + x_1}{2}, \ \frac{y_2 + y_1}{2}$$

$$3, \ -4 = \frac{x_2 + 6}{2}, \ \frac{y_2 + 5}{2}$$

Equating the corresponding coordinates

$$3 = \frac{x_2 + 6}{2} \quad \text{and} \quad -4 = \frac{y_2 + 5}{2}$$

$$6 = x_2 + 6 \quad \text{and} \quad -8 = y_2 + 5$$

$$6 - 6 = x_2 \quad \text{and} \quad -8 - 5 = y_2$$

$$0 = x_2 \quad \text{and} \quad -13 = y_2$$

Q (0, –13) A.

2007/47 UME

What is the value of k if the mid - point of the line joining

(1 – k, –4) and (2, k + 1) is (–k, k)?

A – 3 B. –1 C. –4 D. –2

Solution

Midpoint coordinates = $\frac{x_2 + x_1}{2}, \ \frac{y_2 + y_1}{2}$

$$-k, \ k = \frac{2 + 1 - k}{2}, \ \frac{k + 1 - 4}{2}$$

$$-k, \ k = \frac{3 - k}{2}, \ \frac{k - 3}{2}$$

Equating the corresponding coordinates

$$-k = \frac{3 - k}{2} \quad \text{and} \quad k = \frac{k - 3}{2}$$

$$2k = k - 3$$

$$k = -3 \quad \text{A.}$$

2014/35 UTME

Calculate the mid point of the line segment

y – 4x + 3 = 0 which lies between the

x – axis and y – axis

A $\left(\frac{-3}{2}, \frac{3}{2}\right)$ B $\left(\frac{-2}{3}, \frac{3}{2}\right)$ C $\left(\frac{3}{8}, \frac{-3}{2}\right)$ D $\left(\frac{3}{8}, \frac{3}{2}\right)$

Solution

$$\text{Midpoint} = \frac{x_2 + x_1}{2}, \ \frac{y_2 + y_1}{2}$$

To get the coordinates for x and y, we proceeds as follows

$$y - 4x + 3 = 0$$

$$y = 4x - 3$$

When x = 0, y = 4(0) – 3

i.e x = 0, y = –3 (0, –3)

when y = 0 $\Rightarrow$ 0 = 4x – 3

3 = 4x

¾ = x i.e (¾ , 0)

$$\text{Mid point} = \left(\tfrac{3}{4} + 0\right) \div 2, \quad [0 + (-3)] \div 2$$

$$= \frac{3}{4} \ x \ \frac{1}{2}, \quad \left(\frac{-3}{2}\right)$$

$$= \left(\frac{3}{8}, \frac{-3}{2}\right) \quad \text{C.}$$

2016/39 Neco Exercise 25.1

Find the coordinate of the mid-point of the line joining the points (–1, –1) and (–3, 5)

A (2, 2) B (2, –2) C (–2, –2) D (–2, 2) E (–1, 2)

Exercise 25.2

Find the coordinate of the mid-point of the line joining the points (2, 3) and (–2, 1)

Gradient / slope of a straight line

is given as $\frac{y_2 - y_1}{x_2 - x_1}$

2010/36 UTME

Find the gradient of the line passing through the points P(1, 1) and Q(2, 5)

A 4 B 2 C 3 D 5

$$\text{Gradient} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{5 - 1}{2 - 1}$$

$$= 4 \quad \text{A.}$$

2009/34 UTME

The gradient of the straight line joining the points P(5, –7) and Q(–2, –3) is

A $^1/_2$ B $^2/_5$ C $^{-4}/_7$ D $^{-2}/_3$

Solution

$$\text{Gradient} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{-3 - (-7)}{-2 - 5}$$

$$= \frac{-3 + 7}{-7} = \frac{-4}{7} \quad \text{C.}$$

2014/33 UTME

The gradient of the line joining (x, 4) and (1, 2) is $^1/_2$. Find the value of x

A – 3 B – 5 C 5 D 3

Solution

$$\text{Gradient} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\frac{1}{2} = \frac{2 - 4}{1 - x}$$

$$1 - x = 2(2 - 4)$$

$$1 - x = -4$$

$$1 + 4 = x$$

$$5 = x \quad \text{C.}$$

2008/34 PCE

Find the Gradient of the line joining points P(4, – 1) and Q(–3, – 5)

A $\frac{4}{7}$ B $\frac{7}{4}$ C $\frac{-4}{7}$ D $\frac{-7}{4}$

Solution

Gradient = $\frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{-5-(-1)}{-3-4}$

$= \frac{-5+1}{-7} = \frac{-4}{-7} = \frac{4}{7}$ A.

2006/4 PCE

Find the gradient of the line joining S(5, 6) and R(– 7, – 8)

A $^7/_6$ B – 1 C 1 D – $^7/_6$

Solution

Gradient = $\frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{-8-6}{-7-5}$

$= \frac{-14}{-12} = \frac{7}{6}$ A.

2009/32 UME

What is the value of P if the gradient of the line joining (– 1, P) and (P, 4) is $^2/_3$?

A – 2 B – 1 C 1 D 2

Solution

Gradient = $\frac{y_2 - y_1}{x_2 - x_1}$

$\frac{2}{3} = \frac{4-p}{p-(-1)}$

$\frac{2}{3} = \frac{4-p}{p+1}$

2(p + 1) = 3(4 – p)

2p + 2 = 12 – 3p

2p + 3p = 12 – 2

5p = 10

p = 2 D.

2016/37 Neco Exercise 25.3

Find the gradient of line joining the points (2, 5) and (– 6, 3)

A –$^1/_5$ B – $^1/_4$ C – $^1/_3$ D – $^1/_2$ E $^1/_4$

2009/34 PCE Exercise 25.4

Find the gradient of the straight line passing though the points x(– 1, 3) and y(5, –1)

A 6 B $^2/_3$ C $^{-2}/_3$ D – 4

2016/24 and 25 Exercise 25.5

A straight line passes through the points P (1, 2) and Q(5, 8).

Use this information to answer question **24** and **25**

24. Calculate the gradient of the line PQ

A $^3/_5$ B $^2/_3$ C $^3/_2$ D $^5/_3$

25. Calculate the length PQ

A $4\sqrt{11}$ B $4\sqrt{10}$ C $2\sqrt{17}$ D $2\sqrt{13}$

2015/13 and 14 f/m Exercise 25.6

A line passes through the origin and the point $\left(1\frac{1}{4}, 2\frac{1}{2}\right)$. Use this information to answer question **13** and **14**

13. What is the gradient of the line?

A 1 B 2 C 3 D 4

14. Find the y – coordinate of the line when x = 4

A 2 B 4 C 6 D 8

Equation of a straight line

is given as y = mx + c

Where m is gradient and c is the point where the line cuts the y-axis

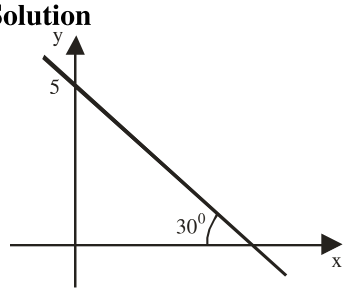
2001/24 f/m

A straight line makes an angle of 30^0 with the positive x – axis and cuts the y – axis at y = 5. find the equation of the straight line

A. $y = \frac{1}{10}x + 5$ B. y = x + 5 C. $\sqrt{3}\,y = -x + 5\sqrt{3}$

D $\sqrt{3}y = x + 5\sqrt{3}$

Solution



From the sketch we have a negative gradient

y = – mx + c

Here tan θ is tan $30^0 = \frac{1}{\sqrt{3}}$ and c is 5,

$y = -\frac{1}{\sqrt{3}}x + 5$

Multiplying through by $\sqrt{3}$

$\sqrt{3}\, y = -x + 5\sqrt{3}$ (C)

1997/34 PCE

What is the y – intercept and the gradient of the line whose equation is $\frac{x}{5} + \frac{y}{3} = \frac{1}{3}$?

A. $^1/_3$, $^3/_5$ B. 1, – $^3/_5$ C. –$^1/_3$, – $^3/_5$ D. –1, $^3/_5$

Solution

First, we rearrange the given equation into y = mx + c format

To clear fractions, multiply through by the Lcm of 5 and 3 i.e 15

$15 \times \frac{x}{5} + \frac{y}{3} \times 15 = \frac{1}{3} \times 15$

3x + 5y = 5

5y = – 3x + 5

$y = -\frac{3}{5}x + 1$

y – intercept is + 1

Gradient is – $^3/_5$ (B)

1998/35 PCE

Find the gradient of the line whose equation is given $(a - b)x + (c - d)y = e - f$

A. $\frac{b - a}{c - d}$ B. $\frac{a - b}{c - d}$ C. $\frac{b - a}{c - d}$ D. $\frac{e - f}{c - d}$

Solution

First, we rearrange the given equation into $y = mx + c$ format

$$(c - d)y = -(a - b)x + (e - f)$$

Divide through the coefficient of y

$$y = -\frac{(a - b)}{c - d}x + \frac{e - f}{c - d}$$

Therefore,

gradient $m = -\frac{(a - b)}{c - d}$

opening up the bracket

$$m = -\frac{a + b}{c - d} \text{ or } \frac{b - a}{c - d} \quad (A)$$

1993/39 PCE

The intercept which the line $2x - 3y - 5 = 0$ makes with the y-axis is of length

A. $^5/_3$ B. $^5/_2$ C. $-^5/_3$ D. $-^5/_2$

Solution

First we rearrange the given equation in $y = mx + c$ format

We are to find C

$$2x - 3y - 5 = 0$$

$$2x - 5 = 3y$$

$$y = \frac{2x}{3} - \frac{5}{3}$$

The intercept at the y-axis is $-^5/_3$ (C)

2005/10 PCE

The equation of the straight line which makes an intercept of –2 with the y-axis and an intercept of 1 with the x-axis is

A. $3x + 2y - 6 = 0$ B. $3x - 2y + 6 = 0$

C. $2x + y - 2 = 0$ D. $2x - y - 2 = 0$

Solution

General form of a straight line is $y = mx + c$; where m is gradient and c is the intercept on the y-axis

The given question only has **c** but no **m**. Though we can simulate the two points needed in the formula

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

At intercept –2 with the y-axis

$x = 0$ and $y = -2$ i.e $(0, -2)$

At intercept 1 with the x-axis

$x = 1$ and $y = 0$ i.e $(1, 0)$

Thus, $m = \frac{-2 - 0}{0 - 1}$

$$= \frac{-2}{-1} = 2$$

The required equation is with $c = -2$ given earlier and $m = 2$

i.e $y = 2x - 2$

Rearranging $2x - y - 2 = 0$ (D)

Equation of a straight line through

x_1, y_1 with gradient m is

$y - y_1 = m(x - x_1)$ **OR**

$$m = \frac{y - y_1}{x - x_1}$$

1992/35 PCE

Find the equation of a straight line of gradient 2 through the point (1, 4)

A. $y - x + 2 = 0$ B. $y - 2x + 4 = 0$ C. $y + 2x + 4 = 0$

D. $y - 2x - 2 = 0$

Solution

Applying the formula

$$m = \frac{y - y_1}{x - x_1}$$

$$2 = \frac{y - 4}{x - 1}$$

cross multiplying

$$2(x - 1) = y - 4$$

$$2x - 2 = y - 4$$

$$0 = y - 4 + 2 - 2x$$

i.e. $y - 2x - 2 = 0$ (D)

Equation of a line passing through two points

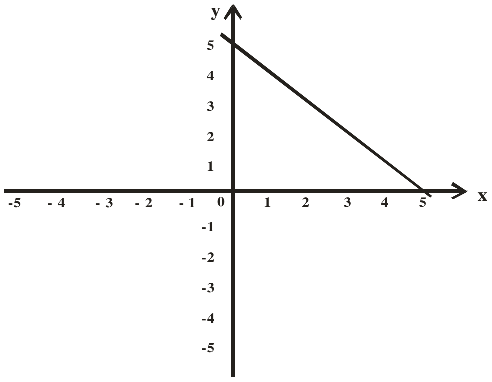
$$m = \frac{y - y_1}{x - x_1}$$

But $m = \frac{y_2 - y_1}{x_2 - x_1}$

Thus

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y - y_1}{x - x_1}$$

2014/34 UTME



In the figure above, what is the equation of the line that passes the y – axis at (0, 5) and passes the x – axis at (5, 0)?

A $y = x - 5$ B $y = -x - 5$ C $y = x + 5$

D $y = -x + 5$

Solution

Formula for equation of a line given **two points** is

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\frac{y - 5}{x - 0} = \frac{0 - 5}{5 - 0}$$

$$\frac{y-5}{x} = \frac{-5}{5}$$

$5(y - 5) = -5x$

$5y - 25 = -5x$

$5y = -5x + 25$

$y = -x + 5$ (D)

2007 / 34 P C E

The equation of a straight line which has intercepts of 3 and –5 respectively on the x and y – axes is

A. $5y - 3x + 15 = 0$ B. $3y + 5x - 15 = 0$

C. $5y + 3x + 15 = 0$ D. $3y - 5x + 15 = 0$

Solution

Intercept at x – axis is 3, then $y = 0$ i.e **(3, 0)**

Intercept at y – axis is –5 then $x = 0$ i.e **(0, –5)**

Formula for equation of a line given **two points** is

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\frac{y-0}{x-3} = \frac{-5-0}{0-3}$$

$$\frac{y}{x-3} = \frac{-5}{-3} \quad \text{i.e} \quad \frac{y}{x-3} = \frac{5}{3}$$

$3y = 5(x - 3)$

$3y = 5x - 15$

$3y - 5x + 15 = 0$ D.

2012/35 f/m

A straight line makes intercepts of –3 and 2 on the x and y axes respectively. Find the equation of the line.

A. $2x + 3y + 6 = 0$ B. $3x - 2y - 6 = 0$

C. $-3x + 2y - 6 = 0$ D. $-2x + 3y - 6 = 0$

Solution

Intercepts at x – axis is –3, then $y = 0$ i.e **(–3, 0)**

Intercepts at y – axis is 2 then $x = 0$ i.e **(0, 2)**

Formula for equation of a line given **two points** is

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\frac{y-0}{x-(-3)} = \frac{2-0}{0-(-3)}$$

$$\frac{y}{x+3} = \frac{2}{3}$$

$3y = 2(x + 3)$

$3y = 2x + 6$

$3y - 2x - 6 = 0$ D.

2009/35 PCE Exercise 25.7

Find the equation of the line joining P(5, –1) to Q(–1, 3)

A. $2x - 3y - 7 = 0$ B. $2x + 3y - 7 = 0$

C. $2x + 3y + 7 = 0$ D. $2x + 3y - 11 = 0$

2012/34 UTME Exercise 25.8

Find the equation of the line through the points (–2, 1) and (– ½, 4)

A. $y = 2x - 3$ B. $y = 2x + 5$

C. $y = 3x - 2$ D. $y = 2x + 1$

2016/32 Neco Exercise 25.9

Find the equation of the straight line which passes through the points X(3, – 4) and Y(– 5, 2)

A $4y = 3x - 7$ B $y = 3x + 2$

C $2y = 3x + 5$ D $3y = -x - 4$ E $4y = 2x + 6$

Parallel lines

If two lines are parallel then $m_1 = m_2$

2007/40 UME

If the lines $3y = 4x - 1$ and $qy = x + 3$ are parallel to each other, the value of q is

A. $\frac{-4}{3}$ B. $\frac{-3}{4}$ C. $\frac{4}{3}$ D. $\frac{3}{4}$

Solution

If two lines are *parallel* then *equal gradient*

Firstly,

$3y = 4x - 1,$ $qy = x + 3$

$y = \frac{4}{3}x - \frac{1}{3}$ $y = \frac{1}{q}x + \frac{3}{q}$

Equating their gradients

$$\frac{4}{3} = \frac{1}{q}$$

$4q = 3$

$q = \frac{3}{4}$ D.

2006/8 UME

PQ and RS are two parallel lines. If the coordinates of P, Q, R, S are (1, q), (3, 2), (3, 4), (5, 2q) respectively, find the value of q.

A. 2 B. 4 C. 1 D. 3

Solution

If two lines are *parallel* then *equal gradient*

Applying the formula for gradient of two points

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\frac{2-q}{3-1} = \frac{2q-4}{5-3}$$

$$\frac{2-q}{2} = \frac{2q-4}{2}$$

Multiply both side by 2

$2 - q = 2q - 4$

$2 + 4 = 2q + q$

$6 = 3q$

$2 = q$ A.

2007/33 PCE

What is the equation of a line parallel to the line $3y - x + 4 = 0$ which passes through the point $(2, -4)$?

A. $3y + x + 14 = 0$ B. $3y - x + 14 = 0$

C. $y + 3x - 14 = 0$ D. $y - 3x - 14 = 0$

Solution

Two lines are parallel then equal gradient

First, we get the gradient M_1 from

$$3y - x + 4 = 0$$
$$3y = x - 4$$
$$y = \frac{1}{3}x - \frac{4}{3}$$

Comparing with $y = mx + c$

$$M_1 = \frac{1}{3}$$

Applying the formula for a line with a given gradient ⅓ and two points $(2, -4)$

$$M = \frac{y - y_1}{x - x_1}$$
$$\frac{1}{3} = \frac{y - (-4)}{x - 2}$$
$$\frac{1}{3} = \frac{y + 4}{x - 2}$$
$$x - 2 = 3(y + 4)$$
$$x - 2 = 3y + 12$$

$3y - x + 14 = 0$ B.

2016/13a

Find the equation of a straight line which passed through the point $(2, -3)$ and is parallel to the line $2x + y = 6$

Solution

If two lines are *parallel* then *equal gradient*

First, we get the gradient M_1 from

$$2x + y = 6$$
$$y = -2x + 6$$

Comparing with $y = mx + c$

Thus, $M_1 = -2$

Applying the formula for a given line with gradient -2 and two points $(2, -3)$

$$m = \frac{y - y_1}{x - x_1}$$
$$-2 = \frac{y - (-3)}{x - 2}$$
$$-2 = \frac{y + 3}{x - 2}$$
$$-2(x - 2) = y + 3$$
$$-2x + 4 = y + 3$$
$$y + 2x - 1 = 0$$

2016/13a

Alternative method

The line equation parallel to $2x + y = 6$ is

$2x + y = k$ at $(2, -3)$

Substituting for $x = 2$ and $y = -3$

$$2x + y = k \text{ becomes}$$
$$2(2) - 3 = k$$
$$4 - 3 = k$$
$$1 = k$$

The line equation is $2x + y = 1$

$$2x + y - 1 = 0$$

2006/ 6 PCE

Find the equation of a line parallel to line $y = 3x + 4$ which passes through the point $(2, -5)$

A. $x - 3y - 11 = 0$ B. $3x - y - 11 = 0$

C. $3x + y - 11 = 0$ D. $x - 3y + 11 = 0$

Solution

If two lines are *parallel* then *equal gradient*

First, we get the gradient M_1 from

$$y = 3x + 4$$

Comparing with $y = mx + x$

$$M_1 = 3$$

Applying the formula for a line, given gradient 3 and two points $(2, -5)$

$$M = \frac{y - y_1}{x - x_1}$$
$$3 = \frac{y - (-5)}{x - 2}$$
$$3 = \frac{y + 5}{x - 2}$$
$$3(x - 2) = y + 5$$

$3x - 6 = y + 5$ thus, $y - 3x + 11 = 0$ B.

2006 / 6 PCE

Alternative method

The line equation parallel to

$y = 3x + 4$ is

$y = 3x + k$ at $(2, -5)$

Substituting for $x = 2$ and $y = -5$

$$y = 3x + k \text{ becomes}$$
$$-5 = 3(2) + k$$
$$-5 = 6 + k$$
$$-5 - 6 = k$$
$$-11 = k$$

The line equation is $y = 3x - 11$ B.

2010/ 37 UTME Exercise 25.10

Find the equation of a line parallel to $y = -4x + 2$ passing through $(2, 3)$

A. $y - 4x + 11 = 0$ B. $y - 4x - 11 = 0$

C. $y + 4x + 11 = 0$ D. $y + 4x - 11 = 0$

2002/32 PCE Exercise 25.11

The equation of a straight line through the point $(-3, -2)$ that is parallel to the line $4x + y - 6 = 0$ is

A. $4x + y + 14 = 0$ B. $4x + y + 12 = 0$

C. $4x + y - 14 = 0$ D. $4x + y - 12 = 0$

2003/18 PCE Exercise 25.12
Find the equation of the line passing through $(4, -2)$ and parallel to the line $x - 3y - 4 = 0$
A. $3y - x + 10 = 0$ B. $y - 3x - 10 = 0$
C. $3y - x - 10 = 0$ D. $y - 3x + 10 = 0$

If intersection of two straight lines is perpendicular then $m_1 m_2 = -1$

2014/36 UTME
Find the equation of the straight line through $(-2, 3)$ and perpendicular to $4x + 3y - 5 = 0$
A. $4x + 5y + 3 = 0$ B. $5x - 2y - 11 = 0$
C. $3x - 4y + 18 = 0$ D. $3x + 2y - 18 = 0$

Solution

Equation of the line through (x_1, y_1) with gradient m is

$$m = \frac{y - y_1}{x - x_1}$$

But two lines are perpendicular when their gradient

$m_1m_2 = -1$

From $4x + 3y - 5 = 0$

$$3y = -4x + 5$$
$$y = \frac{-4}{3}x + \frac{5}{3}$$

Thus, $m_1 = \frac{-4}{3}$

$m_1m_2 = -1$

It follows that $\frac{-4}{3}m_2 = -1$

$$-4m_2 = -3$$
$$m_2 = \frac{3}{4}$$

Equation of line is $\frac{3}{4} = \frac{y - 3}{x + 2}$

$$3(x + 2) = 4(y - 3)$$
$$3x + 6 = 4y - 12$$
$$4y - 3x - 18 = 0 \quad \text{C.}$$

2014/36 UTME

Alternative method

The line equation perpendicular to

$4x + 3y - 5 = 0$ is

$4y - 3x = k$ at $(-2, 3)$

Substituting for $x = -2$ and $y = 3$

$4y - 3x = k$ becomes

$$4(3) - 3(-2) = k$$
$$12 + 6 = k$$
$$18 = k$$

The line equation is $4y - 3x = 18$

$$4y - 3x - 18 = 0$$

i.e $3x - 4y + 18 = 0$ C.

2013/38 f/m
Find the equation of the straight line that passes through $(2, -3)$ and perpendicular to the line $3x - 2y + 4 = 0$
A. $2y - 3x = 0$ B. $3y - 2x + 5 = 0$
C. $3y + 2x + 5 = 0$ D. $2y - 3x - 5 = 0$

Solution

Equation of line through (x_1, y_1) with gradient m is

$$m = \frac{y - y_1}{x - x_1}$$

But two lines are perpendicular, when their gradient

$m_1m_2 = -1$

From $3x - 2y + 4 = 0$

$$3x + 4 = 2y$$
$$y = \frac{3}{2}x + 2$$

Thus, $m_1 = \frac{3}{2}$

But $m_1m_2 = -1$

It follows that $\frac{3}{2}m_2 = -1$

$$m_2 = \frac{-2}{3}$$

Equation of line is $\frac{-2}{3} = \frac{y + 3}{x - 2}$

$$-2(x - 2) = 3(y + 3)$$
$$-2x + 4 = 3y + 9$$
$$3y + 2x + 5 = 0 \quad \text{C.}$$

2013/38 f/m

Alternative method

The line equation perpendicular to

$3x - 2y + 4 = 0$ is

$3y + 2x = k$ at $(2, -3)$

Substituting for $x = 2$, $y = -3$

$3y + 2x = k$ becomes

$$3(-3) + 2(2) = k$$
$$-9 + 4 = k$$
$$-5 = k$$

The equation is $3y + 2x = -5$

$$3y + 2x + 5 = 0 \quad \text{C.}$$

2013/10 f/m
Find the equation of the line that is perpendicular to $2y + 5x - 6 = 0$ and bisects the line joining the points P $(4, 3)$ and Q $(-6, 1)$
A. $y + 5x + 3 = 0$ B. $2y - 5x - 9 = 0$
C. $5y + 2x - 8 = 0$ D. $5y - 2x - 12 = 0$

Solution

Mid point of PQ coordinates $= \frac{1}{2}(x_1 + x_2), \frac{1}{2}(y_2 + y_1)$

$= \frac{1}{2}(-6 + 4), \frac{1}{2}(1 + 3)$

$= (-1, 2)$

The line equation perpendicular to

$2y + 5x - 6 = 0$ is

$2x - 5y = k$ at $(-1, 2)$

Substituting for $x = -1$ and $y = 2$

$2x - 5y = k$ becomes

$2(-1) - 5(2) = k$

$-2 - 10 = k$

$-12 = k$

Thus the new line is $2x - 5y = -12$

$2x - 5y + 12 = 0$

i.e $5y - 2x - 12 = 0$ D.

2013 / 32 UTME

Find the equation of the perpendicular bisector of the line joining P(2, –3) to Q(–5, 1)

A. $8y + 14x + 13 = 0$ B. $8y - 14x + 13 = 0$

C. $8y - 14x - 13 = 0$ D. $8y + 14x - 13 = 0$

Solution

Mid point coordinates $= \frac{1}{2}(x_2 + x_1),\ \frac{1}{2}(y_2 + y_1)$

$= \frac{1}{2}(-5 + 2),\ \frac{1}{2}(1 - 3)$

$= \left(\frac{-3}{2},\ -1\right)$

Equation of line through two points P(2, –3) and Q(–5, 1)

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\frac{y + 3}{x - 2} = \frac{1 + 3}{-5 - 2}$$

$$\frac{y + 3}{x - 2} = -\frac{4}{7}$$

$7(y + 3) = -4(x - 2)$

$7y + 21 = -4x + 8$

$7y = -4x + 8 - 21$

$7y = -4x - 13$

$y = -\frac{4}{7}x - \frac{13}{7}$

The bisector of the line PQ is perpendicular to it

$m_1m_2 = -1$

From $y = -\frac{4}{7}x - \frac{13}{7}$

$m_1 = \frac{-4}{7}$

It follows $\frac{-4}{7}m_2 = -1$

$m_2 = \frac{7}{4}$

Equation of bisector $\left(\frac{-3}{2},\ -1\right)$

$$\frac{7}{4} = \frac{y + 1}{x + \frac{3}{2}}$$

$7(x + \frac{3}{2}) = 4(y + 1)$

$7x + \frac{21}{2} = 4y + 4$

$14x + 21 = 8y + 8$

$8y - 14x - 13 = 0$ (A)

1994/37 PCE Exercise 25.13

Find the equation of the line through (1, 1) perpendicular to $2x - 3y = 4$

A. $3x - 2y = 5$ B. $3x + 2y = 5$ C. $2x - 3y = 5$ D. $2x + 3y = 5$

2005/7 PCE Exercise 25.14

For what value of t are the lines joining points (– 3, 2) and (2, 1) as well as (5, 4) and (6, t) perpendicular ?

A. 9 B. 6 C. 2 D. 1

Ordinary Intersection

2007/44 UME

If the lines $2y - kx + 2 = 0$ and $y + x - \frac{k}{2} = 0$ intersect at (1, –1), find the value of k.

A – 4 B – 3 C – 2 D – 1

Solution

At intersection, the two lines are equal

$2y - kx + 2 = y + x - \frac{k}{2}$

$2y - y - kx - x + 2 + \frac{k}{2} = 0$

$y - kx - x + 2 + \frac{k}{2} = 0$

Substituting for (1, – 2) i.e $x = 1$ and $y = -2$

$-2 - k - 1 + 2 + \frac{k}{2} = 0$

$-1 - k + \frac{1}{2}k = 0$

$-\frac{1}{2}k = 1$

$-k = 2$

$k = -2$ C .

Equation of a straight line through a given angle

is Gradient = tan θ

Also the **Acute angle between two lines is**

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1m_2}\right|$$

2009/24 UME

Find the acute angle between the straight lines

$y = x$ and $y = \sqrt{3}\,x$

A 15^0 B 30^0 C 45^0 D 60^0

Solution

Acute angle between two lines is

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1m_2}\right|$$

$y = x$ $(m_1 = 1)$

$y = \sqrt{3}\,x$ $(m_2 = \sqrt{3}\,)$

$$\tan\theta = \left|\frac{1 - \sqrt{3}}{1 + \sqrt{3}}\right|$$

Rationalizing the surd

$$= \frac{(1-\sqrt{3})\,(1-\sqrt{3})}{(1+\sqrt{3})\,(1-\sqrt{3})}$$

$$= \frac{1-2\sqrt{3}+3}{1-3} = \frac{4-2\sqrt{3}}{2}$$

$\tan\theta = 2 - \sqrt{3}$

$= 2 - 1.732$

$\tan\theta = 0.268$

$\theta = \tan^{-1} 0.268$

$= 15^0$ A.

2012/37 f/m

Find the acute angle between the lines

$2x + y = 4$ and $-3x + y + 7 = 0$

A 40^0 B 44^0 C 45^0 D 54^0

Solution

Acute angle between two lines is

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$$

$2x + y = 4 \Rightarrow y = -2x + 4$ $(m_1 = -2)$

$-3x + y + 7 = 0 \Rightarrow y = 3x - 7$ $(m_2 = 3)$

$$\tan\theta = \left|\frac{-2-3}{1+(-2)(3)}\right|$$

$$= \frac{-5}{1-6} = \frac{-5}{-5} = 1$$

$\tan\theta = 1$

$\theta = \tan^{-1} 1$

$= 45^0$ C.

1997/33 UME

The angle between the positive horizontal axis and a given line is 135^0. Find the equation of the line if it passes through the point (2, 3)

A. $x - y = 1$ B. $x + y = 1$ C. $x + y = 5$ D. $x - y = 5$

Solution

Equation of a line through a point is given as

$$m = \frac{y - y_1}{x - x_2}$$

But gradient (m) = $\tan\theta$

$m = \tan 135^0$

Tan 135^0 falls into the 2^{nd} quadrant with formula (180 – 0) and tangent is negative there

$= -\tan(180 - 135)$

$= -\tan 45$

$= -1$

Substituting

$$-1 = \frac{y-3}{x-2}$$

$-1(x - 2) = y - 3$

$-x + 2 = y - 3$

$y + x = 3 + 2$

$y + x = 5$ (C)

Collinear points

If the three points lie on same straight line then they are collinear, hence same gradient

2014/35 f/m

If the point $(-1, t-1)$, $(t, t-3)$ and $(t-b, 3)$ lie on the same straight line, find the values of t

A $t = -2$ and -3 B $t = 2$ and 3 C $t = -2$ and 3 D $t = 2$ and -3

Solution

If the three points lie on same straight line then they are collinear, hence same gradient

Let $P(-1, t-1)$, $Q(t, t-3)$ and $R(t-6, 3)$

Applying the gradient formula for two point = $\dfrac{y_2 - y_1}{x_2 - x_1}$

Gradient PQ = Gradient QR

$$\frac{t-3-(t-1)}{t-(-1)} = \frac{3-(t-3)}{t-6-t}$$

$$\frac{t-3-t+1}{t+1} = \frac{3-t+3}{-6}$$

$$\frac{-2}{t+1} = \frac{6-t}{-6}$$

$-2 \times (-6) = (t+1)(6-t)$

$12 = t(6-t) + 1(6-t)$

$12 = 6t - t^2 + 6 - t$

$12 = 5t - t^2 + 6$

$t^2 - 5t + 6 = 0$

Factorizing

$t^2 - 2t - 3t + 6 = 0$

$t(t-2) - 3(t-2) = 0$

$(t-3)(t-2) = 0$

$t - 3 = 0$ or $t - 2 = 0$

$t = 3$ or 2 (B)

Application of coordinates geometry and matrix to area of triangle and quadrilaterals

Area of triangle whose vertices are (x_1, y_1), (x_2, y_2), (x_3, y_3)

$$= \frac{1}{2}\begin{vmatrix} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix}$$

1998/11(iii) (Nov)

Given the points P(11, 13), Q(-1, 12) and R(1, -2), find the area of the triangle PQR

Solution

Area of triangle PQR = $\frac{1}{2}\begin{vmatrix} 1 & 1 & 1 \\ 11 & -1 & 1 \\ 13 & 12 & -2 \end{vmatrix}$

$= \frac{1}{2}\{1[(-1)\times(-2) - 12\times1] - 1[11\times(-2) - (13\times1)]$
$\quad + 1[11\times12 - 13\times(-1)\}$

$= \frac{1}{2}\{(2-12) - (-22-13) + (132+13)\}$

$= \frac{1}{2}\{-10 + 35 + 145\}$

= 85 sq units

2009/29 Neco

Find the area of triangle ABC whose vertices are A(0,0) B(3, 4) and C(2, 6)

A 20 sq units B 15 sq units C 10 sq units
D 7 sq units E 5 sq units

Solution

Area of triangle = $\frac{1}{2}\begin{vmatrix} 1 & 1 & 1 \\ 0 & 3 & 2 \\ 0 & 4 & 6 \end{vmatrix}$

$= \frac{1}{2}[1(3\times6 - 4\times2) - 1\times0 + 1\times0]$

$= \frac{1}{2}(10)$

= 5 sq units

Area of quadrilateral whose vertices are $R(x_1, y_1)$, $S(x_2, y_2)$, $T(x_3, y_3)$, $U(x_4, y_4)$

Quad RSTU = $\Delta RST \quad + \quad \Delta STU$

$$= \frac{1}{2}\begin{vmatrix} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix} + \frac{1}{2}\begin{vmatrix} 1 & 1 & 1 \\ x_1 & x_3 & x_4 \\ y_1 & y_3 & y_4 \end{vmatrix}$$

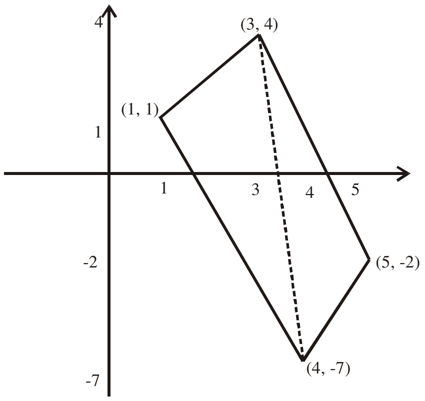
A good sketch to locate the various coordinates will be useful then we pick our values from the beginning lower point going along the boundaries from right to left

Example

Find the area of a quadrilateral with vertices (1, 1), (3, 4), (5, -2), and (4, -7)

Solution

We sketch as:



Area of a quad = $\frac{1}{2}\begin{vmatrix} 1 & 1 & 1 \\ 1 & 4 & 3 \\ 1 & -7 & 4 \end{vmatrix} + \frac{1}{2}\begin{vmatrix} 1 & 1 & 1 \\ 4 & 5 & 3 \\ -7 & -2 & 4 \end{vmatrix}$

$=\{\frac{1}{2}[(16+21) - 1(4-3) + (-7-4)]\} + \frac{1}{2}\{(20+6) -$
$\qquad 1(16+21) + (-8+35)\}$

$= \frac{1}{2}(37-1-11) + \frac{1}{2}(26-37+27)$

$= \frac{1}{2}(41)$

= 20.5 sq units

Chapter twenty six

Matrix

Matrix (matrices for plural) like every other topic in Mathematics is to be covered from the very basics then we advance to the highest point that the syllabus permits. The test on this topic shall be simple but technical as it is with all Mathematics questions- they are not meant to consume your time.

A matrix is a simple mathematical structure that holds numerical information in a rectangular form usually enclosed by bracket

Eg1. A college has its number of students in each of the SS classes classified as follows:

	Male	female
SS1	14	8
SS11	6	12
SS111	21	18

The above information can be represented in matrix form say

Matrix $A = \begin{pmatrix} 14 & 8 \\ 6 & 12 \\ 21 & 18 \end{pmatrix}$

Where the first *row* represents SS1, the second SS11 and so on. Similarly, the first *column* is the number of male students and the second *column* for female students. Thus it is important to retain each element in its original position when dealing with matrix. It is customary to write a matrix in the form

A = $\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{32} \end{pmatrix}$ three by three matrix

$B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$ two by two matrix

The horizontal components
(a_{11} a_{12} a_{13}), (a_{21} a_{22} a_{23}) and

(a_{31} a_{32} a_{33}) are the rows while the vertical components

$\begin{pmatrix} a_{11} \\ a_{21} \\ a_{31} \end{pmatrix}$ *or* $\begin{pmatrix} a_{12} \\ a_{22} \\ a_{32} \end{pmatrix}$ *or* $\begin{pmatrix} a_{13} \\ a_{23} \\ a_{33} \end{pmatrix}$

are the column of the matrix .

Eg2. Identify the position of the elements listed below in 3 by 3 matrix: $a_{22} = 12$ and $a_{31} = 24$

Solution

Element a_{22} being 12 i.e the element on the second row and 2nd column.

$a_{31} = 24$ i.e element on the 3rd row and first column

Class discussion/self test

Try to identify elements in a_{23}, a_{32} and a_{12} of the matrix in example 1 above.

<u>Definitions</u>

A matrix with m rows and n columns is called an **m** by **n** matrix. Where m × n or m by n is called the size or shape of the matrix.

Eg3. Our previous example (i.e Eg1) is a 3 × 2 matrix

Eg4. $B = \begin{pmatrix} 1 & 2 & 3 \\ 5 & 6 & 7 \\ 9 & 10 & 0 \end{pmatrix}$ is a 3×3 matrix

Eg5. $C = \begin{pmatrix} 1 & y & 4 \\ x & 6 & 7 \\ 0 & 0 & 2 \end{pmatrix}$ is also 3×3 matrix

Eg6. $D = \begin{pmatrix} 11 & 25 \\ 1 & 0 \end{pmatrix}$ is a 2×2 matrix

Eg7. $M = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ is a 3×1 matrix (column matrix)

Eg8. $N = \begin{pmatrix} 4 & 5 & 6 & 10 \end{pmatrix}$ is a 1×4 matrix(row matrix)

Matrices are usually denoted by capital letters while their elements are in lower case.

$B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$ *and not* $b = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$

Types of matrix

(i) Zero matrix

This is a type of matrix whose entries are all zeroes

$A_{2\times 2} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ *or* $B_{3\times 3} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

(ii) Square matrix

This is a matrix whose number of rows equal to number of columns. i.e m×n matrix but m = n

$A_{2\times 2} = \begin{pmatrix} 2 & 4 \\ 6 & 8 \end{pmatrix}$ *or* $B_{3\times 3} = \begin{pmatrix} 1 & 4 & 5 \\ 0 & 7 & 11 \\ 13 & 17 & 19 \end{pmatrix}$

(iii) When the elements' **row ≠ column** we say that the matrix is non-square matrix

$C = \begin{pmatrix} 1 & 2 & 5 \\ 3 & 4 & 6 \end{pmatrix}$ $D = \begin{pmatrix} 10 & 15 \\ 21 & 3 \\ 0 & 1 \end{pmatrix}$

(iv) identity matrix – it is denoted by I or I

$I_{2\times 2} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ $I_{3\times 3} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Students should master the above pattern of generalizing the identity matrix for 2×2 and 3 by 3.

Addition & subtraction of matrices

Eg. 1 If $A = \begin{pmatrix} 6 & 4 \\ 12 & 6 \end{pmatrix}$; $B = \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix}$

find (i) A + B (ii) A – B

Solution

i. $A + B = \begin{pmatrix} 6+2 & 4+3 \\ 12+4 & 6+5 \end{pmatrix} = \begin{pmatrix} 8 & 7 \\ 16 & 11 \end{pmatrix}$

ii. $A - B = \begin{pmatrix} 6-2 & 4-3 \\ 12-4 & 6-5 \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 8 & 1 \end{pmatrix}$

Eg. 2 Perform C + D and C – D on the matrices below:

$C = \begin{pmatrix} 3 & 2 & 4 \\ -1 & 0 & 2 \\ -3 & 1 & 6 \end{pmatrix}$ and $D = \begin{pmatrix} -1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 2 & 1 \end{pmatrix}$

Solution

$$C + D = \begin{pmatrix} 3+(-1) & 2+2 & 4+3 \\ -1+0 & 0+1 & 2+1 \\ -3+0 & 1+2 & 6+1 \end{pmatrix} = \begin{pmatrix} 2 & 4 & 7 \\ -1 & 1 & 3 \\ -3 & 3 & 7 \end{pmatrix}$$

$$C - D = \begin{pmatrix} 3-(-1) & 2-2 & 4-3 \\ -1-0 & 0-1 & 2-1 \\ -3-0 & 1-2 & 6-1 \end{pmatrix} = \begin{pmatrix} 4 & 0 & 1 \\ -1 & -1 & 1 \\ -3 & -1 & 5 \end{pmatrix}$$

Eg.3 Part-time teachers/school matrix has the structure:

$$\begin{array}{lc} & \text{Schools} \\ & \begin{array}{cc} A & B \end{array} \\ \begin{array}{l} \text{Teachers} \quad 1 \\ \text{Number} \quad 2 \\ \qquad\qquad\quad 3 \end{array} & \begin{pmatrix} x & x \\ x & x \\ x & x \end{pmatrix} \end{array}$$

and it represents the number of periods taken by a particular teacher in a particular school in a certain week. If W_i is the teacher/school matrix for week i and

$W_1 = \begin{pmatrix} 4 & 0 \\ 0 & 0 \\ 2 & 3 \end{pmatrix}$; $W_2 = \begin{pmatrix} 3 & 1 \\ 2 & 1 \\ 0 & 2 \end{pmatrix}$; $W_3 = \begin{pmatrix} 2 & 0 \\ 2 & 4 \\ 2 & 0 \end{pmatrix}$

find the value of

(a) W_{1+2} (where $W_{1+2} = W_1 + W_2$)

(b) W_{1+2+3} (c) $W_{1+2+3} - W_2$

Solution

(a) $W_{1+2} = W_1 + W_2 = \begin{pmatrix} 4+3 & 0+1 \\ 0+2 & 0+1 \\ 2+0 & 3+2 \end{pmatrix} = \begin{pmatrix} 7 & 1 \\ 2 & 1 \\ 2 & 5 \end{pmatrix}$

(b) $W_{1+2+3} = W_{1+2} + W_3 = \begin{pmatrix} 7+2 & 1+0 \\ 2+2 & 1+4 \\ 2+2 & 5+0 \end{pmatrix} = \begin{pmatrix} 9 & 1 \\ 4 & 5 \\ 4 & 5 \end{pmatrix}$

(c) $W_{1+2+3} - W_2 = \begin{pmatrix} 9-3 & 1-1 \\ 4-2 & 5-1 \\ 4-0 & 5-2 \end{pmatrix} = \begin{pmatrix} 6 & 0 \\ 2 & 4 \\ 4 & 3 \end{pmatrix}$

Multiplication of matrix by a number (scalar multiplication)

Eg.1 If $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ find (i) 5A (ii) – A (iii) –2A

Solution

(i) $5A = \begin{pmatrix} 5\times1 & 5\times2 \\ 5\times3 & 5\times4 \end{pmatrix} = \begin{pmatrix} 5 & 10 \\ 15 & 20 \end{pmatrix}$

(ii) $-A = \begin{pmatrix} -1 & -2 \\ -3 & -4 \end{pmatrix}$

Students/readers to find – 2A

Eg2. If $B = \begin{pmatrix} 0 & -1 \\ -15 & 10 \end{pmatrix}$ find (i) –B (ii) – 3B (iii) – ½ B

Solution

(i) $-B = \begin{pmatrix} 0 & 1 \\ 15 & -10 \end{pmatrix}$ observe that 0 × (-1) = 0 no negative in zero

Students/readers to find – 3B

(iii) $-\tfrac{1}{2}B = \begin{pmatrix} 0 & \tfrac{1}{2} \\ \tfrac{15}{2} & -5 \end{pmatrix}$

Eg3. The following matrix X gives the prices of comparable articles by size and type (in ₦)

$$X = \begin{array}{c} A \\ B \\ C \end{array} \begin{pmatrix} 0.40 & 0.10 \\ 0.30 & 0.50 \\ 0.20 & 0.60 \end{pmatrix}$$

If all prices were increased by 50%, the new prices matrix Y will be?

Solution

Normal prices 100%

Increment 50%

New price = 100 + 50%

= 150%

$= \frac{150}{100} = 1.5$

$$Y = (1.5)\begin{pmatrix} 0.40 & 0.10 \\ 0.30 & 0.50 \\ 0.20 & 0.60 \end{pmatrix}$$

$$= \begin{pmatrix} 1.5\times0.40 & 1.5\times0.10 \\ 1.5\times0.30 & 1.5\times0.50 \\ 1.5\times0.20 & 1.5\times0.60 \end{pmatrix} = \begin{pmatrix} 0.60 & 0.15 \\ 0.45 & 0.75 \\ 0.30 & 0.90 \end{pmatrix}$$

Students/readers poser

If all the prices were decreased by 50%, the new price matrix Z will be?

2014/15 Neco

Given that A = $\begin{pmatrix} -1 & 3 & -4 \\ 2 & -6 & 3 \\ 5 & 2 & -4 \end{pmatrix}$, find 3A

A $\begin{pmatrix} -1 & 3 & -4 \\ 2 & -6 & 3 \\ 5 & 2 & -4 \end{pmatrix}$ B $\begin{pmatrix} 3 & 9 & 12 \\ 6 & 18 & 9 \\ 15 & 6 & 12 \end{pmatrix}$

C $\begin{pmatrix} -3 & 9 & -12 \\ 6 & -18 & 6 \\ 15 & 6 & 12 \end{pmatrix}$ D $\begin{pmatrix} -3 & 9 & -12 \\ 6 & -18 & 9 \\ 15 & 6 & -12 \end{pmatrix}$ E

$\begin{pmatrix} 2 & 6 & -1 \\ 5 & -3 & 6 \\ 8 & 5 & -1 \end{pmatrix}$

Solution

$$3A = 3\begin{pmatrix} -1 & 3 & -4 \\ 2 & -6 & 3 \\ 5 & 2 & -4 \end{pmatrix} = \begin{pmatrix} -3 & 9 & -12 \\ 6 & -18 & 9 \\ 15 & 6 & -12 \end{pmatrix}$$ D.

2001/20 UME Exercise 26.1

If $P = \begin{pmatrix} 3 & -2 & 4 \\ 5 & 0 & 6 \\ 7 & 5 & -1 \end{pmatrix}$ then -2P is

A. $\begin{pmatrix} -6 & 4 & -8 \\ 5 & 0 & 6 \\ 7 & 5 & -1 \end{pmatrix}$ B. $\begin{pmatrix} -6 & 4 & -8 \\ -10 & 0 & 6 \\ -14 & 5 & -1 \end{pmatrix}$

C. $\begin{pmatrix} -6 & -4 & 2 \\ -10 & -2 & -12 \\ -14 & -10 & 2 \end{pmatrix}$ D. $\begin{pmatrix} -6 & 4 & -8 \\ -10 & 0 & -12 \\ -14 & -10 & 2 \end{pmatrix}$

Multiplication of matrix by a matrix

Multiplication of a row matrix by a column matrix can be done by combining the matrix as follows:

$A = \begin{pmatrix} a_1 & a_2 \end{pmatrix}$ and $B = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$

$A \times B = A \bullet B = a_1 b_1 + a_2 b_2$

Eg.1 If $A = \begin{pmatrix} 1 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ find AB

Solution

$AB = 1 \times 3 + 2 \times 4$
$= 3 + 8$
$= 11$

Also if $C = \begin{pmatrix} c_1 & c_2 & c_3 \end{pmatrix}$ and $D = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \end{pmatrix}$

Multiplication of CD is possible

Eg.2 If the matrix X is used to hold the records of sets A, B and C of books in a library and matrix Y to hold the unit cost (₦ 000,000) each set as follows

$$X = \begin{array}{c} \begin{matrix} A & B & C \end{matrix} \\ \begin{pmatrix} 10 & 25 & 5 \end{pmatrix} \end{array} \quad Y = \begin{pmatrix} 2.50 \\ 0.50 \\ 0.20 \end{pmatrix}$$, find XY

$$XY = \begin{pmatrix} 10 & 25 & 5 \end{pmatrix} \bullet \begin{pmatrix} 2.50 \\ 0.50 \\ 0.20 \end{pmatrix}$$

= 10 (2.50) + 25(0.50) + 5(0.20)
= 25 + 12.5 + 1.0
= ₦ (38.5) 000 000
= ₦ 38 500 000

This is the total cost of all the books in the library.
The principle in the example above is to be applied always in multiplying matrices i.e. row by column operation.

Pattern of movement

Eg. 3 If $A = \begin{pmatrix} 2 & 3 \\ 6 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 5 & 7 \\ 9 & 11 \end{pmatrix}$ find AB

Solution

$$A \bullet B = \begin{pmatrix} 2\times5+3\times9 & 2\times7+3\times11 \\ 6\times5+4\times9 & 6\times7+4\times11 \end{pmatrix}$$
$$= \begin{pmatrix} 10+27 & 14+33 \\ 30+36 & 42+44 \end{pmatrix} = \begin{pmatrix} 37 & 47 \\ 66 & 86 \end{pmatrix}$$

Eg.4 If $X = \begin{pmatrix} 1 & 8 \\ 0 & 2 \end{pmatrix}$ and $Y = \begin{pmatrix} 3 & 4 \\ 5 & 6 \end{pmatrix}$ find XY

$$XY = \begin{pmatrix} 1\times3+8\times5 & 1\times4+8\times6 \\ 0\times3+2\times5 & 0\times4+2\times6 \end{pmatrix}$$
$$= \begin{pmatrix} 3+40 & 4+48 \\ 0+10 & 0+12 \end{pmatrix} = \begin{pmatrix} 43 & 52 \\ 10 & 12 \end{pmatrix}$$

Students/readers' poser

If $A = \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 6 \\ 0 & 5 \end{pmatrix}$

find (i) A B (ii) A^2 = (i.e $A^2 = A \times A$)
(iii) B^2 (iv) show that $AB \neq BA$ i.e multiplication of matrix is not commutative, but AA = AA if they are the same

General discussions/observation

Condition under which multiplication is possible with matrix is once number of rows = no of columns. From the list below, identify as many as possible two matrices that fulfill such condition

$A = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}$, $B = \begin{pmatrix} 10 \\ 7 \\ 8 \end{pmatrix}$, $C = \begin{pmatrix} 1 & 0 & 3 \end{pmatrix}$, $D = \begin{pmatrix} 0 \\ 5 \end{pmatrix}$, $E = \begin{pmatrix} 8 & 1 \end{pmatrix}$

$$F=\begin{pmatrix}1&0\\5&6\end{pmatrix},\ G=\begin{pmatrix}1&2&3\\0&4&10\\1&0&5\end{pmatrix},H=\begin{pmatrix}1&0&2\\0&1&5\end{pmatrix},$$

$$J=\begin{pmatrix}1&2\\3&4\\5&6\end{pmatrix}$$

$$K=\begin{pmatrix}15\\3\end{pmatrix},\ L=\begin{pmatrix}0&17\end{pmatrix},\ M=\begin{pmatrix}7&0\\9&1\end{pmatrix},$$

$$N=\begin{pmatrix}4&9&0\\0&2&1\end{pmatrix}$$

some are CA, ED, GA, FD, LK and so on.

2014/12 Neco (Dec)

Given that A = $\begin{pmatrix}1&-3\\4&2\end{pmatrix}$ and B = $\begin{pmatrix}5&1\\-1&0\end{pmatrix}$, evaluate

A × B

A $\begin{pmatrix}8&1\\22&4\end{pmatrix}$ B $\begin{pmatrix}2&1\\18&4\end{pmatrix}$ C $\begin{pmatrix}8&1\\18&4\end{pmatrix}$

D $\begin{pmatrix}6&-2\\3&2\end{pmatrix}$ E $\begin{pmatrix}8&1\\18&-4\end{pmatrix}$

Solution

$$A\times B=\begin{pmatrix}1&-3\\4&2\end{pmatrix}\begin{pmatrix}5&1\\-1&0\end{pmatrix}$$

$$=\begin{pmatrix}1\times5+(-3)\times(-1)&1\times1+(-3)\times0\\4\times5+2\times(-1)&4\times1+2\times0\end{pmatrix}$$

$$=\begin{pmatrix}5+3&1\\20-2&4\end{pmatrix}$$

$$=\begin{pmatrix}8&1\\18&4\end{pmatrix}\quad C.$$

2009/32 f/m

Evaluate $\begin{pmatrix}2&3\\4&1\end{pmatrix}\begin{pmatrix}2\\3\end{pmatrix}$

A (13, 11) B (11, 13) C $\begin{pmatrix}13\\11\end{pmatrix}$ D $\begin{pmatrix}11\\13\end{pmatrix}$

Solution

$$\begin{pmatrix}2&3\\4&1\end{pmatrix}\begin{pmatrix}2\\3\end{pmatrix}=\begin{pmatrix}2\times2+3\times3\\4\times2+1\times3\end{pmatrix}$$

$$=\begin{pmatrix}4+9\\8+3\end{pmatrix}$$

$$=\begin{pmatrix}13\\11\end{pmatrix}\quad C.$$

2015/56 Neco

Calculate the product of the matrix

$(2\quad 3\quad 1)$ and $\begin{pmatrix}5\\6\\7\end{pmatrix}$

A 35 B 23 C 18 D 5 E 2

Solution

$$(2\quad 3\quad 1)\bullet\begin{pmatrix}5\\6\\7\end{pmatrix}=2\times5+3\times6+1\times7$$

$$=10+18+7$$

$$=35\quad A.$$

2002/19 Neco f/m

If A = $\begin{pmatrix}2&1&2\\1&2&-2\\2&-2&-1\end{pmatrix}$, What is A^3 ?

A 3A B 6A C 8A D 9A E 10A

Solution

A^3 is same as $A^2\times A$ and A^2 is $A\times A$

$$A\times A=\begin{pmatrix}2&1&2\\1&2&-2\\2&-2&-1\end{pmatrix}\begin{pmatrix}2&1&2\\1&2&-2\\2&-2&-1\end{pmatrix}$$

$$=\begin{pmatrix}4+1+4&2+2-4&4-2-2\\2+2+-4&1+4+4&2-4+2\\4-2-2&2-4+2&4+4+1\end{pmatrix}$$

$$=\begin{pmatrix}9&0&0\\0&9&0\\0&0&9\end{pmatrix}$$

Then $A^3=A^2\times A$

$$=\begin{pmatrix}9&0&0\\0&9&0\\0&0&9\end{pmatrix}\begin{pmatrix}2&1&2\\1&2&-2\\2&-2&-1\end{pmatrix}$$

$$=\begin{pmatrix}18+0+0&9+0+0&18+0+0\\0+9+0&0+18+0&0-18+0\\0+0+18&0+0-18&0+0-9\end{pmatrix}$$

$$=\begin{pmatrix}18&9&18\\9&18&-18\\18&-18&-9\end{pmatrix}\ \text{i.e. } 9A\ (D)$$

2010/9b

Given that $\begin{pmatrix}2&0&1\\5&-3&1\\0&4&6\end{pmatrix}\begin{pmatrix}1\\m\\r\end{pmatrix}=\begin{pmatrix}k\\2\\26\end{pmatrix}$

find the values of constants k, m and r

Solution

$$\begin{pmatrix}2\times1+0\times m+1\times r\\5\times1+(-3)\times m+1\times r\\0\times1+4\times m+6\times r\end{pmatrix}=\begin{pmatrix}k\\2\\26\end{pmatrix}$$

$$\begin{pmatrix}2+0+r\\5-3m+r\\0+4m+6r\end{pmatrix}=\begin{pmatrix}k\\2\\26\end{pmatrix}$$

Equating rows

$2+0+r=k$ ------ (1)

$5-3m+r=2$ ----- (2)

$0 + 4m + 6r = 26$ ---- (3)

Solving (2) and (3)

$r - 3m = -3$ ---- (a)

$6r + 4m = 26$ ----- (b)

(a) × 4 + (b) × 3

$$\begin{array}{r} 4r - 12m = -12 \\ +\ 18r + 12m = 78 \\ \hline 22r \qquad = 66 \end{array}$$

$r = 3$

Substitute r = 3 into (a)

$3 - 3m = -3$

$3 + 3 = 3m$

$6 = 3m$ and $m = 2$

Substitute r = 3 into (1)

$2 + 3 = k$

$5 = k$

2005/44 PCE Exercise 26.2

If $P = \begin{pmatrix} 1 & 3 \\ -4 & -2 \end{pmatrix}$ and $Q = \begin{pmatrix} 3 & 2 \\ 2 & 1 \end{pmatrix}$,

what is the matrix PQ?

A $\begin{pmatrix} -3 & 5 \\ 8 & 10 \end{pmatrix}$ B $\begin{pmatrix} 9 & 5 \\ -16 & -10 \end{pmatrix}$

C $\begin{pmatrix} -3 & 8 \\ -6 & 2 \end{pmatrix}$ D $\begin{pmatrix} 3 & 6 \\ -8 & -2 \end{pmatrix}$

2007/23 PCE Exercise 26.3

If $R = \begin{pmatrix} 3 & 1 \\ 2 & 5 \end{pmatrix}$ and $5 = \begin{pmatrix} 5 & 2 \\ -2 & 4 \end{pmatrix}$, find RS

A $\begin{pmatrix} 13 & 10 \\ 0 & 24 \end{pmatrix}$ B $\begin{pmatrix} 17 & 10 \\ 20 & 24 \end{pmatrix}$

C $\begin{pmatrix} 13 & 10 \\ 20 & 24 \end{pmatrix}$ D $\begin{pmatrix} 17 & 10 \\ 0 & 24 \end{pmatrix}$

2009/24 PCE Exercise 26.4

If $Q = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$ and $R = \begin{pmatrix} 3 & 2 \\ 1 & 3 \end{pmatrix}$, find QR

A $\begin{pmatrix} 9 & 13 \\ 5 & 8 \end{pmatrix}$ B $\begin{pmatrix} 5 & 6 \\ 13 & 9 \end{pmatrix}$ C $\begin{pmatrix} 8 & 9 \\ 13 & 5 \end{pmatrix}$ D $\begin{pmatrix} 9 & 5 \\ 13 & 8 \end{pmatrix}$

Joint cases of Addition, subtraction and multiplication

2007/25 UME

If $f(x) = 3x - 2I$, $P = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}$ and I is 2 × 2 identity matrix, evaluate f(p)

A $\begin{pmatrix} 6 & 3 \\ -3 & 0 \end{pmatrix}$ B $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ C $\begin{pmatrix} 8 & 3 \\ -3 & 2 \end{pmatrix}$ D $\begin{pmatrix} 4 & 3 \\ -3 & -2 \end{pmatrix}$

Solution

$f(x) = 3x - 2I$ and $P = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}$, $I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$$f(p) = 3\begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix} - 2\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 6 & 3 \\ -3 & 0 \end{pmatrix} - \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 4 & 3 \\ -3 & -2 \end{pmatrix} \quad \text{D.}$$

1994/11

If $P = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$ and $Q = \begin{pmatrix} 1 & -1 \\ 3 & -2 \end{pmatrix}$, find PQ + 3Q

A $\begin{pmatrix} 0 & 3 \\ -6 & 5 \end{pmatrix}$ B $\begin{pmatrix} 0 & -3 \\ -6 & -9 \end{pmatrix}$ C $\begin{pmatrix} 2 & -1 \\ -6 & 4 \end{pmatrix}$ D $\begin{pmatrix} 8 & -7 \\ 12 & -8 \end{pmatrix}$ E $\begin{pmatrix} 2 & -7 \\ -6 & -8 \end{pmatrix}$

Solution

$$PQ + 3Q = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 3 & -2 \end{pmatrix} + 3\begin{pmatrix} 1 & -1 \\ 3 & -2 \end{pmatrix}$$

$$= \begin{pmatrix} 2\times1+1\times3 & 2\times(-1)+1\times(-2) \\ 0\times1+1\times3 & 0\times(-1)+1\times(-2) \end{pmatrix} + \begin{pmatrix} 3 & -3 \\ 9 & -6 \end{pmatrix}$$

$$= \begin{pmatrix} 2+3 & -2-2 \\ 0+3 & 0-2 \end{pmatrix} + \begin{pmatrix} 3 & -3 \\ 9 & -6 \end{pmatrix}$$

$$= \begin{pmatrix} 5 & -4 \\ 3 & -2 \end{pmatrix} + \begin{pmatrix} 3 & -3 \\ 9 & -6 \end{pmatrix} = \begin{pmatrix} 8 & -7 \\ 12 & -8 \end{pmatrix} \quad \text{(D)}$$

2015/8 f/m

Given that $P = \begin{pmatrix} -2 & 1 \\ 3 & 4 \end{pmatrix}$ and $Q = \begin{pmatrix} 5 & -3 \\ 2 & -1 \end{pmatrix}$, find PQ – QP

A $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ B $\begin{pmatrix} 27 & 12 \\ 16 & -15 \end{pmatrix}$ C $\begin{pmatrix} -20 & -6 \\ 12 & -8 \end{pmatrix}$ D $\begin{pmatrix} 11 & 12 \\ 30 & -11 \end{pmatrix}$

Solution

$$PQ = \begin{pmatrix} -2 & 1 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} 5 & -3 \\ 2 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} (-2)\times5+1\times2 & (-2)\times(-3)+1\times(-1) \\ 3\times5+4\times2 & 3\times(-3)+4\times(-1) \end{pmatrix}$$

$$= \begin{pmatrix} -10 + 2 & 6 - 1 \\ 15 + 8 & -9 - 4 \end{pmatrix}$$

$$PQ = \begin{pmatrix} -8 & 5 \\ 23 & -13 \end{pmatrix}$$

$$QP = \begin{pmatrix} 5 & -3 \\ 2 & -1 \end{pmatrix}\begin{pmatrix} -2 & 1 \\ 3 & 4 \end{pmatrix}$$

$$= \begin{pmatrix} 5\times(-2)+(-3)\times3 & 5\times1+(-3)\times4 \\ 2\times(-2)+(-1)\times3 & 2\times1+(-1)\times4 \end{pmatrix}$$

$$= \begin{pmatrix} -10 - 9 & 5 - 12 \\ -4 - 3 & 2 - 4 \end{pmatrix}$$

$$= \begin{pmatrix} -19 & -7 \\ -7 & -2 \end{pmatrix}$$

Thus, PQ – QP = $\begin{pmatrix}-8 & 5\\ 23 & -13\end{pmatrix} - \begin{pmatrix}-19 & -7\\ -7 & -2\end{pmatrix}$

$= \begin{pmatrix}11 & 12\\ 30 & -11\end{pmatrix}$ D.

2009/29 f/m

If P = $\begin{pmatrix}1 & 1\\ 2 & 1\end{pmatrix}$, find ($P^2$ + P)

A $\begin{pmatrix}4 & 3\\ 6 & 1\end{pmatrix}$ B $\begin{pmatrix}4 & 3\\ 6 & 4\end{pmatrix}$ C $\begin{pmatrix}3 & 2\\ 6 & 1\end{pmatrix}$ D $\begin{pmatrix}3 & 2\\ 6 & 4\end{pmatrix}$

Solution

$P^2 = \begin{pmatrix}1 & 1\\ 2 & 1\end{pmatrix}\begin{pmatrix}1 & 1\\ 2 & 1\end{pmatrix}$

$= \begin{pmatrix}1\times1+1\times2 & 1\times1+1\times1\\ 2\times1+1\times2 & 2\times1+1\times1\end{pmatrix}$

$= \begin{pmatrix}1+2 & 1+1\\ 2+2 & 2+1\end{pmatrix} = \begin{pmatrix}3 & 2\\ 4 & 3\end{pmatrix}$

$P^2 + P = \begin{pmatrix}3 & 2\\ 4 & 3\end{pmatrix} + \begin{pmatrix}1 & 1\\ 2 & 1\end{pmatrix}$

$= \begin{pmatrix}4 & 3\\ 6 & 4\end{pmatrix}$ B.

2003/21 f/m

Given that P = $\begin{pmatrix}2 & -1\\ -3 & 3\end{pmatrix}$ and I is a 2 by 2 identity matrix, find P + 2I

A $\begin{pmatrix}5 & -2\\ -6 & 7\end{pmatrix}$ B $\begin{pmatrix}4 & 1\\ -1 & 5\end{pmatrix}$

C $\begin{pmatrix}2 & 1\\ -1 & 3\end{pmatrix}$ D $\begin{pmatrix}4 & -1\\ -3 & 5\end{pmatrix}$

Solution

I a 2×2 identity matrix = $\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$

$P + 2I = \begin{pmatrix}2 & -1\\ -3 & 3\end{pmatrix} + 2\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$

$= \begin{pmatrix}2 & -1\\ -3 & 3\end{pmatrix} + \begin{pmatrix}2 & 0\\ 0 & 2\end{pmatrix}$

$= \begin{pmatrix}4 & -1\\ -3 & 5\end{pmatrix}$ D.

2012/16 f/m

Given that P = $\begin{pmatrix}2 & 1\\ 5 & -3\end{pmatrix}$ and Q = $\begin{pmatrix}4 & -8\\ 1 & -2\end{pmatrix}$, find (2P – Q)

A $\begin{pmatrix}-6 & 17\\ 3 & 1\end{pmatrix}$ B $\begin{pmatrix}-2 & 9\\ 4 & 1\end{pmatrix}$ C $\begin{pmatrix}0 & -6\\ 9 & -8\end{pmatrix}$ D $\begin{pmatrix}0 & 10\\ 9 & -4\end{pmatrix}$

Solution

$2P - Q = 2\begin{pmatrix}2 & 1\\ 5 & -3\end{pmatrix} - \begin{pmatrix}4 & -8\\ 1 & -2\end{pmatrix}$

$= \begin{pmatrix}4 & 2\\ 10 & -6\end{pmatrix} + \begin{pmatrix}-4 & 8\\ -1 & 2\end{pmatrix}$

$= \begin{pmatrix}0 & 10\\ 9 & -4\end{pmatrix}$ D.

2007/24 PCE

If X = $\begin{pmatrix}2 & 0\\ 3 & -1\end{pmatrix}$ and Y = $\begin{pmatrix}4 & 2\\ -1 & 3\end{pmatrix}$, find 2X – Y

A $\begin{pmatrix}8 & 2\\ 5 & -5\end{pmatrix}$ B $\begin{pmatrix}0 & -2\\ 7 & -5\end{pmatrix}$

C $\begin{pmatrix}0 & 2\\ 5 & 1\end{pmatrix}$ D $\begin{pmatrix}-6 & -4\\ 5 & -7\end{pmatrix}$

Solution

$2X - Y = 2\begin{pmatrix}2 & 0\\ 3 & -1\end{pmatrix} - \begin{pmatrix}4 & 2\\ -1 & 3\end{pmatrix}$

$= \begin{pmatrix}4 & 0\\ 6 & -2\end{pmatrix} + \begin{pmatrix}-4 & -2\\ 1 & -3\end{pmatrix}$

$= \begin{pmatrix}0 & -2\\ 7 & -5\end{pmatrix}$ B.

2013/20 Neco f/m

Given that A = $\begin{pmatrix}-3 & 1\\ 4 & 2\end{pmatrix}$ and B = $\begin{pmatrix}2 & -1\\ 3 & 4\end{pmatrix}$

evaluate $\frac{3}{2}$A + B

A $\begin{pmatrix}\frac{5}{2} & -\frac{1}{2}\\ 6 & 3\end{pmatrix}$ B $\begin{pmatrix}-9 & 3\\ 12 & 6\end{pmatrix}$

C $\begin{pmatrix}\frac{-5}{2} & \frac{1}{2}\\ 9 & 7\end{pmatrix}$ D $\begin{pmatrix}\frac{5}{2} & -\frac{1}{2}\\ 9 & 7\end{pmatrix}$ E $\begin{pmatrix}\frac{-5}{2} & \frac{1}{2}\\ -9 & 7\end{pmatrix}$

Solution

$\frac{3}{2}A + B = \frac{3}{2}\begin{pmatrix}-3 & 1\\ 4 & 2\end{pmatrix} + \begin{pmatrix}2 & -1\\ 3 & 4\end{pmatrix}$

$= \begin{pmatrix}\frac{-9}{2} & \frac{3}{2}\\ 6 & 3\end{pmatrix} + \begin{pmatrix}2 & -1\\ 3 & 4\end{pmatrix}$

$= \begin{pmatrix}\frac{-5}{2} & \frac{1}{2}\\ 9 & 7\end{pmatrix}$ C.

2007/27 UME

Evaluate $\begin{pmatrix}3 & -2\\ -7 & 5\end{pmatrix} + 2\begin{pmatrix}-2 & 4\\ 3 & -1\end{pmatrix}$

A $\begin{pmatrix}3 & 4\\ -2 & 6\end{pmatrix}$ B $\begin{pmatrix}-1 & 6\\ -1 & 3\end{pmatrix}$

C $\begin{pmatrix}-1 & 6\\ 1 & 3\end{pmatrix}$ D $\begin{pmatrix}3 & 4\\ 2 & 6\end{pmatrix}$

Solution

$$\begin{pmatrix}3 & -2\\ -7 & 5\end{pmatrix} + 2\begin{pmatrix}-2 & 4\\ 3 & -1\end{pmatrix} = \begin{pmatrix}3 & -2\\ -7 & 5\end{pmatrix} + \begin{pmatrix}-4 & 8\\ 6 & -2\end{pmatrix}$$

$$= \begin{pmatrix}-1 & 6\\ -1 & 3\end{pmatrix} \quad \text{B.}$$

2015/11 Neco

If $P = \begin{pmatrix}3 & 0 & 1\\ 2 & 4 & 3\\ 5 & 0 & 0\end{pmatrix}$ and $Q = \begin{pmatrix}2 & 3 & 1\\ 5 & 1 & 2\\ 3 & 4 & 2\end{pmatrix}$ find 2P – Q

A $\begin{pmatrix}6 & 0 & 2\\ 4 & 8 & 6\\ 10 & 0 & 0\end{pmatrix}$ B $\begin{pmatrix}4 & -3 & 1\\ -1 & 7 & 4\\ 7 & -4 & -2\end{pmatrix}$

C $\begin{pmatrix}4 & 3 & 1\\ 1 & 7 & 4\\ 7 & 4 & 2\end{pmatrix}$ D $\begin{pmatrix}8 & 3 & 3\\ 9 & 9 & 8\\ 13 & 4 & 2\end{pmatrix}$ E $\begin{pmatrix}1 & -3 & 0\\ -3 & 3 & 1\\ 2 & -4 & -2\end{pmatrix}$

Solution

$$2P = 2\begin{pmatrix}3 & 0 & 1\\ 2 & 4 & 3\\ 5 & 0 & 0\end{pmatrix} = \begin{pmatrix}6 & 0 & 2\\ 4 & 8 & 6\\ 10 & 0 & 0\end{pmatrix} \text{ and } -Q = \begin{pmatrix}-2 & -3 & -1\\ -5 & -1 & -2\\ -3 & -4 & -2\end{pmatrix}$$

$$2P - Q = \begin{pmatrix}6-2 & 0-3 & 2-1\\ 4-5 & 8-1 & 6-2\\ 10-3 & 0-4 & 0-2\end{pmatrix} = \begin{pmatrix}4 & -3 & 1\\ -1 & 7 & 4\\ 7 & -4 & -2\end{pmatrix} \quad \text{B.}$$

2001/26 Neco f/m

If $\begin{pmatrix}b & 4 & 1\\ -7 & m & 2\\ 1 & -3 & n\end{pmatrix} + \begin{pmatrix}4 & p & 2\\ -3 & 6 & l\\ a & 1 & 5\end{pmatrix} = \begin{pmatrix}7 & 4 & k\\ c & -2 & 4\\ 4 & j & 6\end{pmatrix}$

then $\begin{pmatrix}m & n & p\\ k & l & j\\ a & b & c\end{pmatrix}$ is

A $\begin{pmatrix}3 & 3 & -10\\ -8 & 1 & 0\\ 3 & 2 & -2\end{pmatrix}$ B $\begin{pmatrix}-8 & 1 & 0\\ 3 & -10 & -2\\ 3 & 3 & 2\end{pmatrix}$

C $\begin{pmatrix}3 & 3 & 3\\ -8 & -2 & -10\\ 1 & 0 & 2\end{pmatrix}$ D $\begin{pmatrix}-8 & 1 & 0\\ 3 & 2 & -2\\ 3 & 3 & -10\end{pmatrix}$ E $\begin{pmatrix}-8 & 3 & 3\\ 1 & 2 & 3\\ 0 & -2 & 10\end{pmatrix}$

Solution

$$\begin{pmatrix}b & 4 & 1\\ -7 & m & 2\\ 1 & -3 & n\end{pmatrix} + \begin{pmatrix}4 & p & 2\\ -3 & 6 & l\\ a & 1 & 5\end{pmatrix} = \begin{pmatrix}7 & 4 & k\\ c & -2 & 4\\ 4 & j & 6\end{pmatrix}$$

Solving column by column

b + 4 = 7 thus b = 3
– 7 – 3 = c thus c = – 10
1 + a = 4 thus a = 3
4 + p = 4 thus p = 0
m + 6 = – 2 thus m = – 8
– 3 + 1 = j thus j = – 2
1 + 2 = k thus k = 3
2 + L = 4 thus L = 2
n + 5 = 6 thus n = 1

Thus $\begin{pmatrix}m & n & p\\ k & l & j\\ a & b & c\end{pmatrix} = \begin{pmatrix}-8 & 1 & 0\\ 3 & 2 & -2\\ 3 & 3 & -10\end{pmatrix}$ D

2012/10 f/m

(a) Write down the matrix A of the linear transformation A (x, y) $\longrightarrow$ (2x – y, – 5x + 3y)

(b) if $B = \begin{pmatrix}3 & 1\\ 5 & 2\end{pmatrix}$

find (i) $A^2 - B^2$ (ii) Matrix $C = B^2A$

(c) What is the relationship between matrix B and matrix C

Solution

A (x, y) $\longrightarrow$ (2x – y, – 5x + 3y)

Coefficients only

Matrix $A = \begin{pmatrix}2 & -1\\ -5 & 3\end{pmatrix}$ i.e $\begin{pmatrix}x_1 & y_1\\ x_2 & y_2\end{pmatrix}$

$$A^2 = \begin{pmatrix}2 & -1\\ -5 & 3\end{pmatrix}\begin{pmatrix}2 & -1\\ -5 & 3\end{pmatrix}$$

$$= \begin{pmatrix}2\times2+(-1)\times(-5) & 2\times(-1)+(-1)\times3\\ (-5)\times2+3\times(-5) & (-5)\times(-1)+3\times3\end{pmatrix}$$

$$= \begin{pmatrix}4+5 & -2-3\\ -10-15 & 5+9\end{pmatrix}$$

$$= \begin{pmatrix}9 & -5\\ -25 & 14\end{pmatrix}$$

$$B^2 = \begin{pmatrix}3 & 1\\ 5 & 2\end{pmatrix}\begin{pmatrix}3 & 1\\ 5 & 2\end{pmatrix}$$

$$= \begin{pmatrix}3\times3+1\times5 & 3\times1+1\times2\\ 5\times3+2\times5 & 5\times1+2\times2\end{pmatrix}$$

$$= \begin{pmatrix}9+5 & 3+2\\ 15+10 & 5+4\end{pmatrix} = \begin{pmatrix}14 & 5\\ 25 & 9\end{pmatrix}$$

$$A^2 - B^2 = \begin{pmatrix}9 & -5\\ -25 & 14\end{pmatrix} - \begin{pmatrix}14 & 5\\ 25 & 9\end{pmatrix}$$

$$= \begin{pmatrix}-5 & -10\\ -50 & 5\end{pmatrix}$$

(ii) Matrix $C = B^2A$

$$= \begin{pmatrix}14 & 5\\ 25 & 9\end{pmatrix}\begin{pmatrix}2 & -1\\ -5 & 3\end{pmatrix}$$

$$= \begin{pmatrix}14\times2+5\times(-5) & 14\times(-1)+5\times3\\ 25\times2+9\times(-5) & 25\times(-1)+9\times3\end{pmatrix}$$

$$= \begin{pmatrix}28-25 & -14+15\\ 50-45 & -25+27\end{pmatrix} = \begin{pmatrix}3 & 1\\ 5 & 2\end{pmatrix}$$

(c) The relationship is matrix B is the same as matrix C

2009/12b f/m ***Adjusted*** **Exercise 26.5**

If $A = \begin{pmatrix}2 & 1\\ 3 & 2\end{pmatrix}$, find $T = A^2 + A + 2I$,

where I is the 2 × 2 units matrix

2013/22 UTME Exercise 26.6

If $P\begin{pmatrix}5 & 3\\ 2 & 1\end{pmatrix}$ and $Q = \begin{pmatrix}4 & 2\\ 3 & 5\end{pmatrix}$, find 2P + Q

A $\begin{pmatrix}7 & 7\\ 8 & 14\end{pmatrix}$ B $\begin{pmatrix}8 & 14\\ 7 & 7\end{pmatrix}$ C $\begin{pmatrix}7 & 7\\ 14 & 8\end{pmatrix}$ D $\begin{pmatrix}14 & 8\\ 7 & 7\end{pmatrix}$

2004/22 PCE Exercise 26.7

If R = $\begin{pmatrix}1 & 4\\3 & 2\end{pmatrix}$ and S = $\begin{pmatrix}0 & 4\\1 & -5\end{pmatrix}$, find 3R + S

A $\begin{pmatrix}3 & 16\\10 & -1\end{pmatrix}$ B $\begin{pmatrix}3 & 16\\10 & 1\end{pmatrix}$ C $\begin{pmatrix}16 & 3\\-10 & 1\end{pmatrix}$ D $\begin{pmatrix}16 & 3\\1 & -10\end{pmatrix}$

2006/40 PCE Exercise 26.8

If P = $\begin{pmatrix}2 & 3\\-3 & 1\end{pmatrix}$ and R = $\begin{pmatrix}3 & -2\\2 & -1\end{pmatrix}$, find the value of 3P – R

A $\begin{pmatrix}3 & -11\\11 & 4\end{pmatrix}$ B $\begin{pmatrix}4 & 9\\-8 & 6\end{pmatrix}$ C $\begin{pmatrix}3 & 11\\-11 & 4\end{pmatrix}$ D $\begin{pmatrix}5 & 4\\4 & -8\end{pmatrix}$

2006/18 UME Exercise 26.9

If x = $\begin{pmatrix}1 & 0 & 1\\2 & -1 & 0\\-1 & 0 & 1\end{pmatrix}$ and y = $\begin{pmatrix}-1 & 1 & 2\\0 & -1 & -1\\2 & -1 & 1\end{pmatrix}$

find 2x – y

A $\begin{pmatrix}3 & -1 & 0\\4 & -3 & -1\\-4 & 1 & 1\end{pmatrix}$ B $\begin{pmatrix}3 & -1 & 0\\4 & -1 & 1\\-4 & 1 & 1\end{pmatrix}$

C $\begin{pmatrix}3 & -1 & 0\\4 & -3 & 1\\-4 & 1 & 1\end{pmatrix}$ D $\begin{pmatrix}3 & -1 & 0\\4 & 1 & 1\\-4 & -1 & 1\end{pmatrix}$

Transpose of a matrix

Defn: The transpose of a matrix A, written A^t is the matrix obtained by writing rows of A in orders as columns

If $A = \begin{pmatrix}1 & 2\\3 & 4\end{pmatrix}$ *then* $A^t = \begin{pmatrix}1 & 3\\2 & 4\end{pmatrix}$

If $B = \begin{pmatrix}6 & 11\end{pmatrix}$ *then* $B^t = \begin{pmatrix}6\\11\end{pmatrix}$

If $C = \begin{pmatrix}0 & 1 & 4\\7 & 10 & 2\\3 & 5 & 16\end{pmatrix}$ *then* $C^t = \begin{pmatrix}0 & 7 & 3\\1 & 10 & 5\\4 & 2 & 16\end{pmatrix}$

2003/29 Neco

Given that A = $\begin{pmatrix}-2 & 4\\1 & -6\end{pmatrix}$ obtain $(A + A^T)^T$

A $\begin{pmatrix}-4 & 8\\2 & -12\end{pmatrix}$ B $\begin{pmatrix}-4 & 2\\8 & -12\end{pmatrix}$

C $\begin{pmatrix}-4 & 5\\5 & -12\end{pmatrix}$ D $\begin{pmatrix}-4 & 5\\-12 & 5\end{pmatrix}$ E $\begin{pmatrix}-2 & 1\\4 & -6\end{pmatrix}$

Solution

$A = \begin{pmatrix}-2 & 4\\1 & -6\end{pmatrix}$ and $A^T = \begin{pmatrix}-2 & 1\\4 & -6\end{pmatrix}$

$A + A^T = \begin{pmatrix}-2 & 4\\1 & -6\end{pmatrix} + \begin{pmatrix}-2 & 1\\4 & -6\end{pmatrix}$

$= \begin{pmatrix}-4 & 5\\5 & -12\end{pmatrix}$

Thus $(A + A^T)^T = \begin{pmatrix}-4 & 5\\5 & -12\end{pmatrix}$ C.

Exercise 26.10

Find the transpose of the matrices below

$P = \begin{pmatrix}4 & 0\\1 & 3\end{pmatrix}$, $Q = \begin{pmatrix}10 & 0 & 1\end{pmatrix}$, $K = \begin{pmatrix}13 & 1 & 0\\15 & 4 & 2\\0 & 1 & 7\end{pmatrix}$

2006/38 PCE Exercise 26.11

Find the transpose of the matrix $\begin{pmatrix}3 & 6\\-5 & -4\end{pmatrix}$

A $\begin{pmatrix}6 & 3\\-5 & 4\end{pmatrix}$ B $\begin{pmatrix}3 & 6\\-4 & -5\end{pmatrix}$ C $\begin{pmatrix}-5 & -4\\6 & 3\end{pmatrix}$ D $\begin{pmatrix}3 & -5\\6 & -4\end{pmatrix}$

Solving simultaneous linear equations with two unknowns using 2×2 matrix

2002/1 f/m

Express

$2x + 5y = 3$

$8x + 7y = 5$

in matrix form

A $\begin{pmatrix}x\\y\end{pmatrix}\begin{pmatrix}2 & 5\\8 & 7\end{pmatrix} = \begin{pmatrix}3\\5\end{pmatrix}$ B $(x, y)\begin{pmatrix}2 & 5\\8 & 7\end{pmatrix} = (3, 5)$

C $\begin{pmatrix}2 & 5\\8 & 7\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}3\\5\end{pmatrix}$ D $\begin{pmatrix}2 & 8\\5 & 7\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}3\\5\end{pmatrix}$

Solution

Since matrix multiplication is not commutative

$2x + 5y = 3$

$8x + 7y = 5$

Can only be represented as

$\begin{pmatrix}2 & 5\\8 & 7\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}3\\5\end{pmatrix}$ (C)

2014/22 UTME

Find y, if $\begin{pmatrix}5 & -6\\2 & -7\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}7\\-11\end{pmatrix}$

A 3 B 2 C 8 D 5

Solution

$\begin{pmatrix}5\times x + (-6)\times y\\2\times x + (-7)\times y\end{pmatrix} = \begin{pmatrix}7\\-11\end{pmatrix}$

$\begin{pmatrix}5x - 6y\\2x - 7y\end{pmatrix} = \begin{pmatrix}7\\-11\end{pmatrix}$

Equating terms

5x – 6y = 7 ---- (1)

2x – 7y = – 11 ---- (2)

Solving the resulting simultaneous equations

(1) × 2 – (2) × 5 to eliminate x

$10x - 12y = 14$

$\underline{-(10x - 35y = -55)}$

$23y = 69$

y = 3 (A)

2004/21 PCE

Given that $\begin{pmatrix}1 & 2\\ k & 1\end{pmatrix}\begin{pmatrix}-1 & 1\\ 2 & 5\end{pmatrix}=\begin{pmatrix}3 & 11\\ 5 & 2\end{pmatrix}$, find k

A – 5 B – 3 C 6 D 7

Solution

$$\begin{pmatrix}1\times(-1)+2\times2 & 1\times1+2\times5\\ k\times(-1)+1\times2 & k\times1+1\times5\end{pmatrix}=\begin{pmatrix}3 & 11\\ 5 & 2\end{pmatrix}$$

$$\begin{pmatrix}-1+4 & 1+10\\ -k+2 & k+5\end{pmatrix}=\begin{pmatrix}3 & 11\\ 5 & 2\end{pmatrix}$$

$$\begin{pmatrix}3 & 11\\ -k+2 & k+5\end{pmatrix}=\begin{pmatrix}3 & 11\\ 5 & 2\end{pmatrix}$$

Equating terms in k

– k + 2 = 5

Or k + 5 = 2

k = 2 – 5 = – 3 (B)

2014/18 Neco f/m

If $\begin{pmatrix}3 & x+2y\\ 3x-y & xy\end{pmatrix}=\begin{pmatrix}3 & 10\\ 2 & 8\end{pmatrix}$ find x and y

A (2, 4) B (2, 5) C (3, 4) D (3, 5) E (5, 2)

Solution

Equating terms

x + 2y = 10 ----- (1)

3x – y = 2 ------ (2)

xy = 8 ------ (3)

Solving (1) and (2); (2) × 2 and add

$$\begin{array}{l} x + 2y = 10\\ (6x-2y=4)\\ \hline 7x \quad = 14\end{array}$$

x = 2

Substitute x = 2 into (1)

2 + 2y = 10 becomes

2y = 10 – 2

2y = 8

y = 4 thus, x, y = 2, 4 (A)

2006/11 UME

Find P, Q, for which $\begin{pmatrix}2p & 8\\ 3 & -5q\end{pmatrix}\begin{pmatrix}1\\ 2\end{pmatrix}=\begin{pmatrix}24\\ -17\end{pmatrix}$

A – 4, – 2 B – 4, 2 C 4, – 2 D 4, 2

Solution

$$\begin{pmatrix}2p\times1+8\times2\\ 3\times1+-5q\times2\end{pmatrix}=\begin{pmatrix}24\\ -17\end{pmatrix}$$

$$\begin{pmatrix}2p+16\\ 3-10q\end{pmatrix}=\begin{pmatrix}24\\ -17\end{pmatrix}$$

Equating terms

3 – 10q = – 17 2p + 16 = 24

3 + 17 = 10q 2p = 24 – 16

20 = 10q 2p = 8

2 = q p = 4

Thus P, Q = 4, 2 (D)

2011/18 f/m

If $\begin{pmatrix}3 & 2\\ 7 & x\end{pmatrix}\begin{pmatrix}2\\ 3\end{pmatrix}=\begin{pmatrix}12\\ 29\end{pmatrix}$

find the value of x

A 5 B 6 C 7 D 8

Solution

$$\begin{pmatrix}3\times2 & +2\times3\\ 7\times2 & +x\times3\end{pmatrix}=\begin{pmatrix}12\\ 29\end{pmatrix}$$

$$\begin{pmatrix}6 & +6\\ 14 & +3x\end{pmatrix}=\begin{pmatrix}12\\ 29\end{pmatrix}$$

Equating terms in x

14 + 3x = 29

3x = 29 – 14

3x = 15

$\frac{3x}{3}=\frac{15}{3}$ thus, x = 5 (A)

2005/12c adjusted

If A = $\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$ and B = $\begin{pmatrix}-3 & 2\\ 1 & 4\end{pmatrix}$

Find C such that : AC = B

Solution

We list the C a 2×2 matrix such that AC = B

$$\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}\begin{pmatrix}a & b\\ c & d\end{pmatrix}=\begin{pmatrix}-3 & 2\\ 1 & 4\end{pmatrix}$$

$$\begin{pmatrix}a+2c & b+2d\\ 3a+4c & 3b+4d\end{pmatrix}=\begin{pmatrix}-3 & 2\\ 1 & 4\end{pmatrix}$$

Equating entries

a + 2c = –3 b + 2d = 2

3a + 4c = 1 3b + 4d = 4

Solving simultaneously

3a + 6c = –9	3b + 6d = 6
– (3a + 4c = 1)	– (3b + 4d = 4)
2c = – 10	2d = 2
c = –5	d = 1
Substituting for c value	Substituting for d value
a = –3–2(–5) i.e 7	b = 2–2(1) i.e 0

Thus C = $\begin{pmatrix}7 & 0\\ -5 & 1\end{pmatrix}$

2009/28 (Nov) f/m Exercise 26.12

If $\begin{pmatrix}5 & x\\ 4 & -1\end{pmatrix}\begin{pmatrix}2\\ y\end{pmatrix}=\begin{pmatrix}4\\ 5\end{pmatrix}$, find (x, y)

A (3, – 2) B (2, 3) C (–2, 3) D (–3, 2)

2006/29 Exercise 26.13

Given that $\begin{pmatrix}2 & -3\\ 1 & 4\end{pmatrix}\begin{pmatrix}-6\\ p\end{pmatrix}=\begin{pmatrix}3\\ -26\end{pmatrix}$,

find the value of P.

A – 8 B – 5 C – 4 D – 3

Determinant of matrix

The determinant of any matrix say B is denoted by $|B|$ or det B is a number computed by the format shown below; depending on the order of the matrix i.e 2×2 or 3×3.If the value of matrix determinant is zero then the matrix is called singular matrix, otherwise it is non-singular and it is only non-singular matrix that has inverse.

2013/10 Neco f/m Exercise 26.14

A matrix is said to be singular if the determinant is

A negative B one C positive

D undefined E zero

2×2 matrix determinant

$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ then,

det A = ad – bc

Eg. (i) If $A = \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}$ find detA

Solution

det A = 3×2 – 1× 4

= 6 – 4 = 2

Eg.(ii) If $D = \begin{pmatrix} 2 & 6 \\ -1 & 4 \end{pmatrix}$ find $|D|$

Solution

$|D|$ = 2 × 4 – 6×(- 1)

= 8 + 6 = 14

2006/19 f/m

Which of the following matrices is a singular matrix?

A $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ B $\begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$ C $\begin{pmatrix} 3 & 8 \\ 6 & 16 \end{pmatrix}$ D $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

Solution

A matrix is singular if det = 0

It is det of $\begin{pmatrix} 3 & 8 \\ 6 & 16 \end{pmatrix}$ = 3 × 16 – 6 × 8

= 48 – 48

= 0 (C)

2014/13 Neco

If x = $\begin{pmatrix} 5 & 3 \\ -2 & -2 \end{pmatrix}$, find the determinant of x

A – 16 B – 4 C 4 D 8 E 16

Solution

Determinant of x = [5 × (– 2) – (– 2) × 3]

= – 10 + 6

= – 4 B

2007/25 PCE

Evaluate $\begin{vmatrix} 2 & 1 \\ 4 & 6 \end{vmatrix}$

A 2 B 4 C 6 D 8

Solution

$\begin{vmatrix} 2 & 1 \\ 4 & 6 \end{vmatrix}$ = 2 × 6 – 4 × 1

= 12 – 4

= 8 (D)

2009/9 (Nov) f/m

If the determinant of the matrix $\begin{pmatrix} x & -1 \\ 2 & 4 \end{pmatrix}$ is – 6, find the value of x

A – 2 B – 1 C 1 D 2

Solution

x × 4 – 2 × (–1) = – 6

4x + 2 = – 6

4x = – 6 – 2

4x = – 8

$\frac{4x}{4} = \frac{-8}{4}$

x = – 2 (A)

2012/23 UMTE

If $\begin{vmatrix} 5 & 3 \\ x & 2 \end{vmatrix} = \begin{vmatrix} 3 & 5 \\ 4 & 5 \end{vmatrix}$, find the value of x

A 3 B 4 C 5 D 7

Solution

$\begin{vmatrix} 5 & 3 \\ x & 2 \end{vmatrix} = \begin{vmatrix} 3 & 5 \\ 4 & 5 \end{vmatrix}$

5 × 2 – 3 × x = 3 × 5 – 5 × 4

10 – 3x = 15 – 20

10 – 3x = – 5

10 + 5 = 3x

$\frac{15}{3} = \frac{3x}{3}$

5 = x (C)

2003/7 f/m

If $\begin{vmatrix} 3-x & 9 \\ -1 & 1+2x \end{vmatrix}$ = 0, find the two possible values of x

A 2 or – 3 B 4 or $\frac{-2}{3}$ C 4 or $\frac{-3}{2}$ D 3 or – 2

Solution

(3 – x) (1 + 2x) – 9×(–1) = 0

3(1 + 2x) – x(1 + 2x) + 9 = 0

3 + 6x – x – $2x^2$ + 9 = 0

5x – $2x^2$ + 12 = 0

$2x^2$ – 5x – 12 = 0

Factorizing

$2x^2$ – 8x + 3x – 12 = 0

2x(x – 4) + 3(x – 4) = 0

(2x + 3)(x – 4) = 0

2x + 3 = 0 or x – 4 = 0

x = $\frac{-3}{2}$ or 4 (C)

2014/23 UTME

If $\begin{vmatrix} -x & 12 \\ -1 & 4 \end{vmatrix}$ = – 12, find x

A 3 B 6 C – 6 D – 2

Solution

[– x × 4 – (–1) × 12] = – 12

– 4x + 12 = – 12

– 4x = – 24

$\frac{-4x}{-4} = \frac{-24}{-4}$

x = 6 (B)

2009/23 PCE

$P = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$, find the value of $|-5P + 6I|$

A 41 B 36 C 31 D – 36

Solution

Here I implies 2 × 2 identity matrix $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$-5P = -5\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ and $6I = 6\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$= \begin{pmatrix} -10 & -5 \\ 0 & -15 \end{pmatrix}$ $= \begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix}$

$-5P + 6I = \begin{pmatrix} -10+6 & -5+0 \\ 0+0 & -15+6 \end{pmatrix}$

$= \begin{pmatrix} -4 & -5 \\ 0 & -9 \end{pmatrix}$

Thus, $|-5P + 6I| = [(-4) \times (-9)] - 0 \times (-5)$
$= 36 - 0$
$= 36$ (B)

2009/23 UME

If $P = \begin{pmatrix} x+3 & x+2 \\ x+1 & x-1 \end{pmatrix}$, evaluate x

if $|P| = -10$

A – 5 B – 2 C 2 D 5

Solution

$|P| = -10$
$(x + 3)(x - 1) - (x + 1)(x + 2) = -10$
$x(x - 1) + 3(x - 1) - [x(x + 2) + 1(x + 2)] = -10$
$x^2 - x + 3x - 3 - [x^2 + 2x + x + 2] = -10$
$x^2 + 2x - 3 - (x^2 + 3x + 2) = -10$
$x^2 + 2x - 3 - x^2 - 3x - 2 = -10$
$-x - 5 = -10$
$-x = -10 + 5$
$-x = -5$
$x = 5$ (D)

2008/25 PCE

If $\begin{vmatrix} 7 & 3 \\ x & x \end{vmatrix} = 24$, find x

A 3 B 4 C 6 D 8

Solution

$\begin{vmatrix} 7 & 3 \\ x & x \end{vmatrix} = 24$

$7x - 3x = 24$
$4x = 24$
$x = \frac{24}{4} = 6$ (C)

2015/28 f/m Exercise 26.15

If $\begin{vmatrix} 1+2x & -1 \\ 6 & 3-x \end{vmatrix} = -3$, find the value of x

A $x = 3, -2$ B $x = 4, \frac{-2}{3}$

C $x = -4, \frac{3}{2}$ D $x = 4, \frac{-3}{2}$

2013/15 f/m Exercise 26.16

Given that $P = \begin{pmatrix} y-2 & y-1 \\ y-4 & y+2 \end{pmatrix}$ and $|P| = -23$, find the value of y

A – 4 B – 3 C – 1 D 2

2010/25 UMTE Exercise 26.17

If $\begin{vmatrix} x & 3 \\ 2 & 7 \end{vmatrix} = 15$, find the value of x

A 3 B 5 C 4 D 2

2009/22 UME Exercise 26.18

If $Q = \begin{pmatrix} 9 & -2 \\ -7 & 4 \end{pmatrix}$, then $|Q|$ is

A – 50 B – 22 C 22 D 50

2016/11 Neco Exercise 26.19

If $\begin{vmatrix} -3 & 3 \\ x & y \end{vmatrix} = 33$, find the value of x + y

A –27 B –24 C –11 D – 9 E – 6

Inverse of matrix (2×2 matrix)

The inverse of a square matrix A denoted by A^{-1} is given by

If $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ then $A^{-1} = \frac{1}{\det A}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

exchange of 'a' and 'd' and negation of 'b' and 'c'

Eg1. Find the inverse of matrix $A = \begin{pmatrix} 4 & 5 \\ 2 & 3 \end{pmatrix}$

Solution

$A^{-1} = \frac{1}{\det A}\begin{pmatrix} 3 & -5 \\ -2 & 4 \end{pmatrix}$

But det A = 3×4 – (5× 2)
=12 – 10 i .e 2

$A^{-1} = \frac{1}{2}\begin{pmatrix} 3 & -5 \\ -2 & 4 \end{pmatrix}$

$= \begin{pmatrix} 3/2 & -5/2 \\ -1 & 2 \end{pmatrix}$

Readers to confirm that $AA^{-1} = I$

It is true for all matrices ie. any matrix multiplied by its inverse equals identity

2010/27 UTME

If $P = \begin{pmatrix} 2 & -3 \\ 1 & 1 \end{pmatrix}$, what is P^{-1}?

A $\begin{pmatrix} \frac{1}{5} & \frac{3}{5} \\ \frac{1}{5} & \frac{2}{5} \end{pmatrix}$ B $\begin{pmatrix} \frac{1}{5} & \frac{3}{5} \\ \frac{-1}{5} & \frac{2}{5} \end{pmatrix}$

C $\begin{pmatrix} \frac{-1}{5} & \frac{3}{2} \\ \frac{-1}{-5} & \frac{5}{2} \end{pmatrix}$ D $\begin{pmatrix} \frac{-1}{5} & \frac{-3}{5} \\ \frac{-1}{5} & \frac{-2}{5} \end{pmatrix}$

Solution

$P^{-1} = \frac{1}{\det P}\begin{pmatrix} 1 & 3 \\ -1 & 2 \end{pmatrix}$

$\det P = [2 \times 1 - 1 \times (-3)]$
$= 2 + 3$
$= 5$

$P^{-1} = \frac{1}{5}\begin{pmatrix} 1 & 3 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} \frac{1}{5} & \frac{3}{5} \\ \frac{-1}{5} & \frac{2}{5} \end{pmatrix}$ B.

2008/24 PCE

Find inverse of $\begin{pmatrix} 8 & -4 \\ -7 & 6 \end{pmatrix}$

A $\begin{pmatrix} \frac{3}{10} & \frac{-1}{5} \\ \frac{-7}{10} & \frac{2}{5} \end{pmatrix}$ B $\begin{pmatrix} \frac{3}{10} & \frac{1}{5} \\ \frac{7}{20} & \frac{2}{5} \end{pmatrix}$ C $\begin{pmatrix} \frac{2}{5} & \frac{-1}{5} \\ \frac{7}{20} & \frac{3}{10} \end{pmatrix}$ D $\begin{pmatrix} \frac{2}{2} & \frac{1}{5} \\ \frac{7}{10} & \frac{3}{10} \end{pmatrix}$

Solution

Inverse of $\begin{pmatrix} 8 & -4 \\ -7 & 6 \end{pmatrix} = \frac{1}{\det}\begin{pmatrix} 6 & 4 \\ 7 & 8 \end{pmatrix}$

But det of $\begin{pmatrix} 8 & -4 \\ -7 & 6 \end{pmatrix} = [8 \times 6 - (-7) \times (-4)]$
$= (48 - 28)$ i.e 20

Inverse of $\begin{pmatrix} 8 & -4 \\ -7 & 6 \end{pmatrix} = \frac{1}{20}\begin{pmatrix} 6 & 4 \\ 7 & 8 \end{pmatrix}$

$= \begin{pmatrix} \frac{6}{20} & \frac{4}{20} \\ \frac{7}{20} & \frac{8}{20} \end{pmatrix} = \begin{pmatrix} \frac{3}{10} & \frac{1}{5} \\ \frac{7}{20} & \frac{2}{5} \end{pmatrix}$. B

2014/20 Neco (Dec)

Find the inverse of the matrix $Q = \begin{pmatrix} 2 & 4 \\ -1 & -3 \end{pmatrix}$

A $\frac{-1}{2}\begin{pmatrix} 3 & -4 \\ 1 & 2 \end{pmatrix}$ B $\frac{-1}{2}\begin{pmatrix} -3 & -4 \\ 1 & 2 \end{pmatrix}$

C $\frac{-1}{2}\begin{pmatrix} 3 & -4 \\ -1 & 2 \end{pmatrix}$ D $\frac{-1}{2}\begin{pmatrix} -3 & -4 \\ 1 & 0 \end{pmatrix}$ E $\frac{-1}{2}\begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}$

Solution

Inverse of $\begin{pmatrix} 2 & 4 \\ -1 & -3 \end{pmatrix} = \frac{1}{\det Q}\begin{pmatrix} -3 & -4 \\ 1 & 2 \end{pmatrix}$

But det $\begin{pmatrix} 2 & 4 \\ -1 & -3 \end{pmatrix} = [2 \times (-3) - 4 \times (-1)]$
$= (-6 + 4)$ i.e -2
$= \frac{-1}{2}\begin{pmatrix} -3 & -4 \\ 1 & 2 \end{pmatrix}$ B.

2003/21 UME

A matrix P has an inverse $P^{-1} = \begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix}$. Find P

A $\begin{pmatrix} -1 & 3 \\ 0 & -1 \end{pmatrix}$ B $\begin{pmatrix} 1 & 3 \\ 0 & -1 \end{pmatrix}$ C $\begin{pmatrix} 1 & -3 \\ 0 & -1 \end{pmatrix}$ D $\begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$

Solution

Taking the inverse of P^{-1} will give P i.e from indices law
$(P^{-1})^{-1} = P^1$ i.e P

inverse of $P^{-1} = \frac{1}{\det P^{-1}}\begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$

But $\det P^{-1} = 1 \times 1 - (0 \times -3)$ $= 1 - 0$ i.e 1

$= \frac{1}{1}\begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$

$= \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$ D

2014/14 f/m Exercise 26.20

If $T = \begin{pmatrix} -2 & -5 \\ 3 & 8 \end{pmatrix}$, find T^{-1}, the inverse of T

A $\begin{pmatrix} -8 & -5 \\ 3 & 2 \end{pmatrix}$ B $\begin{pmatrix} -8 & 5 \\ 3 & -2 \end{pmatrix}$ C $\begin{pmatrix} -8 & -5 \\ -3 & 2 \end{pmatrix}$ D $\begin{pmatrix} -8 & -5 \\ -3 & -2 \end{pmatrix}$

2013/23 UTME Exercise 26.21

Find the inverse of $\begin{pmatrix} 5 & 3 \\ 6 & 4 \end{pmatrix}$

A $\begin{pmatrix} 2 & \frac{3}{2} \\ -3 & \frac{-5}{2} \end{pmatrix}$ B $\begin{pmatrix} 2 & \frac{3}{2} \\ -3 & \frac{5}{2} \end{pmatrix}$

C $\begin{pmatrix} 2 & \frac{-3}{2} \\ -3 & \frac{-5}{2} \end{pmatrix}$ D $\begin{pmatrix} 2 & \frac{-3}{2} \\ -3 & \frac{5}{2} \end{pmatrix}$

Solving simultaneous linear equations with two unknowns using 2×2 matrix (*inverse method*)

If the equation is of the form

$ax + by = c$
$dx + ey = f$

The matrix form is then

$\begin{pmatrix} a & b \\ d & e \end{pmatrix} \times \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} c \\ f \end{pmatrix}$

Readers to verify that the LHS is gotten by matrix multiplication to get to step 1 above

Then solution $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a & b \\ d & e \end{pmatrix}^{-1}\begin{pmatrix} c \\ f \end{pmatrix}$

Eg. 1 Solve the simultaneous equations:
$5x + 9y = -3$
$6x + 2y = 2$ using matrix format

Solution

The matrix expression is

$\begin{pmatrix} 5 & 9 \\ 6 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -3 \\ 2 \end{pmatrix}$

Then solution $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 & 9 \\ 6 & 2 \end{pmatrix}^{-1}\begin{pmatrix} -3 \\ 2 \end{pmatrix}$

$$\boxed{But\begin{pmatrix}5 & 9\\6 & 2\end{pmatrix}^{-1} = \frac{1}{\det}\begin{pmatrix}2 & -9\\-6 & 5\end{pmatrix} = -\frac{1}{44}\begin{pmatrix}2 & -9\\-6 & 5\end{pmatrix}}$$

det = 2×5 – (9×6)
= 10-54 = -44

$$\begin{pmatrix}x\\y\end{pmatrix} = -\frac{1}{44}\begin{pmatrix}2 & -9\\-6 & 5\end{pmatrix}\begin{pmatrix}-3\\2\end{pmatrix}$$

$$= -\frac{1}{44}\begin{pmatrix}-6-18\\18+10\end{pmatrix}$$

$$= -\frac{1}{44}\begin{pmatrix}-24\\28\end{pmatrix}$$

$$\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}\frac{24}{44}\\-\frac{28}{44}\end{pmatrix}$$

i.e $x = \frac{6}{11}$ and $y = -\frac{7}{11}$

1997/(23) UME

Determine x + y if

$$\begin{pmatrix}2 & -3\\-1 & 4\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}-1\\8\end{pmatrix}$$

A. 3 B. 4 C. 7 D. 12

Solution

$$\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}2 & -3\\-1 & 4\end{pmatrix}^{-1}\begin{pmatrix}-1\\8\end{pmatrix}$$

$$\boxed{But\begin{pmatrix}2 & -3\\-1 & 4\end{pmatrix}^{-1} = \frac{1}{\det}\begin{pmatrix}4 & 3\\1 & 2\end{pmatrix} = \frac{1}{5}\begin{pmatrix}4 & 3\\1 & 2\end{pmatrix}}$$

det = 4×2 – (-3 × -1)
= 8-3 = 5

$$\begin{pmatrix}x\\y\end{pmatrix} = \frac{1}{5}\begin{pmatrix}4 & 3\\1 & 2\end{pmatrix}\begin{pmatrix}-1\\8\end{pmatrix}$$

$$= \frac{1}{5}\begin{pmatrix}4\times-1 + 3\times8\\1\times-1 + 2\times8\end{pmatrix}$$

$$= \frac{1}{5}\begin{pmatrix}-4+24\\-1+16\end{pmatrix} = \frac{1}{5}\begin{pmatrix}20\\15\end{pmatrix}$$

$$\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}4\\3\end{pmatrix}$$

Thus x + y = 4 + 3
= 7 (C)

2004/2 Exercise 26.22

Solve the equation $\begin{pmatrix}2 & -3\\1 & 3\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}1\\2\end{pmatrix}$

A. x = –1, y = –$^1/_3$ B. x = 1, y = –$^1/_2$

C. x = 1, y = $^1/_3$ D. x = –1, y = $^1/_2$

2002/2 Exercise 26.23

$$\begin{pmatrix}3 & 0\\3 & 1\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}-6\\-10\end{pmatrix}$$

A. x = 2, y = – 4 B. x = –2, y = 4

C. x = –2, y = –4 D. x = 2, y = 4

2009/10a Neco f/m Exercise 26.24

If A = $\begin{pmatrix}3 & 7\\-4 & 2\end{pmatrix}$

(i) find A^{-1} (ii) Compute $A.A^{-1}$

(iii) Hence, using inverse method, ($x = A^{-1}b$)

solve the equation $\begin{pmatrix}3 & 7\\-4 & 2\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}-1\\1\end{pmatrix}$.

3×3 matrix determinant

$$B = \begin{pmatrix}b_{11} & b_{12} & b_{13}\\b_{21} & b_{22} & b_{23}\\b_{31} & b_{32} & b_{33}\end{pmatrix} \text{ then}$$

$$|B| = +b_{11}\begin{vmatrix}b_{22} & b_{23}\\b_{32} & b_{33}\end{vmatrix} - b_{12}\begin{vmatrix}b_{21} & b_{23}\\b_{31} & b_{33}\end{vmatrix} + b_{13}\begin{vmatrix}b_{21} & b_{22}\\b_{31} & b_{32}\end{vmatrix}$$

$$= +b_{11}(b_{22}b_{33} - b_{23}b_{32}) - b_{12}(b_{21}b_{33} - b_{23}b_{31}) + b_{13}(b_{21}b_{32} - b_{22}b_{31})$$

The sign attached to b_{11}, b_{12} b_{13} on and on is given by the chess board format or

$$\begin{pmatrix}+ & - & +\\- & + & -\\+ & - & +\end{pmatrix}$$

$(-1)^{i+j}$ where i is row and j is column

$b_{11} = (-1)^{1+1} = (-1)^2 = +1$

$b_{12} = (-1)^{1+2} = (-1)^3 = -1$

and so on

2014/24 UTME

Find the value of $\begin{vmatrix}0 & 3 & 2\\1 & 7 & 8\\0 & 5 & 4\end{vmatrix}$

A – 1 B – 2 C 12 D 10

Solution

$$\begin{vmatrix}0 & 3 & 2\\1 & 7 & 8\\0 & 5 & 4\end{vmatrix} = 0\begin{vmatrix}7 & 8\\5 & 4\end{vmatrix} - 3\begin{vmatrix}1 & 8\\0 & 4\end{vmatrix} + 2\begin{vmatrix}1 & 7\\0 & 5\end{vmatrix}$$

= 0 – 3[1× 4 – 0 × 8] + 2[1 × 5 – 0 × 7]
= 0 – 3(4 – 0) + 2(5 – 0)
= 0 – 3(4) + 2(5)
= 0 – 12 + 10 = – 2 B.

2011/10 Neco f/m

Given that A = $\begin{pmatrix}3 & 2 & 1\\6 & 2 & 1\\7 & 5 & 2\end{pmatrix}$

(a) Find the: (i) determinant of A

Solution

$$\text{Det A} = 3\begin{vmatrix}2 & 1\\5 & 2\end{vmatrix} - 2\begin{vmatrix}6 & 1\\7 & 2\end{vmatrix} + 1\begin{vmatrix}6 & 2\\7 & 5\end{vmatrix}$$

= 3[2 × 2 – 5 × 1] – 2[6 × 2 – 7 × 1] + 1[6 × 5 – 7 × 2]
= 3(4 – 5) – 2(12 – 7) + 1(30 – 14)
= 3(– 1) – 2(5) + 1(16)
= – 3 – 10 + 16 = 3

2014/3 Neco f/m

Evaluate $\begin{vmatrix} 1+\cos\theta & 1+\sin\theta & 1 \\ 1-\sin\theta & 1+\cos\theta & 1 \\ 1 & 1 & 1 \end{vmatrix}$

A $1 + \cos\theta$ B $1 + \sin\theta$ C $\sin\theta$ D $\cos\theta$ E 1

Solution

$= 1 + \cos\theta \begin{vmatrix} 1+\cos\theta & 1 \\ 1 & 1 \end{vmatrix} - (1+\sin\theta)\begin{vmatrix} 1-\sin\theta & 1 \\ 1 & 1 \end{vmatrix} +$
$1\begin{vmatrix} 1-\sin\theta & 1+\cos\theta \\ 1 & 1 \end{vmatrix}$

$= 1 + \cos\theta[(1+\cos\theta) \times 1 - 1 \times 1] - (1+ \sin\theta)[(1-\sin\theta) \times 1 - 1\times 1]$
$+ 1[(1 - \sin\theta) \times 1 - 1\times (1 + \cos\theta)$

$= 1+\cos\theta(1+\cos\theta - 1) - (1+\sin\theta)(1- \sin\theta - 1) + (1- \sin\theta - 1-\cos\theta)$

$= 1 + \cos\theta (\cos\theta) - (1 + \sin\theta) (- \sin\theta) + (- \sin\theta - \cos\theta)$

$= \cos\theta + \cos^2\theta + \sin\theta + \sin^2\theta - \sin\theta - \cos\theta$

$= \cos^2\theta + \sin^2\theta$

$= 1$ (by trig identity)

2013/9 Neco f/m

Given that $\begin{vmatrix} 1 & 1 & 0 \\ 0 & x & -1 \\ 1 & 0 & 1 \end{vmatrix} = 2$. Find x

A 3 B 2 C 1 D 0 E – 3

Solution

$1\begin{vmatrix} x & -1 \\ 0 & 1 \end{vmatrix} - 1\begin{vmatrix} 0 & -1 \\ 1 & 1 \end{vmatrix} + 0\begin{vmatrix} 0 & x \\ 1 & 0 \end{vmatrix} = 2$

$1[x \times 1 - 0(-1)] - 1[0 \times 1 - 1 \times (-1)] + 0 = 2$

$1(x - 0) - 1(0 + 1) + 0 = 2$

$1(x) - 1(1) = 2$

$x - 1 = 2$

$x = 3$ A.

2012/24 UTME

Given that I_3 is unit matrix of order 3, find $| I_3 |$

A – 1 B 0 C 1 D 2

Solution

$I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$| I_3 | = 1\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} - 0\begin{vmatrix} 0 & 0 \\ 0 & 1 \end{vmatrix} + 0\begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix}$

$= 1(1 \times 1 - 0 \times 0) - 0 + 0$

$= 1(1)$

$= 1$ C.

2013/9b Neco f/m

Show that the matrix A has no multiplicative inverse,

where $A = \begin{pmatrix} 2 & -4 & 8 \\ 0 & 10 & -10 \\ 6 & 2 & 10 \end{pmatrix}$

Solution

We are to show that $| A | = 0$

$| A | = 2\begin{vmatrix} 10 & -10 \\ 2 & 10 \end{vmatrix} - (-4)\begin{vmatrix} 0 & -10 \\ 6 & 10 \end{vmatrix} + 8\begin{vmatrix} 0 & 10 \\ 6 & 2 \end{vmatrix}$

$= 2[10 \times 10 - 2 \times (-10)] + 4[0 \times 10 - 6 \times (-10)] +$
$8[0 \times 2 - 6 \times 10]$

$= 2(100 + 20) + 4(0 + 60) + 8(0 - 60)$

$= 2(120) + 4(60) + 8(-60)$

$= 240 + 240 - 480$

$= 0$ QED

2015/11b Neco

Evaluate the determinant $\begin{vmatrix} 2 & 3 & -1 \\ 1 & 0 & 2 \\ 3 & -2 & 3 \end{vmatrix}$

Solution

$\begin{vmatrix} 2 & 3 & -1 \\ 1 & 0 & 2 \\ 3 & -2 & 3 \end{vmatrix} = 2\begin{vmatrix} 0 & 2 \\ -2 & 3 \end{vmatrix} - 3\begin{vmatrix} 1 & 2 \\ 3 & 3 \end{vmatrix} - 1\begin{vmatrix} 1 & 0 \\ 3 & -2 \end{vmatrix}$

$= 2[0 \times 3 - (-2) \times 2] - 3[1 \times 3 - 3 \times 2] - 1[1 \times (-2) - 3 \times 0]$

$= 2(0 + 4) - 3(3 - 6) - 1(-2 - 0)$

$= 2(4) - 3(-3) - 1(-2)$

$= 8 + 9 + 2$

$= 19$

2010/26 UTME Exercise 26.25

Evaluate $\begin{vmatrix} 2 & 0 & 5 \\ 4 & 6 & 3 \\ 8 & 9 & 1 \end{vmatrix}$

A – 42 B 102 C 18 D – 102

2009/11b f/m Exercise 26.26

(i) Evaluate $\begin{vmatrix} 2 & -3 & 1 \\ 0 & 1 & -2 \\ 1 & 2 & -3 \end{vmatrix}$

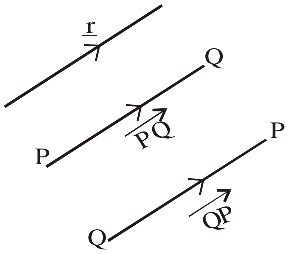
2015/10 bi f/m Exercise 26.27

Evaluate $\begin{vmatrix} 2 & -1 & 2 \\ 1 & 3 & 4 \\ 1 & 2 & 1 \end{vmatrix}$

Chapter twenty seven
Vectors

A vector is a quantity that has both magnitude and direction. It is represented by a small arrowhead on top of a letter showing the direction $\overrightarrow{PQ}$; underlining of letter $\underline{r}$, or boldly written letters **t** are other forms of representing vectors.

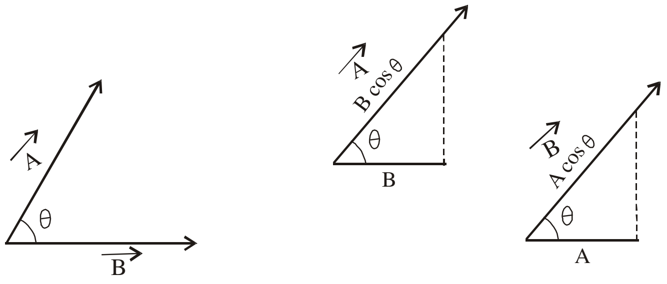
We denote the vector in the direction P to Q by $\overrightarrow{PQ}$ and the vector in the direction Q to P by $\overrightarrow{QP}$



The following terms are relevant to the concept of vector

Scalar product

Suppose we have two vectors forming an angle as shown below



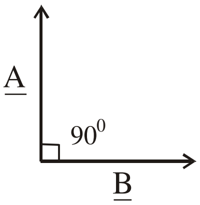
(1) The scalar product or dot product

$\vec{A}.\vec{B} = |AB|\cos\theta$

(2) Cosine of angles between two vectors $\vec{A}$ and $\vec{B}$

$$Cos\theta = \frac{\vec{A}.\vec{B}}{|AB|}$$

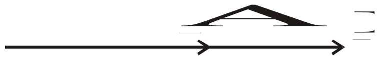
(3) If two vectors $\underline{A}$ and $\underline{B}$ are perpendicular to each other ($\theta = 90^0$) Then $\underline{A} \bullet \underline{B} = 0$



$\underline{A}.\underline{B} = |AB|\cos 90^0$

$\underline{A} \bullet \underline{B} = 0$

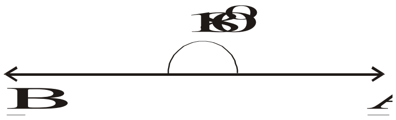
(4) If two vectors have the same direction (i.e $\theta = 0$) Then $\underline{A}.\underline{B} = |AB|$



$\underline{A}.\underline{B} = |AB|\cos 0$

$\underline{A}.\underline{B} = |AB|$

(5) If two vectors are opposite to each other($\theta = 180$) Then $\underline{A}.\underline{B} = -|AB|$

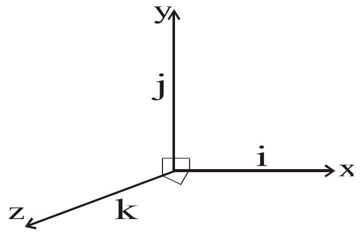


$\underline{A}.\underline{B} = |AB|\cos 180$

$\underline{A}.\underline{B} = -|AB|$

Representation of vectors in three dimensions

A vector $\vec{A}$ has 3 components in x, y, z directions. These directions are perpendicular to each other where i, j, and k are the unit vectors



i . j = $|i|\,|j|\cos 90^0$ i.e 0

Hence we do not multiply unlike terms (vectors) and their coefficients as we have seen that $\cos 90^0 = 0$

Vector product

The vector product or cross product of two vectors $\vec{A}$ and $\vec{B}$ which is denoted as $\vec{A} \times \vec{B}$ and is defined as the product of the magnitude of the vectors $\vec{A}$ and $\vec{B}$ and the sign of their included angle θ

$$\vec{A} \times \vec{B} = |A|\,|B|\ \text{Sin}\theta$$

(i) $Sin\,\theta = \dfrac{\vec{A} \times \vec{B}}{|A|\ |B|}$

(ii) If two vectors $\vec{A}$ and $\vec{B}$ are perpendicular(orthogonal) to each other ($\theta = 90^0$)

$\vec{A} \times \vec{B} = |AB|\ \text{Sin } 90^0$ *since Sin 90⁰ = 1*

$\vec{A} \times \vec{B} = |AB|$

(iii) If two vectors $\vec{A}$ and $\vec{B}$ are collinear or parallel to each other ($\theta = 0^0$)

$\vec{A} \times \vec{B} = |AB|\ \text{Sin } 0^0$ *since Sin 0⁰ = 0*

$\vec{A} \times \vec{B} = 0$

$\vec{A} = a_1\, i + a_2\, j + a_3\, k$

$\vec{B} = b_1 + b_2 j + b_3 k$

Then

$\vec{A} + \vec{B} = (a_1 + b_1)i + (a_2 + b_2)j + (a_3 + b_3)k$

$\vec{A} - \vec{B} = (a_1 - b_1)i + (a_2 - b_2)j + (a_3 - b_3)k$

Also by determinant method

$$\vec{A} \times \vec{B} = \begin{vmatrix} i & j & k \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \text{ same applies to } 2\times2$$

Basic theorem to work with

Two non-zero vectors a and b are parallel if and if $\exists$ a scalar t such that $a = t\,b$

i.e one is a multiple of the other

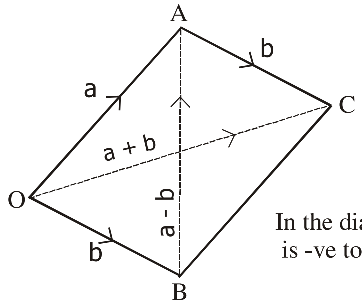
Addition of Vectors

If two vectors $\underline{a}$ and $\underline{b}$ represent the two adjacent sides OA and OB respectively of a parallelogram OACB, then the diagonal OC which passes through their point of intersection will be equal to their vector sum

OC = $\underline{a}$ + $\underline{b}$

And the other diagonal BA is equal to

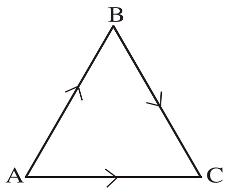
BA = $\underline{a} - \underline{b}$



In the diagonal BA, the movement to A is -ve to B direction and so we have -b

Triangle rule

First, we need to follow the vector arrows as given in the diagram or a triangle with arrows showing the vectors $\overrightarrow{AB}$, $\overrightarrow{BC}$ and $\overrightarrow{AC}$



Here $\overrightarrow{AB}$ follows $\overrightarrow{BC}$ but $\overrightarrow{AC}$ contradicts them

$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$ *deductively*

$\overrightarrow{BC} = \overrightarrow{AC} - \overrightarrow{AB}$ and $\overrightarrow{AB} = \overrightarrow{AC} - \overrightarrow{BC}$

Other definition of terms

Origin vectors are:

2×2 row vector(0, 0) or 3×3 row vector(0, 0, 0)

or 2×2 column vector $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ or 3×3 column vector $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

Unit vector

$$\vec{u} = \frac{1}{|\vec{a}|} \cdot \vec{a}$$

Resultant of vectors

Resultant of vectors is the addition of the vector involved

Projection of vector

Projection of vector **a** on **b**

$P = a \cdot \dfrac{b}{|\vec{b}|}$, Projection of vector **b** on **a** $P = b \cdot \dfrac{a}{|\vec{a}|}$

Magnitude of a vector

The magnitude of the vector **a** = 3i + 2j –7k

$|\mathbf{a}| = \sqrt{3^2 + 2^2 + (-7)^2}$

Sum, subtraction and scalar multiplication in vectors

2013/29 f/m

If s = 3i – j and t = 2i + 3j, find (t + 3s) (t – 3s)

A – 77 B – 71 C – 53 D – 41

Solution

(t + 3s)(t – 3s) = [2i + 3j + 3(3i – j)] [2i + 3j – 3(3i – j)]

= [2i + 3j + 9i – 3j] [2i + 3j – 9i + 3j]

= (11i + 0j) (– 7i + 6j)

= 11×(– 7) + 0×6

= – 77 + 0

= – 77 A.

1994/32 (Nov)

Find the resultant of the vectors

a = 3i

b = –2i – j

c = i + 4j

A 4i + 4j B. 2i +3j C. 6i + 5j

D. – 4i – 4j E – 2i – 3j

Solution

Resultant of the vectors = a + b + c

= 3i – 2i – j + i + 4j

= 2i + 3j (B)

2000/36

Given that $p\begin{pmatrix} 3 \\ 1 \end{pmatrix} + q\begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 8 \\ 6 \end{pmatrix}$, where p and q are constants, find the value of (p + q)

A – 4 B 1 C 3 D 4

Solution

One column matrix (same as vector) problem

3p + 2q = 8 ------ (1)

p + 4q = 6 ------ (2)

(1) × 2 and subtract

6p + 4q = 16

–(p + 4q = 6)

5p = 10

$p = \frac{10}{5}$ i.e 2

Substitute p = 2 into (2)
p + 4q = 6 will become
$4q = 6 – 2$
$4q = 4$
$q = \frac{4}{4}$ i.e 1
Thus p + q = 2 + 1
= 3 (C)

1999/33 (Nov)
Given that a and b are scalars, solve the equation
$a\begin{pmatrix}2\\1\end{pmatrix} + b\begin{pmatrix}4\\-2\end{pmatrix} = \begin{pmatrix}-2\\3\end{pmatrix}$

A. a = - 1 ; b = - 1 B. a = - 1; b = 1
C. a = 1 ; b = - 1 D a = $\frac{4}{5}$; b = $\frac{7}{10}$

Solution
2a + 4b = – 2 --------(1)
a – 2b = 3---------(2)
(2) × 2
2a – 4b = 6
$\underline{-(2a+4b=-2)}$
$-8b = 8$
b = - 1
Substitute b value into (1)
2a + 4(-1) = -2
2a – 4 = - 2
2a = - 2 + 4
2a = 2
a = 1 Thus a = 1 and b = -1(C)

2014/5 NABTEB(Dec) Exercise 27.1
Let P = 2i + 3j, q = i – 2j and r = 4i + 3j,
Find (i) 2p + q (ii) p + q + r
(iii) 3q – 2r (iv) 4p – 3q – 2

2014/50 NABTEB Exercise 27.2
If **a** = $\begin{pmatrix}2\\0\end{pmatrix}$ and **b** = $\begin{pmatrix}-1\\3\end{pmatrix}$, evaluate 2**a** – 3**b**

A $\begin{pmatrix}8\\6\end{pmatrix}$ B $\begin{pmatrix}4\\3\end{pmatrix}$ C $\begin{pmatrix}7\\-9\end{pmatrix}$ D $\begin{pmatrix}2\\-3\end{pmatrix}$

2014/49 NABTEB Exercise 27.3
Find the sum of the vectors $\begin{pmatrix}5\\3\end{pmatrix}$ and $\begin{pmatrix}-1\\3\end{pmatrix}$,

A $\begin{pmatrix}4\\5\end{pmatrix}$ B $\begin{pmatrix}4\\6\end{pmatrix}$ C $\begin{pmatrix}2\\5\end{pmatrix}$ D $\begin{pmatrix}3\\5\end{pmatrix}$

Problems on parallel vectors
2000/17 (Nov)
Given that the vectors 3i – 2j and 4i + Pj are parallel, find the constant P
A –6 B $-{}^{8}/_{3}$ C $-{}^{3}/_{8}$ D ${}^{1}/_{6}$

Solution

Two vectors are parallel then, $\vec{A} \times \vec{B} = 0$

and $\vec{A} \times \vec{B} = \begin{vmatrix} i & j \\ 3 & -2 \\ 4 & p \end{vmatrix}$

$\Rightarrow$ 3p + 8 = 0
$\Rightarrow$ 3p = -8 Thus p = $-{}^{8}/_{3}$ (B)

2003 / 7 (b)
Given that p = $\begin{pmatrix}4\\-3\end{pmatrix}$ and q = $\begin{pmatrix}-12\\5\end{pmatrix}$, find a vector **U** such that **U** is parallel to p + q and is half the size of p + q

Solution

$p + q = \begin{pmatrix}4\\-3\end{pmatrix} + \begin{pmatrix}-12\\5\end{pmatrix} = \begin{pmatrix}-8\\2\end{pmatrix}$

By the theorem:
Two non-zero vectors a and b are parallel if and only if ∃ a scalar t, such that a = t b
Thus, a vector parallel to p + q and half the size of p + q is simply

$\frac{1}{2}(p+q) = \frac{1}{2}\begin{pmatrix}-8\\2\end{pmatrix} = \begin{pmatrix}-4\\1\end{pmatrix}$

2005 / 16(a)
The position vectors of points R, S and T in the X – Y plane are r = 5i + 3j, s = 8i – j and t = 11i – 5j respectively.
(i) Show that, R, S and T are collinear.
(ii) Find scalars k_1 and k_2 such that
15i + 9j = k_1 **r** + k_2 **t**

Solution

(i) $\overrightarrow{RS}$ = **s** – **r**
= (8i – j) – (5i + 3j)
= 3i – 4j

Similarly $\overrightarrow{ST}$ = **t** – **s**
= (11i – 5j) – (8i – j)
= 3i – 4j

$\Rightarrow \overrightarrow{RS}$ and $\overrightarrow{ST}$ are parallel which is possible only when R, S and T lie on one line; hence result

(ii) 15i + 9j = k_1(5i + 3j) + k_2(11i – 5j)
= $5k_1 i + 3k_1 j + 11k_2 i - 5k_2 j$
15i + 9j = $5k_1 i + 11k_2 i + 3k_1 j - 5k_2 j$
Equating terms
15 = $5k_1 + 11k_2$ ------(a) and
9 = $3k_1 - 5k_2$ ----------(b)
a × 3 and b × 5 and subtract
45 = $15k_1 + 33k_2$
$\underline{-(\ 45 = 15k_1 - 25k_2)}$
0 = $58k_2$ $\Rightarrow k_2 = 0$

Substituting k_2 value into (a) 15 = $5k_1$ + 0
$\frac{15}{5} = k_1 \Rightarrow k_1 = 3$

2002/49 Neco
What is the unit vector which is parallel to the vector $7i - 4j$?

Solution

Two vector **a** and **b** are parallel
If **a** = s**b** where S is a scalar
Thus unit vector here

$$= \frac{7i - 4j}{\sqrt{7^2 + (-4)^2}} = \frac{7i - 4j}{\sqrt{65}}$$

Problems on projection of vectors

2004/ 45
The projection of the vector $\underline{a} = 5\,\underline{i} - 2\,\underline{j} + \underline{k}$ on the vector $\underline{b} = 2\,\underline{i} + 3\,\mathrm{j}$.

A $4\sqrt{13}$ B $\frac{5\sqrt{30}}{15}$ C $\frac{5\sqrt{13}}{13}$ D $\frac{4\sqrt{13}}{13}$ E $\frac{\sqrt{30}}{15}$

Solution

P = $\hat{b}.a$

But $\hat{b} = \frac{b}{|b|}$ where $|b|$ is $\sqrt{2^2 + 3^2}$

$$= \frac{1}{\sqrt{13}}(2i + 3j)$$

Thus P ⇒ $\hat{b}.a$ =

$$\frac{1}{\sqrt{13}}\,(2i + 3j).\,(5i - 2j + k)$$

$$= \frac{1}{\sqrt{13}}\,(10 - 6)$$

$$= \frac{1}{\sqrt{13}} \times 4 \quad \text{i.e} \quad \frac{4\sqrt{13}}{13} \quad (D)$$

2000/24 Exercise 27.4
Find the projection of the vector **a** on the vector **b**, if $a = 2\,\underline{i} + 3\underline{j}$ and $b = -\,\underline{i} + 4\underline{j}$

A $\frac{\sqrt{17}}{10}$ B. $10\sqrt{17}$ C $\frac{10\sqrt{17}}{17}$ D. $\frac{10}{17}$ E. $5\sqrt{17}$

Problems on equal vectors

2002/50 Neco
What is the value for which vectors x i + 8j and 5i + $\frac{40}{x}$ j are equal ?
A +5 B +4 C –5 D –9 E –10

Solution

Two vectors are said to be equal if their i – components are equal
In this case
i : i
x : 5 thus x = 5 (A)
Confirm it at the j-component

$\frac{40}{x} = 8$ thus x = 5

Problems on angle between two vectors

2015/16 f/m
If x = i – 3j and y = 6i + j.
Calculate the angle between x and y
A 60^0 B 75^0 C 81^0 D 85^0

Solution

$$\cos\theta = \frac{(i - 3j)\,.\,(6i + j)}{(\sqrt{1^2 + (-3)^2})\ (\sqrt{6^2 + 1^2})}$$

$$\cos\theta = \frac{1\times 6 + (-3)\times 1}{(\sqrt{1+9})\ (\sqrt{36+1})}$$

$$\cos\theta = \frac{6 - 3}{(\sqrt{10})\ (\sqrt{37})}$$

$$\cos\theta = \frac{3}{\sqrt{370}}$$

$\cos\theta = 0.15597$
$\theta = \cos^{-1} 0.15597$
$= 81.026^0 \approx 81^0$ C.

2014/24 f/m
Find the angle between (5i + 3j) and (3i – 5j)
A 180^0 B 90^0 C 45^0 D 0^0

Solution

$$\cos\theta = \frac{(5i + 3j).(3i - 5j)}{(\sqrt{5^2 + 3^2})\ (\sqrt{3^2 + (-5)^2}}$$

$$\cos\theta = \frac{5\times 3 + 3\times(-5)}{(\sqrt{34})\ (\sqrt{34})}$$

$$\cos\theta = \frac{15 - 15}{\sqrt{34\times 34}} = \frac{0}{\sqrt{1156}}$$

$\cos\theta = 0$
$\theta = \cos^{-1} 0$
$\theta = 90^0$ B.

2014/14a f/m
Given that x = 3i – j, y = 2i + kj and the cosine of the angle between x and y is $\frac{\sqrt{5}}{5}$, find the values of the constant k.

Solution

$$\cos\theta = \frac{\vec{x}\,.\,\vec{y}}{|\,xy\,|}$$

$$\frac{\sqrt{5}}{5} = \frac{(3i - j)\,.\,(2i + kj)}{(\sqrt{3^2 + (-1)^2})\ (\sqrt{2^2 + k^2})}$$

$$\frac{\sqrt{5}}{5} = \frac{6 - k}{(\sqrt{10}\ )\ (\sqrt{4 + k^2})}$$

$$\sqrt{5}\ (\sqrt{10})\ (\sqrt{4 + k^2}) = 5(6 - k)$$

Square both sides
$5(10)(4 + k^2) = 25(6 - k)^2$
$2(4 + k^2) = (6 - k)^2$
$8 + 2k^2 = 36 - 12k + k^2$
$k^2 + 12k - 28 = 0$
Factorizing $k^2 - 2k + 14k - 28 = 0$
$k(k - 2) + 14(k - 2) = 0$
$(k - 2)(k + 14) = 0$
$k - 2 = 0$ or $k + 14 = 0$
k = 2 or – 14

2000/16 (Nov)

Given that $|p| = 3$, $|q| = 4$ and $p \cdot q = -6$,

find the angle between p and q.

A. 30^0 B 45^0 C 120^0 D 135^0

Solution

$$\text{Cos}\theta = \frac{p \cdot q}{|p|\ |q|}$$

Substituting for the given values

$$\text{Cos}\theta = \frac{-6}{3\times 4}$$

$\text{Cos}\theta = -0.5$

$\theta = -\text{Cos}^{-1}\ 0.5$

Though it is 60^0 but the minus sign shows that Cosine here is in 2^{nd} quadrant with formula $180 - \theta$

$\theta = 180 - 60$

$= 120^0$ (C)

2004/9

Find the angle between i + 5j and 5i – j

A 30^0 B 45^0 C 60^0 D 90^0

Solution

$$\text{Cos}\ \theta = \frac{(i+5j).(5i-j)}{\left(\sqrt{1^2+5^2}\right)\left(\sqrt{5^2+(-1)^2}\right)}$$

$$\text{Cos}\ \theta = \frac{5-5}{(\sqrt{26})(\sqrt{26})} \quad \text{i.e.} \quad \frac{0}{26} = 0$$

$\text{Cos}\ \theta = 0$

$\theta = \text{Cos}^{-1}\ 0$

$= 90^0$ (D)

2004/40 Neco

If $\underline{m} = 4\underline{i} + 3\underline{j}$ and $\underline{n} = \underline{i} + 2\underline{j}$ what is the cosine of angle between $\underline{m}$ and $\underline{n}$?

A $2\sqrt{5}$ B $\frac{5\sqrt{2}}{2}$ C $\sqrt{2}$ D $\frac{12\sqrt{5}}{25}$ E $\frac{2\sqrt{5}}{5}$

Solution

Cosine of angle between $\underline{m}$ and $\underline{n}$ = $\frac{\underline{m}\ .\ \underline{n}}{|\underline{m}|\ |\underline{n}|}$

$\underline{m}\ .\ \underline{n} = [4\times1+3\times2]$

$= (4 + 6)$

$= 10$

$|\underline{m}| = (4^2+3^2)^{\frac{1}{2}}$ and $|\underline{n}| = (1^2+2^2)^{\frac{1}{2}}$

$= \sqrt{25}$ $\qquad = \sqrt{5}$

$= 5$

Thus, Cosine between $\underline{m}$ and $\underline{n}$ = $\frac{10}{5\sqrt{5}}$

$= \frac{10\sqrt{5}}{25}$

$= \frac{2\sqrt{5}}{5}$ (E)

1996/39

If $\overrightarrow{PQ} = 4\underline{i} + 3\underline{j}$ and $\overrightarrow{QR} = 5\underline{i} + 12\underline{j}$,

Find the cosine of the angle between the two vectors.

A. $\frac{16}{65}$ B. $\frac{13}{13}$ C. $\frac{7}{65}$ D. $\frac{4}{13}$ E. $\frac{56}{65}$

Solution

Cosine of angle between the two vectors = $\frac{\overrightarrow{PQ}.\overrightarrow{QR}}{|\overrightarrow{PQ}||\overrightarrow{QR}|}$

$$= \frac{(4i+3j).(5i+12j)}{\left(\sqrt{4^2+3^2}\right)\left(\sqrt{5^2+12^2}\right)}$$

$$= \frac{20+36}{(\sqrt{25})(\sqrt{169})}$$

$$= \frac{56}{5\times13} \quad \text{i.e} \ \frac{56}{65} \ \text{(E)}$$

2004/ 7 Neco

Given that $\underline{u} = -\underline{i} + 2\underline{j} - 2\underline{k}$ and $\underline{v} = 3\underline{i} + 4\underline{j} - 12\underline{k}$

Find (i) the dot product of $\underline{u}$ and $\underline{v}$

(ii) the angle between $\underline{u}$ and $\underline{v}$

Solution

(i). $\underline{u}\ .\ \underline{v} = [-1\times3+2\times4+(-2\times-12)]$

$= [-3+8+24]$

$= 29$

(ii) $\underline{u} \bullet \underline{v} = |\underline{u}||\underline{v}|\cos\theta$

$$\frac{\underline{u}\ .\ \underline{v}}{|\underline{u}|\ |\underline{v}|} = \cos\theta$$

$|\underline{u}| = [(-1^2)+2^2+(-2)^2]^{\frac{1}{2}}$; $|\underline{v}| = (3^2+4^2+(-12)^2)^{\frac{1}{2}}$

$= (9)^{\frac{1}{2}}$ $\qquad = (169)^{\frac{1}{2}}$

$= 3$ $\qquad = 13$

Thus, $\cos\theta = \frac{29}{3\times13}$ i.e $\frac{29}{39}$

$\cos\theta = 0.7436$

$\theta = \cos^{-1} 0.7436$

$= 41.96^0$

1996/38 (Nov) Exercise 27.5

If p = 2i + j and q = 5i +3j find the cosine of the acute angle between p and q

A. $\frac{13\sqrt{170}}{170}$ B. $\frac{7\sqrt{170}}{170}$ C. $\frac{7}{13}$ D. $\frac{13}{170}$ E. $\frac{7}{170}$

2003/16 Neco Exercise 27.6

If $\underline{a} = 2\underline{i} + 3\underline{j}$ and $\underline{b} = 2i + 4j$, find the cosine of angle between $\underline{a}$ *and* $\underline{b}$

A $\frac{2}{\sqrt{33}}$ B $\frac{4}{\sqrt{65}}$ C $\frac{8}{\sqrt{65}}$ D $\frac{8}{\sqrt{20}}$ E $\frac{4}{\sqrt{13}}$

2002/27 (Nov) Exercise 27.7

Evaluate the angle between **a** = – 4i + 2j and **b** = i – 3j

A 45^0 B 60^0 C 135^0 D 315^0

Problems on vector magnitude

2004/21 Neco (Dec) f/m

Calculate the magnitude of the vector $\bar{a}$ = 4i – 3j + 12k

A 169 B 26 C 19 D 13 E 8

Solution

Magnitude of vector

$\bar{a}$ = 4i – 3j + 12k

$|\bar{a}| = \sqrt{4^2 + (-3)^2 + 12^2}$

$= \sqrt{16+9+144} = \sqrt{169}$ i.e 13(D)

2013/37 f/m

If V = $\begin{pmatrix} -2 \\ 4 \end{pmatrix}$ and U = $\begin{pmatrix} -1 \\ 5 \end{pmatrix}$, find |u + v|

A $3\sqrt{10}$ B $\sqrt{82}$ C $\sqrt{46}$ D $\sqrt{5}$

Solution

U + V = $\begin{pmatrix} -1 \\ 5 \end{pmatrix} + \begin{pmatrix} -2 \\ 4 \end{pmatrix}$

$= \begin{pmatrix} -3 \\ 9 \end{pmatrix}$ i.e – 3i + 9j

$|u + v| = \sqrt{(-3)^2 + 9^2}$

$= \sqrt{9+81}$

$= \sqrt{90} = \sqrt{9\times 10}$

$= 3\sqrt{10}$ A.

2015/23 f/m

Given that r = 2i – j, s = 3i + 5j and t = 6i – 2j, find the magnitude of (2r + s – t)

A $\sqrt{15}$ B 4 C $\sqrt{24}$ D $\sqrt{26}$

Solution

2r + s – t = 2(2i – j) + 3i +5j – (6i – 2j)

= 4i – 2j + 3i + 5j – 6i + 2j

= i + 5j

$|2r + 5 - t| = \sqrt{1^2 + 5^2}$

$= \sqrt{1+25} = \sqrt{26}$ D.

1995/35 (Nov)

Given that **a** = $\begin{pmatrix} 6 \\ 3 \end{pmatrix}$ and **b** = $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$

Evaluate $|a+b| - |a-b|$

A.0 B 14 C. $2\sqrt{5}$ D. $5\sqrt{2}$ E. $10\sqrt{2}$

Solution

a + b and **a – b**

$\begin{pmatrix} 6 \\ 3 \end{pmatrix} + \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 7 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 6 \\ 3 \end{pmatrix} - \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix}$

Next, $|a + b| = \sqrt{7^2 + 1^2}$ and $|a - b| = \sqrt{5^2 + 5^2}$

$= \sqrt{50}$ and $= \sqrt{50}$

Thus, $|a+b| - |a-b| = \sqrt{50} - \sqrt{50}$

= 0 (A)

2002/ 31 Exercise 27.8

If p = (2i + 3j) and q = (i – 7j) then,

A $|q| + |p| = \sqrt{63}$ B $|p+q| = \sqrt{5}$

C $|p-q| = \sqrt{101}$ D $|p-2q| = \sqrt{17}$

Problems on dot product of vectors

1995/31 (Nov)

Evaluate; (8i – 15j) • (8i – 15j)

A. 49 B. 161 C. 189 D. 225 E. 289

Solution

Dot product = (8i – 15j) • (8i – 15j)

$= [8\times 8 + (-15)\times(-15)]$

= 64 + 225

= 289 (E)

2003/4 Neco

If r = i + 2j and q = 2i – 3j, find r • q

A -4 B -3 C $2^1/_2$ D 2 E 4

Solution

r • q = (i + 2j) • (2i – 3j)

= 2 – 6

= – 4 A

2002/48 Neco

The scalar product of vectors –2i – 3j and 4i + 5j is

A 23 B 7 C –7 D – 10 E – 23

Solution

Scalar product of two vectors is same as *dot product*

(–2i – 3j) • (4i + 5j) = (–2 × 4) + (–3 × 5)

= –8 – 15

= – 23 (E)

2005/15

Find the magnitude and direction of the resultant of **a** = 2i + 3j and **b** = 2i + j

A ($4\sqrt{2}$, 030^0) B ($4\sqrt{2}$, 045^0)

C ($2\sqrt{2}$, 045^0) D ($2\sqrt{2}$, 030^0)

Solution

Resultant of ab = a + b

= (2i + 3j) + (2i + j)

= 4i + 4j

magnitude of ab $= \sqrt{4^2 + 4^2}$

$= \sqrt{16+16} = \sqrt{32} = 4\sqrt{2}$

Direction of resultant of ab

$\cos\theta = \dfrac{a\ .\ b}{|a|\ |b|}$

$= \dfrac{(2i+3j).(2i+j)}{\left(\sqrt{2^2+3^2}\right)\left(\sqrt{2^2+1^2}\right)}$

$= \dfrac{4+3}{(\sqrt{13})(\sqrt{5})}$

$\cos\theta = \dfrac{7}{\sqrt{65}}$

$\theta = \cos^{-1} \dfrac{7}{\sqrt{65}}$

$= 29.74^0 \approx 30^0$

Thus magnitude and direction is ($4\sqrt{2}$, 030^0) (A)

1996/38 Exercise 27.9

If $\overrightarrow{PQ} = 2\underline{i} - 3\underline{j}$ and $\overrightarrow{QR} = 3\underline{i} - 5\underline{j}$, find $\overrightarrow{PQ} \bullet \overrightarrow{QR}$

A 21 B. 15 C. 9 D. 1 E. –9

Perpendicular condition of vectors

2003 / 15

Given that p = (6i – 9j) and q = ai + 2j are perpendicular, calculate the value of the constant a.

A – 3 B $\frac{-4}{3}$ C $\frac{3}{4}$ D 3

Solution

Two vectors are perpendicular to each other if their **dot product** equals zero.

$$p \bullet q = 0$$
$$(6i - 9j) \bullet (ai + 2j) = 0$$
$$6 \times a + (-9) \times 2 = 0$$
$$6a - 18 = 0$$
$$6a = 18$$
$$a = 3 \quad (D)$$

2004/42 Neco

If $\underline{a} = 3\underline{i} + 4\underline{j}$ and $\underline{b} = \lambda\underline{j} + 2\underline{k}$ are perpendicular, what is the value of λ ?

A 4 B 0 C $-{}^3/_2$ D $-{}^8/_3$ E – 4

Solution

$\underline{a} \,.\, \underline{b} = 0$ Condition for perpendicularity

$$3\times0 + 4\times\lambda + 0\times2 = 0$$
$$4\lambda = 0$$
$$\lambda = 0 \quad (B)$$

2004/32 Neco (Dec)

Find K such that $\underline{a}$ = 4i – 10j and $\underline{b}$ = Ki + 2j are perpendicular vectors

Solution

$\underline{a} \,.\, \underline{b} = 0$ Condition for perpendicularity

$$(4i - 10j) \bullet (k\,i + 2j) = 0$$
$$4k + (-20) = 0$$
$$4k - 20 = 0$$
$$4k = 20$$
$$\frac{4k}{4} = \frac{20}{4}$$
$$k = 5$$

1994/34

If **p** = 4i + kj and **q** = 2i – 3j are perpendicular, find the value of k, where k is a scalar quantity.

A. 3 B. $\frac{8}{3}$ C. $\frac{2}{3}$ D. $\frac{-3}{2}$ E. – 4

Solution

Two vectors p and q are perpendicular, if

$$p \bullet q = 0$$
$$(4i + kj) \bullet (2i - 3j) = 0$$
$$[4\times2 + (k\times-3)] = 0$$
$$8 - 3k = 0$$
$$8 = 3k$$
$$\tfrac{8}{3} = k \quad (B)$$

1994/7 June

OP, OQ and OR are the vectors 3i + 4j, 3ki + 40j and 8i – 2j respectively, where k is a scalar.If the resultant of OP and OR is perpendicular to OQ, find K .

Solution

First, we find resultant of OP and OR

$$= OP + OR$$
$$= (3i + 4j) + (8i - 2j)$$
$$= (11i + 2j)$$

Next, for two vector to be perpendicular;

their dot product = 0

$$(OP + OR) \bullet OQ = 0$$
$$(11i + 2j) \bullet (3ki + 40j) = 0$$
$$33k + 80 = 0$$
$$33k = -80$$
$$k = -\frac{80}{33}$$

1996/37 (Nov)

For what value of p are the vectors pi – 3j and 3i – 2j perpendicular?

A. -3 B. -2 C. $\frac{2}{3}$ D. 2 E. 3

Solution

Since the two vectors are perpendicular; then

$$(Pi - 3j) \bullet (3i - 2j) = 0$$
$$3p + 6 = 0$$
$$3p = -6$$
$$p = -\tfrac{6}{3} \text{ i.e } -2 \text{ (B)}$$

Problems on Unit vectors

2012/24 f/m

Find the unit vector in the direction of the vector – 12i + 5j

A $\frac{-12}{13}i - \frac{5}{13}j$ B $\frac{-1}{13}i - \frac{5}{13}j$ C $\frac{-12}{13}i + \frac{5}{13}j$

D $\frac{-5}{13}i - \frac{12}{13}j$

Solution

$$\text{Unit vector} = \frac{-12i + 5j}{\sqrt{(-12)^2 + 5^2}}$$
$$= \frac{-12i + 5j}{\sqrt{144 + 25}}$$
$$= \frac{-12i + 5j}{\sqrt{169}}$$
$$= \frac{-12i + 5j}{13} = \frac{-12}{13}i + \frac{5}{13}j \quad C.$$

2013/34 f/m

P and Q are point (3, 1) and (7, 4) respectively

Find the unit vector along PQ.

A $\begin{pmatrix} 4 \\ 3 \end{pmatrix}$ B $\begin{pmatrix} 0.6 \\ 0.8 \end{pmatrix}$ C $\begin{pmatrix} 0.8 \\ 0.6 \end{pmatrix}$ D $\begin{pmatrix} -0.8 \\ -0.6 \end{pmatrix}$

Solution

Vector PQ = q – p

$= (7, 4) – (3, 1)$
$= (4, 3)$ i.e **(4i + 3j)**

Unit vector PQ = $\frac{4i+3j}{\sqrt{4^2+3^2}} = \frac{4i+3j}{\sqrt{25}}$

$= \frac{1}{5}(4i + 3j)$

In column form $= \frac{1}{5}\begin{pmatrix}4\\3\end{pmatrix}$

$= \begin{pmatrix}0.8\\0.6\end{pmatrix}$ C.

1993/31 (Nov)
Find the unit vector in the direction of the resultant of
p = 5i + 3j and q = 7i – 8j

A. $\frac{1}{12}i - \frac{1}{5}j$ B. $\frac{12}{13}i - \frac{5}{13}j$ C. $\frac{2}{13}i + \frac{5}{13}j$
D. $\frac{5}{13}i + \frac{12}{13}j$ E. $\frac{1}{12}i + \frac{1}{5}j$

Solution

Resultant of p and q = p + q
= (5i + 3j) + (7i – 8j)
= (12i – 5j)

Next, unit vector is $= \frac{p+q}{|p+q|}$

$= \frac{12i-5j}{\sqrt{12^2+(-5)^2}}$

$= \frac{12i-5j}{\sqrt{169}}$

$= \frac{1}{13}(12i-5j)$ ie $\frac{12}{13}i - \frac{5}{13}j$ (B)

1996/32
Given $\overrightarrow{OQ} = 4\underset{\sim}{i} - 3\underset{\sim}{j}$, the unit vector in the direction of $\overrightarrow{OQ}$ is

A 20 $\underset{\sim}{i} - 15\underset{\sim}{j}$ B $4\underset{\sim}{i} - 3\underset{\sim}{j}$ C $\frac{1}{5}(4\underset{\sim}{i} - 3\underset{\sim}{j})$
D. $-\frac{1}{5}(4\underset{\sim}{i} + 3\underset{\sim}{j})$ E. $\frac{1}{7}(4\underset{\sim}{i} - 3\underset{\sim}{j})$

Solution

Unit vector in $\hat{OQ}$ the direction of $\overrightarrow{OQ} = \frac{\overrightarrow{OQ}}{|\overrightarrow{OQ}|}$

$= \frac{4\underset{\sim}{i}-3\underset{\sim}{j}}{\sqrt{4^2+(-3)^2}}$

$= \frac{1}{5}(4\underset{\sim}{i}-3\underset{\sim}{j})$ (C)

2000/37
The position vectors of points P and Q are i – j and 2i – 5j respectively. Find a unit vector perpendicular to $\overrightarrow{PQ}$

A. $\pm\frac{(-i + 4j)}{17}$ B $\pm\frac{(i + 4j)}{17}$
C. $\pm\frac{(4i+j)}{\sqrt{17}}$ D. $\pm\frac{(4i-j)}{\sqrt{17}}$

Rule is $\overrightarrow{AB} = \underset{\sim}{b} - \underset{\sim}{a}$
$\overrightarrow{PQ} = (2i – 5j – (i – j)$
$= i – 4j$

Next, Unit vector perpendicular to $\overrightarrow{PQ} = \frac{i-4j}{\sqrt{1^2+(-4)^2}}$

$= \frac{(i-4j)}{\sqrt{17}}$

2000/47 Neco Exercise 27.10
The unit vector in the direction of (3i + 4j) is

A $\frac{1}{6}$(3i + 4j) B $\frac{1}{5}$(3i + 4j) C $\frac{1}{4}$(3i + 4j)
D $\frac{1}{3}$(3i + 4j) E $\frac{1}{2}$(3i + 4j)

1994/34 (Nov) Exercise 27.11
Find the unit vector acting in the direction of 2i – j

A i + j B i – j C $\frac{2i-j}{\sqrt{5}}$ D $\frac{2i-j}{5}$ E $\frac{2i-j}{\sqrt{3}}$

1999/26 (Nov) Exercise 27.12
Find unit vector in the direction of the Resultant of P = (4i – j) and q = (2i – 7j)

A. $\frac{3}{5}i - \frac{4}{5}j$ B. $\frac{1}{6}i - \frac{1}{8}j$ C. $\frac{1}{5}i - \frac{7}{10}j$
D. $\frac{2}{5}i - \frac{1}{10}j$

Problems on Joint cases of vectors properties

1996/14
The position vectors of the points P and Q are p = – i + 3j and q = 2i + 2j respectively. Find
(a) 3p – 4q (b) $|p+q|$ (c) the angle between p and q
(d) the scalars h and k such that hp + kq = 5i + 7j

Solution

(a) 3p – 4q = 3(– i + 3j) – 4(2i + 2j)
= – 3i + 9j – 8i – 8j
= – 11i + j

(b) $|p+q|$
but p + q = – i + 3j + 2i + 2j
= i + 5j

Thus, $|P+q| = \sqrt{1^2+5^2}$
$= \sqrt{26}$

(c) Angle between p and q

cosine of angle b/w p & q = $\frac{p.q}{|p||q|}$

$= \frac{(-i+3j).(2i+2j)}{\left(\sqrt{(-1)^2+3^2}\right)\left(\sqrt{2^2+2^2}\right)}$

$= \frac{-2+6}{(\sqrt{10})(\sqrt{8})}$

$= \frac{4}{4\sqrt{5}}$ i.e $\frac{1}{\sqrt{5}}$

Cosine of angle b/w p and q = 0.4472

Angle b/w p and q = $\cos^{-1}$ 0.4472

$= 63.43^0$

(d) hp + kq = 5i + 7j

h(–i + 3j) + k(2i + 2j) = 5i + 7j

–hi + 3hj + 2ki + 2kj = 5i + 7j

Collect like terms together at LHS

–hi + 2ki + 3hj + 2kj = 5i + 7j

(–h + 2k)i + (3h + 2k)j = 5i + 7j

Equating terms

–h + 2k = 5 ------(1)

– (3h + 2k = 7) -----(2)

– 4h = –2

h = $^1/_2$

Also from (2) 2k = 7 – 3($\frac{1}{2}$)

2k = 7 – ($\frac{3}{2}$)

2k = $\frac{11}{2}$

k = $\frac{11}{4}$

1992/14 (Nov)

The position vector of two points **A** and **B** are **a** = 4i – 2j and **b** = 2i + 3j respectively. Find:

(a) $|3A-2B|$

(b) Scalars α and β such that

8i + 8j = α**A** + β**B**;

(c) the cosine of acute angle between **A** and **B**, leaving your answer in surd form

Solution

(a) $|3A-2B|$

First we find 3A – 2B = 3(4i – 2j) – 2(2i + 3j)

= 12i – 6j – 4i – 6j

= 12i – 4i – 6j – 6j

= 8i – 12j

Next, $|3A-2B| = \sqrt{8^2+(-12^2)}$

$= \sqrt{64+144}$

$= \sqrt{208}$ or 14.42

$= 4\sqrt{13}$

(b) 8i + 8j = α**A** + β**B**;

= α(4i – 2j) + β(2i + 3j)

= 4α i – 2αj + 2βi + 3βj

= 4α i + 2βi – 2αj + 3βj

8i + 8j = (4α +2β)i + (–2α + 3β)j

Equating terms

4α + 2β = 8 -------- (1)

–2α + 3β = 8 ---------(2)

Multiply (2) by 2 and add

– 4α + 6β = 16

4α + 2β = 8

8β = 24 Thus β = 3 and α = $^1/_2$

(C) Cosine of angle b/w A and B = $\frac{A\cdot B}{|A|\,|B|}$

A • B = $[4\times2+(-2\times3)]$

= (8 – 6)

= 2

$|A| = \sqrt{4^2+(-2)^2}$ and $|B| = \sqrt{2^2+3^2}$

$= \sqrt{16+4}$ $= \sqrt{4+9}$

$= \sqrt{20}$ $= \sqrt{13}$

Cosine of angle = $\frac{2}{\sqrt{20}\ \sqrt{13}} = \frac{2}{2\sqrt{5}\times\sqrt{13}}$

$= \frac{1}{\sqrt{65}}$

$= \frac{\sqrt{65}}{65}$ i.e $\frac{1}{65}(\sqrt{65})$

1994/37 (Nov)

If p and q are two parallel vectors in opposite directions, Then which of the following statements is/are true?

I p • q = $|p|\,|q|$

II p • q = $-|p|\,|q|$

III p • q = 0

A II and III only B. I only C. III only
D. I and III only E. II only

Solution

By condition of parallel and opposite vector.
Let the vectors be
p = 3i + 2i and q = – 3i – 2j

Testing

I p • q = $|p|\,|q|$

LHS: (3i +2j) • (–3i – 2j) = – 9 – 4

= – 13

RHS $\left(\sqrt{3^2+2^2}\right)\left(\sqrt{3^2+2^2}\right) = (\sqrt{13})(\sqrt{13})$

= 13 LHS ≠ RHS

II p • q = $-|p|\,|q|$

From the test at **I** we deduce that **II** is true

III p • q = 0

From the test at **I**, we deduce that **III** is not true

The answer is **II only** (E)

1994/7 (Nov)

Given **a** = 2i + 5j and **b** = 4i – 3j, find

(a) The unit vector in the direction of **b**.

(b) The projection of **a** on **b**

(c) | 3**a** – 2**b** |

Solution

(a) Unit vector in the direction of **b** = $\frac{b}{|b|}$

$= \frac{4i-3j}{\sqrt{4^2+3^2}}$

$$= \frac{4i-3j}{\sqrt{25}} \text{ i.e } \frac{1}{5}(4j-3j)$$

(b) Projection **a** on **b** = $\hat{b} \bullet$ a

$$= \frac{b}{|b|} \bullet a$$

$$= \frac{1}{5}(4i-3j)(2i+5j)$$

$$= \frac{1}{5}(8-15) \text{ i.e } -\tfrac{7}{5}$$

(c) First is 3a – 2b = 3(2i + 5j) –2(4i –3j)

= (6i + 15j) + (–8i + 6j)

= –2i + 21j

Thus $|3\mathbf{a} - 2\mathbf{b}| = \sqrt{(-2)^2 + 21^2}$

$= \sqrt{445}$

= 21. 095

2000/22 (Nov)

Given that p = 9i + 12j and q = – 6 i – 8j

Evaluate $|p-q| - \{|p| - |q|\}$

A 10 B 15 C 20 D 30

Solution

First vector Subtraction for p and q

p – q = [9 – (–6)]i + [12 – (–8)] j

= (9 + 6)i + (12 + 8)j

p – q = 15i + 20j

Thus $|p-q| = \sqrt{15^2 + 20^2}$

$= \sqrt{225+400}$

$= \sqrt{625}$ i.e **25**

$|p| = \sqrt{9^2 + 12^2}$

$= \sqrt{81+144}$

$= \sqrt{225}$

= 15

$|q| = \sqrt{(-6)^2 + (-8)^2}$

$= \sqrt{36+64}$

$= \sqrt{100}$

= 10

Therefore $|p-q| - \{|p| - |q|\}$

= 25 – { 15 – 10 }

= 25 – 5

= 20 (C)

2000/15 a (Nov)

If x = 4i – 3j ;

y = 2i + 3j

z = i + j

(i) find z · y ; (ii) find the angle between y and z

(iii) show that x is not perpendicular to z

Solution

(i) z · y = (i + j) · (2i + 3j)

= 1 × 2 + 1 × 3

= 2 + 3

= 5

(ii) Angle between y and z

$$\text{Cos}\theta = \frac{y \cdot z}{|y| \; |z|}$$

$$= \frac{5}{\left(\sqrt{2^2+3^2}\right)\left(\sqrt{1^2+1^2}\right)}$$

$$= \frac{5}{\left(\sqrt{13}\right)\left(\sqrt{2}\right)} = \frac{5}{\sqrt{26}} = \frac{5}{5.099}$$

Cos θ = 0.9806

θ = Cos^{-1} 0.9806 = 11.30^0

(iii) If x is perpendicular to z then $\underline{x} \cdot \underline{z} = 0$

$\underline{x} \cdot \underline{z} \Rightarrow (4i - 3j) \cdot (i + j) = 4 \times 1 + (-3) \times 1$

= 4 – 3

= 1

One is not zero so x is NOT perpendicular to z.

2000/15

Given that p = 2i + 3j; q = i + j ; r = i – 2j ;

Find: (i) $|2p - 3q + r|$;

(ii) The Unit vector in the opposite direction of 2p – 3q + r;

(iii) The angle between p and r, correct to the nearest degree.

Solution

(i) First we do the scalar multiplications

2p – 3q + r = 2(2i + 3j) – 3(i + j) + (i – 2j)

= 4i + 6j – 3i – 3j + i – 2j

Collect like terms together

= 4i – 3i + i + 6j – 3j – 2j

= 2i + j

Thus, $|2p-3q+r| = \sqrt{2^2+1^2}$

$= \sqrt{5}$

(ii) 1st we find unit vector of 2p – 3q + r

$$= \frac{2p-3q+r}{|2p-3q+r|}$$

$$= \frac{2i+j}{\sqrt{5}}$$

Thus, Unit vector in opposite direction of 2p – 3q + r

$$= \frac{-(2i+j)}{\sqrt{5}}$$

$$= \frac{-2i-j}{\sqrt{5}}$$

(iii) angle between p and r is given by

$$\text{Cos } \theta = \frac{p \cdot r}{|p||r|}$$

$$= \frac{(2i+3j) \cdot (i-2j)}{\left(\sqrt{2^2+3^2}\right)\left(\sqrt{1^2+(-2)^2}\right)}$$

$$= \frac{2 \times 1 + 3 \times (-2)}{\left(\sqrt{13}\right)\left(\sqrt{5}\right)}$$

$$\text{Cos } \theta = \frac{-4}{\sqrt{65}}$$

Cos θ = – 0.4961

θ = $60.26^0 \approx 60^0$

Cosine is negative in 2nd & 3rd quadrant

i.e. θ is either 180 – 60 or 180 + 60

= 120^0 or 240^0

2002/7

The position vectors of points P and Q are;

p = (i + 3j)
q = (9i + 8j)

respectively. Find:

(a) $\overrightarrow{PQ}$

(b) correct to the nearest degree, the angle between p and q.

(c) The length of the projection of q on P

Solution

(a) $\overrightarrow{PQ}$ = q – p
= (9i + 8j) – (i + 3j)
= 8i + 5j

(b) Angle between **p** and **q** $\Rightarrow$ Cos θ $= \dfrac{P.Q}{|p||q|}$

$$\text{Cos } \theta = \frac{1\times9+3\times8}{\left(\sqrt{1^2+3^2}\right)\left(\sqrt{9^2+8^2}\right)}$$

$$= \frac{9+24}{\left(\sqrt{10}\right)\left(\sqrt{145}\right)}$$

$$= \frac{33}{\sqrt{1450}}$$

Cos θ = 0.8666

θ = $\cos^{-1}$ 0.8666

= 29.93^0

≈ 30^0 to the nearest degree

(c)Length of projection of q on p = $|q|$ Cos θ

$=\left(\sqrt{9^2+8^2}\right) \times 0.8666$

$= \sqrt{145} \times 0.8666$

= 10.43 units

1997/7

The position vectors of points p and q relative to a fixed point O are

p = 3i + 4j and q = 12i + 5j respectively.

Find: (i) $|2p-3q|$

(ii) the unit vector parallel to $\overrightarrow{PQ}$;

(iii) the length of the projection of p on q.

Solution

(i) $|2p-3q|$

First find 2p – 3q

= 2(3i + 4j) – 3(12i + 5j)
= 6i + 8j – 36i – 15j
= –30i – 7j

Thus $|2p-3p| = \sqrt{(-30)^2 + (-7)^2}$

$= \sqrt{949}$

(ii) $\overrightarrow{PQ}$ = q – p

= (12i + 5j) – (3i + 4j)
= 12i + 5j – 3i – 4j
= 9i + j

Unit vector parallel to PQ

$$= \frac{9i+j}{\sqrt{9^2+1^2}} = \frac{9i+j}{\sqrt{82}}$$

(iii) Length of the projection of p on q = $|p|$ Cos θ

First, we find Cos θ $= \dfrac{P.Q}{|p||q|}$

$$= \frac{(3i+4j).(12i+5j)}{\left(\sqrt{3^2+4^2}\right)\left(\sqrt{12^2+5^2}\right)}$$

$$= \frac{36+20}{(5)(13)}$$

$$\text{Cos } \theta = \frac{56}{65}$$

Thus length of projection of p on q = $|p|$ Cos θ

$$= \left(\sqrt{3^2+4^2}\right)\left(\frac{56}{65}\right)$$

$$= \sqrt{25} \times \frac{56}{65}$$

$$= 5 \times \frac{56}{65}$$

= 4.31 units

1999/32 (Nov) Exercise 27.13

Given that **a**, **b** and **d** are vector quantities and θ the angle between **b** and **d**, which of the following statements is /are true?

I. **a . b = b . a**

II. (**a . b**) + **d = a . b + a .d**

III. **b . d =** $|b|\,|d|$ cosθ

A. I and II B. II and III C. I and III D. I only

1997/38 Exercise 27.14

Which of the following is /are always true of vector P, Q and R ?

I P•Q = Q•P

II P• (Q + R) = P•Q + P•R

III P•Q = O if p is perpendicular to Q

A I only B II only C I and II only
D II and III only E. 1, II and III

2004/10 Exercise 27.15

If p and q are two parallel vectors in the same direction, which of the following statement is/are true ?

I p . q = $|p|\ |q|$

II p . q = – $|p|\ |q|$ III p . q = 0

A I only B I and II only C II and III only D III only

Problems on position vectors

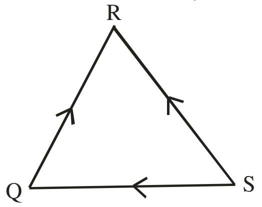
2012/9 f/m

QRS is a triangle such that $\overrightarrow{QR}$ = (3i + 2j) and $\overrightarrow{SR}$ = (– 5i + j). Find $\overrightarrow{SQ}$.

A 8i + j B 2i – j C – 2i – 3j D – 8i – j

Solution

With the aid of the vectors arrow, we sketch as:



By triangle rule

$$\overrightarrow{SR} = \overrightarrow{SQ} + \overrightarrow{QR}$$

$$5i + j = \overrightarrow{SQ} + 3i + 2j$$

$$-5i + j - (3i + 2j) = \overrightarrow{SQ}$$

$$-5i + j - 3i - 2j = \overrightarrow{SQ}$$

$$-8i - j = \overrightarrow{SQ} \qquad \text{D.}$$

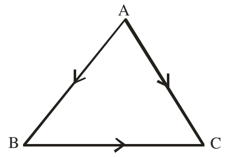
2014/23 f/m

Given that $\overrightarrow{AB} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$ and $\overrightarrow{AC} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$, find $\left|\overrightarrow{BC}\right|$

A. $4\sqrt{2}$ B. $6\sqrt{2}$ C. $2\sqrt{10}$ D. $4\sqrt{10}$

Solution

We draw the diagram based on the arrow flow of the vectors given



$$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$$

$$\begin{pmatrix} 4 \\ 3 \end{pmatrix} + \overrightarrow{BC} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$$

$$\overrightarrow{BC} = \begin{pmatrix} 2 \\ -3 \end{pmatrix} - \begin{pmatrix} 4 \\ 3 \end{pmatrix}$$

$$\overrightarrow{BC} = \begin{pmatrix} -2 \\ -6 \end{pmatrix} \quad \text{i.e } -2i - 9j$$

$$\text{Thus } \left|\overrightarrow{BC}\right| = \sqrt{(-2)^2 + (-6)^2}$$

$$= \sqrt{4 + 36}$$

$$= \sqrt{40} = \sqrt{4\times10} = 2\sqrt{10}$$

1994/36

The position vectors of points P, Q and R relative to an origin O are

$\begin{pmatrix} 8 \\ 4 \end{pmatrix}, \begin{pmatrix} -2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} -3 \\ -5 \end{pmatrix}$ respectively. Find OP-3OQ + 2OR

A. $\begin{pmatrix} 8 \\ -15 \end{pmatrix}$ B. $\begin{pmatrix} 20 \\ -16 \end{pmatrix}$ C. $\begin{pmatrix} 20 \\ 3 \end{pmatrix}$ D. $\begin{pmatrix} 8 \\ -3 \end{pmatrix}$ E. $\begin{pmatrix} -6 \\ -15 \end{pmatrix}$

Solution

OP = P – O

$$= \begin{pmatrix} 8-0 \\ 4-0 \end{pmatrix} \text{ i.e } \begin{pmatrix} 8 \\ 4 \end{pmatrix}$$

Similarly 3OQ = 3Q

$$= \begin{pmatrix} -6 \\ 9 \end{pmatrix}$$

and 2OR = 2R

$$= \begin{pmatrix} -6 \\ -10 \end{pmatrix}$$

$$\text{Hence OP} - 3\text{OQ} + 2\text{OQ} = \begin{pmatrix} 8 \\ 4 \end{pmatrix} - \begin{pmatrix} -6 \\ 9 \end{pmatrix} + \begin{pmatrix} -6 \\ -10 \end{pmatrix}$$

We absorb the minus sign as

$$= \begin{pmatrix} 8 \\ 4 \end{pmatrix} + \begin{pmatrix} 6 \\ -9 \end{pmatrix} + \begin{pmatrix} -6 \\ -10 \end{pmatrix}$$

$$= \begin{pmatrix} 8 \\ -15 \end{pmatrix} \quad \text{A}$$

2003 / 27

Given that $\overrightarrow{OP} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$ and $\overrightarrow{PQ} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$,

where O is the origin, find $\overrightarrow{OQ}$

A $\begin{pmatrix} -9 \\ -4 \end{pmatrix}$ B $\begin{pmatrix} -2 \\ -1 \end{pmatrix}$ C $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ D $\begin{pmatrix} 4 \\ 9 \end{pmatrix}$

Solution

We have a simple rule for vectors $\overrightarrow{AB}$ = b – a

and origin vector 0 in this case is $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$

Thus $\overrightarrow{OP}$ is $\begin{pmatrix} 0-i \\ 0-j \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \end{pmatrix} \Rightarrow \begin{pmatrix} i=-3 \\ j=-5 \end{pmatrix}$ i.e P = $\begin{pmatrix} -3 \\ -5 \end{pmatrix}$

Also $\overrightarrow{PQ}$ .is $\begin{pmatrix} o-i \\ o-j \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix} \Rightarrow \begin{pmatrix} i=-4 \\ j=-9 \end{pmatrix}$ i.e Q = $\begin{pmatrix} -4 \\ -9 \end{pmatrix}$

Hence, $\overrightarrow{OQ}$ is $\begin{pmatrix} 0+4 \\ 0+9 \end{pmatrix} = \begin{pmatrix} i \\ j \end{pmatrix}$ i.e. $\begin{pmatrix} 4 \\ 9 \end{pmatrix}$ (D)

1997/39

The position vector of P and Q are p = i – 3j and q = 4i + 5j respectively. Find the unit vector parallel to $\overrightarrow{PQ}$

A. $\frac{1}{\sqrt{73}}(3i + 8j)$ B. $-\frac{1}{\sqrt{73}}(3i - 8j)$ C. $\frac{1}{\sqrt{73}}(i - 3j)$

D. $\frac{1}{\sqrt{73}}(4i + 5j)$ E. $\frac{1}{\sqrt{89}}(5i + 8j)$

Solution

First we find $\overrightarrow{PQ}$ = q – p

= (4i + 5j) – (i – 3j)

= 4i – i + 5j + 3j

= 3i + 8j

Next, we find unit vector parallel to $\overrightarrow{PQ}$

Recall that two non – zero vectors **b** and **c** are parallel if and only if $\exists$ a scalars.
Such that **b** = S**c**

Thus what we need to do is to get the unit Vector of $\overrightarrow{PQ}$

$$= \frac{\overrightarrow{PQ}}{|\overrightarrow{PQ}|}$$

$$= \frac{3i+8j}{\sqrt{3^2+8^2}}$$

$$= \frac{1}{\sqrt{73}}(3i+8j) \text{ (A)}$$

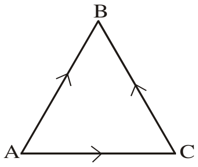
1997/32

Given that $\overrightarrow{AB}$ = 5i + 3j and $\overrightarrow{AC}$ = 2i + 5j, and find $\overrightarrow{CB}$

A. 7i + 8j B. – 3i + 2j C. 3i – 2j
D. 3i + 8j E – 7i – 8j

Solution

Sketching the vector triangle diagram noting the direction of the arrows



From the flow of the arrows

$$\overrightarrow{AC} + \overrightarrow{CB} = \overrightarrow{AB}$$

$$\overrightarrow{CB} = \overrightarrow{AB} - \overrightarrow{AC}$$

= (5i + 3j) – (2i + 5j)
= 5i – 2i + 3j – 5j
= 3i – 2j (C)

2004/7

The position vectors of points P and Q relative to the origin are: **p** = – i – 4j , **q** = 2i – 4j

(a)Find a unit vector parallel to the resultant of **p** and **q**

(b) Calculate $|2p-3q|$, leaving your answer in surd form.

Solution

(a) Resultant of **p** and **q** = **p** + **q**
= (–i – 4j) + 2i – 4j
= – i + 2i – 4j – 4j
= i – 8j

Unit vector parallel to $\overrightarrow{pq} = \frac{i-8j}{\sqrt{65}}$

$$= \frac{1}{\sqrt{65}}(i-8j)$$

(b) $|2p-3q|$

First 2p – 3q = 2(–i – 4j) – 3(2i – 4j)
= - 2i – 8j – 6i + 12j
= - 8i + 4j

Thus, $|2p-3q| = \sqrt{(-8)^2+4^2}$

$$= \sqrt{64+16}$$

$$= \sqrt{80} \text{ i.e } 4\sqrt{5} \text{ in surd form}$$

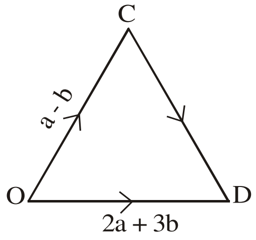
2005 / 7 June

Given that $\overrightarrow{OC}$ = **a** – **b** and $\overrightarrow{OD}$ = **2a** + **3b**,

where **a** = 2i + 3j and **b** = 3i – 2j , find $\overrightarrow{CD}$

Solution

Let us sketch the diagram following the directions of vectors arrow.



By triangle law of vectors
OC + CD = OD
CD = OD – OC
= (2a + 3b) – (a – b)
= 2a + 3b – a + b
= a + 4b
= (2i + 3j) + 4(3i – 2j)
= 2i + 3j + 12i – 8j
= 14i – 5j

2005 / 35

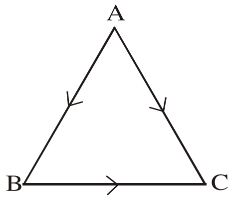
In a triangle ABC, $\overrightarrow{AB}$ = 3i – 4j and $\overrightarrow{AC}$ = –2i – 7j.

Find $\overrightarrow{BC}$.

A - 5i + 11j B - 5i + 3j C 5i – 3j D 5i + 11j

Solution

Let us sketch the diagram following the directions of vectors arrow.



By triangle law for vectors

$$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}$$

$$-2i - 7j = 3i - 4j + \overrightarrow{BC}$$

$$(-2i - 7j) - (3i - 4j) = \overrightarrow{BC}$$

$$-5i - 3j = \overrightarrow{BC}$$

2005 / 16 (b)

The position vectors of points P, Q and R in the x - y plane are $\overrightarrow{OP}$ = 3i + 4j , $\overrightarrow{OQ}$ = 5i + 6kj and $\overrightarrow{OR}$ = 7i + 2j respectively, where k is a scalar. If the resultant of $\overrightarrow{OP}$ and $\overrightarrow{OR}$ is perpendicular to $\overrightarrow{OQ}$, find the value of k .

Solution

First, resultant of $\overrightarrow{OP}$ and $\overrightarrow{OR} = \overrightarrow{OP} + \overrightarrow{OR}$
= (3i + 4j) + (7i + 2j)
= 10i + 6j

Next, resultant of OP and OR perpendicular to $\overrightarrow{OQ}$

Implies $(OP + OR) \bullet (OQ) = 0$

$$(10i + 6j) \bullet (5i + 6kj) = 0$$
$$10 \times 5 + 6 \times 6k = 0$$
$$50 + 36k = 0$$
$$36k = -50$$
$$k = \frac{-50}{36} = \frac{-25}{18}$$

2003/16b

The position vectors of points P, Q and R are
p = 7i + 2j, q = 5i + 5j and r = 3i + 8j respectively
(i) show that P, Q and R lie on a straight line
(ii) find scalars n and m such that 7i – 8j = np + mq

Solution

(i) If P,Q and R lie on a straight line then
the vector $\overrightarrow{PQ}$ is same as the vector $\overrightarrow{QR}$

$\overrightarrow{PQ} = q - p$
$= (5i + 5j) - (7i + 2j)$
$= -2i + 3j$

and $\overrightarrow{QR} = r - q$
$= (3i + 8j) - (5i + 5j)$
$= -2i + 3j$ QED

(ii) p = 7i + 2j and q = 5i + 5j
then np = 7ni + 2nj mq = 5mi + 5mj
But np + mq = 7i – 8j
$\Rightarrow$ 7ni + 2nj + 5mi + 5mj = 7i – 8j
Equate terms in i to i and also do same for j

$$\begin{array}{rl} 7n + 5m = & 7 \\ -(2n + 5m = & -8) \\ \hline 5n = & 15 \end{array}$$
$$n = 3$$

Substitute for n = 3 into 7n + 5m = 7
$$7(3) + 5m = 7$$
$$m = -{}^{14}/_{5}$$

`Problems on vectors and plane shapes`

2014/14b f/m

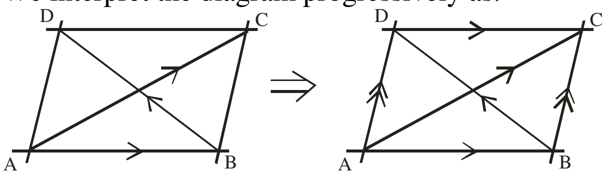
In the quadrilateral ABCD

$\overrightarrow{AB} = \begin{pmatrix}-5\\-1\end{pmatrix}$, $\overrightarrow{AC} = \begin{pmatrix}-6\\-9\end{pmatrix}$ and $\overrightarrow{BD} = \begin{pmatrix}4\\-7\end{pmatrix}$

Show whether or not ABCD is a parallelogram

Solution

We interpret the diagram progressively as:



In ΔADB the arrow at AD will point $\overrightarrow{AD}$, BC follow suit
We are to show that AD = BC

$$\overrightarrow{AD} = \overrightarrow{AB} + \overrightarrow{BD} \quad \text{(triangle law)}$$
$$= \begin{pmatrix}-5\\-1\end{pmatrix} + \begin{pmatrix}4\\-7\end{pmatrix}$$
$$= \begin{pmatrix}-1\\-8\end{pmatrix}$$

Alternatively

In ΔABC; by law of triangle

$$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}$$
$$\begin{pmatrix}-6\\-9\end{pmatrix} = \begin{pmatrix}-5\\-1\end{pmatrix} + \overrightarrow{BC}$$
$$\overrightarrow{BC} = \begin{pmatrix}-6\\-9\end{pmatrix} - \begin{pmatrix}-5\\-1\end{pmatrix}$$
$$= \begin{pmatrix}-6+5\\-9+1\end{pmatrix} = \begin{pmatrix}-1\\-8\end{pmatrix}$$ Thus ABCD is a parallelogram

2006/15b

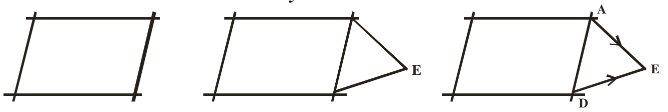
In the pentagon ABCDE, $\overrightarrow{AE} = \begin{pmatrix}2\\0\end{pmatrix}$, $\overrightarrow{DE} = \begin{pmatrix}-1\\2\end{pmatrix}$, $\overrightarrow{DC} = \begin{pmatrix}-2\\-5\end{pmatrix}$ and $\overrightarrow{CB} = \begin{pmatrix}-3\\2\end{pmatrix}$

Show that ABCD is a parallelogram

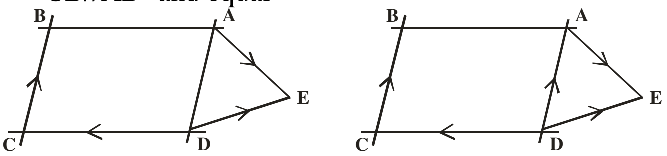
Solution

Let us sketch the diagram progressively as:
Bearing in mind the parallelogram ABCD

Parallelogram *pentagon guided by the vectors arrows*



Since ABCD is a parallelogram
CB//AD and equal



By triangle rule of vectors in ΔADE
Contradiction arrow = sum two flowing/following arrows

$$\overrightarrow{DE} = \overrightarrow{DA} + \overrightarrow{AE} \quad \text{(triangle law)}$$
$$\begin{pmatrix}-1\\2\end{pmatrix} = \overrightarrow{DA} + \begin{pmatrix}2\\0\end{pmatrix}$$
$$\overrightarrow{DA} = \begin{pmatrix}-1\\2\end{pmatrix} - \begin{pmatrix}2\\0\end{pmatrix}$$
$$= \begin{pmatrix}-3\\2\end{pmatrix} \text{ same as } \overrightarrow{CB}$$

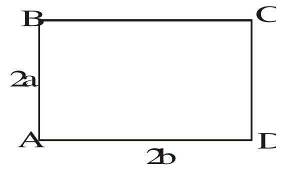
Hence ABCD is a parallelogram

1993/37 (Nov) f/m

ABCD is a rectangle with **AB** = 2a and **AD** = 2b
Find the magnitude and direction of the vector **AB** – **AD**

A. 2a – 2b in the direction **AC**
B. $2\sqrt{(a^2+b^2)}$ in the direction **DB**
C. $2\sqrt{(a^2-b^2)}$ in the direction **BD**
D. $2\sqrt{(a^2+b^2)}$ in the direction **AC**
E. $2\sqrt{(a-b)}$ in the direction **DB**

Solution



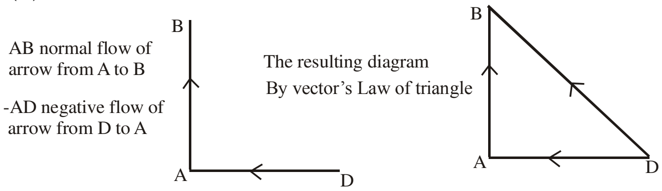
AB – AD = 2a – 2b

Magnitude = $|2a - 2b|$

$= \sqrt{(2a)^2 + (2b)^2}$

$= \sqrt{4a^2 + 4b^2} = \sqrt{4(a^2 + b^2)} = 2\sqrt{(a^2 + b^2)}$

(b) Direction of AB – AD



Thus $2\sqrt{(a^2 + b^2)}$ in the direction of DB (B)

2002/5 Neco f/m

Three vertices A, B and C of a triangle have position vectors a, b and c respectively relative to the origin O, where a = 3i – 2j , b = 6i + j and c = 9i + 4j

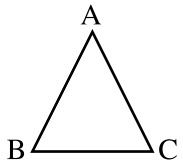
(i) Show that ΔABC is isosceles

(ii) determine the value of cos $B\hat{C}X$ if X is the mid point of $\overrightarrow{AC}$ (give your answer to 3 s.f)

Solution

Isosceles triangle has two equal sides .

Lets sketch as :



The three lengths are :

$\overrightarrow{AB}$ = b – a

= (6i + j) – (3i – 2j)

= 3i + 3j

$\overrightarrow{AC}$ = c – a

= (9i + 4j) – (3i– 2j)

= 6i + 6j

$\overrightarrow{BC}$ = c – b

= (9i + 4j) – (6i + j)

= 3i + 3j

Since $\overrightarrow{AB} = \overrightarrow{BC}$ ΔABC is isosceles

(ii) x = mid point of $\overrightarrow{AC}$

$= \frac{1}{2}$ (6i + 6j)

= 3i + 3j

Next we find x = C – X

= 9i + 4j – (3i + 3j)

= 9i + 4j – 3i – 3j

= 6i + j

Thus cos BCx is the angle b/w $\overrightarrow{BC}$ and $\overrightarrow{\times c}$

$\overrightarrow{BC} \cdot \overrightarrow{\times c} = |\overrightarrow{Bc}|\,|\overrightarrow{\times c}| \cos\theta$

$$\frac{(3i + 3j).(6i + j)}{\left(\sqrt{3^2 + 3^2}\right)\left(\sqrt{6^2 + 1^2}\right)} = \cos\theta$$

$$\frac{18 + 3}{25.81} = \cos\theta$$

$$\frac{21}{25.81} = \cos\theta$$

cosθ = 0.8137

≈ 0.814 to 3 s.f

2005/17 (Nov)

The position vectors of points P, Q , R and S relative to the origin are

$$\begin{pmatrix} -2 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 3 \\ 2 \end{pmatrix} \text{ and } \begin{pmatrix} 1 \\ 4 \end{pmatrix} \text{ respectively.}$$

(a) Show that PQRS is a rectangle

(b) Find the perimeter of the figure

Solution

(a) Let P, Q, R and S be sketched as:

If it is a rectangle then

$\overrightarrow{QP} = \overrightarrow{RS}$ and $\overrightarrow{QR} = \overrightarrow{PS}$

$\overrightarrow{QP}$ = p – q and $\overrightarrow{RS}$ = s – r

$= \begin{pmatrix} -2 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 \\ -1 \end{pmatrix}$ $= \begin{pmatrix} 1 \\ 4 \end{pmatrix} - \begin{pmatrix} 3 \\ 2 \end{pmatrix}$

$= \begin{pmatrix} -2 \\ 2 \end{pmatrix}$ $= \begin{pmatrix} -2 \\ 2 \end{pmatrix}$

$\overrightarrow{QP} = \overrightarrow{RS}$ QED

To confirm, let us try $\overrightarrow{QR} = \overrightarrow{PS}$

$\overrightarrow{QR}$ = r – q and $\overrightarrow{PS}$ = s – p

$= \begin{pmatrix} 3 \\ 2 \end{pmatrix} - \begin{pmatrix} 0 \\ -1 \end{pmatrix}$ $= \begin{pmatrix} 1 \\ 4 \end{pmatrix} - \begin{pmatrix} -2 \\ 1 \end{pmatrix}$

$= \begin{pmatrix} 3 \\ 3 \end{pmatrix}$ $= \begin{pmatrix} 3 \\ 3 \end{pmatrix}$

Verified as $\overrightarrow{QR} = \overrightarrow{PS}$

Thus PQRS is a rectangle

(b) Perimeter of rectangle = 2(L + b)

= 2 (QR + RS)

$= 2\left[\begin{pmatrix} 3 \\ 3 \end{pmatrix} + \begin{pmatrix} -2 \\ 2 \end{pmatrix}\right]$

$= 2\begin{pmatrix} 1 \\ 5 \end{pmatrix}$

Problems on vectors and coordinate geometry

2014/5 f/m

The position vectors of point P, Q and R are 11i + j, $5i + \frac{13}{3}j$ and 2i + 6j respectively

(a) Show that P, Q and R lie on a straight line

(b) Find the ratio $|\overrightarrow{PQ}| : |\overrightarrow{QR}|$

Solution

(a) We are to show that they are collinear i.e they are the same or multiple of each other

$\overrightarrow{PQ} = q - p$

$= 5i + \frac{13}{3}j - (11i + j)$

$= 5i + \frac{13}{3}j - 11i - j$

$= -6i + \frac{10}{3}j$

$\overrightarrow{QR} = r - q$

$= 2i + 6j - \left(5i + \frac{13}{3}j\right)$

$= 2i + 6j - 5i - \frac{13}{3}j$

$= -3i + \frac{5}{3}j$

$\overrightarrow{PR} = r - p$

$= 2i + 6j - (11i + j)$

$= 2i + 6j - 11i - j$

$= -9i + 5j$

rearranged with fraction

$= 3\left(-3i + \frac{5}{3}j\right)$;

$\overrightarrow{PQ}$ is $2\left(-3i + \frac{5}{3}j\right)$ result

(b) $|\overrightarrow{PQ}| : |\overrightarrow{QR}|$

$\left|-6i + \frac{10}{3}j\right| : \left|-3i + \frac{5}{3}j\right|$

$\sqrt{(-6)^2 + \left(\frac{10}{3}\right)^2} : \sqrt{(-3)^2 + \left(\frac{5}{3}\right)^2}$

$\sqrt{36 + \frac{100}{9}} : \sqrt{9 + \frac{25}{9}}$

$\sqrt{\frac{424}{9}} : \sqrt{\frac{106}{9}}$

$\sqrt{\frac{4}{9}(106)} : \sqrt{\frac{1}{9}(106)}$

$\frac{2}{3}\sqrt{106} : \frac{1}{3}\sqrt{106}$

$\frac{2}{3} : \frac{1}{3}$

2 : 1

1998/14a (Nov)

The position vectors of points P, Q and R are:

$\mathbf{p} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ $\mathbf{q} = \begin{pmatrix} 4 \\ -1 \end{pmatrix}$ $\mathbf{r} = \begin{pmatrix} 5 \\ -3 \end{pmatrix}$, respectively

Show that P, Q and R are concurrent

Solution

We are to prove collinearity or parallel properties

i.e $\overrightarrow{PQ} = \overrightarrow{QR}$ or $\overrightarrow{PQ} = t\overrightarrow{QR}$ where t is a scalar

$\overrightarrow{PQ} = \mathbf{q} - \mathbf{p}$

$= \begin{pmatrix} 4 \\ -1 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}$

$\overrightarrow{QR} = \mathbf{r} - \mathbf{q}$

$= \begin{pmatrix} 5 \\ -3 \end{pmatrix} - \begin{pmatrix} 4 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$

Thus $\overrightarrow{PQ} = 2\overrightarrow{QR}$ QED

2003/27 (Nov) ***midpoint, co-ordinate vector***

The position vectors and points P, Q and R are 3i – 5j , –2i + 3j and 4i – 7j respectively.If T is the mid – point of QR, Find $\overrightarrow{PT}$

A - 2i + 3j B 3i – 5j C -2i + 5j D -4i + 6j

Solution

Co-ordinates of mid point $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

Mid point of QR co-ordinates = $\left(\frac{-2+4}{2}, \frac{3-7}{2}\right)$

$= (1, -2)$

i.e T = i – 2j in vectors representation

$\overrightarrow{PT} = T - P$

$= (i - 2j) - (3i - 5j)$

$= i - 3i - 2j + 5j$

$= -2i + 3j$ (A)

2005/8 (Nov)

The position vectors of the points A, B, C, and H are 3i + 2j, 2i + 3j, 3i – 2j, and 8i + 3j respectively. If D is the mid point of BC,

(a)find the position vector of D.

(b)show that $|\overrightarrow{AH}| = 2|\overrightarrow{OD}|$ where O is the origin

Solution

D is mid point of BC i.e b = (2i + 3j) and c = (3i – 2j)

x for D $= \frac{x_2 + x_1}{2}$ y for D $= \frac{y_2 + y_1}{2}$

$= \frac{3+2}{2}$ $= \frac{-2+3}{2}$

$= \frac{5}{2}$ $= \frac{1}{2}$

Thus, position vector D is $\frac{5}{2}i + \frac{1}{2}j$

(b) Show that $|\overrightarrow{AH}| = 2|\overrightarrow{OD}|$

First we find $\overrightarrow{AH}$ and $\overrightarrow{OD}$

$\overrightarrow{AH} = h - a$

$= (8i + 3j) - (3i + 2j)$

$= 5i + j$

and $\overrightarrow{OD} = d - o$

$= \frac{5}{2}i + \frac{1}{2}j - (0i + 0j)$

$= \frac{5}{2}i + \frac{1}{2}j$

Next, $|\overrightarrow{AH}| = \sqrt{5^2 + 1^2} = \sqrt{25+1} = \sqrt{26}$

Also $2|\overrightarrow{OD}| = 2\left(\sqrt{\left(\frac{5}{2}\right)^2 + \left(\frac{1}{2}\right)^2}\right)$

$= 2\left(\sqrt{\frac{25}{4} + \frac{1}{4}}\right)$

$= 2\sqrt{\frac{26}{4}} = \frac{2}{2}\sqrt{26}$ i.e $\sqrt{26}$

Thus, $|\overrightarrow{AH}| = 2|\overrightarrow{OD}| = \sqrt{26}$ QED

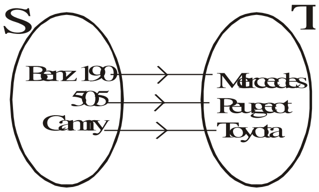
Chapter twenty eight
Mapping

Mapping is simply an association or a relation between two sets.

Let S be the set of cars { Benz 190, 505, Camry} and T the set of car producer { Peugeot, Mercedes Benz, Toyota}, then there is a mapping which associate each car with its producer and we write as

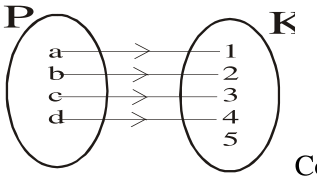
$$f : S \to T \quad \text{or} \quad S \xrightarrow{f} T$$

Also by **mapping diagram** or arrow graph, we can show it as



S = { Benz 190, 505, Camry} is called Domain
T ={ Peugeot, Mercedes Benz, Toyota} is called co domain

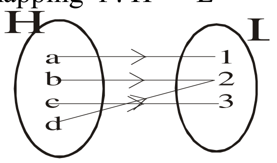
Example MP2 $f : P \to K$ as shown below



Domain Co domain

Note that <u>not all</u> the elements of K are paired or related to P
Domain of the mapping f is {a, b, c, d}
Co domain of the mapping f is{1, 2, 3, 4, 5}
Image or range of the mapping is {**1, 2, 3, 4**}

Example MP2 is a **mapping** and is also a **function**: reason the co domain is a set of real numbers. So you can say that:
A function is a mapping whose co domain is numbers
The mapping $f : H \to L$



is acceptable since every element of the domain H is uniquely paired to an element of the co domain L
Here the co domain is same as Image or range

The relations represented below are not mapping

Fig I

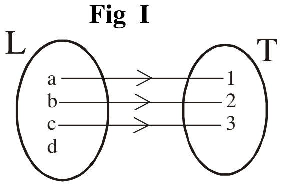
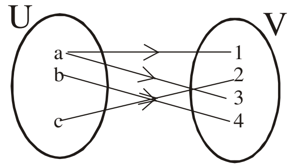


Fig II



Reasons
All the elements in the domain are not paired to the co domain

Reasons
one element **a** in the domain is not uniquely paired i.e paired to more than one element

Example MP5
With each newborn baby associate its weight in grams to the nearest gram. This is a function from the set of newborn babies to the set of natural numbers

Now let us debate it
Will any new baby have weight? *Answer* capital yes
Can any new baby have two or three different weights at the same time? *Answer* capital NO

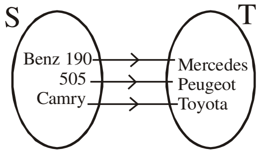
Thus, for a **relation to be a mapping**; it must be that:
- *Every element of the domain has an image in the co domain*
- *The image of every element of the domain is unique*

You will agree that two or three or more new babies can have the same weights

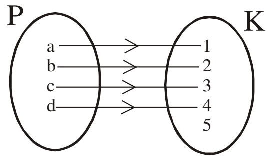
Types of mapping

Injective (one-one)

A mapping $f : S \to T$ is said to be injective or one to one when elements of S are uniquely paired to elements of T or if no element of T corresponds to more than one element of S



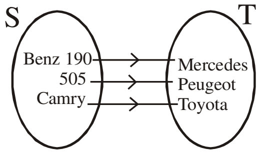
$f : S \to T$ is Injective



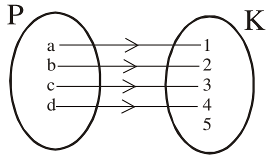
$f : P \to K$ is Injective

Surjective (onto)

A mapping $f : S \to T$ is said to be surjective or onto if every element of T is an image



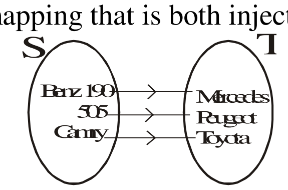
$f : S \to T$ is surjective



$f : P \to K$ is **NOT** surjective

Bijective

A mapping that is both injective and surjective is bijective



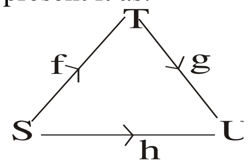
$f : S \to T$ is bijective

composition of mapping

Let $f : S \to T$ and $g : T \to U$ be any function then we can compose then to get a mapping $h : S \to U$, given by

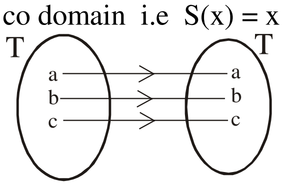
$$h(x) = g(f(x)) \text{ or gof or gf for } x \in S$$

we can represent it as:

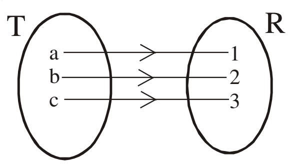


Identity mapping

A mapping $S : T \to T$, where the domain is the same as the co domain i.e $S(x) = x$ for all $x \in T$



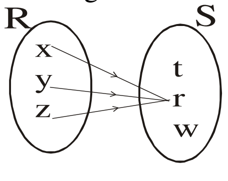
identity mapping
domain = co domain



not identity mapping
domain ≠ co domain

Constant mapping

A mapping T : R→ S, which assigns all the elements of the domain R to a single element in the co domain S

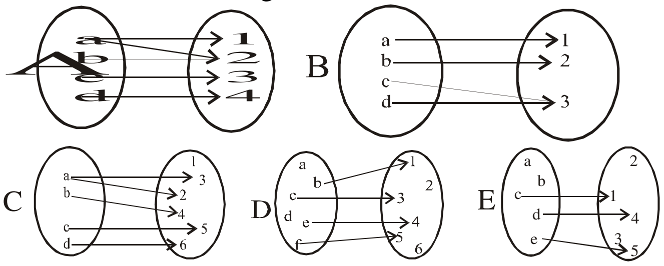


Inverse function

For any function f to have an inverse f^{-1} it must be *bijective i.e the function must be injective and surjective*

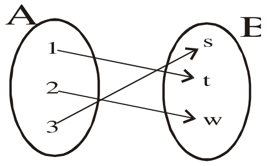
2002/12 Neco Exercise 28.0

Which of the following illustrates a function?



2005/28 Neco Exercise 28.1

Suppose f : A → B be defined in the diagram below.

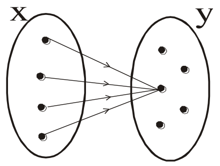


Then f is said to be

A 1 – 1 B. both 1 – 1 and onto C. inverse
D. onto E. orthogonal

2010/3 Neco Exercise 28.2

A mapping f : x → y described in the diagram below refers to what type of mapping?



A Constant B Equality C Identity D One-one
E Onto

Ordered pair mapping

A typical ordered pair mapping is the Euclidean plane which is made up of (x, y); e.g any point on the graph is ordered pair: (3,7) you know that x is 3 and y is 7

Mathematically $X \times Y = \{(x,y) : x \in X, y \in Y\}$

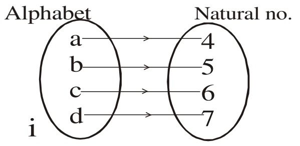
All elements of x make up the domain

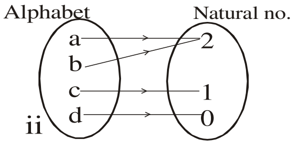
All elements of y make up the co-domain

Example

Identify with reason a mapping from the pairs below:
i{(a,4), (b,5), (c,6), (d,7)} ii.{(a,2), (b,2), (c,1), (d 0)}

Solution. ***Judge by yourself*** **diagrammatically it is:**





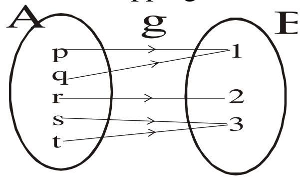
To determine position of a mapping

(Whether it is injective,surjective or both and to locate its domain and co domain)

Here we work based on the given problem; if it has any formula, its application helps to determine whether it is injective, surjective or both and to locate its domain and co domain depending on the question.

2009/26 Neco

Consider the mapping below



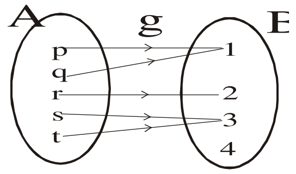
What are the elements in the co-domain of g?

A.{1, 2, 3} B.{g, 1, 2, 3} C.{p, q, g, l}
D.{p, q, r, s, t} E.{p, q, r, s}

Solution

co-domain of g = { 1, 2, 3} i.e set B

Counter illustration



Here co-domain of g = { 1, 2, 3, 4} i.e set B

Here image of g = { **1, 2, 3**} i.e **subset** of B

2003/22 Neco

Find the domain of f : x → $x^2 + 6$,
if the range of f is {6, 7, 10, 15}

A. {0, 1, 2, 3} B. {-1, 0, 1, 2} C. {5, 6, 8, 12}
D. {-1, 0, 1, 2} E. {-2, -1, 0, 1}

Solution

Here we are to reverse the process as

range 6

$x^2 + 6 = 6$, find x

$x^2 = 0$

$x = 0$

range 7

$x^2 + 6 = 7$, find x

$x^2 = 1$

$x = \pm\sqrt{1}$ i.e –1, 1

range 10

$x^2 + 6 = 10$, find x

$x^2 = 4$

$x = \pm\sqrt{4}$ i.e –2, 2

range 15

$x^2 + 6 = 15$

$x^2 = 9$

$x = \pm\sqrt{9}$ i.e –3, 3

The domain set = -3, -2, -1, 0, 1, 2, 3

Thus the domain here = {0, 1, 2, 3}(A) or {-3, -2, -1, 0}

1997/12

Let the function f: w $\to$ R be defined by f : $x^2 - x - 2$, where W = {- 1, 0, 2, 5, 11} and R the set of real numbers. Find the range of f.

A. {- 2, 0, 18, 108}   B. {- 2, 18, 108}
C. {-1, 0, 2, 5, 11}   D. {-2, 0, 18}   E. {0}

Solution

What we are to find here is the image of the function (though identified here as "range of f")

$f(x) = x^2 - x - 2$

Taking the value domain w = {-1, 0, 2, 5, 11} one by one

$f(-1) = (-1)^2 - (-1) - 2$ $= 1 + 1 - 2$ $= 0$	$f(5) = 5^2 - 5 - 2$ $= 25 - 7$ $= 18$
$f(0) = 0^2 - 0 - 2$ $= -2$	$f(11) = 11^2 - 11 - 2$ $= 121 - 13$ $= 108$
$f(2) = 2^2 - 2 - 2$ $= 4 - 4$ $= 0$	

Thus, the required set = {-2, 0, 18, 108} (A)

2004/32

If the domain of f : x $\to$ $x^2 - 2$ is {-2, -1, 0, 1}, find its range.

A{2, 1, 0, -1} B{2, -1, -2} C{0, -1, -2} D{2,1, -2, -1}

Solution

Its "range" here is the result of the values gotten by substituting {-2, -1, 0, 1} into $f(x) = x^2 - 2$

$f(-2) = (-2)^2 - 2$ $= 4 - 2$ i.e 2	$f(0) = 0^2 - 2$ $= 0 - 2$ i.e -2
$f(-1) = (-1)^2 - 2$ $= 1 - 2$ i.e -1	$f(1) = 1^2 - 2$ $= 1 - 2$ i.e -1

The set is = {- 2, - 1, 2} (B)

2007/17 Neco

If f : x $\to$ sec x , find the range ; if the domain is the set $\{x : 0^0 \le x \le 60^0\}$

A. $\{y : 1 \le y \le 2\}$ B. $\{y : \frac{1}{2} \le y \le 1\}$

C. $\{y : 0 \le y \le 1\}$ D. $\{y : -\frac{1}{2} \le y \le \frac{1}{2}\}$

E. $\{y : -1 \le y \le 1\}$

Solution

Here we substitute x value between 0^0 and 60^0

When $x = 0^0$

$$\text{Sec } x = \frac{1}{\text{Cos } x} = \frac{1}{\cos 0^0} = \frac{1}{1} \text{ i.e } 1$$

When $x = 60^0$

$$\text{Sec } x = \frac{1}{\cos 60} = \frac{1}{½} \text{ i.e } 2$$

Introducing y to represent the range and minding our inequalities.

Domain $\{x : 0^0 \le x \le 60^0\}$ becomes range $\{y : 1 \le y \le 2\}$ (A)

2010/1

Find the domain of $f(x) = \dfrac{x}{3-x}$, where $x \in R$, the set of real numbers.

A $\{x : x\in R, x \ne 3\}$ B $\{x : x\in R, x \ne 1\}$
C $\{x : x\in R, x \ne 0\}$ D $\{x : x\in R, x \ne -3\}$

Solution

The domain here will be R – f(x) undefined

i.e R *minus* $3 - x = 0$

i.e R *minus* $x = 3$

i.e R , $x \ne 3$ (A)

Because if x is equal to 3 then f(x) becomes undefined

2010/16 Exercise 28.3

Given that P = {x : x is a factor of 6} is the domain of $g(x) = x^2 + 3x - 5$, find the range of g(x).

A { -1, 5, 13 }   B {5, 13, 49 }   C {1, 2, 3, 6 }
D { -1, 5, 13, 49 }

2002/2 (Nov) Exercise 28.4

Given that the domain of $f(x) = x^2 + 1$ is {-2, -1, 0, 1, 2}, find the range of f.

A.{3, 0, -1} B.{5, 2, 1} C.{3, 1, 5} D.{2, 0, -1}

2009/11 Neco Exercise 28.5

Find the domain of f: x$\to$ x^2+1 if the range is {1, 2, 5, 10}

A. {0, 1, 2, 3}   B. {-3, -2, -1, 0}   C. {-2, -1, 0, 1, 2}
D. {-3, -2, -1, 0, 1, 2, 3}   E. {0, 5, 26, 101}

To determine injectivity and surjectivity

Examples

Which of the following function f defined below is injective, surjective and bijective

(1) f : R $\to$ R, f(x) = 2x+1

(2) f : R $\to$ R, $f(x) = e^x$

(3) f : R $\to$ $^+$R, $f(x) = e^x$

(4) f : R $\to$ R, $f(x) = \dfrac{x^2 + 2}{x^2 + 1}$

Solution

(1) Let a : b$\in$R and suppose that

f(a) = f(b)

Then 2a + 1 = 2b + 1

So 2a = 2b and hence a = b

Thus, f is injective

To show surjectivity

Let y$\in$R be such that

f(x) = y

so 2x + 1 = y

$x = \frac{1}{2}(y - 1) \in R$

Even if we take y = 1 then x = 0$\in$R

Accordingly f is surjective and hence bijective

(2) Let a, b$\in$R such that

f(a) = f(b)

Then, $e^a = e^b$

$Log_e e^a = Log_e e^b \quad || \quad a\,Log_e e = b\,Log_e e$

$\Rightarrow$ a = b Hence f is injective

To show surjectivity

Let $y \in R$ be such that

$f(x) = y$

Then $e^x = y$

Thus $x = Log_e y$

Log of negative numbers do not exist (Range = R)
Hence f is not surjective hence not bijective

(3) However, since we restrict our range to R+ , we conclude here that f is surjective and so it is bijective

(4) Let $a, b \in R$ such that

$f(a) = f(b)$

Then $\dfrac{a^2 + 2}{a^2 + 1} = \dfrac{b^2 + 2}{b^2 + 1}$

You can observe that $a^2 = b^2$ or $a = \pm b$. Accordingly it is possible to have $a \neq b$ when $f(a) = f(b)$.
Thus f is not injective (and so not bijective)

To show surjectivity

Let $y \in R$ be such that

$f(x) = y$

Then , $\dfrac{x^2 + 2}{x^2 + 1} = y$

Solving $x = \pm\sqrt{\dfrac{y-2}{1-y}}$

Observe that $\sqrt{\dfrac{y-2}{1-y}}$ is not defined for all values of y($y = 0$ makes the result becomes $\sqrt{-2}$ which is not real.Thus, f is not surjective

2003/8 Exercise 28.6

Which of the following functions is/are one – to –one?

$f : x \to x^2$, $g : x \to 2x + 1$, $h : x \to \sqrt{x}$

A f only B g only C h only D f and h only

2000/10 theory Exercise 28.7

(a) If $g : x \to \dfrac{2x - 1}{5x + 3}$, for all $x \in R$, the set of real number, find the : (i) domain of g;
(b) Determine whether or not g in 10(a) is onto
(c) find the range of value of x for which $2x^2 + x - 6 < 0$

Value of a function

A function takes another value when the function is redefined; it's redefinition can be real numbers or the multiple of the former variable.

2003/14b

The functions f and g are defined as

$f : x \to 3x - 2$

$g : x \to \dfrac{1}{x}$ ($x \neq 0$)

Evaluate : (i) $f(-2)$ (ii) $g(\frac{1}{2})$

Solve (iii) $f(x) = g(\frac{1}{2})$

(iv) $f(x) = g(x)$

Solution

(i) $f(x) = 3x - 2$

$f(-2) = 3(-2) - 2$
$= -6 - 2$
$= -8$

(ii) $g(x) = \dfrac{1}{x}$

$g(\frac{1}{2}) = 1 \div \dfrac{1}{2}$

$= 1 \times \dfrac{2}{1}$

$= 2$

(iii) $f(x) = g(\frac{1}{2})$

$3x - 2 = 2$
$3x = 4$
$x = {}^4/_3$

(iv) $f(x) = g(x)$

$3x - 2 = \dfrac{1}{x}$

$x(3x - 2) = 1$
$3x^2 - 2x = 1$
$3x^2 - 2x - 1 = 0$

Factorizing

$3x^2 - 3x + x - 1 = 0$
$3x(x - 1) + 1(x - 1) = 0$
$(3x + 1) + (x - 1) = 0$
$3x + 1 = 0$ or $x - 1 = 0$
$3x = -1$ or $x = 1$
$x = -{}^1/_3$ or 1

2012/12 f/m

A polynomial is defined by $f(x + 1) = x^3 + 3x^2 - 4x + 2$, find f(2)

A – 8 B – 2 C 2 D 8

Solution

First, we solve for x in f(x + 1) and f(2)

$x + 1 = 2$
$x = 2 - 1$
$x = 1$

Thus f(2) is obtained by substituting **1** for **x**

$f(2) = 1^3 + 3(1)^2 - 4(1) + 2$
$= 1 + 3 - 4 + 2$
$= 2$ C

1989/16 UME

If $f(x) = 2x^2 - 5x + 3$, find $f(x + 1)$

A. $2x^2 - x$ B. $2x^2 - x + 10$ C. $4x^2 + 3x + 2$ D. $4x^2 + 3x + 12$

Solution

The question says substitute $x + 1$ for x

i.e. $f(x+1) = 2(x+1)^2 - 5(x + 1) + 3$

$= 2(x^2 + 2x + 1) - 5x - 5 + 3$

$= 2x^2 + 4x + 2 - 5x - 5 + 3$

Collect like terms

$= 2x^2 + 4x - 5x - 5 + 3 + 2$

$= 2x^2 - x$ (A)

1988/16 UME

If $g(y) = \frac{y-3}{11} + \frac{11}{y^2 - 9}$ what is $g(y + 3)$?

A. $\frac{y}{11} + \frac{11}{y(y+6)}$ B. $\frac{y}{11} + \frac{11}{y(y+3)}$

C. $\frac{y+30}{11} + \frac{11}{y(y+3)}$ D. $\frac{y+3}{11} + \frac{11}{y(y-6)}$

Solution

The question means substituting $y + 3$ for y

$g(y+3) = \frac{(y+3)-3}{11} + \frac{11}{(y+3)^2 - 9}$

$= \frac{y+3-3}{11} + \frac{11}{(y+3)^2 - 3^2}$

To enable us solve by difference of two squares

$= \frac{y}{11} + \frac{11}{(y+3-3)(y+3+3)}$

$= \frac{y}{11} + \frac{11}{y(y+6)}$ (A)

2014/5 f/m

If $f(x) = 3x^3 + 8x^2 + 6x + k$, and $f(2) = 1$, find the value of k

A – 67 B – 61 C 61 D 67

Solution

$f(x) = 3x^3 + 8x^2 + 6x + k$

$f(2) = 1$

$1 = 3(2)^3 + 8(2)^2 + 6(2) + k$

$1 = 24 + 32 + 12 + k$

$1 = 68 + k$

$-67 = k$ (A)

2013/20

The function $f : R \to R$ is defined by

$$f(x) = \begin{cases} 3x + 2 & : x > 4 \\ 3x - 2 & : x = 4 \\ 5x - 3 & : x < 4 \end{cases}$$

Find $f(4) - f(-3)$

A 28 B –26 C –26 D –28

Solution

f(x) is defined for three conditions here

$x > 4$ i.e 5, 6, 7… $+\infty$

$x = 4$

$x < 4$ i.e 3, 2, 1… $-\infty$

For f(4), only $3x - 2$ fits into it

For f(–3), only $5x - 3$ fits into it

Thus, $f(4) - f(-3) = [3(4) - 2] - [5(-3) - 3]$

$= (12 - 2) - (-15 - 3)$

$= 10 + 18$

$= 28$ (A)

2006/15a

Under the mapping $h(x) = px^2 - qx + 2$, the image of 3 is 14 and the image of –2 is 24. Find:

(i) the values of p and q

(ii) the elements whose image is 4

Solution

(i) $h(x) = px^2 - qx + 2$

When $x = 3$, $h(x) = 14$

$14 = 3^2p - 3q + 2$

$14 = 9p - 3q + 2$

$12 = 9p - 3q$ --------(1)

When $x = -2$, $h(x) = 24$

$24 = (-2)^2p - (-2)q + 2$

$24 = 4p + 2q + 2$

$22 = 4p + 2q$ --------(2)

From (1) divide through by 3

From (2) divide through by2

$4 = 3p - q$

$+ \quad 11 = 2p + q$

$15 = 5p$

$3 = p$

Substitute p value into (1)

$4 = 3p - q$ becomes

$4 = 3(3) - q$

$4 = 9 - q$

$q = 9 - 4 = 5$

(ii) $h(x) = 4$, find x

First we substitute for p and q

$h(x) = px^2 - qx + 2$ *becomes* $h(x) = 3x^2 - 5x + 2$

Thus $4 = 3x^2 - 5x + 2$

$3x^2 - 5x - 2 = 0$

Factorizing $(3x + 1)(x - 2) = 0$

$x = -\frac{1}{3}$ or 2

2007/15a Exercise 28.8

Given that $f(x) = 2x - 1$ and $g(x) = x^2 + 1$

(i) find $f(1 + x)$;

(ii) find the range of value of x for which $f(x) < -3$

(iii) simplify $f(x) - g(x)$

2008/14a Exercise 28.9

The function f is defined as $f : x \to 3x^2 - 5x$

(i) Evaluate $f(-2)$

(ii) Find the values of x for which $f(x) = \frac{-7}{4}$

2009/14c Exercise 28.10

The function f and g are defined as

$f : x \to x - 2$ and $g : x \to 2x^2 - 1$

Solve (i) $f(x) = g\left(-\frac{1}{2}\right)$ (ii) $f(x) + g(x) = 0$

2004/15c (Nov) Exercise 28.11

The functions f and g are given as

$f(x) = x + \frac{3}{x}$ and $g(x) = 2x + 1$

Evaluate : (i) $g(-2)$ (ii) $f(-\frac{1}{2})$

2004/14b Exercise 28.12

Given $f : x \to 3x - 1$ and $g : x \to x^2 + 1$

(i) Evaluate $g(-2) - f(-1)$

(ii) Simplify $f(x) + g(x)$

(iii) Solve $f(x) = g(x)$

Composition of function

If $f(x) = x^2$ and $g(x) = 2x - 1$, find **(i)** $gf(x)$ **(ii)** $fg(x)$

Solution

$gf(x) = g(f(x))$
$= g(x^2)$
$= 2x^2 - 1$

But $fg(x) = f(g(x))$
$= f(2x - 1)$
$= (2x - 1)^2$
$= 4x^2 - 4x + 1$

2015/6 f/m

Given that $f: x \to x^2$ and $g: x \to x + 3$ where $x \in R$ find $fog(2)$

A 25 B 9 C 7 D 5

Solution

$fog(x) = f(x + 3)$
$= (x + 3)^2$
$fog(x) = x^2 + 6x + 9$
$fog(2) = 2^2 + 6(2) + 9$
$= 4 + 12 + 9 = 25$

Alternatively

$fog(2)$
$g(2) = 2 + 3$
$= 5$
$fog(2) = f(5)$
$= 5^2 = 25$

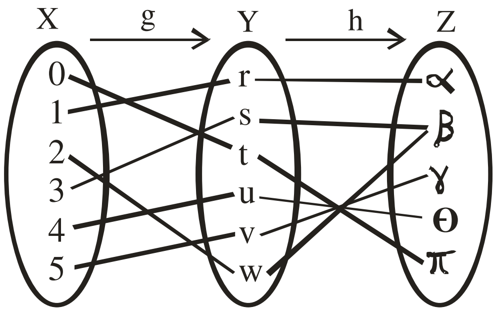
2014/11a f/m

If $f(x) = \frac{x-3}{2x-1}$, $x \neq \frac{1}{2}$ and $g(x) = \frac{x-1}{x+1}$, $x \neq -1$, find gof

Solution

$$gof = g\left(\frac{x-3}{2x-1}\right)$$
$$= \frac{\frac{x-3}{2x-1} - 1}{\frac{x-3}{2x-1} + 1}$$
$$= \frac{x-3-1(2x-1)}{2x-1} \div \frac{x-3+1(2x-1)}{2x-1}$$
$$= \frac{x-3-2x+1}{2x-1} \div \frac{x-3+2x-1}{2x-1}$$
$$= \frac{-x-2}{2x-1} \div \frac{3x-4}{2x-1}$$
$$= \frac{-x-2}{2x-1} \times \frac{2x-1}{3x-4} = \frac{-x-2}{3x-4}$$

2012/25 and 26 f/m



In the diagram, $g : x \to y$ and $h : y \to z$

Use the diagram to answer questions 25 and 26

25. Find $h[g(3)]$

A S B β C $\{s, \beta\}$ D $\{s, w, \beta\}$

Solution

$g(3) = s$

Thus, $h[g(3)] = h(s)$
$= \beta$ B.

26. goh is

A one to one B one to one onto
C a relation D a series

Solution

2012/1 f/m

Two functions g and h are defined on the set R of real numbers by $g : x \to x^2 - 2$ and $h : x \to \frac{1}{x+2}$, $x \neq -2$, find : (a) h^{-1} the inverse of h : (b) goh when $x = -\frac{1}{2}$

Solution

(a) $h : x \to \frac{1}{x+2}$

Let $y = \frac{1}{x+2}$
$y(x + 2) = 1$
$yx + 2y = 1$
$yx = 1 - 2y$
$x = \frac{1-2y}{y}$

Change y to x for proper identification

$h^{-1}(x) = \frac{1-2x}{x}$, $x \neq 0$

(b) $goh\left(-\frac{1}{2}\right) = g\left(\frac{1}{-\frac{1}{2}+2}\right)$ i.e $1 \div \frac{3}{2} = 1 \times \frac{2}{3}$

$= g\left(\frac{2}{3}\right)$

$= \left(\frac{2}{3}\right)^2 - 2 = \frac{4}{9} - 2 = -\frac{14}{9}$

2013/5 f/m Exercise 28.13

If $f(x) = x^2$ and $g(x) = \sin x$, find gof

A $\sin^2 x$ B $\sin x^2$ C $(\sin x)\, x^2$ D $x \sin x$

2006/6 f/m Exercise 28.14

Two functions f and g are defined by $f : x \to 3x - 1$ and $g : x \to 2x^3$, evaluate fg (-2)

A - 49 B - 47 C -10 D - 9

2009/29 (Nov) f/m Exercise 28.15

The functions f and g are defined on the set R, real numbers by $f : x \to 2x^2 - 3$ and $g : x \to 4 - x$.

Find fog.

A $x^2 - 16x + 29$ B $x^2 + 16x - 29$
C $2x^2 - 16x - 29$ D $2x^2 + 16x - 29$

Inverse of a function

If y = f (x) is a function, we obtain the inverse function f^{-1} by solving the equation y = f (x) for x in terms of y then later change y to x

2015/9a f/m

The function $f : x \to x^2 + 1$ and $g : x \to 5 - 3x$ are defined on the set of real numbers, R

(i) state the domain of f^{-1}, the inverse of f

(ii) find $g^{-1}(2)$

Solution

$f : x \to x^2 + 1$

First, we find f^{-1}

Let $y = x^2 + 1$

Next, we make x the subject formula

$y - 1 = x^2$

$\sqrt{y-1} = x$

Change y to x for proper identification

$f^{-1}(x) = \pm\sqrt{x-1}$

The domain here will be R – $f^{-1}(x)$ not real number

$f^{-1}(0) = \pm\sqrt{-1}$ is not real

$f^{-1}(-1) = \pm\sqrt{-1-1} = \pm\sqrt{-2}$ is not real

In fact all $x = R^-$ is not real

Thus domain is $\{x : x \in R^+, x \neq 0, x \neq R^-\}$

i.e $\{x : x \in R^+\}$

(ii) $g^{-1}(2)$

First, we find g^{-1}

$g : x \to 5 - 3x$

Let $y = 5 - 3x$

$3x = 5 - y$

$x = \frac{5-y}{3}$

Change y to x for proper identification

$g^{-1}(x) = \frac{5-x}{3}$

Thus, $g^{-1}(2) = \frac{5-2}{3}$

$= \frac{3}{3} = 1$

2013/35 f/m

If $g(x) = \frac{x+1}{x+2}$, $(x \neq -2)$, find $g^{-1}(2)$

A 3 B 2 C – 2 D – 3

Solution

$g(x) = \frac{x+1}{x+2}$

Let $y = \frac{x+1}{x+2}$

$y(x + 2) = x + 1$

$yx + 2y = x + 1$

$yx - x = 1 - 2y$

$x(y - 1) = 1 - 2y$

$x = \frac{1-2y}{y-1}$

Change y to x for proper identification

$g^{-1}(x) = \frac{1-2x}{x-1}$, $x \neq 1$

$g^{-1}(2) = \frac{1-2(2)}{2\ -\ 1}$

$= \frac{1-4}{1} = -3$ D.

2014/15 and 16 f/m

A function is defined by $h : x \to 2 - \frac{1}{2x-3}$, $x \neq \frac{3}{2}$.

Use the information to answer question 15 and 16

15. Find $h^{-1}(x)$, the inverse of h.

A $h^{-1}(x) = \frac{3x-4}{2x-7}$, $x \neq \frac{7}{2}$ B $h^{-1}(x) = \frac{3x-7}{2x-4}$, $x \neq 2$

C $h^{-1}(x) = \frac{2x-7}{4x-3}$, $x \neq \frac{3}{4}$ D $h^{-1}(x) = \frac{4x-7}{2x-4}$, $x \neq 2$

Solution

$h : x \to 2 - \frac{1}{2x-3}$

Let $y = 2 - \frac{1}{2x-3}$

$\frac{1}{2x-3} = 2 - y$

$1 = (2 - y)(2x - 3)$

$1 = 2(2x - 3) - y(2x - 3)$

$1 = 4x - 6 - 2xy + 3y$

$1 + 6 - 3y = 4x - 2xy$

$7 - 3y = x(4 - 2y)$

$\frac{7-3y}{4-2y} = x$

Change y to x for proper identification

$h^{-1}(x) = \frac{7-3x}{4-2x}$ or $\frac{3x-7}{2x-4}$, $x \neq 2$ B.

16. Find $h^{-1}\left(\frac{1}{2}\right)$

A 6 B $^{11}/_6$ C $^{11}/_4$ D $^5/_3$

Solution

$h^{-1}(x) = \frac{3x-7}{2x-4}$

$h^{-1}\left(\frac{1}{2}\right) = \frac{3\left(\frac{1}{2}\right)-7}{2\left(\frac{1}{2}\right)-4}$

$= \frac{3}{2} - 7 \div \frac{2}{2} - 4$

$= -\frac{11}{2} \div -3$

$= -\frac{11}{2} \times -\frac{1}{3}$

$= \frac{11}{6}$

2015/10 f/m

Given that f: $x \to \frac{2x-1}{x+2}$, $x \neq 2$,

find f^{-1}, the inverse of f

A $f^{-1} : x \to \frac{1+2x}{2-x}$, $x \neq 2$ B $f^{-1} : x \to \frac{1-2x}{x+2}$, $x \neq -2$

C $f^{-1}: x \to \frac{1-2x}{x-2}$, $x \neq 2$ D $f^{-1}: x \to \frac{1+2x}{x+2}$, $x \neq -2$

Solution

$f: x \to \frac{2x-1}{x+2}$, $x \neq 2$

Let $y = \frac{2x-1}{x+2}$

$y(x + 2) = 2x - 1$

$xy + 2y = 2x - 1$

$2y + 1 = 2x - xy$

$2y + 1 = x(2 - y)$

$\frac{2y+1}{2-y} = x$

Change y to x for proper identification

$f^{-1}: x \to \frac{2x+1}{2-x}$, $x \neq 2$ A.

2006/9 f/m Exercise 28.16

If $f(x) = \frac{1}{2-x}$, $x \neq 2$, find $f^{-1}(-\frac{1}{2})$.

A 4   B 0   C -2   D -4

2009/8 (Nov) f/m Exercise 28.17

A function f is defined by f: $x \to 3x - \frac{2}{5}$,

find the inverse of f.

A $\frac{1}{15}(3x + 2)$ B $\frac{1}{15}(3x + 6)$

C $\frac{1}{15}(5x + 2)$ D $\frac{1}{15}(5x + 6)$

2003/35 f/m Exercise 28.18

If $f: x \to \frac{4x-1}{3}$, where $x \in R$, find $f^{-1}(-2)$.

A -3 B $\frac{-5}{4}$ C $\frac{-4}{5}$ D $\frac{-1}{3}$

2010/11b Neco f/m Exercise 28.19

Find the inverse of the function $f(x) = \frac{5x + 11}{2x - 4}$

2000/9 f/m Exercise 28.20

Given that $f: x \to 7 - x$, $x \in R$ (the set of numbers), find f^{-1}

A. $x + 7$ B. $\frac{x}{7}$ C. $-\frac{7}{x}$ D. $7 - x$

2002/25 f/m Exercise 28.21

Given that f; $x \to \frac{1}{2x+1}$, $x \neq -\frac{1}{2}$, find $f^{-1}(x)$

A. $2x + 1$ B. $\frac{1-x}{2x}$, $x \neq 0$

C. $\frac{2x}{1-x}$, $x \neq 1$ D. $2x - 1$

2002/20 f/m Exercise 28.22

If $f(x) = \frac{x-4}{2x-3}$, $(x \neq 4, \neq \frac{3}{2})$, for all $x \in R$, find the value of x for which f^{-1} does not exist.

A. $\frac{1}{2}$ B. $\frac{3}{2}$ C. 2   D. 4

2004/16 Neco (Dec) f/m Exercise 28.23

If $f: x \to 3 - 2x$, find f^{-1}

A. $\frac{x-3}{2}$ B. $x - 1$ C. $3 - 3x$

E. $3x - 3$ D. $2x - 3$

2000/10 theory f/m Exercise 28.24

(a) If $g: x \to \frac{2x-1}{5x+3}$, for all $x \in R$, the set of real number, find the :

(ii) g^{-1} (iii) value of x for which g^{-1} does not exist

2005/33 Neco f/m Exercise 28.25

Let $g: x \to x^3 + 5$ be a function on the set of real number, find the inverse of f(x)

A. $\sqrt{x-5}$ B. $\sqrt[3]{x+5}$ C. $\sqrt[4]{x^2} - 5$

D. $\sqrt[3]{x-5}$ E. $\sqrt{x^2+5}$

Joint cases

1992/10 (Nov) f/m

Two functions f(x) and g(x) are defined on the set of real numbers by $f(x) = 3x^2 - 2$ and $g(x) = x + 3$.

Find: (a) $f(-2)$;   (b) $g^{-1}\left(-\frac{3}{4}\right)$

(c) the value of x for which $f(g(x)) = g(f(x))$

(d) $f^{-1}(g(4))$

Solution

(a) To find $f(-2)$

$f(x) = 3x^2 - 2$ becomes

$f(-2) = 3(-2)^2 - 2$

$= 12 - 2$ i.e. 10

(b) $g^{-1}\left(-\frac{3}{4}\right)$ First, we find g^{-1}

Let $y = x + 3$

Next, we make x subject of formula

$y - 3 = x$

Changing y to x for proper identification

$\Rightarrow g^{-1}(x) = x - 3$

Thus $g^{-1}\left(-\frac{3}{4}\right) = -\frac{3}{4} - 3$

$= -\frac{15}{4}$

(c) $f(g(x)) = g(f(x))$

$f(x + 3) = g(3x^2 - 2)$

$3(x + 3)^2 - 2 = 3x^2 - 2 + 3$

$3(x^2 + 6x + 9) - 2 = 3x^2 + 1$

$3x^2 + 18x + 27 - 2 = 3x^2 + 1$

$24 = -18x$

$x = -\frac{24}{18} = -\frac{4}{3}$

(d) $f^{-1}(g(4))$

First, we find g(4) from $g(x) = x + 3$.

$g(4) = 4 + 3$ i.e 7

Next, $f^{-1}(g(4)) = f^{-1}(7)$, But we find f^{-1} of f(x)

Let $y = 3x^2 - 2$

Next, we make x the subject of formula

$y + 2 = 3x^2$

$\frac{y+2}{3} = x^2$

$\sqrt{\frac{y+2}{3}} = x$

Changing y to x for proper identification

$\Rightarrow f^{-1}(x) = \sqrt{\frac{x+2}{3}}$

Thus ,$f^{-1}(7) = \sqrt{\frac{7+2}{3}} = \pm\sqrt{3}$

1998/4 (Nov) f/m

(a) If $f(x) = \frac{x-1}{x-2}$, $x \neq 1$ and $x \neq 2$, find $f^{-1}(x)$

Solution

Let $y = \frac{x-1}{x-2}$

Next, we make x the subject formula

$y(x-2) = x - 1$

$xy - 2y = x - 1$

Collect terms in x together

$xy - x = 2y - 1$

$x(y-1) = 2y - 1$

$x = \frac{2y-1}{y-1}$

Thus $f^{-1}(x) = \frac{2x-1}{x-1}$

(b) If $f : x \to 2x - 5$ and $g : x \to \sin x$, where x is a real number , find :

(i) (fog)(x) (ii) (gof)$\left(\frac{\pi}{2}\right)$

Solution

(i) (fog) = f(sin x)

= 2(sin x) – 5

= 2 sin x – 5

(ii) (gof) $\left(\frac{\pi}{2}\right)$

First (gof) (x) = g(2x – 5)

= sin(2x – 5)

Thus, (gof) $\frac{\pi}{2} = \sin\left(2 \times \frac{\pi}{2} - 5\right)$

= sin 175

= 0.08716

2002/14 Neco f/m

If $f : x \to x^2 + 2$ and $g : x \to 2x + 2$, find $f^{-1}g(2)$

A. $^1/_2$ B. 2 C. 4 D. 36 E. 38

Solution

To find $f^{-1}g(2)$ first, we resolve f^{-1}

$f(x) = x^2 + 2$

Let $y = x^2 + 2$

Next, we make x subject formula

$y - 2 = x^2$

$\Rightarrow x = \sqrt{y-2}$

$f^{-1}(x) = \sqrt{x-2}$

Next, we find $f^{-1}g(x)$

$= f^{-1}(2x + 2)$

$= \sqrt{2x+2-2}$

$= \sqrt{2x}$

Thus $f^{-1}g(2) = \sqrt{2(2)} = \sqrt{4}$ i.e 2 (B)

2002/15 Neco f/m

If $f : x - 3x$, and $g : x \to 2\sin x$, the formula for gf^{-1} is

A. $\frac{2}{3}\sin x$ B. $2\sin\frac{x}{3}$ C. 2sin 3x D. $3\sin 2\frac{x}{3}$

Solution

To find gf^{-1}, first, we find f^{-1}

f (x) = 3x

Let y = 3x

$\frac{y}{3} = x$

$f^{-1}(x) = \frac{x}{3}$

Thus, $gf^{-1} = g\left(\frac{x}{3}\right)$

$= 2\sin\frac{x}{3}$ (B)

2007/ 26 Neco f/m

If $f : x \to \frac{2}{x}$, $g : x \to x + 1$ and $h : x \to \frac{x}{2}$,

find $f^{-1}ogoh^{-1}(1)$

A. $\frac{1}{3}$ B. $\frac{2}{3}$ C. $\frac{4}{3}$ D. $\frac{3}{2}$ E. 2

Solution

First , we find f^{-1}

$f(x) = \frac{2}{x}$

let $y = \frac{2}{x}$

$x = \frac{2}{y}$

$f^{-1}(x) = \frac{2}{x}$

and h^{-1}

$h(x) = \frac{x}{2}$

let $y = \frac{x}{2}$

$x = 2y$

$h^{-1}(x) = 2x$

First, we find $f^{-1}ogoh^{-1}(x) = f^{-1}og(2x)$

$= f^{-1}(2x + 1)$

$= \frac{2}{2x+1}$

Thus, $f^{-1}ogoh^{-1}(1) = \frac{2}{2(1)+1} = \frac{2}{3}$ (B)

2004/9 Neco f/m Exercise 28.26

Given $f(x) = \frac{1}{x^2 - 1}$, and $g(x) = \frac{x}{x+2}$ where $x \in R$,

find the : (i) inverse of g(x) ;

(ii) expression $g^{-1}f(x)$

(iii) largest domain of f(x) and g(x) respectively

2001/1 Neco f/m Exercise 28.27

(a) State the condition necessary for a function to have inverse.

(b) If $f : x \to \frac{4x-3}{2x+1}$, and $g : x \to 1 + \frac{4}{2x+1}$, find

i. $f^{-1}(x)$ ii. $g^{-1}(x)$ iii. $f^{-1}og^{-1}(0)$ iv. $g^{-1}of^{-1}(0)$

Chapter twenty nine

Remainder, factor theorems and partial fractions

Remainder and factor theorems

These theorems take their roots from long division with numbers.

Now, let us study some long division cases

Evaluate (i) $13 \div 2$ (ii) $12 \div 2$

Solution

(i)

$$\begin{array}{r|l} & 6 \\ \hline 2 & 13 \\ & \underline{12} \\ & 1 \end{array}$$

When 2 divide 13, we get 6 remainder 1

$13 = 2 \times 6 + 1$

divided = divisor × quotient + remainder

(ii)

$$\begin{array}{r|l} & 6 \\ \hline 2 & 12 \\ & \underline{12} \\ & 00 \end{array}$$

When 2 divides 12, we get 6 there is no remainder

Thus 2 is a factor of 12.

In both cases, 2 is their divisor and 6 is their quotient. while 13 and 12 are divided for i and ii respectively.

Remainder theorem becomes factor theorem when the divisor divides the divided without a remainder.

Remainder theorem states that when we divide a polynomial f(x) by another polynomial $ax - b$ and there is a remainder r, then the value of the polynomial f(x) for the value x,

Where $ax - b = 0$

$ax = b$

$x = \frac{b}{a}$

i.e. $f\left(\frac{b}{a}\right) = r$,

Also if $f(c) = r$

Then r is the remainder when f(x) is divided by $x - c$

Meanwhile

if $f\left(\frac{b}{a}\right) = 0$

Then $ax - b$ is a factor of the polynomial f(x)

Also if $f(c) = 0$

Then $x - c$ is a factor of f(x), this is the factor theorem.

2015/9 f/m

Which of the following is a factor of the polynomial

$6x^4 + 2x^3 + 15x + 5$?

A $3x + 1$ B $x + 1$ C $2x + 1$ D $x + 2$

Solution

For ease of working, let us try $x + 1$ and $x + 2$

If $(x + 1)$ is a factor of the polynomial

$x + 1 = 0$

$x = -1$

Put $x = -1$ into the given equation

$6(-1)^4 + 2(-1)^3 + 15(-1) + 5 = 6 - 2 - 15 + 5$

$= -6$

Since the result is not zero, $x + 1$ is not a factor

If $x + 2$ is a factor of the polynomial

$x + 2 = 0$

$x = -2$

Put $x = -2$ into the given equation

$6(-2)^4 + 2(-2)^3 + 15(-2) + 5 =$

$96 - 16 - 30 + 5 = 55$

Since the result is not zero, $x + 2$ is not a factor

If $3x + 1$ is a factor of the polynomial

$3x + 1 = 0$

$3x = -1$

$x = -{}^1/_3$

Put $x = -{}^1/_3$ into the given equation

$6(-{}^1/_3)^4 + 2(-{}^1/_3)^3 + 15(-{}^1/_3) + 5 = \frac{2}{27} - \frac{2}{27} - 5 + 5$

$= 0$ (A)

2014/9 f/m

If $(x + 2)$ and $(3x - 1)$ are factor of $6x^3 + x^2 - 19x + 6$, find the third factor

A $2x - 3$ B $3x + 1$ C $x - 2$ D $3x + 2$

Solution

If $(x + 2)$ is a factor: then

Performing the long division

$$\begin{array}{r|l} & 6x^2 - 11x + 3 \\ \hline (x+2) & 6x^3 + x^2 - 19x + 6 \\ & \underline{-(6x^3 + 12x^2)} \\ & \quad -11x^2 - 19x \\ & \quad \underline{-(-11x^2 - 22x)} \\ & \qquad\qquad 3x + 6 \\ & \qquad\qquad \underline{-(3x + 6)} \\ & \qquad\qquad\; 0 \quad 0 \end{array}$$

To get third factor;

We factorize $6x^2 - 11x + 3 = 6x^2 - 9x - 2x + 3$

$= 3x(2x - 3) - 1(2x - 3)$

$= (3x - 1)(2x - 3)$

$(2x - 3)$ is the third factor A.

2013/9 f/m

Given that $f(x) = 2x^3 - 3x^2 - 11x + 6$ and $f(3) = 0$, factorize f(x)

A $(x - 3)(x - 2)(2x + 2)$ B $(x + 3)(x - 2)(x - 1)$

C $(x - 3)(x + 2)(2x - 1)$ D $(x + 3)(x - 2)(2x - 1)$

Solution

If $f(3) = 0$

Then $x = 3$ and

$x - 3 = 0$

$(x - 3)$ is a factor of f(x) here

To get the other factors, we perform the long division

$$\begin{array}{r|l} & 2x^2 + 3x - 2 \\ \hline (x-3) & 2x^3 - 3x^2 - 11x + 6 \\ & \underline{-(2x^3 - 6x^2)} \\ & \quad 3x^2 - 11x \\ & \quad \underline{-(3x^2 - 9x)} \\ & \qquad\quad -2x + 6 \\ & \qquad\quad \underline{-(-2x + 6)} \\ & \qquad\qquad 0 \quad 0 \end{array}$$

To get the other factors,

we factorize $2x^2 + 3x - 2 = 2x^2 + 4x - x - 2$

$= 2x(x + 2) - 1(x + 2)$

$= (2x - 1)(x + 2)$

Thus f(x) factorize to $(x - 3)(x + 2)(2x - 1)$

2012/10 f/m

If (x + 1) is a factor of the polynomial
$x^3 + px^2 + x + 6 = 0$, find the value of P

A – 8 B – 4 C 4 D 8

Solution

If (x + 1) is a factor, then

$x + 1 = 0$

$x = -1$

Put x = – 1 into the given equation

$(-1)^3 + p(-1)^2 + (-1) + 6 = 0$

$-1 + p - 1 + 6 = 0$

$p + 4 = 0$

$p = -4$ (B)

Remainder cases

2004/35

Find the remainder when $3x^3+5x^2-11x + 4$ is divided by x +3.

A. – 4 B.4 C.1 D. – 1

Solution

Since x + 3 divides the expression

Then f(–3) = remainder

Thus; $3(-3)^3 + 5(-3)^2 - 11(-3) + 4$

$= 3(-27) + 5(9) + 33 + 4$

$= -81 + 45 + 33 + 4$

$= -81 + 82$

$= 1$ (C)

2003/ 26 PCE

Find the remainder when the polynomial
$f(x) = 3x^3 + x^2 - 8x + 4$ is divided by x + 2

A.0 B. – 2 C.2 D.1

Solution

Since x + 2 divides the given expression

Then f(–2) = remainder

$= 3(-2)^3 + (-2)^2 - 8(-2) + 4$

$= 3(-8) + 4 + 16 + 4$

$= -24 + 24$

= 0 Thus the remainder is 0 (A)

1995/14 PCE special case

Find the remainder when $6x^3 - 5x + 4$ is divided by 2x – 1

A.2 ¼ B.2 ½ C.2 ¾ D.3

Solution

Since 2x – 1 divides the expression

Then f(½) = Remainder

$6(\frac{1}{2})^3 - 5(\frac{1}{2}) + 4 = 6(\frac{1}{8}) - \frac{5}{2} + 4$

$= \frac{3}{4} - \frac{5}{2} + 4$

$= \frac{3 - 10 + 16}{4}$

$= \frac{19 - 10}{4}$

$= \frac{9}{4}$ i.e. $2\frac{1}{4}$ (A)

from 2x – 1 = 0

2x = 1

i.e x = ½

2006/29

Find the value of k if the expression $kx^3 + x^2 - 5x - 2$ leaves a remainder 2 when it is divided by 2x + 1.

A.8 B. – 10 C.10 D. – 8

Solution

If 2x + 1 divides the function leaving a remainder 2

then 2x = –1

x = – ½

Then f(– ½) = Remainder

f(– ½) = 2

i.e. $k(-\frac{1}{2})^3 + (-\frac{1}{2})^2 - 5(-\frac{1}{2}) - 2 = 2$

$-\frac{k}{8} + \frac{1}{4} + \frac{5}{2} - 2 = 2$

Multiply through the LCM 8

$-k + 2 + 20 - 16 = 16$

$-k + 22 = 32$

$-k = 10$ i.e $k = -10$ (B)

1994/14 PCE

When a polynomial is divided by c + 2, the quotient is 3c – 3 and the remainder is 5. find the polynomial.

A.$3c^2 +3c -6$ B.$3c^2 +3c -1$ C.$3c^2 +3c +5$ D.$3c^2 -3c +1$

Solution

Recall that

Divided = divisor × quotient + remainder

Polynomial a(x) = divided by b(x) × quotient q(x) + remainder

Polynomial = (c + 2) × (3c – 3) + 5

Opening the brackets

$= 3c^2 - 3c + 6c - 6 + 5$

$= 3c^2 + 3c - 1$ (B)

2007/14 PCE Exercise 29.1

Find the quotient when $x^3 + x^2 - x - 1$ is divided by x – 1.

A $x^2-2x + 1$ B. $x^2 - 2x -1$ C.$x^2 + 2x +1$ D.$x^2 + 2x - 1$

1993/18 PCE Exercise 29.2

What is the remainder when $x^3 + 5x^2 - 3x - 6$ is divided by x + 1 ?

A.6 B.1 C. – 1 D. – 11

2005/ 48 PCE Exercise 29.3

What is the remainder when $5x^3 - 13x + 4$ is divided by x + 2 ?

A. – 4 B.14 C. – 10 D. 4

Partial fractions

Partial fraction is a form of algebraic fractions that follows certain rules in resolving it instead of the usual LCM method of solving fractional problems. The given fraction, which is single; is resolved into (parts) partial fractions depending on the following:

I. To every linear factor of the form $\frac{1}{x-a}$ in the denominator; there corresponds a partial fraction of the form $\frac{A}{x-a}$

II. To every repeated linear factor of the form $(x-a)^r$ in the denominator, there corresponds partial fractions of the form

$$\frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \ldots + \frac{A_r}{(x-a)^r}$$

III. To every quadratic factor of the form $ax^2 + bx + c$ in the denominator, there corresponds a partial fraction of the form $\frac{Ax+B}{ax^2+bx+c}$ if it cannot be factorized. But if it can be factorized we treat as in **I** above; as for repeated quadratic factors, treat as in **II** above

IV. If the degree of the numerator is greater than or equal to that of the dominator, we carry out a division until the degree of the numerator is strictly less than that of the denominator.

In order to determine the constants we introduced in the above rules, we may **substitute** values for x (which will eliminate some of our propose A, B,C…). However, for greater speed and simplicity this may be combined with **equating** coefficients of x, x^2 or x^3 as the case may be

2004/ 26 Neco (Dec)

Resolve into partial fractions $\frac{2x+11}{(x-2)(x+3)}$

A $\frac{2x}{x-2} + \frac{11}{x+3}$ B $\frac{11}{x-2} + \frac{1}{x+3}$

C $\frac{3}{x-2} + \frac{1}{x+3}$ D $\frac{3}{x-2} - \frac{1}{x+3}$ E $\frac{1}{x-2} - \frac{3}{x+3}$

Solution

Let $\frac{2x+11}{(x-2)(x+3)} \equiv \frac{A}{x-2} + \frac{B}{x+3}$

Multiplying by (x – 2) (x + 3) on both sides of the equation

2x + 11 = A(x +3) + B(x – 2) ------***

Put x = 2 into *** to eliminate B

2(2) + 11 = A(2 + 3) + B(0)

15 = 5A

A = 3

Next, to eliminate A, put x = – 3 into ***

2(–3) + 11 = A(0) + B (–3 – 2)

– 6 + 11 = –5B

5 = –5B

it follows that B = –1

Substituting for A and B values into our initial proposition

$$\frac{2x+11}{(x-2)(x+3)} = \frac{3}{x-2} - \frac{1}{x+3} \text{ (D)}$$

2005/5 Neco (Dec)

Express $\frac{x+3}{(x-2)(x+4)}$ *in partial* fractions.

Solution

Let $\frac{x+3}{(x-2)(x+4)} \equiv \frac{A}{x-2} + \frac{B}{x+4}$

Multiplying by $(x-2)(x+4)$ on both sides of the equation

$x + 3 = A(x+4) + B(x-2)$ ----- *

To eliminate B, we put x = 2 into *

2 + 3 = A(2 + 4) + B(2 – 2)

5 = A(6) + B(0)

5 = 6A

A = $\frac{5}{6}$

Next, to eliminate A, we put x = – 4 into *

– 4 + 3 = A (0) + B (– 6)

–1 = – 6B

B = $\frac{1}{6}$

Substituting for A and B values into our initial proposition

$$\frac{x+3}{(x-2)(x+4)} = \frac{5}{6(x-2)} + \frac{1}{6(x+4)}$$

Test your answer

Put x = 1 on both sides.

$$\frac{4}{(-1)(5)} \underline{\underline{\text{required to prove}}} \frac{5}{6(-1)} + \frac{1}{6(5)}$$

$$\frac{4}{-5} \underline{\underline{RTP}} -\frac{5}{6} + \frac{1}{30}$$

$$\underline{\underline{RTP}} \frac{-25+1}{30} = -\frac{24}{30} \text{ i.e } -\frac{4}{5} \text{ QED}$$

1994/1

If $\frac{10x+1}{(x-2)(x+1)} \equiv \frac{K}{x-2} + \frac{3}{x+1}$,find the value of K

A – 3 B 1 C 3 D 7 E 13

Solution

Multiplying both sides by (x – 2) (x + 1)

10x + 1 = K(x + 1) + 3(x – 2)

Since we are to find K, eliminate 3(x – 2) by

Putting x = 2

10(2) +1 = K(3) + 3(0)

21 = 3K

K = 7 (D)

1994/4 a

Resolve $\frac{1}{r(r+2)}$ into partial fractions

Solution

Let $\frac{1}{r(r+2)} \equiv \frac{A}{r} + \frac{B}{r+2}$

Multiplying both sides by r(r + 2)

$1 = A\,(r+2) + B\,r$ ------**

To eliminate B, we put r = 0 into **

$1 = A\,(0+2) + B\,(0)$

$1 = 2A$

$A = \frac{1}{2}$

To eliminate A, we put r = – 2 into **

$1 = A\,(-2+2) + B\,(-2)$

$1 = -2B$

$B = -\frac{1}{2}$

Substituting for A and B value into our initial proposition

$$\frac{1}{r(r+2)} = \frac{1}{2r} - \frac{1}{2(r+2)}$$

2015/7 f/m Exercise 29.4

Given that $\frac{2x}{(x+6)(x+3)} \equiv \frac{P}{x+6} + \frac{Q}{x+3}$, find P and Q

A P = 4 and Q = 2 B P = 2 and Q = 4
C P = 4 and Q = – 2 D P = – 2 and Q = 4

1997/ 11 Exercise 29.5

Given that

$$\frac{4+3x+2x^2}{(1-2x)(1+x)(1-x)} \equiv \frac{p}{1-2x} + \frac{Q}{1-x} + \frac{R}{1+x}$$

Where P, Q and R are constants, find the value of R.

$A\ -1$ **B** $-\frac{1}{2}$ $C\ \frac{1}{2}$ $D\ 1$ $E\ 2$

Cases with quadratic denominator that can be factorized

2005/ 11 a (Nov)

Express $\frac{5-12x}{6x^2+5x+1}$ in partial fractions

Solution

Factorizing the denominator

$$\frac{5-12x}{6x^2+5x+1} = \frac{5-12x}{(3x+1)(2x+1)}$$

Let $\frac{5-12x}{(3x+1)(2x+1)} \equiv \frac{A}{3x+1} + \frac{B}{2x+1}$

Multiplying through by $(3x+1)(2x+1)$

$5-12x = A(2x+1) + B(3x+1)$_____*

Equating constants

5 = A + B ------------ (1)

Equating coefficient of x from *

–12 = 2A + 3B ------------- (2)

From (1) A = 5 – B

Substitute into (2)

–12 = 2(5–B) + 3B

– 12 = 10 – 2B + 3B

– 12 = 10 + B

– 22 = B

Thus A = 5– (–22)

= 27

Substituting for A and B value into our initial proposition

$$\frac{5-12x}{(6x^2+5x+1)} = \frac{27}{3x+1} - \frac{22}{2x+1}$$

You can confirm by putting x = 1 on both sides; you will get

$$-\frac{7}{12} = -\frac{7}{12}$$

2000/10b (Nov)

Resolve into partial fractions $\frac{2-4x}{x(x^2-x-2)}$

Solution

Factorizing $x^2 - x - 2 = (x-2)(x+1)$

Thus, $\frac{2-4x}{x(x^2-2x-2)} = \frac{2-4x}{x(x-2)(x+1)}$

Let $\frac{2-4x}{x(x-2)(x+1)} \equiv \frac{A}{x} + \frac{B}{x-2} + \frac{C}{x+1}$

Multiplying through by x(x – 2)(x + 1)

2 – 4x = A(x – 2)(x + 1) + Bx(x + 1) + Cx(x – 2)__**

To eliminate B & C ; Put x = 0 into **

2 = – 2A

Thus, A = –1

To eliminate C & A ; Put x = 2 into **

2 – 8 = 6B

Thus, B = – 1

To ensure we get C; Put x = –2 into **

2 + 8 = A(–4)(–1) + B(–2)(–1) + C(–2) (–4)

10 = 4A + 2B + 8C

But A is –1 and B is –1 Thus,

10 = –4 – 2 + 8C

16 = 8C

C = 2

Substituting for A, B& C value into our initial proposition

$$\frac{2-4x}{x(x-2)(x+1)} = -\frac{1}{x} - \frac{1}{x-2} + \frac{2}{x+1} \quad \text{Rearranging}$$

$$= \frac{2}{x+1} - \frac{1}{x-2} - \frac{1}{x}$$

2000/1 Neco

Resolve $\frac{4+3x+2x^2}{(1-2x)(1-x^2)}$ into partial fraction

Solution

The (1– x^2) in denominator can factorize into (1– x)(1 + x)

Thus $\frac{4+3x+2x^2}{(1-2x)(1-x^2)} = \frac{4+3x+2x^2}{(1-2x)(1-x)(1+x)}$

Let $\dfrac{4+3x+2x^2}{(1-2x)(1-x^2)} \equiv \dfrac{A}{1-2x} + \dfrac{B}{1-x} + \dfrac{C}{1+x}$

Multiplying through by $(1-2x)(1-x)(1+x)$

$4+3x+2x^2 = A(1-x)(1+x)+B(1-2x)(1+x)+C(1-2x)(1-x)$--**

To eliminate A and C ; Put x = 1 into **

$4+3+2 = A(0) + B(-1)(2) + C(0)$

$9 = -2B$

Thus, $B = -\frac{9}{2}$

To eliminate A and B ; Put x = –1 into **

$4 - 3 + 2 = A(0) + B(0) + C(3)(2)$

$3 = 6C;$

Thus, $C = \frac{1}{2}$

To get A ; Put x = 0 into **

$4 = A + B + C$

But B is $-\frac{9}{2}$ and C is $\frac{1}{2}$ Thus,

$$4 = A - \frac{9}{2} + \frac{1}{2}$$

$4 + \frac{9}{2} - \frac{1}{2} = A$ it follows that A = 8

Substituting for A,B & C values into our initial proposition

$$\frac{4+3x+2x^2}{(1-2x)(1-x)(1+x)} = \frac{8}{1-2x} - \frac{9}{2(1-x)} + \frac{1}{2(1+x)}$$

1994/15 Counter example

Given that $\dfrac{6x+P}{2x^2+7x-15} = \dfrac{4}{x+5} - \dfrac{2}{2x-3}$

Find the value of the constant P

A 20 B 12 C 6 D –10 E –22

Solution

We apply the same partial fraction format: i.e

$$\frac{6x+P}{2x^2+7x-15} = \frac{4}{x+5} - \frac{2}{2x-3}$$

Then multiplying through by (x + 5)(2x – 3)

$6x + P = 4(2x - 3) - 2(x + 5)$

We equate constants to get P

$P = -12 - 10$

P = –22 (E)

1998/18 PCE Counter example

If $\dfrac{3+x}{(x-1)(x-b)}$ express in partial fractions is $\dfrac{5}{x-b} - \dfrac{4}{x-1}$ find b

A. – 2 B. – ½ C. 2 D. 3

Solution

We apply the same partial fractions format: i.e

$$\frac{3+x}{(x-1)(x-b)} = \frac{5}{x-b} - \frac{4}{x-1}$$

Then, multiplying through by (x – 1)(x – b)

$3 + x = 5(x - 1) - 4(x - b)$

Since our target is to find b;

we put x = 1 to eliminate 5(x – 1)

$3 + 1 = 5(1 - 1) - 4(1 - b)$

$4 = 5 \times 0 - 4 + 4b$

$4 = -4 + 4b$

$4 + 4 = 4b$

$8 = 4b$ Thus b = 2 (C)

2008/11a

Express $\dfrac{2x^2-5x+1}{x^3-4x^2+3x}$ in partial fractions.

Solution

Applying factorizing of polynomial to the denominator

We try to make f(x) = 0 by trial and error

$f(1) = (1)^3 - 4(1)^2 + 3(1)$ i.e 0

Thus x – 1 is a factor of $x^3 - 4x^2 + 3x$

Next, we apply long division to get the other factors

$$\begin{array}{r|l} & x^2 - 3x \\ \hline x - 1 & x^3 - 4x^2 + 3x \\ & x^3 - 1x^2 \\ \hline & \quad -3x^2 + 3x \\ & \quad -3x^2 + 3x \\ \hline & \quad\quad 0 \quad\quad 0 \end{array}$$

$x^3 - 4x^2 + 3x$ factorizes into $(x - 1)(x^2 - 3x)$ and $x^2 - 3x$ factorizes into $x(x-3)$.

Thus $x^3 - 4x^2 + 3x$ factorizes into $x(x - 1)(x-3)$.

$$\frac{2x^2-5x+1}{x^3-4x^2+3x} = \frac{2x^2-5x+1}{x(x-1)(x-3)}$$

Let $\dfrac{2x^2-5x+1}{x^3-4x^2+3x} \equiv \dfrac{A}{x} + \dfrac{B}{x-1} + \dfrac{C}{x-3}$

Multiplying through by x(x – 1)(x–3)

$2x^2 - 5x + 1 = A(x - 1)(x-3) + Bx(x-3) + Cx(x - 1)$----***

To eliminate A and C ; Put x = 1 into ***

$2 - 5 + 1 = A(0) + B(-2) + C(0)$

$-2 = -2B$

$B = 1$

To eliminate A and B ; Put x = 3 into ***

$2(3)^2 - 5(3) + 1 = A(0) + B(0) + C(6)$

$4 = 6C$

$C = \frac{2}{3}$

To eliminate B and C ; Put x = 0 into ***

$1 = A(-1)(-3) + B(0) + C(0)$

$1 = 3A$

$A = \frac{1}{3}$

Substituting for A,B & C values into our initial proposition

$$\frac{2x^2-5x+1}{x^3-4x^2+3x} = \frac{1}{3x} + \frac{1}{x-1} + \frac{2}{3(x-3)}$$

PPF/1 Exercise 29.6

Express $\dfrac{3x^2+11x+4}{x^3+4x^2+x-6}$ in partial fractions.

2002/38 Neco Exercise 29.7

Resolve the rational function

$\dfrac{10x+11}{x^2+x-2}$ into partial fractions.

A $\dfrac{3}{x-1} + \dfrac{7}{x+2}$ B $\dfrac{3}{x-1} - \dfrac{7}{x+2}$

C $\dfrac{7}{x+2} - \dfrac{7}{x-1}$ D $\dfrac{7}{x+2} - \dfrac{3}{x-1}$ E $\dfrac{7}{x+2} + \dfrac{7}{x-1}$

2004/5 Neco (Dec) Exercise 29.8

Resolve into partial fractions $\frac{3x^2+4x+2}{x(x^2+3x+2)}$

1995/20 (Nov) Exercise 29.9

Given that $\frac{x+3}{x^2-9x+18} = \frac{p}{x-6} + \frac{Q}{x-3}$

Find the values of the constants P and Q

A P = 4, Q = 3 B P = 3, Q = –2
C P = –3, Q = –1 D P = – 2, Q = 3
E P = – 4, Q = 2

1996/11 Exercise 29.10

If $\frac{3x^2-7}{x^3+2x^2-8x} = \frac{7}{8x} + \frac{P}{x+4} + \frac{Q}{x-2}$, find P + Q

A $\frac{31}{24}$ $B\frac{41}{24}$ $C\frac{17}{8}$ $D\frac{51}{12}$ $E\frac{5}{12}$

Cases with quadratic denominator that has repeated roots

2000/11b

Resolve $\frac{2x^2-x+2}{(x+2)^2(1-2x)}$ into partial fractions

Solution

We proceed as follows

$$\frac{2x^2-x+2}{(x+2)^2(1-2x)} \equiv \frac{A}{x+2} + \frac{B}{(x+2)^2} + \frac{C}{(1-2x)}$$

Repeated root $(x + 2)^2$ was broken down as $(x + 2)$ & $(x + 2)^2$

Multiplying through by $(x + 2)^2(1 – 2x)$

$2x^2 – x + 2 = A(x + 2)(1 – 2x) + B(1 – 2x) + C(x + 2)^2$---**

By observation we eliminate C and A putting x = – 2 into**

$2(– 2)^2 – (–2) + 2 = A(0)(5) + B(5) + C(0)^2$

$12 = 5B$

$\frac{12}{5} = B$

To get A and C let us open up RHS of **

$2x^2 – x + 2 = A(2 – 3x – 2x^2) + B(1 – 2x) + C(x^2 + 4x + 4)$

Equating terms in x^2 from the expansion

$2 = –2A + C$ ---------(1)

Equating constants from the expansion

$2 = 2A + B + 4C$

But B is $\frac{12}{5}$

$2 = 2A + \frac{12}{5} + 4C$

$2 - \frac{12}{5} = 2A + 4C$

$-\frac{2}{5} = 2A + 4C$

$–2 = 10A + 20C$

$–1 = 5A + 10C$ -------(2)

Multiply (1) by 10 and subtract (2)

$–20A + 10C = 20$
$–(5A + 10C = –1)$

$–25A = 21$

$A = -\frac{21}{25}$

Next, we substitute A value into (2)

$5A + 10C = –1$ becomes

$5\left(-\frac{21}{25}\right) + 10C = –1$

$10C = –1 + \frac{21}{5}$

$C = \frac{8}{25}$

Substituting for A,B & C values into our initial proposition

$$\frac{2x^2-x+2}{(x+2)^2(1-2x)} = -\frac{21}{25(x+2)} + \frac{12}{5(x+2)^2} + \frac{8}{25(1-2x)}$$

Example PPF/2

Resolve $\frac{3x^3+3x^2+3x+2}{x^3(x+1)}$ into partial fractions

Solution

Let $\frac{3x^3+3x^2+3x+2}{x^3(x+1)} \equiv \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x^3} + \frac{D}{(x+1)}$

Multiplying through by $x^3(x + 1)$

$3x^3 + 3x^2 + 3x + 2 = Ax^2(x + 1) + Bx(x + 1) + C(x+1) + Dx^3$ ---**

To get C we put x = 0 into **

$2 = C$

To get D we put x = –1 into **

$–3 + 3 – 3 + 2 = –D$

$D = 1$

To enable us equate coefficients, let us expand **

$3x^3+3x^2+3x+2 = (Ax^3 + Ax^2) + (Bx^2 + Bx) + (Cx+C) + Dx^3$

Equating coefficients of x^3 from the expansion

$3 = A + D$

But D is 1 $3 = A + 1$

$A = 2$

Equating coefficients of x^2 from the expansion

$3 = A + B$

But A is 2 $3 = 2 + B$

$B = 1$

Substituting for A,B,C & D values into our initial proposition

$$\frac{3x^3+3x^2+3x+2}{x^3(x+1)} = \frac{2}{x} + \frac{1}{x^2} + \frac{2}{x^3} + \frac{1}{(x+1)}$$

2004/11 Neco Exercise 29.11

(a) Express $\frac{2x+1}{(x+1)(x+2)^2}$ in the form

$\frac{A}{x+1} + \frac{B}{x+2} + \frac{C}{(x+2)^2}$ Where A, B and C are constants

Cases with quadratic denominator that cannot be factorized

2005/6 Neco

Resolve $\frac{3}{(x-1)(x^2-5x+2)}$ into partial fractions

Solution

You will observe that $(x^2 – 5x + 2)$ cannot be factorized

Let $\frac{3}{(x-1)(x^2-5x+2)} \equiv \frac{A}{x-1} + \frac{Bx+C}{x^2-5x+2}$

Multiplying through by $(x – 1)(x^2 – 5x + 2)$

$3 = A(x^2 - 5x + 2) + (Bx + C)(x - 1)$

$3 = A(x^2 - 5x + 2) + Bx(x - 1) + C(x - 1)$------**

To get A, we put x = 1 into**

$3 = A(-2) + B(0) + C(0)$

$3 = -2A$ i.e

$A = -\frac{3}{2}$

To eliminate B, we put x = 0 into**

$3 = 2A - C$

But A is $-\frac{3}{2}$ $\quad 3 = 2(-\frac{3}{2}) - C$

$C = -6$

Equating coefficient of $\mathbf{x^2}$

Note that $Bx(x - 1) = Bx^2 - Bx$ at **

$0 = A + B$

$0 = -\frac{3}{2} + B$

$B = \frac{3}{2}$

$$\text{Thus } \frac{3}{(x-1)(x^2-5x+2)} = \frac{-3}{2(x-1)} + \frac{\frac{3}{2}x - 6}{x^2-5x+2}$$

$$= \frac{3x-12}{2(x^2-5x+2)} - \frac{3}{2(x-1)}$$

2007/36 Neco

Express $\frac{x+3}{x(x^2+2)}$ into partial fractions

A $\frac{3}{2x} + \frac{2-3x}{2(x^2+2)}$ $\quad$ B $\frac{3}{2x} - \frac{2-3x}{2(x^2+2)}$

C $\frac{3}{2x} + \frac{2+3x}{2(x^2+2)}$ $\quad$ D $\frac{3}{2x} - \frac{2+3x}{2(x^2+2)}$ $\quad$ E $\frac{-3}{2x} + \frac{2-3x}{2(x^2+2)}$

Solution

You will observe that $(x^2 + 2)$ cannot be factorized

Let $\frac{x+3}{x(x^2+2)} \equiv \frac{A}{x} + \frac{Bx+C}{x^2+2}$

Multiplying through by $x(x^2 + 2)$

$x + 3 = A(x^2 + 2) + x(Bx + C)$ --------***

Equating constant

$3 = 2A; \quad A = \frac{3}{2}$

C is not among the constants because $x(Bx + C) = Bx^2 + Cx$

Equating coefficient of $\mathbf{x^2}$

$0 = A + B$

$0 = \frac{3}{2} + B$

Thus $B = -\frac{3}{2}$

Equating coefficient of x

$1 = C$

care must be taken to fix B value into our given proposition

$$\frac{x+3}{x(x^2+2)} = \frac{3}{2x} + \frac{(-\frac{3}{2}x+1)}{x^2+2}$$

$$= \frac{3}{2x} + \frac{2-3x}{2(x^2+2)} \qquad \text{A.}$$

PPF/3 Counter example

Resolve $\frac{x^2+4x+1}{x^3+3x^2+4x+2}$ into partial fractions

Solution

A critical look at the denominator shows that

f(-1) = -1 + 3 - 4 + 2 i.e 0

x +1 is a factor

$$\begin{array}{r|l} & x^2 + 2x + 2 \\ \hline x + 1 & x^3 + 3x^2 + 4x + 2 \\ & \underline{x^3 + x^2} \\ & \quad 2x^2 + 4x \\ & \quad \underline{2x^2 + 2x} \\ & \qquad 2x + 2 \\ & \qquad \underline{2x + 2} \\ & \qquad\ 0 \quad 0 \end{array}$$

$x^3 + 3x^2 + 4x + 2$ factorizes into $(x + 1)(x^2 + 2x + 2)$

You will observe that $(x^2 + 2x + 2)$ cannot be factorized

Let $\frac{x^2+4x+1}{x^3+3x^2+4x+2} \equiv \frac{A}{x+1} + \frac{Bx+C}{x^2+2x+2}$

Multiplying through by $(x + 1)(x^2 + 2x + 2)$

$x^2 + 4x + 1 = A(x^2 + 2x + 2) + (Bx + C)(x + 1)$

$x^2 + 4x + 1 = A(x^2 + 2x + 2) + (Bx^2 + Bx) + (Cx + C)$---**

To eliminate B and C we put x = –1 into **

$1 - 4 + 1 = A(1 - 2 + 2) + B(0) + C(0)$

$-2 = A$

To eliminate B, we put x = 0 into **

$1 = A(2) + C(1)$

$1 = 2A + C$

$1 = 2(-2) + C$ Thus C = 5

Equating coefficient of $\mathbf{x^2}$ at **

$1 = A + B$

$1 = -2 + B$ thus, B = 3

$$\text{Thus } \frac{x^2+4x+1}{x^3+3x^2+4x+2} = -\frac{2}{x+1} + \frac{3x+5}{x^2+2x+2}$$

2013/25 f/m

Express $\frac{x^2+x+4}{(1-x)(x^2+1)}$ in partial fractions

Solution

Let $\frac{x^2+x+4}{(1-x)(x^2+1)} \equiv \frac{A}{1-x} + \frac{Bx+c}{x^2+1}$

$x^2 + x + 4 = A(x^2 + 1) + (Bx + C)(1 - x)$

$x^2 + x + 4 = A(x^2 + 1) + Bx - Bx^2 + C - Cx$

$x^2 + x + 4 = Ax^2 + A + Bx + C - Bx^2 - Cx$

By comparing coefficients

$\mathbf{x^2}$: $\quad 1 = A - B$ ---- (1)

$\mathbf{x}$: $\quad 1 = B - C$ ---- (2)

By comparing constants

Constant : $4 = A + C$ ---- (3)

(2) + (3) gives $A + B = 5$ --- (4)

(1) + (4) gives $2A = 6$

$A = 3$

Substituting A value into (1)

$1 = 3 - B$ thus, B = 2

Substituting B value into (2)

$1 = 2 - C$ thus, C = 1

$$\text{Hence } \frac{x^2+x+4}{(1-x)(x^2+1)} = \frac{3}{1-x} + \frac{2x+1}{x^2+1}$$

2014/12 f/m

If $\frac{x+P}{(x-1)(x-3)} \equiv \frac{Q}{x-1} + \frac{2}{x-3}$,

find the value of (P + Q)

A – 2 B – 1 C 0 D 1

Solution

x + P = Q(x – 3) + 2(x – 1)

Put x = 3 to eliminate Q

3 + P = Q(0) + 2(3 – 1)

3 + p = 4

P = 4 – 3 = 1

Equating constants

P = – 3Q – 2

1 = – 3Q – 2

3 = – 3Q

– 1 = Q

Thus P + Q = 1 – 1

= 0 C.

Alternatively : put x = 1 *instead of equating constants*

1 + P = Q(1 – 3) + 0

1 + P = – 2Q

1 + 1 = –2Q

Q = – 1

1992 /11 (Nov) Exercise 29.12

If $\frac{3x^2+4x}{(x-2)(x^2+1)} = \frac{P}{x-2} + \frac{Qx+R}{x^2+1}$, find P + Q + R

A – 1 B 2 C 4 D 5 E 7

Cases with degree of numerator greater than or equal to the denominator

2005/10a Neco (Dec)

Resolve $\frac{x^3+x^2+4x}{x^2+x-2}$ into partial fraction

Solution

The highest degree of x in the numerator is higher than that of the denominator; which is not allowed;

Applying long division

$$\begin{array}{r|l} & \quad x \\ \hline x^2+x-2 & x^3+x^2+4x \\ & -(x^3+x^2-2x) \\ \hline & \qquad\qquad 6x \end{array}$$

Thus $\frac{x^3+x^2+4x}{x^2+x-2} = x + \frac{6x}{x^2+x-2}$

Next, we resolve $\frac{6x}{x^2+x-2}$ into partial fractions

$\frac{6x}{x^2+x-2} \Rightarrow \frac{6x}{(x+2)(x-1)} \equiv \frac{A}{x+2} + \frac{B}{x-1}$

Multiply through by (x + 2)(x – 1)

6x = A(x – 1) + B(x + 2) ------***

To eliminate A; put x = 1 into ***

6 = A(0) + B(3)

6 = 3B

B = 2

To eliminate B ; put x = – 2

6 (–2) = –3A + B(0)

–12 = –3A

A = 4

Hence $\frac{6x}{(x+2)(x-1)} = \frac{4}{x+2} + \frac{2}{x-1}$

Confirm by putting x = 2 and you get 3 = 1+2

Therefore $\frac{x^3+x^2+4x}{x^2+x-2} = x + \frac{4}{x+2} + \frac{2}{x-1}$

2007/37 Neco

If $\frac{x^2+1}{4-x^2} = A + \frac{B}{2-x} + \frac{C}{2+x}$ Where A, B and C are constants, calculate $\frac{A+B}{C}$

A. $\frac{9}{5}$ B. $\frac{4}{5}$ C. $\frac{5}{16}$ D. $\frac{1}{5}$ E. -1

Solution

The degree of x in numerator is equal to that of the denominator; which is not allowed.

Applying long division

$$\begin{array}{r|l} & \quad -1 \\ \hline -x^2+4 & x^2+1 \\ & -(x^2-4) \\ \hline & \qquad 5 \end{array}$$

Thus $\frac{x^2+1}{4-x^2} = -1 + \frac{5}{4-x^2}$

Next, we resolve $\frac{5}{4-x^2}$ into partial fractions

$\frac{5}{4-x^2} \Rightarrow \frac{5}{2^2-x^2}$ $i.e \frac{5}{(2-x)(2+x)}$

Let $\frac{5}{(2-x)(2+x)} \equiv \frac{A}{2-x} + \frac{B}{2+x}$

Multiplying through by (2 – x)(2 + x)

5 = A(2 + x) + B(2 – x) -----**

To eliminate B; put x = 2 into**

5 = 4A + B(0)

5 = 4A;

A = $\frac{5}{4}$

To eliminate A; put x = –2 into**

5 = A(0) + 4B

5 = 4B;

B = $\frac{5}{4}$

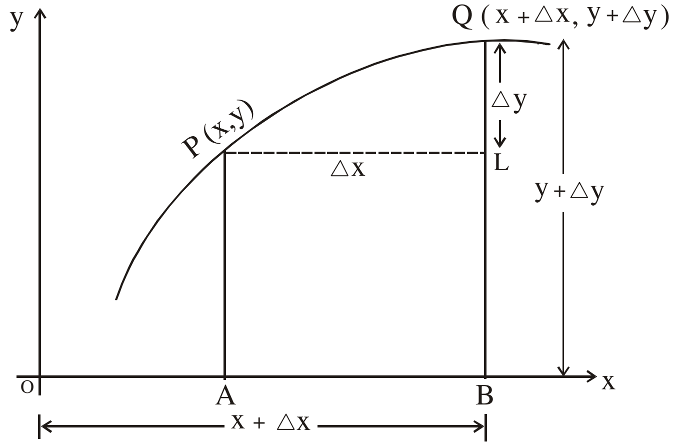
Thus, $\frac{5}{(2-x)(2+x)} = \frac{5}{4(2-x)} + \frac{5}{4(2+x)}$

Therefore $\frac{x^2+1}{4-x^2} = -1 + \frac{5}{4(2-x)} + \frac{5}{4(2+x)}$

and $\frac{A+B}{C} = \frac{-1+\frac{5}{4}}{\frac{5}{4}} = \frac{1}{4} \div \frac{5}{4} = \frac{1}{4} \times \frac{4}{5} ie \frac{1}{5}(D)$

Chapter thirty (Differentiation)

Differentiation by first principle



Consider the graph of a function $y = f(x)$ with points P and Q very close to each other on the curve, where P(x, y) and Q $(x + \Delta x, y + \Delta y)$. Δy being a small increase in y due to a small increase Δx in x .

Here, Δy and Δx are enlarged in the diagram, otherwise, they could not be seen from the diagram,

$QL = \Delta y$ and $PL = \Delta x$

The slope of the chord PQ is $\dfrac{QL}{PL} = \dfrac{\Delta y}{\Delta x}$

As $\Delta x \longrightarrow 0$, Q moves into coincidence with P and the chord PQ produced becomes a tangent at P.

Hence the slope of the tangent at P is the value that $\frac{\Delta y}{\Delta x}$ approaches as Δx approaches zero.

Thus, the slope of the tangent at P is

$$\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$

This value is denoted by the symbol

$$\frac{dy}{dx} \text{ (dee y , dee x)}$$

It is known as the *first differential coefficient of y with respect to x.*. It represents the rate of change of y with respect to x. If y increase as x also increases then $^{dy}/_{dx}$ is also positive whereas if y decreases as x increases then $^{dy}/_{dx}$ is negative.

Derived definition form of first principle

Consider the function

$y = f(x)$ ………..(1)

Deduction from the diagram above ,

Let Δy be a small increase in y due to a small increase Δx in x.

$y + \Delta y = f(x + \Delta x)$……..(2)

Subtracting (2) – (1)

$y + \Delta y - y = f(x + \Delta x) - f(x)$

$\Delta y = f(x + \Delta x) - f(x)$

Then, $\dfrac{\Delta y}{\Delta x} = \dfrac{f(x + \Delta x) - f(x)}{\Delta x}$

Take limits as $\Delta x \longrightarrow 0$

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

is known as the derived definition of $^{dy}/_{dx}$

To <u>differentiate</u> by <u>first principle</u> means to work strictly by the <u>ordinary definition</u> or the <u>derived definition</u> and **<u>not</u>** employing any theorem or result in calculus that will be given later. Our working here shall be based on the <u>ordinary definition.</u>

IPLF 1 Examples

Find the derivative of the following with respect to x using the first principle,

(i) x (ii) x^2 (iii) $2x^2 + x + 2$ (iv) $\dfrac{1}{2x + 1}$

(v) $\sqrt{x - 1}$ (vi) x^n (n is a positive integer)

Solutions

(i) let $y = x$

Then $y + \Delta y = x + \Delta x$

subtracting

$y + \Delta y - y = x + \Delta x - x$

$\Delta y = \Delta x$

$\dfrac{\Delta y}{\Delta x} = \dfrac{\Delta x}{\Delta x} = 1$

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} 1 = 1$$

(ii) Let $y = x^2$

Then $y + \Delta y = (x + \Delta x)^2$

Subtracting

$y + \Delta y - y = (x + \Delta x)^2 - x^2$

$\Delta y = x^2 + 2x\Delta x + (\Delta x)^2 - x^2$

$\Delta y = 2x\Delta x + (\Delta x)^2$

Then $\dfrac{\Delta y}{\Delta x} = \dfrac{2x\Delta x}{\Delta x} + \dfrac{(\Delta x)^2}{\Delta x}$

$\dfrac{\Delta y}{\Delta x} = 2x + \Delta x$

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} (2x + \Delta x)$$

$= 2x$

(iii) Let $y = 2x^2 + x + 2$, then

$y + \Delta x = 2(x + \Delta x)^2 + (x + \Delta x) + 2$

$y + \Delta x = 2[x^2 + 2x\Delta x + (\Delta x)^2] + (x + \Delta x) + 2$

Subtracting

$y + \Delta x - y = 2x^2 + 4x\Delta x + 2(\Delta x)^2 + x + \Delta x + 2 - 2x^2 - x - 2$

$\Delta y = 4x\Delta x + 2(\Delta x)^2 + \Delta x$

Then $\dfrac{\Delta y}{\Delta x} = \dfrac{4x\Delta x}{\Delta x} + \dfrac{2(\Delta x)^2}{\Delta x} + \dfrac{\Delta x}{\Delta x}$

$\dfrac{\Delta y}{\Delta x} = 4x + 2\Delta x + 1$

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = 4x + 1$$

(iv) Let $y = \dfrac{1}{2x + 1}$ then

$y + \Delta y = \dfrac{1}{2(x + \Delta x) + 1}$

Subtracting

$y + \Delta y - y = \dfrac{1}{2(x + \Delta x) + 1} - \dfrac{1}{2x + 1}$

$\Delta y = \dfrac{2x + 1 - 2x - 2\Delta x - 1}{[2(x + \Delta x) + 1]\,[2x + 1]}$

$$\Delta y = \frac{-2\Delta x}{[2(x + \Delta x) + 1]\,[2x + 1]}$$

Next we multiply both sides by $\frac{1}{\Delta x}$

$$\frac{\Delta y}{\Delta x} = \frac{-2}{[2(x + \Delta x) + 1]\,[2x + 1]}$$

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \frac{-2}{[2(x + 0) + 1]\,[2x + 1]}$$

$$= \frac{-2}{(2x + 1)(2x + 1)}$$

$$= \frac{-2}{(2x + 1)^2}$$

(v) Let $y = \sqrt{x - 1}$ then

$$y + \Delta y = \sqrt{(x + \Delta x) - 1}$$

Subtracting

$$y + \Delta y - y = \sqrt{(x + \Delta x) - 1} - \sqrt{x - 1}$$

$$\Delta y = \frac{[\sqrt{(x + \Delta x) - 1} - \sqrt{x - 1}]\,[\sqrt{(x + \Delta x) - 1} + \sqrt{x - 1}\,]}{\sqrt{(x + \Delta x) - 1} + \sqrt{x - 1}}$$

Nothing happened, we just multiplied by $^1/_1$

The top is a surd × its conjugate;

$$\Delta y = \frac{(x + \Delta x - 1) - (x - 1)}{\sqrt{x + \Delta x - 1} + \sqrt{x - 1}}$$

$$\Delta y = \frac{\Delta x}{\sqrt{x + \Delta x - 1} + \sqrt{x - 1}}$$

$$\frac{\Delta y}{\Delta x} = \frac{\Delta x}{(\sqrt{x + \Delta x - 1} + \sqrt{x - 1})\Delta x}$$

$$\Delta y = \frac{1}{\sqrt{x + \Delta x - 1} + \sqrt{x - 1}}$$

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \frac{1}{\sqrt{x - 1} + \sqrt{x - 1}}$$

$$= \frac{1}{2(\sqrt{x - 1}\,)}$$

(vi) Let $y = x^n$ Then

$$y + \Delta y = (x + \Delta x)^n$$

$$= x^n + nx^{n-1}(\Delta x)^1 + \frac{n(n-1)x^{n-2}}{2!}(\Delta x)^2 + \ldots + (\Delta x)^n$$

Subtracting

$$y + \Delta y - y = x^n + nx^{n-1}(\Delta x)^1 + \frac{n(n-1)x^{n-2}}{2!}(\Delta x)^2 + \ldots + (\Delta x)^n - x^n$$

$$\Delta y = nx^{n-1}(\Delta x)^1 + \frac{n(n-1)x^{n-2}}{2!}(\Delta x)^2 + \ldots$$

$\Delta y = nx^{n-1} + \frac{n(n-1)x^{n-2}\Delta x}{2!} + \ldots +$ terms in higher power of Δx.

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = nx^{n-1}$$

Indeed, for any constant n(positive; negative or fractional), if

$$y = x^n$$

Then $$\frac{dy}{dx} = nx^{n-1}$$

General formula: x here is the independent variable but any other variable could be used.

2003/10 (Nov)

Find from the first principles, the derivatives, with respect to x, of $y = 3x^2$

Solution

$$y = 3x^2$$

$$y + \Delta y = 3(x + \Delta x)^2$$

$$y + \Delta y = 3[\,x^2 + 2x\Delta x + (\Delta x)^2\,]$$

$$y + \Delta y = 3x^2 + 6x\Delta x + 3(\Delta x)^2$$

Subtracting

$$y + \Delta y - y = 3x^2 + 6x\Delta x + 3(\Delta x)^2 - 3x^2$$

$$\Delta y = 6x\Delta x + 3(\Delta x)^2$$

Then $$\frac{\Delta y}{\Delta x} = \frac{6x\Delta x}{\Delta x} + \frac{3(\Delta x)^2}{\Delta x}$$

$$\frac{\Delta y}{\Delta x} = 6x + 3\Delta x$$

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = 6x$$

2004/11a Neco (Nov)

Differentiate $\frac{1}{x^3}$ from the first principle

Solution

Let $y = \frac{1}{x^3}$

$$y + \Delta y = \frac{1}{(x + \Delta x)^3}$$

$$= \frac{1}{x^3 + 3x^2\Delta x + 3x(\Delta x)^2 + (\Delta x)^3}$$

Subtracting

$$y + \Delta y - y = \frac{1}{x^3 + 3x^2\Delta x + 3x(\Delta x)^2 + (\Delta x)^3} - \frac{1}{x^3}$$

$$\Delta y = \frac{x^3 - x^3 - 3x^2\Delta x - 3x(\Delta x)^2 - (\Delta x)^3}{[x^3 + 3x^2\Delta x + 3x(\Delta x)^2 + (\Delta x)^3\,]\,[x^3]}$$

$$\Delta y = \frac{-3x^2\Delta x - 3x(\Delta x)^2 - (\Delta x)^3}{[x^3 + 3x^2\Delta x + 3x(\Delta x)^2 + (\Delta x)^3\,]\,[x^3]}$$

$$\frac{\Delta y}{\Delta x} = \frac{-3x^2 - 3x(\Delta x) - (\Delta x)^2}{[x^3 + 3x^2\Delta x + 3x(\Delta x)^2 + (\Delta x)^3\,]\,[x^3]}$$

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \frac{-3x^2}{(x^3)(x^3)}$$

$$= \frac{-3}{x^4}$$

1992 /2 (Nov)

Differentiate $5x - \frac{1}{x^2}$ from the first principles.

Solution

Let $y = 5x - \frac{1}{x^2}$

$$y + \Delta y = 5(x + \Delta x) - \frac{1}{(x + \Delta x)^2}$$

$$y + \Delta y = 5x + 5\Delta x - \frac{1}{x^2 + 2x\Delta x + (\Delta x)^2}$$

Subtracting

$$y + \Delta y - y = 5x + 5\Delta x - \frac{1}{x^2 + 2x\Delta x + (\Delta x)^2} - 5x + \frac{1}{x^2}$$

$$\Delta y = 5\Delta x - \frac{1}{x^2 + 2x\Delta x + (\Delta x)^2} + \frac{1}{x^2}$$

RHS is purely fractional in nature and must be regularized through our basic knowledge in fractional LCM.

$$= \frac{5\Delta x[x^2 + 2x\Delta x + (\Delta x)^2](x^2) - x^2 + x^2 + 2x\Delta x + (\Delta x)^2}{[x^2 + 2x\Delta x + (\Delta x)^2][x^2]}$$

$$= \frac{[5x^2\Delta x + 10x(\Delta x)^2 + 5(\Delta x)^3](x^2) + 2x\Delta x + (\Delta x)^2}{[x^2 + 2x\Delta x + (\Delta x)^2][x^2]}$$

$$\Delta y = \frac{5x^4\Delta x + 10x^3(\Delta x)^2 + 5x^2(\Delta x)^3 + 2x\Delta x + (\Delta x)^2}{[x^2 + 2x\Delta x + (\Delta x)^2][x^2]}$$

$$\frac{\Delta y}{\Delta x} = \frac{5x^4 + 10x^3\Delta x + 5x^2(\Delta x)^2 + 2x + \Delta x}{[x^2 + 2x\Delta x + (\Delta x)^2][x^2]}$$

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \frac{5x^4 + 2x}{[x^2][x^2]}$$

$$= \frac{5x^4 + 2x}{x^4}$$

$$= 5 + \frac{2}{x^3}$$

2015/33 f/m

Given that $y = 4 - 9x$ and $\Delta x = 0.1$, Calculate Δy

A 9.0 B 0.9 C – 0.3 D – 0.9

Solution

$y = 4 - 9x$

$\frac{\Delta y}{\Delta x} = -9$

$\Delta y = -9 \times \Delta x$

$= -9 \times 0.1$

$= -0.9$ (D)

Alternatively

$y = 4 - 9x$

By first principle of differentiation

$y + \Delta y = 4 - 9(x + \Delta x)$

$y + \Delta y = 4 - 9x - 9\Delta x$

Subtracting y from both sides

$y + \Delta y - y = 4 - 9x - 9\Delta x - (4 - 9x)$

$\Delta y = -9\Delta x$

Substituting for Δx value

$\Delta y = -9(0.1)$

$= -0.9$ D.

2011/2b Neco f/m Adjusted Exercise 30.0

From first principles, differentiate $y = x^2$

Basic results (rules) in differentiation achieved by first principle

1. If C is a constant then

$$\frac{d}{dx}(c) = 0$$

2. If **a** is a constant then

$$\frac{d}{dx} a\, f(x) = a \frac{d}{dx} f(x)$$

3. If u, v, w,… are functions of x the **sum**

$$\frac{d}{dx}(u + v + w +, \dots) = \frac{du}{dx} + \frac{dv}{dx} + \frac{dw}{dx} + \dots$$

4. **Product Rule:** If u and v are some functions of x, then

$$\frac{d}{dx}(uv) = V\frac{du}{dx} + U\frac{dv}{dx}$$

5. **Quotient Rule:** If u and v are some functions of x, then

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{V\frac{du}{dx} - U\frac{dv}{dx}}{V^2}$$

6. **Chain Rule:** If y is a function of z and z is a function of x, then

$$\frac{dy}{dx} = \frac{dy}{dz} \times \frac{dz}{dx}$$

7. If y is a function of x then, $\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$

General formula method

For any function $y = ax^n$ (where a and n are constants)

$$\frac{dy}{dx} = n \times a\, x^{n-1}$$

i.e differentiating y with respect to x pronounced dee y-dee x

Examples: Differentiate the following with respect to (wrt) their variable

(i) $y = x^2$ (ii) $y = 3x^5$ (iii) $y = 14x^2$

(iv) $y = -12x^6$ (v) $s = 3t^4$ (vi) $y = 3x$

(vii) $y = x$ (viii) $y = 4x^3$

Solution

(i) $y = x^2$

$\frac{dy}{dx} = 2x^{2-1}$

$= 2x^1$ i.e $2x$

(ii) $y = 3x^5$

$\frac{dy}{dx} = 5 \times 3 \times x^{5-1}$

$= 15x^4$

(iii) $y = 14x^2$

$\frac{dy}{dx} = 2 \times 14 \times x^{2-1}$

$= 28x^1$ i.e $28x$

(iv) $y = -12x^6$

$\frac{dy}{dx} = 6 \times (-12) \times x^{6-1}$

$= -72x^5$

(v) $s = 3t^4$

$\frac{ds}{dt} = 4 \times 3 \times t^{4-1}$ and not $^{dy}/_{dx}$ or $^{dy}/_{dt}$ or $^{dt}/_{ds}$

$= 12t^3$

(vi) $y = 3x$

Here the power of x is one i.e x^1

$\Rightarrow y = 3x^1$

$\frac{dy}{dx} = 1 \times 3 \times x^{1-1}$

$= 3x^0$ *since $x^0 = 1$*

$= 3 \times 1$ i.e 3

(vii) $y = x$

Here the power of x is one i.e x^1

$\Rightarrow y = x^1$

$\frac{dy}{dx} = 1x^{1-1}$

$= 1x^0$ *since $x^0 = 1$*

$= 1 \times 1$ i.e 1

(viii) $y = 4x^3$

$\frac{dy}{dx} = 12x^2$

*Examples **i** to **vii** involves explanation with solution. But **viii** have solution only which is the acceptable standard we shall be following except when the author wants to explain further.*

Further examples

Differentiate the following with respect to their variables.

(1) $y = \frac{1}{x^2}$ (2) $t = \frac{3}{s^6}$ (3) if $y = 10$ find $^{dy}/_{dx}$

(4) if $y = 13x$, find dy/dt

Solution

(1) $y = \frac{1}{x^2}$

$\Rightarrow y = x^{-2}$

$\frac{dy}{dx} = -2x^{-2-1}$

$= -2x^{-3} = \frac{-2}{x^3}$

(2) $t = \frac{3}{s^6} \Rightarrow t = 3s^{-6}$

$\frac{dt}{ds} = -6 \times 3 \times s^{-6-1}$

$= -18s^{-7}$

$= -\frac{18}{s^7}$

(3) $y = 10 \Rightarrow y = 10x^0$

(the variable x was not shown because it has a power of zero)

$\frac{dy}{dx} = 0 \times 10\, x^{0-1}$

$= 0$ (since zero **times** any number is zero)

Generally any constant differential is zero

(4) $y = 13x$, but we are asked to find $^{dy}/_{dt}$

$\frac{dy}{dt} = 0$ since 13x is a constant with respect to t

2007/38 PCE

If $y = \sqrt{x}$, *find* $\frac{dy}{dx}$

A $\frac{1}{2\sqrt{x}}$ B $\frac{2}{\sqrt{x}}$ C $\frac{\sqrt{x}}{2}$ D $\frac{1}{\sqrt{x}}$

Solution

$y = \sqrt{x} \Rightarrow y = x^{1/2}$

$\frac{dy}{dx} = \frac{1}{2}x^{1/2-1} = \frac{1}{2}x^{-1/2}$ i.e $\frac{1}{2\sqrt{x}}$ A.

2001/38 PCE Exercise 30.1

If $y = x^{-\frac{3}{2}}$ then dy/dx is

A. $-{}^2/_3x^{1/2}$ B. $-{}^3/_2x^{-1/2}$ C. $-{}^2/_3x^{-5/2}$ D. $-{}^3/_2x^{-5/2}$

The Derivative of the sum of two (or more) simple functions is the sum of the separate derivatives of the functions: What it implies is shown below

Eg.1 If $y = 4x^3 - 12x^2 + 3x + 12$, find $^{dy}/_{dx}$.

Solution

$\frac{dy}{dx} = 12x^2 - 24x + 3$

2012/37 UTME

If $y = x^2 - \frac{1}{x}$, find $\frac{dy}{dx}$

A $2x - \frac{1}{x^2}$ B $2x + x^2$ C $2x - x^2$ D $2x + \frac{1}{x^2}$

Solution

$y = x^2 - \frac{1}{x}$

$y = x^2 - x^{-1}$

$\frac{dy}{dx} = 2x - (-1)x^{-1-1}$

$= 2x + x^{-2}$

$= 2x + \frac{1}{x^2}$ D.

2014/32 Neco

If $y = 3x^3 + x^2 + 6$, find $\frac{dy}{dx}$ at $x = -2$

A 40 B 32 C 0 D 32 E 40

Solution

$y = 3x^3 + x^2 + 6$

$\frac{dy}{dx} = 9x^2 + 2x$

At $x = -2$

$\frac{dy}{dx} = 9(-2)^2 + 2(-2)$

$= 9 \times 4 - 4$

$= 36 - 4 = 32$ D.

2005/6 PCE

Find the derivative of $f(x) = 2x(x^2 + 3)$

A. $4x^2 + 3$ B. $2(x^2 + 3)$ C. $6(x^2 + 1)$ D. $6(x^2 + 2)$

Solution

We can easily open up the bracket as :

$f(x) = 2x(x^2 + 3)$ becomes
$f(x) = 2x^3 + 6x$
$f^{/}(x) = 6x^2 + 6$ i.e $6(x^2 + 1)$ C

2014/38 UTME Exercise 30.2

If $y = 4x^3 - 2x^2 + x$, find $\frac{dy}{dx}$

A $12x^2 - 2x + 1$ B $12x^2 - 4x + 1$
C $8x^2 - 2x + 1$ D $8x^2 - 4x + 1$

2007/15 UME Exercise 30.3

If $y = (1 + x)^2$, find dy/dx

A. $2x - 1$ B. $x - 1$ C. $2 + 2x$ D. $1 + 2x$

1994/44 PCE Exercise 30.4

Differentiate $y = 3x^2 + 2x + 3$ with respect to x.
A. $6x + 2$ B. $6x^2 - 2$ C. $6x + 3$ D. $6x^2 - 3$

1995/39 PCE Exercise 30.5

Differentiate $3x^2 + {}^2/_x$ with respect to x

A. $6x + 2$ B. $6x - \frac{2}{x}$ C. $6x + \frac{2}{x^2}$ D. $6x - \frac{2}{x^2}$

Second derivative

Differentiation can be repeated as many times as required on a given function. Under this subheading, we shall only discuss 2nd differentiation.

Eg. 1 If $y = 3x^3 - 2x^2 + 4$ find d^2y/dx^2

Solution

$y = 3x^3 - 2x^2 + 4$ is differentiated wrt x, i.e

$\frac{dy}{dx} = 9x^2 - 4x$

$d^2y/dx^2 \Rightarrow$ differentiating the result, i.e

$\frac{d^2y}{dx^2} = 18x - 4$

d^2y/dx^2 is read as (dee squared y dee x squared)

Note: *d^2y/dx^2 is different from $(dy/dx)^2$ as the first is 2nd derivative while the latter is the square of the result of dy/dx*

Sometimes a function can be written as

$f(x) = 2x^2 + 3x + 1$

instead of $y = 2x^2 + 3x + 1$

then we have that its own form of dy/dx will be $f^{/}(x)$.

Hence $f^{/}(x) = 4x + 3$

Also d^2y/dx^2 will be $f^{//}(x) = 4$

Special derivatives' result

Functions	Derivatives'result
lnx or $\log_e x$	**$^1/_x$**
e^x	**e^x**
Sin x	**cos x**
Cos x	**- sin x**
Tan x	**$\sec^2 x$**
Cot x	**- cosec^2 x**
Sec x	**sec x tan x**
Cosec x	**- cosec x cot x**

1995/38 UME

The derivative of cosec x is
A tan x cosec x B. – cot x cosec x C. tan x secx
D. – cot x sec x

Solution

Derivative of cosec x = – cot x cosec x (B)

So – the above derivatives should be part and parcel of us

2014/14 Neco Exercise 30.6

Find the derivative of tan x
A $\cos^2x$ B cosec x C $\sec^2x$
D secant x E sin x

2012/38 UTME Exercise 30.7

Find $\frac{dy}{dx}$ if y = cos x

A sin x B – sin x C tan x D – tan x

Differentiation rules

We have been discussing the general formula so far, there are certain rules to follow when we have 'complex ' or complicated functions to differentiate.

Function of a function (chain rule)

If we have a case like:
Given that $y = (x + 1)^2$ find $^{dy}/_{dx}$
For us to apply the formula
We must first expand i.e
$y = (x + 1)^2$ will become $y = (x + 1)(x + 1)$
$= x^2 + 2x + 1$
Then we differentiate $y = x^2 + 2x + 1$
$\frac{dy}{dx} = 2x + 2$

we can afford to expand it. But if we have
$y = (x + 1)^{10}$ find $^{dy}/_{dx}$
I bet you – you will exhaust the whole of your note to expand it.
The chain rule comes into play here.
Starting
(1) Given that $y = (x + 1)^2$, find $^{dy}/_{dx}$

Solution

Let $z = x + 1$ – the core or bracket content
then our original function will be
$y = z^2$

Then, $\frac{dy}{dx} = \frac{dy}{dz} \times \frac{dz}{dx}$

Please note that z is a dummy variable
It can be changed to u or v etc depending on the author or teacher

Let's go gradually,

$z = x + 1$ then $\frac{dz}{dx} = 1$

Also, $y = z^2$, then $\frac{dy}{dz} = 2z$

Substituting,

$\frac{dy}{dx} = \frac{dy}{dz} \times \frac{dz}{dx}$ will become

$= 2z \times 1$

$= 2z$

Substituting for z value

$= 2(x + 1)$

if we expand. $= 2x + 2$ as gotten earlier.

2010/40 UTME

If $y = (2x + 1)^3$, find $\frac{dy}{dx}$

A $3(2x + 1)^2$ B $3(2x + 1)$ C $6(2x + 1)$ D $6(2x + 1)^2$

Solution

$y = (2x + 1)^3$

Let $z = 2x + 1$ then $\frac{dy}{dz} = 2$

$y = z^3$ then $\frac{dy}{dz} = 3z^2$

Thus $\frac{dy}{dx} = \frac{dy}{dz} \times \frac{dz}{dx}$

$= 3z^2 \times 2$

$= 6(2x + 1)^2$ D.

2009/36 UME

If $y = 3\cos 4x$, $\frac{dy}{dx}$ equals

A $6\sin 8x$ B $-24\sin 4x$ C $12\sin 4x$ D $-12\sin 4x$

Solution

$y = 3\cos 4x$

Let $z = 4x$, $\frac{dz}{dx} = 4$

$y = 3\cos z$ and $\frac{dy}{dz} = -3\sin z$

Thus $\frac{dy}{dx} = \frac{dy}{dz} \times \frac{dz}{dx}$

$= -3\sin z \times 4$

$= -12 \sin z$

$= -12 \sin 4x$ (D)

2006/45 UME

Differentiate $(\cos\theta - \sin\theta)^2$ with respect to θ

A $1 - 2\cos\theta$ B $-2\sin\theta$ C $-2\cos 2\theta$

D $1 - 2\sin 2\theta$

Solution

$y = (\cos\theta - \sin\theta)^2$

Let $Z = \cos\theta - \sin\theta$, $\frac{dz}{d\theta} = -\sin\theta - \cos\theta$

$y = z^2$, $\frac{dy}{dz} = 2z$

$\frac{dz}{d\theta} = \frac{dy}{dz} \times \frac{dz}{d\theta}$

$= 2z \times (-\sin\theta - \cos\theta)$

$= 2(\cos\theta - \sin\theta)(-\sin\theta - \cos\theta)$

$= 2[\cos\theta(-\sin\theta - \cos\theta) - \sin\theta(-\sin\theta - \cos\theta)]$

$= 2[-\cos\theta\sin\theta - \cos^2\theta + \sin^2\theta + \cos\theta\sin\theta]$

$= 2[\sin^2\theta - \cos^2\theta]$

$= 2[(1 - \cos^2\theta) - \cos^2\theta]$

$= 2[1 - 2\cos^2\theta]$ *from trig identity we have*

$= -2(2\cos^2\theta)$

$= -2\cos 2\theta$ C.

2014/14 Neco f/m

Find $\frac{d^2y}{dx^2}$, if $y = e^{5x+3}$

A $5e^{5x+3}$ B $25e^{5x+3}$ C $5\ln(5x + 3)$

D $25\ln(5x + 3)$ E $25[\ln(5x + 3) + e^{5x+3}]$

Solution

$y = e^{5x+3}$

Let $z = 5x + 3$; $\frac{dz}{dx} = 5$

$y = e^z$, $\frac{dy}{dz} = e^z$

$\frac{dy}{dx} = \frac{dy}{dz} \times \frac{dz}{dx}$

$= e^z \times 5$

$= 5e^{5x+3}$

$\frac{d^2y}{dx^2} = \frac{dy}{dz} \times \frac{dz}{dx}$

Let $z = 5x + 3$, $\frac{dz}{dx} = 5$

$y = 5e^z$, $\frac{dy}{dz} = 5e^z$

$\frac{d^2y}{dx^2} = 5e^z \times 5 = 25e^{5x+3}$ B.

2014/13 f/m

Find the derivative of $\sqrt[3]{(3x^3+1)}$ with respect to x

A $\frac{3x}{\sqrt[3]{(3x+1)^2}}$ B $\frac{3x^2}{\sqrt[3]{(3x^3+1)^2}}$

C $\frac{3x}{\sqrt[3]{(3x^2+1)}}$ D $\frac{3x^2}{\sqrt[3]{(3x^2+1)^2}}$

Solution

$y = (3x^3 + 1)^{\frac{1}{3}}$

Let $Z = 3x^3 + 1$ then $\frac{dz}{dx} = 9x^2$

$y = Z^{\frac{1}{3}}$ then $\frac{dy}{dz} = \frac{1}{3}Z^{\frac{1}{3}-1} = \frac{1}{3}Z^{-\frac{2}{3}}$

$\frac{dy}{dx} = \frac{dy}{dz} \times \frac{dz}{dx}$

$= \frac{1}{3}Z^{-\frac{2}{3}} \times 9x^2$

$= \frac{9x^2}{3Z^{\frac{2}{3}}} = \frac{3x^2}{\sqrt[3]{(3x^3+1)^2}}$ B.

2006/47 UME

Differentiate $\left(x^2 - \frac{1}{x}\right)^2$ with respect to x

A $4x^3 - 2 + \frac{2}{x^3}$ B $4x^3 - 2 - \frac{2}{x^3}$

C $4x^3 - 3x + \frac{2}{x}$ D $4x^3 - 3x - \frac{2}{x}$

Solution

$y = \left(x^2 - \frac{1}{x}\right)^2$

Let $z = x^2 - \frac{1}{x}$, $\frac{dz}{dx} = 2x + \frac{1}{x^2}$

$y = z^2$, $\frac{dy}{dz} = 2z$

Thus $\frac{dy}{dx} = \frac{dy}{dz} \times \frac{dz}{dx}$

$= 2z \times \left(2x + \frac{1}{x^2}\right)$

$= 2\left(x^2 - \frac{1}{x}\right)\left(2x + \frac{1}{x^2}\right)$

$= 2\left[2x\left(x^2 - \frac{1}{x}\right) + \frac{1}{x^2}\left(x^2 - \frac{1}{x}\right)\right]$

$= 2[\,2x^3 - 2 + 1 - \frac{1}{x^3}\,]$

$= 2\left(2x^3 - 1 - \frac{1}{x^3}\right)$

$= 4x^3 - 2 - \frac{2}{x^3}$ B.

2009/39 PCE

If $y = \sin 2x - \cos 3x$, find $\frac{dy}{dx}$

A $\cos 2x + \sin 3x$ B $\cos 2x - \sin 3x$
C $2\cos 2x - 3\sin 3x$ D $2\cos 2x + 3\sin 3x$

Solution

$y = \sin 2x - \cos 3x$

$$\frac{dy}{dx} = \frac{d(\sin 2x)}{dx} - \frac{d(\cos 3x)}{dx}$$

$\frac{d(\sin 2x)}{dx}$ as $y = \sin z$

$z = 2x$ and $\frac{dz}{dx} = 2$

$\frac{dy}{dx} = \frac{dy}{dz} \times \frac{dz}{dx}$

$= \cos z \times 2$

$= 2\cos 2x$

$\frac{d(\cos 3x)}{dx}$ as $y = \cos z$

$z = 3x$ and $\frac{dz}{dx} = 3$

$\frac{dy}{dx} = \frac{dy}{dz} \times \frac{dz}{dx}$

$= -\sin z \times 3$

$= -3\sin 3x$

Substituting

Thus $\frac{dy}{dx} = 2\cos 2x - (-3\sin 3x)$

$= 2\cos 2x + 3\sin 3x$ D.

2010/17 Neco f/m Exercise 30.8

Differentiate $\cos\left(\frac{\pi}{2} - 3x\right)$ with respect to x

A $3\cos\left(\frac{\pi}{2} - 3x\right)$ B $3\sin\left(\frac{\pi}{2} - 3x\right)$

C $-3\cos\left(\frac{\pi}{2} - 3x\right)$ D $-\sin\left(\frac{\pi}{2} - 3x\right)$ E $-3\sin\left(\frac{\pi}{2} - 3x\right)$

2005/6 UME Exercise 30.9

Find the derivative of $y = \sin(2x^3 + 3x - 4)$

A $-\cos(2x^3 + 3x - 4)$ B $-(6x^2 + 3)\cos(2x^2 + 3x - 4)$
C $\cos(2x^3 + 3x - 4)$ D $(6x^2 + 3)\cos(2x^3 + 3x - 4)$

2013/35 UTME Exercise 30.10

If $y = (2x + 2)^3$, find $\frac{dy}{dx}$

A $3(2x + 2)$ B $6(2x + 2)^2$ C $3(2x + 2)^2$ D $6(2x + 2)$

2014/39 UTME Exercise 30.11

If $y = \cos 3x$, find $\frac{dy}{dx}$

A $3\sin 3x$ B $-3\sin 3x$ C $^1/_3\sin 3x$ D $-{}^1/_3\sin 3x$

2007/15 UME Exercise 30.12

If $y = (1 + x)^2$, find $\frac{dy}{dx}$

A $2x - 1$ B $x - 1$ C $2 + 2x$ D $1 + 2x$

2013/22 Neco f/m Exercise 30.13

Given that $y = (2x + 3)^{-\frac{1}{2}}$, find $\frac{d^2y}{dx^2}$

A $-(2x + 3)^{-1\frac{1}{2}}$ B $3(2x + 3)^{-2\frac{1}{2}}$

C $-\frac{1}{2}(2x + 3)^{-1\frac{1}{2}}$ D $\frac{3}{4}(2x + 3)^{-2\frac{1}{2}}$ E $2(2x + 3)^{\frac{1}{2}}$

2016/10a Neco Exercise 30.14

Differentiate $y = (2x^2 + 3)^5$ with respect to x.

2016/60 Neco Exercise 30.15

If $\mathbf{y} = (2x^2 + 7)^4$, find $\frac{dy}{dx}$

A. $4x$ B. $4(2x^2 + 7)^3$ C. $4x(2x^2 + 7)^3$
D. $16(2x^2 + 7)^3$ E. $16x(2x^2 + 7)^3$

Product Rule

If y = UV where u and v are some functions of x. then

$$\frac{dy}{dx} = U\frac{dv}{dx} + V\frac{du}{dx}$$

Simply as $\frac{dy}{dx} = Udv + Vdu$

Example 1

If $y = (3x^6 - 10)(x^4 + 30)$, find $^{dy}/_{dx}$

Solution

Let y = uv

i.e $y = (3x^6 - 10)(x^4 + 30)$

Thus, $\frac{dy}{dx} = Udv + Vdu$

$= (3x^6 - 10) \times 4x^3 + (x^4 + 30) \times 18x^5$

$= 12x^9 - 40x^3 + 18x^9 + 540x^5$

$= 30x^9 + 540x^5 - 40x^3$

2000/6 Neco f/m

Find the derivative of $y = (x + 1)(x + 2)$

A. $2x - 3$ B. $2x + 3$ C. $2x + 4$ D. $3x - 2$ E. $3x + 2$

Solution

$y = (x + 1)(x + 2)$

Then $y = uv$

$^{dy}/_{dx} = udv + vdu$

$= (x + 1) \times 1 + (x + 2) \times 1$

$= x + 1 + x + 2$

$= 2x + 3$ (B)

2005/37 Neco fm

If $y = x \sin 2x$, find $^{dy}/_{dx}$

A. $-2\cos 2x + \sin 2x$ B. $-2x \sin \cos x + 2 \sin 2x$
C. $-2x \cos 2x + \sin 2x$ D. $\sin 2x - x\cos 2x$
E. $\sin 2x + 2x \cos 2x$

Solution

$y = x \sin 2x$

$^{dy}/_{dx} = Udv + Vdu$

$= x\, \mathbf{d}(\sin 2x) + \sin 2x \times \mathbf{d}(x)$

Recall that applying chain rule $\frac{d}{dx}(\sin 2x) = 2\cos 2x$

Thus, $^{dy}/_{dx} = x \times 2\cos 2x + \sin 2x \times 1$

$= 2x \cos 2x + \sin 2x$ (E)

Addition is commutative i.e $2 + 3 = 3 + 2$ so,
$2x\cos 2x + \sin 2x$ is same as $\sin 2x + 2x\cos 2x$

2001/38 UME

If $y = x \sin x$, find $^{dy}/_{dx}$ when $x = \frac{\pi}{2}$

A. $-\frac{\pi}{2}$ B. -1 C. 1 D. $\frac{\pi}{2}$

Solution

Let $y = uv$

i.e $y = x \sin x$

Thus, $\frac{dy}{dx} = Udv + Vdu$

$= x \cos x + \sin x \times 1$

$= x \cos x + \sin x$

$\boxed{\text{When } x = \frac{\pi}{2} \text{ means } 90^0}$

$= \frac{\pi}{2} \cos 90^0 + \sin 90^0$

$= \frac{\pi}{2} \times 0 + 1$

$= 1$ (C)

1992/37 UME

If $y = x \sin x$, find d^2y/dx^2

A. $2\cos x - x \sin x$ B. $\sin x + x \cos x$
C. $\sin x - x \cos x$ D. $x \sin x - 2 \cos x$

Solution

The function is in form uv,
we find dy/dx first before d^2y/dx^2

$y = \underset{u}{x} \; \underset{v}{\sin x}$

$\frac{dy}{dx} = Udv + Vdu$

$= x \cos x + \sin x \times 1$

$= x \cos x + \sin x$

Then, d^2y/dx^2 will be differentiating the result.
Where $x \cos x$ is another product rule case.

Thus, $\frac{d^2y}{dx^2} = \frac{d}{dx}(x\cos x) + \frac{d}{dx}(\sin x)$

Let $y = \underset{u}{x} \; \underset{v}{\cos x}$

$d/dx\,(x \cos x) \Rightarrow dy/dx = udv + vdu$

$= x \times (-\sin x) + \cos x \times 1$

$= -x \sin x + \cos x$

Thus, $\frac{d^2y}{dx^2} = -x \sin x + \cos x + \cos x$

$= -x \sin x + 2 \cos x$

$= 2 \cos x - x \sin x$ (A)

2004/37 PCE

If $y = (x^2 + 2x + 4)(x - 2)$, find dy/dx

A. $-3x^2$ B. $3x^2$ C. $3x^4$ D. $2x^4$

Solution

Applying product rule

$y = \underset{u}{(x^2 + 2x + 4)} \; \underset{v}{(x - 2)}$

$\frac{dy}{dx} = Udv + Vdu$

$= (x^2 + 2x + 4) \times 1 + (x - 2)(2x + 2)$

$= (x^2 + 2x + 4) + (2x^2 + 2x - 4x - 4)$

$= x^2 + 2x + 4 + 2x^2 - 2x - 4$

Collect like terms together

$= x^2 + 2x^2 + 2x - 2x + 4 - 4$

$= 3x^2$ (B)

2004/15

Find the derivative of the function $y = 2x^2(2x - 1)$ at the point $x = -1$

A. 18 B. 16 C. -4 D. -6

Solution

Applying product rule

$y = \underset{u}{2x^2} \underset{v}{(2x - 1)}$

$\frac{dy}{dx} = Udv + Vdu$

$= 2x^2 \times 2 + (2x - 1) \times 4x$

$= 4x^2 + 8x^2 - 4x$

$= 12x^2 - 4x$

<u>At point $x = -1$</u>

$12x^2 - 4x = 12(-1)^2 - 4(-1)$

$= 12 + 4$ i.e 16 (B)

2007/12UME Exercise 30.16

If $y = x \cos x$, find dy/dx

A. $\sin x - x\cos x$ B. $\sin x + x\cos x$ C. $\cos x + x\sin x$ D. $\cos x - x\sin x$

2004/13 UME Exercise 30.17

Find the derivative of $(2 + 3x)(1 - x)$ with respect to x.

A. 6 B. -3 C. $1 - 6x$ D. $6x - 1$

2005/6 PCE Exercise 30.18

Find the derivative of $f(x) = 2x(x^2 + 3)$

A. $4x^2 + 3$ B. $2(x^2 + 3)$ C. $6(x^2 + 1)$ D. $6x^2 + 2$

Cases involving product and chain rules

2015/11a Neco

Differentiate $y = x^2(2x+1)^2$ with respect to x

Solution

$y = x^2 (2x+1)^2$

u v

$\frac{dy}{dx} = udv + vdu$

$= x^2\mathbf{dv} + (2x+1)^2 \times 2x$

Here **dv** is a function of function problem

Let $y = (2x+1)^2$

$z = (2x+1)$ then $\frac{dz}{dx} = 2$

$y = z^2$ then $\frac{dy}{dz} = 2z$

Thus **dv** $\Rightarrow \frac{dy}{dx} = \frac{dy}{dz} \times \frac{dz}{dx}$

$= 2z \times 2$

$= 4z$ i.e $4(2x+1)$

Substituting for **dv**

$\frac{dy}{dx} = x^2 \times 4(2x+1) + (2x+1)^2 \times 2x$

$= 4x^2(2x+1) + 2x(2x+1)^2$

$= 2x(2x+1)(2x+2x+1)$

$= 2x(2x+1)(4x+1)$

2009/37 UME

If $S = (2+3t)(5t-4)$, find $\frac{ds}{dt}$ when $t = {}^4/_5$ sec.

A 0 units per sec B 15 units per sec
C 22 units per sec D 26 units per sec

Solution

$S = (2+3t)(5t-4)$

u v

$\frac{ds}{dt} = udv + vdu$

$= (2+3t) \times 5 + (5t-4) \times 3$

$= 10 + 15t + 15t - 12$

$= 30t - 2$

When $t = {}^4/_5$ secs

$\frac{ds}{dt} = 30\left(\frac{4}{5}\right) - 2$

$= 24 - 2$

$= 22$ units per sec (C)

2014/90 f/m

Differentiate $(x-3)(x^2+5)$ with respect to x

Solution

$y = (x-3)(x^2+5)$

u v

$\frac{dy}{dx} = udv + vdu$

$= (x-3) \times 2x + (x^2+5) \times 1$

$= 2x(x-3) + (x^2+5)$

$= 2x^2 - 6x + x^2 + 5$

$= 3x^2 - 6x + 5$

2013/36 UTME

If $y = x \sin x$, find $\frac{dy}{dx}$

A $\cos x + x \sin x$ B $\sin x + x \cos x$
C $\sin x - \cos x$ D $\cos x - x \sin x$

Solution

$y = x \sin x$

u v

$\frac{dy}{dx} = udv + vdu$

$= x \cos x + \sin x \times 1$

$= x \cos x + \sin x$ B.

2001/35 UME Exercise 30.19

Differentiate $(2x+5)^2(x-4)$ with respect to x

A. $4(2x+5)(x-4)$ B. $4(2x+5)(4x-3)$
C. $(2x+5)(2x-13)$ D. $(2x+5)(6x-11)$

Quotient Rule

If $y = \frac{u}{v}$, where u and v are some functions of x, then

$$\frac{dy}{dx} = \frac{vdu - udv}{v^2}$$

It has to do with functions in fraction quoted as:
Bottom dee top minus top dee bottom , all over bottom square

2005/11a Neco f/m

Differentiate with respect to x, the function

$y = \frac{4x^3 + 8}{2x^2 + 3}$

Solution

$y = \frac{u(x)}{v(x)}$ i.e $\frac{4x^3+8}{2x^2+3}$

$\frac{dy}{dx} = \frac{vdu - udv}{v^2}$

$= \frac{(2x^2+3) \times 12x^2 - (4x^3+8)4x}{(2x^2+3)^2}$

$= \frac{24x^4 + 36x^2 - 16x^4 - 32x}{(2x^2+3)^2}$

$= \frac{8x^4 + 36x^2 - 32x}{(2x^2+3)^2}$

2004/18 Neco f/m

Differentiate the function $\frac{4x^4 + x^3 - 5}{4x^2}$ with respect to x.

A. $2x + \frac{5}{2x^3} + \frac{1}{4}$ B. $x^2 + \frac{x}{4} - \frac{5}{4x}$ C. $2x^2 + \frac{5}{2x} + \frac{1}{4x}$

D. $2x - \frac{5}{2x^3} + \frac{1}{2}$ E. $x^2 - \frac{x}{4} - \frac{5}{4x}$

Solution

$y = \frac{u(x)}{v(x)}$ i.e $\frac{4x^4 + x^3 - 5}{4x^2}$

$\frac{dy}{dx} = \frac{vdu - udv}{v^2}$

$= \frac{4x^2(16x^3 + 3x^2) - (4x^4 + x^3 - 5) \times 8x}{(4x^2)^2}$

$= \frac{64x^5 + 12x^4 - 32x^5 - 8x^4 + 40x}{16x^4}$

$= \frac{32x^5}{16x^4} + \frac{4x^4}{16x^4} + \frac{40x}{16x^4}$

$= 2x + \frac{1}{4} + \frac{5}{2x^3}$ (A)

2008/39 PCE

If $y = \frac{\cos x}{\sin x} + 1$, find $\frac{dy}{dx}$

A tan x B cot x C – $\tan^2 x$ D – $\text{cosec}^2 x$

Solution

$$y = \frac{\cos x}{\sin x} + 1$$

By simple fractional operation

$$y = \frac{\cos x + \sin x}{\sin x} \quad \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v du - u dv}{v^2}$$

$$= \frac{\sin x\,(-\sin x + \cos x) - (\cos x + \sin x)\cos x}{(\sin x)^2}$$

$$= \frac{-\sin^2 x + \cos x \sin x - \cos^2 x - \cos x \sin x)}{\sin^2 x}$$

$$= \frac{-\sin^2 x - \cos^2 x}{\sin^2 x}$$

$$= \frac{-(\sin^2 x + \cos^2 x)}{\sin^2 x}$$

By trig identities

$$= -\frac{1}{\sin^2 x} = -\text{cosec}^2 x \qquad \text{D.}$$

2005/11(ii) Exercise 30.20

Differentiate with respect to x:

$$y = \frac{2x - 3}{x^2 + 5}$$

2003/11b (Nov) Exercise 30.21

Find the derivative with respect to x of $y = \frac{2x}{x^2 + 3}$

1996/12 (Nov) Exercise 30.22

If $y = \frac{1 + x}{1 - x}$, find $\frac{dy}{dx}$

A. $\frac{1}{(1-x)^2}$ B. $\frac{2}{(1-x)^2}$ C. $\frac{1+x}{(1-x)^2}$ D. $\frac{x}{(1-x)^2}$

E. $\frac{-2}{(1-x)^2}$

2010/19 Neco Exercise 30.23

Differentiate $\frac{4x^3 + 3x^2 - 5}{5x^2}$ with respect to x

A $\frac{2}{x^3} + \frac{4}{5}$ B $\frac{2}{x^3} + \frac{4x}{5}$ C $\frac{12x^3 + 6x}{10x}$

D $\frac{12x^3 + 6x}{25x^4}$ E $\frac{4x}{5} - \frac{1}{x^2} + \frac{3}{5}$

2012/22 f/m Exercise 30.24

Differentiate $\left(\frac{x}{x+1}\right)$ with respect to x

A $\frac{1}{x+1}$ B $-\frac{1}{(x+1)^2}$ C $\frac{1-x}{(x+1)}$ D $\frac{1}{(x+1)^2}$

Quotient and chain rule

2003/5 Neco

Find the derivative of y with respect to x if $y = \sqrt{\frac{1+x}{1-x}}$

Solution

Applying chain rule

$$y = \sqrt{\frac{1+x}{1-x}} \quad \text{is same as} \quad y = \frac{(1+x)^{1/2}}{(1-x)^{1/2}}$$

Let $U = \frac{1+x}{1-x}$ then $y = U^{1/2}$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$= \frac{1}{2}u^{-\frac{1}{2}} \times \frac{du}{dx}$$

Next we find $\frac{du}{dx}$ by applying Quotient rule

$$\frac{du}{dx} = \frac{(1-x) \times 1 - (1+x) \times (-1)}{(1-x)^2}$$

$$= \frac{1 - x + 1 + x}{(1-x)^2}$$

$$= \frac{2}{(1-x)^2}$$

Thus, $\frac{dy}{dx} = \frac{1}{2}\left(\frac{1+x}{1-x}\right)^{-\frac{1}{2}} \times \frac{2}{(1-x)^2}$

$$= \frac{1}{2}\left(\frac{1-x}{1+x}\right)^{\frac{1}{2}} \times \frac{2}{(1-x)^2}$$

$$= \frac{(1-x)^{\frac{1}{2}}}{(1+x)^{1/2}} \times \frac{1}{(1-x)^2}$$

$$= \frac{1}{(1+x)^{1/2}(1-x)^{3/2}}$$

2000/28 Neco

Differentiate with respect to x, $\frac{\cos x}{\sin 2x}$

A. $-\frac{\sin 3x}{\sin^2 2x}$ B. $\frac{-\sin^2 2x}{\sin 3x \sin^2 2x}$

C. $\frac{-\sin 2x \sin x + 2\cos x \cos 2x}{\sin^2 2x}$

D. $\frac{\sin 2x \cos x + \cos 2x \sin x}{\sin^2 2x \;\; \sin 3x \;\; \sin^2 2x}$

E. $\frac{-[\sin 2x \sin x + 2\cos x \cos 2x]}{\sin^2 2x}$

Solution

Let $y = \frac{\cos x}{\sin 2x}$

Applying quotient rule

$$\frac{dy}{dx} = \frac{\sin 2x \times (-\sin x) - \cos x \times (2\cos 2x)}{(\sin 2x)^2}$$

$$= \frac{-\sin 2x \sin x - 2\cos x \cos 2x}{\sin^2 2x} \quad \text{(E)}$$

1994/10 b (Nov) Exercise 30.25

Differentiate, with respect to x

$y = \dfrac{3 - 2x}{(1 + x)^2}$, simplifying your answer

1993/15 (Nov) Exercise 30.26

If $y = \dfrac{x}{(x^2 - 1)^{1/2}}$ find $\dfrac{dy}{dx}$

A. $\dfrac{2}{(x^2 - 1)^3}$ B. $\dfrac{-1}{(x^2 - 1)^{3/2}}$ C. $\dfrac{1}{(x^2 - 1)^3}$ D. $\dfrac{-2x^2 + 1}{(x^2 - 1)^{3/2}}$

E. $\dfrac{2x^2 - 1}{(x^2 - 1)^{3/2}}$

2001/10 Neco Exercise 30.27

The derivative of $\dfrac{\sin 3x}{e^{-2x}}$ with respect to x is

A. $\dfrac{3\cos 3x + 2\sin 3x}{e^{-2x}}$ B. $\dfrac{3\sin 3x + 2\cos 3x}{e^{-2x}}$

C. $\dfrac{3\cos 3x + 2\sin 3x}{e^{-4x}}$ D. $\dfrac{3\cos 3x - 2\sin 3x}{e^{-2x}}$

E. $\dfrac{3\cos 3x - 2\sin 3x}{e^{-4x}}$

2010/3 Neco Exercise 30.28

Find $\dfrac{dy}{dx}$ if $y = \dfrac{x^3}{\sqrt{2x^2 + 3}}$

Application of Differentiation

Gradient (slope at a given point).

The concept of differentiation is the same as that of gradient or slope. So when we say the gradient or slope of a function at a point we are only referring to the differentiation of that function and substituting for the x value given or among the points given. Other concepts required in this subtopic will be shown as we go

2004/15

The gradient of the curve $y = 3x^2 + 11x + 7$ at the point P(x, y) is – 1. find the co – ordinates of P.

A. (-2, +3) B. (-2, -3) C. (-1, $-^5/_2$) D. (-3, -2)

Solution

$y = 3x^2 + 11x + 7,$

$\dfrac{dy}{dx} = 6x + 11$ i.e gradient

But we are told that gradient is –1

Thus, $^{dy}/_{dx} = -1$

$6x + 11 = -1$

$6x = -12$

$x = -2$

To get y co-ordinates

Substitute x = - 2 into the original equation

$y = 3(-2)^2 + 11(-2) + 7$

$= 12 - 22 + 7$

$= -3$

Co-ordinates of P is (-2, -3) (B)

2002/15

Find co-ordinates of the point at which the gradient of the curve $y = x^2 - x + 4$ is 3

A. (1, 4) B. (2, 6) C. (1, 6) D. (3, 5)

Solution

$y = x^2 - x + 4$

$\dfrac{dy}{dx} = 2x - 1$ same as gradient

But we are told that gradient is 3

Thus, $^{dy}/_{dx} = 3$

$2x - 1 = 3$

$2x = 3 + 1$

$2x = 4$

$x = 2$

To get the y co – ordinates, substitute x = 2 into the original equation.

$y = (2)^2 - 2 + 4$

$= 4 - 2 + 4$

$= 6$

Co – ordinate of the point (2, 6)

1995/2 (Nov)

Find the differential coefficient of the function

$y = \dfrac{1}{x^2 + 9}$ at the point (4, $^1/_{25}$)

A. $-^8/_{625}$ B. $-^4/_{625}$ C. $^2/_{625}$ D. $^4/_{625}$ E. $^8/_{625}$

Solution

Differential coefficient is same as $^{dy}/_{dx}$. Thus

$y = \dfrac{1}{x^2 + 9}$: i.e $y = (x^2 + 9)^{-1}$

Let $z = x^2 + 9$ then $y = z^{-1}$

$\dfrac{dy}{dx} = \dfrac{dy}{dz} \times \dfrac{dz}{dx}$

$= -1z^{-2} \times 2x$ i.e $\dfrac{-2x}{z^2}$

$= \dfrac{-2x}{(x^2 + 9)^2}$

At point (4, $^1/_{25}$), we put x = 4 into our dy/dx result

$= \dfrac{-8}{(25)^2}$ i.e $\dfrac{-8}{625}$ (A)

1995/40 UME

Find the gradient of the curve $y = 2\sqrt{x} - \dfrac{1}{x}$ at the point x = 1

A.0 B.1 C.2 D.3

Solution

$y = 2\sqrt{x} - \dfrac{1}{x} \Rightarrow y = 2x^{1/2} - x^{-1}$

$\dfrac{dy}{dx} = x^{-1/2} + x^{-2}$

$= \dfrac{1}{\sqrt{x}} + \dfrac{1}{x^2}$ which is the gradient

At point x = 1

$= \dfrac{1}{\sqrt{1}} + \dfrac{1}{1^2} = \dfrac{1}{1} + \dfrac{1}{1} = 2$ (C)

2002/17 UME

The slope of the tangent to the curve $y = 3x^2 - 2x + 5$ at the point (1, 6) is

A.6 B.5 C .4 D.1

Solution

$y = 3x^2 - 2x + 5$

$$\frac{dy}{dx} = 6x - 2 \text{ i.e slope}$$
At point (1, 6) ; x = 1
put x = 1 into 6x – 2
6(1) – 2 = 4 (C)

2003/39 UME
Find the slope of the curve $y = 2x^2 + 5x - 3$ at (1, 4) .
A. 4 B.6 C.7 D.9
Solution
$y = 2x^2 + 5x - 3$
$$\frac{dy}{dx} = 4x + 5 \text{ i.e slope}$$
At point (1, 4); x = 1
Put x = 1 into 4x + 5
4(1) + 5 = 9 (D)

1995/42 PCE
Find the points on the curve $y = x^3 + 6x^2 - 15x + 1$ at which the gradient is 0.
A.(- 5, 101), (1, - 7) B.(- 5, -101), (1, - 7)
C. (5, - 101) , (- 1, - 7) D.(5, 101) , (- 1, 7)
Solution
$y = x^3 + 6x^2 - 15x + 1$
$$\frac{dy}{dx} = 3x^2 + 12x - 15$$
Gradient is 0 ⇒ dy/dx = 0 i.e
$3x^2 + 12x - 15 = 0$ divide through by 3
$x^2 + 4x - 5 = 0$ factorising
$x^2 + 5x - x - 5 = 0$
$x (x + 5) - 1 (x + 5) = 0$
$(x + 5) (x - 1) = 0$
$x + 5 = 0$ or $x - 1 = 0$
x = – 5 or 1
Use x values to find y
when x = 1
$y = (1)^3 + 6(1)^2 - 15(1) + 1$
$= -7$
(1, - 7)
when x = - 5 $y = (- 5)^3 + 6 (- 5)^2 - 15(- 5) + 1$
$= - 125 + 150 + 75 + 1$
$= 101$
(- 5 , 101) (A)

2005/3 PCE
At what point is the gradient of the curve $x^2 - 6x + 3$ equal to zero ?
A. – 2 B. 3 C. – 3 D. 2
Solution
$f(x) = x^2 - 6x + 3$
Gradient of curve implies
$f'(x) = 2x - 6$
Gradient = 0 implies $2x - 6 = 0$ solving
$2x = 6$
$x = {}^6/_2$ Thus, x = 3 (B)

2009/11 Exercise 30.29
The gradient of point P on the curve $y = 3x^2 - x + 3$ is 5. Find the coordinates of P.
A. (1, 5) B (1, 7) C (1, 13) D (1, 17)

2006/26 Exercise 30.30
Find the coordinates of the point on the curve $y = x^2 + 4x - 2$, where the gradient is zero.
A (-2, 10) B (-2, 2) C (-2, -2) D (-2, -6)

1994/15 Exercise 30.31
Find the differential coefficient of $y = \frac{-5}{x^2 + 4}$ at the point (1, -1).
A. -2 B. -1 C. $-{}^2/_5$ D. ${}^2/_5$ E. 2

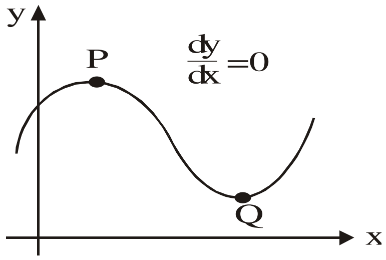
2009/30 (Nov) Exercise 30.32
The equation of a curve is given by $y = \frac{2}{x+3}$
Find its gradient at the point $\left(1, \frac{1}{2}\right)$.
A $-\frac{1}{4}$ B $-\frac{1}{8}$ C $\frac{1}{8}$ D $\frac{1}{4}$

2004/38 PCE Exercise 30.33
The gradient of the curve $y = 3x^2 - 5x + 6$ when x = 1 is
A.1 B.4 C.6 D.7

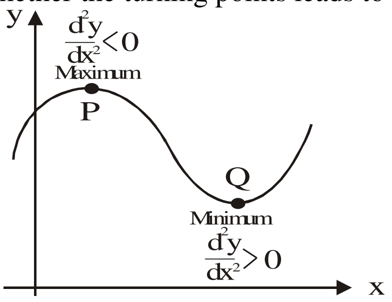
2001/37 PCE Exercise 30.34
The gradient of the curve $y = x^2 + 4x$ at the point (2, 10)
A.0 B.8 C.12 D.24

Turning Points (maximum, & minimum)

At turning points P and Q shown below;



But it is the second differential (d^2y/dx^2) that reveal whether the turning points leads to a maximum or minimum.



The value of x for maximum or minimum point
In this case the question already specifies the turning point to be a minimum or maximum point. Note the use of the phrase 'value of x' in the subsequent questions

2009/38 UME
What value of x will make the function x(4 – x) a minimum?
A 4 B 3 C 2 D 1

Solution

At minimum $\frac{dy}{dx} = 0$

$y = x(4 - x)$

$y = 4x - x^2$

$\frac{dy}{dx} = 4 - 2x$

$\Rightarrow 4 - 2x = 0$

$4 = 2x$

$^4/_2 = ^{2x}/_2$

$2 = x$ (C)

We stopped here as specified by the question "value of x"

2010/42 UMTE

At what value of x does the function $y = -3 - 2x + x^2$ attain a minimum value

A 1 B – 4 C – 1 D 4

Solution

At minimum $\frac{dy}{dx} = 0$

$y = -3 - 2x + x^2$

$\frac{dy}{dx} = -2 + 2x$

$\Rightarrow -2 + 2x = 0$

$2x = 2$

$x = 1$ (A)

We stopped here as specified by the question "value of x"

1998/36 UME

For what value of x does $6\sin(2x - 25)^0$ attain its maximum value in the range $0^0 \le x \le 180^0$?

A. 12 ½ B. 32 ½ C. 57 ½ D. 14 ½

Solution

We let $y = 6\sin(2x - 25)^0$

At min or max; $dy/dx = 0$

Applying function of function rule

$dy/dx = 12\cos(2x - 25)^0$

It follows that

$12\cos(2x - 25) = 0$

which means $\cos(2x - 25) = 0$

it implies $2x - 25 = \cos^{-1} 0$

$2x - 25 = 90$

$2x = 25 + 90$

$x = \frac{115}{2}$ i.e 57 ½ (C)

1996/38 PCE

A swimming pool is treated periodically to control harmful bacterial growth. The concentration of bacteria per cm^3 after t days is given by

$C(t) = 30t^2 - 240t + 500$, In how many days after a treatment will the concentration be minimal ?

A.10 B.8 C.6 D.4

Solution

At minimal, $dc/dt = 0$

In this case, $dc/dt = 60t - 240$

It follows that $60t - 240 = 0$

$60t = 240$

$\frac{60t}{60} = \frac{240}{60}$

$t = 4$ days (D)

1992/19 (Nov) f/m

The maximum point of the curve $y = x - 2\sin x$, $0 \le x < 2\pi$ occurs when x equals

A. $^{\pi}/_6$ B. $^{\pi}/_3$ C. $^{5\pi}/_3$ D. $^{2\pi}/_3$ E. $^{4\pi}/_3$

Solution

At maximum or minimum $^{dy}/_{dx} = 0$

Thus, $y = x - 2\sin x$ will become

$^{dy}/_{dx} = 1 - 2\cos x$

Hence $1 - 2\cos x = 0$

$2\cos x = 1$

$\cos x = ^1/_2$

$\cos x = 0.5$

$x = \cos^{-1} 0.5$

$= 60^0$

The rage is from 0^0 to 2π i.e 360

Next we find other multiples of 60^0 that will give 0.5

120^0, cos 120 = - 0.5 not accepted

180^0 cos 180^0 = -1 not accepted

240^0 cos 240^0 = - 0.5, not accepted

300 cos300^0 = 0.5 accepted

Thus, $x = 60^0$ or 300^0

But for maximum $\frac{d^2y}{dx^2} < 0$

which is $\frac{d^2y}{dx^2} = 2\sin x < 0$

For $2\sin x < 0$, $x = 300^0$ and **not** 60^0

Next convert 300^0 to radians

i.e $\frac{300}{180} \times \pi = ^5/_3\,\pi$ (C)

2010/38 Exercise 30.35

Find the maximum value of $2 + \sin(\theta + 25^0)$

A 1 B 2 C 3 D 4

2003/46 PCE Exercise 30.36

If $y = 3x^2 - 2x + 1$, find the value of x when $dy/dx = 0$

A. $^1/_3$ B. $^1/_2$ C.1 D. $^1/_4$

Cubic function (x^3) case

2012/15 f/m

If $y = x^3 - x^2 - x + 6$, find the value of x at the turning points

A $\frac{1}{2}, -3$ B $\frac{1}{3}, -\frac{1}{2}$ C $1, -\frac{1}{3}$ D $1, \frac{1}{3}$

Solution

At turning point $\frac{dy}{dx} = 0$

$y = x^3 - x^2 - x + 6$

$\frac{dy}{dx} = 3x^2 - 2x - 1$

At turning point: $3x^2 - 2x - 1 = 0$

Factorizing

$3x^2 - 3x + x - 1 = 0$

$3x(x - 1) + 1(x - 1) = 0$

$(3x + 1)(x - 1) = 0$

$3x + 1 = 0$ or $x - 1 = 0$

$3x = -1$ or $x - 1 = 0$

$x = -\frac{1}{3}$ or 1 C.

2009/40 PCE

Find the value of x at which the function $y = x^3 - 2x^2 + x + 4$ is minimum

A $^1/_3$ B 1 C 3 D 4

Solution

At minimum $\frac{dy}{dx} = 0$

$y = x^3 - 2x^2 + x + 4$

$\frac{dy}{dx} = 3x^2 - 4x + 1$

$3x^2 - 4x + 1 = 0$

Factorizing

$3x^2 - 3x - x + 1 = 0$

$3x(x - 1) - 1(x - 1) = 0$

$(3x - 1)(x - 1) = 0$

$x = {}^1/_3$ or 1

To know the minimum point

$\frac{d^2y}{dx^2} = 6x - 4$

The minimum point is at

$\frac{d^2y}{dx^2} > 0$

$6(\mathbf{1}) - 4 > 0$ (B)

2014/8 Neco ***Adjusted***

At what stationary point of x is the function $y = x^3 - 2x^2 + x$ maximum

Solution

At stationary points $\frac{dy}{dx} = 0$

$y = x^3 - 2x^2 + x$

$\frac{dy}{dx} = 3x^2 - 4x + 1$

$\Rightarrow 3x^2 - 4x + 1 = 0$

Factorizing

$3x^2 - 3x - x + 1 = 0$

$3x(x - 1) - 1(x - 1) = 0$

$(3x - 1)(x - 1) = 0$

$3x - 1 = 0$ or $x - 1 = 0$

$3x = 1$ or $x - 1 = 0$

$x = {}^1/_3$ or 1

At maximum $\frac{d^2y}{dx^2} < 0$

$\frac{d^2y}{dx^2} = 6x - 4$

When $x = {}^1/_3$

$6({}^1/_3) - 4 < 0$

Thus $^1/_3$ is maximum

2006/50 UME

Find the value of x for which the function $3x^3 - 9x^2$ is minimum

A 2 B 0 C 5 D 3

Solution

At minimum $\frac{dy}{dx} = 0$

$y = 3x^3 - 9x^2$

$\frac{dy}{dx} = 9x^2 - 18x$

$\Rightarrow 9x^2 - 18x = 0$

Factorizing

$9x(x - 2) = 0$

$9x = 0$ or $x - 2 = 0$

$x = 0$ or $x = 2$

But at minimum $\frac{d^2y}{dx^2} > 0$

$\frac{d^2y}{dx^2} = 18x - 18$

For $18x - 18 > 0$, we check

<u>When x = 0</u>

$18(0) - 18 < 0$

<u>When x = 2</u>

$18(2) - 18 > 0$

Thus min value is 2 A

The value of y for maximum or minimum point

Here we are required to go a step further from finding the value of x for max or min points; by substituting the value of x into the original equation. Though we have two types of cases here:

Quadratic functions (x^2)

Cubic functions (x^3)

Quadratic function case

2005/5 UME

The maximum value of the function $f(x) = 2 + x - x^2$ is

A $\frac{3}{2}$ B $\frac{1}{2}$ C $\frac{9}{4}$ D $\frac{7}{4}$

Solution

At maximum $f'(x) = 0$

$f(x) = 2 + x - x^2$

$f'(x) = 1 - 2x$

$\Rightarrow 1 - 2x = 0$

$1 = 2x$

$^1/_2 = x$

To get maximum value, we substitute $x = {}^1/_2$ into $f(x) = 2 + x - x^2$

$= 2 + {}^1/_2 - \left(\frac{1}{2}\right)^2$

$= 2 + {}^1/_2 - {}^1/_4$

$= \frac{8 + 2 - 1}{4}$

$= \frac{9}{4}$ C.

2015/12 f/m

Find the minimum value of $y = x^2 + 6x - 12$

A – 21 B – 12 C – 6 D – 3

Solution

At minimum value $\frac{dy}{dx} = 0$

$y = x^2 - 6x - 12$

$\frac{dy}{dx} = 2x + 6$

$\Rightarrow 2x + 6 = 0$

$2x = -6$

$x = -\frac{6}{2} = -3$

To get minimum value: we substitute x = – 3 into

$y = x^2 - 6x - 12$

$y = (-3)^2 + 6(-3) - 12$

$= 9 - 18 - 12 = -21$ A.

2003/4 f/m

A blacksmith produced x articles at a total cost of \$ $(200 - 48x + 3x^2)$. If each article is sold at \$ $\frac{3}{5}x$, find:

(a) The value of x for which the blacksmith makes a maximum profit.

(b)The overall maximum profit

Solution

(a) CP Cost price of x articles = \$$(200 - 48x + 3x^2)$

SP Selling price of **x** articles

= selling price of one article × x articles

= \$ $\frac{3}{5}x \times x$

$= \frac{3}{5}x^2$

Profit(P) = Sp – Cp

$= \frac{3}{5}x^2 - (200 - 48x + 3x^2)$

$P = \frac{3}{5}x^2 - 200 + 48x - 3x^2$

At maximum profit $\frac{dp}{dx} = 0$

$\frac{dp}{dx} = \frac{6}{5}x + 48 - 6x$

At maximum profit $\frac{6}{5}x + 48 - 6x = 0$

$6x - 30x = -240$

$x = 10$ articles

(b) For overall maximum profit,
we put x = 10 into our P equation

$P_{\text{overall max,}} = \frac{3}{5}(10)^2 - 200 + 48(10) - 3(10)^2$

$= \frac{300}{5} - 200 + 480 - 300$

$= \$40$

2000/39 PCE

30metres of fencing wire is available to make a rectangular enclosure. Find the maximum area possible.

A.629m^2 B525m^2 C.225m^2 D.125m^2

Solution

Let the length of the rectangle = x

It follows that breath = 30 – x

Area of rectangle = length × breath

= x(30 – x)

Area (A) = $30x - x^2$

For area to be maximum

dA/dx = 0

$\frac{d}{dx}(30x - x^2) = 0$

$30 - 2x = 0$

$30 = 2x$

$\frac{30}{2} = \frac{2x}{2}$

$15 = x$

To get maximum area value

Substitute x = 15 into $30x - x^2$

Maximum area = $30(15) - (15)^2$

$= 450 - 225$

$= 225m^2$ (C)

2009/6 (Nov) Exercise 30.37

Find the minimum value of the function

$y = 2x^2 - 6x + 3$.

A –1.5 B –1.0 C 2.5 D 3.0

2006/27 f/m Exercise 30.38

Find the least value of the function $f(x) = 3x^2 + 18x + 32$

A 5 B 4 C 3 D 2

2001/23 Neco f/m Exercise 30.39

The maximum value of the function $y = 2 + 3x - 4x^2$ is

A.$-\frac{23}{16}$ B.$\frac{5}{16}$ C. $\frac{13}{16}$ D. $\frac{41}{16}$ E. $\frac{59}{16}$

2014/40 UTME Exercise 30.40

Find the minimum value of $y = x^2 - 2x - 3$

A – 1 B – 4 C 4 D 1

Cubic function’s case

2014/15 Neco f/m

Calculate the minimum value of the function

$y = 2x^3 - 6x + 3$

A – 1.5 B – 1.0 C 1.5 D 2.5 E 3.0

Solution

At minimum $\frac{dy}{dx} = 0$

$y = 2x^3 - 6x + 3$

$\frac{dy}{dx} = 6x^2 - 6$

$\Rightarrow 6x^2 - 6 = 0$

$6x^2 = 6$

$x^2 = \frac{6}{6} = 1$

$x = 1$

To get minimum value: we substitute x = 1 into

$y = 2x^3 - 6x + 3$

$= 2(1)^3 - 6(1) + 3$

$= 2 - 6 + 3 = -1$ B.

2008/40 PCE

Find a maximum value of the function $f(x) = x^3 - 12x + 5$

A – 9 B – 2 C 9 D 21

Solution

At maximum $f'(x) = 0$

$f(x) = x^3 - 12x + 5$

$f'(x) = 3x^2 - 12$

Factorizing

$\Rightarrow 3x^2 - 12 = 0$

$3(x^2 - 4) = 0$
$x^2 - 4 = 0$
$x^2 = 4$
$x = \pm 2$
At maximum $f''(x) < 0$
$f''(x) = 6x$
$6x < 0$, When $x = -2$
$6(-2) < 0$
To get maximum value, we put $x = -2$ into
$f(x) = x^3 - 12x + 5$
$f(-2) = (-2)^3 - 12(-2) + 5$
$= -8 + 24 + 5$
$= 21$ D.

1992/39 UME
Obtain a maximum value of the function
$f(x) = x^3 - 12x + 11$
A. -5 B. -2 C. 2 D. 27
Solution
Recall that $f'(x)$ is equivalent of dy/dx
While $f''(x)$ is equivalent of d^2y/dx^2
Already,
$f(x) = x^3 - 12x + 11$
At max or min $f'(x) = 0$
In this case, $f'(x) = 3x^2 - 12$
It follows that
$3x^2 - 12 = 0$
$3x^2 = 12$
$x^2 = \frac{12}{3}$ i.e 4

$x = \pm\sqrt{4}$
$= \pm 2$
To know which of the values of + 2 and – 2 is maximum:
$f''(x) = 6x$
When $x = +2$
$6x = 6(2)$
$= 12 > 0$ minimum point
When $x = -2$
$6x = 6(-2)$
$= -12 < 0$ maximum point
To get maximum value of the function,
We substitute $x = -2$ into $f(x) = x^3 - 12x + 11$
$= (-2)^3 - 12(-2) + 1$

$= -8 + 24 + 11$
$= 27$ (D)

2003/4 Exercise 30.41
Given that $y = x(x + 1)^2$,
calculate, the maximum value of y.
A -2 B 0 C 1 D 2

2014/9a Neco f/m Exercise 30.42
A curve is defined by $f(x) = x^3 - 6x^2 - 15x - 1$
find the:
(i) gradient of the curve at the point where $x = 1$
(ii) maximum and minimum points

Turning points (x,y)
[the value of (x,y) for maximum or minimum point]

2000/6 f/m
If the point p(x, y) is the maximum point on the curve $y = x^3 + 3x^2 - 7$, find the co-ordinates of p.
A. (0, -2) B. (- 2, -3) C. (0, 7) D. (-1, -5)

Solution
At maximum or minimum; $^{dy}/_{dx} = 0$
$y = x^3 + 3x^2 - 7$
$\frac{dy}{dx} = 3x^2 + 6x$
At maximum or minimum; $^{dy}/_{dx} = 0$
Thus, $3x^2 + 6x = 0$
$3x(x + 2) = 0$
$3x = 0$ or $x + 2 = 0$
$x = 0$ or -2
To know which of the values is maximum
$\frac{d^2y}{dx^2} = 6x + 6$
When $x = 0$
$6x + 6 = 6 > 0$ minimum point **not** our target
Next, When $x = -2$
$6x + 6 = -12 + 6$
$= -6 < 0$ maximum point **target gotten**
To get the y co-ordinate; we substitute $x = -2$ into
$y = x^3 + 3x^2 - 7$
i.e $y = (-2)^3 + 3(-2)^2 - 7$
$= -8 + 12 - 7$ i.e -3
P(x, y) is (-2, -3) B

2000/38 PCE
Find the turning points of the function $y = 2x^3 - 6x^2 - 18x + 3$
A.(- 1, 13), (3, - 51) B.(0, 3), (2, - 23)
C.(- 1, - 19), (2, 23) D.(- 1, 19), (3, 51)
Solution
Already, $y = 2x^3 - 6x^2 - 18x + 3$
At turning point $dy/dx = 0$
In this case $dy/dx = 6x^2 - 12x - 18$
It follows that
$6x^2 - 12x - 18 = 0$
Divide through by 6
$x^2 - 2x - 3 = 0$
factorizing
$x^2 - 3x + x - 3 = 0$
$x(x - 3) + 1(x - 3) = 0$
$(x + 1)(x - 3) = 0$
$x + 1 = 0$ or $x - 3 = 0$
$x = -1$ or $+3$
At $x = -1$
$y = 2(-1)^3 - 6(-1)^2 - 18(-1) + 3$
$= -2 - 6 + 18 + 3$
$= -8 + 21$
$= 13$
i.e (x, y) = (- 1, 13)
At $x = +3$
$y = 2(3)^3 - 6(3)^2 - 18(3) + 3$
$= 2(27) - 6(9) - 54 + 3$
$= 54 - 54 - 54 + 3$
$= -51$
i.e (x, y) = (3, - 51) Option A fits in

2002/38 PCE

Find the coordinates of the minimum point for the equation $y = 4t^2 - 40t + 300$

A.(5, 200) B.(5, 100) C.(4,300) D.(4, 100)

Solution

Given $y = 4t^2 - 40t + 300$

At minimum point $dy/dt = 0$

*Note that we did not use dy/dx; s*ince the function is in y and t.

In this case, $dy/dt = 8t - 40$

It follows that

$$8t - 40 = 0$$
$$8t = 40$$
$$\frac{8t}{8} = \frac{40}{8} \quad t = 5$$

The y value is gotten by putting t = 5 into

$y = 4t^2 - 40t + 300$

$y = 4(5)^2 - 40(5) + 300$

$= 4 \times 25 - 200 + 300$

$= 100 + 300 - 200 = 200.$

i.e (t, y) is (5, 200) (A).

2004/39 PCE

The turning point of the curve $y = 5 - 2x - x^2$ occurs at

A.(- 2, 5) B.(- 1 – 6) C.(- 1, 6) D.(1, 6)

Solution

At turning point $dy/dx = 0$

$y = 5 - 2x - x^2$ will become

$dy/dx = -2 - 2x$

At turning point : $dy/dx = 0$

i.e $-2 - 2x = 0$

$$-2x = 2$$
$$\frac{-2x}{-2} = \frac{2}{-2}$$
$$x = -1$$

The y value is gotten by putting x = - 1 into

$y = 5 - 2x - x^2$

$y = 5 - 2(-1) - (-1)^2$

$= 5 + 2 - 1 = 6$

i.e (x, y) = (–1, 6) (C)

1998/41 PCE Exercise 30.43

The turning point of the curve $y = 6 - 4x - x^2$ occurs at

A.(- 2, 10) B.(0, 6) C.(2, - 6) D.(4, -26)

1992/41 PCE Exercise 30.44

The turning point of the curve $y = 5 - 8x - 2x^2$ occurs at

A.(- 2, 13) B.(2, 13) C.(2, - 13) D.(- 2, - 13)

2005/36 f/m Exercise 30.45

Find the stationary point of the function $f(x) = -3x^2 + 3x + 2$

A.($-\frac{11}{4}$, $\frac{9}{4}$) B.($-\frac{3}{2}$, $\frac{1}{2}$) C. ($\frac{1}{2}$, $\frac{11}{4}$) D.($-\frac{1}{2}$, $\frac{9}{4}$)

1994/5(Nov) f/m Exercise 30.46

Determine the turning point of the curve $y = x^2 + \frac{250}{x}$

A. (-5, -25) B.(5, 75) C. (5, 150)
D. (10, 75) E. (5, 0)

Implicit functions

2003/6(b) Neco f/m (Dec)

Find $\frac{dy}{dx}$ of the function $2x^3y^2 - 3xy^2 = 4$

Solution

$2x^3y^2$ is differentiated as product:

$$\frac{d}{dx}(2x^3y^2) = 2x^3\left(\frac{d}{dx}(y^2)\right) + 2y^2\left(\frac{d}{dx}(x^3)\right)$$

Note that: $\frac{d}{dx}(y^2)$ is $\frac{d}{dy}(y^2) \times \frac{dy}{dx}$ i.e $2y\frac{dy}{dx}$

$$\frac{d}{dx}(2x^3y^2) = 2x^3 \times 2y\frac{dy}{dx} + 2y^2 \times 3x^2$$
$$= 4x^3y\frac{dy}{dx} + 6x^2y^2$$

$3xy^2$ is differentiated as product:

$$\frac{d}{dx}(3xy^2) = 3x\left(\frac{d}{dx}(y^2)\right) + 3y^2\frac{dx}{dx}$$

Note the result of $\frac{d}{dx}(y^2)$ as gotten earlier

$$= 3x \times 2y\frac{dy}{dx} + 3y^2$$
$$= 6xy\frac{dy}{dx} + 3y^2$$

4 is a constant, hence its differential is 0

$2x^3y^2 - 3xy^2 = 4$ differentiated becomes

$$4x^3y\frac{dy}{dx} + 6x^2y^2 - \left(6xy\frac{dy}{dx} + 3y^2\right) = 0$$

Opening up bracket we have

$$4x^3y\frac{dy}{dx} + 6x^2y^2 - 6xy\frac{dy}{dx} - 3y^2 = 0$$

Collect terms in dy/dx together

$$4x^3y\frac{dy}{dx} - 6xy\frac{dy}{dx} = 3y^2 - 6x^2y^2$$

Factor out dy/dx

$$\frac{dy}{dx}(4x^3y - 6xy) = 3y^2 - 6x^2y^2$$
$$\frac{dy}{dx} = \frac{3y^2 - 6x^2y^2}{4x^3y - 6xy}$$

2004/2(a) Neco (Dec) f/m

Find $\frac{dy}{dx}$ if $4x^4 + y^3 = 12x^2y$

Solution

$4x^4$ is differentiated directly as

$$\frac{d}{dx}(4x^4) = 16x^3$$

y^3 is differentiated as chain :

$$\frac{d}{dx}(y^3) = \frac{d}{dy}(y^3) \times \frac{dy}{dx} = 3y^2\frac{dy}{dx}$$

$12x^2y$ differentiated as product:

$$\frac{d}{dx}(12x^2y) = 12x^2\frac{dy}{dx} + 12y\frac{d}{dx}(x^2)$$
$$= 12x^2\frac{dy}{dx} + 24xy$$

$4x^4 + y^3 = 12x^2y$ differentiated becomes

$$16x^3 + 3y^2\frac{dy}{dx} = 12x^2\frac{dy}{dx} + 24xy$$

Collect like terms together

$$3y^2\frac{dy}{dx} - 12x^2\frac{dy}{dx} = 24xy - 16x^3$$

Factor out dy/dx in the LHS

$$\frac{dy}{dx}(3y^2 - 12x^2) = 24xy - 16x^3$$

$$\frac{dy}{dx} = \frac{24xy - 16x^3}{3y^2 - 12x^2}$$

2013/26 Neco f/m

If $x^3 + y^3 = 27xy$, find $\frac{dy}{dx}$

A $\frac{x^2+9y}{9x-y^2}$ B $\frac{x^2-9y}{9x+y^2}$

C $\frac{9y-x^2}{y^2-9x}$ D $\frac{3(9y-x^2)}{y^2-9x}$ E $\frac{9y-x^2}{3(y^2-9x)}$

Solution

$$x^3 + y^3 = 27xy$$

$$\frac{d}{dx}(x^3) + \frac{d(y^3)}{dx} = 27y\frac{dx}{dx} + 27x\frac{dy}{dx}$$

$$3x^2 + 3y^2\frac{dy}{dx} = 27y + 27x\frac{dy}{dx}$$

Collect terms in $\frac{dy}{dx}$ together

$$3y^2\frac{dy}{dx} - 27x\frac{dy}{dx} = 27y - 3x^2$$

$$\frac{dy}{dx}(3y^2 - 27x) = 27y - 3x^2$$

$$\frac{dy}{dx} = \frac{27y-3x^2}{3y^2-27x}$$

$$= \frac{3(9y-x^2)}{3(y^2-9x)}$$

$$= \frac{9y-x^2}{y^2-9x} \quad (\text{C})$$

2013/ 3b Neco f/m

Given that $x^2y^2 - 3xy + 4xy^3 = 2$, find $\frac{dy}{dx}$

Solution

$$x^2y^2 - 3xy + 4xy^3 = 2$$

$$x^2\frac{d(y^2)}{dx} + y^2\frac{d(x^2)}{dx} - 3x\frac{dy}{dx} - 3y\frac{dx}{dx} + 4x\frac{d(y^3)}{dx} + 4y^3\frac{dx}{dx} = 0$$

$$x^2\, 2y\frac{dy}{dx} + y^2\, 2x - 3x\frac{dy}{dx} - 3y + 4x\, 3y^2\frac{dy}{dx} + 4y^3 = 0$$

Collect terms in $\frac{dy}{dx}$ together

$$2x^2y\frac{dy}{dx} - 3x\frac{dy}{dx} + 12xy^2\frac{dy}{dx} = -2xy^2 + 3y - 4y^3$$

$$\frac{dy}{dx}(2x^2y - 3x + 12xy^2) = -2xy^2 + 3y - 4y^3$$

$$\frac{dy}{dx} = \frac{-2xy^2+3y-4y^3}{2x^2y-3x+12xy^2}$$

2013/11 f/m Exercise 30.47

Differentiate $x^2 + xy - 5 = 0$ with respect to x

A $\frac{-(2x+y)}{x}$ B $\frac{2x-y}{x}$ C $\frac{-x}{2x+y}$ D $\frac{2x+y}{x}$

2013/2 f/m Exercise 30.48

Calculate the gradient of the curve

$$x^3 + y^3 - 2xy = 11$$

at the point (2, – 1)

Implicit functions in gradient at a point

2014/1 Neco f/m

Find y' at point (1, 1) if $x^2y + y^3 = 2$

A – 1 B – $^1/_2$ C 0 D $^1/_2$ E 1

Solution

$$x^2y + y^3 = 2$$

First, we differentiate implicitly

$$x^2\frac{dy}{dx} + y\frac{d(x^2)}{dx} + 3y^2\frac{dy}{dx} = 0$$

$$x^2\frac{dy}{dx} + 2xy + 3y^2\frac{dy}{dx} = 0$$

Collect terms in $\frac{dy}{dx}$ together

$$x^2\frac{dy}{dx} + 3y^2\frac{dy}{dx} = -2xy$$

$$\frac{dy}{dx}(x^2 + 3y^2) = -2xy$$

$$\frac{dy}{dx} = \frac{-2xy}{x^2+3y^2}$$

At point (1, 1), x is 1 and y is 1

$$\frac{dy}{dx} = \frac{-2(1)(1)}{1^2+3(1)^2}$$

$$= \frac{-2}{1+3} = \frac{-2}{4} = -\frac{1}{2} \quad \text{B.}$$

2000/10 (C) (Nov) f/m

Find the gradient of the curve $y^2x + 3x^2y = 1$, at the point (1, 1)

Solution

First, we differentiate implicitly.

$$y^2 \times \frac{dx}{dx} + x \times \frac{d(y^2)}{dx} + 3x^2\frac{dy}{dx} + 3y \times \frac{d(x^2)}{dx} = 0$$

$$y^2 + x \times 2y\frac{dy}{dx} + 3x^2\frac{dy}{dx} + 3y \times 2x\frac{dx}{dx} = 0$$

$$y^2 + x \times 2y\frac{dy}{dx} + 3x^2\frac{dy}{dx} + 3y \times 2x = 0$$

$$y^2 + 2xy\frac{dy}{dx} + 3x^2\frac{dy}{dx} + 6xy = 0$$

Collect like terms together

$$2xy\frac{dy}{dx} + 3x^2\frac{dy}{dx} = -(y^2 + 6xy)$$

$$\frac{dy}{dx}(2xy + 3x^2) = -(y^2 + 6xy)$$

$$\frac{dy}{dx} = -\frac{(y^2 + 6xy)}{2xy + 3x^2}$$

At point (1, 1) , x is 1 and y is 1

$$dy/dx = -\frac{(1+6)}{2+3} = -^7/_5 \text{ i.e } -1^2/_5$$

1996/16 (Nov) f/m
The gradient of the curve $x^2 - 2xy - 2y^2 - 2x = 0$ at the point $(1, -4)$ is
A. $-^4/_3$ B. $-^4/_7$ C. $-^1/_2$ D. $-^4/_9$ E. $-^1/_8$

Solution

First, we differentiate implicitly.
$$2x - 2x\frac{dy}{dx} - 2y - 4y\frac{dy}{dx} - 2 = 0$$
Collect terms in dy/dx together
$$2x - 2y - 2 = 2x\frac{dy}{dx} + 4y\frac{dy}{dx}$$
$$2x - 2y - 2 = \frac{dy}{dx}(2x + 4y)$$
$$\frac{2x - 2y - 2}{2x + 4y} = \frac{dy}{dx}$$
$$\frac{2(x - y - 1)}{2(x + 2y)} = \frac{dy}{dx}$$
Thus, $\frac{dy}{dx} = \frac{x - y - 1}{x + 2y}$
At point $(1, -4)$; x is 1and y is -4
$$dy/dx = \frac{1 + 4 - 1}{1 - 8}$$
$$= \frac{-4}{7} \text{ (B)}$$

1997/11a f/m
Given the curve $50x^2 + 36xy + 5y^2 = -2$
Evaluate dy/dx at the point $(1, -2)$

Solution

Differentiating implicitly, we have
$$50\frac{d(x^2)}{dx} + 36x\frac{dy}{dx} + 36y\frac{dx}{dx} + 5\frac{d(y^2)}{dx} = 0$$
$$100x + 36x\frac{dy}{dx} + 36y + 10y\frac{dy}{dx} = 0$$
Collect terms in dy/dx together
$$36x\frac{dy}{dx} + 10y\frac{dy}{dx} = -(36y + 100x)$$
$$\frac{dy}{dx}(36x + 10y) = -(36y + 100x)$$
$$\frac{dy}{dx} = -\frac{(36y + 100x)}{(36x + 10y)}$$
At point $(1, -2)$; x is 1 and y is -2
$$dy/dx = -\frac{(-72 + 100)}{36 - 20}$$
$$= -^7/_4 \text{ i.e } -1^3/_4$$

1993/12 (Nov) f/m Exercise 30.49
If $x^2 + 3xy + 2y^2 = 3$, find $^{dy}/_{dx}$ at the point $(-1, 2)$.
A. ¾ B. $^8/_{11}$ C. $^7/_{11}$ D. $^2/_3$ E. $-^4/_5$

IPLF 3 Exercise 30.50
If $ax^2 + 2hxy + by^2 = 0$, find $^{dy}/_{dx}$

IPLF 4 Exercise 30.51
$y^4 + x^2 = x + y^2$, find $^{dy}/_{dx}$ at the point $(-1, 1)$.

2010/4 f/m Exercise 30.52
If $3x^2 + 2y^2 + xy + x - 7 = 0$,
find $\frac{dy}{dx}$ at the point (-2, 1).

Comparing rates of change

If we are given a 'main' function say $y = f(x)$. Then differentiating the main function y w.r.t time t (instead of wrt to x) will have a connecting formula through chain rule as:
$$\frac{dy}{dt} = \frac{dy}{dx}\times\frac{dx}{dt}$$
Where our dy/dx here can be gotten from the main function which is either given directly or mentioned in the question as shown in the examples below:

2000/38 UME
If the volume of a hemisphere is increasing at a steady rate of $18\pi m^3 s^{-1}$, at what rate is its radius changing when it is 6m ?
A. $2.50ms^{-1}$ B. $2.00ms^{-1}$ C.$0.25ms^{-1}$ D.$0.20ms^{-1}$
Solution
The main function here is the vol. of hemisphere $V = {}^2/_3\pi r^3$
But $\frac{dv}{dt} = \frac{dv}{dr}\times\frac{dr}{dt}$ ---------***
Here dv/dt is given as $18\pi m^3s^{-1}$ and dv/dr is gotten from the main function
$V = {}^2/_3\pi r^3$
$dv/dr = 2\pi r^2$
At r = 6m
$2\pi r^2 = 2\pi(6)^2$
$= 72\pi$
Substituting for dv/dt and dv/dr into *** ,
$$18\pi = 72\pi \times \frac{dr}{dt}$$
$$\frac{18\pi}{72\pi} = dr/dt$$
$$dr/dt = ¼ = 0.25ms^{-1} \text{ (C)}$$
The unit is so because length r in metres and time t in seconds. dr/dt means unit metres per second or metres/seconds or metres second^{-1} (ms^{-1})

2002/21 UME
A circle with a radius 5cm has its radius increasing at the rate of $0.2cm\ s^{-1}$.What will be the corresponding increase in the area ?
A. π B.2π C.4π D.5π
Solution
The main function here is Area of a circle $A = \pi r^2$
But $\frac{dA}{dt} = \frac{dA}{dr}\times\frac{dr}{dt}$ ---------***
Here dA/dt is not given.
dA/dr is gotten from the main function
$A = \pi r^2$
$dA/dr = 2\pi r$
When r = 5cm
$2\pi r = 2\pi(5)$
$= 10\pi$
We were given that: $dr/dt = 0.2cms^{-1}$
Substituting for dA/dr and dr/dt into ***,
$$\frac{dA}{dt} = 10\pi \times 0.2$$
$$= 2\pi \text{ (B)}$$

1998/42 PCE

The radius of a circle is increasing at the rate of $0.4cm\ s^{-1}$. Find the rate of increase in $cm^2\ s^{-1}$ of the area when the radius is 5cm.

A.40π B.10π C.5π D.4π

Solution

The main function here is area of circle i.e $A = \pi r^2$

But $\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$

We are to find dA/dt , dr/dt is given as $0.4cms^{-1}$.

But dA/dr can be gotten from the main function

$A = \pi r^2$

$dA/dr = 2\pi r$

When r = 5cm

$2\pi r = 2\pi (5)$

$= 10\pi$

Therefore

$dA/dt = 10\pi \times 0.4$

$= 4\pi$ (D)

2013/37 UTME Exercise 30.53

The radius of a circle is increasing at the rate of $0.02cm5^{-1}$. Find the rate at which the area is increasing when the radius of the circle is 7cm

A $0.35cm^2\ s^{-1}$ B $0.88cm^2\ s^{-1}$

C $0.75cm^2\ s^{-1}$ D $0.53cm^2\ s^{-1}$

2005/3 UME Exercise 30.54

The radius r of a circular disc is increasing at the rate of 0.5cm/secs. At what rate is the area of the disc increasing when its radius is 6cm ?

A.$3\pi cm^2$/sec B.$36\pi cm^2$/sec C.$18\pi cm^2$/sec D.$6\pi cm^2$/sec

2006/25 PCE Exercise 30.55

The radius of an ink drop on a blotting paper expands at the rate of 0.02 cm/sec. Find the rate of change of the area when the radius is 5 cm

A.πcm^2/sec B.$2\pi cm^2$/sec C.$0.2\pi cm^2$/sec D.$0.5\pi cm^2$/sec

2013/24 f/m Exercise 30.56

A circular ink dot on a piece of paper increases its area at the rate of $4mm^2/s$. Find the rate of increase of the radius of the dot when the radius is 8mm (Take $\pi = {}^{22}/_{7}$)

A 0.25mm/s B 0.20mm/s C 0.08mm/s D 0.05mm/s

Curve sketching

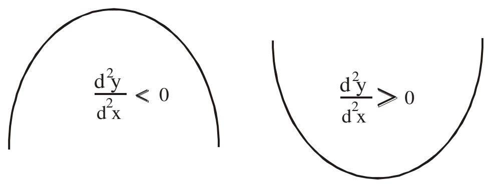
The basic guides in curve sketching are:

Turning points one

At turning points $\frac{dy}{dx} = 0$

Provided NO TWO ROOTS ARE EQUAL in the given function when factorised after differentiation then it could be

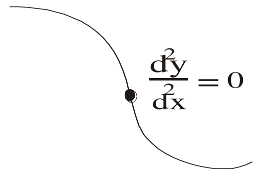
maximum or minimum



Next is the **intercept at the y-axis and x-axis**

Turning points two

If TWO ROOTS ARE EQUAL in the given function when factorised after differentiation then it is an **inflexion**



Note that at the point of inflexion $\frac{d^2y}{dx^2} = 0$

To know the function's pattern before and after the point of inflexion

we get x and y values to the left and right of the point of inflexion say x = -2 and +2 into the main equation to get the corresponding y values. You could try more –3and+3

Example CSK 1

Sketch the curve $y = x^2 - 1$

Solution

$y = x^2 - 1$

At turning points $\frac{dy}{dx} = 0$

$\frac{dy}{dx} \Rightarrow 2x = 0$

x = 0 (*No equal roots here hence no inflexion*)

To determine whether max or min

$\frac{d^2y}{dx^2} = 2 > 0$ min point

Intercept at the y-axis i.e x = 0 in the original equation

$y = 0^2 - 1$

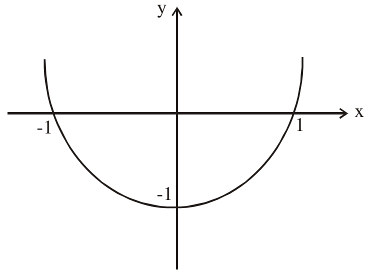
$= -1$ the curve cut the y-axis at -1

Intercept at the x-axis i.e y = 0 in the original equation

$0 = x^2 - 1$

$x = \pm 1$

The curve cut the x-axis at x = -1 and at x = +1 .Sketching



Example CSK 2

Sketch the curve $y = x^3$

Solution

$y = x^3$

At turning points $\frac{dy}{dx} = 0$

$\frac{dy}{dx} \Rightarrow 3x^2 = 0$

$x = 0$ twice (*equal roots present hence inflexion*)

At the point of inflexion $\frac{d^2y}{dx^2} = 0$

i.e $6x = 0$

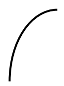
$x = 0$ thus inflexion occurred at $x = 0$

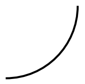
To know the function's pattern before and after the point of inflexion take - 2 and +2

Before inflexion $x = -2$ after inflexion $x = +2$

$y = (-2)^3$ $y = (+2)^3$

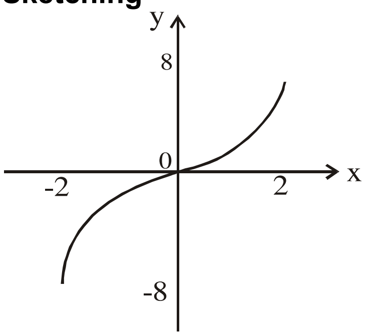
$= -8$ $= +8$





curve going down **curve going up**

Sketching



Example CSK 3

Sketch the curve $y = x^3 - x^2 - 5x$

Solution

$y = x^3 - x^2 - 5x$

At turning points $\frac{dy}{dx} = 0$

$\Rightarrow 3x^2 - 2x - 5 = 0$

$3x^2 + 3x - 5x - 5 = 0$

$3x(x + 1) - 5(x + 1) = 0$

$(3x - 5)(x + 1) = 0$

$x = -1$ or $^5/_3$ (1.7)

(*No equal roots here hence no inflexion*)

To determine which of $x = -1$ or $^5/_3$ (1.7)is max or min

$\frac{d^2y}{dx^2} = 6x - 2$

<u>when $x = -1$</u>

$\frac{d^2y}{dx^2}$ value becomes $-8 < 0$ max point

<u>when $x = {}^5/_3$</u>

$\frac{d^2y}{dx^2}$ value becomes $+8 > 0$ min point

Intercept at the x-axis i.e $y = 0$ in the original equation

$0 = x^3 - x^2 - 5x$

i.e $x(x^2 - x - 5) = 0$

$x = 0$ or $(x^2 - x - 5) = 0$

cannot be factorized $(x^2 - x - 5) = 0$

$$x = \frac{+1 \pm \sqrt{1 - 4 \times 1 \times (-5)}}{2 \times 1}$$

$$= \frac{+1 \pm \sqrt{21}}{2}$$

$$= \frac{1 - 4.6}{2} \text{ or } \frac{1 + 4.6}{2}$$

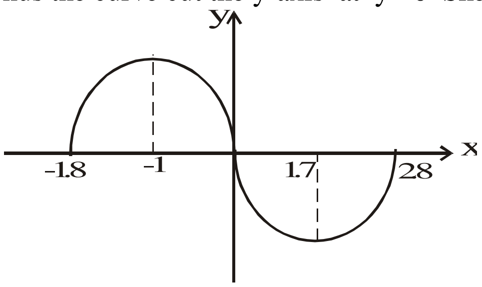
$= -1.8$ or 2.8

Thus the curve cut the x-axis at $x = -1.8$, 0, and 2.8

Intercept at the y-axis i.e $x = 0$ in the original equation

$y = 0^3 - 0^2 - 5(0)$ i.e 0

Thus the curve cut the y-axis at $y = 0$ Sketching



Chapter Thirty One
Integration

Integration is the opposite of differentiation. It is represented by elongated s i.e $\int$

In general : $\int x^n dx = \dfrac{x^{n+1}}{n+1}$

i.e increase the power by one and divide by new power. This is the fundamental formula of integration.

Indefinite Integrals I

Lets us check this out

$y = 10x^3 + 3x$ and $y = 10x^3 + 3$

Now differentiating

$\frac{dy}{dx} = 30x^2 + 3$ while $\frac{dy}{dx} = 30x^2$

To get back our original functions, we integrate i.e reverse the differentiation.

$$\int \frac{dy}{dx} = \int \left(30x^2 + 3\right)$$

cross multiply the dx

$$\int dy = \int \left(30x^2 + 3\right) dx$$

Since there seems to be nothing on the LHS, it means 1 and 1 implies y^0, i.e the variable there. Thus,

$$\frac{y^{0+1}}{0+1} = \int 30x^2 dx + \int 3dx$$

Applying the formula to the RHS

$y = \frac{30x^3}{3} + \frac{3x^{0+1}}{0+1}$

$y = 10x^3 + 3x$

We are back.

Similarly,

$$\int \frac{dy}{dx} = \int 30x^2$$

cross multiply the dx

$$\int dy = \int 30x^2 dx$$

$y = 10x^3$

We are back but we lost something which is the constant + 3. Thus we always add constant "C" or "K" to any indefinite integral like the one shown above. Hence our actual answers are:

$y = 10x^3 + 3x + k$ and $y = 10x^3 + k$

There are some basic integrals we need to know.

i. $\int \cos x dx = \sin x + k$

ii. $\int \sin x dx = -\cos x + k$

iii. $\int \frac{1}{x} dx = \ln x + k$

iv. $\int e^x dx = e^x + k$

Also integral of sum of functions is the sum of the separable integrals i .e

$$\int \left(x^2 + 3x - 5\right) dx = \int x^2 dx + \int 3x dx - \int 5dx$$

We can also relate **i** to **iv** with differentiation for better understanding; since they are opposite.

Differentiation	**integration**
$\frac{d}{dx}(\sin x) = \cos x$	$\int \cos x dx = \sin x + k$
$\frac{d}{dx}(\cos x) = -\sin x$	$\int \sin x dx = -\cos x + k$
$\frac{d}{dx}(\ln x) = \frac{1}{x}$	$\int \frac{1}{x} dx = \ln x + k$
$\frac{d}{dx}(e^x) = e^x$	$\int e^x dx = e^x + k$

1994/41 UME

Integrate $\frac{1-x}{x^3}$ with respect to x

A. $\frac{x - x^2}{x^4} + k$ B. $\frac{4}{x^4} - \frac{3}{x^3} + k$ C. $\frac{1}{x} - \frac{1}{2x^2} + k$ D. $\frac{1}{3x^2} - \frac{1}{2x} + k$

Solution

$$\int \left(\frac{1-x}{x^3}\right) dx = \int \left(\frac{1}{x^3} - \frac{x}{x^3}\right) dx$$

$$= \int \frac{1}{x^3} dx - \int \frac{1}{x^2} dx$$

$$= \int x^{-3} dx - \int x^{-2} dx$$

$$= \frac{x^{-3+1}}{-3+1} - \frac{x^{-2+1}}{-2+1} + k$$

$$= \frac{x^{-2}}{-2} - \frac{x^{-1}}{-1} + k$$

$$= -\frac{1}{2x^2} + \frac{1}{x} + k$$

Rearranging

$= \frac{1}{x} - \frac{1}{2x^2} + k$ (C)

Note: ***All integrals must be simplified as far as possible before integrating as shown above.***

1997/41 UME

Integrate $\frac{1}{x} + \cos x$ with respect to x

A. $\frac{-1}{x^2} + \sin x + k$ B. $\ln x + \sin x + k$ C. $\ln x - \sin x + k$

D. $\frac{-1}{x^2} - \sin x + k$

Solution

$$\int \left(\frac{1}{x} + \cos x\right) dx = \int \frac{1}{x} dx + \int \cos x dx$$

Recall their results.

$= \ln x + \sin x + k$ (B)

1996/40 PCE

If $\frac{dy}{dx} = x^3 - 2x^2 - 3x + 1$, find y

A. $\frac{x^4}{4} - \frac{2x^3}{3} - \frac{3x^2}{2} + x + c$ B. $\frac{x^4}{4} - \frac{2x^3}{3} - \frac{2x^2}{3} + x + c$

C. $4x^4 - \frac{3x^3}{2} - \frac{3x^2}{2} - x + c$ D. $4x^4 - \frac{2x^3}{3} - \frac{3x^2}{2} - x + c$

Solution

$\frac{dy}{dx} = x^3 - 2x^2 - 3x + 1$

Cross multiply dx

$dy = (x^3 - 2x^2 - 3x + 1)dx$

Integrating both sides

$$\int dy = \int (x^3 - 2x^2 - 3x + 1)dx$$

Applying integration formula

$y = \frac{x^4}{4} - \frac{2x^3}{3} - \frac{3x^2}{2} + x + c$ (A)

1996/39 PCE

Simplify $\int \frac{x^2+3x+2}{x+1}dx$

A. $\frac{x^3}{3} + \frac{3x^2}{2} + 2x + c$ B. $\frac{x^3}{3} + 2x + c$ C. $\frac{2x}{2} + 3 + c$

D. $\frac{x^2}{2} + 2x + c$

Solution

First we simplify the numerator by factorization.

$\frac{x^2 + 3x + 2}{x+1} = \frac{(x+2)(x+1)}{x+1}$ will give $x + 2$

Thus, $\int \frac{x^2+3x+2}{x+1}dx = \int (x+2)dx$

$= \frac{x^2}{2} + 2x + c$ (D)

1994/45 PCE

Integrate $2x(2x^2 - 3x + 4)$ with respect to x.

A. $12x^2 - 12x + 8$ B. $x^4 - x^3 + 8x^2 + c$

C. $x^4 - x^3 + x^2 + c$ D. $x^4 - 2x^3 + 4x^2 + c$

Solution

We simplify the function by opening up the bracket,

$$\int 2x(2x^2 - 3x + 4)dx = \int (4x^3 - 6x^2 + 8x)dx$$

Applying integration formula

$= \frac{4x^4}{4} - \frac{6x^3}{3} + \frac{8x^2}{2} + c$

$= x^4 - 2x^3 + 4x^2 + c$ (D)

2005/4 PCE

A function whose derivative is $\frac{1}{x^2}$ has its integral as

A. $-\frac{1}{2x} + k$ B. $\frac{1}{x} + k$ C $-\frac{1}{x} + k$ D. $\frac{2}{x} + k$

Solution

We are simply asked to integrate $^1/x^2$.

Thus $\int \frac{1}{x^2}dx = \int x^{-2}dx$

Applying the formula:

$= \frac{x^{-2+1}}{-2+1} + k$

k or c is just a constant usually added to indefinite integrals

$= \frac{x^{-1}}{-1} + k = \frac{-1}{x} + k$ (C)

2005/39 Neco

Integrate $\tan^2 x$

A $\tan x - x + c$ B $\tan^2 x - 2x + c$ C $x - \tan^2 x + c$

D $x - \tan x + c$ E $-\tan x + 2x + c$

Solution

Applying trig identities

$$\int \tan^2 x dx = \int (\sec^2 x - 1)dx$$

$$= \int \sec^2 x dx - \int dx$$

$= \tan x - x + c$ (A)

2003/23 Neco

Evaluate $\int (e^x - \sec x \tan x)dx$

Solution

$$\int (e^x - \sec x \tan x)dx = \int e^x dx - \int \sec x \tan x dx$$

Applying result of the respective integrals

$= e^x - \sec x + c$

2006/5 UME Exercise 31.0

If $\frac{dy}{dx} = x + \cos x$, find y. A. $\frac{x^2}{2} - \sin x + c$

B. $x^2 - \sin x + c$ C. $\frac{x^2}{2} + \sin x + c$ D. $x^2 + \sin x + c$

1992/42 PCE Exercise 31.1

Integrate $x^2(x+2)$ with respect to x A. $x^4 + x^3 + 4 + c$

B. $\frac{x^3}{4} + \frac{x^4}{3} + 7 + c$ C. $x^3 + x^2 + c$ D. $\frac{x^4}{4} + \frac{2x^3}{3} + c$

Indefinite Integrals II (change of variable)

In some problems, the integral does not fit in directly, in such cases, a **substitution** involving **change of the variable** will help.

Example INTER 1 Solve $\int \cos 2\Theta \, d\Theta$

Solution

Our target will be to change 2θ and $d\theta$

We let $u = 2\theta$

Differentiating u with respect to θ

$\frac{du}{d\theta} = 2$

Cross multiply

$du = 2d\theta$

Making $d\theta$ subject formula

$\frac{du}{2} = d\theta$

Substituting for 2θ and $d\theta$. Our new integral is

$$\int \cos u \frac{du}{2}$$

Factor out the constant. ½

$$= \frac{1}{2}\int \cos u \, du$$

$= ½ \sin u + k$; Replacing u with 2θ $= \frac{1}{2}\sin 2\theta + k$

2002/18 UME

Evaluate $\int \sin 3x dx$

A. $-\frac{1}{3}\cos 3x + c$ B. $\frac{1}{3}\cos 3x + c$ C. $\frac{2}{3}\cos 3x + c$ D. $-\frac{2}{3}\cos 3x + c$

Solution

Our target here will be to change 3x and dx

Let $u = 3x$

Differentiating u with respect to x

$\frac{du}{dx} = 3$

cross multiply

du = 3dx

making dx the subject formula

$\frac{du}{3} = dx$

Substituting for 3x and dx ; our new integral is.

$$\int \sin u \frac{du}{3}$$

Factoring out the constant $^1/_3$; $= \frac{1}{3}\int \sin u\, du$

$= -\frac{1}{3} \cos u + c$

replacing *u* with *3x* $= \frac{-1}{3} \cos 3x + c$ (A)

Note: The constant k is a dummy variable, hence the examiner here used C; any other alphabet can be used in its place. Though C and K are commonly used

2001/34 UME

Evaluate $\int 2(2x-3)^{2/3}\, dx$

A. $\frac{3}{5}(2x-3)^{5/3} + k$ B. $\frac{6}{5}(2x-3)^{5/3} + k$

C. $2x - 3 + k$ D. $2(2x-3) + k$

Solution

We can not expand the bracket as in the case of say

$$\int (2x-3)^2 dx = \int (2x-3)(2x-3)dx$$
$$= \int (4x^2 - 12x + 9)dx$$
$$= \int 4x^2 dx - \int 12x dx + \int 9 dx$$

you know what follows

In this case we have in hand,

$$= 2\int (2x-3)^{2/3} dx$$

We have factored out the constant 2; next let's target 2x - 3 and dx

let u = 2x – 3 ;Differentiating u with respect to x

$\frac{du}{dx} = 2$

Cross multiply

du = 2dx

Making dx the subject formula

$\frac{du}{2} = dx$

Substituting for 2x –3 and dx, our new integral is

$$2\int u^{2/3} \frac{du}{2}$$

Factor out the constant $^1/_2$

$$= \frac{2}{2}\int u^{2/3}\, du$$

$$= \frac{U^{2/3+1}}{^2/_3 + 1} + k$$

$$= \frac{U^{5/3}}{^5/_3} + k$$

$= \frac{3U^{5/3}}{5} + k$; Making a replacement of U = 2x – 3

$= \frac{3}{5}(2x-3)^{5/3} + k$ (A)

2001/40 PCE

Evaluate $\int (\cos 3x + \sin 4x)dx$

A. $\frac{1}{3}\sin 3x + \frac{1}{4}\cos 4x + k$ B. $\frac{1}{3}\sin 3x - \frac{1}{4}\cos 4x + k$

C. $3\sin x + 4\cos x + k$ D. $.\sin^3 x + 4\cos^4 x + k$

Solution

Let's break them into bits

$$= \int \cos 3x\, dx + \int \sin 4x\, dx$$

Changing variables

For $\int \cos 3x\, dx$

Let u = 3x

Differentiating u with respect to x

$\frac{du}{dx} = 3$

Cross multiply

du = 3dx

Making dx subject formula.

$\frac{du}{3} = dx$

Substituting for 3x and dx ; our new integral is

$$\int \cos u \frac{du}{3}$$

Factor out the constant $^1/_3$,

$$= \frac{1}{3}\int \cos u\, du$$

$= \frac{1}{3} \sin U + k$

Next we replace u with 3x

i. e $= \frac{1}{3} \sin 3x + k$

Similarly

$$\int \sin 4x\, dx = -\frac{1}{4}\cos 4x + k$$

Thus, $\int (cox 3x + \sin 4x)dx = \int \cos 3x\, dx + \int \sin 4x\, dx$

$= \frac{1}{3} \sin 3x - \frac{1}{4} \cos 4x + k$ (B)

No need of adding two ks

I HOPE YOU OBSERVE A FORMAT HERE.

$$\int \cos 3x\, dx = \frac{1}{3}\sin 3x + k$$

$$\int \sin 4x\, dx = -\frac{1}{4}\cos 4x + k$$

$\int \cos 5x\, dx = \frac{1}{5}\sin 5x + k$ the pattern goes on

While that of Sine carries a minus sign along as shown below:

$$\int \sin 4x\, dx = -\frac{1}{4}\cos 4x + k$$

$$\int \sin 5x\, dx = -\frac{1}{5}\cos 5x + k$$

$\int \sin 6x\, dx = -\frac{1}{6}\cos 6x + k$ the pattern goes on

1994/20 Exercise 31.2

Using the substitution u = cosx or otherwise, find

$$\int \cos^5 x \sin x dx$$

1996/41 PCE Exercise 31.3

Find the integral of 3cos2x – 2sinx
A. 6sin2x – 2 cosx + c B. – 6sinx – 2cosx + c
C. $^3/_2$ sin2x – 2cosx + c D. $^3/_2$ sinx + 2cosx + c

1999/46 PCE

If $\frac{dy}{dx} = \frac{1}{2}(\sin 4x + \sin 2x + 1)$, find y.

A. $\frac{\cos 4x}{8} + \frac{\cos 2x}{4} + \frac{x}{2} + k$ B. $-2\cos 4x - \cos 2x + \frac{x}{2} + k$

C. $-\frac{\cos 4x}{8} - \frac{\cos 2x}{4} + \frac{x}{2} + k$ D. $2\cos 4x + \cos 2x + \frac{x}{2} + k$

Solution

$\frac{dy}{dx} = \frac{1}{2}(\sin 4x + \sin 2x + 1)$

Cross multiply the dx

$dy = {}^1/_2(\sin 4x + \sin 2x + 1)dx$

Integrating both sides

$$\int dy = \int \frac{1}{2}(\sin 4x + \sin 2x + 1)dx$$

$$= \frac{1}{2}\int(\sin 4x + \sin 2x + 1)dx$$

Applying change of variable format to RHS

$$y = \frac{1}{2}\left[-\frac{1}{4}\cos 4x - \frac{1}{2}\cos 2x + x\right] + k$$

Opening the bracket

$$= -\frac{\cos 4x}{8} - \frac{\cos 2x}{4} + \frac{x}{2} + k \quad (C)$$

If you observe that:

the numerator is a derivative of the denominator

i.e. $\int \frac{g^1(x)}{g(x)}dx = \ln g(x)$

where $g^1(x)$ is the derivative of g(x)

Example INTER 2

$$\int \frac{4x-4}{x^2-2x+5}dx$$

Solution

Observe that $\int \frac{4x-4}{x^2-2x+5}dx = 2\int \frac{2x-2}{x^2-2x+5}dx$

And it can be seen that the numerator is the *derivative* of the denominator

$$2\int \frac{2x-2}{x^2-2x+5}dx = 2\ln(x^2 - 2x + 5)$$

2003/7i

Evaluate $\int_0^1 \frac{x^2}{x^3+1}dx$

Solution

If we differentiate the denominator $x^3 + 1$ it results to $3x^2$ but the numerator has x^2 only
so we introduce $^1/_3$ to take of the coefficient 3 imported

Thus $\int_0^1 \frac{x^2}{x^3+1}dx = \frac{1}{3}\int_0^1 \frac{3x^2}{x^3+1}dx$

$$= \frac{1}{3}\ln(x^3+1)\Big|_0^1$$

Substituting for the upper& lower limits

$$= \frac{1}{3}\log_e(1^3+1) - \frac{1}{3}\log_e(0^3+1)$$

$$= \frac{1}{3}\log_e 2 - \frac{1}{3}\log_e 1$$

$$= \frac{1}{3}\log\frac{2}{1} \quad i.e \quad \frac{1}{3}\log_e 2$$

1996/16 Exercise 31.4

Evaluate $\int_0^1 \frac{2}{1+2x}dx$

$A \log_e 2 \quad B -\frac{1}{2}\log_e 3 \quad C \frac{1}{2}\log_e 3 \quad D \log_e 3 \quad E\ 0$

INTER 3 Exercise 31.5

Evaluate $\int \frac{7x}{3(x^2-3)}dx$

INTER 4 Exercise 31.6

Evaluate $\int \frac{x^3}{2x^4+1}dx$

Definite Integrals

In definite integrals, constant “k” or “c” are not added to our result since the value of the constant is gotten from the data in the question.

Definite integrals are of two kinds:

I. The problems where values of x and y are given to enable us find the value of the constant “k” or “c”

II. The problems, where lower and upper limits are given, thus taking care of the “k” or “c”

Definite Integral I

1998/41 UME

Find the equation of the curve which passes through the point (2,5) and whose gradient at any point is given by 6x – 5
A. $6x^2 - 5x + 5$ B. $6x^2 + 5x + 5$ C. $3x^2 - 5x - 5$ D. $3x^2 - 5x + 3$

Solution

Gradient $\Rightarrow \frac{dy}{dx} = 6x - 5$

$dy = (6x - 5)dx$

integrating both sides

$$\int dy = \int(6x-5)dx$$

$$y = \int 6x\,dx - \int 5dx$$

$$y = \frac{6x^{1+1}}{1+1} - \frac{5x^{0+1}}{0+1}$$

$$y = \frac{6x^2}{2} - \frac{5x^1}{1}$$

$$y = 3x^2 - 5x + k$$

Next we find k value

Point (2,5) means x = 2, y = 5

Substituting x and y values into $y = 3x^2 - 5x + k$

$$5 = 3(2)^2 - 5(2) + k$$
$$5 = 12 - 10 + k$$
$$5 - 2 = k$$
$$k = 3$$

Hence $y = 3x^2 - 5x + 3$ (D)

2000/39 UME

A function f(x) passes through the origin and its first derivation is 3x + 2 . What is f(x)?

A. $y = \frac{3x^2}{2} + 2x$ B. $y = \frac{3x^2}{2} + x$ C. $y = 3x^2 + \frac{x}{2}$ D. $y = 3x^2 + 2x$

Solution

First derivation $\Rightarrow {}^{dy}/_{dx} = 3x + 2$

$$dy = (3x + 2)dx$$

Integrating both sides

$$\int dy = \int (3x+2)dx$$
$$y = \frac{3x^2}{2} + 2x + k$$

Next, we find k value

The origin means (0,0) i.e x = 0, y = 0

Substituting x and y values into $y = \frac{3x^2}{2} + 2x + k$

$$0 = \frac{3(0)^2}{2} + 2(0) + k$$

i.e $k = 0$

Hence $y = \frac{3x^2}{2} + 2x$ (A)

2002/20 UME

If ${}^{dy}/_{dx} = 2x - 3$ and y = 3 when x = 0, find y in terms of x.

A. $x^2 - 3x - 3$ B. $2x^2 - 3x$ C. $x^2 - 3x + 3$ D. $x^2 - 3x$.

Solution

$$\frac{dy}{dx} = 2x - 3$$
$$dy = (2x - 3)dx$$

Integrating, we have

$$\int dy = \int (2x-3)dx$$
$$y = \frac{2x^2}{2} - 3x + k$$
$$y = x^2 - 3x + k$$

Next we find the value of k.

Given : x = 0, y = 3

Substituting x and y values into $y = x^2 - 3x + k$

$$3 = (0)^2 - 3(0) + k$$
$$3 = k$$

Hence $y = x^2 - 3x + 3$ (C)

2002/5 (Nov)

If $\frac{dA}{dr} = 3r^2 - 1$ and A = 2 when r = 1, find A when r = 2

A–10 B –8 C 8 D10

Solution

$$\frac{dA}{dr} = 3r^2 - 1$$
$$dA = (3r^2 - 1)dr$$

Integrating both sides, we have

$$\int dA = \int (3r^2 - 1)dr$$
$$A = \frac{3r^3}{3} - r + c$$
$$= r^3 - r + c$$

To get C value, we substitute for the first set of A and r values

$$2 = (1)^3 - 1 + c$$
$$2 = c$$

Thus $A = r^3 - r + 2$

Next we are to find A,when r = 2

$$A = (2)^3 - 2 + 2$$
$$A = 8 \text{ (C)}$$

1994/4 Exercise 31.7

If $\frac{dA}{dx} = 3x^3 - 1$ and A=1 when x = 0, find A when x = 1

A–10 B –8 C 8 D10

2005/37 Exercise 31.8

Given that $\frac{dy}{dx} = 6x^2 - x + 2$ and y = 5 when x = 1, find an expression for y in terms of x

A. $2y = 4x^3 - x^2 - 4x + 3$ B. $2y = 4x^3 - x^2 + 4x + 3$
C. $2y = 4x^3 + x^2 - 4x + 3$ D. $2y = 4x^3 - x^2 - 4x - 3$

2003/47 Neco Exercise 31.9

If $\frac{dy}{dx} = 4x - 3$ and y = 5 when x = 2, find y in terms of x

A $2x^2 - 3x + 5$ B $2x^2 - 3x + 3$ C $2x^2 - 3x$
D $2x^2 - 3x - 4$ E $4x$

2006/1 UME Exercise 31.10

The gradient of a curve is 2x +7 and the curve passes through point (2,0). Find the equation of the curve.

A. $y = x^2 + 14x + 11$ B. $y = x^2 + 7x + 9$
C. $y = x^2 + 7x - 18$ D. $y = x^2 + 7x + 18$

Definite integral II

2004/16 UME

Evaluate $\int_1^3 (x^2 - 1)dx$.

A. $\frac{2}{3}$ B. $-\frac{2}{3}$ C. $-6^2/_3$ D. $6^2/_3$

Solution

$$\int_1^3 (x^2 - 1)dx = \frac{x^3}{3} - x \Big|_1^3$$

We substitute for upper and lower limits

$$= \frac{(3)^3}{3} - (3) - \left[\left(\frac{1}{3}\right)^3 - (1) \right]$$
$$= \frac{27}{3} - 3 - [\, ^1/_3 - 1 \,]$$
$$= \frac{27 - 9}{3} - \left[\frac{1 - 3}{3} \right]$$
$$= \frac{18}{3} - \left[\frac{-2}{3} \right]$$
$$= \frac{18}{3} + \frac{2}{3} = \frac{20}{3} = 6^2/_3 \text{ (D)}$$

The general pattern is

$$\int_a^b f(x)dx = F(x)\Big|_a^b = F(b) - F(a)$$

Where capital F is the result of the integral and b is the upper limit while a is the lower limit.
Though we will treat the formula pictorially under application in later part of this topic,

2003/37 UME

Evaluate $\int_2^3 (x^2 - 2x)dx$

A. 4 B. 2 C. $\frac{4}{3}$ D. $\frac{1}{3}$

Solution

$$\int_2^3 (x^2 - 2x)dx = \frac{x^3}{3} - \frac{2x^2}{2}\Big|_2^3$$

$$= (\frac{3}{3})^3 - (3)^2 - \left[(\frac{2}{3})^3 - (2)^2 \right]$$

$$= \frac{27}{3} - 9 - \left(\frac{8}{3} - 4 \right)$$

$$= 0 - \left(\frac{8-12}{3} \right)$$

$$= -[-\tfrac{4}{3}] \text{ i.e } +\tfrac{4}{3} \text{ (C)}$$

2000/35 UME

Find the value of $\int_0^\pi \frac{\cos^2\Theta - 1}{\sin^2\Theta} d\Theta$

A. π B. $\frac{\pi}{2}$ C. $-\frac{\pi}{2}$ D. $-\pi$

Solution

$$\int_0^\pi \frac{\cos^2\Theta - 1}{\sin^2\Theta} d\Theta = \int_0^\pi \frac{-\sin^2\Theta}{\sin^2\Theta} d\Theta$$

Recall from $\sin^2\theta + \cos^2\theta = 1$

$\sin^2\theta = 1 - \cos^2\theta$

But $-\sin^2\theta = \cos^2\theta - 1$

$$= -\int_0^\pi d\Theta$$

$$= -\theta \Big|_0^\pi$$

$$= -\pi - [-0] \text{ i.e } -\pi \text{ (D)}$$

1999/37 UME

Evaluate $\int_{-2}^1 (x-1)^2 dx$

A. $-3\tfrac{1}{3}$ B. 7 C. 9 D. 11

Solution

We expand the bracket

$$\int_{-2}^1 (x-1)^2 dx = \int_{-2}^1 (x^2 - 2x + 1)dx$$

$$= \frac{x^3}{3} - \frac{2x^2}{2} + x \Big|_{-2}^1$$

We substitute for upper and lower limits

$$= \frac{(1)^3}{3} - (1)^2 + 1 - [\frac{(-2)^3}{3} - (-2)^2 + (-2)]$$

$$= \frac{1}{3} - 1 + 1 - [\frac{-8}{3} - 4 - 2]$$

$$= \frac{1}{3} - [\frac{-8 - 12 - 6}{3}]$$

$$= \frac{1}{3} - [\frac{-26}{3}]$$

$$= \frac{1}{3} + \frac{26}{3}$$

$$= \frac{27}{3} = 9 \text{ (C)}$$

1998/40 UME

Evaluate $\int_2^\pi (\sec^2 x - \tan^2 x)dx$

A. $\frac{\pi}{2}$ B. $\pi - 2$ C. $\frac{\pi}{3}$ D. $\pi + 2$

Solution

Recall the trig identity : $1 + \tan^2\theta = \sec^2\theta$

Thus $\int_2^\pi (\sec^2 x - \tan^2 x)dx = \int_2^\pi (1 + \tan^2 x - \tan^2 x)dx$

$$= \int_2^\pi 1dx$$

$$= x \Big|_2^\pi$$

$$= \pi - 2 \text{ (B)}$$

1997/42 UME

If $y = x(x^4 + x^2 + 1)$, evaluate $\int_{-1}^1 y\,dx$

A. $\frac{11}{12}$ B. $\frac{11}{16}$ C. $\frac{5}{6}$ D. 0

Solution

First, we expand the bracket

$$\int_{-1}^1 x(x^4 + x^2 + 1)dx = \int_{-1}^1 (x^5 + x^3 + x)dx$$

$$= \frac{x^6}{6} + \frac{x^4}{4} + \frac{x^2}{2}\Big|_{-1}^1$$

$$= (\frac{1}{6})^6 + (\frac{1}{4})^4 + (\frac{1}{2})^2 - \left[(\frac{-1}{6})^6 + (\frac{-1}{4})^4 + (\frac{-1}{2})^2 \right]$$

You will observe that all the powers of –1 in the bracket are even; hence all their values will be positive

$$= \frac{1}{6} + \frac{1}{4} + \frac{1}{2} - \left(\frac{1}{6} + \frac{1}{4} + \frac{1}{2} \right) = 0 \text{ (D)}$$

1992/41 UME

Evaluate the integral $\int_{\frac{\pi}{12}}^{\frac{\pi}{4}} 2\cos 2x\,dx$

A. $-\tfrac{1}{2}$ B. -1 C. $\tfrac{1}{2}$ D. 1

Solution

$$\int_{\frac{\pi}{12}}^{\frac{\pi}{4}} 2\cos 2x\,dx = 2\int_{\frac{\pi}{12}}^{\frac{\pi}{4}} \cos 2x\,dx$$

We change variables 2x and dx

Let u = 2x

Differentiating u with respect to x

$$\frac{du}{dx} = 2$$

$$du = 2dx$$

$$\frac{du}{2} = dx$$

Substituting

$= 2\int_{\frac{\pi}{12}}^{\frac{\pi}{4}} \cos u \frac{du}{2}$

Factor out constant $^1/_2$;

$= \frac{2}{2}\int_{\frac{\pi}{12}}^{\frac{\pi}{4}} \cos u\, du$

$= \text{Sin u} \Big|_{\pi/12}^{\pi/4}$

substituting for u

$= \text{Sin 2x} \Big|_{\pi/12}^{\pi/4}$

$= \sin 2(\pi/4) - \sin 2(\pi/12)$
$= \sin \pi/2 - \text{Sin } \pi/6$
$= \text{Sin } 90^0 - \text{Sin } 30^0$
$= 1 - {}^1/_2$ i.e $^1/_2$ (C)

2003/48 PCE

Evaluate $\int_2^4 (3x^2 - 2x + 1)dx$

A.42 B.64 C.52 D.46

Solution

Applying integration formula

$= \frac{3x^3}{3} - \frac{2x^2}{2} + x \Big|_2^4$
$= (4)^3 - (4)^2 + 4 - [(2)^3 - (2)^2 + (2)]$
$= 64 - 16 + 4 - [8 - 4 + 2]$
$= 64 + 4 - 16 - [8 + 2 - 4]$
$= 68 - 16 - [10 - 4]$
$= 52 - 6$ i.e 46 (D).

2002/40 PCE

Evaluate $\int_0^4 \frac{1}{\sqrt{x}} dx$

A. – 4 B. –2 C.2 D.4

Displaying the power of x properly, we have;

$\int_0^4 \frac{1}{\sqrt{x}} dx = \int_0^4 \frac{1}{x^{1/2}} dx$

$= \int_0^4 x^{-1/2} dx$

integrating, we have

$= \frac{x^{1/2}}{^1/_2} \Big|_0^4$

$= 2x^{1/2} \Big|_0^4$

Substituting for upper and lower limits

$= 2(4)^{1/2} - 2(0)^{1/2} = 2(2) - 0 = 4$ (D)

1999/39 UME

Evaluate $\int_{-1}^2 \left(1 - \frac{1}{x^2}\right)dx$

A.$4^1/_2$ B.$1^7/_8$ C.$1^1/_2$ D $-1^1/_2$

Solution

Displaying the power of x properly;

$\int_{-1}^2 \left(1 - \frac{1}{x^2}\right)dx = \int_{-1}^2 \left(1 - x^{-2}\right)dx$

$= \frac{1x^1}{1} - \frac{x^{-1}}{-1} \Big|_{-1}^2$

$= x + \frac{1}{x} \Big|_{-1}^2$

$= 2 + \frac{1}{2} - \left(-1 + \frac{1}{-1}\right)$
$= 2^1/_2 - (-1 -1)$
$= 2^1/_2 - (- 2)$
$= 2^1/_2 + 2 = 4^1/_2$ (A)

1998/43 PCE

Evaluate $\int_0^{\pi/2} \cos 2x\, dx$

A. –1 B.0 C.1 D.2

Solution

Applying Change of variable format;
We let u = 2x
Differentiating u with respect to x

$\frac{du}{dx} = 2$

du = 2dx

Making dx the subject formula

$\frac{du}{2} = dx$

Substituting for dx and 2x

$\int_0^{\pi/2} \cos 2x\, dx = \int_0^{\pi/2} \cos u \frac{du}{2}$

$= \frac{1}{2}\int_0^{\pi/2} \cos u\, du$

$= \frac{1}{2}\sin u \Big|_0^{\pi/2}$

$= \frac{1}{2}\sin 2x \Big|_0^{\pi/2}$

$= {}^1/_2 \sin 2 ({}^\pi/_2) - {}^1/_2 \sin 2(0)$
$= {}^1/_2 \sin \pi - {}^1/_2 \sin 0$
$= 0$ (B)

2005/4 UME Exercise 31.11

Evaluate $\int_0^{\pi/2} \sin 2x\, dx$

A. $-^1/_2$ B. 1 C. –1 D. 0

2005/1 PCE Exercise 31.12

Evaluate $\int_1^3 (3x^2 - 2x)dx$

A.18 B.17 C.16 D.15

2009/20 Exercise 31.13

Evaluate $\int_1^2 \left(\frac{x^3 - 1}{x^2}\right)dx$

A 0.5 B 1.0 C 1.5 D 2.0

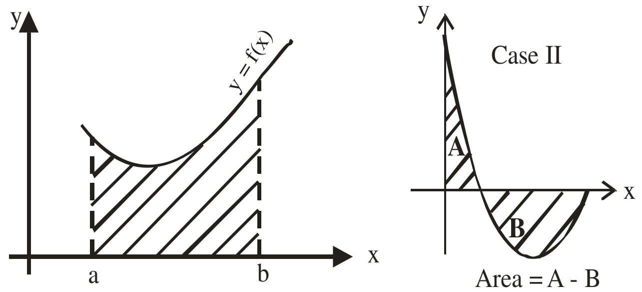
1994/42 UME Exercise 31.14

Evaluate $\int_{-1}^1 (2x+1)^2\, dx$

A. $3^2/_3$ B.4 C.$4^1/_3$ D.$4^2/_3$

Area Under a Curve and Volume

Area under a Curve



The diagram above shows a curve of a function f(x). The area under the curve along the x-axis is denoted by:

$$\int_a^b f(x)dx = F(x)\Big|_a^b = F(b) - F(a)$$

Also F is the integral result of f(x). The unit of measurement is square unit. You will observe that this is an application of definite integrals. Case II is special **Under this subtopic curve sketching is necessary**

Two types of problems exist in areas under a curve:
1. Case where the <u>value of ordinates</u> are <u>not</u> given (No restriction), here the sketched diagram is our guide

2. Case where the <u>value of ordinates</u> are given (there is restriction), here the ordinate given is our guide along with the sketched diagram

1996/11
(a) Sketch the curve $y = 8x - x^2 - 12$
(b) Draw the line of symmetry of the curve
(c) Find the area of the finite region bounded by the curve and the x - axis

Solution

(a) $y = 8x - x^2 - 12$

$$\frac{dy}{dx} = 8 - 2x$$

At turning point $\frac{dy}{dx} = 0$

Thus, $8 - 2x = 0$

$2x = 8$; $x = 4$

Value of y at the turning point

$y = 8(4) - 4^2 - 12$
$= 32 - 16 - 12$ i.e 4

Next we determine whether it is a max or min point

$\frac{d^2y}{dx^2} = -2 < 0$ Hence a maximum point

Intercepts (x – axis) here y is 0

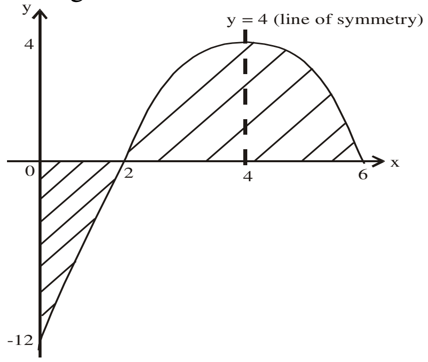
$8x - x^2 - 12 = 0$ is same as

$x^2 - 8x + 12 = 0$
$x^2 - 6x - 2x + 12 = 0$
$x(x - 6) - 2(x - 6) = 0$
$(x - 6)(x - 2) = 0$
$x - 6 = 0$ or $x - 2 = 0$ Thus, x = 6 or 2

Intercept (y – axis) here x is 0

$y = 8(0) - (0)^2 - 12$ i.e -12

Sketching



(c) **Area of bdd region** =

$$-\int_0^2 (8x - x^2 - 12)dx + \int_2^6 (8x - x^2 - 12)dx$$

First $-\int_0^2 (8x - x^2 - 12)dx$ = $-\left\{ \frac{8x^2}{2} - \frac{x^3}{3} - 12x \Big|_0^2 \right\}$

Substituting for upper & lower limits

$$= -\left[4(2^2) - \frac{(2^3)}{3} - 12(2)\right] - \left[4(0^2) - \frac{(0^3)}{3} - 12(0)\right]$$

$$= -\left(-\frac{32}{3}\right) = \frac{32}{3} \text{ Square unit}$$

Secondly $\int_2^6 (8x - x^2 - 12)dx = \frac{8x^2}{2} - \frac{x^3}{3} - 12x \Big|_2^6$

Substituting for upper & lower limits

$$= 4(6^2) - \frac{(6^3)}{3} - 12(6) - \left[4(2^2) - \frac{(2^3)}{3} - 12(2)\right]$$

$$= 144 - \frac{216}{3} - 72 - \left[16 - \frac{8}{3} - 24\right]$$

$$= 0 - \left[-8 - \frac{8}{3}\right]$$

$$= \frac{32}{3} \text{ Square unit}$$

Thus, **Area of bdd region** = $\frac{32}{3} + \frac{32}{3} = \frac{64}{3}$ Square unit

1999/10 (Nov)
Calculate the area of the finite region bounded by the curve $y = x^2 - 7x + 10$ and the x – axis

Solution

$y = x^2 - 7x + 10$

$$\frac{dy}{dx} = 2x - 7$$

At turning point $\frac{dy}{dx} = 0$

Thus, $2x - 7 = 0$

$2x = 7$; $x = 3.5$

Value of y at the turning point

$y = (3.5)^2 - 7(3.5) + 10$
$= 12.25 - 24.5 + 10$ i.e -2.25

Next we determine whether it is a max or min point

$\frac{d^2y}{dx^2} = 2 > 0$ Hence a minimum point

The intercept (x-axis) and here y is 0

$x^2 - 7x + 10 = 0$

$x^2 - 5x - 2x + 10 = 0$

$x(x - 5) - 2(x - 5) = 0$

$(x - 5)(x - 2) = 0$

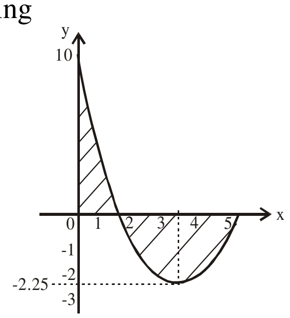
$x = 5$ or 2

Intercept (y – axis) here x is 0

$y = (0)^2 - 7(0) + 10$

$= +10$

Sketching



Area of bdd region =

$$\int_0^2 (x^2 - 7x + 10)dx - \int_2^5 (x^2 - 7x + 10)dx$$

First $\int_0^2 (x^2 - 7x + 10)dx = \left.\frac{x^3}{3} - \frac{7x^2}{2} + 10x\right|_0^2$

Substituting for upper & lower limits

$$= \frac{2^3}{3} - \frac{7(2^2)}{2} + 10(2) - \left[\frac{0^3}{3} - \frac{7(0^2)}{2} + 10(0)\right]$$

$$= \frac{8}{3} - \frac{28}{2} + 20$$

$$= \frac{26}{3} \text{ Square unit}$$

Secondly $\int_2^5 (x^2 - 7x + 10)dx = \left.\frac{x^3}{3} - \frac{7x^2}{2} + 10x\right|_2^5$

Substituting for upper & lower limits

$$= \frac{5^3}{3} - \frac{7(5^2)}{2} + 10(5) - \left[\frac{2^3}{3} - \frac{7(2^2)}{2} + 10(2)\right]$$

$$= \frac{125}{3} - \frac{175}{2} + 50 - \left[\frac{8}{3} - \frac{28}{2} + 20\right]$$

$$= \frac{125}{3} + \frac{8}{3} - \frac{175}{2} + 50 - 20 + 14$$

$$= \frac{250 + 16 - 525 + 264}{6}$$

$$= \frac{5}{6} \text{ square unit}$$

Thus, **Area of bdd region**

$$= \int_0^2 (x^2 - 7x + 10)dx - \int_2^5 (x^2 - 7x + 10)dx$$

$$= \frac{26}{3} - \frac{5}{6}$$

$$= \frac{47}{6} \text{ square unit}$$

2000/7 (Nov)

Calculate, in square units, the area of the finite region bounded by the cure $y = x^2 + 2x - 3$ and the x-axis

A $\frac{14}{3}$ B. $\frac{20}{3}$ C. $\frac{32}{3}$ D. $\frac{35}{3}$

Solution

$y = x^2 + 2x - 3$

$\frac{dy}{dx} = 2x + 2$

At turning point $\frac{dy}{dx} = 0$

Thus, $2x + 2 = 0$

$2x = -2$; $x = -1$

Value of y at the turning point

$y = (-1)^2 + 2(-1) - 3$

$= 1 - 2 - 3$ i.e -4

Next we determine whether it is a max or min point

$\frac{d^2y}{dx^2} = 2 > 0$ Hence a minimum point

the intercept (x-axis) and here y is 0

$x^2 + 2x - 3 = 0$

$x^2 + 3x - x - 3 = 0$

$x(x + 3) - 1(x + 3) = 0$

$(x + 3)(x - 1) = 0$

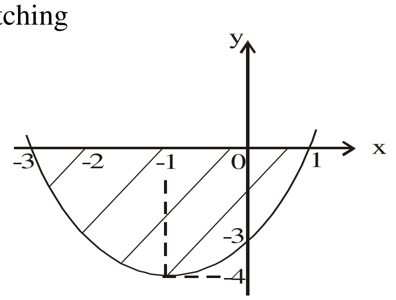
$x = -3$ or 1

Intercept (y – axis) here x is 0

$y = (0)^2 + 2(0) - 3$

$= -3$

Sketching



Area of bdd region $= -\int_{-3}^{1} (x^2 + 2x - 3)dx$

But $\int_{-3}^{1} (x^2 + 2x - 3)dx = \left.\frac{x^3}{3} + \frac{2x^2}{2} - 3x\right|_{-3}^{1}$

Substituting for upper & lower limits

$$= \frac{1}{3} + 1 - 3 - \left[\frac{(-3)^3}{3} + (-3)^2 - 3(-3)\right]$$

$$= \frac{1}{3} - 2 - [-9 + 9 + 9]$$

$$= -\frac{32}{3}$$

Thus **Area of bdd region** $= -\int_{-3}^{1} (x^2 + 2x - 3)dx$

$$= \frac{32}{3} \text{ square units (C)}$$

2001 /11 Neco

Sketch the curve $y = (x+2)(x-2)(x-4)$

(b) Find the maximum and the minimum value of the function to the nearest whole number

(c) Calculate area enclosed by the curve and the x-axis.

Solution

First, we expand to know the function proper.

$(x+2)(x-2)(x-4) = (x-4)[(x+2)(x-2)]$

$= (x-4)(x^2 - 2x + 2x - 4)$

$= (x-4)(x^2 - 4)$

$= x(x^2 - 4) - 4(x^2 - 4)$

$= x^3 - 4x^2 - 4x + 16$

i.e $y = x^3 - 4x^2 - 4x + 16$

Next, we sketch the curve at max or min $\frac{dy}{dx} = 0$

$$\frac{dy}{dx} \Rightarrow 3x^2 - 8x - 4 = 0$$

By quadratic general formula

$$x = \frac{-(-8) \pm \sqrt{(-8)^2 - 4\times 3\times(-4)}}{2\times 3}$$

$$= \frac{8 \pm \sqrt{112}}{6} = 3.1 \text{ or } -0.4$$

Next, we determine which of the turning points is max or min.

$$\frac{d^2y}{dx^2} = 6x - 8$$

<u>When x = 3.1</u>

$6x - 8 = 10.6 > 0$ minimum point

<u>When x = – 0.4</u>

$6x - 8 = -10.4 < 0$ maximum point

Intercept at x – axis here y = 0

$0 = x^3 - 4x^2 - 4x + 16$

i.e $(x+2)(x-2)(x-4) = 0$

$x = -2$ or 2 or 4

Thus ,the curve cuts the x–axis at -2, 2 and 4

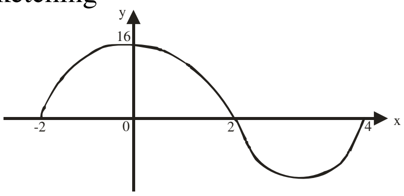
Intercept at y – axis, here x = 0

$y = 0^3 - 4(0)^2 - 4(0) + 16$

$y = 16$

The curve cut the y – axis at y =16.

Sketching



(b) Maximum value of the function

value of x is -0.4

Max value $y = (-0.4)^3 - 4(-0.4)^2 - 4(-0.4) + 16$

$= -0.064 - 0.64 + 1.6 + 16$

$= 17.6 - 0.704$

$= 16.896$

≈ 17

Minimum value of the function

value of x is 3.1

Min value of $y = (3.1)^3 - 4(3.1)^2 - 4(3.1) + 16$

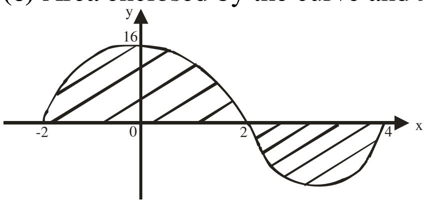
$= 29.791 - 38.44 - 12.4 + 16$

$= 45.791 - 50.84$

$= -5.049$

≈ -5

(c) Area enclosed by the curve and x-axis



$$= \int_{-2}^{2}(x^3 - 4x^2 - 4x + 16)dx + \left[-\int_{2}^{4}(x^3 - 4x^2 - 4x + 16)dx\right]$$

$$= \frac{x^4}{4} - \frac{4x^3}{3} - \frac{4x^2}{2} + 16x\Big|_{-2}^{2} - \left(\frac{x^4}{4} - \frac{4x^3}{3} - \frac{4x^2}{2} + 16x\Big|_{2}^{4}\right)$$

Readers to substitute and complete

2005/11b

(i) Sketch the curve $y = x^2 - 4$

(ii) find the area of finite region enclosed by the curve $y = x^2 - 4$, the x-axis and the ordinates $x = -2$, $x = 3$

Solution

First, we sketch the curve

$y = x^2 - 4$

At max or min $\frac{dy}{dx} = 0$

$$\frac{dy}{dx} \Rightarrow 2x = 0$$

$x = 0$

Value of y at turning point

$y = (0)^2 - 4$

$= -4$

To determine max or min point

$$\frac{d^2y}{dx^2} = 2 > 0 \text{ min point}$$

Intercept at x – axis , here y is 0

$0 = x^2 - 4$

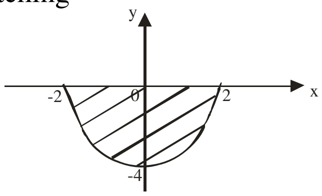
$\Rightarrow x^2 - 2^2 = 0$ thus $(x-2)(x+2) = 0$; $x = 2$ or -2

Intercept at y – axis, here x is 0

$y = (0)^2 - 4$

$= -4$

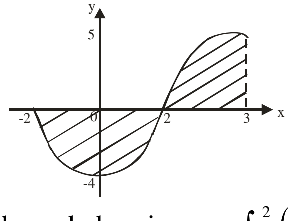
Sketching



The area of bounded region here is between – 2 and 2 but the required area of bounded region is between – 2 and 3. Hence we extend our graph to the point where x = 3 by simple substitution of values.

When x = 3, $y = 3^2 - 4$ i.e. 5

Our new sketch is



Area of bounded region = $-\int_{-2}^{2}(x^2 - 4)dx + \int_{2}^{3}(x^2 - 4)dx$

$= \int_2^3 (x^2 - 4)dx - \int_{-2}^2 (x^2 - 4)dx$

$= \frac{x^3}{3} - 4x \Big|_2^3 - \left[\frac{x^3}{3} - 4x \Big|_{-2}^{2} \right]$

$= \frac{3^3}{3} - 4(3) - \left[\frac{2^3}{3} - 4(2)\right] - \left[\frac{2^3}{3} - 4(2) - \left[\frac{(-2)^3}{3} - 4(-2)\right]\right]$

$= \frac{27}{3} - 12 - \left[\frac{8}{3} - 8\right] - \left[\frac{8}{3} - 8 - \left[\frac{-8}{3} + 8\right]\right]$

$= -3 + \frac{16}{3} - \left[\frac{-16}{3} - \frac{16}{3}\right]$ = 13 square unit

2003/25

Calculate, in square units, the area of the finite region bounded by the curve $y = 1 + x - 2x^2$, the x-axis and the coordinates x = 0 and x = 1

A $^5/_6$ B 1 C $^{13}/_6$ D 4

Solution

First, we sketch $y = 1 + x - 2x^2$

At max or min $\frac{dy}{dx} = 0$,

$\frac{dy}{dx} \Rightarrow 1 - 4x = 0$

x = $^1/_4$ i.e 0.25

Value of y at turning point

$y = 1 + (0.25) - 2(0.25)^2$

$= 1 + 0.25 - 0.125 = 1.125 \approx 1.1$

To determine max or min point

$\frac{d^2y}{dx^2}$ = - 4 < 0 max point

Intercept at x–axis, here y is 0

$0 = 1 + x - 2x^2$

$\Rightarrow 1 + 2x - x - 2x^2 = 0$

$1(1 + 2x) - x(1 + 2x) = 0$

$(1 - x)(1 + 2x) = 0$

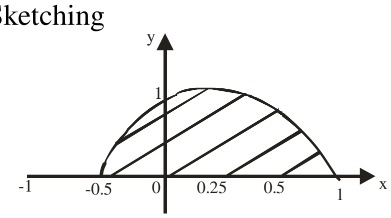
x = 1 or – ½

Intercept y – axis , here x is 0

$y = 1 + 0 - 2(0)^2$

= 1

Sketching



The area of bounded region here is between – 0.5 and 1 but the required area of bounded region is between 0 and 1.

Hence, we don't need to extend graph.

Area of bounded region = $\int_0^1 (1 + x - 2x^2)dx$

$= x + \frac{x^2}{2} - \frac{2x^3}{3}\Big|_0^1$

= 1 + $^1/_2$ – $^2/_3$

= $^5/_6$ square unit

Example AUC 1

Find the area of finite region enclosed by the curve $y = 2\sqrt{x}$ and the lines x = 3 and x = 0

A $4\sqrt{3}$ B $2\sqrt{3}$ C $16\sqrt{3}$ D$18\sqrt{3}$

Solution

$y = 2\sqrt{x}$

$y = 2x^{½}$

$\frac{dy}{dx} = x^{-½}$

At max or min point dy/dx is zero

$\frac{1}{x}$½ = 0

x = 0

Value of y turning point

$y = 2\sqrt{(0)}$ i.e 0

we are heading to a dead end, hence we try sketching the curve using values within the given range x = 0 to x = 3

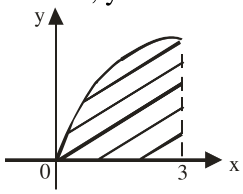
$y = 2\sqrt{x}$

When x = 0, y = 0

When x = 1, y = 2

When x =2, y = 2.8

When x = 3, y = 3.5



From our sketching the curve lies on the upper part of the x–axis within our given range of x = 0 to x = 3

Thus, area of bdd region = $\int_0^3 2x^{\frac{1}{2}}\, dx$

$= \frac{2x^{3/2}}{3/2}\Big|_0^3$

$= \frac{4}{3} x^{\frac{3}{2}}\Big|_0^3$

$= \frac{4(3)^{3/2}}{3} - \frac{4}{3}(0)^{3/2}$

$= \frac{4}{3}(27)^{1/2} = \frac{4 \times 3\sqrt{3}}{3} = 4\sqrt{3}$ (A).

Example AUC 2

Find the area bounded by the curve $y = x^2$, the x-axis and the ordinates x = – 2 and x = 2

Solution

First, we sketch the curve $y = x^2$

At max or min $\frac{dy}{dx} = 0$

$\frac{dy}{dx} \Rightarrow 2x = 0$

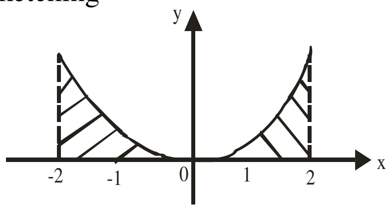
x = 0

Value of y at turning point

$y = (0)^2$ i.e 0

We are heading to a dead end, hence we try sketching the curve using values within the given range x = -2 to x = 2

$y = x^2$
When $x = -2$, $y = (-2)^2$ i.e. 4
When $x = -1$, $y = (-1)^2$ i.e. 1
When $x = 0$ $y = (0)^2$ i.e. 0
When $x = 1$ $y = 1^2$ i.e. 1
When $x = 2$ $y = 2^2$ i.e. 4
Sketching



Area of bounded region = $\int_{-2}^{0} x^2 dx + \int_{0}^{2} x^2 dx$

$= \left.\frac{x^3}{3}\right|_{-2}^{0} + \left.\frac{x^3}{3}\right|_{0}^{2}$

$= \frac{0}{3} - \left[\frac{(-2)^3}{3}\right] + \frac{2^3}{3} - \frac{0}{3}$

$= \frac{8}{3} + \frac{8}{3}$

$= \frac{16}{3} \approx 5.3$ square units

2010/15 Neco Exercise 31.15
Find the area bounded by the curve $y = 2x^2 + 5$, the x-axis and the ordinates $x = 1$, $x = 3$
A $5\frac{2}{3}$ square unit B $27\frac{1}{3}$ square unit
C $28\frac{1}{3}$ square unit D 33 square unit E $38\frac{2}{3}$ square unit

Exercise 31.16
Find the area bounded by the curve $y = 3x^2$ and the x-axis and the ordinates $x = 0$ and $x = 3$

2007/2 Neco Exercise 31.17
Find the area bounded by the curve $y = x^2 - 5x$, the x-axis and the co-ordinates $x = 0$ and $x = 8$ to the nearest square unit
A 4 B 6 C 8 D 11 E 13

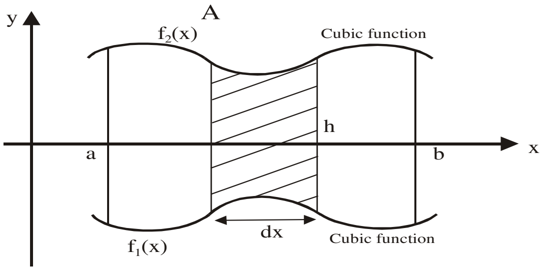
2002/35 (Nov) Exercise 31.18
Calculate in square units, the area of the finite region bounded by the x - axis, y – axis and the line $2y = x + 6$
A 3 B 6 C 9 D 12

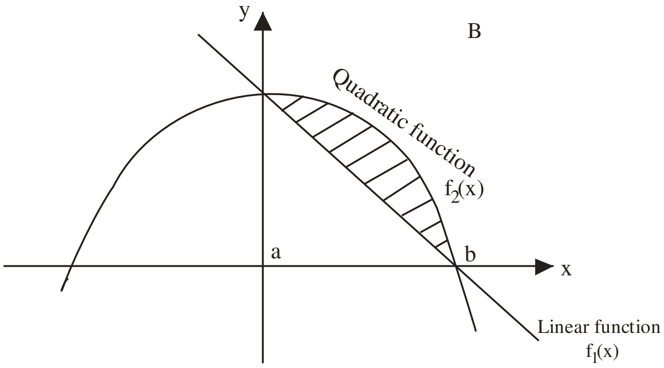
1995/3 (Nov) Exercise 31.19
Calculate in square units, the area of the finite region bounded by the lines $\frac{x}{4} + \frac{y}{5} = 1$, $y = 0$ and $x = 0$

A 5 B10 C 15 D 18 E 20

Exercise 31.20
Find the area of the finite region bounded by the curve $y = x^3 + x^2 - 2x$ and the x-axis

Area bounded between two Curves





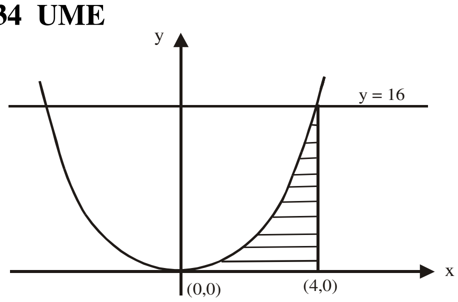
Area between two curves can be treated under:
(i) a case where the sketching can be done by 1st and 2nd differential and solving for maximum or minimum points

(ii) a case where the sketching cannot be done by 1st and 2nd differential since all the results will play around zero hence we sketch by substitution of values (simple range) into the two equations.

Area between two curves $A = \int_a^b \left(f_2(x) - f_1(x)\right)dx$

The order of substituting equation $f_1(x)$ and $f_2(x)$ is based on the uppermost function from the sketch or the presented (diagram)

2000/34 UME



If the diagram above is the graph of $y = x^2$, the shaded area is
A. 64square units B.<u>128</u>square units C.<u>64</u>square units
D.32 square units
3 3

Solution
This is a typical question of area between two curves: Shine your eyes and you will see $y = 16$ (linear equation) and the curve equation which was mentioned $y = x^2$.

Of course, if you err and think it is *area under a curve*, the answer is there too as 64/3.
Here our limits were given in a ,b co-ordinate forms. x = 0 and x = 4;
Recall that the two functions are to be subtracted.
The graph indicates $f_2(x)$ to be y = 16

$$\int_0^4 \left(16 - x^2\right)dx$$

$$= 16x - \frac{x^3}{3}\Big|_0^4$$

$$= 16(4) - \left(\frac{4}{3}\right)^3 - \left(16(0) - \left(\frac{0}{3}\right)^3\right)$$

$$= 64 - \frac{64}{3}$$

$$= \frac{192 - 64}{3}$$

$= \dfrac{128}{3}$ square units (B)

2000/11 (Nov)

a (i) Sketch on the same axes the curve $y = 5x - x^2$ and the line $y = x + 3$

(ii) Find the co-ordinates of their points of intersection

b Find the area of the segment cut off from the curve by the line.

Solution

$y = 5x - x^2$

$$\frac{dy}{dx} = 5 - 2x$$

At turning point $\dfrac{dy}{dx} = 0$

Thus, $5 - 2x = 0$

$5 = 2x$

$x = {}^5/_2$ i.e 2.5

Value of y at the turning point

$y = 5(2.5) - (2.5)^2$

$= 12.5 - 6.25)$ i.e 6.25

Next we determine whether it is a max or min point

$$\frac{d^2y}{dx^2} = -2 < 0 \text{ Hence a maximum point}$$

Intercepts (x – axis) here y is 0

$5x - x^2 = 0$

$x(5 - x) = 0$

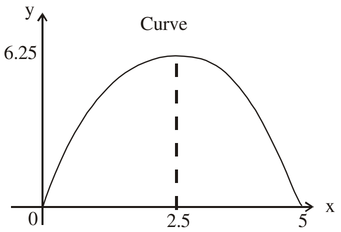
$x = 0$ or $5 - x = 0$

$x = 0$ or 5

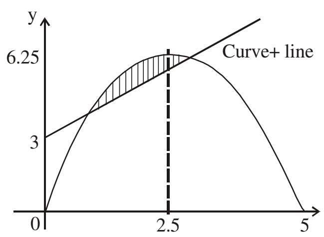
Intercept (y – axis) here x is 0

$y = 5(0) - (0^2)$ i.e 0

Sketching the curve first then fit in the straight line



Incorporating line $y = x + 3$



We only need any two points to sketch a straight line
Here let us take When x = 0 , y = 3 & When x = 1 , y = 4

(iii) Co-ordinates of their points of intersection

$5x - x^2 = x + 3$

$x^2 - 4x + 3 = 0$

$x^2 - 3x - x + 3 = 0$

$(x - 3)(x - 1) = 0$ Thus x = 3 or 1

And they are our upper and lower limits as well
From the sketch the curve is uppermost, hence

$$\text{Area of the segment} = \int_1^3 \left[\left(5x - x^2\right) - (x + 3)\right] dx$$

$$= \int_1^3 \left(4x - x^2 - 3\right)dx$$

$$= \frac{4x^2}{2} - \frac{x^3}{3} - 3x\Big|_1^3$$

$$= \frac{4(3)^2}{2} - \frac{(3)^3}{3} - 3(3) - \left[\frac{4}{2} - \frac{1}{3} - 3\right]$$

$= 18 - 9 - 9 - (-1^1/_3)$

$= 1^1/_3$ square units

2001/36 UME

Find the area bounded by the curves $y = 4 - x^2$ and $y = 2x + 1$.

A. $20^1/_3$ sq. units B. $20^2/_3$ sq. units
C. $10^2/_3$ sq. units D. $10^1/_3$ sq. units

Solution

$y = 4 - x^2$

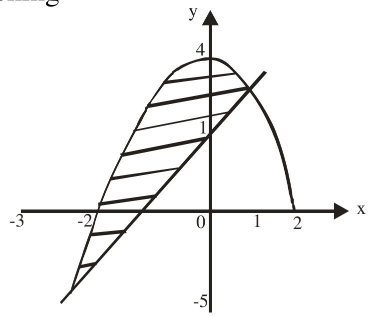
$$\frac{dy}{dx} = -2x$$

At turning point $\dfrac{dy}{dx} = 0$ i.e $-2x = 0$

It will lead us to a dead end
Let us substitute using a simple range of – 3 to 3

$y = 4 - x^2$		$y = 2x + 1$
When	x = -3 , y = -5	x = -3 , y = -5
	x = -2 , y = 0	x = 1 , y = 3
	x = -1 , y = 3	
	x = 0 , y = 4	
	x = 1 , y = 3	
	x = 2 , y = 0	
	x = 3 , y = -5	

Sketching



Here our limits were not given; thus we equate the two equations and solve the resulting quadratic equation.

$4 - x^2 = 2x + 1$

$4 - x^2 - 2x - 1 = 0$

rearranging

$3 - 2x - x^2 = 0$

multiply through by minus

or $2x + 1 - 4 + x^2 = 0$

$x^2 + 2x - 3 = 0$

$x^2 + 2x - 3 = 0$
Factorising
$x^2 + 3x - x - 3 = 0$
$x(x + 3) - 1(x + 3) = 0$
$(x - 1)(x + 3) = 0$
$x = 1$ or -3
The smallest value is –3 and that is the lower limit
Note that from the sketch the curve is uppermost, hence

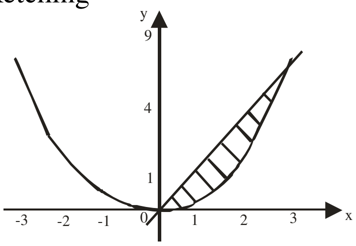
$$\text{Area bounded} = \int_{-3}^{1}\left[\left(4-x^2\right)-\left(2x+1\right)\right]dx$$
$$= \int_{-3}^{1}\left(4-x^2-2x-1\right)dx$$
$$= \int_{-3}^{1}\left(3-2x-x^2\right)dx$$
$$= 3x - \frac{2x^2}{2} - \frac{x^3}{3}\Big|_{-3}^{1}$$
$$= 3x - x^2 - \frac{x^3}{3}\Big|_{-3}^{1}$$
$$= 3(1) - (1)^2 - \left(\frac{1}{3}\right)^3 - \left[3(-3) - (-3)^2 - \left(\frac{-3}{3}\right)^3\right]$$
$$= 3 - 1 - \frac{1}{3} - \left[-9 - 9 - \left(-\frac{27}{3}\right)\right]$$
$$= 2 - \tfrac{1}{3} - (-9)$$
$$= 2 - \tfrac{1}{3} + 9$$
$$= \frac{6 - 1 + 27}{3}$$
$$= \tfrac{32}{3} \text{ i.e } 10\tfrac{2}{3} \text{ sq units} \quad (C)$$

2000/40 PCE
Find the area bounded by the curve $y = x^2$ and the straight line $y = 3x$.
A. 22.5 B. 4.5 C. – 4.5 D. – 22.5

Solution
It is obvious $y = x^2$ differential will play around zero.
Let us substitute using a simple range of – 3 to 3

$y = x^2$	$y = 3x$
When $x = -3$, $y = 9$	When $x = -3$, $y = -9$
$x = -2$, $y = 4$	When $x = 0$, $y = 0$
$x = -1$, $y = 1$	
$x = 0$, $y = 0$	
$x = 1$, $y = 1$	
$x = 2$, $y = 4$	
$x = 3$, $y = 9$	

Sketching



Equating the two curves to get limits of integral
$x^2 = 3x$
$x^2 - 3x = 0$
$x(x - 3) = 0$
$x = 0$ or 3
Note that from the sketch the line is uppermost, Thus

$$\text{Area bounded} = \int_{0}^{3}(3x - x^2)\,dx$$
$$= \frac{3x^2}{2} - \frac{x^3}{3}\Big|_{0}^{3}$$
$$= \frac{27}{2} - \frac{27}{3}$$
$$\frac{81-54}{6} = \frac{27}{6} \text{ i.e. } 4.5 (B)$$

2002/ 9
A fixed point A(m, n) lies on the curve $y = x^2 + 1$ and the line $y = x + 7$
(a) Find: (i) the two possible coordinates of A
(ii) the distances between the two locations of A
(b) Sketch the curve and draw the line
(c) Find the area of the finite region enclosed by the line and the curve.

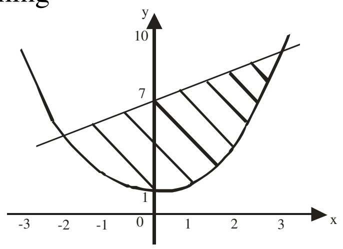
Solution
a . i Points of intersection first
$x^2 + 1 = x + 7$
$x^2 - x - 6 = 0$
$x^2 + 2x - 3x - 6 = 0$
$x(x + 2) - 3(x + 2) = 0$
$(x + 2)(x - 3) = 0$
$x = -2$ or 3
Coordinate of intersection A
When $x = -2$, $y = (-2)^2 - (-2) - 6$ i.e. 0
When $x = 3$, $y = (3)^2 - 3 - 6$ i.e. 0
Thus coordinates of A are (-2, 0) and (3, 0)

(ii) Distances between two points of A
$$= \sqrt{[(3-(-2)]^2 + (0-0)^2}$$
$$= \sqrt{25} \text{ i.e. } 5$$

b. $y = x^2 + 1$
$$\frac{dy}{dx} = 2x$$
it is obvious ,everything here is about zero .Let us work by substitution of values $x = -2$ to 3 say from $x = -3$ to 3

$y = x^2 + 1$	$y = x + 7$
When $x = -3$, $y = 10$	When $x = 0$, $y = 7$
$x = -2$, $y = 5$	$x = 3$, $y = 10$
$x = -1$, $y = 2$	
$x = 0$, $y = 1$	
$x = 1$, $y = 2$	
$x = 2$, $y = 5$	
$x = 3$, $y = 10$	

Sketching



From the sketch, uppermost function is the straight line

c. $$\text{Area enclosed} = \int_{-2}^{3}\left[x + 7 - (x^2 + 1)\right]dx$$
$$= \int_{-2}^{3}(x - x^2 + 6)\,dx$$

$$= \frac{x^2}{2} - \frac{x^3}{3} + 6x\Big|_{-2}^{3}$$

$$= \frac{3^2}{2} - \frac{3^3}{3} + 6(3) - \left[\frac{(-2)^2}{2} - \frac{(-2)^3}{3} + 6(-2)\right]$$

$$= \frac{9}{2} - 9 + 18 - \left[2 + \frac{8}{3} - 12\right]$$

$$= \frac{27}{2} + \frac{22}{3}$$

= 125/6 square units. i.e 20.83 square units

1994 / 9

Given the curves, $y = 4x - x^2$ and $y = x^2 - 2x$

(a) Determine the coordinates of their points of intersection.

(b) Sketch, on the same axes, the two curves.

(c) Find the area of the finite region bounded by the two curves.

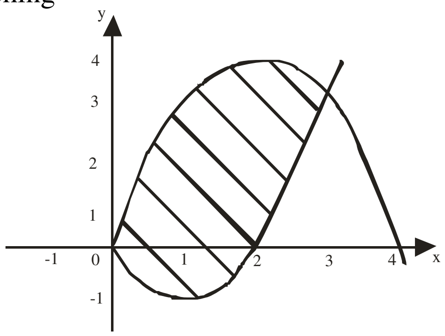
Solution

(a) Their points of intersection:

$$4x - x^2 = x^2 - 2x$$
$$4x - x^2 - x^2 + 2x = 0$$
$$4x - 2x^2 + 2x = 0$$
$$2x - x^2 + x = 0$$
$$3x - x^2 = 0$$
$$x(3 - x) = 0$$
$$x = 0 \text{ or } 3$$

(b) $y = 4x - x^2$	$y = x^2 - 2x$
$\frac{dy}{dx} = 4 - 2x$	$\frac{dy}{dx} = 2x - 2$
At max or min dy/dx = 0	At max or min dy/dx = 0
$4 - 2x = 0$	$2x - 2 = 0$
$x = 2$	$x = 1$
Values of y at turning points	Values of y at turning points
$y = 4(2) - 2^2$	$y = 1^2 - 2(1)$
$= 4$	$= -1$
$\frac{d^2y}{dx^2} = -2 < 0$ Max point	$\frac{d^2y}{dx^2} = 2 > 0$ Min point
Intercept at x-axis here y is 0	**Intercept at x-axis here y is 0**
$4x - x^2 = 0$	$x^2 - 2x = 0$
$x(4 - x) = 0$	$x(x - 2) = 0$
$x = 0$ or 4	$x = 0$ or 2
Intercept at y-axis here x is 0	**Intercept at y-axis here x is 0**
$y = 4(0) - (0)^2$	$y = (0)^2 - 2(0)$
$= 0$	$= 0$

Sketching



Our limits are the points of intersection i.e 0 and 3.

From the sketch, <u>uppermost</u> function is $4x - x^2$

(c) Area of the finite region bounded by the two curves.

$$= \int_0^3 \left[(4x - x^2) - (x^2 - 2x)\right] dx$$

$$= \int_0^3 (3x - x^2)\, dx$$

$$= \frac{3x^2}{2} - \frac{x^3}{3}\Big|_0^3$$

$$= \frac{3(3)^2}{2} - \frac{(3)^3}{3} = \frac{27}{2} - 9 = \frac{9}{2} \text{ square units.}$$

Example ABTC 1 ***Acid test for area between two curves***

Find the area of the figure bounded by the curve $y = \frac{x^5}{16}$ and the line $y = x$ and between $x = -2$ and 2

Solution

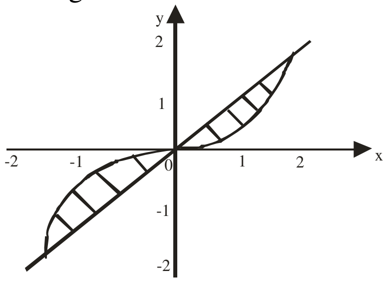
$y = x$	$y = \frac{x^5}{16}$
$\frac{dy}{dx} = 1$	$\frac{dy}{dx} = \frac{5x^4}{16}$
At max or min dy/dx = 0	At max or min dy/dx = 0

It is obvious, everything here is about zero. Let us work by substitution of values x = - 2 to 2

$y = \frac{x^5}{16}$	
x = - 2,	y = - 2
x = - 1.5,	y = - 0.47
x = - 1,	y = -0.0625
x = - 0.5,	y = -0.002
x = 0,	y = 0
x = 0.5,	y = 0.002
x = 1,	y = 0.0625
x = 1.5,	y = 0.47
x = 2,	y = 2

y = x	
x = -2,	y = -2
x = 2	y = 2

Sketching



In the interval 0<x<2 the straight line y = x lies above the curve $y = x^5/16$ but in the interval -2<x<0 their roles are reversed hence

Area of the figure bounded

$$= \int_{-2}^{0}\left(\frac{x^5}{16} - x\right)dx + \int_0^2\left(x - \frac{x^5}{16}\right)dx$$

$$= \frac{x^6}{96} - \frac{x^2}{2}\Big|_{-2}^{0} + \frac{x^2}{2} - \frac{x^6}{96}\Big|_0^2$$

$$= 0 - \left(\frac{64}{96} - \frac{4}{2}\right) + \left(\frac{4}{2} - \frac{64}{96}\right) - 0$$

= 8/3 or $2\frac{2}{3}$ square units

Example ABTC 2 exceptional case

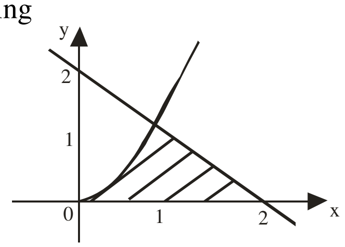
Find the area of the figure bounded by the curve $y = x^2$ and the line $y = 2 - x$ and the x-axis for which $x \geq 0$, $y \geq 0$

Solution

It is obvious $y = x^2$ differential will play around zero. Let us substitute using a simple range but bear in mind the condition $x \geq 0$, $y \geq 0$ i.e. positive axis only. Take 0 to 2

$y = x^2$		$y = 2 - x$	
$x = 0$,	$y = 0$	$x = 0$,	$y = 2$
$x = 0.5$,	$y = 0.25$	$x = 2$,	$y = 0$
$x = 1$,	$y = 1$		
$x = 1.5$,	$y = 2.25$		
$x = 2$,	$y = 4$		

Sketching



The curve and the line meeting points

$$x^2 = 2 - x$$

$$x^2 + x - 2 = 0$$

Factorization

$$x^2 + 2x - x - 2 = 0$$

$$x(x + 2) - 1(x + 2) = 0$$

$$x + 2 = 0 \text{ or } x - 1 = 0$$

$$x = -2 \text{ or } 1$$

Since we are considering $x \geq 0$ and $y \geq 0$, we take 1

Hence they meet at $x = 1$.

The area of $\int_0^1 x^2 dx + \int_1^2 (2 - x)dx$

What happened is the region from $x = 0$ to $x = 1$ is under the curve (x^2) and from $x = 1$ to $x = 2$ that region is under the straight line

$$= \frac{x^3}{3}\Big|_0^1 + 2x - \frac{x^2}{2}\Big|_1^2$$

$$= \tfrac{1}{3} + [\,4 - \tfrac{4}{2} - (2 - \tfrac{1}{2})\,]$$

$$= \tfrac{1}{3} + (2 - 1\tfrac{1}{2})$$

$$= \tfrac{1}{3} + \tfrac{1}{2}$$

$= 5/6$ square unit.

Exercise 31.21

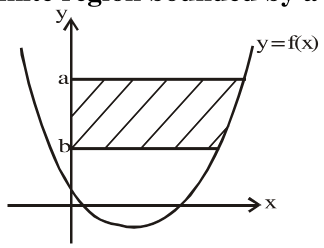
Find the area of the region bounded by the curve $y = 3 - x^2$ and the line $y = -x + 1$ and between $x = 0$ and $x = 2$

2005/2 UME Exercise 31.22

Find the area of the figure bounded by the given pair of curves $y = x^2 - x + 3$ and $y = 3$

A. $\frac{7}{6}$ units(sq) B. $\frac{1}{6}$ units(sq) C. $\frac{17}{6}$ units(sq) D. $\frac{5}{6}$ units(sq)

Area of finite region bounded by a curve & the y-axis



The diagram above shows a curve of a function f(x). The area of the segment cut of by the y-axis at points a and b is denoted by

$$\int_b^a f(y)dy = F(y)\Big|_b^a = F(a) - F(b)$$

Here F is the integral result of f(y).

Example AII 1

Find the area of the finite region bounded by the curve $y = 9x^2 + 1$ and the y-axis at $y = 3$ and $y = 1$

Solution

$$\text{Area } = \int_b^a f(y)dy$$

First we make x the subject formula

$y = 9x^2 + 1$ becomes

$$y - 1 = 9x^2$$

$$\frac{y - 1}{9} = x^2$$

$$\sqrt{\frac{y-1}{9}} = x\text{ ; Thus, } x = \frac{\sqrt{y-1}}{3}$$

$$\text{Area } = \int_1^3 \frac{(y-1)^{1/2}}{3} dy$$

$$= \frac{1}{3}\int_1^3 (y-1)^{1/2} dy$$

$$= \left(\frac{1}{3}\right)\frac{2(y-1)^{3/2}}{3}\Bigg|_1^3$$

$$= \frac{2(y-1)^{3/2}}{9}\Bigg|_1^3$$

$$= \frac{2(2)^{3/2}}{9} - 0$$

$$= \frac{2\sqrt{8}}{9} \text{ i.e } \frac{4\sqrt{2}}{9} \text{ square unit}$$

Example AII 2

Find the area of the finite region bounded by the curve $y = 16x^2$ and the y-axis at $y = 4$ and $y = 0$

Solution

$$\text{Area } = \int_b^a f(y)dy$$

First we make x the subject formula

$y = 16x^2$ becomes

$$\frac{y}{16} = x^2$$

$$\sqrt{\frac{y}{16}} = x \quad \text{Thus, } x = \frac{\sqrt{y}}{4} \text{ i.e } \frac{y^{1/2}}{4}$$

Thus, area $= \int_0^4 \frac{y^{1/2}}{4}\, dy$

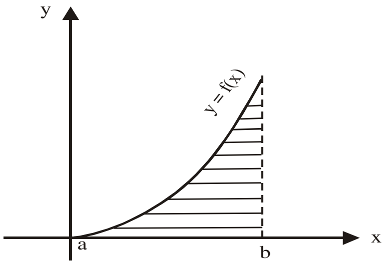
$$= \frac{1}{4}\int_0^4 y^{1/2}\, dy$$

$$= \left(\frac{1}{4}\right) y^{3/2} \div \frac{3}{2} \Big|_0^4$$

$$= \frac{y^{3/2}}{6} \Big|_0^4$$

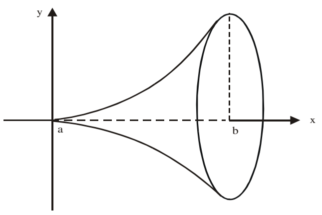
$$= \frac{(4)^{3/2}}{6} - 0 \quad = \frac{4}{3} \text{ square units}$$

VOLUMES



If the area below the curve above is rotated:

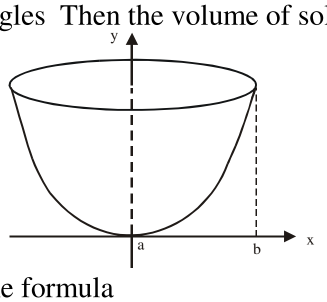
(i) completely about the x - axis or 360^0 or four right angles. Then the volume of solid it generates is



With the formula $V = \int_a^b \pi\, y^2 dx$

Note: y^2 and dx

(ii) Completely about the y – axis or 360^0 or four right angles Then the volume of solid it generates is



With the formula

$\int_a^b \pi\, x^2 dy$ ***Note***: x^2 and dy

The solid figure generated by the two diagrams are called solid of revolution, hence this subtopic is sometimes called *volumes of revolution*

2000/37

A bowl is designed by revolving completely the area enclosed by $y = x^2 - 1$, $y = 0$ and $x \geq 0$ around the y – axis at y =0 and y = 3. What is the volume of this bowl.?

A. 7π cubic units B. $^{15}/_2\, \pi$ cubic units C. 8π cubic units D. $^{17}/_2\, \pi$ cubic units

Solution

Rotation around y – axis, volume is given by:

$$V = \int_a^b \pi\, x^2 dy$$

We first of all make x the subject formula from $y = x^2 - 1$

$$y + 1 = x^2$$

Taking the square root of both sides

$$(y + 1)^{1/2} = x$$

substituting for x in the formula

$$V = \int_0^3 \pi\left[(y+1)^{\frac{1}{2}}\right]^2 dy$$

$$= \pi \left(\frac{y^2}{2} + y\right)\Bigg|_0^3$$

substituting for the upper and lower limits

$$= \pi\left(\frac{3^2}{2} + 3\right) - \pi\left(\frac{(0)^2}{2} - 0\right) = \pi\left(\,^9/_2 + 3\right)$$

$= {}^{15}/_2\, \pi$ cubic units (B)

2003/12a Exercise 31.23

(i) Sketch the graph of $y = x^2$, showing the area in the first quadrant enclosed by the curve and the two lines y = 1 and y = 3

(ii) If the area in a(i) is rotated about the y – axis, find the volume of the solid generated

Logarithms

$x \longrightarrow \log x$

											Differences								
x	0	1	2	3	4	5	6	7	8	9	1	2	3	4	5	6	7	8	9
10	0000	0043	0086	0128	0170	0212	0253	0294	0334	0374	4	8	12	17	21	25	29	33	37
11	0414	0453	0492	0531	0569	0607	0645	0682	0719	0755	4	8	11	15	19	23	26	30	34
12	0792	0828	0864	0899	0934	0969	1004	1038	1072	1106	3	7	10	14	17	21	24	28	31
13	1139	1173	1206	1239	1271	1303	1335	1367	1399	1430	3	6	10	13	16	19	23	26	29
14	1461	1492	1523	1553	1584	1614	1644	1673	1703	1732	3	6	9	12	15	18	21	24	27
15	1761	1790	1818	1847	1875	1903	1931	1959	1987	2014	3	6	8	11	14	17	20	22	25
16	2041	2068	2095	2122	2148	2175	2201	2227	2253	2279	3	5	8	11	13	16	18	21	24
17	2304	2330	2355	2380	2405	2430	2455	2480	2504	2529	2	5	7	10	12	15	17	20	22
18	2553	2577	2601	2625	2648	2672	2695	2718	2742	2765	2	5	7	9	12	14	16	19	21
19	2788	2810	2833	2856	2878	2900	2923	2945	2967	2989	2	4	7	9	11	13	16	18	20
20	3010	3032	3054	3075	3096	3118	3139	3160	3181	3201	2	4	6	8	11	13	15	17	19
21	3222	3243	3263	3284	3304	3324	3345	3365	3385	3404	2	4	6	8	10	12	14	16	18
22	3424	3444	3464	3483	3502	3522	3541	3560	3579	3598	2	4	6	8	10	12	14	15	17
23	3617	3636	3655	3674	3692	3711	3729	3747	3766	3784	2	4	6	7	9	11	13	15	17
24	3802	3820	3838	3856	3874	3892	3909	3927	3945	3962	2	4	5	7	9	11	12	14	16
25	3979	3997	4014	4031	4048	4065	4082	4099	4116	4133	2	3	5	7	9	10	12	14	15
26	4150	4166	4183	4200	4216	4232	4249	4265	4281	4298	2	3	5	7	8	10	11	13	15
27	4314	4330	4346	4362	4378	4393	4409	4425	4440	4456	2	3	5	6	8	9	11	13	14
28	4472	4487	4502	4518	4533	4548	4564	4579	4594	4609	2	3	5	6	8	9	11	12	14
29	4624	4639	4654	4669	4683	4698	4713	4728	4742	4757	1	3	4	6	7	9	10	12	13
30	4771	4786	4800	4814	4829	4843	4857	4871	4886	4900	1	3	4	6	7	9	10	11	13
31	4914	4928	4942	4955	4969	4983	4997	5011	5024	5038	1	3	4	6	7	8	10	11	12
32	5051	5065	5079	5092	5105	5119	5132	5145	5159	5172	1	3	4	5	7	8	9	11	12
33	5185	5198	5211	5224	5237	5250	5263	5276	5289	5302	1	3	4	5	6	8	9	10	12
34	5315	5328	5340	5353	5366	5378	5391	5403	5416	5428	1	3	4	5	6	8	9	10	11
35	5441	5453	5465	5478	5490	5502	5514	5527	5539	5551	1	2	4	5	6	7	9	10	11
36	5563	5575	5587	5599	5611	5623	5635	5647	5658	5670	1	2	4	5	6	7	8	10	11
37	5682	5694	5705	5717	5729	5740	5752	5763	5775	5786	1	2	3	5	6	7	8	9	10
38	5798	5809	5821	5832	5843	5855	5866	5877	5888	5899	1	2	3	5	6	7	8	9	10
39	5911	5922	5933	5944	5955	5966	5977	5988	5999	6010	1	2	3	4	5	7	8	9	10
40	6021	6031	6042	6053	6064	6075	6085	6096	6107	6117	1	2	3	4	5	6	8	9	10
41	6128	6138	6149	6160	6170	6180	6191	6201	6212	6222	1	2	3	4	5	6	7	8	9
42	6232	6243	6253	6263	6274	6284	6294	6304	6314	6325	1	2	3	4	5	6	7	8	9
43	6335	6345	6355	6365	6375	6385	6395	6405	6415	6425	1	2	3	4	5	6	7	8	9
44	6435	6444	6454	6464	6474	6484	6493	6503	6513	6522	1	2	3	4	5	6	7	8	9
45	6532	6542	6551	6561	6571	6580	6590	6599	6609	6618	1	2	3	4	5	6	7	8	9
46	6628	6637	6646	6656	6665	6675	6684	6693	6702	6712	1	2	3	4	5	6	7	7	8
47	6721	6730	6739	6749	6758	6767	6776	6785	6794	6803	1	2	3	4	5	5	6	7	8
48	6812	6821	6830	6839	6848	6857	6866	6875	6884	6893	1	2	3	4	4	5	6	7	8
49	6902	6911	6920	6928	6937	6946	6955	6964	6972	6981	1	2	3	4	4	5	6	7	8
50	6990	6998	7007	7016	7024	7033	7042	7050	7059	7067	1	2	3	3	4	5	6	7	8
51	7076	7084	7093	7101	7110	7118	7126	7135	7143	7152	1	2	3	3	4	5	6	7	8
52	7160	7168	7177	7185	7193	7202	7210	7218	7226	7235	1	2	2	3	4	5	6	7	7
53	7243	7251	7259	7267	7275	7284	7292	7300	7308	7316	1	2	2	3	4	5	6	6	7
54	7324	7332	7340	7348	7356	7364	7372	7380	7388	7396	1	2	2	3	4	5	6	6	7

											Differences								
x	0	1	2	3	4	5	6	7	8	9	1	2	3	4	5	6	7	8	9
55	7404	7412	7419	7427	7435	7443	7451	7459	7466	7474	1	2	2	3	4	5	5	6	7
56	7482	7490	7497	7505	7513	7520	7528	7536	7543	7551	1	2	2	3	4	5	5	6	7
57	7559	7566	7574	7582	7589	7597	7604	7612	7619	7627	1	2	2	3	4	5	5	6	7
58	7634	7642	7649	7657	7664	7672	7679	7686	7694	7701	1	1	2	3	4	4	5	6	7
59	7709	7716	7723	7731	7738	7745	7752	7760	7767	7774	1	1	2	3	4	4	5	6	7
60	7782	7789	7796	7803	7810	7818	7825	7832	7839	7846	1	1	2	3	4	4	5	6	6
61	7853	7860	7868	7875	7882	7889	7896	7903	7910	7917	1	1	2	3	4	4	5	6	6
62	7924	7931	7938	7945	7952	7959	7966	7973	7980	7987	1	1	2	3	3	4	5	6	6
63	7993	8000	8007	8014	8021	8028	8035	8041	8048	8055	1	1	2	3	3	4	5	5	6
64	8062	8069	8075	8082	8089	8096	8102	8109	8116	8122	1	1	2	3	3	4	5	5	6
65	8129	8136	8142	8149	8156	8162	8169	8176	8182	8189	1	1	2	3	3	4	5	5	6
66	8195	8202	8209	8215	8222	8228	8235	8241	8248	8254	1	1	2	3	3	4	5	5	6
67	8261	8267	8274	8280	8287	8293	8299	8306	8312	8319	1	1	2	3	3	4	5	5	6
68	8325	8331	8338	8344	8351	8357	8363	8370	8376	8382	1	1	2	3	3	4	4	5	6
69	8388	8395	8401	8407	8414	8420	8426	8432	8439	8445	1	1	2	2	3	4	4	5	6
70	8451	8457	8463	8470	8476	8482	8488	8494	8500	8506	1	1	2	2	3	4	4	5	6
71	8513	8519	8525	8531	8537	8543	8549	8555	8561	8567	1	1	2	2	3	4	4	5	5
72	8573	8579	8585	8591	8597	8603	8609	8615	8621	8627	1	1	2	2	3	4	4	5	5
73	8633	8639	8645	8651	8657	8663	8669	8675	8681	8686	1	1	2	2	3	4	4	5	5
74	8692	8698	8704	8710	8716	8722	8727	8733	8739	8745	1	1	2	2	3	4	4	5	5
75	8751	8756	8762	8768	8774	8779	8785	8791	8797	8802	1	1	2	2	3	3	4	5	5
76	8808	8814	8820	8825	8831	8837	8842	8848	8854	8859	1	1	2	2	3	3	4	5	5
77	8865	8871	8876	8882	8887	8893	8899	8904	8910	8915	1	1	2	2	3	3	4	4	5
78	8921	8927	8932	8938	8943	8949	8954	8960	8965	8971	1	1	2	2	3	3	4	4	5
79	8976	8982	8987	8993	8998	9004	9009	9015	9020	9025	1	1	2	2	3	3	4	4	5
80	9031	9036	9042	9047	9053	9058	9063	9069	9074	9079	1	1	2	2	3	3	4	4	5
81	9085	9090	9096	9101	9106	9112	9117	9122	9128	9133	1	1	2	2	3	3	4	4	5
82	9138	9143	9149	9154	9159	9165	9170	9175	9180	9186	1	1	2	2	3	3	4	4	5
83	9191	9196	9201	9206	9212	9217	9222	9227	9232	9238	1	1	2	2	3	3	4	4	5
84	9243	9248	9253	9258	9263	9269	9274	9279	9284	9289	1	1	2	2	3	3	4	4	5
85	9294	9299	9304	9309	9315	9320	9325	9330	9335	9340	1	1	2	2	3	3	4	4	5
86	9345	9350	9355	9360	9365	9370	9375	9380	9385	9390	1	1	2	2	3	3	4	4	5
87	9395	9400	9405	9410	9415	9420	9425	9430	9435	9440	0	1	1	2	2	3	3	4	4
88	9445	9450	9455	9460	9465	9469	9474	9479	9484	9489	0	1	1	2	2	3	3	4	4
89	9494	9499	9504	9509	9513	9518	9523	9528	9533	9538	0	1	1	2	2	3	3	4	4
90	9542	9547	9552	9557	9562	9566	9571	9576	9581	9586	0	1	1	2	2	3	3	4	4
91	9590	9595	9600	9605	9609	9614	9619	9624	9628	9633	0	1	1	2	2	3	3	4	4
92	9638	9643	9647	9653	9657	9661	9666	9671	9675	9680	0	1	1	2	2	3	3	4	4
93	9685	9689	9694	9699	9703	9708	9713	9717	9722	9727	0	1	1	2	2	3	3	4	4
94	9731	9736	9741	9745	9750	9754	9759	9763	9768	9773	0	1	1	2	2	3	3	4	4
95	9777	9782	9786	9791	9795	9800	9805	9809	9814	9818	0	1	1	2	2	3	3	4	4
96	9823	9827	9832	9836	9841	9845	9850	9854	9859	9863	0	1	1	2	2	3	3	4	4
97	9868	9872	9877	9881	9886	9890	9894	9899	9903	9908	0	1	1	2	2	3	3	4	4
98	9912	9917	9921	9926	9930	9934	9939	9943	9948	9952	0	1	1	2	2	3	3	4	4
99	9956	9961	9965	9969	9974	9978	9983	9987	9991	9996	0	1	1	2	2	3	3	3	4

Antilogarithms

$x \rightarrow 10^x$

x	0	1	2	3	4	5	6	7	8	9	Differences 1	2	3	4	5	6	7	8	9
·00	1000	1002	1005	1007	1009	1012	1014	1016	1019	1021	0	0	1	1	1	1	2	2	2
·01	1023	1026	1028	1030	1033	1035	1038	1040	1042	1045	0	0	1	1	1	1	2	2	2
·02	1047	1050	1052	1054	1057	1059	1062	1064	1067	1069	0	0	1	1	1	1	2	2	2
·03	1072	1074	1076	1079	1081	1084	1086	1089	1091	1094	0	0	1	1	1	1	2	2	2
·04	1096	1099	1102	1104	1107	1109	1112	1114	1117	1119	0	1	1	1	1	2	2	2	2
·05	1122	1125	1127	1130	1132	1135	1138	1140	1143	1146	0	1	1	1	1	2	2	2	2
·06	1148	1151	1153	1156	1159	1161	1164	1167	1169	1172	0	1	1	1	1	2	2	2	2
·07	1175	1178	1180	1183	1186	1189	1191	1194	1197	1199	0	1	1	1	1	2	2	2	2
·08	1202	1205	1208	1211	1213	1216	1219	1222	1225	1227	0	1	1	1	1	2	2	2	3
·09	1230	1233	1236	1239	1242	1245	1247	1250	1253	1256	0	1	1	1	1	2	2	2	3
·10	1259	1262	1265	1268	1271	1274	1276	1279	1282	1285	0	1	1	1	1	2	2	2	3
·11	1288	1291	1294	1297	1300	1303	1306	1309	1312	1315	0	1	1	1	2	2	2	2	3
·12	1318	1321	1324	1327	1330	1334	1337	1340	1343	1346	0	1	1	1	2	2	2	2	3
·13	1349	1352	1355	1358	1361	1365	1368	1371	1374	1377	0	1	1	1	2	2	2	3	3
·14	1380	1384	1387	1390	1393	1396	1400	1403	1406	1409	0	1	1	1	2	2	2	3	3
·15	1413	1416	1419	1422	1426	1429	1432	1435	1439	1442	0	1	1	1	2	2	2	3	3
·16	1445	1449	1452	1455	1459	1462	1466	1469	1472	1476	0	1	1	1	2	2	2	3	3
·17	1479	1483	1486	1489	1493	1496	1500	1503	1507	1510	0	1	1	1	2	2	2	3	3
·18	1514	1517	1521	1524	1528	1531	1535	1538	1542	1545	0	1	1	1	2	2	2	3	3
·19	1549	1552	1556	1560	1563	1567	1570	1574	1578	1581	0	1	1	1	2	2	3	3	3
·20	1585	1589	1592	1596	1600	1603	1607	1611	1614	1618	0	1	1	1	2	2	3	3	3
·21	1622	1626	1629	1633	1637	1641	1644	1648	1652	1656	0	1	1	2	2	2	3	3	3
·22	1660	1663	1667	1671	1675	1679	1683	1687	1690	1694	0	1	1	2	2	2	3	3	3
·23	1698	1702	1706	1710	1714	1718	1722	1726	1730	1734	0	1	1	2	2	2	3	3	4
·24	1738	1742	1746	1750	1754	1758	1762	1766	1770	1774	0	1	1	2	2	2	3	3	4
·25	1778	1782	1786	1791	1795	1799	1803	1807	1811	1816	0	1	1	2	2	2	3	3	4
·26	1820	1824	1828	1832	1837	1841	1845	1849	1854	1858	0	1	1	2	2	3	3	3	4
·27	1862	1866	1871	1875	1879	1884	1888	1892	1897	1901	0	1	1	2	2	3	3	3	4
·28	1905	1910	1914	1919	1923	1928	1932	1936	1941	1945	0	1	1	2	2	3	3	4	4
·29	1950	1954	1959	1963	1968	1972	1977	1982	1986	1991	0	1	1	2	2	3	3	4	4
·30	1995	2000	2004	2009	2014	2018	2023	2028	2032	2037	0	1	1	2	2	3	3	4	4
·31	2042	2046	2051	2056	2061	2065	2070	2075	2080	2084	0	1	1	2	2	3	3	4	4
·32	2089	2094	2099	2104	2109	2113	2118	2123	2128	2133	0	1	1	2	2	3	3	4	4
·33	2138	2143	2148	2153	2158	2163	2168	2173	2178	2183	0	1	1	2	2	3	3	4	4
·34	2188	2193	2198	2203	2208	2213	2218	2223	2228	2234	1	1	2	2	3	3	4	4	5
·35	2239	2244	2249	2254	2259	2265	2270	2275	2280	2286	1	1	2	2	3	3	4	4	5
·36	2291	2296	2301	2307	2312	2317	2323	2328	2333	2339	1	1	2	2	3	3	4	4	5
·37	2344	2350	2355	2360	2366	2371	2377	2382	2388	2393	1	1	2	2	3	3	4	4	5
·38	2399	2404	2410	2415	2421	2427	2432	2438	2443	2449	1	1	2	2	3	3	4	4	5
·39	2455	2460	2466	2472	2477	2483	2489	2495	2500	2506	1	1	2	2	3	3	4	5	5
·40	2512	2518	2523	2529	2535	2541	2547	2553	2559	2564	1	1	2	2	3	4	4	5	5
·41	2570	2576	2582	2588	2594	2600	2606	2612	2618	2624	1	1	2	2	3	4	4	5	5
·42	2630	2636	2642	2649	2655	2661	2667	2673	2679	2685	1	1	2	2	3	4	4	5	6
·43	2692	2698	2704	2710	2716	2723	2729	2735	2742	2748	1	1	2	3	3	4	4	5	6
·44	2754	2761	2767	2773	2780	2786	2793	2799	2805	2812	1	1	2	3	3	4	4	5	6
·45	2818	2825	2831	2838	2844	2851	2858	2864	2871	2877	1	1	2	3	3	4	5	5	6
·46	2884	2891	2897	2904	2911	2917	2924	2931	2938	2944	1	1	2	3	3	4	5	5	6
·47	2951	2958	2965	2972	2979	2985	2992	2999	3006	3013	1	1	2	3	3	4	5	5	6
·48	3020	3027	3034	3041	3048	3055	3062	3069	3076	3083	1	1	2	3	4	4	5	6	6
·49	3090	3097	3105	3112	3119	3126	3133	3141	3148	3155	1	1	2	3	4	4	5	6	6
·50	3162	3170	3177	3184	3192	3199	3206	3214	3221	3228	1	1	2	3	4	4	5	6	7
·51	3236	3243	3251	3258	3266	3273	3281	3289	3296	3304	1	2	2	3	4	5	5	6	7
·52	3311	3319	3327	3334	3342	3350	3357	3365	3373	3381	1	2	2	3	4	5	5	6	7
·53	3388	3396	3404	3412	3420	3428	3436	3443	3451	3459	1	2	2	3	4	5	6	6	7
·54	3467	3475	3483	3491	3499	3508	3516	3524	3532	3540	1	2	2	3	4	5	6	6	7
·55	3548	3556	3565	3573	3581	3589	3597	3606	3614	3622	1	2	2	3	4	5	6	7	7
·56	3631	3639	3648	3656	3664	3673	3681	3690	3698	3707	1	2	3	3	4	5	6	7	8
·57	3715	3724	3733	3741	3750	3758	3767	3776	3784	3793	1	2	3	3	4	5	6	7	8
·58	3802	3811	3819	3828	3837	3846	3855	3864	3873	3882	1	2	3	4	4	5	6	7	8
·59	3890	3899	3908	3917	3926	3936	3945	3954	3963	3972	1	2	3	4	5	5	6	7	8
·60	3981	3990	3999	4009	4018	4027	4036	4046	4055	4064	1	2	3	4	5	6	6	7	8
·61	4074	4083	4093	4102	4111	4121	4130	4140	4150	4159	1	2	3	4	5	6	7	8	9
·62	4169	4178	4188	4198	4207	4217	4227	4236	4246	4256	1	2	3	4	5	6	7	8	9
·63	4266	4276	4285	4295	4305	4315	4325	4335	4345	4355	1	2	3	4	5	6	7	8	9
·64	4365	4375	4385	4395	4406	4416	4426	4436	4446	4457	1	2	3	4	5	6	7	8	9
·65	4467	4477	4487	4498	4508	4519	4529	4539	4550	4560	1	2	3	4	5	6	7	8	9
·66	4571	4581	4592	4603	4613	4624	4634	4645	4656	4667	1	2	3	4	5	6	7	9	10
·67	4677	4688	4699	4710	4721	4732	4742	4753	4764	4775	1	2	3	4	5	7	8	9	10
·68	4786	4797	4808	4819	4831	4842	4853	4864	4875	4887	1	2	3	4	6	7	8	9	10
·69	4898	4909	4920	4932	4943	4955	4966	4977	4989	5000	1	2	3	5	6	7	8	9	10
·70	5012	5023	5035	5047	5058	5070	5082	5093	5105	5117	1	2	4	5	6	7	8	9	11
·71	5129	5140	5152	5164	5176	5188	5200	5212	5224	5236	1	2	4	5	6	7	8	10	11
·72	5248	5260	5272	5284	5297	5309	5321	5333	5346	5358	1	2	4	5	6	7	9	10	11
·73	5370	5383	5395	5408	5420	5433	5445	5458	5470	5483	1	3	4	5	6	8	9	10	11
·74	5495	5508	5521	5534	5546	5559	5572	5585	5598	5610	1	3	4	5	6	8	9	10	12
·75	5623	5636	5649	5662	5675	5689	5702	5715	5728	5741	1	3	4	5	7	8	9	10	12
·76	5754	5768	5781	5794	5808	5821	5834	5848	5861	5875	1	3	4	5	7	8	9	11	12
·77	5888	5902	5916	5929	5943	5957	5970	5984	5998	6012	1	3	4	5	7	8	10	11	12
·78	6026	6039	6053	6067	6081	6095	6109	6124	6138	6152	1	3	4	6	7	8	10	11	13
·79	6166	6180	6194	6209	6223	6237	6252	6266	6281	6295	1	3	4	6	7	9	10	11	13
·80	6310	6324	6339	6353	6368	6383	6397	6412	6427	6442	1	3	4	6	7	9	10	12	13
·81	6457	6471	6486	6501	6516	6531	6546	6561	6577	6592	2	3	5	6	8	9	11	12	14
·82	6607	6622	6637	6653	6668	6683	6699	6714	6730	6745	2	3	5	6	8	9	11	12	14
·83	6761	6776	6792	6808	6823	6839	6855	6871	6887	6902	2	3	5	6	8	9	11	13	14
·84	6918	6934	6950	6966	6982	6998	7015	7031	7047	7063	2	3	5	6	8	10	11	13	15
·85	7079	7096	7112	7129	7145	7161	7178	7194	7211	7228	2	3	5	7	8	10	12	13	15
·86	7244	7261	7278	7295	7311	7328	7345	7362	7379	7396	2	3	5	7	8	10	12	13	15
·87	7413	7430	7447	7464	7482	7499	7516	7534	7551	7568	2	3	5	7	9	10	12	14	16
·88	7586	7603	7621	7638	7656	7674	7691	7709	7727	7745	2	4	5	7	9	11	12	14	16
·89	7762	7780	7798	7816	7834	7852	7870	7889	7907	7925	2	4	5	7	9	11	13	14	16
·90	7943	7962	7980	7998	8017	8035	8054	8072	8091	8110	2	4	6	7	9	11	13	15	17
·91	8128	8147	8166	8185	8204	8222	8241	8260	8279	8299	2	4	6	8	9	11	13	15	17
·92	8318	8337	8356	8375	8395	8414	8433	8453	8472	8492	2	4	6	8	10	12	14	15	17
·93	8511	8531	8551	8570	8590	8610	8630	8650	8670	8690	2	4	6	8	10	12	14	16	18
·94	8710	8730	8750	8770	8790	8810	8831	8851	8872	8892	2	4	6	8	10	12	14	16	18
·95	8913	8933	8954	8974	8995	9016	9036	9057	9078	9099	2	4	6	8	10	12	15	17	19
·96	9120	9141	9162	9183	9204	9226	9247	9268	9290	9311	2	4	6	8	11	13	15	17	19
·97	9333	9354	9376	9397	9419	9441	9462	9484	9506	9528	2	4	7	9	11	13	15	17	20
·98	9550	9572	9594	9616	9638	9661	9683	9705	9727	9750	2	4	7	9	11	13	16	18	20
·99	9772	9795	9817	9840	9863	9886	9908	9931	9954	9977	2	5	7	9	11	14	16	18	20

Sines of angles

$\theta \rightarrow \sin \theta$

θ	0′ 0·0°	6′ 0·1°	12′ 0·2°	18′ 0·3°	24′ 0·4°	30′ 0·5°	36′ 0·6°	42′ 0·7°	48′ 0·8°	54′ 0·9°	ADD Differences 1′	2′	3′	4′	5′
0°	0·0000	0017	0035	0052	0070	0087	0105	0122	0140	0157	3	6	9	12	15
1	0·0175	0192	0209	0227	0244	0262	0279	0297	0314	0332	3	6	9	12	15
2	0·0349	0366	0384	0401	0419	0436	0454	0471	0488	0506	3	6	9	12	15
3	0·0523	0541	0558	0576	0593	0610	0628	0645	0663	0680	3	6	9	12	15
4	0·0698	0715	0732	0750	0767	0785	0802	0819	0837	0854	3	6	9	12	15
5	0·0872	0889	0906	0924	0941	0958	0976	0993	1011	1028	3	6	9	12	14
6	0·1045	1063	1080	1097	1115	1132	1149	1167	1184	1201	3	6	9	12	14
7	0·1219	1236	1253	1271	1288	1305	1323	1340	1357	1374	3	6	9	12	14
8	0·1392	1409	1426	1444	1461	1478	1495	1513	1530	1547	3	6	9	12	14
9	0·1564	1582	1599	1616	1633	1650	1668	1685	1702	1719	3	6	9	12	14
10°	0·1736	1754	1771	1788	1805	1822	1840	1857	1874	1891	3	6	9	11	14
11	0·1908	1925	1942	1959	1977	1994	2011	2028	2045	2062	3	6	9	11	14
12	0·2079	2096	2113	2130	2147	2164	2181	2198	2215	2233	3	6	9	11	14
13	0·2250	2267	2284	2300	2317	2334	2351	2368	2385	2402	3	6	8	11	14
14	0·2419	2436	2453	2470	2487	2504	2521	2538	2554	2571	3	6	8	11	14
15	0·2588	2605	2622	2639	2656	2672	2689	2706	2723	2740	3	6	8	11	14
16	0·2756	2773	2790	2807	2823	2840	2857	2874	2890	2907	3	6	8	11	14
17	0·2924	2940	2957	2974	2990	3007	3024	3040	3057	3074	3	6	8	11	14
18	0·3090	3107	3123	3140	3156	3173	3190	3206	3223	3239	3	6	8	11	14
19	0·3256	3272	3289	3305	3322	3338	3355	3371	3387	3404	3	5	8	11	14
20°	0·3420	3437	3453	3469	3486	3502	3518	3535	3551	3567	3	5	8	11	14
21	0·3584	3600	3616	3633	3649	3665	3681	3697	3714	3730	3	5	8	11	14
22	0·3746	3762	3778	3795	3811	3827	3843	3859	3875	3891	3	5	8	11	14
23	0·3907	3923	3939	3955	3971	3987	4003	4019	4035	4051	3	5	8	11	14
24	0·4067	4083	4099	4115	4131	4147	4163	4179	4195	4210	3	5	8	11	13
25	0·4226	4242	4258	4274	4289	4305	4321	4337	4352	4368	3	5	8	11	13
26	0·4384	4399	4415	4431	4446	4462	4478	4493	4509	4524	3	5	8	10	13
27	0·4540	4555	4571	4586	4602	4617	4633	4648	4664	4679	3	5	8	10	13
28	0·4695	4710	4726	4741	4756	4772	4787	4802	4818	4833	3	5	8	10	13
29	0·4848	4863	4879	4894	4909	4924	4939	4955	4970	4985	3	5	8	10	13
30°	0·5000	5015	5030	5045	5060	5075	5090	5105	5120	5135	3	5	8	10	13
31	0·5150	5165	5180	5195	5210	5225	5240	5255	5270	5284	2	5	7	10	12
32	0·5299	5314	5329	5344	5358	5373	5388	5402	5417	5432	2	5	7	10	12
33	0·5446	5461	5476	5490	5505	5519	5534	5548	5563	5577	2	5	7	10	12
34	0·5592	5606	5621	5635	5650	5664	5678	5693	5707	5721	2	5	7	10	12
35	0·5736	5750	5764	5779	5793	5807	5821	5835	5850	5864	2	5	7	9	12
36	0·5878	5892	5906	5920	5934	5948	5962	5976	5990	6004	2	5	7	9	12
37	0·6018	6032	6046	6060	6074	6088	6101	6115	6129	6143	2	5	7	9	12
38	0·6157	6170	6184	6198	6211	6225	6239	6252	6266	6280	2	5	7	9	11
39	0·6293	6307	6320	6334	6347	6361	6374	6388	6401	6414	2	4	7	9	11
40°	0·6428	6441	6455	6468	6481	6494	6508	6521	6534	6547	2	4	7	9	11
41	0·6561	6574	6587	6600	6613	6626	6639	6652	6665	6678	2	4	7	9	11
42	0·6691	6704	6717	6730	6743	6756	6769	6782	6794	6807	2	4	6	9	11
43	0·6820	6833	6845	6858	6871	6884	6896	6909	6921	6934	2	4	6	8	11
44	0·6947	6959	6972	6984	6997	7009	7022	7034	7046	7059	2	4	6	8	10

θ	0′ 0·0°	6′ 0·1°	12′ 0·2°	18′ 0·3°	24′ 0·4°	30′ 0·5°	36′ 0·6°	42′ 0·7°	48′ 0·8°	54′ 0·9°	ADD Differences 1′	2′	3′	4′	5′
45°	0·7071	7083	7096	7108	7120	7133	7145	7157	7169	7181	2	4	6	8	10
46	0·7193	7206	7218	7230	7242	7254	7266	7278	7290	7302	2	4	6	8	10
47	0·7314	7325	7337	7349	7361	7373	7385	7396	7408	7420	2	4	6	8	10
48	0·7431	7443	7455	7466	7478	7490	7501	7513	7524	7536	2	4	6	8	10
49	0·7547	7559	7570	7581	7593	7604	7615	7627	7638	7649	2	4	6	8	9
50°	0·7660	7672	7683	7694	7705	7716	7727	7738	7749	7760	2	4	6	7	9
51	0·7771	7782	7793	7804	7815	7826	7837	7848	7859	7869	2	4	5	7	9
52	0·7880	7891	7902	7912	7923	7934	7944	7955	7965	7976	2	4	5	7	9
53	0·7986	7997	8007	8018	8028	8039	8049	8059	8070	8080	2	3	5	7	9
54	0·8090	8100	8111	8121	8131	8141	8151	8161	8171	8181	2	3	5	7	8
55	0·8192	8202	8211	8221	8231	8241	8251	8261	8271	8281	2	3	5	7	8
56	0·8290	8300	8310	8320	8329	8339	8348	8358	8368	8377	2	3	5	6	8
57	0·8387	8396	8406	8415	8425	8434	8443	8453	8462	8471	2	3	5	6	8
58	0·8480	8490	8499	8508	8517	8526	8536	8545	8554	8563	2	3	5	6	8
59	0·8572	8581	8590	8599	8607	8616	8625	8634	8643	8652	1	3	4	6	7
60°	0·8660	8669	8678	8686	8695	8704	8712	8721	8729	8738	1	3	4	6	7
61	0·8746	8755	8763	8771	8780	8788	8796	8805	8813	8821	1	3	4	6	7
62	0·8829	8838	8846	8854	8862	8870	8878	8886	8894	8902	1	3	4	5	7
63	0·8910	8918	8926	8934	8942	8949	8957	8965	8973	8980	1	3	4	5	6
64	0·8988	8996	9003	9011	9018	9026	9033	9041	9048	9056	1	3	4	5	6
65	0·9063	9070	9078	9085	9092	9100	9107	9114	9121	9128	1	2	4	5	6
66	0·9135	9143	9150	9157	9164	9171	9178	9184	9191	9198	1	2	3	5	6
67	0·9205	9212	9219	9225	9232	9239	9245	9252	9259	9265	1	2	3	4	6
68	0·9272	9278	9285	9291	9298	9304	9311	9317	9323	9330	1	2	3	4	5
69	0·9336	9342	9348	9354	9361	9367	9373	9379	9385	9391	1	2	3	4	5
70°	0·9397	9403	9409	9415	9421	9426	9432	9438	9444	9449	1	2	3	4	5
71	0·9455	9461	9466	9472	9478	9483	9489	9494	9500	9505	1	2	3	4	5
72	0·9511	9516	9521	9527	9532	9537	9542	9548	9553	9558	1	2	3	3	4
73	0·9563	9568	9573	9578	9583	9588	9593	9598	9603	9608	1	2	2	3	4
74	0·9613	9617	9622	9627	9632	9636	9641	9646	9650	9655	1	2	2	3	4
75	0·9659	9664	9668	9673	9677	9681	9686	9690	9694	9699	1	1	2	3	4
76	0·9703	9707	9711	9715	9720	9724	9728	9732	9736	9740	1	1	2	3	3
77	0·9744	9748	9751	9755	9759	9763	9767	9770	9774	9778	1	1	2	3	3
78	0·9781	9785	9789	9792	9796	9799	9803	9806	9810	9813	1	1	2	2	3
79	0·9816	9820	9823	9826	9829	9833	9836	9839	9842	9845	1	1	2	2	3
80°	0·9848	9851	9854	9857	9860	9863	9866	9869	9871	9874	0	1	1	2	2
81	0·9877	9880	9882	9885	9888	9890	9893	9895	9898	9900	0	1	1	2	2
82	0·9903	9905	9907	9910	9912	9914	9917	9919	9921	9923	0	1	1	2	2
83	0·9925	9928	9930	9932	9934	9936	9938	9940	9942	9943	0	1	1	1	2
84	0·9945	9947	9949	9951	9952	9954	9956	9957	9959	9960	0	1	1	1	2
85	0·9962	9963	9965	9966	9968	9969	9971	9972	9973	9974	0	0	1	1	1
86	0·9976	9977	9978	9979	9980	9981	9982	9983	9984	9985	0	0	1	1	1
87	0·9986	9987	9988	9989	9990	9990	9991	9992	9993	9993	0	0	0	1	1
88	0·9994	9995	9995	9996	9996	9997	9997	9997	9998	9998	0	0	0	0	0
89	0·9998	9999	9999	9999	9999	1·000	1·000	1·000	1·000	1·000	0	0	0	0	0

Cosines of angles

$\theta \rightarrow \cos\theta$

θ	0′ 0·0°	6′ 0·1°	12′ 0·2°	18′ 0·3°	24′ 0·4°	30′ 0·5°	36′ 0·6°	42′ 0·7°	48′ 0·8°	54′ 0·9°	SUBTRACT Differences 1′	2′	3′	4′	5′
0°	1·000	1·000	1·000	1·000	1·000	1·000	·9999	9999	9999	9999	0	0	0	0	0
1	0·9998	9998	9998	9997	9997	9997	9996	9996	9995	9995	0	0	0	0	0
2	0·9994	9993	9993	9992	9991	9990	9990	9989	9988	9987	0	0	0	1	1
3	0·9986	9985	9984	9983	9982	9981	9980	9979	9978	9977	0	0	1	1	1
4	0·9976	9974	9973	9972	9971	9969	9968	9966	9965	9963	0	0	1	1	1
5	0·9962	9960	9959	9957	9956	9954	9952	9951	9949	9947	0	1	1	1	2
6	0·9945	9943	9942	9940	9938	9936	9934	9932	9930	9928	0	1	1	1	2
7	0·9925	9923	9921	9919	9917	9914	9912	9910	9907	9905	0	1	1	2	2
8	0·9903	9900	9898	9895	9893	9890	9888	9885	9882	9880	0	1	1	2	2
9	0·9877	9874	9871	9869	9866	9863	9860	9857	9854	9851	0	1	1	2	2
10°	0·9848	9845	9842	9839	9836	9833	9829	9826	9823	9820	1	1	2	2	3
11	0·9816	9813	9810	9806	9803	9799	9796	9792	9789	9785	1	1	2	2	3
12	0·9781	9778	9774	9770	9767	9763	9759	9755	9751	9748	1	1	2	3	3
13	0·9744	9740	9736	9732	9728	9724	9720	9715	9711	9707	1	1	2	3	3
14	0·9703	9699	9694	9690	9686	9681	9677	9673	9668	9664	1	1	2	3	4
15	0·9659	9655	9650	9646	9641	9636	9632	9627	9622	9617	1	2	2	3	4
16	0·9613	9608	9603	9598	9593	9588	9583	9578	9573	9568	1	2	2	3	4
17	0·9563	9558	9553	9548	9542	9537	9532	9527	9521	9516	1	2	3	3	4
18	0·9511	9505	9500	9494	9489	9483	9478	9472	9466	9461	1	2	3	4	5
19	0·9455	9449	9444	9438	9432	9426	9421	9415	9409	9403	1	2	3	4	5
20°	0·9397	9391	9385	9379	9373	9367	9361	9354	9348	9342	1	2	3	4	5
21	0·9336	9330	9323	9317	9311	9304	9298	9291	9285	9278	1	2	3	4	5
22	0·9272	9265	9259	9252	9245	9239	9232	9225	9219	9212	1	2	3	4	6
23	0·9205	9198	9191	9184	9178	9171	9164	9157	9150	9143	1	2	3	5	6
24	0·9135	9128	9121	9114	9107	9100	9092	9085	9078	9070	1	2	4	5	6
25	0·9063	9056	9048	9041	9033	9026	9018	9011	9003	8996	1	3	4	5	6
26	0·8988	8980	8973	8965	8957	8949	8942	8934	8926	8918	1	3	4	5	6
27	0·8910	8902	8894	8886	8878	8870	8862	8854	8846	8838	1	3	4	5	7
28	0·8829	8821	8813	8805	8796	8788	8780	8771	8763	8755	1	3	4	6	7
29	0·8746	8738	8729	8721	8712	8704	8695	8686	8678	8669	1	3	4	6	7
30°	0·8660	8652	8643	8634	8625	8616	8607	8599	8590	8581	1	3	4	6	7
31	0·8572	8563	8554	8545	8536	8526	8517	8508	8499	8490	2	3	5	6	8
32	0·8480	8471	8462	8453	8443	8434	8425	8415	8406	8396	2	3	5	6	8
33	0·8387	8377	8368	8358	8348	8339	8329	8320	8310	8300	2	3	5	6	8
34	0·8290	8281	8271	8261	8251	8241	8231	8221	8211	8202	2	3	5	7	8
35	0·8192	8181	8171	8161	8151	8141	8131	8121	8111	8100	2	3	5	7	8
36	0·8090	8080	8070	8059	8049	8039	8028	8018	8007	7997	2	3	5	7	9
37	0·7986	7976	7965	7955	7944	7934	7923	7912	7902	7891	2	4	5	7	9
38	0·7880	7869	7859	7848	7837	7826	7815	7804	7793	7782	2	4	5	7	9
39	0·7771	7760	7749	7738	7727	7716	7705	7694	7683	7672	2	4	6	7	9
40°	0·7660	7649	7638	7627	7615	7604	7593	7581	7570	7559	2	4	6	8	9
41	0·7547	7536	7524	7513	7501	7490	7478	7466	7455	7443	2	4	6	8	10
42	0·7431	7420	7408	7396	7385	7373	7361	7349	7337	7325	2	4	6	8	10
43	0·7314	7302	7290	7278	7266	7254	7242	7230	7218	7206	2	4	6	8	10
44	0·7193	7181	7169	7157	7145	7133	7120	7108	7096	7083	2	4	6	8	10

θ	0′ 0·0°	6′ 0·1°	12′ 0·2°	18′ 0·3°	24′ 0·4°	30′ 0·5°	36′ 0·6°	42′ 0·7°	48′ 0·8°	54′ 0·9°	SUBTRACT Differences 1′	2′	3′	4′	5′
45°	0·7071	7059	7046	7034	7022	7009	6997	6984	6972	6959	2	4	6	8	10
46	0·6947	6934	6921	6909	6896	6884	6871	6858	6845	6833	2	4	6	8	11
47	0·6820	6807	6794	6782	6769	6756	6743	6730	6717	6704	2	4	6	9	11
48	0·6691	6678	6665	6652	6639	6626	6613	6600	6587	6574	2	4	7	9	11
49	0·6561	6547	6534	6521	6508	6494	6481	6468	6455	6441	2	4	7	9	11
50°	0·6428	6414	6401	6388	6374	6361	6347	6334	6320	6307	2	4	7	9	11
51	0·6293	6280	6266	6252	6239	6225	6211	6198	6184	6170	2	5	7	9	11
52	0·6157	6143	6129	6115	6101	6088	6074	6060	6046	6032	2	5	7	9	12
53	0·6018	6004	5990	5976	5962	5948	5934	5920	5906	5892	2	5	7	9	12
54	0·5878	5864	5850	5835	5821	5807	5793	5779	5764	5750	2	5	7	9	12
55	0·5736	5721	5707	5693	5678	5664	5650	5635	5621	5606	2	5	7	10	12
56	0·5592	5577	5563	5548	5534	5519	5505	5490	5476	5461	2	5	7	10	12
57	0·5446	5432	5417	5402	5388	5373	5358	5344	5329	5314	2	5	7	10	12
58	0·5299	5284	5270	5255	5240	5225	5210	5195	5180	5165	2	5	7	10	12
59	0·5150	5135	5120	5105	5090	5075	5060	5045	5030	5015	3	5	8	10	13
60°	0·5000	4985	4970	4955	4939	4924	4909	4894	4879	4863	3	5	8	10	13
61	0·4848	4833	4818	4802	4787	4772	4756	4741	4726	4710	3	5	8	10	13
62	0·4695	4679	4664	4648	4633	4617	4602	4586	4571	4555	3	5	8	10	13
63	0·4540	4524	4509	4493	4478	4462	4446	4431	4415	4399	3	5	8	10	13
64	0·4384	4368	4352	4337	4321	4305	4289	4274	4258	4242	3	5	8	11	13
65	0·4226	4210	4195	4179	4163	4147	4131	4115	4099	4083	3	5	8	11	13
66	0·4067	4051	4035	4019	4003	3987	3971	3955	3939	3923	3	5	8	11	14
67	0·3907	3891	3875	3859	3843	3827	3811	3795	3778	3762	3	5	8	11	14
68	0·3746	3730	3714	3697	3681	3665	3649	3633	3616	3600	3	5	8	11	14
69	0·3584	3567	3551	3535	3518	3502	3486	3469	3453	3437	3	5	8	11	14
70°	0·3420	3404	3387	3371	3355	3338	3322	3305	3289	3272	3	5	8	11	14
71	0·3256	3239	3223	3206	3190	3173	3156	3140	3123	3107	3	6	8	11	14
72	0·3090	3074	3057	3040	3024	3007	2990	2974	2957	2940	3	6	8	11	14
73	0·2924	2907	2890	2874	2857	2840	2823	2807	2790	2773	3	6	8	11	14
74	0·2756	2740	2723	2706	2689	2672	2656	2639	2622	2605	3	6	8	11	14
75	0·2588	2571	2554	2538	2521	2504	2487	2470	2453	2436	3	6	8	11	14
76	0·2419	2402	2385	2368	2351	2334	2317	2300	2284	2267	3	6	8	11	14
77	0·2250	2233	2215	2198	2181	2164	2147	2130	2113	2096	3	6	9	11	14
78	0·2079	2062	2045	2028	2011	1994	1977	1959	1942	1925	3	6	9	11	14
79	0·1908	1891	1874	1857	1840	1822	1805	1788	1771	1754	3	6	9	11	14
80°	0·1736	1719	1702	1685	1668	1650	1633	1616	1599	1582	3	6	9	12	14
81	0·1564	1547	1530	1513	1495	1478	1461	1444	1426	1409	3	6	9	12	14
82	0·1392	1374	1357	1340	1323	1305	1288	1271	1253	1236	3	6	9	12	14
83	0·1219	1201	1184	1167	1149	1132	1115	1097	1080	1063	3	6	9	12	14
84	0·1045	1028	1011	0993	0976	0958	0941	0924	0906	0889	3	6	9	12	14
85	0·0872	0854	0837	0819	0802	0785	0767	0750	0732	0715	3	6	9	12	14
86	0·0698	0680	0663	0645	0628	0610	0593	0576	0558	0541	3	6	9	12	15
87	0·0523	0506	0488	0471	0454	0436	0419	0401	0384	0366	3	6	9	12	15
88	0·0349	0332	0314	0297	0279	0262	0244	0227	0209	0192	3	6	9	12	15
89	0·0175	0157	0140	0122	0105	0087	0070	0052	0035	0017	3	6	9	12	15

Tangents of angles

$\theta \rightarrow \tan\theta$

θ	0′ 0·0°	6′ 0·1°	12′ 0·2°	18′ 0·3°	24′ 0·4°	30′ 0·5°	36′ 0·6°	42′ 0·7°	48′ 0·8°	54′ 0·9°	ADD Differences 1′	2′	3′	4′	5′
0°	0·0000	0017	0035	0052	0070	0087	0105	0122	0140	0157	3	6	9	12	15
1	0·0175	0192	0209	0227	0244	0262	0279	0297	0314	0332	3	6	9	12	15
2	0·0349	0367	0384	0402	0419	0437	0454	0472	0489	0507	3	6	9	12	15
3	0·0524	0542	0559	0577	0594	0612	0629	0647	0664	0682	3	6	9	12	15
4	0·0699	0717	0734	0752	0769	0787	0805	0822	0840	0857	3	6	9	12	15
5	0·0875	0892	0910	0928	0945	0963	0981	0998	1016	1033	3	6	9	12	15
6	0·1051	1069	1086	1104	1122	1139	1157	1175	1192	1210	3	6	9	12	15
7	0·1228	1246	1263	1281	1299	1317	1334	1352	1370	1388	3	6	9	12	15
8	0·1405	1423	1441	1459	1477	1495	1512	1530	1548	1566	3	6	9	12	15
9	0·1584	1602	1620	1638	1655	1673	1691	1709	1727	1745	3	6	9	12	15
10°	0·1763	1781	1799	1817	1835	1853	1871	1890	1908	1926	3	6	9	12	15
11	0·1944	1962	1980	1998	2016	2035	2053	2071	2089	2107	3	6	9	12	15
12	0·2126	2144	2162	2180	2199	2217	2235	2254	2272	2290	3	6	9	12	15
13	0·2309	2327	2345	2364	2382	2401	2419	2438	2456	2475	3	6	9	12	15
14	0·2493	2512	2530	2549	2568	2586	2605	2623	2642	2661	3	6	9	12	16
15	0·2679	2698	2717	2736	2754	2773	2792	2811	2830	2849	3	6	9	13	16
16	0·2867	2886	2905	2924	2943	2962	2981	3000	3019	3038	3	6	9	13	16
17	0·3057	3076	3096	3115	3134	3153	3172	3191	3211	3230	3	6	10	13	16
18	0·3249	3269	3288	3307	3327	3346	3365	3385	3404	3424	3	6	10	13	16
19	0·3443	3463	3482	3502	3522	3541	3561	3581	3600	3620	3	7	10	13	16
20°	0·3640	3659	3679	3699	3719	3739	3759	3779	3799	3819	3	7	10	13	17
21	0·3839	3859	3879	3899	3919	3939	3959	3979	4000	4020	3	7	10	13	17
22	0·4040	4061	4081	4101	4122	4142	4163	4183	4204	4224	3	7	10	14	17
23	0·4245	4265	4286	4307	4327	4348	4369	4390	4411	4431	3	7	10	14	17
24	0·4452	4473	4494	4515	4536	4557	4578	4599	4621	4642	4	7	11	14	18
25	0·4663	4684	4706	4727	4748	4770	4791	4813	4834	4856	4	7	11	14	18
26	0·4877	4899	4921	4942	4964	4986	5008	5029	5051	5073	4	7	11	15	18
27	0·5095	5117	5139	5161	5184	5206	5228	5250	5272	5295	4	7	11	15	18
28	0·5317	5340	5362	5384	5407	5430	5452	5475	5498	5520	4	8	11	15	19
29	0·5543	5566	5589	5612	5635	5658	5681	5704	5727	5750	4	8	12	15	19
30°	0·5774	5797	5820	5844	5867	5890	5914	5938	5961	5985	4	8	12	16	20
31	0·6009	6032	6056	6080	6104	6128	6152	6176	6200	6224	4	8	12	16	20
32	0·6249	6273	6297	6322	6346	6371	6395	6420	6445	6469	4	8	12	16	20
33	0·6494	6519	6544	6569	6594	6619	6644	6669	6694	6720	4	8	13	17	21
34	0·6745	6771	6796	6822	6847	6873	6899	6924	6950	6976	4	9	13	17	21
35	0·7002	7028	7054	7080	7107	7133	7159	7186	7212	7239	4	9	13	18	22
36	0·7265	7292	7319	7346	7373	7400	7427	7454	7481	7508	5	9	14	18	23
37	0·7536	7563	7590	7618	7646	7673	7701	7729	7757	7785	5	9	14	18	23
38	0·7813	7841	7869	7898	7926	7954	7983	8012	8040	8069	5	9	14	19	24
39	0·8098	8127	8156	8185	8214	8243	8273	8302	8332	8361	5	10	15	20	24
40°	0·8391	8421	8451	8481	8511	8541	8571	8601	8632	8662	5	10	15	20	25
41	0·8693	8724	8754	8785	8816	8847	8878	8910	8941	8972	5	10	16	21	26
42	0·9004	9036	9067	9099	9131	9163	9195	9228	9260	9293	5	11	16	21	27
43	0·9325	9358	9391	9424	9457	9490	9523	9556	9590	9623	6	11	17	22	28
44	0·9657	9691	9725	9759	9793	9827	9861	9896	9930	9965	6	11	17	23	29

θ	0′ 0·0°	6′ 0·1°	12′ 0·2°	18′ 0·3°	24′ 0·4°	30′ 0·5°	36′ 0·6°	42′ 0·7°	48′ 0·8°	54′ 0·9°	ADD Differences 1′	2′	3′	4′	5′
45°	1·0000	0035	0070	0105	0141	0176	0212	0247	0283	0319	6	12	18	24	30
46	1·0355	0392	0428	0464	0501	0538	0575	0612	0649	0686	6	12	18	25	31
47	1·0724	0761	0799	0837	0875	0913	0951	0990	1028	1067	6	13	19	25	32
48	1·1106	1145	1184	1224	1263	1303	1343	1383	1423	1463	7	13	20	27	33
49	1·1504	1544	1585	1626	1667	1708	1750	1792	1833	1875	7	14	21	28	34
50°	1·1918	1960	2002	2045	2088	2131	2174	2218	2261	2305	7	14	22	29	36
51	1·2349	2393	2437	2484	2527	2572	2617	2662	2708	2753	8	15	23	30	38
52	1·2799	2846	2892	2938	2985	3032	3079	3127	3175	3222	8	16	24	31	39
53	1·3270	3319	3367	3416	3465	3514	3564	3613	3663	3713	8	16	25	33	41
54	1·3764	3814	3865	3916	3968	4019	4071	4124	4176	4229	9	17	26	34	43
55	1·4281	4335	4388	4442	4496	4550	4605	4659	4715	4770	9	18	27	36	45
56	1·4826	4882	4938	4994	5051	5108	5166	5224	5282	5340	10	19	29	38	48
57	1·5399	5458	5517	5577	5637	5697	5757	5818	5880	5941	10	20	30	40	50
58	1·6003	6066	6128	6191	6255	6319	6383	6447	6512	6577	11	21	32	43	53
59	1·6643	6709	6775	6842	6909	6977	7045	7113	7182	7251	11	23	34	45	56
60°	1·7321	7391	7461	7532	7603	7675	7747	7820	7893	7966	12	24	36	48	60
61	1·8040	8115	8190	8265	8341	8418	8495	8572	8650	8728	13	26	38	51	64
62	1·8807	8887	8967	9047	9128	9210	9292	9375	9458	9542	14	27	41	55	68
63	1·9626	9711	9797	9883	9970	0057	0145	0233	0323	0413	15	29	44	58	73
64	2·0503	0594	0686	0778	0872	0965	1060	1155	1251	1348	16	31	47	63	78
65	2·1445	1543	1642	1742	1842	1943	2045	2148	2251	2355	17	34	51	68	85
66	2·2460	2566	2673	2781	2889	2998	3109	3220	3332	3445	18	37	55	73	92
67	2·3559	3673	3789	3906	4023	4142	4262	4383	4504	4627	20	40	60	79	99
68	2·4751	4876	5002	5129	5257	5386	5517	5649	5782	5916	22	43	65	87	108
69	2·6051	6187	6325	6464	6605	6746	6889	7034	7179	7326	24	47	71	95	119
70°	2·7475	7625	7776	7929	8083	8239	8397	8556	8716	8878	26	52	78	104	131
71	2·9042	9208	9375	9544	9714	9887	0061	0237	0415	0595	29	58	87	116	145
72	3·0777	0961	1146	1334	1524	1716	1910	2106	2305	2506	32	64	96	129	161
73	3·2709	2914	3122	3332	3544	3759	3977	4197	4420	4646	36	72	108	144	180
74	3·4874	5105	5339	5576	5816	6059	6305	6554	6806	7062	41	81	122	163	204
75	3·7321	7583	7848	8118	8391	8667	8947	9232	9520	9812	46	93	139	186	232
76	4·0108	0408	0713	1022	1335	1653	1976	2303	2635	2972					
77	4·3315	3662	4015	4374	4737	5107	5483	5864	6252	6646					
78	4·7046	7453	7867	8288	8716	9152	9594	0045	0504	0970					
79	5·1446	1929	2422	2924	3435	3955	4486	5026	5578	6140					
80°	5·6713	7297	7894	8502	9124	9758	0405	1066	1742	2432					
81	6·3138	3859	4596	5350	6122	6912	7720	8548	9395	0264					
82	7·1154	2066	3002	3962	4947	5958	6996	8062	9158	0285	differences unreliable				
83	8·1443	2636	3863	5126	6427	7769	9152	0579	2052	3572					
84	9·514	9·677	9·845	10·02	10·20	10·39	10·58	10·78	10·99	11·20					
85	11·43	11·66	11·91	12·16	12·43	12·71	13·00	13·30	13·62	13·95					
86	14·30	14·67	15·06	15·46	15·89	16·35	16·83	17·34	17·89	18·46					
87	19·08	19·74	20·45	21·20	22·02	22·90	23·86	24·90	26·03	27·27					
88	28·64	30·14	31·82	33·69	35·80	38·19	40·92	44·07	47·74	52·08					
89	57·29	63·66	71·62	81·85	95·49	114·6	143·2	191·0	286·5	573·0					

Logarithms of sines

$\theta \rightarrow \log \sin \theta$

θ	0′ 0·0°	6′ 0·1°	12′ 0·2°	18′ 0·3°	24′ 0·4°	30′ 0·5°	36′ 0·6°	42′ 0·7°	48′ 0·8°	54′ 0·9°	ADD Differences 1′	2′	3′	4′	5′
0°	$-\infty$	$\bar{3}$·2419	5429	7190	8439	9408	$\bar{0}$200	$\bar{0}$870	$\bar{1}$450	$\bar{1}$961					
1	$\bar{2}$·2419	2832	3210	3558	3880	4179	4459	4723	4971	5206					
2	$\bar{2}$·5428	5640	5842	6035	6220	6397	6567	6731	6889	7041					
3	$\bar{2}$·7188	7330	7468	7602	7731	7857	7979	8098	8213	8326					
4	$\bar{2}$·8436	8543	8647	8749	8849	8946	9042	9135	9226	9315	16	32	48	64	80
5	$\bar{2}$·9403	9489	9573	9655	9736	9816	9894	9970	$\bar{0}$046	$\bar{0}$120	13	26	39	52	65
6	$\bar{1}$·0192	0264	0334	0403	0472	0539	0605	0670	0734	0797	11	22	33	44	55
7	$\bar{1}$·0859	0920	0981	1040	1099	1157	1214	1271	1326	1381	10	19	29	38	48
8	$\bar{1}$·1436	1489	1542	1594	1646	1697	1747	1797	1847	1895	8	17	25	34	42
9	$\bar{1}$·1943	1991	2038	2085	2131	2176	2221	2266	2310	2353	8	15	23	30	38
10°	$\bar{1}$·2397	2439	2482	2524	2565	2606	2647	2687	2727	2767	7	14	20	27	34
11	$\bar{1}$·2806	2845	2883	2921	2959	2997	3034	3070	3107	3143	6	12	19	25	31
12	$\bar{1}$·3179	3214	3250	3284	3319	3353	3387	3421	3455	3488	6	11	17	23	28
13	$\bar{1}$·3521	3554	3586	3618	3650	3682	3713	3745	3775	3806	5	11	16	21	26
14	$\bar{1}$·3837	3867	3897	3927	3957	3986	4015	4044	4073	4102	5	10	15	20	24
15	$\bar{1}$·4130	4158	4186	4214	4242	4269	4296	4323	4350	4377	5	9	14	18	23
16	$\bar{1}$·4403	4430	4456	4482	4508	4533	4559	4584	4609	4634	4	9	13	17	21
17	$\bar{1}$·4659	4684	4709	4733	4757	4781	4805	4829	4853	4876	4	8	12	16	20
18	$\bar{1}$·4900	4923	4946	4969	4992	5015	5037	5060	5082	5104	4	8	11	15	19
19	$\bar{1}$·5126	5148	5170	5192	5213	5235	5256	5278	5299	5320	4	7	11	14	18
20°	$\bar{1}$·5341	5361	5382	5402	5423	5443	5463	5484	5504	5523	3	7	10	14	17
21	$\bar{1}$·5543	5563	5583	5602	5621	5641	5660	5679	5698	5717	3	6	10	13	16
22	$\bar{1}$·5736	5754	5773	5792	5810	5828	5847	5865	5883	5901	3	6	9	12	15
23	$\bar{1}$·5919	5937	5954	5972	5990	6007	6024	6042	6059	6076	3	6	9	12	15
24	$\bar{1}$·6093	6110	6127	6144	6161	6177	6194	6210	6227	6243	3	6	8	11	14
25	$\bar{1}$·6259	6276	6292	6308	6324	6340	6356	6371	6387	6403	3	5	8	11	13
26	$\bar{1}$·6418	6434	6449	6465	6480	6495	6510	6526	6541	6556	3	5	8	10	13
27	$\bar{1}$·6570	6585	6600	6615	6629	6644	6659	6673	6687	6702	2	5	7	10	12
28	$\bar{1}$·6716	6730	6744	6759	6773	6787	6801	6814	6828	6842	2	5	7	9	12
29	$\bar{1}$·6856	6869	6883	6896	6910	6923	6937	6950	6963	6977	2	4	7	9	11
30°	$\bar{1}$·6990	7003	7016	7029	7042	7055	7068	7080	7093	7106	2	4	6	9	11
31	$\bar{1}$·7118	7131	7144	7156	7168	7181	7193	7205	7218	7230	2	4	6	8	10
32	$\bar{1}$·7242	7254	7266	7278	7290	7302	7314	7326	7338	7349	2	4	6	8	10
33	$\bar{1}$·7361	7373	7384	7396	7407	7419	7430	7442	7453	7464	2	4	6	8	10
34	$\bar{1}$·7476	7487	7498	7509	7520	7531	7542	7553	7564	7575	2	4	6	7	9
35	$\bar{1}$·7586	7597	7607	7618	7629	7640	7650	7661	7671	7682	2	4	5	7	9
36	$\bar{1}$·7692	7703	7713	7723	7734	7744	7754	7764	7774	7785	2	3	5	7	9
37	$\bar{1}$·7795	7805	7815	7825	7835	7844	7854	7864	7874	7884	2	3	5	7	8
38	$\bar{1}$·7893	7903	7913	7922	7932	7941	7951	7960	7970	7979	2	3	5	6	8
39	$\bar{1}$·7989	7998	8007	8017	8026	8035	8044	8053	8063	8072	2	3	5	6	8
40°	$\bar{1}$·8081	8090	8099	8108	8117	8125	8134	8143	8152	8161	1	3	4	6	7
41	$\bar{1}$·8169	8178	8187	8195	8204	8213	8221	8230	8238	8247	1	3	4	6	7
42	$\bar{1}$·8255	8264	8272	8280	8289	8297	8305	8313	8322	8330	1	3	4	6	7
43	$\bar{1}$·8338	8346	8354	8362	8370	8378	8386	8394	8402	8410	1	3	4	5	7
44	$\bar{1}$·8418	8426	8433	8441	8449	8457	8464	8472	8480	8487	1	3	4	5	6

θ	0′ 0·0°	6′ 0·1°	12′ 0·2°	18′ 0·3°	24′ 0·4°	30′ 0·5°	36′ 0·6°	42′ 0·7°	48′ 0·8°	54′ 0·9°	ADD Differences 1′	2′	3′	4′	5′
45°	$\bar{1}$·8495	8502	8510	8517	8525	8532	8540	8547	8555	8562	1	2	4	5	6
46	$\bar{1}$·8569	8577	8584	8591	8598	8606	8613	8620	8627	8634	1	2	4	5	6
47	$\bar{1}$·8641	8648	8655	8662	8669	8676	8683	8690	8697	8704	1	2	3	5	6
48	$\bar{1}$·8711	8718	8724	8731	8738	8745	8751	8758	8765	8771	1	2	3	4	6
49	$\bar{1}$·8778	8784	8791	8797	8804	8810	8817	8823	8830	8836	1	2	3	4	5
50°	$\bar{1}$·8843	8849	8855	8862	8868	8874	8880	8887	8893	8899	1	2	3	4	5
51	$\bar{1}$·8905	8911	8917	8923	8929	8935	8941	8947	8953	8959	1	2	3	4	5
52	$\bar{1}$·8965	8971	8977	8983	8989	8995	9000	9006	9012	9018	1	2	3	4	5
53	$\bar{1}$·9023	9029	9035	9041	9046	9052	9057	9063	9069	9074	1	2	3	4	5
54	$\bar{1}$·9080	9085	9091	9096	9101	9107	9112	9118	9123	9128	1	2	3	4	5
55	$\bar{1}$·9134	9139	9144	9149	9155	9160	9165	9170	9175	9181	1	2	3	3	4
56	$\bar{1}$·9186	9191	9196	9201	9206	9211	9216	9221	9226	9231	1	2	3	3	4
57	$\bar{1}$·9236	9241	9246	9251	9255	9260	9265	9270	9275	9279	1	2	2	3	4
58	$\bar{1}$·9284	9289	9294	9298	9303	9308	9312	9317	9322	9326	1	2	2	3	4
59	$\bar{1}$·9331	9335	9340	9344	9349	9353	9358	9362	9367	9371	1	1	2	3	4
60°	$\bar{1}$·9375	9380	9384	9388	9393	9397	9401	9406	9410	9414	1	1	2	3	4
61	$\bar{1}$·9418	9422	9427	9431	9435	9439	9443	9447	9451	9455	1	1	2	3	3
62	$\bar{1}$·9459	9463	9467	9471	9475	9479	9483	9487	9491	9495	1	1	2	3	3
63	$\bar{1}$·9499	9503	9507	9510	9514	9518	9522	9525	9529	9533	1	1	2	3	3
64	$\bar{1}$·9537	9540	9544	9548	9551	9555	9558	9562	9566	9569	1	1	2	2	3
65	$\bar{1}$·9573	9576	9580	9583	9587	9590	9594	9597	9601	9604	1	1	2	2	3
66	$\bar{1}$·9607	9611	9614	9617	9621	9624	9627	9631	9634	9637	1	1	2	2	3
67	$\bar{1}$·9640	9643	9647	9650	9653	9656	9659	9662	9666	9669	1	1	2	2	3
68	$\bar{1}$·9672	9675	9678	9681	9684	9687	9690	9693	9696	9699	0	1	1	2	2
69	$\bar{1}$·9702	9704	9707	9710	9713	9716	9719	9722	9724	9727	0	1	1	2	2
70°	$\bar{1}$·9730	9733	9735	9738	9741	9743	9746	9749	9751	9754	0	1	1	2	2
71	$\bar{1}$·9757	9759	9762	9764	9767	9770	9772	9775	9777	9780	0	1	1	2	2
72	$\bar{1}$·9782	9785	9787	9789	9792	9794	9797	9799	9801	9804	0	1	1	2	2
73	$\bar{1}$·9806	9808	9811	9813	9815	9817	9820	9822	9824	9826	0	1	1	2	2
74	$\bar{1}$·9828	9831	9833	9835	9837	9839	9841	9843	9845	9847	0	1	1	1	2
75	$\bar{1}$·9849	9851	9853	9855	9857	9859	9861	9863	9865	9867	0	1	1	1	2
76	$\bar{1}$·9869	9871	9873	9875	9876	9878	9880	9882	9884	9885	0	1	1	1	2
77	$\bar{1}$·9887	9889	9891	9892	9894	9896	9897	9899	9901	9902	0	1	1	1	1
78	$\bar{1}$·9904	9906	9907	9909	9910	9912	9913	9915	9916	9918	0	1	1	1	1
79	$\bar{1}$·9919	9921	9922	9924	9925	9927	9928	9929	9931	9932	0	0	1	1	1
80°	$\bar{1}$·9934	9935	9936	9937	9939	9940	9941	9943	9944	9945	0	0	1	1	1
81	$\bar{1}$·9946	9947	9949	9950	9951	9952	9953	9954	9955	9956	0	0	1	1	1
82	$\bar{1}$·9958	9959	9960	9961	9962	9963	9964	9965	9966	9967	0	0	1	1	1
83	$\bar{1}$·9968	9968	9969	9970	9971	9972	9973	9974	9975	9975	0	0	1	1	1
84	$\bar{1}$·9976	9977	9978	9978	9979	9980	9981	9981	9982	9983	0	0	0	0	1
85	$\bar{1}$·9983	9984	9985	9985	9986	9987	9987	9988	9988	9989	0	0	0	0	0
86	$\bar{1}$·9989	9990	9990	9991	9991	9992	9992	9993	9993	9994	0	0	0	0	0
87	$\bar{1}$·9994	9994	9995	9995	9996	9996	9996	9996	9997	9997	0	0	0	0	0
88	$\bar{1}$·9997	9998	9998	9998	9998	9999	9999	9999	9999	9999	0	0	0	0	0
89	$\bar{1}$·9999	9999	$\bar{0}$000	$\bar{0}$000	$\bar{0}$000	$\bar{0}$000	$\bar{0}$000	$\bar{0}$000	$\bar{0}$000	$\bar{0}$000	0	0	0	0	0

Logarithms of tangents

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θ	0′ 0·0°	6′ 0·1°	12′ 0·2°	18′ 0·3°	24′ 0·4°	30′ 0·5°	36′ 0·6°	42′ 0·7°	48′ 0·8°	54′ 0·9°	ADD Differences 1′	2′	3′	4′	5′
0°	$-\infty$	$\bar{3}$·2419	5429	7190	8439	9409	$\bar{0}$200	$\bar{0}$870	$\bar{1}$450	$\bar{1}$962					
1	$\bar{2}$·2419	2833	3211	3559	3881	4181	4461	4725	4973	5208					
2	$\bar{2}$·5431	5643	5845	6038	6223	6401	6571	6736	6894	7046					
3	$\bar{2}$·7194	7337	7475	7609	7739	7865	7988	8107	8223	8336					
4	$\bar{2}$·8446	8554	8659	8762	8862	8960	9056	9150	9241	9331	16	32	48	64	81
5	$\bar{2}$·9420	9506	9591	9674	9756	9836	9915	9992	$\bar{0}$068	$\bar{0}$143	13	26	40	53	66
6	$\bar{1}$·0216	0289	0360	0430	0499	0567	0633	0699	0764	0828	11	22	34	45	56
7	$\bar{1}$·0891	0954	1015	1076	1135	1194	1252	1310	1367	1423	10	20	29	39	49
8	$\bar{1}$·1478	1533	1587	1640	1693	1745	1797	1848	1898	1948	9	17	26	35	43
9	$\bar{1}$·1997	2046	2094	2142	2189	2236	2282	2328	2374	2419	8	16	23	31	39
10°	$\bar{1}$·2463	2507	2551	2594	2637	2680	2722	2764	2805	2846	7	14	21	28	35
11	$\bar{1}$·2887	2927	2967	3006	3046	3085	3123	3162	3200	3237	6	13	19	26	32
12	$\bar{1}$·3275	3312	3349	3385	3422	3458	3493	3529	3564	3599	6	12	18	24	30
13	$\bar{1}$·3634	3668	3702	3736	3770	3804	3837	3870	3903	3935	6	11	17	22	28
14	$\bar{1}$·3968	4000	4032	4064	4095	4127	4158	4189	4220	4250	5	10	16	21	26
15	$\bar{1}$·4281	4311	4341	4371	4400	4430	4459	4488	4517	4546	5	10	15	20	25
16	$\bar{1}$·4575	4603	4632	4660	4688	4716	4744	4771	4799	4826	5	9	14	19	23
17	$\bar{1}$·4853	4880	4907	4934	4961	4987	5014	5040	5066	5092	4	9	13	18	22
18	$\bar{1}$·5118	5143	5169	5195	5220	5245	5270	5295	5320	5345	4	8	13	17	21
19	$\bar{1}$·5370	5394	5419	5443	5467	5491	5516	5539	5563	5587	4	8	12	16	20
20°	$\bar{1}$·5611	5634	5658	5681	5704	5727	5750	5773	5796	5819	4	8	12	15	19
21	$\bar{1}$·5842	5864	5887	5909	5932	5954	5976	5998	6020	6042	4	7	11	15	19
22	$\bar{1}$·6064	6086	6108	6129	6151	6172	6194	6215	6236	6257	4	7	11	14	18
23	$\bar{1}$·6279	6300	6321	6341	6362	6383	6404	6424	6445	6465	3	7	10	14	17
24	$\bar{1}$·6486	6506	6527	6547	6567	6587	6607	6627	6647	6667	3	7	10	13	17
25	$\bar{1}$·6687	6706	6726	6746	6765	6785	6804	6824	6843	6863	3	7	10	13	16
26	$\bar{1}$·6882	6901	6920	6939	6958	6977	6996	7015	7034	7053	3	6	9	13	16
27	$\bar{1}$·7072	7090	7109	7128	7146	7165	7183	7202	7220	7238	3	6	9	12	15
28	$\bar{1}$·7257	7275	7293	7311	7330	7348	7366	7384	7402	7420	3	6	9	12	15
29	$\bar{1}$·7438	7455	7473	7491	7509	7526	7544	7562	7579	7597	3	6	9	12	15
30°	$\bar{1}$·7614	7632	7649	7667	7684	7701	7719	7736	7753	7771	3	6	9	12	14
31	$\bar{1}$·7788	7805	7822	7839	7856	7873	7890	7907	7924	7941	3	6	9	11	14
32	$\bar{1}$·7958	7975	7992	8008	8025	8042	8059	8075	8092	8109	3	6	8	11	14
33	$\bar{1}$·8125	8142	8158	8175	8191	8208	8224	8241	8257	8274	3	5	8	11	14
34	$\bar{1}$·8290	8306	8323	8339	8355	8371	8388	8404	8420	8436	3	5	8	11	14
35	$\bar{1}$·8452	8468	8484	8501	8517	8533	8549	8565	8581	8597	3	5	8	11	13
36	$\bar{1}$·8613	8629	8644	8660	8676	8692	8708	8724	8740	8755	3	5	8	11	13
37	$\bar{1}$·8771	8787	8803	8818	8834	8850	8865	8881	8897	8912	3	5	8	10	13
38	$\bar{1}$·8928	8944	8959	8975	8990	9006	9022	9037	9053	9068	3	5	8	10	13
39	$\bar{1}$·9084	9099	9115	9130	9146	9161	9176	9192	9207	9223	3	5	8	10	13
40°	$\bar{1}$·9238	9254	9269	9284	9300	9315	9330	9346	9361	9376	3	5	8	10	13
41	$\bar{1}$·9392	9407	9422	9438	9453	9468	9483	9499	9514	9529	3	5	8	10	13
42	$\bar{1}$·9544	9560	9575	9590	9605	9621	9636	9651	9666	9681	3	5	8	10	13
43	$\bar{1}$·9697	9712	9727	9742	9757	9773	9788	9803	9818	9833	3	5	8	10	13
44	$\bar{1}$·9848	9864	9879	9894	9909	9924	9939	9955	9970	9985	3	5	8	10	13

θ	0′ 0·0°	6′ 0·1°	12′ 0·2°	18′ 0·3°	24′ 0·4°	30′ 0·5°	36′ 0·6°	42′ 0·7°	48′ 0·8°	54′ 0·9°	ADD Differences 1′	2′	3′	4′	5′
45°	0·0000	0015	0030	0045	0061	0076	0091	0106	0121	0136	3	5	8	10	13
46	0·0152	0167	0182	0197	0212	0228	0243	0258	0273	0288	3	5	8	10	13
47	0·0303	0319	0334	0349	0364	0379	0395	0410	0425	0440	3	5	8	10	13
48	0·0456	0471	0486	0501	0517	0532	0547	0562	0578	0593	3	5	8	10	13
49	0·0608	0624	0639	0654	0670	0685	0700	0716	0731	0746	3	5	8	10	13
50°	0·0762	0777	0793	0808	0824	0839	0854	0870	0885	0901	3	5	8	10	13
51	0·0916	0932	0947	0963	0978	0994	1010	1025	1041	1056	3	5	8	10	13
52	0·1072	1088	1103	1119	1135	1150	1166	1182	1197	1213	3	5	8	10	13
53	0·1229	1245	1260	1276	1292	1308	1324	1340	1356	1371	3	5	8	11	13
54	0·1387	1403	1419	1435	1451	1467	1483	1499	1516	1532	3	5	8	11	13
55	0·1548	1564	1580	1596	1612	1629	1645	1661	1677	1694	3	5	8	11	14
56	0·1710	1726	1743	1759	1776	1792	1809	1825	1842	1858	3	5	8	11	14
57	0·1875	1891	1908	1925	1941	1958	1975	1992	2008	2025	3	6	8	11	14
58	0·2042	2059	2076	2093	2110	2127	2144	2161	2178	2195	3	6	9	11	14
59	0·2212	2229	2247	2264	2281	2299	2316	2333	2351	2368	3	6	9	12	14
60°	0·2386	2403	2421	2438	2456	2474	2491	2509	2527	2545	3	6	9	12	15
61	0·2562	2580	2598	2616	2634	2652	2670	2689	2707	2725	3	6	9	12	15
62	0·2743	2762	2780	2798	2817	2835	2854	2872	2891	2910	3	6	9	12	15
63	0·2928	2947	2966	2985	3004	3023	3042	3061	3080	3099	3	6	9	13	16
64	0·3118	3137	3157	3176	3196	3215	3235	3254	3274	3294	3	6	10	13	16
65	0·3313	3333	3353	3373	3393	3413	3433	3453	3473	3494	3	7	10	13	17
66	0·3514	3535	3555	3576	3596	3617	3638	3659	3679	3700	3	7	10	14	17
67	0·3721	3743	3764	3785	3806	3828	3849	3871	3892	3914	4	7	11	14	18
68	0·3936	3958	3980	4002	4024	4046	4068	4091	4113	4136	4	7	11	15	19
69	0·4158	4181	4204	4227	4250	4273	4296	4319	4342	4366	4	8	12	15	19
70°	0·4389	4413	4437	4461	4484	4509	4533	4557	4581	4606	4	8	12	16	20
71	0·4630	4655	4680	4705	4730	4755	4780	4805	4831	4857	4	8	13	17	21
72	0·4882	4908	4934	4960	4986	5013	5039	3066	5093	5120	4	9	13	18	22
73	0·5147	5174	5201	5229	5256	5284	5312	5340	5368	5397	5	9	14	19	23
74	0·5425	5454	5483	5512	5541	5570	5600	5629	5659	5689	5	10	15	20	25
75	0·5719	5750	5780	5811	5842	5873	5905	5936	5968	6000	5	10	16	21	26
76	0·6032	6065	6097	6130	6163	6196	6230	6264	6298	6332	6	11	17	22	28
77	0·6366	6401	6436	6471	6507	6542	6578	6615	6651	6688	6	12	18	24	30
78	0·6725	6763	6800	6838	6877	6915	6954	6994	7033	7073	6	13	19	26	32
79	0·7113	7154	7195	7236	7278	7320	7363	7406	7449	7493	7	14	21	28	35
80°	0·7537	7581	7626	7672	7718	7764	7811	7858	7906	7954	8	16	23	31	39
81	0·8003	8052	8102	8152	8203	8255	8307	8360	8413	8467	9	17	26	35	43
82	0·8522	8577	8633	8690	8748	8806	8865	8924	8985	9046	10	20	29	39	49
83	0·9109	9172	9236	9301	9367	9433	9501	9570	9640	9711	11	22	34	45	56
84	0·9784	9857	9932	$\bar{0}$008	$\bar{0}$085	$\bar{0}$164	$\bar{0}$244	$\bar{0}$326	$\bar{0}$409	$\bar{0}$494	13	26	40	53	66
85	1·0580	0669	0759	0850	0944	1040	1138	1238	1341	1446	16	32	48	64	81
86	1·1554	1664	1777	1893	2012	2135	2261	2391	2525	2663					
87	1·2806	2954	3106	3264	3429	3599	3777	3962	4155	4357					
88	1·4569	4792	5027	5275	5539	5819	6119	6441	6789	7167					
89	1·7581	8038	8550	9130	9800	$\bar{0}$591	$\bar{1}$561	$\bar{2}$810	$\bar{4}$571	7581					

Logarithms of tangents

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θ	0′ 0·0°	6′ 0·1°	12′ 0·2°	18′ 0·3°	24′ 0·4°	30′ 0·5°	36′ 0·6°	42′ 0·7°	48′ 0·8°	54′ 0·9°	ADD Differences 1′	2′	3′	4′	5′
0°	$-\infty$	$\bar{3}$·2419	5429	7190	8439	9409	$\bar{0}$200	$\bar{0}$870	$\bar{1}$450	$\bar{1}$962					
1	$\bar{2}$·2419	2833	3211	3559	3881	4181	4461	4725	4973	5208					
2	$\bar{2}$·5431	5643	5845	6038	6223	6401	6571	6736	6894	7046					
3	$\bar{2}$·7194	7337	7475	7609	7739	7865	7988	8107	8223	8336					
4	$\bar{2}$·8446	8554	8659	8762	8862	8960	9056	9150	9241	9331	16	32	48	64	81
5	$\bar{2}$·9420	9506	9591	9674	9756	9836	9915	9992	$\bar{0}$068	$\bar{0}$143	13	26	40	53	66
6	$\bar{1}$·0216	0289	0360	0430	0499	0567	0633	0699	0764	0828	11	22	34	45	56
7	$\bar{1}$·0891	0954	1015	1076	1135	1194	1252	1310	1367	1423	10	20	29	39	49
8	$\bar{1}$·1478	1533	1587	1640	1693	1745	1797	1848	1898	1948	9	17	26	35	43
9	$\bar{1}$·1997	2046	2094	2142	2189	2236	2282	2328	2374	2419	8	16	23	31	39
10°	$\bar{1}$·2463	2507	2551	2594	2637	2680	2722	2764	2805	2846	7	14	21	28	35
11	$\bar{1}$·2887	2927	2967	3006	3046	3085	3123	3162	3200	3237	6	13	19	26	32
12	$\bar{1}$·3275	3312	3349	3385	3422	3458	3493	3529	3564	3599	6	12	18	24	30
13	$\bar{1}$·3634	3668	3702	3736	3770	3804	3837	3870	3903	3935	6	11	17	22	28
14	$\bar{1}$·3968	4000	4032	4064	4095	4127	4158	4189	4220	4250	5	10	16	21	26
15	$\bar{1}$·4281	4311	4341	4371	4400	4430	4459	4488	4517	4546	5	10	15	20	25
16	$\bar{1}$·4575	4603	4632	4660	4688	4716	4744	4771	4799	4826	5	9	14	19	23
17	$\bar{1}$·4853	4880	4907	4934	4961	4987	5014	5040	5066	5092	4	9	13	18	22
18	$\bar{1}$·5118	5143	5169	5195	5220	5245	5270	5295	5320	5345	4	8	13	17	21
19	$\bar{1}$·5370	5394	5419	5443	5467	5491	5516	5539	5563	5587	4	8	12	16	20
20°	$\bar{1}$·5611	5634	5658	5681	5704	5727	5750	5773	5796	5819	4	8	12	15	19
21	$\bar{1}$·5842	5864	5887	5909	5932	5954	5976	5998	6020	6042	4	7	11	15	19
22	$\bar{1}$·6064	6086	6108	6129	6151	6172	6194	6215	6236	6257	4	7	11	14	18
23	$\bar{1}$·6279	6300	6321	6341	6362	6383	6404	6424	6445	6465	3	7	10	14	17
24	$\bar{1}$·6486	6506	6527	6547	6567	6587	6607	6627	6647	6667	3	7	10	13	17
25	$\bar{1}$·6687	6706	6726	6746	6765	6785	6804	6824	6843	6863	3	7	10	13	16
26	$\bar{1}$·6882	6901	6920	6939	6958	6977	6996	7015	7034	7053	3	6	9	13	16
27	$\bar{1}$·7072	7090	7109	7128	7146	7165	7183	7202	7220	7238	3	6	9	12	15
28	$\bar{1}$·7257	7275	7293	7311	7330	7348	7366	7384	7402	7420	3	6	9	12	15
29	$\bar{1}$·7438	7455	7473	7491	7509	7526	7544	7562	7579	7597	3	6	9	12	15
30°	$\bar{1}$·7614	7632	7649	7667	7684	7701	7719	7736	7753	7771	3	6	9	12	14
31	$\bar{1}$·7788	7805	7822	7839	7856	7873	7890	7907	7924	7941	3	6	9	11	14
32	$\bar{1}$·7958	7975	7992	8008	8025	8042	8059	8075	8092	8109	3	6	8	11	14
33	$\bar{1}$·8125	8142	8158	8175	8191	8208	8224	8241	8257	8274	3	5	8	11	14
34	$\bar{1}$·8290	8306	8323	8339	8355	8371	8388	8404	8420	8436	3	5	8	11	14
35	$\bar{1}$·8452	8468	8484	8501	8517	8533	8549	8565	8581	8597	3	5	8	11	13
36	$\bar{1}$·8613	8629	8644	8660	8676	8692	8708	8724	8740	8755	3	5	8	11	13
37	$\bar{1}$·8771	8787	8803	8818	8834	8850	8865	8881	8897	8912	3	5	8	10	13
38	$\bar{1}$·8928	8944	8959	8975	8990	9006	9022	9037	9053	9068	3	5	8	10	13
39	$\bar{1}$·9084	9099	9115	9130	9146	9161	9176	9192	9207	9223	3	5	8	10	13
40°	$\bar{1}$·9238	9254	9269	9284	9300	9315	9330	9346	9361	9376	3	5	8	10	13
41	$\bar{1}$·9392	9407	9422	9438	9453	9468	9483	9499	9514	9529	3	5	8	10	13
42	$\bar{1}$·9544	9560	9575	9590	9605	9621	9636	9651	9666	9681	3	5	8	10	13
43	$\bar{1}$·9697	9712	9727	9742	9757	9773	9788	9803	9818	9833	3	5	8	10	13
44	$\bar{1}$·9848	9864	9879	9894	9909	9924	9939	9955	9970	9985	3	5	8	10	13

θ	0′ 0·0°	6′ 0·1°	12′ 0·2°	18′ 0·3°	24′ 0·4°	30′ 0·5°	36′ 0·6°	42′ 0·7°	48′ 0·8°	54′ 0·9°	ADD Differences 1′	2′	3′	4′	5′
45°	0·0000	0015	0030	0045	0061	0076	0091	0106	0121	0136	3	5	8	10	13
46	0·0152	0167	0182	0197	0212	0228	0243	0258	0273	0288	3	5	8	10	13
47	0·0303	0319	0334	0349	0364	0379	0395	0410	0425	0440	3	5	8	10	13
48	0·0456	0471	0486	0501	0517	0532	0547	0562	0578	0593	3	5	8	10	13
49	0·0608	0624	0639	0654	0670	0685	0700	0716	0731	0746	3	5	8	10	13
50°	0·0762	0777	0793	0808	0824	0839	0854	0870	0885	0901	3	5	8	10	13
51	0·0916	0932	0947	0963	0978	0994	1010	1025	1041	1056	3	5	8	10	13
52	0·1072	1088	1103	1119	1135	1150	1166	1182	1197	1213	3	5	8	10	13
53	0·1229	1245	1260	1276	1292	1308	1324	1340	1356	1371	3	5	8	11	13
54	0·1387	1403	1419	1435	1451	1467	1483	1499	1516	1532	3	5	8	11	13
55	0·1548	1564	1580	1596	1612	1629	1645	1661	1677	1694	3	5	8	11	14
56	0·1710	1726	1743	1759	1776	1792	1809	1825	1842	1858	3	5	8	11	14
57	0·1875	1891	1908	1925	1941	1958	1975	1992	2008	2025	3	6	8	11	14
58	0·2042	2059	2076	2093	2110	2127	2144	2161	2178	2195	3	6	9	11	14
59	0·2212	2229	2247	2264	2281	2299	2316	2333	2351	2368	3	6	9	12	14
60°	0·2386	2403	2421	2438	2456	2474	2491	2509	2527	2545	3	6	9	12	15
61	0·2562	2580	2598	2616	2634	2652	2670	2689	2707	2725	3	6	9	12	15
62	0·2743	2762	2780	2798	2817	2835	2854	2872	2891	2910	3	6	9	12	15
63	0·2928	2947	2966	2985	3004	3023	3042	3061	3080	3099	3	6	9	13	16
64	0·3118	3137	3157	3176	3196	3215	3235	3254	3274	3294	3	6	10	13	16
65	0·3313	3333	3353	3373	3393	3413	3433	3453	3473	3494	3	7	10	13	17
66	0·3514	3535	3555	3576	3596	3617	3638	3659	3679	3700	3	7	10	14	17
67	0·3721	3743	3764	3785	3806	3828	3849	3871	3892	3914	4	7	11	14	18
68	0·3936	3958	3980	4002	4024	4046	4068	4091	4113	4136	4	7	11	15	19
69	0·4158	4181	4204	4227	4250	4273	4296	4319	4342	4366	4	8	12	15	19
70°	0·4389	4413	4437	4461	4484	4509	4533	4557	4581	4606	4	8	12	16	20
71	0·4630	4655	4680	4705	4730	4755	4780	4805	4831	4857	4	8	13	17	21
72	0·4882	4908	4934	4960	4986	5013	5039	5066	5093	5120	4	9	13	18	22
73	0·5147	5174	5201	5229	5256	5284	5312	5340	5368	5397	5	9	14	19	23
74	0·5425	5454	5483	5512	5541	5570	5600	5629	5659	5689	5	10	15	20	25
75	0·5719	5750	5780	5811	5842	5873	5905	5936	5968	6000	5	10	16	21	26
76	0·6032	6065	6097	6130	6163	6196	6230	6264	6298	6332	6	11	17	22	28
77	0·6366	6401	6436	6471	6507	6542	6578	6615	6651	6688	6	12	18	24	30
78	0·6725	6763	6800	6838	6877	6915	6954	6994	7033	7073	6	13	19	26	32
79	0·7113	7154	7195	7236	7278	7320	7363	7406	7449	7493	7	14	21	28	35
80°	0·7537	7581	7626	7672	7718	7764	7811	7858	7906	7954	8	16	23	31	39
81	0·8003	8052	8102	8152	8203	8255	8307	8360	8413	8467	9	17	26	35	43
82	0·8522	8577	8633	8690	8748	8806	8865	8924	8985	9046	10	20	29	39	49
83	0·9109	9172	9236	9301	9367	9433	9501	9570	9640	9711	11	22	34	45	56
84	0·9784	9857	9932	$\bar{0}$008	$\bar{0}$085	$\bar{0}$164	$\bar{0}$244	$\bar{0}$326	$\bar{0}$409	$\bar{0}$494	13	26	40	53	66
85	1·0580	0669	0759	0850	0944	1040	1138	1238	1341	1446	16	32	48	64	81
86	1·1554	1664	1777	1893	2012	2135	2261	2391	2525	2663					
87	1·2806	2954	3106	3264	3429	3599	3777	3962	4155	4357					
88	1·4569	4792	5027	5275	5539	5819	6119	6441	6789	7167					
89	1·7581	8038	8550	9130	9800	$\bar{0}$591	$\bar{1}$561	$\bar{2}$810	$\bar{4}$571	7581					

Reciprocals

$x \longrightarrow \frac{1}{x}$

x	0	1	2	3	4	5	6	7	8	9	SUBTRACT Differences 1	2	3	4	5	6	7	8	9
1·0	1·000	9901	9804	9709	9615	9524	9434	9346	9259	9174	9	18	28	37	46	55	64	74	83
1·1	0·9091	9009	8929	8850	8772	8696	8621	8547	8475	8403	8	15	23	31	38	46	53	61	69
1·2	0·8333	8264	8197	8130	8065	8000	7937	7874	7813	7752	7	13	20	26	33	39	46	52	59
1·3	0·7692	7634	7576	7519	7463	7407	7353	7299	7246	7194	6	11	17	22	28	33	39	44	50
1·4	0·7143	7092	7042	6993	6944	6897	6849	6803	6757	6711	5	10	14	19	24	29	33	38	43
1·5	0·6667	6623	6579	6536	6494	6452	6410	6369	6329	6289	4	8	13	17	21	25	29	33	38
1·6	0·6250	6211	6173	6135	6098	6061	6024	5988	5952	5917	4	7	11	15	18	22	26	29	33
1·7	0·5882	5848	5814	5780	5747	5714	5682	5650	5618	5587	3	7	10	13	16	20	23	26	29
1·8	0·5556	5525	5495	5464	5435	5405	5376	5348	5319	5291	3	6	9	12	15	17	20	23	26
1·9	0·5263	5236	5208	5181	5155	5128	5102	5076	5051	5025	3	5	8	11	13	16	18	21	24
2·0	0·5000	4975	4950	4926	4902	4878	4854	4831	4808	4785	2	5	7	10	12	14	17	19	21
2·1	0·4762	4739	4717	4695	4673	4651	4630	4608	4587	4566	2	4	7	9	11	13	15	17	19
2·2	0·4545	4525	4505	4484	4464	4444	4425	4405	4386	4367	2	4	6	8	10	12	14	16	18
2·3	0·4348	4329	4310	4292	4274	4255	4237	4219	4202	4184	2	4	5	7	9	11	13	14	16
2·4	0·4167	4149	4132	4115	4098	4082	4065	4049	4032	4016	2	3	5	7	8	10	12	13	15
2·5	0·4000	3984	3968	3953	3937	3922	3906	3891	3876	3861	2	3	5	6	8	9	11	12	14
2·6	0·3846	3831	3817	3802	3788	3774	3759	3745	3731	3717	1	3	4	6	7	9	10	11	13
2·7	0·3704	3690	3676	3663	3650	3636	3623	3610	3597	3584	1	3	4	5	7	8	9	11	12
2·8	0·3571	3559	3546	3534	3521	3509	3497	3484	3472	3460	1	2	4	5	6	7	9	10	11
2·9	0·3448	3436	3425	3413	3401	3390	3378	3367	3356	3344	1	2	3	5	6	7	8	9	10
3·0	0·3333	3322	3311	3300	3289	3279	3268	3257	3247	3236	1	2	3	4	5	6	8	9	10
3·1	0·3226	3215	3205	3195	3185	3175	3165	3155	3145	3135	1	2	3	4	5	6	7	8	9
3·2	0·3125	3115	3106	3096	3086	3077	3067	3058	3049	3040	1	2	3	4	5	6	7	8	9
3·3	0·3030	3021	3012	3003	2994	2985	2976	2967	2959	2950	1	2	3	4	4	5	6	7	8
3·4	0·2941	2933	2924	2915	2907	2899	2890	2882	2874	2865	1	2	3	3	4	5	6	7	8
3·5	0·2857	2849	2841	2833	2825	2817	2809	2801	2793	2786	1	2	2	3	4	5	6	6	7
3·6	0·2778	2770	2762	2755	2747	2740	2732	2725	2717	2710	1	2	2	3	4	5	5	6	7
3·7	0·2703	2695	2688	2681	2674	2667	2660	2653	2646	2639	1	1	2	3	4	4	5	6	6
3·8	0·2632	2625	2618	2611	2604	2597	2591	2584	2577	2571	1	1	2	3	3	4	5	5	6
3·9	0·2564	2558	2551	2545	2538	2532	2525	2519	2513	2506	1	1	2	3	3	4	4	5	6
4·0	0·2500	2494	2488	2481	2475	2469	2463	2457	2451	2445	1	1	2	2	3	4	4	5	5
4·1	0·2439	2433	2427	2421	2415	2410	2404	2398	2392	2387	1	1	2	2	3	3	4	5	5
4·2	0·2381	2375	2370	2364	2358	2353	2347	2342	2336	2331	1	1	2	2	3	3	4	4	5
4·3	0·2326	2320	2315	2309	2304	2299	2294	2288	2283	2278	1	1	2	2	3	3	4	4	5
4·4	0·2273	2268	2262	2257	2252	2247	2242	2237	2232	2227	1	1	2	2	3	3	4	4	5
4·5	0·2222	2217	2212	2208	2203	2198	2193	2188	2183	2179	0	1	1	2	2	3	3	4	4
4·6	0·2174	2169	2165	2160	2155	2151	2146	2141	2137	2132	0	1	1	2	2	3	3	4	4
4·7	0·2128	2123	2119	2114	2110	2105	2101	2096	2092	2088	0	1	1	2	2	3	3	4	4
4·8	0·2083	2079	2075	2070	2066	2062	2058	2053	2049	2045	0	1	1	2	2	3	3	3	4
4·9	0·2041	2037	2033	2028	2024	2020	2016	2012	2008	2004	0	1	1	2	2	2	3	3	4
5·0	0·2000	1996	1992	1988	1984	1980	1976	1972	1969	1965	0	1	1	2	2	2	3	3	4
5·1	0·1961	1957	1953	1949	1946	1942	1938	1934	1931	1927	0	1	1	2	2	2	3	3	3
5·2	0·1923	1919	1916	1912	1908	1905	1901	1898	1894	1890	0	1	1	1	2	2	3	3	3
5·3	0·1887	1883	1880	1876	1873	1869	1866	1862	1859	1855	0	1	1	1	2	2	2	3	3
5·4	0·1852	1848	1845	1842	1838	1835	1832	1828	1825	1821	0	1	1	1	2	2	2	3	3

x	0	1	2	3	4	5	6	7	8	9	SUBTRACT Differences 1	2	3	4	5	6	7	8	9
5·5	0·1818	1815	1812	1808	1805	1802	1799	1795	1792	1789	0	1	1	1	2	2	2	3	3
5·6	0·1786	1783	1779	1776	1773	1770	1767	1764	1761	1757	0	1	1	1	2	2	2	3	3
5·7	0·1754	1751	1748	1745	1742	1739	1736	1733	1730	1727	0	1	1	1	1	2	2	2	3
5·8	0·1724	1721	1718	1715	1712	1709	1706	1704	1701	1698	0	1	1	1	1	2	2	2	3
5·9	0·1695	1692	1689	1686	1684	1681	1678	1675	1672	1669	0	1	1	1	1	2	2	2	3
6·0	0·1667	1664	1661	1658	1656	1653	1650	1647	1645	1642	0	1	1	1	1	2	2	2	3
6·1	0·1639	1637	1634	1631	1629	1626	1623	1621	1618	1616	0	1	1	1	1	2	2	2	2
6·2	0·1613	1610	1608	1605	1603	1600	1597	1595	1592	1590	0	1	1	1	1	2	2	2	2
6·3	0·1587	1585	1582	1580	1577	1575	1572	1570	1567	1565	0	0	1	1	1	1	2	2	2
6·4	0·1562	1560	1558	1555	1553	1550	1548	1546	1543	1541	0	0	1	1	1	1	2	2	2
6·5	0·1538	1536	1534	1531	1529	1527	1524	1522	1520	1517	0	0	1	1	1	1	2	2	2
6·6	0·1515	1513	1511	1508	1506	1504	1502	1499	1497	1495	0	0	1	1	1	1	2	2	2
6·7	0·1493	1490	1488	1486	1484	1481	1479	1477	1475	1473	0	0	1	1	1	1	2	2	2
6·8	0·1471	1468	1466	1464	1462	1460	1458	1456	1453	1451	0	0	1	1	1	1	2	2	2
6·9	0·1449	1447	1445	1443	1441	1439	1437	1435	1433	1431	0	0	1	1	1	1	2	2	2
7·0	0·1429	1427	1425	1422	1420	1418	1416	1414	1412	1410	0	0	1	1	1	1	1	2	2
7·1	0·1408	1406	1404	1403	1401	1399	1397	1395	1393	1391	0	0	1	1	1	1	1	2	2
7·2	0·1389	1387	1385	1383	1381	1379	1377	1376	1374	1372	0	0	1	1	1	1	1	2	2
7·3	0·1370	1368	1366	1364	1362	1361	1359	1357	1355	1353	0	0	1	1	1	1	1	2	2
7·4	0·1351	1350	1348	1346	1344	1342	1340	1339	1337	1335	0	0	1	1	1	1	1	1	2
7·5	0·1333	1332	1330	1328	1326	1325	1323	1321	1319	1318	0	0	1	1	1	1	1	1	2
7·6	0·1316	1314	1312	1311	1309	1307	1305	1304	1302	1300	0	0	1	1	1	1	1	1	2
7·7	0·1299	1297	1295	1294	1292	1290	1289	1287	1285	1284	0	0	0	1	1	1	1	1	1
7·8	0·1282	1280	1279	1277	1276	1274	1272	1271	1269	1267	0	0	0	1	1	1	1	1	1
7·9	0·1266	1264	1263	1261	1259	1258	1256	1255	1253	1252	0	0	0	1	1	1	1	1	1
8·0	0·1250	1248	1247	1245	1244	1242	1241	1239	1238	1236	0	0	0	1	1	1	1	1	1
8·1	0·1235	1233	1232	1230	1229	1227	1225	1224	1222	1221	0	0	0	1	1	1	1	1	1
8·2	0·1220	1218	1217	1215	1214	1212	1211	1209	1208	1206	0	0	0	1	1	1	1	1	1
8·3	0·1205	1203	1202	1200	1199	1198	1196	1195	1193	1192	0	0	0	1	1	1	1	1	1
8·4	0·1190	1189	1188	1186	1185	1183	1182	1181	1179	1178	0	0	0	1	1	1	1	1	1
8·5	0·1176	1175	1174	1172	1171	1170	1168	1167	1166	1164	0	0	0	1	1	1	1	1	1
8·6	0·1163	1161	1160	1159	1157	1156	1155	1153	1152	1151	0	0	0	1	1	1	1	1	1
8·7	0·1149	1148	1147	1145	1144	1143	1142	1140	1139	1138	0	0	0	1	1	1	1	1	1
8·8	0·1136	1135	1134	1133	1131	1130	1129	1127	1126	1125	0	0	0	1	1	1	1	1	1
8·9	0·1124	1122	1121	1120	1119	1117	1116	1115	1114	1112	0	0	0	1	1	1	1	1	1
9·0	0·1111	1110	1109	1107	1106	1105	1104	1103	1101	1100	0	0	0	1	1	1	1	1	1
9·1	0·1099	1098	1096	1095	1094	1093	1092	1090	1089	1088	0	0	0	0	1	1	1	1	1
9·2	0·1087	1086	1085	1083	1082	1081	1080	1079	1078	1076	0	0	0	0	1	1	1	1	1
9·3	0·1075	1074	1073	1072	1071	1070	1068	1067	1066	1065	0	0	0	0	1	1	1	1	1
9·4	0·1064	1063	1062	1060	1059	1058	1057	1056	1055	1054	0	0	0	0	1	1	1	1	1
9·5	0·1053	1052	1050	1049	1048	1047	1046	1045	1044	1043	0	0	0	0	1	1	1	1	1
9·6	0·1042	1041	1039	1038	1037	1036	1035	1034	1033	1032	0	0	0	0	1	1	1	1	1
9·7	0·1031	1030	1029	1028	1027	1026	1025	1024	1022	1021	0	0	0	0	1	1	1	1	1
9·8	0·1020	1019	1018	1017	1016	1015	1014	1013	1012	1011	0	0	0	0	1	1	1	1	1
9·9	0·1010	1009	1008	1007	1006	1005	1004	1003	1002	1001	0	0	0	0	0	1	1	1	1

Squares

$x \rightarrow x^2$

											Differences								
x	0	1	2	3	4	5	6	7	8	9	1	2	3	4	5	6	7	8	9
10	1000	1020	1040	1061	1082	1103	1124	1145	1166	1188	2	4	6	8	10	13	15	17	19
11	1210	1232	1254	1277	1300	1323	1346	1369	1392	1416	2	5	7	9	11	14	16	18	21
12	1440	1464	1488	1513	1538	1563	1588	1613	1638	1664	2	5	7	10	12	15	17	20	22
13	1690	1716	1742	1769	1796	1823	1850	1877	1904	1932	3	5	8	11	13	16	19	22	24
14	1960	1988	2016	2045	2074	2103	2132	2161	2190	2220	3	6	9	12	14	17	20	23	26
15	2250	2280	2310	2341	2372	2403	2434	2465	2496	2528	3	6	9	12	15	19	22	25	28
16	2560	2592	2624	2657	2690	2723	2756	2789	2822	2856	3	7	10	13	16	20	23	26	30
17	2890	2924	2958	2993	3028	3063	3098	3133	3168	3204	3	7	10	14	17	21	24	28	31
18	3240	3276	3312	3349	3386	3423	3460	3497	3534	3572	4	7	11	15	18	22	26	30	33
19	3610	3648	3686	3725	3764	3803	3842	3881	3920	3960	4	8	12	16	19	23	27	31	35
20	4000	4040	4080	4121	4162	4203	4244	4285	4326	4368	4	8	12	16	20	25	29	33	37
21	4410	4452	4494	4537	4580	4623	4666	4709	4752	4796	4	9	13	17	21	26	30	34	39
22	4840	4884	4928	4973	5018	5063	5108	5153	5198	5244	4	9	13	18	22	27	31	36	40
23	5290	5336	5382	5429	5476	5523	5570	5617	5664	5712	5	9	14	19	23	28	33	38	42
24	5760	5808	5856	5905	5954	6003	6052	6101	6150	6200	5	10	15	20	24	29	34	39	44
25	6250	6300	6350	6401	6452	6503	6554	6605	6656	6708	5	10	15	20	25	31	36	41	46
26	6760	6812	6864	6917	6970	7023	7076	7129	7182	7236	5	11	16	21	26	32	37	42	48
27	7290	7344	7398	7453	7508	7563	7618	7673	7728	7784	5	11	16	22	27	33	38	44	49
28	7840	7896	7952	8009	8066	8123	8180	8237	8294	8352	6	11	17	23	28	34	40	46	51
29	8410	8468	8526	8585	8644	8703	8762	8821	8880	8940	6	12	18	24	29	35	41	47	53
30	9000	9060	9120	9181	9242	9303	9364	9425	9486	9548	6	12	18	24	30	37	43	49	55
31	9610	9672	9734	9797	9860	9923	9986	–	–	–	6	13	19	25	31	38	44	50	57
31	–	–	–	–	–	–	–	1005	1011	1018	1	1	2	3	3	4	5	5	6
32	1024	1030	1037	1043	1050	1056	1063	1069	1076	1082	1	1	2	3	3	4	5	5	6
33	1089	1096	1102	1109	1116	1122	1129	1136	1142	1149	1	1	2	3	3	4	5	5	6
34	1156	1163	1170	1176	1183	1190	1197	1204	1211	1218	1	1	2	3	3	4	5	6	6
35	1225	1232	1239	1246	1253	1260	1267	1274	1282	1289	1	1	2	3	4	4	5	6	6
36	1296	1303	1310	1318	1325	1332	1340	1347	1354	1362	1	1	2	3	4	4	5	6	7
37	1369	1376	1384	1391	1399	1406	1414	1421	1429	1436	1	2	2	3	4	5	5	6	7
38	1444	1452	1459	1467	1475	1482	1490	1498	1505	1513	1	2	2	3	4	5	5	6	7
39	1521	1529	1537	1544	1552	1560	1568	1576	1584	1592	1	2	2	3	4	5	6	6	7
40	1600	1608	1616	1624	1632	1640	1648	1656	1665	1673	1	2	2	3	4	5	6	6	7
41	1681	1689	1697	1706	1714	1722	1731	1739	1747	1756	1	2	2	3	4	5	6	7	7
42	1764	1772	1781	1789	1798	1806	1815	1823	1832	1840	1	2	3	3	4	5	6	7	8
43	1849	1858	1866	1875	1884	1892	1901	1910	1918	1927	1	2	3	3	4	5	6	7	8
44	1936	1945	1954	1962	1971	1980	1989	1998	2007	2016	1	2	3	4	5	5	6	7	8
45	2025	2034	2043	2052	2061	2070	2079	2088	2098	2107	1	2	3	4	5	5	6	7	8
46	2116	2125	2134	2144	2153	2162	2172	2181	2190	2200	1	2	3	4	5	6	7	7	8
47	2209	2218	2228	2237	2247	2256	2266	2275	2285	2294	1	2	3	4	5	6	7	8	9
48	2304	2314	2323	2333	2343	2352	2362	2372	2381	2391	1	2	3	4	5	6	7	8	9
49	2401	2411	2421	2430	2440	2450	2460	2470	2480	2490	1	2	3	4	5	6	7	8	9
50	2500	2510	2520	2530	2540	2550	2560	2570	2581	2591	1	2	3	4	5	6	7	8	9
51	2601	2611	2621	2632	2642	2652	2663	2673	2683	2694	1	2	3	4	5	6	7	8	9
52	2704	2714	2725	2735	2746	2756	2767	2777	2788	2798	1	2	3	4	5	6	7	8	9
53	2809	2820	2830	2841	2852	2862	2873	2884	2894	2905	1	2	3	4	5	6	7	9	10
54	2916	2927	2938	2948	2959	2970	2981	2992	3003	3014	1	2	3	4	5	7	8	9	10

											Differences								
x	0	1	2	3	4	5	6	7	8	9	1	2	3	4	5	6	7	8	9
55	3025	3036	3047	3058	3069	3080	3091	3102	3114	3125	1	2	3	4	6	7	8	9	10
56	3136	3147	3158	3170	3181	3192	3204	3215	3226	3238	1	2	3	5	6	7	8	9	10
57	3249	3260	3272	3283	3295	3306	3318	3329	3341	3352	1	2	3	5	6	7	8	9	10
58	3364	3376	3387	3399	3411	3422	3434	3446	3457	3469	1	2	4	5	6	7	8	9	10
59	3481	3493	3505	3516	3528	3540	3552	3564	3576	3588	1	2	4	5	6	7	8	10	11
60	3600	3612	3624	3636	3648	3660	3672	3684	3697	3709	1	2	4	5	6	7	8	10	11
61	3721	3733	3745	3758	3770	3782	3795	3807	3819	3832	1	2	4	5	6	7	9	10	11
62	3844	3856	3869	3881	3894	3906	3919	3931	3944	3956	1	2	4	5	6	7	9	10	11
63	3969	3982	3994	4007	4020	4032	4045	4058	4070	4083	1	3	4	5	6	8	9	10	11
64	4096	4109	4122	4134	4147	4160	4173	4186	4199	4212	1	3	4	5	6	8	9	10	12
65	4225	4238	4251	4264	4277	4290	4303	4316	4330	4343	1	3	4	5	7	8	9	10	12
66	4356	4369	4382	4396	4409	4422	4436	4449	4462	4476	1	3	4	5	7	8	9	11	12
67	4489	4502	4516	4529	4543	4556	4570	4583	4597	4610	1	3	4	5	7	8	9	11	12
68	4624	4638	4651	4665	4679	4692	4706	4720	4733	4747	1	3	4	5	7	8	10	11	12
69	4761	4775	4789	4802	4816	4830	4844	4858	4872	4886	1	3	4	6	7	8	10	11	13
70	4900	4914	4928	4942	4956	4970	4984	4998	5013	5027	1	3	4	6	7	8	10	11	13
71	5041	5055	5069	5084	5098	5112	5127	5141	5155	5170	1	3	4	6	7	9	10	11	13
72	5184	5198	5213	5227	5242	5256	5271	5285	5300	5314	1	3	4	6	7	9	10	11	13
73	5329	5344	5358	5373	5388	5402	5417	5432	5446	5461	1	3	4	6	7	9	10	12	13
74	5476	5491	5506	5520	5535	5550	5565	5580	5595	5610	1	3	4	6	7	9	10	12	13
75	5625	5640	5655	5670	5685	5700	5715	5730	5746	5761	2	3	5	6	8	9	11	12	14
76	5776	5791	5806	5822	5837	5852	5868	5883	5898	5914	2	3	5	6	8	9	11	12	14
77	5929	5944	5960	5975	5991	6006	6022	6037	6053	6068	2	3	5	6	8	9	11	12	14
78	6084	6100	6115	6131	6147	6162	6178	6194	6209	6225	2	3	5	6	8	9	11	13	14
79	6241	6257	6273	6288	6304	6320	6336	6352	6368	6384	2	3	5	6	8	10	11	13	14
80	6400	6416	6432	6448	6464	6480	6496	6512	6529	6545	2	3	5	6	8	10	11	13	14
81	6561	6577	6593	6610	6626	6642	6659	6675	6691	6708	2	3	5	7	8	10	11	13	15
82	6724	6740	6757	6773	6790	6806	6823	6839	6856	6872	2	3	5	7	8	10	12	13	15
83	6889	6906	6922	6939	6956	6972	6989	7006	7022	7039	2	3	5	7	8	10	12	13	15
84	7056	7073	7090	7106	7123	7140	7157	7174	7191	7208	2	3	5	7	8	10	12	14	15
85	7225	7242	7259	7276	7293	7310	7327	7344	7362	7379	2	3	5	7	9	10	12	14	15
86	7396	7413	7430	7448	7465	7482	7500	7517	7534	7552	2	3	5	7	9	10	12	14	16
87	7569	7586	7604	7621	7639	7656	7674	7691	7709	7726	2	3	5	7	9	10	12	14	16
88	7744	7762	7779	7797	7815	7832	7850	7868	7885	7903	2	4	5	7	9	11	12	14	16
89	7921	7939	7957	7974	7992	8010	8028	8046	8064	8082	2	4	5	7	9	11	13	14	16
90	8100	8118	8136	8154	8172	8190	8208	8226	8245	8263	2	4	5	7	9	11	13	14	16
91	8281	8299	8317	8336	8354	8372	8391	8409	8427	8446	2	4	5	7	9	11	13	15	16
92	8464	8482	8501	8519	8538	8556	8575	8593	8612	8630	2	4	6	7	9	11	13	15	17
93	8649	8668	8686	8705	8724	8742	8761	8780	8798	8817	2	4	6	7	9	11	13	15	17
94	8836	8855	8874	8892	8911	8930	8949	8968	8987	9006	2	4	6	8	9	11	13	15	17
95	9025	9044	9063	9082	9101	9120	9139	9158	9178	9197	2	4	6	8	10	11	13	15	17
96	9216	9235	9254	9274	9293	9312	9332	9351	9370	9390	2	4	6	8	10	12	14	15	17
97	9409	9428	9448	9467	9487	9506	9526	9545	9565	9584	2	4	6	8	10	12	14	16	18
98	9604	9624	9643	9663	9683	9702	9722	9742	9761	9781	2	4	6	8	10	12	14	16	18
99	9801	9821	9841	9860	9880	9900	9920	9940	9960	9980	2	4	6	8	10	12	14	16	18

Square roots from 1 to 10

$x \longrightarrow \sqrt{x}$

x	0	1	2	3	4	5	6	7	8	9	Differences 1	2	3	4	5	6	7	8	9
1·0	1·000	1·005	1·010	1·015	1·020	1·025	1·030	1·034	1·039	1·044	0	1	1	2	2	3	3	4	4
1·1	1·049	1·054	1·058	1·063	1·068	1·072	1·077	1·082	1·086	1·091	0	1	1	2	2	3	3	4	4
1·2	1·095	1·100	1·105	1·109	1·114	1·118	1·123	1·127	1·131	1·136	0	1	1	2	2	3	3	4	4
1·3	1·140	1·145	1·149	1·153	1·158	1·162	1·166	1·171	1·175	1·179	0	1	1	2	2	3	3	3	4
1·4	1·183	1·187	1·192	1·196	1·200	1·204	1·208	1·212	1·217	1·221	0	1	1	2	2	2	3	3	4
1·5	1·225	1·229	1·233	1·237	1·241	1·245	1·249	1·253	1·257	1·261	0	1	1	2	2	2	3	3	4
1·6	1·265	1·269	1·273	1·277	1·281	1·285	1·288	1·292	1·296	1·300	0	1	1	2	2	2	3	3	3
1·7	1·304	1·308	1·312	1·315	1·319	1·323	1·327	1·330	1·334	1·338	0	1	1	2	2	2	3	3	3
1·8	1·342	1·345	1·349	1·353	1·357	1·360	1·364	1·368	1·371	1·375	0	1	1	1	2	2	3	3	3
1·9	1·378	1·382	1·386	1·389	1·393	1·396	1·400	1·404	1·407	1·411	0	1	1	1	2	2	3	3	3
2·0	1·414	1·418	1·421	1·425	1·428	1·432	1·435	1·439	1·442	1·446	0	1	1	1	2	2	2	3	3
2·1	1·449	1·453	1·456	1·460	1·463	1·466	1·470	1·473	1·477	1·480	0	1	1	1	2	2	2	3	3
2·2	1·483	1·487	1·490	1·493	1·497	1·500	1·503	1·507	1·510	1·513	0	1	1	1	2	2	2	3	3
2·3	1·517	1·520	1·523	1·526	1·530	1·533	1·536	1·539	1·543	1·546	0	1	1	1	2	2	2	3	3
2·4	1·549	1·552	1·556	1·559	1·562	1·565	1·568	1·572	1·575	1·578	0	1	1	1	2	2	2	3	3
2·5	1·581	1·584	1·587	1·591	1·594	1·597	1·600	1·603	1·606	1·609	0	1	1	1	2	2	2	3	3
2·6	1·612	1·616	1·619	1·622	1·625	1·628	1·631	1·634	1·637	1·640	0	1	1	1	2	2	2	2	3
2·7	1·643	1·646	1·649	1·652	1·655	1·658	1·661	1·664	1·667	1·670	0	1	1	1	2	2	2	2	3
2·8	1·673	1·676	1·679	1·682	1·685	1·688	1·691	1·694	1·697	1·700	0	1	1	1	1	2	2	2	3
2·9	1·703	1·706	1·709	1·712	1·715	1·718	1·720	1·723	1·726	1·729	0	1	1	1	1	2	2	2	3
3·0	1·732	1·735	1·738	1·741	1·744	1·746	1·749	1·752	1·755	1·758	0	1	1	1	1	2	2	2	3
3·1	1·761	1·764	1·766	1·769	1·772	1·775	1·778	1·780	1·783	1·786	0	1	1	1	1	2	2	2	3
3·2	1·789	1·792	1·794	1·797	1·800	1·803	1·806	1·808	1·811	1·814	0	1	1	1	1	2	2	2	2
3·3	1·817	1·819	1·822	1·825	1·828	1·830	1·833	1·836	1·839	1·841	0	1	1	1	1	2	2	2	2
3·4	1·844	1·847	1·849	1·852	1·855	1·857	1·860	1·863	1·866	1·868	0	1	1	1	1	2	2	2	2
3·5	1·871	1·874	1·876	1·879	1·882	1·884	1·887	1·889	1·892	1·895	0	1	1	1	1	2	2	2	2
3·6	1·897	1·900	1·903	1·905	1·908	1·911	1·913	1·916	1·918	1·921	0	1	1	1	1	2	2	2	2
3·7	1·924	1·926	1·929	1·931	1·934	1·937	1·939	1·942	1·944	1·947	0	1	1	1	1	2	2	2	2
3·8	1·949	1·952	1·955	1·957	1·960	1·962	1·965	1·967	1·970	1·972	0	1	1	1	1	2	2	2	2
3·9	1·975	1·977	1·980	1·982	1·985	1·988	1·990	1·993	1·995	1·998	0	1	1	1	1	2	2	2	2
4·0	2·000	2·003	2·005	2·008	2·010	2·013	2·015	2·017	2·020	2·022	0	0	1	1	1	1	2	2	2
4·1	2·025	2·027	2·030	2·032	2·035	2·037	2·040	2·042	2·045	2·047	0	0	1	1	1	1	2	2	2
4·2	2·049	2·052	2·054	2·057	2·059	2·062	2·064	2·066	2·069	2·071	0	0	1	1	1	1	2	2	2
4·3	2·074	2·076	2·078	2·081	2·083	2·086	2·088	2·091	2·093	2·095	0	0	1	1	1	1	2	2	2
4·4	2·098	2·100	2·102	2·105	2·107	2·110	2·112	2·114	2·117	2·119	0	0	1	1	1	1	2	2	2
4·5	2·121	2·124	2·126	2·128	2·131	2·133	2·135	2·138	2·140	2·142	0	0	1	1	1	1	2	2	2
4·6	2·145	2·147	2·149	2·152	2·154	2·156	2·159	2·161	2·163	2·166	0	0	1	1	1	1	2	2	2
4·7	2·168	2·170	2·173	2·175	2·177	2·179	2·182	2·184	2·186	2·189	0	0	1	1	1	1	2	2	2
4·8	2·191	2·193	2·195	2·198	2·200	2·202	2·205	2·207	2·209	2·211	0	0	1	1	1	1	2	2	2
4·9	2·214	2·216	2·218	2·220	2·223	2·225	2·227	2·229	2·232	2·234	0	0	1	1	1	1	2	2	2
5·0	2·236	2·238	2·241	2·243	2·245	2·247	2·249	2·252	2·254	2·256	0	0	1	1	1	1	2	2	2
5·1	2·258	2·261	2·263	2·265	2·267	2·269	2·272	2·274	2·276	2·278	0	0	1	1	1	1	2	2	2
5·2	2·280	2·283	2·285	2·287	2·289	2·291	2·294	2·296	2·298	2·300	0	0	1	1	1	1	2	2	2
5·3	2·302	2·304	2·307	2·309	2·311	2·313	2·315	2·317	2·320	2·322	0	0	1	1	1	1	2	2	2
5·4	2·324	2·326	2·328	2·330	2·332	2·335	2·337	2·339	2·341	2·343	0	0	1	1	1	1	1	2	2

x	0	1	2	3	4	5	6	7	8	9	Differences 1	2	3	4	5	6	7	8	9
5·5	2·345	2·347	2·350	2·352	2·354	2·356	2·358	2·360	2·362	2·364	0	0	1	1	1	1	1	2	2
5·6	2·366	2·369	2·371	2·373	2·375	2·377	2·379	2·381	2·383	2·385	0	0	1	1	1	1	1	2	2
5·7	2·388	2·390	2·392	2·394	2·396	2·398	2·400	2·402	2·404	2·406	0	0	1	1	1	1	1	2	2
5·8	2·408	2·410	2·412	2·415	2·417	2·419	2·421	2·423	2·425	2·427	0	0	1	1	1	1	1	2	2
5·9	2·429	2·431	2·433	2·435	2·437	2·439	2·441	2·443	2·445	2·447	0	0	1	1	1	1	1	2	2
6·0	2·450	2·452	2·454	2·456	2·458	2·460	2·462	2·464	2·466	2·468	0	0	1	1	1	1	1	2	2
6·1	2·470	2·472	2·474	2·476	2·478	2·480	2·482	2·484	2·486	2·488	0	0	1	1	1	1	1	2	2
6·2	2·490	2·492	2·494	2·496	2·498	2·500	2·502	2·504	2·506	2·508	0	0	1	1	1	1	1	2	2
6·3	2·510	2·512	2·514	2·516	2·518	2·520	2·522	2·524	2·526	2·528	0	0	1	1	1	1	1	2	2
6·4	2·530	2·532	2·534	2·536	2·538	2·540	2·542	2·544	2·546	2·548	0	0	1	1	1	1	1	2	2
6·5	2·550	2·551	2·553	2·555	2·557	2·559	2·561	2·563	2·565	2·567	0	0	1	1	1	1	1	2	2
6·6	2·569	2·571	2·573	2·575	2·577	2·579	2·581	2·583	2·585	2·587	0	0	1	1	1	1	1	2	2
6·7	2·588	2·590	2·592	2·594	2·596	2·598	2·600	2·602	2·604	2·606	0	0	1	1	1	1	1	2	2
6·8	2·608	2·610	2·612	2·613	2·615	2·617	2·619	2·621	2·623	2·625	0	0	1	1	1	1	1	2	2
6·9	2·627	2·629	2·631	2·632	2·634	2·636	2·638	2·640	2·642	2·644	0	0	1	1	1	1	1	2	2
7·0	2·646	2·648	2·650	2·651	2·653	2·655	2·657	2·659	2·661	2·663	0	0	1	1	1	1	1	2	2
7·1	2·665	2·667	2·668	2·670	2·672	2·674	2·676	2·678	2·680	2·681	0	0	1	1	1	1	1	1	2
7·2	2·683	2·685	2·687	2·689	2·691	2·693	2·694	2·696	2·698	2·700	0	0	1	1	1	1	1	1	2
7·3	2·702	2·704	2·706	2·707	2·709	2·711	2·713	2·715	2·717	2·719	0	0	1	1	1	1	1	1	2
7·4	2·720	2·722	2·724	2·726	2·728	2·729	2·731	2·733	2·735	2·737	0	0	1	1	1	1	1	1	2
7·5	2·739	2·740	2·742	2·744	2·746	2·748	2·750	2·751	2·753	2·755	0	0	1	1	1	1	1	1	2
7·6	2·757	2·759	2·760	2·762	2·764	2·766	2·768	2·769	2·771	2·773	0	0	1	1	1	1	1	1	2
7·7	2·775	2·777	2·778	2·780	2·782	2·784	2·786	2·787	2·789	2·791	0	0	1	1	1	1	1	1	2
7·8	2·793	2·795	2·796	2·798	2·800	2·802	2·804	2·805	2·807	2·809	0	0	1	1	1	1	1	1	2
7·9	2·811	2·812	2·814	2·816	2·818	2·820	2·821	2·823	2·825	2·827	0	0	1	1	1	1	1	1	2
8·0	2·828	2·830	2·832	2·834	2·835	2·837	2·839	2·841	2·843	2·844	0	0	1	1	1	1	1	1	2
8·1	2·846	2·848	2·850	2·851	2·853	2·855	2·857	2·858	2·860	2·862	0	0	1	1	1	1	1	1	2
8·2	2·864	2·865	2·867	2·869	2·871	2·872	2·874	2·876	2·877	2·879	0	0	1	1	1	1	1	1	2
8·3	2·881	2·883	2·884	2·886	2·888	2·890	2·891	2·893	2·895	2·897	0	0	1	1	1	1	1	1	2
8·4	2·898	2·900	2·902	2·903	2·905	2·907	2·909	2·910	2·912	2·914	0	0	1	1	1	1	1	1	2
8·5	2·915	2·917	2·919	2·921	2·922	2·924	2·926	2·927	2·929	2·931	0	0	1	1	1	1	1	1	2
8·6	2·933	2·934	2·936	2·938	2·939	2·941	2·943	2·944	2·946	2·948	0	0	1	1	1	1	1	1	2
8·7	2·950	2·951	2·953	2·955	2·956	2·958	2·960	2·961	2·963	2·965	0	0	1	1	1	1	1	1	2
8·8	2·966	2·968	2·970	2·972	2·973	2·975	2·977	2·978	2·980	2·982	0	0	1	1	1	1	1	1	2
8·9	2·983	2·985	2·987	2·988	2·990	2·992	2·993	2·995	2·997	2·998	0	0	1	1	1	1	1	1	2
9·0	3·000	3·002	3·003	3·005	3·007	3·008	3·010	3·012	3·013	3·015	0	0	0	1	1	1	1	1	1
9·1	3·017	3·018	3·020	3·022	3·023	3·025	3·027	3·028	3·030	3·032	0	0	0	1	1	1	1	1	1
9·2	3·033	3·035	3·036	3·038	3·040	3·041	3·043	3·045	3·046	3·048	0	0	0	1	1	1	1	1	1
9·3	3·050	3·051	3·053	3·055	3·056	3·058	3·059	3·061	3·063	3·064	0	0	0	1	1	1	1	1	1
9·4	3·066	3·068	3·069	3·071	3·072	3·074	3·076	3·077	3·079	3·081	0	0	0	1	1	1	1	1	1
9·5	3·082	3·084	3·085	3·087	3·089	3·090	3·092	3·094	3·095	3·097	0	0	0	1	1	1	1	1	1
9·6	3·098	3·100	3·102	3·103	3·105	3·106	3·108	3·110	3·111	3·113	0	0	0	1	1	1	1	1	1
9·7	3·115	3·116	3·118	3·119	3·121	3·123	3·124	3·126	3·127	3·129	0	0	0	1	1	1	1	1	1
9·8	3·131	3·132	3·134	3·135	3·137	3·139	3·140	3·142	3·143	2·145	0	0	0	1	1	1	1	1	1
9·9	3·146	3·148	3·150	3·151	3·153	3·154	3·156	3·158	3·159	3·161	0	0	0	1	1	1	1	1	1

Square roots from 10 to 100

$x \longrightarrow \sqrt{x}$

x	0	1	2	3	4	5	6	7	8	9	Differences 1	2	3	4	5	6	7	8	9
10	3·162	3·178	3·194	3·209	3·225	3·240	3·256	3·271	3·286	3·302	2	3	5	6	8	9	11	12	14
11	3·317	3·332	3·347	3·362	3·376	3·391	3·406	3·421	3·435	3·450	1	3	4	6	7	9	10	12	13
12	3·464	3·479	3·493	3·507	3·521	3·536	3·550	3·564	3·578	3·592	1	3	4	6	7	8	10	11	13
13	3·606	3·619	3·633	3·647	3·661	3·674	3·688	3·701	3·715	3·728	1	3	4	5	7	8	10	11	12
14	3·742	3·755	3·768	3·782	3·795	3·808	3·821	3·834	3·847	3·860	1	3	4	5	7	8	9	11	12
15	3·873	3·886	3·899	3·912	3·924	3·937	3·950	3·962	3·975	3·988	1	3	4	5	6	8	9	10	11
16	4·000	4·012	4·025	4·037	4·050	4·062	4·074	4·087	4·099	4·111	1	2	4	5	6	7	9	10	11
17	4·123	4·135	4·147	4·159	4·171	4·183	4·195	4·207	4·219	4·231	1	2	4	5	6	7	8	10	11
18	4·243	4·254	4·266	4·278	4·290	4·301	4·313	4·324	4·336	4·347	1	2	3	5	6	7	8	9	10
19	4·359	4·370	4·382	4·393	4·405	4·416	4·427	4·438	4·450	4·461	1	2	3	5	6	7	8	9	10
20	4·472	4·483	4·494	4·506	4·517	4·528	4·539	4·550	4·561	4·572	1	2	3	4	6	7	8	9	10
21	4·583	4·594	4·604	4·615	4·626	4·637	4·648	4·658	4·669	4·680	1	2	3	4	5	6	8	9	10
22	4·690	4·701	4·712	4·722	4·733	4·743	4·754	4·765	4·775	4·785	1	2	3	4	5	6	7	8	9
23	4·796	4·806	4·817	4·827	4·837	4·848	4·858	4·868	4·879	4·889	1	2	3	4	5	6	7	8	9
24	4·899	4·909	4·919	4·930	4·940	4·950	4·960	4·970	4·980	4·990	1	2	3	4	5	6	7	8	9
25	5·000	5·010	5·020	5·030	5·040	5·050	5·060	5·070	5·079	5·089	1	2	3	4	5	6	7	8	9
26	5·099	5·109	5·119	5·128	5·138	5·148	5·158	5·167	5·177	5·187	1	2	3	4	5	6	7	8	9
27	5·196	5·206	5·215	5·225	5·235	5·244	5·254	5·263	5·273	5·282	1	2	3	4	5	6	7	8	9
28	5·292	5·301	5·310	5·320	5·329	5·339	5·348	5·357	5·367	5·376	1	2	3	4	5	6	7	7	8
29	5·385	5·394	5·404	5·413	5·422	5·431	5·441	5·450	5·459	5·468	1	2	3	4	5	5	6	7	8
30	5·477	5·486	5·495	5·505	5·514	5·523	5·532	5·541	5·550	5·559	1	2	3	4	4	5	6	7	8
31	5·568	5·577	5·586	5·595	5·604	5·612	5·621	5·630	5·639	5·648	1	2	3	3	4	5	6	7	8
32	5·657	5·666	5·675	5·683	5·692	5·701	5·710	5·718	5·727	5·736	1	2	3	3	4	5	6	7	8
33	5·745	5·753	5·762	5·771	5·779	5·788	5·797	5·805	5·814	5·822	1	2	3	3	4	5	6	7	8
34	5·831	5·840	5·848	5·857	5·865	5·874	5·882	5·891	5·899	5·908	1	2	3	3	4	5	6	7	8
35	5·916	5·925	5·933	5·941	5·950	5·958	5·967	5·975	5·983	5·992	1	2	2	3	4	5	6	7	8
36	6·000	6·008	6·017	6·025	6·033	6·042	6·050	6·058	6·066	6·075	1	2	2	3	4	5	6	7	7
37	6·083	6·091	6·099	6·107	6·116	6·124	6·132	6·140	6·148	6·156	1	2	2	3	4	5	6	7	7
38	6·164	6·173	6·181	6·189	6·197	6·205	6·213	6·221	6·229	6·237	1	2	2	3	4	5	6	6	7
39	6·245	6·253	6·261	6·269	6·277	6·285	6·293	6·301	6·309	6·317	1	2	2	3	4	5	6	6	7
40	6·325	6·332	6·340	6·348	6·356	6·364	6·372	6·380	6·387	6·395	1	2	2	3	4	5	6	6	7
41	6·403	6·411	6·419	6·427	6·434	6·442	6·450	6·458	6·465	6·473	1	2	2	3	4	5	5	6	7
42	6·481	6·488	6·496	6·504	6·512	6·519	6·527	6·535	6·542	6·550	1	2	2	3	4	5	5	6	7
43	6·557	6·565	6·573	6·580	6·588	6·595	6·603	6·611	6·618	6·626	1	2	2	3	4	5	5	6	7
44	6·633	6·641	6·648	6·656	6·663	6·671	6·678	6·686	6·693	6·701	1	2	2	3	4	5	5	6	7
45	6·708	6·716	6·723	6·731	6·738	6·745	6·753	6·760	6·768	6·775	1	1	2	3	4	4	5	6	7
46	6·782	6·790	6·797	6·804	6·812	6·819	6·826	6·834	6·841	6·848	1	1	2	3	4	4	5	6	7
47	6·856	6·863	6·870	6·878	6·885	6·892	6·899	6·907	6·914	6·921	1	1	2	3	4	4	5	6	7
48	6·928	6·935	6·943	6·950	6·957	6·964	6·971	6·979	6·986	6·993	1	1	2	3	4	4	5	6	6
49	7·000	7·007	7·014	7·021	7·029	7·036	7·043	7·050	7·057	7·064	1	1	2	3	4	4	5	6	6
50	7·071	7·078	7·085	7·092	7·099	7·106	7·113	7·120	7·127	7·134	1	1	2	3	4	4	5	6	6
51	7·141	7·148	7·155	7·162	7·169	7·176	7·183	7·190	7·197	7·204	1	1	2	3	4	4	5	6	6
52	7·211	7·218	7·225	7·232	7·239	7·246	7·253	7·259	7·266	7·273	1	1	2	3	3	4	5	6	6
53	7·280	7·287	7·294	7·301	7·308	7·314	7·321	7·328	7·335	7·342	1	1	2	3	3	4	5	5	6
54	7·349	7·355	7·362	7·369	7·376	7·382	7·389	7·396	7·403	7·410	1	1	2	3	3	4	5	5	6

x	0	1	2	3	4	5	6	7	8	9	Differences 1	2	3	4	5	6	7	8	9
55	7·416	7·423	7·430	7·436	7·443	7·450	7·457	7·463	7·470	7·477	1	1	2	3	3	4	5	5	6
56	7·483	7·490	7·497	7·503	7·510	7·517	7·523	7·530	7·537	7·543	1	1	2	3	3	4	5	5	6
57	7·550	7·556	7·563	7·570	7·576	7·583	7·589	7·596	7·603	7·609	1	1	2	3	3	4	5	5	6
58	7·616	7·622	7·629	7·635	7·642	7·649	7·655	7·662	7·668	7·675	1	1	2	3	3	4	5	5	6
59	7·681	7·688	7·694	7·701	7·707	7·714	7·720	7·727	7·733	7·740	1	1	2	3	3	4	4	5	6
60	7·746	7·752	7·759	7·765	7·772	7·778	7·785	7·791	7·797	7·804	1	1	2	3	3	4	4	5	6
61	7·810	7·817	7·823	7·829	7·836	7·842	7·849	7·855	7·861	7·868	1	1	2	3	3	4	4	5	6
62	7·874	7·880	7·887	7·893	7·899	7·906	7·912	7·918	7·925	7·931	1	1	2	3	3	4	4	5	6
63	7·937	7·944	7·950	7·956	7·962	7·969	7·975	7·981	7·987	7·994	1	1	2	3	3	4	4	5	6
64	8·000	8·006	8·012	8·019	8·025	8·031	8·037	8·044	8·050	8·056	1	1	2	2	3	4	4	5	6
65	8·062	8·068	8·075	8·081	8·087	8·093	8·099	8·106	8·112	8·118	1	1	2	2	3	4	4	5	6
66	8·124	8·130	8·136	8·142	8·149	8·155	8·161	8·167	8·173	8·179	1	1	2	2	3	4	4	5	5
67	8·185	8·191	8·198	8·204	8·210	8·216	8·222	8·228	8·234	8·240	1	1	2	2	3	4	4	5	5
68	8·246	8·252	8·258	8·264	8·270	8·276	8·283	8·289	8·295	8·301	1	1	2	2	3	4	4	5	5
69	8·307	8·313	8·319	8·325	8·331	8·337	8·343	8·349	8·355	8·361	1	1	2	2	3	4	4	5	5
70	8·367	8·373	8·379	8·385	8·390	8·396	8·402	8·408	8·414	8·420	1	1	2	2	3	4	4	5	5
71	8·426	8·432	8·438	8·444	8·450	8·456	8·462	8·468	8·473	8·479	1	1	2	2	3	4	4	5	5
72	8·485	8·491	8·497	8·503	8·509	8·515	8·521	8·526	8·532	8·538	1	1	2	2	3	3	4	5	5
73	8·544	8·550	8·556	8·562	8·567	8·573	8·579	8·585	8·591	8·597	1	1	2	2	3	3	4	5	5
74	8·602	8·608	8·614	8·620	8·626	8·631	8·637	8·643	8·649	8·654	1	1	2	2	3	3	4	5	5
75	8·660	8·666	8·672	8·678	8·683	8·689	8·695	8·701	8·706	8·712	1	1	2	2	3	3	4	5	5
76	8·718	8·724	8·729	8·735	8·741	8·746	8·752	8·758	8·764	8·769	1	1	2	2	3	3	4	5	5
77	8·775	8·781	8·786	8·792	8·798	8·803	8·809	8·815	8·820	8·826	1	1	2	2	3	3	4	4	5
78	8·832	8·837	8·843	8·849	8·854	8·860	8·866	8·871	8·877	8·883	1	1	2	2	3	3	4	4	5
79	8·888	8·894	8·899	8·905	8·911	8·916	8·922	8·927	8·933	8·939	1	1	2	2	3	3	4	4	5
80	8·944	8·950	8·955	8·961	8·967	8·972	8·978	8·983	8·989	8·994	1	1	2	2	3	3	4	4	5
81	9·000	9·006	9·011	9·017	9·022	9·028	9·033	9·039	9·044	9·050	1	1	2	2	3	3	4	4	5
82	9·055	9·061	9·066	9·072	9·077	9·083	9·088	9·094	9·099	9·105	1	1	2	2	3	3	4	4	5
83	9·110	9·116	9·121	9·127	9·132	9·138	9·143	9·149	9·154	9·160	1	1	2	2	3	3	4	4	5
84	9·165	9·171	9·176	9·182	9·187	9·192	9·198	9·203	9·209	9·214	1	1	2	2	3	3	4	4	5
85	9·220	9·225	9·230	9·236	9·241	9·247	9·252	9·257	9·263	9·268	1	1	2	2	3	3	4	4	5
86	9·274	9·279	9·284	9·290	9·295	9·301	9·306	9·311	9·317	9·322	1	1	2	2	3	3	4	4	5
87	9·327	9·333	9·338	9·343	9·349	9·354	9·359	9·365	9·370	9·375	1	1	2	2	3	3	4	4	5
88	9·381	9·386	9·391	9·397	9·402	9·407	9·413	9·418	9·423	9·429	1	1	2	2	3	3	4	4	5
89	9·434	9·439	9·445	9·450	9·455	9·460	9·466	9·471	9·476	9·482	1	1	2	2	3	3	4	4	5
90	9·487	9·492	9·497	9·503	9·508	9·513	9·518	9·524	9·529	9·534	1	1	2	2	3	3	4	4	5
91	9·539	9·545	9·550	9·555	9·560	9·566	9·571	9·576	9·581	9·586	1	1	2	2	3	3	4	4	5
92	9·592	9·597	9·602	9·607	9·613	9·618	9·623	9·628	9·633	9·638	1	1	2	2	3	3	4	4	5
93	9·644	9·649	9·654	9·659	9·664	9·670	9·675	9·680	9·685	9·690	1	1	2	2	3	3	4	4	5
94	9·695	9·701	9·706	9·711	9·716	9·721	9·726	9·731	9·737	9·742	1	1	2	2	3	3	4	4	5
95	9·747	9·752	9·757	9·762	9·767	9·772	9·778	9·783	9·788	9·793	1	1	2	2	3	3	4	4	5
96	9·798	9·803	9·808	9·813	9·818	9·823	9·829	9·834	9·839	9·844	1	1	2	2	3	3	4	4	5
97	9·849	9·854	9·859	9·864	9·869	9·874	9·879	9·884	9·889	9·894	1	1	1	2	3	3	4	4	5
98	9·900	9·905	9·910	9·915	9·920	9·925	9·930	9·935	9·940	9·945	0	1	1	2	2	3	3	4	4
99	9·950	9·955	9·960	9·965	9·970	9·975	9·980	9·985	9·990	9·995	0	1	1	2	2	3	3	4	4

1.1 $\frac{37}{56}$
1.2 $^2/_3$
1.3 $^4/_9$
1.4 47
1.5 $1^9/_{10}$
1.6 $^5/_{36}$
1.7 $^2/_3$
1.8 $1^6/_7$
1.9 $^1/_{40}$
1.10 $7^7/_{30}$
1.11 $3^2/_{39}$
1.12 $2^1/_2$
1.13 $^{17}/_{15}$
1.14 $-^{19}/_5$
1.15 Total profit ₦ 28000
Ade ₦ 9333.3 Tayo ₦ 16333.3
Uche ₦ 2333.3
1.16 $^2/_5$
1.16b $^{11}/_{60}$

1.17 II and III only
1.18 0.0059
1.19 0.17
1.20 3.80
1.21 0.0100
1.22 0.00005
1.23 5.10
1.24 0.98%
1.25 $3^1/_8$%
1.26 $5^5/_9$%
1.27 2.6%

2.1 A 36yrs B 18yrs C 9yrs
2.2 8:12:21
2.3 2:3:4
2.4 ₦ 27,500
2.5 ₦384.62, ₦ 961.54, ₦1153.84
2.6 $\frac{17}{1000}$
2.7 ₦ 4.00
2.8 ₦3,200 and Ade ₦ 900, Chi ₦1,100
2.9 434
2.9b ₦1,600
2.10 Le1,800 at 7% and Le1,200 at 8%
2.11 2730 – 2000 = ₵730
2.12 £3,634.2
2.13 (a) GH₵ 560, (b) GH₵ 7.5
2.14 ₦3,690.00 to 3 s.f
2.15 i. ₦ 458,750 ii. ₦ 5,734.375
iii. ₦ 65,098.96
2.16 GH₵ 6,352.94
2.17 ₦500,000.00
2.18 ₦ 450.00
2.19 30.4%
2.20 10% profit
2.21 10 plates
2.22 gain : 25%
2.23 2%
2.24 GH₵ 38 (14%)
2.25 SP = ₦ 3,280.00
2.26 ₦3,200.00
2.27 i. D4,000 ii. D4,940 iii. D940.00
2.28 ₦2,500.00

2.29 4%
2.30 2yrs
2.31 ₦2138.40
2.32 $7^1/_2$%
2.33 $16^2/_3$ %
2.34 ₦ 189.00
2.35 2%
2.36 ₦ 225.00
2.37 ₦ 420.00
2.38 10%

2.39 ₦ 420.00
2.40 ₦10,123.2
2.41 ₦ 56.19

3.1 $5^2/_3$
3.2 x = 1
3.3 3
3.4 – 4
3.5 x = 8
3.6 – 5
3.7 $15^1/_2$
3.8 5/24
3.9 5 = x
3.10 x = 7
3.11 5
3.12 x = 12

3.13 y + 1
3.14 –10
3.15 –14
3.16 5
3.17 (a + c)(b – d)
3.18 4xy(ab – ac + d)
3.19 (3a + 4b)(2x – 3y)
3.20 (x – 3y)(x – 2)
3.21 (n + q)
3.22 (x + b)
3.23 3(7x – y)(7x + y)
3.24 3(7x – y)(7x + y)
3.25 (5a – 3b)(5a + 3b)
3.26 1133
3.27 (3r + 4s)
3.27b 7x(x + 2)
3.28 (n – 3)(n + 27)
3.29 (1 – x)
3.30 (5 – x)(7 + x)
3.31 (2a – 3b)(a + 5b)
3.32 – 6
3.33 $24a^2x^3y^3$
3.34 $^1/_2(a + b + c)$
3.35 –2(2x – y)
3.36 $\frac{x-4}{2x-3}$
3.37 6(3k –1)
3.38 a + 2b / a – 3b
3.39 –m–n
3.40 $\frac{4x-13}{(x+2)(2x-3)}$
3.41 $\frac{x-6}{12}$
3.42 $-\frac{4x}{4-x^2}$
3.43 $\frac{2-3x}{4-x^2}$
3.44 $\frac{1}{x+1}$
3.45 $\frac{4y-x}{y}$
3.46 $\frac{-(x+4)}{x^2-3x+2}$
3.47 $\frac{4-x}{(x-1)(x+2)}$
3.48 $\frac{3x+2}{6}$
3.49 $\frac{5a}{6}$
3.50 $^8/_{125}$
3.51 0
3.51b 1
3.52 $\frac{12x+5}{5x-12}$

3.53 x = –3
3.54 x = $^3/_2$ or – $^3/_2$
3.55 x = $^4/_3$ or –3
3.56 x = 1 or –1
3.57 x = 2 or 7

3.58 $y = \frac{mn}{x - z^2 m}$
3.59 **(i)** $n = \sqrt{5m\left(\frac{m}{2} - p\right)}$ **ii.** ≈ 26.8 to 3sf
3.60 $x = \frac{\sqrt{9y^2N^2 - 4m}}{2}$
3.61 $u = \sqrt{v^2 - \frac{2Eg}{m}}$
3.62 $k = \frac{H(T^2-1)}{T^2 - T}$
3.63 $r = 100\left(\sqrt[t]{\frac{v}{p}} - 1\right)$
3.64 $a = \frac{2A - bh}{h}$

3.65 $p = \dfrac{q(hr-1)}{ht-1}$

3.66 $a = \dfrac{bx}{x-y}$

3.67 $p = \dfrac{2qr - rs}{6}$

3.68 $u = \dfrac{2s-r}{s-2v}$

3.69 $v = \dfrac{fu}{u-f}$

3.70 $m = \dfrac{dhp}{t-dh}$

3.71 $n = \dfrac{m}{mt-1}$

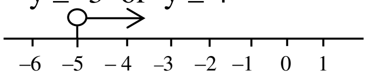
4.1 4
4.2 10
4.3 30
4.4 7
4.5 $x = 3, y = 4$
4.6 $x = 8, y = -3$
4.7 $y = -4$
4.8 $x = 5, y = -1$
4.9 $x = 1, y = -1$
4.10 $x = 5, y = -3$
4.11 2^2
4.12 4
4.13 4
4.14 $g = -1$

4.15 $x^2 - 2x - 15 = 0$
4.16 $x^2 - 8x + 15 = 0$
4.17 $2x^2 - 5x + 2 = 0$
4.18 $6x^2 + x - 2 = 0$
4.19 $4x^2 - 4x - 3 = 0$
4.20 $2x^2 + 3x - 14 = 0$
4.21 (-1, 3)
4.22 –1
4.23 1
4.24 5 or -2
4.25 $x = 1^2/_3$ or $x = -^1/_2$
4.26 $-3,\ ^9/_2$
4.27 $-9,\ ^2/_3$
4.28 2 or -5
4.29 $-^3/_2$
4.30 $1,\ -^1/_5$
4.31 x = 0 or 2
4.32 y = 0 or 9
4.33 0
4.34 16
4.35 p = -6 and k = 36
4.36 400
4.37 $2 \pm \dfrac{\sqrt{6}}{2}$
4.38 $x = \frac{3}{2}$ or 5
4.39 –3 and –1

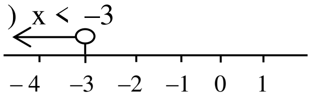
4.40 7
4.41 $x = 1$ or $^2/_3$
4.42 $x = ^3/_2,\ y = -^3/_2$

4.43 $x \geq -11$
4.44 $x > \dfrac{9}{17}$
4.45 $x \leq -47$
4.46 $x < 2^1/_4$
4.47 $x < -^1/_2$
4.48 $x \geq 1^1/_2$
4.49 $x \geq -12$
4.50 $x \leq ^{11}/_7$

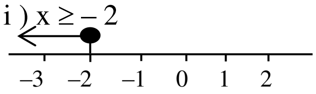
4.51 $-3 \leq x < 4$
4.52 $-3 \leq x \leq 4$
4.53 $x \geq 3$ or $x \geq -4$
4.54 $y \leq -3$ or $y \leq 4$

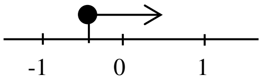
4.55 

4.56 (i) $x < -3$

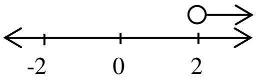
(ii) 

4.57 (i) $x \geq -2$

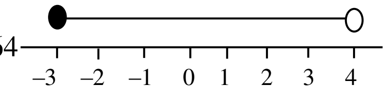
(ii) 

4.58 

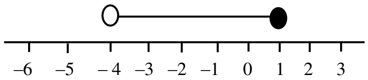
4.59 $x < 3$

4.60 

4.61 $-4 < x \leq 3$
4.62 $-2 < x \leq 3$
4.63 $-5 < x \leq 2$

4.64 

4.65 $-3 < x \leq 5$

4.66 

4.67 $-5 \leq x < 1$

5.1 7^{-2}
5.2 $3^{-\frac{25}{6}}$
5.3 4
5.4 $2^1/_2$
5.5 $^1/_3$
5.6 76
5.7 $^{10}/_9$
5.8 $8a^{-2}$
5.9 $^1/_9$
5.10 1

5.11 $3^{-20x+12}$
5.12 $^3/_2$
5.13 $2^1/_2$
5.14 $x = ^5/_2$ i.e 2.5
5.15 $x = ^5/_4$
5.16 n = 3
5.17 x = 1
5.18 x = –2
5.19 x = 2
5.20 x = –1
5.21 $2^{\frac{1}{3}}$
5.22 **a** x = 2 and y = 1
5.22 b $x = 2,\ y = -^1/_2$
5.23 x = 1
5.24 2, 1
5.25 x = –1 or 1
5.26 x = 1
5.27 $x = ^1/_5$
5.28 $x = ^5/_9$

5.29 $^3/_4$
5.30 $1^1/_2$
5.31 0 ($\log_{10}1 = 0$)
5.32 $^1/_2$
5.33 9
5.34 $3^3/_{10}$ ($^{33}/_{10}$)
5.35 $-^1/_2$ ($-^2/_4$)
5.36 $3^{\frac{1}{3}}$
5.37 $-1^1/_2$ ($-\frac{3}{2}$)
5.38 1.5441
5.39 p = 8
5.40 12
5.41 $^{10}/_8 = Q$ i.e $^5/_4 = 1^1/_4$
5.42 $x = 3^3$ i.e 27
5.43 2 = x
5.44 $^6/_{25}$
5.45 $\log y = 3\log 2 + \frac{1}{2}\log m$
5.46 $1^5/_9$
5.47 y = –1 , x = –68 **or** y = 4, x = 12
5.48 x = 16 , y = 4
5.49 $\pm\sqrt{10}$
5.50 $m = \frac{1}{4}$ or 32
5.51 n = -1 or -5

6.1 0.06425
6.2 5.02×10^{-4}
6.3 1.6256×10^{-2}
6.3b 3.18×10^9
6.4 6.02×10^{-9}
6.5 5.4×10^{-3}
6.6 2.23×10^3
6.7 9.687×10^2
6.8 1.055×10^2
6.9 9.778×10^{-2}
6.10 1.368×10^2

6.11 10^6
6.12 8.4×10^{-3}
6.13 7×10^4
6.14 740.97
6.15 17.57
6.16 0.0369 to 3sf
6.17 121.0 to 3sf

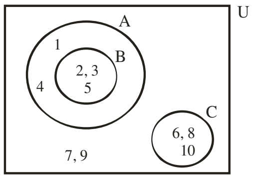
7.1 $35_{ten} = 100011_{ten}$
7.2 $37_{ten} = 100101_{two}$
7.3 1111_{two}
7.4 0.275_8
7.5 11201.100_{three}
7.6 0.011_{two}
7.7 2100.12_{four}
7.8 $20^7/_{9\ ten}$ or 29.78_{ten}
7.9 107_8
7.10 7
7.10b
7.10c
7.11 seven
7.12 2
7.13 101101_{two}
7.14 n = 6
7.15 x = 6
7.16 x = 8
7.17 y = 1
7.18 4 = y and x = 6
7.19 7 = y and x = 6
7.20 200
7.21
7.22 108
7.23 141_{five}
7.24 34_{five}, 21_{six}, 22_{three}

7.25 $1 + 3\sqrt{6}$
7.26 $2\sqrt{3}$
7.27 $\frac{5\sqrt{3}}{3}$
7.28 $\sqrt{3} - \sqrt{2}$
7.29 $17 - 12\sqrt{2}$
7.30 $\frac{1}{3}\sqrt{3}$
7.31 $10\sqrt{2}$
7.32 2
7.33 $1^1/_3$

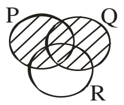
8.1 13
8.2 9
8.3 6
8.4 y = 9
8.5 4.0
8.6 13
8.7 7
8.8 mode
8.9 III only
8.10 ii. 5 – 2 = 3
8.11
8.12 3
8.13 1.2
8.14 1.5
8.15 1.1
8.16 1.41
8.17 2.76
8.18 1.4
8.19 i. ₦12 ii.₦14, ₦15, 3.55
8.20 43
8.21 1
8.22 i. 5.47 ii. 5 iii. 5
8.23 ***54*** is 7 ***55*** is 5 ***56*** is 7
8.24 9

8.25 ***20*** is 4.2 ***21*** is 3 ***22*** is 6
8.26 28.75, 29.25, 26.75, 22.75
8.27 28, 25

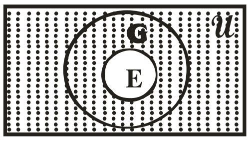
9.0 { Multiples of three }
9.1 Their intersection is an empty set
9.2 {2, 5, 7, 8, 10, 11, 13, 14}
9.3 **a.** $A \cap \phi$ i.e ϕ **b.** {2, 3, 4, 6, 7, 8, 9}
9.4 {0, 6}
9.5 P = {–5, –3, 2, 4}
9.6 {s, p, m, e}
9.7 **i.** {3, 8} **ii.** {1, 2, 4, 5, 6, 7, 9, 10}
9.8 **i.** {10, 12, 14} **ii.** $n(X^1 \cap Y) = 1$
9.9 {3, 5, 7}
9.10 Q
9.11 {4, 6}
9.12



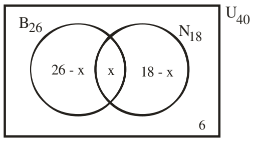
ii. $A \cap C = \{\ \}$ **iii.** $A \cap B^1 = (1, 4\}$
9.13



9.14 $A \cap (B^1 \cap C)$
9.15 $R^1 \cup S$
9.16

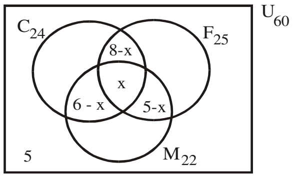


9.17 $(A \cup B)^1$
9.18 $(A \cap C) \cap B^1$
9.19 $P^1 \cap Q \cap R^1$
9.20 36 boys
9.21



i. x = 10 **ii.** 16 **iii.** 8

9.22 x = 5
9.23 1
9.24 12
9.25 i. 4 ii. 10 iii. 13 iv. 20
9.26 i. 18 ii. 17 iii. 101
9.27 3
9.28 15
9.29 7
9.30 i. 8 ii. 18
9.31



b.i x = 3, b ii. 13 b. iii 15 b. iv 14

9.32

10.1 1 – x
10.2 $^3/_{36}$ i.e $^1/_{12}$
10.3 i. $^6/_{36}$ i.e $^1/_6$ ii. $^{26}/_{36}$ i.e $^{13}/_{18}$
10.4 bi. $^5/_{36}$ bii. $^5/_6$ biii. $^{35}/_{36}$
10.5 $^7/_{20}$
10.6 $^1/_3$
10.7 $^4/_{13}$
10.8 i. $\frac{480}{973}$ ii. $\frac{495}{973}$
10.9 $^9/_{10}$
10.10 $^1/_{20}$
10.11 b $^1/_{12}$ c $^5/_9$
10.12 58 is $^{13}/_{15}$ 59 is $^{11}/_{12}$ 60 is $^2/_{15}$
10.13 0
10.14 i. $^{11}/_{36}$ ii. $^5/_9$ iii. $^1/_6$
10.15 $^3/_5$
10.16 $^4/_{25}$
10.17 $^2/_5$
10.18 $^1/_3$
10.19 $^1/_6$
10.20 $^5/_{36}$

11.1 p = $^{29}/_2$ and q = 18
11.2 x = 1, y = 10
11.3 x = 8 and y = 13
11.4 41
11.5 3
11.6 $U_n = 18 - 7n$
11.7 2(3n – 1)
11.8 3n + 2
11.9 73
11.10 –44

11.11 i. $a = -5$ and $d = 16$ ii. 331
11.12 2
11.13 0
11.14 (i) 3 (ii) 0
(iii) $n^2 - 9n - 90 = 0$, $n = 15$
11.15 $n(3n + 2)$
11.16 $\frac{1}{2}n(n + 1)$
11.17 6144
11.18 45
11.19 $81\sqrt{2}$
11.20 $\sqrt{2}$
11.21 ± 2
11.22 $^1/_2$
11.23 3
11.24 $r = 3$, $a = {}^4/_9$, $S_5 = {}^{484}/_9$
11.25 $r = {}^1/_3$, $S_5 = {}^{242}/_{81}$
11.26 $r = 2$, $T_5 = {}^{640}/_7$
11.27 8
11.28 128
11.29 10
11.30 $-{}^1/_4$
11.31 **i.** 268 **ii a.** $n = 7$ **ii b.** $10 = n$
11.32 50
11.33 127
11.34 $1 + (-2)^n$
11.35 $5n - 2$

12.1 R = 2.84 ohms
12.2 $p = 12$
12.3 $B = 4$
12.4 216
12.5 $22^1/_2$km
12.6 80
12.7 0.05v
12.8 $p = 2$
12.9 $^1/_6$
12.10 $Q = 150$
12.11 $Q = 5$
12.12 $z = 4$
12.13 18.9
12.14 (i) $A = \dfrac{4B}{9C^2}$ (ii) $A = 64$
12.15 $2^1/_2$ kg/cm^3
12.16 $\dfrac{C_1}{C_2} = \dfrac{2}{9}$
12.17 $E = \dfrac{4R}{d^2}$ a. 36
b. 4.899 c. 9.27%
12.18 $P = aQ + \dfrac{b}{\sqrt{Q}}$
12.19 x is partly constant and partly varies as y
12.20 (i) $y = \dfrac{48}{5} + \dfrac{8}{5}x^2$
(ii) $y = 35.2$
12.21 $y = 4x + 3$
12.22 **i.** $w = 26 - \dfrac{32}{t^2}$ **ii.** $t = {}^2/_3$
12.23 **i.** $c = 830 + \dfrac{1}{5}n$ **ii.** GH₵ 930
12.24 Le1600.00
12.25 45km/h,

13.1 $\dfrac{\sqrt{2}-\sqrt{6}}{4}$
13.2 $\dfrac{2}{\sqrt{2}}$
13.3 1
13.4 5
13.5 $^1/_2$
13.6 $-\tan 60^0$
13.7 3
13.8 210^0 and 330^0
13.9 $-\dfrac{\sqrt{2}}{2}$
13.10 -1
13.11 150^0 and 210^0
13.12 ½
13.13 250^0, 290^0
13.14 $\theta = 7.5^0$
13.15 $x = 40^0$
13.16 $^{169}/_{25}$
13.17 $^7/_{25}$
13.18 $^9/_{41}$
13.19 $\dfrac{b^2}{a^2}$
13.20 $^{31}/_{20}$
13.21 $^{12}/_{13}$
13.22 $^1/_3$
13.23 11
13.24 $\dfrac{4\sqrt{13}}{13}$
13.25 $^{31}/_{12}$
13.26 $^{15}/_{17}$
13.27 13^0
13.28 $x = 3 \pm 2\sqrt{2}$
13.29 60^0 and 300^0
13.30 60^0
13.31 **i.** 1.3383 **ii.** 0.3839
13.32 $\dfrac{14\sqrt{3}}{3}$ m
13.33 x = 21.65m to 2 d.p
13.34 x = 12.2 cm
13.35 10m = x
13.36 $6\sqrt{3}$
13.37 12m
13.38 10.95cm
13.39 29^0
13.40 $10(3 + \sqrt{3})$cm

14.1 $5\sqrt{3}$ m
14.2 14.5m = x
14.3 53.13^0
14.4 $\theta = 36.87^0$
14.5 $\sqrt{55}$ m
14.6 48.3m to 1 d.p
14.7 ≈ 38.16m to 2 d.p
14.8 1000m
14.9 69.3m
14.10 y = 260m to 3 s.f
14.11 ≈ 373m
14.12 x = 7.15m to 2 d.p
14.13 ≈ 10^0 to the nearest degree
14.14 (alternate angles) 72^0
14.15 $21\sqrt{3}$ m
14.16 i. 150m ii.913m
14.17 38.9m
14.18 9.60m

15.1 60^0
15.2 30^0
15.3 100^0
15.4 270^0
15.5 70^0
15.6 120^0
15.7 $x = 49^0$, $y = 311^0$
15.8 42^0
15.9 60^0
15.10 17^0
15.11 216^0
15.12 70^0
15.13 33^0
15.14 64^0
15.15 $m = 50^0$, $n = 60^0$
15.16 70^0
15.17 122^0
15.18 70^0

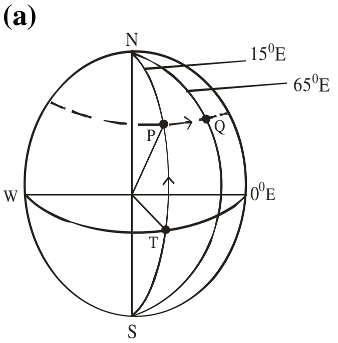
15.19 $^1/_3$
15.20 32.31km
15.21 3.78cm
15.22 11.59cm
15.23 140^0
15.24 230^0
15.25 220^0
15.26 230^0
15.27 158^0
15.28 25m
15.29 073.32^0
15.30 36.9^0
15.31 17km
15.32 Option B
15.33 25km
15.34 781.02km
15.35 13m

15.36 240^0
15.37 8.7km
15.38 184^0
15.39 19.08km
15.40 **(i)** PR = 276.80km
Total distance covered
= 200 + 250 + 276.80
≈ 727.8km to 3s.f
(ii) $74.26^0 \approx 74^0$ to the nearest degree

15.41 **i** XZ = 49.29km ≈ 49.3km to 1 d.p
(ii) $239.6^0 \approx 240^0$

15.42 13km
15.43 a i: 6.93km a ii : 10km
(b) 12.2km
15.44 (i) 32km (ii) 167^0

16.1 7000km
16.2 140^0
16.3 3200km
16.4 $\frac{\pi R}{2}\cos 15^0$
16.5 **(a)**



(b) i Thus latitude of P is 34^0N
(b) ii. ≈ 4632km to 4 s.f

16.6
(i) x^0 E ≈ 43^0 to nearest degree
(ii) θ ≈ 45^0 to the nearest degree

16.7 i. 6131km ii. 5587km

17.1 (a) Mean = 14
(b) M.D ≈ 4.67 to 2dp
(c) S.D = 5.26
17.2 Mean = 59.05
17.3 mean = 78.75 S.D ≈ 7.95
17.4 mean ≈ 1.2 m
17.5 Mode 51.86 ≈ 52kg

18.1 60^0
18.2 60
18.3 **20** ₦60,000 **21** ₦200,000
18.4 JSS 1 = 48^0 JS 2 = 56^0
JS 3 = 96^0
SS 1 = 64^0 SS 2 = 52^0 SS 3 = 44^0
(b) $\frac{8}{45}$

18.5 A = 100^0 B = 80^0 C = 50^0
D = 70^0 E = 60^0
18.6 Japan = 36^0 U.S.S.R. = 90^0
Canada = 62^0 U.K. = 74^0
USA = 54^0 China = 44^0

18.7 **4** 20 **5** $^{11}/_{20}$
18.8

18.9 b 10.5 + 3 = 13.5 c $^{7}/_{10}$
18.10

Interval	Tally	freq	class boundaries
21 – 25	~~IIII~~ II	7	20.5 – 25.5
26 – 30	~~IIII~~ ~~IIII~~ I	11	25.5 – 30.5
31 – 35	~~IIII~~ IIII	9	30.5 – 35.5
36 – 40	~~IIII~~ IIII	9	35.5 – 40.5
41 – 45	~~IIII~~ I	6	40 .5 – 45.5
46 – 50	~~IIII~~ III	8	45.5 – 50.5

(c) 25.5 + 3.2 = 28.7 year

18.11

Age (year)	class boundaries	frequency
10 – 14	9.5 – 14.5	7
15 – 19	14.5 – 19.5	18
20 – 24	19.5 – 24.5	25
25 – 29	24.5 – 29 .5	17
30 – 34	29.5 – 34.5	9
35 – 39	34.5 – 39.5	4

(b) ii 19.5 + 2.2 = 21.7
(c) $\frac{5}{16}$

18.12

Class int.	Tally	f	cf	C/boundaries
21 – 30	II	2	2	20.5 – 30.5
31 – 40	III	3	5	30.5 – 40.5
41 – 50	~~IIII~~ I	6	11	40.5 – 50.5
51 – 60	~~IIII~~ ~~IIII~~ I	11	22	50.5 – 60.5
61 – 70	~~IIII~~ IIII	9	31	60.5 – 70.5
71 – 80	~~IIII~~ ~~IIII~~	10	41	70.5 – 80.5
81 – 90	~~IIII~~	5	46	80.5 – 90.5
91 – 100	IIII	4	50	90.5 – 100.5

(b) (i) 64.5 (ii) 76.5 – 52.5 = 24
(c) Pr (> 60kg) = $\frac{28}{50} = \frac{14}{25}$

18.13

Height	Class boundaries	f	cf
45 – 49	44.5 – 49.5	10	10
50 – 54	49.5 – 54.5	36	46
55 – 59	54.5 – 59.5	64	110
60 – 64	59.5 – 64.5	52	162
65 – 69	64.5 – 69.5	28	190
70 – 74	69.5 – 74.5	10	200

(iii) 65.5 – 55 = 10.5

18.14

Marks	Class boundaries	f	cf
30 – 39	29.5 – 39.5	11	11
40 – 49	39.5 – 49.5	25	36
50 – 59	49.5 – 59.5	45	81
60 – 69	59.5 – 69.5	15	96
70 – 79	69.5 – 79.5	10	106
80 – 89	79.5 – 89.5	14	120

(b) (i) 54.5 (ii) 65.5
(c) Pr(50 and 60) = $\frac{96-36}{120} = \frac{60}{120} = \frac{1}{2}$

18.15 linear

18.16
18.17 $y = ax^2$
18.18 $x^2 - 3x - 1 = 0$
18.19 (a) x = 1 (b) Slope = 8
(c) x = – 0.6 or 2.6 (d) 0.3 < x < 1.7
Hint : $2x^2 - 4x + 1 < 0$ is $2x^2 - 4x + 3 < 2$

18.20
Intermediates: $^1/_2$, $^3/_2$, $^5/_2$, $^7/_2$
Hint : 0, **0.5**, **1.5**, 2, **2.5**, 3, **3.5**, 4
(a) 72.2cm
(b) Ball reaches ground
when y = 0, t = 4s
(c) At t = 3, vel = 36m/s

18.21
(a) 0 and 4
(b) x = 2 and y = 12
(c) x = 2
(d) – 1 and 3.3
18.22
(d)(i) – 1.55 and 2.55
(ii) x = 1,
(iii) $0 \le x \le 2$

18.23
(a) $3x^2 - 5x + 4 = 0$ has no roots
(b) x = – 0.3 or 2
(c) – 1.9 < x<3.5
Hint $3x^2 - 5x < 20$ is $3x^2 - 5x + 4 < 24$
18.24 (c) i – 0.2 and 1.85 ii. y = 0.8
18.25
18.25 **b**

18.26 y = 4 cos x
18.27 (c) 15^0 and 165^0
18.28
18.29
18.30 y = 2cos x
18.31
18.32 $y + x \le 3$, $x \ge 0$, $y \ge 0$
18.33

19.1 The area of P is equal to the area of Q
19.2 Three angles
19.3 ΔTYS (AAS)
19.4 12cm
19.5 8cm
19.6 13.33cm
19.7 4.5cm
19.8 9cm
19.9 4.0cm
19.10 4cm
19.11
19.12 Corresponding angles are equal
19.13 4.6cm
19.14 20 = n
19.15 15 = n
19.16 6 = n
19.17 19

19.18 Octagon
19.19 90^0
19.20 130^0
19.21 150^0
19.22 128^0
19.23 124^0
19.24
19.25 60^0
19.26 35 Ans 58^0 36 Ans 130^0
19.27 25^0
19.28 II and III only
19.29 60^0
19.30 I & II only ($\angle$s in same segment)
19.31 90^0
19.32 $m = \frac{n}{2}$
19.33 19^0
19.34 45^0
19.35 50^0
19.36 120^0
19.37 65^0
19.38 100^0
19.39 130
19.40 90^0
19.41 112^0
19.42 38^0
19.43 107^0
19.44 72^0
19.45 56^0
19.46 53^0
19.47 122^0
19.48 77^0
19.49 72^0
19.50 65^0
19.51 16cm
19.52 **35** ans 12.5, 7.7 **36** ans15^0

20.1 13cm
20.2 $233m^2$
20.3 6cm × 5cm
20.4 x = 1
20.4b L 100m and W 80m
20.5 $242.00cm^2$
20.6 2112cm
20.7 $122\ ^4/_7\ cm^2$
20.8 $16\sqrt{3}\ cm^2$
20.9 $54cm^2$
20.10 h = 35.42cm
20.10b **4cm**
20.11 ***17*** 3.35cm ***18*** $27cm^2$
20.11b 482.96cm *and* $2.24\times10^5\ cm^2$
20.12 $96cm^2$
20.13 i.Rhombus, ii. AB = 17cm, iii. $240cm^2$
20.14 5.0
20.15 16.13cm
20.16 44cm
20.17 πcm
20.18 36.7cm
20.19 108^0
20.20 45.71cm
20.21 36.9cm
20.22 61cm
20.23 101.33cm
20.24 61cm
20.25 $4.19cm^2$
20.26 $18\ ^6/_7\ cm^2$
20.27 25cm
20.28 $7\sqrt{3}$ cm
20.29 (a) 10.2848cm (b) 6.129cm
20.30 4.2cm
20.31 8cm
20.32 17.32 cm
20.33 7.36cm
20.34 1.2cm
20.35 4.0cm
20.36 **a** 9.78cm **b** $10.09cm^2$
20.36b 10cm
20.37 $154cm^2$
20.38 $34cm^2$
20.39 $346.5m^2$

20.40 $5425cm^2$
20.41 **i** = $30cm^2$ **ii** $\approx 103cm^2$ to 3s.f **iii** $\approx$ 20.4cm to 3s.f
20.42 (i) $\approx 58cm^2$ (ii) 30cm
20.43 8 : 1
20.44 (a) $135.3cm^2$ (b) 47.5cm
20.45 $707cm^2$

21.1 $2,816cm^3$
21.2 $629cm^3$
21.3 11 litres
21.4 $4400cm^3$
21.5 $1484.7cm^3$
21.6 (i) 4cm (ii) $62.9cm^3$
21.7 $643m^3$
21.8 $749\pi m^2$
21.9 7cm
21.10 3.5cm
21.10 b $55cm^3$
21.11 2cm
21.12 i. 4cm ii. h = 11.3cm vol $190cm^3$
21.13 $462cm^3$
21.14 i.13cm ii. $204.29cm^2$ iii. $314.29cm^3$
21.14b r = 3.5cm, h = 9.90cm Vol = $127cm^3$
21.15 $12\pi cm^3$
21.16 $38cm^3$
21.17 $2425.5cm^3$
21.18 7cm
21.19 3.50cm
21.20 $132cm^2$
21.21 $240cm^3$
21.22 $495cm^3$
21.23 7cm
21.24 $96cm^2$
21.25 $104cm^2$
21.26 1.68litres
21.26b depth = 0.95m, 7800litres would be needed
21.27 $53.08m^3$
21.28 $336cm^3$
21.29 a. *12cm* b. *67^0* c.*$192cm^3$*
21.30 $92cm^2$
21.31 a. $360cm^2$ b. $400cm^3$
21.32 $84cm^3$
21.33 i. 13.82cm ii. 67.13^0 iii. 276.4cm iv. 289.6cm
21.34 16cm
21.35 21.67m
21.36 8.44 litres
21.37 $522.67cm^2$
21.38 r = 7cm
21.39 $33\ \pi cm^2$
21.40 3.16m
21.41 3.32cm
21.42 680*litres*
21.43 61cm
21.44 76.3cm
21.45 4.9cm
21.46 3,011pipes
21.47 a. $1440cm^2$ b. $4667cm^3$

22.1
22.2
22.3
22.4 4.8cm and 9.2cm
22.5 PM = 4.3cm
22.6 PC = 7.6cm
22.7
22.8
22.9
22.10
22.11
22.12 4.3cm and 1.3cm
22.13
22.14
22.15

23.1
23.2
(b) i. $(6 \otimes 5) \oplus (3 \otimes 2) = 2 \oplus 6 = 1$
ii. n = 5
23.3 (i) $6 \otimes 2 \otimes 5 = (6 \otimes 2) \otimes 5 = 5 \otimes 5 = 4$
(ii) n = 5 (iii) n = 1

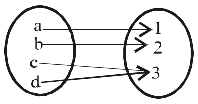
24.1 john is not older than me
24.2 option C
24.3 $p \wedge q$

25.1 $(-2, 2)$
25.2 $(0, 2)$
25.3 $^1/_4$
25.4 $^{-2}/_3$
25.5 ***24 ans*** $^3/_2$ ***25 ans*** $2\sqrt{13}$
25.6 ***13 Ans*** 2 ***14 Ans*** $y = 8$
25.7 $2x + 3y - 7 = 0$
25.8 $y = 2x + 5$
25.9 $4y = -3x - 7$
25.10 $y + 4x - 11 = 0$
25.11 $4x + y + 14 = 0$
25.12 $3y - x + 10 = 0$
25.13 $3x + 2y = 5$
25.14 $t = 9$

26.1 $\begin{pmatrix} -6 & 4 & -8 \\ -10 & 0 & -12 \\ -14 & -10 & 2 \end{pmatrix}$
26.2 $\begin{pmatrix} 9 & 5 \\ -16 & -10 \end{pmatrix}$
26.3 $\begin{pmatrix} 13 & 10 \\ 0 & 24 \end{pmatrix}$
26.4 $\begin{pmatrix} 9 & 13 \\ 5 & 8 \end{pmatrix}$
26.5 $T = \begin{pmatrix} 11 & 5 \\ 15 & 11 \end{pmatrix}$
26.6 $\begin{pmatrix} 14 & 8 \\ 7 & 7 \end{pmatrix}$
26.7 $\begin{pmatrix} 3 & 16 \\ 10 & 1 \end{pmatrix}$
26.8 $\begin{pmatrix} 3 & 11 \\ -11 & 4 \end{pmatrix}$
26.9 $\begin{pmatrix} 3 & -1 & 0 \\ 4 & -1 & 1 \\ -4 & 1 & 1 \end{pmatrix}$
26.10
26.11 $\begin{pmatrix} 3 & -5 \\ 6 & -4 \end{pmatrix}$
26.12 $(-2, 3)$
26.13 -5
26.14 zero
26.15 $x = 4, \frac{-3}{2}$
26.16 -3
26.17 3
26.18 22
26.19 -11
26.20 $\begin{pmatrix} -8 & -5 \\ 3 & 2 \end{pmatrix}$
26.21 $\begin{pmatrix} 2 & \frac{-3}{2} \\ -3 & \frac{5}{2} \end{pmatrix}$
26.22 $x = 1, y = {}^1/_3$
26.23 $x = -2, y = -4$
26.24
26.25 -102
26.26 7
26.27 -15

27.1
27.2 $\begin{pmatrix} 7 \\ -9 \end{pmatrix}$
27.3 $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$
27.4 $\frac{10\sqrt{17}}{17}$
27.5 $\frac{13\sqrt{170}}{170}$
27.6 $\frac{8}{\sqrt{65}}$
27.7 135^0
27.8 $|p-q| = \sqrt{101}$
27.9 21
27.10 $\frac{1}{5}(3i + 4j)$
27.11 $\frac{2i - j}{\sqrt{5}}$
27.12 $\frac{3}{5}i - \frac{4}{5}j$
27.13 I and II
27.14 1, II and III are true
27.15 $p \cdot q = |p||q|$ **only**

28.0



28.1 both 1 – 1 and onto 28.2 constant
28.3 {-1, 5, 13, 49}
28.4 {5, 2, 1}
28.5 {-3, -2, -1, 0, 1, 2, 3}
28.6 g only
28.7 a.i R, $x \neq -{}^3/_5$
(b) Not onto (c) $-2<x<{}^3/_2$
28.8
28.9
28.10
28.11 i. -3 ii $-6^1/_2$
28.12 i. 9 ii. $x^2 + 3x$ iii. $x = 1$ or 2
28.13 $\sin x^2$
28.14 -49
28.15 $2x^2-16x+29$
28.16 4
28.17 $\frac{1}{15}(5x+2)$
28.18 $-{}^5/_4$
28.19 $\frac{4x+11}{2x-5}$
28.20 $7 - x$
28.21 $\frac{1-x}{2x}$
28.22 $^1/_2$
28.23 $\frac{3-x}{2}$
28.24 ii $3x+1/2-5x$ iii $x = {}^2/_5$
28.25 $\sqrt[3]{x-5}$
28.26 i $2x/1-x$ ii $2/x^2-2$
iii f(x) largest domain in R
is $x \neq \pm1$, $g(x) \neq -2$

28.27 a bijective function
b i $x+3/4-2x$ b ii $5-x/2x-2$
b iii $^1/_{18}$ b iv $-{}^{17}/_2$

29.1 $x^2 + 2x + 1$
29.2 1
29.3 -10
29.4 $P = 4$ and $Q = -2$
29.5 $\frac{1}{2}$
29.6 $A = \frac{3}{2}$ $B = 2$ $C = -\frac{1}{2}$
29.7 $\frac{7}{x-1} + \frac{3}{x+2}$
29.8 $\frac{1}{x} + \frac{3}{x+2} - \frac{1}{x+1}$
29.9 $P = 3$, $Q = -2$
29.10 $P = {}^{41}/_{92}$ and $Q = {}^5/_{96}$
$P + Q = {}^{17}/_{64}$
29.11 $A = -1$, $B = 1$, $C = 3$
29.12 5

30.1 $-{}^3/_2x^{-5/2}$
30.2 $12x^2 - 4x + 1$
30.3 $2 + 2x$
30.4 $6x + 2$
30.5 $6x - \frac{2}{x^2}$
30.6 $\sec^2 x$
30.7 $-\sin x$
30.8 $3\sin\left(\frac{\pi}{2} - 3x\right)$
30.9 $(6x^2 + 3)\cos(2x^3 + 3x - 4)$
30.10 $6(2x + 2)^2$
30.11 $-3\sin 3x$
30.12 $2 + 2x$
30.13 $3(2x + 3)^{-2\frac{1}{2}}$
30.14 $20x(2x^2 + 3)^4$
30.15 $16x(2x^2 + 7)^3$
30.16 $\cos x - x\sin x$
30.17 $1 - 6x$
30.18 $6(x^2 + 1)$
30.19 $(2x + 5)(6x - 11)$
30.20 $(-2x^2 + 6x+10)/(x^2 + 5)^2$
30.21 $(6 - 2x^2)/(x^2 + 3)^2$

30.22 $\frac{2}{(1-x)^2}$
30.23 $\frac{2}{x^3} + \frac{4}{5}$
30.24 $\frac{1}{(x+1)^2}$
30.25 $\frac{2x^2-6x-8}{(1+x)^4}$
30.26 $-\frac{1}{(x^2-1)^{3/2}}$
30.27 $\frac{3\cos 3x+2\sin 3x}{e^{-2x}}$
30.28 $(4x^4 + 9x^2)/(2x^2 + 3)^{3/2}$
30.29 (1, 5)
30.30 (–2, – 6)
30.31 $^2/_5$
30.32 $-^1/_8$
30.33 1
30.34 8
30.35 3
30.36 $^1/_3$
30.37 –1.5
30.38 5
30.39 $^{41}/_{16}$
30.40 – 4
30.41 0
30.42 i. -24 ii.Min pt (5, -101)
Max pt (-1, 7)
30.43 (-2, 10)
30.44 (-2, 13)
30.45 ($^1/_2$, $^{11}/_4$)
30.46 (5, 75)
30.47 $\frac{-(2x+y)}{x}$
30.48 14
30.49 $-^4/_5$
30.50 $-\frac{(ax+hy)}{hx+hy}$
30.51 $^3/_2$ i.e $1^1/_2$
30.52 5
30.53 $0.88cm^2s^{-1}$
30.54 $6\pi cm^2/sec$
30.55 $0.2\pi cm^2/sec$
30.56 0.08mm/s

31.0 $\frac{x^2}{2}$ + sinx + c
31.1 $\frac{x^4}{4} + \frac{2x^3}{3}$ + c
31.2 $-\frac{1}{6}\cos^6 x$ + c
31.3 $^3/_2$ sinx + 2cosx + c
31.4 $Log_e 3$
31.5 $^7/_6 \ln(x^2 - 3)$
31.6 $^1/_8 \ln(x^4 + 1)$
31.7 8
31.8 $2y = 4x^3 - x^2 + 4x + 3$
31.9 $2x^2 - 3x + 3$
31.10 $y = x^2 + 7x - 18$
31.11 1
31.12 18
31.13 1.0
31.14 $4^2/_3$
31.15 $27^1/_3$ square units
31.16 27square units
31.17 11sq units
31.18 9
31.19 10
31.20 37/12 sq units
31.21
31.22
31.23